

Appendix E



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY.DOCKET NO./TITLE	REQUEST ID
14/324,990	07/07/2014	Netlist, Inc.	129980-5043US	67121

Acknowledgement of Loss of Entitlement to Entity Status Discount

The entity status change request below filed through Private PAIR on 07/06/2018 has been accepted.

CERTIFICATIONS:

Change of Entity Status:

✕ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be notification of loss of entitlement to small or micro entity status, as applicable.

This portion must be completed by the signatory or signatories making the entity status change in accordance with 37 CFR 1.4(d)(4).

Signature:	/Jamie J. Zheng/
Name:	Jamie J. Zheng
Registration Number:	51167



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/324,990	05/19/2015	9037809	NETL.045C2	1022

79141 759 4/29/2015
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA, Assignee (with 37 CFR 1.172 Interest);
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Customer No. 79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

14324990 07/07/2014 Hyun Lee NETL.045C2 1022

TITLE OF INVENTION:

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional SMALL \$480 \$0 \$0 \$480 04/23/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
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1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a **Customer Number is required.**

2. For printing on the patent front page, list

(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,

1 JAMIE J. ZHENG, Ph.D, Esq.

(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

☒ Issue Fee

☐ Publication Fee (No small entity discount permitted)

☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

☐ A check is enclosed.

☐ Payment by credit card. Form PTO-2038 is attached.

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 50-5963 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27.

☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /Jamie J. Zheng/

Date April 17, 2015

Typed or printed name Jamie J. Zheng

Registration No. 51,167

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14324990			
Filing Date:	07-Jul-2014			
Title of Invention:	Memory Module With Circuit Providing Load Isolation and Noise Reduction			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.045C2			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl Issue Fee	2501	1	480	480

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				480

Electronic Acknowledgement Receipt

EFS ID:	22101612
Application Number:	14324990
International Application Number:	
Confirmation Number:	1022
Title of Invention:	Memory Module With Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C2
Receipt Date:	18-APR-2015
Filing Date:	07-JUL-2014
Time Stamp:	01:10:08
Application Type:	Utility under 35 USC 111(a)

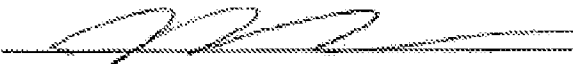
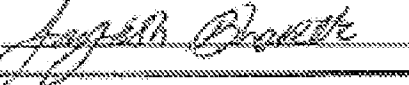
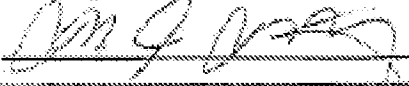
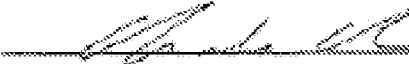
Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$480
RAM confirmation Number	6310
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	2015-04-17_Issue_Fee_Transmission_NETL045C2.pdf	74178	no	2
			d0348d21bcb707a7b5832f975e8c00503add09d9		
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30379	no	2
			b6544cf00b75740dba3aaa2ae7fa128108d0b441		
Warnings:					
Information:					
Total Files Size (in bytes):			104557		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Customer No. 79141
Attorney Docket No.: NETL045C2

DECLARATION FOR UTILITY OR DESIGN APPLICATION	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
DECLARATION	
As a below named inventor, I hereby declare that: This declaration is directed to: ☐ The attached application, or ☒ United States application or PCT international application number <u>14/324,990</u> filed on <u>July 7, 2014</u> .	
The above-identified application was made or authorized to be made by me.	
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.	
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.	
Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."	
First Inventor (Legal Name): <u>Hyun Lee</u>	
Signature: <u></u>	Date: <u>3-24-2015</u>
Second Inventor (Legal Name): <u>Jayesh R. Bhakta</u>	
Signature: <u></u>	Date: <u>3-24-2015</u>
Third Inventor (Legal Name): <u>Jeffery C. Solomon</u>	
Signature: _____	Date: _____
Fourth Inventor (Legal Name): <u>Mario Jesus Martinez</u>	
Signature: <u></u>	Date: <u>3-24-2015</u>
Fifth Inventor (Legal Name): <u>Chi She Chen</u>	
Signature: <u></u>	Date: <u>3-24-2015</u>

**DECLARATION FOR
UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

**CIRCUIT PROVIDING LOAD ISOLATION AND NOISE
REDUCTION**

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application
number 14/324,990 filed on July 7, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001
by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application
unless that person has reviewed and understands the contents of the application, including the claims, and is aware
of the duty to disclose to the Office all information known to the person to be material to patentability as defined in
§ 1.56."

First Inventor (Legal Name): Hyun Lee

Signature: _____

Date: _____

Second Inventor (Legal Name): Jayesh R. Bhakta

Signature: Jayesh Bhakta

Date: 3-24-2015

Third Inventor (Legal Name): Jeffery C. Solomon

Signature: Jeffery Solomon

Date: 04-06-2015

Fourth Inventor (Legal Name): Mario Jesus Martínez

Signature: Mario Jesus Martinez

Date: 3-24-2015

Fifth Inventor (Legal Name): Chi She Chen

Signature: Chi She Chen

Date: 3-24-2015

Electronic Acknowledgement Receipt

EFS ID:	22050500
Application Number:	14324990
International Application Number:	
Confirmation Number:	1022
Title of Invention:	Memory Module With Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C2
Receipt Date:	14-APR-2015
Filing Date:	07-JUL-2014
Time Stamp:	01:53:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Oath or Declaration filed	2015-03-24_Declaration_NETL045C2.pdf	567329 3bc3fbfa9565252cb049792c19bf747753650b4f	no	2

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):

567329

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 01/23/2015
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

BANSAL, GURTEJ

ART UNIT

PAPER NUMBER

2139

DATE MAILED: 01/23/2015

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/324,990

07/07/2014

Hyun Lee

NETL.045C2

1022

TITLE OF INVENTION: Circuit Providing Load Isolation and Noise Reduction

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	04/23/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 01/23/2015
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/324,990	07/07/2014	Hyun Lee	NETL.045C2	1022

TITLE OF INVENTION: Circuit Providing Load Isolation and Noise Reduction

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	04/23/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
BANSAL, GURTEJ	2139	711-149000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted: ☐ Issue Fee ☐ Publication Fee (No small entity discount permitted) ☐ Advance Order - # of Copies _____	4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above) ☐ A check is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).
---	---

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/324,990	07/07/2014	Hyun Lee	NETL.045C2	1022
79141	7590	01/23/2015	EXAMINER	
The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306			BANSAL, GURTEJ	
			ART UNIT	PAPER NUMBER
			2139	
DATE MAILED: 01/23/2015				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

<i>Notice Requiring Inventor's Oath or Declaration</i>	Application No. 14/324,990	Applicant(s) Hyun Lee	
	Examiner BANSAL, GURTEJ	Art Unit 2139	

This notice is an attachment to the Notice of Allowability (PTOL-37), or the Notice of Allowability For A Design Application (PTOL-37D).

An inventor's oath or declaration in compliance with 37 CFR 1.63 or 1.64 executed by or with respect to each inventor has not yet been submitted.

An oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each inventor (for any inventor for which a compliant oath, declaration, or substitute statement has not yet been submitted) **MUST** be filed no later than the date on which the issue fee is paid. See 35 U.S.C. 115(f). Failure to timely comply will result in ABANDONMENT of this application.

A properly executed inventor's oath to declaration has not been received for the following inventor(s):

If applicant previously filed one or more oaths, declarations, or substitute statements, applicant may have received an informational notice regarding deficiencies therein.

The following deficiencies are noted:

INFORMAL ACTION PROBLEMS

- A properly executed inventor's oath or declaration has not been received for the following inventor(s): **Hyun Lee, Jayesh R. Bhakta, Jeffrey C. Solomon, Mario Jesus Martinez, and Chi-She Chen.**

Applicant may submit the inventor's oath or declaration at any time before the Notice of Allowance and Fee(s) Due, PTOL-85, is mailed.

Questions relating to this Notice should be directed to the Application Assistance Unit at 571-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.** Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/324,990	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2139	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to preliminary amendment dated 01/16/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|---|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>01/16/2015</u> | 6. ☐ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date <u>20150116</u> . | |

/GURTEJ BANSAL/
Primary Examiner, Art Unit 2139

<i>Examiner-Initiated Interview Summary</i>	Application No. 14/324,990	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2139	

All participants (applicant, applicant's representative, PTO personnel):

(1) GURTEJ BANSAL. (3) ____.

(2) JAMIE ZHENG (Reg. No. 51,167). (4) ____.

Date of Interview: 14 January 2015.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☐102 ☒103 ☐Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: Haddock.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Examiner indicated that the 3 sets of claims originally presented would need to be restricted and an election would need to be made. Examiner further discussed the limitations of the elected claims in view of the prior art. Examiner suggested various amendments to the claims which would distinguish the claims from the prior art. Applicant agreed to make these amendments in a preliminary amendment in order place the claims in condition for allowance.

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139	
--	--



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Memory modules with voltage regulation and level translation

BG Hazelzet, MW Kellogg, JW Nagel - US Patent 5,757,712, 1998 - Google Patents

... For example, synchronous DRAMs (SDRAMs) and static random access **memories** (SRAMs) can be ... Sep 18, 2000, Sep 23, 2003, Intel Corporation, **Memory module** employing a ... Intel Corporation, Buffering and interleaving data transfer between a chipset and **memory modules**. ...

Cited by 73 Related articles All 2 versions Cite Save

High-speed backplane

GS La Rue - US Patent 6,081,430, 2000 - Google Patents

... the **impedance** locally so the average line **impedance matches** the backplane line **impedance**, and by ... High capacity **memory module** with higher density and improved manufacturability. ... US6631083, Jul 23, 2001, Oct 7, 2003, Intel Corporation, Systems with **modules** and clocking ...

Cited by 49 Related articles All 2 versions Cite Save

Method for calibrating semiconductor devices using a common calibration reference and a calibration circuit

G Braun, S Muff, H Ruckerbauer - US Patent 6,958,613, 2005 - Google Patents

... For dynamic random access **memories** (DRAMs) based on a double data rate (DDR) standard, the ... PCs, workstations and servers, eg double data rate dual in line **memory modules** (DDR-DIMMs) ... require as many **memory** chips as possible to be integrated on the **memory module**. ...

Cited by 29 Related articles All 4 versions Cite Save

High bandwidth memory interface

P Gillingham, B Millar - US Patent 6,510,503, 2003 - Google Patents

... the controller, it is possible to mix buffered and unbuffered **modules** in the same system and achieve consistent timing between all **memories**. ... deeper address space it is also possible to buffer the data busses and connect more than one **memory** device on each **module** to the ...

Cited by 177 Related articles All 3 versions Cite Save

Defect tolerant integrated circuit subsystem for communication between a module and a bus controller in a wafer-scale integrated circuit system

FC Hsu, WY Leung - US Patent 5,592,632, 1997 - Google Patents

... bits of the communication address received in a packet is used to select one of the four **modules**. ... are used in the OTP register 1510 to hold the communication address of the **module** for system ... Any nonvolatile **memory** elements such as EPROM and EEPROM can also be used. ...

Cited by 102 Related articles All 2 versions Cite Save

Memory with large number of memory modules

W Leung, FC Hsu - 2001 - Google Patents

... a length of said sense amplifier cache line is programmable by setting a cache-line-size bit in a register of said **memory modules**. ... A method for activating only a portion of a bus communicating directly between a communicating **module** in a plurality of functional **modules** and a ...

Cited by 54 Related articles All 4 versions Cite Save

Printed circuit board (PCB) including channeled capacitive plane structure

L Jones, TW Steigerwald - US Patent 5,912,809, 1999 - Google Patents

... A main **memory** 1025, typically assembled from a plurality of dynamic random access **memory** ("DRAM") **modules**, is connected to ... BIOS 1062 and a system BIOS program code 1055 including a power-on self-test (POST) **module** are stored in the BIOS **memory** 1024 and ...

Cited by 121 Related articles All 2 versions Cite Save

Electromagnetic interference isolator with common mode choke

LF Harpham - US Patent 5,825,259, 1998 - Google Patents

... In this arrangement, the load across each series connected transformer of the constant **impedance** choke is one half of the split primary winding of the **isolation** transformer. ... **Memory module** having series-connected printed **circuit** boards. ... Ethernet **module**. ...

Cited by 54 Related articles All 2 versions Cite Save

Digital signal processing assembly and test method

JD Koontz, DF Coffin - US Patent 6,181,004, 2001 - Google Patents

... 88 and the bottom side signal layer 124 so that the processors and **memories** on both ... not limited

to connecting the surface mount pads 150 within a processor footprint or **memory** footprint to ... error is detected, the entire **module** 50 can be removed and a new **module** 50 inserted ...
Cited by 68 Related articles All 2 versions Cite Save

Low noise microstrip voltage controlled oscillator
DC Buck, RD Hess, EC Niehenke - US Patent 4,310,809, 1982 - Google Patents
... This achieves the desired admittance **matching** function previously described. ... preferred embodiment of the invention is an oscillator **circuit** which includes an amplifying **circuit** having a negative input **impedance** within the ... **Memory module** having **memory** devices on two sides. ...
Cited by 34 Related articles All 2 versions Cite Save

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	1	of	12	Attorney Docket Number	NETL.045C2

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	4218740	08-19-1980	MOTOROLA INC	
	2.	4249253	02-03-1981	Gentili et al.	
	3.	4368515	01-11-1983	Nielsen	
	4.	4392212	07-05-1983	Miyasaka et al.	
	5.	4571676	02-18-1986	Mantellina et al.	
	6.	4592011	05-27-1986	Mantellina et al.	
	7.	4633429	12-30-1986	Lewandowski et al.	
	8.	4670748	06-02-1987	Williams	
	9.	4706214	11-10-1987	Mitsubishi Denki Kabushiki Kaisha	
	10.	4866603	09-02-1989	Chiba	
	11.	4958322	09-18-1990	Kosugi et al.	
	12.	4961172	10-02-1990	Shubat et al.	
	13.	4961204	10-02-1990	Hitachi Ltd	
	14.	4980850	12-25-1990	Morgan	
	15.	5060188	10-22-1991	Zulian et al.	
	16.	5247643	09-21-1993	Shottan	
	17.	5272664	12-21-1993	Silicon Graphics Inc	
	18.	5345412	09-06-1994	Shiratsuchi	
	19.	5357478	10-18-1994	Mitsubishi Electric Corp	
	20.	5388072	02-07-1995	IBM	
	21.	5388240	02-01-1995	IBM Corporation	
	22.	5392252	02-21-1995	Visi Technology Inc	
	23.	5426753	06-20-1995	Moon	
	24.	5463590	10-31-1995	Watanabe	
	25.	5483497	01-09-1996	Mochizuki et al.	
	26.	5485589	01-16-1996	Dell Usa Lp	
	27.	5495435	02-27-1996	Sugahara	
	28.	5513135	04-30-1996	Dell et al.	
	29.	5532954	07-02-1996	Bechtolsheim et al.	
	30.	5541448	07-30-1996	Carpenter	
	31.	5572691	11-05-1996	Koudmani	
	32.	5581498	12-03-1996	Ludwig et al.	
	33.	5590071	12-31-1996	Kolor et al.	

EXAMINER

/Gurtej Bansal/ (01/16/2015)

DATE CONSIDERED

01/16/2015

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
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Sheet	2	of	12	Attorney Docket Number	NETL.045C2

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		Number-Kind Code ² (if known)			
	34.	5602999	02-11-1997	Hyatt; Gilbert P.	
	35.	5638534	06-10-1997	Samsung Electronics Co Ltd	
	36.	5655153	08-05-1997	EMC CORP	
	37.	5699542	12-16-1997	Mehta et al.	
	38.	5702984	12-30-1997	Bertin et al.	
	39.	5703826	12-30-1997	Hush et al.	
	40.	5745914	04-28-1998	Connolly et al.	
	41.	5764590	06-09-1998	Mitsubishi Electric Corp	
	42.	5784705	07-21-1998	Leung, Wingyu	
	43.	5784705	07-21-1998	Jr Albert Pesak	
	44.	5802395	09-01-1998	Connolly et al.	
	45.	5805520	09-08-1998	Anglada et al.	
	46.	5822251	10-13-1998	Bruce et al.	
	47.	5909388	06-01-1999	Siemens Ag	
	48.	5926827	07-20-1999	Dell et al.	
	49.	5926839	07-20-1999	IBM	
	50.	5953215	09-14-1999	Karabatsos, Chris	
	51.	5953280	09-14-1999	Matsui	
	52.	5958025	09-01-1999	Nippon Electric Co	
	53.	5959930	09-28-1999	Sakurai	
	54.	5963464	10-05-1999	Dell et al.	
	55.	5966736	10-12-1999	Gittinger et al.	
	56.	5974493	10-26-1999	Mitsubishi Electric Corp	
	57.	6011710	01-04-2000	Wiggers	
	58.	6018787	01-25-2000	Ip	
	59.	6044032	03-28-2000	Li	
	60.	6070217	05-30-2000	Connolly et al.	
	61.	6070227	05-30-2000	Rokicki	
	62.	6097652	08-01-2000	Roh	
	63.	6108745	08-22-2000	Gupta et al.	
	64.	6115278	09-05-2000	Deneroff et al.	
	65.	6134638	10-17-2000	Olarig et al.	
	66.	6151271	11-21-2000	Lee	

EXAMINER

/Gurtej Bansal/ (01/16/2015)

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				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	3	of	12	Attorney Docket Number	NETL.045C2

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Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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				Art Unit	2184
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				Examiner Name	BANSAL, GURTEJ
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	249.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1014 to Corrected Petition for Inter Partes Review, "Standard Dictionary of Electrical and Electronics Terms," IEEE 1988, filed on September 22, 2014.	
	250.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-00882, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.	
	251.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-00882, Exhibit 1007 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 22, 2014.	
	252.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-00882, Patent Owner's Preliminary Response Pursuant to 37 C.F.R. § 42.107, filed October 7, 2014	
	253.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-00882, Decision – Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014	
	254.	Inter Partes Review of U.S. Patent No. 8,081,536, IPR Case No. IPR2014-00883, Corrected Petition for Inter Partes Review of U.S. Patent No. 8,081,536, filed on July 8, 2014.	
	255.	Inter Partes Review of U.S. Patent No. 8,081,536, IPR Case No. IPR2014-00883, Exhibit 1011 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 21, 2014.	
	256.	Inter Partes Review of U.S. Patent No. 8,081,536, IPR Case No. IPR2014-00883, Patent Owner's Preliminary Response Pursuant to 37 C.F.R. § 42.107, filed October 7, 2014	
	257.	Inter Partes Review of U.S. Patent No. 8,081,536, IPR Case No. IPR2014-00883, Decision – Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014	
	258.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.	
	259.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 1007 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 22, 2014.	
	260.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Patent Owner's Preliminary Response Pursuant to 37 C.F.R. § 42.107, filed October 7, 2014	
	261.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Decision – Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014	

EXAMINER

/Gurtej Bansal/ (01/16/2015)

DATE CONSIDERED 01/16/2015

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	10	of	12	Attorney Docket Number	NETL.045C2

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	262.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 3001 to Decision – Institution of Inter Partes Review, Excerpts from IEEE Dictionary, issued December 16, 2014	
	263.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 3002 to Decision – Institution of Inter Partes Review, Excerpts from IEEE Dictionary, issued December 16, 2014	
	264.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 3003 to Decision – Institution of Inter Partes Review, Excerpts from Oxford English Dictionary, issued December 16, 2014	
	265.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 3004 to Decision – Institution of Inter Partes Review, Excerpts from Oxford English Dictionary, issued December 16, 2014	
	266.	McCluskey, Edward J., <i>Logic Design Principles with Emphasis on Testable Semicustom Circuits</i> , Prentice Hall, 1986, pp. 104-107 and 119-120.	
	267.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.'s Invalidity Contentions, dated June 6, 2014	
	268.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	269.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F.1-F.5 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	270.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G.1-G.6 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	271.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	272.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Diablo Technologies, Inc.'s Invalidity Contentions, dated June 6, 2014.	
	273.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits D-1 to D6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	274.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F-1 to F-5 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	

EXAMINER

/Gurtej Bansal/ (01/16/2015)

DATE CONSIDERED 01/16/2015

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	11	of	12	Attorney Docket Number	NETL.045C2

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	275.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G-1 to G-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	276.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	277.	Altera, ACEX iK, Programmable Logic Device Family, Data Sheet, May 2003, Ver 3.4	
	278.	Horowitz, "The Art of Electronics," Cambridge Univ. Press, 1989, selected pages	
	279.	Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379	
	280.	Jacob, Bruce L.; "Synchronous DRAM Architectures, Organizations, and Alternative Technologies". University of Maryland, Dec. 10, 2002.	
	281.	JEDEC Standard No. 21-C Section 4.5.7, 168 Pin Registered SDRAM DIMM Family, Release 7, October, 2001	
	282.	JEDEC 21-C, Section 4.6.1, 278 Pin Buffered SDRAM DIMM Family, June 1997	
	283.	JEDEC Standard No. 21-C Section 4.1.2.5, Appendix E, "Specific PD's for Synchronous DRAM (SDRAM)," pp. 1-25, May 2003	
	284.	JEDEC Standard, "Fully Buffered DIMM (FBDIMM): DFX Design for Validation and Test," JESD82-28, February 2008	
	285.	Anonymous. (December 1996). "Applications Note: Understanding DRAM Operation," IBM, 10 pages.	
	286.	Non-final office action, U.S. Patent Application No. 13/411,344, July 7, 2014, 19 pages	
	287.	Non-Final Office Action, U.S. Pat. App. 13/412,243, January 2, 2014, 20 pages	
	288.	Non-final office action, U.S. Patent Application No. 13/288,850, October 11, 2013, 24 pages	
	289.	Non-final office action, U.S. Patent Application No. 13/411,344, December 31, 2013, 28 pages	
	290.	Non-final office action, U.S. Patent Application No. 13/473,413, November 17, 2011, 46 pages	
	291.	Response to non-final office action dated October 11, 2013 for U.S. Patent Application No. 13/288,850, filed January 13, 2014, 15 pages	
	292.	Response to non-final office action dated 12/31/2013 for U.S. Patent Application No. 13/411,344, filed March 31, 2014, 12 pages	
	293.	Notice of Allowance, U.S. Pat. App. 12/504,131, February 12, 2013, 52 pages	
	294.	Non-Final Office Action, dated January 2, 2014, for U.S. Pat. App. No. 13/287,042, filed November 1, 2011, 42 pages	
	295.	Response to Non-Final Office Action dated January 2, 2014, filed April 2, 2014, for U.S. Pat. App. No. 13/287,042, filed November 1, 2011, 12 pages	
	296.	Office Action mailed April 2, 2014, for Japanese Patent Application No. 2012-520662 and English translation thereof, 7 pages	

EXAMINER /Gurtej Bansal/ (01/16/2015)

DATE CONSIDERED 01/16/2015

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				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	12	of	12	Attorney Docket Number	NETL.045C2

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	297.	Miles J. Murdocca et al., "Principles of Computer Architecture", Prentice Hall, 2000, pp. 249-251	
	298.	Intel® 6400/6402 Advanced Memory Buffer Datasheet, published October 2006.	
	299.	JEDEC "JEDEC Standard: Double Data Rate (DDR) SDRAM Specification", JESD79C Mar. 2003.	
	300.	JEDEC Standard No. 21-C, 4.20-2-168 Pin, PC133 SDRAM Registered Design Specification, Revision 1.4, Release 11a, Feb. 2002.	
	301.	Jedec Standard No. 21-C, DDR SDRAM PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification, Revision 1.3, Release 11b, Jan. 2002	
	302.	Jedec Standard, "Definition of the SSTV16859 2.5 V 13-Bit to 26-Bit SSTL.sub.--2 Registered Buffer for Stacked DDR DIMM Applications," JESD82-4B, May 2003.	
	303.	Lee et al. "A banked-promotion translation lookaside buffer system," Journal of Systems Architecture, vol. 47, 2002, pp. 1065-1078.	
	304.	Lee et al. "An on-chip cache compression technique to reduce decompression overhead and design complexity." Journal of Systems Architecture, vol. 46, 2000, pp. 1365-1382.	
	305.	JEDEC Standard JESD79D, "Double Data Rate (DDR) SDRAM Specification," published Jan. 2004 (Feb. 2004)	

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	23828	variable with impedance	US-PGPUB; USPAT	OR	ON	2015/01/17 20:40
L2	9321	bi\$1directional near3 port	US-PGPUB; USPAT	OR	ON	2015/01/17 20:40
L3	10	L2 with L1	US-PGPUB; USPAT	OR	ON	2015/01/17 20:40
L4	12	match\$3 with impedance with circuit with port same memory	US-PGPUB; USPAT	OR	ON	2015/01/17 20:47
L5	1082	match\$3 with impedance with circuit same memory	US-PGPUB; USPAT	OR	ON	2015/01/17 21:19
L6	526	match\$3 with impedance with circuit with memory	US-PGPUB; USPAT	OR	ON	2015/01/17 21:20
L7	13701	match\$3 with impedance with circuitwith rank same memory	US-PGPUB; USPAT	OR	ON	2015/01/17 21:20
L8	1	match\$3 with impedance with circuit with rank same memory	US-PGPUB; USPAT	OR	ON	2015/01/17 21:20
S1	13	US-4368515-\$.DID. OR US-4571676-\$.DID. OR US-4592011-\$.DID. OR US-4633429-\$.DID. OR US-4670748-\$.DID. OR WO-2006055497-\$.DID. OR US-2008285-\$.DID. OR US-7201419-\$.DID. OR US-3412243-\$.DID. OR US-2201420-\$.DID. OR US-2504131-\$.DID. OR US-1201142-\$.DID. OR US-1201112-\$.DID. OR US-2006299-\$.DID.	US-PGPUB; USPAT	OR	OFF	2015/01/16 18:18
S2	2951	G06F12/02.cpc. G06F13/4022.cpc.	US-PGPUB; USPAT	OR	OFF	2015/01/16 18:27
S3	9321	bi\$1directional near3 port	US-PGPUB; USPAT	OR	ON	2015/01/16 18:49
S4	6984	ranks with memory	US-PGPUB; USPAT	OR	ON	2015/01/16 18:49
S5	71	S3 and S4	US-PGPUB; USPAT	OR	ON	2015/01/16 18:49
S6	25363	select\$3 with (rank module) with memory	US-PGPUB; USPAT	OR	ON	2015/01/16 19:02
S7	57	S5 and S6	US-	OR	ON	2015/01/16

			PGPUB; USPAT			19:02
S8	161	impedence with match\$3 with circuit	US- PGPUB; USPAT	OR	ON	2015/01/16 19:18
S9	0	S7 and S8	US- PGPUB; USPAT	OR	ON	2015/01/16 19:19
S10	0	impedence with match\$3 with circuit same memory with port	US- PGPUB; USPAT	OR	ON	2015/01/16 19:41
S11	0	impedence with variable same memory with port	US- PGPUB; USPAT	OR	ON	2015/01/16 19:41
S12	75	(isolat\$3 disconnect\$3) with memory with rank	US- PGPUB; USPAT	OR	ON	2015/01/16 19:58
S13	0	S8 and S12	US- PGPUB; USPAT	OR	ON	2015/01/16 20:08
S14	97	match\$3 with impedence with signal	US- PGPUB; USPAT	OR	ON	2015/01/16 20:09
S15	0	S12 and S14	US- PGPUB; USPAT	OR	ON	2015/01/16 20:10
S16	2	match\$3 with impedence with signal with port	US- PGPUB; USPAT	OR	ON	2015/01/16 20:20
S17	1	"5134311".pn.	US- PGPUB; USPAT	OR	ON	2015/01/16 20:22
S18	852	711/149.ccls.	US- PGPUB; USPAT	OR	OFF	2015/01/16 20:48
S19	11	cloudera.as.	US- PGPUB; USPAT	OR	OFF	2015/01/16 21:44
S20	21456	variable with impedance	US- PGPUB; USPAT	OR	OFF	2015/01/16 21:56
S21	28985	match\$3 with impedance with signal	US- PGPUB; USPAT	OR	ON	2015/01/16 21:56
S22	18	S7 and S21	US- PGPUB; USPAT	OR	ON	2015/01/16 21:58
S23	365	S6 and S21	US- PGPUB; USPAT	OR	ON	2015/01/16 22:20
S24	30958	match\$3 with impedance with circuit	US- PGPUB; USPAT	OR	ON	2015/01/16 22:22
S25	1313	match\$3 with impedance with circuit with port	US- PGPUB; USPAT	OR	ON	2015/01/16 22:22
S26	3	S25 and S7	US-	OR	ON	2015/01/16

				PGPUB; USPAT			22:22
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EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L9	1489	G06F12/02.cpc. G06F13/4022.cpc.	USPAT; UPAD	OR	OFF	2015/01/17 21:20
L10	10009	switching.clm. and impedance.clm.	USPAT; UPAD	OR	OFF	2015/01/17 21:21
L11	9	L9 and L10	USPAT; UPAD	OR	OFF	2015/01/17 21:21

1/ 17/ 2015 9:22:11 PM
C:\ Users\ gbansal\ Documents\ EAST\ Workspaces\ 14324990.wsp

Search Notes 	Application/Control No. 14324990	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2139

CPC- SEARCHED		
Symbol	Date	Examiner
G06F12/02.cpc. G06F13/4022.cpc.	01/16/2015	GB

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
Inventor Search in eDAN	01/16/2015	GB
Text Search in EAST (See Attached)	01/16/2015	GB
NPL Search in Google Scholar (See Attached)	01/16/2015	GB

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
G06F	12/02; 13/4022	01/16/2015	GB

	/GURTEJ BANSAL/ Primary Examiner.Art Unit 2139
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Issue Classification 	Application/Control No. 14324990	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2139

CPC						
Symbol					Type	Version
G11C		5		02	F	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
				
				

NONE		Total Claims Allowed: 20	
(Assistant Examiner)	(Date)		
/GURTEJ BANSAL/ Primary Examiner.Art Unit 2139	01/16/2015	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 14324990	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2139

☐ Claims renumbered in the same order as presented by applicant															
☐ CPA															
☐ T.D.															
☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	18	17												
10	2	19	18												
11	3	9	19												
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6	14														
7	15														
17	16														

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	20	
/GURTEJ BANSAL/ Primary Examiner.Art Unit 2139	01/16/2015	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1



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BIB DATA SHEET

CONFIRMATION NO. 1022

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
14/324,990	07/07/2014	711	2139	NETL.045C2
RULE				
APPLICANTS Netlist, Inc., Irvine, CA, Assignee (with 37 CFR 1.172 Interest); INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Jeffrey C. Solomon, Irvine, CA; Mario Jesus Martinez, Laguna Niguel, CA; Chi-She Chen, Walnut, CA;				
** CONTINUING DATA ***** This application is a CON of 13/412,243 03/05/2012 PAT 8782350 which is a CON of 12/422,853 04/13/2009 PAT 8154901 which claims benefit of 61/044,839 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,801 04/14/2008				
** FOREIGN APPLICATIONS *****				
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 07/18/2014				
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /GURTEJ BANSAL/ Acknowledged Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 9 TOTAL CLAIMS 12 INDEPENDENT CLAIMS 3
ADDRESS The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES				
TITLE Circuit Providing Load Isolation and Noise Reduction				
FILING FEE RECEIVED 800	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees	
			☐ 1.16 Fees (Filing)	
			☐ 1.17 Fees (Processing Ext. of time)	
			☐ 1.18 Fees (Issue)	
			☐ Other _____	
☐ Credit				

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant	:	Netlist
App. No.	:	14/324,990
Filed	:	July 7, 2014
For	:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	:	BANSAL, GURTEJ
Art Unit	:	2189
Conf. No.	:	1022
Docket No.	:	NETL.045C2

PRELIMINARY AMENDMENT**Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

Applicant submits the following amendments and remarks in the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 9 of this paper.

In the Specification:

Please replace the title with the following:

MEMORY MODULE WITH CIRCUIT PROVIDING LOAD ISOLATION AND NOISE
REDUCTION

In the Claims:

1. (Currently Amended) A memory module operable to communicate ~~signals~~ data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

a plurality of memory devices on the printed circuit board, the plurality of memory devices being arranged in multiple ranks, including a first rank and a second rank, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus;

a circuit coupled between the at least one connector and the plurality of memory devices, ~~the circuit capable of selectively coupling a one or more first memory devices among the plurality of memory devices to the system memory controller via the at least one connector while isolating one or more second memory devices among the plurality of memory devices from the system memory controller;~~ the circuit comprising a first set of ports coupled programmable impedance matching circuits corresponding to respective ones of the plurality of edge connections, the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a programmable impedance matching circuit ~~circuits including having~~ at least one variable impedance that can be programmed to match a characteristic impedance of a corresponding signal path in the system, the circuit further comprising a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports coupled to the second rank, the circuit further comprising a switching sub-circuit configured to operatively couple the first subset of ports to respective ones of the set of bidirectional ports and to isolate the second subset of ports from the set of bidirectional ports in response to a set of address and control signals from the system memory controller, the set of address and control signals causing the memory module to select the first rank and not the second rank for memory access by the system memory controller; and

control circuitry configured to control the switching sub-circuit-selective coupling ~~in the circuit~~, and to ~~program~~ the programmable impedance matching circuits in response to one or more signals in the address and control signals.

2. (Currently Amended) The memory module of claim 1, wherein ~~the circuit comprises a set of ports coupled to the plurality of memory devices, and second programmable impedance matching circuits corresponding to respective ones of the set of ports, the second programmable impedance matching circuits being each port in the~~ second set of ports includes an impedance matching circuit that is pre-set for matching impedance associated with a respective one of the plurality of memory devices.

3. (Currently Amended) The memory module of claim 2, wherein the first set of ports includes a set of uni-directional ports in addition to the set of bidirectional ports ~~a number of ports in the set of ports is larger than a number of edge connections in the edge connector.~~

4. (Currently Amended) The memory module of claim 2, wherein the circuit further comprises signal waveform reshaping circuits corresponding to respective ones of the first set of ports.

5. (Currently Amended) The memory module of claim 1, wherein each of the first ~~first~~ programmable impedance matching circuits comprises at least one network of programmable resistors that is dynamically controllable.

6. (Original) The memory module of claim 1, wherein the circuit further comprises self-adjusting damper circuits corresponding to respective ones of the plurality of edge connections.

7. (Original) The memory module of claim 1, wherein the circuit further comprises signal waveform reshaping circuits corresponding to respective ones of the plurality of edge connections.

8. (Currently Amended) ~~A memory module operable to communicate signals with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:~~

~~a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;~~

~~a plurality of memory devices on the printed circuit board;~~

~~a circuit coupled between the at least one connector and the plurality of memory devices, the circuit capable of selectively coupling one or more first memory devices among the plurality of memory devices to the system memory controller via the at least one connector while isolating one or more second memory devices among the plurality of memory devices from the system memory controller, The memory module of claim 1, wherein the circuit is further capable of reshaping data signals between the system memory controller and the one or more first memory devices in the first rank; and~~

~~wherein the control circuitry is further configured to control the selective coupling and the reshaping of the data signals in response to one or more signals in the address and control signals.~~

9. (Original) The memory module of claim 8, wherein the circuit comprises first dynamically programmable repeater/amplifier elements corresponding to respective ones of the plurality of edge connections.

10. (Currently Amended) The memory module of claim 8, wherein the circuit comprises ~~a set of ports coupled to the plurality of memory devices, and second~~ dynamically programmable repeater/amplifier elements corresponding to respective ones of the second set of ports.

11. (Currently Amended) The memory module of claim 10, wherein a number of ports in the second set of ports is larger than a number of edge connections in the edge connector.

12. (Currently Amended) The memory module of claim 1, wherein the circuit is further configured to provide a known static value to each port in the second subset of

ports while the second subset of ports are being isolated from the set of bidirectional ports.~~A memory module operable to communicate signals with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:~~

~~a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus;~~

~~a plurality of memory devices on the printed circuit board;~~

~~a circuit coupled to the at least one connector, the circuit including a first set of ports coupled to one or more first memory devices among the plurality of memory devices, a second set of ports coupled to one or more second memory devices among the plurality of memory devices, and a switching circuit capable of selectively coupling the one or more first memory devices to the system memory controller via the at least one connector while isolating the one or more second memory devices from the system memory controller, the circuit being further configured to provide a known static value to the second set of ports; and~~

~~control circuitry configured to control the selective coupling in the circuit.~~

13. (New) The memory module of claim 1, wherein the number of bi-directional ports in the first set of ports corresponds to a bit width of the memory bus.

14. (New) The memory module of claim 1, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.

15. (New) The memory module of claim 1, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices.

16. (New) A circuit mountable on a memory module, the memory module operable to communicate data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, the memory module including a plurality of memory devices on a printed circuit board, the plurality of memory devices being arranged in multiple ranks, including a first rank and a second

rank, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus, the printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board, the circuit comprising:

- a first set of ports coupled to respective ones of the plurality of edge connections, the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a programmable impedance matching circuit having at least one variable impedance that can be programmed to match a characteristic impedance of a corresponding signal path in the system;

- a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports coupled to the second rank; and

- a switching sub-circuit configured to operatively couple the first subset of ports to respective ones of the set of bidirectional ports and to isolate the second subset of ports from the set of bidirectional ports in response to a set of address and control signals from the system memory controller, the set of address and control signals causing the memory module to select the first rank and not the second rank for memory access by the system memory controller.

17. (New) The circuit of claim 16, wherein each port in the second set of ports includes an impedance matching circuit that is pre-set for matching impedance associated with a respective one of the plurality of memory devices.

18. (New) The circuit of claim 16, wherein the circuit is further configured to provide a known static value to each port in the second subset of ports while the second subset of ports are being isolated from the set of bidirectional ports.

19. (New) The circuit of claim 6, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.

20. (New) The circuit of claim 16, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices.

REMARKS

Claims 1-5, 8, and 10-12 are amended. New claims 13-20 are presented. No new matter is added. After entry of this amendment, the pending claims are 1-20.

The Examiner is invited to call the undersigned attorney at (650) 273-6008, if a telephone call could help resolve any issues.

Respectfully submitted,

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(54) **Integrated circuit I/O using a high performance bus interface**

(57) The present invention includes a semiconductor device, the semiconductor device comprising connection means adapted to connect the semiconductor device to an external bus which is a part of a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to the external bus, wherein the external bus includes a plurality of bus lines for carrying substantially all address, data, control and device select information needed by the semiconductor device for communication with substantially every other semiconductor device connected to the external bus, and wherein

the external bus has substantially fewer bus lines than the number of bits in a single address, wherein each said bus line is a terminated transmission line, and wherein the external bus carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to the semiconductor device, a bus line driver capable of producing a low-voltage-swing signal on one of said terminated transmission lines, wherein the semiconductor device is a semiconductor transceiver device to interface multiple units to the bus, wherein the multiple units are DRAMs.

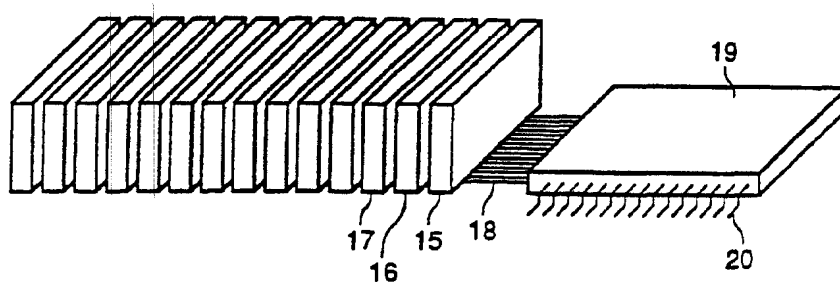


FIG. 3

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DescriptionFIELD OF THE INVENTION

[0001] An integrated circuit bus interface for computer and video systems is described which allows high speed transfer of blocks of data, particularly to and from memory devices, with reduced power consumption and increased system reliability. A new method of physically implementing the bus architecture is also described.

BACKGROUND OF THE INVENTION

[0002] Semiconductor computer memories have traditionally been designed and structured to use one memory device for each bit, or small group of bits, of any individual computer word, where the word size is governed by the choice of computer. Typical word sizes range from 4 to 64 bits. Each memory device typically is connected in parallel to a series of address lines and connected to one of a series of data lines. When the computer seeks to read from or write to a specific memory location, an address is put on the address lines and some or all of the memory devices are activated using a separate device select line for each needed device. One or more devices may be connected to each data line but typically only a small number of data lines are connected to a single memory device. Thus data line 0 is connected to device(s) 0, data line 1 is connected to device(s) 1, and so on. Data is thus accessed or provided in parallel for each memory read or write operation. For the system to operate properly, every single memory bit in every memory device must operate dependably and correctly.

[0003] To understand the concept of the present invention, it is helpful to review the architecture of conventional memory devices. Internal to nearly all types of memory devices (including the most widely used Dynamic Random Access Memory (DRAM), Static RAM (SRAM) and Read Only Memory (ROM) devices), a large number of bits are accessed in parallel each time the system carries out a memory access cycle. However, only a small percentage of accessed bits which are available internally each time the memory device is cycled ever make it across the device boundary to the external world.

[0004] Referring to Fig. 1, all modern DRAM, SRAM and ROM designs have internal architectures with row (word) lines 5 and column (bit) lines 6 to allow the memory cells to tile a two dimensional area 1. One bit of data is stored at the intersection of each word and bit line. When a particular word line is enabled, all of the corresponding data bits are transferred onto the bit lines. Some prior art DRAMs take advantage of this organization to reduce the number of pins needed to transmit the address. The address of a given memory cell is split into two addresses, row and column, each of which can be multiplexed over a bus only half as wide as the memory cell address of the prior art would have required.

COMPARISON WITH PRIOR ART

[0005] Prior art memory systems have attempted to solve the problem of high speed access to memory with limited success. U.S. Patent No. 3,821,715 (Hoff et. al.), was issued to Intel Corporation for the earliest 4-bit micro-processor. That patent describes a bus connecting a single central processing unit (CPU) with multiple RAMs and ROMs. That bus multiplexes addresses and data over a 4-bit wide bus and uses point-to-point control signals to select particular RAMs or ROMs. The access time is fixed and only a single processing element is permitted. There is no block-mode type of operation, and most important, not all of the interface signals between the devices are bused (the ROM and RAM control lines and the RAM select lines are point-to-point).

[0006] In U.S. Patent No. 4,315,308 (Jackson), a bus connecting a single CPU to a bus interface unit is described. The invention uses multiplexed address, data, and control information over a single 16-bit wide bus. Block-mode operations are defined, with the length of the block sent as part of the control sequence. In addition, variable access-time operations using a "stretch" cycle signal are provided. There are no multiple processing elements and no capability for multiple outstanding requests, and again, not all of the interface signals are bused.

[0007] In U.S. Patent No. 4,449,207 (Kung, et. al.), a DRAM is described which multiplexes address and data on an internal bus. The external interface to this DRAM is conventional, with separate control, address and data connections.

[0008] In U.S. Patent Nos. 4,764,846 and 4,706,166 (Go), a 3-D package arrangement of stacked die with connections along a single edge is described. Such packages are difficult to use because of the point-to-point wiring required to interconnect conventional memory devices with processing elements. Both patents describe complex schemes for solving these problems. No attempt is made to solve the problem by changing the interface.

[0009] In U.S. Patent No. 3,969,706 (Proebsting, et. al.), the current state-of-the-art DRAM interface is described. The address is two-way multiplexed, and there are separate pins for data and control (RAS, CAS, WE, CS). The number of pins grows with the size of the DRAM, and many of the connections must be made point-to-point in a memory system using such DRAMs.

[0010] There are many backplane buses described in the prior art, but not in the combination described or having the

features of this invention. Many backplane buses multiplex addresses and data on a single bus (e.g., the NU bus). ELXSI and others have implemented split-transaction buses (U.S. Patent No. 4,595,923 and 4,481,625 (Roberts)). ELXSI has also implemented a relatively low-voltage-swing current-mode ECL driver (approximately 1 V swing). Address-space registers are implemented on most backplane buses, as is some form of block mode operation.

[0011] Nearly all modern backplane buses implement some type of arbitration scheme, but the arbitration scheme used in this invention differs from each of these. U.S. Patent Nos. 4,837,682 (Culler), 4,818,985 (Ikeda), 4,779,089 (Theus) and 4,745,548 (Blahut) describe prior art schemes. All involve either log N extra signals, (Theus, Blahut), where N is the number of potential bus requestors, or additional delay to get control of the bus (Ikeda, Culler). None of the buses described in patents or other literature use only bused connections. All contain some point-to-point connections on the backplane. None of the other aspects of this invention such as power reduction by fetching each data block from a single device or compact and low-cost 3-D packaging even apply to backplane buses.

[0012] The clocking scheme used in this invention has not been used before and in fact would be difficult to implement in backplane buses due to the signal degradation caused by connector stubs. U.S. Patent No. 4,247,817 (Heller) describes a clocking scheme using two clock lines, but relies on ramp-shaped clock signals in contrast to the normal rise-time signals used in the present invention.

[0013] In U.S. Patent No. 4,646,279 (Voss), a video RAM is described which implements a parallel-load, serial-out shift register on the output of a DRAM. This generally allows greatly improved bandwidth (and has been extended to 2, 4 and greater width shift-out paths.) The rest of the interfaces to the DRAM (RAS, CAS, multiplexed address, etc.) remain the same as for conventional DRAMS.

[0014] One object of the present invention is to use a new bus interface built into semiconductor devices to support high-speed access to large blocks of data from a single memory device by an external user of the data, such as a microprocessor, in an efficient and cost-effective manner.

[0015] Another object of this invention is to provide a clocking scheme to permit high speed clock signals to be sent along the bus with minimal clock skew between devices.

[0016] Another object of this invention is to allow mapping out defective memory devices or portions of memory devices.

[0017] Another object of this invention is to provide a method for distinguishing otherwise identical devices by assigning a unique identifier to each device.

[0018] Yet another object of this invention is to provide a method for transferring address, data and control information over a relatively narrow bus and to provide a method of bus arbitration when multiple devices seek to use the bus simultaneously.

[0019] Another object of this invention is to provide a method of distributing a high-speed memory cache within the DRAM chips of a memory system which is much more effective than previous cache methods.

[0020] Another object of this invention is to provide devices, especially DRAMs, suitable for use with the bus architecture of the invention.

SUMMARY OF INVENTION

[0021] The present invention includes a memory subsystem comprising at least two semiconductor devices, including at least one memory device, connected in parallel to a bus, where the bus includes a plurality of bus lines for carrying substantially all address, data and control information needed by said memory devices, where the control information includes device-select information and the bus has substantially fewer bus lines than the number of bits in a single address, and the bus carries device-select information without the need for separate device-select lines connected directly to individual devices.

[0022] Referring to Fig. 2, a standard DRAM 13, 14, ROM (or SRAM) 12, microprocessor CPU 11, I/O device, disk controller or other special purpose device such as a high speed switch is modified to use a wholly bus-based interface rather than the prior art combination of point-to-point and bus-based wiring used with conventional versions of these devices. The new bus includes clock signals, power and multiplexed address, data and control signals. In a preferred implementation, 8 bus data lines and an AddressValid bus line carry address, data and control information for memory addresses up to 40 bits wide. Persons skilled in the art will recognize that 16 bus data lines or other numbers of bus data lines can be used to implement the teaching of this invention. The new bus is used to connect elements such as memory, peripheral, switch and processing units.

[0023] In the system of this invention, DRAMs and other devices receive address and control information over the bus and transmit or receive requested data over the same bus. Each memory device contains only a single bus interface with no other signal pins. Other devices that may be included in the system can connect to the bus and other non-bus lines, such as input/output lines. The bus supports large data block transfers and split transactions to allow a user to achieve high bus utilization. This ability to rapidly read or write a large block of data to one single device at a time is an important advantage of this invention.

[0024] The DRAMs that connect to this bus differ from conventional DRAMs in a number of ways. Registers are

provided which may store control information, device identification, device-type and other information appropriate for the chip such as the address range for each independent portion of the device. New bus interface circuits must be added and the internals of prior art DRAM devices need to be modified so they can provide and accept data to and from the bus at the peak data rate of the bus. This requires changes to the column access circuitry in the DRAM, with only a minimal increase in die size. A circuit is provided to generate a low skew internal device clock for devices on the bus, and other circuits provide for demultiplexing input and multiplexing output signals.

[0025] High bus bandwidth is achieved by running the bus at a very high clock rate (hundreds of MHz). This high clock rate is made possible by the constrained environment of the bus. The bus lines are controlled-impedance, doubly-terminated lines. For a data rate of 500 MHz, the maximum bus propagation time is less than 1 ns (the physical bus length is about 10 cm). In addition, because of the packaging used, the pitch of the pins can be very close to the pitch of the pads. The loading on the bus resulting from the individual devices is very small. In a preferred implementation, this generally allows stub capacitances of 1-2 pF and inductances of 0.5 - 2 nH. Each device 15, 16, 17, shown in Figure 3, only has pins on one side and these pins connect directly to the bus 18. A transceiver device 19 can be included to interface multiple units to a higher order bus through pins 20.

[0026] A primary result of the architecture of this invention is to increase the bandwidth of DRAM access. The invention also reduces manufacturing and production costs, power consumption, and increases packing density and system reliability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027]

Figure 1 is a diagram which illustrates the basic 2-D organization of memory devices.

Figure 2 is a schematic block diagram which illustrates the parallel connection of all bus lines and the serial Reset line to each device in the system.

Figure 3 is a perspective view of a system of the invention which illustrates the 3-D packaging of semiconductor devices on the primary bus.

Figure 4 shows the format of a request packet.

Figure 5 shows the format of a retry response from a slave.

Figure 6 shows the bus cycles after a request packet collision occurs on the bus and how arbitration is handled.

Figure 7 shows the timing whereby signals from two devices can overlap temporarily and drive the bus at the same time.

Figure 8 shows the connection and timing between bus clocks and devices on the bus.

Figure 9 is a perspective view showing how transceivers can be used to connect a number of bus units to a transceiver bus. Figure 10 is a block and schematic diagram of input/output circuitry used to connect devices to the bus.

Figure 11 is a schematic diagram of a clocked sense-amplifier used as a bus input receiver.

Figure 12 is a block diagram showing how the internal device clock is generated from two bus clock signals using a set of adjustable delay lines.

Figure 13 is a timing diagram showing the relationship of signals in the block diagram of Figure 12.

Figure 14 is timing diagram of a preferred means of implementing the reset procedure of this invention.

Figure 15 is a diagram illustrating the general organization of a 4 Mbit DRAM divided into 8 subarrays.

DETAILED DESCRIPTION

[0028] The present invention is designed to provide a high speed, multiplexed bus for communication between processing devices and memory devices and to provide devices adapted for use in the bus system. The invention can also be used to connect processing devices and other devices, such as I/O interfaces or disk controllers, with or without memory devices on the bus. The bus consists of a relatively small number of lines connected in parallel to each device on the bus. The bus carries substantially all address, data and control information needed by devices for communication with other devices on the bus. In many systems using the present invention, the bus carries almost every signal between every device in the entire system. There is no need for separate device-select lines since device-select information for each device on the bus is carried over the bus. There is no need for separate address and data lines because address and data information can be sent over the same lines. Using the organization described herein, very large addresses (40 bits in the preferred implementation) and large data blocks (1024 bytes) can be sent over a small number of bus lines (8 plus one control line in the preferred implementation).

[0029] Virtually all of the signals needed by a computer system can be sent over the bus. Persons skilled in the art recognize that certain devices, such as CPUs, may be connected to other signal lines and possibly to independent buses, for example a bus to an independent cache memory, in addition to the bus of this invention. Certain devices, for

example cross-point switches, could be connected to multiple, independent buses of this invention. In the preferred implementation, memory devices are provided that have no connections other than the bus connections described herein and CPUs are provided that use the bus of this invention as the principal, if not exclusive, connection to memory and to other devices on the bus.

[0030] All modern DRAM, SRAM and ROM designs have internal architectures with row (word) and column (bit) lines to efficiently tile a 2-D area. Referring to Fig. 1, one bit of data is stored at the intersection of each word line 5 and bit line 6. When a particular word line is enabled, all of the corresponding data bits are transferred onto the bit lines. This data, about 4000 bits at a time in a 4 MBit DRAM, is then loaded into column sense amplifiers 3 and held for use by the I/O circuits.

[0031] In the invention presented here, the data from the sense amplifiers is enabled 32 bits at a time onto an internal device bus running at approximately 125 MHz. This internal device bus moves the data to the periphery of the devices where the data is multiplexed into an 8-bit wide external bus interface, running at approximately 500 MHz.

[0032] The bus architecture of this invention connects master or bus controller devices, such as CPUs, Direct Memory Access devices (DMAs) or Floating Point Units (FPUs), and slave devices, such as DRAM, SRAM or ROM memory devices. A slave device responds to control signals; a master sends control signals. Persons skilled in the art realize that some devices may behave as both master and slave at various times, depending on the mode of operation and the state of the system. For example, a memory device will typically have only slave functions, while a DMA controller, disk controller or CPU may include both slave and master functions. Many other semiconductor devices, including I/O devices, disk controllers, or other special purpose devices such as high speed switches can be modified for use with the bus of this invention.

[0033] Each semiconductor device contains a set of internal registers, preferably including a device identification (device ID) register, a device-type descriptor register, control registers and other registers containing other information relevant to that type of device. In a preferred implementation, semiconductor devices connected to the bus contain registers which specify the memory addresses contained within that device and access-time registers which store a set of one or more delay times at which the device can or should be available to send or receive data.

[0034] Most of these registers can be modified and preferably are set as part of an initialization sequence that occurs when the system is powered up or reset. During the initialization sequence each device on the bus is assigned a unique device ID number, which is stored in the device ID register. A bus master can then use these device ID numbers to access and set appropriate registers in other devices, including access-time registers, control registers, and memory registers, to configure the system. Each slave may have one or several access-time registers (four in a preferred embodiment). In a preferred embodiment, one access-time register in each slave is permanently or semi-permanently programmed with a fixed value to facilitate certain control functions. A preferred implementation of an initialization sequence is described below in more detail.

[0035] All information sent between master devices and slave devices is sent over the external bus, which, for example, may be 8 bits wide. This is accomplished by defining a protocol whereby a master device, such as a microprocessor, seizes exclusive control of the external bus (i.e., becomes the bus master) and initiates a bus transaction by sending a request packet (a sequence of bytes comprising address and control information) to one or more slave devices on the bus. An address can consist of 16 to 40 or more bits according to the teachings of this invention. Each slave on the bus must decode the request packet to see if that slave needs to respond to the packet. The slave that the packet is directed to must then begin any internal processes needed to carry out the requested bus transaction at the requested time. The requesting master may also need to transact certain internal processes before the bus transaction begins. After a specified access time the slave(s) respond by returning one or more bytes (8 bits) of data or by storing information made available from the bus. More than one access time can be provided to allow different types of responses to occur at different times.

[0036] A request packet and the corresponding bus access are separated by a selected number of bus cycles, allowing the bus to be used in the intervening bus cycles by the same or other masters for additional requests or brief bus accesses. Thus multiple, independent accesses are permitted, allowing maximum utilization of the bus for transfer of short blocks of data. Transfers of long blocks of data use the bus efficiently even without overlap because the overhead due to bus address, control and access times is small compared to the total time to request and transfer the block.

Device Address Mapping

[0037] Another unique aspect of this invention is that each memory device is a complete, independent memory subsystem with all the functionality of a prior art memory board in a conventional backplane-bus computer system. Individual memory devices may contain a single memory section or may be subdivided into more than one discrete memory section. Memory devices preferably include memory address registers for each discrete memory section. A failed memory device (or even a subsection of a device) can be "mapped out" with only the loss of a small fraction of the memory, maintaining essentially full system capability. Mapping out bad devices can be accomplished in two ways, both compatible with this

invention.

[0038] The preferred method uses address registers in each memory device (or independent discrete portion thereof) to store information which defines the range of bus addresses to which this memory device will respond. This is similar to prior art schemes used in memory boards in conventional backplane bus systems. The address registers can include a single pointer, usually pointing to a block of known size, a pointer and a fixed or variable block size value or two pointers, one pointing to the beginning and one to the end (or to the "top" and "bottom") of each memory block. By appropriate settings of the address registers, a series of functional memory devices or discrete memory sections can be made to respond to a contiguous range of addresses, giving the system access to a contiguous block of good memory, limited primarily by the number of good devices connected to the bus. A block of memory in a first memory device or memory section can be assigned a certain range of addresses, then a block of memory in a next memory device or memory section can be assigned addresses starting with an address one higher (or lower, depending on the memory structure) than the last address of the previous block.

[0039] Preferred devices for use in this invention include device-type register information specifying the type of chip, including how much memory is available in what configuration on that device. A master can perform an appropriate memory test, such as reading and writing each memory cell in one or more selected orders, to test proper functioning of each accessible discrete portion of memory (based in part on information like device ID number and device-type) and write address values (up to 40 bits in the preferred embodiment, 10^{12} bytes), preferably contiguous, into device address-space registers. Non-functional or impaired memory sections can be assigned a special address value which the system can interpret to avoid using that memory.

[0040] The second approach puts the burden of avoiding the bad devices on the system master or masters. CPUs and DMA controllers typically have some sort of translation look-aside buffers (TLBs) which map virtual to physical (bus) addresses. With relatively simple software, the TLBs can be programmed to use only working memory (data structures describing functional memories are easily generated). For masters which don't contain TLBs (for example, a video display generator), a small, simple RAM can be used to map a contiguous range of addresses onto the addresses of the functional memory devices.

[0041] Either scheme works and permits a system to have a significant percentage of non-functional devices and still continue to operate with the memory which remains. This means that systems built with this invention will have much improved reliability over existing systems, including the ability to build systems with almost no field failures.

Bus

[0042] The preferred bus architecture of this invention comprises 11 signals: BusData[0:7]; AddrValid; Clk1 and Clk2; plus an input reference level and power and ground lines connected in parallel to each device. Signals are driven onto the bus during conventional bus cycles. The notation "Signal[i:j]" refers to a specific range of signals or lines, for example, BusData[0:7] means BusData0, BusData1, ..., BusData7. The bus lines for BusData[0:7] signals form a byte-wide, multiplexed data/address/control bus. AddrValid is used to indicate when the bus is holding a valid address request, and instructs a slave to decode the bus data as an address and, if the address is included on that slave, to handle the pending request. The two clocks together provide a synchronized, high speed clock for all the devices on the bus. In addition to the bused signals, there is one other line (ResetIn, ResetOut) connecting each device in series for use during initialization to assign every device in the system a unique device ID number (described below in detail).

[0043] To facilitate the extremely high data rate of this external bus relative to the gate delays of the internal logic, the bus cycles are grouped into pairs of even/odd cycles. Note that all devices connected to a bus should preferably use the same even/odd labeling of bus cycles and preferably should begin operations on even cycles. This is enforced by the clocking scheme.

Protocol and Bus Operation

[0044] The bus uses a relatively simple, synchronous, split-transaction, block-oriented protocol for bus transactions. One of the goals of the system is to keep the intelligence concentrated in the masters, thus keeping the slaves as simple as possible (since there are typically many more slaves than masters). To reduce the complexity of the slaves, a slave should preferably respond to a request in a specified time, sufficient to allow the slave to begin or possibly complete a device-internal phase including any internal actions that must precede the subsequent bus access phase. The time for this bus access phase is known to all devices on the bus - each master being responsible for making sure that the bus will be free when the bus access begins. Thus the slaves never worry about arbitrating for the bus. This approach eliminates arbitration in single master systems, and also makes the slave-bus interface simpler.

[0045] In a preferred implementation of the invention, to initiate a bus transfer over the bus, a master sends out a request packet, a contiguous series of bytes containing address and control information. It is preferable to use a request packet containing an even number of bytes and also preferable to start each packet on an even bus cycle.

[0046] The device-select function is handled using the bus data lines. AddrValid is driven, which instructs all slaves to decode the request packet address, determine whether they contain the requested address, and if they do, provide the data back to the master (in the case of a read request) or accept data from the master (in the case of a write request) in a data block transfer. A master can also select a specific device by transmitting a device ID number in a request packet. In a preferred implementation, a special device ID number is chosen to indicate that the packet should be interpreted by all devices on the bus. This allows a master to broadcast a message, for example to set a selected control register of all devices with the same value.

[0047] The data block transfer occurs later at a time specified in the request packet control information, preferably beginning on an even cycle. A device begins a data block transfer almost immediately with a device-internal phase as the device initiates certain functions, such as setting up memory addressing, before the bus access phase begins. The time after which a data block is driven onto the bus lines is selected from values stored in slave access-time registers. The timing of data for reads and writes is preferably the same; the only difference is which device drives the bus. For reads, the slave drives the bus and the master latches the values from the bus. For writes the master drives the bus and the selected slave latches the values from the bus.

[0048] In a preferred implementation of this invention shown in Figure 4, a request packet 22 contains 6 bytes of data -- 4.5 address bytes and 1.5 control bytes. Each request packet uses all nine bits of the multiplexed data/address lines (AddrValid 23 + BusData[0:7] 24) for all six bytes of the request packet. Setting 23 AddrValid = 1 in an otherwise unused even cycle indicates the start of an request packet (control information). In a valid request packet, AddrValid 27 must be 0 in the last byte. Asserting this signal in the last byte invalidates the request packet. This is used for the collision detection and arbitration logic (described below). Bytes 25-26 contain the first 35 address bits, Address[0:35]. The last byte contains AddrValid 27 (the invalidation switch) and 28, the remaining address bits, Address[36:39], and BlockSize [0:3] (control information).

[0049] The first byte contains two 4 bit fields containing control information, AccessType[0:3], an op code (operation code) which, for example, specifies the type of access, and Master[0:3], a position reserved for the master sending the packet to include its master ID number. Only master numbers 1 through 15 are allowed - master number 0 is reserved for special system commands. Any packet with Master[0:3] = 0 is an invalid or special packet and is treated accordingly.

[0050] The AccessType field specifies whether the requested operation is a read or write and the type of access, for example, whether it is to the control registers or other parts of the device, such as memory. In a preferred implementation, AccessType[0] is a Read/Write switch: if it is a 1, then the operation calls for a read from the slave (the slave to read the requested memory block and drive the memory contents onto the bus); if it is a 0, the operation calls for a write into the slave (the slave to read data from the bus and write it to memory). AccessType[1:3] provides up to 8 different access types for a slave. AccessType[1:2] preferably indicates the timing of the response, which is stored in an access-time register, AccessRegN. The choice of access-time register can be selected directly by having a certain op code select that register, or indirectly by having a slave respond to selected op codes with pre-selected access times (see table below). The remaining bit, AccessType[3] may be used to send additional information about the request to the slaves.

[0051] One special type of access is control register access, which involves addressing a selected register in a selected slave. In the preferred implementation of this invention, AccessType[1:3] equal to zero indicates a control register request and the address field of the packet indicates the desired control register. For example, the most significant two bytes can be the device ID number (specifying which slave is being addressed) and the least significant three bytes can specify a register address and may also represent or include data to be loaded into that control register. Control register accesses are used to initialize the access-time registers, so it is preferable to use a fixed response time which can be preprogrammed or even hard wired, for example the value in AccessReg0, preferably 8 cycles. Control register access can also be used to initialize or modify other registers, including address registers.

[0052] The method of this invention provides for access mode control specifically for the DRAMs. One such access mode determines whether the access is page mode or normal RAS access. In normal mode (in conventional DRAMS and in this invention), the DRAM column sense amps or latches have been precharged to a value intermediate between logical 0 and 1. This precharging allows access to a row in the RAM to begin as soon as the access request for either inputs (writes) or outputs (reads) is received and allows the column sense amps to sense data quickly. In page mode (both conventional and in this invention), the DRAM holds the data in the column sense amps or latches from the previous read or write operation. If a subsequent request to access data is directed to the same row, the DRAM does not need to wait for the data to be sensed (it has been sensed already) and access time for this data is much shorter than the normal access time. Page mode generally allows much faster access to data but to a smaller block of data (equal to the number of sense amps). However, if the requested data is not in the selected row, the access time is longer than the normal access time, since the request must wait for the RAM to precharge before the normal mode access can start. Two access-time registers in each DRAM preferably contain the access times to be used for normal and for page-mode accesses, respectively.

[0053] The access mode also determines whether the DRAM should precharge the sense amplifiers or should save the contents of the sense amps for a subsequent page mode access. Typical settings are "precharge after normal

access" and "save after page mode access" but "precharge after page mode access" or "save after normal access" are allowed, selectable modes of operation. The DRAM can also be set to precharge the sense amps if they are not accessed for a selected period of time.

[0054] In page mode, the data stored in the DRAM sense amplifiers may be accessed within much less time than it takes to read out data in normal mode (~10-20 nS vs. 40-100 nS). This data may be kept available for long periods. However, if these sense amps (and hence bit lines) are not precharged after an access, a subsequent access to a different memory word (row) will suffer a precharge time penalty of about 40-100 nS because the sense amps must precharge before latching in a new value.

[0055] The contents of the sense amps thus may be held and used as a cache, allowing faster, repetitive access to small blocks of data. DRAM-based page-mode caches have been attempted in the prior art using conventional DRAM organizations but they are not very effective because several chips are required per computer word. Such a conventional page-mode cache contains many bits (for example, 32 chips x 4Kbits) but has very few independent storage entries. In other words, at any given point in time the sense amps hold only a few different blocks or memory "locales" (a single block of 4K words, in the example above). Simulations have shown that upwards of 100 blocks are required to achieve high hit rates (>90% of requests find the requested data already in cache memory) regardless of the size of each block. See, for example, Anant Agarwal, et. al., "An Analytic Cache Model," ACM Transactions on Computer Systems, Vol. 7 (2), pp. 184-215 (May 1989).

[0056] The organization of memory in the present invention allows each DRAM to hold one or more (4 for 4MBit DRAMS) separately- addressed and independent blocks of data. A personal computer or workstation with 100 such DRAMs (i.e. 400 blocks or locales) can achieve extremely high, very repeatable hit rates (98-99% on average) as compared to the lower (50-80%), widely varying hit rates using DRAMS organized in the conventional fashion. Further, because of the time penalty associated with the deferred precharge on a "miss" of the page-mode cache, the conventional DRAM-based page-mode cache generally has been found to work less well than no cache at all.

[0057] For DRAM slave access, the access types are preferably used in the following way:

AccessType[1:3]	Use	AccessTime
0	Control Register Access	Fixed, 8[AccessReg0]
1	Unused	Fixed, 8[AccessReg0]
2-3	Unused	AccessReg1
4-5	Page Mode DRAM access	AccessReg2
6-7	Normal DRAM access	AccessReg3

Persons skilled in the art will recognize that a series of available bits could be designated as switches for controlling these access modes. For example:

AccessType[2] = page mode/normal switch

AccessType[3] = precharge/save-data switch

BlockSize[0:3] specifies the size of the data block transfer. If BlockSize[0] is 0, the remaining bits are the binary representation of the block size (0-7). If BlockSize[0] is 1, then the remaining bits give the block size as a binary power of 2, from 8 to 1024. A zero-length block can be interpreted as a special command, for example, to refresh a DRAM without returning any data, or to change the DRAM from page mode to normal access mode or vice-versa.

BlockSize[0:2]	Number of Bytes in Block
0-7	0-7 respectively
8	8
9	16
10	32
11	64
12	128
13	256
14	512
15	1024

Persons skilled in the art will recognize that other block size encoding schemes or values can be used.

[0058] In most cases, a slave will respond at the selected access time by reading or writing data from or to the bus over bus lines BusData[0:7] and AddrValid will be at logical 0. In a preferred embodiment, substantially each memory

access will involve only a single memory device, that is, a single block will be read from or written to a single memory device.

Retry Format

[0059] In some cases, a slave may not be able to respond correctly to a request, e.g., for a read or write. In such a situation, the slave should return an error message, sometimes called a N(o)ACK(nowledge) or retry message. The retry message can include information about the condition requiring a retry, but this increases system requirements for circuitry in both slave and masters. A simple message indicating only that an error has occurred allows for a less complex slave, and the master can take whatever action is needed to understand and correct the cause of the error.

[0060] For example, under certain conditions a slave might not be able to supply the requested data. During a page-mode access, the DRAM selected must be in page mode and the requested address must match the address of the data held in the sense amps or latches. Each DRAM can check for this match during a page-mode access. If no match is found, the DRAM begins precharging and returns a retry message to the master during the first cycle of the data block (the rest of the returned block is ignored). The master then must wait for the precharge time (which is set to accommodate the type of slave in question, stored in a special register, PreChargeReg), and then resend the request as a normal DRAM access (AccessType = 6 or 7).

[0061] In the preferred form of the present invention, a slave signals a retry by driving AddrValid true at the time the slave was supposed to begin reading or writing data. A master which expected to write to that slave must monitor AddrValid during the write and take corrective action if it detects a retry message. Figure 5 illustrates the format of a retry message 28 which is useful for read requests, consisting of 23 AddrValid=1 with Master[0:3] = 0 in the first (even) cycle. Note that AddrValid is normally 0 for data block transfers and that there is no master 0 (only 1 through 15 are allowed). All DRAMs and masters can easily recognize such a packet as an invalid request packet, and therefore a retry message. In this type of bus transaction all of the fields except for Mastery[0:3] and AddrValid 23 may be used as information fields, although in the implementation described, the contents are undefined. Persons skilled in the art recognize that another method of signifying a retry message is to add a DataInvalid line and signal to the bus. This signal could be asserted in the case of a NACK.

Bus Arbitration

[0062] In the case of a single master, there are by definition no arbitration problems. The master sends request packets and keeps track of periods when the bus will be busy in response to that packet. The master can schedule multiple requests so that the corresponding data block transfers do not overlap.

[0063] The bus architecture of this invention is also useful in configurations with multiple masters. When two or more masters are on the same bus, each master must keep track of all the pending transactions, so each master knows when it can send a request packet and access the corresponding data block transfer. Situations will arise, however, where two or more masters send a request packet at about the same time and the multiple requests must be detected, then sorted out by some sort of bus arbitration.

[0064] There are many ways for each master to keep track of when the bus is and will be busy. A simple method is for each master to maintain a bus-busy data structure, for example by maintaining two pointers, one to indicate the earliest point in the future when the bus will be busy and the other to indicate the earliest point in the future when the bus will be free, that is, the end of the latest pending data block transfer. Using this information, each master can determine whether and when there is enough time to send a request packet (as described above under Protocol) before the bus becomes busy with another data block transfer and whether the corresponding data block transfer will interfere with pending bus transactions. Thus each master must read every request packet and update its bus-busy data structure to maintain information about when the bus is and will be free.

[0065] With two or more masters on the bus, masters will occasionally transmit independent request packets during the same bus cycle. Those multiple requests will collide as each such master drives the bus simultaneously with different information, resulting in scrambled request information and neither desired data block transfer. In a preferred form of the invention, each device on the bus seeking to write a logical 1 on a BusData or AddrValid line drives that line with a current sufficient to sustain a voltage greater than or equal to the high-logic value for the system. Devices do not drive lines that should have a logical 0; those lines are simply held at a voltage corresponding to a low-logic value. Each master tests the voltage on at least some, preferably all, bus data and the AddrValid lines so the master can detect a logical '1' where the expected level is '0' on a line that it does not drive during a given bus cycle but another master does drive.

[0066] Another way to detect collisions is to select one or more bus lines for collision signalling. Each master sending a request drives that line or lines and monitors the selected lines for more than the normal drive current (or a logical value of ">1"), indicating requests by more than one master. Persons skilled in the art will recognize that this can be implemented with a protocol involving BusData and AddrValid lines or could be implemented using an additional bus line.

[0067] In the preferred form of this invention, each master detects collisions by monitoring lines which it does not drive to see if another master is driving those lines. Referring to Fig. 4, the first byte of the request packet includes the number of each master attempting to use the bus (Master[0:3]). If two masters send packet requests starting at the same point in time, the master numbers will be logical "or"ed together by at least those masters, and thus one or both of the masters, by monitoring the data on the bus and comparing what it sent, can detect a collision. For instance if requests by masters number 2 (0010) and 5 (0101) collide, the bus will be driven with the value Master[0:3]=7 (0010 + 0101 = 0111). Master number 5 will detect that the signal Master[2] = 1 and master 2 will detect that Master[1] and Master[3] = 1, telling both masters that a collision has occurred. Another example is masters 2 and 11, for which the bus will be driven with the value Master[0:3]=11 (0010 + 1011 = 1011), and although master 11 can't readily detect this collision, master 2 can. When any collision is detected, each master detecting a collision drives the value of AddrValid 27 in byte 5 of the request packet 22 to 1, which is detected by all masters, including master 11 in the second example above, and forces a bus arbitration cycle, described below.

[0068] Another collision condition may arise where master A sends a request packet in cycle 0 and master B tries to send a request packet starting in cycle 2 of the first request packet, thereby overlapping the first request packet. This will occur from time to time because the bus operates at high speeds, thus the logic in a second-initiating master may not be fast enough to detect a request initiated by a first master in cycle 0 and to react fast enough by delaying its own request. Master B eventually notices that it wasn't supposed to try to send a request packet (and consequently almost surely destroyed the address that master A was trying to send), and, as in the example above of a simultaneous collision, drives a 1 on AddrValid during byte 5 of the first request packet 27 forcing an arbitration. The logic in the preferred implementation is fast enough that a master should detect a request packet by another master by cycle 3 of the first request packet, so no master is likely to attempt to send a potentially colliding request packet later than cycle 2.

[0069] Slave devices not need to detect a collision directly, but they must wait to do anything irrecoverable until the last byte (byte 5) is read to ensure that the packet is valid. A request packet with Master[0:3] equal to 0 (a retry signal) is ignored and does not cause a collision. The subsequent bytes of such a packet are ignored.

[0070] To begin arbitration after a collision, the masters wait a preselected number of cycles after the aborted request packet (4 cycles in a preferred implementation), then use the next free cycle to arbitrate for the bus (the next available even cycle in the preferred implementation). Each colliding master signals to all other colliding masters that it seeks to send a request packet, a priority is assigned to each of the colliding masters, then each master is allowed to make its request in the order of that priority.

[0071] Figure 6 illustrates one preferred way of implementing this arbitration. Each colliding master signals its intent to send a request packet by driving a single BusData line during a single bus cycle corresponding to its assigned master number (1-15 in the present example). During two-byte arbitration cycle 29, byte 0 is allocated to requests 1-7 from masters 1-7, respectively, (bit 0 is not used) and byte 1 is allocated to requests 8-15 from masters 8-15, respectively. At least one device and preferably each colliding master reads the values on the bus during the arbitration cycles to determine and store which masters desire to use the bus. Persons skilled in the art will recognize that a single byte can be allocated for arbitration requests if the system includes more bus lines than masters. More than 15 masters can be accommodated by using additional bus cycles.

[0072] A fixed priority scheme (preferably using the master numbers, selecting lowest numbers first) is then used to prioritize, then sequence the requests in a bus arbitration queue which is maintained by at least one device. These requests are queued by each master in the bus-busy data structure and no further requests are allowed until the bus arbitration queue is cleared. Persons skilled in the art will recognize that other priority schemes can be used, including assigning priority according to the physical location of each master.

System Configuration/Reset

[0073] In the bus-based system of this invention, a mechanism is provided to give each device on the bus a unique device identifier (device ID) after power-up or under other conditions as desired or needed by the system. A master can then use this device ID to access a specific device, particularly to set or modify registers of the specified device, including the control and address registers. In the preferred embodiment, one master is assigned to carry out the entire system configuration process. The master provides a series of unique device ID numbers for each unique device connected to the bus system. In the preferred embodiment, each device connected to the bus contains a special device-type register which specifies the type of device, for instance CPU, 4 MBit memory, 64 MBit memory or disk controller. The configuration master should check each device, determine the device type and set appropriate control registers, including access-time registers. The configuration master should check each memory device and set all appropriate memory address registers.

[0074] One means to set up unique device ID numbers is to have each device to select a device ID in sequence and store the value in an internal device ID register. For example, a master can pass sequential device ID numbers through shift registers in each of a series of devices, or pass a token from device to device whereby the device with the token

reads in device ID information from another line or lines. In a preferred embodiment, device ID numbers are assigned to devices according to their physical relationship, for instance, their order along the bus.

[0075] In a preferred embodiment of this invention, the device ID setting is accomplished using a pair of pins on each device, ResetIn and ResetOut. These pins handle normal logic signals and are used only during device ID configuration. On each rising edge of the clock, each device copies ResetIn (an input) into a four-stage reset shift register. The output of the reset shift register is connected to ResetOut, which in turn connects to ResetIn for the next sequentially connected device. Substantially all devices on the bus are thereby daisy-chained together. A first reset signal, for example, while ResetIn at a device is a logical 1, or when a selected bit of the reset shift register goes from zero to non-zero, causes the device to hard reset, for example by clearing all internal registers and resetting all state machines. A second reset signal, for example, the falling edge of ResetIn combined with changeable values on the external bus, causes that device to latch the contents of the external bus into the internal device ID register (Device[0:7]).

[0076] To reset all devices on a bus, a master sets the ResetIn line of the first device to a "1" for long enough to ensure that all devices on the bus have been reset (4 cycles times the number of devices -- note that the maximum number of devices on the preferred bus configuration is 256 (8 bits), so that 1024 cycles is always enough time to reset all devices.) Then ResetIn is dropped to "0" and the BusData lines are driven with the first followed by successive device ID numbers, changing after every 4 clock pulses. Successive devices set those device ID numbers into the corresponding device ID register as the falling edge of ResetIn propagates through the shift registers of the daisy-chained devices. Figure 14 shows ResetIn at a first device going low while a master drives a first device ID onto the bus data lines BusData[0:3]. The first device then latches in that first device ID. After four clock cycles, the master changes BusData[0:3] to the next device ID number and ResetOut at the first device goes low, which pulls ResetIn for the next daisy-chained device low, allowing the next device to latch in the next device ID number from BusData[0:3]. In the preferred embodiment, one master is assigned device ID 0 and it is the responsibility of that master to control the ResetIn line and to drive successive device ID numbers onto the bus at the appropriate times. In the preferred embodiment, each device waits two clock cycles after ResetIn goes low before latching in a device ID number from BusData[0:3].

[0077] Persons skilled in the art recognize that longer device ID numbers could be distributed to devices by having each device read in multiple bytes from the bus and latch the values into the device ID register. Persons skilled in the art also recognize that there are alternative ways of getting device ID numbers to unique devices. For instance, a series of sequential numbers could be clocked along the ResetIn line and at a certain time each device could be instructed to latch the current reset shift register value into the device ID register.

[0078] The configuration master should choose and set an access time in each access-time register in each slave to a period sufficiently long to allow the slave to perform an actual, desired memory access. For example, for a normal DRAM access, this time must be longer than the row address strobe (RAS) access time. If this condition is not met, the slave may not deliver the correct data. The value stored in a slave access-time register is preferably one-half the number of bus cycles for which the slave device should wait before using the bus in response to a request. Thus an access time value of '1' would indicate that the slave should not access the bus until at least two cycles after the last byte of the request packet has been received. The value of AccessReg0 is preferably fixed at 8 (cycles) to facilitate access to control registers.

[0079] The bus architecture of this invention can include more than one master device. The reset or initialization sequence should also include a determination of whether there are multiple masters on the bus, and if so to assign unique master ID numbers to each. Persons skilled in the art will recognize that there are many ways of doing this. For instance, the master could poll each device to determine what type of device it is, for example, by reading a special register then, for each master device, write the next available master ID number into a special register.

ECC

[0080] Error detection and correction ("ECC") methods well known in the art can be implemented in this system. ECC information typically is calculated for a block of data at the time that block of data is first written into memory. The data block usually has an integral binary size, e.g. 256 bits, and the ECC information uses significantly fewer bits. A potential problem arises in that each binary data block in prior art schemes typically is stored with the ECC bits appended, resulting in a block size that is not an integral binary power.

[0081] In a preferred embodiment of this invention, ECC information is stored separately from the corresponding data, which can then be stored in blocks having integral binary size. ECC information and corresponding data can be stored, for example, in separate DRAM devices. Data can be read without ECC using a single request packet, but to write or read error-corrected data requires two request packets, one for the data and a second for the corresponding ECC information. ECC information may not always be stored permanently and in some situations the ECC information may be available without sending a request packet or without a bus data block transfer.

[0082] In a preferred embodiment, a standard data block size can be selected for use with ECC, and the ECC method will determine the required number of bits of information in a corresponding ECC block. RAMs containing ECC information

can be programmed to store an access time that is equal to: (1) the access time of the normal RAM (containing data) plus the time to access a standard data block (for corrected data) minus the time to send a request packet (6 bytes); or (2) the access time of a normal RAM minus the time to access a standard ECC block minus the time to send a request packet. To read a data block and the corresponding ECC block, the master simply issues a request for the data immediately followed by a request for the ECC block. The ECC RAM will wait for the selected access time then drive its data onto the bus right after (in case (1) above)) the data RAM has finished driving out the data block. Persons skilled in the art will recognize that the access time described in case (2) above can be used to drive ECC data before the data is driven onto the bus lines and will recognize that writing data can be done by analogy with the method described for a read. Persons skilled in the art will also recognize the adjustments that must be made in the bus-busy structure and the request packet arbitration methods of this invention in order to accommodate these paired ECC requests.

[0083] Since this system is quite flexible, the system designer can choose the size of the data blocks and the number of ECC bits using the memory devices of this invention. Note that the data stream on the bus can be interpreted in various ways. For instance the sequence can be 2^m data bytes followed by 2^m ECC bytes (or vice versa), or the sequence can be 2^k iterations of 8 data bytes plus 1 ECC byte. Other information, such as information used by a directory-based cache coherence scheme, can also be managed this way. See, for example, Anant Agarwal, et al., "Scaleable Directory Schemes for Cache Consistency," 15th International Symposium on Computer Architecture, June 1988, pp. 280-289. Those skilled in the art will recognize alternative methods of implementing ECC schemes that are within the teachings of this invention.

Low Power 3-D Packaging

[0084] Another major advantage of this invention is that it drastically reduces the memory system power consumption. Nearly all the power consumed by a prior art DRAM is dissipated in performing row access. By using a single row access in a single RAM to supply all the bits for a block request (compared to a row-access in each of multiple RAMs in conventional memory systems) the power per bit can be made very small. Since the power dissipated by memory devices using this invention is significantly reduced, the devices potentially can be placed much closer together than with conventional designs.

[0085] The bus architecture of this invention makes possible an innovative 3-D packaging technology. By using a narrow, multiplexed (time-shared) bus, the pin count for an arbitrarily large memory device can be kept quite small - on the order of 20 pins. Moreover, this pin count can be kept constant from one generation of DRAM density to the next. The low power dissipation allows each package to be smaller, with narrower pin pitches (spacing between the IC pins). With current surface mount technology supporting pin pitches as low as 20 mils, all off-device connections can be implemented on a single edge of the memory device. Semiconductor die useful in this invention preferably have connections or pads along one edge of the die which can then be wired or otherwise connected to the package pins with wires having similar lengths. This geometry also allows for very short leads, preferably with an effective lead length of less than 4 mm. Furthermore, this invention uses only bused interconnections, i.e., each pad on each device is connected by the bus to the corresponding pad of each other device.

[0086] The use of a low pin count and an edge-connected bus permits a simple 3-D package, whereby the devices are stacked and the bus is connected along a single edge of the stack. The fact that all of the signals are bused is important for the implementation of a simple 3-D structure. Without this, the complexity of the "backplane" would be too difficult to make cost effectively with current technology. The individual devices in a stack of the present invention can be packed quite tightly because of the low power dissipated by the entire memory system, permitting the devices to be stacked bumper-to-bumper or top to bottom. Conventional plastic-injection molded small outline (SO) packages can be used with a pitch of about 2.5 mm (100 mils), but the ultimate limit would be the device die thickness, which is about an order of magnitude smaller, 0.2-0.5 mm using current wafer technology.

Bus Electrical Description

[0087] By using devices with very low power dissipation and close physical packing, the bus can be made quite short, which in turn allows for short propagation times and high data rates. The bus of a preferred embodiment of the present invention consists of a set of resistor-terminated controlled impedance transmission lines which can operate up to a data rate of 500 MHz (2 ns cycles). The characteristics of the transmission lines are strongly affected by the loading caused by the DRAMs (or other slaves) mounted on the bus. These devices add lumped capacitance to the lines which both lowers the impedance of the lines and decreases the transmission speed. In the loaded environment, the bus impedance is likely to be on the order of 25 ohms and the propagation velocity about $c/4$ (c = the speed of light) or 7.5 cm/ns. To operate at a 2 ns data rate, the transit time on the bus should preferably be kept under 1 ns, to leave 1 ns for the setup and hold time of the input receivers (described below) plus clock skew. Thus the bus lines must be kept quite short, under about 8 cm for maximum performance. Lower performance systems may have much longer lines, e.g. a 4 ns bus

may have 24 cm lines (3 ns transit time, 1 ns setup and hold time).

[0088] In the preferred embodiment, the bus uses current source drivers. Each output must be able to sink 50 mA, which provides an output swing of about 500 mV or more. In the preferred embodiment of this invention, the bus is active low. The unasserted state (the high value) is preferably considered a logical zero, and the asserted value (low state) is therefore a logical 1. Those skilled in the art understand that the method of this invention can also be implemented using the opposite logical relation to voltage. The value of the unasserted state is set by the voltage on the termination resistors, and should be high enough to allow the outputs to act as current sources, while being as low as possible to reduce power dissipation. These constraints may yield a termination voltage about 2V above ground in the preferred implementation. Current source drivers cause the output voltage to be proportional to the sum of the sources driving the bus.

[0089] Referring to Fig.7, although there is no stable condition where two devices drive the bus at the same time, conditions can arise because of propagation delay on the wires where one device, A 41, can start driving its part of the bus 44 while the bus is still being driven by another device, B 42 (already asserting a logical 1 on the bus). In a system using current drivers, when B 42 is driving the bus (before time 46), the value at points 44 and 45 is logical 1. If B 42 switches off at time 46 just when A 41 switches on, the additional drive by device A 41 causes the voltage at the output 44 of A 41 to drop briefly below the normal value. The voltage returns to its normal value at time 47 when the effect of device B 42 turning off is felt. The voltage at point 45 goes to logical 0 when device B 42 turns off, then drops at time 47 when the effect of device A 41 turning on is felt. Since the logical 1 driven by current from device A 41 is propagated irrespective of the previous value on the bus, the value on the bus is guaranteed to settle after one time of flight (t_f) delay, that is, the time it takes a signal to propagate from one end of the bus to the other. If a voltage drive was used (as in ECL wired-ORing), a logical 1 on the bus (from device B 42 being previously driven) would prevent the transition put out by device A 41 being felt at the most remote part of the system, e.g., device 43, until the turnoff waveform from device B 42 reached device A 41 plus one time of flight delay, giving a worst case settling time of twice the time of flight delay.

25 Clocking

[0090] Clocking a high speed bus accurately without introducing error due to propagation delays can be implemented by having each device monitor two bus clock signals and then derive internally a device clock, the true system clock. The bus clock information can be sent on one or two lines to provide a mechanism for each bused device to generate an internal device clock with zero skew relative to all the other device clocks. Referring to Figure 8, in the preferred implementation, a bus clock generator 50 at one end of the bus propagates an early bus clock signal in one direction along the bus, for example on line 53 from left to right, to the far end of the bus. The same clock signal then is passed through the direct connection shown to a second line 54, and returns as a late bus clock signal along the bus from the far end to the origin, propagating from right to left. A single bus clock line can be used if it is left unterminated at the far end of the bus, allowing the early bus clock signal to reflect back along the same line as a late bus clock signal.

[0091] Figure 8b illustrates how each device 51, 52 receives each of the two bus clock signals at a different time (because of propagation delay along the wires), with constant midpoint in time between the two bus clocks along the bus. At each device 51, 52, the rising edge 55 of Clock1 53 is followed by the rising edge 56 of Clock2 54. Similarly, the falling edge 57 of Clock1 53 is followed by the falling edge 58 of Clock2 54. This waveform relationship is observed at all other devices along the bus. Devices which are closer to the clock generator have a greater separation between Clock1 and Clock2 relative to devices farther from the generator because of the longer time required for each clock pulse to traverse the bus and return along line 54, but the midpoint in time 59, 60 between corresponding rising or falling edges is fixed because, for any given device, the length of each clock line between the far end of the bus and that device is equal. Each device must sample the two bus clocks and generate its own internal device clock at the midpoint of the two.

[0092] Clock distribution problems can be further reduced by using a bus clock and device clock rate equal to the bus cycle data rate divided by two, that is, the bus clock period is twice the bus cycle period. Thus a 500 MHz bus preferably uses a 250 MHz clock rate. This reduction in frequency provides two benefits. First it makes all signals on the bus have the same worst case data rates -- data on a 500 MHz bus can only change every 2 ns. Second, clocking at half the bus cycle data rate makes the labeling of the odd and even bus cycles trivial, for example, by defining even cycles to be those when the internal device clock is 0 and odd cycles when the internal device clock is 1.

Multiple Buses

[0093] The limitation on bus length described above restricts the total number of devices that can be placed on a single bus. Using 2.5 mm spacing between devices, a single 8 cm bus will hold about 32 devices. Persons skilled in the art will recognize certain applications of the present invention wherein the overall data rate on the bus is adequate but memory or processing requirements necessitate a much larger number of devices (many more than 32). Larger systems can easily be built using the teachings of this invention by using one or more memory subsystems, designated primary

bus units, each of which consists of two or more devices, typically 32 or close to the maximum allowed by bus design requirements, connected to a transceiver device.

Referring to Figure 9, each primary bus unit can be mounted on a single circuit board 66, sometimes called a memory stick. Each transceiver device 19 in turn connects to a transceiver bus 65, similar or identical in electrical and other respects to the primary bus 18 described at length above. In a preferred implementation, all masters are situated on the transceiver bus so there are no transceiver delays between masters and all memory devices are on primary bus units so that all memory accesses experience an equivalent transceiver delay, but persons skilled in the art will recognize how to implement systems which have masters on more than one bus unit and memory devices on the transceiver bus as well as on primary bus units. In general, each teaching of this invention which refers to a memory device can be practiced using a transceiver device and one or more memory devices on an attached primary bus unit. Other devices, generically referred to as peripheral devices, including disk controllers, video controllers or I/O devices can also be attached to either the transceiver bus or a primary bus unit, as desired. Persons skilled in the art will recognize how to use a single primary bus unit or multiple primary bus units as needed with a transceiver bus in certain system designs.

[0094] The transceivers are quite simple in function. They detect request packets on the transceiver bus and transmit them to their primary bus unit. If the request packet calls for a write to a device on a transceiver's primary bus unit, that transceiver keeps track of the access time and block size and forwards all data from the transceiver bus to the primary bus unit during that time. The transceivers also watch their primary bus unit, forwarding any data that occurs there to the transceiver bus. The high speed of the buses means that the transceivers will need to be pipelined, and will require an additional one or two cycle delay for data to pass through the transceiver in either direction. Access times stored in masters on the transceiver bus must be increased to account for transceiver delay but access times stored in slaves on a primary bus unit should not be modified.

[0095] Persons skilled in the art will recognize that a more sophisticated transceiver can control transmissions to and from primary bus units. An additional control line, TrncvrRW can be bused to all devices on the transceiver bus, using that line in conjunction with the AddrValid line to indicate to all devices on the transceiver bus that the information on the data lines is: 1) a request packet, 2) valid data to a slave, 3) valid data from a slave, or 4) invalid data (or idle bus). Using this extra control line obviates the need for the transceivers to keep track of when data needs to be forwarded from its primary bus to the transceiver bus - all transceivers send all data from their primary bus to the transceiver bus whenever the control signal indicates condition 2) above. In a preferred implementation of this invention, if AddrValid and TrncvrRW are both low, there is no bus activity and the transceivers should remain in an idle state. A controller sending a request packet will drive AddrValid high, indicating to all devices on the transceiver bus that a request packet is being sent which each transceiver should forward to its primary bus unit. Each controller seeking to write to a slave should drive both AddrValid and TrncvrRW high, indicating valid data for a slave is present on the data lines. Each transceiver device will then transmit all data from the transceiver bus lines to each primary bus unit. Any controller expecting to receive information from a slave should also drive the TrncvrRW line high, but not drive AddrValid, thereby indicating to each transceiver to transmit any data coming from any slave on its primary local bus to the transceiver bus. A still more sophisticated transceiver would recognize signals addressed to or coming from its primary bus unit and transmit signals only at requested times.

[0096] An example of the physical mounting of the transceivers is shown in Figure 9. One important feature of this physical arrangement is to integrate the bus of each transceiver 19 with the original bus of DRAMs or other devices 15, 16, 17 on the primary bus unit 66. The transceivers 19 have pins on two sides, and are preferably mounted flat on the primary bus unit with a first set of pins connected to primary bus 18. A second set of transceiver pins 20, preferably orthogonal to the first set of pins, are oriented to allow the transceiver 19 to be attached to the transceiver bus 65 in much the same way as the DRAMs were attached to the primary bus unit. The transceiver bus can be generally planar and in a different plane, preferably orthogonal to the plane of each primary bus unit. The transceiver bus can also be generally circular with primary bus units mounted perpendicular and tangential to the transceiver bus.

[0097] Using this two level scheme allows one to easily build a system that contains over 500 slaves (16 buses of 32 DRAMs each). Persons skilled in the art can modify the device ID scheme described above to accommodate more than 256 devices, for example by using a longer device ID or by using additional registers to hold some of the device ID. This scheme can be extended in yet a third dimension to make a second-order transceiver bus, connecting multiple transceiver buses by aligning transceiver bus units parallel to and on top of each other and busing corresponding signal lines through a suitable transceiver. Using such a second-order transceiver bus, one could connect many thousands of slave devices into what is effectively a single bus.

Device Interface

[0098] The device interface to the high-speed bus can be divided into three main parts. The first part is the electrical interface. This part includes the input receivers, bus drivers and clock generation circuitry. The second part contains the address comparison circuitry and timing registers. This part takes the input request packet and determines if the request

is for this device, and if it is, starts the internal access and delivers the data to the pins at the correct time. The final part, specifically for memory devices such as DRAMs, is the DRAM column access path. This part needs to provide bandwidth into and out of the DRAM sense amps greater than the bandwidth provided by conventional DRAMs. The implementation of the electrical interface and DRAM column access path are described in more detail in the following sections. Persons skilled in the art recognize how to modify prior-art address comparison circuitry and prior-art register circuitry in order to practice the present invention.

Electrical Interface - Input/Output Circuitry

[0099] A block diagram of the preferred input/output circuit for address/data/control lines is shown in Figure 10. This circuitry is particularly well-suited for use in DRAM devices but it can be used or modified by one skilled in the art for use in other devices connected to the bus of this invention. It consists of a set of input receivers 71, 72 and output driver 76 connected to input/output line 69 and pad 75 and circuitry to use the internal clock 73 and internal clock complement 74 to drive the input interface. The clocked input receivers take advantage of the synchronous nature of the bus. To further reduce the performance requirements for device input receivers, each device pin, and thus each bus line, is connected to two clocked receivers, one to sample the even cycle inputs, the other to sample the odd cycle inputs. By thus de-multiplexing the input 70 at the pin, each clocked amplifier is given a full 2 ns cycle to amplify the bus low-voltage-swing signal into a full value CMOS logic signal. Persons skilled in the art will recognize that additional clocked input receivers can be used within the teachings of this invention. For example, four input receivers could be connected to each device pin and clocked by a modified internal device clock to transfer sequential bits from the bus to internal device circuits, allowing still higher external bus speeds or still longer settling times to amplify the bus low-voltage-swing signal into a full value CMOS logic signal.

[0100] The output drivers are quite simple, and consist of a single NMOS pulldown transistor 76. This transistor is sized so that under worst case conditions it can still sink the 50 mA required by the bus. For 0.8 micron CMOS technology, the transistor will need to be about 200 microns long. Overall bus performance can be improved by using feedback techniques to control output transistor current so that the current through the device is roughly 50 mA under all operating conditions, although this is not absolutely necessary for proper bus operation. An example of one of many methods known to persons skilled in the art for using feedback techniques to control current is described in Hans Schumacher, et al., "CMOS Subnanosecond True-ECL Output Buffer," J. Solid State Circuits, Vol. 25 (1), pp. 150-154 (Feb. 1990). Controlling this current improves performance and reduces power dissipation. This output driver which can be operated at 500 MHz, can in turn be controlled by a suitable multiplexer with two or more (preferably four) inputs connected to other internal chip circuitry, all of which can be designed according to well known prior art.

[0101] The input receivers of every slave must be able to operate during every cycle to determine whether the signal on the bus is a valid request packet. This requirement leads to a number of constraints on the input circuitry. In addition to requiring small acquisition and resolution delays, the circuits must take little or no DC power, little AC power and inject very little current back into the input or reference lines. The standard clocked DRAM sense amp shown in Figure 11 satisfies all these requirements except the need for low input currents. When this sense amp goes from sense to sample, the capacitance of the internal nodes 83 and 84 in Figure 11 is discharged through the reference line 68 and input 69, respectively. This particular current is small, but the sum of such currents from all the inputs into the reference lines summed over all devices can be reasonably large.

[0102] The fact that the sign of the current depends upon on the previous received data makes matters worse. One way to solve this problem is to divide the sample period into two phases. During the first phase, the inputs are shorted to a buffered version of the reference level (which may have an offset). During the second phase, the inputs are connected to the true inputs. This scheme does not remove the input current completely, since the input must still charge nodes 83 and 84 from the reference value to the current input value, but it does reduce the total charge required by about a factor of 10 (requiring only a 0.25V change rather than a 2.5V change). Persons skilled in the art will recognize that many other methods can be used to provide a clocked amplifier that will operate on very low input currents.

[0103] One important part of the input/output circuitry generates an internal device clock based on early and late bus clocks. Controlling clock skew (the difference in clock timing between devices) is important in a system running with 2 ns cycles, thus the internal device clock is generated so the input sampler and the output driver operate as close in time as possible to midway between the two bus clocks.

[0104] A block diagram of the internal device clock generating circuit is shown in Figure 12 and the corresponding timing diagram in Figure 13. The basic idea behind this circuit is relatively simple. A DC amplifier 102 is used to convert the small-swing bus clock into a full-swing CMOS signal. This signal is then fed into a variable delay line 103. The output of delay line 103 feeds three additional delay lines: 104 having a fixed delay; 105 having the same fixed delay plus a second variable delay; and 106 having the same fixed delay plus one half of the second variable delay. The outputs 107, 108 of the delay lines 104 and 105 drive clocked input receivers 101 and 111 connected to early and late bus clock inputs 100 and 110, respectively. These input receivers 101 and 111 have the same design as the receivers described

above and shown in Fig. 11. Variable delay lines 103 and 105 are adjusted via feedback lines 116, 115 so that input receivers 101 and 111 sample the bus clocks just as they transition. Delay lines 103 and 105 are adjusted so that the falling edge 120 of output 107 precedes the falling edge 121 of the early bus clock, Clock1 53, by an amount of time 128 equal to the delay in input sampler 101. Delay line 108 is adjusted in the same way so that falling edge 122 precedes the falling edge 123 of late bus clock, Clock2 54, by the delay 128 in input sampler 111.

[0105] Since the outputs 107 and 108 are synchronized with the two bus clocks and the output 73 of the last delay line 106 is midway between outputs 107 and 108, that is, output 73 follows output 107 by the same amount of time 129 that output 73 precedes output 108, output 73 provides an internal device clock midway between the bus clocks. The falling edge 124 of internal device clock 73 precedes the time of actual input sampling 125 by one sampler delay. Note that this circuit organization automatically balances the delay in substantially all device input receivers 71 and 72 (Fig. 10), since outputs 107 and 108 are adjusted so the bus clocks are sampled by input receivers 101 and 111 just as the bus clocks transition.

[0106] In the preferred embodiment, two sets of these delay lines are used, one to generate the true value of the internal device clock 73, and the other to generate the complement 74 without adding any inverter delay. The dual circuit allows generation of truly complementary clocks, with extremely small skew. The complement internal device clock is used to clock the 'even' input receivers to sample at time 127, while the true internal device clock is used to clock the 'odd' input receivers to sample at time 125. The true and complement internal device clocks are also used to select which data is driven to the output drivers. The gate delay between the internal device clock and output circuits driving the bus is slightly greater than the corresponding delay for the input circuits, which means that the new data always will be driven on the bus slightly after the old data has been sampled.

DRAM Column Access Modification

[0107] A block diagram of a conventional 4 MBit DRAM 130 is shown in Figure 15. The DRAM memory array is divided into a number of subarrays 150-157, for example, 8. Each subarray is divided into arrays 148, 149 of memory cells. Row address selection is performed by decoders 146. A column decoder 147A, 147B, including column sense amps on either side of the decoder, runs through the core of each subarray. These column sense amps can be set to precharge or latch the most-recently stored value, as described in detail above. Internal I/O lines connect each set of sense-amps, as gated by corresponding column decoders, to input and output circuitry connected ultimately to the device pins. These internal I/O lines are used to drive the data from the selected bit lines to the data pins (some of pins 131-145), or to take the data from the pins and write the selected bit lines. Such a column access path organized by prior art constraints does not have sufficient bandwidth to interface with a high speed bus. The method of this invention does not require changing the overall method used for column access, but does change implementation details. Many of these details have been implemented selectively in certain fast memory devices, but never in conjunction with the bus architecture of this invention.

[0108] Running the internal I/O lines in the conventional way at high bus cycle rates is not possible. In the preferred method, several (preferably 4) bytes are read or written during each cycle and the column access path is modified to run at a lower rate (the inverse of the number of bytes accessed per cycle, preferably 1/4 of the bus cycle rate). Three different techniques are used to provide the additional internal I/O lines required and to supply data to memory cells at this rate. First, the number of I/O bit lines in each subarray running through the column decoder 147 is increased, for example, to 16, eight for each of the two columns of column sense amps and the column decoder selects one set of columns from the "top" half 148 of subarray 150 and one set of columns from the "bottom" half 149 during each cycle, where the column decoder selects one column sense amp per I/O bit line. Second, each column I/O line is divided into two halves, carrying data independently over separate internal I/O lines from the left half 147A and right half 147B of each subarray (dividing each subarray into quadrants) and the column decoder selects sense amps from each right and left half of the subarray, doubling the number of bits available at each cycle. Thus each column decode selection turns on n column sense amps, where n equals four (top left and right, bottom left and right quadrants) times the number of I/O lines in the bus to each subarray quadrant (8 lines each $\times 4 = 32$ lines in the preferred implementation). Finally, during each RAS cycle, two different subarrays, e.g. 157 and 153, are accessed. This doubles again the available number of I/O lines containing data. Taken together, these changes increase the internal I/O bandwidth by at least a factor of 8. Four internal buses are used to route these internal I/O lines. Increasing the number of I/O lines and then splitting them in the middle greatly reduces the capacitance of each internal I/O line which in turn reduces the column access time, increasing the column access bandwidth even further.

[0109] The multiple, gated input receivers described above allow high speed input from the device pins onto the internal I/O lines and ultimately into memory. The multiplexed output driver described above is used to keep up with the data flow available using these techniques. Control means are provided to select whether information at the device pins should be treated as an address, and therefore to be decoded, or input or output data to be driven onto or read from the internal I/O lines.

[0110] Each subarray can access 32 bits per cycle, 16 bits from the left subarray and 16 from the right subarray. With

8 I/O lines per sense-amplifier column and accessing two subarrays at a time, the DRAM can provide 64 bits per cycle. This extra I/O bandwidth is not needed for reads (and is probably not used), but may be needed for writes. Availability of write bandwidth is a more difficult problem than read bandwidth because over-writing a value in a sense-amplifier may be a slow operation, depending on how the sense amplifier is connected to the bit line. The extra set of internal I/O lines provides some bandwidth margin for write operations.

[0111] Persons skilled in the art will recognize that many variations of the teachings of this invention can be practiced that still fall within the claims of this invention which follow.

[0112] It follows a list of further embodiments of the invention:

Embodiment 1. A memory subsystem comprising

two memory devices connected in parallel to a bus,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said memory devices,

said control information including device-select information,

said bus containing substantially fewer bus lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual memory devices.

Embodiment 2. The memory subsystem with the features of embodiment 1 wherein said bus contains at least 8 bus lines adapted to carry at least 16 address bits and at least 8 data bits.

Embodiment 3. The memory subsystem with the features of embodiment 1 wherein said bus also includes parallel lines for clock and power.

Embodiment 4. A system comprising

a memory subsystem with the features of embodiment 1 wherein each bus of said memory subsystem is connected to its own transceiver device,

a transceiver bus connecting said transceiver devices, and

a means for transferring information between each of said buses of said memory subsystems and said transceiver bus, whereby memory subsystems may be integrated into a larger system having more memory than an individual memory subsystem.

Embodiment 5. The system with the features of embodiment 4 having a plurality of memory subsystems.

Embodiment 6. The system with the features of embodiment 4 further comprising a master device connected to said transceiver bus.

Embodiment 7. The system with the features of embodiment 6 wherein said master device is selected from the group consisting of a central processing unit, a floating point unit and a direct memory access unit.

Embodiment 8. The system with the features of embodiment 4 further comprising a peripheral device connected to the transceiver bus, said peripheral device adapted for connection to other devices not on the bus.

Embodiment 9. The system with the features of embodiment 8 wherein said peripheral device is selected from the group consisting of an I/O interface port, a video controller and a disk controller.

Embodiment 10. The system with the features of embodiment 5 wherein said transceiver bus is in a different plane than the plane of the bus of each of said memory subsystems.

Embodiment 11. The system with the features of embodiment 5 wherein the bus of each memory subsystem lies substantially in a subsystem bus plane and said transceiver bus lies substantially in a plane orthogonal to said subsystem bus plane.

Embodiment 12. The system with the features of embodiment 4 having at least two transceiver buses, each transceiver bus having a plurality of memory subsystem buses connected through a first transceiver to said transceiver bus, each of said transceiver buses being further connected to a second transceiver adapted to interface to a second-order transceiver bus, whereby each transceiver bus is connected through said second transceiver to form a second-order transceiver bus unit.

Embodiment 13. A semiconductor subsystem bus for interconnecting semiconductor devices comprising

a plurality of semiconductor devices connected in parallel to a bus, at least one of said semiconductor devices being a memory device or a transceiver device which in turn is connected to a memory subsystem,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said semiconductor devices,

said control information including semiconductor device-select information,

said bus containing substantially fewer bus lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual semiconductor devices, and

at least one modifiable register in each of the semiconductor devices on said bus, said modifiable registers being accessible from said bus, whereby the subsystem can be configured using signals transmitted on said bus.

Embodiment 14. The semiconductor subsystem bus with the features of embodiment 13 wherein one type of modifiable

register is an access-time register designed to store a time delay after which a device may take some specified action on said bus.

Embodiment 15. The semiconductor subsystem bus with the features of embodiment 13 further comprising a semiconductor device having at least two access-time registers and

one of said access-time registers is permanently programmed to contain a fixed value and at least one of said access-time registers can be modified by information carried on said bus.

Embodiment 16. The semiconductor subsystem bus with the features of embodiment 13 further comprising a memory device having at least one discrete memory section and also having a modifiable address register adapted to store memory address information which corresponds to each said discrete memory section.

Embodiment 17. The semiconductor subsystem bus with the features of embodiment 16 wherein said memory address information comprises a pointer to said discrete memory section.

Embodiment 18. The semiconductor subsystem bus with the features of embodiment 16 wherein said discrete memory section has a top and a bottom and said memory address information comprises pointers to said top and said bottom.

Embodiment 19. The semiconductor subsystem bus with the features of embodiment 16 wherein said memory address information comprises

a pointer to said discrete memory section and

a range value indicating the size of said discrete memory section.

Embodiment 20. The semiconductor subsystem bus with the features of embodiment 16 wherein said address registers of each of said discrete memory sections of each of said memory devices connected to said bus are set to contain memory address information that is different for each discrete memory section and such that the highest memory address in each discrete memory section is one less than the lowest memory address in another discrete memory section, whereby memory may be organized into one or a small number of contiguous memory blocks.

Embodiment 21. The semiconductor subsystem bus with the features of embodiment 16 further comprising a means for testing each of said discrete memory sections of each of said memory devices for proper function, and

for each non-functional discrete memory section, a means for setting at least one address register which corresponds to said discrete memory section to indicate that said discrete memory section is non-functional,

for each functional discrete memory section, a means for setting at least one address register which corresponds to said discrete memory section to contain such corresponding address information.

Embodiment 22. The semiconductor subsystem bus with the features of embodiment 21 wherein said address registers corresponding to said discrete memory sections are set to provide one contiguous memory block within the subsystem.

Embodiment 23. The semiconductor subsystem bus with the features of embodiment 13 wherein one of said modifiable registers is a device identification register which can be modified to contain a value unique to that semiconductor device.

Embodiment 24. The semiconductor subsystem bus with the features of embodiment 23 wherein said device identification register is set to contain a unique value which is a function of the physical position of that semiconductor device either along said bus or in relationship to other semiconductor devices or said bus.

Embodiment 25. A bus subsystem comprising

two semiconductor devices connected in parallel to a bus, wherein one of said semiconductor devices is a master device, said master device including a means for initiating bus transactions,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said devices,

said control information including device-select information,

said bus containing substantially fewer lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual devices on said bus, whereby said master device initiates bus transactions which transfer information between said semiconductor devices on said bus.

Embodiment 26. The bus subsystem with the features of embodiment 25 wherein one of said semiconductor devices is a memory device connected to said bus, said memory device having at least one discrete memory section and also having a modifiable address register adapted to store memory address information which corresponds to each said discrete memory section.

Embodiment 27. The bus subsystem with the features of embodiment 26 wherein one of said semiconductor devices comprises a transceiver device connected in parallel to said bus and connected in parallel to a memory device on a bus other than said bus.

Embodiment 28. The bus subsystem with the features of embodiment 26 further including a means for said master device to request said memory device to prepare for a bus transaction by sending a request packet along said bus, said memory device and said master device each having a device-internal means to prepare to begin said bus transaction during a device-internal phase and further having a bus access means to effect said bus transaction during a bus access phase, said request packet including

a sequence of bytes containing address and control information,

said control information including information about the requested bus transaction and about the access time, which corresponds to a number of bus cycles, which needs to intervene before beginning said bus-access phase, and said address information pointing to at least one memory location within one of said discrete memory sections of said memory device.

Embodiment 29. The bus subsystem with the features of embodiment 28 wherein said memory device includes a means to read said control information and initiate said device-internal means at a time so as to complete said device-internal phase within said access time and begin said bus access phase after said number of bus cycles.

Embodiment 30. The bus subsystem with the features of embodiment 28 wherein said control information comprises an op code.

Embodiment 31. The bus subsystem with the features of embodiment 30 wherein said memory device includes sense amplifiers adapted to hold a bit of information or to precharge after a selected time and a means to transfer a data block during a data block transfer either reading data from said memory device or writing data into said memory device, and wherein said op code instructs said memory device to activate a response means, said response means including a means to

initiate a data block transfer,
select the size of said data block,
select the time to initiate said data block transfer,
access a control register, including reading from or writing to said control register,
precharge said sense amplifiers after each of said data block transfers is complete,
hold a bit of information in each of said sense amplifiers after each of said data block transfers is complete, or
select normal or page-mode access.

Embodiment 32. The bus subsystem with the features of embodiment 31 wherein said data block transfer comprises a read from or a write to memory within a single memory device.

Embodiment 33. The bus subsystem with the features of embodiment 28 further comprising a means for said master device to send control information to a specific one of said semiconductor devices on said bus by including in said request packet a device identification number unique to said semiconductor device.

Embodiment 34. The bus subsystem with the features of embodiment 28 further comprising a means for said master device to send control information to a selected one of said discrete memory portions by including in said request packet a specific memory address.

Embodiment 35. The bus subsystem with the features of embodiment 28 further comprising a means for said master device to send control information to substantially all semiconductor devices on said bus by including in said request packet a special device identification number which is recognized by said semiconductor devices.

Embodiment 36. The bus subsystem with the features of embodiment 28 wherein said control information specifies directly or indirectly the number of bus cycles for said master device and said memory device to wait before beginning said bus access phase.

Embodiment 37. The bus subsystem with the features of embodiment 36 wherein, for a data block transfer, said master device and said memory device use the same access time and same data block size regardless of whether said data block transfer is a read or write operation.

Embodiment 38. The bus subsystem with the features of embodiment 28 wherein said control information further includes a block-size value that encodes and specifies the size of the block of data to be transferred.

Embodiment 39. The bus subsystem with the features of embodiment 38 wherein said block-size value is encoded as a linear value for relatively small block sizes values and is encoded as a logarithmic value for relatively larger block sizes.

Embodiment 40. The bus subsystem with the features of embodiment 38 wherein said block-size value is encoded using four bits, and where the encoded value is

Encoded Value	Block Size (Bytes)
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	16

(continued)

Encoded Value	Block Size (Bytes)
10	32
11	64
12	128
13	256
14	512
15	1024

Embodiment 41. The bus subsystem with the features of embodiment 26 wherein said memory device is a DRAM device containing

a plurality of sense amplifiers,

a means to hold said sense amplifiers in an unmodified state after a read or write operation, leaving the device in page mode,

a means to precharge said sense amplifiers and

a means for selecting whether to precharge said sense amplifiers or to hold said sense amplifiers in an unmodified state.

Embodiment 42. The bus subsystem with the features of embodiment 28 wherein said request packet comprises an even number of bytes.

Embodiment 43. The bus subsystem with the features of embodiment 28 further including a means for generating and controlling a plurality of bus cycles, during which said bus carries said address, data and control information, and wherein alternate said bus cycles are designated odd cycles and even cycles, respectively, and wherein said request packet begins only on an even cycle.

Embodiment 44. The bus subsystem with the features of embodiment 28 further including a means for generating ECC information corresponding to a block of data and a means for using said ECC information to correct errors in storing or reading said block of data, wherein said ECC information may be stored separately from said block of data.

Embodiment 45. The bus subsystem with the features of embodiment 44 further comprising at least two of said memory devices wherein said ECC information and said corresponding block of data are stored in a first and a second said memory device, respectively, and said master device includes a means to write or read said block of data with error correction by sending separate ones of said request packets for said ECC information and for said corresponding block of data.

Embodiment 46. A bus subsystem comprising

a memory device and a master device connected in parallel on a bus,

a means for said master device to send a request packet and initiate a bus transaction and

a means for said master device to keep track of current and pending bus transactions,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said memory devices,

said bus containing substantially fewer lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual devices on said bus, whereby said master device initiates bus transactions which transfer information between devices on said bus and collisions on said bus are avoided because said master device avoids initiating bus transactions which would conflict with current or pending bus transactions.

Embodiment 47. The bus subsystem with the features of embodiment 46 having at least two of said master devices and including

a collision detecting means whereby a first said master device sending a first said request packet can detect a second said master device sending one of said colliding request packets, where one of said said colliding request packet may be sent simultaneous with the initial sending of or overlapping the sending of said first request packet, and

an arbitration means whereby said first and said second master devices select a priority order in which each of said master devices will be allowed to access said bus sequentially .

Embodiment 48. The bus subsystem with the features of embodiment 47 wherein each of said master devices has a master ID number and each of said request packets includes a master ID position which is a predetermined number of bits in a predetermined position in said request packet, and wherein said collision detection means comprises

a means included in each master device for sending a request packet including said master ID number of said master device in said master ID position of said request packet and

a means to detect a collision and invoke said arbitration means if any master device detects any other master ID number in said master ID position.

Embodiment 49. The bus subsystem with the features of embodiment 47 wherein each of said master devices includes

a means for sending a request packet,
a means for driving a selected bus line or lines during at least one selected bus cycle while said request packet is being sent,
a means for monitoring said selected bus line or lines to see if a said master device is sending a colliding request packet and
a means for informing all other master devices that a collision has occurred and for invoking said arbitration means.

Embodiment 50. The bus subsystem with the features of embodiment 47 wherein each of said master devices includes
a means, when sending a request packet, to drive a selected bus line or lines with a certain current during at least one
selected bus cycle,

a means for monitoring said selected bus line or lines for a greater than normal current to see if another master device
is driving that line or lines,

a means for detecting said greater than normal current, and
a means for informing all said master devices that a collision has occurred and for invoking said arbitration means.

Embodiment 51. The bus subsystem with the features of embodiment 47 wherein said arbitration means comprises

a means for initiating an arbitration cycle,

a means for allocating a single bus line to each master device during at least one selected bus cycle relative to the start
of said arbitration cycle,

a means for allocating each master device to a single bus line during one of said selected bus cycles if there are more
master devices than available bus lines,

a means for each of said master devices which sent a colliding request packet to drive said bus line allocated to said
master device during said selected bus cycle, and

a means in at least one of said master devices for storing information about which master devices sent a colliding request
packet,

whereby said master devices can monitor selected bus lines during said arbitration cycle and identify each said master
device which sent a colliding request packet.

Embodiment 52. The bus subsystem with the features of embodiment 47 wherein said arbitration means comprises

a means included in a first one of said master devices which sent colliding request packets for identifying each of said
master devices which sent colliding request packets,

a means for assigning a priority to each said master device which sent a colliding request packet, and

a means for allowing each said master device which sent a colliding request packet to access the bus sequentially
according to that priority.

Embodiment 53. The bus subsystem with the features of embodiment 52 wherein said priority is based on the physical
location of each of said master devices.

Embodiment 54. The bus subsystem with the features of embodiment 52 wherein said priority is based on said master
ID number of said master devices.

Embodiment 55. The bus subsystem with the features of embodiment 52 wherein each of said master devices includes
a means, when sending a colliding request packet, for deciding which master device can send the next request packet
in what order or at what time, whereby no master device may send a new request packet until responses to each pending
request packet have been completed or scheduled.

Embodiment 56. A bus subsystem comprising

a plurality of semiconductor devices connected in parallel to a bus,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by
said semiconductor devices,

said control information including device-select information,

said bus containing substantially fewer lines than the number of bits in a single address,

said bus carrying said device-select information without the need for separate device-select lines connected directly to
individual semiconductor devices,

said semiconductor devices including a reset means having an input and an output, the output of the reset means of
one semiconductor device being connected to the input of the reset means of the next semiconductor device in series.

Embodiment 57. The bus subsystem with the features of embodiment 56 further including system reset means comprising
a means for generating a first and a second reset signal,

a means for passing said first reset signal to a first of said semiconductor devices and then to subsequent ones of said
semiconductor devices in series and

a means for passing a second reset signal to said first semiconductor device and then to said subsequent semiconductor
devices in series,

said bus subsystem including one of said semiconductor devices containing

a device identification register adapted to contain a number unique to said semiconductor device within said bus sub-
system,

a device identification register setting means, and

a device reset means for resetting said semiconductor device to some desired, known reset state in response to said

first reset signal and for setting said device identification register in response to said second reset signal, whereby said bus subsystem can be reset to a known reset state with a unique device identification value in said device identification register of each of said semiconductor devices.

Embodiment 58. The bus subsystem with the features of embodiment 57 wherein said desired, known reset state is where all registers in the semiconductor device are cleared and the state machines are reset.

Embodiment 59. The bus subsystem with the features of embodiment 57 wherein said device identification register setting means comprises

a means for detecting said second reset signal,

a means for reading a device identification number from said bus lines at a specific time relative to said second reset signal and

a means for storing said device identification number in said device identification register of said semiconductor device.

Embodiment 60. The bus subsystem with the features of embodiment 57 wherein said second reset signal comprises multiple pulse sequences and wherein said device identification setting means includes

a means for interpreting said pulse sequences as a device identification number and

a means for storing said device identification number in said device identification register of said semiconductor device.

Embodiment 61. The bus subsystem with the features of embodiment 57 wherein said device reset means comprises an n -stage shift register capable of storing n -bit values, wherein said device reset means interprets a specific value in said shift register as said first reset signal and interprets a specific value in said shift register as said second reset signal.

Embodiment 62. The bus subsystem with the features of embodiment 57 wherein one of said semiconductor devices is a master device, said master device including a means for generating said first and said second reset signals.

Embodiment 63. The bus subsystem with the features of embodiment 57 wherein one of said semiconductor devices is a master device, said master device including

a master ID register,

a means for assigning a master ID number to said master device and

a means for storing said master ID number in said master ID register.

Embodiment 64. The bus subsystem with the features of embodiment 63 further comprising a second one of said master devices, and a means for a first one of said master devices to assign a master ID number to substantially all other said master devices, whereby said first master device assigns one of said master ID numbers to each of said master devices on said bus subsystem and each said master device stores said assigned master ID number in said master ID register.

Embodiment 65. The bus subsystem with the features of embodiment 57 wherein one of said semiconductor devices includes a device-type register adapted to contain an identifier characteristic of that type of semiconductor device, and one or more modifiable registers, at least one of which is an access-time register adapted for storing access times.

Embodiment 66. The bus subsystem with the features of embodiment 65 wherein one of said semiconductor devices is a master device having

a means for selecting a semiconductor device,

a means for reading said device-type register of said selected semiconductor device,

a means for determining the device type of said selected semiconductor device,

a means for determining access-time values appropriate for said selected semiconductor device and for storing said access-time values in said access-time registers of said selected semiconductor device, and

a means for selecting and storing other values appropriate for said selected semiconductor device in corresponding registers of said selected semiconductor device,

whereby said master device can select a semiconductor device, determine what type it is, and set said access-time and other registers to contain appropriate values.

Embodiment 67. The bus subsystem with the features of embodiment 66 further comprising a memory device having at least one discrete memory section and at least one modifiable address register adapted to store memory address information which corresponds to each of said discrete memory sections, and

said master device further comprising a means for selecting and testing each of said discrete memory sections and a means for storing address information in said address registers corresponding to each of said discrete memory sections, whereby said master device can test all said discrete memory sections and assign unique address values thereto.

Embodiment 68. A bus subsystem comprising

two semiconductor devices connected in parallel to a bus, one of said semiconductor devices being a master device, said bus including a plurality of bus data lines for carrying substantially all address, data and control information needed by said semiconductor devices,

said control information including device-select information,

said bus containing substantially fewer of said bus data lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual semiconductor devices,

wherein all of said bus data lines are terminated transmission lines and all of said address, data and control information

is carried on said bus data lines as a sequential series of bits in the form of low-voltage-swing signals.

Embodiment 69. The bus subsystem with the features of embodiment 68 further comprising a semiconductor device including a current-mode driver connected to drive one of said bus data lines.

Embodiment 70. The bus subsystem with the features of embodiment 69 further comprising a semiconductor device having a means to measure the voltage of said low-voltage-swing signals on a selected one of said bus data lines, whereby said semiconductor device can determine whether zero, one, or more than one of said current-mode drivers are driving said selected bus data line.

Embodiment 71. The bus subsystem with the features of embodiment 70 further comprising a semiconductor device having

a plurality of input receivers connected to one of said bus data lines, and a selection means for selecting said input receivers one by one to sense and store, one at a time, the bits of said sequential series of bits.

Embodiment 72. The bus subsystem with the features of embodiment 70 further comprising a semiconductor device having two input receivers connected to one of said bus data lines.

Embodiment 73. A bus subsystem comprising two semiconductor devices connected in parallel to a bus having a first and a second end, said bus including a bus clock line, said bus clock line having first and second ends corresponding to said first and second ends of said bus, respectively, a clock generator connected to said first end of said bus clock line to generate early bus clock signals with a normal rise time, and

signal return means at said second end of said bus clock line to return said early bus clock signals to said first end of said bus as corresponding late bus clock signals,

whereby each of said early bus clock signals will propagate from said clock generator along said clock line starting from said first end to said second end of said bus and then return at a later time to said first end of said bus as a corresponding late bus clock signal, whereby each semiconductor device on said bus can detect said early bus clock signals and said corresponding late bus clock signals.

Embodiment 74. The bus subsystem with the features of embodiment 73 further comprising a first and a second said bus clock line having first and second ends at said first and said second ends of said bus, respectively, wherein said signal return means directly connects said second ends of said first and said second bus clock lines whereby each of said early bus clock signals will propagate from said clock generator at said first end of said bus along said first bus clock line to said second end of said bus and then return on said second bus clock line to said first end of said bus as one of said corresponding late bus clock signals.

Embodiment 75. The bus subsystem with the features of embodiment 73 wherein said signal return means comprises said first bus clock line without a line terminator at said second end thereof whereby each of said early bus clock signals reaching said second end of said first bus clock line will be reflected back along said first bus clock line as said corresponding late bus clock signals.

Embodiment 76. The bus subsystem with the features of embodiment 73 further comprising a means for operating said bus in bus cycles timed to have a certain bus cycle frequency and a corresponding bus cycle period and

a means for operating said clock generator with a period of twice the bus cycle period.

Embodiment 77. The bus subsystem with the features of embodiment 76 wherein said bus cycle frequency is greater than approximately 50 MHz and less than or equal to approximately 500 MHz.

Embodiment 78. The bus subsystem with the features of embodiment 73 further including a semiconductor device having an internal device clock generating means to derive the midpoint time between said early and corresponding late bus clock signals and to generate an internal device clock synchronized to said midpoint time.

Embodiment 79. The bus subsystem with the features of embodiment 73 further including a semiconductor device having a low-skew clock generator circuit comprising

a first delay line having an input, an output and a basic delay and means for synchronizing the output of said first delay line with said early bus clock signal,

a second delay line having said basic delay plus a variable delay, said second delay line having an output and a means for synchronizing the output of said second delay line with said late bus clock signal, and

a third delay line having a third delay and a means to set said third delay midway between the delays of said first and second delay lines, said third delay line having an output which provides an internal device clock signal synchronized to a time halfway between said early and said late bus clock signals.

Embodiment 80. The bus subsystem with the features of embodiment 73 wherein said early and said late bus clock signals are low-voltage-swing signals that transition cyclically between low and high logical values, and further including a semiconductor device having a low-skew clock generator circuit comprising

a DC amplifier to convert said early and said late bus clock signals into full-swing logic signals,

a first variable delay line having a first variable delay and an input and an output, the input of said first variable delay

line being connected to said DC amplifier

a first, a second and a third additional delay line, each having an input and an output, the input of each of said additional delay lines being connected to the output of said first delay line,

said first additional delay line having a fixed delay,

5 said second additional delay line having said fixed delay plus a second variable delay, and

said third additional delay line having said fixed delay plus one half of said second variable delay,

a first clocked input receiver connected to sample said early bus clock signal and gated by said output of said first additional delay line,

10 a means for adjusting said first variable delay so said first clocked input receiver samples said early bus clock signal just as said early bus clock signal transitions,

a second clocked input receiver connected to sample said late bus clock signal and gated by said output of said second additional delay line,

a means for adjusting said second variable delay so said second clocked input receiver samples said late bus clock signal just as said late bus clock signal transitions,

15 whereby said output of said third additional delay line is synchronized to a time halfway between said outputs of said first and said second additional delay lines, and said output of said third additional delay line provides an internal device clock signal.

Embodiment 81. The bus subsystem with the features of embodiment 80 further comprising a semiconductor device having

20 a first one of said low-skew clock generator circuits which generates a "true" internal device clock signal and

a second one of said low-skew clock generator circuits connected to generate a "complement" internal device clock signal synchronized with but opposite in logical value to said "true" internal device clock signal.

Embodiment 82. A DRAM device designed to be connected to an external bus having a plurality of bus lines for carrying substantially all address, data and control information needed by said DRAM device as a sequential series of bits, said

25 control information including device-select information, said external bus containing substantially fewer said bus lines than the number of bits in a single address, and said bus carrying device-select information without the need for separate device-select lines connected directly to said DRAM device, said DRAM device comprising

an array of memory cells connected in rows and columns, each of said memory cells adapted to store one of said bits, a row address selection means for selecting one of said rows,

30 a column sense amp connected to each of said columns, each of said column sense amps adapted to latch one of said bits as a binary logical value or to precharge to a selected state,

a column decoding means connected to each of said column sense amps for selecting a plurality of said column sense amps for inputting one of said bits to or outputting one of said bits from said memory cells,

35 an internal I/O bus having a plurality of internal I/O lines wherein each of said internal I/O lines is connected to a plurality of said column sense amps, and

a plurality of bus connection means designed to connect said internal I/O lines to said external bus,

whereby a selected bit of said sequential series of bits can be transferred from said external bus to a selected one of said memory cells or said bit contained in a selected one of said memory cells can be transferred to said external bus.

Embodiment 83. The DRAM device with the features of embodiment 82 further comprising

40 an output driver connected to one said bus connection means,

an output multiplexer having an output connected to said output driver and a plurality of inputs, each of said inputs being connected to one of said internal I/O lines, and

a control means to select whether said output driver can drive said external bus,

45 whereby a plurality of memory cells are selected using said row address selection means and said column decoding means and a plurality of bits contained in said plurality of memory cells are output through said column sense amps to said internal I/O bus to said output multiplexer to said output driver to said external bus.

Embodiment 84. The DRAM device with the features of embodiment 82 further comprising

a plurality of input receivers connected to one of said bus data lines and to said internal I/O bus,

50 a selection means for selecting said input receivers one by one to sense and store, one at a time, the bits of said sequential series of bits, and

a control means to select whether an input receiver can drive said internal I/O bus, whereby a bit of said sequential series of bits is input from said external bus through one of said input receivers to one of said internal I/O lines to one of said column sense amps to one of said memory cells.

Embodiment 85. The DRAM device with the features of embodiment 82 further comprising

55 a first and a second half-array of said memory cells wherein each said row of said array of said memory cells is subdivided into two parts,

a first and a second one of said internal I/O buses connected to said column sense amps in said first and said second half-arrays, respectively, and

a column decoder means to gate selected ones of said column sense amps connected to said memory cells in a selected row of said first and said second half-arrays simultaneously.

Embodiment 86. The DRAM device with the features of embodiment 85 wherein said column decoder means selects sixteen column sense amps at a time.

Embodiment 87. The DRAM device with the features of embodiment 82 wherein said external bus operates at a certain speed and wherein said DRAM device includes four of said internal I/O buses, each of which operates at one-fourth the speed of said external bus.

Embodiment 88. The DRAM device with the features of embodiment 82 further comprising
a means for precharging one of said column sense amps to a precharged state from which a binary logical value can quickly be loaded into said column sense amp,
if said column sense amp contains a binary logical value, a means for latching the logical value currently contained in said column sense amp and
a means for instructing said DRAM device to precharge said column sense amp or latch said binary logical value in said column sense amp.

Embodiment 89. The DRAM device with the features of embodiment 88 further comprising a means for instructing said DRAM device to precharge said column sense amp without further instruction whenever said row address selection means selects a different one of said rows.

Embodiment 90. The DRAM device with the features of embodiment 88 further comprising a means for instructing said DRAM device to precharge said column sense amp without further instruction at a first or a second preselected time after latching the latest said binary logical value, said first preselected time being long enough for said DRAM to latch said binary logical value into said column sense amp and transfer said binary logical value into memory or onto one of said internal I/O lines, and said second preselected time being a variable which can be stored in said DRAM device whereby said DRAM can latch a binary logical value into said column sense amp for transferring said binary logical value into or out of a selected said memory cell, then precharge to allow a faster subsequent read or write.

Embodiment 91. A package containing
a semiconductor die having a side, circuitry and a plurality of connecting areas positioned along or near said side, spaced at a selected pitch and connected to said circuitry,
said package comprising a plurality of bus connecting means for connecting to a plurality of external bus lines, each of said external bus lines corresponding to one of said connecting areas, each of said bus connecting means being
positioned on a first side of said package,
connected to one said external bus line and to said corresponding connecting area on said semiconductor die, and spaced at a pitch substantially identical to said selected pitch of said connecting areas,
whereby each of said external bus lines can be connected to said corresponding connecting area on said semiconductor die by bus connection means positioned along a single side of said package.

Embodiment 92. The package with the features of embodiment 91 further comprising a plurality of said bus connecting means wherein each of said bus connecting means includes
a pin adapted for connection to one of said external bus lines and
a wire connecting said pin to one of said connecting areas on said semiconductor die,
said wire having an effective lead length less than about 4 millimeters and wherein the effective lead length of said wire of each of said bus connection means for said package is approximately equal.

Embodiment 93. A plurality of packages with the features of embodiment 91 wherein at least two of said semiconductor die are memory devices, each of said packages being generally flat, having a top and a bottom, and wherein said packages are physically secured adjacent and parallel to each other in a stack,
where a first one of said packages is adjacent to a second one of said packages in said stack, said top of said first package is substantially aligned with said bottom of said second package, and
said bus connecting means of each of said packages are substantially aligned and are lying substantially in a plane.

Embodiment 94. The plurality of packages with the features of embodiment 93 further comprising a plurality of stacks wherein each of said bus connecting means can be electrically connected to corresponding said bus connecting means in each of said stacks.

Embodiment 95. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and
at least one modifiable identification register accessible to said bus through said connection means, whereby data may

be transmitted to said register via said bus and enable said device thereafter to be uniquely identified.

Embodiment 96. The semiconductor device with the features of embodiment 95 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 97. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and at least one modifiable register to hold device address information, said modifiable register accessible to said bus through said connection means, whereby data may be transmitted to said register via said bus which enables said device thereafter to respond to a predetermined range of addresses.

Embodiment 98. The semiconductor device with the features of embodiment 97 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 99. The semiconductor device with the features of embodiment 98 wherein said memory device has at least one discrete memory section and also has at least one modifiable address register adapted to store memory address information which corresponds to each said discrete memory section.

Embodiment 100. The semiconductor device with the features of embodiment 99 wherein said memory address information comprises a pointer to said discrete memory section.

Embodiment 101. The semiconductor device with the features of embodiment 100 wherein said discrete memory section has a top and a bottom and said memory address information comprises pointers to said top and said bottom.

Embodiment 102. The semiconductor device with the features of embodiment 100 wherein said memory address information comprises

a pointer to said discrete memory section and

a range value indicating the size of said discrete memory section.

Embodiment 103. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data and control information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and at least one modifiable access-time register accessible to said bus through said connection means, whereby data may be transmitted to said register via said bus which establishes a predetermined amount of time that said semiconductor device thereafter must wait before using said bus in response to a request.

Embodiment 104. The semiconductor device with the features of embodiment 103 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 105. The semiconductor device with the features of embodiment 103 further comprising at least two access-time registers and one of said access-time registers is permanently programmed to contain a fixed value and at least one of said access-time registers can be modified by information carried on said bus.

Embodiment 106. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, and wherein each said bus line is a terminated transmission line, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and

a bus line driver capable of producing a low-voltage-swing signal on one of said terminated transmission lines.

Embodiment 107. The semiconductor device with the features of embodiment 106 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 108. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying

substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said bus further including at least one bus clock line for carrying early and late bus clock signals, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and
an internal device clock generating means which generates an internal device clock synchronized to a time halfway between said early and said late bus clock signals.

embodiment 109. The semiconductor device with the features of embodiment 108 wherein said bus further includes a first and a second one of said bus clock lines, said first bus clock line carries said early bus clock signal and said second bus clock line carries said late bus clock signal, said semiconductor device further comprising a means to detect said early bus clock signal on said first bus clock line and a means to detect said late bus clock signal on said second bus clock line.

Embodiment 110. The semiconductor device with the features of embodiment 109 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 111. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying as a sequential series of bits substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus,
a plurality of input receivers connected to one of said bus data lines and
a selection means for selecting said input receivers one by one to sense and store, one at a time, the bits of said sequential series of bits.

Embodiment 112. The semiconductor device with the features of embodiment 111 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 113. The semiconductor device with the features of embodiment 112 wherein two input receivers are connected to one of said bus lines.

Embodiment 114. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said bus system includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said system bus,
an internal input/output bus within said semiconductor device having more lines than said system bus, and
a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus.

Embodiment 115. The semiconductor device with the features of embodiment 114 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 116. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said system bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said system bus,
an internal input/output bus within said semiconductor device having more lines than said system bus,
a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus, and

at least one modifiable identification register accessible to said system bus through said connection means, whereby data may be transmitted to said register via said system bus and which enables said device thereafter to be uniquely identified.

Embodiment 117. The semiconductor device with the features of embodiment 116 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 118. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said system bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said system bus,

an internal input/output bus within said semiconductor device having more lines than said system bus,

a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus, and

at least one modifiable register to hold device address information, said modifiable register accessible to said system bus through said connection means, whereby data may be transmitted to said register via said system bus which enables said device thereafter to respond to a predetermined range of addresses.

Embodiment 119. The semiconductor device with the features of embodiment 118 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 120. The semiconductor device with the features of embodiment 119 wherein said memory device has at least one discrete memory section and also has at least one modifiable address register adapted to store memory address information which corresponds to each said discrete memory section.

Embodiment 121. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said system bus includes a plurality of bus lines for carrying substantially all address, data and control information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said system bus,

an internal input/output bus within said semiconductor device having more lines than said system bus,

a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus, and

at least one modifiable access-time register accessible to said system bus through said connection means, whereby data may be transmitted to said register via said system bus which establishes a predetermined amount of time that said semiconductor device thereafter must wait before using said system bus in response to a request.

Embodiment 122. The semiconductor device with the features of embodiment 121 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 123. The semiconductor device with the features of embodiment 121 further comprising at least two access-time registers and one of said access-time registers is permanently programmed to contain a fixed value and at least one of said access-time registers can be modified by information carried on said system bus.

Embodiment 124. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, wherein said address, data, control and device-select information is carried over said bus in the form of request packets and bus transactions, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus,

a means to receive said request packets over said bus,

a means to decode information in said request packets, and

a means to respond to said information in said request packets.

Embodiment 125. The semiconductor device with the features of embodiment 124 wherein said means to decode information in said request packet further comprises

a means to identify and decode said control information in said request packet,
a means to identify and decode said device-select information in said request packet,
a means to identify and decode said address information in said request packet and
a means to determine whether said control information or said address information instructs said semiconductor device
5 to begin a response.

Embodiment 126. The semiconductor device with the features of embodiment 124 wherein each of said bus transactions
is carried out in response to said address and said control information in one of said request packets, and wherein said
means to identify and decode information in said request packets includes a means to identify a sequence of bytes on
said bus as one of said request packets containing said address and said control information, said control information
10 including information about the type of said bus transaction being requested and the access time which needs to intervene
before beginning said bus transaction over said bus and said address and said control information includes device-
select information instructing one or more said semiconductor devices to respond to said address and said control
information.

Embodiment 127. The semiconductor device with the features of embodiment 124 further comprising
15 a plurality of sense amplifiers adapted to precharge to a selected state or to latch a bit of information,
a means to hold said sense amplifiers in an unmodified state after latching one of said bits of information,
a means to precharge said sense amplifiers and
a means for selecting whether said semiconductor device should precharge said sense amplifiers or should hold said
sense amplifiers in an unmodified state.

Embodiment with the features of embodiment 128. The semiconductor device 124 wherein said means to respond to
20 said information, where said information is control information, further comprises a means to
transfer a data block during a data block transfer, further including a means to
read data from said semiconductor device and
write data into said semiconductor device, and
25 initiate a data block transfer,
transfer a data block of a selected size,
transfer a data block at a selected time,
access a control register, including a means to read from or write to said control register, or
select normal or page-mode access.

Embodiment 129. The semiconductor device with the features of embodiment 124 further comprising a means to respond
30 to said information in said request packet if said information includes a device identification number unique to said
semiconductor device.

Embodiment 130. The semiconductor device with the features of embodiment 124 further comprising a means to respond
to said information in said request packet if said information includes a special device identification number which calls
35 for said semiconductor device to respond.

Embodiment 131. The semiconductor device with the features of embodiment 124 further comprising a means to respond
to said information in said request packet if said information includes an address unique to said semiconductor device.
Embodiment 132. The semiconductor device with the features of embodiment 124 further comprising a means to interpret
said control information and decode the time to wait before beginning said bus transaction over said bus.

40 Embodiment 133. The semiconductor device with the features of embodiment 124 further comprising a means to interpret
said control information and decode the size of a data block to transfer during one of said bus transactions.

Embodiment 134. The semiconductor device with the features of embodiment 124, 125, 126, 127, 128, 129, 130, 131,
132 or 133 wherein said semiconductor device is a memory device which connects substantially only to said bus and
sends and receives substantially all address, data and control information over said bus.

45 Embodiment 135. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of
semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying
substantially all address, data, control and device-select information needed by said semiconductor device for commu-
nication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus
lines than the number of bits in a single address, and carries device-select information for said semiconductor device
50 without the need for a separate device-select line connected directly to said individual semiconductor device, wherein
said address, data, control and device-select information is carried over said bus in the form of request packets and bus
transactions, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus,
a means to encode address and control information in said request packets and
55 a means to send said request packets over said bus.

Embodiment 136. The semiconductor device 135 further comprising a means to request a bus transaction wherein each
of said bus transactions is carried out in response to said address and said control information in one of said request
packets, and wherein said means to encode information in said request packets includes a means to mark a sequence

of bytes on said bus as one of said request packets, said control information including information about the type of said bus transaction being requested and the access time which needs to intervene before beginning said bus transaction over said bus and said address and said control information includes device-select information instructing one or more said semiconductor devices to respond to said address and said control information.

Embodiment 137. The semiconductor device with the features of embodiment 135 wherein one or more of said plurality of semiconductor devices has a unique device identification number, said semiconductor device further comprising a means to send control information to a specific one of said plurality of semiconductor devices by including in said request packet a selected said device identification number.

Embodiment 138. The semiconductor device with the features of embodiment 135 wherein each of said plurality of semiconductor devices is adapted to respond to a special device identification number, said semiconductor device further comprising a means to send control information to each of said plurality of semiconductor devices by including in said request packet said special device identification number.

Embodiment 139. The semiconductor device with the features of embodiment 135 wherein one or more of said plurality of semiconductor devices is a memory device having a plurality of addresses, said semiconductor device further comprising a means to send control information to a specific address or range of addresses in one of said plurality of semiconductor devices by including said specific address or range of addresses in said request packet.

Embodiment 140. The semiconductor device with the features of embodiment 135 wherein at least one of said request packets is a request packet requesting a bus transaction which is followed by a corresponding one of said bus transactions, said semiconductor device further comprising a means to encode said control information to specify directly or indirectly the time between the end of said request packet requesting a bus transaction and said corresponding bus transaction over said bus.

Embodiment 141. The semiconductor device with the features of embodiment 140 wherein one type of said bus transactions is a transfer of a data block, said semiconductor device further comprising a means to encode said control information to specify the size of said data block to transfer.

Embodiment 142. The semiconductor device with the features of embodiment 140 further comprising a means to keep track of current and pending bus transactions, whereby collisions on said bus are avoided because said semiconductor device avoids initiating bus transactions which would conflict with current or pending bus transactions.

Embodiment 143. The semiconductor device with the features of embodiment 135 wherein said semiconductor device is a first master device and one of said plurality of semiconductor devices is a second master device, further comprising a collision detecting means whereby said first master device when sending a first one of said request packets can detect said second master device sending a colliding one of said request packets, where said colliding request packet may be sent simultaneous with the initial sending of or overlapping the sending of said first request packet, and an arbitration means whereby said first and said second master devices select a priority order in which each of said master devices will be allowed to access said bus sequentially.

Embodiment 144. The semiconductor device with the features of embodiment 143 wherein said semiconductor device is a master device and at least one of said plurality of semiconductor devices is a master device, each of said master devices has a master ID number and each of said request packets includes a master ID position which is a predetermined number of bits in a predetermined position in said request packet, and wherein said collision detection means comprises a means for said semiconductor device to send its master ID number in said request packet and a means to detect a collision and invoke said arbitration means if said semiconductor device detects any other master ID number in said master ID position.

Embodiment 145. The semiconductor device with the features of embodiment 144 wherein said system bus architecture includes a means for carrying information on said bus during bus cycles, said semiconductor device further comprising a means for driving a selected bus line or lines during at least one selected bus cycle while sending each said request packet,

a means for monitoring said selected bus line or lines to see if another said master device is sending one of said colliding request packets and

a means for informing all said master devices that a collision has occurred and for invoking said arbitration means.

Embodiment 146. The semiconductor device with the features of embodiment 145 further comprising a means, when sending a request packet, for driving a selected bus line or lines with a certain current during at least one selected bus cycle,

a means for monitoring said selected bus line or lines for a greater than normal current to see if another said master device is driving that line or lines,

a means for detecting said greater than normal current, and

a means for informing all said master devices that a collision has occurred and for invoking said arbitration means.

Embodiment 147. The semiconductor device with the features of embodiment 143 wherein said arbitration means comprises

a means for initiating an arbitration cycle,

a means for allocating a single bus line to each said master device during at least one selected bus cycle relative to the start of said arbitration cycle,

a means for allocating each said master device to a single bus line during one of said selected bus cycles if there are more master devices than available bus lines,

a means for each of said master devices which sent one of said colliding request packets to drive said bus line allocated to said master device during said selected bus cycle, and

a means in at least one of said master devices for storing information about which master devices sent one of said colliding request packets,

whereby said master devices can monitor selected bus lines during said arbitration cycle and identify each said master device which sent one of said colliding request packets.

Embodiment 148. The semiconductor device with the features of embodiment 143 wherein said arbitration means comprises

a means for identifying each of said master devices which sent one of said colliding request packets,

a means for assigning a priority to each said master device which sent one of said colliding request packets, and

a means for allowing each said master device which sent one of said colliding request packets to access the bus sequentially according to that priority.

Embodiment 149. The semiconductor device with the features of embodiment 143 wherein said priority is based on the physical location of each of said master devices.

Embodiment 150. The semiconductor device with the features of embodiment 143 wherein said priority is based on said master ID number of said master devices.

Claims

1. A semiconductor device, the semiconductor device comprising:

connection means adapted to connect the semiconductor device to an external bus which is a part of a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to the external bus, wherein the external bus includes a plurality of bus lines for carrying substantially all address, data, control and device select information needed by the semiconductor device for communication with substantially every other semiconductor device connected to the external bus, and wherein the external bus has substantially fewer bus lines than the number of bits in a single address, wherein each bus line of the plurality of bus lines is a terminated transmission line, and wherein the external bus carries device-select information for the semiconductor device without the need for a separate device-select line connected directly to the semiconductor device, a bus line driver capable of producing a low-voltage-swing signal on one of the terminated transmission lines,

wherein the semiconductor device is a semiconductor transceiver device to interface multiple units to the bus, wherein the multiple units are DRAMs.

2. The semiconductor device of claim 1,

wherein the bus line driver is capable of producing a low-voltage-swing signal having a voltage swing of not more than 1 Volt on one of said terminated transmission lines.

3. The semiconductor device of claim 1 or 2,

wherein the bus line driver is capable of producing a low-voltage-swing signal having a voltage swing of at least 500 mV on one of said terminated transmission lines.

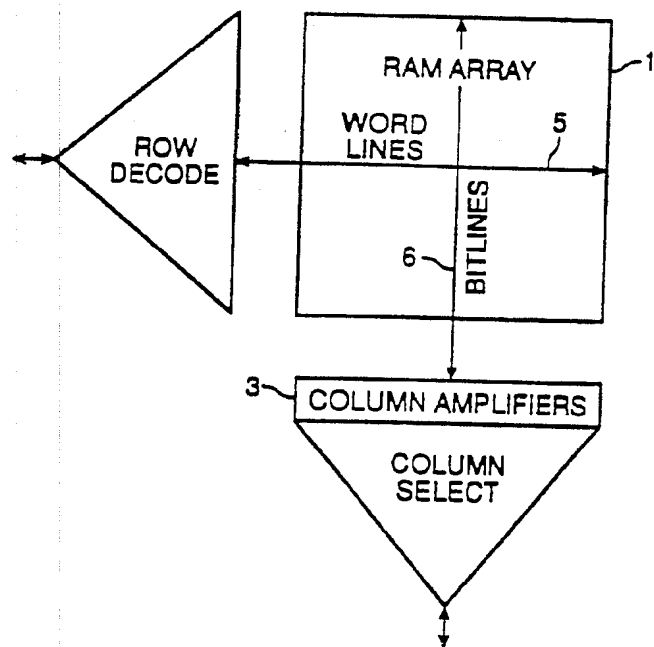


FIG. 1

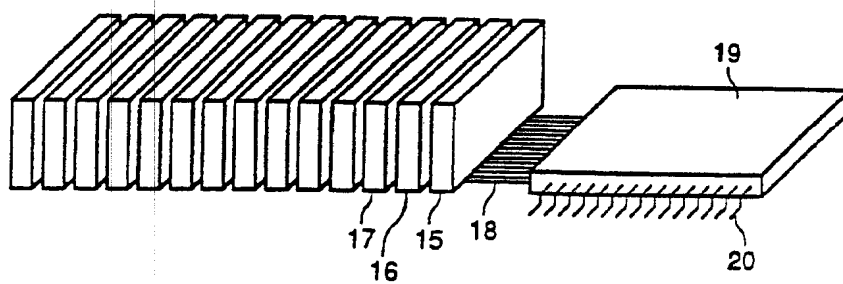


FIG. 3

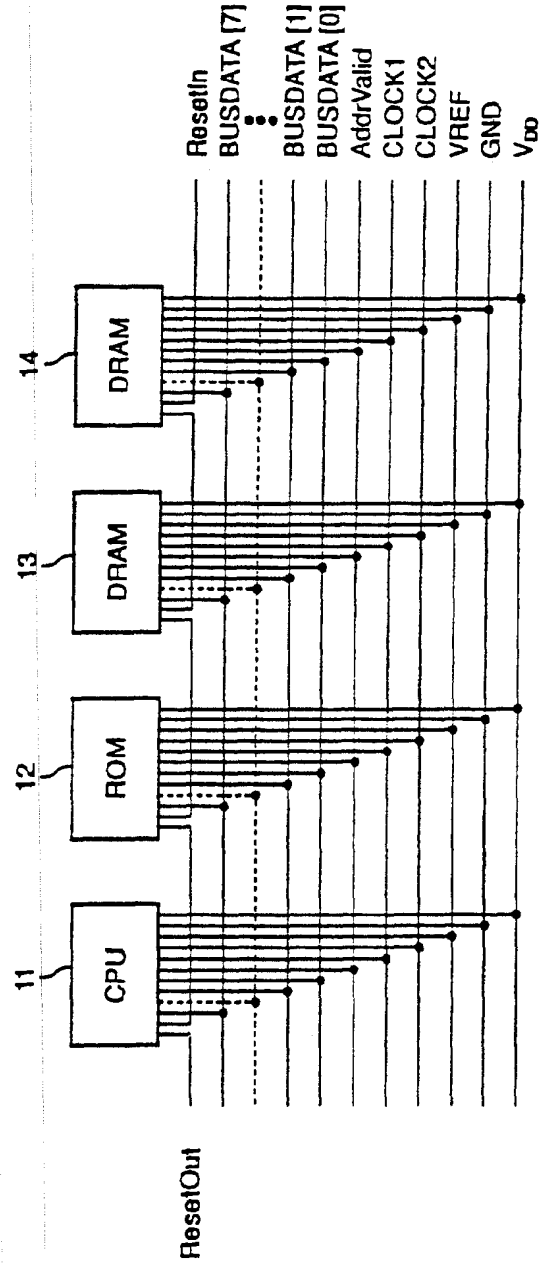


FIG. 2

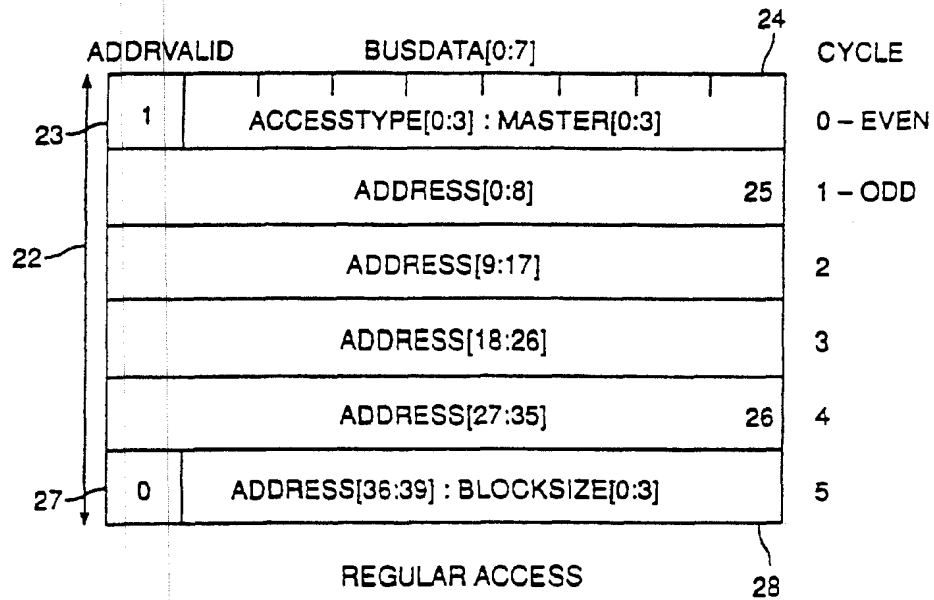


FIG. 4

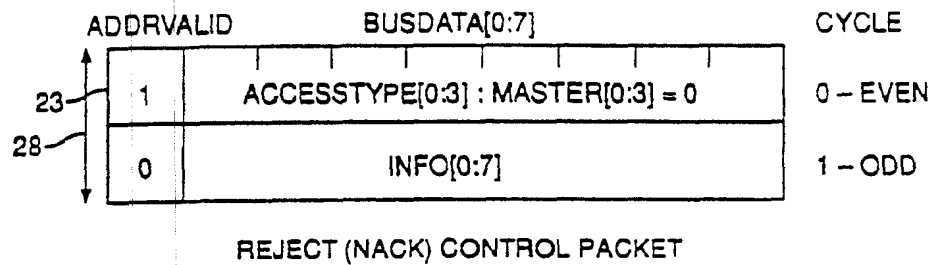


FIG. 5

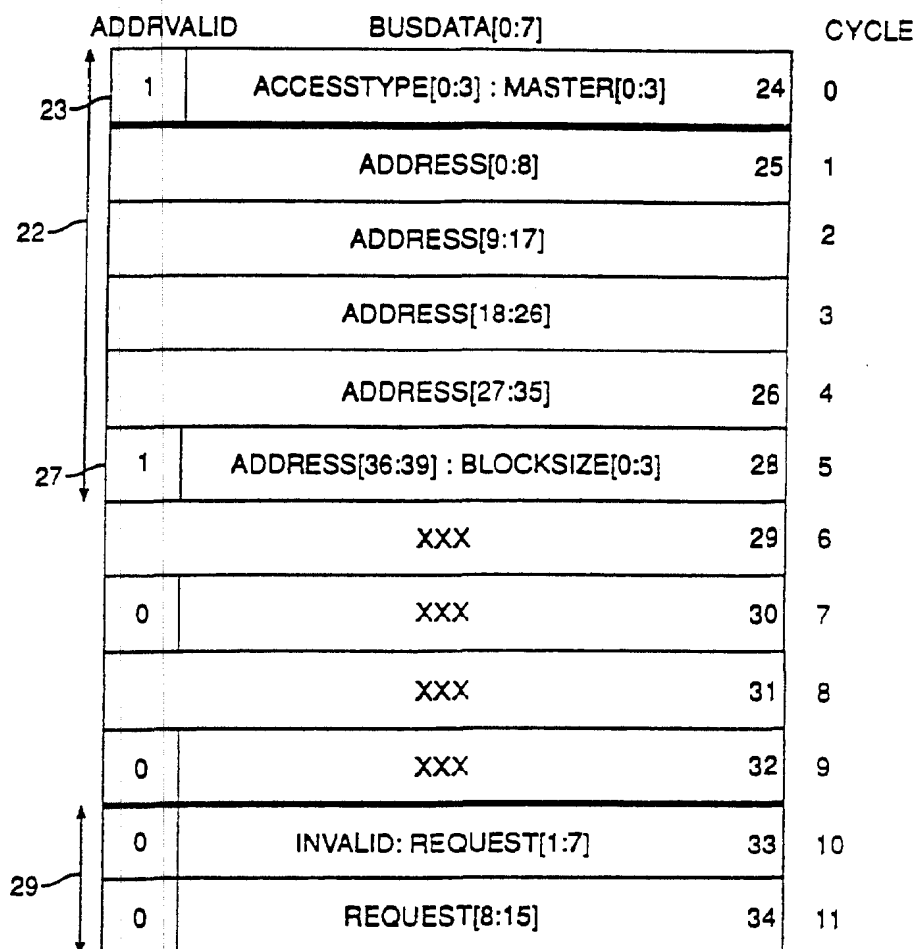


FIG. 6



FIG. 7a

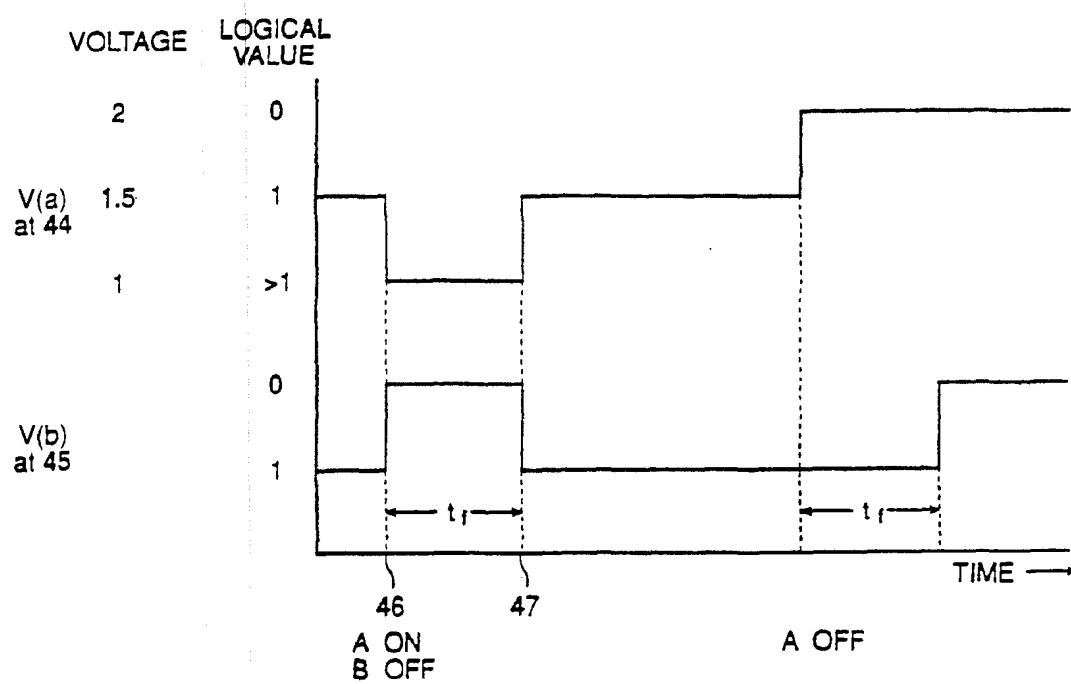


FIG. 7b

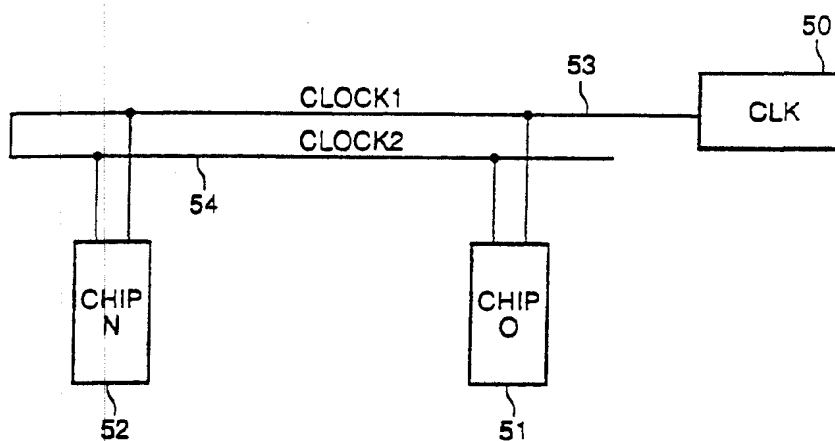


FIG. 8a

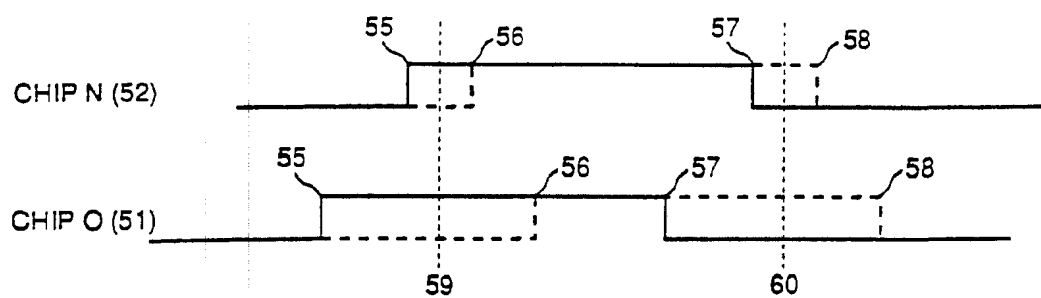


FIG. 8b

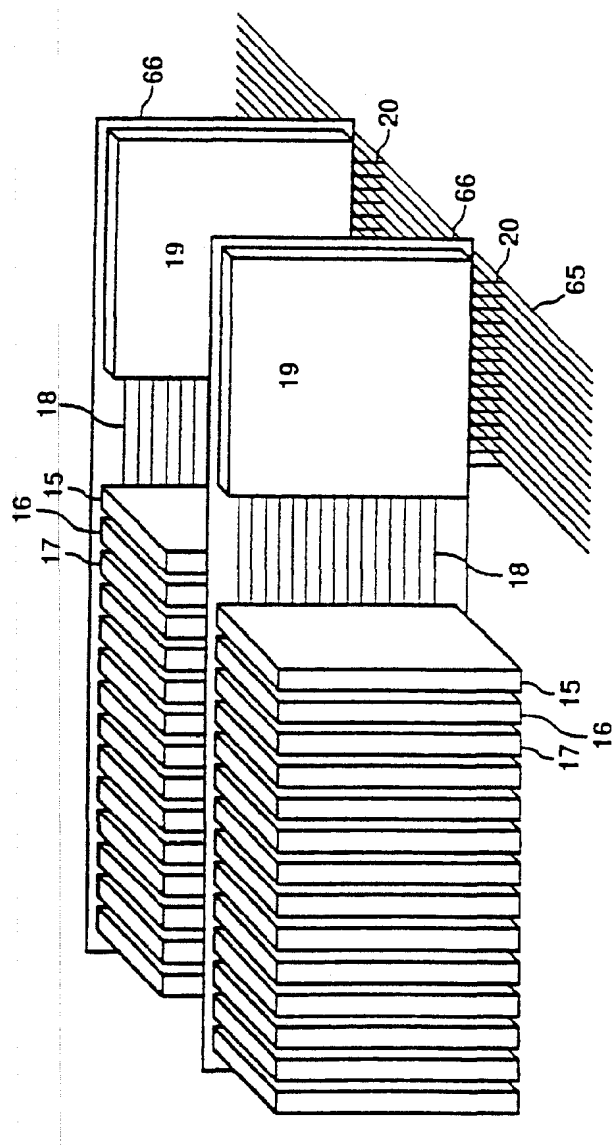


FIG. 9

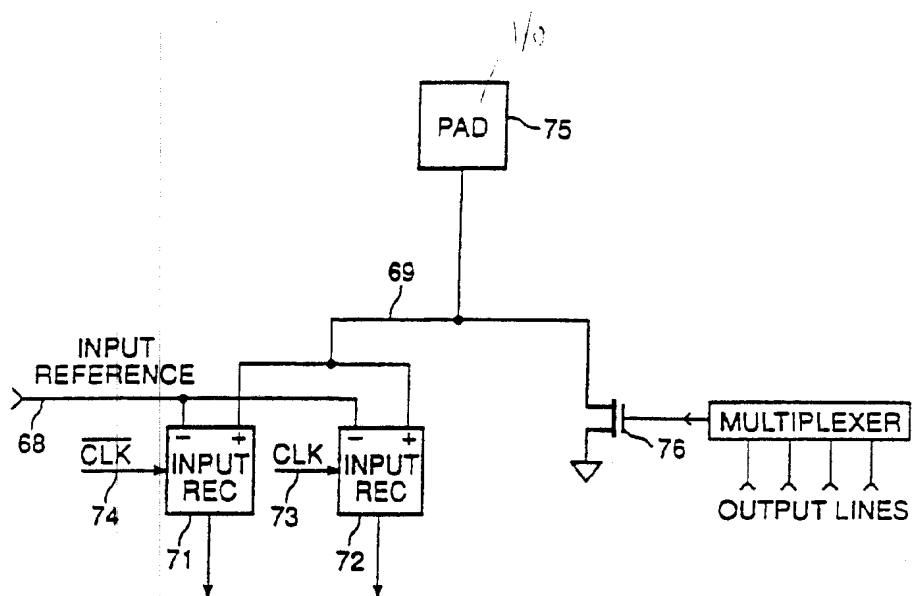


FIG. 10

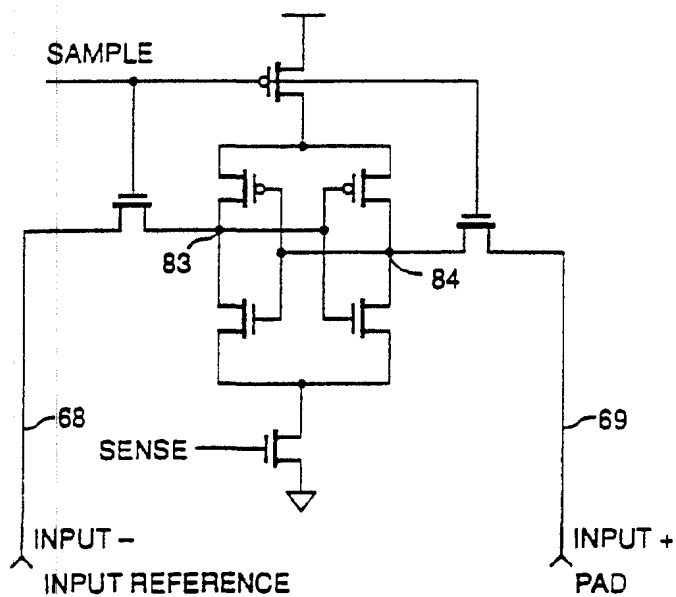


FIG. 11

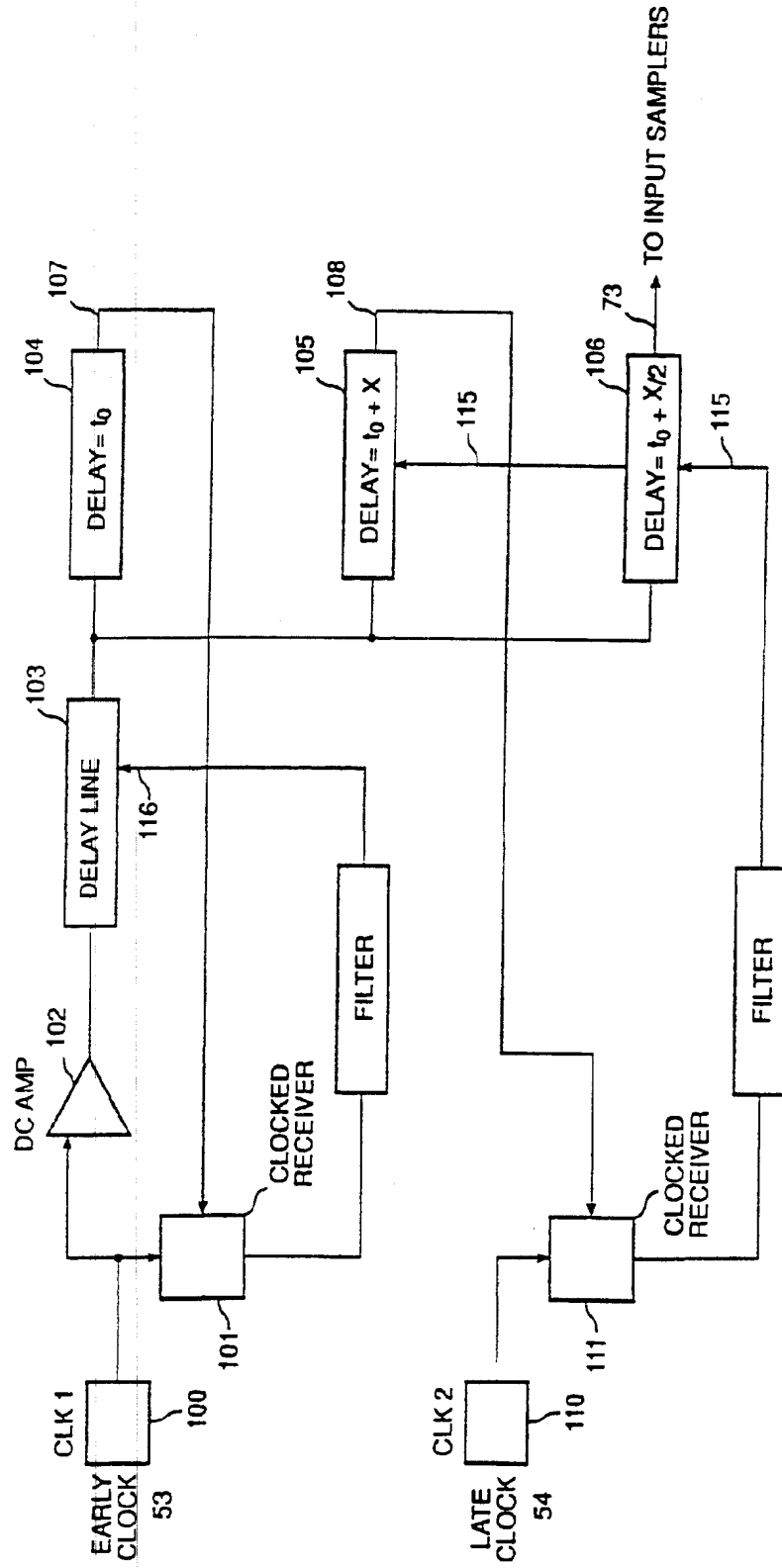


FIG. 12

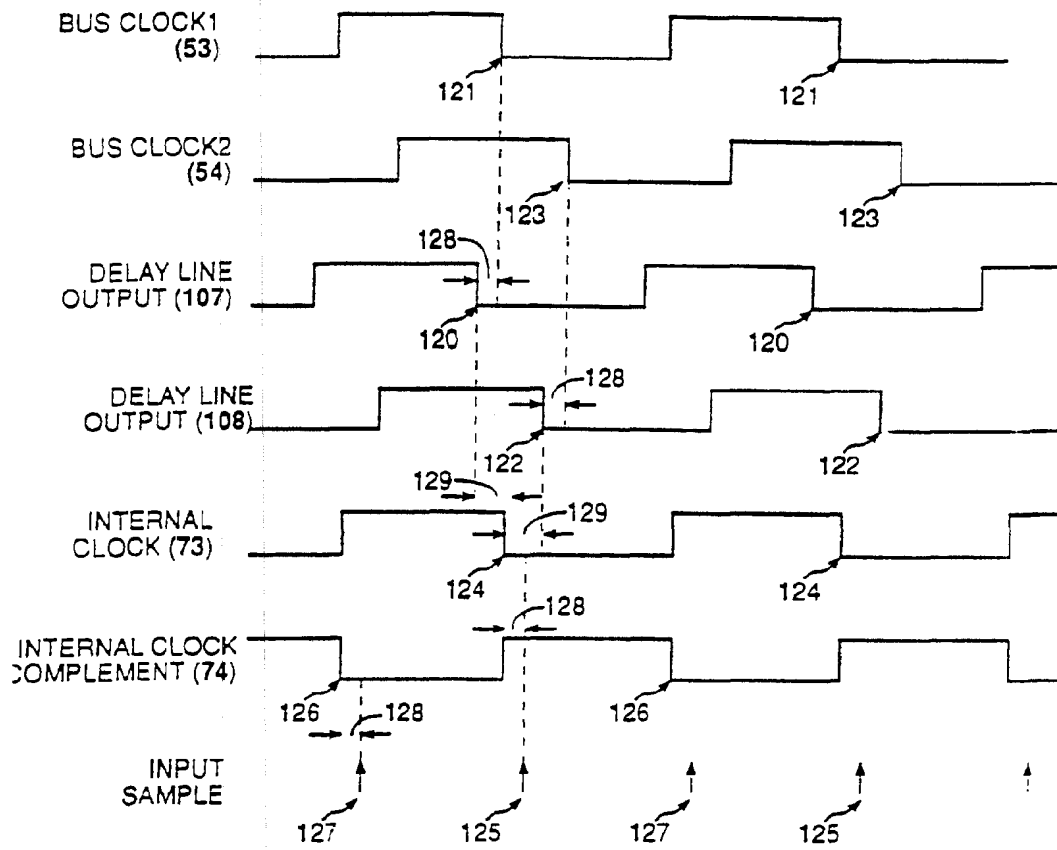


FIG. 13

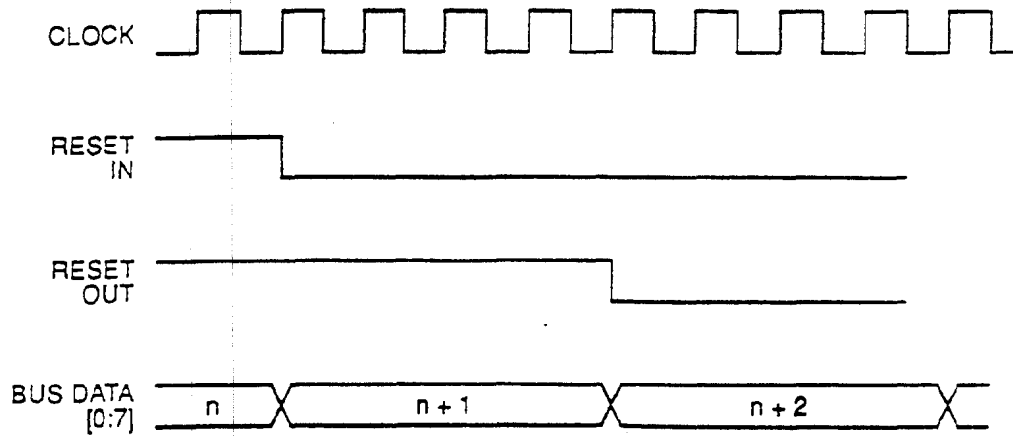


FIG. 14

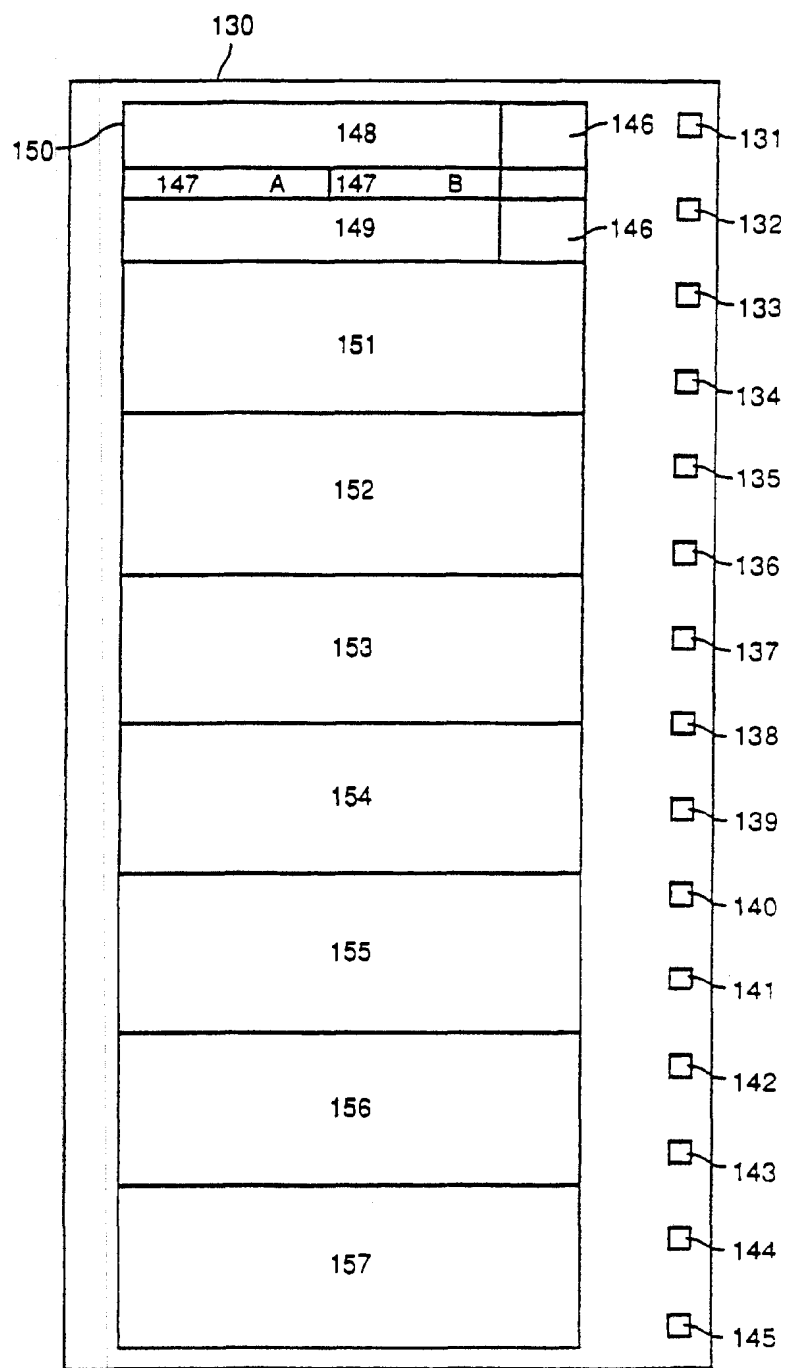


FIG. 15

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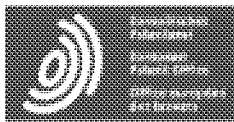
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Espacenet

Bibliographic data: JP2000311485 (A) — 2000-11-07

**MEMORY MODULE SYSTEM CONNECTING ONLY SELECTED MEMORY MODULE
TO DATA LINE AND DATA INPUT/OUTPUT METHOD UTILIZING THE SYSTEM**

No documents available for this priority number.

Inventor(s): KIN CHIKYOKU ± (KIN CHIKYOKU)

Applicant(s): SAMSUNG ELECTRONICS CO LTD ± (SAMSUNG ELECTRONICS
CO LTD)

Classification: - international: **G06F12/00; G06F12/06; G06F13/16; G11C11/401;
G11C11/407; G11C5/00; G11C7/10;** (IPC1-
7): G06F13/16; G11C11/401; G11C11/407
- cooperative: **G11C5/04; G11C7/1006**

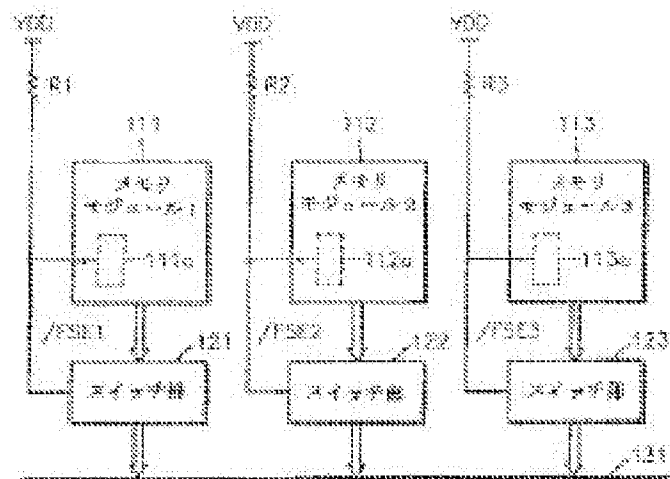
**Application
number:** JP20000072737 20000315

**Priority
number(s):** KR19990012027 19990407

**Also
published as:** JP4319320 (B2) GB2348722 (A) GB2348722 (B) US6526473 (B1)
TW475110 (B) KR20000065601 (A) FR2792105 (A1)
FR2792105 (B1) DE19960005 (A1) less

Abstract of JP2000311485 (A)

PROBLEM TO BE SOLVED: To improve data write and read rates by generating a connection control signal by activation while data having burst length specified by a write mode is inputted. **SOLUTION:** As resistors R1-R3 arranged at the outside of memory modules 1-3, 111-113 are made high resistance value, voltage levels of first to third connection control signals /FSE1-/FSE3 are decided by an internal signal of the memory module 1-3, 111-113 connected to the resistors, but when an internal signal is in a floating state, voltage levels of the first to third connection control signal /FSE1-/FSE3 are made external power source voltage VDD through the resistors R1-R3, that is, made a high state.; Therefore, when the module 111 is selected, second, third connection control signals /FSE2, /FSE3 are made non-activated and second, third switches 122, 124 are turned off, loads of data pins of modules 2, 3, 112, 113 are separated from a data base line 131.





US006526473B1

(12) **United States Patent**
Kim

(10) **Patent No.:** **US 6,526,473 B1**
(45) **Date of Patent:** **Feb. 25, 2003**

(54) **MEMORY MODULE SYSTEM FOR CONTROLLING DATA INPUT AND OUTPUT BY CONNECTING SELECTED MEMORY MODULES TO A DATA LINE**

(75) Inventor: **Chi-wook Kim**, Suwon (KR)

(73) Assignee: **Samsung Electronics Co., Ltd.**, Suwon (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **09/440,728**

(22) Filed: **Nov. 16, 1999**

(30) **Foreign Application Priority Data**

Apr. 7, 1999 (KR) 99-12027

(51) **Int. Cl.⁷** **G06F 12/02**

(52) **U.S. Cl.** **711/105; 711/5; 711/111; 711/209; 365/230.03; 710/131**

(58) **Field of Search** **711/5, 105, 111, 711/209; 365/230.03, 230.08; 710/131**

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Assistant Examiner—Jasmine Song

(74) *Attorney, Agent, or Firm*—F. Chau & Associates, LLP

(57) **ABSTRACT**

A memory module system for connecting only selected memory modules to a data line to control data input and output is disclosed. The memory module system has a multiplicity of memory modules for outputting data to a data bus line, and more particularly, only the memory modules outputting data is electrically connected to the data bus line in response to activation of a predetermined connection control signal. The connection control signal has an activation width corresponding to a burst length of the output data. Only selected memory modules are connected to the data line during the data burst length, so that load per data pin is minimized, to thereby improve speed of writing and reading data.

27 Claims, 13 Drawing Sheets

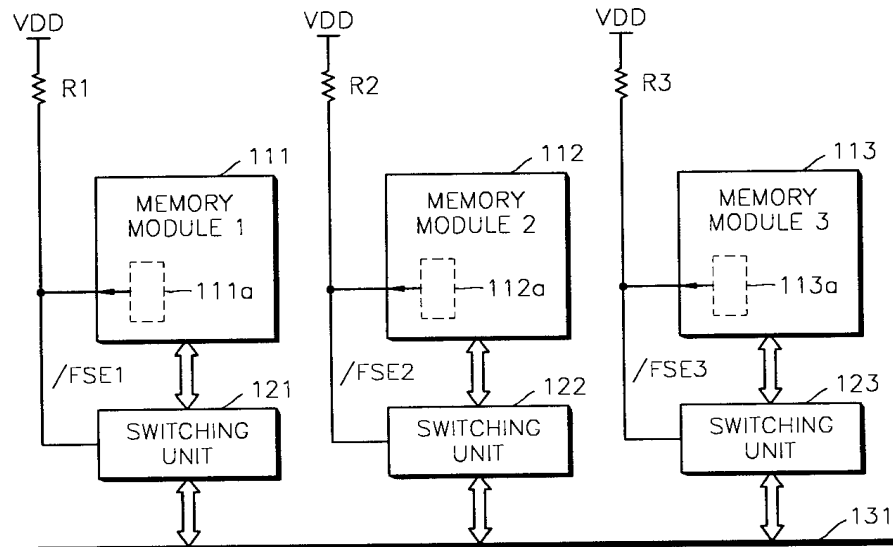


FIG. 1

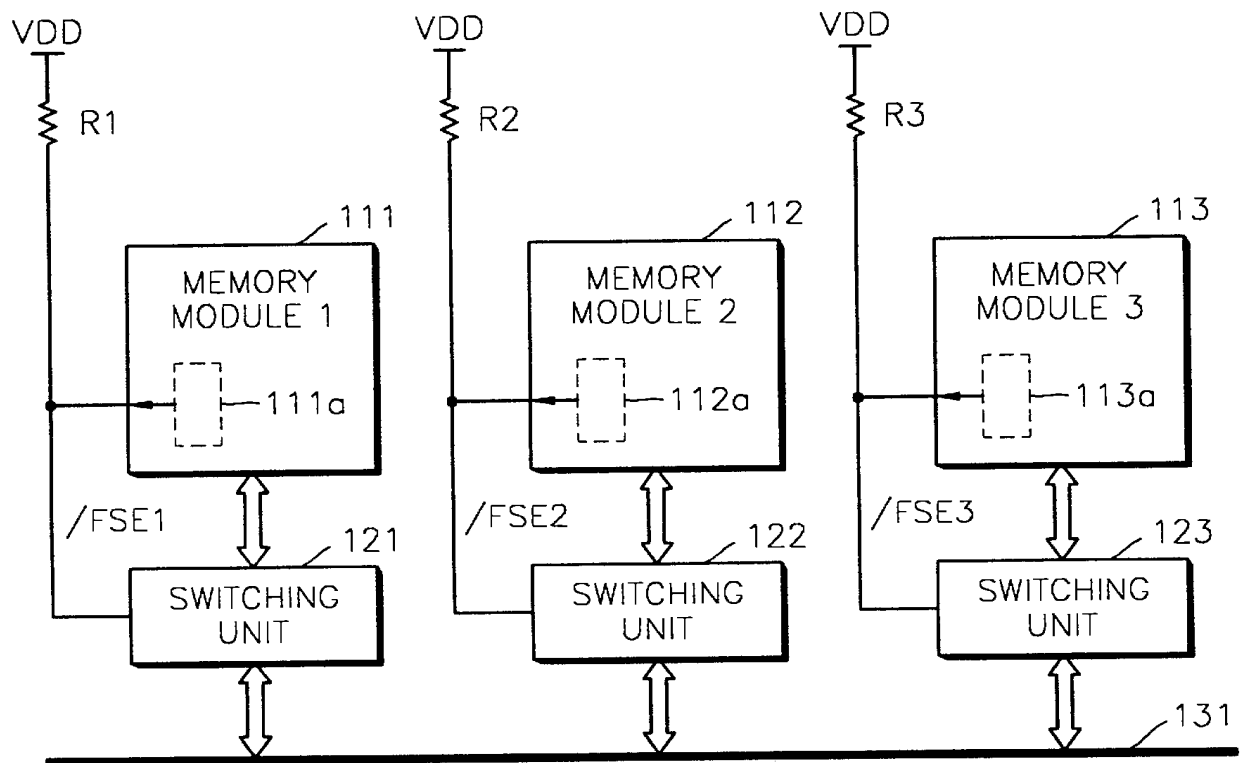


FIG. 2

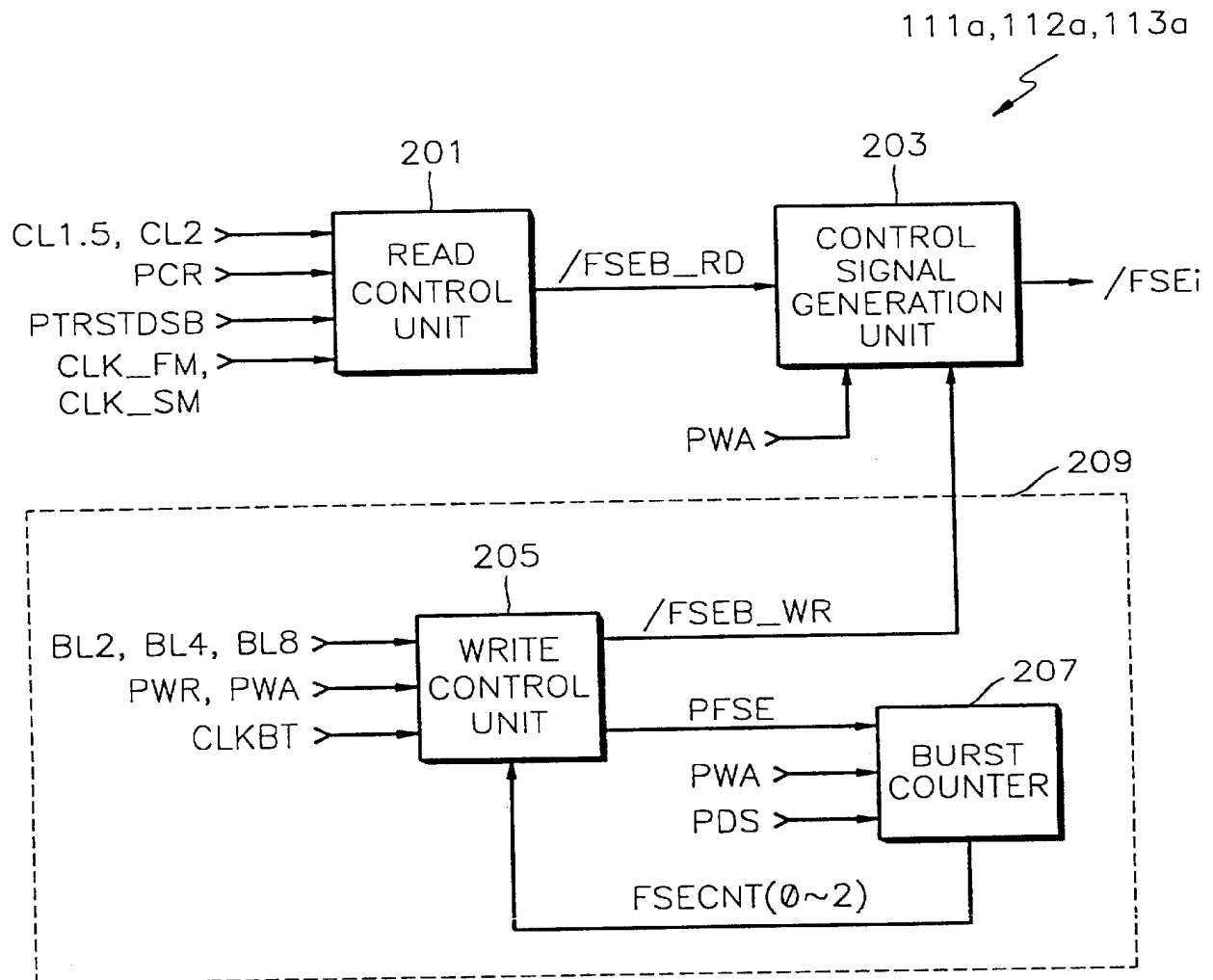


FIG. 3

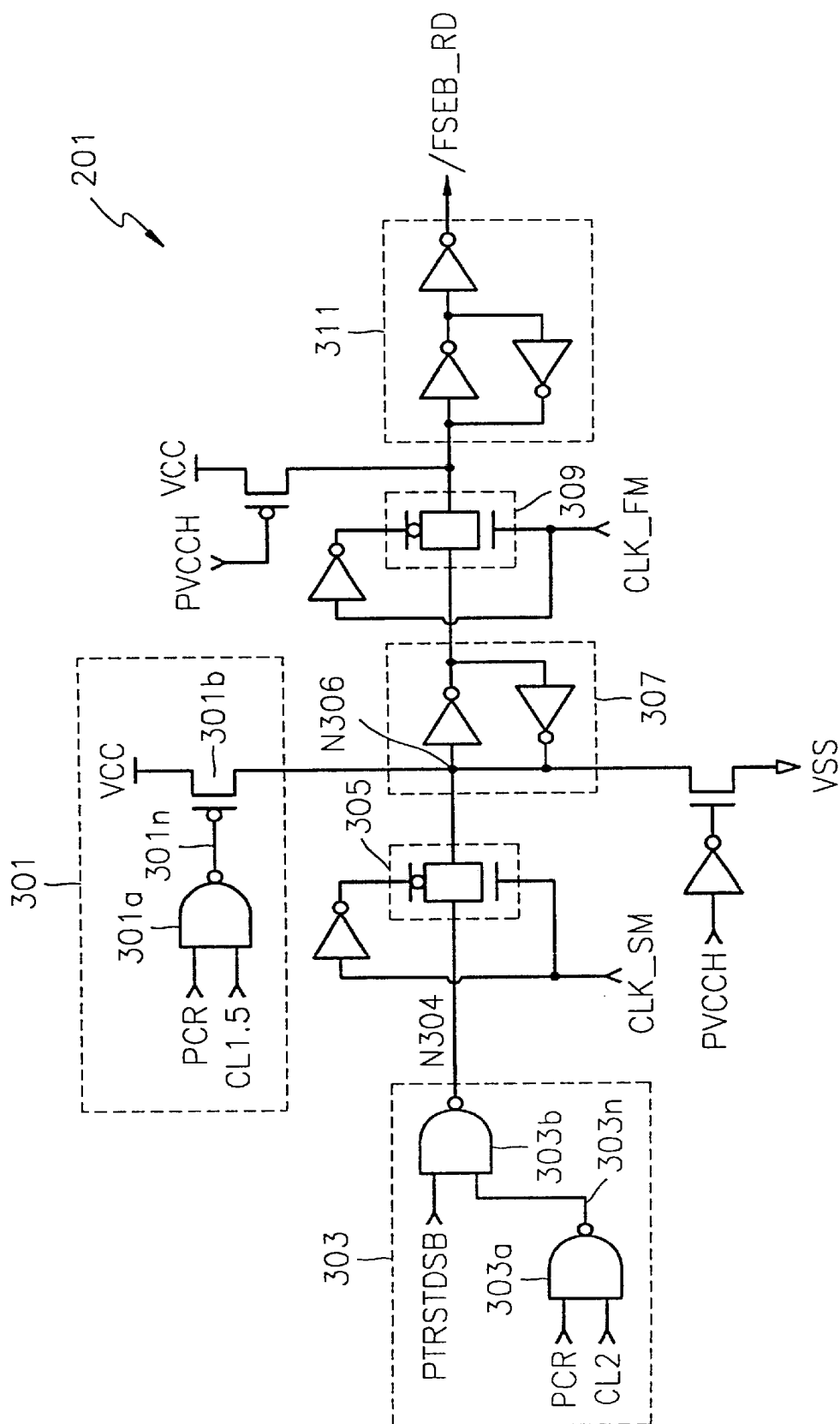


FIG. 4

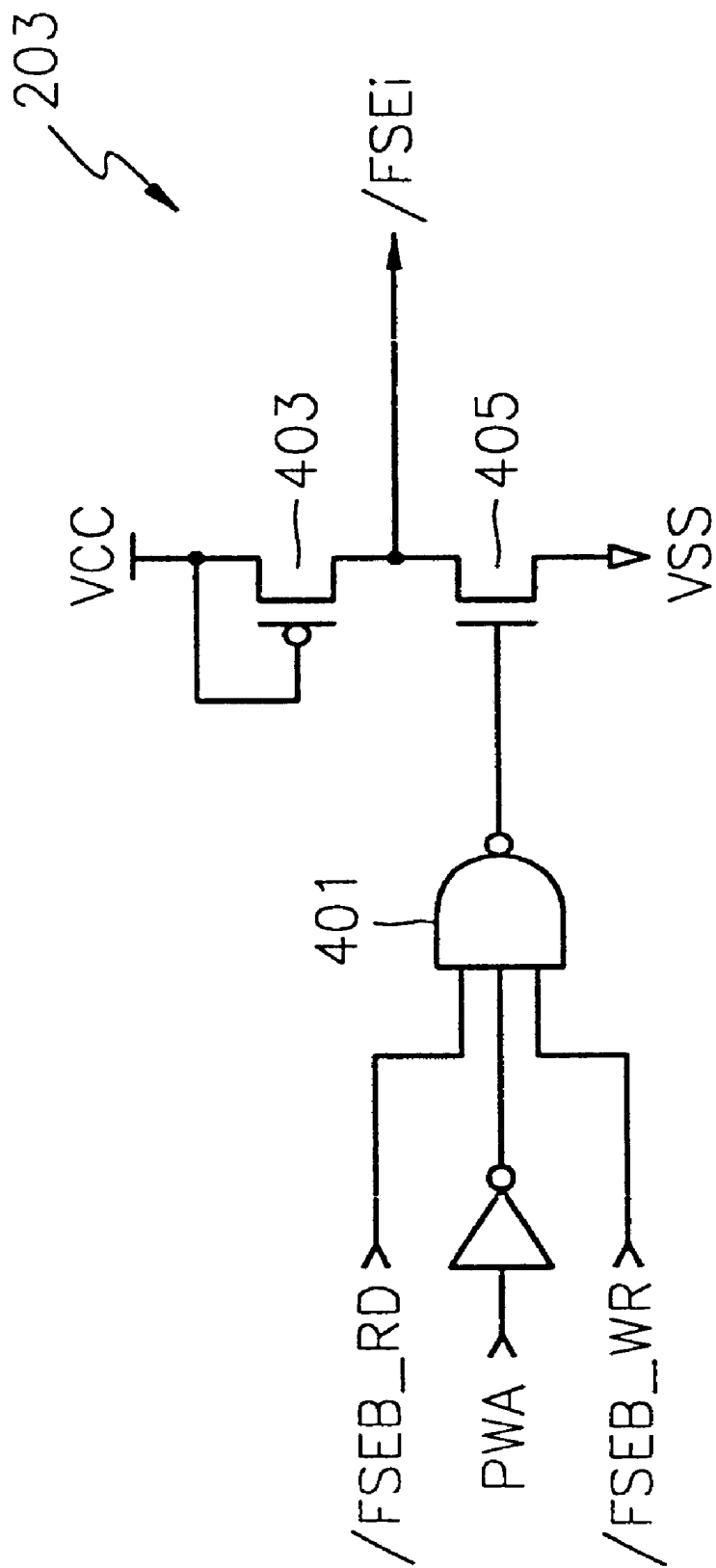


FIG. 5

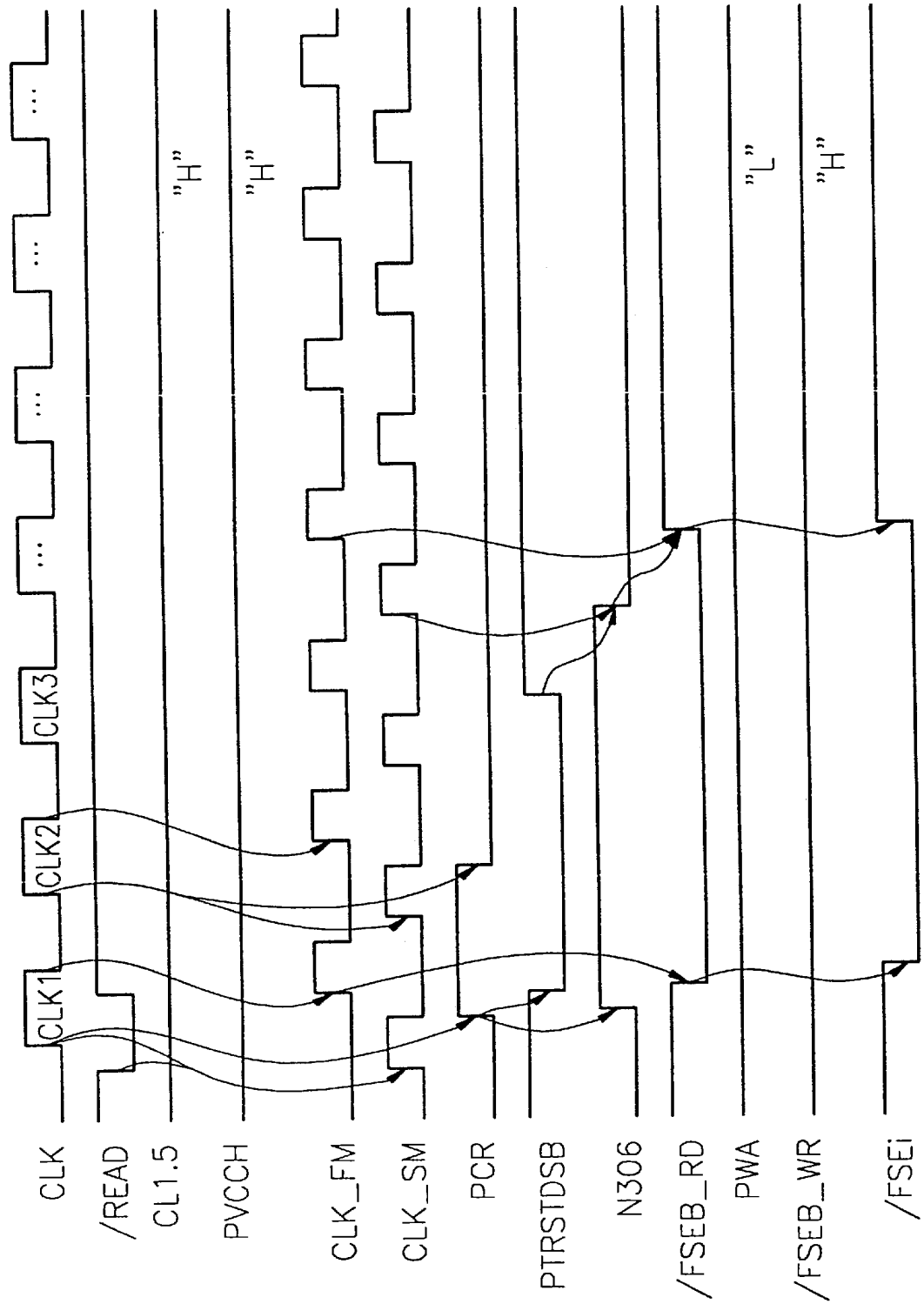


FIG. 6

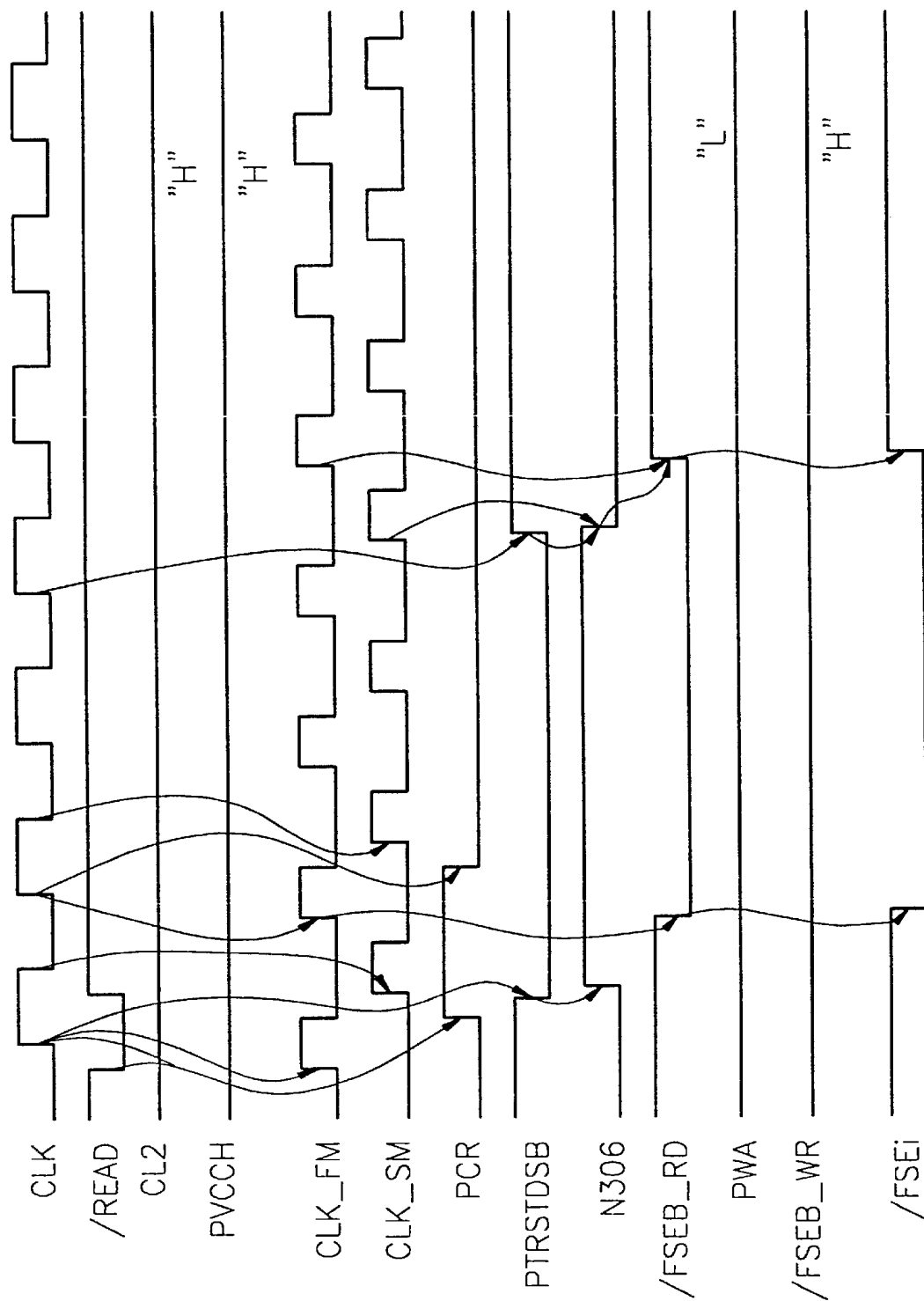


FIG. 7

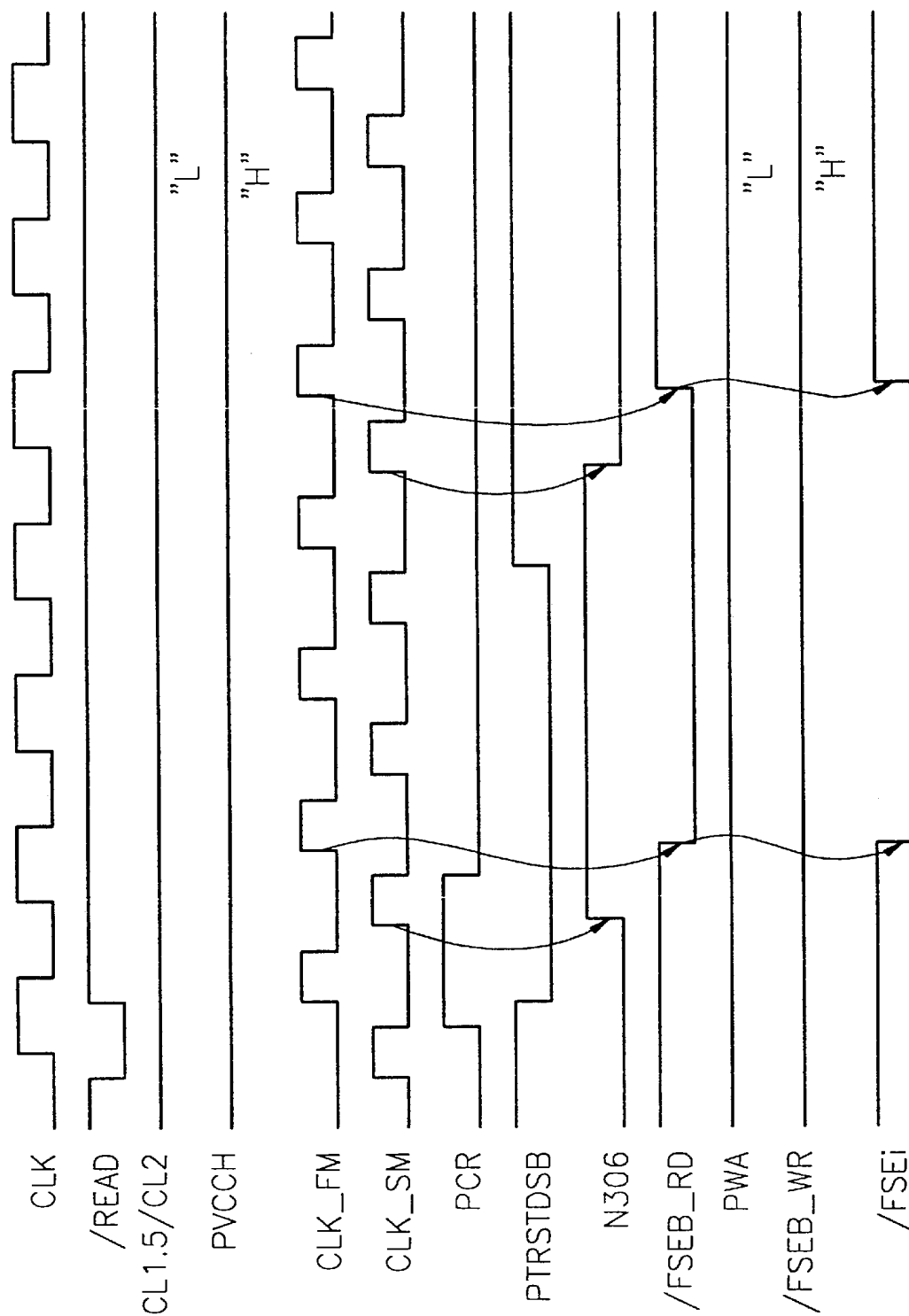


FIG. 8

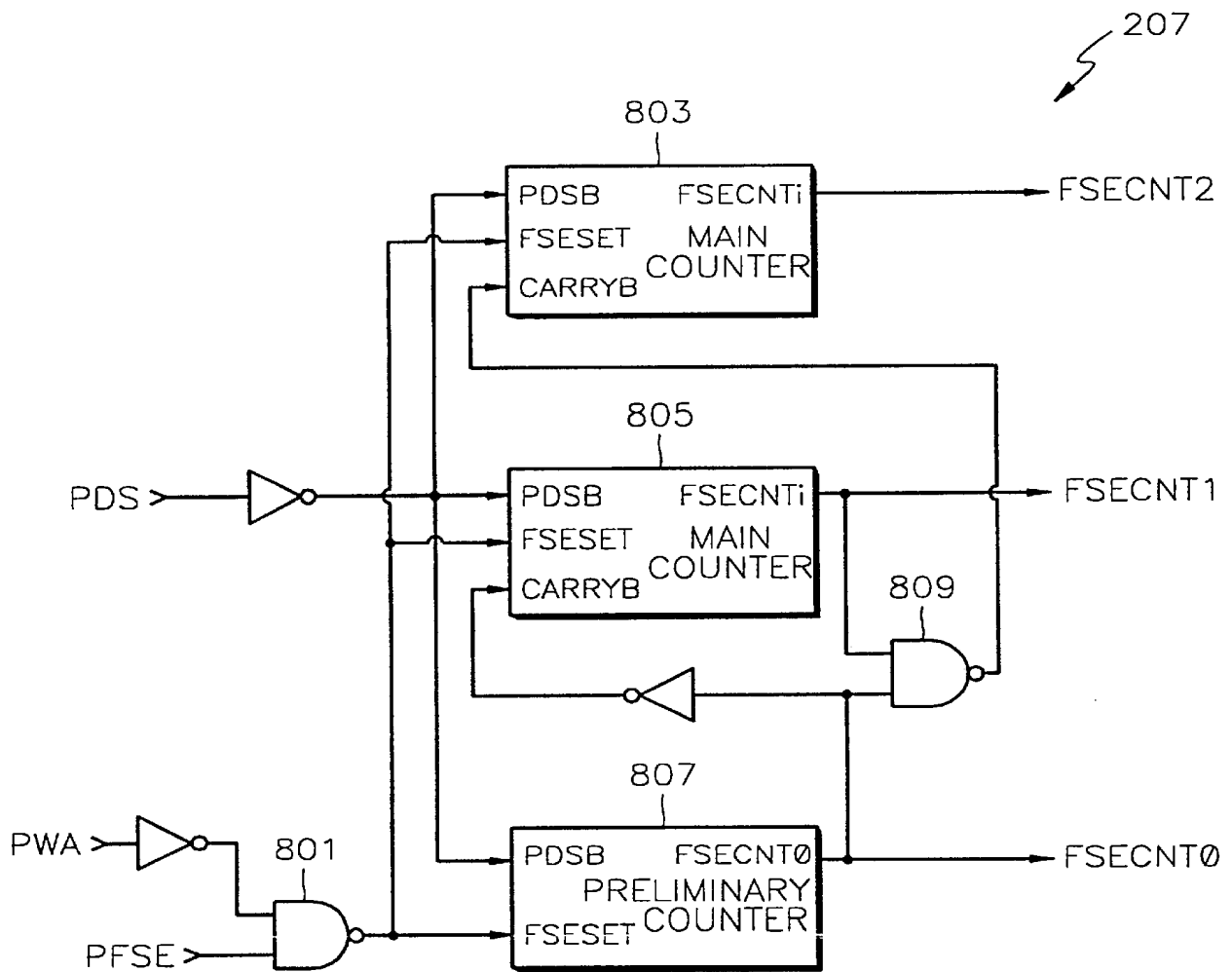


FIG. 9

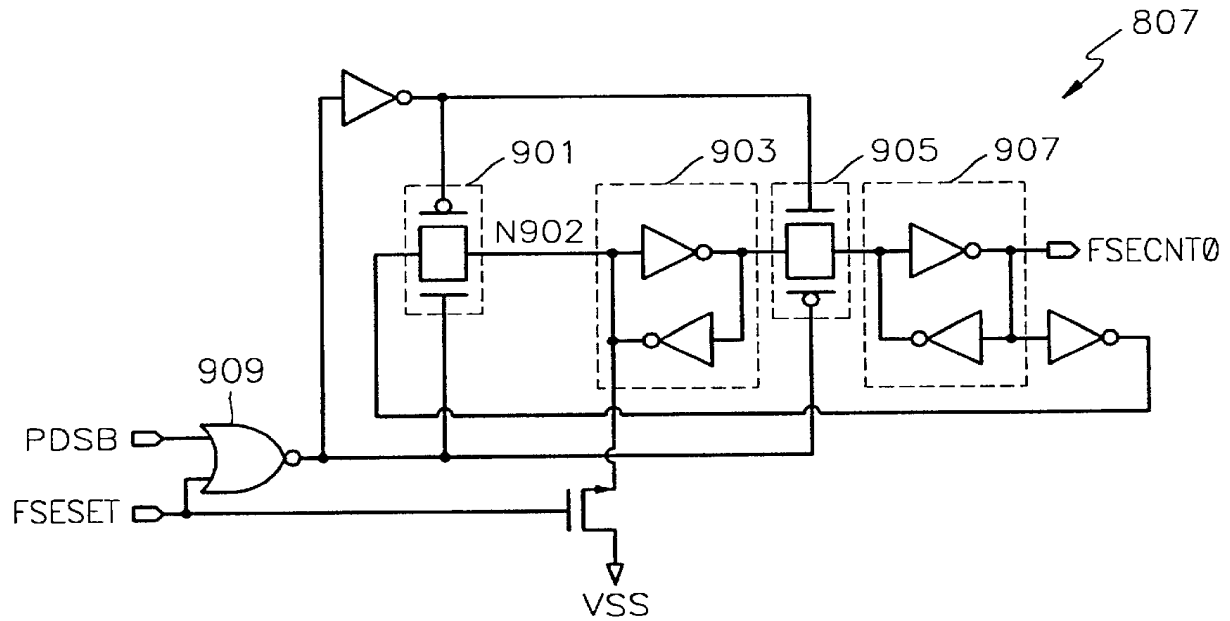


FIG. 10

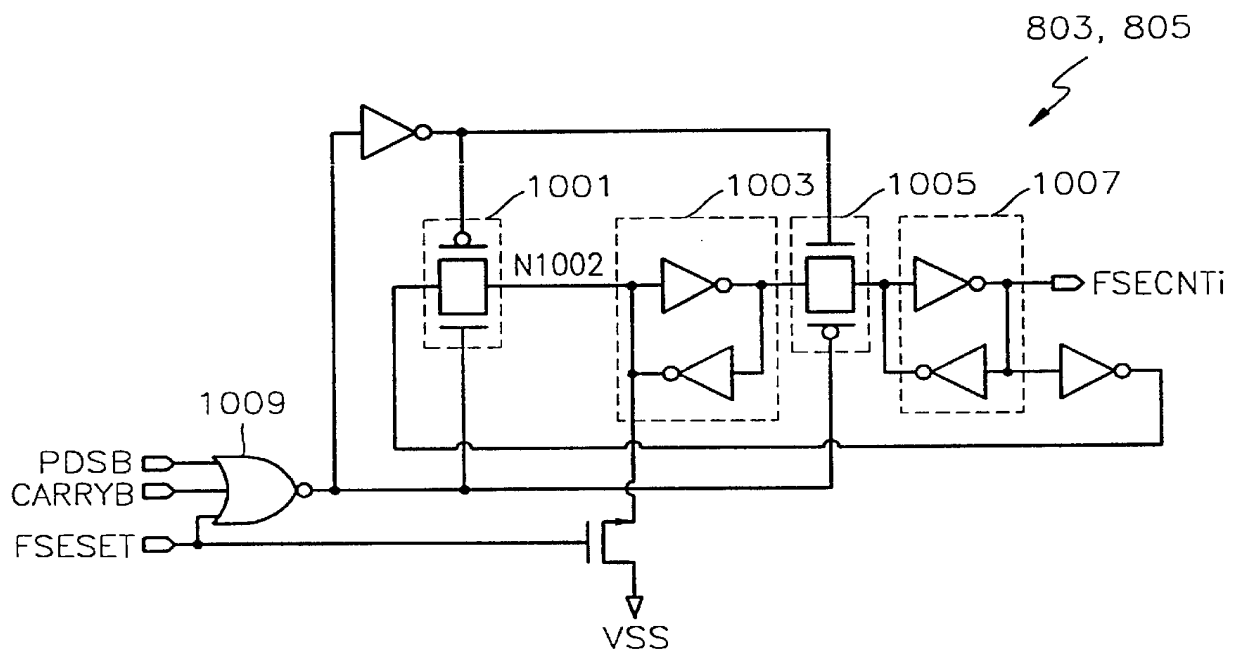


FIG. 11

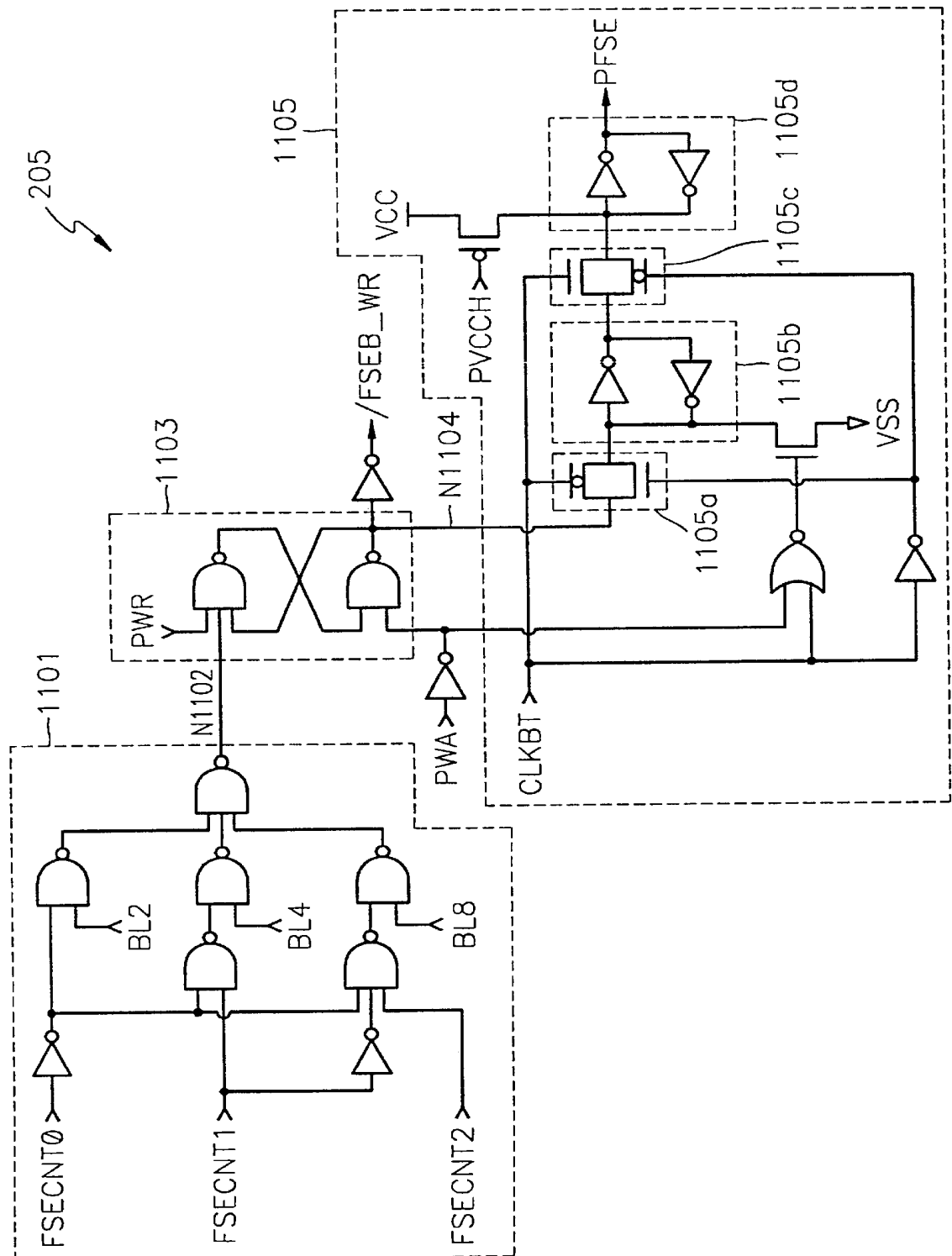


FIG. 12

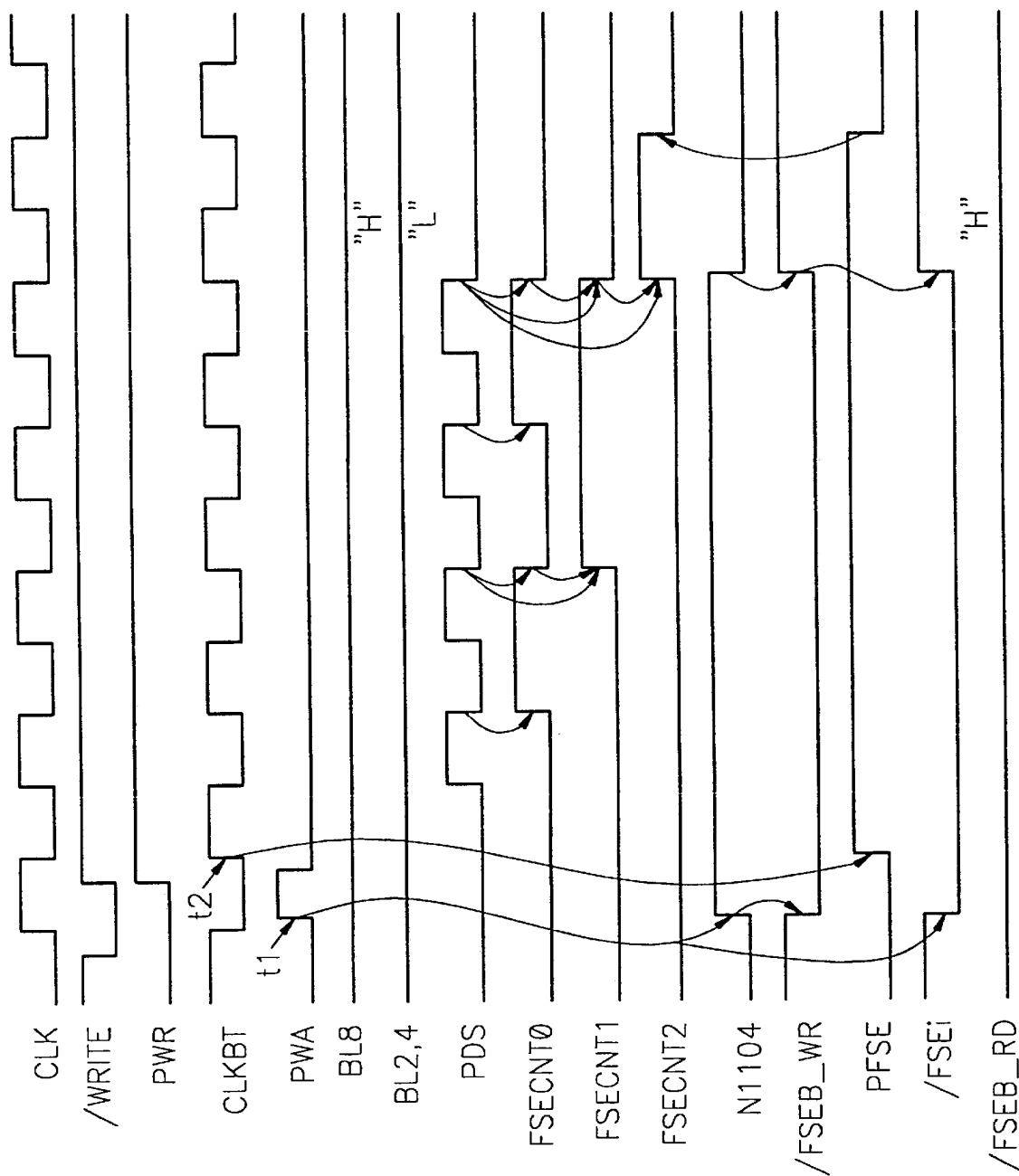


FIG. 13

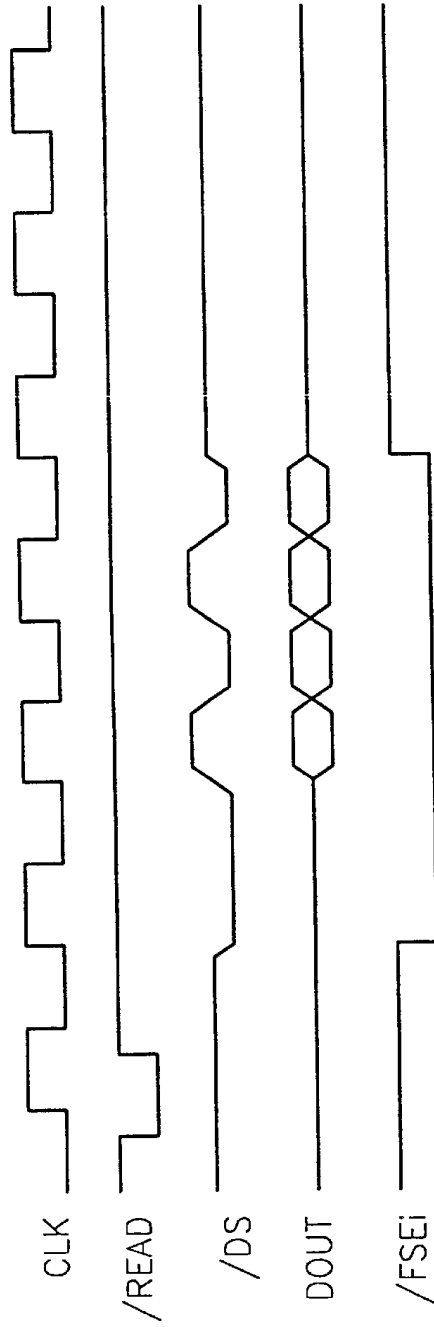


FIG. 14

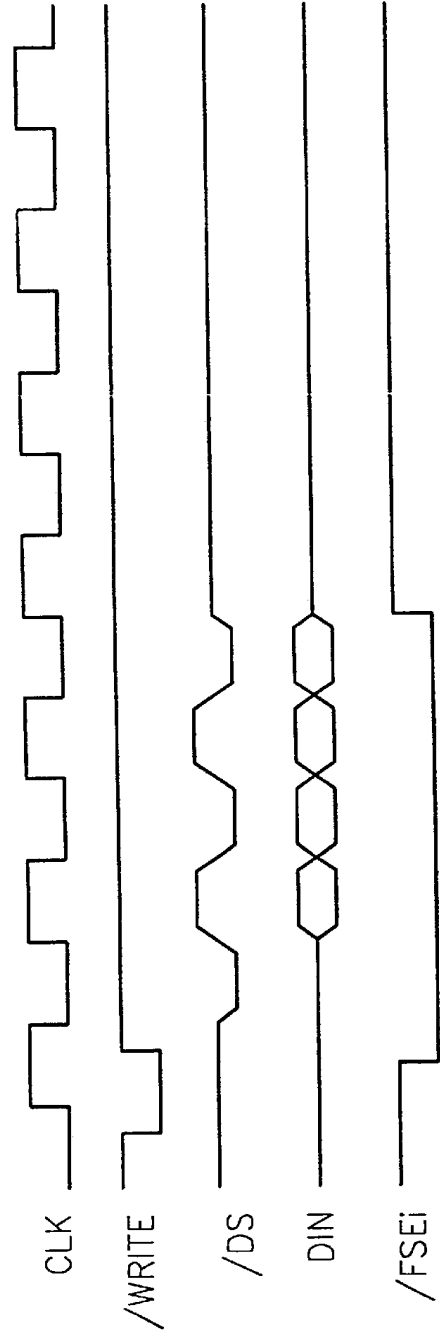
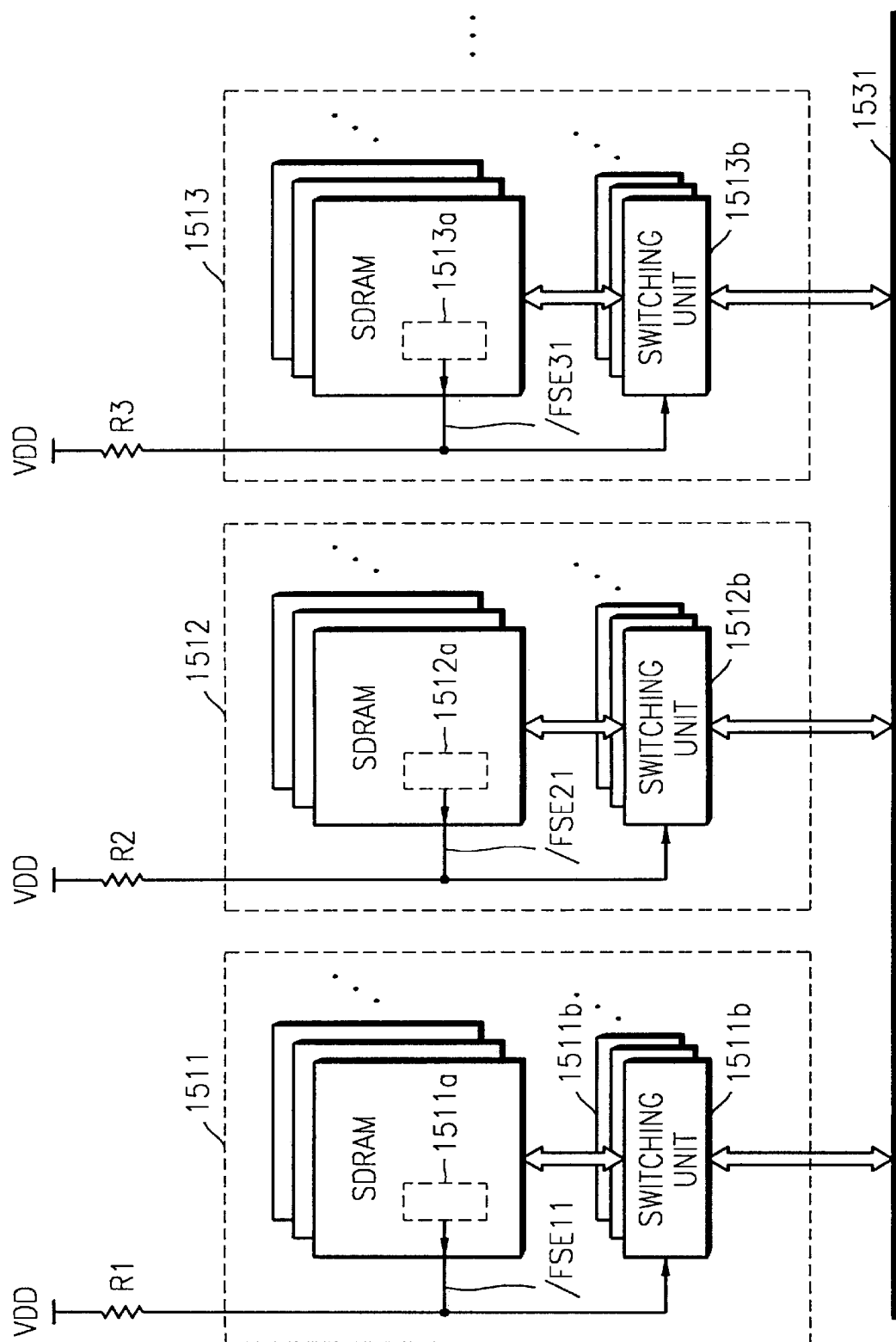


FIG. 15



MEMORY MODULE SYSTEM FOR CONTROLLING DATA INPUT AND OUTPUT BY CONNECTING SELECTED MEMORY MODULES TO A DATA LINE

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a system with memory modules for controlling data input and output, and more particularly, to a system having a multiplicity of memory modules for controlling data input and output from and to a common data line by connecting selected memory modules to the common data line.

2. Description of the Related Art

A semiconductor memory device such as a dynamic random access memory (DRAM) can be used as a main memory in a system such as a computer. A memory using semiconductor memory devices generally has memory modules, such as dual in-line memory modules (DIMM) and single in-line memory modules (SIMM). A memory may be required to include a plurality of memory modules for operations of a system. In such a system, each of the memory modules uses a common data line. For instance, assuming that a main memory has a capacity of 32M bytes and is made with DIMMs each of which has a capacity of 8M bytes and made of four 16M (1M×16) DRAMs, the main memory of 32M bytes requires four slots, i.e., four DIMMs. Since the number of data bus lines is 64 and the number of data pins per DIMM is 64, the main memory of 32M bytes requires a multiplicity of data pins. Thus, four DIMMs constituting the main memory of 32M bytes use a common data bus line of a system. As the capacity of the main memory is increased, the necessity of using a common data line is increased.

The capacitance per data pin of a DIMM module is 20~25 pF. Thus, when the 32M byte memory is composed of four DIMMs of 1M×64, the capacitance per data pin becomes approximately 80~100 pF. Thus, the capacitance per data pin is so great that an increase in the load of an output driver increases power consumption, and that speed for writing and reading data is remarkably reduced.

SUMMARY OF THE INVENTION

It is an object of the present invention to provide a memory module with minimized load capacitance per data pin to improve speed of writing and reading data and of the memory module.

To achieve the above and other objects, according to an aspect of the present invention, there is provided a memory module system for controlling data output to a data bus line, the system comprising a multiplicity of memory modules each of which includes a control signal generation circuit for generating a connection control signal having a width of activation corresponding to a burst length of output data, wherein the control signal generation circuit has a read control unit for activating input data during output of data of a predetermined burst length in a read mode to generate a read control signal, the read control unit receiving CAS latency information, a read command signal and burst length information of read data, and a multiplicity of switching units for electrically connecting output terminals of the memory modules to the data bus line in response to activation of the connection control signal. In the memory module system according to the present invention, the memory

modules for receiving outputting data are electrically connected to the data bus line in response to activation of the connection control signals. The control signal generation circuit further comprises a write information generation unit.

The write information generation unit receives a write command signal and burst length information of write data, and it is activated in a write mode during inputting data of a predetermined burst length to generate a write control signal.

In a memory module system for controlling data input and output according to the present invention, selected memory modules are connected to a data line during a data burst length in which data is input and output. Thus, load is minimized per data pin, so that speed of writing and reading data is improved.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other objects and advantages of the present invention will become more apparent by describing in detail a preferred embodiment thereof with reference to the attached drawings in which:

FIG. 1 is a block diagram schematically showing a system having a multiplicity of memory modules according to a first embodiment of the present invention;

FIG. 2 is a block diagram of one of the control signal generation circuits of FIG. 1;

FIG. 3 is a circuit diagram of the read control unit of FIG. 2;

FIG. 4 is a circuit diagram of the control signal generation unit of FIG. 2;

FIGS. 5 through 7 are timing diagrams of control and data signals in FIGS. 3 and 4 in a read mode having CAS latencies of 1.5, 2 and 2.5, respectively;

FIG. 8 is a circuit diagram of the burst counter of FIG. 2; FIG. 9 is a circuit diagram of the preliminary counter of FIG. 8;

FIG. 10 is a detailed circuit diagram of one of the main counters of FIG. 8;

FIG. 11 is a circuit diagram of the write control unit of FIG. 2;

FIG. 12 is a timing diagram of control and data signals related to the burst counter of FIG. 2;

FIG. 13 is a timing diagram of external signals of memory modules in a read mode of a DDR mode in a system of the present invention;

FIG. 14 is a timing diagram of external signals of memory modules in a write mode of a DDR mode in a system of the present invention; and

FIG. 15 is a block diagram schematically showing a system having a multiplicity of memory modules according to a second embodiment of the present invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

Referring to FIG. 1, a first embodiment of a memory module system according to the present invention has a multiplicity of memory modules. For convenience, three memory modules, i.e., first through third memory modules 111, 112 and 113, are shown. The first through third memory modules 111, 112 and 113 are selected, so that output data is transferred to a common data bus line 131 through switching units 121, 122 and 123.

Each of the switching units includes switches, through which data is output from a corresponding memory module.

Also, the data output through the switches is transferred to the data line bus **131**. Each of the switching units **121**, **122** and **123** is turned on when data is input to and output from the corresponding memory module. However, when data is not input to or output from the corresponding memory module, each of the switching units **121**, **122** and **123** is turned off. For instance, assuming that the first memory module **111** is selected, to input or output data to or from the first memory module **111**, a first connection control signal /FSE1 is activated to logic "low", so that the first switching unit **121** is turned on. Thus, output data of the first memory module **111** is transferred to the data bus line **131** through the first switching unit **121**. At this time, second and third connection control signals /FSE2 and /FSE3 are deactivated to logic "high" by an external power supply voltage VDD applied through resistors **R2** and **R3**, respectively. The resistors **R2** and **R3** may be external to the memory modules **112** and **113**.

In detail, the resistors **R1** to **R3** have relatively high resistances. Thus, voltage levels of the first through third connection control signals /FSE1, /FSE2 and /FSE3 are determined by the first through third memory modules **111**, **112** and **113**, respectively. However, when signals to determine the voltage levels of the connection control signals /FSE1, /FSE2 and /FSE3 in the respective memory modules are in floating state, the voltage levels of the first through third connection control signals /FSE1, /FSE2 and /FSE3 become an external power supply voltage VDD, i.e., logic "high", applied through the resistors **R1** to **R3** connected to the respective memory modules. That is, when the first memory module **111** is selected, the second and the third connection control signals /FSE2 and /FSE3 become logic "high". The second and the third connection control signals /FSE2 and /FSE3 deactivated to the "high" then turn off the second and the third switching units **122**, **123**. Thus, load of data pins of the second and third memory modules **112** and **113** is separated from the data bus line **131**. In other words, memory modules other than the selected memory module are separated from the data line **131**, to thereby reduce load of the data bus line **131**.

Preferably, each of the first through third connection control signals /FSE1, /FSE2 and /FSE3 is generated by the corresponding memory modules **111**, **112** and **113**. Control signal generation circuits **111a**, **112a** and **113a** which are included in the corresponding memory modules **111**, **112** and **113** generate the first through third connection control signals /FSE1, /FSE2 and /FSE3, respectively. Also, activation intervals of the first through third connection control signals /FSE1, /FSE2 and /FSE3 are preferably determined by burst lengths of input or output data. Also, the first through third connection control signals /FSE1, /FSE2 and /FSE3 may be provided from a control unit (not shown) which transfers the input data from the outside of the memory modules.

FIG. 2 is a block diagram illustrating a preferred embodiment of a control signal generation circuit according to the present invention. As described above, the control signal generation circuits **111a**, **112a** and **113a** each having the same configuration as in FIG. 2 are included in the memory modules **111**, **112** and **113**, respectively. For convenience, the first through third connection control signals /FSE1, /FSE2 and /FSE3 are indicated by /FSEi.

Referring to FIG. 2, the control signal generation circuit includes a read control unit **201**, a control signal generation unit **203**, a write control unit **205** and a burst counter **207**. The read control unit **201** receives column address strobe (CAS) latency signals **CL1.5** and **CL2**, a read command

signal **PCR** and a read strobe signal **PTRSTD**, and generates a read control signal /FSEB_RD. The CAS latency signals **CL1.5** and **CL2** include /CAS latency information, and the read command signal **PCR** introduces a read command for a read mode. The read strobe signal **PTRSTD** includes information on a burst length of read data, for example, information on burst signals **BL2**, **BL4** and **BL8** input to the write control unit **205**. Also, the read control signal /FSEB_RD is activated in response to the burst length of the read data in the read mode. The read control unit **201** will be described in detail with reference to FIG. 3.

The control signal generation unit **203** supplies the connection control signal /FSEi in response to the read control signal /FSEB_RD and a write control signal /FSEB_WR. The write control signal /FSEB_WR supplied by the write control unit **205**, is activated for an interval corresponding to a burst length of data input in a write mode. Thus, the connection control signal /FSEi is activated in response to the read control signal /FSEB_RD in the read mode, and also activated in response to the write control signal /FSEB_WR in the write mode. The control signal generation unit **203** will be described in detail, with reference to FIG. 4.

The write control unit **205** receives burst signals **BL2**, **BL4** and **BL8** and first and second write command signals **PWA** and **PWR**, and generates a write control signal /FSEB_WR and a counter control signal **PFSE**. Here, the burst signals **BL2**, **BL4** and **BL8** include information on burst lengths of data in a write mode. The first write command signal **PWA** is a pulse indicating that a memory module enters into a write mode, in response to a write command. The second write command signal **PWR** indicates that a memory module is in a write mode.

The write control signal /FSEB_WR which is activated for an interval corresponding to the data burst length in which data is input in the write mode, is supplied to the control signal generation unit **203**. Also, the counter control signal **PFSE** which is supplied to the burst counter **207**, is activated in response to an inverted clock signal **CLKBT** which is firstly generated after generating the write command signal, and deactivated after receiving a predetermined number of data. The inverted clock signal **CLKBT** is an inverted signal of an external clock signal **CLK** (not shown). The write control unit **205** will be described in detail with reference to FIG. 11.

The burst counter **207** is enabled in response to the activation of the counter control signal **PFSE**. The burst counter **207** receives a first write command signal **PWA** and a data strobe signal **PDS**, and counts the number of input data. Also, a counting signal **FSECNTi** (where, $i=0$ to 2) indicating the number of input data is supplied to the write control unit **205**. The data strobe signal **PDS** indicating an input of data, is formed by an external data strobe signal of a memory module. The data strobe signal **PDS** indicates an input or output of one of the data by one transition in a double data rate (DDR). The memory module system of the present invention will be described in the DDR mode. The number of input data can be indicated by a combination of the counting signals **FSECNTi**, where $i=0$ to 2 . The burst counter **207** will be described in detail with reference to FIG. 8.

A control signal generation circuit which generates the connection control signal /FSEi can be realized with the read control unit **201** and the control signal generation unit **203** of FIG. 2 in a read mode. A write information generation unit

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209 for activating the write control signal /FSEB_WR for an interval corresponding to the burst length of data input may be realized in a write mode with the write control unit 205 and the burst counter 207. Thus, the control signal generation circuit for generating the connection control signal /FSEi may be realized in the write mode with the write information generation unit 209 and the control signal generation unit 203 of FIG. 2.

FIG. 3 is a circuit diagram of a preferred embodiment of the read control unit 201 in FIG. 2. Referring to FIG. 3, the read control unit 201 includes a first latency response unit 301, a second latency response unit 303, a first transfer unit 309, a first latch unit 307, a second transfer unit 305 and a second latch unit 311.

The first latency response unit 301 is enabled in response to a CAS latency shorter than a predetermined latency. According to a preferred embodiment of the present invention, the first latency response unit 301 is enabled, when the CAS latency is 1.5. The first latency response unit 301 includes a logic gate 301a and a drive transistor 301b. The logic gate 301a is enabled by activation of CL1.5 indicating the CAS latency of 1.5, and responds to a read command signal PCR. The read command signal PCR is activated in response to a first clock signal from generation of a read command, and is maintained in the activated state during a period of one clock. The drive transistor 301b is gated by an output signal 301n of the logic gate 301a. Thus, the transistor 301b outputs an output signal to the first latch unit 307 in response to activation of the read command signal PCR. Preferably, the logic gate 301a is a NAND gate, and the transistor 301b is a PMOS transistor.

When the CAS latency is two or more, the second latency response unit 303 is enabled. An output signal N304 of the second latency response unit 303 maintains a logic "high" state by activation of a read strobe signal PTRSTDSB. The second latency response unit 303 includes logic gates 303a and 303b. The logic gate 303a is enabled by activation of CL2 indicating the CAS latency of 2, and responds to the read command signal PCR. Also, the logic gate 303b is enabled by the output signal 303n of the logic gate 303a, and responds to the read strobe signal PTRSTDSB. Preferably, the logic gates 303a and 303b are NAND gates. The second transfer unit 305 transfers the output signal N304 of the second latency response unit 303 in response to a first clock signal CLK_SM.

The first latch unit 307 latches either the output signal of the first latency response unit 301 or the output signal N304 of the second latency response unit 303 transferred by the second transfer unit 305. The first transfer unit 309 transfers signals latched by the first latch unit 307 in response to a second clock signal CLK_FM. The second latch unit 311 latches a signal N306 transferred by the first transfer unit 309, and outputs the latched signal as a read control signal /FSEB_RD.

Here, the first and the second clock signals CLK_SM and CLK_FM are pulse signals generated in response to rising and falling transitions of the external clock signal CLK. The first and the second clock signals CLK_SM and CLK_FM which control data output timing, have timing with respect to the external clock signal CLK as shown in FIGS. 5 through 7.

FIG. 4 is a circuit diagram illustrating a preferred embodiment of the control signal generation unit 203 of FIG. 2. Referring to FIG. 4, the control signal generation unit 203 includes a NAND gate 401, a PMOS transistor 403 and an NMOS transistor 405. The NAND gate 401 receives an

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inverted signal of the read control signal /FSEB_RD, an inverted signal of the write control signal /FSEB_WR and the first write command signal PWA. That is, the NAND gate 401 is enabled by the first write command signal PWA, and acts as a logic gate responding to the read control signal /FSEB_RD or the write control signal /FSEB_WR.

The NMOS transistor 405 is gated by an output signal of the NAND gate 401. Thus, the NMOS transistor 405 acts for driving connection control signal in response to the read control signal /FSEB_RD or the write control signal /FSEB_WR.

The PMOS transistor 403 has source and gate terminals which are in common connected. A drain terminal of the PMOS transistor 403 and a drain terminal of the NMOS transistor 405 are in common connected to generate the connection control signal /FSEi.

In a precharge state, the read control signal /FSEB_RD and the write control signal /FSEB_WR are both logic "high" and the first write command signal PWA is logic low. Thus, the output signal of the NAND gate 401 becomes low and the NMOS transistor 405 is turned off. The connection control signal /FSEi becomes logic "high" in response to an external power supply voltage VDD applied through a resistor outside a memory module.

An embodiment of a control signal generation circuit for generating a connection control signal can be realized with the read control unit 201 in FIG. 3 and the control signal generation unit 203 in FIG. 4 in a read mode.

FIGS. 5 through 7 are timing diagrams of control signals and data signals in FIGS. 3 and 4 in a read mode in which CAS latencies are 1.5, 2, and 2.5, respectively. Assuming that a data burst length is 4, and the memory modules operate in a DDR mode, the operation of the read control unit 201 when the CAS latency is 1.5, will FIG. 5 be described with reference to FIGS. 3, 4 and 5.

Referring to FIG. 5, a logic state of the CAS latency signal CL1.5 is high, and a logic state of the CAS latency signal CL2 is low. Activation of /READ signal to low state indicates a generation of a read command. The read command signal PCR is activated for one clock period in response to a rising edge of the first clock pulse CLK1 after the generation of the read command. Thus, the output signal N306 of the first latency response unit 301 becomes logic "high" in response to the read command signal PCR, and is latched to high by the first latch unit 307. That is, when the read command is generated, the output signal N306 of the first latency response unit 301 is latched to high by the first latch unit 307, regardless of a level of the clock signal.

The read strobe signal PTRSTDSB is activated to logic "low" in response to activation of the read command signal, and is activated to logic "high" after a period of time corresponding to the data burst length. The output signal N304 of the second latency response unit 303 is transferred to the first latch unit 307 by the second transfer unit 305 turned on in response to the first clock signal CLK_SM. Thus, when the read strobe signal PTRSTDSB is activated to logic "low", the input signal at the terminal N306 of the first latch unit 307 is maintained at logic "high" even when the read command signal PCR is deactivated to logic "low".

Data latched by the first latch unit 307 is transferred to the second latch unit 311 by the first transfer unit 309 turned on in response to the second clock signal CLK_FM. The output signal of the second latch unit 311 is connected to an output terminal of the read control signal /FSEB_RD. Thus, the read control signal /FSEB_RD is activated to logic "low" in response to the generation of the read command, and then

deactivated to high in a period of time corresponding to the burst length. That is, a time period of the activation of the read control signal /FSEB_RD corresponds to that of the burst length.

In a read mode, the first write command signal PWA is maintained at logic "low", and the write control signal /FSEB_WR is maintained at logic "high". Thus, in the read mode, activation of the connection control signal /FSEi depends on activation of the read control signal /FSEB_RD.

In a preferred embodiment of the present invention, when the CAS latency is 1.5, the input signal N306 of the first latch unit 307 is activated regardless of the clock signal, and the activation is maintained by the read strobe signal PTRSTDSB. When the read control signal /FSEB_RD is activated in response to the first clock signal CLK_SM, the connection control signal /FSEi is not activated until a first data burst is read, despite of the CAS latency of 1.5.

The operation of the read control unit 201 with the CAS latency of 2 will now be described, with reference to FIGS. 3, 4 and 6. A logic state of the CAS latency signal CL2 is high, and a logic state of the CAS latency signal CL1.5 is low. Thus, the output signal N306 of the first latency response unit 301 does not respond to the read command signal PCR.

The output signal N304 of the second latency response unit 303 is activated to high in response to an activation of the read command signal PCR, and is transferred to the first latency unit 307 in response to an activation of the first clock signal CLK_SM. The logic state of the input signal N306 of the first latch unit 307 is maintained at high in response to activations of both the read strobe signal PTRSTDSB and the first clock signal CLK_SM.

Thus, the read control signal /FSEB_RD, similar to the case of the CAS latency of 1.5, is activated to low in response to a generation of the read command, and deactivated to high after a period corresponding to the burst length. That is, the read control signal /FSEB_RD is activated for an interval corresponding to the burst length.

The operation of the read control unit 201 when the CAS latency is 2 or more, i.e., the logic states of the CAS latency signals CL1.5 and CL2 are low, will now be described with reference to FIGS. 3, 4 and 7. The output signal N304 of the second latency response unit 303 becomes high after the read strobe signal PTRSTDSB is activated to low. The output signal N304 of the second latency response unit 303 is transferred to the first latch unit 307 in response to activation of the first clock signal CLK_SM. When the read strobe signal PTRSTDSB is deactivated to high, the input signal N306 of the first latch unit 307 becomes low. Thus, the read control signal /FSEB_RD is activated or deactivated by the read strobe signal PTRSTDSB, unlike the case in which the CAS latency is 1.5 or 2.

As described above, the control signal generation circuit according to a preferred embodiment of the present invention generates a connection control signal activated in response to the data burst length in which data is read from a memory module in a read mode. Also, only the memory module selected by the connection control signal is connected to a data line during the data burst length in which data is output.

FIG. 8 is a circuit diagram illustrating an embodiment of the burst counter 207 of FIG. 2. In FIG. 8, a burst counter capable of counting eight data bursts is shown. The burst counter 207 shown in FIG. 8 includes one preliminary counter 807 and two main counters 803 and 805. The preliminary counter 807 and the main counters 803 and 805

are reset by activation of a counter control signal PFSE. That is, when a write command is generated so that the counter control signal PFSE is activated to high, the preliminary counter 807 and the main counters 803 and 805 are reset. Output signals of the preliminary counter 807 and the main counters 803 and 805, i.e., first through third counting signals FSECNT0~2 are reset to low.

As described above, a data strobe signal PDS indicates inputs of external data during rising and falling transitions in a DDR mode. Structure and operation of the burst counter 207 in a DDR mode will now be described.

In an embodiment of the present invention, data is input during rising and falling edges of the data strobe signal PDS, and the data is continuously input when a phase transition of the data strobe signal PDS is repeated. The first counting signal FSECNT0 performs the phase transition at every falling edge of the input data strobe signal. Also, when carries CARRYBs of the main counters 803 and 805 are activated to low, the second and the third counting signals FSECNT1~2 repeat the phase transition every two inputs of data.

Referring to FIG. 8, the carry CARRYB of the main counter 805 is activated to low, when the first counting signal FSECNT0 is activated to high. The carry CARRYB of the main counter 803 is activated to low, when the first and the second counting signals FSECNT0~1 are high.

The preliminary counter 807 and the main counters 803 and 805 of FIG. 8 will now be described with reference to FIGS. 9 and 10, respectively. FIG. 9 is a detailed circuit diagram illustrating an embodiment of the preliminary counter 807 of FIG. 8. Referring to FIG. 9, the preliminary counter 807 includes first and second transfer units 901 and 905, first and second latch units 903 and 907, and a NOR gate 909. The NOR gate 909 performs an OR operation on a reset signal FSESET and an inverted signal PDSB of the data strobe signal PDS and inverts the OR-operated signals. The reset signal FSESET is connected to the counter control signal PFSE. Thus, when the reset signal FSESET is disabled to low, the preliminary counter 807 is enabled to respond to the phase transition of the inverted signal PDSB. That is, when a phase of the inverted signal PDSB is transited from high to low, the first transfer unit 901 is turned on. When the phase of the inverted signal PDSB is transited from low to high, the second transfer unit 905 is turned on. Whenever the inverted signal PDSB is deactivated, the first counting signal FSECNT0 repeats the phase transition.

FIG. 10 is a detailed circuit diagram illustrating an embodiment of the main counters 803 and 805 of FIG. 8. The main counters 803 and 805 may have the same structure as shown in FIG. 10. For convenience, output signals of the main counters 803 and 805, i.e., second and third counting signals FSECNT1~2, are indicated by a counting signal FSECNTi.

A main counter has a structure and operation similar to those of the preliminary counter 807 of FIG. 9. However, a NOR gate 1009 with three input terminals is used, to which a reset signal FSESET, an inverted signal PDSB and a carry CARRYB are provided. Thus, when the reset signal FSESET and the carry CARRYB are low, the main counter is enabled in response to a phase transition of the inverted signal PDSB. That is, when the reset signal FSESET and the carry CARRYB are low, the counting signal FSECNTi repeats the phase transition whenever the inverted signal PDSB is deactivated.

FIG. 11 is a detailed circuit diagram illustrating an embodiment of the write control unit 205 of FIG. 2. Refer-

ring to FIG. 11, the write control unit 205 includes a counting response unit 1101, a latch unit 1103 and a counter control signal generation unit 1105. The counting response unit 1101 indicates a data burst of input data. That is, a phase of an output signal N1102 is transited when a predetermined number of data is input.

When the first write command signal PWA is activated to high, the latch unit 1103 is enabled to respond to the output signal N1102 of the counting response unit 1101. When the second write command signal PWR becomes high, an output signal N1104 of the latch unit 1103 is latched to high. At this time, a write control signal /FSEB_WR connected to an output terminal inverting the output signal N1104 of the latch unit 1103 is activated to low. When the first write command signal PWA becomes low, the output signal N1104 of the latch unit 1103 is maintained at high. When the second write command signal PWR is high and the predetermined number of data is input so that the output signal N1102 of the counting response unit 1101 is transited to low, the output signal N1104 of the latch unit 1103 becomes low, and the write control signal /FSEB_WR is deactivated to high.

The counter control signal generation unit 1105 includes first and second transfer units 1105a and 1105 and first and second latches 1105b and 1105d. The first transfer unit 1105a transfers the output signal N1104 of the latch unit 1103 in response to a falling edge of an inverted clock signal CLKBT. The first latch unit 1105b latches a signal transferred by the first transfer unit 1105a. The second transfer unit 1105c transfers an output signal of the first latch 1105b in response to a rising edge of the inverted clock signal CLKBT. Also, the second latch 1105d latches a signal transferred by the second transfer unit 1105c, and generates a counter control signal PFSE. Here, the inverted clock signal CLKBT is an inverted signal of an external clock signal CLK.

FIG. 12 is a timing diagram of signals related to the burst counter illustrated in FIGS. 2 and 8 through 11 in a write mode. FIG. 12 shows a case of a data burst of 8 in a DDR mode. Operation of the write control unit 205, the burst counter 207, the preliminary counter 807 and the main counters 803 and 805 of the burst counter 207 will be described with reference to FIG. 12.

The data burst is 8, that is, burst signal BL8 is high and burst signals BL2 and BL4 are low. The DDR mode indicates that data is input in every phase transition of the data strobe signal PDS. When the first write command signal PWA becomes high (t1), the output signal N104 of the latch unit 1103 is latched to high, and the write control signal /FSEB_WR is activated to low. When the second write command signal PWR is activated to high and activation and deactivation of the data strobe signal PDS are performed four times, i.e., 8 data are input, both the first and second counting signals FSECNT0~1 become low and the third counting signal FSECNT2 becomes high. At this time, the phase of the output signal N1102 of the counting response unit 1101 is transited to low from high, and the output signal N1104 of the latch unit 1103 becomes low. The write control signal /FSEB_WR is deactivated to high. The counter control signal PFSE is deactivated to low in response to a rising edge of the inverted clock signal CLKBT. The preliminary counter 807 and the main counters 803 and 805 are reset, so that the first through third counting signals FSECNT0~2 become low.

Referring to FIG. 4 again, a connection control signal /FSEi is activated and deactivated in response to the write control signal /FSEB_WR in a write mode. During the write mode the read control signal /FSEB_RD is deactivated to high.

FIG. 13 is a timing diagram of external signals of memory modules when reading data in a DDR mode in a system of the present invention. Referring to FIG. 13, the CAS latency is 2, and the burst length is 4. When a predetermined preamble interval, i.e., 1 clock cycle, passes after a read command is input, i.e., a signal /READ is activated, an external data strobe signal /DS is activated to low. The connection control signal /FSEi is also enabled to low, so that switching units 121, 122 and 123 (see FIG. 1) connected to the activated memory modules are turned on. First, data are output from the memory modules, and then the output data are transferred to the data line 131 (see FIG. 1). When a postamble interval, i.e., a half clock, passes after the last data of a data burst is output, the external data strobe signal /DS becomes high impedance state. At this time, the connection control signal /FSEi of low is disabled to high, to thereby turn off the switching units 121, 122 and 123.

FIG. 14 is a timing diagram of external signals of memory modules in a write mode of a DDR mode in a system of the present invention. In FIG. 14, the burst length is 4. In a ¼ clock cycle after a write command is input, i.e., a signal /WRITE is activated, the connection control signal /FSEi is enabled to low. Activated memory modules receive data corresponding to the burst length from a control unit (not shown) through the data bus line 131 (see FIG. 1). When data holding time passes after last data is received, the connection control signal /FSEi of low is disabled to high to turn off the corresponding switching units 121, 122 and 123.

As described above, a control signal generation circuit according to a preferred embodiment of the present invention generates a connection control signal activated during an interval corresponding to the burst length in which data is written in a write mode. Only the memory module selected by the connection control signal is connected to a data line during the data burst length in which data is input.

FIG. 15 is a block diagram illustrating a second embodiment of the memory module system according to the present invention. Structure and operation of the memory module system according to the second embodiment are similar to those of the first embodiment. Switching units 1511b, 1512b and 1513b corresponding to the switching units 121, 122 and 123 of FIG. 1 are installed in memory modules 1511, 1512 and 1513, respectively.

As shown in FIG. 15, a multiplicity of SDRAMs (synchronous DRAMs) are installed in each of the memory modules. Also, a multiplicity of switching units each corresponding to each of the SDRAMs are installed in each of the memory modules. Connection control signals /FSE11, /FSE21 and /FSE31 controlling the switching units are generated by the SDRAMs included in the memory modules. For convenience, the connection control signals /FSE11, /FSE21 and /FSE31 are shown for controlling the switching units 1511b, 1512b and 1513b, respectively, each of which connects with a first SDRAM in a memory module. In FIG. 15, although not shown, switching units connected to other SDRAMs exist, and connection control signals for controlling the switching units are generated from the corresponding SDRAMs. A switching unit may preferably have as many switches as output terminals for outputting data from an SDRAM corresponding to the switching unit. The data output through the multiple switches are transferred to a data bus line 1531. Structure and operation of control signal generation circuits 1511a, 1512a and 1513a of the second embodiment in FIG. 15 are the same as those of the control signal generation circuits 111a, 112a and 113a of the first embodiment in FIGS. 2 through 14.

Having described preferred embodiments of a memory module system of the present invention, it is noted that

modification and variations can be made by persons skilled in the art in light of the above teachings. For example, the switches which are arranged outside the SDRAM may be arranged in the SDRAM. Also, other integrated memory circuit devices such as DRAMs or SRAMs may be used instead of SDRAMs. In a DDR mode, the counter circuit of the first embodiment of the present invention counts eight data by using one preliminary counter and two main counters. However, the number of main counters may be increased to increase the number of counted data.

Also, in a memory module system of the present invention, only selected memory modules are connected to a data line during a time period of data burst length in which data is input and output, so that load per data pin is minimized and data writing and reading speed is improved.

What is claimed is:

1. A memory module system for controlling data output to a data bus, the system comprising:

a multiplicity of memory modules each of which includes a control signal generation circuit for generating a connection control signal having a width of activation corresponding to a burst length of output data, wherein the control signal generation circuit has a read control unit for activating input data during output of data of a predetermined burst length in a read mode to generate a read control signal, the read control unit receiving CAS latency information, a read command signal and burst length information of read data; and

a multiplicity of switching units for electrically connecting output terminals of the memory modules to the data bus in response to activation of the connection control signal;

wherein the read control unit comprises:

a first latency response unit for being enabled in a first CAS latency;
a second latency response unit for being enabled in a second CAS latency;
a first latch unit for latching one of an output signal of the first latency response unit and an output signal of the second latency response unit; and
a first transfer unit for transferring a latched signal by the first latch unit to generate the read control signal in response to a first internal clock signal that is synchronized with a first external clock signal.

2. The memory module system of claim 1, wherein the first CAS latency is 1.5, and the second CAS latency is 2.

3. The memory module system of claim 2, wherein the connection control signal is activated in response to activation of a read strobe signal including the burst length information in case of a third CAS latency that is greater than the first and the second CAS latencies.

4. The memory module system of claim 1, wherein the first latency response unit comprises:

a logic gate responding to activation of the first CAS latency and a read command signal; and
a drive transistor for driving an output signal to the first latch unit in response to an output signal of the logic gate.

5. The memory module system of claim 1, wherein the second latency response unit comprises a logic gate responding to a predetermined read strobe signal which is activated during a period corresponding to the burst length of data.

6. The memory module system of claim 1, wherein the memory modules are synchronized with a second external clock signal, and the read control unit further comprises:

a second transfer unit for transferring the output signal of the second latency response unit in response to a

second internal clock signal synchronized with the second external clock signal; and

a second latch unit for latching a signal transferred by the first transfer unit to generate the read control signal.

7. The memory module system of claim 1, wherein the connection control signal is activated by a read strobe signal including the first CAS latency and the burst length information.

8. The memory module system of claim 1, wherein the control signal generation circuit further comprising a control signal generation unit for generating the connection control signal in response to activation of the read control signal output from the read control unit.

9. The memory module system of claim 8, wherein the control signal generation unit comprises:

a logic gate responding to the read control signal; and
a drive transistor for activating the connection control signal in accordance with the activation of the read control signal, the drive transistor being gated by an output signal of the logic gate.

10. A memory module system for controlling data output to a data bus, comprising:

a multiplicity of memory modules; and
a multiplicity of switching units for electrically connecting only memory module outputting data to the data bus in response to activation of a connection control signal, wherein the connection control signal is generated in the memory module outputting data and has a width of activation corresponding to a burst length of the output data;

wherein the connection control signal, with respect to a first and a second CAS latencies, is activated in response to a CAS latency and a read command signal, and the activation is maintained by a read strobe signal including the burst length information, and the connection control signal, with respect to a third CAS latency longer than the first and the second CAS latencies, is activated in response to activation of the read strobe signal.

11. A memory module system for controlling data input from a data bus, comprising:

a multiplicity of memory modules each of which includes a control signal generation circuit for generating a connection control signal having an activation width corresponding to a burst length of input data, wherein the control signal generation circuit includes a write information generation unit for receiving a write command signal and burst length information of write data, the write information generation unit being activated in a write mode during inputting data of a predetermined burst length to generate a write control signal; and

a multiplicity of switches for electrically connecting input terminals of the memory modules to the data bus in response to activation of the connection control signal;

wherein the write information generation unit comprises:
a write control unit for receiving the write command signal and for supplying the write control signal and a counter control signal; and

a burst counter for counting the number of external input data to supply the counted data to the write control unit, wherein the burst counter is enabled in response to activation of the counter control signal;

wherein the write control unit comprises:

a counting response unit for generating an output signal in response to input of data of a predetermined burst length in response to an output signal of the burst counter;

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- a latch unit for being enabled in response to the write command signal, for latching the output signal of the counting response unit, and for supplying the write control signal; and
- a counting control signal generation unit for being enabled in response to the write command signal and for inputting the write control signal to supply the counter control signal.

12. The memory module system of claim 11, wherein the memory modules are synchronized with an external clock signal, and the latch unit comprises:

- a first logic gate for being enabled by a first write command signal to generate the write control signal; and
 - a second logic gate for being enabled by a second write command signal and for responding to an output signal of the counting response unit;
- wherein the data of the burst length is input, and then the write control signal is deactivated by the output signal of the counting response unit.

13. The memory module system of claim 11, wherein the counting control signal generation unit comprises:

- a first transfer unit for transferring an output signal of the latch unit in response to a first transition of the external clock signal;
 - a first latch for latching a signal transferred by the first transfer unit;
 - a second transfer unit for transferring the signal latched by the first latch in response to a second transition of the external clock signal; and
 - a second latch for latching the signal transferred by the first transfer unit to generate the counter control signal;
- wherein a direction of the first transition is opposite to that of the second transition.

14. The memory module system of claim 11, wherein the control signal generation circuit further comprises a control signal generation unit for being activated in response to the write command signal, and the activation is maintained by the write control signal.

15. The memory module system of claim 14, wherein the control signal generation unit comprises:

- a logic gate responding to the write information signal; and
- a drive transistor for activating the connection control signal in response to the activation of the write control signal, the drive transistor being gated by an output signal of the logic gate.

16. A semiconductor memory device comprising:

- a multiplicity of integrated memory circuit devices including a control signal generation circuit for generating a connection control signal having an activation width corresponding to a burst length of output data, wherein the control signal generation circuit comprises a read control unit for activating input data during output of data of a predetermined burst length in a read mode to generate a read control signal, the read control unit receiving CAS latency information, a read command signal and burst length information of data to be read; and

at least one switching unit for connecting output terminals of the integrated memory circuit devices where data is output and a data bus in response to activation of the connection control signal;

wherein the read control unit comprises:

- a first latency response unit for being enabled in a first CAS latency;

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a second latency response unit for being enabled in a second CAS latency;

a first latch unit for latching one of an output signal of the first latency response unit and an output signal of the second latency response unit; and

a first transfer unit for transferring a latched signal by the first latch unit to generate the read control signal in response to a first internal clock signal that is synchronized with a first external clock signal.

17. The semiconductor memory device of claim 16, wherein the first CAS latency is 1.5, and the second CAS latency is 2.

18. The semiconductor memory device of claim 16, wherein the first latency response unit comprises:

- a logic gate responding to the activation of the first CAS latency and a predetermined read command signal; and
- a drive transistor for driving an output signal to the first latch unit in response to an output signal of the logic gate.

19. The semiconductor memory device of claim 16, wherein the second latency response unit comprises a logic gate responding to a predetermined read strobe signal, wherein activation of the read strobe signal is maintained during a period corresponding to a burst length of minimum data.

20. The semiconductor memory device of claim 16, wherein the semiconductor memory device is synchronized with a second external clock signal, and the read control further comprises:

- a second transfer unit for transferring the output signal of the second latency response unit to the first latch unit in response to a second internal clock signal synchronized with the second external clock signal; and
- a second latch unit for latching a signal transferred by the first transfer unit to generate the read control signal.

21. The semiconductor memory device of claim 16, wherein activation of the read control signal is maintained by a read strobe signal including the first CAS latency and the burst length information.

22. The semiconductor memory device of claim 21, wherein the read control signal, with respect to a third CAS latency longer than the first and the second CAS latencies, is activated in response to activation of the read strobe signal including the burst length information.

23. A semiconductor memory device comprising:

- a multiplicity of the integrated memory circuits including at least one control signal generation circuit for generating a connection control signal having an activation width corresponding to a burst length of input data, wherein the control signal generation circuit includes a write information generation unit for receiving burst length information of write command and write data, the control signal generation circuit being activated during input of data of a predetermined burst length in a write mode to generate a write control signal; and

at least one switching unit for connecting input terminals of the integrated memory circuits where data is input and a data bus in response to activation of the connection control signal;

wherein the write information generation unit comprises:

- a write control unit for receiving the write command signal to supply the write control signal and a counter control signal; and

a burst counter for counting the number of external input data to supply the counted data to the write control unit, wherein the burst counter is enabled in response to activation of the counter control signal;

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wherein the write control unit comprises:

- a counting response unit for generating an output signal responding to input of data having a predetermined burst length in response to an output signal of the burst counter;
- a latch unit for being enabled in response to the write command signal to latch the output signal of the counting response unit, and for supplying the write control signal; and
- a counting control signal generation unit for being enabled in response to the write command signal and for inputting the write control signal to supply the counter control signal.

24. The semiconductor memory device of claim **23**, wherein the semiconductor memory device is synchronized with an external clock signal, and the latch unit comprises:

- a first logic gate for being enabled by a first write command signal to generate the write control signal; and
- a second logic gate for being enabled by a second write command signal to respond to an output signal of the counting response unit;

wherein the write control signal is deactivated by the output signal of the counting response unit after input of data having a predetermined burst length.

25. The semiconductor memory device of claim **23**, wherein the counting control signal generation unit comprises:

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a first transfer unit for transferring an output signal of the latch unit in response to a first transition of the external clock signal;

a first latch for latching a signal transferred by the first transfer unit;

a second transfer unit for transferring a signal latched by the first latch in response to a second transition of the external clock signal; and

a second latch for latching a signal transferred by the first transfer unit, to generate the counter control signal; wherein a direction of the first transition is opposite to that of the second transition.

26. The semiconductor memory device of claim **23**, wherein the control signal generation circuit further comprises a control signal generation unit for generating the connection control signal activated in response to the write command signal and for maintaining the activation in accordance with the write control signal.

27. The semiconductor memory device of claim **26**, wherein the control signal generation unit comprises:

a logic gate responding to the write information signal; and

a drive transistor for activating the connection control signal in accordance with activation of the write control signal, the drive transistor being gated by an output signal by the logic gate.

* * * * *

PATENT ABSTRACTS OF JAPAN

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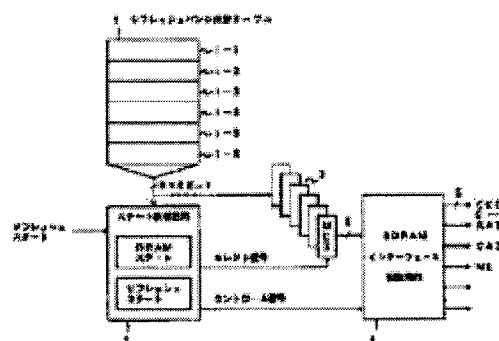
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(54) MEMORY CONTROL DEVICE

(57)Abstract:

PROBLEM TO BE SOLVED: To always suppress the amount of the current flowing at the time of refreshing synchronous DRAMs (dynamic random access memory 0) bellow a prescribed value regardless of kinds and the number of synchronous DRAMs being used.

SOLUTION: The content of refresh control flags is successively checked from the register 1-1 of the first stage in a refresh bank specifying table 1 and refreshings are executed only to synchronous banks whose executions of refreshings are specified by the refresh control flags. Thus, the number of synchronous DRAM banks whose refreshings are executed simultaneously and the combination of the banks at that time become to be arbitrarily stipulated by setting the pattern of refresh control flags to the refresh bank specifying table 1 and it is made possible to execute the refreshing of plural synchronous DRAM banks while dividing the refreshing into one cycle or more.



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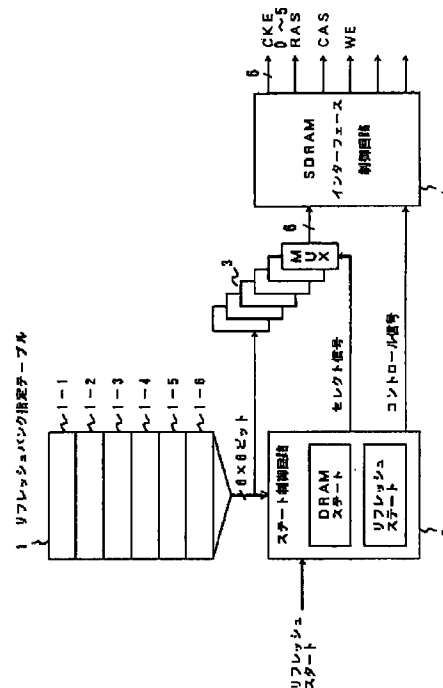
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(54)【発明の名称】 メモリ制御装置

(57)【要約】

【課題】使用されているシンクロナスDRAMの種類や数によらずに、シンクロナスDRAMのリフレッシュ時に流れる電流量を常に所定値以下に抑える。

【解決手段】リフレッシュバンク指定テーブル1の中の1段目のレジスタ1-1から順にリフレッシュ制御フラグの内容が調べられ、そのリフレッシュ制御フラグによってリフレッシュの実行が指定されているシンクロナスDRAMバンクに対してのみのリフレッシュが実行される。このため、リフレッシュバンク指定テーブル1に対してどのようなリフレッシュ制御フラグのパターンを設定するかによって、同時にリフレッシュが実行されるシンクロナスDRAMバンクの数およびその時のバンクの組み合わせを任意に規定できるようになり、複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて実行することが可能となる。



【特許請求の範囲】

【請求項1】 1以上のシンクロナスDRAMを各々が有する複数のシンクロナスDRAMバンクをアクセス制御するメモリ制御装置において、前記複数のシンクロナスDRAMバンクの数に対応する複数のレジスタから構成され、各レジスタには、前記シンクロナスDRAMバンク毎にリフレッシュを行うか否かを指定する複数のリフレッシュ制御情報が設定されているリフレッシュバンク指定テーブルと、リフレッシュサイクルの要求にตอบสนองして、前記リフレッシュバンク指定テーブルの複数段のレジスタを1段ずつ順番に選択し、その選択したレジスタに設定されているリフレッシュ制御情報によってリフレッシュの実行が指定されているシンクロナスDRAMバンクに対してのみリフレッシュを実行させるリフレッシュサイクル制御手段とを具備し、同時にリフレッシュが実行されるシンクロナスDRAMバンクを任意に組み合わせることにより、前記複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて実行できるようにしたことを特徴とするメモリ制御装置。

【請求項2】 前記リフレッシュサイクル制御手段は、前記リフレッシュ制御情報によってリフレッシュの実行が指定されているシンクロナスDRAMバンクそれぞれに対応するクロックイネーブル信号をアクティブにして、リフレッシュコマンドを発行することを特徴とする請求項1記載のメモリ制御装置。

【請求項3】 前記リフレッシュサイクル制御手段は、前記選択したレジスタに設定されているリフレッシュ制御情報が全てリフレッシュの非実行を示すとき、後段のレジスタを参照せずにリフレッシュサイクルの実行を終了することを特徴とする請求項1記載のメモリ制御装置。

【請求項4】 コンピュータシステムの主記憶として使用される複数のシンクロナスDRAMバンクをアクセス制御するメモリ制御装置において、前記コンピュータシステムに実装可能なシンクロナスDRAMバンクの最大数に対応する複数のレジスタから構成され、各レジスタには、前記シンクロナスDRAMバンク毎にリフレッシュを行うか否かを指定する複数のリフレッシュ制御情報が設定されているリフレッシュバンク指定テーブルと、リフレッシュサイクルの要求にตอบสนองして、前記リフレッシュバンク指定テーブルの複数段のレジスタを1段ずつ順番に選択し、その選択したレジスタに設定されているリフレッシュ制御情報によってリフレッシュの実行が指定されているシンクロナスDRAMバンクに対してのみリフレッシュを実行させるリフレッシュサイクル制御手段とを具備し、同時にリフレッシュが実行されるシンクロナスDRAM

バンクを任意に組み合わせることにより、前記複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて実行できるようにしたことを特徴とするメモリ制御装置。

【請求項5】 1以上のDRAMを各々が有する複数のDRAMバンクをアクセス制御するメモリ制御装置において、前記複数のDRAMバンクの数に対応する複数のレジスタから構成され、各レジスタには、前記DRAMバンク毎にリフレッシュを行うか否かを指定する複数のリフレッシュ制御情報が設定されているリフレッシュバンク指定テーブルと、リフレッシュサイクルの要求にตอบสนองして、前記リフレッシュバンク指定テーブルの複数段のレジスタを1段ずつ順番に選択し、その選択したレジスタに設定されているリフレッシュ制御情報によってリフレッシュの実行が指定されているDRAMバンクに対してのみリフレッシュを実行させるリフレッシュサイクル制御手段とを具備し、同時にリフレッシュが実行されるDRAMバンクを任意に組み合わせることにより、前記複数のDRAMバンクのリフレッシュを1以上のサイクルに分けて実行できるようにしたことを特徴とするメモリ制御装置。

【発明の詳細な説明】**【0001】**

【発明の属する技術分野】この発明はシンクロナスDRAMをアクセス制御するメモリ制御装置に関し、特に1以上のシンクロナスDRAMを各々が含む複数のシンクロナスDRAMバンクをアクセス制御するメモリ制御装置に関する。

【0002】

【従来の技術】近年、非同期DRAMに代わる高速DRAMとして、シンクロナスDRAMが注目されている。シンクロナスDRAMは、外部クロックに同期した入出力回路構成、コマンド形式のアクセス、バースト転送による連続アクセス、2バンク構成などの特徴を持つ。

【0003】このシンクロナスDRAMをコンピュータシステムの主記憶として利用すれば、例えばCPUとシンクロナスDRAMのクロックの共通化によりアクセス時のロスを少なくでき、またバースト転送を利用することによってCPUをノーウェイトで動作させること等を実現できる。

【0004】従来の非同期DRAMへのアクセスを行う場合、RAS#信号をアサートしローアドレスをDRAMに与えた後、CAS#信号をアサートしカラムアドレスをDRAMに与える事でアクセスを行った。

【0005】これに対し、シンクロナスDRAMへのアクセスを行う場合は、クロックイネーブルとなるCKE信号をアクティブにした状態で、ACT(バンク・アクティブ)コマンドにより、ローアドレスをシンクロナス

DRAMに与えた後、リード／ライトコマンドによりコラムアドレスをシンクロナスDRAMに与える事でアクセスを行なう。シンクロナスSDRAMの上記コマンド受け取りは、そのシンクロナスDRAMに入力されたクロックCLKに同期して行われる。

【0006】ACT (バンク・アクティブ) コマンドは非同期DRAMにおけるRAS#信号の立ち下がりに相当し、リード／ライトコマンドは非同期DRAMにおけるCAS#信号の立ち下がりに相当する。シンクロナスDRAMは、ACT (バンク・アクティブ) コマンドの入力にตอบสนองしてアクティブ状態となり、以降、プリチャージコマンドが入力されるまでそのアクティブ状態を維持する。

【0007】また、シンクロナスDRAMのリフレッシュを行う場合は、対象となるシンクロナスDRAMがアクティブ状態の時は、PRC (プリチャージ) コマンドにより、非アクティブ状態にし、tRP時間 (PRE to ACTIVE Command Period) 後、CKE信号をアクティブにした状態で、CBR (CAS Before RAS Refresh) コマンドにより、シンクロナスDRAM内部リフレッシュ動作を起動した後、tRC時間 (REF to REF/ACTIVE Command Period) 時間待つ事で可能となる。

【0008】しかしながら、このようなシンクロナスDRAMは、その高速動作のための構造上、非同期DRAMに比べてその消費電流は大きい。このため、シンクロナスDRAMを主記憶として利用した場合には、そのリフレッシュ時に多大な電流が流れることになり、システム上の電源回路の容量を大きくする必要がある。

【0009】リフレッシュ時に大電流が流れるのは、次の理由による。すなわち、主記憶は、複数のシンクロナスDRAMチップを各々が含む複数のシンクロナスDRAMバンクから構成されるが、リフレッシュ時にはこれら全てのバンクのチップそれぞれに対して同時にリフレッシュ動作が行われる。したがって、主記憶全体では、シンクロナスDRAM1チップ当たりのリフレッシュ時のピーク電流の全チップ数倍もの電流が流れることになる。

【0010】

【発明が解決しようとする課題】上述したように、従来では、全てのシンクロナスDRAMバンクのチップそれぞれに対して同時にリフレッシュ動作が行われるので、リフレッシュ時に多大な電流が流れるという問題があった。

【0011】この発明はこのような点に鑑みてなされたものであり、使用されているシンクロナスDRAMの種類や数によらずに、シンクロナスDRAMのリフレッシュ時に流れる電流量を常に所定値以下に抑えることができるメモリ制御装置を提供することを目的とする。

【0012】

【課題を解決するための手段】この発明は、1以上のシンクロナスDRAMを各々が有する複数のシンクロナスDRAMバンクをアクセス制御するメモリ制御装置において、前記複数のシンクロナスDRAMバンクの数に対応する複数のレジスタから構成され、各レジスタには、前記シンクロナスDRAMバンク毎にリフレッシュを行うか否かを指定する複数のリフレッシュ制御情報が設定されているリフレッシュバンク指定テーブルと、リフレッシュサイクルの要求にตอบสนองして、前記リフレッシュバンク指定テーブルの複数段のレジスタを1段ずつ順番に選択し、その選択したレジスタに設定されているリフレッシュ制御情報によってリフレッシュの実行が指定されているシンクロナスDRAMバンクに対してのみリフレッシュを実行させるリフレッシュサイクル制御手段とを具備し、同時にリフレッシュが実行されるシンクロナスDRAMバンクを任意に組み合わせることにより、前記複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて実行できるようにしたことを特徴とする。

【0013】このメモリ制御装置においては、リフレッシュサイクルの実行が要求されると、まず、リフレッシュバンク指定テーブルの中の1段目のレジスタが選択される。そして、その1段目のレジスタに設定されているリフレッシュ制御情報によってリフレッシュの実行が指定されているシンクロナスDRAMバンクに対してリフレッシュが実行される。この後、2段目以降のレジスタについても、リフレッシュの実行を指定するリフレッシュ制御情報が含まれている場合には、そのリフレッシュ制御情報に対応するバンクに対してのリフレッシュが実行される。このため、リフレッシュバンク指定テーブルに対してどのようなリフレッシュ制御情報のパターンを設定するかによって、同時にリフレッシュが実行されるシンクロナスDRAMバンクの数およびその時のバンクの組み合わせを任意に規定できるようになり、複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて実行することが可能となる。

【0014】したがって、使用されているシンクロナスDRAMの種類や数によらずに、シンクロナスDRAMのリフレッシュ時に流れるピーク電流値を常に所定値以下に抑えることができる。

【0015】

【発明の実施の形態】以下、図面を参照してこの発明の実施形態を説明する。図1には、この発明の一実施形態に係るシンクロナスDRAMコントローラの構成が示されている。このシンクロナスDRAMコントローラは、コンピュータシステムに主記憶として設けられた複数のシンクロナスDRAM (SDRAM) バンクをアクセス制御するためのものであり、図示のように、リフレッシュバンク指定テーブル1、ステート制御回路2、セレクト

タ3、SDRAMインターフェース制御回路4を備えている。

【0016】ここでは、コンピュータシステムに実装可能な最大シンクロナスDRAMバンク数が6個であり、システムボード上には図2に示されているように4個のシンクロナスDRAMバンク0～3が実装されており、残りの2個のシンクロナスDRAMバンクは拡張用として使用される場合を例示して説明する。各シンクロナスDRAMバンクにはクロックイネーブル信号CKEが個別に割り当てられるので、SDRAMインターフェース制御回路4からは6本のクロックイネーブル信号CKEが発生されることになる。

【0017】リフレッシュバンク指定テーブル1は、システムのCPUによってリードライト可能な6段のレジスタ1-1～1-6から構成されており、レジスタ1-1～1-6それぞれのビット0～5にはシンクロナスDRAMバンク0～5に対応するリフレッシュ制御フラグが設定される。ここで、リフレッシュ制御フラグの値が“1”時は、そのフラグのビット位置に対応するシンクロナスDRAMバンクがリフレッシュの対象であることが指定される。

【0018】ステート制御回路2は、各シンクロナスDRAMバンクの状態の遷移を管理するDRAMステート回路と、リフレッシュサイクルの状態遷移を管理するリフレッシュステート回路とを含んでおり、リフレッシュバンク指定テーブル1に設定されたリフレッシュ制御フラグの読み取り、セクタ3の選択動作の制御、SDRAMインターフェース制御回路4によるリフレッシュサイクルの実行制御を行う。

【0019】このステート制御回路2は、リフレッシュタイマなどから定期的に発生されるリフレッシュスタート要求にตอบสนองして、リフレッシュバンク指定テーブル1の1段目のレジスタ1-1から順番に1段ずつ選択し、その選択したレジスタに設定されているリフレッシュ制御フラグの内容を調べる。6ビットのリフレッシュ制御フラグの中に“1”のリフレッシュ制御フラグが含まれている場合には、セレクト信号を発生してそのフラグをセクタ3に選択させる。また、SDRAMインターフェース制御回路4にコントロール信号を供給して、リフレッシュサイクルの実行を制御する。

【0020】セクタ3は6ビットも01ビットのセクタであり、6本のCKE信号それぞれに対応して設けられている。すなわち、各セクタ3は、6個のレジスタ1-1～1-6それぞれの同一ビット位置に設定されている6ビットのリフレッシュ制御フラグを入力し、そのうちの1ビットを選択する。したがって6ビットのリフレッシュ制御フラグの中に“1”のリフレッシュ制御フラグが含まれている場合には、6個のセクタ3からは、その6ビットのリフレッシュ制御フラグの値がそのまま出力されることになる。

【0021】SDRAMインターフェース制御回路4は、ステート制御回路2からのコントロール信号に従って、各SDRAMバンクへの制御信号を生成する。この図1の構成においては、リフレッシュスタート要求が入力されると、まず、リフレッシュバンク指定テーブル1の中の1段目のレジスタ1-1の内容がステート制御回路2によって参照される。その1段目のレジスタ1-1に設定されている6ビットのリフレッシュ制御フラグの中に“1”のリフレッシュ制御フラグが含まれている場合には、ステート制御回路2からのセレクト信号により、6個のセクタ3それぞれが1段目のレジスタ1-1に対応するビット位置の入力を選択する。そして、ステート制御回路2の制御の下、“1”のリフレッシュ制御フラグに対応するシンクロナスDRAMバンクのリフレッシュがSDRAMインターフェース制御回路4によって同時実行される。この後、2段目以降のレジスタについても順次同様の処理が行われ、リフレッシュの実行を指定する“1”のリフレッシュ制御フラグが含まれている場合には、そのリフレッシュ制御フラグに対応するシンクロナスDRAMバンクに対してのリフレッシュが実行される。

【0022】このようにして、1段目から6番目までのレジスタ1-1～1-6の内容が順次読み込まれて、“1”のリフレッシュ制御フラグに対応するシンクロナスDRAMバンクに対してリフレッシュが実行される。もし、読み込んだ6ビットのリフレッシュ制御フラグが全て“0”であった場合には、読み込んでないレジスタが残っていても、その時点でリフレッシュサイクルは終了される。

【0023】このため、リフレッシュバンク指定テーブル1に対してどのようなリフレッシュ制御フラグのパターンを設定するかによって、同時にリフレッシュが実行されるシンクロナスDRAMバンクの数およびその時のバンクの組み合わせを任意に規定できるようになり、複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて（最大で6サイクルに分割される）実行することが可能となる。

【0024】次に、図3乃至図6を参照して、リフレッシュバンク指定テーブル1に設定される6×6ビットのリフレッシュ制御フラグの具体的なパターンについて説明する。

【0025】図3のパターンでは、1段のレジスタ毎に“1”のリフレッシュ制御フラグが互いに異なるビット位置に1つずつ設定されている。この場合、シンクロナスDRAMバンクのリフレッシュは1バンク単位で6回に分けて行われる。したがって、リフレッシュ時のピーク電流は1バンクに属するチップの合計分だけで済み、全てのバンクを同時にリフレッシュする場合の1/6にピーク電流を低減できる。

【0026】図4のパターンでは、1段目のレジスタ1

ー1の全てのビット位置に“1”のリフレッシュ制御フラグが設定されている。この場合、全てのバンクが同時にリフレッシュされ、それが終了した時点でリフレッシュサイクルが終了される。このパターンは、システムに用意された電源回路の容量が十分大きい場合に好適なものであり、リフレッシュサイクルに要する時間を低減することができる。

【0027】図5のパターンでは、1段目から3段目までのレジスタ1-1~1-3それぞれの互いに異なるビット位置に“1”のリフレッシュ制御フラグが2つずつ設定されている。この場合、シンクロナスDRAMバンクのリフレッシュは2バンク単位で3回に分けて行われ、3回目のリフレッシュが終了した時点でリフレッシュサイクルが終了される。したがって、リフレッシュ時のピーク電流は2バンクに属するチップの合計分だけで済み、全てのバンクを同時にリフレッシュする場合の1/3にピーク電流を低減できる。

【0028】図6のパターンは、標準実装されているバンク0~3と、拡張スロットを介して増設されたバンク4、5のバンク構成やチップの種類が異なり、バンク0~3に比べ、バンク4、5に大きなピーク電流が流れる場合に好適なパターンである。すなわち、このパターンにおいては、バンク0~3については同時にリフレッシュされ、その後、バンク4、5のリフレッシュが2回に分けて行われる。

【0029】つぎに、図7および図8を参照して、リフレッシュ制御動作の手順を具体的に説明する。図7はステート制御回路2によって管理されているリフレッシュサイクルの状態遷移を示す状態遷移図であり、また図8はリフレッシュサイクルの動作を示すタイミングチャートである。

【0030】まず、システム起動時に、リフレッシュバンク指定テーブル1のレジスタに対してリフレッシュ制御フラグの設定がCPUによって行われる。ここでは、図3のリフレッシュ制御パターンが設定される場合を想定する。

【0031】リフレッシュスタート要求が発行されると、ステート制御回路2は、まず、SDRAM状態ステート回路からアクティブなバンクの有無を調べる。いずれかのバンクがアクティブな時は、IDLEサイクルからPRCサイクルへ進み、そのアクティブなバンク（図8ではバンク1）に対してプリチャージコマンド（RAS#=0, CAS#=1, WE#=0）を送り、tRP時間確保後、CHECKサイクルへ進む。またアクティブなバンクが無い時は、PRCサイクルに移行せずに、CHECKサイクルに直接進む。

【0032】CHECKサイクルでは、レジスタ1-1の内容がチェックされる。もし、6ビット全てが“0”ならば、リフレッシュを実行せずにIDLEサイクルに戻る。

【0033】この例では、バンク0に対応するリフレッシュ制御フラグが“1”であるので、それがセクタ3を介してSDRAMインターフェース制御回路4に送られる。SDRAMインターフェース制御回路4は、CBRコマンドを発行する。このコマンドは最後のバンクについてのリフレッシュが完了するまで保持される。

【0034】ステートはCBR00サイクルへ進み、“1”のリフレッシュ制御フラグに対応するバンク（ここでは、バンク0）のCKE0信号がアクティブにされる。次にCBR01サイクルへ進み、チップセレクト信号CSがアクティブにされる。これにより、バンク0に対してリフレッシュコマンドが送り出される。

【0035】また、このCBR01サイクルでは、レジスタ1-2の内容のチェックも同時に行われる。もし、6ビット全てが“0”であれば、tRCサイクルに進みtRC時間確保後、IDLEサイクルに戻りリフレッシュサイクルを終了する。

【0036】“1”にセットされたビットがあれば（この例では、バンク1に対応するリフレッシュ制御フラグが“1”）、CBR10サイクルへ進む。このサイクルでは、バンク1に対応するCKE1信号がアクティブにされる。またこの時、CBR00においてアクティブにしたCKE0信号をインアクティブにする。次にCBR11サイクルへ進み、チップセレクト信号CSがアクティブにされる。これにより、バンク1に対してリフレッシュコマンドが送り出される。また、このCBR11サイクルでは、レジスタ1-3の内容のチェックも同時に行われる。以降、同様にして、バンク2~5についてのリフレッシュが行われる。

【0037】以上説明したように、この実施形態においては、リフレッシュバンク指定テーブル1の中の1段目のレジスタ1-1から順にリフレッシュ制御フラグの内容が調べられ、そのリフレッシュ制御フラグによってリフレッシュの実行が指定されているシンクロナスDRAMバンクに対してのみのリフレッシュが実行される。このため、リフレッシュバンク指定テーブル1に対してどのようなリフレッシュ制御フラグのパターンを設定するかによって、同時にリフレッシュが実行されるシンクロナスDRAMバンクの数およびその時のバンクの組み合わせを任意に規定できるようになり、複数のシンクロナスDRAMバンクのリフレッシュを1以上のサイクルに分けて実行することが可能となる。なお、この様なリフレッシュ制御は非同期のDRAMバンクに対して適用しても良い。これにより、DRAMのさらなる低消費電力化を図ることができる。

【0038】

【発明の効果】以上説明したように、この発明のメモリ制御装置によれば、同時にリフレッシュが実行されるシンクロナスDRAMバンクの数およびその時のバンクの組み合わせを任意に規定できるので、使用されているシ

ンクロナスDRAMの種類や数によらずに、シンクロナスDRAMのリフレッシュ時に流れる電流量を常に所定値以下に抑えることが可能となる。

【図面の簡単な説明】

【図1】この発明の一実施形態に係るSDRAMコントローラの構成を示すブロック図。

【図2】同実施形態のSDRAMコントローラによって制御されるSDRAMバンクの構成を示す図。

【図3】同実施形態のSDRAMコントローラのリフレッシュバンク指定テーブルに設定されるリフレッシュ制御フラグの第1パターンを説明するための図。

【図4】同実施形態のSDRAMコントローラのリフレッシュバンク指定テーブルに設定されるリフレッシュ制御フラグの第2パターンを説明するための図。

【図5】同実施形態のSDRAMコントローラのリフレ

ッシュバンク指定テーブルに設定されるリフレッシュ制御フラグの第3パターンを説明するための図。

【図6】同実施形態のSDRAMコントローラのリフレッシュバンク指定テーブルに設定されるリフレッシュ制御フラグの第4パターンを説明するための図。

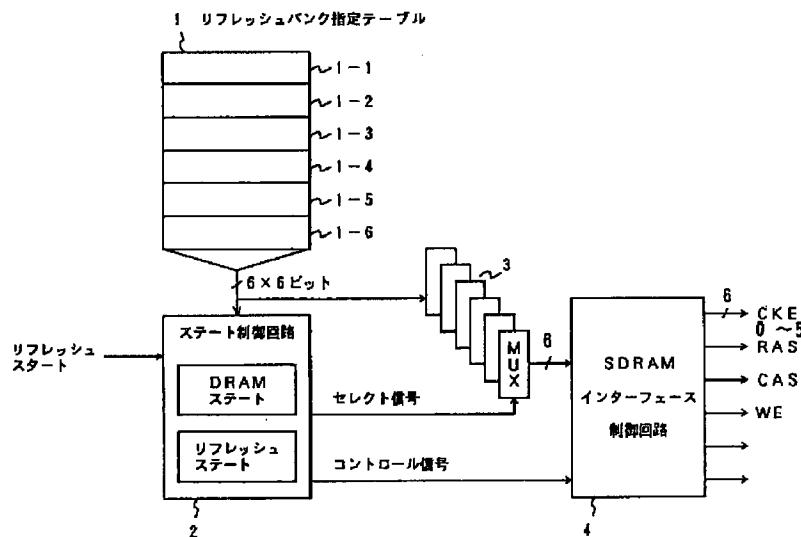
【図7】同実施形態のSDRAMコントローラによって管理されているリフレッシュサイクルの状態遷移を示す図。

【図8】同実施形態のSDRAMコントローラによって実行されるリフレッシュ制御動作を示すタイミングチャート。

【符号の説明】

1…リフレッシュバンク指定テーブル、1-1～1-6…レジスタ、2…ステート制御回路、4…SDRAMインターフェース制御回路。

【図1】



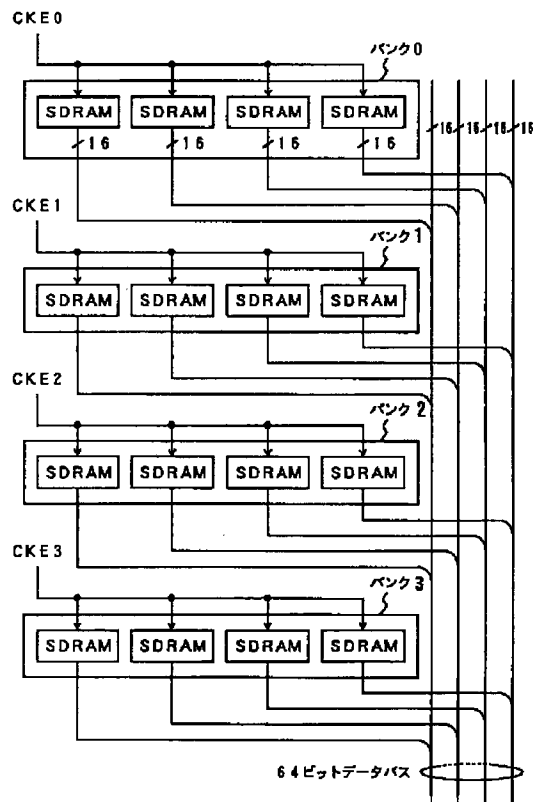
【図3】

バンク0	1	2	3	4	5	
1	0	0	0	0	0	1-1
0	1	0	0	0	0	1-2
0	0	1	0	0	0	1-3
0	0	0	1	0	0	1-4
0	0	0	0	1	0	1-5
0	0	0	0	0	1	1-6

【図4】

バンク0	1	2	3	4	5	
1	1	1	1	1	1	1-1
0	0	0	0	0	0	1-2
0	0	0	0	0	0	1-3
0	0	0	0	0	0	1-4
0	0	0	0	0	0	1-5
0	0	0	0	0	0	1-6

【図2】



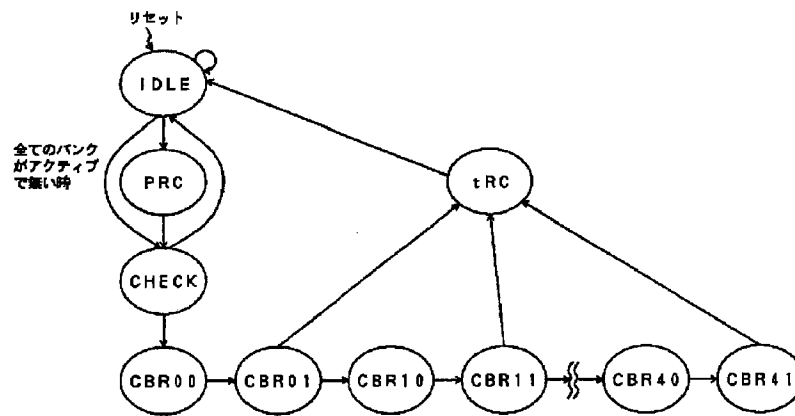
【図5】

バンク0	1	2	3	4	5	
1	0	0	1	0	0	1-1
0	1	0	0	1	0	1-2
0	0	1	0	0	1	1-3
0	0	0	0	0	0	1-4
0	0	0	0	0	0	1-5
0	0	0	0	0	0	1-6

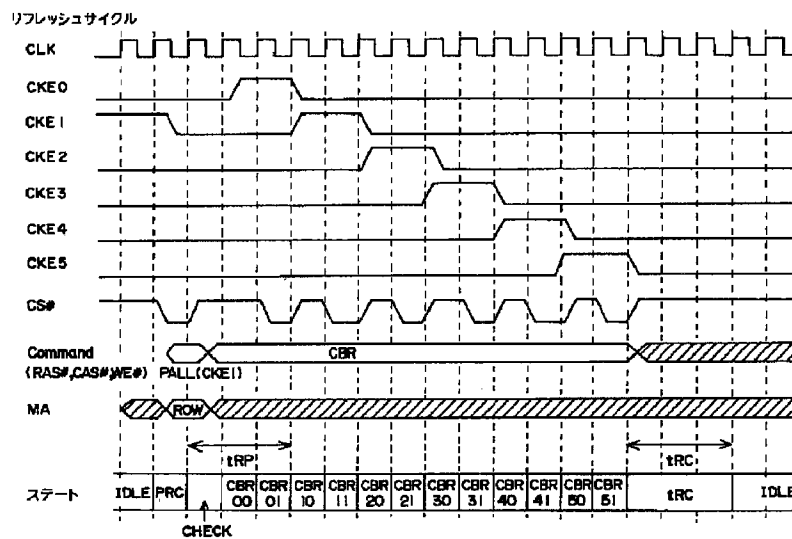
【図6】

バンク0	1	2	3	拡張メモリ		
				4	5	
1	1	1	1	0	0	1-1
0	0	0	0	1	0	1-2
0	0	0	0	0	1	1-3
0	0	0	0	0	0	1-4
0	0	0	0	0	0	1-5
0	0	0	0	0	0	1-6

【図7】



【図8】



*** NOTICES ***

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- 1.This document has been translated by computer. So the translation may not reflect the original precisely.
- 2.**** shows the word which can not be translated.
- 3.In the drawings, any words are not translated.

CLAIMS

[Claim(s)]

[Claim 1]In a memory controller which carries out access control of two or more synchronous DRAM banks where each has one or more Synchronous DRAMs,

A memory controller characterized by dividing refreshment of two or more aforementioned synchronous DRAM banks into one or more cycles, and enabling it to perform it by having the following and combining arbitrarily synchronous DRAM banks where refreshment is performed simultaneously.

The refreshment bank appointed table where it comprises two or more registers corresponding to the number of two or more aforementioned synchronous DRAM banks, and two or more refreshment control information which specifies whether it refreshes for said every synchronous DRAM banks is set to each register.

Answer a demand of a refresh cycle and it chooses two or more steps of registers [one step of] of the aforementioned refreshment bank appointed table at a time in order, A refresh cycle control means which performs refreshment only to synchronous DRAM banks where execution of refreshment is specified by refreshment control information set as the selected register.

[Claim 2]The aforementioned refresh cycle control means activates a clock enable signal corresponding to each synchronous DRAM banks where execution of refreshment is specified by the aforementioned refreshment control information, The memory controller according to claim 1 publishing a refresh command.

[Claim 3]The memory controller according to claim 1 when the aforementioned refresh cycle control means shows [all refreshment control information set as a register which chose / aforementioned] un-performing of refreshment, wherein it ends execution of a refresh cycle, without referring to a latter register.

[Claim 4]In a memory controller which carries out access control of two or more synchronous DRAM banks used as main memory of a computer system,

A memory controller characterized by dividing refreshment of two or more aforementioned synchronous DRAM banks into one or more cycles, and enabling it to perform it by having the following and combining arbitrarily synchronous DRAM banks where refreshment is performed simultaneously.

Comprise two or more registers corresponding to the maximum number of synchronous DRAM banks which can be mounted in the aforementioned computer system, and to each register, The

refreshment bank appointed table where two or more refreshment control information which specifies whether it refreshes for said every synchronous DRAM banks is set up.

Answer a demand of a refresh cycle and it chooses two or more steps of registers [one step of] of the aforementioned refreshment bank appointed table at a time in order, A refresh cycle control means which performs refreshment only to synchronous DRAM banks where execution of refreshment is specified by refreshment control information set as the selected register.

[Claim 5]In a memory controller which carries out access control of two or more DRAM banks where each has one or more DRAMs,

A memory controller characterized by dividing refreshment of two or more aforementioned DRAM banks into one or more cycles, and enabling it to perform it by having the following and combining arbitrarily DRAM banks where refreshment is performed simultaneously.

The refreshment bank appointed table where it comprises two or more registers corresponding to the number of two or more aforementioned DRAM banks, and two or more refreshment control information which specifies whether it refreshes for every aforementioned DRAM banks is set to each register.

Answer a demand of a refresh cycle and it chooses two or more steps of registers [one step of] of the aforementioned refreshment bank appointed table at a time in order, A refresh cycle control means which performs refreshment only to DRAM banks where execution of refreshment is specified by refreshment control information set as the selected register.

DETAILED DESCRIPTION

[Detailed Description of the Invention]

[0001]

[Field of the Invention]This invention relates to the memory controller which carries out access control of two or more synchronous DRAM banks where each contains especially one or more Synchronous DRAMs about the memory controller which carries out access control of the Synchronous DRAM.

[0002]

[Description of the Prior Art]In recent years, the Synchronous DRAM attracts attention as high-speed DRAM replaced with asynchronous DRAM. A Synchronous DRAM has the characteristics, such as input output circuit composition in sync with an external clock, access of command format, continuous access by burst transmission, and 2 bank configurations.

[0003]If this Synchronous DRAM is used as main memory of a computer system, For example, it is realizable by being able to lessen the loss at the time of access by communalization of the clock of CPU and a Synchronous DRAM, and using burst transmission to operate CPU by nowait etc.

[0004]When access to the conventional asynchronous DRAM is performed, after asserting the RAS# signal and giving a row address to DRAM, it accessed by asserting a CAS# signal and giving a column address to DRAM.

[0005]On the other hand, when performing access to a Synchronous DRAM, where the CKE signal used as clock enabling is activated, after giving a row address to a Synchronous DRAM by the ACT (bank - active) command, it accesses by giving a column address to a Synchronous DRAM with Read/write command. The above-mentioned command receipt of synchronous SDRAM is performed synchronizing with the clock CLK input into the Synchronous DRAM.

[0006]the ACT (bank - active) command is equivalent to falling of the RAS# signal in asynchronous DRAM, and Read/write command is equivalent to falling of the CAS# signal in asynchronous DRAM. a Synchronous DRAM will answer the input of the ACT (bank - active) command, will be in an active state, and henceforth, it maintains the active state until a precharge command is input.

[0007]When refreshing a Synchronous DRAM, When the target Synchronous DRAM is bitter taste tape shape voice, Where it used non-bitter taste tape shape voice and a CKE signal is made into AKUTIPU after tRP time (PRE to ACTIVE Command Period) by the PRC (precharge) command, By the CBR (CAS Before RASRefresh) command, after starting Synchronous DRAM internal refreshment operation, it becomes possible by waiting tRC time (REF to REF/ACTIVE Command Period) time.

[0008]However, compared with asynchronous DRAM, the consumed electric current of such a Synchronous DRAM is large on the structure for the high-speed operation. For this reason, when a Synchronous DRAM is used as main memory, current great at the time of that refreshment will flow, and it is necessary to carry out capacity of the power supply circuit on a system largely.

[0009]It is based on the following Reason that a high current flows at the time of refreshment. That is, although main memory comprises two or more synchronous DRAM banks where each contains two or more synchronous DRAM chips, at the time of refreshment, refresh operation is simultaneously performed to each chip of all of a bank of these. Therefore, in the whole main memory, all the chip number double thing current of the peak current at the time of refreshment per Synchronous DRAM 1 chip will flow.

[0010]

[Problem to be solved by the invention]As mentioned above, since refresh operation was simultaneously performed to each chip of all the synchronous DRAM banks, by the former, there was a problem that current great at the time of refreshment flowed.

[0011]The present invention is made in view of such a point, and is a thing.

The purpose is to provide a memory controller which can always hold down a current amount which flows at the time of refreshment of a Synchronous DRAM to below a specified value, without being based on a kind and number of Synchronous DRAMs.

[0012]

[Means for solving problem]In the memory controller with which this invention carries out access control of two or more synchronous DRAM banks where each has one or more Synchronous DRAMs, Comprise two or more registers corresponding to the number of two or more aforementioned synchronous DRAM banks, and to each register, The refreshment bank appointed table where two or more refreshment control information which specifies whether it refreshes for every above-mentioned synchronous DRAM banks is set up, Answer the demand of a refresh cycle and it chooses two or more steps of registers [one step of] of the aforementioned refreshment bank appointed table at a time in order, The refresh cycle control means which performs refreshment only to the synchronous DRAM banks where execution of refreshment is specified by the refreshment control information set as the selected register is provided, Refreshment of two or more aforementioned synchronous DRAM banks is divided into one or more cycles, and it enabled it to perform it by combining arbitrarily the synchronous DRAM banks where refreshment is performed simultaneously.

[0013]In this memory controller, a demand of execution of a refresh cycle will choose the 1st step of register in the refreshment bank appointed table first. And refreshment is performed to the

synchronous DRAM banks where execution of refreshment is specified by the refreshment control information set as the 1st step of the register. Then, when the refreshment control information which specifies execution of refreshment is included also about the register after the 2nd step, refreshment to the bank corresponding to that refreshment control information is performed. For this reason, by of what kind of refreshment control information a pattern is set up to the refreshment bank appointed table. The number of the synchronous DRAM banks where refreshment is performed simultaneously, and the combination of the bank at that time can be arbitrarily specified now, and it becomes possible to divide refreshment of two or more synchronous DRAM banks into one or more cycles, and to perform it.

[0014]Therefore, the peak current value which flows at the time of refreshment of a Synchronous DRAM can always be held down below to a specified value, without being based on the kind and number of Synchronous DRAMs which are used.

[0015]

[Mode for carrying out the invention]Hereafter, the embodiment of this invention is described with reference to Drawings. The composition of the synchronous DRAM controller concerning one embodiment of this invention is shown in Fig.1. Like [it is for this synchronous DRAM controller carrying out access control of two or more Synchronous DRAM (SDRAM) banks provided by the computer system as main memory, and] a graphic display, It has the refreshment bank appointed table 1, the State control circuit 2, the selector 3, and the SDRAM interface control circuit 4.

[0016]Here, the number of the maximum synchronous DRAM banks which can be mounted in a computer system is six pieces, On the system board, four synchronous DRAM banks 0-3 are mounted as shown in Fig.2, and the two remaining synchronous DRAM banks illustrate and describe the case where it is used as an object for extension. Since the clock enable signal CKE assigns individually each synchronous DRAM banks and it is divided, the six clock enable signals CKE will be generated from the SDRAM interface control circuit 4.

[0017]The refreshment bank appointed table 1 is constituted from six steps of registers 1-1 to 1-6 in which read/write is possible by CPU of the system, the register 1-1 to 1-6 -- the refreshment control flag corresponding to synchronous DRAM banks 0-5 is set to each of pits 0-5. Here, it is specified that the value of a refreshment control flag is an object of refreshment of the synchronous DRAM banks corresponding to the Flag's bit position at the time of "1."

[0018]The DRAM State circuit where the State control circuit 2 manages transition of the state of each synchronous DRAM banks, The refreshment State circuit which manages the change state of a refresh cycle is included, Reading of the fresh control flag set as the refreshment bank appointed table 1, control of the selection operation of the selector 3, and execution control of the refresh cycle by the SDRAM interface control circuit 4 are performed.

[0019]This State control circuit 2 answers the refreshment start demand periodically generated from refresh timers etc., It chooses one step at a time in an order from the 1st step of register 1-1 of the refreshment bank appointed table 1, and the contents of the refreshment control flag set as the selected register are investigated. When the refreshment control flag of "1" is contained in a 6-bit refreshment control flag, a select signal is generated and the Flag is made to choose it as the selector 3. A control signal is supplied to the SDRAM interface control circuit 4, and execution of a refresh cycle is controlled.

[0020]The selector 3 is 6-bit a selector of 1 bit of to(es), and is provided corresponding to each six CKE signals. namely, each selector 3 -- the six registers 1-1 to 1-6 -- the 6-bit refreshment control flag set as each identical bit position is input, and 1 bit of them is chosen. Therefore,

when the refreshment control flag of "1" is contained in a 6-bit refreshment control flag, from the six selectors 3, the value of the 6-bit refreshment control flag will be outputted as it is.

[0021]The SDRAM interface control circuit 4 generates the control signal to each SDRAM bank according to the control signal from the State control circuit 2. In the composition of this Fig.1, if a refreshment start demand is input, refer to the contents of the 1st step of register 1-1 in the refreshment bank appointed table 1 for the State control circuit 2 first. When the refreshment control flag of "1" is contained in the 6-bit refreshment control flag set as the 1st step of the register 1-1, the select signal from the State control circuit 2 -- the six selectors 3 -- each chooses the input of the bit position corresponding to the 1st step of register 1-1. And the concurrency of the refreshment of the synchronous DRAM banks corresponding to the refreshment control flag of "1" is carried out by the SDRAM interface control circuit 4 under control of the State control circuit 2. Then, same processing is sequentially performed also about the register after the 2nd step, and when the refreshment control flag of "1" which specifies execution of refreshment is contained, refreshment to the synchronous DRAM banks corresponding to that refreshment control flag is performed.

[0022]Thus, the contents of the register 1-1 to 1-6 from the 1st step to the 6th are read sequentially, and refreshment is performed to the synchronous DRAM banks corresponding to the refreshment control flag of "1." When all of the read 6-bit refreshment control flag are "0", even if the register which has not been read remains, a refresh cycle is ended at the time.

[0023]For this reason, by of what kind of refreshment control flag a pattern is set up to the refreshment bank appointed table 1. The number of the synchronous DRAM banks where refreshment is performed simultaneously, and the combination of the bank at that time can be arbitrarily specified now, It becomes possible to divide refreshment of two or more synchronous DRAM banks into one or more cycles, and to perform it (divided into six cycles at the maximum).

[0024]Next, with reference to Fig.3 thru/or Fig.6, it describes about the specific pattern of the 6X6-bit refreshment control flag set as the refreshment bank appointed table 1.

[0025]In the pattern of Fig.3, the refreshment control flag of "1" is set as every one mutually different bit position for one step of every register. In this case, refreshment of synchronous DRAM banks is performed in 6 steps by 1 bank unit. Therefore, the peak current at the time of refreshment requires only a totaled part of the chip belonging to one bank, and can reduce peak current to one sixth in the case of refreshing all the banks simultaneously.

[0026]In the pattern of Fig.4, the refreshment control flag of "1" is set to all the bit positions of the 1st step of register 1-1. In this case, all the banks are refreshed simultaneously, and a refresh cycle is ended when it is completed. This pattern is preferable when the capacity of the power supply circuit prepared for the system is sufficiently large, and it can reduce the time which a refresh cycle takes.

[0027]the pattern of Fig.5 -- the register 1-1 to 1-3 from the 1st step to the 3rd step -- every two refreshment control flags of "1" are set to the bit position where each differs mutually. In this case, refreshment of synchronous DRAM banks is performed in 3 steps by 2 bank units, and a refresh cycle is ended when the 3rd refreshment is completed. Therefore, the peak current at the time of refreshment requires only a totaled part of the chip belonging to two banks, and can reduce peak current to one third in the case of refreshing all the banks simultaneously.

[0028]The pattern of Fig.6 is a preferable pattern, when the bank configuration of the banks 0-3 by which standard mounting is carried out, and the banks 4 and 5 extended via the expansion slot differs from the kind of chip and big peak current flows into the banks 4 and 5 compared with

the banks 0-3. That is, in this pattern, about the banks 0-3, it refreshes simultaneously, and refreshment of the banks 4 and 5 is performed in 2 steps after that.

[0029]Next, with reference to Fig.7 and Fig.8, the procedure of a refreshment control action is described specifically. Fig.7 is a state transition diagram showing the change state of the refresh cycle managed by the State control circuit 2, and Fig.8 is a timing chart which shows operation of a refresh cycle.

[0030]First, setting out of a refreshment control flag is performed by CPU to the register of the refreshment bank appointed table 1 at the time of a system startup. Here, the case where the refreshment control pattern of Fig.3 is set up is assumed.

[0031]If a refreshment start demand is published, the State control circuit 2 will investigate the existence of an AKUTEIPU bank from a SDRAM state State circuit first. When AKUTEIPU [banks / one of], a precharge command (RAS#=0, CAS#=1, WE#=0) is sent from an IDLE cycle to PRC SAIKURUHE progress and its active bank (Fig.8 bank 1), and they are after tRP time reservation and CHECK SAIKURUHE ****. When there is no AKUTEIPU bank, it proceeds to a CHECK cycle directly, without shifting to a PRC cycle.

[0032]The contents of the register 1-1 are checked in a CHECK cycle. if -- all six pits -- "0" -- if it becomes, it will return to an IDLE cycle, without performing refreshment.

[0033]In this example, since the refreshment control flag corresponding to the bank 0 is "1", it is sent to the SDRAM interface control circuit 4 via the selector 3. The SDRAM interface control circuit 4 publishes the CBR command. This command is held until the refreshment about the last bank is completed.

[0034]CKE0 signal of the bank (here bank 0) corresponding to CBR00 SAIKURUHE progress and the refreshment control flag of "1" in the State is made into AKUTEIPU. Next, CBR01 SAIKURUHE progress and chip select signal CS are made into AKUTEIPU. Thereby, a refresh command is sent out to the bank 0.

[0035]In this 01 cycle CBR, the check of the contents of the register 1-2 is also performed simultaneously. If 6 bits of all are "0", it proceeds to a tRC cycle, and after tRC time reservation, it will return to an IDLE cycle and a refresh cycle will be ended.

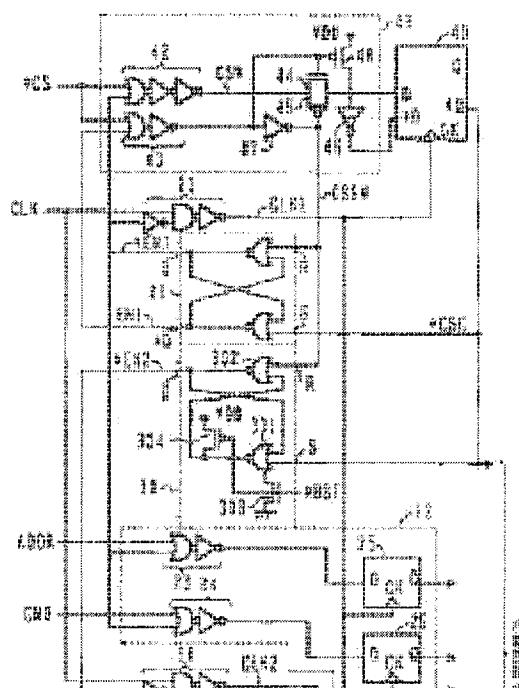
[0036]If there is a pit set to "1" (the refreshment control flag corresponding to the bank 1 is "1" at this example), it will be CBR10 SAIKURUHE ****. In this cycle, CKE1 signal corresponding to the bank 1 is made into AKUTEIPU. At this time, CKE0 signal activated in CBR00 is made inactive. Next, CBR11 SAIKURUHE progress and chip select signal CS are made into AKUTEIPU. Thereby, a refresh command is sent out to the bank 1. In this 11 cycle CBR, the check of the contents of the register 1-3 is also performed simultaneously. Henceforth, refreshment about the banks 2-5 is performed similarly.

[0037]In [as described above] this embodiment, The contents of the refreshment control flag are investigated sequentially from the register 1-1 of one-step Sakka in the refreshment bank appointed table 1, and refreshment only to the synchronous DRAM banks where execution of refreshment is specified by the refreshment control flag is performed. For this reason, by of what kind of refreshment control flag a pattern is set up to the refreshment bank appointed table 1. The number of the synchronous DRAM banks where refreshment is performed simultaneously, and the combination of the bank at that time can be arbitrarily specified now, and it becomes possible to divide refreshment of two or more synchronous DRAM banks into one or more cycles, and to perform it. Such refreshment control may be applied to asynchronous DRAM banks. Thereby, further power consumption reduction of DRAM can be attained.

[0038]

[Effect of the Invention] Since the number of the synchronous DRAM banks where refreshment is performed simultaneously, and the combination of the bank at that time can be arbitrarily specified according to the memory controller of this invention as described above, It becomes possible to always hold down the current amount which flows at the time of refreshment of a Synchronous DRAM to below a specified value, without being based on the kind and number of Synchronous DRAMs which are used.

[Translation done.]

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G11C7/10W; G11C7/10W7; G11C7/22**Application number:** JP19990092246 19990331**Priority number (s):** JP19990092246 19990331**Also published as:** JP4187346 (B2) US6188641 (B1)**Abstract of JP2000285674 (A)**

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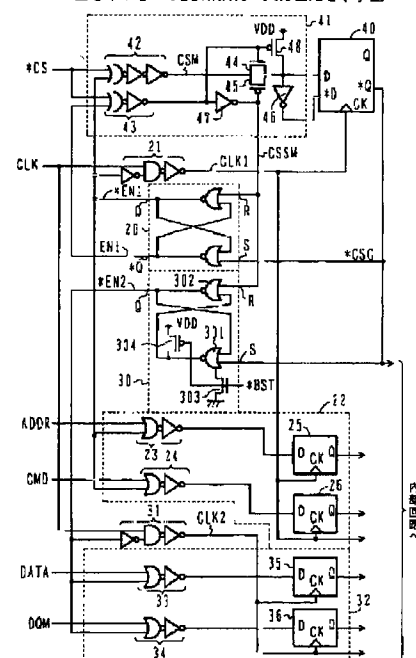
(54) 【発明の名称】 同期型半導体記憶装置

(57) 【要約】

【課題】外部配線数を増加することなく入力回路での無駄な消費電力を低減する。

【解決手段】コマンドCMDを内部クロックCLK 1に同期してラッチするコマンド入力回路24及び26と、ラッチされたコマンドをデコードするコマンドデコーダと、チップセレクト信号*CSの活性化にตอบสนองしてイネーブル信号EN 1を活性化させ、チップセレクト信号*CSの不活性化にตอบสนองして、外部クロックCLKの次のパルスが到来した後にイネーブル信号EN 1を不活性化にし、チップセレクト信号*CSを外部クロックCLKで同期化して内部チップセレクト信号*CSGを生成するチップセレクト信号入力回路41、40及び20と、イネーブル信号EN 1が活性である間、外部クロックCLKにตอบสนองして内部クロックCLK 1を動作させるクロック入力回路21と、内部チップセレクト信号*CSGを内部クロックCLK 1で同期化してコマンドデコーダ活性化信号を生成するDフリップフロップとを備えている。

図1中の1つのSDRAM内の入力回路を示す図



【特許請求の範囲】

【請求項1】 外部コマンドを内部クロックに同期してラッチするコマンド入力回路と、ラッチされた該コマンドをデコードするコマンドデコーダとを備え、デコード結果に応じた動作を行う同期型半導体記憶装置において、さらに、外部チップセレクト信号の活性化にตอบสนองしてイネブル信号を活性化させ、該外部チップセレクト信号の不活性化にตอบสนองして、外部クロックの次のパルスが到来した後に該イネブル信号を不活性にし、該外部チップセレクト信号を該外部クロックで同期化して内部チップセレクト信号を生成するチップセレクト信号入力回路と、該イネブル信号が活性である間、該外部クロックにตอบสนองして該内部クロックを動作させるクロック入力回路と、該内部チップセレクト信号を該内部クロックで同期化して、該コマンドデコーダを活性化させる信号を生成するデコードイネブル回路と、を有することを特徴とする同期型半導体記憶装置。

【請求項2】 上記チップセレクト信号入力回路は、上記外部チップセレクト信号の不活性状態を上記内部クロックに同期して保持し第1状態になり、その状態を上記内部チップセレクト信号として出力する第1フリップフロップと、該第1フリップフロップの出力を遅延させて上記イネブル信号を生成する遅延回路と、を有することを特徴とする請求項1記載の同期型半導体記憶装置。

【請求項3】 上記チップセレクト信号入力回路はさらに、上記外部チップセレクト信号の活性化を検出するタイミング回路を有し、上記遅延回路は、上記第1フリップフロップの上記第1状態にตอบสนองして第1状態になり、該活性化の検出にตอบสนองして第2状態になる第2フリップフロップであることを特徴とする請求項2記載の同期型半導体記憶装置。

【請求項4】 上記タイミング回路は、上記イネブル信号が活性であるときに上記外部チップセレクト信号を通す第1論理ゲートと、該イネブル信号が不活性であるときに該外部チップセレクト信号を通すことにより該外部チップセレクト信号の活性化を検出する第2論理ゲートと、該第1論理ゲートの出力端と上記第1フリップフロップのデータ入力端との間に接続され、該第2論理ゲートの出力でオン/オフ制御される転送ゲートと、該第1フリップフロップのデータ入力端と電源電位との間に接続され、該転送ゲートのオン/オフ制御と連動してオフ/オン制御されるスイッチング素子と、を有することを特徴とする請求項3記載の同期型半導体記憶装置。

【請求項5】 上記チップセレクト信号入力回路は、上記外部チップセレクト信号を上記外部クロックに同期して保持しその状態を上記内部チップセレクト信号として出力する第1フリップフロップと、該第1フリップフロップの出力を遅延させて上記イネブル信号を生成する遅延回路と、を有することを特徴とする請求項1記載の同期型半導体記憶装置。

【請求項6】 上記第1フリップフロップは、第1及び第2データ入力端、クロック入力端並びに第1及び第2ノードを備え、該クロック入力端が第1レベルのとき該第1及び第2ノードがプリチャージされ、該クロック入力端が第2レベルのとき該第1及び第2データ入力端の信号レベルに応じてそれぞれ該第1及び第2ノードがフローティング状態になり又はディスチャージされるダイナミック型フリップフロップと、該第1及び第2ノードの一方がディスチャージされているときその状態が書き込まれ、該第1及び第2ノードがプリチャージされているとき該書き込まれた状態を保持するラッチ回路と、を有することを特徴とする請求項5記載の同期型半導体記憶装置。

【請求項7】 上記チップセレクト信号入力回路はさらにタイミング回路を有し、該タイミング回路は、上記外部チップセレクト信号の入力端子と上記第1データ入力端との間に接続され、上記外部チップセレクト信号が活性状態のときオンになるように上記第1フリップフロップの出力でオン/オフ制御される転送ゲートと、該第1データ入力端と電源電位との間に接続され、該電源電位はこれを該第1データ入力端に供給すると上記第1ノードがフローティング状態になる電位であり、該転送ゲートのオン/オフ制御と連動してオフ/オン制御されるスイッチング素子と、を有することを特徴とする請求項6記載の同期型半導体記憶装置。

【請求項8】 上記タイミング回路はさらに、上記外部チップセレクト信号が活性化した時、上記外部クロックが上記第2レベルの場合に上記第1レベルになるのを待って上記第2データ入力端のレベルを反転する同期化回路を有することを特徴とする請求項7記載の同期型半導体記憶装置。

【請求項9】 上記遅延回路は、上記第1フリップフロップの状態の変化にตอบสนองして状態が変化する第2フリップフロップであることを特徴とする請求項5乃至8のいずれか1つに記載の同期型半導体記憶装置。

【請求項10】 入力端が上記第2フリップフロップと並列接続され、第2イネブル信号を出力し、バースト信号が活性であるときに該第2イネブル信号が強制的に活性状態になる第3フリップフロップと、該第2イネブル信号が活性である間、上記外部クロック

クにตอบสนองして第2内部クロックを動作させる第2クロック入力回路と、
該第2イネーブル信号が活性である期間、該第2クロックに同期してデータを入力するデータ入力回路と、
をさらに有することを特徴とする請求項3又は9記載の同期型半導体記憶装置。

【請求項11】 タイマーからの定期的なタイムアップ信号にตอบสนองして自動リフレッシュ動作を行う自動リフレッシュ回路をさらに有することを特徴とする請求項1乃至10のいずれか1つに記載の同期型半導体記憶装置。

【発明の詳細な説明】

【0001】

【発明の属する技術分野】本発明は、クロックに同期して信号を入出力する同期型半導体記憶装置に関する。

【0002】

【従来の技術】DRAMは、SRAMに比し記憶密度が高いが消費電力が大きいので、その低減化が要求されている。

【0003】図9は、SDRAMコントローラ10と複数のSDRAM11～14との間の配線接続図である。

【0004】SDRAMコントローラ10は、供給されるアドレスの上位2ビットをデコードしてチップセレクト信号*CS1～*CS4を生成し、該上位2ビットを除いたアドレスを行アドレスと列アドレスに分けて時分割多重化するとともに、行アドレス及び列アドレスに対応してそれぞれロウアドレスストロブ信号RAS及びコラムアドレスストロブ信号CASを生成する。これらRAS及びCASは、ライトイネーブル信号WEとともに多ビットのコマンドCMDを構成している。SDRAMコントローラ10は、SDRAM11～14のいずれかを選択する前に、外部クロックCLKとの論理積で内部クロックを生成するためのクロックイネーブル信号CKEを高レベルにする。

【0005】SDRAM11～14は、SDRAMコントローラ10を介して不図示のマイクロプロセッサに接続されている。SDRAMコントローラ10のチップセレクト信号出力端子*CS1～*CS4はそれぞれ、SDRAM11～14のチップセレクト信号入力端子*CSに接続され、SDRAMコントローラ10の外部クロックCLK、コマンドCMD、データDATA、アドレスADDR及びクロックイネーブル信号CKEの出力端子はSDRAM11～14の対応する端子に共通に接続されている。

【0006】チップセレクト信号*CS1～*CS4が全て高レベルでSDRAM11～14が非選択のときには、クロックイネーブル信号CKEが低レベルにされてSDRAM11～14内の内部クロックが停止し、入力を内部クロックに同期化させる入力回路が不活性になる。これにより、SDRAM11～14の低消費電力化が図られている。

【0007】

【発明が解決しようとする課題】しかし、図10に示す如く、例えばチップセレクト信号*CS2～*CS4を高レベルに維持してチップセレクト信号*CS1のみ低レベルにすることによりSDRAM11～14のうちSDRAM11のみを選択する場合には、その前にクロックイネーブル信号CKEが高レベルにされて、SDRAM11～14内の内部クロックが動作し入力回路が活性化されるので、非選択のSDRAM12～14において無駄な電力が消費される。

【0008】これを避けるために、SDRAM11～14に異なるクロックイネーブル信号を供給すれば、配線数が増加して複雑になる。

【0009】本発明の目的は、このような問題点に鑑み、外部配線数を増加することなく入力回路での無駄な消費電力を低減することが可能な同期型半導体記憶装置を提供することにある。

【0010】

【課題を解決するための手段及びその作用効果】請求項1では、外部コマンドを内部クロックに同期してラッチするコマンド入力回路と、ラッチされた該コマンドをデコードするコマンドデコーダとを備え、デコード結果に応じた動作を行う同期型半導体記憶装置において、さらに、例えば図8に示す如く、外部チップセレクト信号の活性化にตอบสนองしてイネーブル信号を活性化させ、該外部チップセレクト信号の不活性化にตอบสนองして、外部クロックの次のパルスが到来した後に該イネーブル信号を不活性にし、該外部チップセレクト信号を該外部クロックで同期化して内部チップセレクト信号を生成するチップセレクト信号入力回路と、該イネーブル信号が活性である間、該外部クロックにตอบสนองして該内部クロックを動作させるクロック入力回路と、該内部チップセレクト信号を該内部クロックで同期化して、該コマンドデコーダを活性化させる信号を生成するデコードイネーブル回路(40N)とを有する。

【0011】この同期型半導体記憶装置によれば、該内部チップセレクト信号が活性である時に、該内部クロックに同期してコマンドデコーダを活性化させる信号が生成され、コマンドデコーダのデコード結果に応じた動作が行われるので、従来のクロックイネーブル信号を用いる必要がない。また、外部チップセレクト信号が不活性であればイネーブル信号が不活性であるので、非選択の同期型半導体記憶装置のクロック入力回路及びコマンド入力回路が不活性となって、その消費電力が低減される。

【0012】請求項2の同期型半導体記憶装置では、請求項1において例えば図3に示す如く、上記チップセレクト信号入力回路は、上記外部チップセレクト信号の不活性状態を上記内部クロック(CK1)に同期して保持し第1状態になり、その状態を上記内部チップセレクト

信号として出力する第1フリップフロップ(40)と、該第1フリップフロップの出力を遅延させて上記イネーブル信号を生成する遅延回路(20)とを有する。

【0013】この同期型半導体記憶装置によれば、外部チップセレクト信号の不活性化にตอบสนองして、外部クロックの次のパルスが到来した後に該イネーブル信号を不活性にすることを確実に行うことができる。

【0014】請求項3の同期型半導体記憶装置では、請求項2において、上記チップセレクト信号入力回路はさらに、上記外部チップセレクト信号の活性化を検出するタイミング回路を有し、上記遅延回路は、上記第1フリップフロップの上記第1状態にตอบสนองして第1状態になり、該活性化の検出にตอบสนองして第2状態になる第2フリップフロップである。

【0015】この同期型半導体記憶装置によれば、外部チップセレクト信号が不活性になるとイネーブル信号が不活性になって、クロック入力回路及びコマンド入力回路が不活性となるので、選択された同期型半導体記憶装置の消費電力も低減される。

【0016】請求項4の同期型半導体記憶装置では、請求項3において例えば図3に示す如く、上記タイミング回路(41)は、上記イネーブル信号(EN1)が活性であるときに上記外部チップセレクト信号(*CS)を通す第1論理ゲート(42)と、該イネーブル信号が不活性であるときに該外部チップセレクト信号を通すことにより該外部チップセレクト信号の活性化を検出する第2論理ゲート(43)と、該第1論理ゲートの出力端と上記第1フリップフロップのデータ入力端との間に接続され、該第2論理ゲートの出力でオン/オフ制御される転送ゲート(44及び45)と、該第1フリップフロップのデータ入力端と電源電位との間に接続され、該転送ゲートのオン/オフ制御と連動してオフ/オン制御されるスイッチング素子(41)とを有する。

【0017】この同期型半導体記憶装置によれば、外部チップセレクト信号が不活性になったとき、イネーブル信号はまだ活性であるので第1論理ゲートを通り、第1フリップフロップにこの不活性状態が保持されて、イネーブル信号が不活性になる。これにより転送ゲートがオフ、スイッチング素子がオンになって、転送ゲートの入力側と出力側の信号の衝突が避けられ、かつ、スイッチング素子のオンにより第1フリップフロップの状態が維持される。イネーブル信号の不活性により第1論理ゲートが不活性になっているので、外部チップセレクト信号が活性化しても第1フリップフロップの状態は変化しない。しかし、イネーブル信号が不活性であるとき外部チップセレクト信号は第2論理ゲートを通るので、イネーブル信号が活性化すると第2論理ゲートの出力にตอบสนองして第2フリップフロップが第2状態に遷移し、これによりイネーブル信号が活性になる。

【0018】請求項5の同期型半導体記憶装置では、請

求項1において、上記チップセレクト信号入力回路は、例えば図6に示す如く、上記外部チップセレクト信号を上記外部クロックに同期して保持しその状態を上記内部チップセレクト信号として出力する第1フリップフロップ(40)と、該第1フリップフロップの出力を遅延させて上記イネーブル信号を生成する遅延回路(20)とを有する。

【0019】請求項6の同期型半導体記憶装置では、請求項5において例えば図4に示す如く、上記第1フリップフロップは、第1及び第2データ入力端(D及び*D)、クロック入力端(CK)並びに第1及び第2ノード(N01及びN02)を備え、該クロック入力端が第1レベルのとき該第1及び第2ノードがプリチャージされ、該クロック入力端が第2レベルのとき該第1及び第2データ入力端の信号レベルに応じてそれぞれ該第1及び第2ノードがフローティング状態になり又はディスチャージされるダイナミック型フリップフロップ(50)と、該第1及び第2ノードの一方がディスチャージされているときその状態が書き込まれ、該第1及び第2ノードがプリチャージされているとき該書き込まれた状態を保持するラッチ回路(40)とを有する。

【0020】この同期型半導体記憶装置によれば、ダイナミック型フリップフロップが用いられているので、スタティック型の場合よりも消費電力が低減される。

【0021】請求項7の同期型半導体記憶装置では、請求項6において例えば図6に示す如く、上記チップセレクト信号入力回路はさらにタイミング回路(41A)を有し、該タイミング回路は、上記外部チップセレクト信号の入力端子と上記第1データ入力端との間に接続され、上記外部チップセレクト信号が活性状態のときオンになるように上記第1フリップフロップの出力でオン/オフ制御される転送ゲート(44及び45)と、該第1データ入力端と電源電位との間に接続され、該電源電位はこれを該第1データ入力端に供給すると上記第1ノードがフローティング状態になる電位であり、該転送ゲートのオン/オフ制御と連動してオフ/オン制御されるスイッチング素子(81)とを有する。

【0022】この同期型半導体記憶装置によれば、外部チップセレクト信号が不活性になると転送ゲートがオフ、スイッチング素子がオンになって、第1フリップフロップ内で第1及び第2ノードがフローティング状態になり、クロック入力端に外部クロックパルス列が供給されても第1及び第2ノード上の電荷がディスチャージされないで、消費電力が低減され、また、転送ゲートの入力側と出力側の信号の衝突が避けられる。

【0023】請求項8の同期型半導体記憶装置では、請求項7において、上記タイミング回路はさらに、上記外部チップセレクト信号が活性化した時、上記外部クロックが上記第2レベルの場合に上記第1レベルになるのを待って上記第2データ入力端のレベルを反転する同期化

回路(82)を有する。

【0024】この同期型半導体記憶装置によれば、外部チップセレクト信号の活性化により外部クロックに同期せずにコマンド入力回路が活性化されて誤動作するのが防止される。

【0025】請求項9の同期型半導体記憶装置では、請求項5乃至8のいずれか1つにおいて、上記遅延回路は、上記第1フリップフロップの状態の変化に応答して状態が変化する第2フリップフロップ(20)である。

【0026】請求項10の同期型半導体記憶装置では、請求項3又は9において、入力端が上記第2フリップフロップと並列接続され、第2イネーブル信号を出力し、バースト信号が活性であるときに該第2イネーブル信号が強制的に活性状態になる第3フリップフロップ(30)と、該第2イネーブル信号が活性である間、上記外部クロックに反応して第2内部クロックを動作させる第2クロック入力回路(31)と、該第2イネーブル信号が活性である期間、該第2クロックに同期してデータを入力するデータ入力回路(32)とをさらに有する。

【0027】この同期型半導体記憶装置によれば、バースト動作が終了すると第2イネーブル信号が不活性になるので、バーストモードにおいても消費電力低減効果が得られる。

【0028】請求項11の同期型半導体記憶装置では、請求項1乃至10のいずれか1つにおいて例えば図8に示す如く、タイマーからの定期的なタイムアップ信号に反応して自動リフレッシュ動作を行う自動リフレッシュ回路(96、92、91及び94)をさらに有する。

【0029】この同期型半導体記憶装置によれば、外部チップセレクト信号により選択されていないチップは自動的にパワーダウン状態になるとともに自動的にリフレッシュが行われるので、非選択チップに対しては一切の注意を払う必要がなくなり、これにより、コマンド供給はアクセスしたいチップに対してのみ行えばよく、このような同期型半導体記憶装置を用いたシステム設計が容易になるとともに、アクセスしたいチップに対するコマンド供給可能な時間が増えて、システムの高速度が可能となる。

【0030】

【発明の実施の形態】以下、図面を参照して本発明の実施形態を説明する。

【0031】[第1実施形態] 図1は、本発明の第1実施形態に係る、SDRAMコントローラ10AとSDRAM11A~14Aとの間の配線接続図であり、図9に対応している。図2は、図1の回路の動作を示す概略タイミングチャートである。

【0032】SDRAMコントローラ10Aは、クロックイネーブル信号CKEを出力せず、また、図2に示す如く時点t1及びt3でのコマンドCMD1及びCMD2の発行の1クロックサイクル前のクロックCLK立ち

上がり時点t0及びt2でチップセレクト信号*CS1が低レベル(活性)になっている点で図9のSDRAMコントローラ10と異なり、他はSDRAMコントローラ10と同一である。

【0033】SDRAM11A~14Aは互いに同一構成であり、クロックイネーブル信号の外部端子がないことと、チップセレクト信号入力端子*CSの活性化(低レベル)に反応して図2に示すようなイネーブル信号を内部で生成し、これに反応して、内部クロックを動作させ入力回路を活性化する点で、図9のSDRAM11~14と異なる。SDRAM11A~14Aの他の点は、図9のSDRAM11~14と同一である。

【0034】1クロックサイクル前にチップセレクト信号*CSを活性化しておくのは、次のクロックサイクルの内部クロック立ち上がりでコマンドCMD等をラッチするのを確実にを行うためである。図2では、SDRAM11Aのみ選択され、SDRAM12A~14A内のイネーブル信号ENが低レベル(不活性)でこれらの内部クロックが停止し、これらの入力回路が不活性になってコマンドCMD等が該入力回路に取り込まれない。

【0035】図3は、図1中の1つのSDRAM内の入力回路を示す。図3では簡単化のために、アドレスADDR、コマンドCMD、データDATA及びデータマスクDQMがいずれも1ビットである場合を示している。多ビットの場合には、各ビットに対応した入力回路が並列して存在する。

【0036】チップセレクト信号*CSを外部クロックCLKの立ち上がり同期化したイネーブル信号*EN1は、RSフリップフロップ20に保持され、その非反転出力端Qから取り出されてクロック入力回路21並びにアドレス・コマンド入力回路22のオアゲート23及び24の一方の入力端に供給される。クロック入力回路21、オアゲート23及び24の他方の入力端にはそれぞれ外部クロックCLK、アドレスADDR及びコマンドCMDが供給される。オアゲート23及び24の出力はアドレス・コマンド入力回路22のDフリップフロップ25及び26のデータ入力端Dに供給され、Dフリップフロップ25及び26のクロック入力端CKにはクロック入力回路21の出力が内部クロックCLK1として供給される。

【0037】イネーブル信号*EN1が高レベルのとき、クロック入力回路21の出力は外部クロックCLKの変化によらず低レベルとなり、オアゲート23及び24の出力はそれぞれアドレスADDR及びコマンドCMDの変化によらず高レベルとなるので、アドレス・コマンド入力回路22は不活性状態になる。イネーブル信号*EN1が低レベルのとき、外部クロックCLK、アドレスADDR及びコマンドCMDはそれぞれクロック入力回路21、オアゲート23及び24を通るので、アドレス・コマンド入力回路22は活性状態になる。Dフリ

ップフロップ25及び26の出力は内部回路に供給される。

【0038】ここで、SDRAM11がバーストモードのとき、同一行アドレスに対し、列アドレスが不図示のカウantaに保持された後、そのカウントが内部クロックでインクリメントされてデータDATA及びデータマスクDQMが読み込まれる。このとき、アドレス・コマンド入力回路22は不活性である。そこで、RSフリップフロップ20とは別に、イネーブル信号*EN2を保持するRSフリップフロップ30が備えられている。

【0039】RSフリップフロップ30は、ノアゲート301とノアゲート302とがクロス結合されてRSフリップフロップ20と同一になっており、さらに、ノアゲート301のグランド電位供給端がnMOSTランジスタ303を介してグランド線に接続され、ノアゲート301の出力端がpMOSTランジスタ304を介して電源電位VDDに接続されている。nMOSTランジスタ303及びpMOSTランジスタ304のゲートにはバースト信号*BSTが供給されている。ノアゲート301は、nMOSTランジスタ303がオフでセット入力端Sが高レベルのとき、出力がハイインピーダンス状態になる。バーストモードのとき、バースト信号*BSTが低レベルになって、nMOSTランジスタ303及びpMOSTランジスタ304がそれぞれオフ及びオンになり、セット入力端Sが高レベルであっても非反転出力端Qが低レベルに固定される。バースト信号*BSTが高レベルのとき、nMOSTランジスタ303及びpMOSTランジスタ304がそれぞれオン及びオフになって、ノアゲート301が活性になる。

【0040】RSフリップフロップ30の非反転出力端Qから取り出されたイネーブル信号*EN2は、クロック入力回路31並びにデータ入力回路32のオアゲート33及び34の一方の入力端に供給される。クロック入力回路31、オアゲート33及び34の他方の入力端にはそれぞれ外部クロックCLK、データDATA及びデータマスクDQMが供給される。オアゲート33及び34の出力はデータ入力回路32のDフリップフロップ35及び36のデータ入力端Dに供給され、Dフリップフロップ35及び36のクロック入力端CKにはクロック入力回路31の出力が内部クロックCLK2として供給される。

【0041】イネーブル信号*EN2が高レベルのとき、クロック入力回路31の出力は外部クロックCLKの変化によらず低レベルとなり、オアゲート33及び34の出力はそれぞれデータDATA及びデータマスクDQMの変化によらず高レベルとなるので、データ入力回路32は不活性状態になる。イネーブル信号*EN2が低レベルのとき、外部クロックCLK、データDATA及びデータマスクDQMはそれぞれクロック入力回路31、オアゲート33及び34を通るので、データ入力回

路32は活性状態になる。Dフリップフロップ35及び36の出力は内部回路に供給される。

【0042】RSフリップフロップ20及び30のセット入力端Sには、Dフリップフロップ40の反転出力端*Qからチップセレクト信号*CSSが供給される。Dフリップフロップ40の相補データ入力端D及び*Dには、チップセレクト信号*CSがタイミング回路41を介して供給される。Dフリップフロップ40は、チップセレクト信号*CSが高レベルになった後、外部クロックCLKの最初の立ち上がりに同期してRSフリップフロップ20及び30をセットするためのものである。

【0043】チップセレクト信号*CSの高レベルをタイミング回路41で反転したものが内部クロックCLK1の立ち上がりでDフリップフロップ40に保持されると、チップセレクト信号*CSSが高レベルとなってRSフリップフロップ20がセットされ、イネーブル信号*EN1が高レベルになり、クロック入力回路21が閉じて内部クロックCLK1が停止する。このため、チップセレクト信号*CSが低レベルに遷移してもこれがDフリップフロップ40に保持されないで、Dフリップフロップ40の非反転出力端QでRSフリップフロップ20をリセットすることができない。また、外部クロックCLKが高レベルのときに、チップセレクト信号*CSの立ち下がりでもRSフリップフロップ20をリセットすれば、クロック入力回路21、オアゲート23及び24が活性化されると同時に、この高レベルがクロック入力回路21を通して内部クロックCLK1が立ち上がるので、アドレス・コマンド入力回路22の動作が1クロックサイクル早くなってしまう。

【0044】そこで、構成要素42～48を備えたタイミング回路41でこのような問題を解決している。

【0045】タイミング回路41では、ノアゲート42及びオアゲート43の一方の入力端にチップセレクト信号*CSが供給され、他方の入力端にそれぞれイネーブル信号*EN1及びRSフリップフロップ20の反転出力端*Qからのイネーブル信号EN1が供給される。ノアゲート42及びオアゲート43はそれぞれ、チップセレクト信号*CSの不活性化（高レベルへの遷移）及び活性化（低レベルへの遷移）を検出するためのものである。ノアゲート42の出力は、nMOSTランジスタ44とpMOSTランジスタ45の並列接続を介し、一方ではDフリップフロップ40のデータ入力端Dに供給され、他方ではさらにインバータ46を介してDフリップフロップ40のデータ入力端Dに供給される。nMOSTランジスタ44のゲートにはオアゲート43の出力が供給され、pMOSTランジスタ45のゲートにはオアゲート43の出力がインバータ47を介して供給される。オアゲート43の出力はさらに、pMOSTランジスタ48のゲートに供給される。pMOSTランジスタ48は、電源電位VDDとDフリップフロップ40のデ

ータ入力端Dとの間に接続されている。インバータ47の出力は、チップセレクト信号CSSMとしてRSフリップフロップ20及び30のリセット入力端に供給される。

【0046】上記構成において、ノアゲート42が不活性（CSSMが高レベル）でオアゲート43が活性の状態においてチップセレクト信号*CSが低レベルに遷移すると、オアゲート43の出力が低レベルに遷移し、pMOSTランジスタ48がオンになってDフリップフロップ40のデータ入力端D及び*Dがそれぞれ高レベル及び低レベルに遷移し、また、インバータ47の出力が高レベルに遷移してRSフリップフロップ20及び30がリセットされる。nMOSTランジスタ44及びpMOSTランジスタ45は、pMOSTランジスタがオンのときチップセレクト信号CSSMが低レベルであるので、nMOSTランジスタ44及びpMOSTランジスタ45をオフにして衝突を避けるためのものである。

【0047】チップセレクト信号入力回路は、タイミング回路41、Dフリップフロップ40及びRSフリップフロップ20、30で構成されている。

【0048】次に、上記の如く構成された入力回路の動作を、図5を参照して説明する。

【0049】図5において、図2と異なりチップセレクト信号*CSが複数クロックサイクル低レベルを維持しているのは、次の理由による。すなわち、SDRAMでは、あるバンクにコマンドを供給した後、次のクロックサイクルで他のバンクにコマンドを供給したり、また、同一バンクに対して1ビット読み出しコマンドを連続して供給する場合があるからである。図8について後述するように、クロックCLKの立ち上がりの1クロックサイクル前にチップセレクト信号*CSが活性化していれば、すなわち内部チップセレクト信号*CSSCが活性のとき、内部クロックの立ち上がりに同期して、コマンドCMDのデコードが行われ、その結果に応じた動作が行われる。

【0050】(t1) イネーブル信号*EN1が低レベルであるので、チップセレクト信号*CSに対しノアゲート42がインバータとして機能する。チップセレクト信号*CSが低レベルであるので、ノアゲート42の出力CSSMは高レベルになっている。イネーブル信号*EN1及び*EN2が低レベルであるので、クロック入力回路21、オアゲート23、24、クロック入力回路31、オアゲート33及び34が活性状態になっていて、Dフリップフロップ40、アドレス・コマンド入力回路22及びデータ入力回路32も活性状態になっている。また、イネーブル信号EN1が高レベルであるので、オアゲート43が不活性、すなわちその出力がチップセレクト信号*CSの変化によらず高レベルになっていて、nMOSTランジスタ44及びpMOSTランジスタ45がオン、pMOSTランジスタ48がオフとなって、

Dフリップフロップ40のデータ入力端D及び*Dにそれぞれ高レベル及び低レベルが供給される。インバータ47の出力CSSM及びDフリップフロップ40の反転出力端*Qは低レベルであり、RSフリップフロップ20及び30のリセット入力端R及びセット入力端Sは低レベルになっている。バースト信号*BSTが低レベル（バーストモード）であり、nMOSTランジスタ303及びpMOSTランジスタ304はそれぞれオフ及びオンになっている。

【0051】(t2) この状態からチップセレクト信号*CSが立ち上がり、ノアゲート42の出力CSSMが低レベルに遷移し、Dフリップフロップ40のデータ入力端D及び*Dがそれぞれ低レベル及び高レベルになる。

【0052】(t3) 次に外部クロックCLKが立ち上がり、内部クロックCLK1が立ち上がり、Dフリップフロップ40の入力が保持され、その反転出力端*Qからのチップセレクト信号*CSSCが高レベルに遷移する。チップセレクト信号*CSSCの高レベルは、RSフリップフロップ20内のノアゲート201の出力EN1を低レベルに反転させ、次いでノアゲート202の出力*EN1を高レベルに反転させるので、RSフリップフロップ20は遅延回路としても機能している。この信号伝搬遅延により、CLK1のパルスがクロック入力回路21を通った後に、イネーブル信号*EN1が高レベルに遷移してクロック入力回路21が不活性になる。

【0053】CLK1の立ち上がりに同期して、アドレスADDR及びコマンドCMDがアドレス・コマンド入力回路22のDフリップフロップに保持される。*BSTが低レベルであるので、このアドレスADDRは列アドレスであり、コマンドCMDはリードコマンド又はライトコマンドである。

【0054】イネーブル信号*EN1が高レベルに遷移すると、クロック入力回路21、オアゲート23及び24が不活性になり、内部クロックCLK1の動作が停止してDフリップフロップ40及びアドレス・コマンド入力回路22も不活性になる。また、ノアゲート42が不活性になると同時にオアゲート43が活性になり、後に生ずるチップセレクト信号*CSの立ち下がりには、ノアゲート42を通らずにオアゲート43を通る。オアゲート43の出力は高レベルを維持している。

【0055】バースト信号*BSTが低レベルであるので、チップセレクト信号*CSSCが高レベルに遷移してもイネーブル信号*EN2は低レベルを維持、すなわちクロック入力回路31及びデータ入力回路32は活性状態を維持する。これによりバーストモードでの動作が行われ、データDATA及びデータマスクDQMの内容に応じたデータが内部に書き込まれる。Dフリップフロップ36の出力が高レベルのとき、Dフリップフロップ35の出力は内部回路でマスクされ、指定アドレスのメモ

リセルへの書き込みは行われない。Dフリップフロップ36の出力が低レベルのとき、このマスクが解除されてDフリップフロップ35の出力が指定アドレスのメモリセルへ書き込まれる。

【0056】(t4)バーストモードが終了してバースト信号*BSTが高レベルに遷移し、これによりRSフリップフロップ30のセット入力端Sに供給されている高レベルが有効になり、イネーブル信号*EN2が高レベルとなって、クロック入力回路31、オアゲート33及び34が不活性化され、内部クロックCLK2の動作が停止する。その結果、不図示のコラムアドレス系回路が不活性になる。

【0057】(t5)チップセレクト信号*CSが低レベルに遷移して、オアゲート43及びインバータ47の出力がそれぞれ低レベル及び高レベルに遷移し、nMOSTランジスタ44及びpMOSTランジスタ45がオフ、pMOSTランジスタ48がオンになる。これにより、チップセレクト信号CSSMが高レベルに遷移して、RSフリップフロップ20及び30がリセットされ、イネーブル信号*EN1及び*EN2が低レベルに遷移してノアゲート42、クロック入力回路21、23、24、クロック入力回路31、オアゲート33及び34が活性化され、内部クロックCLK1及びCLK2が動作を開始し、ノアゲート42、Dフリップフロップ40、アドレス・コマンド入力回路22及びデータ入力回路32が活性状態になる。ノアゲート42の活性化によりチップセレクト信号CSMが高レベルに遷移する。

【0058】このような動作により、内部クロックCLK1及びCLK2が停止していてもチップセレクト信号*CSの活性化(立ち上がり)にตอบสนองしてイネーブル信号*EN1及び*EN2が活性化されて、内部クロックCLK1及びCLK2の動作が再開される。

【0059】(t6)Dフリップフロップ40のデータ入力端D及*Dがそれぞれ高レベル及び低レベルであるので、内部クロックCLK1の立ち上がりに同期してDフリップフロップ40がセットされ、チップセレクト信号*CSが低レベルに遷移する。これにより、イネーブル信号EN1が高レベルに遷移してオアゲート43が不活性になり、その出力が高レベルに遷移してnMOSTランジスタ44及びpMOSTランジスタ45がオン、pMOSTランジスタ48がオフになり、またチップセレクト信号CSSMが低レベルに遷移する。

【0060】内部クロックCLK1の立ち上がり時点では内部クロック*CSが高レベルであるので、コマンドCMDのデコードは行われない。

【0061】(t7)バースト信号*BSTが低レベルに遷移して、時点t1と同じ状態になる。

【0062】本第1実施形態によれば、クロックCLKの立ち上がりの1クロックサイクル前にチップセレクト信号*CSが活性(低レベル)であれば内部クロックC

LK1が動作しており、内部チップセレクト信号*CSが活性のとき、内部クロックCLK1の立ち上がりでコマンドCMD等が内部回路に取り込まれてこれに応じた動作が行われる。また、チップセレクト信号*CSが不活性(高レベル)になってから、1クロックパルスの後に内部クロックCLK1が停止して、アドレス・コマンド入力回路22が不活性になり、さらに、バーストモードによるデータ取り込み終了後に内部クロックCLK2が停止してデータ入力回路32が不活性になるので、選択されたSDRAM11においても入力回路の消費電力が低減される。しかも、非選択のSDRAM11ではチップセレクト信号*CSが高レベルであるので、内部クロックが停止して入力回路の消費電力が低減される。クロックイネーブル信号CKEを用いずにこのような動作が自動的に行われるので、低消費電力化が達成されるとともに、複数のSDRAM11を用いたシステムの制御及び設計が容易になる。

【0063】図4は、図3中のDフリップフロップ40の構成例を示す。この回路自体は公知であるが、クロックのパルス列による無駄な電力消費を示すためにこの回路の構成をその動作とともに説明する。

【0064】Dフリップフロップ40は、入力段のダイナミックフリップフロップ50と、その一対の出力の各々を相補信号化してダイナミックフリップフロップ50をフィードバック制御する相補化回路60と、回路60の出力により状態が制御されるラッチ回路70とからなる。

【0065】ダイナミックフリップフロップ50では、クロック入力端CKが低レベルのとき、pMOSTランジスタ51A及び51Bがオン、nMOSTランジスタ52がオフとなって、ノードNO1及びNO2へのプリチャージが行われる。

【0066】増幅回路53は、フリップフロップ型であり、pMOSTランジスタ51A及び51Bにそれぞれ並列接続されたpMOSTランジスタ54A及び54Bと、pMOSTランジスタ54A及び54Bにそれぞれ直列接続されたnMOSTランジスタ55A及び55Bとからなり、これらがクロス結合されている。上記ノードNO1及びNO2へのプリチャージによりpMOSTランジスタ54A及び54Bがオフ、nMOSTランジスタ55A及び55Bがオンで、増幅回路53が不活性状態になっている。

【0067】nMOSTランジスタ55AのソースとnMOSTランジスタ52のドレインとの間にはnMOSTランジスタ56Aと57Aとが並列接続され、nMOSTランジスタ55BのソースとnMOSTランジスタ52のドレインとの間にはnMOSTランジスタ56Bと57Bとが並列接続されている。ノードNO1は相補化回路60のインバータ61Aを介してnMOSTランジスタ57Aのゲートに接続され、ノードNO2は相補

化回路60のインバータ61Bを介してnMOSTランジスタ57Bのゲートに接続されている。したがって、ノードNO1及びNO2がプリチャージされている状態では、インバータ61A及び61Bの出力が低レベルとなってnMOSTランジスタ57A及び57Bがオフになっている。nMOSTランジスタ56A及び56Bのゲートにはそれぞれ、データ入力端D及び*Dからの信号が供給される。データ入力端D及び*Dの信号レベルが互いに逆の時、nMOSTランジスタ56A及び56Bは一方がオン、他方がオフであり、データ入力端Dが高レベルのときにはnMOSTランジスタ56Aがオン、nMOSTランジスタ56Bがオフになっている。

【0068】上記プリチャージ状態において、ラッチ回路70では、nMOSTランジスタ71A、pMOSTランジスタ71B、72A及びnMOSTランジスタ72Bが全てオフになっており、その前の状態が、インバータ73と74とからなる保持回路に保持される。

【0069】この状態から外部クロックCLKが立ち上がると、pMOSTランジスタ51A及び51Bがオフ、nMOSTランジスタ52がオンになり、ノードNO1上の電荷がnMOSTランジスタ55A、56A及び52を通してグランド線へ引き抜かれ、その電位が低下し、ノードNO2とNO1の電位差が増幅回路53で増幅され、インバータ61Aの出力が高レベルに遷移してnMOSTランジスタ57Aがオンになり、ノードNO1からの電荷引き抜きが助長される。また、nMOSTランジスタ71A及びpMOSTランジスタ72Aがオンになって、出力端Q及び*Qがそれぞれ高レベル及び低レベルになる。

【0070】逆にデータ入力端Dが低レベルの場合には、nMOSTランジスタ56A及び57Aがオフ、nMOSTランジスタ56B及び57Bがオン、pMOSTランジスタ71B及びnMOSTランジスタ72Bがオンになって、出力端Q及び*Qがそれぞれ低レベル及び高レベルになる。

【0071】次に、クロック入力端CKが低レベルになると、上記プリチャージが行われ、pMOSTランジスタ71B、nMOSTランジスタ71A、pMOSTランジスタ72A及びnMOSTランジスタ72Bがオフになって出力端Q及び*Qの電位が保持される。

【0072】データ入力端Dが高レベルの状態でクロック入力端CKにパルス列が供給されると、出力端Q及び*Qはそれぞれ高レベル及び低レベルで一定であるが、ノードNO1のプリチャージ及びディスチャージが繰り返されて、無駄な電力が消費される。図5では内部クロックCK1及びCK2のパルスが不要になるとそのクロックが停止するので、消費電力が低減される。

【0073】クロック入力端CKが高レベルの状態でデータ入力端Dが変化しても、増幅回路53の増幅動作によりその出力が保持されるので、Dフリップフロップ4

0の出力も一定に保持される。

【0074】〔第2実施形態〕図6は、本発明の第2実施形態のSDRAM内の入力回路を示し、図3に対応している。

【0075】この回路では、図3のタイミング回路41の代わりにタイミング回路41Aを用い、外部クロックCLKをDフリップフロップ40のクロック入力端CKに直接供給し、Dフリップフロップ40の非反転出力端QをRSフリップフロップ20及び30のセット入力端Sに接続し、Dフリップフロップ40の反転出力端*QをRSフリップフロップ20及び30のリセット入力端Rに接続している。

【0076】タイミング回路41Aでは、チップセレクト信号*CSの駆動能力がツルーバッファゲート80で増幅され、その出力*CSMが、nMOSTランジスタ44とpMOSTランジスタ45との並列接続を介してDフリップフロップ40のデータ入力端Dに供給される。データ入力端Dとグランド線の間にはnMOSTランジスタ81が接続されている。nMOSTランジスタ44のゲートはDフリップフロップ40の非反転出力端Qに接続され、pMOSTランジスタ45及びnMOSTランジスタ81のゲートはDフリップフロップ40の反転出力端*Qに接続されている。これにより、出力端Q及び*Qがそれぞれ低レベル及び高レベルの状態では、nMOSTランジスタ44及びpMOSTランジスタ45がオン、nMOSTランジスタ81がオフになり、低レベルのチップセレクト信号*CSがツルーバッファゲート80、nMOSTランジスタ44及びpMOSTランジスタ45を介してDフリップフロップ40のデータ入力端Dに供給される。

【0077】この状態からチップセレクト信号*CSが高レベルに遷移すると、外部クロックCLKの立ち上がりにより同期してこれがDフリップフロップ40に保持され、出力端Q及び*Qがそれぞれ高レベル及び低レベルに遷移して、nMOSTランジスタ44及びpMOSTランジスタ45がオフ、nMOSTランジスタ81がオンになり、Dフリップフロップ40のデータ入力端Dが低レベルになる。図4において、クロック入力端CKが低レベルのとき、pMOSTランジスタ51A及び51Bがオン、nMOSTランジスタ52がオフになってノードNO1及びNO2がプリチャージされ、これによりnMOSTランジスタ55A及び55Bがオン、nMOSTランジスタ57A及び57Bがオフになっている。データ入力端D及び*Dが共に低レベルであれば、nMOSTランジスタ56A及び56Bがオフであるので、外部クロックCLKが高レベルになってもディスチャージが行われず、外部クロックCLKのパルス列がDフリップフロップ40のクロック入力端CKDに供給されても、無駄な電力消費が防止される。

【0078】チップセレクト信号*CSが低レベルに遷

移する場合には、上記nMOSTランジスタ81のオンによりDフリップフロップ40のデータ入力端Dが既に低レベルになっているので、データ入力端*Dのみ高レベルに遷移させればよい。この時、外部クロックCLKが高レベルであれば、図4においてノードNO2上の電荷がnMOSTランジスタ55B、56B及び52を通過して引き抜かれるので、直ちにDフリップフロップ40の出力が反転し、誤動作する。

【0079】この問題を解決するために、ツルーバッファゲート80の出力*CSMは、立ち下がり同期化インバータ82を介してDフリップフロップ40のデータ入力端*Dに接続されている。立ち下がり同期化インバータ82は、インバータ46の入力を外部クロックCLKの立ち下がりに同期して反転させるためのものであり、ツルーバッファゲート80の出力端とDフリップフロップ40のデータ入力端*Dとの間にインバータ46が接続され、その電源入力端がpMOSTランジスタ83を介して電源電位VDDに接続され、pMOSTランジスタ83のゲートが外部クロックCLKに接続されている。外部クロックCLKが低レベルのときには、pMOSTランジスタ83がオンであるので、立ち下がり同期化インバータ82は通常のインバータとして機能する。外部クロックCLKが高レベルのときには、pMOSTランジスタ83がオフであり、インバータ46の出力が入力の変化にตอบสนองしないので、外部クロックCLKが高レベルになる前のインバータ46の出力レベルを保持するために、インバータ46の出力端にインバータ84と85とが環状結合された保持回路が接続されている。

【0080】図6の回路の他の構成は、図3のそれと同一である。

【0081】図7は、図6の回路の動作を示すタイミングチャートである。

【0082】図7では、時点t1の前にチップセレクト信号*CSが低レベルで、これがDフリップフロップ40に保持され、その出力端Q及び*Qからのチップセレクト信号*CSC及びCSCがそれぞれ低レベル及び高レベルになっていて、RSフリップフロップ20及び30がリセット状態である。したがって、イネーブル信号*EN1及び*EN2が低レベルで、クロック入力回路21、オアゲート23、24、クロック入力回路31、オアゲート33及び34が活性状態であり、内部クロックCLK1及びCLK2が動作し、アドレス・コマンド入力回路22及びデータ入力回路32が活性状態になっている。また、Dフリップフロップ40の出力端Q及び*Qがそれぞれ低レベル及び高レベルであるので、nMOSTランジスタ44及びpMOSTランジスタ45はオン、nMOSTランジスタ81はオフになっている。Dフリップフロップ40の出データ入力端D及び*Dがそれぞれ低レベル及び高レベルであるので図4のnMOSTランジスタ56A及び56Bがそれぞれオフ及びオ

ンになっている。

【0083】(t1)外部クロックCLKの立ち上がりにより、図4においてクロック入力端CKが高レベルに遷移し、ノードNO2の電荷がnMOSTランジスタ55B、56B及び52を通過してグラウンド線へ引き抜かれる。これに対し、nMOSTランジスタ56A及び57Aがオフであるので、ノードNO1は高レベルを維持している。

【0084】(t2)チップセレクト信号*CSが高レベルに遷移し、ツルーバッファゲート80の出力*CSMが高レベルに遷移し、図4中のノードNO3が高レベルに遷移して、nMOSTランジスタ56Aがオンになる。nMOSTランジスタ55Aがオフであるので、ノードNO1は高レベルを維持している。

【0085】(t3)外部クロックCLKが低レベルに遷移し、pMOSTランジスタ83がオンになって、チップセレクト信号*CSMがインバータ46で反転され、図4中のノードNO4が低レベルになってnMOSTランジスタ56Bがオフになる。

【0086】(t4)内部クロックCLK1の立ち上がりに同期して、アドレスADDR及びコマンドCMDがアドレス・コマンド入力回路22のDフリップフロップに保持される。

【0087】外部クロックCLKの立ち上がりによりチップセレクト信号*CSの高レベルがDフリップフロップ40に保持され、出力端Q及び*Qがそれぞれ高レベル及び低レベルに遷移する。これにより、RSフリップフロップ20がセットされてイネーブル信号*EN1が高レベルに遷移し、クロック入力回路21、オアゲート23及び24が不活性になる。RSフリップフロップ30については、バースト信号*BSTが低レベルであるので、イネーブル信号*EN2の低レベルが維持され、データ入力回路32は活性状態を維持する。また、nMOSTランジスタ44及びpMOSTランジスタ45がオフ、nMOSTランジスタ81がオンになり、図4中のノードNO3が低レベルとなってnMOSTランジスタ56Aがオフになる。

【0088】(t5)外部クロックCLKの立ち下がりにより、図4においてpMOSTランジスタ51A及び51Bがオン、nMOSTランジスタ52がオフになり、ノードNO1がプリチャージされ、増幅回路53が不活性になる。この状態では、nMOSTランジスタ56A、57A、56B及び57Bがオフであるので、外部クロックCLKが動作してもノードNO1及びNO2のディスチャージが行われず、Dフリップフロップ40での無駄な電力消費が防止される。

【0089】(t6)バーストモードが終了してバースト信号*BSTが高レベルに遷移し、これによりRSフリップフロップ30のセット入力端Sに供給されている高レベルが有効になり、イネーブル信号*EN2が高レ

ベルとなって、クロック入力回路31、オアゲート33及び34が不活性化され、内部クロックCLK2の動作が停止する。その結果、不図示のコラムアドレス系回路が不活性になる。

【0090】(t7)チップセレクト信号*CSが立ち下がり、ツルババッファゲート80の出力*CSMも立ち下がる。このとき、外部クロックCLKが高レベルでnMOSTランジスタ52がオンであっても、pMOSTランジスタ83がオフであるのでインバータ46の出力がハイインピーダンス状態になってノードNO4は低レベルを維持し、nMOSTランジスタ56Bはオフを維持し、Dフリップフロップ40はセット状態を維持する。したがって、チップセレクト信号*CSの立ち下がりにより外部クロックCLKに同期せずにアドレス・コマンド入力回路22が活性化されて誤動作するのが防止される。

【0091】(t8)外部クロックCLKの立ち下がりによりpMOSTランジスタ83がオンになってインバータ46が活性化され、ノードNO4が高レベルに遷移し、nMOSTランジスタ56Bがオンになる。これと同時にnMOSTランジスタ52がオフになるので、ノードNO2は高レベルを維持する。

【0092】(t9)外部クロックCLKの立ち上がりによりpMOSTランジスタ51A及び51Bがオフ、nMOSTランジスタ52がオンになって、ノードNO2の電荷がnMOSTランジスタ55B、56B及び52を通過してグラウンド線に引き抜かれ、ノードNO1とNO2の電位差が増幅回路53で増幅されてノードNO2が低レベルに遷移する。これにより、出力端Q及び*Qがそれぞれ低レベル及び高レベルに遷移し、RSフリップフロップ20及び30がリセットされて、イネーブル信号*EN1及び*EN2が低レベルとなり、クロック入力回路21、23、24、クロック入力回路31、オアゲート33及び34が活性化される。

【0093】時点t9は図5中の時点t6に相当し、内部クロックCLK1が停止している点で図5の場合と異なる。しかし、次のクロックCLKの立ち上がりでは図5及び図7のいずれについても、内部チップセレクト信号*CSが活性で内部クロックCLK1が立ち上がるので、図8について後述するように、コマンドCMDのデコード開始時点は同じになる。

【0094】(t10)バースト信号*BSTが低レベルに遷移して、時点t1の前と同じ状態になる。

【0095】本第2実施形態についても、上記第1実施形態と同様の効果が得られる。

【0096】〔第3実施形態〕図8は、本発明の第3実施形態のSDRAM11Bの概略構成を示すブロック図である。

【0097】入力回路90は、図3又は図6に示す回路を含んでいる。図3及び図6では簡単化のためにアドレ

スADDR、コマンドCMD、データDATA及びデータマスクDQMが1ビットである場合を示しているが、入力回路90ではこれらがいずれ也多ビットである。また、図3及び図6ではデータ出力回路が省略されているが、入力回路90にはデータ出力回路が含まれている。

【0098】ロウアドレス切換回路91は、アービター92からの制御信号に応答してアドレス入力回路93又はリフレッシュアドレス発生回路94からのロウアドレスの一方を選択し、ロウデコーダ95へ供給する。アービター92は、先に活性化された入力を受け付け、同時の場合には、リフレッシュタイム96から定期的に供給されるタイムアップ信号の活性化を優先的に受け付ける。アービター92は、タイムアップ信号の活性化を受け付けた場合、ロウアドレス切換回路91に対しリフレッシュアドレス発生回路94の出力を選択させる。

【0099】デコードイネーブル回路としてのDフリップフロップ40Nのデータ入力端D及びクロック入力端にはそれぞれ、入力回路90の*CS入力回路及びクロック入力回路から内部チップセレクト信号*CS及び内部クロックCLK1が供給され、Dフリップフロップ40Nの反転出力端Qからイネーブル信号EN3がコマンドデコーダに供給される。コマンド入力回路97からの信号がコマンドデコーダ98に供給され、イネーブル信号EN3が活性（高レベル）のとき、アクティベイトコマンド、リードコマンド、ライトコマンド又はプリチャージコマンド等のコマンドのデコードが行われる。すなわち、上述のように、内部チップセレクト信号*CSが活性のとき内部クロックCLK1の立ち上がりでコマンドCMDのデコードが行われる（コマンドデコーダ98が活性化される）。、アクティベイトコマンドはアービター92に供給され、アービター92はこれを受け付けると、ロウアドレス切換回路91に対しアドレス入力回路93の出力を選択させる。ロウデコーダ95の出力によりメモリセル100内のワード線が選択されてそのセル行の内容がセンスアンプに保持される。

【0100】コラムアドレス切換回路101は、コマンドデコーダ98の出力がリードコマンド又はライトコマンドであることを示している場合には、アドレス入力回路93からのアドレスを列アドレスとして受け取り、バーストモードのときにはこれをバーストアドレス発生回路103のカウンタにロードさせる。このカウンタは、クロック入力回路からの内部クロックCLK2でインクリメントされる。コラムアドレス切換回路101は、バーストモードでない場合にはアドレス入力回路93の出力を選択し、バーストモードの場合にはバーストアドレス発生回路103の出力を選択して、コラムデコーダ102に供給する。コラムデコーダ102の出力によりメモリセル100内のコラムが選択される。

【0101】書き込みの場合には、入力回路90のデータ入力回路からのデータがリード・ライトアンプ104

を介してメモリセル100内の選択された行及び列のセルに書き込まれ、読み出しの場合には、この選択されたメモリセルの内容がリード・ライトアンプ104及び入力回路90のデータ出力回路を介して外部に取り出される。

【0102】アービター92を用いてアドレス入力回路93とリフレッシュアドレス発生回路94の出力の一方をロウアドレス切換回路91で選択するのは、非同期型RAMでは公知であるがSDRAMでは用いられていない。

【0103】本第3実施形態では、上記第1実施形態又は第2実施形態のような入力回路90を用い、かつ、リフレッシュタイマ96及アービター92を用いており、チップセレクト信号*CSにより選択されていないチップは自動的にパワーダウン状態になるとともに自動的にリフレッシュが行われるので、非選択チップに対しては一切の注意を払う必要がなくなる。これにより、コマンド供給はアクセスしたいチップに対してのみ行えばよく、このようなSDRAMを用いたシステム設計が容易になるとともに、アクセスしたいチップに対するコマンド供給可能な時間が増えて、システムの高速化が可能となる。

【図面の簡単な説明】

【図1】本発明の第1実施形態に係る、SDRAMコントローラと複数のSDRAMとの間の配線接続図である。

【図2】図1の回路の動作を示す概略タイミングチャートである。

【図3】図1中の1つのSDRAM内の入力回路を示す図である。

【図4】図3中のDフリップフロップの構成例を示す図である。

【図5】図3の回路の動作を示すタイミングチャートである。

【図6】本発明の第2実施形態のSDRAM内の入力回路を示す図である。

【図7】図6及び図4の回路の動作を示すタイミングチャートである。

【図8】本発明の第3実施形態のSDRAMを示す概略ブロック図である。

【図9】従来の、SDRAMコントローラと複数のSD

RAMとの間の配線接続図である。

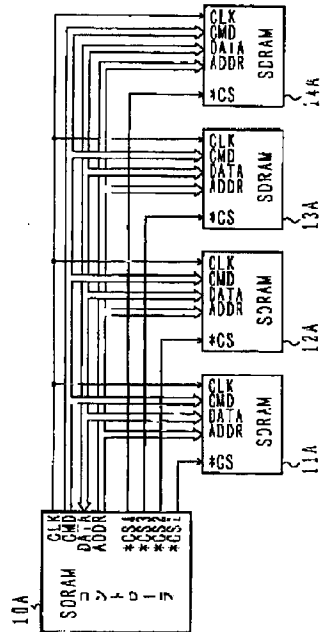
【図10】図9の回路の動作を示す概略タイミングチャートである。

【符号の説明】

10、10A SDRAMコントローラ
11～14、11A～14A SDRAM
20、30 RSフリップフロップ
21、23、24、31、33、34、43 オアゲート
22 アドレス・コマンド入力回路
25、26、35、36、40 Dフリップフロップ
301、302、42 ノアゲート
304、45、48、51A、51B、54A、54B、71B、72A、45、83 pMOSTランジスタ
303、44、52、55A、55B、56A、56B、57A、57B、71A、72B、44、81 nMOSTランジスタ
32 データ入力回路
41、41A タイミング回路
46、47、61A、61B、73、74、84、85 インバータ
50、50A ダイナミックフリップフロップ
53 増幅回路
60 相補化回路
70 ラッチ回路
80 ツルババッファゲート
82 立ち下がり同期化インバータ
90 入力回路
*CS、CSSM、CSM、*CSC チップセレクト信号
CLK 外部クロック
CLK1、CLK2 内部クロック
EN1、*EN1、*EN2 イネーブル信号
*BST バースト信号
ADDR アドレス
CMD コマンド
DATA データ
DQM データマスク
NO1～NO4 ノード

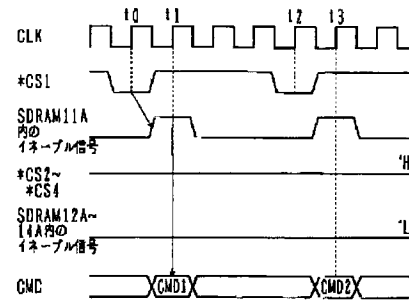
【図1】

本発明の第1実施形態に係る、SDRAMコントローラと複数のSDRAMとの間の配線接続図



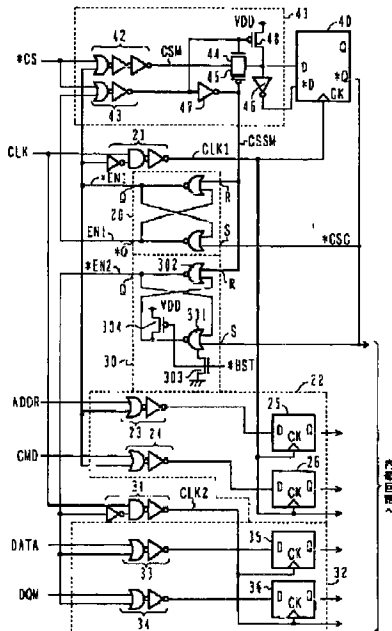
【図2】

図1の回路の動作を示す概略タイミングチャート



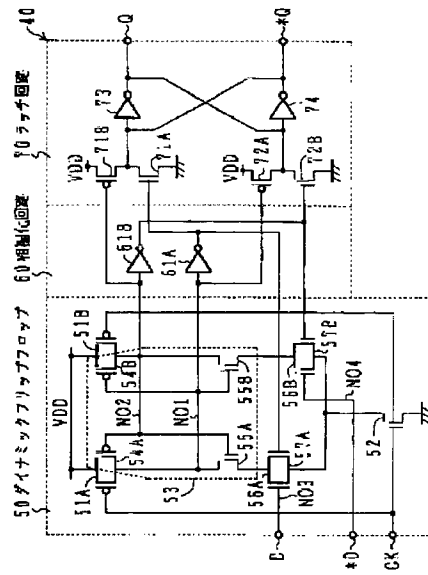
【図3】

図1中の1つのSDRAM内の入力回路を示す図



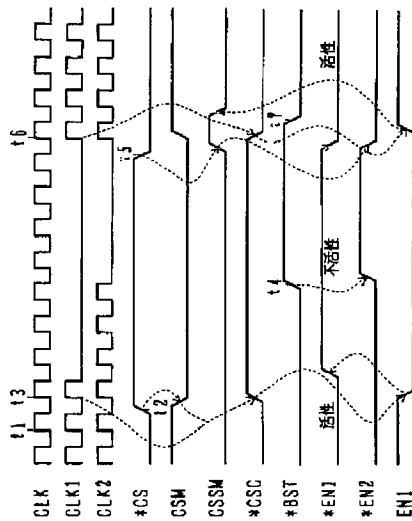
【図4】

図3中のDフリップフロップの構成例を示す図



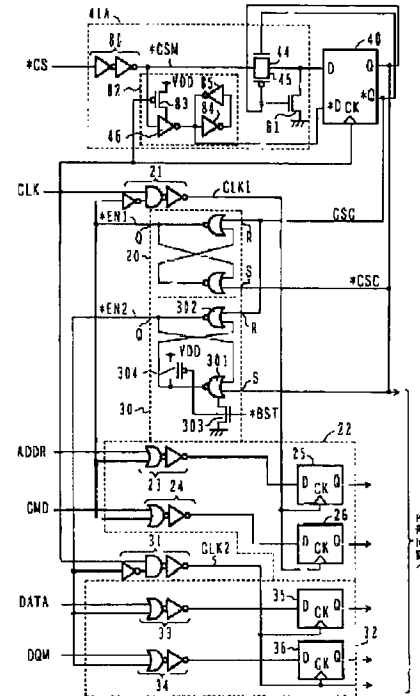
【図5】

図3の回路の動作を示すタイミングチャート



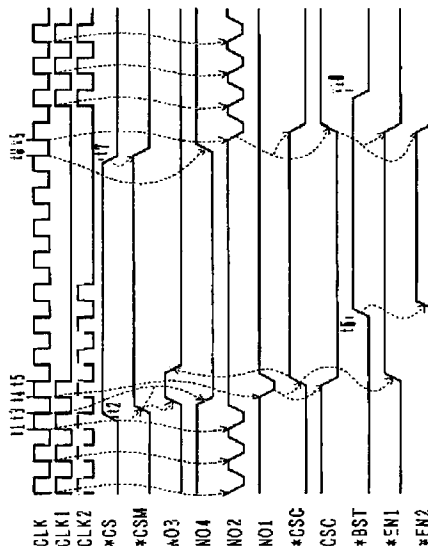
【図6】

本発明の第2実施形態のSDRAM内の入力回路を示す図



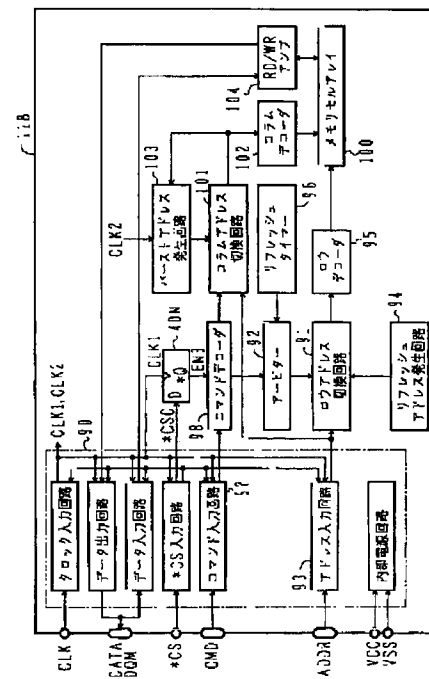
【図7】

図6及び図4の回路の動作を示すタイミングチャート



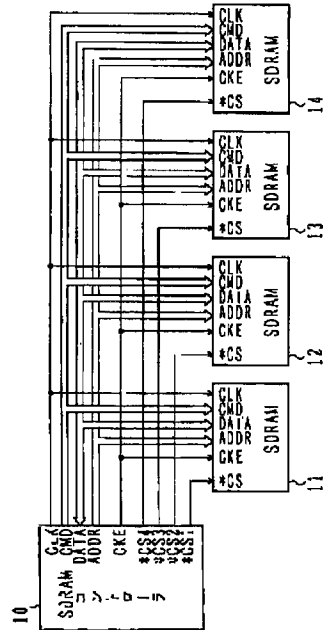
【図8】

本発明の第3実施形態のSDRAMを示す概略ブロック図



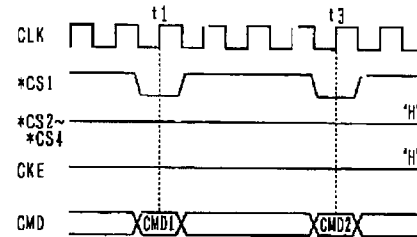
【図9】

従来の、SDRAMコントローラと複数のSDRAMとの間の配線接続図



【図10】

図9の回路の動作を示す概略タイミングチャート



**Espacenet****Bibliographic data: JP2002184176 (A) — 2002-06-28****SDRAM CONTROL CIRCUIT AND SDRAM MODULE****Inventor(s):** KIKUCHI MASASHI ± (KIKUCHI MASASHI)**Applicant(s):** NEC TOHOKU LTD ± (NEC TOHOKU LTD)**Classification:** - international: **G11C11/401; G11C11/407; G11C5/00;** (IPC1-7): G11C11/401; G11C11/407
- European:**Application number:** JP20000378746 20001213**Priority number (s):** JP20000378746 20001213**Abstract of JP2002184176 (A)**

PROBLEM TO BE SOLVED: To provide a memory module being inexpensive, practical, and large capacity by constituting a memory module (DIMM) using an inexpensive SDRAM put to practical use now. **SOLUTION:** A control circuit outputting plural new chip selecting signal from an address signal, a clock enable-signal, and a chip selecting signal is formed to a SDRAM module, and the number of SDRAM being controllable is increased. Consequently, four inexpensive 16 Mbit× 4 bit.SDRAM can be used instead of expensive 32 Mbit× 8 bit.SDRAM.

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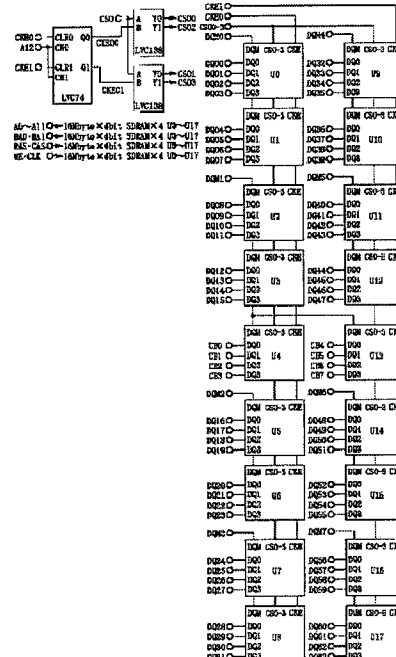
Fターム(参考) 5B024 AA15 BA21 CA16

(54) 【発明の名称】 SDRAM制御回路およびSDRAMモジュール

(57) 【要約】

【課題】 年々SDRAMの容量増が要求されているが、これに応えるためには、より大容量のSDRAMを開発し、モジュール化して使用することが望ましい。しかし、大容量のSDRAMの開発には膨大な開発費、長期の開発期間などの問題がある。

【解決手段】 アドレス信号とクロックイネーブル信号とチップセレクト信号から複数の新たなチップセレクト信号を出力する制御回路をSDRAMモジュールに形成し、制御可能なSDRAMの数を増やす。この結果、高価な32Mbit×8bit・SDRAMに代えて4個の安価な16Mbit×4bit・SDRAMを使用できる。



【特許請求の範囲】

【請求項1】 SDRAMモジュールの制御信号を生成するSDRAM制御回路であって、アドレス信号とクロックイネーブル信号を入力させ新たな複数のクロックイネーブル信号を出力する第1の回路と、該第1の回路から出力されたクロックイネーブル信号とチップセレクト信号を入力させ複数の新たなチップセレクト信号を出力する第2の回路とを備えることを特徴とするSDRAM制御回路。

【請求項2】 SDRAMモジュールは複数の第1のSDRAM部を備え、各第1のSDRAM部は該第1のSDRAM部より小さいメモリ容量の第2のSDRAMの複数個によって構成される請求項1記載のSDRAM制御回路。

【請求項3】 前記第2のSDRAMには第2の回路が出力するチップセレクト信号がそれぞれ入力される請求項2記載のSDRAM制御回路。

【請求項4】 アドレス信号はバンクアドレス信号である請求項1、2、または3記載のSDRAM制御回路。

【請求項5】 複数の第1のSDRAM部を備えるSDRAMモジュールであって、前記第1のSDRAM部は該SDRAMより小さいメモリ容量の第2のSDRAMの複数個によって構成され、アドレス信号を入力し前記第2のSDRAMを制御する信号を出力するSDRAM制御回路を備えることを特徴とするSDRAMモジュール。

【請求項6】 SDRAM制御回路の出力する信号は、前記第2のSDRAMを制御するチップセレクト信号である請求項5記載のSDRAMモジュール。

【請求項7】 SDRAM制御回路は、アドレス信号とクロックイネーブル信号を入力させ新たな複数のクロックイネーブル信号を出力する第1の回路と、該第1の回路から出力されたクロックイネーブル信号とチップセレクト信号を入力させ複数の新たなチップセレクト信号を出力する第2の回路とを備える請求項5記載のSDRAMモジュール。

【請求項8】 前記SDRAMモジュールはデュアル・インライン・メモリー・モジュールである請求項5、6または7記載のSDRAMモジュール。

【発明の詳細な説明】

【0001】

【発明の属する技術分野】本発明はシンクロナス・ダイナミック・ランダム・アクセス・メモリー (Synchronous Dynamic Random Access Memory、以下SDRAMと記載する) 制御回路、およびSDRAMを使用したメモリー・モジュールに関する。

【0002】

【従来の技術】SDRAMは、CPUやプロセッサモジュールなどのコンピュータあるいは通信ユニットシステ

ムのメモリ部に使用されている。SDRAMは一般的にシステムに合ったメモリ容量が必要であるが、年々、アクセスデータや通信速度は増大する傾向にあり、SDRAMの容量増が要求されている。

【0003】図5に、SDRAMを複数組み合わせ、容量の増加を図ったSDRAMモジュールの例として、容量512MByteのデュアル・インライン・メモリー・モジュール (Dual In-Line Memory Module、以下DIMM) のブロック図を示す。このDIMMにおいては、18個の32Mbit×8bitのSDRAM (図中のU0～U17、ただしこのうち2個はチェックビット用) に、それぞれ、13信号のアドレス信号 (A0～A12)、およびチップセレクト信号 (CS0) とクロックイネーブル信号 (CKE0～CKE1) が1信号ずつ供給され制御されている。さらにクロック信号 (CLK)、バンクアドレス信号 (BA0～BA1)、ロウアドレスストローブ信号 (RAS)、カラムアドレスストローブ信号 (CAS)、ライトイネーブル信号 (WE) が入力する。また入出力信号として、データマスク信号 (DQM0～DQM7)、データ信号 (DQ00～DQ63)、チェックビット信号 (CB0～CB7) が供給される。

【0004】

【発明が解決しようとする課題】上記SDRAM容量増の要求に応えるためには、より大容量のSDRAMを開発し、モジュール化して使用することが望ましい。しかし、大容量のSDRAMの開発には膨大な開発費、長期の開発期間などの問題がある。さらに、できあがったシステムは高価で実用的ではなくなるおそれもある。また、メモリだけでなく、これを制御するLSI等の開発も必要であり、ますます開発期間が長期化し、かつ高価なシステムになるという問題もある。

【0005】本発明の目的は、現段階で実用化されている安価なSDRAMを使用してメモリーモジュール (DIMM) を構成することにより、安価で実用的な大容量のメモリーモジュールを提供することにある。具体的には、DIMMを構成する各SDRAMについて、このSDRAMの容量より小容量のSDRAMを複数使用して、同じ容量のDIMMを形成することを目的としている。

【0006】

【課題を解決するための手段】本発明のSDRAM制御回路は、SDRAMモジュールを制御するSDRAM制御回路であって、アドレス信号とクロックイネーブル信号を入力させ新たな複数のクロックイネーブル信号を出力する第1の回路と、該第1の回路から出力されたクロックイネーブル信号とチップセレクト信号を入力させ複数の新たなチップセレクト信号を出力する第2の回路とを備える。

【0007】また、本発明のSDRAMモジュールは、

複数の第1のSDRAM部を備え、それぞれの第1のSDRAMは該SDRAMより小さいメモリ容量の複数のSDRAMによって構成され、アドレス信号が入力され上記小さいメモリ容量のSDRAMを制御する信号を出力するSDRAM制御回路を備える。

【0008】上記発明により、現在入手可能な安価な小容量SDRAMを用いて大容量のSDRAMモジュールが構成できる。

【0009】

【発明の実施の形態】図1は、本発明のSDRAM制御回路とSDRAMモジュール（具体的にはDIMM）の例を示すブロック図である。図1の本発明のDIMMにおいては、右側の回路部分が18個の32Mbit×8bit・SDRAM(U0~U17)1からなる部分であり、左側の回路部分が右側のSDRAMを制御するSDRAM制御回路である。本SDRAMモジュールも、制御するため入力信号としては上述の従来例と同様の信号を用いる。しかし本発明では、SDRAM(U0~U17)のそれぞれが、図2で示されるような4個の16Mbit×4bit・SDRAM(U0-0~U0-3)2によって形成されている。1個の16Mbit×4bit・SDRAM2は、チップセレクト信号(CS)が1信号、アドレス信号(A0~A11)が12信号必要である。

【0010】すなわち、従来の32Mbit×8bit・DRAM1を使用したDIMMにおいて用いられたA12のアドレス信号は、カラムアドレス番号12を指定、制御する信号であり、本発明の16Mbit×4bit・SDRAM2を制御する場合には不要となる。このため本発明のSDRAMモジュールでは、従来のDIMMよりもアドレス信号が1信号余分となる。この余分なA12信号を使用して、16Mbit×4bitの4個のSDRAMを制御するチップセレクト信号(CS00~CS03)がSDRAM制御回路によって作られる。

【0011】まず、A12信号がクロックイネーブル信号(CKE0、CKE1)と共にD-フリップフロップ回路(LVC74)に入力され、処理(クリア・プリセット)されて、新たなクロックイネーブル信号(CKE00、CKE01)が出力される。このクロックイネーブル信号(CKE00、CKE01)は、チップセレクト信号(CS0)と共に2個のデコード回路(LVC138)にそれぞれ入力され、該回路にて処理(デコード)され、新しいチップセレクト信号(CS00~CS03)の4信号が出力される。かくして得られたチップセレクト信号(CS00~CS03)は、該モジュールに入力するチップセレクト信号(CS0)、アドレス信号(A12)、クロックイネーブル信号(CKE)と同期して動作する信号であり、図2の16Mbit×4bit・SDRAM(U0-0~U0-3)2のチップセ

レクト信号となって、これらSDRAMのリード/ライト、リフレッシュ、プリチャージなどの制御を行う。図2の他のSDRAM(U1~U17)でも同様である。なお、上述のチップセレクト信号(CS00~CS03)を出力するD-フリップフロップ回路とデコード回路は、高速で出力遅延の少ないゲートアレイでも構成できる。

【0012】従って、これらの信号によって72個の16Mbit×4bit・SDRAM2が制御できるようになり、図5のモジュールと同じ容量である512MByteのDIMMが構成可能となる。

【0013】次に、本SDRAMモジュールの動作について、図3(a)~(f)のタイミングチャートを用いて説明する。各信号はCLK信号に同期して制御される。

【0014】図3(a)では、D-フリップフロップ回路(LVC74)において、CLR端子に入力されるクロックイネーブル信号(CKE)がハイレベルで、CLK端子に入力されるアドレス信号(A12)がローレベルとなる場合、該回路の出力端子からは、ローレベルのクロックイネーブル信号(CKE0)が出力される。このローレベルのクロックイネーブル信号(CKE0)とチップセレクト信号(CS0)がデコード回路(LVC138)に入力され、チップセレクト信号(CS0)がローレベルの場合、出力端子にはCS0信号と同期したローレベルのチップセレクト信号(CS00)が出力され、SDRAM(U0-0)はアクティブとなり、またハイレベルのチップセレクト信号(CS02)が出力され続け、SDRAM(U0-2)はスタンバイ状態となる。

【0015】図3(b)では、D-フリップフロップ回路(LVC74)のCLR端子に入力されるクロックイネーブル信号(CKE)がハイレベルで、CLK端子に入力されるアドレス信号(A12)がハイレベルになると、それ以降CKE信号がローレベルになるまで、該回路から出力される新たなクロックイネーブル信号(CKE0)はハイレベルで出力される。このハイレベルのクロックイネーブル信号(CKE0)とチップセレクト信号(CS0)がデコード回路(LVC138)に入力され、チップセレクト信号(CS0)がローレベルになると、該デコード回路から出力される新たなチップセレクト信号(CS00)はハイレベルで出力され続け、SDRAM(U0-0)はスタンバイ状態となり、チップセレクト信号(CS02)はCS0信号と同期したローレベルの信号となり、SDRAM(U0-2)はアクティブとなる。

【0016】図3(c)では、D-フリップフロップ回路(LVC74)の入力信号と出力信号は図3(a)と同じである。このとき、デコード回路(LVC138)に入力されるチップセレクト信号(CS0)がハイレベ

ルのままであるとき、出力されるCS00信号およびCS02信号はハイレベルのままである。

【0017】図3(d)では、D-フリップフロップ回路(LVC74)の入力信号と出力信号は図3(b)と同じである。このとき、デコード回路(LVC138)に入力されるチップセレクト信号(CS0)がハイレベルのままであるとき、出力されるCS00信号およびCS02信号はハイレベルのままである。

【0018】図3(e)、図3(f)では、D-フリップフロップ回路(LVC74)に入力するクロックイネーブル信号(CKE)がローレベルならば、アドレス信号(A12)がローレベルであり、またはハイレベルになっても、出力される新たなクロックイネーブル信号(CKE0)はローレベルである。さらに同様にデコード回路(LVC138)に入力されるチップセレクト信号(CS0)がハイレベルのままであるとき、出力されるCS00信号およびCS02信号はハイレベルのままである。もう一方のデコード回路(LVC138)から出力されるCS01、CS03についても上述のタイムチャートと同様に動作する。

【0019】このD-フリップフロップ回路(LVC74)とデコード回路(LVC138)を備えた制御回路によって作り出されたチップセレクト信号(CS00~CS03)を使用し、通常DIMMで使用している制御信号と組み合わせることで、SDRAMのリード/ライト、その他の制御を行うようにしている。

【0020】なお、上述のD-フリップフロップ回路(LVC74)に入力させる信号として、アドレス信号(A12)に代えて、バンクアドレス信号(BA0)を使用することができる。図4にこの場合の制御回路部の構成を示す。BA0信号を用いても新しいCS00~C

SO3信号によってSDRAMを制御することができるが、BA信号を持たないSDRAM部品も使い、DIMMにおけるSDRAM部品において使用できる部品の範囲が広がる。

【0021】上記16Mbit×4bitのSDRAM部品はCSP(Chip Size Package)構造となっており、4段のスタック品となって1つのSDRAMを形成している。

【0022】

【発明の効果】このように、本発明では、チップセレクト信号(CS)、クロックイネーブル信号(CKE)、アドレス信号(A12)を使用し、新たなチップセレクト信号(CS00~CS03)という4出力の制御信号を作り出すことによって、制御できるSDRAMを増加させ、小容量のSDRAMを使用し、大容量のDIMMを構成することが可能となる。また32Mbit×8bitのSDRAM部品に比べ、16Mbit×4bitのSDRAM部品の4個分はかなり安価であるので、大容量のDIMMを安価に構成できる。

【図面の簡単な説明】

【図1】本発明が適用されたDIMMの一例を示す構成図。

【図2】DIMMを構成する32Mbit×8bit・SDRAMの構成図。

【図3】DIMMの制御信号のタイムチャート。

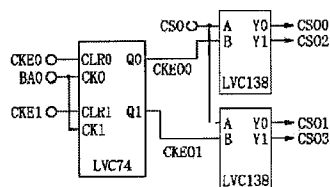
【図4】バンクアドレス信号が入力するDIMM制御部の構成図。

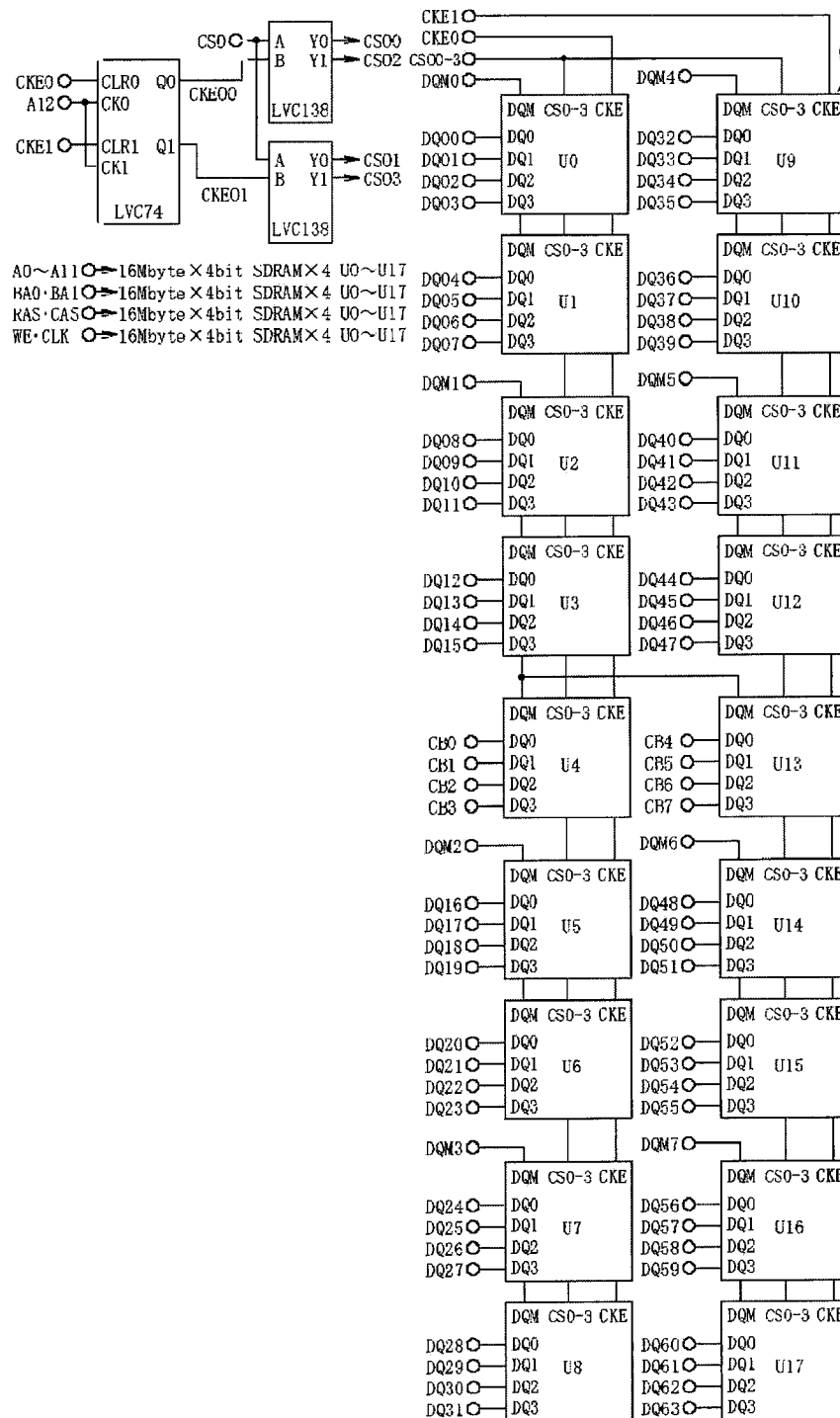
【図5】従来のDIMMの構成図。

【符号の説明】

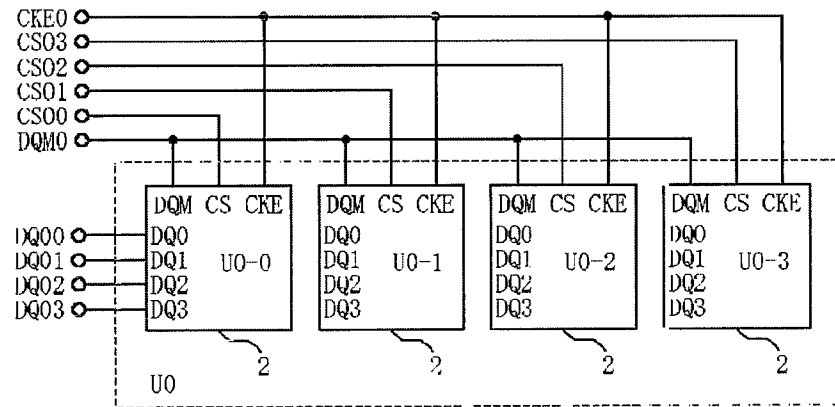
- 1 32Mbit×8bit・SDRAM
- 2 16Mbit×4bit・SDRAM

【図4】



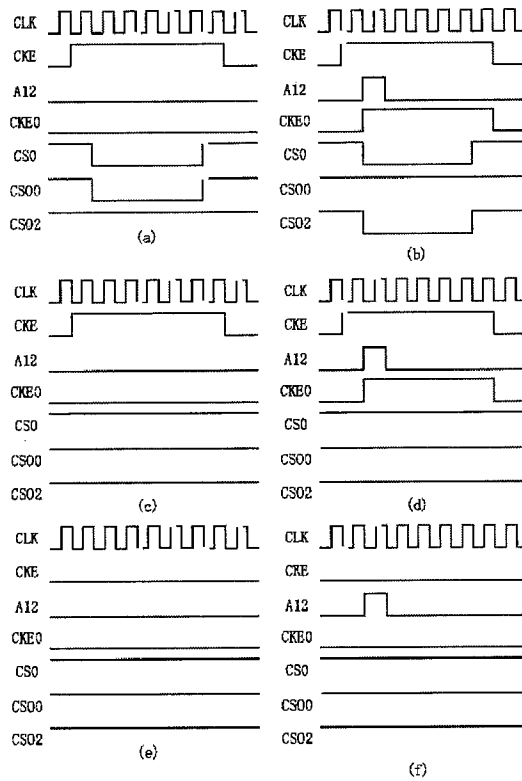


【図2】

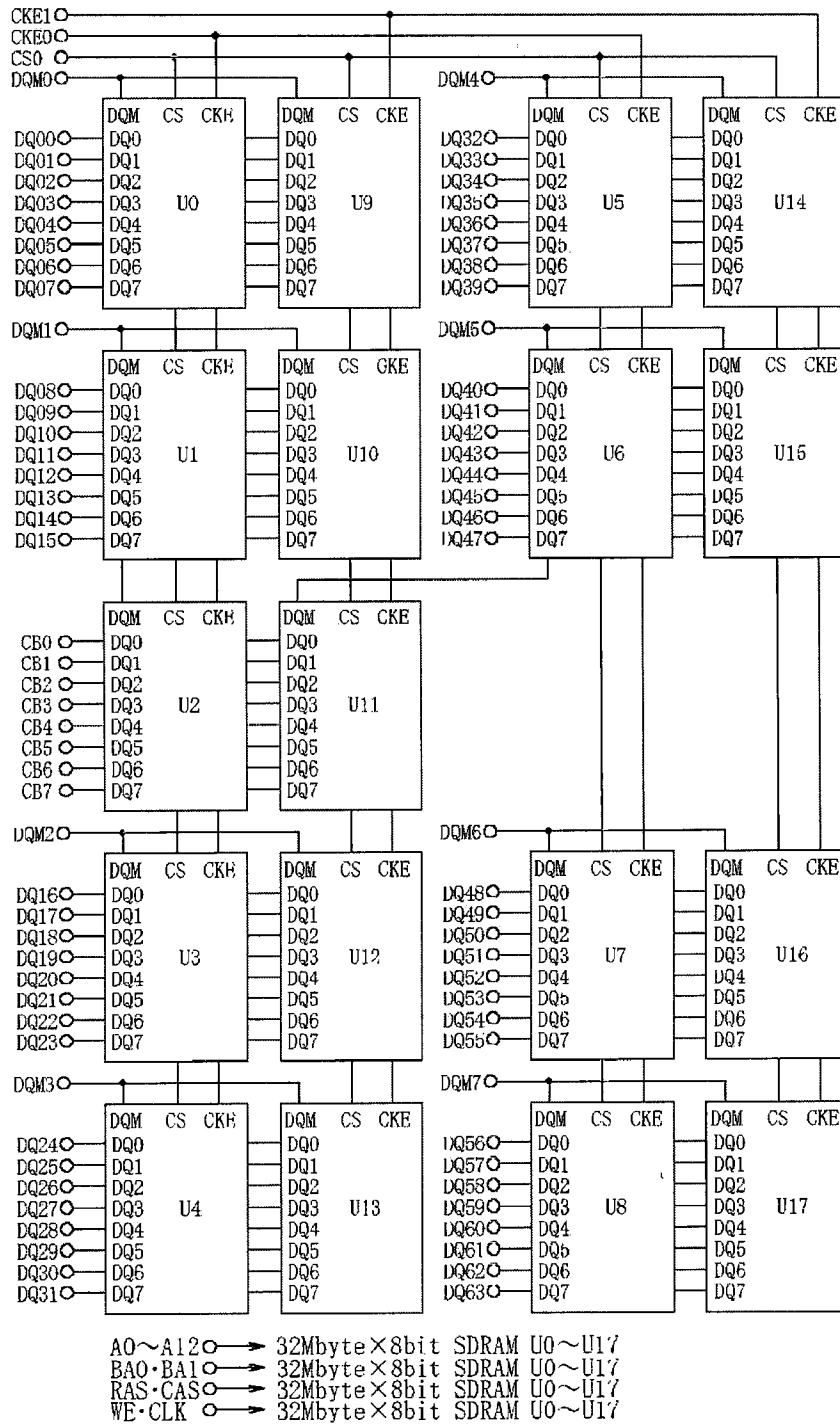


AO~A11 → 16Mbyte×4bit SDRAM U0-0~U0-3
 BA0·BA1 → 16Mbyte×4bit SDRAM U0-0~U0-3
 RAS·CAS → 16Mbyte×4bit SDRAM U0-0~U0-3
 WE·CLK → 16Mbyte×4bit SDRAM U0-0~U0-3

【図3】



【図5】



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(22) Filing Date: December 13, 2000

20 (71) Applicant: 000222060
NEC Tohoku Ltd., 1 Karakai, Ichinoseki-shi, Iwate-ken

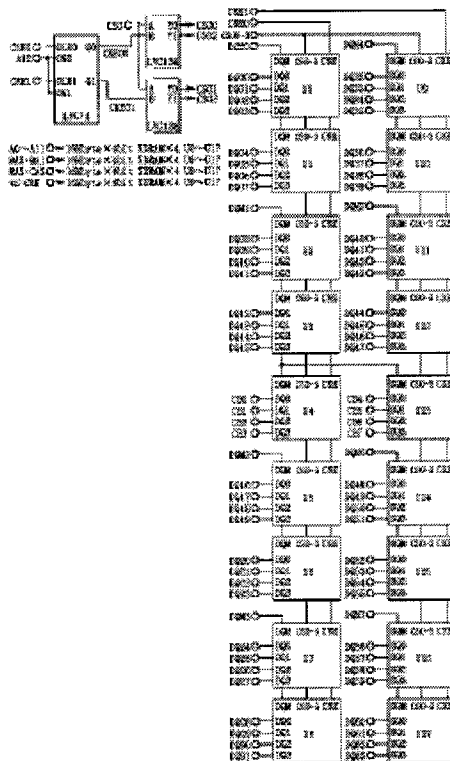
(72) Inventor: Masashi KIKUCHI
c/o NEC Tohoku Ltd., 1 Karakai, Ichinoseki-shi, Iwate-ken

25 (74) Representative: 100082935 Naoki KYOMOTO,
Patent Attorney (and two others)

F-Terms (reference) 5B024 AA15 BA21 CA16

5 [Problem] Every year sees a demand for SDRAMs having
greater capacity, and in order to respond to this
demand it would be desirable to develop SDRAMs of
greater capacity and to modularize these for use.
However, the development of large-capacity SDRAMs
10 entails problems such as very large development costs
and a long development period.

20



[Patent Claims]

[Claim 1] An SDRAM control circuit for generating a control signal for an SDRAM module, characterized in that it comprises: a first circuit which receives as
5 input an address signal and a clock enable signal and outputs a plurality of new clock enable signals; and a second circuit which receives as input a clock enable signal and a chip select signal output from said first circuit and outputs a plurality of new chip select
10 signals.

[Claim 2] The SDRAM control circuit as claimed in claim 1, wherein the SDRAM module comprises a plurality of first SDRAM parts and each first SDRAM part is composed
15 of a plurality of second SDRAM parts having a smaller memory capacity than said first SDRAM parts.

[Claim 3] The SDRAM control circuit as claimed in claim 2, wherein the chip select signals output by the second
20 circuit are input to the second SDRAMs, respectively.

[Claim 4] The SDRAM control circuit as claimed in claim 1, 2 or 3, wherein the address signal is a bank address
25 signal.

[Claim 5] An SDRAM module comprising a plurality of first SDRAM parts, characterized in that the first SDRAM parts are composed of a plurality of second SDRAM parts having a smaller memory capacity than said
30 SDRAMs, and comprise an SDRAM control circuit which receives as input an address signal and outputs a signal for controlling the second SDRAMs.

[Claim 6] The SDRAM module as claimed in claim 5, in
35 which the signal output by the SDRAM control circuit is a chip select signal for controlling the second SDRAMs.

[Claim 7] The SDRAM module as claimed in claim 5, in which the SDRAM control circuit comprises: a first

circuit which receives as input an address signal and a clock enable signal and outputs a plurality of new clock enable signals; and a second circuit which receives as input a clock enable signal and a chip select signal output from said first circuit and outputs a plurality of new chip select signals.

[Claim 8] The SDRAM module as claimed in claim 5, 6 or 7, which is a dual in-line memory module.

10

[Detailed Description of the Invention]

[0001]

[Technical Field of the Invention] The present invention relates to a Synchronous Dynamic Random Access Memory (referred to below as an "SDRAM") control circuit, and to a memory module employing SDRAMs.

15

[0002]

[Prior Art] SDRAMs are used in memories of computers such as CPUs and processor modules, or communication unit systems. SDRAMs generally require a memory capacity which is compatible with the system, but as access data and communication speed tend to increase year on year, there is a need for increased SDRAM capacity.

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[0003] Figure 5 is a block diagram showing a 512 Mbyte-capacity dual in-line memory module (referred to below as a "DIMM") as one example of an SDRAM module comprising a plurality of combined SDRAMs, whereof the capacity is to be increased. This DIMM is supplied with and controlled by one each of the following signals: 13 address signals (A0 - A12), and a chip select signal (CS0) and clock enable signals (CKE0 - CKE1). A clock signal (CLK), bank address signals (BA0 - BA1), row-address strobe signal (RAS), column-address strobe signal (CAS), and a write enable signal are also input. Furthermore, data mask signals (DQM0 - DQM7), data

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signals (DQ00 - DQ63) and check bit signals (CB0 - CB7) are supplied.

[0004]

5 [Problem to be Resolved by the Invention] In order to respond to the abovementioned need for increased SDRAM capacity, it would be desirable to develop SDRAMs of greater capacity and to modularize these for use. However, the development of large-capacity SDRAMs
10 entails problems such as very large development costs and a long development period. In addition, finished systems are costly and risk becoming impractical. Furthermore, there is also a need for development such as LSI for controlling these systems, so there are
15 problems in terms of development periods becoming longer and the costs of the systems increasing.

[0005] The aim of the present invention lies in providing an inexpensive, practical and large-capacity
20 memory module by constructing a memory module (DIMM) using inexpensive SDRAMs which are currently utilized. Specifically, for each SDRAM which forms part of the DIMM, a plurality of SDRAMs having a smaller capacity than the capacity of said SDRAMs are employed, with the
25 aim of forming a DIMM having the same capacity.

[0006]

[Means for Resolving the Problem] The SDRAM control circuit according to the present invention is an SDRAM
30 control circuit for controlling an SDRAM module, which comprises: a first circuit which receives as input an address signal and a clock enable signal and outputs a plurality of new clock enable signals; and a second circuit which receives as input a clock enable signal
35 and a chip select signal output from said first circuit and outputs a plurality of new chip select signals.

[0007] Furthermore, the SDRAM module according to the present invention comprises a plurality of first SDRAM

parts, where each of the first SDRAMs is composed of a plurality of SDRAMs having a smaller memory capacity than said SDRAMs, and comprises an SDRAM control circuit which receives as input an address signal and
5 outputs a signal for controlling the SDRAMs having a small memory capacity.

[0008] According to the present invention described above, a large-capacity SDRAM can be constructed using
10 inexpensive, small-capacity SDRAMs which are currently available.

[0009]

[Mode of Embodiment of the Invention] Figure 1 is a
15 block diagram showing an example of the inventive SDRAM control circuit and SDRAM module (a DIMM, to be specific). In the DIMM according to the present invention shown in Figure 1, the circuit portion on the right-hand side comprises eighteen 32 Mbit x 8 bit
20 SDRAMs (U0 - U17) 1, and the circuit portion on the left-hand side is an SDRAM control circuit for controlling the SDRAMs on the right-hand side. The SDRAM module also employs the same signals as described in the conventional example above as input signals for
25 control. However, according to the present invention, each of the SDRAMs (U0 - U17) is formed by four 16 Mbit x 4 bit SDRAMs (U0-0 to U0-3) 2 shown in Figure 2. One 16 Mbit x 4 bit SDRAM 2 requires one chip select signal (CS) and 12 address signals (A0 - A11).

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[0010] That is to say, the A12 address signal used in a conventional DIMM employing the 32 Mbit x bit DRAMs 1 is a control signal for designating the column address number 12, and is not required to control the 16 Mbit x
35 4 bit SDRAMs 2 according to the present invention. In the SDRAM module according to the present invention, there is therefore one extra address signal compared with a conventional DIMM. Using this extra A12 signal, chip select signals (CS00 - CS03) for controlling the

four 16 Mbit $\times$ 4 bit SDRAMs are produced by means of the SDRAM control circuit.

[0011] The A12 signal is first of all input to a D-flip
5 flop circuit (LVC74) together with clock enable signals
(CKE0, CKE1), processing (clear/preset) is carried out,
and new clock enable signals (CKE00, CKE01) are output.
The clock enable signals (CKE00, CKE01) are input to
two decoder circuits (LVC138) together with a chip
10 select signal (CS0), respectively, this circuit carries
out processing (decoding), and four new chip select
signals (CS00 - CS03) are output. The chip select
signals (CS00 - CS03) obtained in this way are signals
which operate synchronously with the chip select signal
15 (CS0), address signal (A12) and clock enable signal
(CKE) which are input to said module, and constitute
chip select signals of the 16 Mbit $\times$ 4 bit SDRAMs (U0-0
to U0-3) 2 shown in Figure 2, controlling read/write,
refresh, precharge and the like for these SDRAMs. The
20 same also applies to the other SDRAMs (U1 - U17) shown
in Figure 2. It should be noted that the D-flip flop
circuit and decoder circuit which output the
abovementioned chip select signals (CS00 - CS03) may
also comprise gate arrays which provide high speed and
25 little delay in output.

[0012] The seventy-two 16 Mbit $\times$ 4 bit SDRAMs 2 can
therefore be controlled by means of these signals, and
it is possible to construct a 512 Mbyte-DIMM having the
30 same capacity as the module shown in Figure 5.

[0013] The operation of this SDRAM module will be
described next with the aid of the timing charts shown
in Figures 3(a) - (f). The signals are controlled
35 synchronously with the CLK signal.

[0014] In Figure 3(a), when the clock enable signal
(CKE) input to the CLR terminal is high level and the
address signal (A12) input to the CLK terminal is low

level, the output terminal of the D-flip flop circuit (LVC74) outputs the low-level clock enable signal (CKEO). This low-level clock enable signal (CKEO) and the chip select signal (CS0) are input to the decoder
5 circuit (LVC138), and when the chip select signal (CS0) is low level, the low-level chip select signal (CS00) synchronous with the CS0 signal is output at the output terminal, the SDRAM (U0-0) becomes active, and a high-level chip select signal (CS02) continues to be output,
10 and the SDRAM (U0-2) is in a standby state.

[0015] In Figure 3(b), when the clock enable signal (CKE) input to the CLR terminal of the D-flip flop circuit (LVC74) is high level and the address signal
15 (A12) input to the CLK terminal is high level, the new clock enable signal (CKEO) output from said circuit is subsequently output at high level until the CKE signal becomes low level. When the high-level clock enable signal (CKEO) and the chip select signal (CS0) are
20 input to the decoder circuit (LVC138) and the chip select signal (CS0) becomes low level, the new chip select signal (CS00) output from said decoder circuit continues to be output at high level, and the SDRAM (U0-0) is in a standby state, while the chip select
25 signal (CS02) is a low-level signal synchronous with the CS0 signal, and the SDRAM (U0-2) becomes active.

[0016] In Figure 3(c), the input signal and output signal of the D-flip flop circuit (LVC74) are the same
30 as in Figure 3(a). In this case, when the chip select signal (CS0) input to the decoder circuit (LVC138) remains high level, the CS00 signal and CS02 signal which are output remain high level.

35 [0017] In Figure 3(d), the input signal and output signal of the D-flip flop circuit (LVC74) are the same as in Figure 3(b). In this case, when the chip select signal (CS0) input to the decoder circuit (LVC138)

remains high level, the CS00 signal and CS02 signal which are output remain high level.

[0018] In Figure 3(e) and Figure 3(f), if the clock
5 enable signal (CKE) input to the D-flip flop circuit
(LVC74) is low level, the address signal (A12) is low
level, and even if it is high level, the new clock
enable signal (CKEO) which is output is low level.
Likewise, if the chip select signal (CS0) input to the
10 decoder circuit (LVC138) remains at high level, the
CS00 signal and CS02 signal which are output remain at
high level. The system operates in the same way as in
the time charts described above for CS01, CS03 which
are output from the other decoder circuit (LVC138).

15

[0019] The SDRAMs are read/write controlled and
controlled in other ways by using the chip select
signals (CS00 - CS03) produced by means of the control
circuit provided with the D-flip flop circuit (LVC74)
20 and decoder circuit (LVC138) and combining these
signals with control signals conventionally used by a
DIMM.

[0020] It should be noted that a bank address signal
25 (BA0) may be used instead of the address signal (A12)
as a signal which is input to the abovementioned D-flip
flop circuit (LVC74). Figure 4 shows the construction
of a control circuit in this case. The SDRAMs can still
be controlled by means of the new CS00 - CS03 signals
30 if the BA0 signal is used, and SDRAM components which
do not have a BA signal may be used, so the range of
components which can be used for the SDRAM components
in the DIMM can be broadened.

35 [0021] The abovementioned 16 Mbit $\times$ 4 bit SDRAM
components have a CSP (chip size package) structure
arranged in a four-level stack to form one SDRAM.

[0022]

[Advantage of the Invention] In this way, according to the present invention, four output control signals, namely the new chip select signals (CS00 - CS03) are produced using a chip select signal (CS), clock enable signal (CKE), and address signal (A12), thereby increasing the SDRAMs which can be controlled, and small-capacity SDRAMs are used, and it is possible to construct a large-capacity DIMM. Furthermore, four 16 Mbit $\times$ 4 bit SDRAM components are fairly inexpensive compared with 32 Mbit $\times$ 8 bit SDRAM components, and therefore a large-capacity DIMM can be constructed inexpensively.

[Brief Description of the Figures]

[Figure 1] is a structural diagram showing one example of a DIMM in which the present invention has been applied;

[Figure 2] is a structural diagram of 32 Mbit $\times$ 8 bit SDRAMs which form a DIMM;

[Figure 3] is a time chart of DIMM control signals;

[Figure 4] is a structural diagram of a DIMM control unit to which a bank address signal is input; and

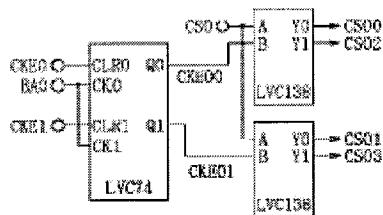
[Figure 5] is a structural diagram of a conventional DIMM.

[Key to Symbols]

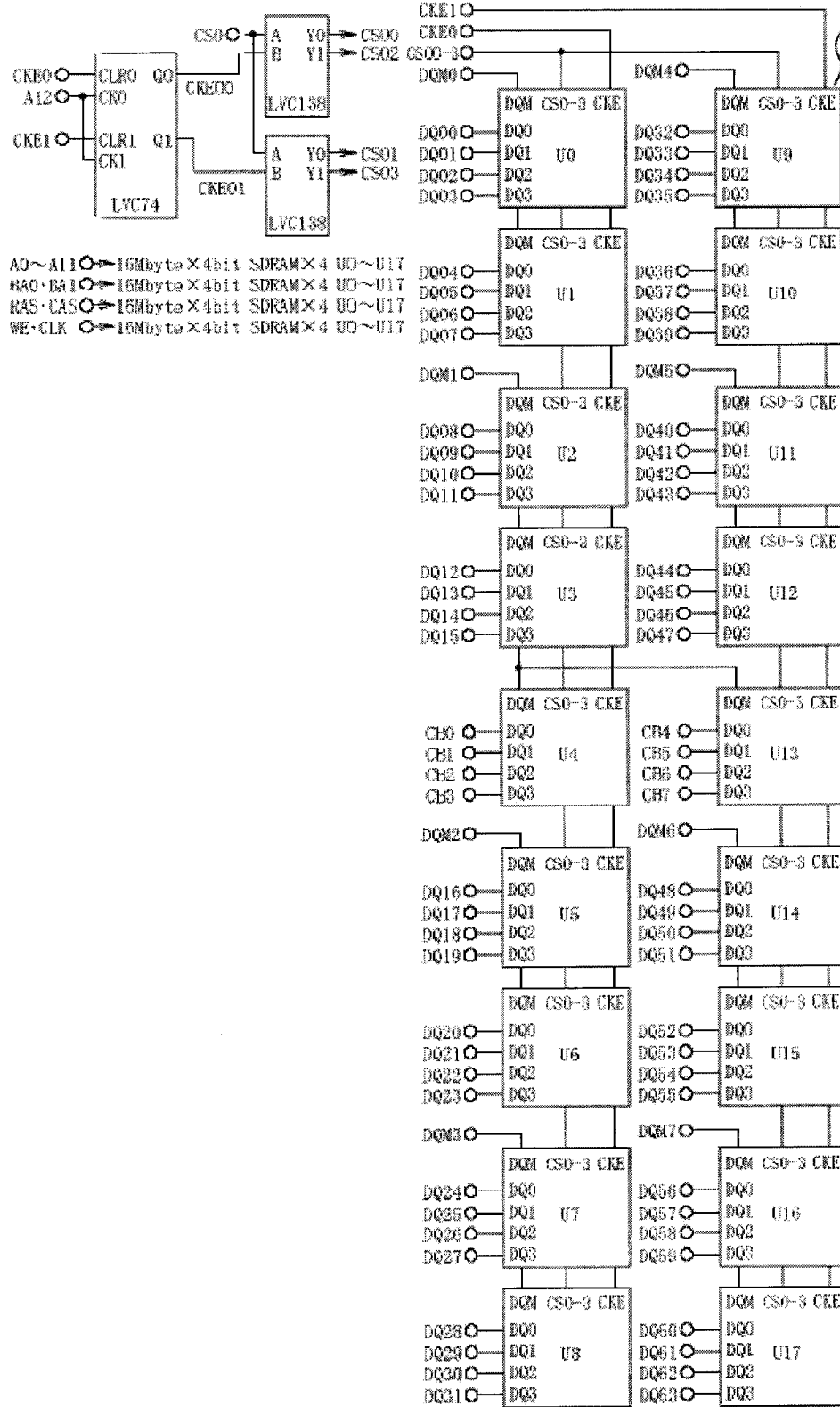
1 32 Mbit $\times$ 8 bit SDRAM

2 16 Mbit $\times$ 4 bit SDRAM

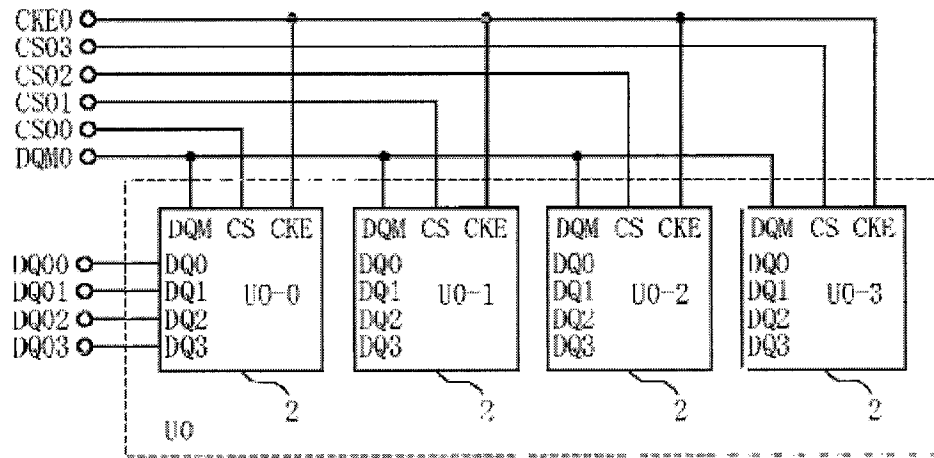
[Figure 4]



[Figure 1]

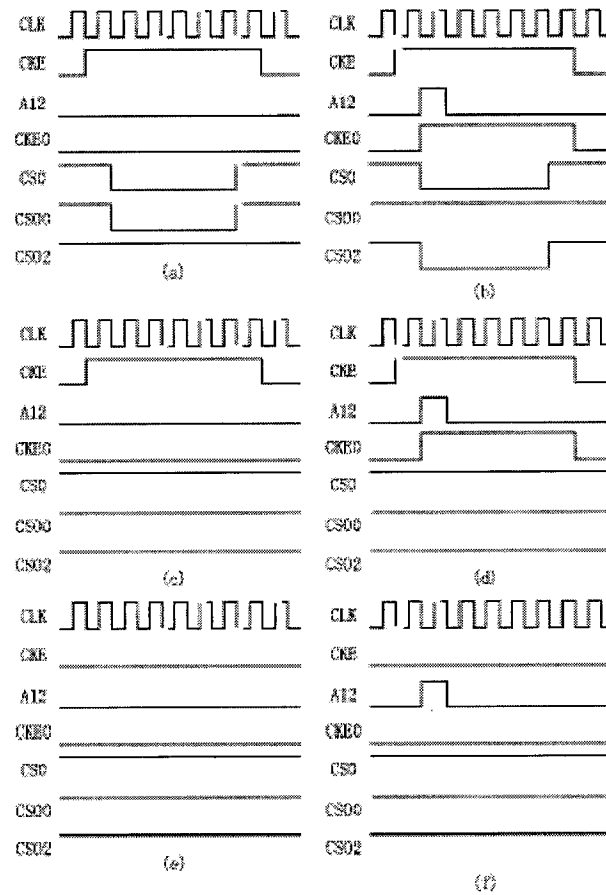


[Figure 2]

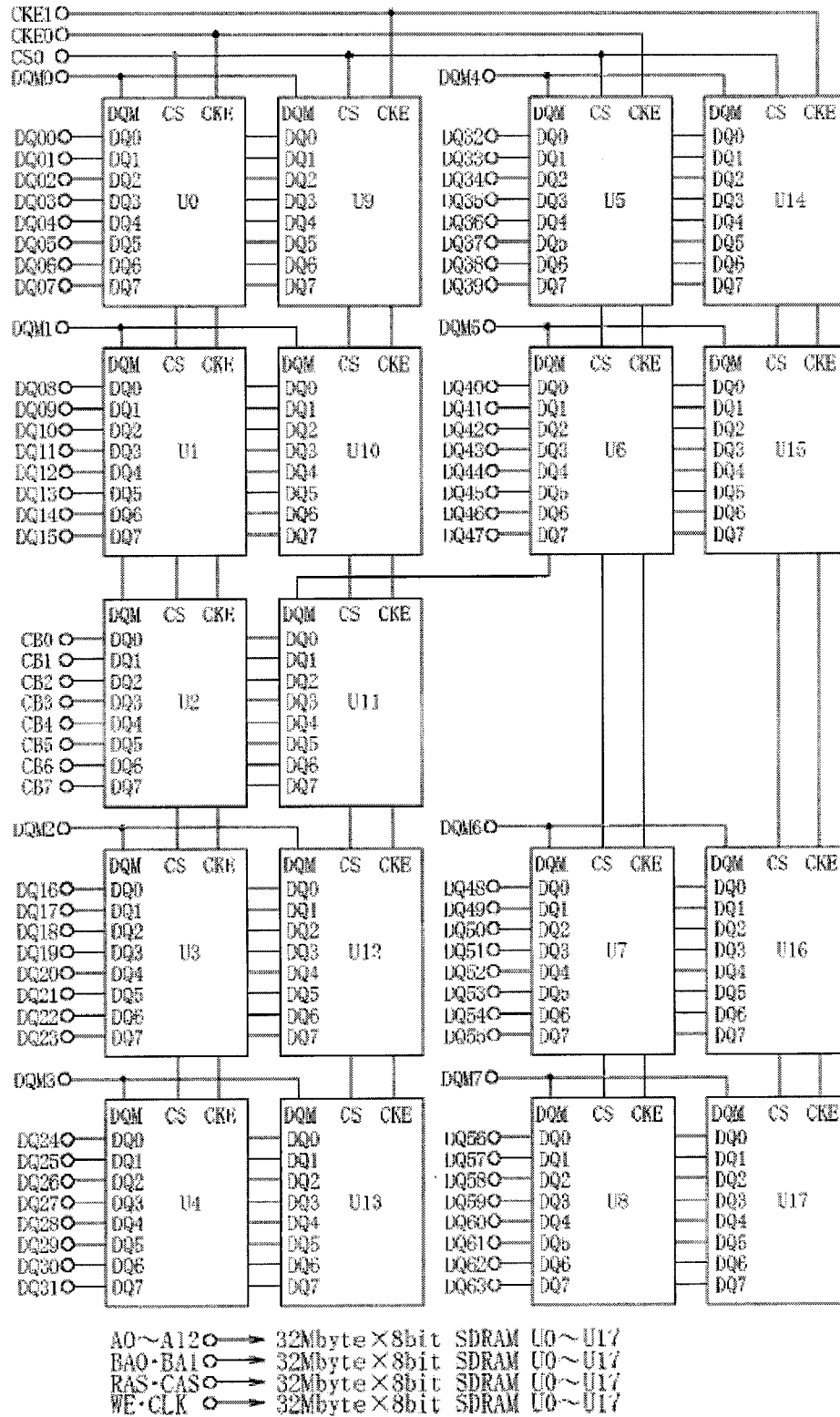


A0~A11 → 16Mbyte×4bit SDRAM U0-0~U0-3
 BA0·BA1 → 16Mbyte×4bit SDRAM U0-0~U0-3
 RAS·CAS → 16Mbyte×4bit SDRAM U0-0~U0-3
 WE·CLK → 16Mbyte×4bit SDRAM U0-0~U0-3

[Figure 3]



[Figure 5]



**Espacenet****Bibliographic data: JP2003007963 (A) — 2003-01-10****SEMICONDUCTOR MEMORY DEVICE AND ITS MANUFACTURING METHOD**

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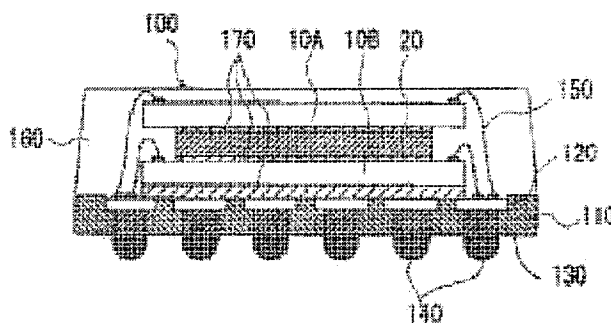
Classification: - international: *G11C16/02; G11C29/00; G11C29/04; H01L25/065; H01L25/07; H01L25/18*; (IPC1-7): G11C16/02; G11C29/00; H01L25/065; H01L25/07; H01L25/18

- European:

Application number: JP20010186288 20010620

Priority number (s): JP20010186288 20010620

Abstract of JP2003007963 (A)



PROBLEM TO BE SOLVED: To overcome such a problem of a conventional multichip stack type semiconductor memory that since a plurality of chip enable terminals are provided, the number of terminals is increased as compared with a one chip semiconductor device when the semiconductor memory is regarded as one semiconductor device and thereby the package size is increased and compatibility with conventional products is lost. **SOLUTION:** The semiconductor memory device comprises a first terminal part (CAD) being connected with any one of first or second power supply voltage terminal, and chip selection control circuits (17, 20) for controlling internal chip selection signal based on the state of the first terminal part and the state of any one or more than one of a plurality of address input terminal parts (e.g. the input pad of a most significant bit A22). The chip selection control circuit controls the internal chip selection signal (CEB) to valid or invalid state based on any one or more than one bit of address signals inputted under a set state of the first terminal part and a state where an external chip selection signal is validated.

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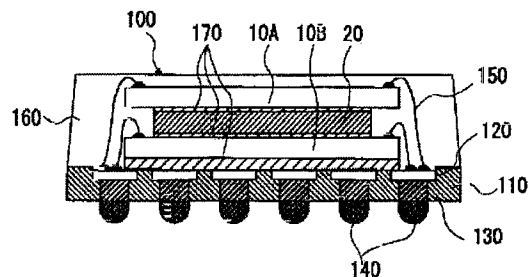
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(54) 【発明の名称】 半導体記憶装置および製造方法

(57) 【要約】

【課題】 従来のマルチチップスタック型の半導体記憶装置は、チップイネーブル用端子が複数個あるため、これを1つの半導体装置として見たときにはワンチップの半導体装置に比べて端子数が増えることとなり、パッケージが大型化するとともに、従来の製品との互換性がなくなるという課題があった。

【解決手段】 第1の電源電圧端子または第2の電源電圧端子のいずれかに接続される第1端子部 (C A D) と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部 (例えば最上位ビット A 2 2 の入力パッド) の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路 (1 7, 2 0) とを備え、該チップ選択制御回路は、上記第1端子部の設定状態と、外部からのチップ選択信号が有効にされた状態で入力されたアドレス信号のいずれか1または2以上のビットとに基づいて内部のチップ選択信号 (C E B) を有効状態または無効状態に制御するように構成した。



【特許請求の範囲】

【請求項1】 第1の電源電圧端子または第2の電源電圧端子のいずれかに接続される第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、該チップ選択制御回路は、上記第1端子部の設定状態と、外部からのチップ選択信号が有効にされた状態で入力されたアドレス信号のいずれか1または2以上のビットとに基づいて内部のチップ選択信号を有効状態または無効状態に制御するように構成されていることを特徴とする半導体記憶装置。

【請求項2】 複数の半導体メモリチップが1つのパッケージに収納され各チップの端子部は対応するもの同士がパッケージの対応する端子に共通に接続されてなる半導体記憶装置であって、上記複数の半導体メモリチップは、各々異なる状態に設定可能な第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、上記複数の半導体メモリチップ内の各チップ選択制御回路は、上記第1端子部が各々異なる状態に設定されている場合には、外部からのチップ選択信号が有効にされた状態で入力されたアドレス信号が同一であってもいずれか1のチップにおいて内部のチップ選択信号を有効にするように構成されていることを特徴とする半導体記憶装置。

【請求項3】 複数の半導体メモリチップが1つのパッケージに収納され、データ入出力端子部以外の各チップの端子部は対応するもの同士がパッケージの対応する端子に共通に接続されてなる半導体記憶装置であって、上記複数の半導体メモリチップは、各々異なる状態に設定可能な第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、上記複数の半導体メモリチップ内の各チップ選択制御回路は、上記第1端子部が同一の状態に設定されかつアドレス入力端子部のうち所定の1または2以上の端子部が同一の電位に固定されている場合には、外部からのチップ選択信号が有効にされたことに応じてそれぞれ内部のチップ選択信号を有効にするように構成されていることを特徴とする半導体記憶装置。

【請求項4】 上記アドレス信号のいずれか1または2以上のビットは、アドレスの最上位ビットまたは最上位側2ビット以上であることを特徴とする請求項2または3に記載の半導体記憶装置。

【請求項5】 複数の半導体メモリチップが1つのパッケージに収納され、該パッケージには各チップに対応してデータ入出力用の端子が設けられ各チップのデータ入

出力端子部はパッケージの対応する上記端子に別々に接続されてなるとともに、パッケージには所定の端子が設けられている半導体記憶装置であって、上記複数の半導体メモリチップは、各々異なる状態に設定可能な第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、上記複数の半導体メモリチップ内の各チップ選択制御回路は、上記第1端子部が同一の状態に設定されかつアドレス入力端子部のうち所定の1または2以上の端子部が同一の電位に固定されている場合には、外部からのチップ選択信号が有効にされたことに応じてそれぞれ内部のチップ選択信号を有効にするように構成され、上記複数の半導体メモリチップのいずれかの上記第1端子部は上記パッケージの上記所定の端子に接続され、他の半導体メモリチップの上記第1端子部は上記パッケージのいずれかの電源電圧端子に接続されていることを特徴とする半導体記憶装置。

【請求項6】 上記複数の半導体メモリチップは、それぞれ同一方向に端子部が現われるように積層され、各チップ間はこれらのチップの上記端子部よりも内側の部位に介在された接着剤層によって互いに接合されていることを特徴とする請求項5に記載の半導体記憶装置。

【請求項7】 複数の半導体メモリチップが1つのパッケージに収納され各チップの端子部は対応するもの同士がパッケージの対応する端子に共通に接続され、上記複数のメモリチップ毎に各々異なる状態に設定可能な第1端子部を有してなる半導体記憶装置の製造方法であって、前工程終了後にウェハ状態で上記複数の半導体メモリチップのテストを行ない、該テストにより良品と判定された半導体メモリチップに対してウェハ状態でトリミング処理を行なった後、ウェハを切断して各チップに分割し、上記各半導体メモリチップの上記第1端子部に対する設定および各チップの端子部とパッケージの端子との接続を行なってからパッケージに封止することを特徴とする半導体記憶装置の製造方法。

【請求項8】 上記第1端子部に対する設定は、該第1端子部をパッケージに設けられているいずれかの電源電圧端子にボンディングワイヤで接続する処理であることを特徴とする請求項7に記載の半導体記憶装置の製造方法。

【請求項9】 上記複数の半導体メモリチップはそれぞれ同一方向に端子部が現われるように積層され、上記ボンディングワイヤで接続する処理は各チップを接着剤で接合する処理の後に各チップ毎に行なわれることを特徴とする請求項8に記載の半導体記憶装置の製造方法。

【請求項10】 上記ウェハ状態でのトリミング処理と共に上記テストにより検出された不良メモリセルを置換する救済処理を行なうことを特徴とする請求項7に記載の半導体記憶装置の製造方法。

【発明の詳細な説明】

【0001】

【発明の属する技術分野】本発明は、半導体記憶装置さらには複数の半導体メモリチップが1つのパッケージに収納されたマルチチップスタック型の半導体記憶装置に適用して有効な技術に関し、例えばフラッシュメモリのような不揮発性半導体記憶装置の大容量化に利用して有効な技術に関する。

【0002】

【従来の技術】従来、半導体集積回路の分野では、1つのパッケージ内に複数の半導体チップを積層収納して実装密度の向上を図るようにした技術があり、半導体メモリにおいてもかかる技術により大容量化を図るようにしたものが実用化されつつある。このようなマルチチップスタック型の半導体記憶装置においては、アドレス信号や制御信号など大部分の入力信号に関しては、パッケージの端子を複数のチップ間で共有することで端子数を減らすことができる。また、データを入出力するI/O端子についてもチップ間で共有することによりパッケージの端子数を減らすことができる。

【0003】しかしながら、I/O端子を共有させた場合には、各チップを別々に選択して所望のチップに対してデータのリード／ライトを行なえるようにするために、一般にチップイネーブル信号／CEと呼ばれるチップ選択信号に関しては、チップごとにパッケージの別の端子を割り当てそれぞれのチップのCE信号用のパッドとパッケージのCE信号用端子とをボンディングワイヤ等で接続するように構成せざるを得なかった。

【0004】

【発明が解決しようとする課題】そのため、従来のマルチチップスタック型の半導体記憶装置は、チップイネーブル用端子が複数個あるため、これを1つの半導体装置として見たときにはワンチップの半導体装置に比べて端子数が増えることとなり、パッケージが大型化するとともに、従来の製品との互換性がなくなる。つまり、チップイネーブル用端子を複数個有しているため、メモリを使用したシステムを新たに開発したり、従来のメモリチップに代えてマルチチップスタック型の半導体記憶装置を用いたメモリモジュールを構成したりする場合に、すでにある基板の設計データの利用が図れず、新たに基板を設計し直す必要が生じる。

【0005】また、1つのパッケージに複数のCE信号用端子があるとアドレス信号をデコードして複数のチップイネーブル信号のいずれか1つを有効レベルにするためのデコード回路を外付け回路として設けたり、メモリをアクセスするためアドレスを出力する装置の側で複数のチップイネーブル信号を生成して出力できるように構成したりしなくてはならず、ユーザの設計負担が大きくなるといった不具合があった。

【0006】そこで、チップイネーブル端子を2つのチ

ップで共有し、アドレス信号（最上位ビット）によってチップを選択できるようにしたマルチチップスタック型の半導体記憶装置の発明が提案されている（特願平11-207701号）。ただし、この先願発明においては、マルチチップスタック型の半導体記憶装置の概略構成およびデバイス構造は開示されているものの、アドレス最上位ビットによっていずれかのチップを選択できるようにする具体的な回路構成や具体的なボンディングの仕方など製造技術については開示されていない。

【0007】この発明の目的は、チップイネーブル端子を複数のチップで共有し、アドレス信号によってチップを選択できるようにしたマルチチップスタック型の半導体記憶装置を提供することにある。

【0008】この発明の他の目的は、チップイネーブル端子を複数のチップで共有し、アドレス信号によってチップを選択できるようにしたマルチチップスタック型の半導体記憶装置の有効な製造方法を提供することにある。

【0009】この発明の前記ならびにそのほかの目的と新規な特徴については、本明細書の記述および添付図面から明らかになるであろう。

【0010】

【課題を解決するための手段】本願において開示される発明のうち代表的なものの概要を説明すれば、下記のとおりである。すなわち、本発明に係る半導体記憶装置は、第1の電源電圧端子または第2の電源電圧端子のいずれかに接続される第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、該チップ選択制御回路は、上記第1端子部の設定状態と、外部からのチップ選択信号が有効にされた状態で入力されたアドレス信号のいずれか1または2以上のビットとに基づいて内部のチップ選択信号を有効状態または無効状態に制御するように構成したものである。

【0011】上記した手段によれば、複数のメモリチップを組み合わせたときに、第1端子部の設定状態を変えることでアドレス信号に応じてチップが選択される記憶装置として動作したり、第1端子部の設定状態とアドレス端子部の設定状態の組合せにより入出力ビット数が異なる記憶装置としても動作したりすることができるようになる。

【0012】また、複数の半導体メモリチップが1つのパッケージに収納され各チップの端子部は対応するもの同士がパッケージの対応する端子に共通に接続されてなる半導体記憶装置において、上記複数の半導体メモリチップは、各々異なる状態に設定可能な第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備

え、上記複数の半導体メモリチップ内の各チップ選択制御回路は、上記第1端子部が各々異なる状態に設定されている場合には、外部からのチップ選択信号が有効にされた状態で入力されたアドレス信号が同一であってもいずれか1のチップにおいて内部のチップ選択信号を有効にするように構成した。

【0013】これにより、複数のメモリチップを組み合わせたときに、第1端子部の設定状態を変えることでアドレス信号に応じてチップが選択される記憶装置として動作することができ、チップが複数個あってもチップ選択信号は共通化することができ、装置にとっての外部端子数を減らすことができるとともに、通常の1個のメモリチップと同一に扱うことができるため、メモリを使用したシステムを開発する場合に既存の設計基板を利用することができるようになる。

【0014】さらに、複数の半導体メモリチップが1つのパッケージに収納され、データ入出力端子部以外の各チップの端子部は対応するもの同士がパッケージの対応する端子に共通に接続されてなる半導体記憶装置において、上記複数の半導体メモリチップは、各々異なる状態に設定可能な第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、上記複数の半導体メモリチップ内の各チップ選択制御回路は、上記第1端子部が同一の状態に設定されかつアドレス入力端子部のうち所定の1または2以上の端子部が同一の電位に固定されている場合には、外部からのチップ選択信号が有効にされたことに応じてそれぞれ内部のチップ選択信号を有効にするように構成した。

【0015】これにより、複数のメモリチップを組み合わせたときに、各チップの第1端子部の設定状態と所定のアドレス入力端子部の設定状態を一致させることにより、チップが複数個あっても入出力データのビット数が2倍である通常の1個のメモリチップと同一に扱うことができるため、メモリを使用したシステムを開発する場合に既存の設計基板を利用することができるようになる。

【0016】また、望ましくは、上記アドレス信号のいずれか1または2以上のビットは、アドレスの最上位ビットまたは最上位側から2ビット以上とする。これにより、従来のメモリチップのマトリクス構成やアドレスデコードの構成を変えずにアドレス信号に応じてチップが選択されるマルチチップスタック型の半導体記憶装置を実現することができる。

【0017】さらに、本願の他の発明は、複数の半導体メモリチップが1つのパッケージに収納され、該パッケージには各チップに対応してデータ入出力用の端子が設けられ各チップのデータ入出力端子部はパッケージの対応する上記端子に別々に接続されてなるとともに、パッ

ケージには所定の端子が設けられている半導体記憶装置において、上記複数の半導体メモリチップは、各々異なる状態に設定可能な第1端子部と、該第1端子部の状態と複数のアドレス入力端子部のうちいずれか1または2以上の端子部の状態に基づいて内部のチップ選択信号を制御するチップ選択制御回路とを備え、上記複数の半導体メモリチップ内の各チップ選択制御回路は、上記第1端子部が同一の状態に設定されかつアドレス入力端子部のうち所定の1または2以上の端子部が同一の電位に固定されている場合には、外部からのチップ選択信号が有効にされたことに応じてそれぞれ内部のチップ選択信号を有効にするように構成され、上記複数の半導体メモリチップのいずれかの上記第1端子部は上記パッケージの上記所定の端子に接続され、他の半導体メモリチップの上記第1端子部は上記パッケージのいずれかの電源電圧端子に接続されるようにした。これにより、ユーザは上記所定の端子に印加される電圧を適宜設定することによって、当該半導体記憶装置の入出力データのビット数を選択することができる。

【0018】また、望ましくは、上記複数の半導体メモリチップは、それぞれ同一方向に端子部が現われるように積層され、各チップ間はこれらのチップの上記端子部よりも内側の部位に介在された接着剤層によって互いに接合されるようにする。これにより、各チップの端子部とパッケージ側の対応する端子との電気的接続をボンディングワイヤで行なう場合に、それぞれのチップの端子部が同一方向に現われるように積層してもワイヤボンディング処理を行なうことができる。

【0019】本発明に係る半導体記憶装置の製造方法は、複数の半導体メモリチップが1つのパッケージに収納され各チップの端子部は対応するもの同士がパッケージの対応する端子に共通に接続されてなる半導体記憶装置の製造方法において、前工程終了後にウェハ状態で上記複数の半導体メモリチップのテストを行ない、該テストにより良品と判定された半導体メモリチップに対してウェハ状態でトリミング処理を行なった後、ウェハを切断して各チップに分割し、上記各半導体メモリチップの上記第1端子部に対する設定および各チップの端子部とパッケージの端子との接続を行なってからパッケージに封止するようにしたものである。これにより、状態を設定する必要があるメモリチップを使用しながらその端子設定を、各チップの端子部とパッケージの端子とを接続する工程で行なうことができるため、従来の半導体メモリのプロセスを全く変更することなくマルチチップスタック型の半導体記憶装置を製造することができる。上記トリミング処理とともに不良ビットの救済処理を行うようにしても良い。

【0020】また、上記第1端子部に対する設定は、該第1端子部をパッケージに設けられているいずれか一の電源電圧端子にボンディングワイヤで接続する処理とす

る。ワイヤボンディング技術はきわめて信頼性の高い技術であるので、これを用いて第1端子部に対する設定を行なうことにより、設定ミスを防止できるとともに、従来の半導体メモリのプロセスを全く変更することなくマルチチップスタック型の半導体記憶装置を製造することができる。

【0021】さらに、上記複数の半導体メモリチップはそれぞれ同一方向に端子部が現われるように積層され、上記ボンディングワイヤで接続する処理は各チップを接着剤で接合する処理の後に各チップ毎に行なわれるようにする。これにより、それぞれのチップの端子部が同一方向に現われるように積層しても既存の装置を用いてワイヤボンディング処理を行なうことができる。

【0022】

【発明の実施の形態】以下、本発明の好適な実施例を図面に基いて説明する。図1は、本発明を適用したマルチチップスタック型半導体記憶装置のパッケージ構造を、また図2は回路のブロック構成図を示す。

【0023】図1に示されているように、この実施例のマルチチップスタック型半導体記憶装置は、2個のメモリチップ10A、10Bが積層されてパッケージ100内に封入されている。この実施例でパッケージ100内に封入されるメモリチップ10A、10Bは、例えば電氣的にデータを書き込むことが可能でありかつデータを所定の単位で電氣的に一括消去可能なフラッシュメモリのような不揮発性メモリである。

【0024】特に制限されるものでないが、メモリチップ10Aと10Bは、これらのチップよりも小さな絶縁体からなるスペーサ20を挟んで互いに電氣的に絶縁された状態で積層されている。スペーサ20を設けているのは、2つのチップはサイズが同じであるため、直接重ねると下のチップのボンディングパッド部分が隠れてしまいワイヤボンディングができなくなるためである。メモリチップ10Aおよび10Bとスペーサ20とはペレット接着剤170により結合される。メモリチップ10A、10Bは、1つのメモリセルに1ビットのデータを記憶可能な2値メモリであっても良いし、1つのメモリセルに2ビット以上のデータを記憶可能ないわゆる多値メモリであっても良い。

【0025】110はセラミックなどからなる絶縁基板で、この絶縁基板110の一面(内面)にはチップの外延に沿って複数のボンディング用電極端子120が所定の間隔で配置され、また各ボンディング用電極端子120に対応して絶縁基板110には当該基板を貫通するように埋設された導電性プラグ130が設けられている。これらの導電性プラグ130は、一方の端部(図では上端)が上記いずれか一つのパッド120の下面に接触され、また他端すなわち基板110の他面(外面)に露出した部位(図では下端)には、半田ボール140がそれぞれ溶着されている。メモリチップ10Bは絶縁基板1

10上にペレット接着剤170により結合される。

【0026】また、上記基板側のボンディング用電極端子120にボンディングワイヤ150の一端が結合され、該ボンディングワイヤ150の他端はメモリチップ10A、10Bの対応するボンディングパッドに結合されて、基板側のボンディング用電極端子120とメモリチップ10A、10Bの対応するボンディングパッドとが電氣的に接続されている。そして、絶縁基板110上にメモリチップ10Aと10Bがボンディングワイヤ150で接続された状態で樹脂160によりモールドされてパッケージ100が構成されている。

【0027】図2に示されているように、各メモリチップ10A、10Bは、複数のメモリセルがマトリクス状に配置されたメモリアレイ11と、入力されたX系アドレスをデコードしてメモリアレイ11内の一本のワード線を選択レベルにするXデコーダ12と、入力されたY系アドレスをデコードしてメモリアレイ11内の対応するビット線を選択するYデコーダ13と、入力されたX系アドレスおよびY系アドレスを取り込んで上記Xデコーダ12およびYデコーダ13に供給するアドレスバッファ14と、選択されたビット線に読み出された信号を増幅したり書込みデータに応じてビット線の電位を与えるセンスアンプ&書込み回路15と、読出しデータを出力したり書込みデータをチップ外部から取り込む入出力回路16と、チップ外部から入力されるチップイネーブル信号/CEを取り込んで内部制御信号を生成するチップイネーブル制御回路17と、ライトイネーブル信号/WEやリセット信号RES、アウトイネーブル信号/OEなどの外部制御信号を取り込む入力バッファ18と、取り込まれた制御信号に応じてチップ内部の制御信号を生成して内部を制御するコントロール回路19などにより構成されている。

【0028】チップイネーブル信号/CEは当該チップが選択状態にされることを示す信号、ライトイネーブル信号/WEは当該チップが書込み状態にされることを示す信号、リセット信号RESはチップ内部をリセット状態にする信号、アウトイネーブル信号/OEは読出しデータ信号が出力される状態にあることを示す信号である。これらの制御信号およびアドレス信号が入力されるパッドのうち、以下に述べる2つのパッド(CAD, A22)以外のパッドに関しては、メモリチップ10A、10Bの対応するパッド同士が、それぞれボンディングワイヤでパッケージに設けられている対応する電極端子に共通に接続されている。また、各チップのデータ入力用パッドI/O0~I/O15もパッケージに設けられている対応する電極端子に共通に接続されている。

【0029】この実施例のメモリチップ10A、10Bには、通常のメモリには設けられていないチップアドレスデータ設定用パッドCADがそれぞれ設けられている。また、各メモリチップ10A、10Bには、チップ

外部から供給される例えば23ビットのようなアドレス信号A0～A22のうち最上位ビットA22と、上記チップアドレスデータ設定用パッドCADに印加される電位とに応じて、上記チップイネーブル制御回路17を有効にするアドレスビット比較回路20がそれぞれ設けられている。そして、メモリチップ10Aのチップアドレスデータ設定用パッドCADとメモリチップ10Bのチップアドレスデータ設定用パッドCADは、それぞれ一方がパッケージの第1の電源電圧端子Vccに、また他方がパッケージの第2の電源電圧端子Vssに接続される。

【0030】図3には、チップイネーブル制御回路17とアドレスビット比較回路20の具体的な構成例が示されている。チップイネーブル制御回路17は、チップイネーブル信号／CEとリセット信号RESを入力とするANDゲートG1と、該ANDゲートG1の出力を反転してアドレスビット比較回路20に供給するインバータG2と、ANDゲートG1の出力信号とアドレスビット比較回路20からの信号とに基づいて内部チップイネーブル信号CEBを生成するNANDゲートG3とからなる。

【0031】一方、アドレスビット比較回路20は、チップイネーブル制御回路17の上記インバータG2の出力信号とアドレス最上位ビットA22とを入力とするNORゲートG11と、上記チップアドレスデータ設定用パッドCADに印加される電位に応じた信号を生成するインバータG12と、該インバータG12の出力信号と上記NORゲートG11の出力信号を入力とするイクスクルーシブORゲートG13と、該イクスクルーシブORゲートG13の出力を反転してチップイネーブル制御回路17のNANDゲートG3に供給するインバータG24とからなる。

【0032】以下、図3の回路の動作を説明する。リセット信号RESはハイレベルが有効レベルの信号であり、リセット信号RESがロウレベルのときにチップイネーブル制御回路17はチップイネーブル信号／CEを有効にする。すなわち、リセット信号RESがロウレベルのときにANDゲートG1の出力がチップイネーブル信号／CEに応じて変化する。チップイネーブル信号／CEは、ロウレベルが有効レベルつまりチップ選択状態を示す信号であり、リセット信号RESとチップイネーブル信号／CEが共にロウレベルのときに、ANDゲートG1の出力がハイレベルとなる。

【0033】そして、ANDゲートG1の出力はインバータG2により反転されてアドレスビット比較回路20のNORゲートG11に供給されるため、リセット信号RESとチップイネーブル信号／CEが共にロウレベルのときに、NORゲートG11の他方の入力であるアドレス最上位ビットA22に応じてNORゲートG11が変化する。リセット信号RESまたはチップイネーブル

信号／CEのいずれか一方がハイレベルのときは、NORゲートG11の出力はロウレベルに固定される。また、リセット信号RESまたはチップイネーブル信号／CEのいずれか一方がハイレベルのときは、チップイネーブル制御回路17の出力段のNANDゲートG3の出力がハイレベルに固定され、内部チップイネーブル信号CEBはチップ非選択状態を示すこととなる。

【0034】リセット信号RESとチップイネーブル信号／CEが共にロウレベルのとき、アドレスビット比較回路20のNORゲートG11はアドレス最上位ビットA22を入力信号とするインバータとして動作し、アドレス最上位ビットA22がロウレベルのときにはNORゲートG11の出力がハイレベルに、またアドレス最上位ビットA22がハイレベルのときにはNORゲートG11の出力がロウレベルになる。そして、このNORゲートG11の出力信号を入力とするイクスクルーシブORゲートG13は排他的論理和回路であるので、チップアドレスデータ設定用パッドCADに印加される電位に応じた信号を出力するバッファG12の出力論理レベルとNORゲートG11の出力論理レベルが同じである時はイクスクルーシブORゲートG13の出力はロウレベルに、またG11とG12の出力論理レベルが異なる時はイクスクルーシブORゲートG13の出力はハイレベルとなる。

【0035】そして、イクスクルーシブORゲートG13の出力はインバータG14により反転してチップイネーブル制御回路17の出力段のNANDゲートG3に供給されるため、G11とG12の出力論理レベルが異なる時はNANDゲートG3の出力である内部チップイネーブル信号CEBはチップ非選択状態を示すハイレベルにされる。一方、G11とG12の出力論理レベルが同じである時は、NANDゲートG3がインバータとして動作し、その出力である内部チップイネーブル信号CEBは、NANDゲートG3の他方の入力信号（このときハイレベル）に応じてチップ選択状態を示すロウレベルにされる。

【0036】以上のように、図3の回路はアドレス最上位ビットA22とチップアドレスデータ設定用パッドCADに印加される電位に応じて内部チップイネーブル信号CEBをチップ選択状態にしたり、非選択状態にしたりする。次の表1に、図3の実施例回路を搭載したメモリチップを2個使用した半導体記憶装置におけるアドレス最上位ビットA22とチップアドレスデータ設定用パッドCADに印加される電位との組合せと、チップの動作状態との関係を整理して示す。

【0037】

【表1】

	CAD	A22	内部状態	メモリ構成
(1)	Vss	H	非選択	×16
		L	選択	
(2)	Vcc	H	選択	
		L	非選択	×32
(3)	Vcc	Vcc	同時選択	

【0038】表1に示されているように、チップアドレスデータ設定用パッドCADに印加される電位がVssであるチップ(1)は、アドレス最上位ビットA22に応じて、A22がハイレベルのときは／CEのいかんにかかわらず内部チップイネーブル信号CEBがハイレベルとなってチップは非選択状態となり、A22がロウレベルのときは／CEがロウレベルとされると内部チップイネーブル信号CEBがロウレベルとなってチップは選択状態となる。

【0039】一方、チップアドレスデータ設定用パッドCADに印加される電位がVccであるチップ(2)は、アドレス最上位ビットA22に応じて、A22がハイレベルのときは／CEがロウレベルとされると内部チップイネーブル信号CEBがロウレベルとなってチップは選択状態となり、A22がロウレベルのときは／CEのいかんにかかわらず内部チップイネーブル信号CEBがハイレベルとなってチップは非選択状態となる。

【0040】さらに、表1の(3)のように、アドレス最上位ビットA22とチップアドレスデータ設定用パッドCADに印加される電位が共にVcc（または共にVss）のときは、チップは外部からのチップイネーブル信号／CEに応じて内部チップイネーブル信号CEBが同じように変化される。そのため、2チップともチップアドレスデータ設定用パッドCADに印加される電位と、アドレス最上位ビットA22が入力されるべき端子をVccに固定することにより、2つのチップを同時に選択状態にさせることができる。

【0041】従って、1つのパッケージに封入される2つのチップのデータ入出力パッドに対応してそれぞれ別のデータ入出力用電極端子をパッケージに設けるとともに、チップアドレスデータ設定用パッドCADとアドレス最上位ビットA22が入力されるべきパッドをVccに固定することにより、2倍のデータ幅を有するメモリ(×2n)として動作させることができる。例えば、2つのチップがそれぞれ64Mビットの記憶容量を有し16ビットのデータを並列入出力するものであれば、128Mビットの記憶容量を有し32ビットのデータを並列入出力する半導体記憶装置として構成することができる。

【0042】ただし、この場合、チップのアドレスA22入力用パッドはパッケージ内でVcc端子に接続されるので、パッケージにはアドレス最上位ビットA22に対応した電極端子は不要である。図2に示すような構成を有するチップは、マルチチップとしてではなく1つの

チップを1つのパッケージに収納した半導体記憶装置として構成した場合には、アドレス最上位ビットA22とチップアドレスデータ設定用パッドCADに印加される電位を共にパッケージ内部でVccに固定することにより、64Mビットの記憶容量を有し16ビットのデータを並列入出力する通常の半導体記憶装置として構成することもできる。

【0043】一方、表1(1)、(2)の場合は、2つのチップがそれぞれ64Mビットの記憶容量を有し16ビットのデータを並列入出力するものであれば、128Mビットの記憶容量を有し16ビットのデータを並列入出力する半導体記憶装置として構成され、各チップはアドレス最上位ビットA22によって選択的にアクセスされる。

【0044】なお、図3の実施例においては、チップイネーブル制御回路17からアドレスビット比較回路20のNORゲートG11にチップイネーブル信号／CEに応じた制御信号を、インバータG2により送ってアドレスA22の取り込みを制御しているが、この信号は必ずしも必要なものではなく、省略することも可能である。この制御信号によりNORゲートG11を制御することで、チップ非選択状態においてNORゲートG11の入力であるアドレスA22が変化することで内部回路に貫通電流が流れて消費電量が増加するのを防止する効果がある。

【0045】従って、アドレスA22以外のビットA0～A21をチップ内部に取り込むバッファに関しても、同様にインバータG2によりチップ非選択状態では動作しないように制御するのが望ましい。また、図3の実施例においては、チップイネーブル制御回路17にリセット信号RESに応じてチップイネーブル信号／CEの入力を制御するANDゲートG1が設けられているが、このANDゲートG1も省略することが可能である。

【0046】図4には、チップイネーブル制御回路17とアドレスビット比較回路20の他の構成例が示されている。図4は、4つのチップを1組として半導体記憶装置を構成する場合に適した回路構成例である。この実施例においては、2つのチップアドレスデータ設定用パッドCAD1、CAD2と、アドレス信号A23の入力端子が設けられている。そして、これに対応して、アドレスビット比較回路20には、論理ゲートG11～G13と同様な論理ゲートG21～G23が設けられているとともに、インバータG14の代わりにANDゲートG24が設けられている。そして、NORゲートG21にはアドレス信号A23とチップイネーブル制御回路17のインバータG2の出力が入力され、イクスクルーシブORゲートG23にはNORゲートG21の出力とチップアドレスデータ設定用パッドCAD2に印加される電位に応じた信号を生成するインバータG22の出力が入力されている。

【0047】さらに、イクスクルーシブORゲートG13とG23の出力がANDゲートG24に入力されている。チップイネーブル制御回路17は、図3の実施例と同じである。この実施例においては、4つのチップにおいて、それぞれチップアドレスデータ設定用パッドCAD1、CAD2に印加される電圧Vcc、Vssの組合せが異なるように設定（ボンディングワイヤの接続）がなされることにより、アドレス信号A22、A23に応じていずれか1つのチップの内部イネーブル信号CEBがロウレベルにされて、4つのうち1つのチップのみが選択状態とされる。ただし、図3の実施例と同様に、すべてのチップのチップアドレスデータ設定用パッドCA

D1、CAD2とアドレス信号A22、A23の入力パッドに印加される電圧をVccとすることにより、4倍のデータ幅を有するメモリとして動作させることができる。

【0048】表2に、図4の実施例回路を搭載したメモリチップを4個使用した半導体記憶装置におけるアドレス最上位ビットA22、A23とチップアドレスデータ設定用パッドCAD1、CAD2に印加される電位との組合せと、チップの動作状態との関係を整理して示す。

【0049】

【表2】

CAD1	CAD2	A22	A23	内部状態	メモリ構成
Vss	Vss	H	H	非選択	×16
		L	H	非選択	
		H	L	非選択	
		L	L	選択	
Vcc	Vss	H	H	非選択	
		L	H	非選択	
		H	L	選択	
		L	L	非選択	
Vss	Vcc	H	H	非選択	
		L	H	選択	
		H	L	非選択	
		L	L	非選択	
Vcc	Vcc	H	H	選択	
		L	H	非選択	
		H	L	非選択	
		L	L	非選択	
Vcc	Vcc	Vcc	Vcc	同時選択	×64

【0050】図5には、図3に示されているアドレスビット比較回路20の他の構成例が示されている。図3においては、アドレス信号の最上位ビットA22とチップアドレスデータ設定用パッドCADの状態とを比較するイクスクルーシブORゲートG13を用いているが、図5においては、イクスクルーシブORゲートG13の代わりにセクタSELを設け、このセクタSELをチップアドレスデータ設定用パッドCADに印加される電位に応じた信号を生成するバッファG12の出力で制御するように構成されている。

【0051】また、図5の実施例では、図3のNORゲートG11の代わりにORゲートG11'を用いている。ORゲートG11'の入力はNORゲートG11の入力と同じく、アドレスA22とチップイネーブル制御回路17からの信号であり、ORゲートG11'はチップイネーブル制御回路17からの信号に応じてアドレスA22を通過させたり遮断したりする伝送ゲートとして作用する。

【0052】上記セクタSELのデータ入力端子には、ORゲートG11'の出力と、該ORゲートG11'の出力をインバータG16で反転した信号が入力されている。このようにして、パッドCADの電位に応じてORゲートG11'を通過したアドレス最上位ビット

A22またはその反転信号をセクタSELで選択してチップイネーブル制御回路17に供給することにより、同一のアドレスA22が入力された場合でも、パッドCADの電位に応じて一方は内部チップイネーブル信号CEBが有効レベルとされ、他方は内部チップイネーブル信号CEBが無効レベルとされる。その結果、アドレスA22によってチップを選択することができるようになる。

【0053】なお、図5の実施例においては、パッドCADに印加される電位に応じた信号を生成するバッファG12の出力でセクタSELを制御してアドレスA22またはその反転信号のいずれか一方を選択するようにしているが、逆にアドレスA22すなわちORゲートG11'の出力でセクタSELを制御してバッファG12の出力またはその反転信号のいずれか一方を選択して出力するように構成しても良い。

【0054】また、本実施例のアドレスビット比較回路20は、4つのチップを1組として半導体記憶装置を構成する場合に適した図4に示されている実施例に対しても適用することができる。具体的には、図4のイクスクルーシブORゲートG13、G23の代わりにセクタを用いてアドレスビット比較回路20を構成すれば良い。

【0055】図6は、本発明の他の実施例を示す。この実施例は、図1および図2に示されている2つのフラッシュメモリチップを積層した半導体記憶装置の変形例であって、フラッシュメモリチップ10A、10Bの構成は、図2と同一である。図2と異なるのは、パッケージにビット構成切換え端子BCが設けられメモリチップ10Aと10Bのうち一方のチップ（例えば10A）のチップアドレス設定用端子CADがこのビット構成切換え端子BCに接続されている点と、パッケージのデータ入出端子が16本（I/O0～I/O15）でなく32本（I/O0～I/O31）とされ一方のチップ（例えば10A）のデータ入出力パッドはパッケージのデータ入出端子I/O0～I/O15に接続され、他方のチップ（例えば10B）のデータ入出力パッドはパッケージのデータ入出端子I/O16～I/O31に接続されている点である。

【0056】この実施例の半導体記憶装置は、ユーザが16ビット並列入出力のメモリとしても32ビット並列入出力のメモリとしても利用できる点に特徴がある。具体的には、メモリチップ10Aのチップアドレス設定用端子CADが接続されているビット構成切換え端子BCを電源電圧V_{ss}に接続すると、図2の実施例の半導体記憶装置と同様にアドレス最上位ビットA22によってチップが選択され16ビットのデータを並列に入出力するメモリとして動作する。従って、この場合には、パッケージのデータ入出端子I/O0～I/O15とI/O16～I/O31は、同一の16ビットのバスに共通に接続される。

【0057】一方、メモリチップ10Aのチップアドレス設定用端子CADが接続されているビット構成切換え端子BCを電源電圧V_{cc}に接続すると、2つのチップはチップイネーブル信号CEによって同時に選択状態とされ、32ビットのデータを並列に入出力するメモリとして動作する。従って、この場合には、パッケージのデータ入出端子I/O0～I/O15とI/O16～I/O31は、32ビットのバスの別の信号線に接続される。

【0058】図7および図8は、本発明のさらに他の実施例を示す。この実施例は、図1および図2に示されている2つのフラッシュメモリチップを積層した半導体記憶装置の発展形態であって、積層された2つのフラッシュメモリチップ10A、10Bの上にさらにスタティックRAMチップ10Cを積層したものである。フラッシュメモリチップ10A、10Bの構成は、図2と同一である。この実施例では、スタティックRAMチップ10Cのサイズがフラッシュメモリチップ10A、10Bのサイズよりも小さくフラッシュメモリチップ10Aの上にスタティックRAMチップ10Cを重ねてもフラッシュメモリチップ10Aのボンディングパッドが隠れないため、フラッシュメモリチップ10Aの保護膜の上にベ

レット接着剤170によりスタティックRAMチップ10Cが接合されている。

【0059】これらのチップ10A、10B、10Cに設けられているパッドのうちライトイネーブル信号/WEが入力されるパッドおよびアドレス信号A0～A18の入力パッドは、図7に示すように、ボンディングワイヤ150を介してパッケージ100に設けられている対応する電極端子120に共通に接続されている。

【0060】また、図8に示すように、各チップのデータ入力用パッドI/O0～I/O15もパッケージに設けられている対応する端子に共通に接続されている。チップイネーブル信号/CEに関しては、フラッシュメモリチップ10A、10Bのパッドは前記実施例と同様に、パッケージに設けられている対応する端子PCE-Fに共通に接続されているが、SRAMチップ10Cのチップイネーブル信号/CEが入力されるパッドは、パッケージに設けられている専用の端子PCE-Sに接続されている。

【0061】スタティックRAMチップ10Cは、メモリアレイ11、Xデコーダ12、Yデコーダ13、アドレスバッファ14、センスアンプ&書込み回路15、データ入出力回路16、ライトイネーブル信号/WEなど制御信号の入力バッファ18、コントロール回路19などを備えた汎用のSRAMと同じ構成にされており、前記実施例のフラッシュメモリチップ10A、10Bに設けられているようなチップアドレスデータ設定用パッドCADやチップイネーブル制御回路17、アドレスビット比較回路20は設けられていない。

【0062】本実施例の半導体記憶装置は1つのパッケージ内に2つのフラッシュメモリと1つのSRAMが積層収納されているため、別々のパッケージに収納されているものを使用する場合に比べてシステムの実装密度をさらに高めることができ、装置の小型化を図ることができる。

【0063】次に、上記実施例のマルチチップスタック型の半導体記憶装置の開発から製造までの手順の一例を、図9を用いて説明する。まず、前記実施例で説明したチップアドレスデータ設定用パッドCADやチップイネーブル制御回路17、アドレスビット比較回路20を設けたメモリチップの論理設計とシミュレーションによる論理動作の確認を行なう（ステップS200）。次に、上記論理設計データに基づいてレイアウトツールと呼ばれる設計支援プログラムを用いてレイアウト設計を行なう（ステップS201）。それから、このレイアウト設計データに基づいてプロセスに使用するマスクを作成する（ステップS202）。

【0064】次に、上記マスクを用いて半導体ウェハ上に複数のメモリチップを形成する前工程を行なう（ステップS203）。そして、テストによりウェハ状態で各チップのパッドにプローブを当てて試験するウェハテス

トを行なう(ステップS204)。このウェハテストで不良と判定されたチップはマーキングが付されて各チップに切断されてから不良品として廃棄される。一方、ステップS204で良品と判定されたチップは、冗長回路による欠陥ビットの救済情報や内部電源回路の電圧値のトリミング情報をチップ上に設けられているヒューズや不揮発性記憶素子に設定する処理が行なわれる(ステップS205)。

【0065】ウェハ上のすべてのチップについてプローブ検査および救済等の処理が終了すると、ウェハは各チップに切断される(ステップS206)。それから、チップを絶縁基板上に搭載して、チップのパッドと対応する基板側の電極端子とを接続するワイヤボンディングと樹脂でモールドするパッケージング処理を行なう(ステップS207)。このボンディング&パッケージング処理では、先ず下側のメモリチップ10Bを絶縁基板110上に接着剤ペレット170で接合してワイヤボンディングを行なった後、上側のメモリチップ10Aを接着剤ペレット170で下側のメモリチップ10B上に接合してこのチップ10Aに対してワイヤボンディングを行なうようにするのが良い。

【0066】そして、この実施例においては、上記ワイヤボンディングの際に、各チップのチップアドレスデータ設定用パッドCADをパッケージに設けられているいずれか一方の電源電圧端子に接続する処理を行なう。これにより、チップアドレスデータ設定用パッドCADに対する設定処理を何ら新たな工程を追加することなく行なうことができる。また、上側のメモリチップ10Aを接着剤ペレット170で下側のメモリチップ10B上に接合してから2つのチップ10A、10Bに対してまとめてワイヤボンディングを行なおうとすると、上のチップが邪魔になって下のチップに対するワイヤボンディングが困難になるが、下のチップ10Bに対するワイヤボンディングを行なった後、上側のメモリチップ10Aを接合してワイヤボンディングを行なうことにより、既存装置でワイヤボンディング処理を行なうことができる。

【0067】なお、チップアドレスデータ設定用パッドCADをいずれかの電源電圧端子に接続する方法としては、ワイヤボンディングの他に、予めチップアドレスデータ設定用パッドCADを電源電圧VccまたはVssに接続するヒューズを設けておいて、パッケージング処理の前にヒューズを切断する方法が考えられるが、ヒューズの場合にはチップアドレスデータ設定用パッドCADを電源電圧Vccに接続したチップとVssに接続したチップの2つが生じるので、各チップを個々に管理する必要があるが、前記実施例のようにパッケージング処理のワイヤボンディングで設定するようにすれば、チップを別々に管理する必要がないので、その分コストの低減が可能となる。

【0068】上記パッケージング処理の後、各デバイ

スをテストでテストする出荷前テストが実行される(ステップS208)。このテストで不良と判定されたチップはマーキングが付されて後の選別工程で不良品として廃棄される一方、ステップS208で良品と判定されたデバイスは製品として出荷される。

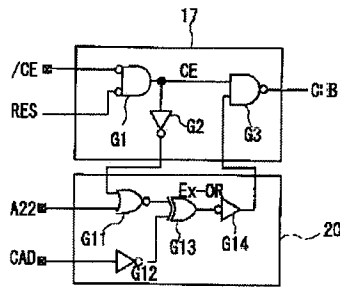
【0069】図10には、第1の実施例(図1および図2参照)の半導体記憶装置の応用例としてのメモリカードの構成を示す。図10において、CDは絶縁材料で形成されたカード本体、FLM0~FLM3は該カード本体に内蔵された不揮発性記憶装置、CNTはデータバスDB、アドレスバスABおよび制御バスCBを介してこれらの不揮発性記憶装置FLM0~FLM3に接続されたコントローラである。不揮発性記憶装置FLM0~FLM3は各々図2に示すような構成を有する2つのチップを内蔵し、または1のチップのみを内蔵していても3以上のチップを内蔵していても良い。コントローラCNTはシリアル入出力端子SIOを介して外部のマイクロプロセッサなどの制御装置と接続されて、制御装置からの指令に応じて、カード内部の不揮発性記憶装置FLM0~FLM3に対するデータの書込みや読出しを行なう。

【0070】コントローラCNTは各不揮発性記憶装置FLM0~FLM3に対してそれぞれ専用の選択信号/CE0、/CE1、/CE2、/CE3を生成していずれか1つを選択状態にする。各不揮発性記憶装置FLM0~FLM3は、図2の実施例で説明したように、一方メモリチップ(10A)のチップアドレスデータ設定用パッドCADはパッケージの第1の電源電圧Vssに接続され、他方のメモリチップ(10B)のチップアドレスデータ設定用パッドCADはパッケージの第2の電源電圧端子Vccに接続されており、アドレス最上位ビット(例えばA22)に応じていずれか一方のチップが選択状態にされる。

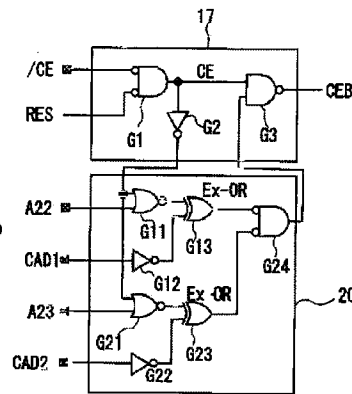
【0071】この応用例からも分かるように、前記実施例の半導体記憶装置を使用すると、従来の半導体メモリと全く同等に扱えしかも記憶容量が2倍のメモリカードを構成することができる。なお、この応用例のメモリカードにおいては、図1に示されている絶縁基板110を、各記憶装置FLM0~FLM3とコントローラCNTに共通の基板として構成して、全体を樹脂等でモールドして1つのパッケージに封入しても良い。つまり、図1に示されているような構造を有する半導体記憶装置として製造されたものを使用してメモリカードを構成するとパッケージが2重構造となるが、各記憶装置FLM0~FLM3とコントローラCNTを共通の基板上に搭載すれば、1つのパッケージに封入させることができる。

【0072】以上本発明者によってなされた発明を実施例に基づき具体的に説明したが、本発明は上記実施例に限定されるものではなく、その要旨を逸脱しない範囲で種々変更可能であることはいうまでもない。例えば図1

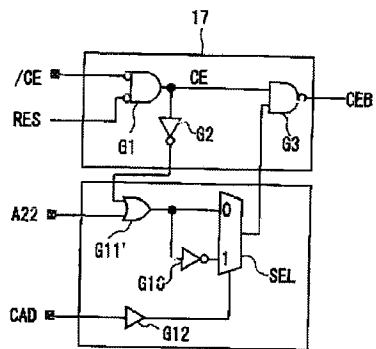
【図3】



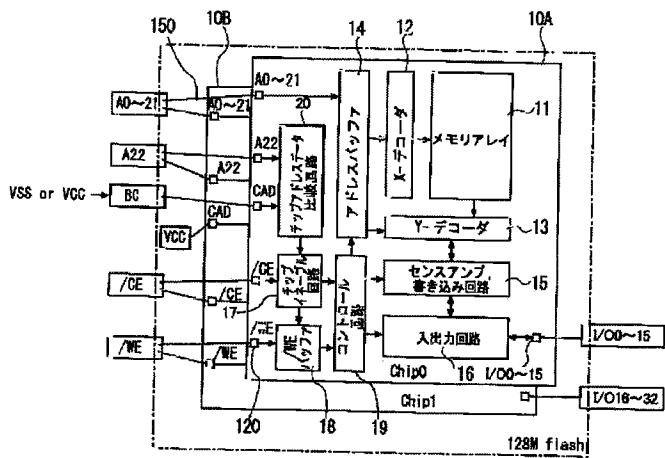
【図4】



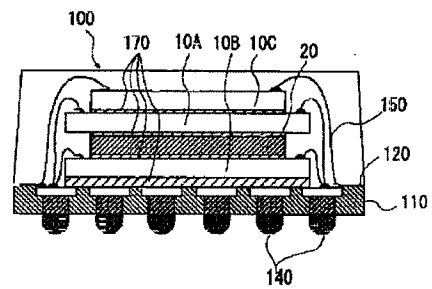
【図5】



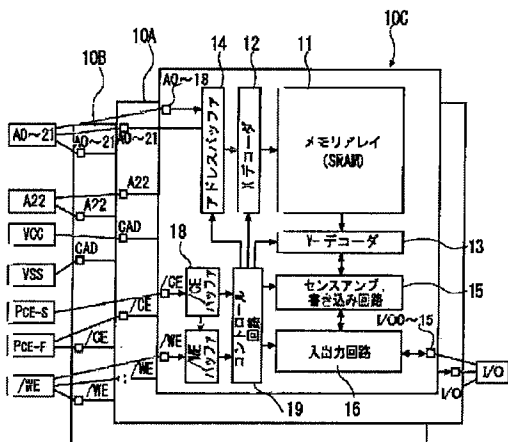
【図6】



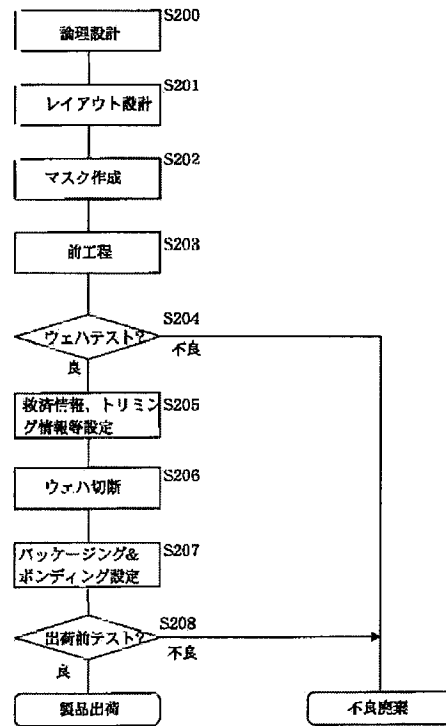
【図7】



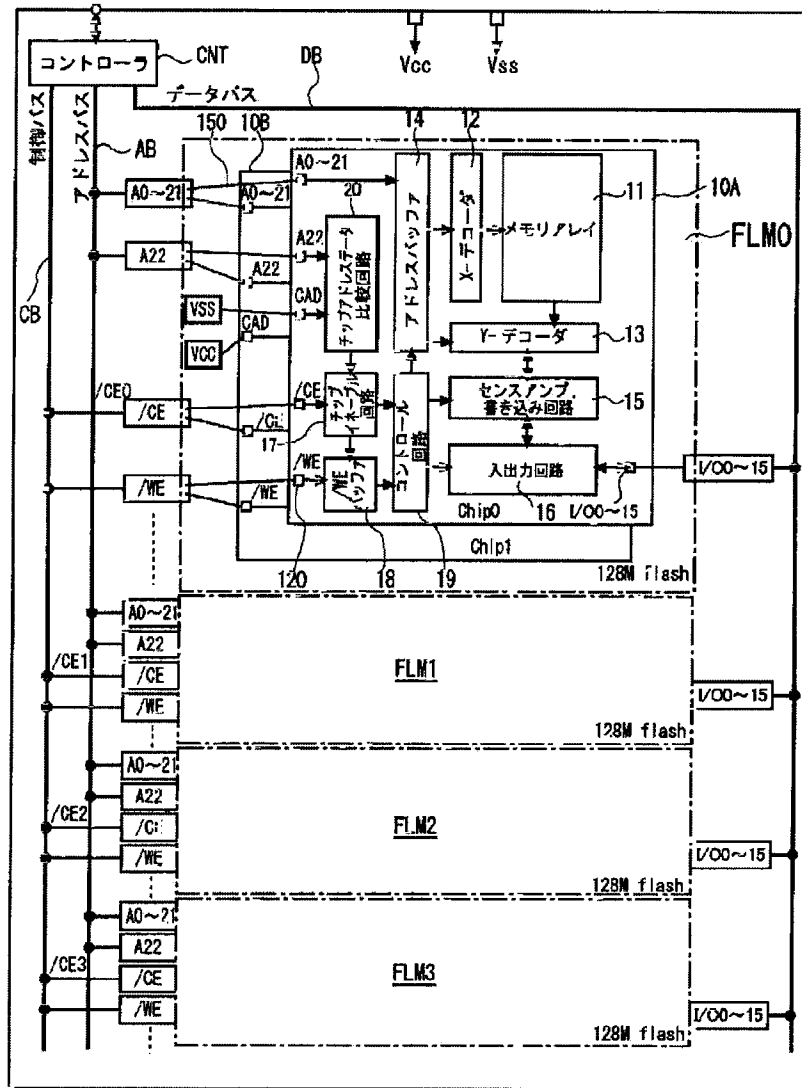
【図8】



【図9】



【図10】



フロントページの続き

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CC13 CC21 CC31 GG00 GG06

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1.This document has been translated by computer. So the translation may not reflect the original precisely.

2.**** shows the word which can not be translated.

3.In the drawings, any words are not translated.

CLAIMS

[Claim(s)]

[Claim 1]The 1st terminal area connected to either a first source voltage terminal or a second source voltage terminal, It has a chip-select control circuit which controls Uchibe's chip selection signal based on a state of two or more terminal areas among [1 / any] a state of this 1st terminal area, and two or more address input terminal areas, This chip-select control circuit An established state of the 1st terminal area of the above, A semiconductor memory device constituting so that a chip selection signal from the outside may control Uchibe's chip selection signal to an effective state or an invalid state based on any 1 or two or more bits of an address signal input in the state where it validated.

[Claim 2]Have the following and each chip-select control circuit in two or more above-mentioned semiconductor memory chips, When the 1st terminal area of the above is set as the state of differing respectively, A semiconductor memory device constituting so that Uchibe's chip selection signal may be validated in any 1 chip, even if an address signal input where a chip selection signal from the outside is validated is the same.

The 1st terminal area that two or more semiconductor memory chips are stored by one package, and a terminal area of each chip is a semiconductor memory device to which it comes to connect corresponding things common to a terminal in which a package corresponds, and can be set as the state where two or more above-mentioned semiconductor memory chips differ respectively. A chip-select control circuit which controls Uchibe's chip selection signal based on a state of two or more terminal areas among [1 / any] a state of this 1st terminal area, and two or more address input terminal areas.

[Claim 3]Have the following and each chip-select control circuit in two or more above-mentioned semiconductor memory chips, When the 1st terminal area of the above is set as the same state and 1 predetermined or two or more terminal areas are fixed to the same electric potential among address input terminal areas, It embraces that a chip selection signal from the outside was validated, and is each the semiconductor memory device constituting so that an internal chip selection signal may be validated.

What two or more semiconductor memory chips are stored by one package, and correspond in a terminal area of each chip other than a data input/output terminal part is the semiconductor memory devices which it comes to connect common to a terminal whose package corresponds,

The 1st terminal area that can be set as the state where two or more above-mentioned semiconductor memory chips differ respectively.

A chip-select control circuit which controls an internal chip selection signal based on a state of two or more terminal areas among [1 / any] a state of this 1st terminal area, and two or more address input terminal areas.

[Claim 4]The semiconductor memory device according to claim 2 or 3, wherein any 1 or two or more bits of a most significant bit [of an address] or top side are 2 bits or more. [of the above-mentioned address signal]

[Claim 5]Have the following and each chip-select control circuit in two or more above-mentioned semiconductor memory chips, When the 1st terminal area of the above is set as the same state and 1 predetermined or two or more terminal areas are fixed to the same electric potential among address input terminal areas, It is constituted so that an internal chip selection signal may be validated according to a chip selection signal from the outside having been validated, respectively, A semiconductor memory device, wherein one 1st terminal area of the above of two or more above-mentioned semiconductor memory chips is connected to the above-mentioned predetermined terminal of the above-mentioned package and the 1st terminal area of the above of other semiconductor memory chips is connected to one source voltage terminal of the above-mentioned packages.

While two or more semiconductor memory chips are stored by one package, and a terminal for data input/outputs is provided by this package corresponding to each chip and coming independently to connect a data input/output terminal part of each chip to the above-mentioned terminal in which a package corresponds, The 1st terminal area that is the semiconductor memory device with which a predetermined terminal is provided by package, and can be set as the state where two or more above-mentioned semiconductor memory chips differ respectively. A chip-select control circuit which controls an internal chip selection signal based on a state of two or more terminal areas among [1 / any] a state of this 1st terminal area, and two or more address input terminal areas.

[Claim 6]The semiconductor memory device according to claim 5 laminating two or more above-mentioned semiconductor memory chips so that a terminal area may appear in a uniform direction, respectively, and being mutually joined between each chip by adhesion agent layer by which it was interposed between parts inside the above-mentioned terminal area of these chips.

[Claim 7]Two or more semiconductor memory chips are stored by one package, and, as for a terminal area of each chip, corresponding things are connected common to a terminal in which a package corresponds, It is a manufacturing method of a semiconductor memory device which has the 1st terminal area that can be set as the state of differing respectively for two or more above-mentioned memory chips of every, Two or more above-mentioned semiconductor memory chips are tested by a wafer state after an end of a previous process, After performing trimming treatment by a wafer state to a semiconductor memory chip judged by this test to be a good article, A manufacturing method of a semiconductor memory device sealing in a package after cutting a wafer, dividing into each chip and making setting out to the 1st terminal area of the above of each above-mentioned semiconductor memory chip, and connection between a terminal area of each chip, and a terminal of a package.

[Claim 8]A manufacturing method of the semiconductor memory device according to claim 7, wherein setting out to the 1st terminal area of the above is processing which connects this 1st

terminal area to any 1 source voltage terminal currently provided by package by a bonding wire.
[Claim 9]A manufacturing method of the semiconductor memory device according to claim 8, wherein processing which two or more above-mentioned semiconductor memory chips are laminated, respectively so that a terminal area may appear in a uniform direction, and is connected by the above-mentioned bonding wire is performed for every chip after processing which joins each chip with adhesives.

[Claim 10]A manufacturing method of the semiconductor memory device according to claim 7 performing relief processing which replaces a defective memory cell detected by the above-mentioned test with trimming treatment in the above-mentioned wafer state.

DETAILED DESCRIPTION

[Detailed Description of the Invention]

[0001]

[Field of the Invention]A semiconductor memory device and also two or more semiconductor memory chips apply the present invention to the multichip stack type semiconductor memory device stored by one package, and concerning effective technology, it is used for large scale-ization of a nonvolatile semiconductor memory like a flash memory, and relates to effective technology.

[0002]

[Description of the Prior Art]Conventionally, in the field of a semiconductor integrated circuit, there is technology which carries out lamination storage of two or more semiconductor chips into one package, and aimed at improvement in packaging density, and what attained large scale-ization by this technology also in semiconductor memory is being put in practical use. In such a multichip stack type semiconductor memory device, a terminal number can be reduced by sharing the terminal of a package among two or more chips about most input signals, such as an address signal and a control signal. The terminal number of a package can be reduced by sharing between chips also about the I/O terminal which outputs and inputs data.

[0003]However, when an I/O terminal is made to share. In order to choose each chip independently and to be able to perform read/write of data to a desired chip, about the chip selection signal generally called a chip enable signal/CE, It could not but constitute so that another terminal of a package might be assigned for every chip and the pad for CE signals of each chip and the terminal for CE signals of a package might be connected by a bonding wire etc.

[0004]

[Problem to be solved by the invention]Therefore, a conventional multichip stack type semiconductor memory device, While a terminal number will increase compared with the semiconductor device of a one chip and a package is enlarged when this is seen as one semiconductor device since there are two or more terminals for chip enable, compatibility with the conventional product is lost. That is, since it has two or more terminals for chip enable, newly develop the system which uses a memory, or. When it replaces with the conventional memory chip and constitutes the memory module using a multichip stack type semiconductor memory device, use of the design data of a certain substrate cannot already be aimed at, but it will be necessary to newly redesign a substrate.

[0005]Provide the decoder circuit for decoding an address signal and making any one of two or more of the chip enable signals into a valid level, if one package has two or more terminals for CE signals as an external circuit, or, In order to access a memory, it had to constitute so that two or more chip enable signals could be generated and outputted by the equipment side which outputs an address, and there was fault that a user's design burden became largely.

[0006]Then, two chips share a chip enable terminal and invention of the multichip stack type semiconductor memory device which enabled it to choose a chip by an address signal (most significant bit) is proposed (Tokuganheil-207701). However, in this prior invention, although multichip stack type the schematic structure and device structure of the semiconductor memory device are disclosed, It does not disclose about production technology, such as a specific circuit configuration which enables it to choose one of chips by the address most significant bit, and the method of specific bonding.

[0007]The purpose of this invention shares a chip enable terminal between two or more chips, and there is in providing the multichip stack type semiconductor memory device which enabled it to choose a chip by an address signal.

[0008]Other purposes of this invention share a chip enable terminal between two or more chips, and there are in providing the effective manufacturing method of the multichip stack type semiconductor memory device which enabled it to choose a chip by an address signal.

[0009>About the above of this invention and the other purposes, and the new characteristics, it will become clear from description and the accompanying drawing of this Description.

[0010]

[Means for solving problem]It will be as follows if the outline of a typical thing is described among invention disclosed in an application concerned. Namely, the 1st terminal area by which the semiconductor memory device concerning the present invention is connected to either a first source voltage terminal or a second source voltage terminal, It has a chip-select control circuit which controls an internal chip selection signal based on the state of two or more terminal areas among [1 / any] the state of this 1st terminal area, and two or more address input terminal areas, This chip-select control circuit is constituted so that an internal chip selection signal may be controlled to an effective state or an invalid state based on any 1 or two or more bits of the address signal input where the chip selection signal from the outside is validated, the established state of the 1st terminal area of the above, and.

[0011]When two or more memory chips are combined according to the above-mentioned means, It can operate as memory storage with which a chip is chosen according to an address signal by changing the established state of the 1st terminal area, or can operate now also as memory storage from which the input-and-output number of bits differs with the combination of the established state of the 1st terminal area, and the established state of an address terminal part.

[0012]In the semiconductor memory device to which it comes to connect those to which two or more semiconductor memory chips are stored by one package, and the terminal area of each chip corresponds common to the terminal in which a package corresponds, The 1st terminal area that can be set as the state where two or more above-mentioned semiconductor memory chips differ respectively, It has a chip-select control circuit which controls an internal chip selection signal based on the state of two or more terminal areas among [1 / any] the state of this 1st terminal area, and two or more address input terminal areas, Each chip-select control circuit in two or more above-mentioned semiconductor memory chips, When the 1st terminal area of the above was set as the state of differing respectively, even if the address signal input where the chip selection signal from the outside is validated was the same, it constituted so that an internal chip

selection signal might be validated in any 1 chip.

[0013]By this, when two or more memory chips are combined, it can operate as memory storage with which a chip is chosen according to an address signal by changing the established state of the 1st terminal area, Even if there are two or more chips, while being able to commonalize a chip selection signal and being able to reduce the number of external terminals for equipment, Since it can treat identically to one usual memory chip, when developing the system which uses a memory, the existing design board can be used.

[0014]In the semiconductor memory device to which two or more semiconductor memory chips are stored by one package, and, as for the terminal area of each chip other than a data input/output terminal part, it comes to connect corresponding things common to the terminal in which a package corresponds, The 1st terminal area that can be set as the state where two or more above-mentioned semiconductor memory chips differ respectively, It has a chip-select control circuit which controls an internal chip selection signal based on the state of two or more terminal areas among [1 / any] the state of this 1st terminal area, and two or more address input terminal areas, Each chip-select control circuit in two or more above-mentioned semiconductor memory chips, When the 1st terminal area of the above was set as the same state and 1 predetermined or two or more terminal areas were fixed to the same electric potential among address input terminal areas, it constituted so that an internal chip selection signal might be validated according to the chip selection signal from the outside having been validated, respectively.

[0015]By this, when two or more memory chips are combined, by making the established state of the 1st terminal area of each chip, and the established state of a predetermined address input terminal area correspond, Since the number of bits of an I/O data can treat identically to double the one usual memory chip even if there are two or more chips, when developing the system which uses a memory, the existing design board can be used.

[0016]Any 1 or two or more bits of the above-mentioned address signal shall be 2 bits or more from the most significant bit [of an address], or top side desirably. The multichip stack type semiconductor memory device with which a chip is chosen by this according to an address signal without changing the mat composition of the conventional memory chip and the composition of an address decoder is realizable.

[0017]As for invention of everything but an application concerned, two or more semiconductor memory chips are stored by one package, While the terminal for data input/outputs is provided by this package corresponding to each chip and coming independently to connect the data input/output terminal part of each chip to the above-mentioned terminal in which a package corresponds, In a package, a predetermined terminal in the semiconductor memory device currently provided two or more above-mentioned semiconductor memory chips, It has a chip-select control circuit which controls an internal chip selection signal based on the state of two or more terminal areas among [1 / any] the 1st terminal area that can be set as the state of differing respectively, and the state of this 1st terminal area and two or more address input terminal areas, Each chip-select control circuit in two or more above-mentioned semiconductor memory chips, When the 1st terminal area of the above is set as the same state and 1 predetermined or two or more terminal areas are fixed to the same electric potential among address input terminal areas, embracing that the chip selection signal from the outside was validated -- each -- it being constituted so that an internal chip selection signal may be validated, and it being connected to the above-mentioned predetermined terminal of the above-mentioned package, and one 1st terminal area of the above of two or more above-mentioned semiconductor memory chips, The

1st terminal area of the above of other semiconductor memory chips was connected to one source voltage terminal of the above-mentioned packages. Thereby, the user can choose the number of bits of the I/O data of the semiconductor memory device concerned by setting up suitably the voltage applied to the above-mentioned predetermined terminal.

[0018]Desirably, two or more above-mentioned semiconductor memory chips are laminated so that a terminal area may appear in a uniform direction, respectively, and it is mutually joined between each chip by the adhesion agent layer by which it was interposed between the parts inside the above-mentioned terminal area of these chips. When this performs the electrical link of the terminal area of each chip, and the corresponding terminal by the side of a package by a bonding wire, wirebonding processing can be performed, even if it laminates so that the terminal area of each chip may appear in a uniform direction.

[0019]In the manufacturing method of the semiconductor memory device to which it comes to connect those to which, as for the manufacturing method of the semiconductor memory device concerning the present invention, two or more semiconductor memory chips are stored by one package, and the terminal area of each chip corresponds common to the terminal in which a package corresponds, Two or more above-mentioned semiconductor memory chips are tested by a wafer state after the end of a previous process, After performing trimming treatment by a wafer state to the semiconductor memory chip judged by this test to be a good article, A wafer is cut and it divides into each chip, and after making setting out to the 1st terminal area of the above of each above-mentioned semiconductor memory chip, and connection between the terminal area of each chip, and the terminal of a package, it is made to seal in a package. Since the terminal setting out can be performed at the process of connecting the terminal area of each chip, and the terminal of a package while this uses the memory chip which needs to set up a state, A multichip stack type semiconductor memory device can be manufactured without completely changing the process of the conventional semiconductor memory. It may be made to perform relief processing of a defective bit with the above-mentioned trimming treatment.

[0020]Setting out to the 1st terminal area of the above is considered as the processing which connects this 1st terminal area to any 1 source voltage terminal currently provided by the package by a bonding wire. Since wirebonding technology is very reliable technology, while being able to prevent a setting error by performing setting out to the 1st terminal area using this, A multichip stack type semiconductor memory device can be manufactured without completely changing the process of the conventional semiconductor memory.

[0021]Two or more above-mentioned semiconductor memory chips are laminated, respectively so that a terminal area may appear in a uniform direction, and the processing connected by the above-mentioned bonding wire is made to be performed after the processing which joins each chip with adhesives for every chip. Even if this laminates so that the terminal area of each chip may appear in a uniform direction, wirebonding processing can be performed using existing equipment.

[0022]

[Mode for carrying out the invention]Hereafter, the preferable working example of the present invention is described based on Drawings. Fig.2 shows the Brock configuration diagram of a circuit for the package structure of the multichip stack type semiconductor memory device with which Fig.1 applied the present invention again.

[0023]The two memory chips 10A and 10B are laminated, and the multichip stack type semiconductor memory device of this working example is enclosed in the package 100 as shown in Fig.1. The memory chips 10A and 10B enclosed in the package 100 in this working example

can write in data electrically, for example, and are nonvolatile memory like the flash memory in which collective erasure is possible electrically in a predetermined unit about data.

[0024]Although not restricted in particular, the memory chips 10A and 10B are laminated in the state where it was mutually insulated from these chips electrically on both sides of the spacer 20 which consists of a small insulator. The spacer 20 is provided because the bonding pad part of Shimo's chip hides and wirebonding becomes impossible, when it piles up directly, since two chips have the same size. The memory chips 10A and 10B and the spacer 20 are combined by the pellet adhesives 170. The memory chips 10A and 10B may be the binary memories which can memorize 1-bit data to one memory cell, and may be what is called multi level memories that can memorize data of 2 bits or more to one memory cell.

[0025]110 is an insulating substrate which consists of ceramics etc., and two or more electrode terminals 120 for bonding are arranged at the predetermined intervals along with the extent of a chip at the whole surface (inner surface) of this insulating substrate 110, Corresponding to each electrode terminal 120 for bonding, the conductive plug 130 embedded so that the substrate concerned might be penetrated is provided by the insulating substrate 110. The solder ball 140 is welded on the part (a figure lower end) to which one end (a figure upper end) was contacted on the lower surface of any one above-mentioned pad 120, and the other end 110, i.e., a substrate, on the other hand (outer surface), exposed these conductive plugs 130, respectively. The memory chip 10B is combined by the pellet adhesives 170 on the insulating substrate 110.

[0026]The end of the bonding wire 150 is combined with the electrode terminal 120 for bonding by the side of the above-mentioned substrate, The other end of this bonding wire 150 is combined with the bonding pad in which the memory chips 10A and 10B correspond, and the electrode terminal 120 for bonding by the side of a substrate and the bonding pad in which the memory chips 10A and 10B correspond are electrically connected. And where the memory chips 10A and 10B are connected by the bonding wire 150 on the insulating substrate 110, a mold is carried out with the resin 160, and the package 100 is constituted.

[0027]As shown in Fig.2 each memory chip 10A and 10B, The memory array 11 by which two or more memory cells are arranged at matrix form, Input X decoder 12 which decodes an address X system and makes a selection level one word line in the memory array 11, The Y decoder 13 which chooses the input bit line which decodes an address Y system, and with which it corresponds in the memory array 11, input X system -- an address and Y system -- with the address buffer 14 which incorporates an address and is supplied to above-mentioned X decoder 12 and the Y decoder 13, The sense amplifier & writing circuit 15 which gives the electric potential of a bit line corresponding to write data in amplifying the signal read to the selected bit line ****, The input output circuit 16 which outputs read-out data or incorporates write data from chip outside, The chip enable control circuit 17 which incorporates the chip enable signal/CE input from the chip outside, and generates an internal control signal, and the input buffer 18 which incorporates external control signals, such as a write enable signal/WE, reset-signal RES, an out enable signal / OE, It is constituted by the control circuit 19 etc. which generate the control signal inside a chip according to the incorporated control signal, and control an inside.

[0028]The signal with which a chip enable signal/CE shows that the chip concerned is made into a selective state, They are a signal which shows that the chip concerned writes in a write enable signal/WE, and it changes it into a state, a signal with which reset-signal RES makes the inside of a chip a reset state, and a signal which shows that an out enable signal / OE is in the state where a read-out data signal is outputted. About pads other than two pads (CAD, A22) described

below among the pads into which these control signals and address signals are input, The pads to which the memory chips 10A and 10B correspond are connected common to the corresponding electrode terminal currently provided by the package by the bonding wire, respectively. Pad I/O0 for data input of each chip - I/O15 are connected common to the corresponding electrode terminal currently provided by the package.

[0029]Pad CAD for chip address information setting out which is not provided is provided by the usual memory at the memory chips 10A and 10B of this working example, respectively. For example, each memory chip 10A and 10B is supplied from the chip outside, among the address signals A0-A22 like 23 bits The most significant bit A22, According to the electric potential applied to the above-mentioned pad CAD for chip address information setting out, the address bit comparison circuit 20 which validates the above-mentioned chip enable control circuit 17 is provided, respectively. and pad CAD for chip address information setting out of the memory chip 10A and pad CAD for chip address information setting out of the memory chip 10B -- respectively -- on the other hand, the first source voltage terminal Vcc of a package -- another side is connected to the second source voltage terminal Vss of a package.

[0030]The specific configuration example of the chip enable control circuit 17 and the address bit comparison circuit 20 is shown in Fig.3. AND gate G1 to which the chip enable control circuit 17 considers a chip enable signal/CE, and reset-signal RES as an input, It consists of NAND gate G3 which generates the internal chip enable signal CEB based on the inverter G2 which reverses the output of this AND gate G1, and is supplied to the address bit comparison circuit 20, and the output signal of AND gate G1 and the signal from the address bit comparison circuit 20.

[0031]NOR gate G11 to which the address bit comparison circuit 20, on the other hand, considers the output signal and the address most significant bit A22 of the above-mentioned inverter G2 of the chip enable control circuit 17 as an input, The inverter G12 which generates the signal according to the electric potential applied to the above-mentioned pad CAD for chip address information setting out, It consists of IKUSUKURUSHIBU OR gate G13 which considers the output signal of this inverter G12, and the output signal of above-mentioned NOR gate G11 as an input, and the inverter G24 which reverses the output of this IKUSUKURUSHIBU OR gate G13, and is supplied to NAND gate G3 of the chip enable control circuit 17.

[0032]Hereafter, operation of the circuit of Fig.3 is described. The high level of reset-signal RES is a signal of a valid level, and when reset-signal RES is a low level, the chip enable control circuit 17 validates a chip enable signal/CE. That is, when reset-signal RES is a low level, the output of AND gate G1 changes according to a chip enable signal/CE. A low level is a signal which shows a valid level, i.e., a chip-select state, and when both reset-signal RES, and a chip enable signal/CE are low levels, it becomes high-level outputting a chip enable signal/CE of AND gate G1.

[0033]And since it is reversed with the inverter G2 and the output of AND gate G1 is supplied to NOR gate G11 of the address bit comparison circuit 20, When both reset-signal RES, and a chip enable signal/CE are low levels, NOR gate G11 changes according to the address most significant bit A22 which is an input of another side of NOR gate G11. When either one of reset-signal RES, or a chip enable signal/CE is high-level, the output of NOR gate G11 is fixed to a low level. When either one of reset-signal RES, or a chip enable signal/CE is high-level, the output of NAND gate G3 of the output stage of the chip enable control circuit 17 will be fixed high-level, and, as for the internal chip enable signal CEB, a chip non selection state will be

shown.

[0034]When both reset-signal RES, and a chip enable signal/CE are low levels, NOR gate G11 of the address bit comparison circuit 20 operates as an inverter which makes an input signal the address most significant bit A22, When the address most significant bit A22 is a low level, in the output of NOR gate G11, when the address most significant bit A22 is high-level, the output of NOR gate G11 becomes a low level high-level. And since IKUSUKURUSHIBU OR gate G13 which considers the output signal of this NOR gate G11 as an input is an exclusive "or" circuit, When the output logical level of the buffer G12 and the output logical level of NOR gate G11 which output the signal according to the electric potential applied to pad CAD for chip address information setting out are the same, the output of IKUSUKURUSHIBU OR gate G13 to a low level, When the output logical levels of G11 and G12 differ, the output of IKUSUKURUSHIBU OR gate G13 becomes high-level.

[0035]And since it is reversed with the inverter G14 and the output of IKUSUKURUSHIBU OR gate G13 is supplied to NAND gate G3 of the output stage of the chip enable control circuit 17, When the output logical levels of G11 and G12 differ, the internal chip enable signal CEB which is an output of NAND gate G3 is made into the high level which shows a chip non selection state. On the other hand, when the output logical level of G11 and G12 is the same, NAND gate G3 operates as an inverter and the internal chip enable signal CEB which is that output is made into the low level which shows a chip-select state according to the input signal (at this time high level) of another side of NAND gate G3.

[0036]As mentioned above, the circuit of Fig.3 changes the internal chip enable signal CEB into a chip-select state according to the electric potential applied to the address most significant bit A22 and pad CAD for chip address information setting out, or changes it into a non selection state. The relation between the combination of the address most significant bit A22 in the semiconductor memory device which uses two MEMORIPPU which carries the working-example circuit of Fig.3 in the next table 1, and the electric potential applied to pad CAD for chip address information setting out, and the operating state of a chip is arranged and shown.

[0037]

[Table 1]

[0038]The electric potential applied to pad CAD for chip address information setting out the chip (1) which is Vss as shown in Table 1, According to the address most significant bit A22, when A22 is high-level, the internal chip enable signal CEB becomes high-level regardless of /CE, and a chip will be in a non selection state, If /CE is made into a low level when A22 is a low level, the internal chip enable signal CEB will serve as a low level, and a chip will be in a selective state.

[0039]On the other hand, the electric potential applied to pad CAD for chip address information setting out the chip (2) which is Vcc, According to the address most significant bit A22, if /CE is made into a low level when A22 is high-level, the internal chip enable signal CEB will serve as a low level, and a chip will be in a selective state, When A22 is a low level, regardless of /CE, the internal chip enable signal CEB becomes high-level, and a chip will be in a non selection state.

[0040]As shown in (3) of Table 1, when both the electric potential applied to the address most

significant bit A22 and pad CAD for chip address information setting out is Vcc(s) (or both Vss), In a chip, according to the chip enable signal/CE from the outside, the internal chip enable signal CEB changes in a similar manner. Therefore, two chips can be simultaneously made into a selective state by fixing to Vcc the electric potential by which both chips are applied to pad CAD for chip address information setting out, and the terminal into which the address most significant bit A22 should be input.

[0041]Therefore, while providing the respectively different electrode terminal for data input/outputs in a package corresponding to the data input/output pad of two chips enclosed with one package, It can be made to operate as a memory (x2n) which has twice as many data width as this by fixing to Vcc the pad into which pad CAD for chip address information setting out and the address most significant bit A22 should be input. For example, if two chips have a storage capacity which is 64M bit, respectively and carry out the parallel input/output of the 16-bit data, it can constitute as a semiconductor memory device which has a 128M bit storage capacity and carries out the parallel input/output of the 32-bit data.

[0042]However, since the pad for address A22 input of a chip is connected to a Vcc terminal within a package in this case, in a package, the electrode terminal corresponding to the address most significant bit A22 is unnecessary. When constituted as a semiconductor memory device which stored one chip in one package as multichip, the chip which has composition as shown in Fig.2, By fixing to Vcc both the electric potential applied to the address most significant bit A22 and pad CAD for chip address information setting out inside a package, it can also constitute as a usual semiconductor memory device which has a 64M bit storage capacity and carries out the parallel input/output of the 16-bit data.

[0043]On the other hand, in the case of Table 1 (1) and (2), if two chips have a storage capacity which is 64M bit, respectively and carry out the parallel input/output of the 16-bit data, It is constituted as a semiconductor memory device which has a 128M bit storage capacity and carries out the parallel input/output of the 16-bit data, and each chip is selectively accessed by the address most significant bit A22.

[0044]Although the control signal according to a chip enable signal/CE is sent to NOR gate G11 of the address bit comparison circuit 20 with the inverter G2 from the chip enable control circuit 17 and incorporation of the address A22 is controlled in the working example of Fig.3, This signal is not necessarily required and omitting is also possible. It is effective in preventing consumption electrical quantity from penetration current flowing into an internal circuit because the address A22 which is an input of NOR gate G11 changes in a chip non selection state, and increasing by controlling NOR gate G11 by this control signal.

[0045]Therefore, it is desirable to control also about the buffer which incorporates the bits A0-A21 other than address A22 into the inside of a chip not to operate in the state of chip non selection with the inverter G2 similarly. In the working example of Fig.3, although AND gate G1 which controls the input of a chip enable signal/CE according to reset-signal RES to the chip enable control circuit 17 is provided, it is possible to also omit this AND gate G1.

[0046]Other configuration examples of the chip enable control circuit 17 and the address bit comparison circuit 20 are shown in Fig.4. Fig.4 is the example of a circuit configuration for which it was suitable when four chips were made into 1 set and a semiconductor memory device was constituted. In this working example, two pad CAD1 for chip address information setting out, CAD2, and the input terminal of the address signal A23 are provided. And while the logic gates G11-G13 and the same logic gates G21-G23 are provided by the address bit comparison circuit 20 corresponding to this, AND gate G24 is provided instead of the inverter G14. And the

output of the inverter G2 of the address signal A23 and the chip enable control circuit 17 is input into NOR gate G21, The output of NOR gate G21 and the output of the inverter G22 which generates the signal according to the electric potential applied to pad CAD2 for chip address information setting out are input into IKUSUKURUSHIBU OR gate G23.

[0047]The output of IKUSUKURUSHIBU OR gates G13 and G23 is input into AND gate G24. The chip enable control circuit 17 is the same as the working example of Fig.3. By making setting out (connection of a bonding wire) so that the combination of the voltage VccVss applied to pad CAD1 for chip address information setting out and CAD2, respectively may differ in four chips in this working example, It has responded to the address signal A22 and A23, the internal enable signal CEB of a gap or one chip is made into a low level, and only one chip is made into a selective state among four. However, it can be made to operate as a memory which has one 4 times the data width of this by setting to Vcc pad CAD1 for chip address information setting out of all the chips, CAD2, the address signal A22, and voltage applied to the input pad of A23 like the working example of Fig.3.

[0048]The relation between the combination of the address most significant bit A22 in the semiconductor memory device which uses four MEMORIPPU which carries the working-example circuit of Fig.4 in Table 2, A23, and the electric potential applied to pad CAD1 for chip address information setting out and CAD2, and the operating state of a chip is arranged and shown.

[0049]

[Table 2]

[0050]Other configuration examples of the address bit comparison circuit 20 shown in Fig.3 are shown in Fig.5. Although IKUSUKURUSHIBU OR gate G13 which compares the most significant bit A22 of an address signal with the state of pad CAD for chip address information setting out is used in Fig.3, In Fig.5, the selector SEL is provided instead of IKUSUKURUSHIBU OR gate G13, and it is constituted so that it may control by the output of the buffer G12 which generates the signal according to the electric potential applied to pad CAD for chip address information setting out in this selector SEL.

[0051]In the working example of Fig.5, OR gate G11' is used instead of NOR gate G11 of Fig.3. The input of OR gate G11' is a signal from the address A22 and the chip enable control circuit 17 as well as the input of NOR gate G11, OR gate G11' acts as a transmission gate passed

the address A22 or intercepted according to the signal from the chip enable control circuit 17.

[0052]The signal which reversed the output of OR gate G11' and the output of this OR gate G11' with the inverter G16 is input into the data input terminal of the above-mentioned selector SEL. Thus, by choosing the address most significant bit A22 which passed OR gate G11' according to the electric potential of pad CAD, or its inversion signal by the selector SEL, and supplying the chip enable control circuit 17, Even when the same address A22 is input, as for one side, the internal chip enable signal CEB is made into a valid level according to the electric potential of pad CAD, and, as for another side, let the internal chip enable signal CEB be an invalid level. As a result, a chip can be chosen now with the address A22.

[0053]Although the selector SEL is controlled by the output of the buffer G12 which generates the signal according to the electric potential applied to pad CAD in the working example of [Fig.5](#) and he is trying to choose either one of the address A22 or its inversion signal, Conversely, the selector SEL may be controlled by the output of the address A22, i.e., OR gate G11', and it may constitute so that the output of the buffer G12 or either of the inversion signal may be chosen and outputted.

[0054]The address bit comparison circuit 20 of this example is applicable also to the working example shown in the [Fig.4](#) which was suitable when four chips were made into 1 set and a semiconductor memory device was constituted. What is necessary is specifically, to use a selector instead of IKUSUKURUSHIBU OR gate [of [Fig.4](#)] G13, and G23, and just to constitute the address bit comparison circuit 20.

[0055][Fig.6](#) shows other working examples of the present invention. This working example is a modification of the semiconductor memory device which laminated two flash memory chips shown in [Fig.1](#) and [Fig.2](#), and the composition of the flash memory chips 10A and 10B is the same as that of [Fig.2](#).The point that, as for differing from [Fig.2](#), bit configuration change terminal BC is provided by the package, and terminal CAD for chip address selections of one chip (for example, 10A) is connected to this bit configuration change terminal BC among the memory chips 10A and 10B, While is carried out to 32 (I/O0-I/O31) instead of 16 (I/O0-I/O15), and, as for the data input/output pad of a chip (for example, 10A), the data ON appearance terminal of a package is connected to data ON appearance terminal I/O0 of a package - I/O15, The data input/output pad of the chip (for example, 10B) of another side is a point connected to data ON appearance terminal I/O16 of a package - I/O31.

[0056]The semiconductor memory device of this working example has the characteristics in the point which a user can use also as a memory of a 32-bit parallel input/output also as a memory of a 16-bit parallel input/output. If bit configuration change terminal BC to which terminal CAD for chip address selections of the memory chip 10A is connected is specifically connected to the power supply voltage Vss, It operates as a memory which a chip is chosen by the address most significant bit A22 like the semiconductor memory device of the working example of [Fig.2](#), and outputs and inputs in parallel the data which is 16 bits. Therefore, data ON appearance terminal I/O0 of a package - I/O15, I/O16 - I/O31 are connected common to the 16-bit same Buss in this case.

[0057]On the other hand, if bit configuration change terminal BC to which terminal CAD for chip address selections of the memory chip 10A is connected is connected to the power supply voltage Vcc, Two chips are simultaneously made a selective state by chip enable signal CE, and it operates as a memory which outputs and inputs 32-bit data in parallel. Therefore, data ON appearance terminal I/O0 of a package - I/O15, I/O16 - I/O31 are connected to Buss's 32-bit another signal wire in this case.

[0058]Fig.7 and Fig.8 show the working example of further others of the present invention. This working example is a development form of the semiconductor memory device which laminated two flash memory chips shown in Fig.1 and Fig.2, and laminates the static RAM chip 10C further on the two laminated flash memory chips 10A and 10B. The composition of the flash memory chips 10A and 10B is the same as that of Fig.2. In this working example, Even if the size of the static RAM chip 10C piles up the static RAM chip 10C on the flash memory chip 10A smaller than the size of the flash memory chips 10A and 10B, in order that the bonding pad of the flash memory chip 10A may not hide, The static RAM chip 10C is joined by the pellet adhesives 170 on the protective film of the flash memory chip 10A.

[0059]The pad and the input pad of the address signals A0-A18 into which a write enable signal/WE is input among the pads currently provided by these chips 10A, 10B, and 10C, As shown in Fig.7, it is connected to the corresponding electrode terminal 120 currently provided by the package 100 via the bonding wire 150 in common.

[0060]As shown in Fig.8, pad I/O0 for data input of each chip - I/O15 are connected common to the corresponding terminal currently provided by the package. Although the pad of the flash memory chips 10A and 10B is connected common to corresponding terminal PCE-F currently provided by the package like the aforementioned working example about the chip enable signal/CE, The pad into which the chip enable signal/CE of the SRAM chip 10C are input is connected to terminal PCE-S for exclusive use currently provided by the package.

[0061]The static RAM chip 10C, The input buffer 18 of control signals, such as the memory array 11, X decoder 12, the Y decoder 13, the address buffer 14, the sense amplifier & writing circuit 15, the data output circuit 16, a write enable signal/WE, the control circuit 19, etc. It has the same composition as general-purpose SRAM which it had, and pad CAD for chip address information setting out, the chip enable control circuit 17, and the address bit comparison circuit 20 which are provided by the flash memory chips 10A and 10B of the aforementioned working example are not provided.

[0062]Since lamination storage of two flash memories and the one SRAM is carried out into one package, the semiconductor memory device of this example can raise the packaging density of a system further compared with the case where what is stored by the separate package is used, and can attain the miniaturization of equipment.

[0063]Next, an example of the procedure from development of the multichip stack type semiconductor memory device of the above-mentioned working example to manufacture is described using Fig.9. First, the logic operation by the logic design and simulation of a memory chip which provided pad CAD for chip address information setting out, the chip enable control circuit 17, and the address bit comparison circuit 20 which were described in the aforementioned working example is checked (Step S200). Next, a layout design is performed using the design support program called a layout tool based on the above-mentioned logic design data (Step S201). And the mask used for a process based on this layout design data is created (Step S202).

[0064]Next, the previous process which forms two or more memory chips on a semiconductor wafer using the above-mentioned mask is performed (Step S203). And the wafer tests which put a probe in a wafer state to the pad of each chip with a circuit tester, and are examined are done (Step S204). The chip judged that is poor at these wafer tests is discarded as a defective article, after marking is attached and being cut by each chip. On the other hand, processing which sets the relieving information of a defective bit and the trimming information of the pressure value of an internal electrical power source circuit according [the chip judged at Step S204 to be a good article] to a redundant circuit as the fuse currently provided on the chip or a nonvolatile memory

element is performed (Step S205).

[0065]A wafer is cut by each chip after processing of probe inspection, relief, etc. is completed about all the chips on a wafer (Step S206). And a chip is carried in an insulating substrate and packaging processing which carries out a mold to wirebonding which connects the pad of a chip and the electrode terminal by the side of a corresponding substrate by resin is performed (Step S207). In this bonding & packaging processing, After joining the lower memory chip 10B with the adhesives pellet 170 on the insulating substrate 110 first and performing wirebonding, It is good to join the upper memory chip 10A on the lower memory chip 10B with the adhesives pellet 170, and to be made to perform wirebonding to this chip 10A.

[0066]And in this working example, processing which connects pad CAD for chip address information setting out of each chip to one which is provided by the package of source voltage terminals in the case of the above-mentioned wirebonding is performed. Thereby, setting processing to pad CAD for chip address information setting out can be performed, without adding a new process in any way. If it collects to the two chips 10A and 10B and tries to perform wirebonding after joining the upper memory chip 10A on the lower memory chip 10B with the adhesives pellet 170, Although the upper chip becomes obstructive and wirebonding to a lower chip becomes difficult, After performing wirebonding to the lower chip 10B, an existing apparatus can perform wirebonding processing by joining the upper memory chip 10A and performing wirebonding.

[0067]As a method of connecting pad CAD for chip address information setting out to one of source voltage terminals, Although the fuse which connects pad CAD for chip address information setting out other than wirebonding to the power supply voltage Vcc or Vss previously is provided and how to cut a fuse before packaging processing can be considered, Since two, the chip connected to the power supply voltage Vcc and the chip connected to Vss, produce pad CAD for chip address information setting out in the case of a fuse, it is necessary to manage each chip separately but, and. Since it is not necessary to manage a chip independently if it is made to set up by wirebonding of packaging processing like the aforementioned working example, reduction of the part cost is attained.

[0068]After the above-mentioned packaging processing, the test before shipment which tests each device with a circuit tester is performed (Step S208). While marking is attached and the chip judged that is poor by this test is discarded as a defective article by a next sorting process, the device judged at Step S208 to be a good article is shipped as a product.

[0069]The composition of the memory card as an application of the semiconductor memory device of a first working example (refer to Fig.1 and Fig.2) is shown in Fig.10. A card body in which CD was formed by the insulating material in Fig.10, a nonvolatile storage with which FLM0-FLM3 were built in this card body, CNT is the controller connected to these nonvolatile storages FLM0-FLM3 via the data bus DB, the address bus AB, and the control bus CB. The nonvolatile storages FLM0-FLM3 may contain two chips which have composition as respectively shown in Fig.2, may contain only the chip of 1, or may contain three or more chips. The controller CNT is connected with control devices, such as an external microprocessor, via serial input/output terminal SIO, and the writing and read-out of data to the nonvolatile storages FLM0-FLM3 inside a card are performed according to the instructions from a control device.

[0070]The controller CNT is generating a selection signal /CE0 for exclusive use, /CE1, /CE2, and /CE3 to each nonvolatile storage FLM0-FLM3, respectively, and makes a gap or one a selective state. As described in the working example of Fig.2, each nonvolatile storage FLM0-FLM3. On the other hand, pad CAD for chip address information setting out of a memory chip

(10A) is connected to the first power supply voltage V_{ss} of a package, Pad CAD for chip address information setting out of the memory chip (10B) of the other is connected to the second source voltage terminal V_{cc} of a package, it has responded to the address most significant bit (for example, A22), and a gap or one chip is made into a selective state.

[0071] If the semiconductor memory device of the aforementioned working example is used so that this application may also show, it can treat on a par with the conventional semiconductor memory completely, and the memory card it is moreover twice whose storage capacity of this can be constituted. In the memory card of this application, the insulating substrate 110 shown in Fig.1 may be constituted as a substrate common to each memory storage FLM0-FLM3 and the controller CNT, the mold of the whole may be carried out by resin etc., and it may enclose with one package. That is, if a memory card is constituted using what was manufactured as a semiconductor memory device which has structure as shown in Fig.1, a package will serve as double structure, but. If each memory storage FLM0-FLM3 and the controller CNT are carried on a common substrate, it can be made to enclose with one package.

[0072] Although invention made by the inventor above was specifically described based on the working example, it cannot be overemphasized that it can change variously in the range which the present invention is not limited to the above-mentioned working example, and does not deviate from the summary. For example, in the working example of Fig.1, although two or more memory chips described about what is enclosed with the package by the laminating condition, also when two or more memory chips are enclosed with the package in the state where it arranged horizontally on the insulating substrate of one sheet, it can apply. The address bit for selection of a chip may not be limited to the address most significant bit, and may be, other bits, for example, least significant bit etc., etc.

[0073] Although described in the above description about the multichip stack type semiconductor memory device which makes a subject the flash memory which is the field of the invention which became the background about invention mainly made by the inventor, EEPROMs other than a flash memory can be used for the semiconductor memory device etc. which were built in.

[0074]

[Effect of the Invention] It will be as follows if the effect acquired by the typical thing among invention disclosed in an application concerned is described simply. Namely, while two or more chips share a chip enable terminal, being able to choose a chip now by an address signal and being able to reduce the number of external terminals as a semiconductor memory device by this, Since it can treat on a par with one usual semiconductor memory, when developing the system which uses a memory, the multichip stack type semiconductor memory device which can use the existing design board can be realized. It can manufacture without changing the manufacturing process of the conventional semiconductor memory for such a semiconductor memory device, or converting a manufacturing installation.

[Translation done.]



Espacenet

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MEMORY CONTROL DEVICE

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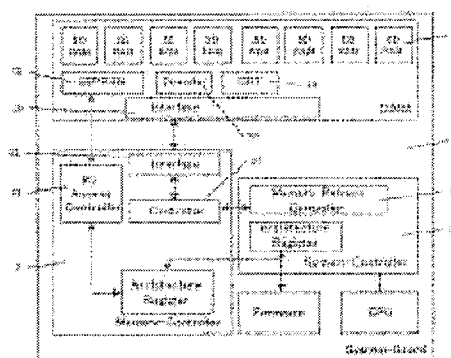
Application number: JP20060223567 20060818

Priority number (s): JP20060223567 20060818

Also published as: US2008046631 (A1)

Abstract of JP2008046989 (A)

PROBLEM TO BE SOLVED: To provide a technique for increasing the number of ranks of a memory module by a small change in configuration. ; **SOLUTION:** A memory control device which accesses the memory module having a plurality of ranks includes: an interface part having a select signal line for a select signal for rank selection and an address signal line for an address signal indicating an address on a selected rank; and a control part which transmits the signal for rank selection through a part of an address signal line and the select signal line. ; **COPYRIGHT:** (C) 2008,JPO&INPIT



システム全体図

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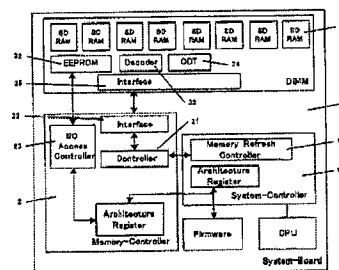
(54) 【発明の名称】 メモリ制御装置

(57) 【要約】

【課題】少ない構成変更でメモリモジュールのランク数を増加させる技術を提供する。

【解決手段】複数のランクを有するメモリモジュールにアクセスするメモリ制御装置であって、ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有するインターフェイス部と、前記アドレス信号線の一部及びセレクト信号線を介して前記ランクを選択する信号を送信する制御部とを備えた。

【選択図】図3



システム全体図

【特許請求の範囲】**【請求項1】**

複数のランクを有するメモリモジュールにアクセスするメモリ制御装置であって、
ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有するインターフェイス部と、
前記アドレス信号線の一部及びセレクト信号線を介して前記ランクを選択する信号を送信する制御部と、
を備えたメモリ制御装置。

【請求項2】

前記制御部が、前記セレクト信号線数よりも多い数のランクを選択する信号を送信する請求項1に記載のメモリ制御装置。

【請求項3】

前記インターフェイス部が、DDR2インターフェイスである請求項1又は2に記載のメモリ制御装置。

【請求項4】

前記制御部からの命令に基づいて、前記ランク数に応じたメモリリフレッシュ処理を行うメモリリフレッシュコントロール回路を備えた請求項1から3の何れかに記載のメモリ制御装置。

【請求項5】

前記制御部は、異ランク間に対するアクセスが連続した場合に、該アクセス間に所定の待機時間を挿入する請求項1から4の何れかに記載のメモリ制御装置。

【請求項6】

前記制御部が、前記メモリモジュールのランク数及び／又はアクセス命令の種別に応じて、各ランクが備えるODTをそれぞれ制御する請求項1から5の何れかに記載のメモリ制御装置。

【請求項7】

複数のランクに区分され、各ランクが選択的にアクセス可能に接続されたメモリ郡と、
ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有するインターフェイス部と、
前記インターフェイス部を介してアクセス命令を受信した際のアドレス信号線を介して得た信号及びセレクト信号に基づいて前記ランクを選択してアクセスさせるデコード部と、
を備えたメモリモジュール。

【請求項8】

前記セレクト信号線数よりも多い数のランクを備えた請求項7に記載のメモリモジュール。

【請求項9】

前記デコード部をレジスタ間に配置した請求項7又は8に記載のメモリモジュール。

【請求項10】

複数のランクのメモリモジュールと、
前記メモリモジュールに格納される情報を用いて処理を実行するプロセッサと、
前記メモリモジュールへのアクセスを制御するメモリ制御装置と、を有する情報処理装置において、
前記メモリ制御装置が、
メモリモジュールのランクを選択するためのセレクト信号を介するセレクト信号線と、
選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線とが接続されるインターフェイス部と、
前記アドレス信号及び前記セレクト信号から、ランクを選択する信号を送信する制御部と、を備えた情報処理装置。

【発明の詳細な説明】

【技術分野】

【0001】

本発明は、大容量のメモリモジュールを実現する技術に関する。

【背景技術】

【0002】

バイトスライス型メモリコントローラを搭載したサーバ等のコンピュータの構成を例として図1に示す。

装置を制御するシステムコントローラ100に、CPU、メモリコントローラ101が接続されている。各メモリコントローラ101に少なくとも1つのDIMM(Dual Inline Memory Module)が接続され、4つのメモリコントローラが同期してメモリアksesを行う。(図1)

【0003】

図2は、該DIMMの概略構成図であり、DDR2インターフェイスを用いた2ランク(RANK)のDIMMの例を示している。

図2のDIMMは、1ランクに18個のSDRAM(Synchronous Dynamic Random Access Memory)を備え、2ランク有しているので、DIMM全体でのSDRAM総数は36個となっている。

複数のランクで構成されたDIMMは、ランク数分のチップセレクト信号を用い、各ランクの読み書きを選択的に行っている。現在、デファクトスタンダードの一つとなっているDDR2インターフェイスでは、チップセレクト信号線2本、アドレス信号線がA0～A15の16本となっているため、このインターフェイスを採用した場合、ランク数は最大2であり、2ランクで16本のアドレス信号線を用いて読み書きできる最大の容量は16GBである。

【0004】

また、本願発明に関連する先行技術として、例えば、下記の特許文献1に開示される技術がある。

【特許文献1】特開2003-7963号公報

【発明の開示】

【発明が解決しようとする課題】

【0005】

(1)問題点1

従来のDDR2インターフェイスでは、制御できるランク数が2ランクまでとなっており、装置に搭載されるメモリ容量を増やすには、DIMMアクセスチャンネルを増やすか、DIMMに搭載される各SDRAMの容量を増加させるしか方法がなかった。

しかし、SDRAMの容量の上限は、製造技術に依存するため、任意に上限を超えて容量を増加できるものではない。

【0006】

また、ランク数を増やす場合、1ランク毎にチップセレクト信号線1本を追加する必要があり、メモリコントローラのピン数を増やすなどボードの再開発も行わなくてはならず、現実的ではない。

【0007】

(2)問題点2

公知例「特開2002-184176」では、2ランクから4ランクに増量する例が提案されているが、DDR2のような高速インターフェイスには適用できない。また、ランク数を増やした場合、メモリコントローラ側を変更しなければ、公知例に挙げられたDIMMの制御ができないはずであるが、この制御については開示されておらず、実現性に乏しいという問題があった。

【0008】

例えば、ランクを増やすと言うことは、物理的なSDRAMが増えるということなので、増えたSDRAMを初期化するための回路が必要であるし、メモリリフレッシュの回路

も必要である。

【0009】

そこで本発明は、少ない構成変更でメモリモジュールのランク数を増加させる技術を提供する。

【課題を解決するための手段】

【0010】

上記課題を解決するため、本発明は、以下の構成を採用した。

即ち、本発明のメモリ制御装置は、

複数のランクを有するメモリモジュールにアクセスする装置であって、

ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有するインターフェイス部と、

前記アドレス信号線の一部及びセレクト信号線を介して前記ランクを選択する信号を送信する制御部と、を備えた。

【0011】

また、本発明の情報処理装置は、

複数ランクのメモリモジュールと、

前記メモリモジュールに格納される情報を用いて処理を実行するプロセッサと、

前記メモリモジュールへのアクセスを制御するメモリ制御装置と、を有する情報処理装置において、

前記メモリ制御装置が、

メモリモジュールのランクを選択するためのセレクト信号を介するセレクト信号線と、選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線とが接続されるインターフェイス部と、

前記アドレス信号及び前記セレクト信号から、ランクを選択する信号を送信する制御部と、を備えた。

【0012】

前記制御部は、前記セレクト信号線数よりも多い数のランクを選択する信号を送信しても良い。

【0013】

前記インターフェイス部は、DDR2インターフェイスであっても良い。

【0014】

前記メモリ制御部は、前記制御部からの命令に基づいて、前記ランク数に応じたメモリリフレッシュ処理を行うメモリリフレッシュコントロール回路を備えても良い。

【0015】

前記制御部は、異ランク間に対するアクセスが連続した場合に、該アクセス間に所定の待機時間を挿入しても良い。

【0016】

前記制御部は、前記メモリモジュールのランク数及び／又はアクセス命令の種別に応じて、各ランクが備えるODTをそれぞれ制御しても良い。

【0017】

また、本発明のメモリモジュールは、

複数ランクに区分され、各ランクが選択的にアクセス可能に接続されたメモリ郡と、

ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有するインターフェイス部と、

前記インターフェイス部を介してアクセス命令を受信した際のアドレス信号線を介して得た信号及びセレクト信号に基づいて前記ランクを選択してアクセスさせるデコード部と、を備えた。

【0018】

前記メモリモジュールは、前記セレクト信号線数よりも多い数のランクを備えても良い。

。

【0019】

前記メモリモジュールは、前記デコード部をレジスタ間に配置しても良い。

【発明の効果】

【0020】

本発明によれば、少ない構成変更でメモリモジュールのランク数を増加させる技術を提供できる。

【発明を実施するための最良の形態】

【0021】

以下、図面を参照して本発明を実施するための最良の形態について説明する。以下の実施の形態の構成は例示であり、本発明は実施の形態の構成に限定されない。

【0022】

§ 1. 全体構成

図3は、本発明の一実施例に係る情報処理システムの概略図である。

本例の情報処理システム10は、システムボード上に、CPUや、システムコントローラ1、メモリコントローラ2、メモリモジュール3を備えた汎用のコンピュータである。

情報処理システム10は、メモリコントローラ（メモリ制御装置）2やメモリモジュール3を複数備えても良い。例えば図1と同様に各メモリコントローラ2に少なくとも1つのDIMMが接続され、4つのメモリコントローラ2が同期してメモリアクセスを行う。

【0023】

§ 2. 各構成部分の説明

本実施形態のメモリモジュール3は、図4に示すように、DDR2インターフェイスを利用したDIMMであり、以下DIMM3とも称する。該DIMM3は、1ランクに18個のSDRAM31を持ち、これを4ランク、即ち72個のSDRAM31（メモリ群）を備えている。

該DIMM3は、1GBのSDRAM2つ分で従来の1ランク相当を構成しており、計8GBの容量を有している。

【0024】

DIMM3上には記憶素子としてのSDRAM31の他に、EEPROM32、デコード回路（デコード部）33、ODT（On Die Termination）34、インターフェイス35を備えている。

【0025】

EEPROM32は、高速伝送実現の為にアドレス信号線等をラッチするレジスタや、PLL、DIMMのスペックを記録している。

【0026】

ODT34は、ランク毎のSDRAMに搭載されたバス終端抵抗であり、メモリコントローラ2からON/OFF制御される。

【0027】

インターフェイス部35は、ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有するDDR2インターフェイスである。

【0028】

デコード回路33は、前記インターフェイス部35を介してアクセス命令を受信した際のアドレス信号線を介して得た信号及びランクセレクト信号に基づいて、増加したランク2、3を含めたランクセレクト信号を生成することにより、前記ランクを選択及びアクセスを可能にしている。

【0029】

システムコントローラ1は、CPU及び各メモリコントローラ2を制御する。システムコントローラ1は各メモリコントローラ2に対し、同期制御とメモリアクセスを命令する。各メモリコントローラ2はシステムコントローラ1からの命令に従い、メモリアクセスを行う。

【0030】

システムコントローラ1は、メモリリフレッシュコントロール回路13を備えている。

【0031】

メモリコントローラ2は、制御部21や、インターフェイス部22、EEPROMアクセスコントロール回路23を備えている。

【0032】

制御部21は、前記アドレス信号線の一部及びセレクト信号線を介して前記ランクを選択する信号を送信するなど、DIMM3に対するアクセス制御を行う。

【0033】

インターフェイス部22は、ランクを選択するためのセレクト信号を介するセレクト信号線及び選択されたランク上のアドレスを示すアドレス信号を介するアドレス信号線を有したDDR2インターフェイスである。

【0034】

各メモリコントローラ2は、図1と同様にDDR2バスインターフェイスを利用したDIMMバスチャンネルを4つ搭載し、各DIMMバスチャンネルには2つのDIMMスロットがバス接続されている。

【0035】

図5にDIMMに搭載されるSDRAM31のダイヤグラムを示す。

メモリコントローラ2から接続されるコマンド信号(／RAS、／CAS、／WE)と／CSによりSDRAM31のセレクト及び動作が決定される。

【0036】

アドレス選択用として接続される信号はA0～A14、BA0～BA2があり、BA0～BA2は合計8つのメモリバンクの中の1つを選択し、A0～A14はバンク内の記憶セルの選択に使用される。

【0037】

記憶セルの選択には、A0～A14を時分割した、ロウアドレスとカラムアドレスの2つのアドレスが使用される。記憶セル内のデータは、4本のデータ信号(DQ)で入出力される。DQ信号は、データストローブ信号対(DQS、／DQS)によりラッチされる。

【0038】

DIMM3上に搭載されるEEPROM32には、コマンドのタイミングやランクなどの構成情報が記録されている。メモリコントローラ2のI2Cアクセスコントロール回路23は、I2Cと呼ばれるシリアル規格に従ってこの情報を読み出し、DIMM3をコントロールする為の初期化をファームウェアに従って行う。

【0039】

SDRAM31は、記憶保持の為のメモリリフレッシュ機能を備えている。メモリリフレッシュはメモリリードやメモリライトと同様に、メモリコントローラ2からの命令により行われる。

【0040】

メモリコントローラ2とSDRAM31間は、DDR2インターフェイスを利用したバスで構成されている。バス上にはスロット及びランクによりドット接続される。

【0041】

システムコントローラ1はCPUからの物理アドレスをメモリアドレスに変更し、リード／ライトコマンドと共に各メモリコントローラ2に送出する。メモリライトの際には、ライトデータがメモリコントローラ2に送出される。

【0042】

上記のメモリアドレスとは、メモリインターフェイス上のSDRAM郡のランクを選択するチップセレクト信号(CS[3:0])、SDRAM内のBANKを選択するバンクアドレス(BA[2:0])、BANK内の記憶セル郡を選択するロウアドレスと、ロウアドレスで選択された記憶セル郡からデータビットを選択するカラムアドレスで構成される。

【0043】

メモリコントローラ2は初期化動作時にI2Cアクセスコントロール回路23がEEPROM32を参照し、搭載されたDIMM3の構成やコマンドタイミングといったスペックを知ることができる。

【0044】

本実施形態の初期化動作の例を図3を用いて説明する。情報処理システム10への電源投入後にファームウェアが起動され、DIMM3のEEPROM32から情報の読み出しを行う。読み出された情報を元にメモリコントローラ2内のアーキテクチャレジスタをセットする。例えば、DIMM3搭載情報によりメモリマップの決定、動作タイミングの決定、ランク数の決定を行う。また、構成情報を元にDIMM3そのものの初期化や、DIMM3用の設定レジスタのセット、ODT (On Die Termination) 34の抵抗値の設定を行う。

【0045】

本実施形態では、制御部21が増加させたランク2, 3用のレジスタセットを備えており、参照したランク数が4であった場合には、通常どおりランク0, 1の初期化を行うのと共に、このレジスタセットを用いてランク2, 3の初期化も行う。

【0046】

また、情報処理システム10におけるメモリリフレッシュの例を図8に示す。前述した初期化動作により、システムコントローラ内のアーキテクチャレジスタにリフレッシュ間隔が設定される。この値はメモリコントローラ2に搭載されるDIMM3のランク総数によって決定される。システムコントローラ1内のリフレッシュカウンタが、設定されたリフレッシュ間隔になるとメモリコントローラ2を介してリフレッシュ命令がDIMM3に対して発行される。リフレッシュはランク毎に発行され、対象アドレスはDIMM内で自動的に決定される。

【0047】

そして、メモリコントローラ2がシステムコントローラ1からリード/ライトコマンドを受信した場合、該メモリコントローラ2のシーケンサによってDIMM3に対するリード/ライトを行い、データの送受信を行う。

【0048】

§3. メモリコントローラ2及びDIMM3の特徴部分の説明

本実施形態のDIMM3では、A0～A14の15本持っているアドレス信号線のうち、最上位の1本 (A14) をチップセレクト信号の生成に用いる為、図6に示すようにデコード回路33を備えている。

【0049】

デコード回路33は、アドレス信号線A14からの信号と既存のチップセレクト信号線2本からのセレクト信号に基づき、図7の真理値表に示す4つの状態のチップセレクト信号を生成する。

【0050】

即ち、アドレス信号線A14の信号がアサートされておらず、チップセレクト信号CS0がアサートされた場合にはランク0が選択され、チップセレクト信号CS1がアサートされた場合にはランク1が選択される。また、アドレス信号線A14の信号がアサートされており、チップセレクト信号CS0がアサートされた場合にはランク2が選択され、チップセレクト信号CS1がアサートされた場合にはランク3が選択される。

【0051】

ここで使用されるアドレス信号線A14は、DIMM3に入力されるアドレスの最上位ビットを示すアドレス信号を伝送する為の既存の線であるので、メモリコントローラ2とDIMM3の間の伝送路上に信号の増減が発生せず、ボードの再開発が必要ない。

【0052】

本実施形態のDIMM3には1GBのSDRAMが搭載されており、アドレス信号線はA0～A13しか使用しないため、最上位ビットであるA14をチップセレクトに使用し

でもシステムとして問題ない。

【0053】

図9にD I M M 3とDDR 2インターフェイスで規定されているメモリ空間イメージの比較を示す。

空間イメージ4 1は、DDR 2インターフェイスで規定されているもので、2 G BのS D R A Mを搭載した2ランク構成とすることで、最大8 G Bの容量が得られる。

【0054】

これに対し、本実施形態のD I M M 3は、1 G BのS D R A Mを搭載した4ランク構成で8 G Bの容量を得ている。このため、ロウアドレスの表現にA 0～A 1 3、カラムアドレスの表現にA 0～A 9、A 1 1を使用しており、A 1 4は未使用である。

【0055】

従って、メモリコントローラ2の制御部2 1は、システムコントローラ1からD I M M 3に対するリード／ライトコマンドを受信した場合、選択するランクに応じて前記真理値表のとおりA 1 4、C S 0、C S 1に信号をアサートする。

【0056】

なお制御部2 1は、初期化時にE E P R O M 3 2から読み出した情報に従い、アクセスするメモリモジュールが、本実施形態の4ランク構成であればA 1 4、C S 0、C S 1を用いてランク選択の為の信号をアサートし、2ランク構成であれば規定どおりC S 0、C S 1を用いてランク選択の為の信号をアサートしてA 1 4をアドレス信号線として使用する。

【0057】

これにより、本実施形態のシステムボードは、DDR 2インターフェイスに従う2 G BのS D R A Mを搭載した2ランクのD I M Mとの互換性を維持しつつ、1 G BのS D R A Mを搭載した4ランクのD I M M 3も利用可能にしている。

【0058】

なお、本実施形態では、既存の2ランク構成よりもランク数を増加させているため、図1 0 (A)のようにランク0、1に対するメモリリフレッシュのフローを図1 0 (B)のように2 F l o wにして行う。

【0059】

メモリコントローラ2内の制御部2 1は、増加させたランク数に応じたシーケンスを備え、ランクを増加させたD I M M 3に対するリフレッシュコマンドを受けた場合、メモリリフレッシュの処理をランク数分増加させて各ランクのメモリリフレッシュを行わせる。

【0060】

例えば、図1 1に示すパターン1のように、ランク0、1に対して、P A L LからもR A Sを待ってR E Fまでの処理のフロー1を行った後に同様のフロー2をランク2、3に行う。しかし、この場合、メモリリフレッシュにかかる時間がランク数に応じて単純に増加してしまい、リード／ライトのアクセスを圧迫してしまう可能性がある。

【0061】

このため、パターン2のように、ランク0、1に対するフロー1のP A L Lの後、ランク2、3に対するフロー2のP A L Lを行い、メモリリフレッシュ処理を並行して行うのが望ましい。

【0062】

また、本実施形態のように、ランク数を増加させた場合、スタブの増加も考えられるが、制御部2 1が、ランク数に応じて、所定ランクのO D Tをオン又はオフに制御することで抑えることができる。また、このO D Tのオン／オフ制御は、リード／ライト等の命令の種別に応じて行っても良い。このとき、何れのランクのO D Tをオンにし、何れのランクのO D Tをオフにするかは、E E P R O M 3 2の内容で判断するかファームウェアに予め設定しておけば良い。

【0063】

更に、増設ランクのデコード回路3 3の為に回路段数が増え、伝送特性が悪化して伝送

速度の高速化の妨げになることが考えられるが、本実施形態では、図12に示すようにデコード回路33をレジスタで挟みこみ、この伝送特性を改善している。

【0064】

このように、レジスタ43の直後にデコード回路33を配置し、デコード回路33の直後にレジスタ44を配置したことにより、デコード回路33の直前で信号のタイミングが揃い、デコード回路33の直後でも信号のタイミングが揃うので、伝送特性の悪化が防止できる。

【0065】

また、DIMM3は、ランクを増加したことによってバスファイト (Bus Fight) が増える可能性がある。

【0066】

メモリコントローラ2の制御部21は、同一ランク内の前半部分と後半部分のアクセスが連続した場合、図13に示すパターン1のように、連続してアクセスを行う。しかし、異なるランクに対するアクセスが連続した場合、図13に示すパターン2のように、該アクセス間にウェイトを挿入する。

【0067】

即ち、制御部21は、増設したランク2、3も含め各ランクへのアクセスを排他的に管理し、異なるランクに対するアクセスの間には、ウェイトを挿入する。

【0068】

以上のように本実施形態では、既存信号線を用いてランク選択の為の信号を送信するので、ランク数を増やしても従来のインターフェイスを用いることが可能となった。

【0069】

従って、SDRAMの容量増加が望めない場合であっても、ランクを増やすことでDIMMの容量を増やすことが出来るようになった。例えば、現在入手が困難な2GBのSDRAMを用いなくても1GBのSDRAMで高容量のDIMMを達成できる。このため、普及している安価なSDRAMを選択可能で、高容量のDIMMを安価に製造しやすい。

【0070】

また、増やしたランクに対する初期化やリフレッシュの制御が可能となった。

DIMMに渡されるアドレス自体はそのまま使用できるので、CPUからの物理アドレスの制御に変更を加えなくともDIMMの容量を増やすことが可能となった。

【0071】

更に、情報処理システムに搭載するDIMMの数を増やさずともメモリ総容量を増やすことが出来るようになる為、システムの開発コストを軽減できる。

【図面の簡単な説明】

【0072】

【図1】 関連技術の概略図

【図2】 関連技術のメモリ構成を示す図

【図3】 情報処理装置の全体図

【図4】 本実施形態のメモリ構成を示す図

【図5】 SDRAMの説明図

【図6】 デコード部の説明図

【図7】 ランク選択のための真理値表

【図8】 メモリリフレッシュ処理の説明図

【図9】 メモリ空間イメージを示す図

【図10】 メモリリフレッシュ処理の説明図

【図11】 メモリリフレッシュ処理の説明図

【図12】 デコード回路の説明図

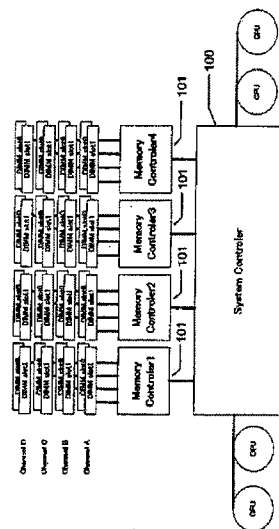
【図13】 バスファイト防止の説明図

【符号の説明】

【0073】

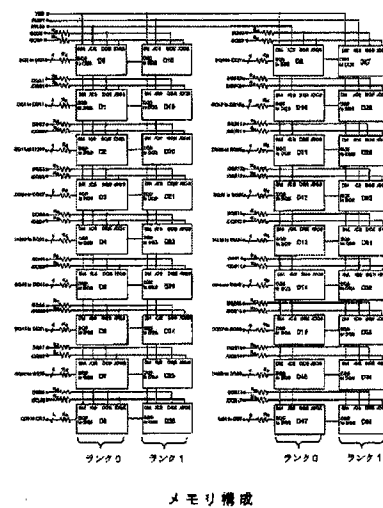
- 10 情報処理システム
- 1 システムコントローラ
- 13 メモリリフレッシュコントロール回路
- 2 メモリコントローラ
- 21 制御部
- 22 インターフェイス部
- 23 I2Cアクセスコントロール回路
- 3 メモリモジュール (DIMM)
- 31 SDRAM
- 32 EEPROM
- 33 デコード回路 (デコード部)
- 34 ODT (On Die Termination)
- 35 インターフェイス

【図1】



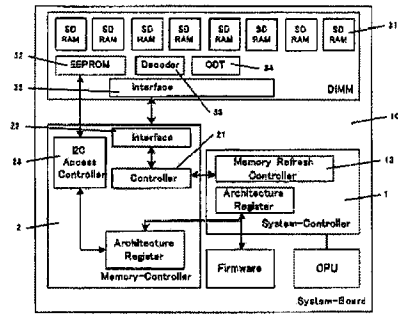
関連技術

【図2】



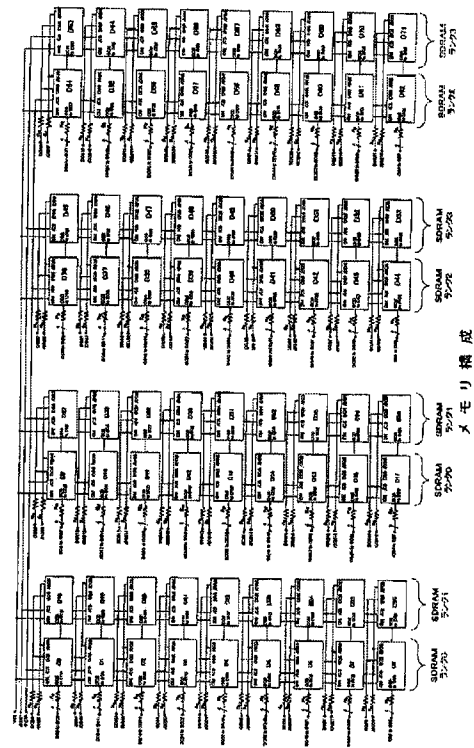
メモリ構成

【図3】



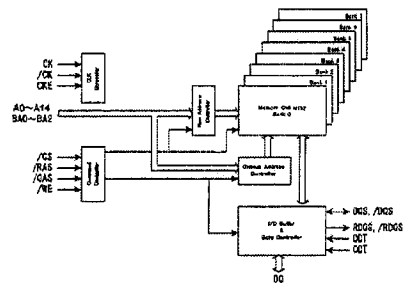
システム全体図

【図4】



メモリ構成

【図5】



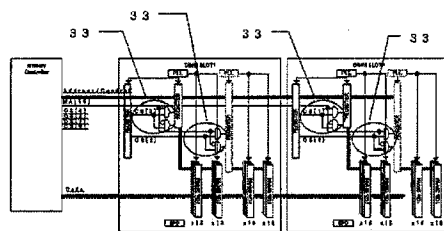
SDRAMの説明図

【図7】

	A14	CS1	CS0
RANK0	0	0	1
RANK1	0	1	0
RANK2	1	0	1
RANK3	1	1	0

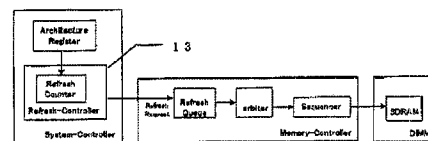
真環境表

【図6】



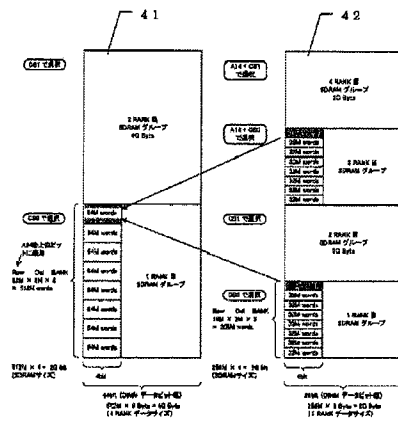
デコード部の説明図

【図8】



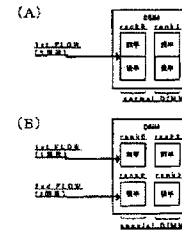
メモリフレッシュ処理の説明図

【図9】



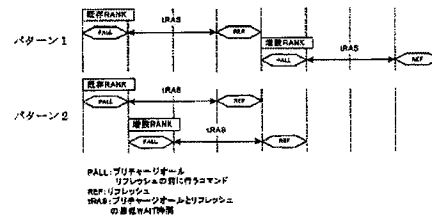
メモリ空間イメージを示す図

【図10】



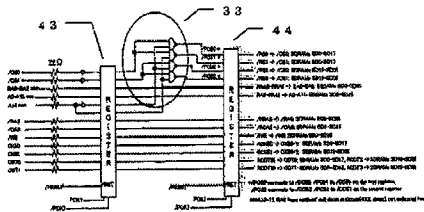
メモリリフレッシュ処理の説明図

【図11】



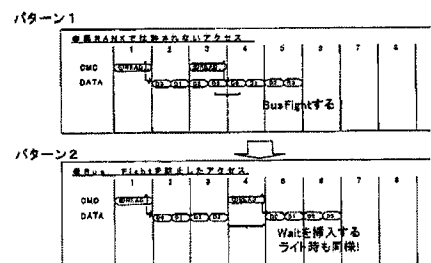
メモリリフレッシュ処理の説明図

【図12】



デコード回路の説明図

【図13】



バスファイト防止の説明図

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5M024 AA54 EE02 JJ03 LL20 PP01 PP07 PP10

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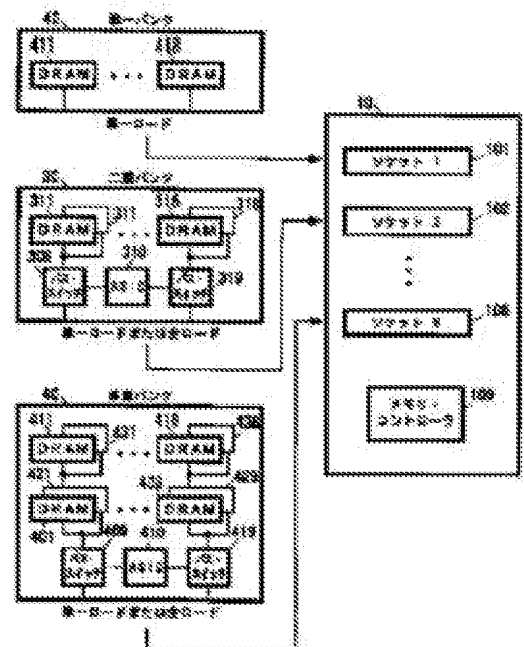
Priority number : 96 676609 Priority date : 08.07.1996 Priority country : US

(54) MEMORY MODULE

(57)Abstract:

PROBLEM TO BE SOLVED: To minimize a data line load by providing a bus switch between a system and a RAM and then comprising a logic circuit to produce a signal for enabling this switch into an ASIC.

SOLUTION: DRAM chips 301 to 304 and DRAM chips 311 to 314 are connected to a bus switch 309, while DRAM chips 305 to 308 and DRAM chips 315 to 318 are connected to a bus switch 319. A couple of bus switches are connected to ASIC 310 which is connected to a RAS pin and a CAS pin. A data bus load can be minimized by forming a bank DIMM 30 or DIMM 40 by utilizing bus switches 309, 319 or bus switches 409, 419.



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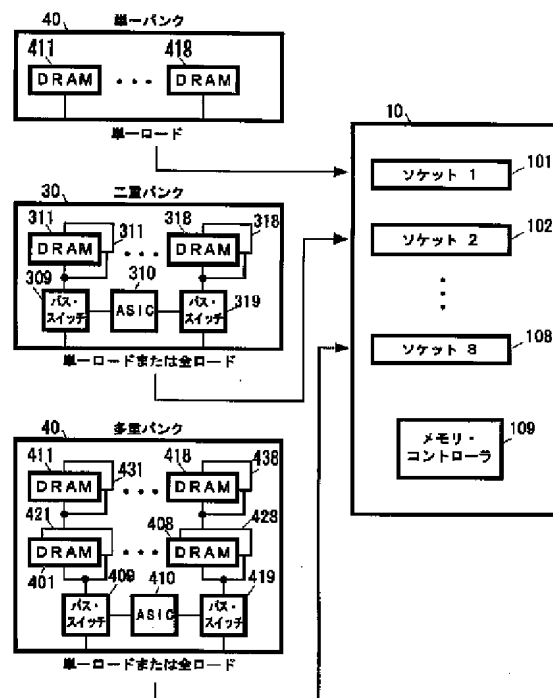
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(54)【発明の名称】 メモリ・モジュール

(57)【要約】

【課題】 複数のDRAMを有する高密度モジュール上のデータ線負荷を最小にし、データ線容量性負荷に起因する性能低下なしに、システムの最大のメモリ密度を可能にすること。

【解決手段】 システム・ボード10上のコネクタ101~108と相互接続されるタブを有するプリント回路カード20~40と、プリント回路カードに取り付けられた複数のバンクのランダム・アクセス・メモリ・デバイス301~318、401~438と、選択されたランダム・アクセス・メモリ・デバイスと接続された、イネーブル信号に応答する1つまたは複数のバス・スイッチ309、319、409、419と、システムの行アドレス・ストロブ信号および列アドレス・ストロブ信号にตอบสนองして、1つまたは複数のバス・スイッチのイネーブル信号を生成する、論理手段とを含む。



【特許請求の範囲】

【請求項1】システム・ボード上のコネクタと相互接続される接点タブを有し、カード・データ・バスを有するプリント回路カードと、

前記プリント回路カードに取り付けられ、前記カード・データ・バスに接続された、複数バンクのランダム・アクセス・メモリ・デバイスと、

前記複数バンクのランダム・アクセス・メモリ・デバイスのうちの選択されたものが前記カード・データ・バスを介して接続された、イネーブル信号に応答する、1つまたは複数のバス・スイッチと、

システムの実行アドレス・ストロブ信号および列アドレス・ストロブ信号に応答して、前記1つまたは複数のスイッチの前記イネーブル信号を生成する、論理手段とを含む、メモリ・モジュール。

【請求項2】前記1つまたは複数のバス・スイッチおよび前記論理手段が、前記プリント回路カードに取り付けられることを特徴とする、請求項1に記載の高密度メモリ・モジュール。

【請求項3】第2のプリント回路カードを含み、前記第1および第2のプリント回路カードが異なる数のメモリ・バンクを有することを特徴とする、請求項1に記載の高密度メモリ・モジュール。

【請求項4】前記1つまたは複数のバス・スイッチのそれぞれが、前記カード・データ・バスの信号線1本につき1つの複数の電界効果トランジスタ(FET)と、前記イネーブル信号に応答して前記FETをバイアスするための手段とを含むことを特徴とする、請求項1に記載の高密度メモリ・モジュール。

【発明の詳細な説明】**【0001】**

【発明の属する技術分野】本発明は、全般的にはコンピュータ用の高密度メモリ・モジュールに関し、具体的には、最小限のメモリ・バス負荷を有する多重バンク・メモリ・モジュールに関する。

【0002】

【従来の技術】SIMM(シングル・インライン・メモリ・モジュール)やDIMM(デュアル・インライン・メモリ・モジュール)などの高密度メモリ・モジュールは、多くの応用分野で追加のシステム性能を達成できるので、ハイ・エンド・パーソナル・コンピュータ(PC)、ネットワーク・サーバおよびワークステーションで需要がある。しかし、最大システム密度は、下記の1つまたは複数によって不自然に制限されることがしばしばである。

・システムが有するメモリ・モジュール「スロット」の数が限られている。

・システムが有するメモリ「バンク」の数が(メモリ・コントローラからの選択信号線の不足が原因で)限られている。

・高密度メモリ・チップ(たとえば64メガビット(Mb)、256Mbおよびそれ以上)は、非常に高価であるか、簡単に入手できないか、その両方である。

・高密度メモリ・チップは、システムのメモリ・インターフェース電圧より低い動作電圧を有する場合がある。

・標準ランダム・アクセス・メモリ・デバイス(RAM)(ダイナミックRAM(DRAM)、シンクロナスDRAM(SDRAM)およびスタティックRAM(SRAM)を含む)の積層化/立体化または大型モジュールの使用は、システム限界をはるかに超えるデータ線容量をもたらす。以前の解決は、最後の問題以外の上記の問題のすべてについて提供されてきたが、これらの解決は、データ線負荷の問題が原因で応用性が制限されている。メモリ拡張を制限されているシステムで安価なRAMチップをよりよく利用するために必要なものは、データ線の容量性負荷を最小にし、その結果、RAMチップのバンクを有する大型メモリ・モジュールをシステムに追加できるようにする方法である。

【0003】

【発明が解決しようとする課題】したがって、本発明の目的は、データ線ごとに複数のランダム・アクセス・メモリ・デバイスを有する高密度モジュールでのデータ線負荷を最小にするための解決を提供することである。

【0004】本発明のもう1つの目的は、データ線の容量性負荷に起因する性能低下なしに、密度を制限されているシステムのメモリ密度を最大にすることである。

【0005】

【課題を解決するための手段】本発明によれば、データ線容量を許容可能なシステム限界まで減らす解決法には、2つの部分がある。第1の部分には、インライン・バス・スイッチを有するメモリ・モジュールの設計が含まれる。バス・スイッチは、モジュール・タブ(システム)とランダム・アクセス・メモリ・デバイス(RAM)の間に置かれ、高インピーダンス(オフ)またはアクティブ(活動)状態のいずれかになる。高インピーダンス状態の時には、メモリ・モジュールの実効負荷は、ビット・スイッチ・デバイスの負荷になる。アクティブ(たとえば、リード/ライト・サイクル)の時には、RAM負荷の他に、最小限のキャパシタンス/抵抗がメモリ・バスに追加される。一時に1つのメモリ・モジュールだけがアクティブになる。

【0006】本発明による解決の第2の部分は、バス・スイッチをイネーブルする信号を生成する論理回路を特定用途向け集積回路(ASIC)に組み込むことである。バス・スイッチは、メモリ・モジュールへの行アドレス・ストロブ(RAS)選択線の立ち上がりエッジでアクティブになり、RAS選択線または列アドレス・ストロブ(CAS)選択線のうちの最後の線がインアクティブ(非活動)状態になるまで活動状態にとどまり、これによって、高速ページ・モード(FPM)動作

とEDO（拡張データ出力）動作の両方がサポートされる。ASICの回路は、システムのRAS選択線とCAS選択線をデコードし、信号を駆動してバス・スイッチをイネーブルすることによって、この作業を実行する。論理回路は、リード／ライト・サイクル、ROR（RAS only refresh）サイクル、CBR（CAS before RAS refresh）サイクルおよびヒドン・リフレッシュ（hidden refresh）サイクルを区別し、それ相応にバス・スイッチをアクティブにしなければならない。CBRサイクルが発生する時には、バス・スイッチはインアクティブのままになる。ヒドン・リフレッシュ・サイクルでは、バス・スイッチはアクティブにされ、リフレッシュ・サイクルが完了するまでアクティブのままになる。これは、リード動作またはライト動作が存在する可能性があるため必要である。

【0007】

【発明の実施の形態】本発明を、DIMMを使用する特定の実施例に関して説明するが、当業者であれば、SIMMなどの他のメモリ・モジュールを本発明の実施に使用できることを諒解するであろう。このようなモジュールは、通常は、コンピュータ・システム基板上のコネクタに挿入するための接点タブを有するプリント回路カードとして実施される。

【0008】ここで図面、具体的には図1を参照すると、8つのメモリ・モジュール・ソケット101～108とメモリ・コントローラ109を有する通常のシステム・ボード10が示されている。たとえば8つの4MbのDRAMチップ201～208を有する通常の単一バンクDIMM 20は、通常はメモリ・コントローラ109に最も近いメモリ・モジュール・ソケット101から始めて、メモリ・モジュール・ソケット101～108のすべてに単一バンクDIMM 20が挿入されるまで順番に、メモリ・モジュール・ソケット101～108のうちの1つに挿入することができる。したがって、この例では、システム・ボードは、4メガバイト（MB）の最小構成と32MBの最大構成を有する。

【0009】本発明によれば、二重バンクDIMM 30または多重バンクDIMM 40によって、単一バン

クDIMM 20を置換することができる。二重バンクDIMM 30の場合、バス・スイッチ309および319に接続された、8つのDRAMチップ301～308の第1のグループと、8つのDRAMチップ311～318の第2のグループがある。DRAMチップ301～304およびDRAMチップ311～314はバス・スイッチ309に接続され、DRAMチップ305～308およびDRAMチップ315～318はバス・スイッチ319に接続される。2つのバス・スイッチは、ASIC 310に接続され、ASIC 310は、このモジュールのRASピンおよびCASピンに接続される。この概念は、多重バンクDIMM 40の場合に拡張され、この例では、DRAMチップの4つのグループすなわち、DRAMチップ401～408、DRAMチップ411～418、DRAMチップ421～428およびDRAMチップ431～438がある。DRAMチップ401～404、DRAMチップ411～414、DRAMチップ421～424およびDRAMチップ431～434は、バス・スイッチ409に接続され、DRAMチップ405～408、DRAMチップ415～418、DRAMチップ425～428およびDRAMチップ435～438は、バス・スイッチ419に接続される。二重バンクDIMM 30の場合と同様に、バス・スイッチ409および419は、ASIC 410に接続され、ASIC 410は、このモジュールのRASピンおよびCASピンに接続される。

【0010】二重バンクDIMM 30を使用する場合、システムの最小構成は8MBに増加し、最大構成は64MBに増加する。多重バンクDIMM 40を使用すると、図示の4バンクのDRAMチップの例では、最小構成は16MB、最大構成は128MBに増加する。より高密度のデバイス（たとえば16Mb、64Mbなど）を使用すると、最大メモリ密度のかかなりの増加がもたらされるはずである。これは、下の表からわかるように、最小限のデータ・バス負荷で達成される。

【0011】

【表1】

データ・バス負荷の比較*（最大値）			
	1バンク	2バンク	4バンク
従来技術	8	16	32
本発明	8	9	11

* DIMM当たりのDRAM等価負荷の数として表す（通常は7 pF/DRAM）

【0012】これは、二重バンクDIMM 30の場合にはバス・スイッチ309および319、多重バンクD

IMM 40の場合にはバス・スイッチ409および419を使用することによって達成される。ASIC 3

10および410は、バス・スイッチとシステム・ボードの間のインターフェースを提供する。

【0013】ここで図2に移ると、バス・スイッチの一般化された論理図が示されている。バス・スイッチには、信号線ごとに1つずつ、複数のFETが含まれる。1つまたは複数のインバータ50が、FETをオンにバイアスするのに使用される。インバータ50への入力BEN(NはBEの反転を表す)は、バス・スイッチのイネーブル入力である。イネーブル入力が行うになると、インバータの出力はハイになり、その結果、正バイアスがFETのゲートに印加され、FETが導通する。このバス・スイッチを介する遅延は、1ナノ秒(ns)未満であり、これによって、正しいメモリ動作が保証される。

【0014】図3のブロック図からわかるように、ASIC 60(図1のASIC 310および410に対応する)は、システムのRAS信号(SYS_RAS)とCAS信号(SYS_CAS)を受け取り、バス・スイッチ61および62のイネーブル入力へのRC_SELECT信号を生成する。バス・スイッチ61および62は、カード・データ・バス63とプリント回路カードの接点タブ64の間のインターフェースを提供する。接点タブ64はシステム・ボードのソケット1~8に挿入されるモジュール20、30、40のプリント回路カードのエッジに設けられている。

【0015】RC_SELECT信号は、アクティブ・ロウの信号であり、独自の機能を有する。通常のメモリ・リード・サイクルおよびメモリ・ライト・サイクルの間、RC_SELECT信号は、図4に示されるように、システムのRAS選択線のアクティブ(立ち下がり)エッジでアクティブにされ、システムのRAS選択線とCAS選択線の両方がインアクティブになるまでアクティブのままになる。

【0016】ヒドン・リフレッシュ・サイクル中は、図5からわかるように、ヒドン・リフレッシュ・サイクルの性質が原因で、RC_SELECT信号はアクティブになる。ヒドン・リフレッシュ・サイクルの第1の部分は、RASが立ち上がってから立ち下がる(CBRリフレッシュ・サイクルの開始)までシステムのCAS選択線がアクティブのままになるメモリ・サイクルである。RC_SELECT論理回路は、CAS外乱をフィルタリングし、インアクティブのままになる。

【0017】CBRリフレッシュ中は、図6からわかるように、システムのCAS選択線が、システムのRAS選択線がアクティブになるより前にアクティブになる。RC_SELECT信号は、CBRサイクル中はインアクティブのままになる。

【0018】RORサイクル中は、図7からわかるように、システムのRAS選択線がアクティブになり、RC_SELECT信号上に駆動される。バス・スイッチが

アクティブになっても、データがその時点で駆動されていないので、システム動作には影響しない。

【0019】所与の応用分野で必要であれば、追加回路を追加して、RORサイクル中にバス・スイッチがアクティブにならないようにすることができる。

【0020】図8のASICの内部ロジックには、システムのCAS選択線CAS_N(Nは反転を表す)のアクティブ(すなわち立ち下がり)エッジでシステムRAS選択線(ANY_RAS)の状態をラッチするD型のフリップフロップ70が含まれる。ANY_RAS信号は、ANDゲート71によって生成され、バッファ72を介してフリップフロップ70のD入力に供給されるが、ANY_CAS信号は、ANDゲート73によって生成され、インバータ74を介してフリップフロップ70のクロック入力に供給される。フリップフロップ70の出力は、インバータ76を介して2入力NANDゲート75に接続され、マルチプレクサ77の入力「0」にも接続される。インバータ76の出力は、信号CBR_BIT_Nである。信号ANY_RASは、インバータ78によって反転され、インバータ78の出力は、2入力NANDゲート75のもう一方の入力に接続される。2入力NANDゲート75の出力は、信号RC_ACTIVEであり、マルチプレクサ77の入力「1」に接続される。フリップフロップ70は、システムのRAS選択線とCAS選択線がインアクティブであり、フリップフロップ70の「Q」出力が論理「1」である時に、NANDゲート79の出力によってリセットされる。マルチプレクサ77の選択線は、4つのゲート(インバータ80と3つのバッファ81~83)によって遅延されたシステムのCAS選択線DELAY_CASによって選択される。これは、フリップフロップ70が信号ANY_RASの状態をラッチする時間を与えるために行われる。

【0021】各サイクルの初期の開始時に、マルチプレクサ77は、「1」入力位置にセットされる。これによって、システムのRAS選択線をすばやく流すことができるようになる。システムのCAS選択線がアクティブになった時には、マルチプレクサ77は、「0」入力位置に切り替わり、これによって、フリップフロップ70の出力が選択される。

【0022】図1に戻って、本発明の実装は、DRAMチップのバンク群を1つのプリント回路カードに取り付け、ASICとバス・スイッチを別のキャリアに取り付けることによって、変更することができる。たとえば、ASICとバス・スイッチは、システム・ボード10に直接に取り付けることができるが、これにはシステム・ボードの変更が必要である。代替案は、DRAMカードを受けるコネクタを備えた別のプリント回路カードにASICとバス・スイッチを取り付けることである。どちらの代替案でも、DRAMチップだけでDRAMカード

を製造することができる。

【図面の簡単な説明】

【図1】システム・メモリ・ボードと、従来の単一バンクD I M Mと本発明による多重バンクD I M Mの両方を示すブロック図である。

【図2】図1に示された本発明の実施例に使用されるバス・スイッチを一般化した論理ブロック図である。

【図3】図1の実施例でのA S C Iとバス・スイッチの関係を示すブロック図である。

【図4】通常のリード／ライト・サイクルの場合の、R C _ S E L E C T出力信号を生成するA S I C論理回路の動作を示すタイミング図である。

【図5】ヒドン・リフレッシュの場合の、R C _ S E L E C T出力信号を生成するA S I C論理回路の動作を示すタイミング図である。

【図6】C B Rの場合の、R C _ S E L E C T出力信号を生成するA S I C論理回路の動作を示すタイミング図である。

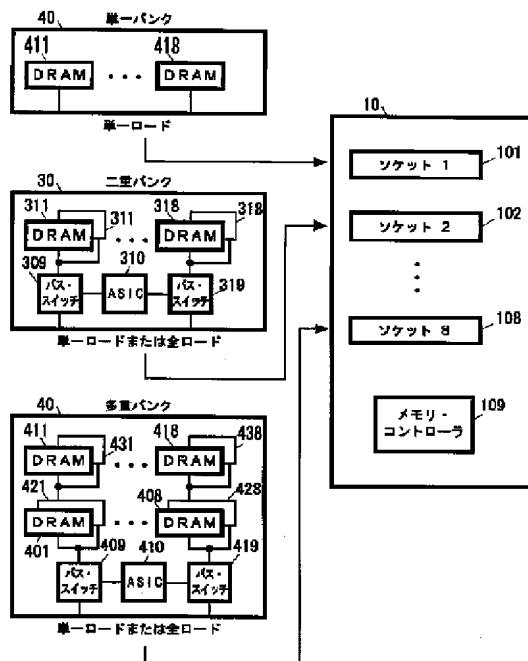
【図7】R O Rの場合の、R C _ S E L E C T出力信号を生成するA S I C論理回路の動作を示すタイミング図である。

【図8】本発明の好ましい実施例によるA S C Iの論理回路を示す論理ブロック図である。

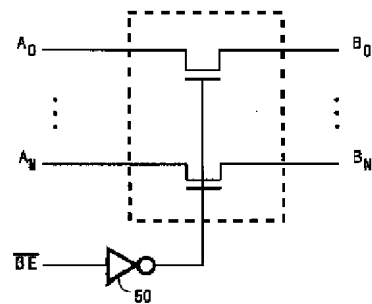
【符号の説明】

- 60 ASIC
- 61 バス・スイッチ
- 62 バス・スイッチ
- 63 カード・データ・バス
- 64 タブ
- 72 バッファ
- 73 ANDゲート
- 74 インバータ
- 75 2入力NANDゲート
- 76 インバータ
- 77 マルチプレクサ
- 78 インバータ
- 79 NANDゲート
- 80 インバータ
- 81 バッファ
- 82 バッファ
- 83 バッファ

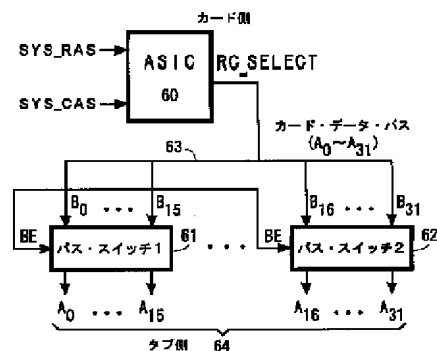
【図1】



【図2】



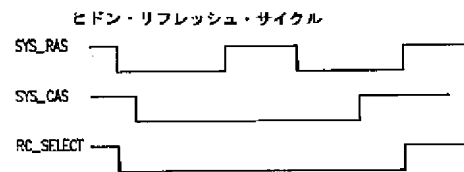
【図3】



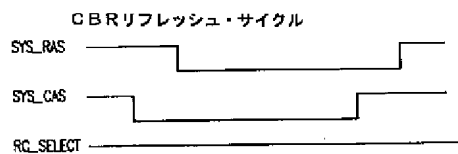
【図4】



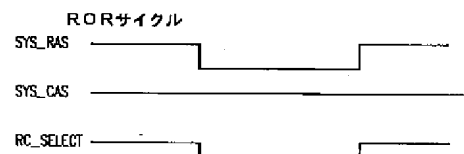
【図5】



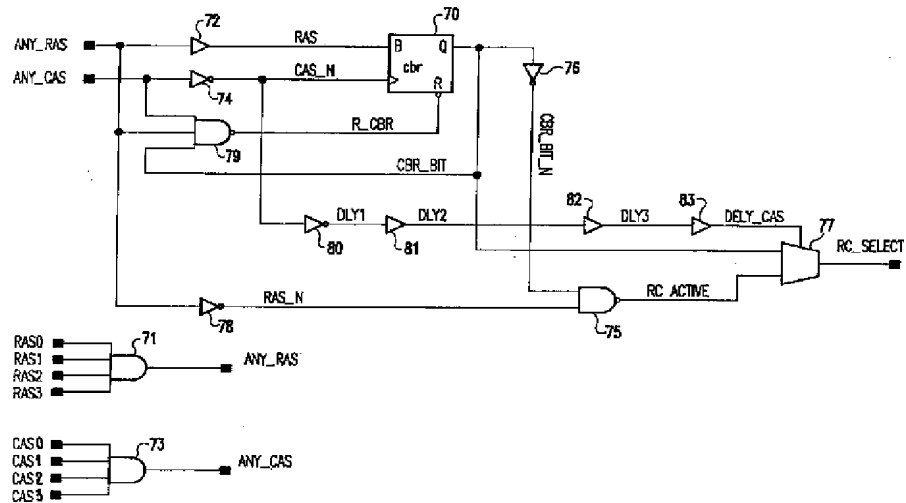
【図6】



【図7】



【図8】



【手続補正書】

【提出日】平成9年7月23日

【手続補正1】

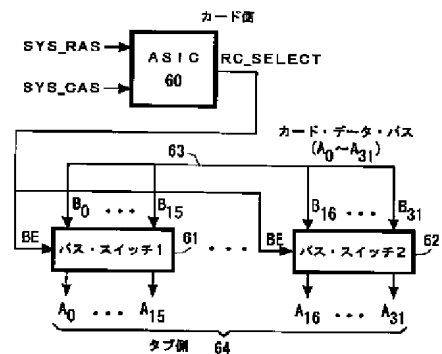
【補正対象書類名】図面

【補正対象項目名】図3

【補正方法】変更

【補正内容】

【図3】



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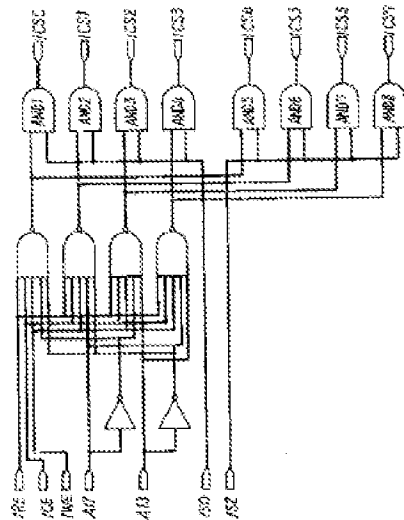
25

(54) [Title of the Invention] Memory module

(57) [Abstract]

[Problem] The aim is to provide a memory module which
5 achieves lower costs, reduced power consumption and low
temperature rises using current-generation SDRAMs, by
resolving a conventional problem, namely that if the
capacity of an SDRAM memory module is increased, there
is excessively large power consumption or an
10 excessively large rise in temperature with current-
generation $\times 1$ configuration SDRAMs unless next-
generation SDRAMs are present.

[Means of Resolution] It is possible to reduce power
15 consumption and produce low temperature rises which
cannot be achieved using only next-generation SDRAMs,
by adopting a configuration composed of a plurality of
banks comprising a plurality of current-generation
SDRAMs, and a bank control unit for converting drive
20 signals from outside the module, which are sent in
order to control the plurality of banks, to signals for
controlling the plurality of banks.



[Patent Claims]

[Claim 1] A memory module composed of: a plurality of banks comprising a plurality of current-generation SDRAMs for achieving increased memory capacity with
5 current-generation SDRAMs, which cannot be achieved using only next-generation SDRAMs; and a bank control unit for converting drive signals from outside the module, which are sent in order to control the plurality of banks, to signals for controlling the
10 plurality of banks.

[Claim 2] The memory module as claimed in claim 1, composed of: a plurality of banks which are installed using an installation method enabling a plurality of
15 current-generation SDRAMs to be installed in multiple stages, such as TCP (tape carrier packaging); and a bank control unit for converting drive signals from outside the module, which are sent in order to control the plurality of banks, to signals for controlling the
20 plurality of banks.

[Claim 3] The memory module as claimed in claim 1 or 2, in which the bank control unit is a one-chip unit.

25 [Detailed Description of the Invention]

[0001]

[Technical Field of the Invention] The present invention relates to a memory module which is used in a computer or the like.

30

[0002]

[Prior Art] Known configurations for memory modules which are used in computers and the like have been disclosed in JEDEC documents in the past. Figure 4
35 shows a conventional memory module structure, which is a 128 MB SDRAM module comprising sixteen 64 Mbit, ×4 configuration SDRAMs.

[0003]

[Problem to be Resolved by the Invention] However, there are problems with memory modules having the abovementioned conventional configuration in that it is necessary to use 64 Mbit SDRAMs (referred to here as
5 "next-generation" components), and compared with configurations employing 16 Mbit SDRAMs (referred to here as "current-generation" components), configurations employing next-generation components are more costly until the value of SDRAMs is bit crossed.
10 There are further problems in that when use is made of $\times 1$ bit configuration current-generation components, the number of SDRAMs driven at the same time increases so a greater amount of power is consumed, and the heat generated by DRAMs causes the temperature of the module
15 to rise.

[0004] The aim of the present invention is to resolve these kinds of conventional problems by providing a large-capacity memory module which employs a multibit
20 configuration of current-generation components, and therefore makes it possible to reduce costs compared with using next-generation components, and also makes it possible to reduce the number of DRAMs driven at the same time compared with a system employing a current-
25 generation $\times 1$ bit configuration, so power consumption is reduced and temperature rises are lowered.

[0005]

[Means for Resolving the Problem] In order to resolve
30 the problem, the memory module according to the present invention is composed of: a plurality of banks comprising a plurality of current-generation SDRAMs for achieving increased memory capacity with current-generation SDRAMs, which cannot be achieved using only
35 next-generation SDRAMs; and a bank control unit for converting drive signals from outside the module, which are sent in order to control the plurality of banks, to signals for controlling the plurality of banks.

[0006]

By virtue of the configuration according to the present invention, it is possible to obtain a large capacity memory module at lower cost than a system employing
5 temperature next-generation components, and with lower temperature rises than a system employing a $\times 1$ configuration with current-generation components, so it is a simple matter to install a large capacity memory in a set such as a computer, and an inexpensive memory
10 module can be obtained.

[0007] Furthermore, it is also possible to reduce problematic temperature rises if the DRAMs are installed using an installation method enabling
15 multiple stage installation, such as TCP (tape carrier packaging), and therefore the installation density can be increased, the size of the module can be reduced, and it is possible to obtain a compact, large-capacity memory module.

20

[0008]

[Mode of Embodiment of the Invention] The invention disclosed in claim 1 of the present invention is a memory module composed of: a plurality of banks
25 comprising a plurality of current-generation SDRAMs for achieving increased memory capacity with current-generation SDRAMs, which cannot be achieved using only next-generation SDRAMs; and a bank control unit for converting drive signals from outside the module, which
30 are sent in order to control the plurality of banks, to signals for controlling the plurality of banks, so an action is afforded such that it is possible to obtain a large capacity memory module at lower cost than a system employing temperature next-generation
35 components, and with lower temperature rises than a system employing a $\times 1$ configuration with current-generation components, so it is a simple matter to install a large capacity memory in a set such as a

computer, and an inexpensive memory module can be obtained.

[0009] The invention disclosed in claim 2 is the memory
5 module as claimed in claim 1, composed of: a plurality
of banks which are installed using an installation
method enabling a plurality of current-generation DRAMs
to be installed in multiple stages, such as TCP (tape
10 carrier packaging); and a bank control unit for
converting drive signals from outside the module, which
are sent in order to control the plurality of banks, to
signals for controlling the plurality of banks, so an
action is afforded such that the memory module can be
made compact.

15 [0010] The invention disclosed in claim 3 is the
memory module as claimed in claim 1 or 2, in which the
bank control unit is a one-chip unit, so an action is
afforded such that it is possible to obtain a compact
20 memory module.

[0011] A mode of embodiment of the present invention
will be described below with the aid of Figures 1 to 3.
Figure 1 is a block diagram showing a plurality of
25 banks composed of a plurality of multibit DRAMs
according to a first mode of embodiment of the present
invention; Figure 2 is a block diagram of the bank
control unit for converting drive signals from outside
the module into signals for controlling the plurality
30 of banks; and Figure 3 is a timing chart showing the
operation of the bank control unit in Figure 2. In
Figure 1, D0-D63 are SDRAMs in four banks, where D0 -
D15 form bank 0, D16 - D31 form bank 1, D32 - D47 form
bank 2, and D48 - D63 form bank 3.

35 [0012] The operation of this mode of embodiment will be
described below with the aid of Figures 1, 2 and 3.
Drive signals /RE, /CE, /WE, A12, A13 etc. from outside
the module are input to the memory module. The bank

control unit shown in Figure 2 identifies SDRAM bank active commands (when /RE; "L", /CE; "H", /WE; "H") using the signals /RE, /CE, /WE, decodes A12, A13, performs bank selection, and sets /CS (chip select
5 signal), which corresponds to a selected bank, to active ("L").

[0013] Period (1) in Figure 3 involves active commands, as described above, and A12 is "L" and A13 is "L", so
10 two from among /CS0 to 7, namely /CS0 and /CS4 are set to "L" in order to select bank 0 which is composed of SDRAMs D0 - D15 shown in Figure 1, but the other banks are in an inactive state.

15 [0014] Period (2) in Figure 3 involves write commands to the SDRAMs, but writing is performed only to the selected bank 0 and the other banks 1 to 3 do not operate. Period (3) in Figure 3 involves precharge commands to the SDRAMs, and precharging is performed
20 only for the bank 0 and the other banks do not operate, in the same way as in period (2).

[0015] In the same way subsequently, in period (4) in Figure 3, only bank 1 is active, in period (7) in
25 Figure 3, bank 2 is active, and in period (10) in Figure 3, bank 3 is active. As a result, it is possible to achieve equivalent operation to that of a system in which a 64 bit memory is made up of the conventional 64 Mbit SDRAMs shown in Figure 4. Furthermore, the banks
30 are active one at a time, so in other words the active SDRAMs constitute 1/4 of the total number of SDRAMs in the memory module. Power consumption for the inactive SDRAMs is close to the power consumed during standby and is very low compared with the power consumed during
35 operation. It is therefore possible for the memory module to operate at approximately 1/4 power consumption.

[0016]

[Advantage of the Invention] The memory module obtained in the manner described above produces an advantageous effect in that low power consumption (low temperature rises) which can be achieved by using 64 Mbit SDRAMs
5 can be achieved using 16 Mbit SDRAMs, so costs can be reduced.

[0017] It should be noted that in this mode of embodiment, the current-generation components were 16
10 Mbit SDRAMs while the next-generation components were 64 Mbit SDRAMs, but the same effect can still be achieved if the generations are advanced so that the current-generation components are 64 Mbit SDRAMs while the next-generation components are 256 Mbit SDRAMs.

15

[Detailed Description of the Invention]

[Figure 1] is a block diagram showing the configuration of the plurality of banks composed of a plurality of multibit DRAMs, in accordance with a mode of embodiment
20 of the present invention;

[Figure 2] is a block diagram of the bank control unit for converting drive signals from outside the module into signals for controlling the plurality of banks, in accordance with a mode of embodiment of the present
25 invention;

[Figure 3] is a timing chart showing the operation of the bank control unit in Figure 2, in accordance with a mode of embodiment of the present invention; and

[Figure 4] is a block diagram of a memory module, which
30 is a conventional 128 MB SDRAM module comprising sixteen 64 Mbit, ×4 configuration SDRAMs.

[Key to Symbols]

D0 - D63 SDRAM (synchronous DRAM)

35

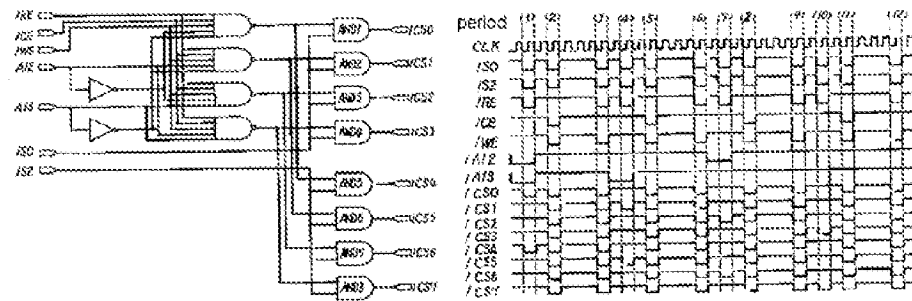
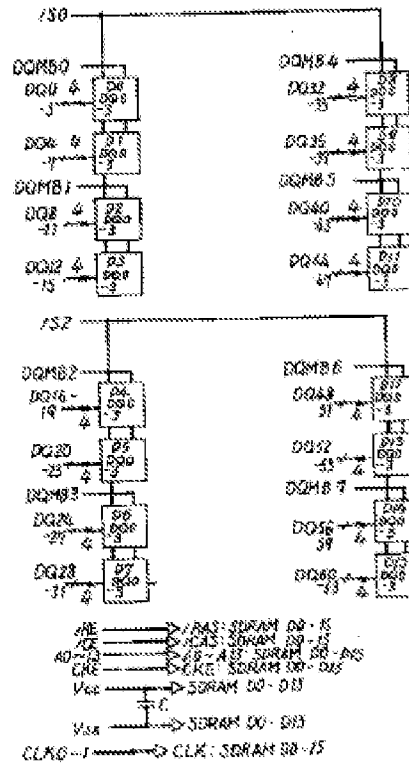


Figure 2
Figure 2
AT211

[Figure 4]





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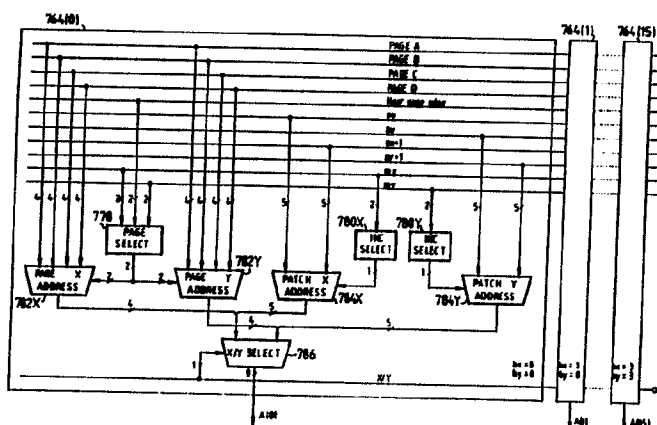
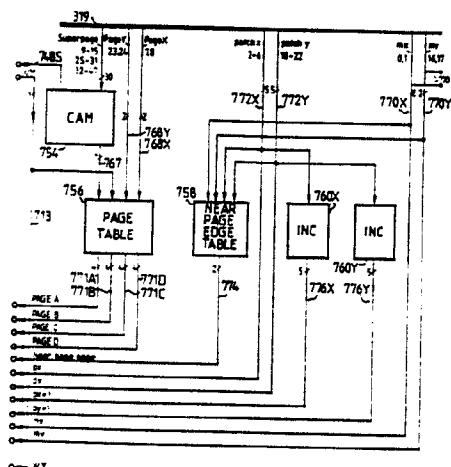
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(54) Title: VIRTUAL MEMORY SYSTEM



(57) Abstract

In a demand-paged virtual memory system, the pages are arranged in the virtual memory space in groups. In order to translate an address from the virtual address space to a physical memory address space, the virtual group address component is input to a contents addressable memory (CAM) (745) which outputs a group code (767), and the group code and virtual page address component (768X,Y) are input to a RAM page table (750) which outputs the page address. When the physical memory capacity is substantially smaller than the virtual address space, the CAM provides a large saving in page table size. In the case where the data-elements provide a plural-dimensional representation, for example as in pixel data, the pages include data elements which are contiguous in each of the plural dimensions in order to reduce the amount of page-swapping between the physical memory and a paging memory. The data-elements in the physical memory are accessible in parallel as contiguous patches. In order to allow parallel access of patches which are not aligned with respect to a page and therefore may fall partly on the edge of one page and partly on the edge of a page which is contiguous therewith in the virtual memory but not in the physical memory, the addressing means includes means (758) to determine whether a patch to be accessed has data-elements in different pages, and means (756, 778, 782X,Y) to modify the addresses to the physical memory.

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VIRTUAL MEMORY SYSTEM

The present invention relates to a demand-paged virtual memory system.

More particularly, the present invention is concerned with a memory system for storing an array of data elements, such as pixel data, which form in a virtual memory space contiguous aligned pages of contiguous data elements and comprising a first memory, or paging memory, for storing the data-array, a second memory, such as a video RAM, for storing part of the data-array, and means for transferring data-elements page-by-page between the first and second memories.

In a simple non-demand-paged system, the whole of the second memory could be filled with contiguous pages of the data elements. In this case, in order to translate between a page address in the virtual memory address space and a page address in the second memory address space, all that would be required would be to know the virtual page address of page zero in the second memory, and to use this as an offset between a second memory page address and the corresponding virtual memory page address. However, such a simple system is inflexible, and requires the whole content of the second memory to be swapped each time a new page is to be transferred from the first memory to the second memory.

In a more flexible system, that is a demand-paged system, pages of data-elements are transferred from the first memory to the second memory as required and placed in any available page location in the second memory. As a result, pages which are contiguous in the virtual memory address space are not necessarily stored contiguously in the second memory address space, and it is therefore necessary to provide a means to keep track of the pages in the second memory and to translate between the virtual page address and the corresponding second memory page address. This means could be provided by a RAM page table which has an address for each page in the virtual memory space and which stores

at that address in the page table a flag indicating whether that page is also stored in the second memory and if so also the address in the second memory of that page.

In the system with which the preferred embodiment of this invention is concerned, the virtual memory space contains 2^{34} pages (or 16 Gigapages) and the second memory space contains 2^8 (or 256) pages. Therefore, if a page table as described above were used in a system of such size, the page table would require a RAM of 242 bits (or 4 Terabits) for the addresses and a RAM of 2^{34} (or 16 Gigabits) for the flags.

A first aspect of the present invention is concerned with providing an address translation means which does not require tables of such massive size.

In the system in accordance with the first aspect of the present invention, the virtual memory address space is considered to be formatted as aligned groups of the pages, with the pages in each group being aligned and contiguous, and the translating means comprises a contents addressable memory (CAM) which receives the group component of the virtual address and outputs a group code corresponding thereto, and a page table which receives the group code and the page component of the virtual address and outputs the corresponding page address in the second memory address space.

Comparing the system of the first aspect of the invention with the example given above, if for example each group is chosen to contain 2^4 pages, then the virtual memory space will contain 2^{30} groups (or 1 Gigagroup). A CAM may then be chosen which has, say 2^7 (or 128) addresses, each with a 30 bit capacity. Thus the CAM will have a 30-bit input and a 7-bit group code output. The page table will therefore be addressed by 11 bits (comprising the 7-bit group code and then 4-bit page component) and will output the 8-bit page address in the second memory. Accordingly, the CAM stores 30×2^7 (or 3840 bits), and the page table stores $2^{11} \times 2^8$ bits (or $\frac{1}{2}$ Megabit), which in combination compare favourably with the 4 Terabits required by the earlier described system.

It will be appreciated that, for good advantage, the number of group codes which can be stored in the CAM is preferably substantially less than the number of the groups which can be stored in the first memory, or alternatively stated that each group code has a number of bits (e.g. 7) which is substantially less than the number of bits (e.g. 30) of the virtual group component.

In the preferred embodiment, the CAM is operable to compare the received virtual group component with the contents at the addresses of the CAM and to output as the group code the address of the CAM whose content matches the virtual group component. In this case, in order to flag locations in the CAM at which a virtual group component is not stored the CAM contents preferably have a bit (e.g. msb) which is not used in the virtual group component, and which is set at any location in the CAM at which a virtual group component is not stored.

In the preferred embodiment, the CAM is preferably operable to supply a group fault signal if a match is not made with the received virtual group component, and the system preferably further comprises processor means operable in response to the group fault signal to store the received virtual group component at an available location in the CAM. In this case the processor means is preferably operable in response to the group fault signal to determine whether there is any available location in the CAM, and if not to select a virtual group component to delete from the CAM, and to determine whether any pages of data-elements of the selected group are stored in the second memory, and if so to cause the transfer means to transfer those pages of data-elements from the second memory to the first memory. In order to do this, the system preferably further comprises means for determining the least recently used group having a page stored in the second memory, and the processor means is operable to select the virtual group component of the determined least recently used group as the virtual group component for deletion from the CAM.

In a conventional demand-paged virtual memory system, the virtual memory address space is one-dimensional, and the locations in each page

are adjacent in one dimension. The present invention is concerned more particularly, but not exclusively, concerned with processing of a data-array in which the data-elements form a plural (such as 2- or 3-) dimensional representation and in which the relative positions of the data-elements in the array are significant, in addition to their values, for example as in pixel data or vector data. In this case, if processing concerns a plurality of adjacent data-element arranged in the one dimension of the address space, there is a high probability that these data-elements will belong to the same page. However if processing concerns data-elements which are adjacent in some other direction, they will belong to different pages. Accordingly, in the case where the second memory is of substantially smaller capacity than the first memory, a vast amount of page swapping may be required between the first and second memories.

A second aspect of the present invention is concerned with a demand-paged virtual memory system in which the amount of page-swapping, when dealing with plural-dimensional data-arrays, is reduced. In accordance with this second aspect of the present invention, the pages which are transferred between the memories include data-elements which are contiguous in each of the plural dimensions of the representation provided by the data elements. For example, in the case of pixel data representing a two-dimensional image, each transferred page of pixel data is a two-dimensional portion of the image, for example of 128 pixels x 128 pixels. Similarly, in the case of three-dimensional pixel data, each transferred page would be a three dimensional portion of the image, for example of 32 pixels x 32 pixels x 16 pixels.

Preferably, the virtual address space and the second memory address space are plural dimensional and the addresses include components in each of the plural dimensions of the data-array.

In order to increase memory access speed and processing speed, it is desirable that a set or "patch" of contiguous data-element locations in the second memory can be accessed in parallel, for example so that the parallel-accessed data-elements can be processed in parallel by respective

ones of a set of processors. Conveniently, these patches are aligned with respect to the page so that the patches form a sub-array in each page. Despite this, the memory is preferably accessible so that a non-aligned patch can be accessed having some data-elements in one aligned patch and other data-elements in one or more adjacent aligned patches. Throughout most of the page, the data-elements of such misaligned patches will all be in the same page, but for patches on the edge of the page, some of the data-elements of the misaligned patch may belong to one or more different page. As described above with respect to the first aspect of the invention, contiguous pages in the virtual address space are not necessarily stored contiguously in the second memory address space, and therefore these misaligned patches on the page boundaries present a problem, because the page components of the addresses in the second memory of some of the data-elements in the patch will be entirely different to those of others in the patch.

In order to deal with this problem, in accordance with a third aspect of the present invention, the means for addressing the second memory to access in parallel a patch of data-elements which is not necessarily an aligned patch includes page edge means for determining whether the patch to be accessed includes data-elements in different pages in the second memory, and means to modify the addresses to the second memory in response to such a determination.

Preferably, the address of a patch to be accessed in the second memory has an aligned page address component, an aligned patch address component and a misalignment component indicating a misalignment of the patch to be accessed relative to an aligned patch having an address defined by the aligned page and aligned patch components, and in this case the addressing means preferably includes page determining means operable to determine, upon addressing, an aligned page address component for a basic aligned page for the patch to be accessed and an aligned page address component of at least one other page which is contiguous with the basic aligned page in the virtual address space.

In the case where the data-array is a plural-dimensional representation having D-dimensions, the page determining means is

preferably operable to determine the aligned page addresses in the second memory for all aligned pages which differ by one from the basic aligned page address in one direction in each of the D-dimensions in the virtual address space and in any combinations of those one directions.

In order to deal with the problem that not all required pages may be stored in the second memory, the system preferably further comprises page fault determining means for determining, upon addressing, whether the or each aligned page which is to be accessed is not stored in the second memory, the transferring means being operable to transfer the missing page from the first memory to the second memory in response to such a determination. Preferably, the page fault determining means is operable to indicate a page fault if the basic page is not stored in the second memory, or if any said other aligned page is not stored in the second memory and that page is required to be accessed. Preferably, the page edge determining means is operable to determine when the aligned patch address component has a maximum value for a patch in a page and the misalignment component together with said maximum value indicate that the patch to be accessed extends outside the page defined by the aligned page address component.

Preferred embodiments of the invention will now be described, by way of non-limiting example, with reference to the drawings, in which Figures 4 to 16, 20 and 21 are of major relevance to the present invention, and in which:

Figure 1 is a high-level schematic illustration of a computer system embodying the invention;

Figures 2 and 3 illustrations of modified forms of the system of Figure 1;

Figure 4 is an illustration in greater detail of a renderer employed in the systems of Figures 1 to 3;

Figure 5 is an illustration in greater detail of a front-end processor

board employed in the systems of Figures 1 to 3;

• Figure 6A and 6B show how patches of pixel data are made up;

• Figures 7A and 7B show how pages of patch data, and groups or 'superpages' of page data are made up;

Figure 8 is a schematic illustration of a physical image memory and the address lines therefor, used in the renderer of Figure 4;

Figure 9A is a 3-D representation of an aligned patch of data within a single page in the image memory;

Figure 9B is a 2-D representation of a page, showing the patch of Figure 9A;

Figure 10A is a 3-D representation of a non-aligned patch of data within a single page in the image memory;

Figure 10B is a 2-D representation of a page, showing the patch of Figure 10A;

Figure 11A is a 2-D representation of four pages in a virtual memory, showing a non-aligned patch which crosses the page boundaries and an enlargement of the circled part of the page boundary intersection;

Figure 11B is a 2-D representation of the physical memory illustrating locations of the four pages shown in Figure 11A;

• Figure 11C is a 3-D representation of the non-aligned patch of Figure 11A;

• Figure 12 is a truth table showing how page selection is made for patches which cross page boundaries;

Figure 13 shows two truth tables for selecting, respectively, X and Y patch address incrementation;

Figure 14 is a schematic illustration in greater detail of part of the renderer of Figure 4;

Figure 15 is a schematic illustration in greater detail of an address translator of Figure 14;

Figure 16 is an illustration of the operation of a contents addressable memory used in the address translator of Figure 15;

Figure 17 is a schematic illustration in greater detail of a read surface shifter used Figure 14;

Figure 18 shows in greater detail an array of multiplexers forming part of the surface shifter of Figure 17;

Figure 19 illustrates the translation made by the surface shifter of Figure 17;

Figure 20 is an illustration of the operation a least-recently-used superpage table which may be used with the address translator of Figure 15;

Figure 21 is a schematic diagram showing a page fault table which may be used with the address translator of Figure 15;

Figure 22 is a schematic diagram of an exchange and grid processor of the renderer of Figure 4;

Figure 23 is a flow diagram illustrating the operation of the processors and a priority encoder of the grid processor of Figure 22;

Figure 24 is a table giving an example of the operation of the priority encoder of Figure 22;

Figure 25 illustrates the correlation between aligned memory cells and two levels of a patch in a 16-bit split patch system;

Figures 26 and 27 show how pages of patch data, and superpages of page data are made up in a 16-bit split patch system;

Figures 28 and 29 correspond to Figures 26 and 27 respectively in an 8-bit split patch system;

Figures 30A to C shows modifications of parts of the address translator of Figure 15 used in the split patch system;

Figure 31 is a table to explain the operation of a funnel shifter used in the circuit of Figure 30A;

Figures 32 and 33 illustrate non-aligned split patches in a 16-bit and an 8-bit patch system, respectively;

Figure 34 shows a further modification of part of the address translator of Figure 15 used in the split patch system;

Figures 35A and 35B are tables which illustrate the operation of look-up tables in the circuit of Figure 34;

Figures 36A and 36B shows modifications of a near-page-edge table of Figure 15A used in the split patch system;

Figure 37 illustrates, in part, a modification to the exchange and grid processor of Figure 22 used in the split patch system;

Figures 38 and 39 are tables which illustrate the operation of further tables in a further modification of part of the address translator of Figure 15;

Figure 40 shows the further modification to Figure 15;

Figure 41 shows a modification to Figure 8 which is made in addition

to the modification shown in Figure 40;

Figure 42 is a representation of the VRAM memory space, showing how pages of data are rendered in one section of the memory and then copied to another monitoring section of the memory;

Figure 43 shows a circuit for determining which pages need not be copied from the rendering section to the monitoring section and to the virtual memory;

Figure 44 illustrates the setting and resetting of flags in a table of the circuit of Figure 43;

Figure 45A to 45C are flow diagrams illustrating the copying operations and Figure 45D shows the notation used in Figures 45A to 45C;

Figure 46 is a circuit diagram of a modification to the exchange of Figure 22;

Figure 47A to 47C are simplified forms of the circuit of Figure 46 when operating in three different modes;

Figure 48 shows a modification to part of the flow diagram at Figure 23;

Figure 49 is a schematic diagram of the processors and a microcode memory, with one of the processors shown in detail;

Figures 50A to 50D illustrate three images (Figs 50A to C) which are processed to form a fourth image (Fig. 50D); and

Figure 51 is a system diagram showing in particular a page filing system.

HARDWARE OVERVIEW

Figures 1 to 3 show three different hardware configurations of computer systems embodying the invention. Referring firstly to Figure 1, a host computer 10 has its own backplane in the form of a VME bus 12 which provides general purpose communications between various circuit boards of the computer, such as processor, memory and disk controller boards. To this known configuration, and within a standard housing 14 for the computer 10, there is added a board on which is provided a renderer 16 and a video processor 18, a Futurebus+ 20, and a front-end board 22. The renderer 16 is connected to the VME bus 12 and the Futurebus+ 20, and also communicates with the video processor 18, which in turn drives an external colour monitor 24 having a high-resolution of, for example, 1280 X 1024 pixels. The front-end board 22 is also connected to the Futurebus+ 20 and can communicate with a selection of peripherals, which are illustrated collectively by the block 26. The configuration of Figure 1 is of use when the host computer 10 has a VME backplane 12 and there is sufficient room in the computer housing 14 for the renderer 16, video processor 18, Futurebus+ 20 and front-end board 22, and may be used, for example, with a 'Sun Workstation'.

In the case where the computer housing 14 is physically too small, or where the host computer 10 does not have a VME or Futurebus+ backplane, the configuration of Figure 2 may be employed. In Figure 2, a separate housing 28 is used for the renderer 16, video processor 18, front-end board 22 and Futurebus+ 20, as described above, together with a VME bus 12 and a remote interface 30. In the host computer housing 14, a host interface 32 is connected to the backplane 34 of the host computer 10, which may be of VME, Qbus, Sbus, Multibus II, MCA, PC/AT, etc. format. The host interface 32 and remote interface 30 are connected by an asynchronous differential bus 36 which provides reliable communication despite the physical separation of the host and remote interfaces. The configuration of Figure 2 is appropriate when the host computer 10 is, for example, an 'Apple Mackintosh', 'Sun Sparkstation', 'IBM-PC', or Du Pont Pixel Systems bRISC'.

In the event that a host computer becomes available which has a Futurebus+ backplane and sufficient space in its housing for the additional components, then the configuration of Figure 3 may be employed. In Figure 3, the renderer 16 and the front-end board 22 are directly connected to the Futurebus+ backplane 20.

The general functions of the elements shown in Figures 1 to 3 will now be described in more detail. The host computer 10 supplies data in the form of control information, high level commands and parameters therefor to the renderer 16 via the VME backplane (Figure 1), via the backplane 34, host and remote interfaces 32, 30 and the VME bus 12 (Figure 2), or via the Futurebus+ backplane 20 (Figure 3). Some of this data may be forwarded to the front-end board via the Futurebus+ 20 (Figures 1 and 2), or sent direct via the Futurebus+ backplane 20 (Figure 3) to the front-end board 22.

The Futurebus+ 20 serves to communicate between the renderer 16 and the front-end processor 22 and is used, in preference to a VME bus or the like, in view of its high bit width of 128 bits and its high bandwidth of about 500 to 800 Mbytes/s.

As will be described in greater detail below, the renderer 16 includes an image memory, part of which is mapped to the monitor 24 by the video processor 18, and the renderer serves to perform image calculations and rendering, that is the drawing of polygons in the memory, in accordance with the commands and parameter supplied by the host computer 10 or the front-end board 22.

The front-end board 22 serves a number of functions. It includes a large paging RAM, which also interfaces with external disk storage, to provide a massive paging memory, and pages of image data can be swapped between the paging RAM and the image memory of the renderer 16 via the Futurebus+ 20. The front-end board also has a powerful floating-point processing section which can be used for graphics transformation and shading operations. Furthermore, the front-end board

may provide interfacing with peripherals such as a video camera or recorder, monitor, MIDI audio, microphone, SCSI disk and RS 232.

Overall, therefore, the renderer 16, video processor 18 and front-end board 22 can accelerate pixel handling aspects of an application, and also accelerate other computation intensive aspects of an application.

The renderer 16 and video processor 18 will now be described in greater detail with reference to Figure 4, which shows the main elements of the renderer 16 and the main data and address pathways.

The renderer 16 includes a 32-bit internal bus 300, a VME interface 301 which interfaces between the VME bus 12 (Figure 1) or the remote interface 30 (Figure 2) and the internal bus 300, and a Futurebus+ interface 302 which interfaces between the Futurebus+ 20 and the internal bus 300. Also connecting to the internal bus 300 are a control processor 314 implemented by an Intel 80960i, an EPROM 303, 4 or 16 Mbyte of DRAM 304, a real time clock and an I/O block 306 including a SCSI ports. The functions of the control processor 314 and the associated DRAM 304 and EPROM 303 are (a) to boot-up and configure the system; (b) to provide resource allocation for local PRAM 318, 322 of address and grid processors 310, 312 (described in detail below) to ensure that there is no memory space collision; (c) to control the loading of microcode into microcode memories 307, 308 (described below); (d) to run application specific remote procedure calls (RPCs); and (e) to communicate via the I/O block 306 with a diagnostics port of the host computer 10 to enable diagnostics information to be displayed on the monitor 24. The DRAM 304 can also be used as a secondary image page store for the VRAM 700 described below.

The renderer 16 also includes an address processing section 309 comprising an address broadcast bus 311 to which are connected 64 kbyte of global GRAM 316, a data/instruction cache 313 which also connects to the internal bus 300, an internal bus address generator 315 which also connects to the internal bus 300, an address processor 310 with 16 kbyte

of local PRAM 318, and a sequencer 317 for the address processor 310 which receives microcode from a microcode memory 307. The address processor 310 also connects to a virtual address bus 319. The main purpose of the address processing section 309 is to generate virtual addresses which are placed on the virtual address bus under control of microcode from the microcode memory 307.

Also included in the renderer 16 is an address translator 740 (described in further detail below) which receives the virtual addresses on the virtual address bus 319 and translates them into physical addresses of data in the video RAM 700, if the required data is present, or interrupts the address processor 310 to cause the required data to be swapped in from the paging RAM 304 or other page stores on the external buses, if the required data is not present in the VRAM 700.

The renderer 16 furthermore includes a data processing section 321 which is somewhat similar to the address processing section 309 and comprises a data broadcast bus 323, to which are connected 64 kbyte of global GRAM 324, a diagnostics register 325 which also connects to the internal bus 300 and which may be used instead of the I/O block 306 to send diagnostics information to the host computer 10, an internal bus address generator 327 which also connects to the internal bus 300, a grid processor 312 having sixteen processors each with 8 kbyte of local PRAM 322, and a sequencer 329 for the grid processor 312 which receives microcode from a microcode memory 308. The processors of the grid processor 312 also connect to a data bus 331. The main purpose of the data processing section 321 is to receive data on the data bus 331, process the data under control of microcode from the microcode memory 308, and to put the processed data back onto the data bus 331.

The physical VRAM 700 connects with the data bus 331 via an exchange 326 which is described in detail below, but which has the main purposes of shuffling the order of the sixteen pixels read from or written to the VRAM 700 at one time, as desired, to enable any of the sixteen processors in the grid processor 312 to read from or write to any of the

sixteen addressed locations in the VRAM 700 and to enable any of the sixteen processors to transfer pixel data to any other of the sixteen processors.

The last main element of the renderer 16 is a bidirectional FIFO 332 connecting between the broadcast buses 311, 323 of the address and data processing sections 309, 321, which enables virtual addresses to be transferred directly between these two sections.

The front-end board 22 will now be described in greater detail with reference to Figure 5.

The front-end board 22 has an internal bus 502 which communicates with the Futurebus+ 20. A paging memory section 504 is connected to the internal bus 502 and comprises a large paging RAM 506 of, for example, 4 to 256 Mbytes capacity which can be used in conjunction with the DRAM 304 of the renderer, a paging memory control processor 508, and connections to, for example, two external high speed IPI-2 disk drives 510 (one of which is shown) each of which may have a capacity of, for example, 4 Gbytes, and a data communication speed of 50 Mbytes/s, or two external SCSI drives. The paging RAM 506 enables an extremely large amount of pixel data to be stored and to be available to be paged into the renderer 32 as required, and the fast disk 510 enables even more pixel data to be available ready to be transferred into the paging RAM 506.

Floating point processing is provided by 1 to 4 Intel 80860 processors 516, each rated at 80 MFlops peak. The general purpose processing power can be used on dedicated tasks such as geometric pipeline processing, or to accelerate any part of an application which is compute-intensive, such as floating point fast Fourier transforms. Each of the floating point processors 516 has a 128KByte secondary cache memory 518 in addition to its own internal primary cache memory.

The front-end board 22 may also, if desired, include a broadcast

standard 24-bit frame grabber connected to the internal bus 502 and having a video input 514 and output 516 for connection to video camera or television-type monitor.

The front-end board 22 may also, if desired, include an input/output processor 520 which provides interfacing with MIDI on line 522, SCSI disk on line 524, at least one mouse on line 526, RS232 on line 528, and audio signals on line 530 via a bi-directional digital/analogue convertor 532.

VIDEO RAM AND ADDRESSING THEREOF

Now that an overview of the hardware of the whole system has been set out, the image memory configuration will be described in more detail.

As mentioned above, the VRAM has a of 16 Mbyte capacity. The system is capable of operating with 32-bit pixels, and therefore the image memory has a capacity of $16M \times 8/32 = 4$ Mpixels. As illustrated in Figures 6A and 6B, pixels are arranged in 4×4 groups referred to as 'patches'. Figures 6A and 6B show, respectively, two-and one-dimensional notations for designating a pixel in a patch, as will be used in the following description. In turn, as illustrated in Figure 7A, the patches are arranged in 32×32 groups referred to as 'pages'. Furthermore, as illustrated in Figure 7B, the pages are arranged in 4×4 groups referred to as 'superpages'. The VRAM therefore has a capacity of 4 Mpixels, or 256k patches, or 256 complete pages, or 16 complete superpages. However, not all pages of a particular superpage need be stored in the memory at any one time, and support is provided for pages from parts of up to 128 different superpages to be stored in the physical memory at the same time.

The VRAM 700 and addressing lines therefor are shown schematically in Figure 8. Each small cube 702 in Figure 8 represents a 32-bit pixel. The pixels are arranged in 512 pixel $\times$ 512 pixel banks B(0) -

B(15) lying in the XY plane, and these pixel banks are 16 pixels deep (in the P direction). A line of 16 pixels in the P direction provides an aligned patch 704. The pixels in each bank are addressable as to X address by a respective one of 16 9-bit X address lines AX(0) to AX(15) and are addressable as to Y address by a respective one of 16 9-bit Y address lines AY(0) to AY(15). The Y and X addresses are sequentially supplied on a common set of 16 9-bit address lines A(0) to A(15), with the Y addresses being supplied first and latched in a set of 16 9-bit Y latch groups 706-0 to 706-15 each receiving a row address strobe (RAS) signal on 1-bit line 708, and the X addresses then being supplied and latched in a set of 16 9-bit X latch groups, 707-0 to 707-15 each receiving a respective column address strobe signal CAS(0) to CAS(15) on lines 709(0) to 709(15), respectively.

The memory for each XY bank of pixels (512 pixels x 512 pixels x 1 pixel) is implemented using eight video-RAM (VRAM) chips 710, each 256 K (4-bit) nibbles. Each chip provides a one-eighth thick slice of each pixel bank, whereby $8 \times 16 = 128$ chips are required. Each Y latch group and X latch group comprises eight latches (shown in detail for Y latch group 706(1) and X latch group 707(1) and a respective one of the X and Y latches is provided on each VRAM chip 710.

In this specification, the banks of memory will sometimes be referred to by the bank number B(0) to B(15) and at other times by a 2-dimensional bank address (bx,by) with the correlation between the two being as follows:

<u>Bank Number</u>	<u>(bx,by)</u>	<u>Bank Number</u>	<u>(bx,by)</u>
B(0)	(0,0)	B(8)	(0,2)
B(1)	(1,0)	B(9)	(1,2)
B(2)	(2,0)	B(10)	(2,2)
B(3)	(3,0)	B(11)	(3,2)
B(4)	(0,1)	B(12)	(0,3)
B(5)	(1,1)	B(13)	(1,3)
B(6)	(2,1)	B(14)	(2,3)
B(7)	(3,1)	B(15)	(3,3)

When a location in the memory 700 is to be accessed, a patch of 16 pixels is made available for reading or writing at one time. If the Y address and X address for all of the VRAMs 710 is the same, then an "aligned" patch of pixels (such as patch 704) will be accessed. However, it is desirable that access can be made to patches of sixteen pixels which are not aligned, but where various pixels in the patch to be accessed are derived from two or four adjacent aligned patches.

It will be appreciated that access to an aligned patch in memory is more straightforward than access to a non-aligned patch, because for an aligned patch the (x,y) address of each pixel in the different XY planes of memory as shown in Figure 8 is the same. Furthermore, the (x,y) address of each pixel in the patch is equal to the bank address (bx,by) of the memory cell from which that pixel is derived. Referring to Figures 9A and 9B, an aligned patch "a" having a patch address (12, 16) in a page "A" having a page address (8, 6) is shown, as an example. The pixels in the aligned patch all have the same address in the sixteen XY banks of the memory, as represented in Figure 9A, and when displayed would produce a 4 X 4 patch of pixels offset from the page boundaries by an integral number of patches, as represented in Figure 9B. In the particular example the absolute address of the aligned patch in the memory would be $(8 \times 32 + 12, 6 \times 32 + 16) = (268, 208)$.

If, however, a patch "p" is non-aligned, and has a misalignment $(mx, my) = (2, 1)$, for example, from the previously considered aligned patch "a" at patch address (12, 16) in page A at page address (8, 6), then some of the pixels of patch "p" will need to be derived from three other aligned patches "b", "c" and "d" having patch addresses $(12 + 1, 16)$, $(12, 16 + 1)$ and $(12 + 1, 16 + 1)$, or (13, 16), (12, 17) and (13, 17) in page A at page address (8, 6). This situation is represented in Figures 10A and 10B. The absolute address of these patches "b", "c" and "d" in the VRAM 700 are (269, 208), (268, 209) and (269, 209); respectively.

A further problem which arises in accessing a non-aligned patch "p" is that the (x,y) address of each pixel in the patch "p" does not correspond to

the bank address (bx,by) in the memory from which that pixel is derived. In the particular example, the following pixel derivations and translations are required.

Address (x,y) of pixel in non-aligned <u>patch "p"</u>	Aligned patch (px,py) from which pixel <u>is derived</u>	Bank address (bx,by) from which pixel <u>is derived</u>	Translation required from bank address (bx,by) to pixel address (x,y) in <u>patch "p"</u>
(0,0)	a (12,16)	(2,1)	(-2,-1)
(1,0)	a (12,16)	(3,1)	(-2,-1)
(2,0)	b (13,16)	(0,1)	(-2,-1) mod 4
(3,0)	b (13,16)	(1,1)	(-2,-1) mod 4
(0,1)	a (12,16)	(2,2)	(-2,-1)
(1,1)	a (12,16)	(3,2)	(-2,-1)
(2,1)	b (13,16)	(0,2)	(-2,-1) mod 4
(3,1)	b (13,16)	(1,2)	(-2,-1) mod 4
(0,2)	a (12,16)	(2,3)	(-2,-1)
(1,2)	a (12,16)	(3,3)	(-2,-1)
(2,2)	b (13,16)	(0,3)	(-2,-1) mod 4
(3,2)	b (13,16)	(1,3)	(-2,-1) mod 4
(0,3)	c (12,17)	(2,0)	(-2,-1) mod 4
(1,3)	c (12,17)	(3,0)	(-2,-1) mod 4
(2,3)	d (13,17)	(0,0)	(-2,-1) mod 4
(3,3)	d (13,17)	(1,0)	(-2,-1) mod 4

From the right hand column above, it will be noted that the translation from the bank address (bx,by) to the corresponding address in the non-aligned patch is constant for a particular non-aligned patch and in particular is equal to the negative of the misalignment (mx,my) of the non-aligned patch "p" from the base aligned patch "a", all translations being in modulus arithmetic with the modulus equal to the patch dimension.

Yet another further complication arises with non-aligned patches, and that is that the patch may extend across the boundary between two or four pages. To provide flexibility, not all pages which make up an image and which are contiguous in the virtual address space need to be stored in the VRAM at one time, and pages are swapped between the paging memory and the VRAM as required. This results in those pages making up an image which are in the VRAM not necessarily being stored adjacent each other in the VRAM, but possibly being scattered in non-contiguous areas of the VRAM.

For example, Figure 11A represents four contiguous pages A, B, C, D in the virtual address space. When these pages are swapped into the physical memory 700, they may be scattered at, for example, page addresses (8,6), (4,8), (12, 12) and (6,10) in the VRAM, as represented in Figure 11B. Now, if it is desired to access a non-aligned patch "p" whose base aligned patch "a" in page A has an x or y patch address of 31, then the non-aligned patch "p" may extend into page B, page C or pages B, C and D, depending on the direction of the misalignment. In the example shown specifically in Figure 11, the patch "p" to be accessed has a misalignment $(mx, my) = (2, 1)$ relative to base aligned patch "a" having patch address $(px, py) = (31, 31)$ in page A having page address (8,6) in the VRAM. It will be appreciated that, in addition to translating accessed pixels between their bank addresses (bx, by) and the addresses (x, y) in the non-aligned patch as described above with reference to Figure 10, it is also necessary to determine the various pages B, C, D which are to be accessed in addition to the basic page A and furthermore to determine the addresses in these other pages B, C, D of the aligned patches to be accessed, it being noted in the example that although the aligned patch "a" in page A has a patch address of (31,31), different patch addresses need to be used in other the pages B, C, D, that is (0,31), (31,0) and (0,0), respectively. The following table sets out, for each of the pixels in the patch "p" to be accessed: the page and patch address of the aligned patch from which that pixel is derived; the translation necessary from the patch address of the basic patch "a" in page A to the patch address of the patch from which the pixel is derived; the bank address from which the pixel is derived; and the translation necessary from this latter address to

the address of the pixel in the patch "p".

Address (x,y) of pixel in non-aligned <u>patch "p"</u>	Page and aligned patch from which pixel is derived <u>P/(px,py)</u>	Translation necessary from address of patch "a" to address (px,py) of patch from which pixel is <u>derived</u>	Bank address (bx,by) of <u>pixel</u>	Translation necessary from bank address (bx,by) to pixel address (x,y) in <u>patch "p"</u>
(0,0)	A/(31,31)a	(0,0)	(2,1)	(-2,-1)
(1,0)	A/(31,31)a	(0,0)	(3,1)	(-2,-1)
(2,0)	B/(0,31)b	(1,0) mod 32	(0,1)	(-2,-1) mod 4
(3,0)	B/(0,31)b	(1,0) mod 32	(1,1)	(-2,-1) mod 4
(0,1)	A/(31,31)a	(0,0)	(2,2)	(-2,-1)
(1,1)	A/(31,31)a	(0,0)	(3,2)	(-2,-1)
(2,1)	B/(0,31)b	(1,0) mod 32	(0,2)	(-2,-1) mod 4
(3,1)	B/(0,31)b	(1,0) mod 32	(1,2)	(-2,-1) mod 4
(0,2)	A/(31,31)a	(0,0)	(2,3)	(-2,-1)
(1,2)	A/(31,31)a	(0,0)	(3,3)	(-2,-1)
(2,2)	B/(0,31)b	(1,0) mod 32	(0,3)	(-2,-1) mod 4
(3,2)	B/(0,31)b	(1,0) mod 32	(1,3)	(-2,-1) mod 4
(0,3)	C/(31,0)c	(0,1) mod 32	(2,0)	(-2,-1) mod 4
(1,3)	C/(31,0)c	(0,1) mod 32	(3,0)	(-2,-1) mod 4
(2,3)	D/(0,0)d	(1,1) mod 32	(0,0)	(-2,-1) mod 4
(3,3)	D/(0,0)d	(1,1) mod 32	(1,0)	(-2,-1) mod 4

A representation of the locations of the pixels in the four aligned patches is shown in Figure 11C.

In the example, the basic patch "a" has a patch address (px,py) of (31,31) and the non-aligned patch "p" to be accessed has a misalignment (mx,

my) of (2,1) relative to the basic patch "a". In the general case of a base patch address (px, py), where $0 \leq px, py \leq 31$, and a misalignment (mx, my), where $0 \leq mx, my \leq 3$, the table of Figure 12 sets out which page A, B, C or D should be used when accessing a pixel at bank address (bx,by), where $0 \leq bx, by \leq 3$, in dependence upon bx, by, mx, my, px and py, and the table of Figure 13 sets out the X patch address px, or $px + 1 \bmod 4$, and the Y patch address py, or $py + 1 \bmod 4$, which should be used in order to obtain the address of the aligned patch a, b, c or d to be accessed, in dependence upon bx, by, mx and my. The increment is calculated using modular arithmetic of base 32. It is also to be noted that for all pixels where $(mx, my) \neq (0,0)$, a translation of $(-mx, -my)$ is required between the bank address (bx,by) from which the pixel is derived and the address (x,y) of the pixel in the non-aligned patch "p".

Having described various addressing functions which it is required to be performed, there now follows a description in greater detail of the apparatus for performing these functions.

As described above with reference to Figure 4, the VRAM 700 is addressed by the address processor 310 via the address translator 740, communicates data with the grid processor 312 via the exchange 326 and provides data to the video processor 34. A greater degree of detail of the address translator, VRAM and exchange is shown in Figure 14.

The address translator 740 receives a 48-bit virtual address on bus 319 of a patch origin address. The translator determines whether the required page(s) to access the addressed patch are resident in the VRAM physical memory 700. If not, a page or superpage fault is flagged on line 748, as will be described in detail below. However, if so, the address translator determines the addresses in the sixteen XY banks of the physical memory of the sixteen pixels making up the patch, and addresses the memory 700 firstly with the Y addresses on the sixteen sets of 9-bit lines A(0) to A(15) and then with the X addresses on these lines. The X and Y addresses are generated under control of the X/Y select signal on line 713.

The exchange 326 includes a read surface shifter 742 and a write surface shifter 744. Pixel data is transferred, during a read operation, from the memory 700 to the read surface shifter 742 by a set of sixteen 32-bit data lines $D''(0)$ to $D''(15)$, and, during a write operation, from the write surface shifter 744 to the memory 700 by the same data lines $D''(0)$ to $D''(15)$. The read and write surface shifters 742, 744 receive 4-bit address data from the address translator on line 770, consisting of the least significant two bits of the X and Y address data. This data represents the misalignment (mx, my) of the accessed patch "p" from the basic aligned patch "a". The purpose of the surface shifters is re-order the pixel data in non-aligned patches, that is to apply the translation $(-mx, -my)$ when reading and an opposite translation (mx, my) when writing. Pixel data to be written is supplied by a crossbar 327 forming part of the exchange 326 to the write surface shifter 744, and pixel data which has been read is supplied by the read surface shifter 742 to the crossbar 327, on the 512-bit line 750 made up of a set of 16 32-bit lines. The write surface shifter also receives on line 745 16-bit write enable signals $WE(0) - WE(15)$ from the crossbar 327 one for each pixel, and the write surface shifter 744 re-organises these signals in accordance with the misalignment (mx, my) of the patch "p" to be accessed to provide the sixteen column write enable signals $WE''(0)$ to $WE''(15)$. Each of these signals is then ANDed with a common CAS signal on line 709 to form sixteen CAS signals $CAS(0)$ to $CAS(15)$, one for each of the sixteen banks of memory. This enables masking of pixels within a patch during writing, taking into account any misalignment of the patch.

The address translator 740 will now be described in more detail primarily with reference to Figure 15. The translator 740 includes as shown, a contents addressable memory (CAM) 754, a page address table 756, a near-page-edge table 758, and X and Y incrementers 760X, 760Y. The translator 740 also includes sixteen sections 764(0) to 764(15), one for each output address line $A(0)$ to $A(15)$, and thus for each memory bank $B(0)$ to $B(15)$.

The translator 740 receives a 48-bit virtual address of the origin $(0,0)$ pixel of a patch on the bus 319. It will therefore be appreciated that up to 2^{48} (i.e. 281, 474, 976, 710, 656) different pixels can be addressed. Many

formats of the 48-bit address can be employed, but the following example will be considered in detail.

	<u>BITS</u>	<u>IDENTITY</u>
LSB	0,1	X misalignment (mx) of patch (p) to be accessed relative to basic aligned patch (a)
	2-6	X address (px) of aligned patch (a) in page A
	7,8	X address of page A
	9-15	X portion of superpage address
	16,17	Y misalignment (my) of patch (p) to be accessed relative to basic aligned patch (a)
	18-22	Y address (py) of aligned patch (a) in page A
	23,24	Y address of page A
	25-31	Y portion of superpage address
MSB	32-47	Image ID portion of superpage address

The bits identifying the superpage (i.e. bits 9 to 15, 25 to 31 and 32 to 47) are supplied to the CAM 754. The CAM 754 is an associative memory device which compares the incoming 30-bit word with all of the words held in its memory array, and if a match occurs it outputs the location or address in the memory of the matching value on line 767. The CAM 754 has a capacity of 128 32-bit words. Thirty of these bits are used to store the virtual address of a superpage which is registered in the CAM 754. Thus up to 128 superpages can be registered in the CAM. One of the other bits is used to flag any location in the CAM which is unused. The remaining bit is spare. Figure 16 illustrates how the CAM 754 operates. Upon input of a 30-bit superpage address, e.g. 01234569 (hex), this input value is compared with each of the contents of the CAM. If a match is found and provided the unused flag is not set, the address in the CAM of the match is output, e.g. 1 in the illustration. If no match is found with the contents at any of the 128 addresses of the CAM, then a superpage fault is flagged on line 748S, and the required superpage is then set up in the CAM in the manner described in

detail later.

Referring back to Figure 15, the 7-bit superpage identification output from the CAM 754 on line 767 is used as part of an address for the page address table 756, implemented by a 4k word x 16-bit SRAM. The remaining 5 bits of the address for the page table 756 are made up by: bits 7, 8, 23 and 24 of the virtual address which identify the page within a superpage; and an X/Y select signal on line 713. The page table 756 has registered therein the X and Y page addresses in the VRAM 700 of: a) the basic page A in which the pixel to be accessed is located; b) the page B which is to the right of the page A in the virtual address space; c) the page C which is above the page A in the virtual address space; and d) the page D which is to the right of page C and above page B in the virtual address space, and these addresses are output on lines 771A to 771D, respectively. If these pages A to D are required, but are not stored in the VRAM 700 and thus are not registered in the page table 756, then a page fault is flagged on a line 748p (as described below with reference to Figure 21) and the required page of data is then swapped into the VRAM 700 in the manner described in detail below. However, if all of the pages A to D which may possibly need to be accessed are stored, their addresses are made available on the lines 771A to 771D to all of the sections 764(0) to 764(15), the Y or X addresses being output depending on the state of the X/Y select signal on line 713.

Bits 2 to 6 and 18 to 22 of the virtual address are also supplied to each of the sections 764(0) to 764(15) on lines 772X and 772Y. These denote the patch address (px, py). The X and Y patch addresses together with bits 0, 1, 16 and 17 of the virtual address (which indicate the misalignment mx, my of the patch p to be accessed) are also supplied to the near-page-edge table 758, implemented using combinatorial logic, which provides a 2-bit output to the sections 764(0) to 764(15) on line 774, with one bit being high only if the patch X address px is 31 and the X misalignment mx is greater than zero and the other bit being high only if the patch Y address py is 31 and the Y misalignment my is greater than zero.

Furthermore, the X and Y patch addresses (px, py) are also supplied

to the X and Y incrementers 760X, 760Y, and these incrementers supply the incremented values $px + 1, \text{ mod } 32$ and $py + 1, \text{ mod } 32$, to each of the sections 764(0) to 764(15) on lines 776X, 776Y.

The four bits 0,1, 16 and 17 giving the misalignment mx and my are also supplied to the sections 764(0) to 764(15) on lines 770X, 770Y and are also supplied to the surface shifters 742, 744 on line 770.

Each section 764(0) to 764(15) comprises: a page selection logic circuit 778; X and Y increment select logic circuits 780X 780Y; X and Y 4:1 4-bit page address multiplexers 782X, 782Y; X and Y 2:1 5-bit patch address multiplexers 784X, 784Y; and a 2:1 9-bit address selection multiplexer 786.

The page selection logic circuit 778 implemented using combinatorial logic, provides a 2-bit signal to the page address multiplexers 782X,Y to control which page address A, B, C or D to use. The page selection logic circuit 778 performs this selection by being configured to act as a truth table which corresponds to the table of Figure 12. The circuit 778 receives the 2-bit signal on line 774 from the near-page-edge table 758 and this determines which of the four columns of the table of Figure 12 to use. The circuit 778 also receives the misalignment (mx, my) on lines 770X, 770Y, and this data in combination with which section 764(0) to 764(15) (and thus which bx and by applies) determines which of the four rows in Figure 12 to use. The X and Y page address multiplexers 782X, 782Y therefore supply appropriate page address as four bits to complementary inputs of the X/Y address selection multiplexer 786.

The increment selection logic circuits 780X, 780Y, which are implemented using combinatorial logic, receive the respective X and Y misalignments mx, my and provide respective 1-bit signals to control the patch address multiplexers 784X, 784Y. The increment selection circuits perform this selection by being configured to act as truth tables which correspond to the upper and lower parts, respectively, of the table of Figure 13. It will be noted that selection depends upon the misalignment mx or my in combination with the bx or by position of the memory bank (and thus

which of the sections 764(0) to 764(15) is being considered). The X and Y patch address multiplexers 784X, 784Y therefore output the appropriate 5-bit patch addresses px or $px + 1 \pmod{32}$ and py or $py + 1 \pmod{32}$ which are combined with the X and Y page addresses at the inputs to the X/Y selection multiplexers 786. This latter multiplexer receives as its control signal the X/Y selection signal on line 713 and therefore outputs the 9-bit X or Y address appropriate to the particular section 764(0) to 764(15).

The address translator 740 therefore deals with the problems described above of addressing pixels from different aligned patches a, b, c, d in the memory 700 when a patch "p" to be accessed is misaligned, and of addressing pixels from different pages A, B, C, D in the memory 700 when a patch "p" to be accessed extends across the boundary of the basic page A.

It is still necessary also to perform a translation of the pixel positions in the accessed patch of $(-mx, -my)$ if reading, or (mx, my) if writing. This is performed by the surface shifter 742 for reading and the surface shifter 744 for writing. The read surface shifter 742 will now be described with reference to Figures 17 and 18.

The read surface shifter 742 comprises a pair of 4×4 32-bit barrel shifters, 788X, 788Y. The X barrel shifter 788X has four banks 790X(0) to 790X(3) of multiplexers arranged in one direction, and the outputs of the X barrel shifter 788X are connected to the inputs of the Y barrel shifter 788Y, which has four banks 790Y(0) to 790Y(3) of multiplexers arranged in the orthogonal direction. As control signals, the X and Y barrel shifters 788X, Y receive the X and Y misalignments mx , my , respectively.

One of the banks of multiplexers 790X(0) is shown in greater detail in Figure 18, and comprises four 32-bit 4:1 multiplexers 792(0) to 792(3). The data from bank (0,0) is supplied to inputs 0, 3, 2 and 1, respectively, of the multiplexers 792(0) to 792(3). The data from bank (1,0) is supplied to inputs 1, 0, 3 and 2, respectively, of the multiplexers 792(0) to 792(3). The data from bank (2,0) is supplied to the inputs 2, 1, 0 and 3, respectively, of the multiplexers 792(0) and 792(3). The remaining data from bank (3,0) is

supplied to the remaining inputs 3, 2, 1, 0, respectively, of the multiplexers 792(0) to 792(3). The other banks of multiplexers 790X(1) to 790X(3) in the X barrel shifter 788X are similarly connected, and the banks 790Y(0) to 790Y(3) in the Y barrel shifter 788Y are also similarly connected. It will therefore be appreciated that the read surface shifter performs a translation with wrap-around in the -X direction of mx positions and a translation with wrap-around in the -Y direction of my positions as shown in Figure 19.

As shown in the drawings, the write surface shifter 744 may be provided by a separate circuit to the read surface shifter. In this case the write surface shifter is configured similarly to the read surface shifter, except that the inputs 1 and 3 to the multiplexers 792 in the barrel shifter banks are transposed. This results in translations of +mx and +my in the X and Y directions, rather than -mx and -my for the read surface shifter. The part of the write surface shifter which operates on the write enable signals WE(0) to WE(15) is identical to the part which operates on the data signals, except that the signals are 1-bit, rather than 32-bit.

As an alternative to employing separate circuits for the read and write surface shifters 742, 744, a single circuit may be employed, with appropriate data routing switches, and in this case translation provided by the surface shifter may be switched between (-mx, -my) and (+mx, +my), in dependence upon whether the memory is being read or written, as described with reference to Figures 46 and 47.

As mentioned above, if a required superpage is not registered in the CAM 754, then a superpage fault is flagged, on line 748S. This superpage fault is used to interrupt the address processor 310, which is programmed to perform a superpage interrupt routine as follows. Firstly, the address processor checks whether the CAM 754 has any space available for a new superpage to be registered. If not, the address processor selects a registered superpage which is to be abandoned in the manner described below and causes the, or each, page of that superpage which is stored in the VRAM 700 to be copied to its appropriate location in the paging memory.

The registration of that superpage is then cancelled from the CAM 754. Secondly, the new superpage is registered in the CAM 754 at the, or one of the, available locations.

In order to select which superpage to abandon, a determination is made as to which is the least recently used (LRU) superpage which is registered in the CAM 754. To do this, a 128 x 16-bit LRU table 802 is provided, as illustrated in Figure 20. Each of the 128 addresses represents a respective one of the superpages registered in the CAM 754. The 7-bit superpage identification output from the CAM 754 on line 767 is used to address the LRU table 802 each time the superpage identification changes, as detected by the change detector 804. The change detector 804 also serves to increment a 16-bit counter 806, and the content of the counter 806 is written to the addressed location in the LRU table 802.

Accordingly, for all of the registered superpages, the LRU table contains an indication of the order in which those superpages were last used. When registering a new superpage in the CAM 754, the address processor 310 checks the contents of the LRU table 802 to determine which superpage has the lowest count and in that way decides which superpage to abandon.

As also mentioned above, if the required pages of the registered superpage are not stored in the VRAM 700, a page fault is flagged, on line 748P. The page fault generator is shown in Figure 21, and comprises a page fault table 794 constituted by a 2k x 4-bit SRAM, a set of three AND gates 796B, C, D and an OR gate 798. The page fault table 794 is addressed by the 7-bit superpage identity code on line 767, and by the X and Y page addresses on line 768X, Y. At each address, the page fault table 794 contains a 4-bit flag in which the bits denote whether the basic addressed page A and the pages B, C and D, respectively, to the right, above, and to the right and above, page A are stored in the VRAM 700. The page B flag is ANDed by gate 796B with the bit of the near-page-edge signal on line 774 denoting whether the patch "p" to be accessed extends across the boundary between pages A and B. Similarly, the page C flag is ANDed by gate 796C with the bit of the near-page-edge signal on line 774 denoting whether the

patch "p" to be accessed extends across the boundary between pages A and C. Furthermore, the page D flag is ANDed by gate 796D with both bits of the near-page-edge signal, which in combination denote whether the patch "p" to be accessed extends in page D above page B and to the right of page C. The outputs of the three AND gate 796B, C, D and the page A flag are then ORed by the OR gate 798, the output of which provides the page fault flag on line 748P.

From the above, it will be appreciated that a page fault is always generated if the basic page A is not stored in the VRAM, but if page B, C or D is not stored in the memory, a page fault will be generated in response thereto only if the respective page B, C or D will be used, as indicated by the two bits of the near-page-edge signal on line 774.

The page fault signal on line 748P is used to interrupt the address processor 310. The address processor then searches a table in its PRAM 318 for a spare page address in the VRAM 700, causes the required page to be swapped into the VRAM at the spare page address, and update the table in its PRAM 318.

GRID PROCESSOR AND EXCHANGE

As described above with reference to Figure 4, in the operation of the preferred embodiment, the exchange 326 and the VRAM 700 communicate in patches of sixteen pixels of data, each pixel having 32 bits. Furthermore, the grid processor 312 has sixteen processors, each of which processes pixel data and communicates with the exchange 326. Also, the grid processor 312 and the address processor 310 can communicate address data via the FIFO 332.

The exchange 326 includes a crossbar 377, and a logical implementation of the crossbar 377 and of the grid processor 312 is shown in more detail in Figure 22. As shown, the crossbar 377 comprises sixteen 16:1 32-bit data multiplexers 602(0) to 602(15); sixteen 16:1 1-bit write enable multiplexers 603(0) to 603(15); a 512-bit bidirectional FIFO 604 for pixel

data; and a 16-bit bidirectional FIFO 605 for the write enable signals. During a read operation, the 16 pixels of a 4 x 4 patch are supplied from the VRAM 700 (Figure 8) via the read surface shifter 742 and via the FIFO 604 as data D(0) to D(15) to the sixteen inputs of each data multiplexer 602(0) to 602(15). During a write operation, the data multiplexers 602(0) to 602(15) supply data D(0) to D(15) via the FIFO 604 and the write surface shifter 744 to the VRAM and the write enable multiplexers 603(0) to 603(15) supply write enable signals WE(0) to WE(15) via the FIFO 605 to the write surface shifter 744. The FIFOs 604, 605 and also the FIFO 332 are employed so that the grid processor 312 does not need to be stalled to take account of different access speeds of the VRAM 700 in dependence upon whether page-mode or non-page-mode access is taking place.

Each of the data multiplexers 602(0) to 602(15) is associated with a respective one of sixteen processors 606(0) to 606(15) and communicates therewith respective data signals D'(0) to D'(15), which are logically 32 bits, but which in practice may be implemented physically as 16 bits, with appropriate multiplexing techniques. The data signals D'(0) to D'(15) are also supplied to respective parts of the bus 331. Also, each of the write enable multiplexers 603(0) to 603(15) is associated with a respective one of the sixteen processors 606(0) to 606(15) which supply respective 1-bit write enable signals WE'(0) to WE'(15) to the write enable multiplexers. Each processor 606(0) to 606(15) provides a logical control signal CO(0) to CO(15) to control both its associated data multiplexer 602 and write enable multiplexer 603. Thus, during writing to the memory, any processor may provide any respective one of the data signals by providing the number 0 to 15 of the required data signal as its control signal to its data and write enable multiplexers. Furthermore, during reading from the memory, any processor may read any of the data signals by providing the number 0 to 15 of the required data signal to its data multiplexer. Thus, there is no restriction on data being processable only relative to a particular processor, and each processor can select and control the routing of its own data.

The crossbar 377 shown in Figure 22 is simplified for reasons of clarity, and shows, for example, bi-directional multiplexers, which in

practice are difficult to implement. A modified form of the exchange, incorporating the crossbar and the surface shifters, is shown in Figure 46.

The exchange of Figure 46 comprises sixteen sections, of which one typical section 326(i) is shown for simplicity. The data $D''(i)$ from the memory is supplied via a buffer BA(i) and register RA(i) to one input of a 2:1 multiplexer SA(i) acting as a two-way switch. The output of the switch SA(i) is fed to an input i of the surface shifter 743 which performs surface shifting for read and for write. The corresponding output i of the surface shifter 743 is fed to one input of a multiplexer switch SB(i) and is also fed back to the data $D''(i)$ input via a register RB(i) and a tri-state buffer BB(i). The output of the switch SB(i) is input to a FIFO(i), the output of which forms the other input of switch SA(i) and is also fed to one input of a further switch SC(i). The set of sixteen data lines D(0) to D(15) connect the exchange sections 326(0) to 326(15) and the output of switch SC(i) is connected to data line D(i). In the general case, the output of each switch SC(0) to SC(15) is connected to the data line of the same number.

The sixteen inputs of a 16:1 multiplexer MUX(i) are connected to the data lines D(0) to D(15), and the output of the multiplexer MUX(i) is connected via a register RC(i) and a tri-state buffer BC(i) to the respective processor PROC(i) via the data line $D'(i)$. The output of the multiplexer MUX(i) is also connected to the other input of switch SB(i). Furthermore, the data line $D'(i)$ from the processor PROC(i) is also connected via a buffer BD(i) and a register RD(i) to the other input of the switch SC(i).

The control signal CO(i) for the multiplexer MUX(i) is provided by a switch SD(i) which can select between a hardwired value i or the output of a register RE(i) which receives its input from the output of the register RD(i).

Also, control signals CSB, CSC, CSD and CBC are supplied to the multiplexer switches SB(0) to (15), the multiplexer switches SC(0) to (15), the multiplexer switches SD(0) to SD(15), the tri-state buffers BC(0) to (15) from the microcode memory 308 (Figure 4) of the processing section 321. Furthermore, control signals CSA, CBB and CSS derived from the microcode

memory 307 of the address processing section 309 are supplied to the multiplexer switches SA(0) to (15), the tristate buffers BB(0) to (15) and the surface shifter 743.

The exchange 326 of Figure 46 is operable in three modes. In a read mode, the processors PROC(0) to PROC(15) can read the memory; in a write mode, they can write to the memory; and in a transfer mode, they can transfer pixel data between each other. The values of the control signals for these three modes are as follows:

	CSA	CSB	CSC	CSD	CBB	CBC	CSS
READ	0	0	0	0/1	0	1	0
WRITE	1	1	1	0/1	1	0	1
TRANSFER	X	X	1	1	X	0/1	X

It should be noted that the control signal CSD can select between a "straight-through" mode in which each multiplexer MUX(i) selects its input i and thus data D(i), or a "processor-selection" mode in which it selects an input j and thus data D(j) in accordance with the value j which the processor has loaded into the register RE(i).

The effective configuration of a generalised one of the exchange sections 326(i) of Figure 46 in the read mode is shown in Figure 47A. In this configuration, the data path from the data line D'(i) is via the register RA(i) to the surface shifter 743. In the read mode, the surface shifter applies a shift of (-mx, -my) (mod. 4) to the data paths. From the surface shifter, the data path continues via the FIFO(i) to the data line D(i). The multi-plexer MUX(i) can select if CSD=0 the straight-through path in which its output is D(i), or if CSD=1 the processor selection path in which its output is D(j) where j is the value loaded into the register RE(i). The output data passes via the register RC(i) as data D'(i) to the processor PROC(i).

The effective configuration of the exchange section 326(i) in the write mode is shown in Figure 47B. The data D'(i) from the respective

processor PROC(i) passes via the register RD(i) to the data line D(i). The multiplexer MUX(i) can select, if CSD=0, the straight-through path in which its output is D(i), or if CSD=1 the processor selection path in which its output is D(j) where j is the value loaded into the register RE(i). The output data passes via the FIFO(i) to the surface shifter 743. In the write mode, the surface shifter applies a shift (+mx,+my) (mod. 4) to the data paths. From the surface shifter, the data path continues via the register RB(i) as data D''(i) to the VRAM 700.

It should be noted from Figures 46 and 47B that, in the write mode, the write-enable signal follows the same path WE'(i) to WE(i) to WE''(i) as the data signal path D'(i) to D(i) to D''(i). Thus these paths are logically 33 bits made up from 32 bits for the data signal and 1 bit for the write-enable signal.

In the transfer mode, the effective configuration of the exchange section 326(i) is as shown in Figure 47C. In this configuration the control signal CSD to the switch SD(i) is set to 1 so that the multiplexer MUX(i) receives as its control signal the value j loaded into the register RE(i). There are four phases to a transfer. In the first phase the processors output the values j of the data D(j) which they wish to receive as the lowest four bits of their data lines, and these values j are clocked into the registers RD(i). In the second phase, the processors output the data to be transferred out, and this data is clocked into the registers RD(i), while the values j are clocked out of the registers RD(i) and into the registers RE(i), thus setting the multiplexers MUX(i) to receive the data on the respectively selected lines D(j). In the third phase, the data in the registers RD(i) is clocked out onto the lines D and each multiplexer MUX(i) receives and outputs the data on respectively selected line D(j). In the fourth phase, the outputs of the multiplexers are clocked into the registers RC(i) and the tristate buffers BC(i) are enabled so that the processors can transfer in the data from the buffers BC(i). Thus, each processor PROC(i) receives the data (j) from the processor PROC(j) which was selected by the processor PROC(i) by its output value j in the first phase.

Referring back to Figure 22, the processors 606(0) are connected to the data broadcast bus 323 and to a priority encoder 614 having 16 sections and which is associated with the sequencer 329. The processors 606(0) to 606(15) communicate address data with the data broadcast bus 323 and the FIFO 332 connects the data broadcast bus 323 with the address processor 310. The processors 606(0) to 606(15) can also supply respective "unsatisfied" signals US(0) to US(15) and respective "X waiting" signals XW(0) to XW(15) to the respective sections of the priority encoder and can receive respective "process enable" signals EN(0) to EN(15) from the respective sections of the priority encoder 614. Lastly, the priority encoder 614 has a sequencer enable (SE) output on line 618 to the sequencer 329 which controls the sequence of processing of a series of microcode instructions by the processors 606.

The purpose of the priority encoder 614 is to provide high efficiency in the accessing by the processors 606 of the memory 700. In order to do this, the encoder 614 and processors perform the following process, which is shown in the flow diagram of Figure 23. In Figure 23, the left-hand three columns contain steps which are taken by the processors 606(0)....606(i)....606(15), or PROC(0)....PROC(i)....PROC(15), in parallel with each other, the right-hand column contains steps performed by the priority encoder.

At the beginning of each microcode instruction, there are a series of initialisation steps 620 to 628. In steps 620 to 625, those processors which require access to the memory set (1) their respective unsatisfied signals US and reset (0) their X waiting signals XW, and those processors which do not require access reset (0) their unsatisfied signals US and their X waiting signals XW. In steps 626, 628, the priority encoder resets (0) the process enable signals EN for all of the processors and also resets (0) the sequencer enable signal SE.

After initialisation, the priority encoder 614 checks through the XW signals, starting with XW(0) in step 630 to find any processor which is X waiting, and if a match is found (step 632) at a processor, designated

PROC(q), then the routine proceeds to step 640. If a match is not found, however, in step 632, then the priority encoder checks through the US signals, starting with US(0) in step 634 to find a processor which is unsatisfied, and if a match is found (step 636) for a processor, designated PROC(q), then the routine proceeds to step 640. If a match is not found, however, in step 636, then this indicates that all processors are satisfied, and accordingly the microcode program can proceed. Therefore, the sequence enable signal SE is set in step 638, and the routine terminates.

In step 640, the process enable signal EN(q) for the selected processor PROC(q) is set. In steps 642, each processor determines whether it is unsatisfied, and if not exits the subroutine of steps 642 to 654. For any processor which is unsatisfied, then in steps 644, that processor determines whether it is the selected processor, and if so supplies, in step 645, to the data broadcast bus 323 as (xq, yq) the virtual address of the base pixel (0,0) of the patch of pixel data which it wishes to process. This address is supplied via the FIFO 332 to the address processor 310, which in response accesses the appropriate locations in the memory 700, swapping in and out pages of pixel data, if required, as described above.

Then, in steps 646, each unsatisfied processor determines whether the y address yi of its required patch of pixel data is equal to the y address yq of the patch which is being accessed. If not, then the processor exits the subroutine of steps 642 to 654. If, however, yi = yq, then in step 648 the processor determines whether the X address xi of its required patch of pixel data is equal to the X address xq of the patch which is being accessed. If so, then the processor resets (0) its unsatisfied signal US(i) and X waiting signal XW(i) in step 650, and accesses the memory for read or write, as appropriate, in step 652. The processor then exits the subroutine of steps 642 to 654. If, in step 648, xi <> xq then in step 654 the X waiting signal XW(i) is set (1), and then the subroutine is exited.

Upon exit from the subroutine of steps 642 to 654 of all processors PROC(0) to PROC(15), the routine proceeds to step 656, where the priority encoder resets (0) the process enable signal EN(q) for the selected processor.

The routine then loops back to step 630.

It will be appreciated from the above that (A) the lowest numbered processor (an arbitrary choice) which is unsatisfied is selected and given access to the memory initially, together with any other processors which require access to the same address as that selected processor. Then, (B) of any remaining unsatisfied processors which require access to the same y address as the selected processor, the lowest numbered processor is given access, together with any other processors requiring the same address. Then, (C) of any remaining unsatisfied processors which require access to the same y address as the last satisfied processor, the lowest numbered processor is given access, together with any other processors which require access to the same address. Step C is repeated, if necessary, and then steps A and B are repeated until all of the processors have been satisfied. The next microcode instruction sequence is then processed.

As example of the operation of the priority encoder and processors in accessing the memory will now be described with reference to the table of Figure 24. In the example, PROC(0) to (3) and (8) to (11) require access to the patches having the base pixel X and Y addresses listed in column 660 of the table, the addresses being in hexadecimal notation. Thus, after the initialisation routine, JS(0) to (3) and (8) to (11) are set to 1 and the other JS signals and the XW signals are reset to 0, as shown in column 662.

In the first loop of the main routine, PROC(0) is selected, i.e. $q = 0$, and thus accesses the memory at (1234, 1234). Because PROC(1) requires the same address, it also becomes satisfied, i.e. $US(1) = 0$, and accesses the memory at (1234, 1234). Furthermore, because PROC(2) and PROC(10) require the same Y address as PROC(0), they become X waiting, i.e. $XW(2) = XW(10) = 1$. This is shown in column 664.

In the next loop of the main routine, PROC(2) is found to be X waiting $XW(2) = 1$, and thus PROC(2) is selected, i.e. $q = 2$. Therefore PROC(2) becomes satisfied, ($US(2) = XW(2) = 0$), as shown in column 666, and accesses the memory at (1235, 1234).

In the next loop of the routine, PROC(10) is found to be X waiting ($XW(10) = 1$), and thus PROC(10) is selected, i.e. $q = 10$. Therefore PROC(10) becomes satisfied, ($US(10) = XW(10) = 0$), as shown in column 668, and accesses the memory at (1236, 1234).

In the next loop of the routine, no processor is found to be X waiting, and PROC(3) is found to be the first completely unsatisfied processor, i.e. $US(3) = 1$, $YS(3) = 0$. Therefore PROC(3) is selected ($q = 3$), becomes satisfied ($US(3) = XW(3) = 0$) and accesses the memory at (1235, 1235). Also because PROC(11) has the same Y address as PROC(3), PROC(11) becomes X waiting, i.e. $XW(11) = 1$, as shown in column 670.

In the next loop of the routine, PROC(11) is found to be the only X waiting processor, ($US(11) = XW(11) = 1$). Therefore, PROC(11) is selected ($q = 11$), becomes satisfied ($US(11) = XW(11) = 0$) and accesses the memory at (1236, 1235), as shown in column 672.

In the next loop of the routine, PROC(8) is found to be the first unsatisfied processor ($US(8) = 1$). Therefore, PROC(8) is selected ($q = 8$), becomes satisfied, and accesses the memory at (1234, 1236). Furthermore, because PROC(9) requires the same address, it also becomes satisfied ($US(9) = 0$) and accesses the memory.

During the next loop of the routine, no processors are found to be unsatisfied, and therefore the sequence enable signal SE is set and the next microcode instruction is processed.

By using the priority encoder as described above, processors which require access to the same patch can access that patch simultaneously. Furthermore, when a plurality of processors require access to different patches having the same Y address, their accesses are made immediately one after the other, in "page mode". Therefore the address translator does not need to re-latch the Y address(es) in the Y address latches 706(0) to (15) (Figures 8 and 14) between such accesses. Thus, a considerable improvement in performance is achieved as compared with a case where the

processors PROC(0) to (15) access their required patches one at a time, sequentially and without reference to any similarity between the addresses to be accessed.

In the system described above, up to sixteen pixels in a patch are processed in parallel by sixteen processors. Preferably, the system is also arranged so that a group of patches, for example, up to 32 patches, are processed in series in order to reduce pipeline start and finish overheads. In this case, the method of operation may be modified, as compared with that shown in Figure 23, in order to increase efficiency, by permitting any processor requiring access to, say, a j th pixel in the group to request that pixel without firstly waiting for all the other processors to complete access to their $(j-1)$ th pixels in the group. To do this, between steps 623 and 630 in Figure 23, for each processor the step "set address of first required pixel in group as (x_i, y_i) " is included for each processor PROC(i). Furthermore, steps 650 and 652 for each processor as shown in Figure 23 are replaced by the steps shown in Figure 48. In step 682, the memory is accessed at address x_i, y_i for the particular processor PROC(i). Then, in step 684, it is determined whether or not the processor PROC(i) requires access to a further pixel in the group, and if not in step 686, the unsatisfied flag US(i) and the X waiting flag XW(i) are both reset to 0, similarly to step 650 in Figure 23. However, if so in step 684, then in step 688 the processor PROC(i) sets the address of the next required pixel as (x_i, y_i) . Then, in step 690, it is determined whether or not the new y address y_i is equal to the Y address y_q of the last accessed pixel. If so, then in step 692, the X waiting flag XW(i) is set to 1, whereas if not, then in step 694, the X waiting flag XW(i) is reset to 0. After steps 692 or 694, the routine proceeds to step 656 as in Figure 23. It will therefore be appreciated that, once any processor has accessed a pixel in its series of required thirty-two pixels, it can immediately make itself ready to access the next pixel in its series, irrespective of how many of their required thirty-two pixels each of the other processors has accessed. This therefore makes good use of the page mode accessing of the VRAM in which a series of pixels with the same Y address are accessed without the need to re-latch the Y address between each access.

A problem which can arise with the modification of Figure 48 is that some of the processors can inordinately race ahead of others of the processors in accessing their thirty-two pixels. For example, in the case where the processors require access to many different Y addresses, it may arise that PROC(0) accesses all of its thirty-two required pixels first, then PROC(1) accesses its thirty-two pixels, and so on. In order to obviate this problem, the following further modification may be made. Basically, access is permitted with the following order of priority: (a) of highest priority, processors which require access in page-mode (i.e. with the same Y address as the last access) are arbitrated for access; (b) of second priority, processors which have progressed least through their series of thirty-two accesses are arbitrated for access; and (c) of lowest priority there is arbitrary selection of any processors still requiring access. This is achieved by maintaining in a register file of each processor a respective local pointer LP(i) indicating which of its 32 accesses it is waiting for, a common low watermark pointer WM for all the processors, and a common high watermark pointer HP for all the processors. Furthermore, the significance of each unsatisfied flag US(i) in Figures 23 and 48 is modified so that US(i)=1 only if LP(i)=WM and the processor PROC(i) is unsatisfied. The process of Figures 23 and 48 is then modified as follows. In the initialisation steps 622 to 625, the additional steps are included of resetting to zero LP(i) and WM in the register files of all processors, and setting HP to the number of accesses in the series, usually 31. The step 642 in Figure 23 is replaced by "LP(i)<HP?". Furthermore, accompanying step 682 in Figure 48, where a processor accesses the memory, it also increments its local pointer to LP(i)+1. This then has the effect of dealing with priorities "a" and "b" described above. In order to deal with the priority "c" above, an additional decision is included between steps 636 and 638 in Figure 23. If the low watermark pointer WM is less than the high pointer HP, then the low watermark pointer in each of the processor register files is incremented to WM+1, and the process loops back to step 630. However, if WM=31 the process proceeds to step 638. From the above, it will be appreciated that the low watermark pointer is always less than or equal to the lowest local pointer LP(i). When there is no page mode, only those processors whose local pointer LP(i) is equal to the low watermark pointer WM are initially

involved in the access arbitration. If there are none, the watermark pointer is incremented, unless it is equal to HP.

SPLIT-LEVEL PATCHES

It will be noted from the above that the memory is capable of storing pixel data of 32 bits and that the grid processor is capable of processing pixel data logically of at least 32 bits. In some applications, pixel data having a resolution as great as 32 bits is not needed, and all that may be required is 16-bit or 8-bit pixel data. In such cases it is possible to use only 16 or 8 bits of the 32 bits available for each pixel, but this would then result in the VRAM not being used to its full capacity, and pages of pixel data would need to be swapped between the VRAM and the paging memory more often than is necessary.

It may therefore be considered expedient to split the whole image memory into two for 16-bit data, or four for 8-bit data, and thus overlay whole pages of data one on top of another. This would make available the whole capacity of the VRAM, but would suffer from the disadvantage that severe complications would arise when swapping, for example, just one page of 16-bit or 8-bit data between the VRAM and page memory, because it would be necessary to select only half or a quarter of the stored data for transfer from the VRAM to the paging memory, and it would be necessary to mask off half or three-quarters of the VRAM when transferring a page of data from the paging memory to the VRAM.

There now follows a description of an arrangement which avoids these problems associated with transfer of 16-bit or 8-bit data between the VRAM and paging memories.

In essence, the data is overlaid so that at no single address for each of the 128 VRAMs 710 does there exist data for more than one page. This is achieved by overlaying the 8- or 16-bit pixel data in units of a pixel, or more preferably units of a patch, as described below.

Referring to Figure 25, an aligned set of memory cells C(0) to C(127), one from each VRAM chip, and each 4 bits wide, is shown. In the 32-bit arrangement described above, these cells form an aligned patch of 4 x 4 pixels.

In the 16-bit patch-overlay modification, these cells form two layers L(0), L(1) of a 8 x 4 patch. L(0) is provided by C(0) to (3), C(8) to (11), C(16) to (19).... C(120) to (123). L(1) is provided by the remaining cells C(4) to (7), C(12) to (15), C(20) to (27).... C(124) to (127). When the image represented by the two layers of the patch is to be displayed, layer L(0) is displayed immediately to the left of the layer L(1), as shown in Figure 25.

In the 8-bit patch-overlay modification the cells form four layers L(0) to (3) of 16 pixel x 4 pixel patch. The layers are provided by the cells as follows:

Layer L(0) : C(0), C(1), C(8), C(9) C(120), C(121)
 Layer L(1) : C(2), C(3), C(10), C(11) C(122), C(123)
 Layer L(2) : C(4), C(5), C(12), C(13) C(124), C(125)
 Layer L(3) : C(6), C(7), C(14), C(15) C(126), C(127)

When the image represented by the four layers of a patch is to be displayed, the layers are displayed left to right in the order L(0), L(1), L(2), L(3).

A different address format needs to be employed when using 16-bit and 8-bit overlayed patches as compared with that used for the more straightforward 32-bit case, and is given in the table below:

	BITS OF VIRTUAL ADDRESS		
	32-bit mode	16-bit mode	8-bit mode
X misalignment	0,1	0,1	0,1
Level	-	2	2,3
X patch address	2-6	3-7	4-8
X page address	7,8	8	-
X portion of superpage address	9-15	9-15	9-15
Y misalignment	16,17	16,17	16,17
Y patch address	18-22	18-22	18-22
Y page address	23,24	23,24	23,24
Y portion of superpage address	25-31	25-31	25-31
Image ID portion of superpage address	32-47	32-47	32-47

It will be noted that, between the different modes, there is no change of identity of the sixteen bits representing the image ID (32-47), the sixteen bits representing the Y address (16-31), the seven bits representing the X portion of the superpage address (9-15), and the two X misalignment bits (0,1). The X patch address is, however, represented by bits 2-6 for 32-bit mode, by bits 3-7 for 16-bit mode, and by bits 4-8 for 8-bit mode. This makes available bit 2 in the 16-bit mode, and the two bits 2 and 3 in the 8-bit mode, to provide the level data, and leaves only one bit 8 in the 16-bit mode, and no bits in the 8-bit mode, for the X page address.

The patch and page arrangements and the address notations used for them are represented in Figures 26-29. Figure 26 shows the arrangement of patches in a single 16-bit page, and Figure 27 shows the arrangement of 8 pages in one complete 16-bit superpage. Figure 28 shows the arrangement of patches in a single 8-bit page, and Figure 29 shows the arrangement of 4 pages in one complete 8-bit superpage.

A number of complications arise when dealing with 16- or 8-bit

data. Firstly, it is necessary to ensure that the X bits of the address are used in the proper manner. In order to do this, the supply of data from the virtual address bus 319 to the page table 756, near page edge table 758 and X patch address incrementor 760X as shown in Figure 15A is modified as shown in Figure 30A. As before, the page Y bits, 23,24 are fed directly to the page table 756 and the patch Y bits, 18-22 are fed directly to the patch Y address multiplexers 784Y, etc. However, the X bits 2-8 (which form the page and patch X addresses in the 32-bit version) are input to a funnel shifter 812. The shift provided by the funnel shifter is controlled by a mode select signal MS on line 814 which is generated by a separate circuit in response to image header information provided prior to an image or graphics processing operation and which indicates whether the pixel data is 32-, 16- or 8-bit. The funnel shifter provides a page X address of up to two bits, a 5-bit patch X address, and the level data L of up to two bits. The relationship between the inputs to and outputs from the funnel shifter 812 is shown in the table of Figure 31, and it will be noted that it corresponds to the required shifting derivable from the table set out above.

The next complication arises due to the need to present the 16- or 8-bit pixels to the grid processor during reading such that the appropriate 16 or 8 bits of each pixel will be processed and not the remaining irrelevant 16 or 24 bits. This complication is overcome by supplying, during a read operation, all 32 bits from a location in the memory to the grid processor, together with shift data ZSFT in response to which the grid processor shifts the read pixel data by an amount corresponding to the ZSFT data, and then processes predetermined bits of the shift data, e.g. bits 0-15 for 16-bit processing, or bits 0-7 for 8-bit processing.

A further complication arises due to the possibility of a read patch of data not being aligned with the patch level boundaries. This complication is overcome in a somewhat similar manner to that described above with respect to 32-bit patches not being aligned with the patch boundaries in the memory. To illustrate the above, reference is made to Figure 32, which shows a 16-bit patch p in which the base pixel is in level

$L = 1$ of base patch a at $(12,16)$ and is misaligned $(mx,my) = (2,1)$. The address of the patch p in its respective page would therefore be $(px,py) = (12,16)$; $L = 1$; $(mx,my) = (2,1)$. It will be seen that, because patch p has a non-zero x misalignment, $mx > 0$, part of the patch is at the other level $L = 0$, and furthermore because both $mx > 0$ and the level of the base pixel is 1, part of the patch p is in another aligned patch b having patch address $(13,16)$. Furthermore, because also the y misalignment $my > 0$, the patch p also extends into aligned patches c and d at patch addresses $(12,17)$ and $(13,17)$ respectively and at levels 1 and 0, respectively. The determination of the further aligned patch addresses b, c, d is performed by the patch x and y address multiplexers 784 X,Y and the patch y address increment select tables 780 Y described above with reference to Figure 15 and by a modified form of the patch X address increment select table 780X which is responsive to the level data L and the mode select signal MS in addition to the X misalignment mx , as shown in Figure 30B. The modified table 780X provides a 1-bit output to the X patch address multiplexer 784X in accordance with the truth table set out in Figure 30C.

The amount of shifting $ZSFT$ required for each pixel in the grid processor so that each pixel occupies bits 0-15 in 16-bit mode and bits 0-7 in 8-bit mode at the grid processor is determined as follows. It will be appreciated from viewing the 16-bit example of Figure 32, that the pixels labelled 6, 7, 10, 11, 14, 15, 2 and 3 will require $ZSFT$ of 16 bits and that the pixels labelled 4, 5, 8, 9, 12, 13, 0 and 1 require zero $ZSFT$. This is specific to the case where the x misalignment $mx = 2$ and the base level is 1. It will be appreciated that for the general case of a misalignment mx , where $0 \leq mx \leq 3$ and a level $L = 0$ or 1, the required $ZSFT$ for a pixel at an X location x relative to the base pixel of the patch p will be 0 bits if $mx + x < 4$ and $L = 0$, or if $mx + x > 3$ and $L = 1$, and will be 16 bits if $mx + x > 3$ and $L = 0$, or if $mx + x < 4$ and $L = 1$.

As a further illustration, reference is made to Figure 33, which shows an 8-bit patch p which has an address in its respective page of $(px,py) = (12,16)$; $L = 1$; $(mx,my) = (2,1)$. In this case, the pixels labelled 6,

7, 10, 11, 14, 15, 2 and 3 require a ZSFT of 8 bits, and the pixels labelled 4, 5, 8, 9, 12, 13, 0 and 1 require a ZSFT of 16 bits. In the general case of a misalignment mx , where $0 \leq mx \leq 3$, and a level L where $0 \leq L \leq 3$, the required ZSFT for a pixel at an X location x relative to the base pixel of the patch will be zero bits if $mx + x < 4$ and $L = 0$, or if $mx + x > 3$ and $L = 3$; will be 8 bits if $mx + x < 4$ and $L = 1$ or if $mx + x > 3$ and $L = 0$; will be 16 bits if $mx + x < 4$ and $L = 2$; or if $mx + x > 3$ and $L = 1$; and will be 24 bits if $mx + x < 4$ and $L = 3$, if $mx + x > 3$ and $L = 2$.

In order to provide the required ZSFT value for each pixel, the circuit of Figure 15 includes the addition shown in Figure 34, in addition to being modified as described above with reference to Figures 30 and 31. The level value L and also the bits 0,1 of the virtual address for the misalignment mx are supplied as addresses to four ZSFT tables 818a to d implemented using combinational logic. The ZSFT tables 818 also receive the mode select signal MS on line 814 and have three sections for 32-, 16- and 8-bit operation which are selected in dependence upon the MS signal. The ZSFT table 818a supplies the ZSFT values ZSFT(0), (4), (8), (12) corresponding to data D(0), (4), (8), (12) supplied from the read surface shifter 742 to the exchange 326; ZSFT table 818b supplies ZSFT (1), (5), (9), (13) for data D(1), (5), (9), (13); ZSFT table 818c supplies ZSFT (2), (6), (10), (14) for data D(2), (6), (10), (14); and ZSFT table 818d supplies ZSFT (3), (7), (11), (15) for data D(3), (7), (11) and (15). It will therefore be appreciated that the four ZSFT tables 818a to d correspond to pixels having X addresses of $x = 0$, $x = 1$, $x = 2$ and $x = 3$, respectively, in the patch p relative to the base pixel of the patch p .

The table set out in Figure 35A defines the values of ZSFT stored in the ZSFT tables 818a to d for different input misalignments mx , levels L and modes (8-, 16- or 32-bit) and in dependence upon the x value for the particular ZSFT table. As a further example, Figure 35B sets out the values of ZSFT for the particular ZSFT table 818b ($x = 1$) for all possible values of mx , L and mode. In these tables, the ZSFT values of 0, 1, 2, 3 represent a required shift of 0, 8, 16 and 24 bits respectively.

A further complication which arises when dealing with 8 or 16 bit data is that the X near-page-edge signal no longer needs to be dependent solely upon whether or not $4px + mx > 124$, but is also dependent upon the mode selected and the level data L. The X near-page-edge signal is set only if the highest X patch address is designated (i.e. $px = 31$), and if the highest level data is designated (i.e. $L = 1$ in 16-bit mode, or $L = 3$ in 8-bit mode), and if the misalignment mx is non-zero. Accordingly, the near-page-edge table 758 shown in Figure 15A is modified as shown in Figure 36A so as to receive the mode select signal MS on line 814 and the level signal L, in addition to the patch address (px, py) and the misalignment (mx, my) . The modified table 758 of Figure 36A produces X and Y values NPE_x and NPE_y of the 2-bit NPE signal as shown by the table set out in Figure 36B.

As described above, during reading, ZSFT data ZSFT(0) to (15) is supplied to the crossbar 327 with the pixel data D(0) to (15). Also, as described earlier with respect to Figure 22, each processor PROC(0) to (15) is capable of reading any of the data D(0) to (15). It is therefore necessary to ensure that the ZSFT data appropriate to the selected pixel data is supplied each processor. Figure 37 shows a modification to the crossbar 377 and part of the grid processor arrangement of Figure 22 for a generalised processor PROC(i) where $0 \leq i \leq 15$. The modified arrangement is similar to the arrangement of Figure 15 except in the following respects. Firstly, a 16 x 2-bit ZSFT FIFO 678 is provided to receive ZSFT(0) to (15). The output of the ZSFT FIFO 678 is supplied to each of sixteen 16:1 2-bit multiplexers 680(0) to 680(15). The 2-bit outputs of the ZSFT multiplexers 680(0) to 680(15) are supplied to the respective processors PROC(0) to PROC(15) as signals ZSFT'(0) to (15). The ZSFT multiplexers are controlled by the same logical control signals CO(0) to CO(15) as the associated data and write enable multiplexers. It will therefore be appreciated that each processor receives the appropriate ZSFT data for the pixel data which it selects and can then shift the received pixel data by 0, 8, 16 or 24 bits in dependence upon the value 0, 1, 2 or 3 of the received ZSFT data so that the received pixel data then always occupies the first 8 bits of the processor's input register in 8-bit

mode, or the first 16 bits of the input register in 16-bit mode.

It will be appreciated that the arrangement of the multiplexers and FIFOs shown in Figure 37 may be modified in a similar manner to the modification of Figure 22 which is described above with reference to Figures 46 and 47.

A further complication which arises when dealing with 16-bit or 8-bit pixel data is that, during writing to the memory 700, only the appropriate 16 or 8 bits should be written, and the remaining 16 or 24 should not be overwritten. For example, referring to Figure 32, during writing of the patch p as shown, the memory cells which are to store the 16-bit pixels labelled 6, 7, 10, 11, 14, 15, 2 and 3 need to have bits 16 to 31 written, with writing of bits 0 to 15 disabled, and the memory cells which are to store the pixels labelled 4, 5, 8, 9, 12, 13, 0 and 1 need to have bits 0 to 15 written, with bits 16 to 31 being disabled. As a further example, referring to Figure 33, the memory cells which are to store the 8-bit pixels labelled 6, 7, 10, 11, 14, 15, 2 and 3 need to have bits 8 to 15 written, with bits 0 to 7 and 16 to 31 being disabled, and the memory cells which are to store the pixels labelled 4, 5, 8, 9, 12, 13, 0 and 1 need to have bits 16 to 23 written with bits 0 to 15 and 24 to 31 disabled.

In order to deal with this complication, the circuit of Figure 40 is employed, which provides partial write enable signals PWEa to PWEd for the memory banks having x addresses of $bx = 0, 1, 2$ and 3 , respectively. The circuit of Figure 40 comprises four PWE tables 822a to d for the values $bx = 0$ to 3 , respectively. Each PWE table 822 is provided with the bits 0,1 of the virtual address on bus 319 indicating the X misalignment mx , the value L from the circuit 820 of Figure 30, and the mode select MS signal on line 814. The PWE tables contain the data as set out in Figure 38 and therefore a table having a particular value of bx can provide the 4-bit value PWE in dependence upon the input values of mx , L and MS.

In addition to adding the circuit of Figure 40, the connections to the X latch groups 707(0) to (15) (see Figures 8 and 14) are modified as

shown in Figure 41. The column address strobe CAS signal is still ANDed with the write enable signals WE⁽⁰⁾ to (15) to produce the signals CAS(0) to (15) and the addresses A(0) to (15) are also applied to respective groups 707(0) to 707(15) of X latches. The various bits of the partial write enable signals PWE(a) to (d) are connected to write enable inputs of the X latches for the cells 0 to 127 as follows:

<u>PWE bits</u>	<u>Cells</u>
PWE(a) bit 0	0, 1, 32, 33, 64, 65, 96, 97
bit 1	2, 3, 34, 35, 66, 67, 98, 99
bit 2	4, 5, 36, 37, 68, 69, 100, 101
bit 3	6, 7, 38, 39, 70, 71, 102, 103
PWE(b) bit 0	8, 9, 40, 41, 72, 73, 104, 105
bit 1	10, 11, 42, 43, 74, 75, 106, 107
bit 2	12, 13, 44, 45, 76, 77, 108, 109
bit 3	14, 15, 46, 47, 78, 79, 110, 111
PWE(c) bit 0	16, 17, 48, 49, 80, 81, 112, 113
bit 1	18, 19, 50, 51, 82, 83, 114, 115
bit 2	20, 21, 52, 53, 84, 85, 116, 117
bit 3	22, 23, 54, 55, 86, 87, 118, 119
PWE(d) bit 0	24, 25, 56, 57, 88, 89, 120, 121
bit 1	26, 27, 58, 59, 90, 91, 122, 123
bit 2	28, 29, 60, 61, 92, 93, 124, 125
bit 3	30, 31, 62, 63, 94, 95, 126, 127

It will therefore be appreciated that, during writing in the 8-bit or 16-bit mode, only the relevant memory cells are write enabled, and the remaining cells are disabled.

It will be recalled that, in 8-bit mode, the data is processed as the first 8 bits of their 32-bit capacity by the processors, and in 16-bit mode as the first 16 bits. Therefore, in order to ensure that, upon writing, the processors can write to bits 8 to 31 of the memory in 8-bit mode, or bits 16 to 31 of the memory in 16-bit mode, prior to writing, each processor

which is to write duplicates, in 8-bit mode, the pixel data of locations 0 to 7 in its output register at bit locations 8 to 15, 16 to 23 and 24 to 31 of the output register, and duplicates, in 16-bit mode, the pixel data of bit locations 16 to 31. Accordingly, when the enabled bits of the pixel data are written to the memory, the complete data for the pixel is written.

FLAGGING OF MODIFIED PAGES

Referring to Figure 42, it is convenient that a predetermined section 830 of the VRAM 700 is always mapped to the monitor 40, and for simplicity the section will be considered between page addresses (0,0) and (7,7) giving a total mapped area of $8 \times 8 \times 31 \times 31 \times 4 \times 4 = 1$ Mpixel. It is also convenient that images are rendered in another section of the VRAM 700, and for simplicity the section 832 between pages addresses (8,8) and (15,15) will be considered. Then, periodically, the data of the rendering section 832 is copied to the monitoring section, for display on the monitor. It will be appreciated that the data for some pixels may not change between one copying operation and the next, and indeed it can arise that no pixel data changes between two successive copying operations. If these unchanged pixels are unnecessarily copied from the rendering section to the monitoring section, then the performance of system is impaired.

In order to overcome this problem, it may be considered expedient to flag each pixel which is modified during a rendering operation and to copy only those pixels which have been flagged. However, this would require an inordinate amount of memory to store the flags and would require an excessive amount of flag setting, testing and resetting, which would degrade the system performance. In the arrangement described below, therefore, pages which have been changed, or dirtied, in a rendering operation are flagged, and only the flagged dirty pages are copied to the monitoring section of the memory.

It will furthermore be appreciated that, if a page of pixel data is copied from the paging memory to the VRAM, and that if that page is not

modified, or dirtied, in the VRAM, then there is no need to copy that page of data back to the paging memory when the time comes to replace that page in the VRAM with a different page from the paging memory. Accordingly, in the arrangement described in detail below, a flag is set when a page is dirtied in any rendering operation while it is in the VRAM, and when the page is to be replaced, it is copied back to the paging memory only if the flag is set.

It should be noted that pixel data in the VRAM is processed in patches and that a non-aligned patch may extend across a page boundary. Therefore the arrangement described below also includes for each page, dirty flags for the pages B, C and D, as shown in Figure 11, to the right, above and the right and above, of the page A in question. It should be noted that if page A has a virtual page address (PX,PY) then pages B, C and D have virtual page addresses (PX + 1, PY), (PX, PY + 1) and (PX + 1, PY + 1), respectively.

Referring to Figure 43, a dirty-page table 834 is provided by a 2K SRAM which is addressed by the 7-bit superpage identification on line 767 from the CAM 754, and the 2-bit page X address and 2-bit page Y address from the virtual address bus 314 on lines 768X,Y. The eight data bits at each location in the table 834 are assigned as follows:

bit 7	Page A dirty swap	dsA
bit 6	Page B dirty swap	dsB
bit 5	Page C dirty swap	dsC
bit 4	Page D dirty swap	dsD
bit 3	Page A dirty render	drA
bit 2	Page B dirty render	drB
bit 1	Page C dirty render	drC
bit 0	Page D dirty render	drD

Bits 0 to 2 and 4 to 6 of the dirty page data are supplied to respective OR gates 836 (0) to (2) and 836 (4) to (6). At gates 836 (6) and (2), the signals dsB and drB are ORed with the near-page-edge X signal NPEX. At gates 836 (5) and (1), dsC and drC are ORed with the near-page-edge Y signal NPEY, and at gates 836 (4) and (0), the signals dsD and drD are ORed with an ANDed form of the near page edge X and Y signals on line 774. The six bits output from the OR gates, together with a pair of high bits, representing the new signals dsA and drA, are then passed via a register 838 for writing back into the dirty page table 834 under control of a dirty pages write-enable signal DWE on line 840. The 8-bit data line of the dirty page table 834 is also multiplexed onto the 48-bit virtual address bus 319, and the address processor is operable (a) to reset the appropriate dirty swap bits and set the appropriate dirty render bits when a new page is swapped from the paging memory to the VRAM, (b) to set the appropriate dirty swap bits and dirty render bits for a page when rendering operation is carried out on that page, (c) to test the appropriate dirty swap bits for a page when that page is to be replaced by a different page in the VRAM, and (d) to test the appropriate dirty render bits for a page when that page is to be copied from the rendering section to the monitoring section of the VRAM and to reset the dirty render bits.

An example of the operation of the dirty page arrangement will now be described with reference to Figures 42 and 43, the table of Figure 44 and the flow diagrams of Figure 45. Suppose that 4 pages P, Q, R, S of pixel data at (X,Y) page addresses (0,0), (1,0), (2,0), (3,0) in the same superpage are copied into the VRAM at contiguous page addresses (8,8),

(8,9), (8,10), (8,11), and that the superpage has an identification code of 25 in the CAM 754. Suppose also that the rendering section 832 between page addresses (8,8) and (15,15) in the VRAM is copied over to the monitoring section 830 between pages addresses (0,0) and (7,7) in the memory. Suppose also that three rendering operations are carried out in the rendering section, the first rendering operation affecting page Q, the second operation affecting page P and including a misaligned patch which extends into page Q, and the third operation affecting pages Q and S; the pages P to S then being replaced by four other pages.

The dirty page data for pages P to S will be located at addresses 400 ($= 25 \times 16 + 0 + 0$), 401, 402 and 403 in the dirty page table 840. Referring to Figures 44 and 45A, when page P is copied from the paging memory into the VRAM, it is treated as page A for the purposes of Figure 45A. In step 842 bit 7 (dsA) of the dirty flag for page A is reset and bit 3 (drA) of the dirty flag for page A is set. In step 844, the address processor 310 determines whether there is a page B' stored in the physical memory, that is the page to the left of page A. If so, in step 846, bit 6 (dsB) and bit 2 (drB) of the dirty flag for page B' are reset and set respectively. Similar steps 848, 850 and 852, 854 are carried out for pages C' and D', that is pages below and to the left and below of page A in the paging memory. Then, in step 856, page A is copied from the paging memory of the VRAM. The process of Figure 45A is then repeated for pages Q, R & S. It will therefore be appreciated that the dirty flags for pages P to S attain the state as shown in column 902 of Figure 44.

The monitoring section 830 of the VRAM is then to be updated, the address processor 310 carries out the process shown in Figure 45B. In the loop of steps 858 and 860, all of the pages of the rendering section which may possibly need to be copied are selected one-by-one. In step 862, bit 3 (drA) of the selected page (A) is tested, and if set page A is copied to the monitoring section in step 864, and in step 866 bit 3 (drA) for page A, bit 2 (drB) for page B' to the left of page A, bit 1 (drC) for page C' below page A and bit 0 (drD) for page D' to the left and below page A are reset. In step 868, bit 2 (drB) of page A is tested, and if set page B relative to page A is copied to the monitoring section in step 870, and in step 872 bit

2 (drB) of page A and bit 3 (drA) for page B to the right of page A, bit 0 (drD) for page C below page A, and bit 1 (drC) for page E below and to the right of page A are reset. Somewhat similar steps 874 to 884 are performed for bits 1 and 0 (drC, drD), as shown in Figure 45B, and if set the respective page C or D is copied to the monitoring section and various bits are reset as shown. It will therefore be appreciated that when this process is carried out with the dirty flags in the state as shown in column 902 of Figure 44, all four pages P to S are copied to the monitoring section of the VRAM, and the dirty flags attain the states as shown in column 904.

In the first rendering operation, page Q only is modified, and it will therefore be appreciated that the circuit of Figure 43 serves to set bit 7 (dsA) and bit 3 (drA) for page Q, as shown in column 906 of Figure 44.

The monitoring section of the VRAM is again updated in accordance with the process of Figure 45B. The only dirty render flag bit set is drA for page Q, and therefore only page Q is copied, and the bit drA for page Q is reset, as shown in column 908.

In the second rendering operation, page P is modified, and also a misaligned patch in page P modifies page Q. As a result, bits 7, 6, 3 and 2 (dsA, dsB, drA, drB) of the page P dirty flag are set, as shown in column 910. Because bits drA and drB for page P are set, pages P and Q are copied to the monitoring section by the process of Figure 45B, and bits 3 and 2 (drA, drB) for page P are then reset, as shown in column 912.

In the third rendering operation, pages Q and S are modified. As a result, bits 7 and 3 (dsA, drA) of the page S flag are set; bit 3 (drA) of the page Q flag is set, and bit 7 (dsA) of the page Q flag remains set, as shown in column 914. Because bits 3 (drA) of pages Q and S are set, pages Q and S are copied to the monitoring section of the VRAM, and these bits are then reset, as shown in column 916.

When the pages P to S are to be replaced, the address processor performs the process of Figure 45C for each selected page to be replaced.

In step 886, a copy flag is reset. Then in step 888, it is determined whether bit 7 (dsA) for page A is set, and if so in step 889 that bit is reset and the copy flag is set. Steps 888 and 889 are then repeated as steps 890 to 895 for bits 6, 5 and 4 (dsB, dsC, dsD) respectively of the dirty page flags for pages B', C' and D' relative to page A. Then in steps 896 and 897, if the copy flag has been set, page A is copied to the paging memory.

Referring back to column 916 of Figure 44, it will be appreciated that as a result of performing the process of Figure 45C for page P, this page is copied to the paging memory because dsA is set for page P (step 888). This is then reset (step 889). Page Q is copied to the paging memory because dsA is set for page Q (step 888). Even if it were not, page Q would be copied because dsB is set for page P (step 890). The flag bits dsA for page Q and dsB for page P are also reset (steps 889 and 891). Page R is not copied because none of dsA for page R (step 888), dsB for page Q (step 890), and dsC and dsD for the pages below, and below and to the left, of page R (step 892 and 894) are set. Page S is copied because dsA is set for page S (step 888). This bit is then reset (step 889). Accordingly, pages P, R and S are copied back to paging memory, and the flags attain the status shown in column 918 of Figure 44.

CONDITIONAL PROCESSING

The processors 606(0) to (15) of the grid processor 312 described above are arranged basically as a SIMD array, SIMD standing for 'Single Instruction - Multiple Data' and meaning that all of the processors receive the same instruction and apply it to their own particular data elements. This can be an efficient and simple way of obtaining good performance from a parallel-processing machine, but it does assume that all of the data elements need exactly the same instruction sequence. However, the processors are preferably arranged, as described below, to be able to deal with conditional instructions. Further detail of such an arrangement is shown in Figure 49.

Figure 49 shows three of the processors PROC 0, PROC i and PROC 15, with PROC i being shown in greater detail, their PRAMs 322(0),

(i), (15), the microcode memory 308 and the processing section broadcast bus 323. The microcode memory 308 supplies microcode instructions of about 90 bits to each respective instruction decode logic (IDL) circuit 100 in each of the processors. The same microcode instruction is supplied to each processor. The instruction decode logic is provided by a gate array which decodes the 90 bit instruction to provide about 140 control bits to various elements in the respective processor including an arithmetic logic unit ALU 102, a 32-bit pixel accumulator (pa) 104, a 1-bit condition accumulator (ca) 106 and a status select circuit 108 which is provided by a gate array. The ALU 102 connects with the data bus D' via the exchange 326 to the VRAM 700, the pa 104 and a stack of pixel registers p0 to pn in the PRAM 322. The main data paths for pixel data are from the data bus D' to the ALU 102 and the pa 104; from the pa 104 to the ALU 102, the data bus D' and selected pixel registers p0 to pn; from the ALU 102 to the data bus D' and the pa 104; and from selected pixel registers p0 to pn to the ALU 102. Various status bits are output from the ALU 102 to the status select circuit 108, such as a "negative" bit, a "zero" bit and an "overflow" bit. Some of these status bits are also fed out externally. Also, external status bits such as the EN flag (see Figures 22, 23) are fed in to the status select circuit 108. Under control of the IDL 100, the status select circuit 108 can select a respective status bit and output it to the ca 106. The ca 106 is associated with a stack of condition registers c0 to cn in the PRAM 322. The ca 106 also connects to the IDL 100 and provides the write enable output WE' of the processor. The main paths for condition and status bits are: from the ALU 102 to the status select circuit 108 and to the external outputs; from the external inputs to the status select circuit 108; from the status select circuit 108 to the ca 106; from the ca 106 to the condition stack registers c0 to cn, the write enable output WE' and the ALU 102; and from the condition stack registers c0 to cn to the ca 106.

The 1-bit input from the ca 106 to the IDL 100 is important. This input condition bit enables the IDL 100 to modify the control outputs from the IDL 100 in dependence upon the value of the condition bit, and accordingly the arrangement provides direct support for microcode instructions from the microcode memory 308 to the IDL 100 which in

high-level language would be represented by, for example, if (condition) then (operation X) else (operation Y). As an example, reference is made to Figures 50A to 50D. Suppose that the VRAM 700 contains three images: image A of Figure 50A which in this simple example is a rectangle of horizontal lines; image B of Figure 50B which is a rectangle of vertical lines; and image C of Figure 50C which is a mask in which the upper-left and lower-right corners are black (say pixel values of 0) and the remainder is white (say pixel values of $(2^{32}-1)$). In the example, it is desired to combine images A and B using image C as a mask to form an output image D such that image A appears where the mask image C is black and image B appears where the mask image C is white. The process performed by the processors under control of the microcode instructions from the microcode memory 308 to perform this operation can be considered, using high-level pseudo-language, to be as follows:

1. For each patch (x,y) in the rectangle:
2. If pixel in rectangle, ca = 1, else ca = 02.
3. c0 = ca
4. pa = A(x,y)
5. p0 = pa
6. pa = B(x,y)
7. p1 = pa
8. pa = C(x,y)
9. ca = zero-status (pa)
10. If ca = 1 then pa = p0 else pa = p1
11. ca = c0
12. D(x,y) = pa
13. Next patch

In the above, steps 1 and 13 set up a loop for each patch (x,y) having its origin in the rectangle. For each patch, each processor PROC 0 to PROC 15 will process a different pixel in the patch. In step 2 a test is made to determine whether the particular processor's pixel in the patch is in the rectangle, and if so the ca 106 is set, otherwise it is reset. This value of ca will form the write-enable signal WE'. In step 3, this value which is stored in the ca 106 is put onto the condition stack in c0 and an

associated condition stack pointer is modified accordingly. In step 4, the value of the processor's pixel in the current selected patch in image A is loaded into the pa 104, and in step 5 is transferred to the p0 register. Similarly in step 6, the value of the processor's pixel in the current selected patch in image B is loaded into the pa 104, and in step 7 is transferred to the p1 register. In step 8, the value of the processor's pixel in the current selected patch in the mask image C is loaded into the pa 104, and then in step 9 the zero status bit of the ALU 102 is selected by the status select circuit 108 and is loaded into the ca 106. Thus, if the pixel in the mask image is black, the ca 106 value becomes 1, and if it is white, the ca 106 value becomes 0. The next step 10 is a conditional instruction "If ca = 1 then pa = p0 else pa = p1". The IDL 100 modifies this instruction in dependence upon the value in the ca 106 so that it becomes simply "pa = p0" or "pa = p1" and the modified instruction is used by the processor. In step 11, the signal which was put onto the condition stack at c0 in step 3 is pulled off the stack and placed in the ca 106 in order to constitute the write enable signal WE' and the condition stack pointer is modified accordingly. Lastly, in step 12, the pixel value in the pa 104 is transferred out to the image D at the appropriate pixel position for the processor in the current selected patch.

As a result of the above operations carried out by the processors on the pixels of all of the patches in the rectangle, an image D is formed as shown in Figure 50D.

In the above simple example, the condition stack c0 to cn was used simply to store the initially generated value which will form the write enable signal, and only one register in the stack was employed. By virtue of the provision of more than one register in the condition stack, nesting of the conditional instructions is permitted.

PAGE FILING SYSTEM

As described above, pages of data can be swapped between the VRAM 700, on the one hand, and the paging memory comprising the DRAM 304 (Figure 4), and the paging RAM 504 and fast disk 510 (Figure

5), on the other hand. There now follows a description of how pages are handled as between the VRAM and the paging memory, with reference to the system diagram of Figure 51.

The total system is based on a distributing operating system denoted by the triangle 200. Part of this system constitutes a host page manager module 202 running on the processor 10 of the host computer. Another part constitutes a front-end page manager module 204 running on the i960 control processor 508 of the front-end board 22 and handling the paging RAM 504 and fast disk 510. A further part constitutes a renderer page manager module 206 running on the i960 control processor 314 of the renderer board 16 and handling the VRAM 700 and the DRAM 304. Each of these page manager modules 202, 204, 206 can make a request R to any other module for a page P of image data specified by the virtual page address (VPA) consisting of the following bits of the virtual address:

32-47	Image ID component
25-31	Y superpage component
23, 24	Y page component
9-15	X superpage component
7, 8	X page component

In response, the module to which a request R is made determines whether it is responsible for the requested page, and if so it transfers the page of data P and responsibility therefor to the requesting module, but if not it indicates to the requesting module that it is not responsible for the requested page.

To give two examples of how the filing system would be used, suppose that the page fault table 794 (Figure 21) of the renderer has generated a page fault in respect of a particular page, this page fault is handled by the renderer page manager module 206. Firstly, the module 206 checks with itself whether the required page is stored in the renderer DRAM 304, and if so swaps the page of data into the VRAM 700. If not, the module 206 checks with the front-end page manager module 204 whether it is responsible for the page, and, if so, the page of data is

swapped from the RAM 506 or disk 510, as appropriate, into the VRAM 700. If the front-end module 204 is not responsible, the renderer module 206 asks the host module 202 for the page of data, which is then swapped into the VRAM 700. As another example, suppose that the system is to be closed down and a complete image is to be saved to disk 510. Such saving of an image is handled by the front-end module 204. For each page in the image the module 204 firstly checks with itself whether it is responsible for that page. If it is and the page is already stored on the disk 510, it stays there, and if the page in question is stored in the front-end RAM 506 the data of that page is copied to the disk 510. If the module 204 is not responsible, it checks with the renderer module 206 whether the renderer module has responsibility for the page, and, if so, the page of data is copied from the VRAM 700 or DRAM 304 of the renderer to the disk 510. If not, the front-end module 204 requests the page in question from the host module 202, and the page of data is transferred to the disk 510.

In order to keep track of the pages for which they are responsible, the front-end module 204 and the renderer module 206 each maintain a table 208, 210 containing a list of the virtual page addresses of the pages, and against each address an indication of the location of that page. For example, the location data in the front-end table 208 would comprise an indication of whether the page is in the RAM 506 or on the disk 510. If in the RAM 506, the physical address of that page in the RAM would be included, and if on the disk 510, an indication of the location on the disk would be included. The location data for each virtual page address in the renderer table 210 may contain an indication of whether the page is in the DRAM 304 or the VRAM 700 and the physical address of the page in the respective memory. In the case of a page in the VRAM 700, the physical address of the page need not necessarily be kept in the table 210, because this address can be determined by the module 206 from the CAM 754 and the page table 756 (Figure 15A) of the address translator 740, and indeed it is not necessary for the table 210 to include the virtual page address of the pages in the VRAM 700, because the module can check whether a page is present by referring to the CAM 754 and page table 756 and testing whether or not a page fault is generated.

An important feature of the filing system, in the preferred embodiment, is that the host page manager module 202 is not responsible for the storage of whole pages of data. The host module 202 is used when an image is initially created. The image is specified by the host processor 10 as being of a particular dimension, size, bit width (see Figures 25 to 41) and background colour. In response, the system software 200 allocates to that image the next available image ID (bits 32 to 47 of the virtual address). Until any rendering operations or copying operations are carried out on the image, the colour of every pixel in the new image is the background colour, and the host module 202 therefore merely sets up a table 212 containing the virtual page address of the or each page required in the new image, and against the or each page address the table 212 contains the 32-bit background colour of the image. There is no need for this 32-bit word of data for the page to be expanded into a full page of data, for example 16k words, until the page is transferred to the control of one of the other modules 204, 206. Accordingly, when one of the other modules requests a page from the host module 202, the host module 202 determines from its table 212 the 32-bit background colour of that page, and then repeatedly sends that 32-bit word to the requesting module, once for each pixel in the page.

In the above description, it is assumed that only one of the modules 202, 204, 206 has responsibility for any given page at any given time and that when a page of data is transferred from one module to another, the sending module cancels the entry for that page from its table 212, 208, 210 and that the receiving module makes an entry in its table for the page. It will be appreciated that the dirty page-swap scheme described with reference to Figures 42 to 45 above will not be effective if the filing system operates in this way, because when, for example, a page is swapped from the disk 510 to the VRAM 700, the entry for that page is cancelled from the table 208 of the front-end module 204, and so even if the page is not dirtied in the VRAM 700, it would be necessary to swap all of the data-elements of the page back to the disk 510.

The filing system described above may be modified so that it works in conjunction with the dirty page-swap scheme, by including against each

virtual page address in each table 208, 210, 212 a bit indicating whether that page is current. The operation of each module 202, 204, 206 is then modified so that when a module has responsibility for a page, the current bit is set to 1 and when responsibility is transferred to a different module the current bit is reset to zero. Furthermore, when a page which has not be dirtied is to be swapped out of the VRAM 700, the renderer module 206 polls the other modules 202, 204 to check which has an entry in its table for the page with the current bit reset, and instructs that module to set the current bit, obviating the need to copy all of the data-elements for that page from the renderer module to the other module.

In the above arrangement, a single word representing the image background colour is stored for each new image. Rather than storing a single word, a few words may be stored, for example as a patch, and representing, for example, a pattern which is to be repeated in the new image.

MODIFICATIONS AND DEVELOPMENTS

Although preferred embodiments of the invention have been described above, it will be appreciated that many modifications and developments may be made within the scope of the invention. To take a few examples, the non-split-level patches, pages and superpages described above are two-dimensional and have a pixel resolution of 32-bits, a patch size of 4 pixels x 4 pixels, a page size of 32 patches x 32 patches, and a superpage size of 4 pages x 4 pages. It will be appreciated that the system may be configured so as to operate for example with one- or three-dimensional patches, and/or pages and/or superpages, with patches, pages and superpages of different sizes, and with different pixel resolutions. Furthermore, the system may be arranged to operate selectably in different configurations through appropriate use of funnel shifters, switches and the like. In the above description, examples of specific sizes of the memories have been given, but it will be appreciated that other sizes may be used. In the split level patch system, division into two and four in the X direction has been illustrated, but it will be appreciated that other divisors may alternatively or selectably be

employed, that division in other directions may alternatively or selectably be employed, and that division on a pixel basis rather than a patch basis may alternatively or selectably be employed. The dirty page facility described above deals with copying between the rendering section and monitoring section of the VRAM and also with swapping between the VRAM and the paging memory, but it will be appreciated that either of these two features may be employed without the other. In the page filing system, the page manager modules are run on specific processors, but it will be appreciated that each page manager module may be run on different processors, and that the modules may be combined.

RELATED APPLICATIONS

This application is one of a series of eight applications filed the same day under filing numbers PCT/GB90/ , PCT/GB90/ , PCT/GB90/ , PCT/GB90/ , PCT/GB90/ , PCT/GB90/ , PCT/GB90/ and PCT/GB90/ , and with agent's references PCT 98 1, PCT 99 1, PCT 100 1, PCT 101 1, PCT 102 1, PCT 103 1, PCT 104 1 and PCT 105 1, and the disclosures of the other applications are incorporated herein by reference as if printed below in full.

CLAIMS

1. A memory system for storing an array of data elements which form in a virtual memory space aligned groups (Fig. 7B) of contiguous aligned pages (Fig. 7A) of contiguous data-elements, comprising:
 - a first memory (304, 506, 510) for storing the data-array;
 - a second memory (700) for storing part of the data-array;
 - means (326, 331, 302, 20) for transferring data-elements page-by-page between the first and second memories such that contiguous pages of the data-elements in the virtual memory address space are not necessarily stored contiguously in the address space of the second memory; and
 - means (740) for translating a group component (bits 9-15, 25-31, 32-47) and a page component (bits 7, 8, 23, 24) of the address of a page in the virtual memory space into the corresponding page address (PAGE A) of that page in the second memory address space, the translating means comprising a contents addressable memory (CAM) (754) which receives the virtual group component and outputs a group code (767) corresponding thereto, and a page table (756) which receives the group code and the virtual page component and outputs the corresponding page address in the second memory address space.
2. A system as claimed in claim 1, wherein the number of group codes which can be stored by the CAM is substantially less than the number of the groups which can be stored in the first memory.
3. A system as claimed in claim 1 or 2, wherein each group code has a number of bits (e.g. 7) which is substantially less than the number of bits (e.g. 30) of the virtual group component.
4. A system as claimed in any preceding claim, wherein the CAM is operable to compare the received virtual group component with the contents at the addresses of the CAM and to output as the group code the address of the CAM whose contents matches the virtual group component.

5. A system as claimed in claim 4, wherein the CAM contents have a bit (e.g. msb) which is not used in the virtual group component, and wherein said bit is set at any location in the CAM at which a virtual group component is not stored.

6. A system as claimed in any preceding claim, wherein the CAM is operable to supply a group fault signal (7485) if a match is not made with the received virtual group component, and further comprising processor means (310) operable in response to the group fault signal to store the received virtual group component at an available location in the CAM.

7. A system as claimed in claim 6, wherein the processor means (310) is operable in response to the group fault signal to determine whether there is any available location in the CAM, and if not to select a virtual group component to delete from the CAM, and to determine whether any pages of data-elements of the selected group are stored in the second memory, and if so to cause the transfer means to transfer those pages of data-elements from the second memory to the first memory.

8. A system as claimed in claim 7, further comprising means (Fig. 20) for determining the least recently used group having a page stored in the second memory, the processor means (310) being operable to select the virtual group component of the determined least recently used group as the virtual group component for deletion from the CAM.

9. A memory system for storing an array of data-elements which form in a virtual memory space contiguous aligned pages (Fig. 7A) of contiguous data-elements, comprising:

a first memory (304, 506, 510) for storing the data-array;

a second memory (700) of smaller capacity than the first memory for storing some of the pages of the data-elements; and

means (310, 304, 508) operable to address locations in the first and second memories and to transfer data-elements between the addressed locations page-by page;

wherein the data-elements provide a plural-dimensional

representation, and wherein the transferred pages include data-elements which are contiguous in each of the plural dimensions.

10. A system as claimed in claim 9, wherein the addressing means is operable to address the second memory with addresses which include components in each of said plural dimensions.

11. A memory system for storing an array of data-elements which form in a virtual memory space contiguous aligned pages (A, B, C, D) of contiguous aligned patches (a, b, c, d) of contiguous data-elements, comprising:

- a first memory (304, 506, 510) for storing the data array;
- a second memory (700) for storing part of the data-array;
- means (326, 331, 302, 20) for transferring data-elements page-by-page between the first and second memories such that contiguous pages of the data-elements in the virtual memory address space (Fig. 11A) are not necessarily stored contiguously in the address space of the second memory (Fig. 11B); and

- means (310, 740) for addressing the second memory to access in parallel a patch (p) of data-elements which is not necessarily such an aligned patch, the addressing means including page edge means (758) for determining whether the patch to be accessed includes data-elements in different pages in the second memory, and means (756, 778, 782X, Y) to modify the addresses to the second memory in response to such a determination.

12. A system as claimed in claim 11, wherein the address of a patch (p) to be accessed in the second memory has an aligned page address component (PAGE A), an aligned patch address component (px,py) and a misalignment component indicating a misalignment (mx,my) of the patch (p) to be accessed relative to an aligned patch (a) having an address defined by the aligned page and aligned patch components.

13. A system as claimed in claim 12, wherein the addressing means includes page determining means (756) operable to determine, upon addressing, an aligned page address component for a basic aligned page

(PAGE A) for the patch (p) to be accessed and an aligned page address component of at least one other page (PAGE B, C, D) which is contiguous with the basic aligned page (PAGE A) in the virtual address space.

14. A system as claimed in claim 13, wherein the data-array is a plural-dimensional representation having D-dimensions, and wherein the page determining means (756) is operable to determine the aligned page addresses in the second memory for all aligned pages which differ by one from the basic aligned page address in one direction in each of the D-dimensions in the virtual address space and in any combinations of those one directions.

15. A system as claimed in claim 13 or 14, further comprising page fault determining means (794) for determining, upon addressing, whether the or each aligned page which is to be accessed is not stored in the second memory, the transferring means being operable to transfer the missing page from the first memory to the second memory in response to such a determination.

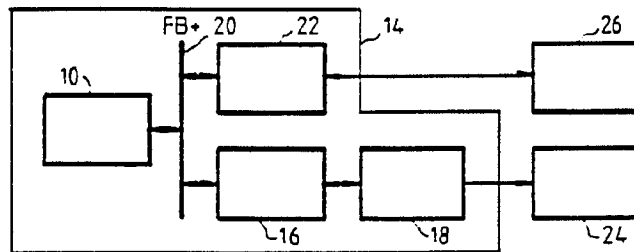
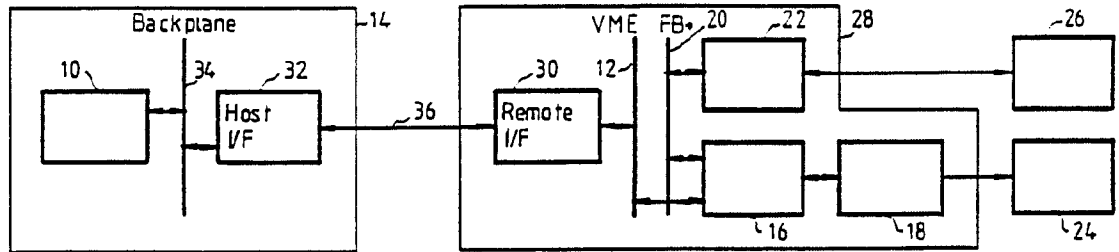
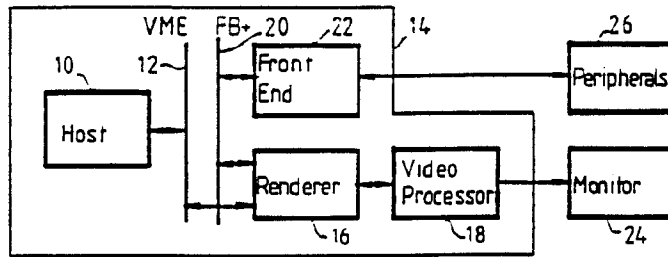
16. A system as claimed in claim 15, wherein the page fault determining means is operable to indicate a page fault (748P) if the basic page is not stored in the second memory (794 output A), or if any said other aligned page is not stored in the second memory (794 output B, C, D) and (796 B, C, D) that page is required to be accessed.

17. A system as claimed in any of claims 12 to 16, wherein said page edge determining means (758) is operable to determine when the aligned patch address component (px,py) has a maximum value (e.g. 31) for a patch in a page and the misalignment component (mx,my) together with said maximum value indicate that the patch (p) to be accessed extends outside the page (PAGE A) defined by the aligned page address component.

18. A system as claimed in any preceding claim, wherein the data-elements are image pixel data.

19. A system as claimed in any preceding claim, wherein the second memory is a video RAM (700).

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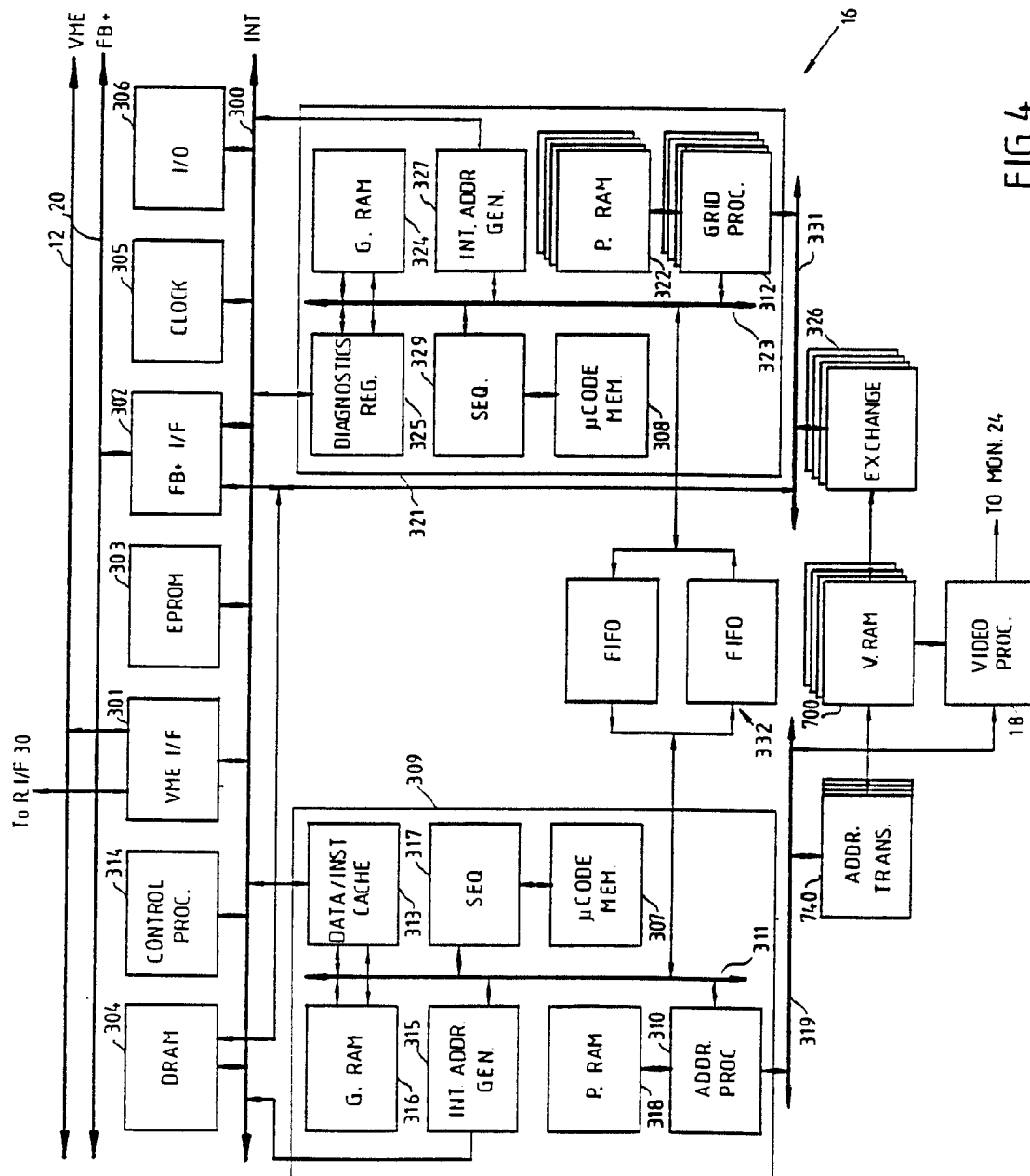


FIG. 4

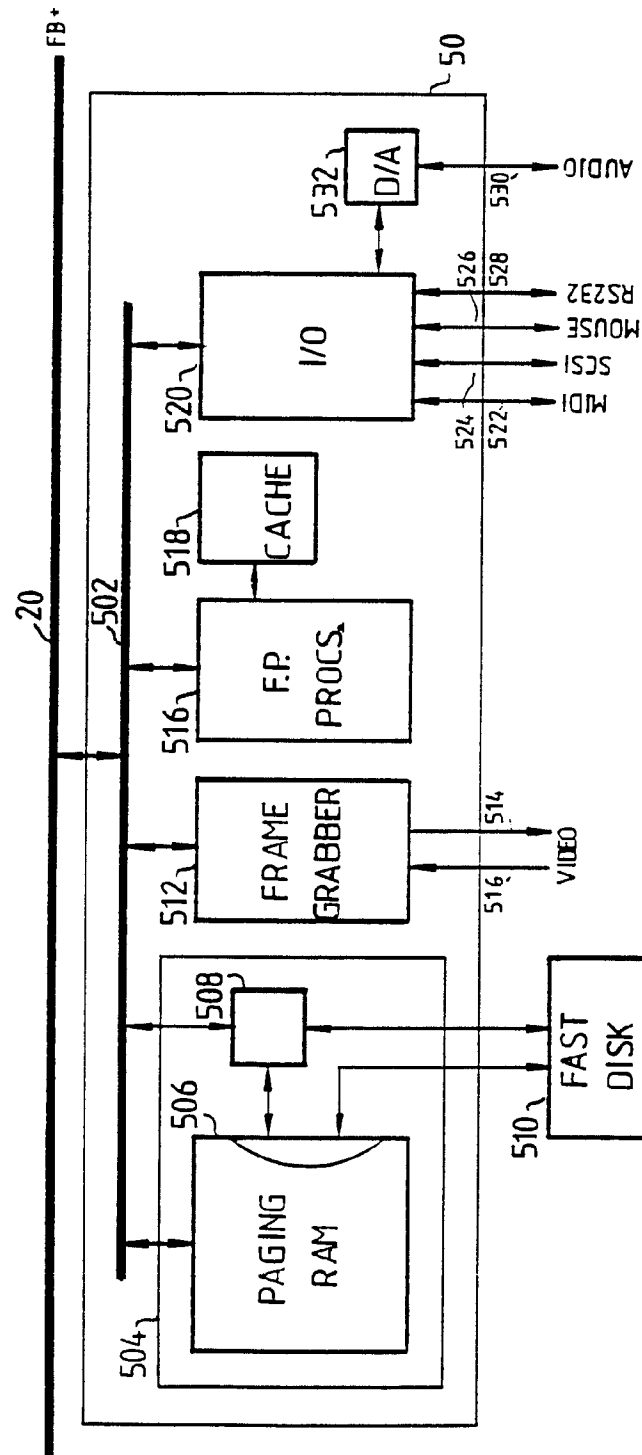
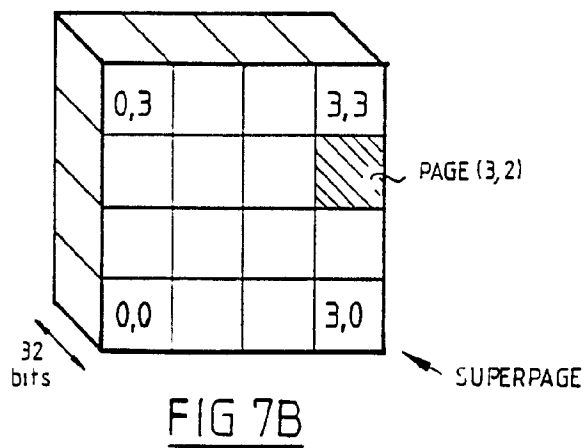
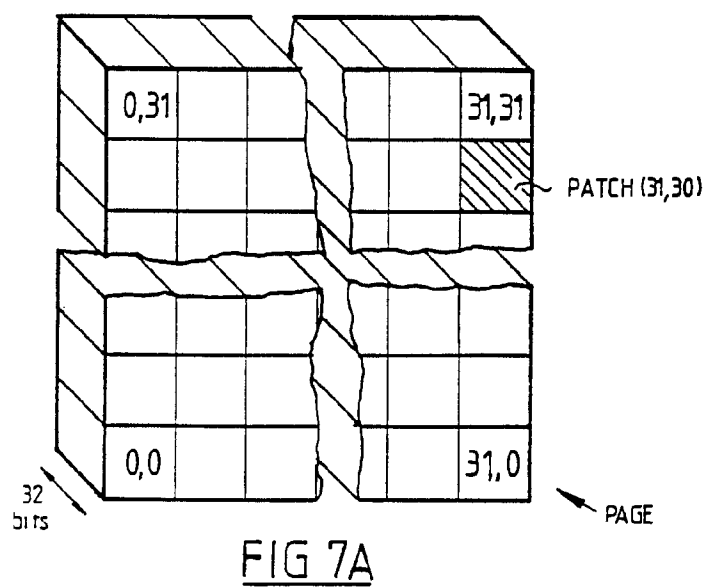
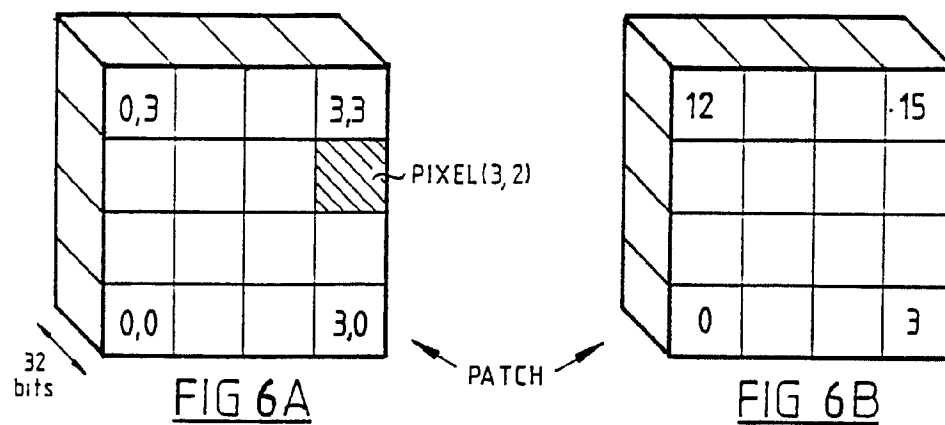
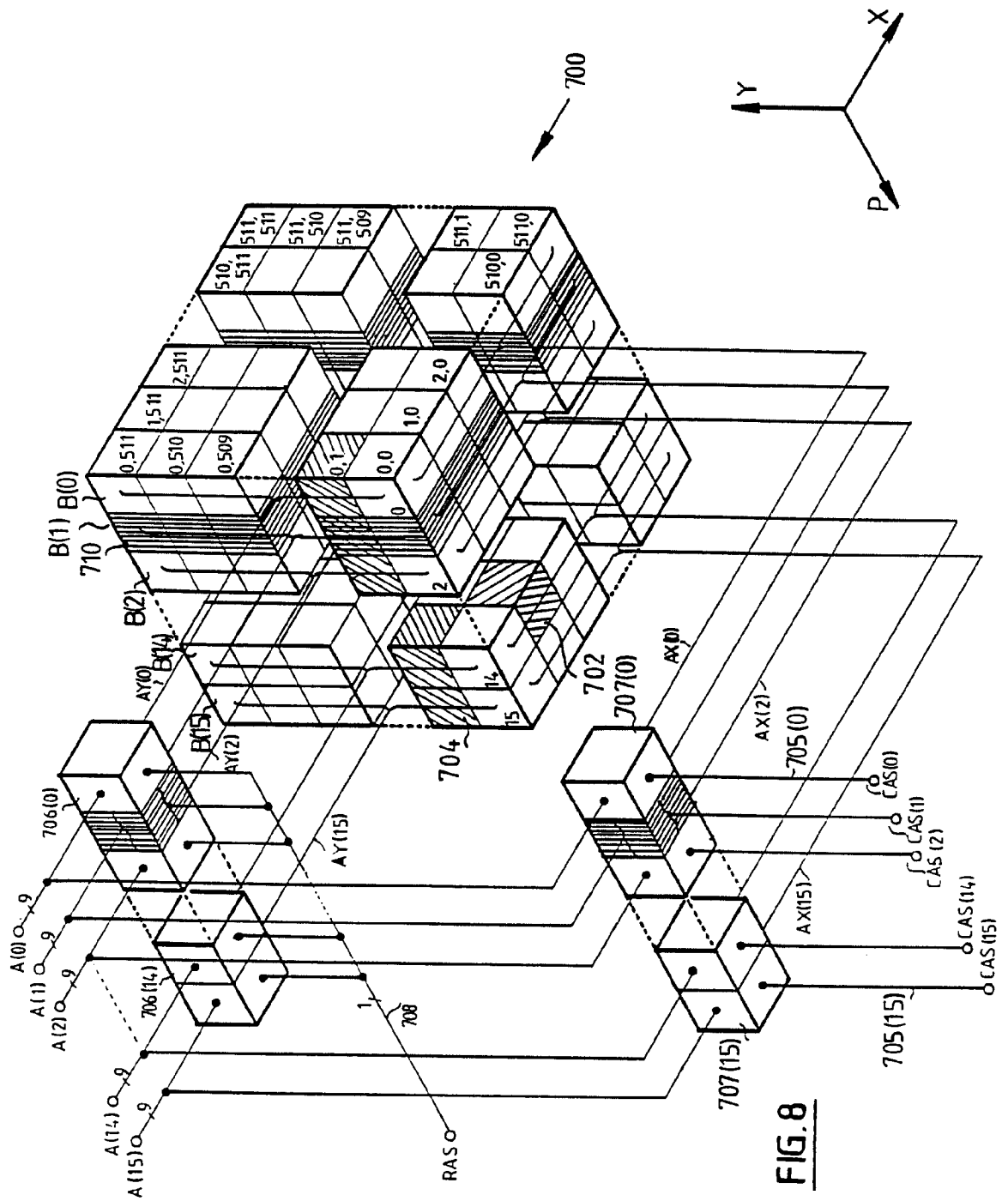


FIG. 5

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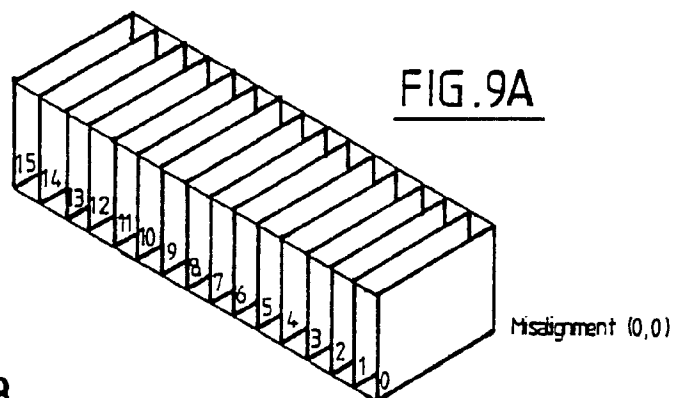
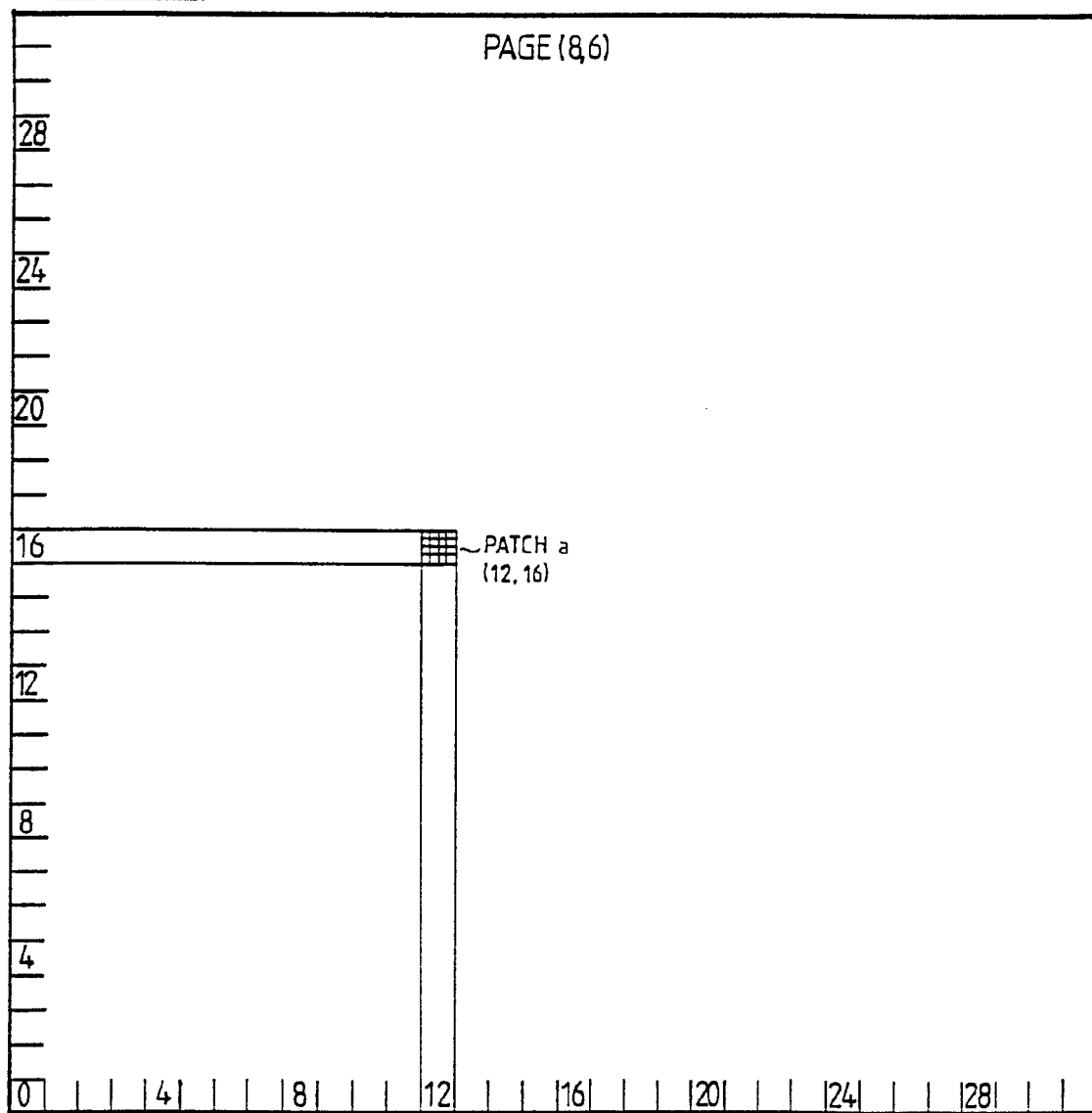
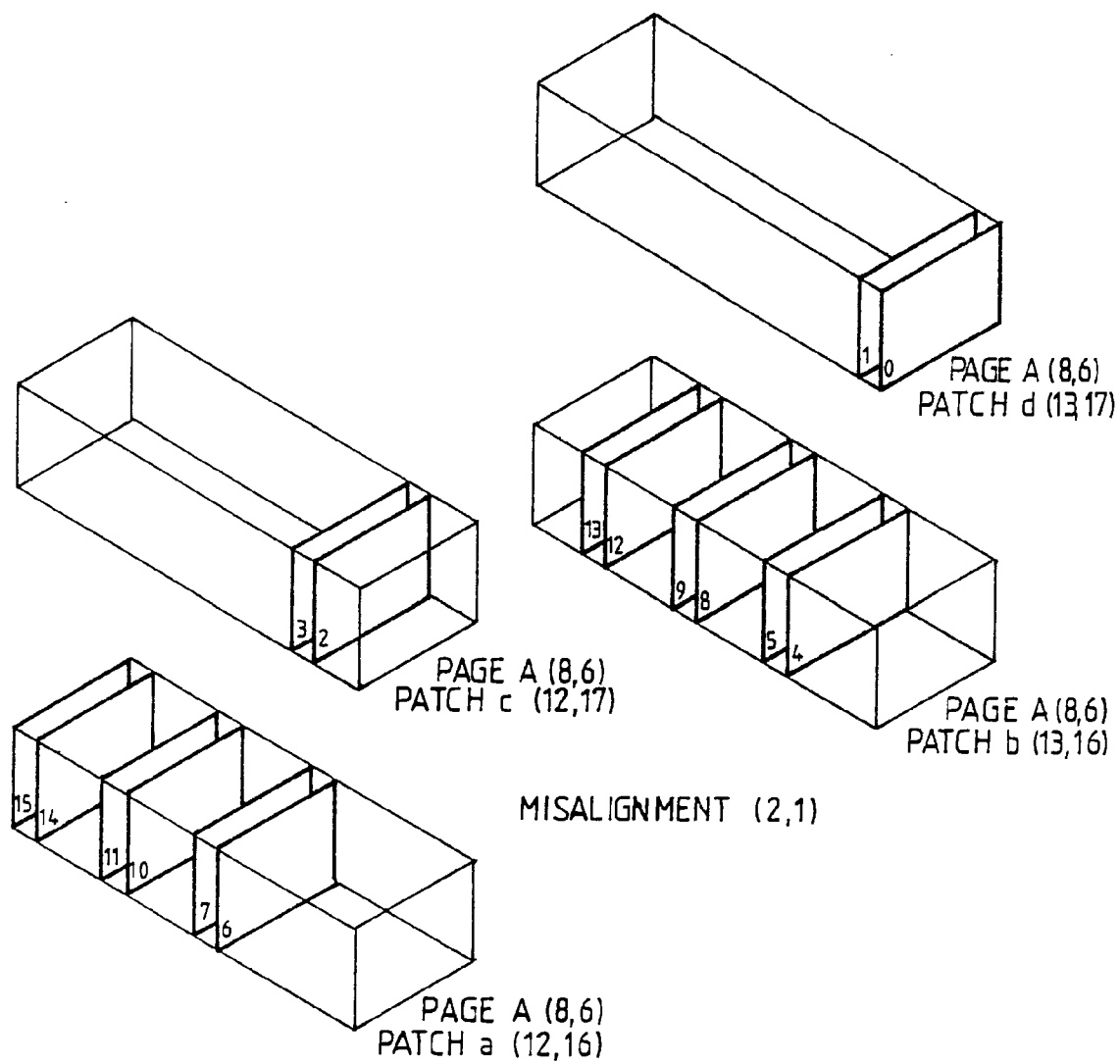


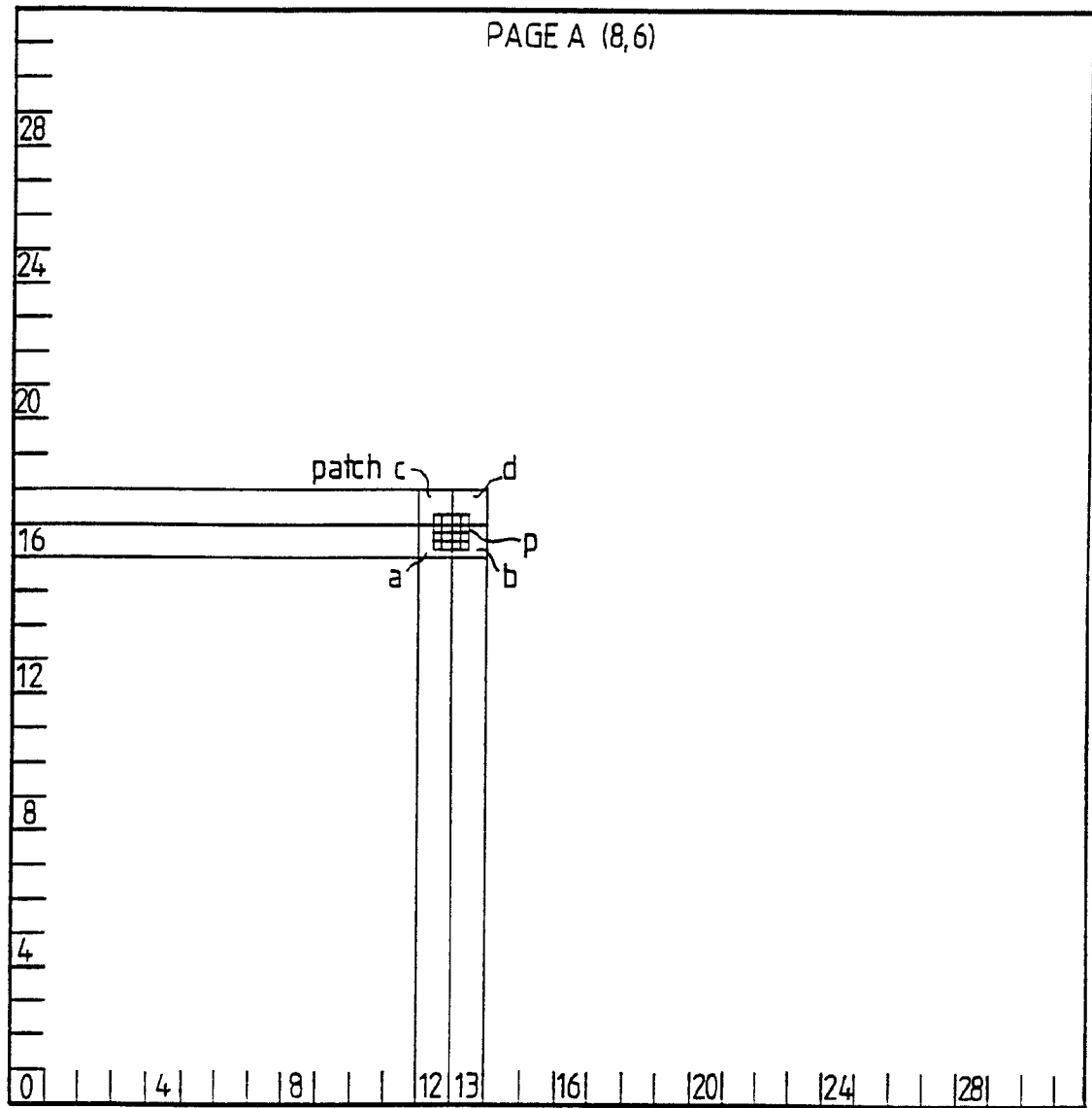
FIG. 9B



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FIG. 10A

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FIG. 10 B

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FIG.11A

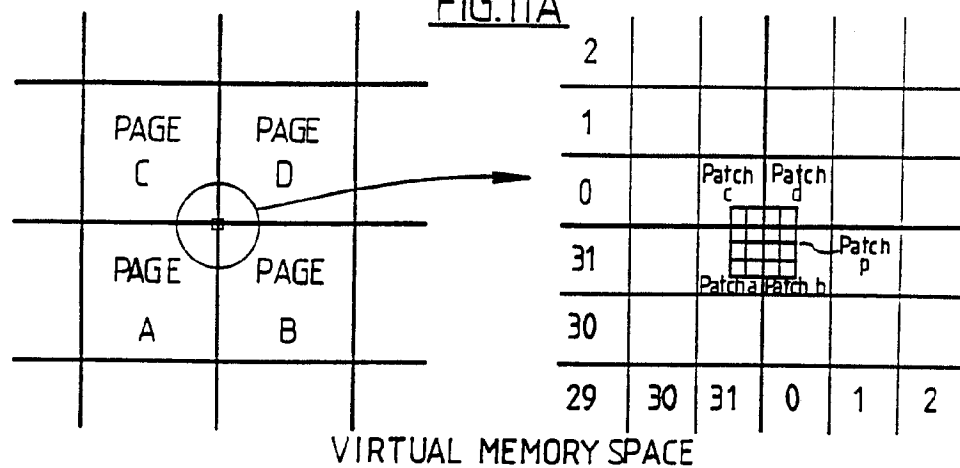
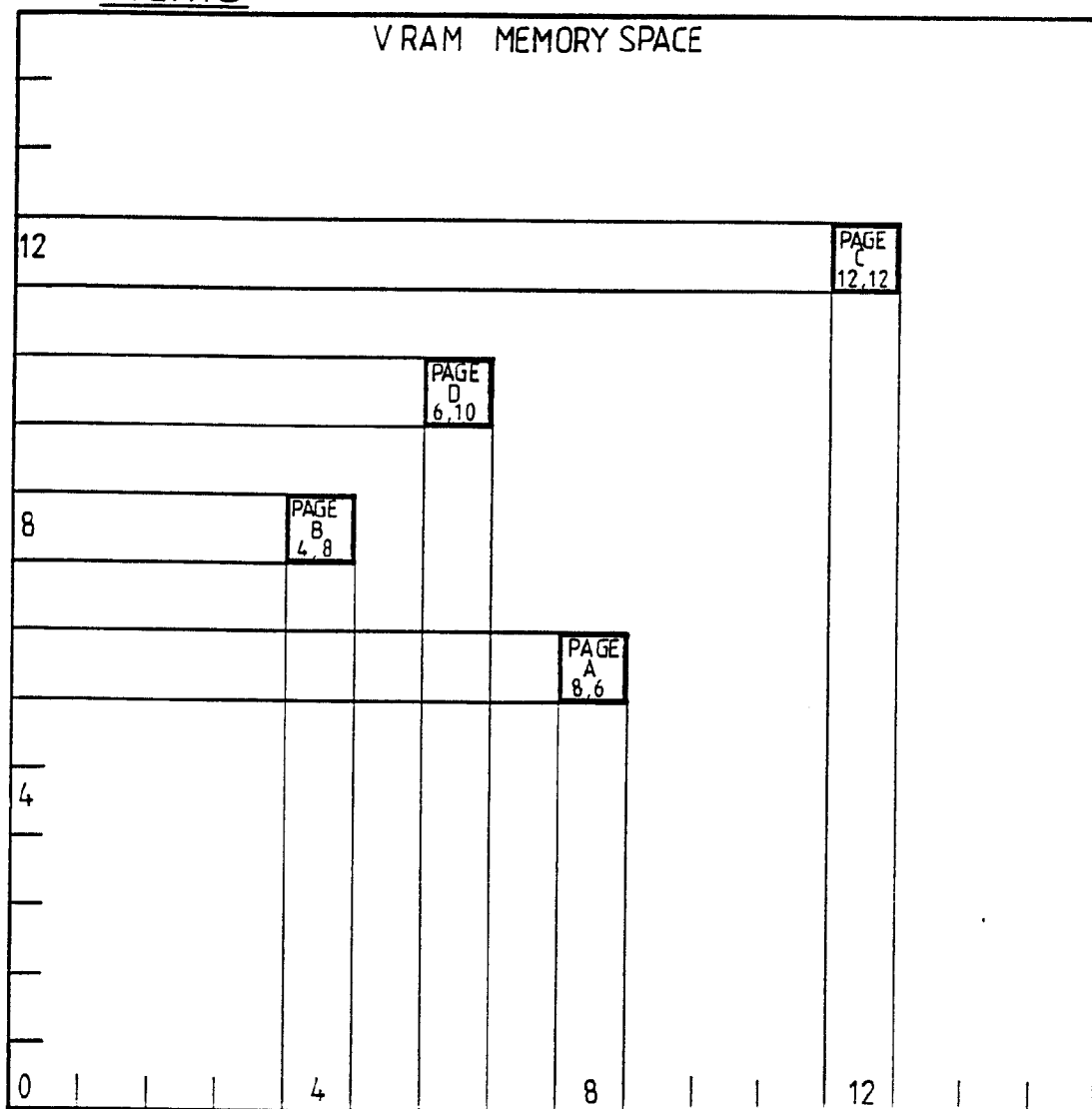
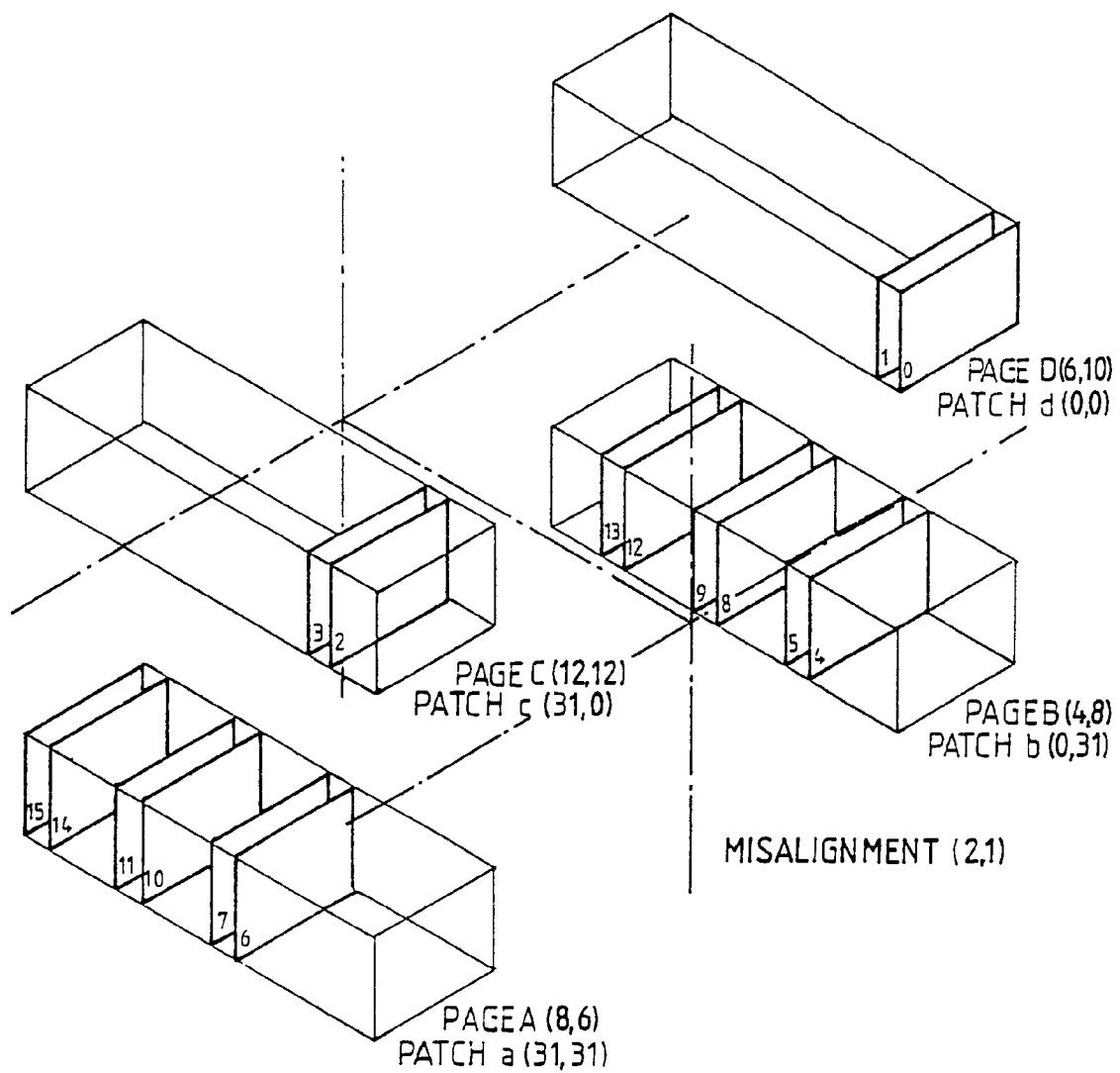


FIG.11B



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FIG.11C

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PAGE SELECTION

	IF $4px+mx < 125$ & $4py+my < 125$	IF $4px+mx > 124$ & $4py+my < 125$	IF $4px+mx < 125$ & $4py+my > 124$	IF $4px+mx > 124$ & $4py+my > 124$
IF $bx \geq mx$ & $by \geq my$	A	A	A	A
IF $bx < mx$ & $by \geq my$	A	B	A	B
IF $bx \geq mx$ & $by < my$	A	A	C	C
IF $bx < mx$ & $by < my$	A	B	C	D

FIG. 12

PATCH ADDRESS
INCREMENT SELECTION
X ADDRESS

CONDITION	PATCH X ADDRESS
$bx \geq mx$	px
$bx < mx$	$px+1$

Y ADDRESS

CONDITION	PATCH Y ADDRESS
$by \geq my$	py
$by < my$	$py+1$

FIG. 13

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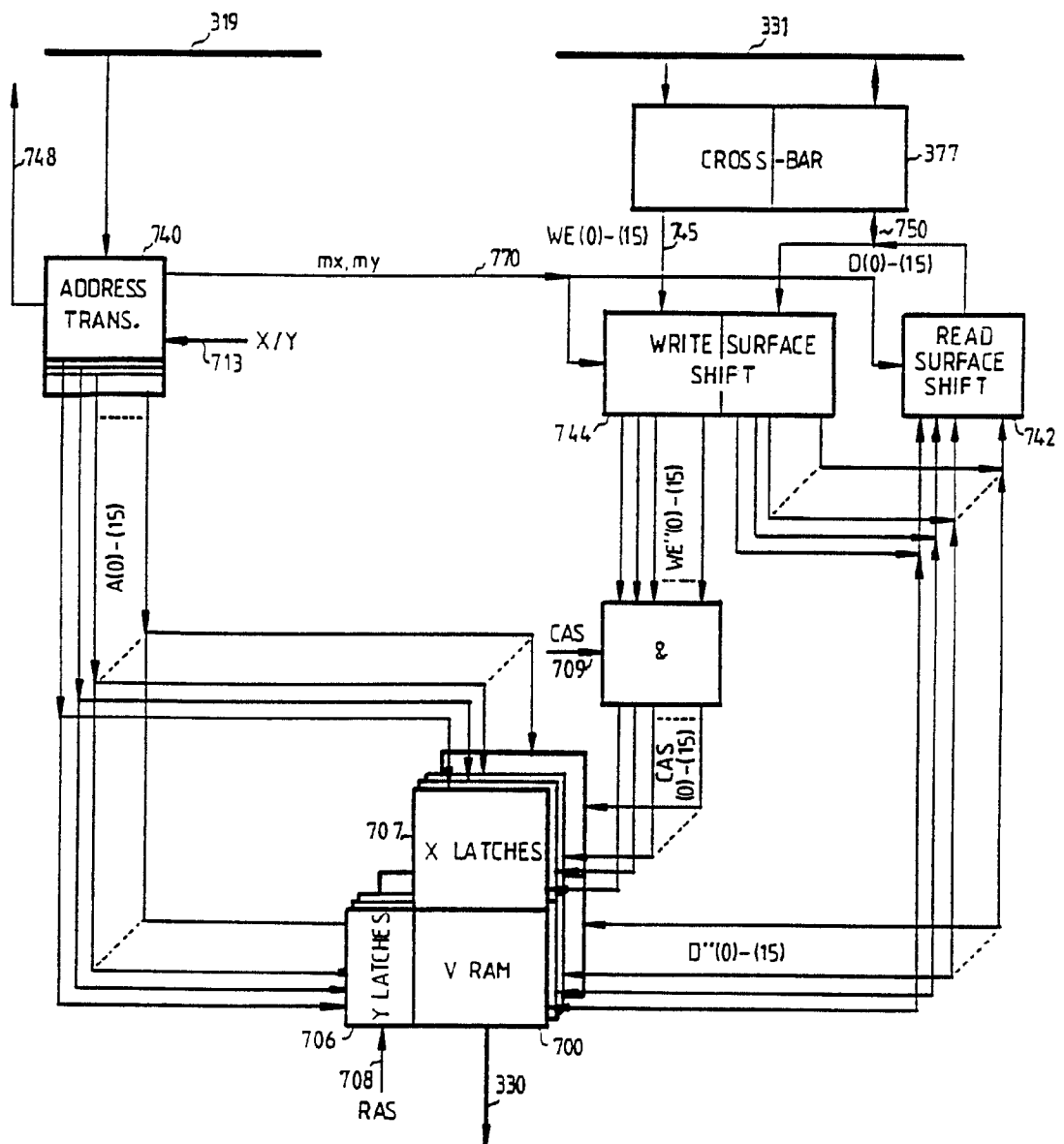


FIG. 14

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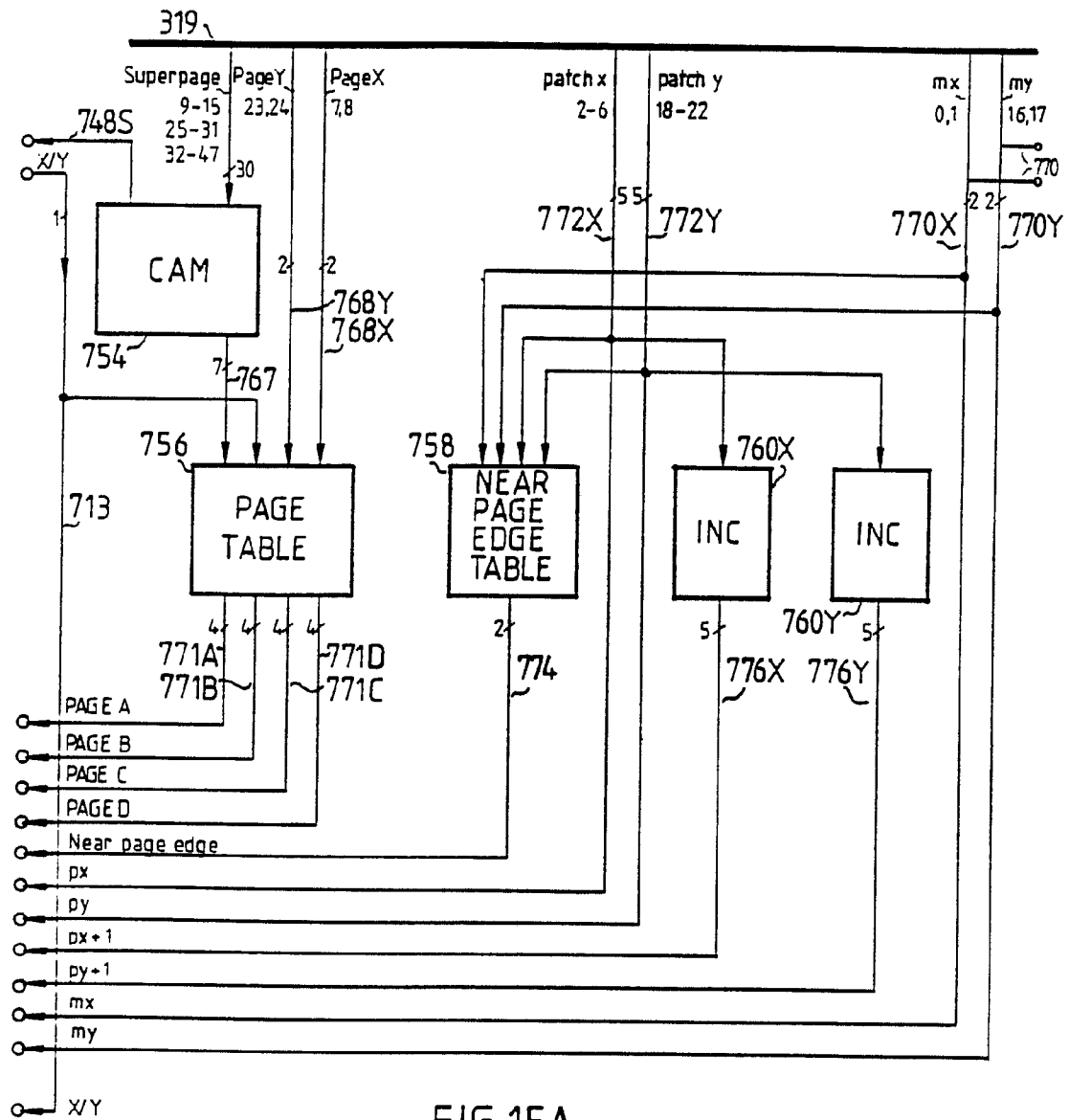


FIG. 15A

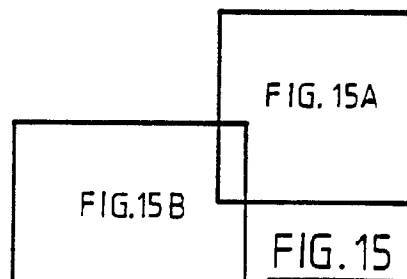
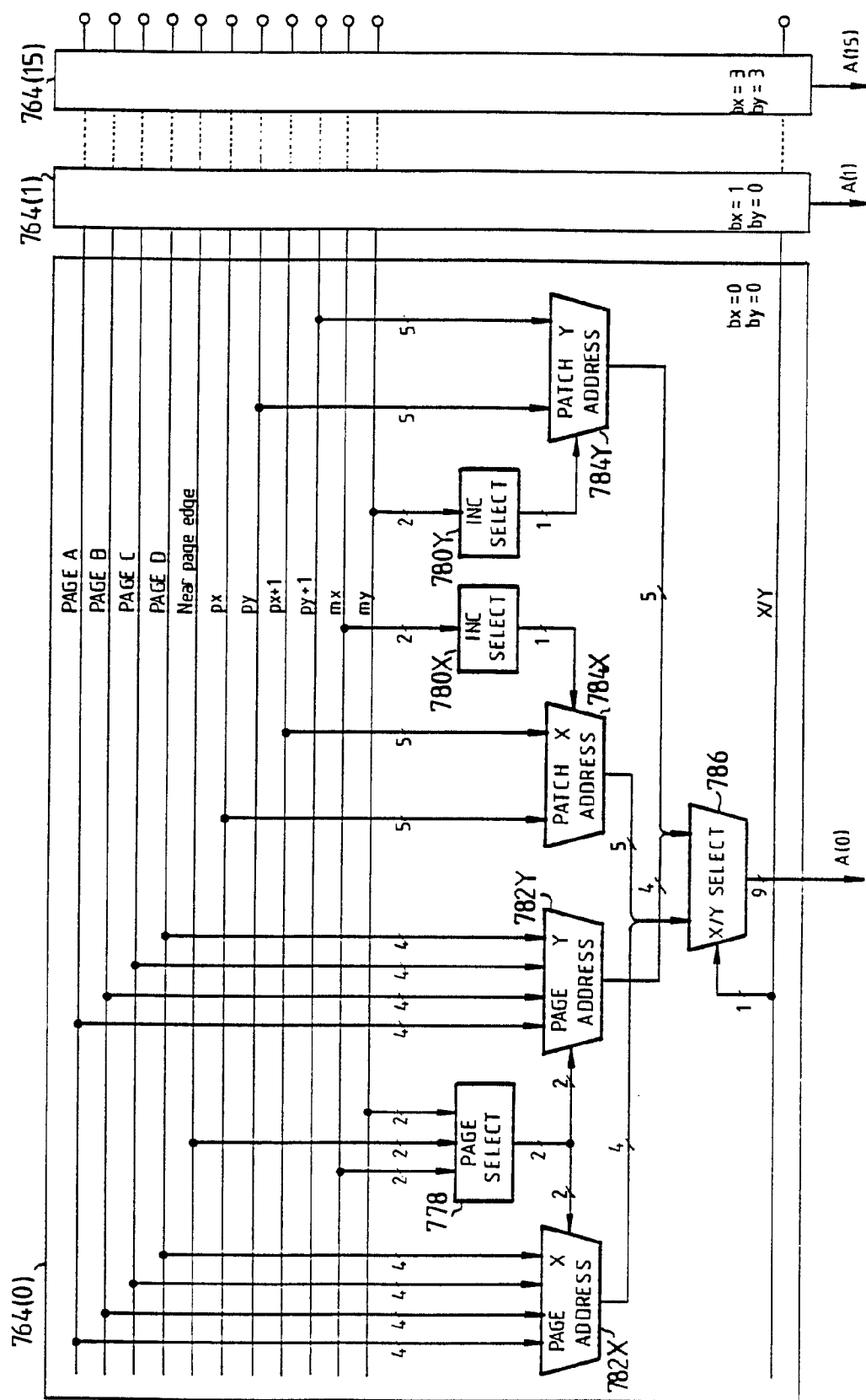


FIG. 15B



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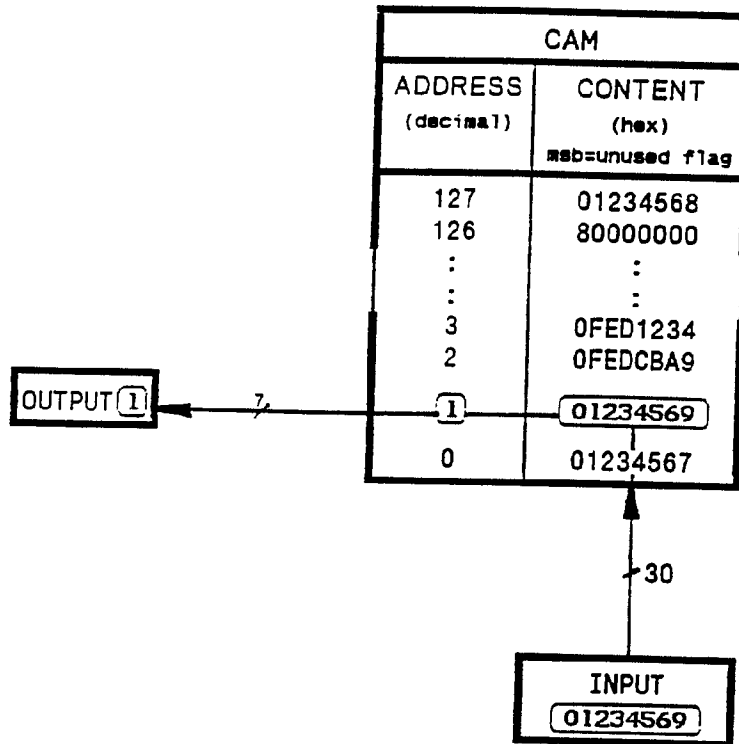
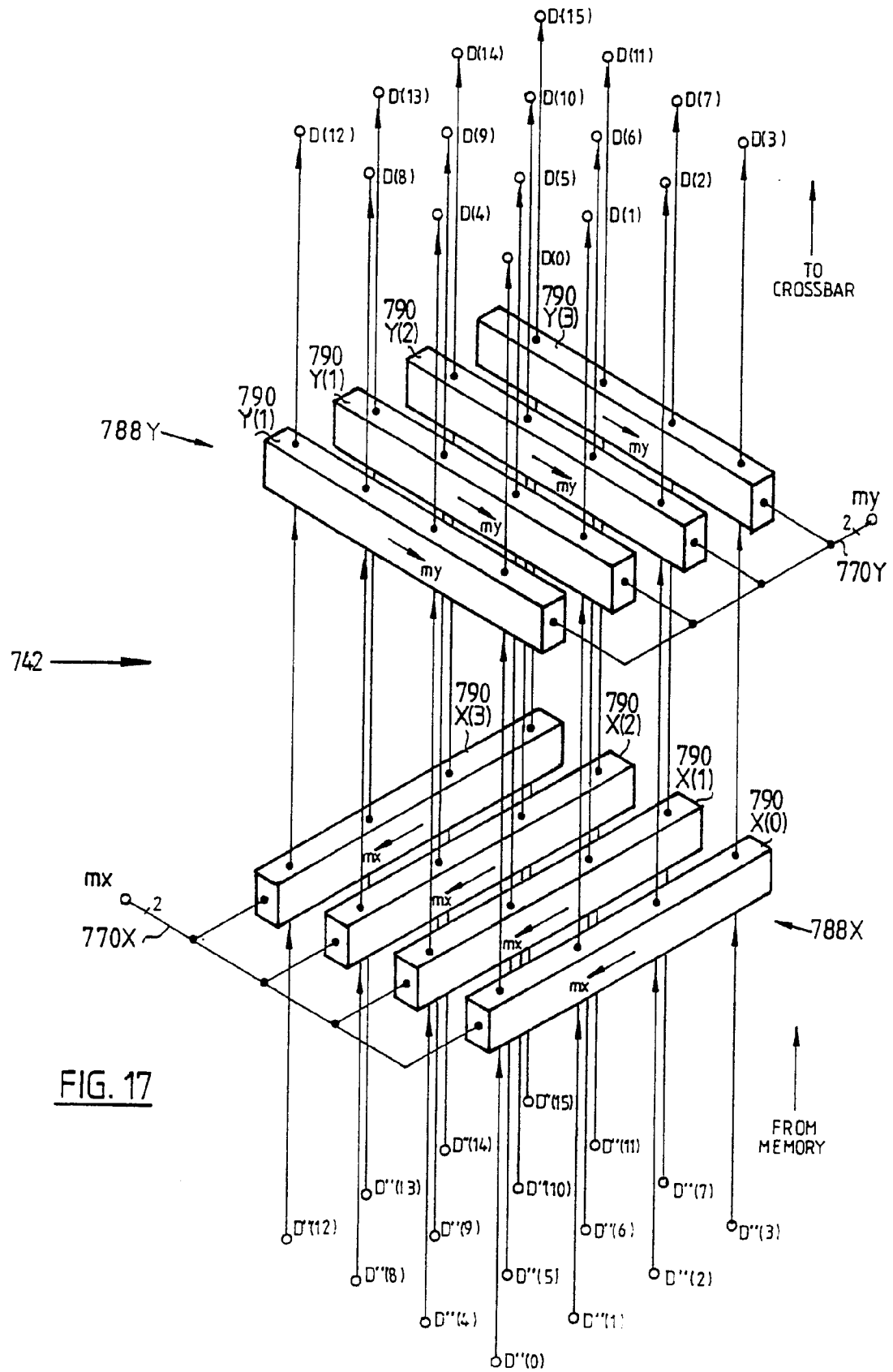


FIG. 16

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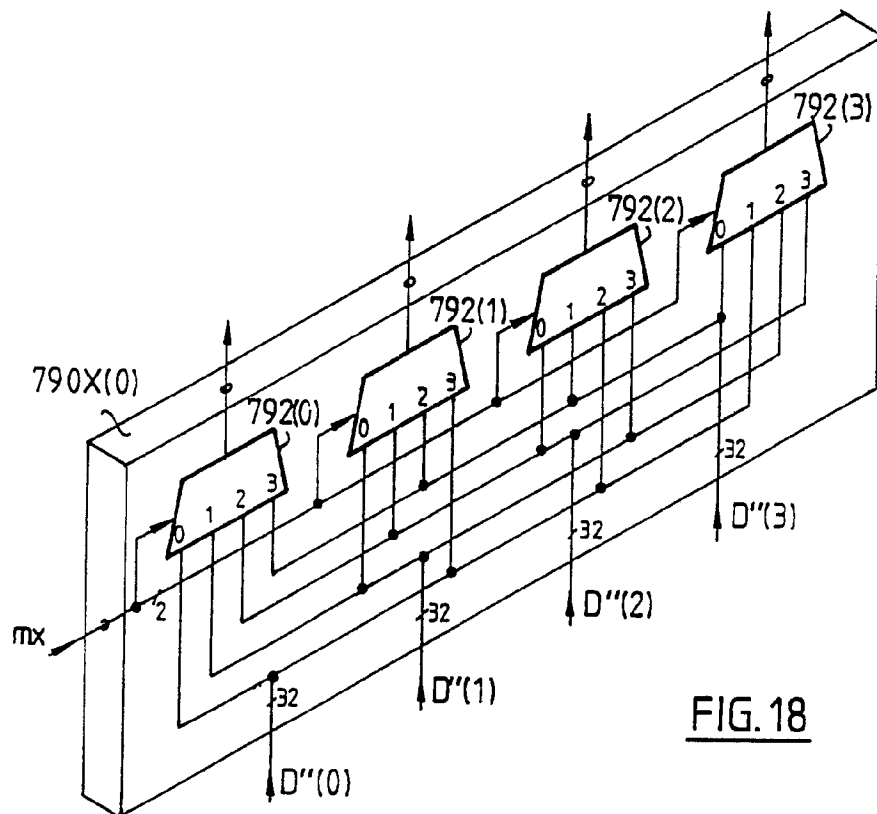


FIG. 18

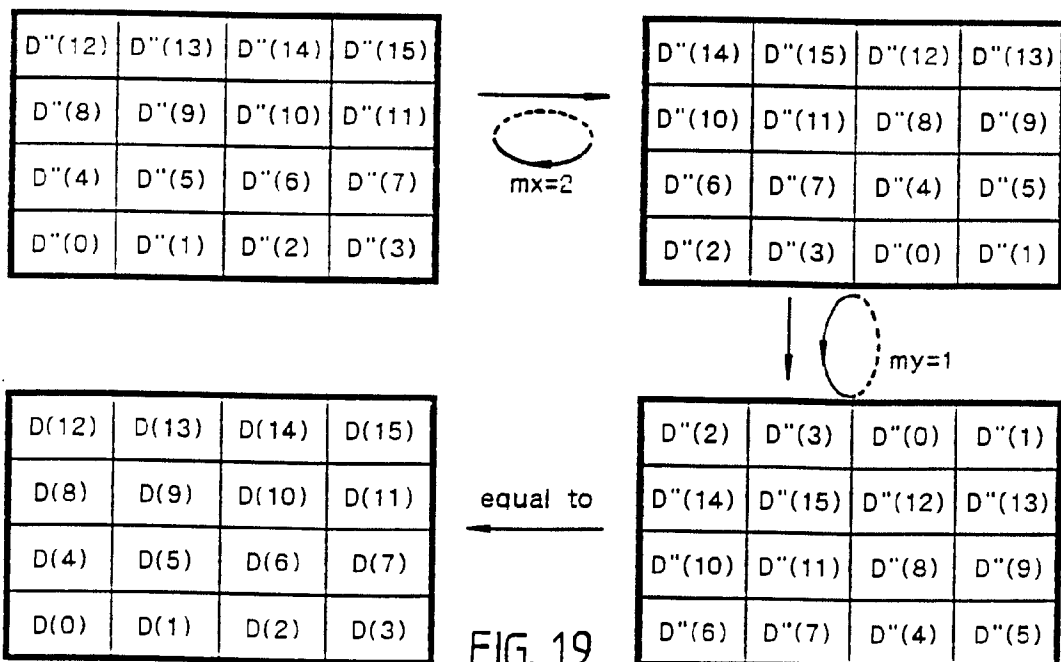


FIG. 19

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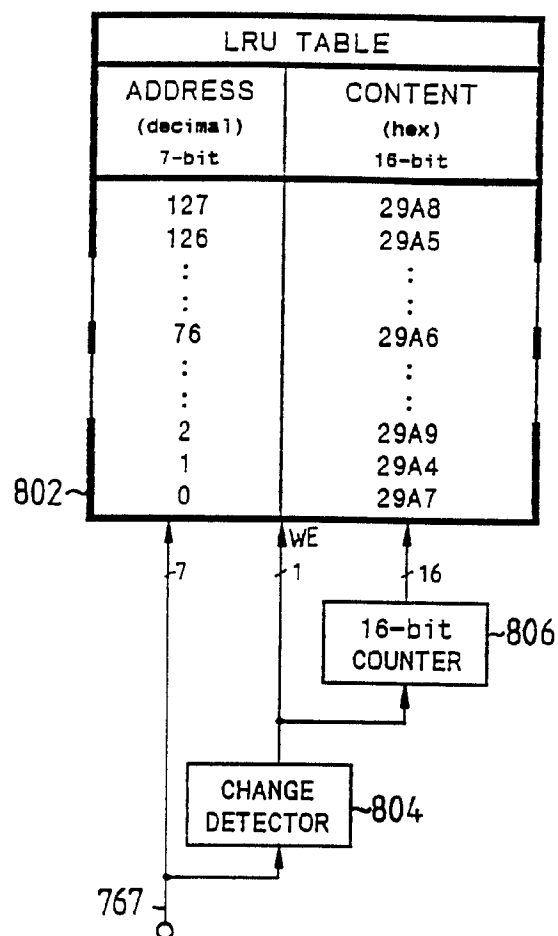
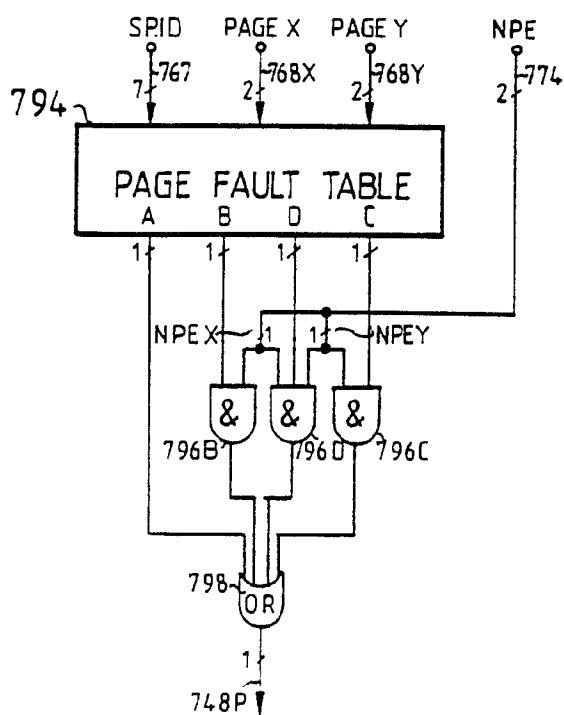
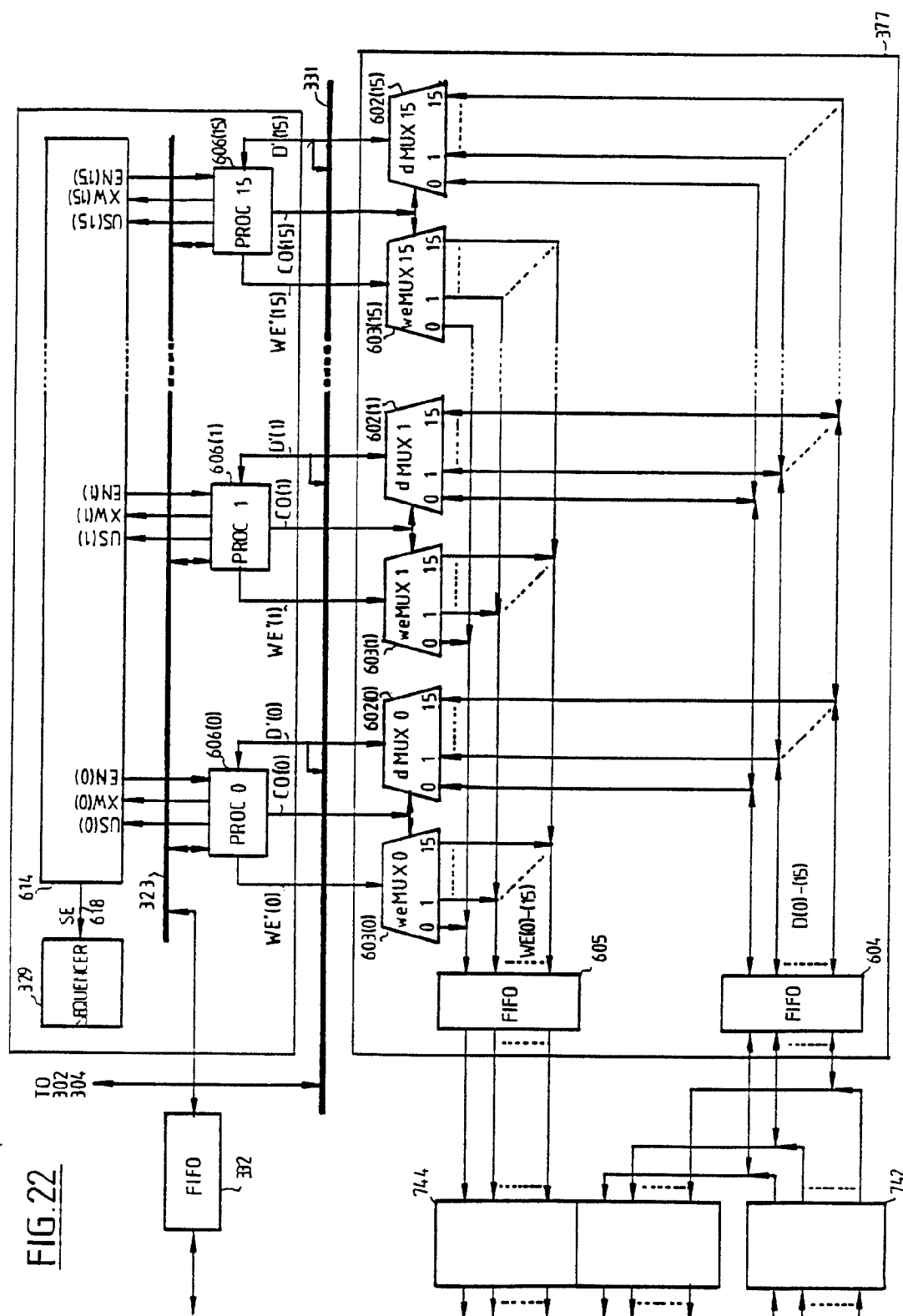
FIG. 20FIG. 21

FIG. 22



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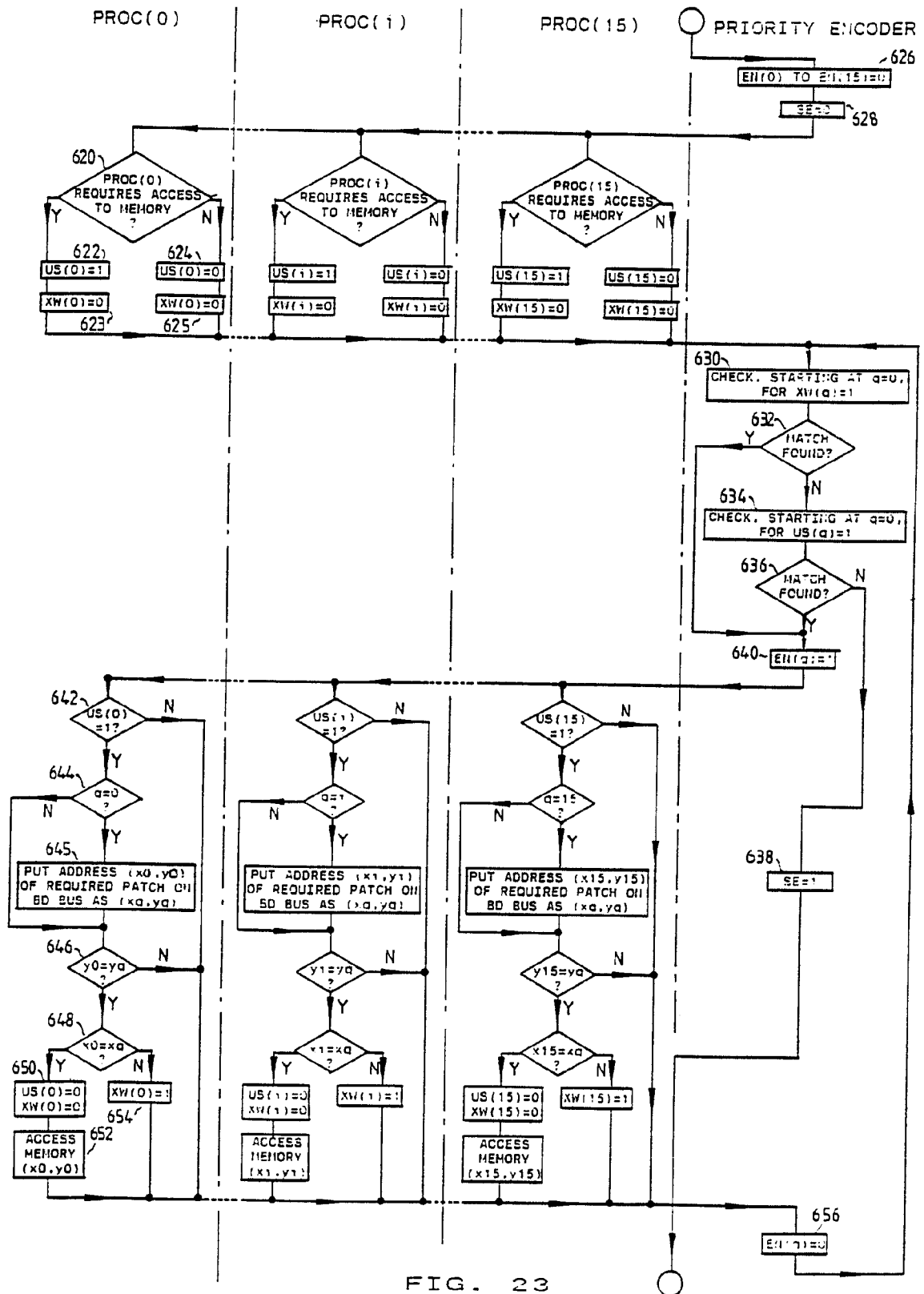


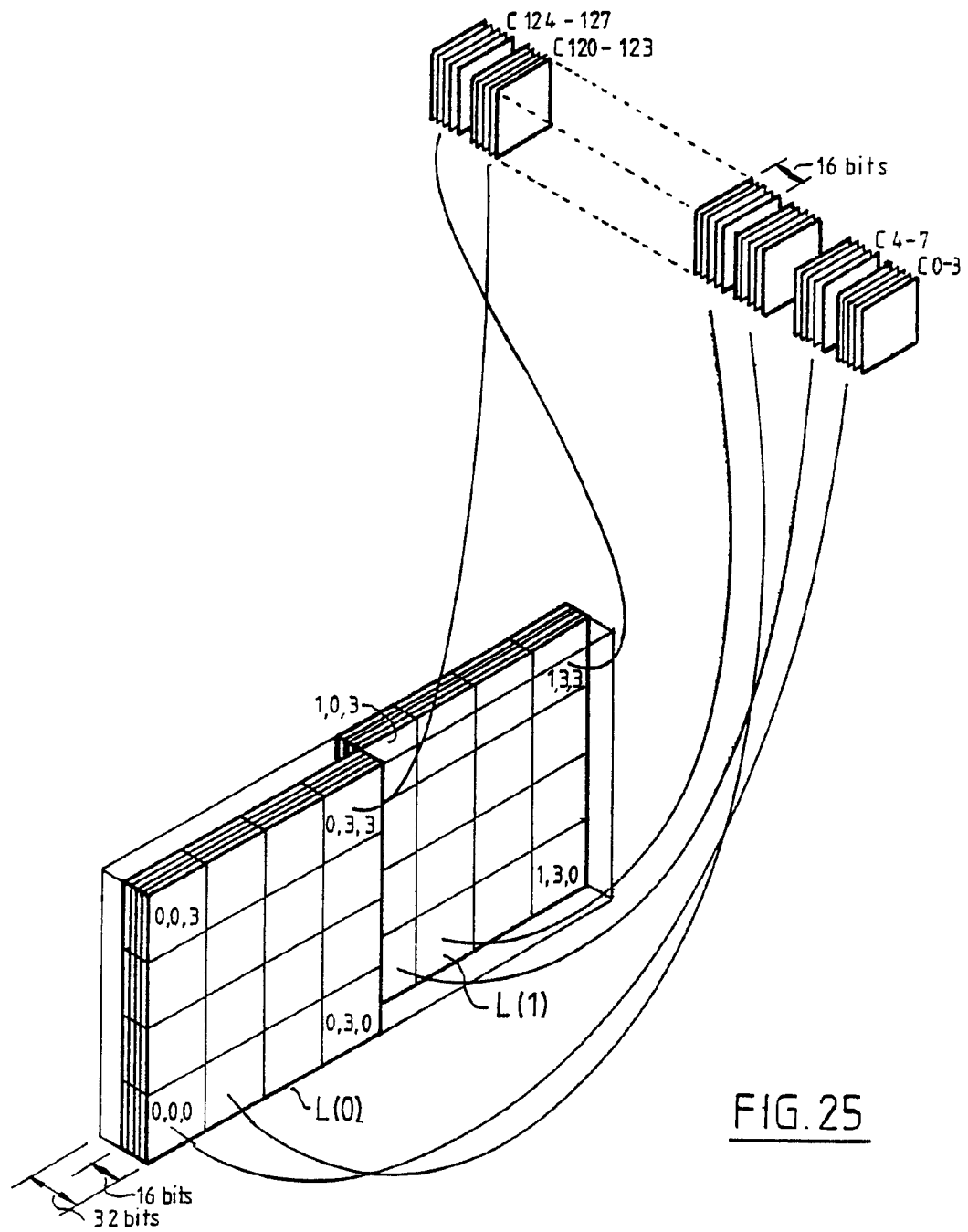
FIG. 23

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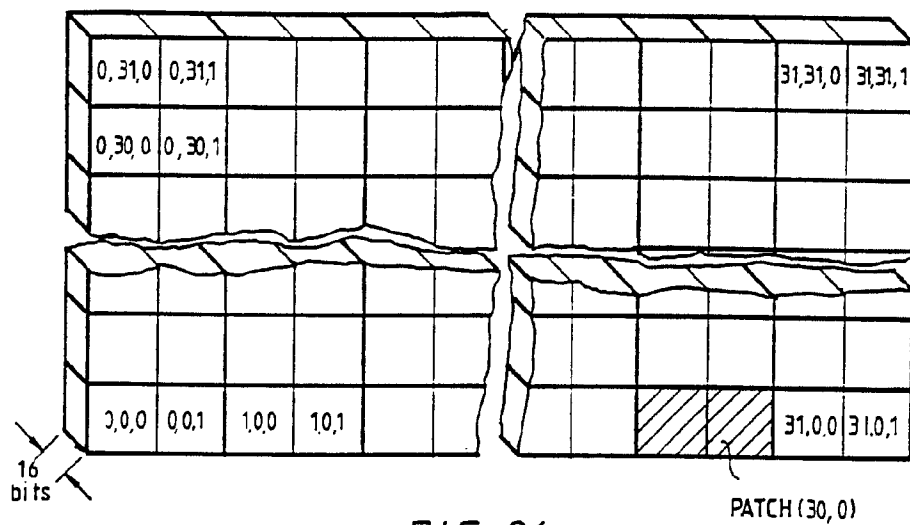
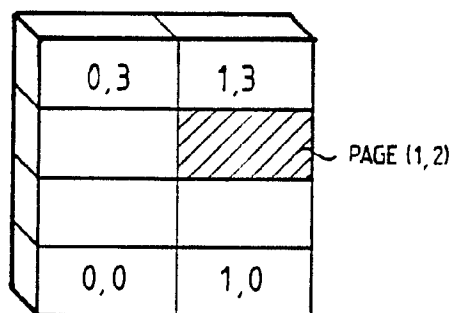
i	660		662		664		666		668		670		672		674	
	xi hex	yi hex	Init. US XW		q=0 US XW		q=2 US XW		q=10 US XW		q=3 US XW		q=11 US XW		q=8 US XW	
0	1234	1234	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1234	1234	1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1235	1234	1	0	1	1	0	0	0	0	0	0	0	0	0	0
3	1235	1235	1	0	1	0	1	0	1	0	0	0	0	0	0	0
4	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	1234	1236	1	0	1	0	1	0	1	0	1	0	1	0	0	0
9	1234	1236	1	0	1	0	1	0	1	0	1	0	1	0	0	0
10	1236	1234	1	0	1	1	1	1	0	0	0	0	0	0	0	0
11	1236	1235	1	0	1	0	1	0	1	0	1	1	0	0	0	0
12	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ADDRESSES READ					x0,y0 x1,y1		x2,y2		x10,y10		x3,y3		x11,y11		x8,y8 x9,y9	

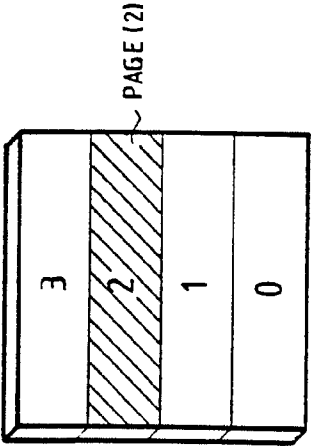
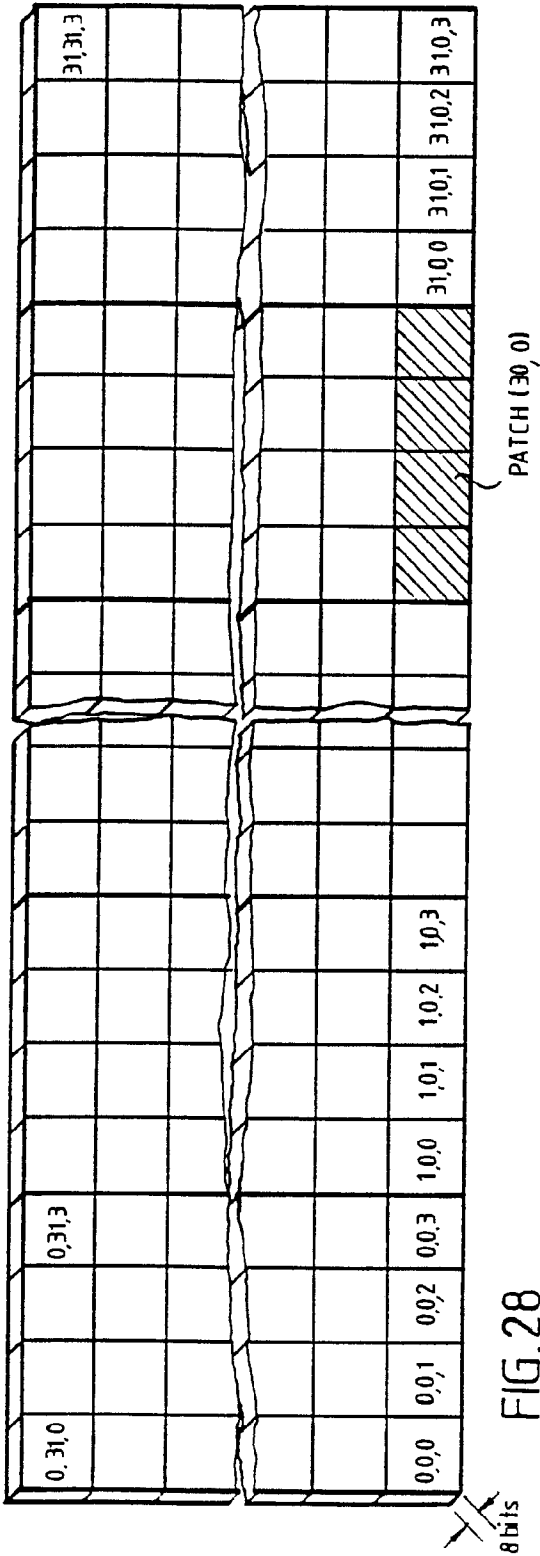
FIG. 24

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FIG. 26FIG. 27



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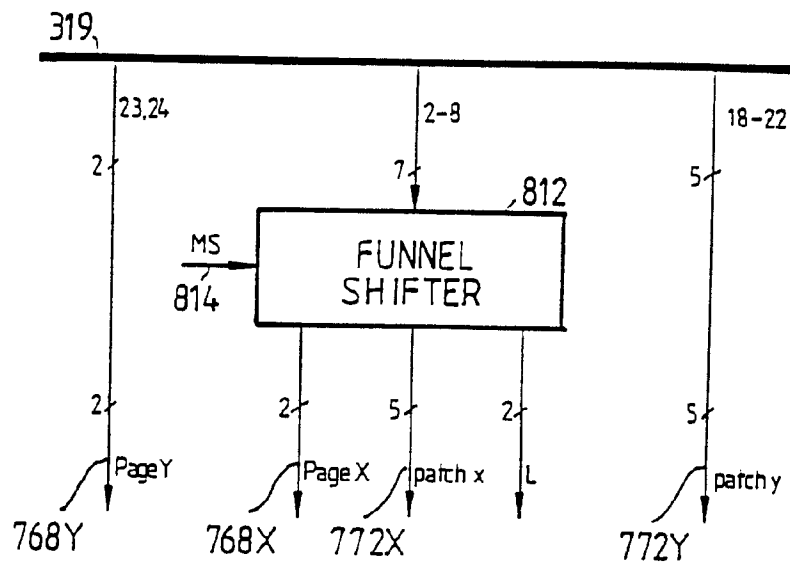


FIG. 30A

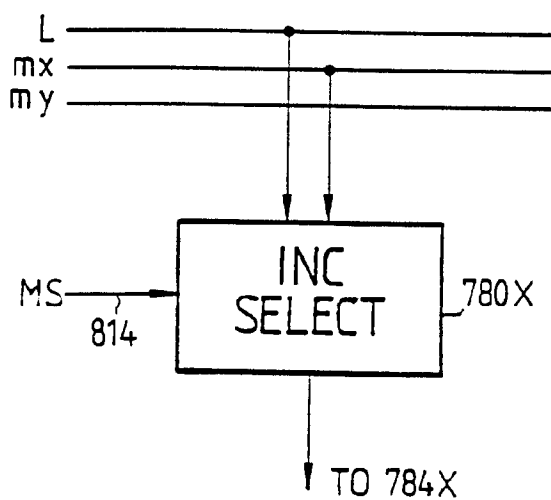


FIG. 30B

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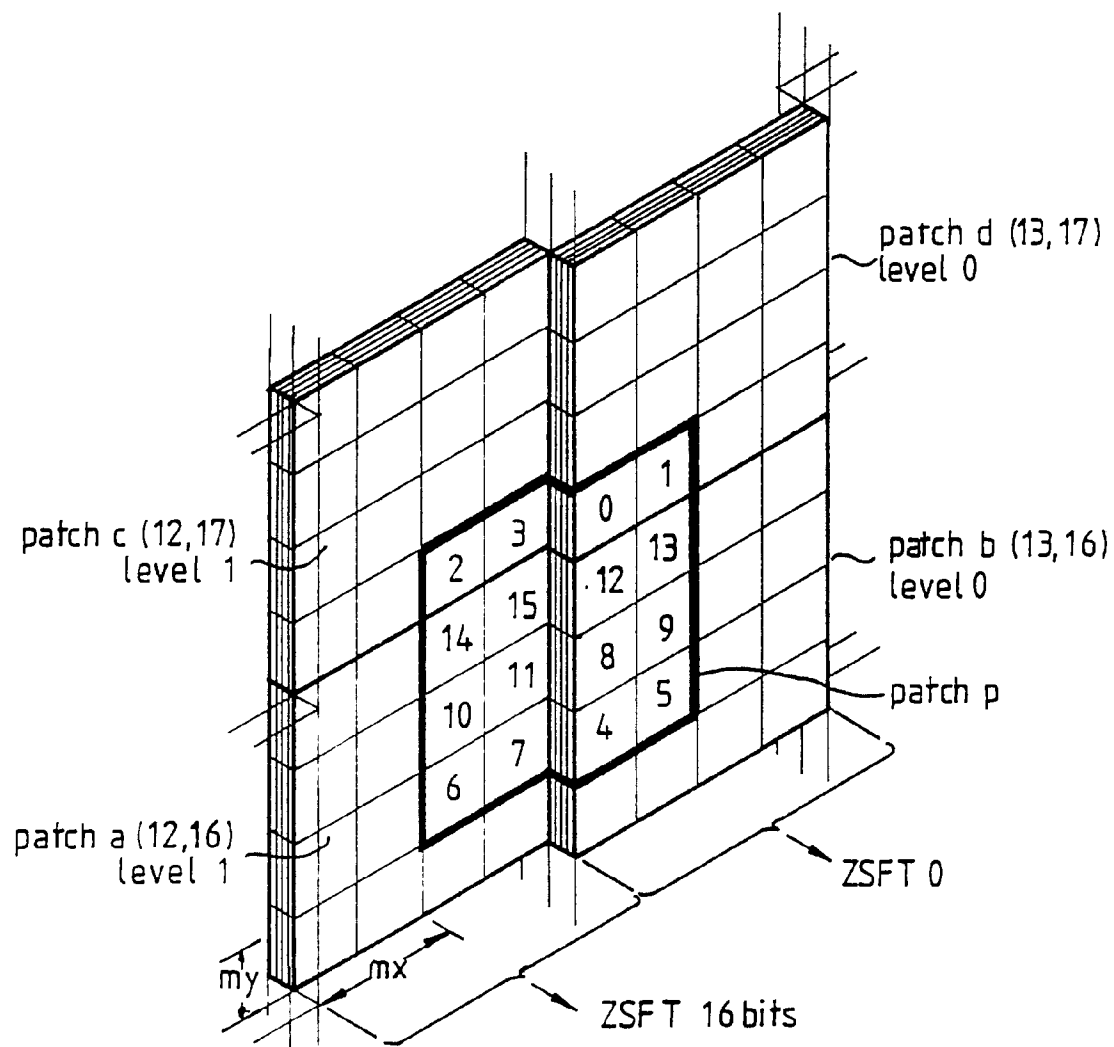
CONDITION			PATCH X ADDRESS
32-bit L ignored	16-bit L=0 or 1	8-bit $0 \leq L \leq 3$	
$bx \geq mx$	L=0 OR $bx \geq mx$	$L < 3$ OR $bx \geq mx$	px
$bx < mx$	L=1 AND $bx < mx$	L=3 AND $bx < mx$	px+1

FIG. 30C

VIRTUAL ADDRESS BITS							
MODE	8	7	6	5	4	3	2
32-bit	PAGE PX (2)		patch px (5)				
16-bit	PX (1)	px (5)					L (1)
8-bit	px (5)					L (2)	

FIG. 31

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FIG. 32

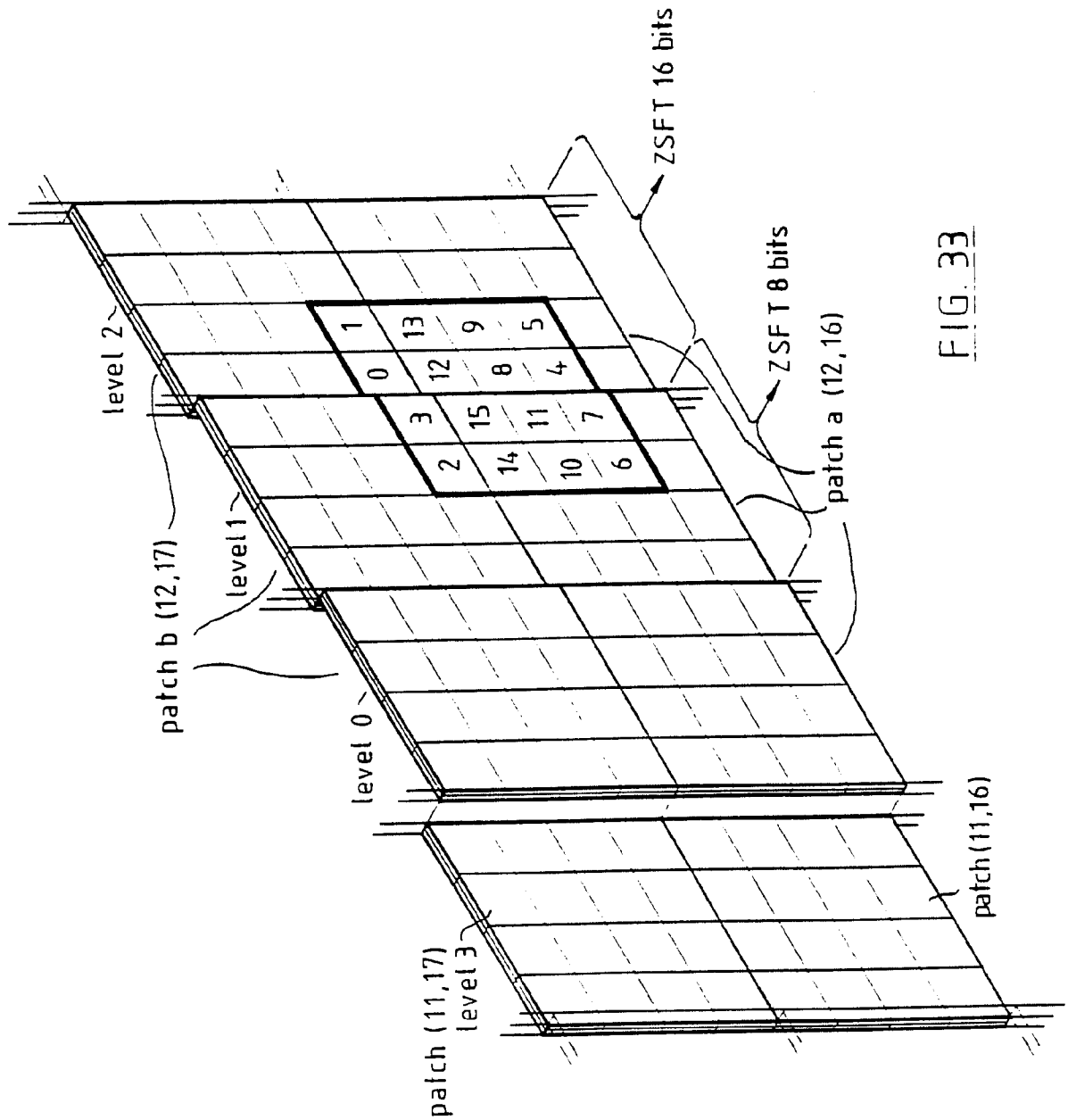


FIG. 33

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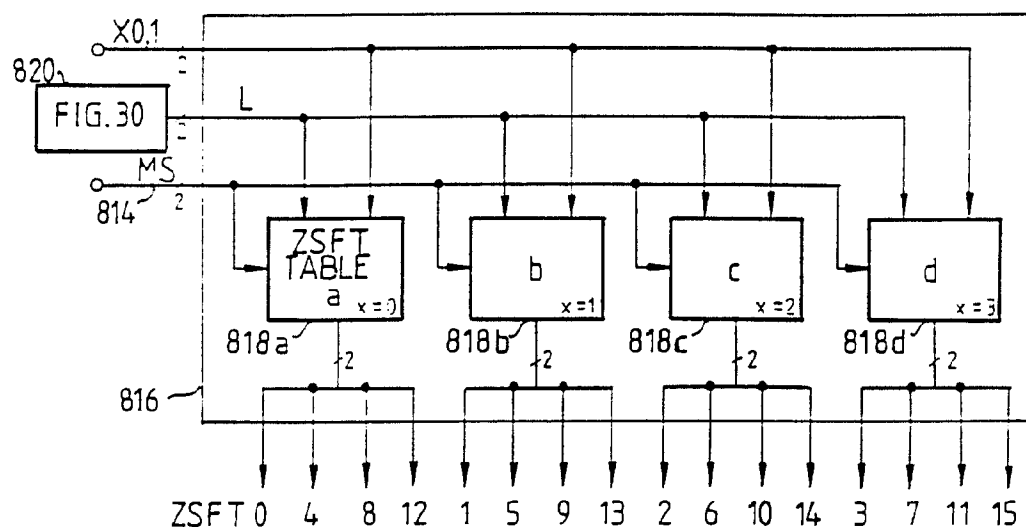


FIG. 34

GENERAL ZSFT TABLE

	8-bit MODE				16-bit MODE		32-bit
	IF L=0	IF L=1	IF L=2	IF L=3	IF L=0	IF L=1	
IF $mx+x+4$	0	1	2	3	0	2	ALWAYS
IF $mx+x+3$	1	2	3	0	2	0	0

FIG. 35A

ZSFT TABLE 818b (x=1)

ZSFT (1), (5), (9), (13)

mx \ L	8-bit MODE				16-bit MODE		32-bit
	0	1	2	3	0	1	
0	0	1	2	3	0	2	ALWAYS 0
1	0	1	2	3	0	2	
2	0	1	2	3	0	2	
3	1	2	3	0	2	0	

FIG. 35B

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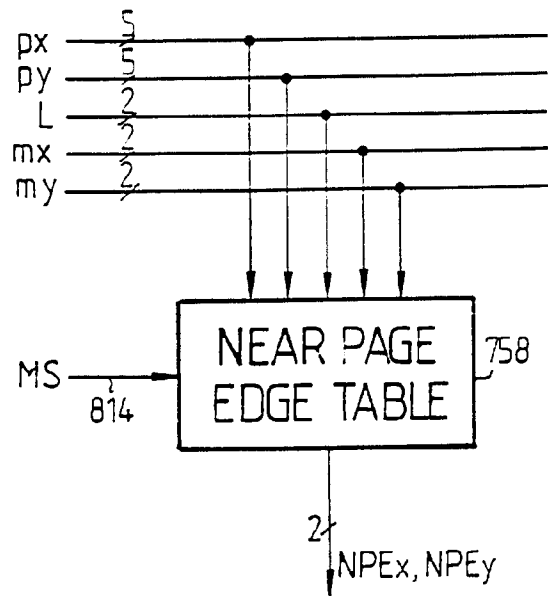


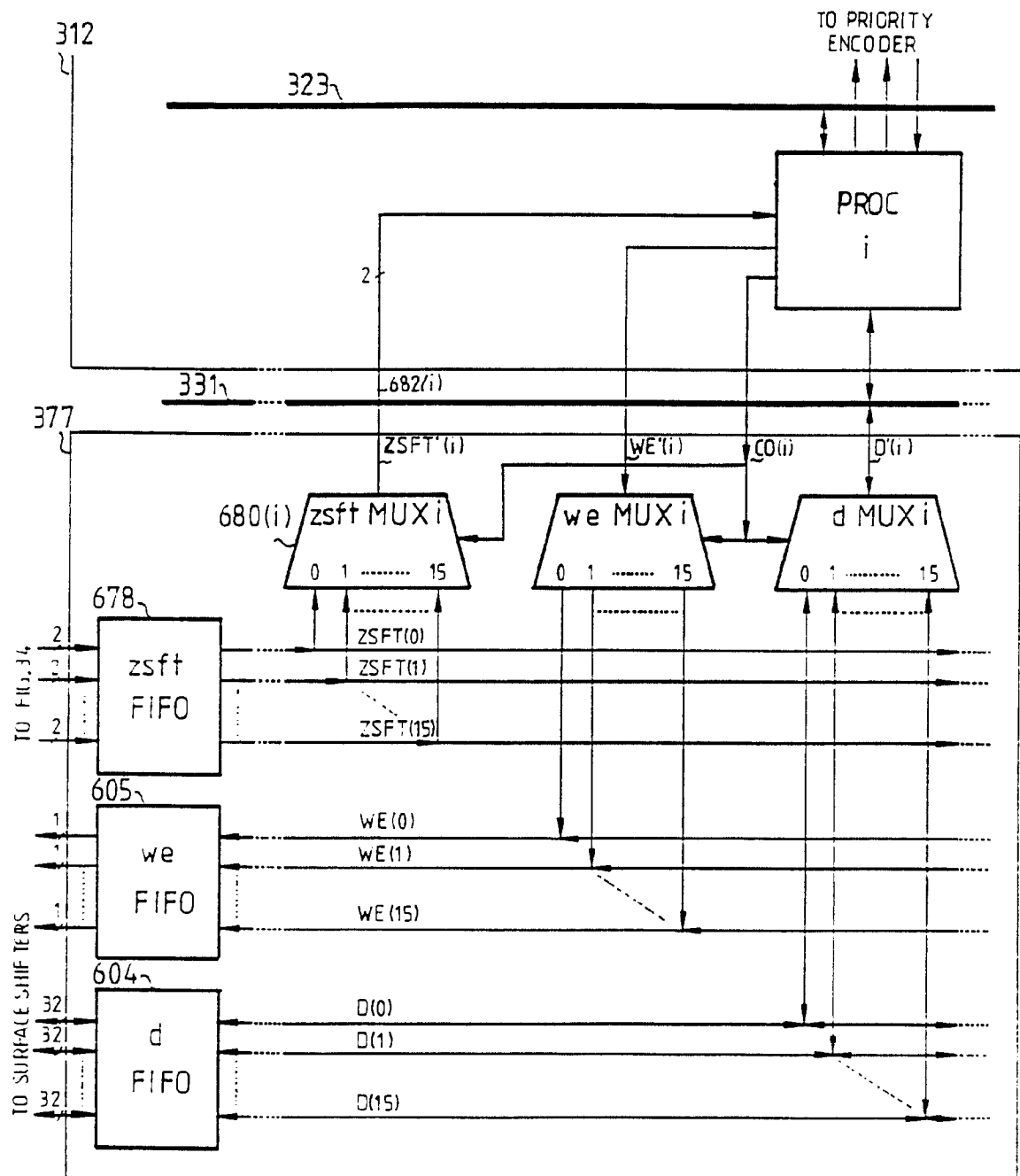
FIG. 36A

CONDITION			
32-bit L=0	16-bit L=0 or 1	8-bit 0≤L≤3	
$4px+(4L)+mx<125$	$8px+4L+mx<253$	$16px+4L+mx<509$	NPE _x =0
$4px+(4L)+mx>124$	$8px+4L+mx>252$	$16px+4L+mx>508$	NPE _x =1
$4py+my<125$	$4py+my<125$	$4py+my<125$	NPE _y =0
$4py+my>124$	$4py+my>124$	$4py+my>124$	NPE _y =1

FIG. 36B

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FIG. 37



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GENERAL PWE TABLE

	8-bit MODE				16-bit MODE		32-bit
	IF L=0	IF L=1	IF L=2	IF L=3	IF L=0	IF L=1	
IF $bx \geq mx$	0001	0010	0100	1000	0011	1100	ALWAYS 1111
IF $bx < mx$	0010	0100	1000	0001	1100	0011	

FIG. 38

PWE TABLE ($bx=2$)

mx \ L	8-bit MODE				16-bit MODE		32-bit
	0	1	2	3	0	1	
0	0001	0010	0100	1000	0011	1100	ALWAYS 1111
1	0001	0010	0100	1000	0011	1100	
2	0001	0010	0100	1000	0011	1100	
3	0010	0100	1000	0001	1100	0011	

FIG. 39

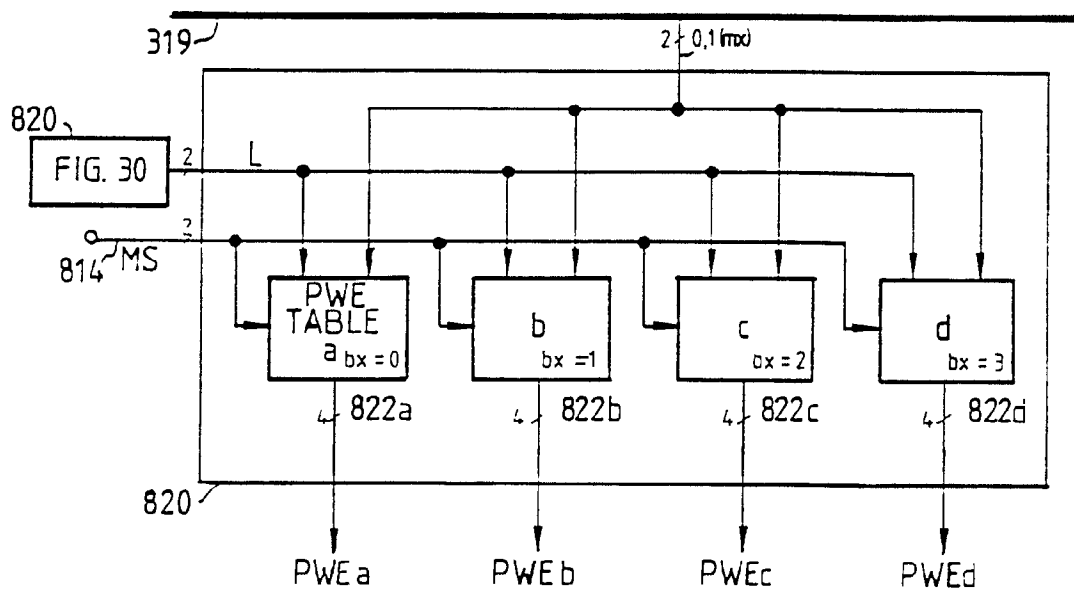
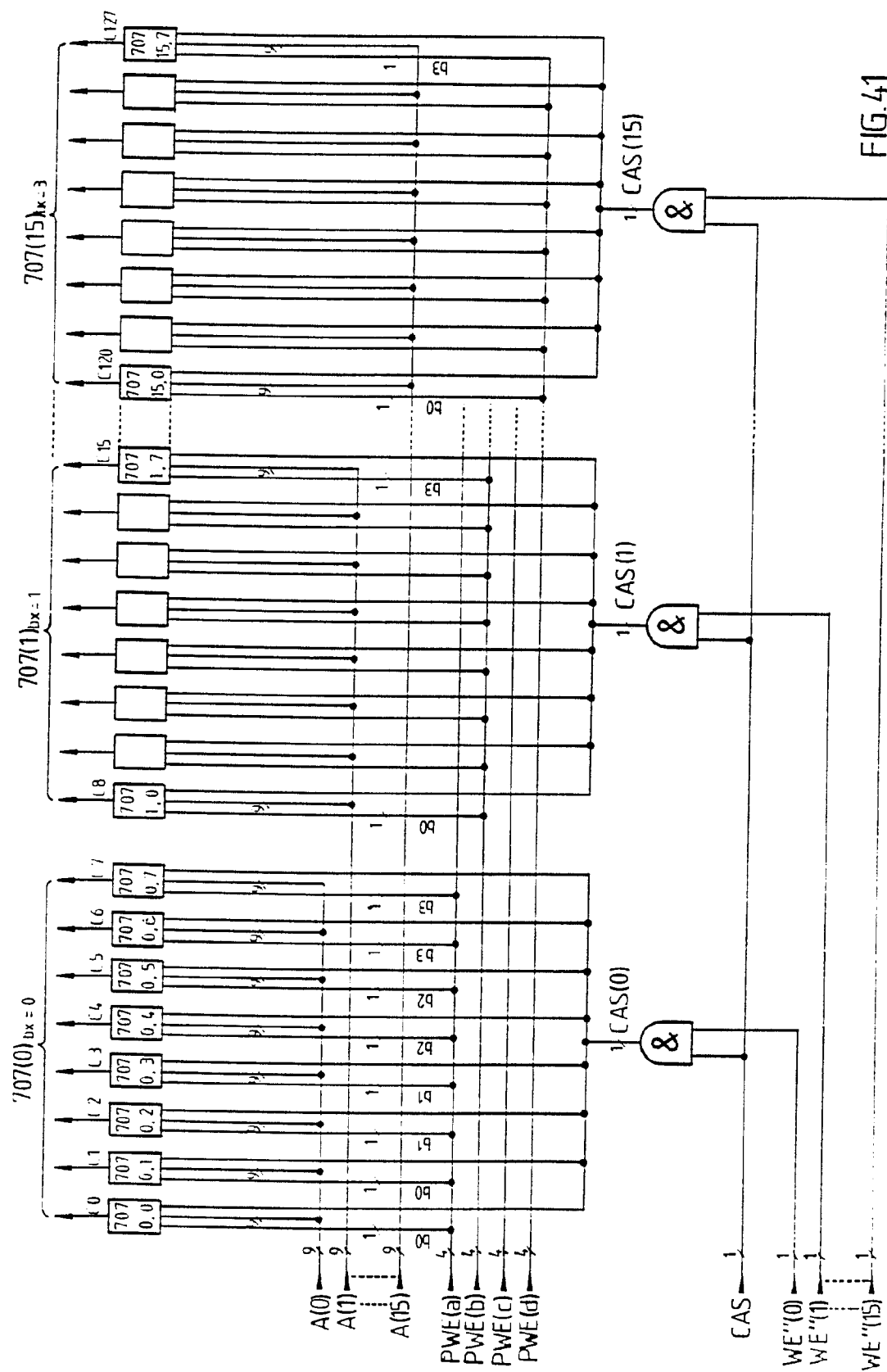
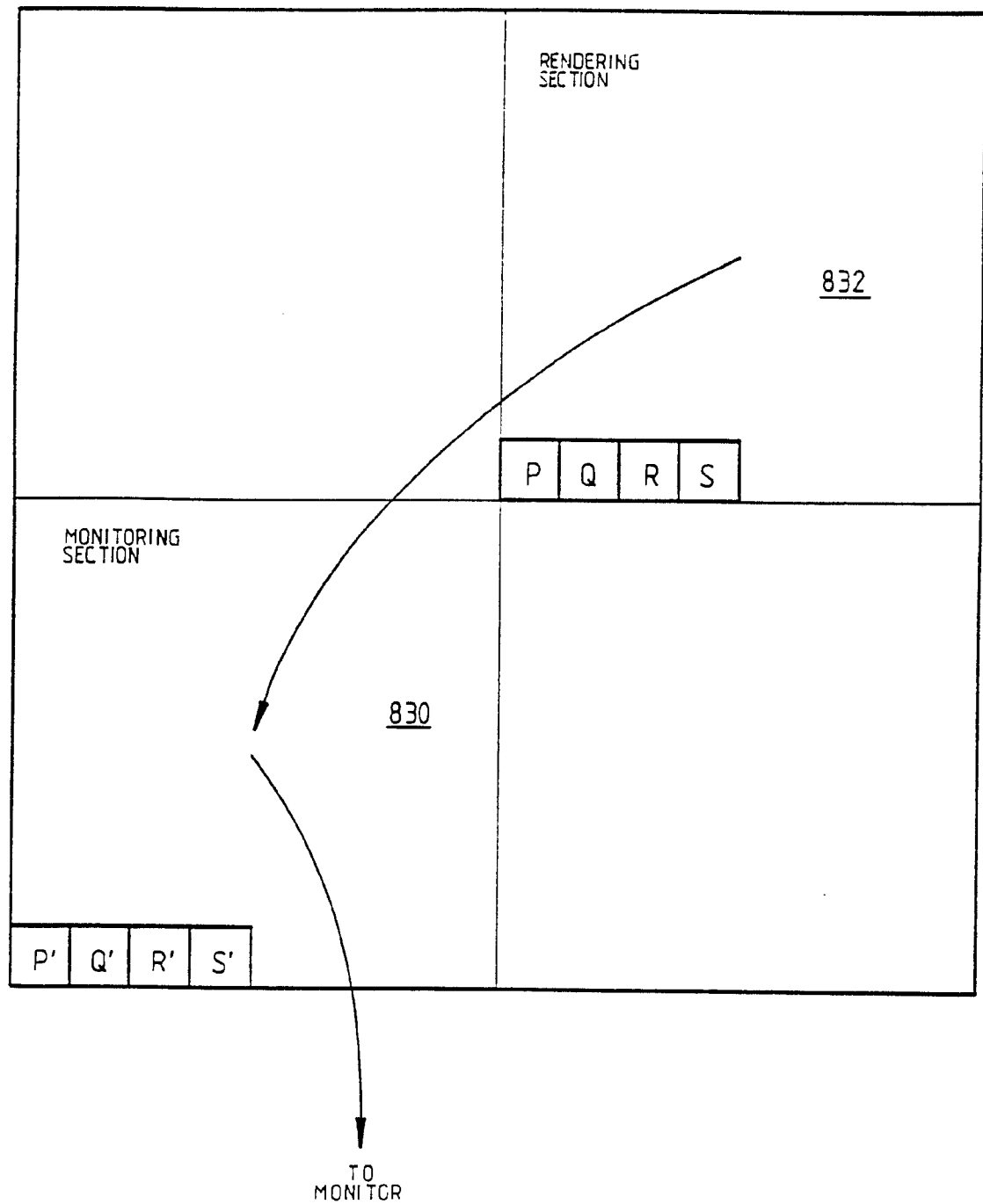


FIG. 40

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FIG. 42

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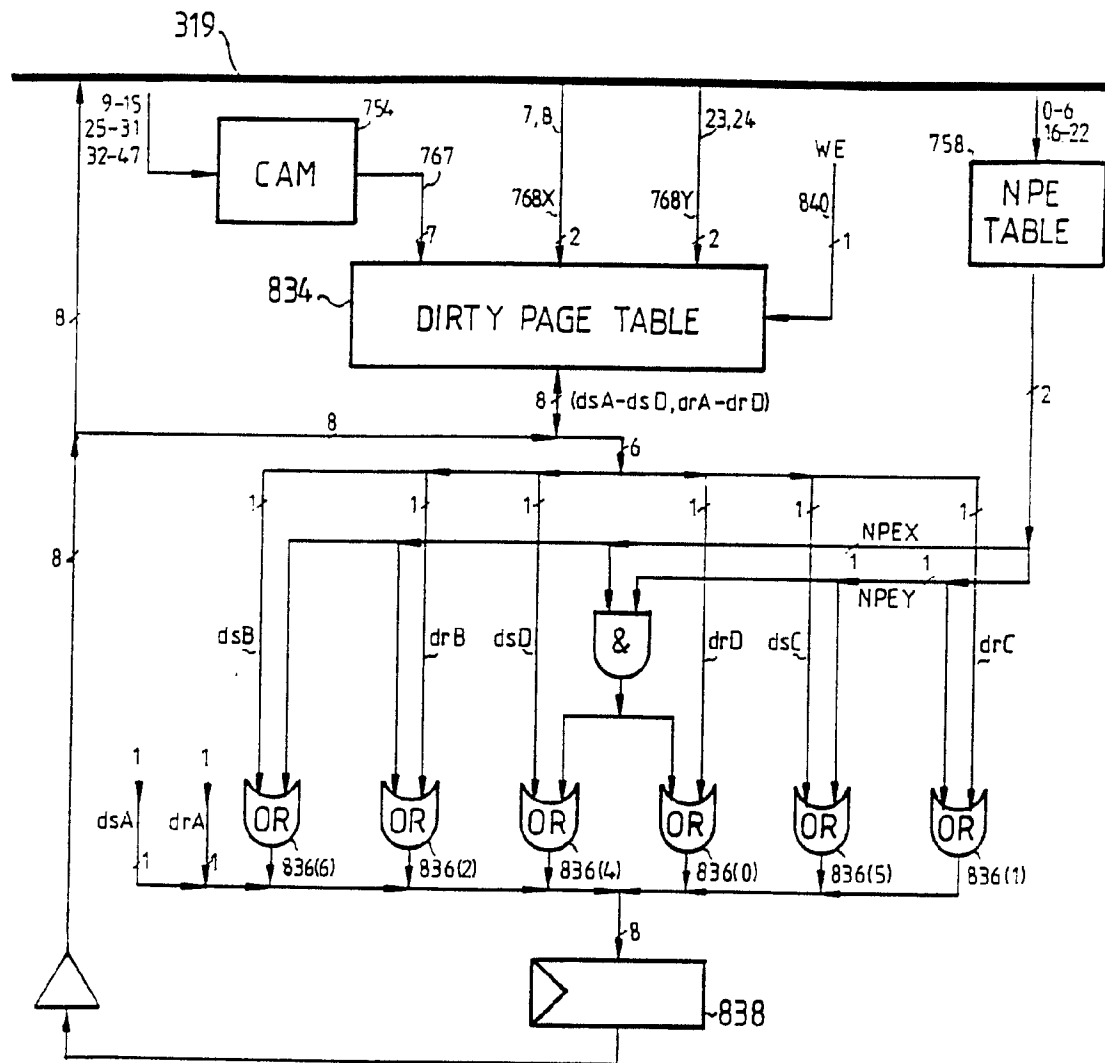
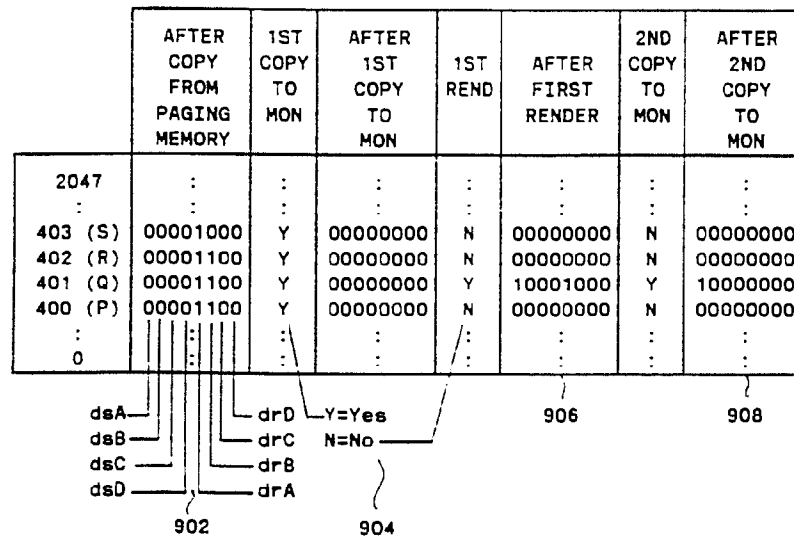


FIG. 43

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	2ND REND	AFTER SECOND RENDER	3RD COPY TO MON	AFTER 3RD COPY TO MON	3RD REND	AFTER THIRD RENDER	4TH COPY TO MON	AFTER 4TH COPY TO MON	COPY BACK TO PAGING MEMORY	AFTER COPY BACK
2047	:	:	:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:	:	:	:
403 (S)	N	00000000	N	00000000	Y	10001000	Y	10000000	Y	00000000
402 (R)	N	00000000	N	00000000	N	00000000	N	00000000	N	00000000
401 (Q)	M	10000000	Y	10000000	Y	10001000	Y	10000000	Y	00000000
400 (P)	Y	11001100	Y	11000000	N	11000000	N	11000000	Y	00000000
:	:	:	:	:	:	:	:	:	:	:
0	:	:	:	:	:	:	:	:	:	:
	M=	910		912		914		916		918

Mis-ali-gned

FIG. 44

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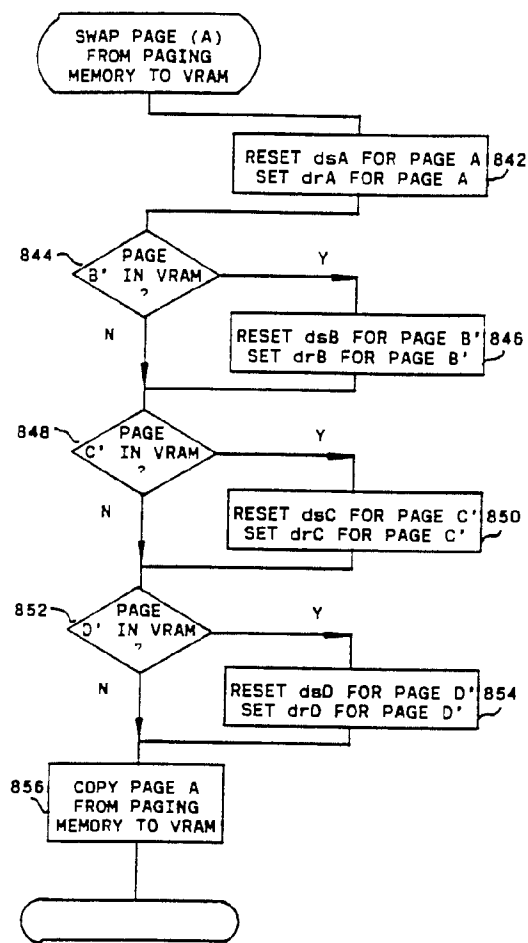


FIG. 45A

PAGE A AT (PX,PY)
 PAGE B AT (PX+1,PY); PAGE B' AT (PX-1,PY)
 PAGE C AT (PX,PY+1); PAGE C' AT (PX,PY-1)
 PAGE D AT (PX+1,PY+1); PAGE D' AT (PX-1,PY-1)
 PAGE E AT (PX+1,PY-1); PAGE E' AT (PX-1,PY+1)

PAGE E'	PAGE C	PAGE D
PAGE B'	PAGE A	PAGE B
PAGE D'	PAGE C'	PAGE E

FIG. 45D

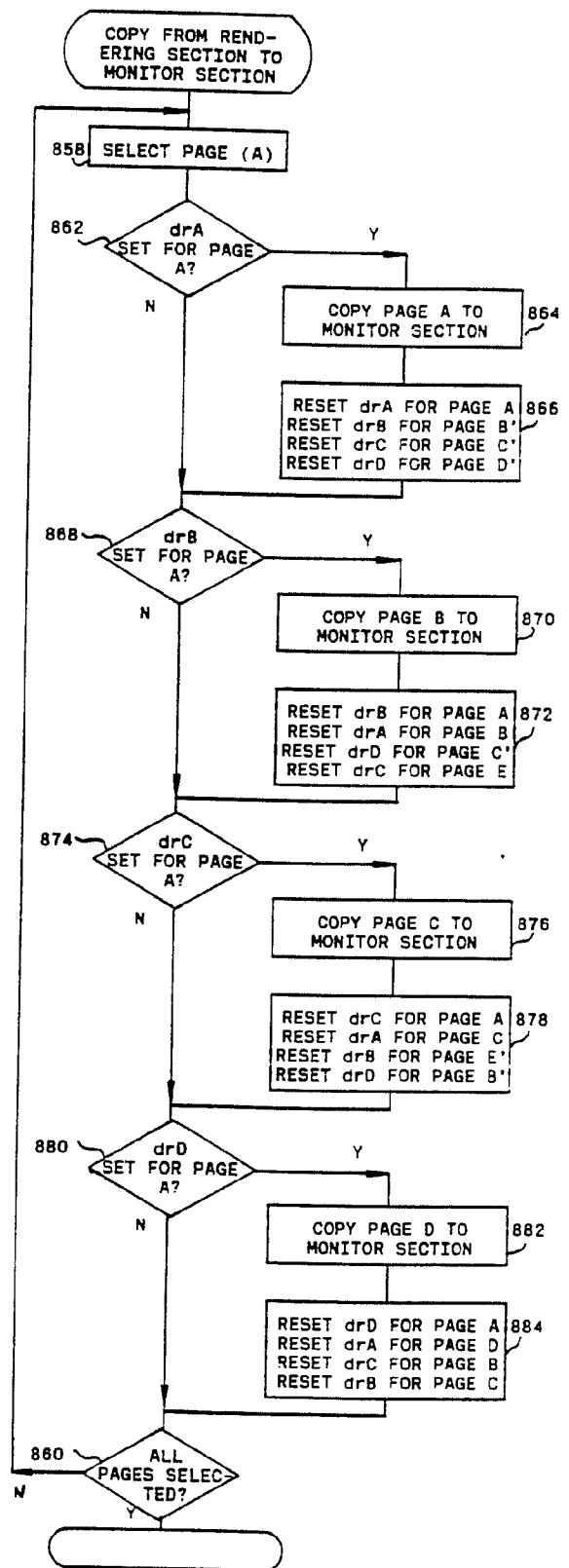


FIG. 45B

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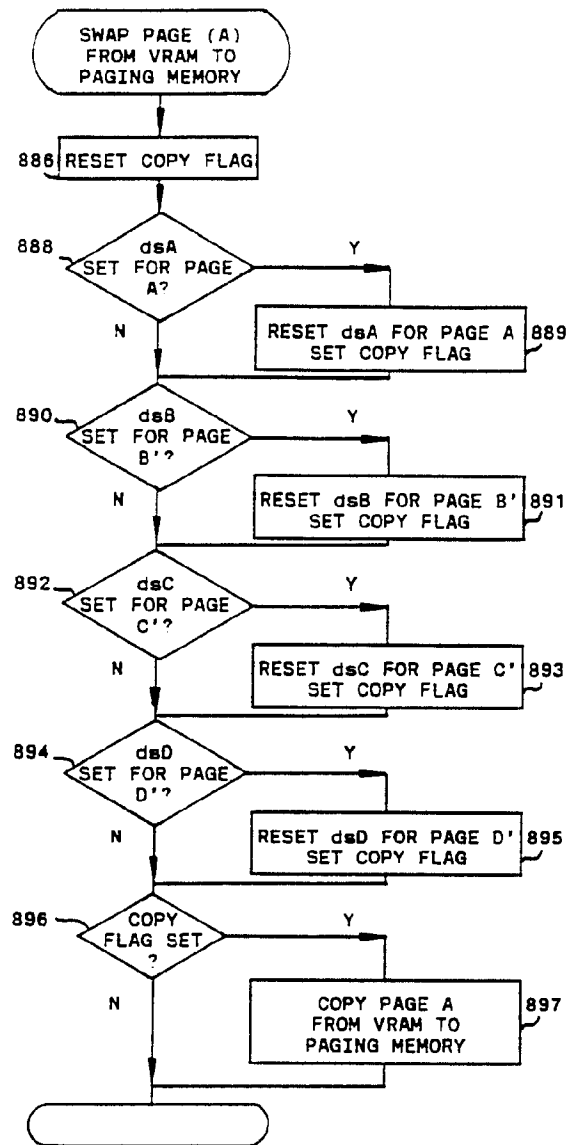


FIG. 45C

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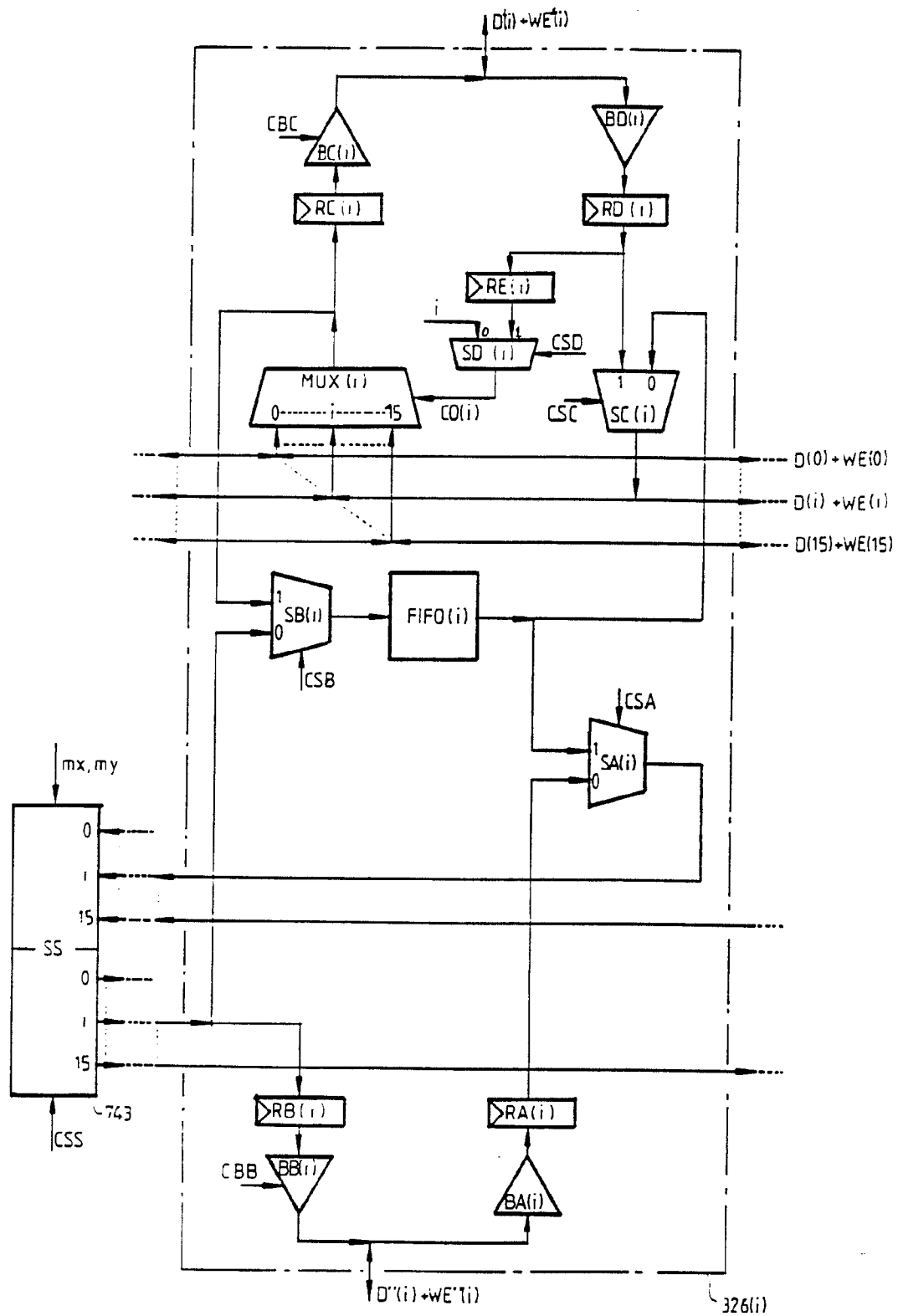


FIG. 46

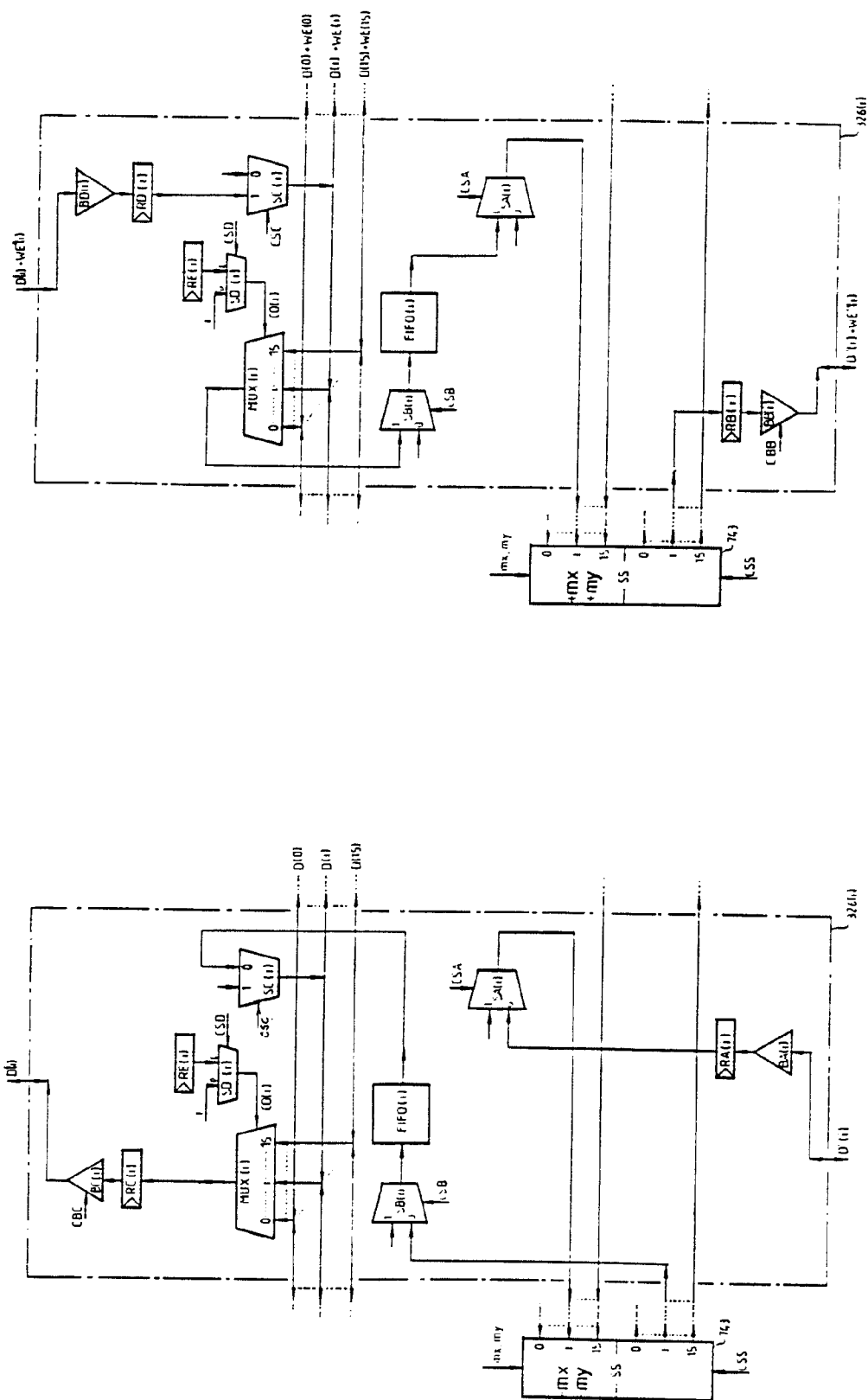


FIG. 47B

FIG. 47A

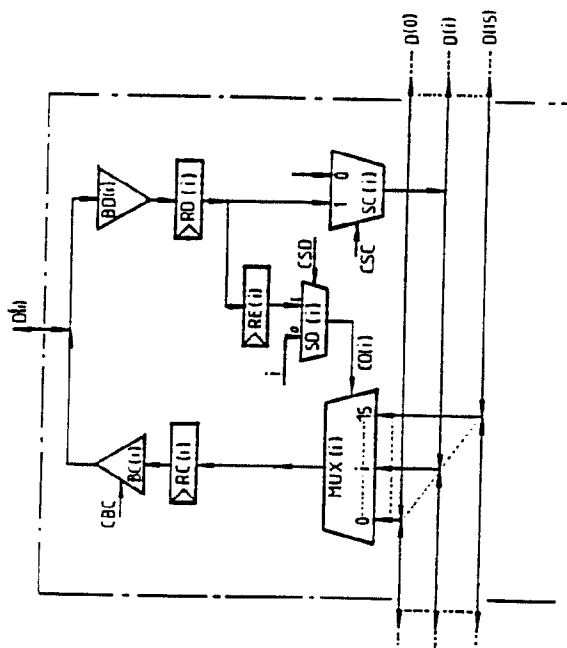


FIG. 47C

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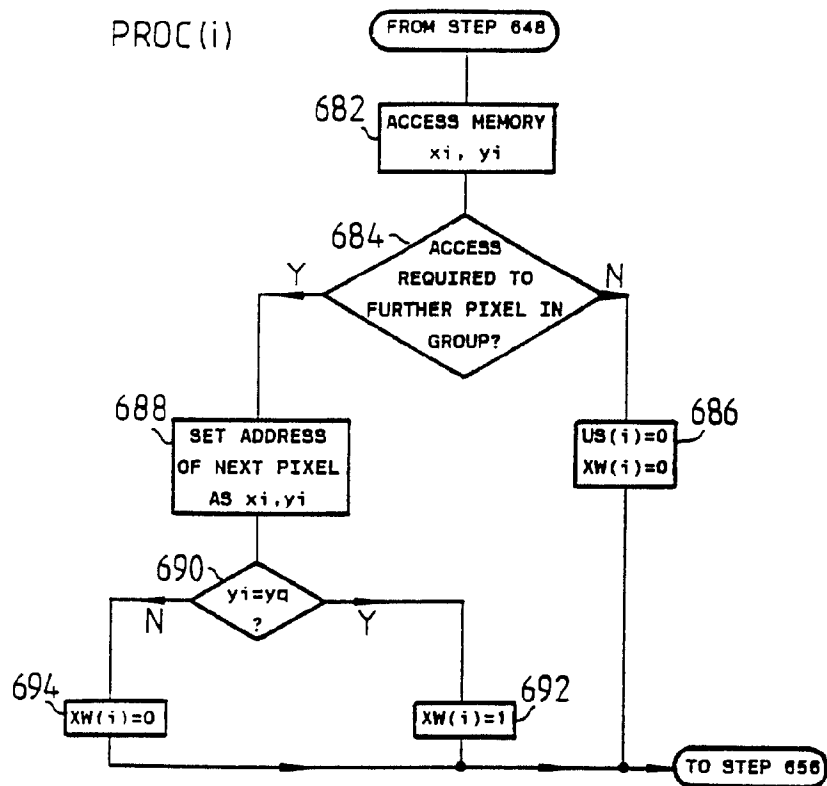


FIG. 48

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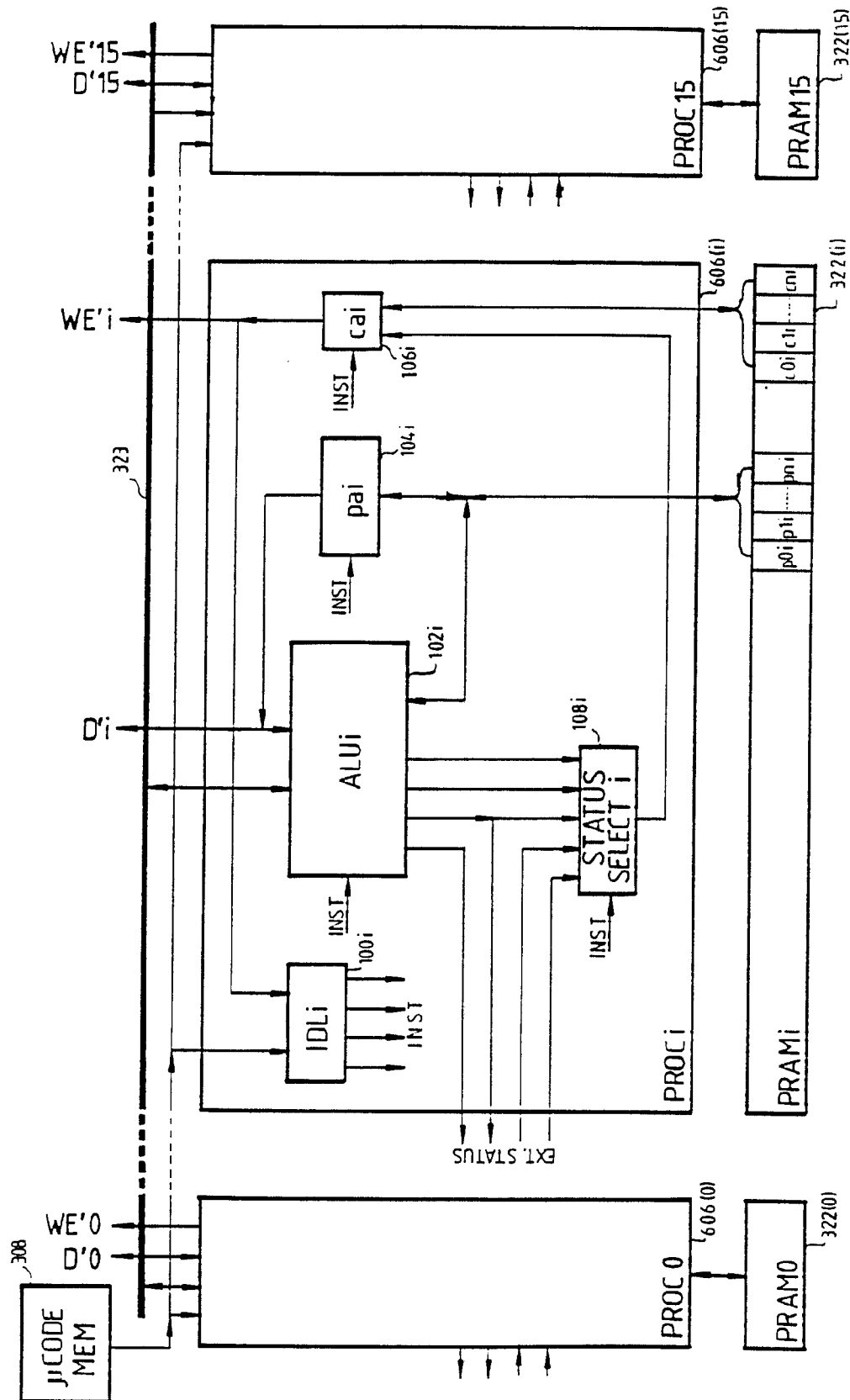


FIG. 49

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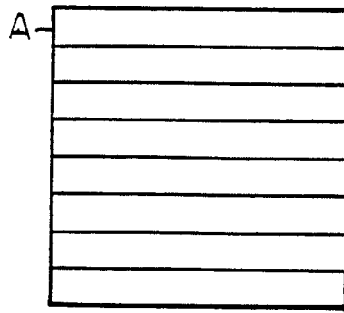


FIG. 50A

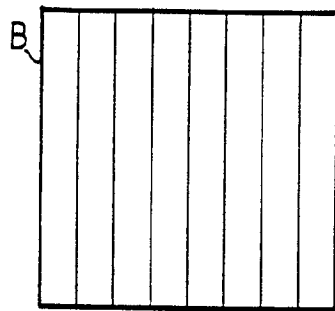


FIG. 50B

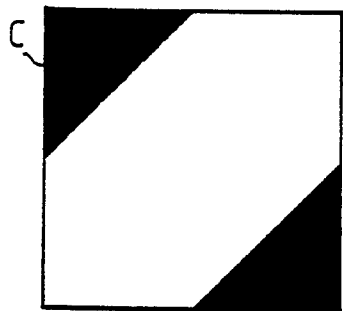


FIG. 50C

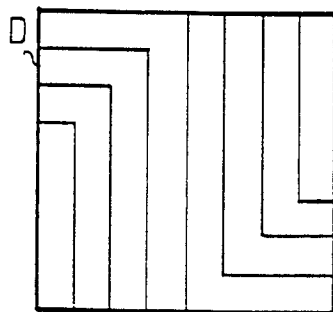


FIG. 50D

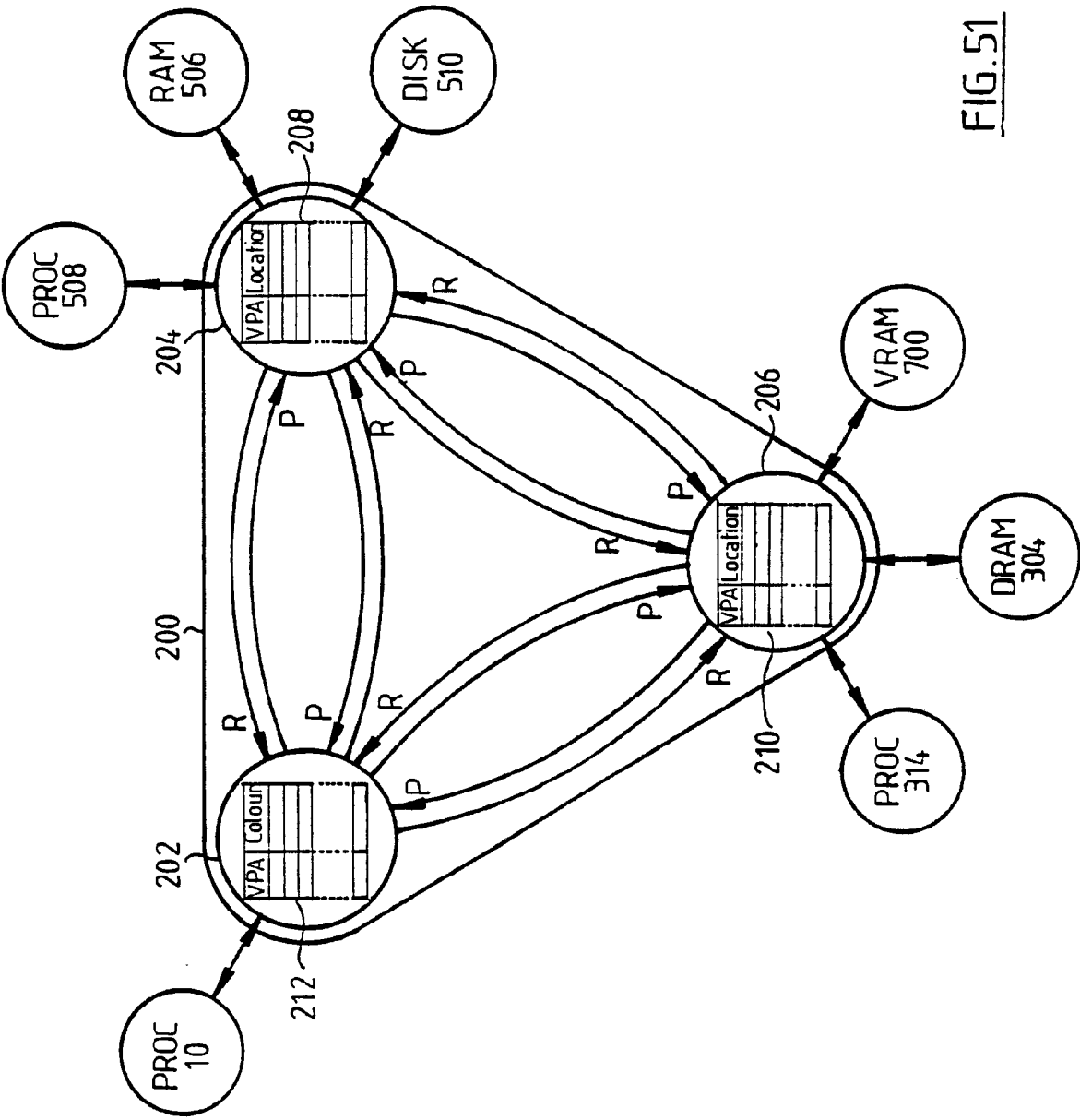
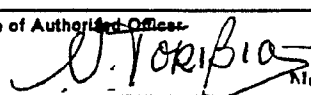


FIG. 51

INTERNATIONAL SEARCH REPORT

International Application No PCT/GB 90/01213

I. CLASSIFICATION OF SUBJECT MATTER (if several classification symbols apply, indicate all) ⁶		
According to International Patent Classification (IPC) or to both National Classification and IPC IPC ⁵ : G 06 F 12/10, G 09 G 1/16		
II. FIELDS SEARCHED		
Minimum Documentation Searched ⁷		
Classification System	Classification Symbols	
IPC ⁵	G 06 F, G 09 G	
Documentation Searched other than Minimum Documentation to the Extent that such Documents are Included in the Fields Searched ⁸		
III. DOCUMENTS CONSIDERED TO BE RELEVANT⁹		
Category ⁹	Citation of Document, ¹¹ with indication, where appropriate, of the relevant passages ¹²	Relevant to Claim No. ¹³
A	IEEE Micro, vol. 8, no. 2, April 1988, IEEE, (New York, US), Shinya Kimura et al.: "Implementation of the V60/70 and its FRM function", pages 22-36 see figure 2; page 25, left-hand column, lines 3-1.	1
--		
A	EP, A, 0254270 (HONEYWELL BULL) 27 January 1988 see figure 1; abstract	1
--		
A	EP, A, 0338416 (IBM) 25 October 1989 see figure 1; abstract	18,19

⁹ Special categories of cited documents: ¹⁰ "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier document but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step "Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art. "&" document member of the same patent family		
IV. CERTIFICATION		
Date of the Actual Completion of the International Search		Date of Mailing of this International Search Report
12th April 1991		27. 05. 91
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EUROPEAN PATENT OFFICE		 Nuria TORIBIO

**ANNEX TO THE INTERNATIONAL SEARCH REPORT
ON INTERNATIONAL PATENT APPLICATION NO.**

GB 9001213
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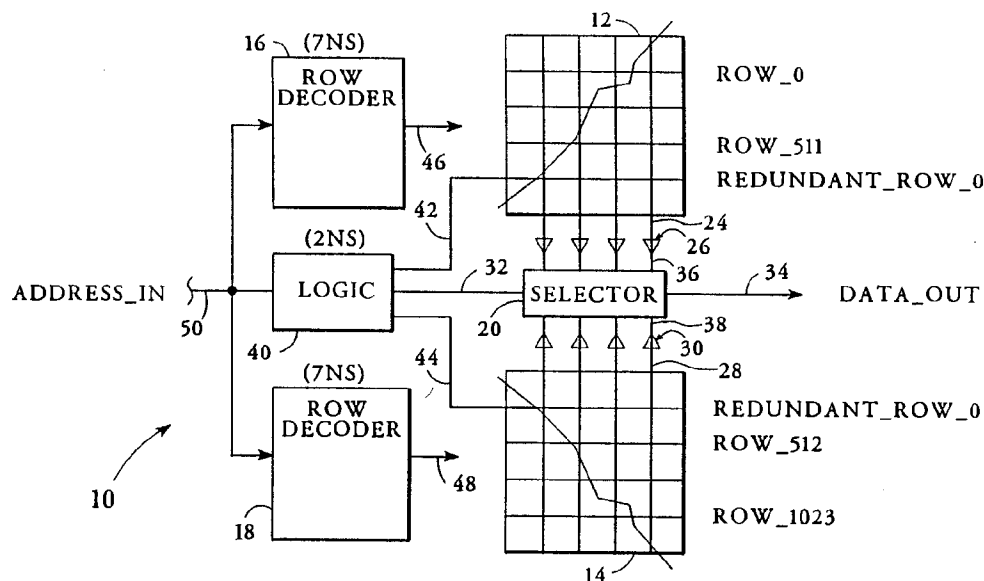
This annex lists the patent family members relating to the patent documents cited in the above-mentioned international search report.
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Patent document cited in search report	Publication date	Patent family member(s)	Publication date
EP-A- 0254270	27-01-88	US-A- 4813002	14-03-89
		AU-B- 597697	07-06-90
		AU-A- 7598087	28-01-88
		JP-A- 63044257	25-02-88
EP-A- 0338416	25-10-89	JP-A- 2012523	17-01-90



INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(51) International Patent Classification ⁵ : G11C 7/00	A1	(11) International Publication Number: WO 94/07242 (43) International Publication Date: 31 March 1994 (31.03.94)
(21) International Application Number: PCT/US93/04231 (22) International Filing Date: 5 May 1993 (05.05.93) (30) Priority data: 949,421 21 September 1992 (21.09.92) US (71) Applicant: ATMEL CORPORATION [US/US]; 2125 O'Nel Drive, San Jose, CA 95131 (US). (72) Inventors: PATHAK, Saroj ; 12998 Robleda Road, Los Al- toso Hills, CA 94022 (US). ROSENDALE, Glen, A. ; 2250 Monroe Street, Apt. #268, Santa Clara, CA 95050 (US). PAYNE, James, E. ; 125 Brook Lane, Boulder Creek, CA 95005 (US).		(74) Agent: SCHNECK, Thomas; Schneck & McHugh, P.O. Box 2-E, San Jose, CA 95109-0005 (US). (81) Designated States: DE, GB, JP, KR, European patent (AT, BE, CH, DE, DK, ES, FR, GB, GR, IE, IT, LU, MC, NL, PT, SE). Published <i>With international search report.</i> <i>With amended claims and statement.</i>

(54) Title: HIGH SPEED REDUNDANT MEMORY**(57) Abstract**

A memory circuit (10) in which redundant cell groups are located in a second memory bank (14; 88) to replace any defective cell groups present in a first memory bank (12; 86). Each bank has its own address decoders (16, 18; 90, 92, 94, 96) and read/write circuits. When an address corresponding to a defective cell group is received, the first bank is enabled as usual, while a fast logic circuit (40; 100) enables the second bank, so that both banks are simultaneously enabled. A signal (32; 102) is transmitted by the logic circuit to an output selector (20; 98) where data from a redundant cell group in the second bank is selected for output (34; 84). The time needed to detect an address corresponding to a defective cell group, is masked by the longer time required by the decoders and sense amplifiers of the respective banks.

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ES	Spain			VN	Viet Nam
FI	Finland				

Description
High Speed Redundant Memory

Technical Field

5 The invention relates to monolithic integrated memory circuits, particularly those adapted for reliable operation through the use of redundant memory cells.

Background Art

10 Many of today's large-scale ("LSI") and very-large-scale ("VLSI") integrated circuit memories would be rejected during manufacture if the presence of a few defective memory cells could not be dealt with in an effective manner. One common solution is to provide spare memory
15 cells and some means of selecting these to replace the defective cells. U.S. Pat. No's 4,947,375 and 4,908,798 are typical examples. In such an arrangement, post-production wafer testing is used to identify the addresses of defective cells. At the factory, these addresses are programmed into
20 a cell selection logic circuit, typically independent of the normal address decoders.

 During normal operation this independent selection circuit compares a received memory address with the factory programmed addresses of the previously identified groups of
25 defective cells. Typically, when a defective group of cells is addressed, signals are generated by the circuit to deselect the defective cells and to substitute a group of spare replacement cells. In this way a memory having such a redundant architecture can operate reliably despite the
30 presence of a few defective memory cells.

 The prior art typically places the spare memory cells in one of two locations, either in a separate region of the same array as the defective cells (U.S. Pat. No's 4,947,375, 5,058,069, 5,058,070 and 5,058,071 are examples
35 of this approach), or in a separate array dedicated for that purpose (U.S. Pat. No's 4,908,798 and 5,033,024 are examples of this latter approach). These devices, while quite valuable in saving an otherwise defective chip, sacrifice speed

because the delay resulting from address decoding, redundant group selection, and defective normal group deselection, is added to other delays typically associated with the speed of memory circuits. The deselection time in particular results
5 in a slowing of memory speed.

Two recent U.S. patents have attacked the speed problem in interesting ways. Kondo, in U.S. Pat. No. 4,918,662, teaches a redundant structure in which the spare memory cells are placed in the same array and use the same
10 sense amplifiers as the normal memory cells. Kondo recognizes the slowing effect of the stray capacitance of an overly long normal cell group selection-inhibit-signal. Kondo partitions this length among many shorter inhibit signal paths, thereby reducing stray capacitance and reducing
15 the "in line" delays. In a completely different approach, Busch et al., in U.S. Pat. No. 4,992,984, disclose a memory module made up of a plurality of identical memory "chips," each having several arrays of storage elements. Factory testing identifies arrays within each chip which are good
20 and those arrays which contain defects. A memory address is applied to all chips in common. A control circuit inside each chip responds to previously programmed addresses corresponding to known defective arrays. The control circuit in the chip having the defective array and the control circuit
25 in the chip having the replacement array both respond to an address corresponding to a known defective array. Both arrays are addressed, but the output data of the replacement array is enabled, while the output data of the defective array is disabled.

30

Summary of the Invention

The invention is a monolithic integrated circuit memory which contains at least two banks of normal data cells. These cells, which can be SRAMs, DRAMs, PROMs,
35 EPROMs, EEPROMs or flash cells, are arranged in rows and columns, and a few of the normal data cells may be defective. Each bank has its own address decoders and the banks

can be simultaneously enabled. The integrated circuit memory has address lines for receiving a memory address and at least one data output line. The address space is partitioned among the memory banks so that normally only one bank
5 responds to a given memory address.

Each bank has redundant rows or columns which contain one or more memory elements which are programmed with data in a defective element of another bank. In accord with the invention, an entire row or column of one bank is substituted for the row or column containing the defective element of another bank. The redundant row or column is read
10 by the same sense amplifier which reads the rows or columns of the non-redundant elements in the memory bank.

A programmable logic device is programmed to receive all memory address signals and to respond to addresses with a bad memory element. Rather than deselect a bad row or column in the first memory bank, the logic device selects the redundant row or column in the second memory bank and then directs the sense amplifier output from the redundant
15 row or column as the memory output. An output from the sense amplifier with the defective element is available, but is not selected. A selector circuit, operating as a multiplexer, is connected to both memory banks and responds to the logic device for selection of an output from an appropriate memory bank. In the case of an address for a non-defective cell in a first memory bank, the output from a sense amplifier associated with the first bank is selected.
20 In the case of an address for a defective cell in the first memory bank, the output from a sense amplifier associated with the second bank is selected.
25 30

The logic circuit operates at a speed faster than memory bank decoders and sense amplifiers combined so that redundant row or column data is available faster than normal data. Since the defective row or column is not deselected, only ignored by the selector circuit, the delay normally
35 present with deselection of a defective row or column is not present and memory output data is delayed by only an amount required by the logic circuit and the selector switch.

Since a switch can be very fast, the total delay of the logic device and the selector can be comparable, or faster, than the speed of a normal decoder. In this way, the use of a redundant memory element is transparent with respect to
5 time delay and no time delay is experienced from deselection of a row or column containing a defective element.

The key ingredients of the present invention include placing a backup row or column for a defective memory bank address into a different memory bank, simultaneously
10 selecting both the defective and backup row or column, and selecting the outputs of the memory bank containing the backup row or column when a defective element is addressed. A fast logic device and a fast selector switch is used, compared to the speed of an address decoder. In a read-only
15 memory application, the correct data must exist in the backup row so that when both defective and redundant rows are selected, the correct data will be available at the sense amplifiers of the bank containing the redundant row. When writing is done, the data is written into the backup rows or
20 columns, and later will be retrieved from the backup row or column.

Brief Description of the Drawings

Fig. 1 is a schematic block diagram showing the
25 present invention in which redundant rows may be substituted.

Fig. 2 is pictorial version of Fig. 1 showing selection of a redundant row.

Fig. 3 is a schematic block diagram showing the
30 present invention in which redundant row or columns may be substituted.

Best Mode for Carrying Out the Invention

With reference to Fig. 1 there is shown a high
35 speed redundant memory 10 capable of reliable operation in the presence of a small number of defective memory cells.

The memory contains first and second memory banks, 12 and 14, each bank includes memory elements which are ar-

ranged in rows and columns. Each memory bank has its own row decoder 16, 18 and each column has its own sense amplifiers, 26, 30. Selector 20 is used to select the output of either first bank sense amplifiers along lines 36 or second bank sense amplifiers along lines 38 depending upon the level of a select signal 32. Selector 20 provides the selected data as memory output data on lines 34.

Each bank includes at least one redundant row of memory elements which share the sense amplifiers for the bank but are not selectable by the row decoder for the bank. Redundant_Row_0 is shown in the first bank 12 and is selectable by logic 40 using a signal carried on line 42. Redundant_Row_1 is shown in the second bank 14 and is selectable by a signal carried on line 44.

The normal rows of each memory bank are shown as Row_0 through Row_511 for first bank 12 and Row_512 through Row_1023 for second bank 14. The normal rows of first bank 12 are selectable by row decoder 16 output signals carried on lines 46, while the normal rows of second bank 14 are selectable by row decoder 18 output signals carried on lines 48. Row select lines 46, 48 are connected to the individual rows of the respective memory banks, generally indicated by the arrows.

Redundant memory 10 receives memory address signals on lines 50. Appropriate subsets of these memory signals are connected as inputs to row decoders 16, 18.

A factory programmable logic device 40 contains a map of bad cells derived from factory probe testing. The map lists the addresses of the bad cells and a corresponding replacement row. The logic circuit is made of non-volatile memory devices, such as PROMs, EPROMs, EEPROMs or flash cells. The speed of logic circuit 40 is about 2 nanoseconds, low compared to the speed of address decoder 16 or 18 which is about 7 nanoseconds. The address decoder 20 has a speed of several nanoseconds. The combined speed of the logic circuit is comparable or less than the speed of an address decoder and associated sense amplifier. When the logic circuit 40 receives memory address signals from lines

50 for bad cells it produces a predetermined signal carried on lines 32, 42 and 44.

In operation, a first predetermined memory address signal corresponding to the address of a defective row cell in second bank 14 is decoded by logic circuit 40 to generate a signal carried on line 42 to select Redundant_Row_0 in first bank 12. This occurs because logic circuit 40 has been programmed with the address of the bad element and responds to the bad address. The logic circuit 40 also generates a signal carried on line 32 to cause selector 20 to select the output of the sense amplifiers 26 of first bank 12 for presentation on memory data output lines 34.

Another operational example is as follows. A second predetermined memory address signal corresponding to the address of a defective row cell in first bank 12 is decoded by logic circuit 40 to generate a signal carried on line 44 to select Redundant_Row_1 in second bank 14. The same predetermined memory signal is decoded by logic circuit 40 to generate a signal carried on line 32 to cause selector 20 to select the output of the sense amplifiers 30 of second bank 14 for presentation on memory data output lines 34.

Fig. 2 depicts the embodiment shown in Fig. 1 in a pictorial fashion to emphasize the basic operation of the invention. In this example, a memory address is received on lines 50 which corresponds to a cell in Defective_Row_K in first bank 12. Defective_Row_K is shown being selected by an output of row decoder 16. Programmable logic circuit 40 receives and responds to the memory address by generating a row selection signal on line 44 connected to and selecting Redundant_Row_1 in second bank 14 at the same time Defective_Row_K is being selected. The contents of each selected row is available at the outputs of the respective sense amplifiers, 26, 30. The outputs of second bank sense amplifiers 30 are selected by selector 20 for output on lines 34 as the memory output data. Programmable logic circuit 40 generates an output select signal on line 32 for this purpose. The row decoder 18 for second bank 14 has no active output signal because the normal rows in second bank 14 do not re-

spond to addresses found the address space of first bank 12.

Fig. 3 depicts a high speed redundant memory 80 capable of reliable operation in the presence of a few bad cells in defective rows or columns. The memory has first and second memory banks, 86 and 88 respectively. Each bank may have bad cells which define a defective row or column and includes at least one redundant row, depicted as Redundant_Row_0 and Redundant_Row_1, and one redundant column, 116 and 118. Each bank has its own row decoder, 90 and 94, and its own column decoder, 92 and 96. A factory programmable logic circuit 100 receives a memory address on lines 82 and generates select signals 102, 104 and 106 in response to programmed addresses. All columns in each bank share a common sense amplifier, 108 and 110. Select signal 102 is used by selector 98 to select the output of one sense amplifier as the single output bit Data_Out on line 84. Select signal 104 is made active in response to the programmed address of a known defective row or column and is applied through programmable switch 112 to select either Redundant_Row_0 or redundant column 116. Similarly, select signal 106 may be programmably connected via switch 114 to select Redundant_Row_1 or redundant column 118. The normal rows and columns of each bank partition the memory address space such that only a single row and column in one bank will be selected at one time.

When the address of a known defective cell group is received on lines 82, both memory banks are enabled and the defective row or column is selected by its address decoders producing a defective output at the bank sense amplifier. Simultaneously, select signal 104 or 106 selects a redundant row or column and a good output is available at the sense amplifier of the bank containing the redundant cell group. Select signal 102 is made active in response to the memory address to select the good sense amplifier output and connect the good output to line 84.

Without loss of generality, the structures depicted in Figs. 1, 2 and 3 can be extended, by means well known to those having ordinary skill in the art, to include

a redundant memory having both read and write capability.

Other embodiments are contemplated. The backup for bank A can be located in bank B, the backup for B located in C, for C in D, and for D in A, so long as both the
5 bank of the defective row and its backup bank are simultaneously enabled and the output of the backup bank selected. Each memory bank may be a portion of a chip, a chip or a laser module, such as a SIMM or SIP.

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Claims

1. A high-speed redundant memory, comprising:
 - 5 a memory having first and second banks of storage cells, each bank having its own address decoders and at least one sense amplifier, each bank having redundant memory cell groups which are not selectable by the address decoders;
 - 10 logic circuit means for receiving and decoding a memory address which corresponds with a defective cell in one bank and for generating a logic signal in response thereto having an address which corresponds to a redundant cell group in the other bank;
 - 15 switch means responsive to the logic signal to enable said other bank, said switch means connected to the first and second banks for using the logic signal to select an output signal of a sense amplifier of the other bank as the memory output signal.
- 20 2. The redundant memory of claim 1 wherein the redundant memory cell group comprises at least one row.
3. The redundant memory of claim 1 wherein the redundant
25 memory cell group comprises at least one column.
4. The redundant memory of claim 1 wherein the redundant cell group has a redundant sense amplifier.
- 30 5. The redundant memory of claim 1 wherein the logic circuit means comprises non-volatile memory cells.
6. The redundant memory of claim 1 wherein the redundant
35 memory cell groups include at least one redundant row and at least one redundant column.

7. The redundant memory of claim 1 wherein said storage cells are selected from the group of SRAM, DRAM, PROM, EPROM, EEPROM or flash memory cells.

5 8. The redundant memory of claim 1 comprising pairs of memory banks such that the redundant storage cell groups of each bank of a pair serve as substitutes for defective cells in the other bank of the pair, and the decoding of a memory address corresponding to a defective cell group in either
10 bank of the pair results in at least both banks of the pair being simultaneously enabled and wherein the output of the at least one sense amplifier of the bank having the selected replacement group is output by the memory as output data.

15 9. A high-speed redundant memory for operation in the presence of defective memory elements, comprising:

a plurality of memory banks each with an array of memory elements with regular addresses and each array having its own address decoding means and having redundant elements
20 with addresses whose contents may be read via each bank's associated sense amplifiers,

logic means for recognizing the address of a defective memory element with a regular address in a first memory bank and for selecting a redundant element with an
25 address in a second memory bank, and selecting outputs of the sense amplifiers of the second memory bank when the address of a defective memory element is recognized; and

a selector means responsive to the logic means for outputting data from a redundant memory element in the second memory bank when an address of a defective memory element in the first memory bank is received and otherwise outputting data from a memory element with an address in the first memory bank, said first and second memory banks simultaneously enabled when an address of a defective memory element
30 is received, the combined speed of the logic means being faster than said decoder means and associated sense amplifiers.
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10. A method for addressing a redundant memory, comprising the steps of:

providing a memory having first and second banks, each having its own address decoders and sense amplifiers
5 and having redundant memory cells which are not selectable by the address decoders;

receiving and decoding a memory address which corresponds to a defective cell group in a first bank and generating a signal in response thereto;

10 while the first bank is enabled to read the defective cell group, using the signal (i) to select a redundant cell group in the second bank, (ii) to simultaneously enable the second bank, and (iii) to select an output signal of the sense amplifiers of the second bank; and

15 outputting the selected output signal of the second bank as memory output data.

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AMENDED CLAIMS

[received by the International Bureau on 30 November 1993 (30.11.93) ;
original claims 1, 9 and 10 amended ; new claim 11 added ;
other claims unchanged (3 pages)]

1. A high-speed redundant memory, comprising:
 - a memory having first and second banks of
5 storage cells, each bank having an address decoder and
at least one sense amplifier, each bank having redundant
memory cell groups which are not selectable by the
address decoder of a respective bank;
logic circuit means for receiving and decoding
10 a memory address which corresponds with a defective cell
in one bank and for generating a logic signal in response
thereto having an address which corresponds to a redun-
dant cell group in the other bank, the logic circuit
means operating at a speed faster than the address decod-
15 er and sense amplifier combined of either of said first
and second banks,
switch means connected to the first and second
banks and to the logic circuit means for using the logic
signal of the logic circuit means to select an output
20 signal of a sense amplifier of the other bank as the
memory output signal.
- 25 2. The redundant memory of claim 1 wherein the redundant
memory cell group comprises at least one row.
3. The redundant memory of claim 1 wherein the redundant
memory cell group comprises at least one column.
4. The redundant memory of claim 1 wherein the redundant
30 cell group has a redundant sense amplifier.
5. The redundant memory of claim 1 wherein the logic
circuit means comprises non-volatile memory cells.
- 35 6. The redundant memory of claim 1 wherein the redundant
memory cell groups include at least one redundant row and
at least one redundant column.

7. The redundant memory of claim 1 wherein said storage cells are selected from the group of SRAM, DRAM, PROM, EPROM, EEPROM or flash memory cells.

5 8. The redundant memory of claim 1 comprising pairs of memory banks such that the redundant storage cell groups of each bank of a pair serve as substitutes for defective cells in the other bank of the pair, and the decoding of
10 a memory address corresponding to a defective cell group in either bank of the pair results in at least both banks of the pair being simultaneously enabled and wherein the output of the at least one sense amplifier of the bank having the selected replacement group is output by the memory as output data.

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9. A high-speed redundant memory for operation in the presence of defective memory elements, comprising:

a plurality of memory banks each with an array of memory elements with regular addresses and each array
20 having an address decoding means and having redundant elements with addresses whose contents may be read via sense amplifiers associated with each bank,

logic means for recognizing the address of a defective memory element with a regular address in a
25 first memory bank and for selecting a redundant element with an address in a second memory bank, and

a selector means responsive to the logic means for outputting data from a redundant memory element in the second memory bank when an address of a defective
30 memory element in the first memory bank is received and otherwise outputting data from a memory element with an address in the first memory bank, said first and second memory banks simultaneously enabled when an address of a defective memory element is received, the speed of the
35 logic means being faster than the combined speed of said address decoding means and associated sense amplifiers of any memory bank.

10. A method for addressing a redundant memory, comprising the steps of:

5 addressing a memory having first and second banks, each having an address decoder and sense amplifiers and having redundant memory cells which are not selectable by the address decoders of a respective bank, addressing a logic circuit with a memory address which corresponds to a defective cell group in a first bank and generating a signal from the logic circuit
10 in response thereto in a second amount of time less than the first amount of time; while the first bank is enabled to read the defective cell group, using the signal from the logic circuit (i) to select a redundant cell group in the second bank, (ii) to simultaneously enable the second bank
15 and (iii) to select an output signal of the sense amplifiers of the second bank, and outputting the selected output signal of the second bank as memory output data.

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11. The memory of claim 1 wherein said logic circuit means is a programmable logic device.

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STATEMENT UNDER ARTICLE 19

In response to the International Search Report, claims 1, 9 and 10 were rewritten to point out that the logic means operates faster than the combined speed of the decoder and sense amplifiers, thereby distinguishing applicant's invention from the prior art cited in the International Search Report. In addition, a new claim 11 has been added to point out that the applicant achieves increased speed with a programmable logic device.

Applicant predicates patentability on the basis that the redundant memory element is transparent with respect to time delay in that no time delay is experienced from deselection of a row or column containing a defective element. Applicant achieves this result by having the logic circuit operate at a speed faster than both the memory decoders and sense amplifiers combined. This configuration makes available redundant row or column data faster than normal data.

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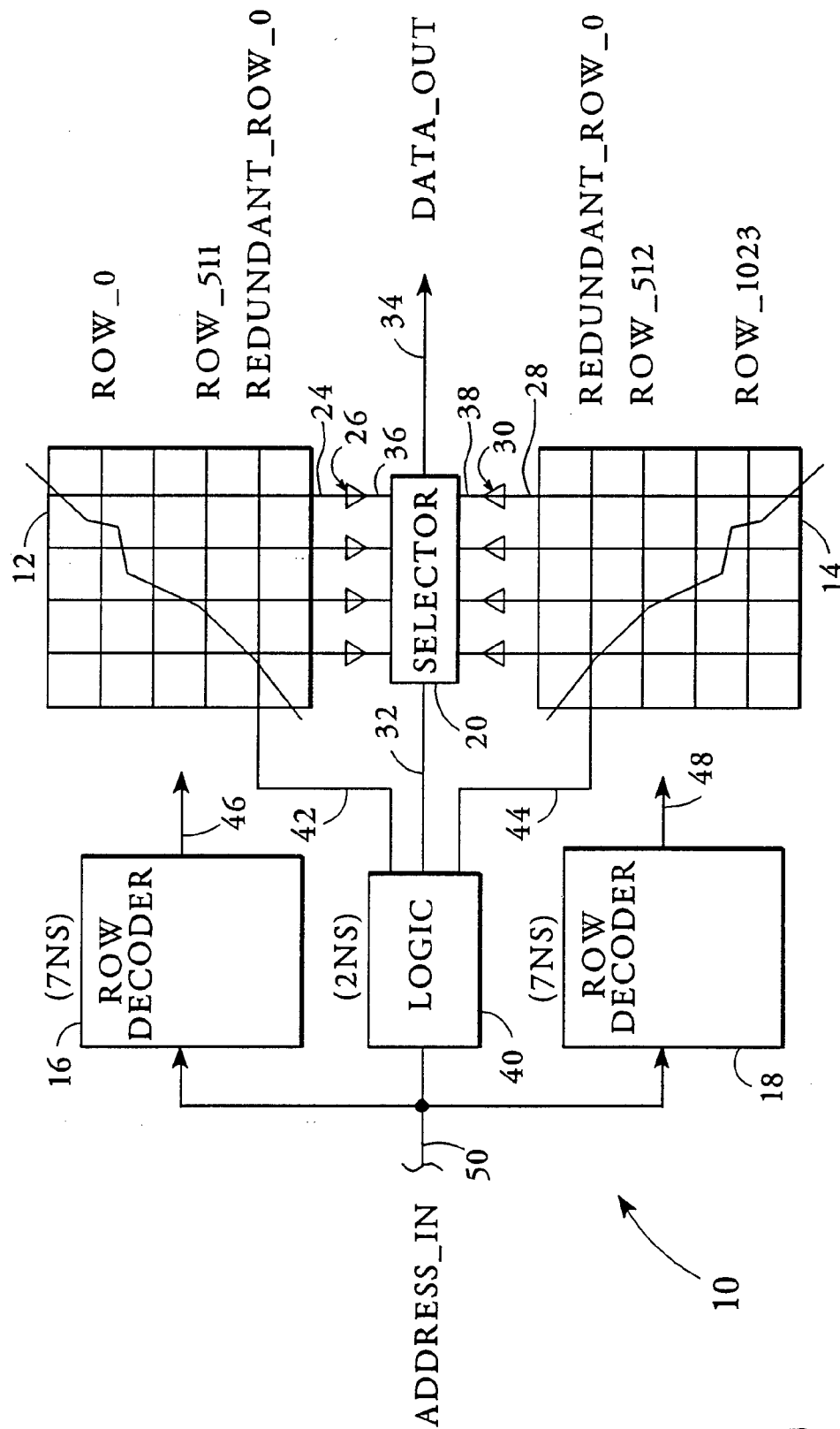


FIG. 1

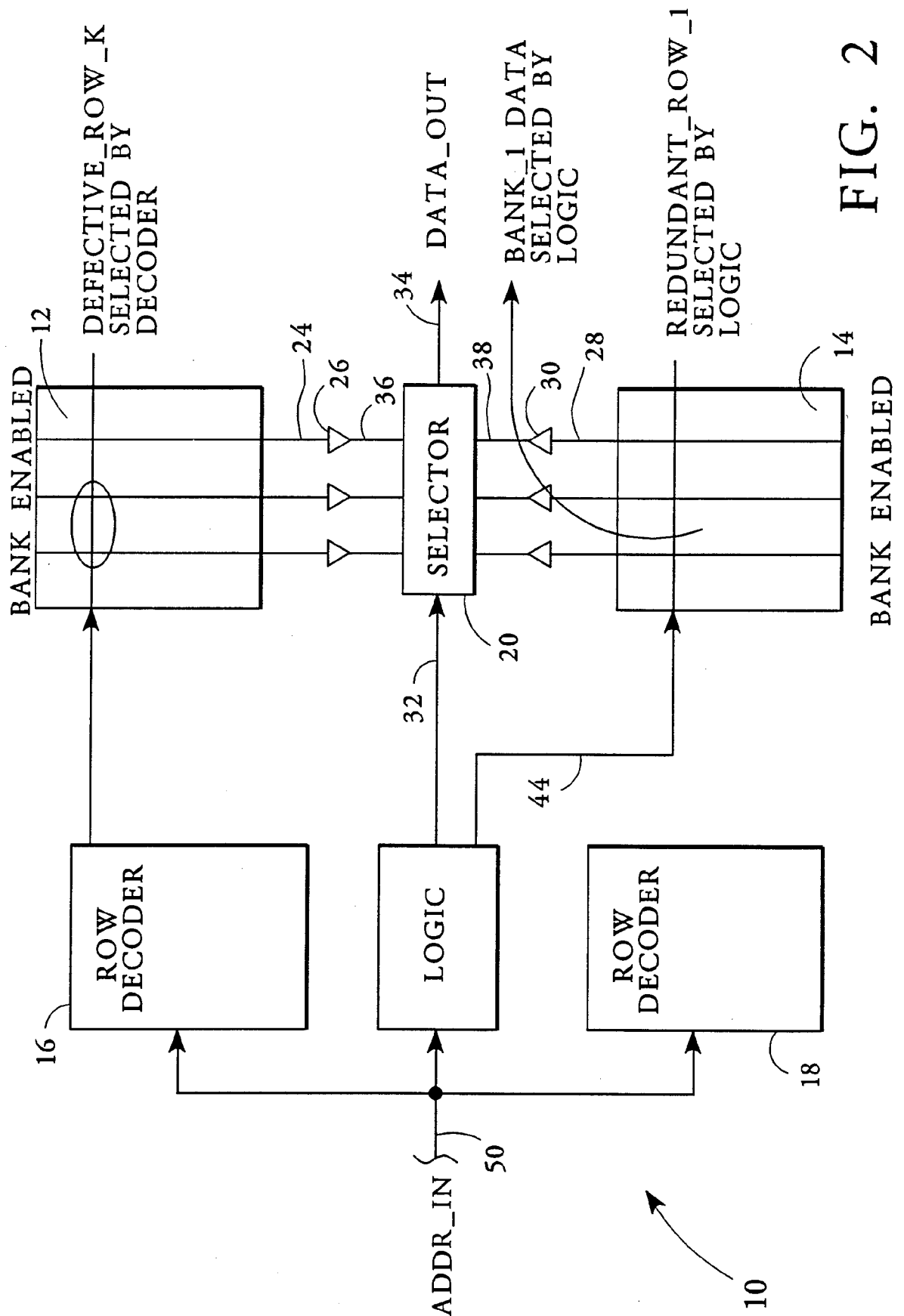
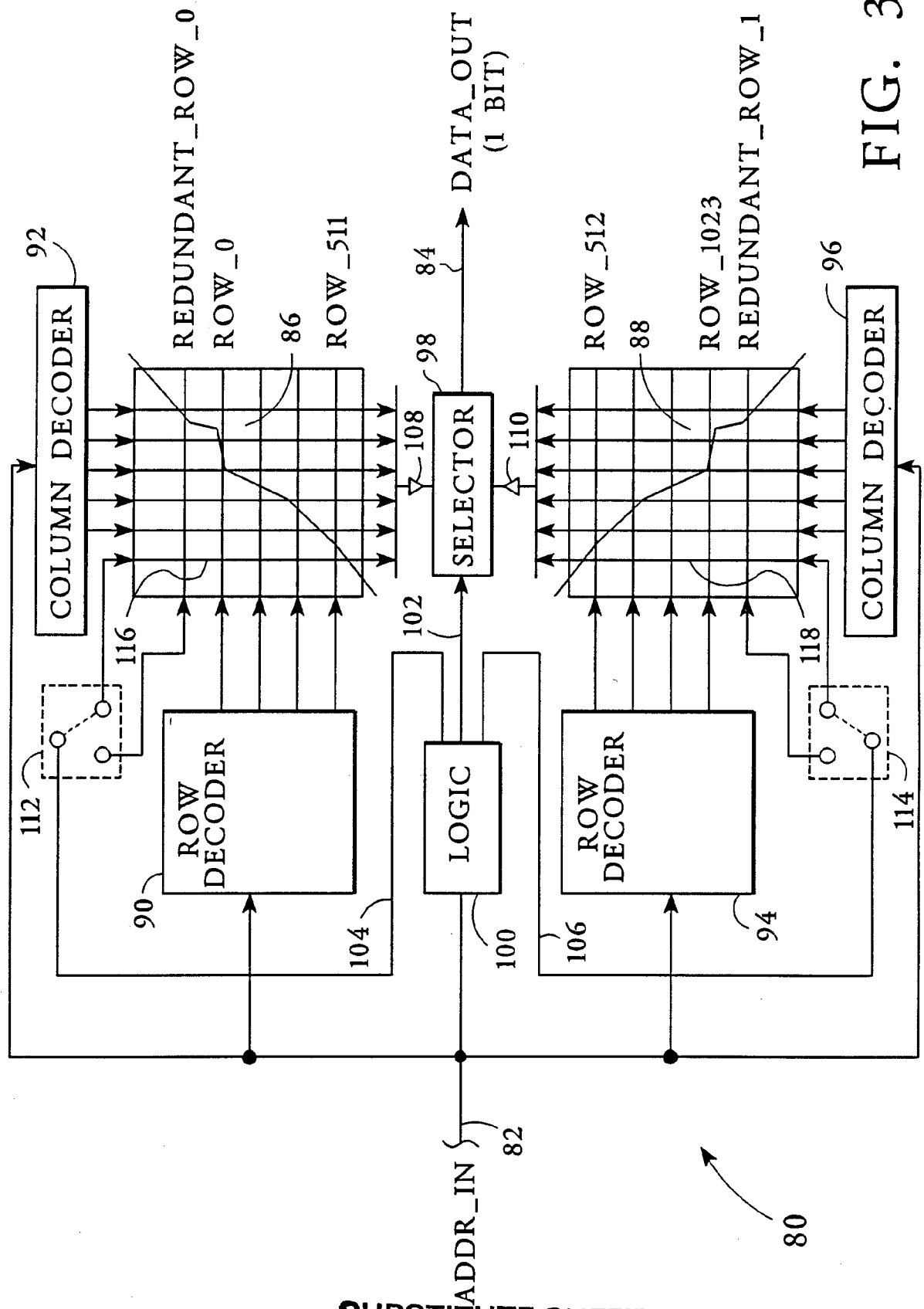


FIG. 2



INTERNATIONAL SEARCH REPORT

International application No.
PCT/US93/04231

A. CLASSIFICATION OF SUBJECT MATTER

IPC(5) :G11C 7/00

US CL :365/200, 201, 230.03; 371/10.2, 10.3

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

U.S. : NONE

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

APS (MEMORY, STORAGE, CELL, ARRAY, MATRIX, BLOCK, REDUNDANT, SPARE, ADDRESS DECODER, SELECTOR, SENSE AMPLIFIER, NONVOLATILE)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X --- Y	IBM J. RES. DEVELOP., VOL. 24, NO. 3, MAY 1980, FITZGERALD ET AL., "CIRCUIT IMPLEMENTATION OF FUSIBLE REDUNDANT ADDRESSES ON RAMs FOR PRODUCTIVITY ENHANCEMENT", PP. 291-298	1-7, 10 ----- 8, 9
Y	US, A, 5,126,973 (GALLIA ET AL.) 30 JUN 1992 see entire document.	8, 9

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:	
"A" document defining the general state of the art which is not considered to be part of particular relevance	"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
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"P" document published prior to the international filing date but later than the priority date claimed	

Date of the actual completion of the international search

22 July 1993

Date of mailing of the international search report

28 JUL 1993

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(21) International Application Number: PCT/US95/07062 (22) International Filing Date: 1 June 1995 (01.06.95) (30) Priority Data: 08/253,499 3 June 1994 (03.06.94) US (71) Applicant: INTEL CORPORATION [US/US]; 2200 Mission College Boulevard, Santa Clara, CA 95052 (US). (72) Inventors: MILLS, Duane, R.; 103 Casselman Street, Folsom, CA 95630 (US). DIPERT, Brian, Lyn; 9840 Burling Street, Sacramento, CA 95827 (US). SAMBANDAN, Sachidanandan; 101 Waterboro Square, Folsom, CA 95630 (US). MCCORMICK, Bruce; 8012 Chris Court, Granite Bay, CA 95746 (US). PASHLEY, Richard, D.; 77710 Haley Drive, Roseville, CA 95661 (US). (74) Agents: TAYLOR, Edwin, H. et al.; Blakely, Sokoloff, Taylor & Zafman, 7th floor, 12400 Wilshire Boulevard, Los Angeles, CA 90025-1026 (US). (81) Designated States: AM, AT, AT (Utility model), AU, BB, BG, BR, BY, CA, CH, CN, CZ, CZ (Utility model), DE, DE (Utility model), DK, DK (Utility model), EE, ES, FI, FI (Utility model), GB, GE, HU, IS, JP, KE, KG, KP, KR, KZ, LK, LR, LT, LU, LV, MD, MG, MN, MW, MX, NO, NZ, PL, PT, RO, RU, SD, SE, SG, SI, SK, SK (Utility model), TJ, TM, TT, UA, UG, UZ, VN, European patent (AT, BE, CH, DE, DK, ES, FR, GB, GR, IE, IT, LU, MC, NL, PT, SE), OAPI patent (BF, BJ, CF, CG, CI, CM, GA, GN, ML, MR, NE, SN, TD, TG), ARIPO patent (KE, MW, SD, SZ, UG). Published <i>Without international search report and to be republished upon receipt of that report.</i>		
(54) Title: FLASH MEMORY BASED MAIN MEMORY		
(57) Abstract A flash memory chip that can be switched into four different read modes is described. Computer systems and hierarchies that exploit these modes are also described. In the first read mode, 'asynchronous flash mode, the flash memory is read as a standard flash memory. In this mode, the reading of the contents of a first address must be completed before a second address to be read can be specified. In the second read mode, synchronous flash mode, a clock signal is provided to the flash chip and a series of addresses belonging to a data burst are specified, one address per clock tick. Then, the contents stored at the addresses specified for the burst are output sequentially during subsequent clock ticks in the order in which the addresses were provided. Alternately, if a single address is provided to the flash chip when it is in the synchronous mode, the subsequent addresses for the burst will be generated within the flash chip and the data burst will then be provided as output from the flash chip. In the third read mode, asynchronous DRAM (dynamic random access memory) mode, the flash memory emulates DRAM. Thus, row and column addresses are strobed into the flash memory using row and column address strobe signals. The flash memory then converts the row and column addresses internally into a single address and provides as output the data stored at that single address. Furthermore, although the flash memory does not need an extended precharge period or to be refreshed, when in the asynchronous DRAM mode, the flash memory responds to precharge periods and refresh cycles as would a DRAM. Therefore, when in the asynchronous DRAM mode, the flash memory can be controlled by a standard DRAM controller. In the fourth read mode, synchronous DRAM mode, the features of the second and third modes are combined to yield a flash memory that emulates a synchronous DRAM. SYSTEM <u>100</u> <pre> graph LR 110[MICROPROCESSOR 110] <--> BUS 120 130[FLASH MEMORY 130] </pre>		

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FLASH MEMORY BASED MAIN MEMORY

FIELD OF THE INVENTION

The present invention pertains to the field of the architecture of computer systems. More particularly, the present invention relates to computer systems that use a large-block erasable non-volatile semiconductor memory as main memory.

BACKGROUND OF THE INVENTION

As modern computer programs have become increasingly more sophisticated, modern personal computer systems have also had to become more sophisticated in order to accommodate these computer programs. Computer programs are made up of a larger number of code instructions than they once were and on average, require access to larger files of data that are read from, and written to, when executing the programs.

Typically, the heart of a personal computer system is a central processing unit (CPU) that resides on a microprocessor chip. New microprocessor chips that operate at increasingly high operating speeds are constantly being developed in order to permit personal computers to execute the larger programs in a timely manner. Usually, these microprocessor chips are developed using CMOS (complementary metal-oxide semiconductor) technology. The greatest amount of power consumption for CMOS chips occurs on the leading and trailing edges of clock pulses (i.e. when a clock signal transitions from a low voltage state to a higher voltage state and vice versa).

When the operating speed of the microprocessor is increased, the number of clock pulses in a particular time period increases thereby increasing the power consumption of the microprocessor during this time period. Furthermore, more heat is generated by the microprocessor and must be dissipated in order to prevent the damage of components within the computer system.

Both power consumption and heat dissipation pose serious problems when designing a personal computer system. This is

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especially true in the case of mobile computers that are typically powered by batteries. The more power that the computer consumes, the less time that the computer can operate off of a given sized battery. Therefore, as the operating speed of the computer is increased, a designer is faced with several unattractive alternatives.

If the same sized batteries are used, then the effective operating time for the computer system must decrease when the operating speed is increased. On the other hand, if the effective operating time is to remain constant then it is necessary to either add additional batteries, thereby increasing the bulk and weight of the computer, or to use an exotic and therefore expensive battery technology (or both).

The trend in mobile computers is towards smaller, faster, less expensive and lighter units. Thus, the need to add additional batteries, or more expensive batteries is a significant disadvantage. This disadvantage is exacerbated by the need to add cooling fans, or to implement other cooling techniques, in order to dissipate the additional heat that is generated by the high speed microprocessors.

Additionally, because the microprocessors are operating at a higher speed, they can execute more instructions in a given amount of time, and therefore can also process a greater amount of data during that period. A bottle neck has developed in computer systems having fast microprocessors that can prevent the higher speed of the microprocessor to be utilized effectively. This bottle neck is the bus (or buses) that provide instructions for the microprocessor to execute and the data that the microprocessor will use when executing the instructions.

If the next instruction to be executed is not available when the microprocessor needs it, then the microprocessor must wait idly (i.e. insert wait cycles) while the required instruction is retrieved and provided to the microprocessor. Furthermore, if the next instruction to be executed requires data that is not immediately available to the microprocessor, the microprocessor must also idle until the data has been retrieved. During this idle time, the microprocessor clock continues to toggle

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thereby needlessly consuming power and generating heat that must be dissipated.

In order to decrease the frequency with which the microprocessor encounters these wait cycles, many modern high performance microprocessors have a small internal cache, called a primary cache. Instructions that are likely to be executed and data that is likely to be needed by the executing instructions are stored in the internal cache so that they may be accessed immediately by the CPU of the microprocessor.

The sequential nature of computer programs is such that when a particular instruction within the program is executed, it is highly probable that the next instruction to be executed will be the instruction that follows the currently executing instruction. Therefore, when an instruction is to be executed, the cache is checked to determine whether a copy of the required instruction is immediately available within the cache. If a copy of the required instruction is stored within the cache (called a cache hit), then the copy of the instruction can be supplied to the CPU immediately from the cache and there is no need for the CPU to wait while the instruction is retrieved to the microprocessor chip from wherever it is stored in the computer system.

On the other hand, if a copy of the required instruction is not stored within the cache (called a cache miss), then the CPU must wait while the instruction is retrieved to the microprocessor chip from wherever it is stored within the computer system. Actually, rather than only retrieving the next instruction to be executed, a cache line is formed by retrieving the next instruction to be executed and a certain number of instructions following the next instruction to be executed. That way, if the subsequent instructions are in fact required to be executed, they will be immediately available to the CPU from within the cache line of the cache. Because of the sequential nature of programs, the benefits of caching also applies to data used by the programs.

Because the internal cache is filled a cache line at a time, many microprocessors can accept data in a burst mode. In a typical burst read, the microprocessor specifies the first address of the data or instructions to be read into a cache line. Then, the data or instructions that are stored at the addresses of the cache line are sent sequentially from where they are stored within the computer system to the microprocessor.

Frequently the internal cache of the microprocessor is formed using static random access memory (SRAM). Because each SRAM cell is formed by six to eight transistors, there is only room on a microprocessor chip for a relatively small SRAM cache. Furthermore, SRAM is volatile meaning that SRAM retains the information stored as long as there is enough power to run the device. If power is removed, the contents of the SRAM cache are lost.

Some microprocessors are dynamic, meaning that if power is removed from them, when power is restored they cannot return directly to the state they were in when the power was removed. When power is restored the microprocessor must be reinitialized, and at least some of the processing progress previously made will probably be lost.

Other microprocessors are static, meaning that they can be placed in an energy saving deep powerdown mode, and then be returned relatively quickly to the state they were in immediately before they entered the deep powerdown mode.

As mentioned earlier, data and instructions are stored within the computer system and provided to the microprocessor over one (or more) bus systems. Because most types of relatively fast random access memory are both volatile and relatively expensive, a typical computer system stores code and data on relatively inexpensive, nonvolatile memory store such as a floppy disk or hard disk.

The typical computer system also has a main memory made of volatile memory because the nonvolatile memory has a relatively slow access speed. When a program is to be executed, the computer system uses a technique known as shadowing to copy the code and

data required to execute the program from the slow nonvolatile memory to the faster volatile memory. The shadow copy in the main memory is then used to execute the program. If any changes are made to the shadow copy during the course of the program execution, the shadow copy can be copied back to the slower nonvolatile memory, when the program finishes execution. Furthermore, because an unexpected power failure will cause the contents of the volatile main memory to be lost, it is common to save intermediate results generated during the course of execution of the program.

The most common form of main memory is dynamic random access memory (DRAM). DRAM is more commonly used than SRAM, even though it is slower than SRAM because DRAM can hold approximately four times as much data as a SRAM of the same complexity.

DRAMs store information in integrated circuits that contain capacitors. Because capacitors lose their charge over time, DRAMs must be controlled by logic that causes the DRAM chips to continuously "refresh" (recharge). When a DRAM is being refreshed, it cannot be read from, or written to, by the microprocessor. Thus, if the microprocessor must access the DRAM while it is being refreshed, one or more wait states occur.

In some computer systems, SRAM is used as main memory in place of DRAM. One advantage of using SRAM as main memory is that SRAM is relatively faster to access than DRAM. Furthermore, because SRAM does not need to be refreshed, it is always available for access by the microprocessor, thereby eliminating the DRAM associated need for the microprocessor to include wait states when accesses are attempted while the DRAM is being refreshed. Moreover, the lack of a refresh requirement simplifies designing a computer system having SRAM based main memory because one does not have to worry about controlling refresh cycles. In fact, a simple battery back-up can be supplied to preserve the contents of the SRAM in the event of a power failure. Of course, if the battery back-up fails, the contents of the SRAM main memory will be lost.

Rather than building a main memory completely from SRAM, it is more common to implement the main memory using DRAM, and then to supplement the DRAM based main memory with a SRAM based external cache memory (i.e. a cache memory that is external to the microprocessor chip). Because the external cache is not contained on the microprocessor chip, it can typically be made to store more data and instructions than can be stored by the internal cache. Because the external cache is not located on the microprocessor chip, however, it must supply the data and instructions to the microprocessor using one of the buses that often form bottlenecks for data and instructions entering and leaving the microprocessor chip.

A high speed microprocessor chip typically interfaces with the rest of the computer system using one or two high speed buses. The first of these buses is a relatively high speed asynchronous bus called a main memory bus. The second of these buses is a relatively high speed synchronous bus called a local bus. The typical operating speed of main memory and local buses is in the range of 16 to 33 MHz and the trend is towards increasingly faster buses.

Although most microprocessors can interface directly with a main memory bus, some microprocessors do not provide an external interface to a local bus. These microprocessors typically interface with a relatively slow speed synchronous bus called an expansion bus. The typical operating speed of an expansion bus is in the range of 8 to 12 MHz.

The main memory (or DRAM) bus is used by the microprocessor chip to access main memory. Usually, rather than interfacing directly to the DRAM chips, the microprocessor is coupled to a DRAM controller chip that, in turn, is coupled to the DRAM chip or chips. The DRAM controller controls accesses to the DRAM chips initiated by the microprocessor. The DRAM controller also controls overhead maintenance such as the refresh cycles for periodically refreshing the DRAM contents. Some microprocessors have the DRAM controller built directly into them. Frequently, the DRAM or SRAM chips are contained in surface-mount packages and several

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DRAMs or SRAMs are attached to a small circuit board to form what is called a Single In-line Memory Module (SIMM). One can then relatively easily modify the total amount (or the access speed) of main memory in a computer system by simply swapping one type of SIMM for another. A SRAM based external cache may also be coupled to the microprocessor through the DRAM bus.

If a computer system has a local bus, then the microprocessor can access devices coupled to the local bus at a relatively fast speed. Thus, high bandwidth devices such as graphics adapter cards and fast input/output devices are typically coupled directly to the local bus. Sometimes the external cache is coupled to the local bus rather than to the DRAM bus. It is also possible to supplement (or replace) the main memory on the main memory bus by coupling DRAM to the local bus using a DRAM controller designed to interface with the local bus.

Each device coupled to the local bus has an associated capacitive load. As the load on the local bus is increased, the maximum operating speed for the local bus decreases and the power required to drive the bus increases. Therefore, one device coupled to the local bus can be a peripheral bus bridge from the local bus to another bus called a high speed peripheral bus (e.g. a peripheral component interconnect (PCI) bus). The bus bridge isolates the load of the devices coupled to the high speed peripheral bus from the high speed local bus.

Another device coupled to the local bus is typically an expansion bus bridge that couples the high performance local bus to a lower performance expansion bus. The low bandwidth components of the computer system are then coupled to the lower performance expansion bus. One type of device that is typically coupled to the expansion bus uses flash memory. Flash memory typically is a high-density, nonvolatile, read-write memory. Examples of flash memory based devices include BIOS ROM and hard disk substitutes.

Flash memories differ from conventional EEPROMs (electrically erasable programmable read only memories) with respect to erasure. Conventional EEPROMs use a select transistor for

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individual byte erase control. Flash memories, on the other hand, achieve much higher density with single transistor cells. For a typical flash memory array, a logical "one" means that few if any electrons are stored on a floating gate associated with a bit cell. A logical "zero" means that many electrons are stored on the floating gate associated with the bit cell. Each bit of the flash memory array cannot be overwritten from a logical zero state to a logical one state without a prior erasure. During a flash erase operation, a high voltage is supplied to the sources of every memory cell in a block or in the entire chip simultaneously. This results in a full array or a full block erasure.

After a flash memory array has been erased, a logical one is stored in each bit cell of the flash memory array. Each single bit cell of the flash memory array can then be programmed (overwritten) from a logical one to a logical zero, given that this entails simply adding electrons to a floating gate that contains the intrinsic number of electrons associated with the erased state. Program operations for flash memories are also referred to as write operations.

The read operation associated with a typical flash memory array closely resembles the read operation associated with other read-only memory devices. A read operation for a typical high speed flash memory array takes on the order of 80 nanoseconds (nS). Write and erase operations for a flash memory array are, however, significantly slower. Typically, an erase operation takes on the order of one second. A write operation for a single word of a flash memory array takes on the order of 10 microseconds.

British patent document no. GB 2 251 324 A, published July 1, 1992, describes a computer system that uses flash memory. The patent document discloses various architectures to incorporate a flash memory into a computer system. One architecture referred to therein is a variable file structure. For the variable file structure, computer code is stored contiguously in flash memory, allowing a CPU to execute computer code directly from the flash memory array without the need for RAM. A direct mapped variable file structure is described

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that allows direct code execution from all of the flash memory array. A page mapped variable file structure is also described that allows direct code execution from a portion of the flash memory array. Thus, flash memory can serve as the main memory within portable computers, providing user functions similar to those of disk-based systems.

A ROM-executable DOS is available commercially and provides several benefits to both system manufacturers and ultimately end users. First, because most of the operating system is composed of fixed code, the amount of system RAM required to execute DOS is reduced from 50K to 15K, thereby conserving system space and power. Secondly, DOS can now be permanently stored in, and executed from, a single ROM-type of device such as flash memory. This enables systems to be provided that are ready to run right out of the box. Lastly, users enjoy "instant on" performance because the traditional disk-to-DRAM boot function and software downloading steps are eliminated.

For example, by storing application software and operating system code in a Resident Flash Array (RFA), users enjoy virtually instant-on performance and in-place code execution. An RFA also protects against software obsolescence because, unlike ROM, it is in-system updatable. Resident software, stored in flash rather than disk, extends battery life and increases system reliability.

Because erasing and writing data to flash memory is a distinctly different operation than rewriting information to a disk, new software techniques have been developed to allow flash to emulate disk functionality. File management software such as Microsoft's Flash File System (FFS) allows Flash Memory components and flash cards to emulate the file storage capabilities of disk. Microsoft's FFS transparently handles data swaps between flash blocks similar to the way MS-DOS handles swaps between disk sectors. Under FFS, the user can input a MS-DOS or Windows command without regard for whether a flash memory or magnetic disk is installed in the system. Flash filing systems make the management of flash memory devices

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completely transparent to the user. Flash filing systems similar to the Microsoft FFS are available or are being developed for other operating systems besides DOS and Windows.

Flash Memory is exceptionally well-suited to serve as a solid-state disk or a cost-effective and highly reliable replacement for DRAMs and battery-backed static RAMs. Its inherent advantages over these technologies make it particularly useful in portable systems that require the utmost in low power, compact size, and ruggedness while maintaining high performance and full functionality.

Flash memory, however, typically has an asynchronous interface wherein an address to be read is specified and then, a set time later, the contents stored at the specified address are output from the flash chip. It is only after the data has been output from the flash chip that the next address to be read can be sent to the flash chip. A high speed bus like the local bus can run at 33 MHz wherein every cycle of the bus takes about 30 nS. A typical high performance flash chip, on the other hand, has a read access time of about 80 nS. Hence, if flash is to be used as main memory, every single memory access to flash involves wait states and zero wait state back to back burst read cycles from flash cannot be supported. This is true for other devices having a read latency similar to that of flash memory. Thus, using prior art technology, it is not practical to use these memories as main memory for a high speed microprocessor.

SUMMARY AND OBJECTS OF THE INVENTION

Therefore, one object of the present invention is to provide an efficient memory hierarchy based on non-volatile memory versus volatile memory wherein both data and applications are stored in random access nonvolatile memory and further wherein applications are executed directly from the random access non-volatile memory.

It is also an object of this invention is to enable flash memory to operate in an optimal synchronous fashion with any synchronous bus.

It is also an object of this invention is to enable flash memory to operate in an optimal synchronous fashion with any synchronous bus to provide a low cost, low power alternative to volatile main memory, and to eliminate the time required to transfer code and data from the hard disk to the main memory.

It is also an object of this invention to enable flash memory to operate in an optimal synchronous fashion with any synchronous bus so that the CPU can execute programs directly out of the flash memory without any degradation in performance when compared to volatile memory based main memory.

It is also an object of this invention to enable flash memory to operate in an optimal synchronous fashion with any synchronous bus and to thereby eliminate the need to incorporate costly memory subsystem designs such as interleaving into the system.

It is also an object of this invention to enable flash memory to operate in an optimal synchronous fashion with any synchronous bus and to thereby support back to back burst cycles and thus ensure that cache line fills are performed in a quick and optimal fashion.

It is also an object of this invention to enable flash memory to operate in an optimal asynchronous fashion with any asynchronous main memory bus.

It is also an object of this invention to enable flash memory to operate in an optimal asynchronous fashion with any asynchronous main memory bus to provide a low cost, low power alternative to volatile memory based main memory and to also eliminate the time

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required to transfer code and data from the hard disk to the main memory.

It is also an object of this invention to enable flash memory to operate in an optimal asynchronous fashion with any asynchronous main memory bus such that the CPU can execute programs directly out of the flash memory without any degradation in performance when compared to volatile memory.

It is also an object of this invention to enable flash memory to operate in an optimal asynchronous fashion with any asynchronous main memory bus and to eliminate the need to have custom controllers.

It is also an object of this invention to enable flash memory to operate in an optimal asynchronous fashion with any asynchronous main memory bus to provide a glueless interface to the existing main memory controller and thus reduces cost and loading on the local bus.

A flash memory chip that can be switched into four different read modes is described. Computer systems and hierarchies that exploit these modes are also described. In the first read mode, asynchronous flash mode, the flash memory is read as a standard flash memory. In this mode, the reading of the contents of a first address must be completed before a second address to be read can be specified.

In the second read mode, synchronous flash mode, a clock signal is provided to the flash chip and a series of addresses belonging to a data burst are specified, one address per clock tick. Then, the contents stored at the addresses specified for the burst are output sequentially during subsequent clock ticks in the order in which the addresses were provided. Alternately, if a single address is provided to the flash chip when it is in the synchronous mode, the subsequent addresses for the burst will be generated within the flash chip and the data burst will then be provided as output from the flash chip.

In the third read mode, asynchronous DRAM (dynamic random access memory) mode, the flash memory emulates DRAM. Thus, row and column addresses are strobed into the flash memory

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using row and column address strobe signals. The flash memory then converts the row and column addresses internally into a single address and provides as output the data stored at that single address. Furthermore, although the flash memory does not need an extended precharge period or to be refreshed, when in the asynchronous DRAM mode, the flash memory responds to precharge periods and refresh cycles as would a DRAM. Therefore, when in the asynchronous DRAM mode, the flash memory can be controlled by a standard DRAM controller.

In the fourth read mode, synchronous DRAM mode, the features of the second and third modes are combined to yield a flash memory that emulates a synchronous DRAM. Thus, addresses to be read as a data burst are specified by strobing row and column addresses into the flash memory using RAS and CAS signals. The data of the data burst is then provided sequentially as output from the flash memory on subsequent clock ticks.

Other objects, features, and advantages of the present invention will be apparent from the accompanying drawings and from the detailed description which follows below.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar elements, and in which:

Figure 1 illustrates a computer system having a microprocessor that is coupled to a flash main memory by a high speed bus;

Figure 2 illustrates computer system that combines a static microprocessor with a flash main memory to form a low power consumption, but high performance, computer system;

Figure 3 illustrates a flash memory subsystem that interlaces several asynchronous flash memory units to support synchronous back to back data burst read cycles;

Figure 4 is a timing diagram that illustrates a read cycle for an asynchronous flash memory;

Figure 5 illustrates a state transition diagram for the flash memory subsystem of Figure 3;

Figure 6 illustrates a block diagram of a flash memory integrated circuit having a synchronous flash interface to support synchronous back to back data burst read cycles from internal banks of flash arrays;

Figure 7 is a timing diagram that illustrates a back to back data burst read cycle for the flash memory having the synchronous flash interface;

Figure 8 is a block diagram of a computer system illustrating a generic synchronous flash interface flash memory unit coupled to a microprocessor using a high speed synchronous bus and bus specific glue logic;

Figure 9 is a block diagram of a flash memory integrated circuit having an asynchronous main memory interface;

Figure 10 is a timing diagram that illustrates a column and row address multiplexed asynchronous read cycle for a flash memory having an asynchronous main memory interface;

Figure 11 is a timing diagram that illustrates a refresh cycle for a flash memory having an asynchronous main memory interface;

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Figure 12 is a block diagram of an asynchronous main memory interface single in-line memory module for the flash memory integrated circuit having the asynchronous main memory interface;

Figure 13 is a block diagram of a computer system that uses a dynamic random access memory controller to access a flash memory based asynchronous main memory interface single in-line memory module;

Figure 14 is a block diagram of a flash memory integrated circuit having a synchronous main memory interface; and

Figure 15 is a block diagram of a computer system that uses a synchronous dynamic random access memory controller to access a flash memory based synchronous main memory interface single in-line memory module.

DETAILED DESCRIPTION

Figure 1 illustrates a computer system 100 wherein microprocessor 110 is coupled to flash memory 130 by high speed bus 120. In computer system 100, flash memory 130 is used to replace some, if not all, of volatile main memory (not shown). Thus, flash memory 120 is used in lieu of volatile main memory such as dynamic random access memory (DRAM) or static random access memory (SRAM) to provide a high speed non-volatile main memory for computer system 100.

A computer hierarchy based upon volatile main memory loses all information in main memory when power is turned off. A flash-based non-volatile main memory, however, reduces or eliminates the lengthy process of obtaining information from disk when power is turned on. Therefore flash main memory based computer system 100 has higher system performance when a program is initially executed than would a volatile main memory based computer system.

Furthermore, by using flash memory as a main store, the duplication of shadowing information on both disk and RAM is no longer necessary thereby reducing memory cost by eliminating memory duplication.

Moreover, by having information stored in non-volatile flash memory, power consumption is reduced because battery backup of volatile memory is eliminated and because disk accesses are minimized or eliminated.

Figure 2 illustrates a computer system 200 wherein static microprocessor 210 is coupled to flash main memory 230 and battery-backed SRAM 240 by high speed bus 220. Bus bridge 280 couples high speed bus 280 to low speed bus 260. Slow non-volatile memory 250 is coupled to low speed bus 260.

In a typical modern volatile main memory based computer system, there will be four to eight megabytes of volatile main memory, of which, approximately half will be used to store an operating system, executable code for applications and non-changing data files. The rest of the main memory typically contains data files

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(and executable code for infrequently used applications) that are read from and written to frequently. The former type of information is stored in what is commonly referred to as "read mostly" files and the latter type of information is stored in what is commonly referred to as "read/write" files. In computer system 200, flash memory 230 is used to store the read mostly files and SRAM 240 is used to store the read/write files that are currently in use.

The benefits obtainable by using a computer system having a flash memory based main memory architecture of system 200 is best illustrated by example. Suppose that a user had a computer and frequently used the computer in a windows operating system environment to run a word processor application (to write letters). Further suppose that the user occasionally used the computer to play video games.

Because most computer operating systems (and applications) are designed to be executed from random access memory (i.e. memory that can be read from and written to), no effort is made to divide their memory address space into read only and read/write areas. In actuality, very little of the memory address space will be written to. Most of the memory address space will be read from to provide code instructions and data that infrequently changes.

ROM-able programs are written to provide code that can be executed from read-only memory (ROM). In ROM-able programs, the code is carefully written to ensure that most of the memory address space is read-only. If there are any parts of the program memory address space that must be written to, these parts are contained in a separate read/write memory address space (that can be saved to a read/write memory device). ROM-able versions of Windows, the underlying Disk Operating System (DOS), and popular applications such as word processing programs are currently commercially available.

Thus, the user of computer system 200 can benefit by storing the read-only parts of ROM-able DOS, ROM-able Windows and a ROM-able word processing program in flash memory 230. The read/write

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parts of these programs and the rest of the applications and files can be stored in slow non-volatile memory 250.

Then, when the user starts computer system 200 to write a letter, the read-only parts of the DOS, Windows and the word processor will be resident in flash memory 230. If these programs need any read/write files (e.g. system configuration information), these (typically small) files will quickly be loaded from slow non-volatile memory 250 to battery backed SRAM 240 over bus bridge 280 and low speed bus 260. In short, the major parts of the DOS, Windows and word processing programs will be resident in non-volatile flash memory 230 and can be executed in place directly from non-volatile main memory 230 as soon as the system powers up. Therefore, it will appear to the user as though computer system 200 starts up instantly with the word processor loaded and ready to write the letter.

The user can then retrieve into SRAM 240 a file containing a previously written letter to modify (from where it was stored in slow non-volatile memory 250). Alternately, the user can open a new file in the memory address space of SRAM 240 and can use the new file to start an entirely new letter.

After working on the letter for a while, suppose the user decides to play one of the computer games stored in slow non-volatile memory 250. In such a case, the program associated with the selected game will be loaded into SRAM 250 and the game will begin.

If the user wishes to take a break, the user can suspend the game and leave computer system 200 running. While the user is gone, static microprocessor 210 will sense user inactivity and send a powerdown signal to flash memory 230 over powerdown signal line 270. Then static microprocessor 210 and flash memory 230 will enter a state of low power consumption.

Flash memory 230 and slow non-volatile memory 250 are both non-volatile. Additionally, flash memory 230 will draw little power when in its energy saving powerdown mode and non-volatile memory 250 will not have to be accessed while the user is away. Furthermore, because microprocessor 210 is a static microprocessor, it

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will draw very little power but will retain its state when it enters its deep powerdown mode. Moreover, the read/write files (including the executable code for the game) stored in battery backed SRAM 240 will not be lost.

Later, when the user returns, the game can be continued almost instantly from the place where it was suspended. This is because, even though very little power will be consumed while the user is away, the state of computer system 200 will be retained.

Furthermore, if the user desires to upgrade the DOS or Windows operating systems or to upgrade the program stored in flash main memory 230, this will be easy to do because of the ability to write to flash memory. The erasing and then programming operations required to write to flash memory will mean that it will take longer to store programs in flash memory 230 than would be the case where the main memory was volatile memory such as DRAM. Installation of a program, however, is performed infrequently and is usually a relatively prolonged process anyway. Therefore, the extra time required to install a program into non-volatile flash main memory 230 will usually be insignificant when compared to the benefit of having the program instantly available for execution from non-volatile flash memory 230 whenever system 200 is started.

The ability to write to flash main memory 230 further means that the user can install a different program in place of (or in addition to) the word processing program currently stored in flash main memory 230. Therefore, the user can switch to a rival word processing program by installing the rival word processing program in flash main memory 230. Alternately, the user can, for example, install a spreadsheet or graphics design program in flash memory 230 in place of the word processing program that is currently stored there.

Moreover, provided there is sufficient room in the flash memory 230, the user can install additional applications there. Thus, for example, if the user has a particular computer game that is the current favorite, it can be installed to reside in the flash memory 230.

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Note that a program can be installed in flash memory 230, even if the program is not ROM-able. As described previously, because flash memory can be written to, a flash file system will permit standard applications to execute in place from flash memory 230. The asymmetric read/write nature of flash will, however, cause a performance penalty to be incurred when standard (i.e. non-ROM-able) programs are executed in place from flash main memory 230.

There are approaches that can be used, however, to mitigate these performance problems. One approach is for the operating system to mark the memory address space of the application as being read-only while the application is executing. Then, if the application attempts to write to flash main memory 230, an interrupt will occur and an error handler can cause the write to be redirected to SRAM memory 240. Later, when there is a lull in processing, the information can be written from SRAM 240 to flash 230 as a background operation. In effect, a portion of SRAM 240 can be used as a write cache for flash main memory 230.

One benefit of using a portion of SRAM 240 as a write cache is that applications stored in flash 230 do not need to be ROM-able. Another benefit is that the write areas of the application memory address space will probably be relatively well localized. In the fortunate case where this is so, any subsequent read or write operations to the write cache will benefit from the faster access of SRAM 240. Also, because it is frequently the case that the information to be written back to the application memory space is ephemeral (i.e. not to be saved from one execution of the application to the next), any ephemeral information in the write cache when the application terminates will not have to be written back to the flash memory 230.

From the above example, one can see that, when compared to a computer system that has the traditional volatile main memory based architecture, there are many beneficial and synergistic effects that stem from having the non-volatile main memory computer system architecture of computer system 200.

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First, computer 200 can start instantly when power is first applied. Thus, there is no need to spend time and power loading the operating system and frequently used applications such as the word processor into main memory; they are already there and virtually instantly available.

Second, there is no need to store a copy of an operating system (e.g. DOS and Windows) or to store frequently used application programs (e.g. the word processing program) on slow non-volatile memory 250. Thus, slow non-volatile memory 250 can be smaller than would be required if shadowing of these files was required.

Third, using techniques that will be described below in greater detail, unlike asynchronous (volatile) DRAM, a burst read can be made to flash memory 230. Thus, a cache line for internal cache 215 of microprocessor 210 can be filled from flash memory 220 faster than would be the case if asynchronous DRAM based main memory was used.

Fourth, SRAM 240 can be written to, and read from, faster than can volatile DRAM. Thus, a cache line of internal cache 215 of microprocessor 210 can be filled from SRAM 240 faster than would be the case if a DRAM based main memory was used. Also, programs executing from SRAM 240 can be accessed, and hence executed, faster than would be the case if a DRAM based memory was used.

Fifth, a portion of SRAM 240 can be used to provide an external cache for flash memory 220 thereby further enhancing the execution speed.

Sixth, a lower performance microprocessor can be used to provide the same level of performance as would be the case if a DRAM based main memory was used. This is because the low bandwidth constraint of a volatile DRAM based main memory caused by the need to load the operating system and programs will be eliminated. The operating system and programs are instead stored in non-volatile flash main memory 230. This is also because in system 200 there will be a better match between the data and program instruction needs of microprocessor 210 and the ability of the flash

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230/SRAM 240 based main memory to supply them than would be the case with a DRAM main memory system. This will result in fewer idle wait states for microprocessor 210. Frequently, microprocessor 210 will be able to operate with zero wait states. Therefore, even though there will be fewer cycles in which microprocessor 210 can do work because it is a lower performance microprocessor, fewer of the cycles that are available to the microprocessor will be wasted while the microprocessor is waiting for code or data.

Seventh, because microprocessor 210 can have a lower clock rate for a given level of performance, microprocessor 210 will be less expensive, have fewer heat dissipation problems and use less power than if a DRAM based main memory that required that a faster processor to be used.

Eighth, because microprocessor 210 and flash 230 can enter a deep powerdown mode until they are needed, further energy savings will result.

Thus, a non-volatile read/write general purpose computer hierarchy can be achieved with the architecture described above when the dominant portion of main (execute) memory is flash memory.

Depending upon the requirements of system 200, archival storage (i.e. slow nonvolatile memory 250) may or may not be present in the system. If it is present, the needs of system 200 can govern the technology used to provide it (i.e. mechanical, optical or solid state). Also, depending upon the particular needs of the system 200, battery-backed SRAM 240 can alternately be replaced by non-volatile RAM (NVRAM), electrically erasable programmable ROM (EEPROM), DRAM, SRAM, or battery-backed DRAM.

The underlying theme of each of the above configurations for system 200 is that the CPU of microprocessor 210 is fetching code and/or data from random access non-volatile (flash) memory 230 and writing back different data and/or programs on an as-needed basis. As described previously, the new information may or may not be written to the same place in main memory. A RAM based system is

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inherently volatile. A ROM based main memory system, although non-volatile, does not provide the capability to write back to the ROM.

Flash is not a symmetrical technology as is the DRAM, i.e. it does not allow for byte alterability at read speeds. Therefore, to modify a file or program in one embodiment, the portion to be altered is moved into byte alterable memory (e.g. NVRAM, EEPROM, SRAM or DRAM), modified, and then written back to flash main memory. This alterable memory can be separate memory chips and/or integrated onto a CPU or microprocessor. Data that is not being altered is read directly by the CPU from Flash. This configuration essentially eliminates disk down load for both read and unaltered data to provide faster read performance than a disk based system. For writes, the write back is comparable in performance to disk. Because the read cycle is faster, however, a higher overall system performance is provided.

The key to each of these configurations is that a flash memory is used as a main memory and is read at a speed that approximates, or surpasses, the read access speed of DRAM. There are basically four possible ways that this can be accomplished. The first way is to provide a hardware controller that interleaves read operations to more than one conventional asynchronous flash main memory chip.

Multiple Chip Interleave for Asynchronous Flash Main Memory

Figure 3 illustrates a flash memory sub-system wherein four flash memory integrated circuits 320 (#1 through #4) are interlaced under the control of a controller unit to provide a burst read capability. Although Figure 3 shows four flash integrated circuits 320, it is to be understood that if the data width of each individual integrated circuit 320 is not that of a double-word, an appropriate number of flash integrated circuits 320 would be provided in parallel with each of the circuits 320 of the figure to provide a double word tdata width. In this example, the controller is implemented using fully programmable gate array (FPGA) 310. In alternate embodiments, the interlace control is provided using alternate technologies.

Figure 4 illustrates the read cycle for a typical asynchronous flash memory such as flash memory integrated circuits 320 of Figure 3.

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From Figure 4 it can be seen that before and after a read cycle the data lines (DOUT) of the flash memory are tri-stated (indicated in the figure as OPEN). This means that the data lines have been placed into a high impedance state.

The read cycle of Figure 4 begins with the transition of output enable (OE#) and chip enable (CE#) signals from low voltage (V_L) to high (V_H). Next, if it is not already high, a write enable (WE#) signal is transitioned high to signify a read operation and the address to be read is placed on the address pins (A₀ through A₁₉) of the flash chip. The data stored at the address specified is then retrieved internally from the flash array. The read process takes a known amount of time that is constant. Once that amount of time has elapsed, the OE# signal is brought low and the data that was read from the specified address is driven by the flash chip out onto the data lines (DOUT). The read cycle ends when the CE# and OE# signals are again brought high (perhaps to begin the next read cycle).

Note in Figure 4 that a new address to be read cannot be specified until the previous read cycle has completed. Therefore, a single standard asynchronous flash memory unit cannot be interlaced to support rapid, back to back read cycles. One must instead wait for the first read cycle to complete before a next read cycle can be begun.

The total access time of a standard asynchronous flash component (TACC) can be divided into two parts. First, there is the time to decode addresses and chip enable, select the correct bits in the flash array to be read and then sense their stored data (TACC-TOE). Second, there is the time required to drive this information onto the data bus through the output buffers (TOE). Once a single address for a data burst read is known, the rest of the addresses of the data burst read are also known. Therefore, referring again to Figure 3, interlacing exploits this property of a data burst to amortize the first access time (TACC-TOE) across two or more flash chips to provide a burst that has an improved average read access time.

Consider the case of a 20 MHz synchronous local bus (i.e. each clock cycle takes approximately 50 nS). At that speed, a typical

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microprocessor will have a maximum burst read capability of 2-1-1-1 bursts. That means that when the microprocessor reads a burst of data, the burst will be made of four double-words of data that are each part of the same cache line. Furthermore, once the microprocessor specifies the first address of the burst, it will be unable to accept the contents stored at that address for two cycles (approximately 100 nS in this example). Then, the microprocessor will be able to accept the respective contents of the second, third and fourth addresses of the cache line during the next three clock cycles.

If flash memory chips 320 each have a (TACC-TOE) less than 2 clock cycles (approximately 85 nS) and a (TOE) less than 40 nS, then flash memory sub-system 300 will be able to provide a burst read that matches the maximum burst read attainable by the microprocessor. This is because, as soon as FPGA 310 receives the first address of the data burst to be read, it will enable all four flash memory chips 320 and order each chip 320 to read one of the addresses of the data burst in parallel with the other chips 320. After the first two cycles, the data stored at the first address of the data burst will be available from flash 320 #1. Then, on each successive cycle thereafter, the data stored at the next address of the data burst will be available (from flash 320 #2, then #3 and finally #4).

This means that in this example flash memory sub-system 300 will be able to provide a data read burst of four double-words of data during a five clock cycle period (approximately 250 nS) thereby providing an average read time of approximately 50 nS. If interleaving were not used (e.g. if only one flash memory chip 320 was used), the read for each address would take three cycles, (i.e. two cycles for (TACC-TOE) and one for (TOE)) and each read would have to wait for any earlier read to complete. Using the numbers presented above, each address of the data "burst" to a single asynchronous flash 320 would take 150 nS. Therefore, without interlacing, the fastest burst speed attainable would be a 3-3-3-3 data burst that would take 600 nS to complete (4 times 150).

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Clock signal CLK coordinates the operation of flash memory sub-system 300. In sub-system 300, four byte flash memory pages are selected by common addresses A21 through A2 of the AX through A0 memory space. The higher order addresses (AX through A22) decode the 4 Mbyte flash array in the system memory map. Addresses A1 and A0, along with the RD# (read) signal input to FPGA 310, cause FPGA 310 to generate output enable signals (OE#) for flash memory components 320 (#1 through #3). Therefore, in one embodiment, system addresses of flash sub-system 300 are associated with flash memory components 320 as follows:

System Address Address	Component	Component
0	#1	0
1	#2	0
2	#3	0
3	#4	0
4	#1	1
5	#2	1
etc. . . .		

A data burst series of read accesses from memory sub-system 300 will proceed as follows:

First, the processor will signal that it is to read data from system address 0. Signal CE# will enable flash memories 320 #1 through #4 and signal ADDR will provide them with address 0. FPGA 310 will decode system address bits A1 and A0 and, because they are both zero, will enable the OE# signal for flash component 320 #1. Because this is the first access to the four byte page, the access will take 150 nS (i.e., three 50 nS clock cycles).

Second, the processor, executing sequential code, will read from system address 1. Components 320 #1 through #4 will remain enabled, and address (ADDR) inputs to them will remain as zeros. FPGA 310 will then decode the "01" on system address bits A1 and A0, and enable the OE# line for component 320 #2. Because the data for component 320 #2 has already been selected and sensed, the access

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time for component 320 #2, therefore, will only have a 50 nS delay (i.e., one clock cycle).

Then, similar 50 nS accesses will follow for system addresses 2 and 3, reading from components #3 and #4, respectively.

If the next data burst begins with address 4, the processor will then read from system address 4 on its next cycle. This will change system address A2 from zero to one thereby incurring another full 150 nS delay for this access from component 320 #1. This is because a new 4-byte page, and therefore, new data from each of the flash memories 320 is to be read. Once again, however, subsequent accesses to system addresses 5, 6, and 7 will each only have a 50 nS duration.

FPGA 310 decodes lower addresses and, correspondingly, generates OE# signals to flash memory devices 320 within an interleaving page. The earlier example was for a four-byte page, so addresses A1 and A0 are decoded. For a two-byte page, only address A0 would be examined, whereas an eight-byte page would use addresses A2 through A0.

FPGA 310 will examine all upper address bits (i.e. AX to A22 in this example), to determine first if the memory subsystem 300 is being accessed, and then if access to the same page is occurring, and adjust wait states back to the processor accordingly (via the READY signal output). This function is accomplished by the comparison logic sub-block 340 of Figure 3.

Furthermore, FPGA 310 will distinguish between a read or write to flash memories 320, and adjust wait states accordingly. Figure 5 illustrates a state transition diagram for hardware interleaving that is implemented in FPGA logic 310.

State transition diagram 500 begins in initial state 510 wherein the FPGA is waiting for an access to a flash memory 320 of sub-system 300. When the FPGA detects the beginning of an access to the flash memory 320 of sub-system 300, a test is made in state 520 to determine whether the access is to the same four-byte page as was the previous access. If the access is to the same four-byte page as was the previous

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access, then transition 524 is taken to state 530 and a test is made to determine whether the access is a read operation.

If it is determined in state 530 that the access is a read operation, then the information to be read is already available and a transition is made on path 534 to ready state 560 and the read is performed. After the read is performed, a transition is made from ready state 560 back to initial state 510 and the next access is awaited.

Returning now to state 530, if it is determined in state 530 that the access is not a read operation, then a transition is made on path 532 to wait state(s) generation state 550 and an appropriate number of wait states are performed while an erase is performed (if needed) and the information is programmed.

Returning now to state 520, if it is determined in state 520 that the access is not to the same four-byte page as the previous access, then a transition is made on path 522 to state 540 and a new address is latched for the page compare. Then, in state 550, an appropriate number of wait states are performed while the new four-byte page is accessed.

Note that the flash memory writes cannot take advantage of interleaving as can reads because there are no TACC or TOE equivalents for write operations. Although memory sub-system 300 supports back-to-back read bursts and reduces the average read access time for sequential accesses, it increases the complexity of the system hardware required over that required for access to a conventional asynchronous flash memory chip.

As described previously, however, a write cache or software interleaving techniques can be used to maximize program and erase performance of memory sub-system 300. Furthermore, in the case where a ROM-able program is being executed in place from flash memory sub-system 300, there will not be any writes to the flash memory during the execution of the program. In such a case, the slower write performance of the flash memory will only be a factor on the rare occasion that a new ROM-able version of the program is installed.

Synchronous Flash Interface (SFI) for Flash Main Memory

Besides requiring the additional system complexity of FPGA 310, a memory subsystem such as subsystem 300 will have to have at least two flash memory chips 320 to interleave between (four in the example of Figure 3). Furthermore, if additional flash memory chips are to be added to sub-system 300, they too must (at least) be added in pairs. Moreover, each flash memory chip added to sub-system 300 is enabled simultaneously and, hence sub-system 300 consumes a large amount of power. Alternately, an entire synchronous flash interface can be incorporated onto a single flash memory chip.

Figure 6 illustrates a block diagram of a synchronous flash interface (SFI) flash memory integrated circuit 600 that incorporates a complete synchronous flash interface in a single flash memory chip. The synchronous flash interface provides a means of converting the synchronous cycles that appear from a controller to the asynchronous protocol that the flash core understands. The device is internally split into two banks and the least significant address is used to distinguish between the two banks. In short the device is interleaved internally. The user can direct back to back cycles to the alternate banks and, after an initial delay equal to an asynchronous flash access time, data will output onto the output bus at a rate that is approximately equivalent to half the asynchronous flash access time. Hence, the device using this interface creates an average access time for sequential read accesses that is significantly less than the access time of an asynchronous flash device.

A clock input is a part of the interface. An address latch enable pin is present to indicate that a valid address is present on the address bus. All the external operations of the device are synchronized to the rising edge of the clock. ALE# is sampled on the rising edge of the clock and, if it is sampled valid, the address that is present on the address bus is latched into the part and an access commences. The user can alternately strobe addresses into the device every clock cycle. After an initial delay the data corresponding to these addresses is output on the data bus in synchronism with the rising edge of the

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clock. The user can cycle the device at frequencies as high as 33 MHz and can expect back to back burst cycles to be performed without any wait states in between. Also part of the design is a programmable latency that enables the user to define the clock cycle latency between address strobe and valid data.

In Figure 6, the flash array of SFI 600 is divided into two banks, flash bank A 610, and flash bank B 620. Mode register 680 can be set to enable or disable the SFI mode. Thus, when mode register 680 is set, signal interlace enable (IE) enables interlace control logic and address transition detection (ATD) logic 570, transistor-to-transistor logic (TTL) stages 672, bank select logic 674 and bus logic 640. When SFI is enabled, interlace control 670 and bank select logic 674 operate to interlace read (and write) operations between flash bank A 610 and flash bank B 620 in a manner that is analogous to the way that FPGA 310 interlaced accesses to flash chips 320 in sub-memory system 300 of Figure 3. Thus, a single SFI flash chip 600 can support back-to-back read bursts when mode register 680 has been set to SFI mode.

In the SFI mode, an address to be accessed is presented to TTL stages 672 on address buses A19 through A0 along with a clock pulse (CLK) and an address latch enable signal (ALE#). Output enable and write enable signals are also provided to TTL stages 672. Addresses A19 through A1 are then provided from TTL stages 672 to bank A address latch 630 and bank B address latch 632. Signals CLK, ALE# and A0 are provided to interlace control logic and ATD logic 670.

Based on the low order bit of the address (A0), interface control logic 670 controls bank A address latch 630 and bank B address latch 632 using signals address latch enable A (ALEA) and address latch enable B (ALEB), respectively. Thus, interface control logic 670 thereby steers the remaining higher order bits of the address to be accessed to flash bank A 610 or flash bank B 620. Signals address transition detected A (ATDA) and address transition detected B (ATDB) output from interlace control logic and address transition detection logic 670 prepare flash array banks A 610 and B 620, respectively, to be read.

Data multiplexer (MUX) 650 receives as input signal S/A OUT A from the sense amplifiers of flash bank A 610 when bank A 610 is read. Data MUX 650 also receives as input signal S/A OUT B from the sense amplifiers of flash bank B 620 when bank B is read. Bank select logic 674 receives signals CLK, ALE# and A0 as input and provides output signal bank select (BS). Signal BS is used to control data MUX 650 so that it interlaces the data read from banks A 610 and B 620 and steers the data into output buffer 660. Output buffer 660 then drives the data read out onto data input/output (I/O) pads 690 of SFI flash chip 600.

Thus, if a read access is being performed when SFI flash 600 is in the SFI mode, the next address to be accessed can be begun while the preceding access is being performed. If the next address to be read belongs to the bank that is not presently being read, then the next address can be steered to the appropriate bank (A 610 or B 620) while the output from the previous access is being latched into output buffer 660 and is being driven onto data I/O pads 690. Also part of the design is a programmable latency that enables the user to define the clock cycle latency between address strobe and the presentation of valid data on data I/O pads 690.

Alternately, more than two flash banks can be used in SFI chip 600 provided that an appropriate number of address bits are used to select the proper bank. Thus, if four banks are interlaced, two address bits would be used to select among the banks.

Figure 7 is a timing diagram that illustrates a sustained read burst cycle for SFI flash 600 of Figure 6. In Figure 7, synchronizing clock signal CLK is operating at 33 MHz, therefore each pulse (T1 through T8) of signal CLK is 30 nS long. The read burst begins on the rising edge of clock pulse T1 when the first address to be read (ADDR 0) is placed on the address pins and address latch enable signal ALE# is brought low to indicate that a valid address is ready to be read. Note that the low order bit A0 of address ADDR 0 is low on the rising edge of clock pulse T1. Therefore, address ADDR 0 belongs to bank A and bank A will be read to retrieve the data stored at address ADDR 0.

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The read burst continues on the rising edge of clock pulse T2 when the second address to be read (ADDR 1) is placed on the address pins. In one embodiment, address latch enable signal ALE# is brought high and then low to indicate that a valid address is ready to be read. Alternately, address latch enable signal ALE# is simply held low to indicate that another valid address is ready to be read. Note that the low order bit A0 of address ADDR 1 is high on the rising edge of clock pulse T2. Therefore, address ADDR 1 belongs to bank B and bank B will be read to retrieve the data stored at address ADDR 1. Because addresses ADDR 0 and ADDR 1 belong to different flash banks, both address locations can be read concurrently.

The read burst continues on the rising edge of clock pulse T3 when the third address to be read (ADDR 2) is placed on the address pins and address latch enable signal ALE# is held low to indicate that another valid address is ready to be read. Note that during period T3 the read from ADDR 0 of bank A (begun during period T1) completes. Thus, during period T3, bank select signal BS is brought low to steer the data read from ADDR 0 to the output buffer where it is latched. The data of ADDR 0 is then driven out onto the data I/O pads of the SFI chip at the beginning of pulse T4.

Meanwhile, because the low order bit A0 of address ADDR 2 is low on the rising edge of clock pulse T3, address ADDR 2 belongs to bank A and bank A will be read to retrieve the data stored at address ADDR 2. The bank A output from address ADDR 0 has been latched into the output buffer, so the reading of ADDR 2 of bank A can begin. Furthermore, addresses ADDR 2 and ADDR 1 belong to different banks, therefore both address locations can be read concurrently.

The burst read can continue indefinitely (or at least until the address space of SFI flash chip 600 is exhausted) with a new address being input on each clock pulse and the data for that address being output two pulses later.

Referring again to Figure 6, in order for SFI chip 600 to be truly compatible with the high speed synchronous bus with which it communicating, bus logic block 640 receives and provides bus specific

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signals for SFI 600. Thus, the particular high speed synchronous bus may require, for example, that a parity bit be received with every write operation and then sent again as a handshake. In one embodiment, interface logic 600 will latch and examine all bits written to flash memory 600 during write operations and provide this parity data during the post-write "handshake". Alternately, by examining data during write attempts to flash memory 600, interface logic 640 can use the same parity algorithm used by the high speed synchronous bus to generate the required parity information for the post-write "handshake".

Furthermore, the bus protocol may be such that the parity bit may need to be sent from SFI 600 when data is read. Because flash memory does not require that parity bits be stored with data, bus logic 640 would use the same parity algorithm used by the high speed synchronous bus to generate the necessary parity bit when data is read. Thus, during flash memory 600 read operations, logic 640 will derive parity bits from the output of flash memory 600 for transfer to the high speed synchronous bus. Therefore, by examining data both during write attempts to flash memory 600 and read attempts from flash memory 600, interface logic 640 will generate the required parity information.

When mode register 680 is not set to indicate SFI mode, the synchronized bank interlacing control functionality of interlace control 670 is disabled and SFI flash chip 600 will function as would a standard asynchronous flash memory chip. In such a case, the CLK and ALE# signals will be ignored as will the bus specific input signals to bus logic 640. Instead signals CE# and OE# are used to access SFI flash 600 using an asynchronous read signal of the type depicted in Figure 4.

In one embodiment, mode register 680 is implemented using content addressable memory. Alternately, in another embodiment there is no mode register 680 and the SFI functionality of chip 600 cannot be disabled and SFI chip 600 will not function as a standard asynchronous flash memory.

In one embodiment, the SFI flash memory 600 requires one address per word/byte read. Some burst processor buses, on the other hand, initiate a multiple-read burst access by providing a start address, and then repeatedly signaling for reads with no additional address information provided. Alternately, to accommodate these processors, SFI flash chip 600 will latch the start address of a burst, and using on-chip counters, will generate subsequent addresses throughout the remainder of the burst read sequence. In such a case, SFI chip 600 will contain a full-address latch, and a multi-bit counter that matches the addressable space in SFI flash memory array 600. After being provided the start address, the counter logic will auto-increment (in linear or non-linear order, depending on the bus) and feed addresses to the SFI flash memory banks. This allows a multiple-burst access of indeterminate length to occur at highest system performance.

Figure 8 illustrates a computer system 800 that uses an alternate generic SFI chip approach. In Figure 8, SFI flash chip 860 has a generic synchronous flash memory interface. Glue logic 830 is a separate integrated circuit chip that is placed between microprocessor 810 and SFI flash chip 860 to provide the circuitry required to interface SFI chip 860 to high speed synchronous bus 820.

One benefit of the approach of Figure 8 is that providing a generic interface control 870 within SFI flash chip 860 permits a burst read of banks A 880 and B 890. Thus, rather than requiring pairs of asynchronous flash chips in order to perform interlacing, a flash main memory sub-system can be expanded in units of a single chip. At the same time, when compared to FPGA 310 of Figure 3, the design of SFI glue logic 830 is simpler because glue logic 830 does not have to include the generic synchronous interface control logic 870 that would be required for any SFI chip 860.

Furthermore, a single SFI chip 860 can be used to interface with a large number of different synchronous buses because the bus specific control logic 850 is implemented within glue logic 830 instead of within SFI chip 860.

An additional benefit to this approach is that, when compared to a standard asynchronous flash chip, SFI chip 860 need only have two additional pins (i.e. CLK and ALE#). Any other signals that are required for synchronous bus 820 are provided by glue logic 830 in a manner that is transparent to SFI chip 860. Typically, a flash chip is mounted in a small package such as a thin small outline package (TSOP). Because of the small size of the package, a standard asynchronous flash chip will use almost all of the pins available to the package. When providing a generic synchronous flash interface, the two additional pins required are available even within constraints imposed by mounting the SFI flash chip 860 in a small package.

Note that glue logic 830 includes burst logic 840. This is because, as mentioned earlier, different microprocessors 810 use different burst ordering sequences.

A common sequence is the linear addressing sequence. In the linear addressing sequence, a data burst increments sequentially from the beginning address of the burst. Typically, in this sequence, the data burst will begin with the lowest address in the cache line being read and will increment through the addresses of the cache line.

The other burst sequence is a non-linear addressing sequence. In this sequence, the burst read does not necessarily have to begin with the lowest address of a cache line. If the burst does begin with the lowest address of the cache line, then the non-linear addressing sequence behaves as does the linear sequence. If, on the other hand, the burst begins on an address other than the lowest address of the cache line, the subsequent addresses of the burst are generated in a manner that guarantees that every address of the cache line will be accessed and that they will be accessed in a way that interleaves between odd and even address banks.

Burst logic 840 takes a first data address for a data burst and generates the rest of the addresses of the burst in accordance to whatever addressing sequence is required by microprocessor 810.

Note that because both the linear and non-linear burst sequences interleave between odd and even banks, SFI chip 860 will

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interleave between banks regardless of the sequence generated by burst logic 840 (be it linear or non-linear).

Asynchronous Main Memory Interface (AMMI) for
Flash Main Memory

The synchronous flash interface (SFI) supports back to back data bursts thereby permitting flash memory to be used as main memory. There may be occasions, however, where system constraints prevent the use of the synchronous flash interface to a high speed synchronous bus. One such case is where there are already several components that are coupled to the high speed synchronous bus. In such a case, the additional coupling of the SFI chip (or its glue logic controller chip) will increase the capacitive load of the high speed synchronous bus to the point where it can no longer support a desired level of performance. Another such case is where the SFI chip requires a glue logic controller chip and the cost constraints imposed when designing the system prohibit the additional expense of this chip.

Furthermore, many portable systems do not have a local bus (i.e. do not have a high speed synchronous bus). Typically, in these systems, the CPU is integrated with the peripheral device controllers onto a single microprocessor chip. In such systems, only an asynchronous main memory bus and a relatively slow expansion bus are available within the system.

Asynchronous flash memory as it exists today must interface within these systems to the expansion bus through a custom controller. Bus cycles on the expansion bus are typically slow. Therefore, because each expansion bus cycle is typically longer than the asynchronous access time of a flash memory, the synchronous flash interface provides no real benefit when interfacing with the expansion bus.

An Asynchronous Main Memory Interface (AMMI) permits flash memory to interface to the asynchronous high speed main memory bus in any computer system. In effect, the AMMI permits a flash memory chip to emulate an asynchronous DRAM chip. In

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addition to providing a low cost, low power flash main memory, the AMMI also eliminates the time required to transfer programs or data from the hard disk to the DRAM by permitting nonvolatile flash memory to be used as main memory in place of volatile DRAM. Thus, as was the case with SFI, the CPU can execute programs directly out of a flash main memory without any degradation in performance when compared to volatile main memory.

Because the AMMI permits a flash chip to emulate a DRAM chip, the AMMI eliminates the need to have custom controllers. It provides a glueless interface to the existing DRAM controller of a system and thus reduces cost and loading on the local bus.

The AMMI improves system performance and provides an alternative to hard disks in portable computer systems. It also provides for faster execution out of flash with minimal change to the system architecture. Moreover, the system performance can be increased without requiring any additional peripheral logic.

Figure 9 illustrates a block diagram of one embodiment of a flash memory chip having an asynchronous main memory interface. In the embodiment illustrated in Figure 9, flash chip 900 can operate as a standard asynchronous flash memory chip or can emulate a DRAM. The particular addressing mode to be used is determined by whether mode register 910 has been set or not. Thus, if mode register 910 has been set to signify that an asynchronous main memory interface is desired, AMMI flash chip 900 will emulate a DRAM chip. On the other hand, if mode register 910 has not been set to signify that an asynchronous main memory interface is desired, AMMI flash chip 900 will not emulate a DRAM chip. In the latter case, the DRAM emulation circuitry of AMMI chip 900 would be disabled and AMMI flash chip 900 will operate as would a standard asynchronous flash memory chip. In one embodiment, mode register 910 is implemented using content addressable memory.

Figure 10 illustrates a read cycle of an AMMI flash chip 900 when it is emulating a DRAM. Because the AMMI flash chip 900 is

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emulating a DRAM chip, Figure 10 also illustrates a read cycle for a typical DRAM chip.

The asynchronous main memory interface provides a means of interpreting DRAM cycles and converting these cycles to a protocol that can be used to access a flash memory array. To reduce the pin count on a typical DRAM device, rather than using a single 20 bit address, as does flash, a DRAM multiplexes a ten bit row address and a ten bit column addresses. Address signal multiplexing for a DRAM is accomplished by providing two additional pins not found on a prior art asynchronous flash device.

Thus, in a DRAM, a Row Address Strobe (RAS#) pin and a Column Address Strobe (CAS#) pin are used to control the strobing in and latching of respective row and column addresses. Typically, these addresses are latched on the falling edges of these strobes and are used to access specific bits inside the DRAM array. Because AMMI flash memory chip 900 can emulate a DRAM chip, AMMI flash chip 900 has row and address pins for accepting row and column address strobe signals, respectively.

Some DRAM devices use row and column addresses that are not ten bits. For example, the row and column addresses of a DRAM device may only require eight bits to be expressed. Throughout the following discussion ten bit row and column addresses will be used. It is to be understood, however, that AMMI flash 900 can be practiced using other sized row and column addresses.

In Figure 10 it can be seen that the read cycle for an AMMI flash chip 900 that is emulating a DRAM begins with the RAS# and CAS# signals both high and the multi-bit data output signal (DOUT) open (tri-stated). The ten bit row address is then placed on the A₀ through A₉ address pins of the flash chip 900 and the RAS# signal is brought low to indicate that a valid address is to be strobed into the AMMI flash. (Note that in the case of an eight bit row address, the eight bit row address would be placed on the A₀ through A₇ address pins of the flash chip.)

After the row address has been strobed into AMMI flash chip 900, the RAS# signal continues to be held low for a predetermined period of time. In a standard DRAM, the RAS# signal is held high at the end of the read cycle prior to the next read cycle to precharge. Of course, because flash memory is not based on the capacitor-based storage technology of the DRAM, the flash array does not need the lengthy precharge time required by a DRAM. On the other hand, because AMMI flash 900 is emulating DRAM, it must react as would a DRAM.

Once the row address has been latched within AMMI flash chip 900, a read operation is signified by transitioning a WRITE# signal from low to high. Then, the ten bit column address is asserted on the A0 through A9 address pins of AMMI flash chip 900. Next, the CAS# signal is asserted and the column address is latched within AMMI flash chip 900. The ten bit row address and the ten bit column address are then combined within AMMI flash chip 900 to form a twenty bit address used to specify a location within the flash array to be read. Once read, the contents of the twenty bit address location are driven onto the data pins of AMMI flash chip 900 as multi-bit signal data out (DOUT). Then, after a CAS# goes high, the multi-bit signal data out (DOUT) is tri-stated again.

Meanwhile, after the row address has been strobed into AMMI flash chip 900 and the RAS# has been held low long enough for a row precharge to occur (if a DRAM was being accessed), the RAS# signal transitions back to high. The CAS# signal also transitions back to high after the column address has been strobed into AMMI flash chip 900.

Referring again to Figure 9, the mode value stored within mode register 910 is provided to control logic 920. One output of control logic 920 is an address multiplexer select (AMS) signal. Signal AMS controls address multiplexer (mux) 930. Address mux 930, in turn, steers the ten bit address that is asserted on address pins A0 through A9 of AMMI flash 900 into the proper portion of address latch 940.

Address latch 940 is divided into a ten bit wide high portion 942 and a ten bit wide low portion 944 for storing, respectively, the ten high and low order bits of flash address 950. Demultiplexed address 950 is a twenty bit address used to address flash array 960. Another output from control logic 920 is a multi-bit address latch control (ALC) signal that controls the latching of addresses within the high portion 942 and the low portion 944 of address latch 940.

The mode value stored within mode register 910 is also provided to control bypass multiplexer (MUX) 955. The inputs to bypass multiplexer 955 are demultiplexed address 950 from address latch 940 and the twenty bit external flash address asserted on address pins A₀ through A₁₉ of the flash chip 900 when flash chip 900 is operating as a standard asynchronous flash chip. The output from bypass multiplexer 900 is flash address 957 that is provided to flash array 960.

Thus, when the AMMI flash chip 900 is operating as a standard asynchronous flash chip, a twenty bit flash address will be asserted on address pins A₀ through A₁₉ of the flash chip 900. Signal 915 from mode register 910 will set bypass multiplexer 955 to steer the twenty bit flash address asserted on address pins A₀ through A₁₉ of the flash chip 900 to flash array 960 as flash address 957.

Alternately, when AMMI chip 900 is emulating a DRAM chip, address pins A₀ through A₉ of flash chip 900 will be multiplexed. First, a ten bit row address will be placed on address pins A₀ through A₉ of flash chip 900 and the RAS# signal will indicate that the row address is ready to be strobed into the flash chip. Then, a ten bit column address will be placed on address pins A₀ through A₉ of flash chip 900 and the CAS# signal will indicate that the column address is ready to be strobed into the flash chip.

Note that in the DRAM emulation mode, no address signals will be placed on address pins A₁₀ through A₁₉ of the flash chip. Therefore, when the row address is to be strobed into AMMI flash device 900, control logic 920 will set address mux 920 to steer the ten bit row address that is asserted on address pins A₀ through A₉ of the

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flash chip into the low portion 944 of address latch 940. Signal ALC will then cause the ten bit row address that is asserted on address pins A₀ through A₉ of flash chip 900 to be latched by low portion 944 of address latch 940.

Later, when the CAS# signal indicates that the column address is to be strobed into AMMI flash device 900, control logic 920 will set address mux 920 to steer the ten bit column address that is asserted on address pins A₀ through A₉ of the flash chip into the high portion 942 of address latch 940. Signal ALC will then cause the ten bit column address that is asserted on address pins A₀ through A₉ of the flash chip to be latched by high portion 942 of address latch 940. In this way, the ten bit row address will be combined with the ten bit column address internally within AMMI flash memory 900 to form a twenty bit demultiplexed flash address 950 that is stored in latch 940 and can be used to address the flash array 960. Signal 915 from mode register 910 will set bypass multiplexer 955 to steer the twenty bit demultiplexed flash address 950 stored in address latch 940 to flash array 960 as flash address 957.

Regardless of whether the AMMI chip 900 is operating as a standard flash chip or is emulating a DRAM chip, once a complete twenty bit flash address 950 has been provided to flash array 960, the contents stored within flash array 960 at address 950 are then retrieved from flash array 960 and sent over bus 970 to data latch 980. Data latch control (DLC) signal from control logic 920 controls data latch 980. At the proper time, data latch 980 drives the data that has been read from flash array 960 out onto the data bus 990 (and data pins of AMMI flash 900). Then, at the completion of the read cycle, control logic 920 causes data latch 980 to tri-state. Other DRAM read accesses such as fast page mode are handled similarly.

In an alternate embodiment, AMMI flash chip 900 is a dedicated DRAM emulation device and cannot function as a standard flash chip. In this alternate embodiment, mode register 910 and address pins A₁₀ through A₁₉ of the flash chip are not required. This

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embodiment enables a lower pin count interface to AMMI chip 900 and is especially attractive to space constrained design situations.

In addition, control logic 920 contains logic that is used to interpret DRAM cycles that are not needed for flash 900 when it is emulating a DRAM. As mentioned before, unlike DRAM, flash memory does not need to have an extensive precharge or the various refresh cycles commonly required by DRAM. Figure 11 illustrates a typical refresh cycle for an AMMI flash chip that is emulating a DRAM. Because an AMMI flash chip that is emulating a DRAM will behave as would a DRAM chip, Figure 11 illustrates a typical refresh cycle for a DRAM.

Note that the refresh cycle of Figure 11 begins as does the read cycle of Figure 10. Thus, at the beginning of the refresh cycle, the column address strobe signal (CAS#) is brought high and a row address is placed on address pins A₀ through A₉ of AMMI flash chip 900 while the row address strobe signal (RAS#) is brought low.

Unlike the read cycle of Figure 10, however, during the refresh cycle of Figure 11, after the row address (of the row to be refreshed) has been strobed into AMMI flash memory chip 900, no column address is strobed into the flash chip. Instead, the RAS# signal is held low for a period sufficiently long to permit the row specified to be refreshed (if a DRAM was being refreshed) and the CAS# signal is held high throughout this period. During the refresh cycle, the data pins of AMMI flash chip 900 remain open (tri-stated).

Referring again to Figure 9, when in the DRAM emulation mode, control logic 920 will identify a refresh cycle pattern and hold data latch 980 tri-stated throughout the refresh cycle. Other standard refresh cycles (e.g. CAS before RAS (CBR) refresh, self-refresh and hidden refresh cycles) are well known in the art and handled similarly. In this manner, a glueless interface from the AMMI flash chip 900 to a DRAM controller is achieved.

Alternately, some of the inefficiency associated with emulation of a DRAM can be eliminated while the benefits of DRAM emulation are retained by controlling access to AMMI flash chip 900 using a

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DRAM controller that has been modified to compensate for the fact that some, if not all, of the "DRAM" that it is controlling is actually AMMI flash 900 that is emulating DRAM. In such a case, the multiplexing of row and column addresses to the AMMI flash chip 900 would be retained. The DRAM controller would be modified, however, so that the DRAM controller will not force AMMI flash chip 900 to wait for a lengthy (and unnecessary) row precharge period to complete. The DRAM controller would also be modified to eliminate refresh cycles for the AMMI flash memory 900. That way the DRAM controller will not force AMMI flash chip 900 to be tied up during lengthy (and unnecessary) refresh cycles. Thus, in the case of a programmable DRAM controller, the DRAM controller can be programmed to generate DRAM control signals that have been optimized to access an AMMI flash chip 900. An example of a suitable programmable DRAM controller is described in U. S. Patent Number 4, 785,428, Programmable Memory Array Control Signals, Bajwa et al., issued November 15, 1988.

Some programmable DRAM controllers permit more than one type of DRAM to be controlled simultaneously by the same controller even when each DRAM type being controlled has different timing cycles than the others. This is accomplished by having more than one bank of DRAM and permitting the controller to have different access cycle timings programmed for each bank. The DRAM are then grouped so that all of the DRAM in a particular bank are of the same type.

Thus, in an embodiment where a programmable controller can be programmed on a bank-by-bank basis, the AMMI flash memory 900 is placed in one bank and standard DRAM is placed in another bank. Then, the programmable DRAM controller can be programmed to control the DRAM in the DRAM bank using standard DRAM timing cycles. The programmable controller can also be programmed to control the bank of AMMI flash memory 900 that is emulating DRAM in a way that has been optimized to meet the needs of AMMI flash memory 900.

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Alternately, rather than using a programmable controller, a special non-programmable DRAM controller can be provided that has been optimized for access to AMMI flash memory 900. This special non-programmable controller could either control AMMI flash memory 900 exclusively, or could also control one or more banks of conventional DRAM.

In yet another alternate embodiment, a DRAM controller that has been designed to interface to a high speed synchronous bus (i.e. a local bus) can be used to control access to AMMI flash memory 900.

Programming an AMMI flash chip 900 that is emulating DRAM is performed in a fashion similar to writing to DRAM. As is the case with a read operation, the control logic 920 interprets the commands on the control pins of the flash chip and issues them in accordance to the protocol required by asynchronous main memory flash device 900.

Because programming (and erasing) flash memory takes longer than writing to DRAM, approaches similar to those mentioned earlier can be used to minimize the impact of the time required to program an AMMI flash device 900. For example, flash file system software can insert wait states whenever a write to AMMI flash 900 is required. The number of writes to the AMMI flash 900 memory can be significantly reduced by using the AMMI flash device 900 to store ROM-able programs and data that are infrequently changed and by supplementing the AMMI flash memory 900 with a memory technology that can be written to quickly. Alternately, a write cache can be provided to store information that is to be programmed into the AMMI flash 900. The programming of the AMMI flash 900 memory can then be performed in the background during what would otherwise be idle periods.

Figure 12 illustrates a scheme for creating a DRAM-compatible SIMM using one or more AMMI flash memory chips. This scheme allows AMMI SIMM 1200 containing an AMMI flash memory chip 900 to plug directly into a SIMM socket intended for DRAM. Thus, a standard DRAM controller can control AMMI SIMM 1200. The

AMMI SIMM enables the system to access data and direct-execute code stored in the flash memory 900 as if it were stored in DRAM while requiring no system hardware modification.

Figure 12 illustrates an AMMI SIMM 1200 that is populated with a single AMMI flash chip 900. In an alternate embodiment, AMMI SIMM 1200 is populated with two or more AMMI flash memory chips 900.

An advantage of the AMMI flash chip 900 is that if a DRAM controller exists in the system, the hardware interface to the AMMI flash memory chip 900 is dramatically simplified by the ability of the AMMI chip to emulate a DRAM. AMMI SIMM 1200 further simplifies the addition of flash main memory into a computer system. This is because AMMI SIMM 1200 allows a DRAM controller to interface, not only to individual flash memory components, but also to an industry-standard SIMM containing one or more AMMI flash devices 900. All additional circuitry required for flash memory compatibility is included directly on the AMMI SIMM 1200.

When compared to a similar density DRAM chip, an AMMI flash memory chip 900 can have several extra pins. For example, the AMMI flash chip 900 may have a RY/BY# output pin to indicate when the chip is ready to be accessed and when the chip cannot be accessed because it is busy (for example, when the flash array is being programmed or when a flash block is being erased). Therefore, RY/BY# output pin of AMMI flash chip 900 on SIMM 1200 is simply not connected (i.e. is "floated"). Instead of using the RY/BY# pin to determine when AMMI flash chip 900 is ready or busy, system software is used to poll component status registers within chip 900 to determine the program/erase status of the chip.

When it is emulating a DRAM, AMMI flash chip 900 will multiplex row and addresses onto address pins A0 through A9 when they are strobed into chip 900. Therefore, address pins A10 through A19 are not used by AMMI flash chip 900 when it is part of AMMI SIMM 1100. Thus, address pins A10 through A19 of AMMI flash chip

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900 are not connected when AMMI flash 900 is populating AMMI SIMM 1200.

The RAS# and CAS# signals are both provided directly to AMMI flash chip 900. They are also provided to read/write logic 1210 along with the DRAM WRITE# signal. The WRITE# signal indicates whether a read operation or a write operation is to be performed. Logic 1210 of AMMI SIMM 1200 translates the RAS#, CAS# and WRITE# signals provided to AMMI SIMM 1200 into appropriately timed output enable (OE#) and write enable (WE#) signals for AMMI flash chip 900.

Although not all flash memory chips require an external programming voltage (Vpp) supply, in the embodiment of Figure 12, AMMI flash chip 900 requires an external Vpp (e.g. 12 Volts). Vpp is not provided to a SIMM as part of the standard SIMM interface. System operating voltage Vcc (e.g. 3.3 or 5 Volts), however, is provided to a SIMM as part of a standard SIMM interface. Therefore, in the embodiment of Figure 12, Vpp converter 1220 is provided as part of AMMI SIMM 1200 to take the Vcc system operating voltage and convert it to a Vpp for AMMI flash chip 900. Suitable Vpp voltage converters 1220 are well known in the art and are commercially available. These voltage converters 1220 typically operate as DC-to-DC converters or as voltage pumps.

The WP# input of flash chip 900 is tied to the flash memory Vcc input. Vcc is also provided directly to AMMI chip 900.

Finally, the PWD# input for AMMI flash memory 900 is coupled to an on-SIMM programming voltage monitoring device 1230 such as the Maxim MAX705, manufactured by Maxim Integrated Products, Sunnyvale, California. The Vpp monitor circuit 1230 will hold AMMI flash memory 900 in Deep Powerdown mode unless the supply voltage is within tolerance, thereby protecting AMMI flash memory 900 from unintended writing or erasure.

This scheme allows AMMI SIMM 1200 containing AMMI flash memory 900 to use a Vpp program/erase voltage that is not part of a standard SIMM interface and yet also to plug directly into a SIMM

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socket intended for DRAM. Of course, in an alternate embodiment wherein AMMI flash memory chip 900 does not require the external supply of Vpp, AMMI SIMM 1200 will not require Vpp converter 1220. Vpp monitor 1230 would then be used to monitor Vcc.

For an embodiment wherein the SIMM pinout permits the addition of generate purpose input/output signals, one can switch off Vpp when not programming or erasing. One can also control signals RP# to provide power management and WP# to control block lock and unlock.

Figure 13 is a block diagram that illustrates the main memory organization of a computer system that uses AMMI flash SIMMs as main memory. In computer system 1300, microprocessor 1310 includes on-board DRAM controller 1330. DRAM controller 1330, in turn, is coupled to DRAM SIMM 1360 and AMMI flash SIMM 1350 by main memory bus 1320. DRAM SIMM 1360 and AMMI flash SIMM 1350 together form part of the main memory for microprocessor 1310.

When microprocessor 1310 is to access code or data stored in DRAM SIMM 1360 or AMMI flash SIMM 1350, DRAM controller 1330 of microprocessor 1310 will place an access request on main memory bus 1320. The access request will be made to DRAM SIMM 1360 or AMMI flash SIMM 1350 according to the asynchronous protocol of main memory bus 1320. Because AMMI flash SIMM 1350 contains one or more AMMI flash devices that are emulating an asynchronous DRAM device, DRAM controller 1330 will be able to access AMMI flash SIMM 1350 using a DRAM access protocol (i.e. by strobing row and column addresses). DRAM controller 1330 will also generate DRAM refresh cycles and provide precharge delays for AMMI flash SIMM 1350 even though a flash device in AMMI flash SIMM 1350 would not need them.

Furthermore, in computer system 1300, microprocessor 1310 is coupled to DRAM controller 1335 by high speed synchronous local bus 1325. DRAM controller 1335, in turn, is coupled to DRAM SIMM 1365 and AMMI flash SIMM 1355 by main memory bus 1345. DRAM

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SIMM 1365 and AMMI flash SIMM 1355 together form part of the main memory for microprocessor 1310.

When microprocessor 1310 is to access code or data stored in DRAM SIMM 1365 or AMMI flash SIMM 1355, microprocessor 1310 will place an access request on local bus 1325. The access request will be made according to the synchronous protocol of local bus 1325. DRAM controller 1335 will then interpret the access request and then make the access request to DRAM SIMM 1365 or AMMI flash SIMM 1355 using the protocol of main memory bus 1345. Because AMMI flash SIMM 1355 contains one or more AMMI flash devices that are emulating an asynchronous DRAM device, DRAM controller 1330 will be able to access AMMI flash SIMM 1355 using a DRAM access protocol (i.e. by strobing row and column addresses). DRAM controller 1335 will also generate DRAM refresh cycles and provide a precharge delay for AMMI flash SIMM 1350 even though a flash device in AMMI flash SIMM 1355 would not need them.

Secondary cache 1375 is also coupled to local bus 1325. If microprocessor 1310 attempts to access data or instructions from DRAM SIMMs 1360 or 1365 or AMMI flash SIMMs 1350 or 1355 that are available in cache 1375, the access will be made to cache 1375 instead.

Note that alternate embodiments of computer system 1300 exist wherein computer system 1300 uses a subset of the main memory capability depicted in Figure 13. Therefore, in one embodiment, microprocessor 1300 will address all of its main memory through main memory bus 1320. Alternately, microprocessor 1300 will address all of its main memory through local bus 1325. In such a case, microprocessor 1310 need not include on-board DRAM controller 1330.

In yet another embodiment, secondary cache 1375 is not available in system 1300. Furthermore, alternate embodiments exist wherein all of the main memory controlled by DRAM controller 1330 or DRAM controller 1535, is comprised of AMMI flash SIMMs 1350 or 1355, respectively. Moreover, in yet another embodiment, rather than

being coupled directly to microprocessor 1310, local bus 1325 is coupled to main memory bus 1320 by a bus bridge.

Synchronous Main Memory Interface (SMMI)
for Flash Main Memory

As processors have become faster, asynchronous DRAM has become increasingly unable to supply data and instructions to these processors at a speed fast enough to meet the processing speed sustainable by the processors. One recent solution has been the introduction of synchronous DRAM. Synchronous DRAM uses an external clock signal to synchronize interleaving within a DRAM integrated circuit and thereby provide a faster burst access than could be provided asynchronously.

Figure 14 illustrates a block diagram of a flash memory integrated circuit that is able to emulate a synchronous DRAM integrated circuit chip. In the embodiment depicted in Figure 14, synchronous main memory interface (SMMI) flash memory unit 1400 combines the internal flash bank interlacing features of SFI flash 600 (Figure 6) with the row and column address multiplexing of AMMI flash memory 900 (Figure 9).

In the embodiment of SMMI flash 1400 depicted in Figure 14, SMMI flash 1400 functions in one of four modes as determined by the setting of mode register 1480. In the first mode, asynchronous flash mode, the flash memory 1400 is read as a standard flash memory. In this mode, the reading of the contents of a first address must be completed before a second address to be read can be specified.

In the second mode, synchronous flash mode, a clock signal CLK is provided to SMMI flash chip 1400 and a series of addresses belonging to a data burst are specified, one address per clock tick. Then, the contents stored at the addresses specified for the burst are output sequentially during subsequent clock ticks in the order in which the addresses were provided. Alternately, if a single address is provided to SMMI flash chip 1400 when it is in the synchronous mode, the subsequent addresses for the burst will be generated within

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SMMI flash chip 1400 and the data burst will then be provided as output from the flash chip.

In the third mode, asynchronous DRAM (dynamic random access memory) mode, SMMI flash memory 1400 emulates asynchronous DRAM. Thus, row and column addresses are strobed into flash memory 1400 using row (RAS#) and column (CAS#) address strobe signals. SMMI flash memory 1400 then converts the row and column addresses internally into a single address and provides as output the data stored at that single address.

Furthermore, although the SMMI flash memory 1400 does not need an extended precharge period or to be refreshed, when in the asynchronous DRAM mode, the flash memory 1400 responds to precharge periods and refresh cycles as would an asynchronous DRAM. Therefore, when in the asynchronous DRAM mode, SMMI flash memory 1400 can be controlled by a standard DRAM controller.

Finally, in the fourth mode, synchronous DRAM mode, the features of the second and third modes are combined to yield a flash memory that emulates a synchronous DRAM. Thus, addresses to be read as a data burst are specified by strobing row and column addresses into the flash memory using RAS# and CAS# signals. The data of the data burst is then provided sequentially as output from the flash memory on subsequent clock ticks.

In an alternate embodiment, SMMI flash 1400 only has one mode, synchronous DRAM emulation mode, or has synchronous DRAM mode and only one or two of the other modes.

In the four mode embodiment depicted in Figure 14, mode register 1480 has a multi-bit output mode enable signal ME that reflects the current mode of SMMI flash 1400. Signal ME is provided to bank select logic 1474, bus logic 1440, TTL stages 1472 and logic and ATD control block 1470. These circuits control the function of SMMI flash 1400 differently according to the current mode of SMMI flash device 1400.

Thus, bus logic 1440 will function as does bus logic 640 of Figure 6 when SMMI flash device 1400 is operating in synchronous flash

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mode and will be disabled in the other modes. Note that in an alternate embodiment wherein SMMI flash device 1400 is operating as a generic synchronous flash device, bus logic 1440 will not be part of chip 1400, but instead will be provided off chip.

When SMMI flash device 1400 is operating synchronously, in either the synchronous flash or synchronous DRAM emulation modes, TTL stages 1472, bank select logic 1474 and logic and ATD circuit 1470 will control the interlacing of accesses to flash banks A 1410 and B 1420. In an alternate embodiment, interlacing will be performed among more than two flash banks within SMMI flash device 1400 when it is placed in these synchronous modes.

Similarly, when SMMI flash device 1400 is emulating a DRAM, in either the asynchronous or synchronous DRAM emulation modes, TTL stages 1472, bank select logic 1474 and logic and ATD circuit 1470 will control the multiplexing of row and column addresses by address multiplexers 1492 and 1494 and the tri-stating of output buffer 1460.

A SMMI flash based synchronous SIMM can be formed using the techniques described above in connection with AMMI SIMM 1200 of Figure 12. This would be accomplished by populating AMMI SIMM 1200 with one or more SMMI flash devices 1400 (rather than AMMI flash devices 900) and by providing a clock signal to each SMMI flash device.

Figure 15 is a block diagram that illustrates the main memory organization of a computer system that uses SMMI flash SIMMs as main memory. In computer system 1500, microprocessor 1510 includes on-board synchronous DRAM controller 1530. Synchronous DRAM controller 1530, in turn, is coupled to synchronous DRAM SIMM 1560 and SMMI flash SIMM 1550 by synchronous main memory bus 1520. Synchronous DRAM SIMM 1560 and SMMI flash SIMM 1550 together form part of the main memory for microprocessor 1510.

When microprocessor 1510 is to access code or data stored in synchronous DRAM SIMM 1560 or SMMI flash SIMM 1550, synchronous DRAM controller 1530 of microprocessor 1510 will place

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an access request on synchronous main memory bus 1520. The access request will be made to synchronous DRAM SIMM 1560 or SMMI flash SIMM 1550 according to the synchronous protocol of synchronous main memory bus 1520. Because SMMI flash SIMM 1550 contains one or more SMMI flash devices that are emulating a synchronous DRAM device, synchronous DRAM controller 1530 will be able to access SMMI flash SIMM 1550 using a synchronous DRAM access protocol (i.e. by strobing row and column addresses in conjunction with providing a clock signal). DRAM controller 1530 will also generate DRAM refresh cycles and provide precharge delays for SMMI flash SIMM 1550 even though a flash device in SMMI flash SIMM 1550 would not need them.

Furthermore, in computer system 1500, microprocessor 1510 is coupled to synchronous DRAM controller 1535 by high speed synchronous local bus 1525. Synchronous DRAM controller 1535, in turn, is coupled to synchronous DRAM SIMM 1565 and SMMI flash SIMM 1555 by synchronous main memory bus 1545. Synchronous DRAM SIMM 1565 and SMMI flash SIMM 1555 together form part of the main memory for microprocessor 1510.

When microprocessor 1510 is to access code or data stored in synchronous DRAM SIMM 1565 or SMMI flash SIMM 1555, microprocessor 1510 will place an access request on local bus 1525. The access request will be made according to the synchronous protocol of local bus 1525. Synchronous DRAM controller 1535 will then interpret the access request and then make the access request to synchronous DRAM SIMM 1565 or SMMI flash SIMM 1555 using the synchronous protocol of synchronous main memory bus 1545. Because SMMI flash SIMM 1555 contains one or more SMMI flash devices that are emulating a synchronous DRAM device, synchronous DRAM controller 1530 will be able to access SMMI flash SIMM 1555 using a synchronous DRAM access protocol (i.e. by strobing row and column addresses and by providing a clock signal). Synchronous DRAM controller 1535 will also generate synchronous DRAM refresh cycles and provide a precharge delay for SMMI flash

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SIMM 1550 even though a flash device in SMMI flash SIMM 1555 would not need them.

Secondary cache 1575 is also coupled to local bus 1525. If microprocessor 1510 attempts to access data or instructions from synchronous DRAM SIMMs 1560 or 1565 or SMMI flash SIMMs 1550 or 1555 that are available in cache 1575, the access will be made to cache 1575 instead.

Note that alternate embodiments of computer system 1500 exist wherein computer system 1500 uses a subset of the main memory capability depicted in Figure 15. Therefore, in one embodiment, microprocessor 1500 will address all of its main memory through main memory bus 1520. Alternately, microprocessor 1500 will address all of its main memory through local bus 1525. In such a case, microprocessor 1510 need not include on-board synchronous DRAM controller 1530.

In yet another embodiment, secondary cache 1575 is not available in system 1500. Furthermore, alternate embodiments exist wherein all of the main memory controlled by synchronous DRAM controller 1530 or synchronous DRAM controller 1535, is comprised of SMMI flash SIMMs 1550 or 1555, respectively. Moreover, in yet another embodiment, rather than being coupled directly to microprocessor 1510, local bus 1525 is coupled to main memory bus 1520 by a bus bridge.

In the foregoing specification the invention has been described with reference to specific exemplary embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the invention as set forth in the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative rather than restrictive sense.

CLAIMS

What is claimed is:

1. A computer system comprising:
a microprocessor;
a high speed bus;
a nonvolatile main memory, the non-volatile main memory providing first data to the microprocessor through the high speed bus.
2. The computer system as set forth in claim 1, wherein the non-volatile main memory provides the first data to the microprocessor through the high speed bus with zero microprocessor wait states.
3. The computer system as set forth in claim 1, wherein the nonvolatile main memory is flash memory.
4. The computer system as set forth in claim 1, wherein the high speed bus is a synchronous bus.
5. The computer system as set forth in claim 1, wherein the high speed bus is an asynchronous bus.
6. The computer system as set forth in claim 1, wherein the high speed bus is a local bus.
7. The computer system as set forth in claim 1, wherein the high speed bus is a main memory bus.
8. The computer system as set forth in claim 1, wherein the first data includes program instructions to be executed by the microprocessor.
9. The computer system as set forth in claim 1, wherein the microprocessor has an internal cache, the internal cache storing data for use by the microprocessor.
10. The computer system as set forth in claim 1, wherein the microprocessor is a static microprocessor.
11. The computer system as set forth in claim 1, wherein the nonvolatile main memory can be placed into an energy saving deep powerdown state.

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12. The computer system as set forth in claim 11, wherein the microprocessor can order the nonvolatile memory to enter and to leave the deep powerdown state.

13. The computer system as set forth in claim 1, further comprising a volatile main memory, the volatile main memory providing second data to the microprocessor through the high speed bus.

14. The computer system as set forth in claim 13, wherein the second data includes program instructions to be executed by the microprocessor.

15. The computer system as set forth in claim 13, wherein the volatile main memory is battery backed.

16. The computer system as set forth in claim 13, wherein the volatile main memory is static random access memory.

17. The computer system as set forth in claim 13, wherein the volatile main memory is dynamic random access memory.

18. The computer system as set forth in claim 13, wherein the volatile main memory operates as a write cache for third data to be stored in the nonvolatile main memory.

19. The computer system as set forth in claim 13, wherein the volatile main memory operates as an external cache for the first data stored in the nonvolatile main memory.

20. The computer system as set forth in claim 1, wherein the computer system further comprises nonvolatile store memory coupled to the nonvolatile main memory.

21. The computer system as set forth in claim 1, wherein the computer system further comprises:

- a low speed bus;

- a bus bridge coupling the high and low speed buses; and

- a nonvolatile store memory coupled to the low speed bus.

22. The computer system as set forth in claim 1, wherein the first data has a first part and a second part, the nonvolatile main memory further comprising:

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a plurality of nonvolatile memory units, the plurality of nonvolatile memory units including first and second memory units, the first part of the first data being stored in the first nonvolatile memory unit and the second part of the first data being stored in the second nonvolatile memory unit;

an interlace control unit, the interlace control unit controlling access by the microprocessor to the plurality of nonvolatile memory units, if the first data is to be provided to the microprocessor, the interlace controller causing the first nonvolatile memory unit to provide the first part of the first data to the microprocessor and the interlace controller causing the second nonvolatile memory unit to begin retrieving the second part of the first data while the first nonvolatile memory unit is providing the first part of the first data to the microprocessor, the interlace controller then causing the second nonvolatile memory unit to provide the second part of the first data to the microprocessor after the first nonvolatile memory unit has provided the first part of the first data to the microprocessor.

23. The computer system as set forth in claim 22, further comprising a clock, the clock providing a clock signal to the interlace controller, the interlace controller causing the first nonvolatile memory unit to provide the first part of the first data to the microprocessor on a first pulse of the clock and the interlace controller causing the second nonvolatile memory unit to provide the second part of the first data to the microprocessor on a second pulse of the clock.

24. The computer system as set forth in claim 23, wherein the first part of the first data has an associated first address and the microprocessor requests that the first data be provided to the microprocessor by providing the first address to the interlace controller on a third pulse of the clock, the third pulse occurring before the first and second pulses of the clock.

25. The computer system as set forth in claim 24, wherein an address latch enable signal is provided to the interlace controller on

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the third pulse to indicate that the first address is being provided to the interlace controller.

26. The computer system as set forth in claim 24 wherein the second part of the first data has an associated second address and the interlace controller calculates the second address based upon the first address.

27. The computer system as set forth in claim 26, wherein the interlace controller calculates the second address by incrementing the first address by a predetermined amount.

28. The computer system as set forth in claim 27, wherein the predetermined amount is one.

29. The computer system as set forth in claim 26, wherein the interlace controller calculates the second address by decrementing the first address by a predetermined amount.

30. The computer system as set forth in claim 29, wherein the predetermined amount is one.

31. The computer system as set forth in claim 24, wherein the second part of the first data has an associated second address and a request that the second part of the first data be provided to the microprocessor is made by providing the second address to the interlace controller on a fourth pulse of the clock, the fourth pulse occurring after the third pulse of the clock.

32. The computer system as set forth in claim 31, wherein an address latch enable signal is provided to the interlace controller on the fourth pulse to indicate that the second address is being provided to the interlace controller.

33. The computer system as set forth in claim 22, wherein the first and second parts of the first data belong to a single cache line.

34. The computer system as set forth in claim 22, wherein the first nonvolatile memory unit is a first integrated circuit and the second nonvolatile memory units is a second integrated circuit.

35. The computer system as set forth in claim 34, wherein the interlace controller is a third integrated circuit.

36. The computer system as set forth in claim 22, wherein the first and second nonvolatile memory units are separate banks of a first integrated circuit.

37. The computer system as set forth in claim 36, wherein the interlace controller is a second integrated circuit.

38. The computer system as set forth in claim 36, wherein the interlace controller is part of the first integrated circuit.

39. The computer system as set forth in claim 22, wherein the first part of the first data has an associated first row address and first column address and wherein the microprocessor initiates access to the first part of the first data by providing the first row and column addresses to the interlace controller.

40. The computer system as set forth in claim 39, wherein the first row address is provided during a row address strobe pulse and the first column address is provided during a column address strobe pulse.

41. The computer system as set forth in claim 39, wherein the second part of the first data has an associated second row address and second column address and wherein the microprocessor initiates access to the second part of the first data by providing the second row and column addresses to the interlace controller.

42. The computer system as set forth in claim 41, wherein the second row address is provided during a row address strobe pulse and the second column address is provided during a column address strobe pulse.

43. The computer system as set forth in claim 22, wherein the first data has an associated row address and column address and wherein the microprocessor initiates access to the first data by providing the row and column addresses to the nonvolatile main memory.

44. The computer system as set forth in claim 43, wherein the row address is provided during a row address strobe pulse and the column address is provided during a column address strobe pulse.

45. A memory device comprising:

a plurality of nonvolatile memory banks including first and second banks;

an interlace controller, the interlace controller controlling access to the plurality of nonvolatile memory banks, the interlace controller causing the first bank to provide first data and causing the second bank to begin retrieving second data while the first bank is providing the first data, the interlace controller then causing the second bank to provide the second data after the first bank has provided the first data.

46. The memory device as set forth in claim 45, wherein a clock signal is provided to the interlace controller, the interlace controller causing the first bank to provide the first data on a first pulse of the clock and the interlace controller causing the second bank to provide the second data on a second pulse of the clock.

47. The memory device as set forth in claim 46, wherein the first data has an associated first address specified to the interlace controller on a third pulse of the clock, the third pulse occurring before the first and second pulses of the clock.

48. The memory device as set forth in claim 47, wherein an address latch enable signal is provided to the interlace controller on the third pulse to indicate that the first address is being provided to the interlace controller.

49. The memory device as set forth in claim 47 wherein the second data has an associated second address and the interlace controller calculates the second address based upon the first address.

50. The memory device as set forth in claim 49, wherein the interlace controller calculates the second address by incrementing the first address by a predetermined amount.

51. The memory device as set forth in claim 50, wherein the predetermined amount is one.

52. The memory device as set forth in claim 49, wherein the interlace controller calculates the second address by decrementing the first address by a predetermined amount.

53. The memory device as set forth in claim 52, wherein the predetermined amount is one.

54. The memory device as set forth in claim 47, wherein the second data has an associated second address specified to the interlace controller on a fourth pulse of the clock, the fourth pulse occurring after the third pulse of the clock.

55. The memory device as set forth in claim 54, wherein an address latch enable signal is provided to the interlace controller on the fourth pulse to indicate that the second address is being provided to the interlace controller.

56. The memory device as set forth in claim 45, wherein the first and second data belong to a single cache line.

57. The memory device as set forth in claim 45, wherein the interlace controller can enter a noninterlace mode, when in the noninterlace mode, the interlace controller will not permit a second access to the plurality of nonvolatile memory banks to begin until a first access to the plurality of nonvolatile memory banks has completed.

58. The memory device as set forth in claim 57, wherein the interlace controller can enter an address multiplexing mode, when in the address multiplexing mode, a particular portion of one of the plurality of nonvolatile memory banks is specified to the interlace controller by a row and column address pair.

59. The memory device as set forth in claim 58, wherein the row address is provided during a row address strobe pulse and the column address is provided during a column address strobe pulse.

60. The memory device as set forth in claim 58, wherein the interlace controller combines the row and column address pair to form a bank address, the interlace controller then uses the bank address to specify the particular portion of one of the plurality of nonvolatile memory banks to access.

61. The memory device as set forth in claim 58, wherein the first data has an associated first row address and first column address

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pair and wherein the interlace controller initiates access to the first data upon receiving the first row and column addresses.

62. The memory device as set forth in claim 61, wherein the first row address is received during a row address strobe pulse and the first column address is received during a column address strobe pulse.

63. The memory device as set forth in claim 61, wherein the interlace controller combines the first row and column address pair to form a first bank address, the interlace controller then uses the bank address to specify that the first data of the first bank is to be accessed.

64. The memory device as set forth in claim 61, wherein the second data has an associated second row address and second column address pair and wherein the interlace controller initiates access to the second data by upon receiving the second row and column addresses.

65. The memory device as set forth in claim 64, wherein the second row address is received during a row address strobe pulse and the second column address is received during a column address strobe pulse.

66. The memory device as set forth in claim 65, wherein the interlace controller combines the second row and column address pair to form a second bank address, the interlace controller then uses the second bank address to specify the second data of the second bank is to be accessed.

67. The memory device as set forth in claim 45, wherein the interlace controller can enter an address multiplexing mode, when in the address multiplexing mode, a particular portion of one of the plurality of nonvolatile memory banks is specified to the interlace controller by a row and column address pair.

68. The memory device as set forth in claim 67, wherein the row address is provided during a row address strobe pulse and the column address is provided during a column address strobe pulse.

69. The memory device as set forth in claim 68, wherein the interlace controller combines the row and column address pair to form a bank address, the interlace controller then uses the bank

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address to specify the particular portion of one of the plurality of nonvolatile memory banks to access.

70. The memory device as set forth in claim 45, wherein the first data is identified by a first address and wherein the memory device further comprises a first address latch, the interlace controller controlling the first address latch and causing the first address latch to latch the first address prior to the interlace controller causing the first bank to provide the first data.

71. The memory device as set forth in claim 70, wherein the second data is identified by a second address and wherein the memory device further comprises a second address latch, the interlace controller controlling the second address latch and causing the second address latch to latch the second address prior to the interlace controller causing the second bank to provide the second data.

72. The memory device as set forth in claim 45, wherein the first data is identified by a first address and wherein the interlace controller further comprises an address transition detector, when the address transition detector detects an address transition to an address of the first bank, the interlace controller causes the first bank to prepare to provide data.

73. The memory device as set forth in claim 72, wherein the second data is identified by a second address and wherein when the address transition detector detects an address transition to an address of the second bank, the interlace controller causes the second bank to prepare to provide data.

74. The memory device as set forth in claim 45, further comprising:

- an output buffer, the output buffer storing output data; and
- a bank selector, the bank selector selecting the first bank to provide to the output buffer the first data as the output data when the first bank is providing the first data and the bank selector selecting the second bank to provide to the output buffer the second data as the output data when the second bank is providing the second data.

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75. The memory device as set forth in claim 45, further comprising bus specific logic, the bus specific logic receiving bus specific input signals and providing bus specific output signals in response.

76. The memory device as set forth in claim 45, further comprising a parity generator, the parity generator generating a parity signal based upon the first data when the first data is provided.

77. The memory device as set forth in claim 45, wherein the parity generator generates a parity signal based upon the second data when the second data is provided.

78. The memory device as set forth in claim 45, wherein at least one of the plurality of nonvolatile memory banks is comprised of flash memory.

79. A memory device, comprising:

a nonvolatile memory bank;

an address demultiplexer, the address demultiplexer receiving a multiplexed row and column address pair and combining the row and column address to form a bank address, the bank address being used to access the memory bank.

80. The memory device as set forth in claim 79, wherein the nonvolatile memory bank is a flash memory bank.

81. The memory device as set forth in claim 79, wherein the nonvolatile memory bank is a flash memory and the memory device emulates a dynamic random access memory (DRAM) device.

82. The memory device as set forth in claim 79, wherein the address demultiplexer comprises:

an address latch, the address latch having a row address latch and a column address latch;

an address multiplexer, when the address multiplexer receives the row address, the address multiplexer provides the row address to the row address latch to be latched therein, when the address multiplexer receives the column address, the address multiplexer provides the column address to the column address latch to be latched therein;

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once the row and column addresses have been latched in the address latch, the address latch provides the row and column addresses to the memory bank as the bank address used to access the memory bank.

83. The memory device as set forth in claim 79, wherein the row address is provided during a row address strobe pulse and the column address is provided during a column address strobe pulse.

84. The memory device as set forth in claim 83, wherein the address demultiplexer comprises:

an address latch, the address latch having a row address latch and a column address latch;

an address multiplexer, during the row address strobe pulse, the address multiplexer receives the row address and provides the row address to the row address latch to be latched therein, and during the column address strobe pulse, the address multiplexer receives the column address and provides the column address to the column address latch to be latched therein;

once the row and column addresses have been latched in the address latch, the address latch provides the row and column addresses to the memory bank as the bank address used to access the memory bank.

85. The memory device as set forth in claim 79, wherein the address demultiplexer can enter a nonmultiplexed addressing mode, when in the nonmultiplexed addressing mode, the address demultiplexer receives a bank address and uses the bank address to access the memory bank.

86. The memory device as set forth in claim 79, wherein the nonvolatile memory bank does not require refreshing and wherein the address demultiplexer further comprises a refresh cycle detector to detect a refresh cycle.

87. The memory device as set forth in claim 86, wherein the memory device further comprises a tri-statable latch coupled to the memory bank and to the refresh cycle detector, the latch adapted to hold data for an access to the memory bank, and further wherein if

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the refresh cycle detector detects a refresh cycle, the refresh cycle detector causes the latch to tri-state during the refresh cycle.

88. A memory sub-system comprising:

at least two nonvolatile memory devices, each memory device for storing a corresponding portion of a page of data;

an interlace controller coupled to the nonvolatile memory devices and controlling an interlaced read of the page of data, if the interlace controller detects a read operation to the page, the interlace controller enables the memory devices and causes each memory device to read the corresponding portion of the page, the interlace controller then sequentially enabling each memory device to output the corresponding portion of the page.

89. The memory sub-system as set forth in claim 88 wherein each memory device is a flash memory device.

90. The memory sub-system as set forth in claim 88, wherein there are a plurality of pages of data including a first and second page and wherein the interlace controller further comprises a page transition detector to detect a transition from a read operation for the first page to a read operation for the second page and if the transition is detected, the page transition detector causes the interlace controller to control the reading of the second page from the memory devices.

91. The memory sub-system as set forth in claim 90, wherein the page transition detector detects every page transition from one page of the plurality of pages to another page of the plurality of pages and if the transition is detected, the page transition detector causes the interlace controller to control the reading from the memory devices of the page to which the transition was made.

92. The memory sub-system as set forth in claim 88, wherein the interlace controller receives a clock signal comprising a series of clock pulses and wherein the interlace controller sequentially enables each memory device to output the corresponding portion of the page during a different one of the pulses of the clock signal.

93. The memory sub-system as set forth in claim 92, wherein each of the portions of the page has a corresponding address, wherein

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the interlace controller receives a different address of the corresponding addresses during a corresponding different one of the clock cycles and wherein the interlace controller then causes the each of the portions of the page to be output in the order in which the addresses were received.

94. The memory sub-system as set forth in claim 93, wherein the interlace controller receives an address latch enable pulse when the different address of the corresponding addresses is to be received.

95. The memory sub-system as set forth in claim 93, wherein if the interlace controller does not receive addresses for every portion of the page, the interlace controller will only cause the portions of the page to be read that correspond to the addresses received.

96. A system comprising:

a high speed synchronous bus having a data burst access bus protocol;

a nonvolatile memory device having at least two banks of nonvolatile memory and a generic interlace controller to control interlaced accesses to the banks according to a generic data burst access memory protocol;

a glue logic device coupled between the high speed synchronous bus and the nonvolatile memory device, the glue logic receiving the burst access bus protocol from the high speed bus of a burst access to the nonvolatile memory, converting the burst access bus protocol from the high speed bus to the generic data burst access memory protocol, and then providing the converted generic data burst access memory protocol to the memory device.

97. The system as set forth in claim 96, wherein at least one of the banks of nonvolatile memory is flash memory.

98. The system as set forth in claim 96, wherein the data burst access bus protocol uses a linear addressing sequence.

99. The system as set forth in claim 96, wherein the data burst access bus protocol uses a non-linear addressing sequence.

100. The system as set forth in claim 96, wherein the glue logic device further comprising bus specific logic, the bus specific logic

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receiving bus specific input signals from the bus and providing bus specific output signals to the bus in response.

101. The system as set forth in claim 96, further comprising a parity generator, the parity generator generating a parity signal based upon the data burst being accessed.

102. A memory module comprising:

a nonvolatile memory;

a memory module board upon which the nonvolatile memory is mounted, the memory module board, if the memory module board provides the nonvolatile memory with a multiplexed row and column address pair indicating a memory location within the nonvolatile memory to be accessed, the nonvolatile memory accessing the memory location indicated.

103. The memory module set forth in claim 102, wherein the nonvolatile memory is comprised of at least one flash memory device.

104. The memory module as set forth in claim 102, wherein the nonvolatile memory emulates dynamic random access memory.

105. The memory module as set forth in claim 102, wherein the nonvolatile memory emulates asynchronous dynamic random access memory.

106. The memory module as set forth in claim 102, wherein the nonvolatile memory emulates synchronous dynamic random access memory.

107. The memory module as set forth in claim 102, further comprising a voltage converter coupled to the nonvolatile memory and to the memory module board, the voltage converter converting an operating voltage from the memory module board into a programming voltage and providing the programming voltage to the nonvolatile memory.

108. The memory module as set forth in claim 107, wherein the nonvolatile memory can be disabled and wherein the memory module further comprises a programming voltage monitor coupled to the nonvolatile memory and to the voltage converter, the voltage

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monitor monitoring the programming voltage from the voltage converter programming voltage and if the programming voltage provided to the nonvolatile memory is not within a predetermined voltage range, the voltage monitor disabling the nonvolatile memory.

109. The memory module as set forth in claim 108, wherein the nonvolatile memory is disabled by entering a deep powerdown mode.

110. The memory module as set forth in claim 102, wherein the memory module is a single in-line memory module.

111. A system comprising:

a nonvolatile memory module having a nonvolatile memory device that emulates a dynamic random access memory (DRAM) device; and

a DRAM controller, coupled to the nonvolatile memory module and controlling access to the nonvolatile memory module.

112. The system as set forth in claim 111, wherein the nonvolatile memory device is a flash memory device.

113. The system as set forth in claim 111, wherein the nonvolatile memory module comprises a plurality of nonvolatile memory devices.

114. The system as set forth in claim 111, wherein the nonvolatile memory device emulates an asynchronous DRAM device.

115. The system as set forth in claim 111, wherein the nonvolatile memory device emulates a synchronous DRAM device.

116. The system of claim 111, further comprising a volatile memory module having a volatile memory device, the volatile memory module coupled to, and controlled by, the DRAM controller.

117. The system as set forth in claim 111, further comprising a high speed bus, coupled to the DRAM controller and indicating to the DRAM controller the accesses to the nonvolatile memory to be controlled by the DRAM controller.

118. The system as set forth in claim 117, wherein the high speed bus is an asynchronous bus.

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119. The system as set forth in claim 117, wherein the high speed bus is a synchronous bus.

120. The system as set forth in claim 117, wherein the high speed bus is a main memory bus.

121. The system as set forth in claim 117, wherein the high speed bus is a local bus.

122. The system as set forth in claim 117, further comprising a memory cache coupled to the high speed local bus.

123. The system as set forth in claim 117, further comprising a microprocessor coupled to the high speed local bus.

124. A method of accessing a nonvolatile memory device, the method comprising the steps of:

providing a first address to the memory device, the first address specifying a first location within the nonvolatile memory device to be read;

receiving a series of output data from the memory device, the output data from the series of output data including first data stored at the first address.

125. The method as set forth in claim 124, further comprising the step of:

providing a clock signal comprising a series of output clock pulses to the memory device, each output clock pulse signifying to the memory device when a different one of the series of output data is to be received from the memory device.

126. The method as set forth in claim 125, further comprising the step of:

providing an address latch signal to the memory device when the first address is provided to the memory device to indicate to the memory device that the first address is being provided.

127. The method as set forth in claim 124, further comprising the steps of:

providing a clock signal to the memory device, the clock signal comprising a series of address input clock pulses followed by a series of output clock pulses, each input clock pulse of the series of input clock pulses being provided with a

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corresponding address specifying a location in the memory device to be read, each output clock pulse signifying to the memory device when a different one of the series of output data is to be received from the memory device.

128. The method as set forth in claim 127, further comprising the steps of:

providing an address latch signal to the memory device with each address input clock pulse of the series of address input clock pulses to indicate to the memory device that the address corresponding to the input clock pulse is being provided.

129. The method as set forth in claim 124, wherein the step of providing the first address to the memory device further comprises the steps of:

providing a row address to the memory device;

providing a column address to the memory device, the row and column address together indicating the first address to the memory device.

130. The method as set forth in claim 129, further comprising the step of:

providing a row address strobe to the memory device when providing the row address to the memory device, to indicate to the memory device that the row address is being provided.

131. The method as set forth in claim 130, further comprising the step of:

providing a column address strobe to the memory device when providing the column address to the memory device, to indicate to the memory device that the column address is being provided.

132. The method as set forth in claim 129, further comprising the step of:

prior to providing the row address strobe to the memory device, indicating to the memory device that the memory device is to enter a multiplexed address mode.

133. The method as set forth in claim 124, further comprising the step of:

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prior to providing the first address to the memory device, indicating to the memory device that the memory device is to enter a synchronous address mode.

134. A method of accessing a nonvolatile memory device, comprising the steps of:

providing a row address to the memory device;

providing a column address to the memory device;

receiving output data from the memory device, the output data being contents stored by the memory device at a location corresponding to the row and column address provided to the memory device.

135. The method as set forth in claim 134, further comprising the step of providing a row address strobe to the memory device when the row address is provided to the memory device to indicate that the row address is being provided.

136. The method as set forth in claim 134, further comprising the step of providing a column address strobe to the memory device when the column address is provided to the memory device to indicate that the column address is being provided.

137. The method as set forth in claim 134, further comprising the step of:

receiving a series of additional output data from the memory device, the additional output data from the series of output data being received after the output data corresponding to the row and column address provided to the device.

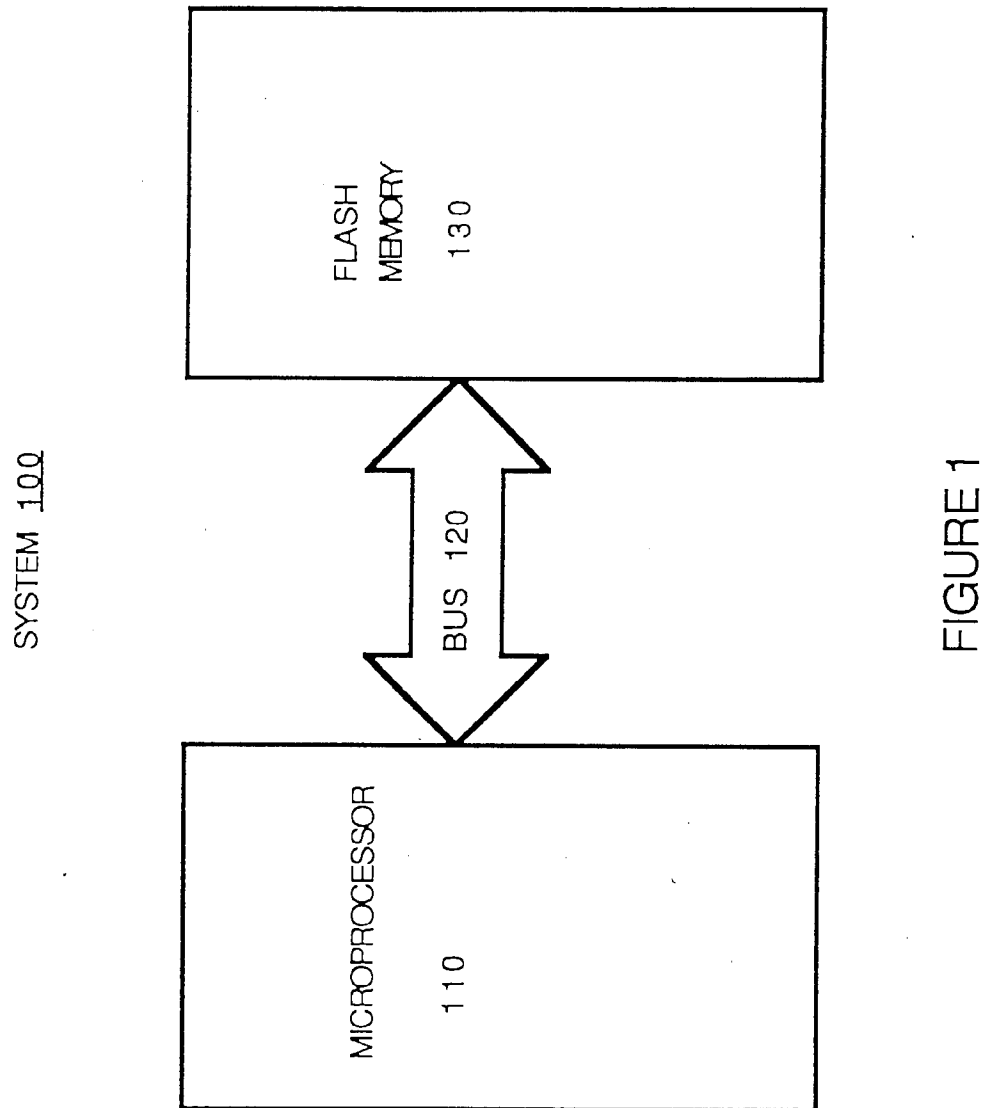
138. The method as set forth in claim 137, further comprising the step of:

providing a clock signal comprising a series of output clock pulses to the memory device, each output clock pulse signifying to the memory device when a different one of the series of output data is to be received from the memory device.

139. The method as set forth in claim 138, further comprising the step of:

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prior to providing the row address to the memory device, indicating to the memory device that the memory device is to enter a synchronous address mode.



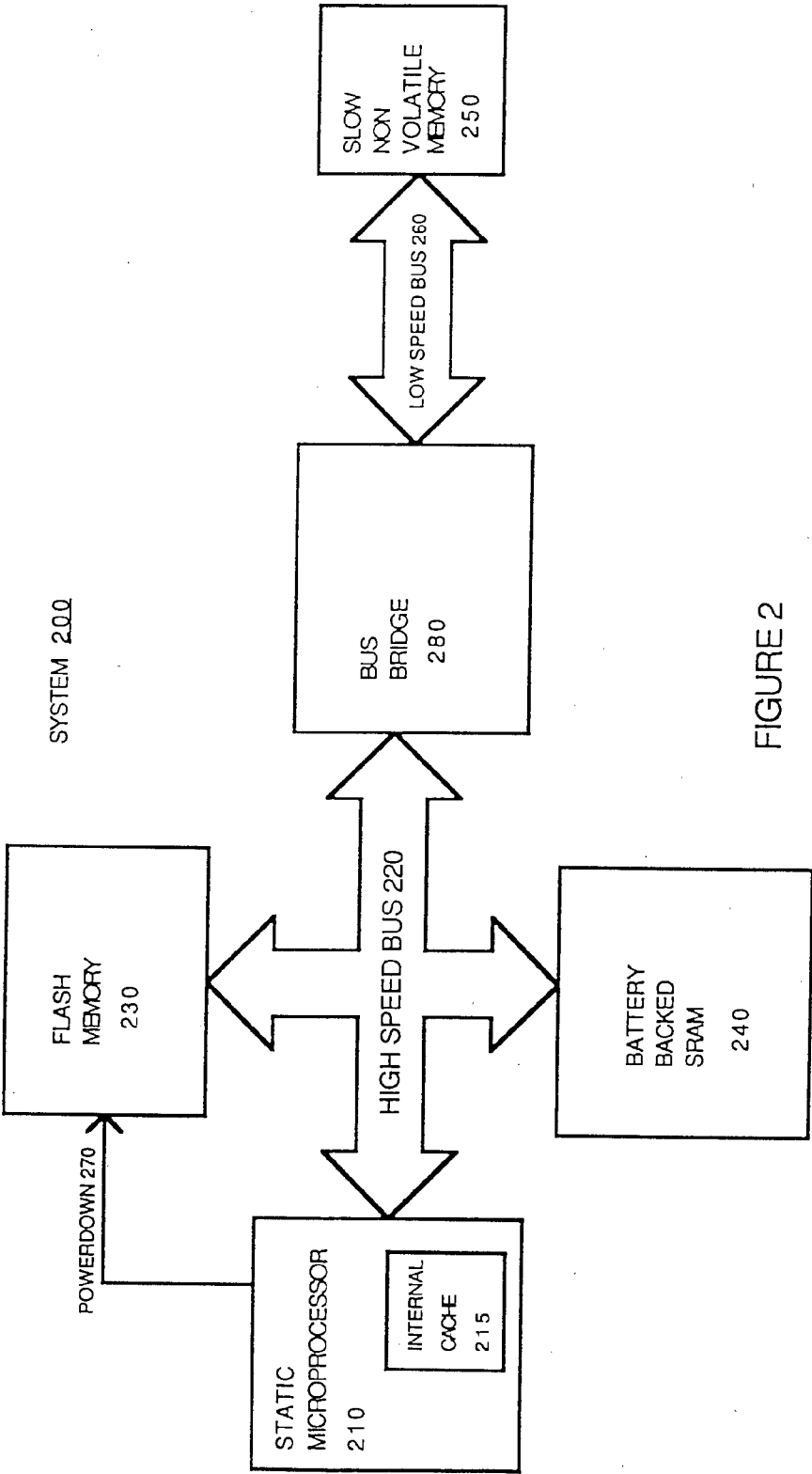
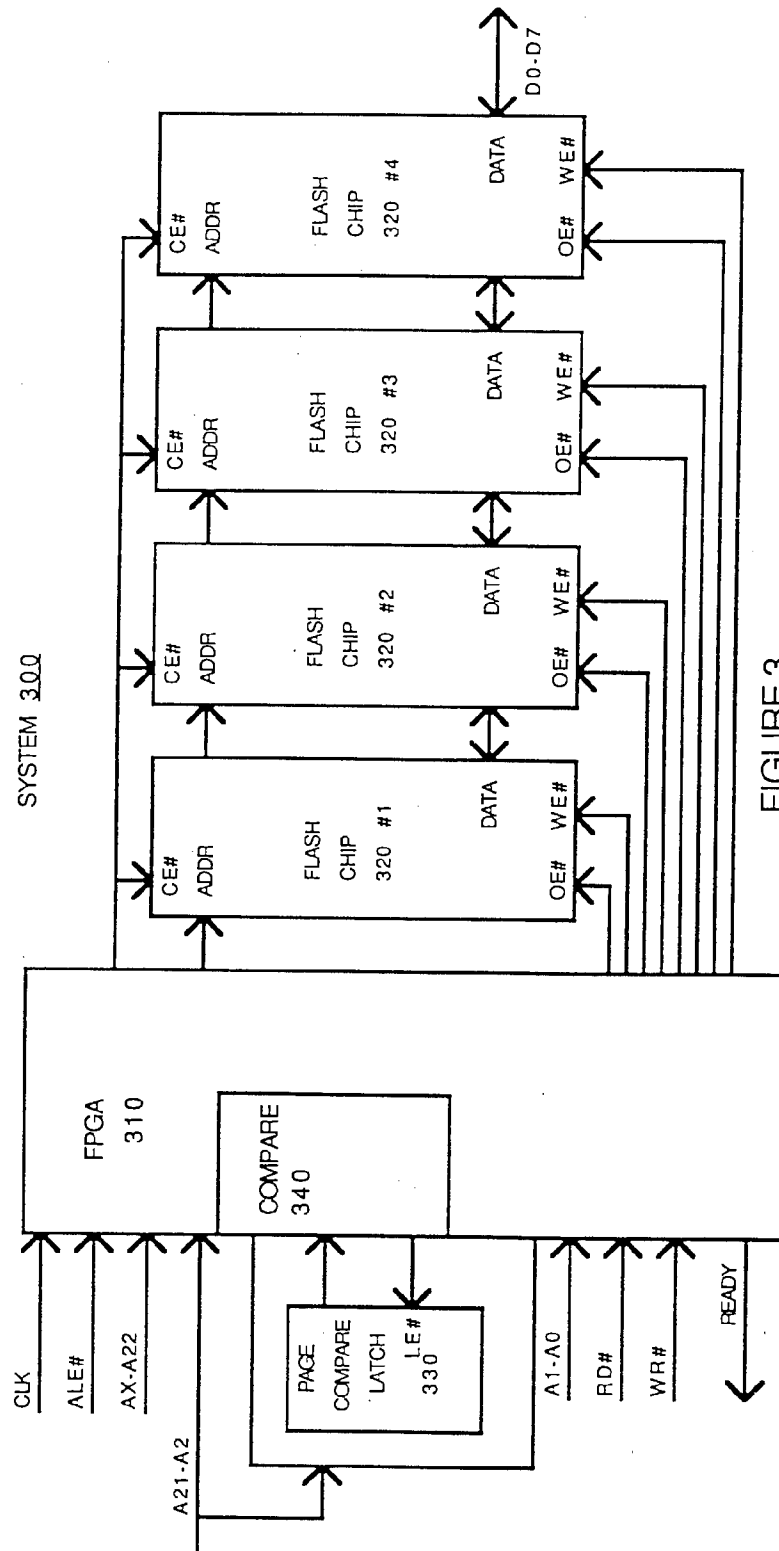


FIGURE 2



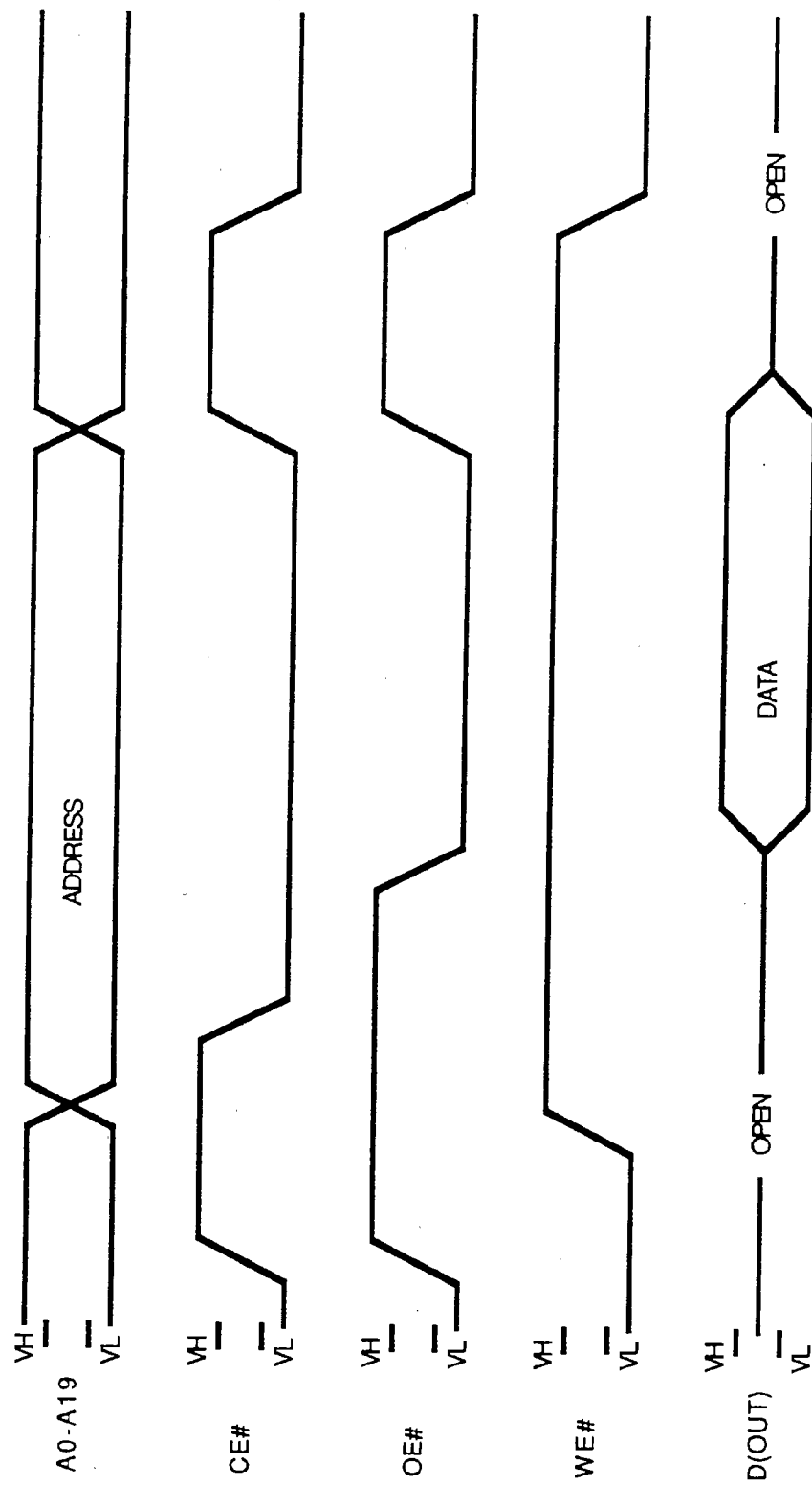


FIGURE 4

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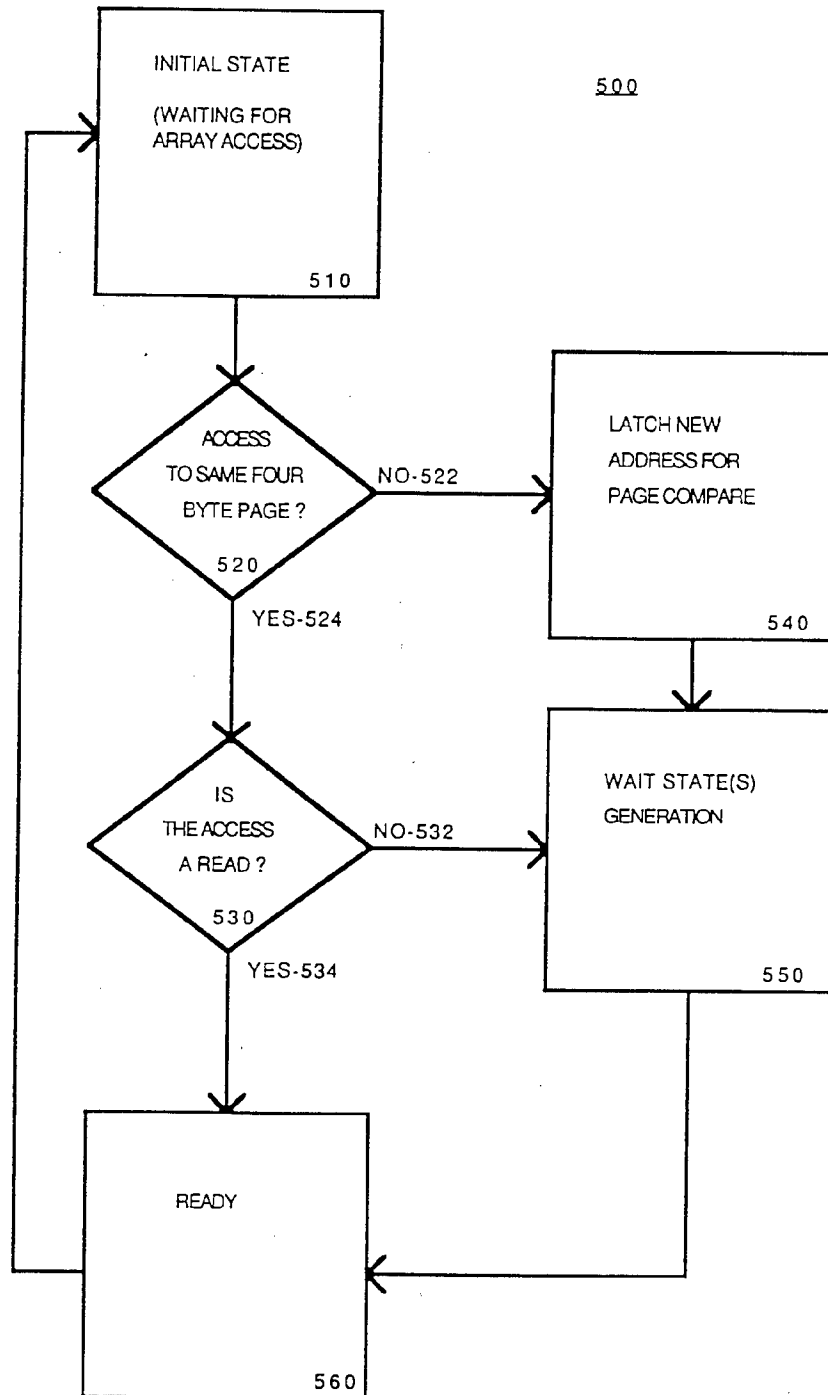


FIGURE 5

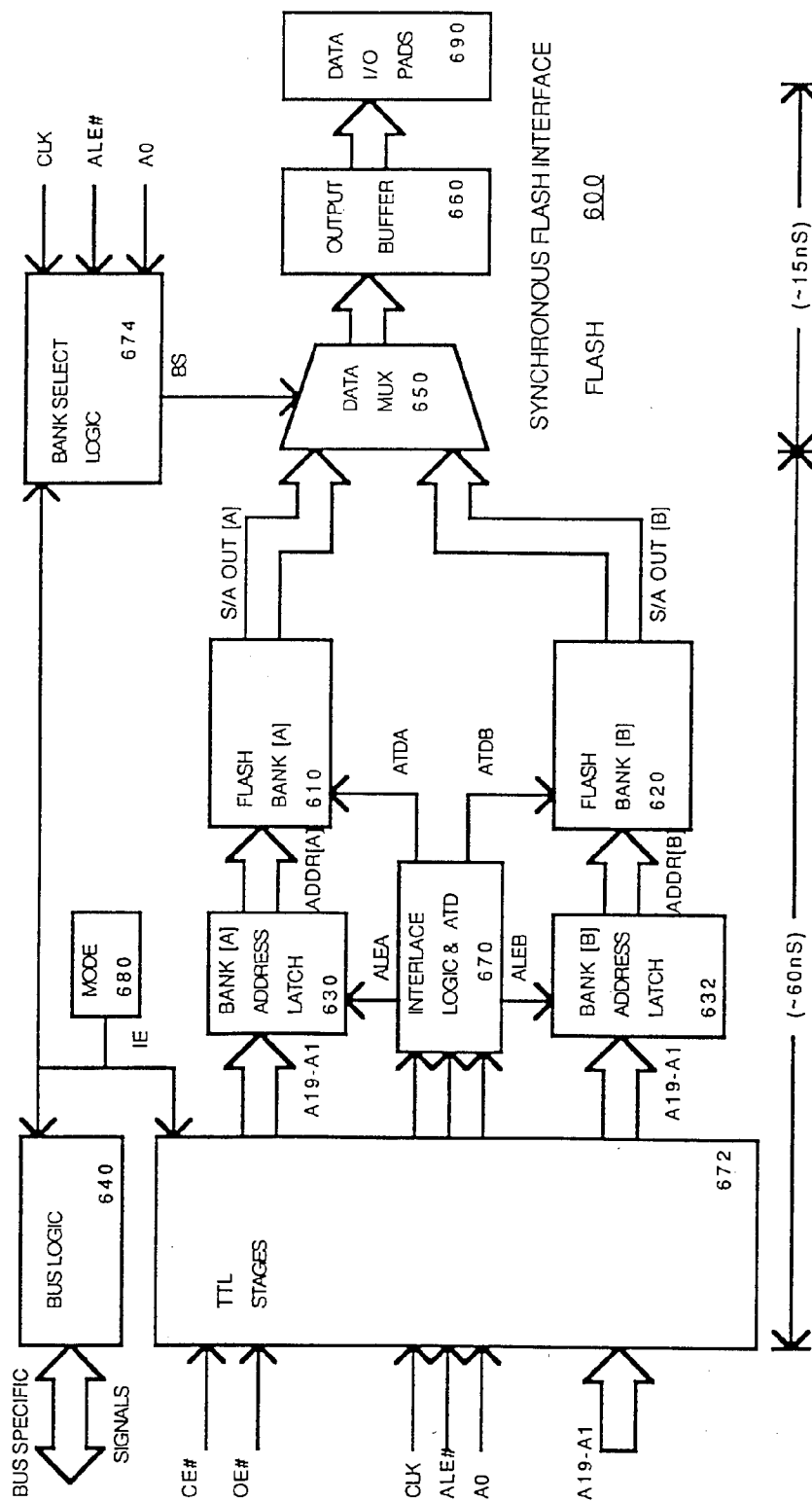


FIGURE 6

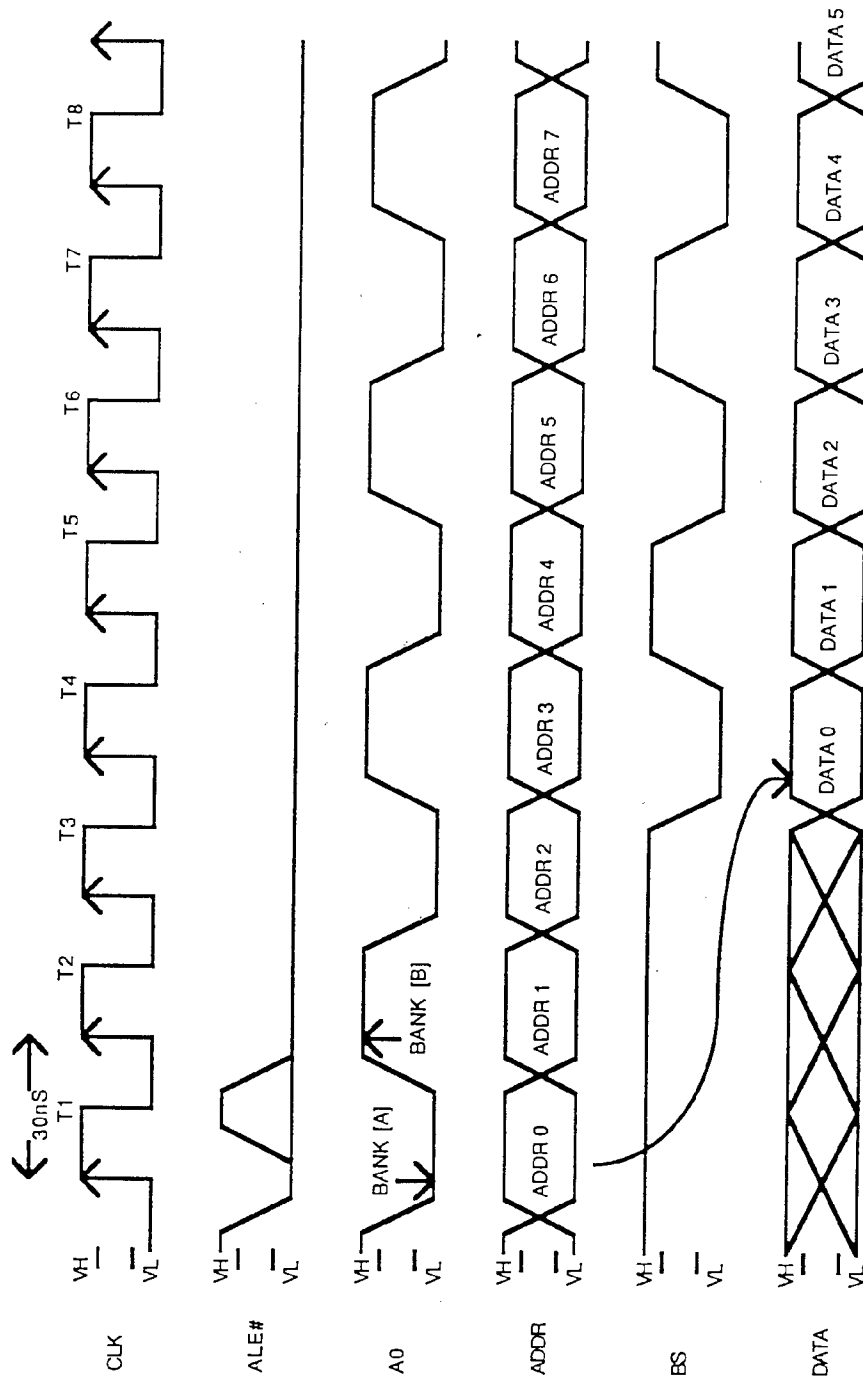


FIGURE 7

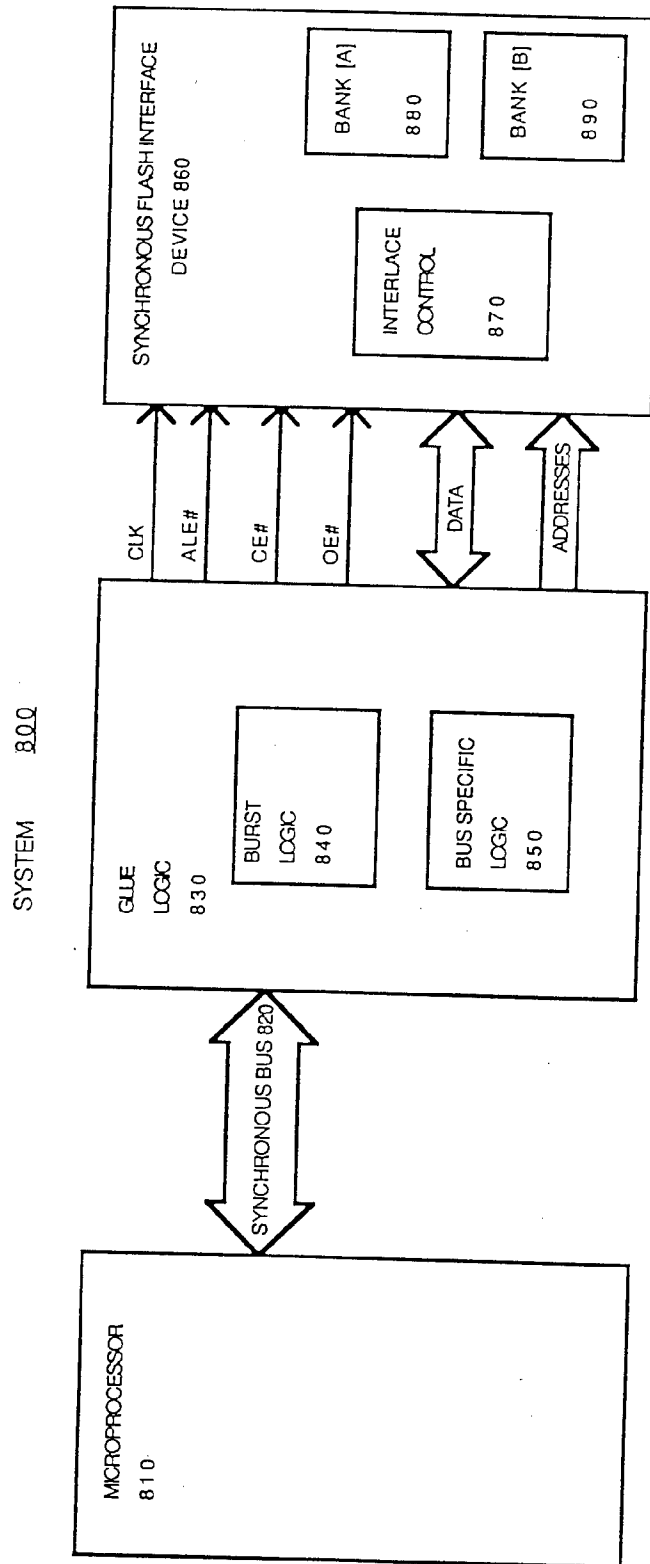


FIGURE 8

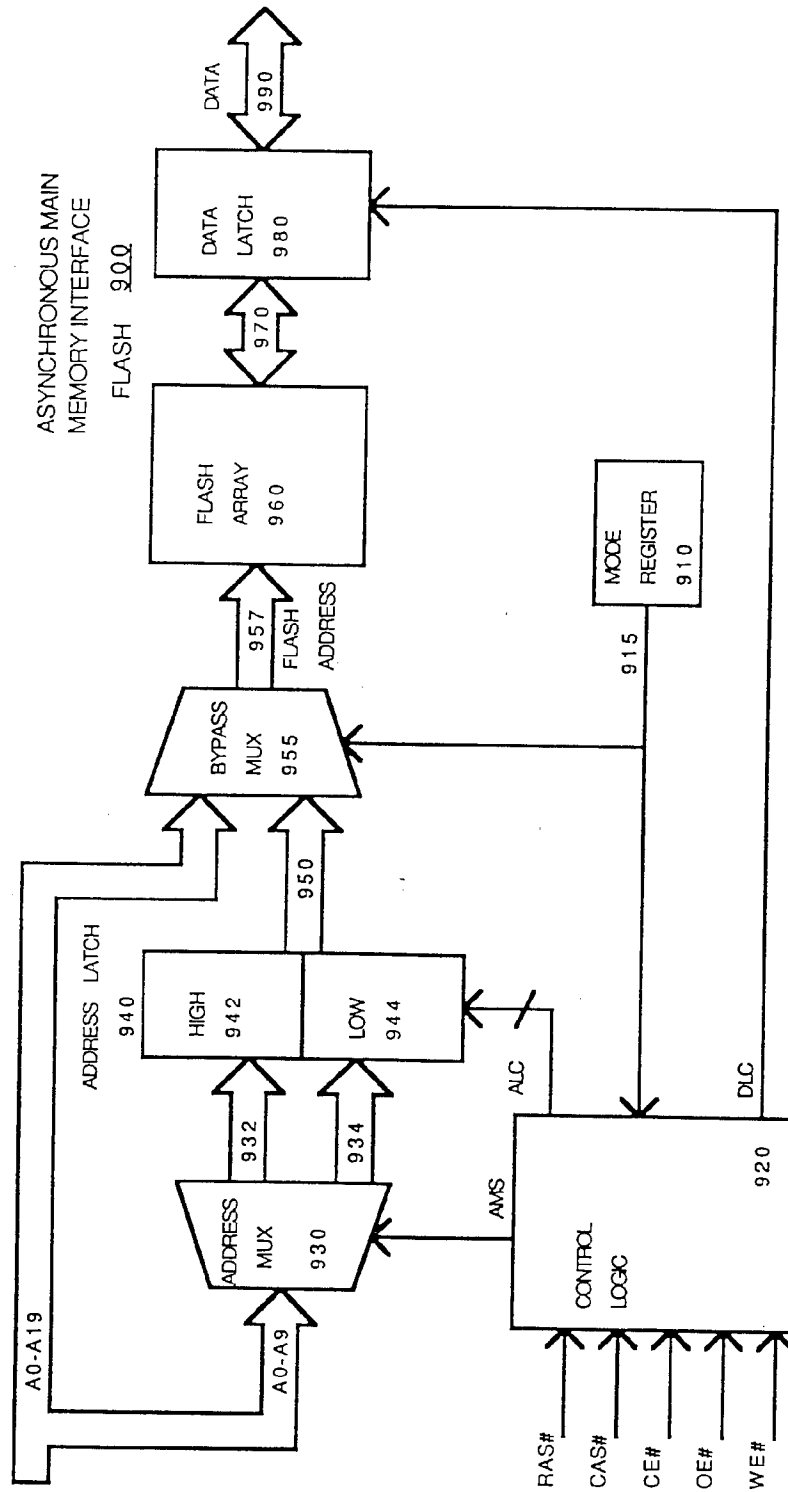


FIGURE 9

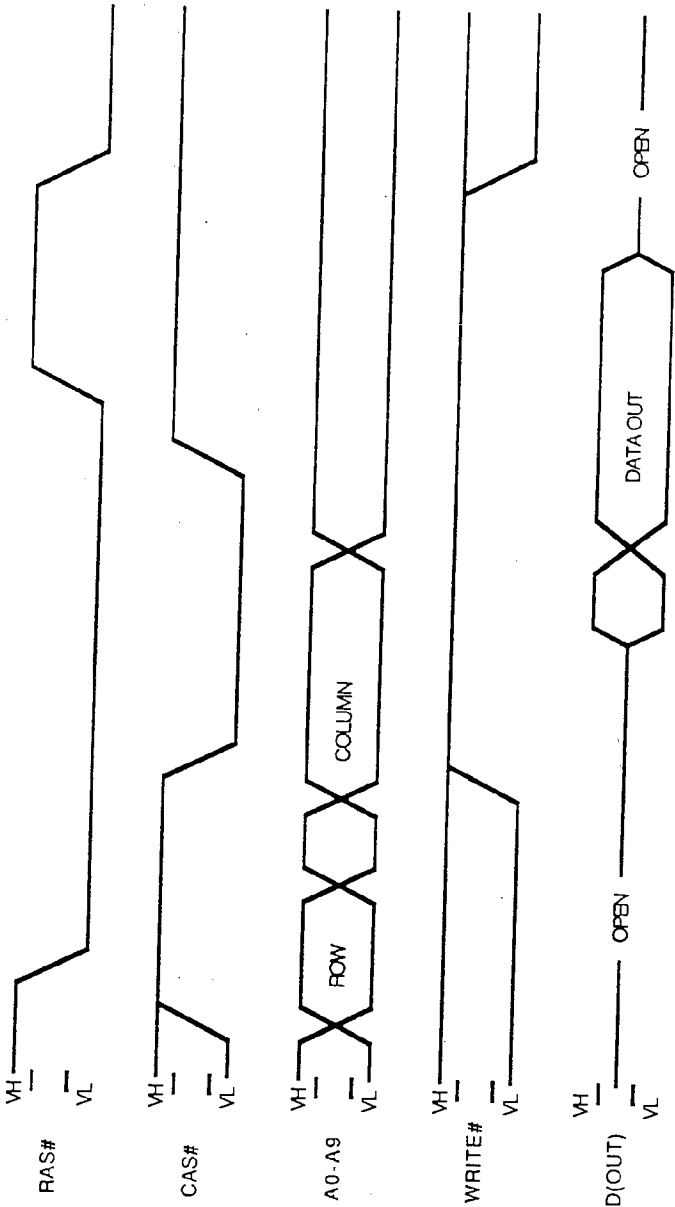


FIGURE 10

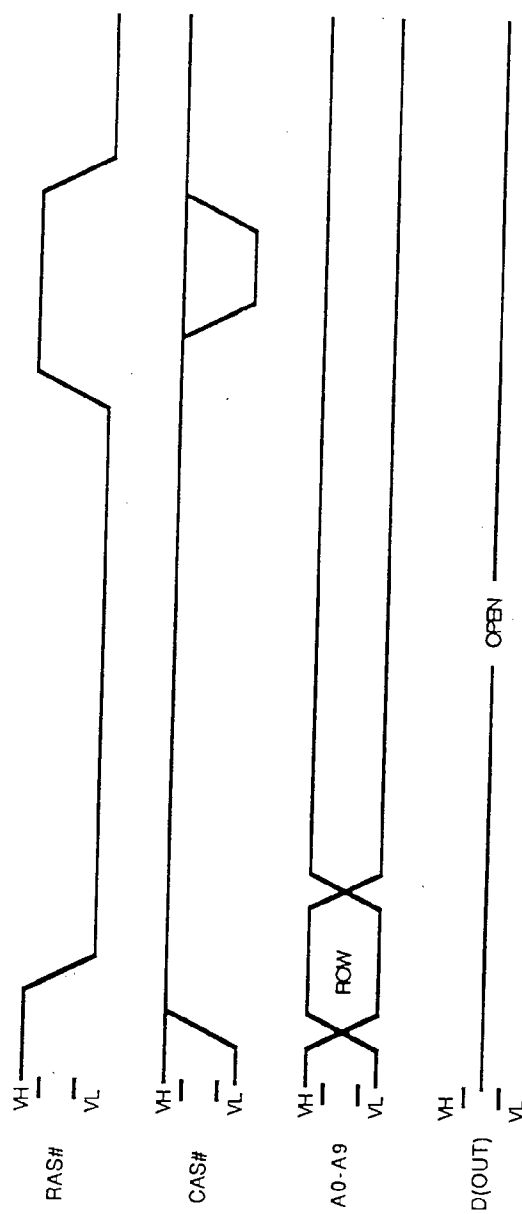
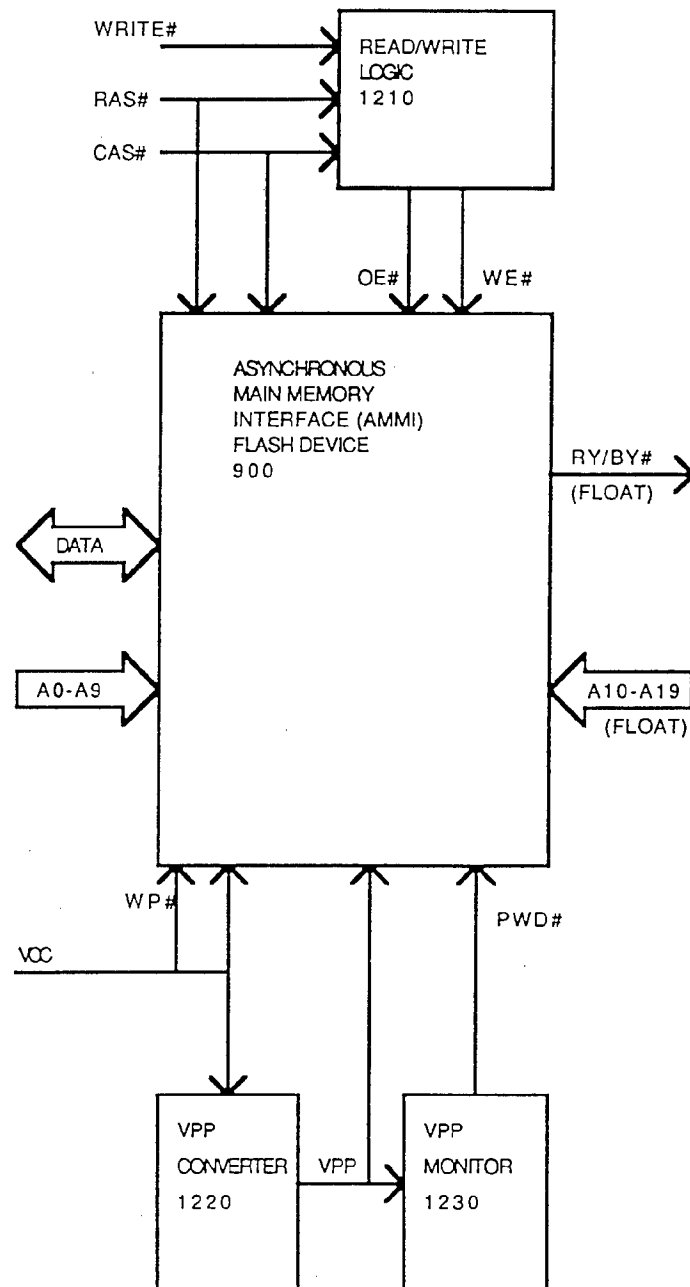


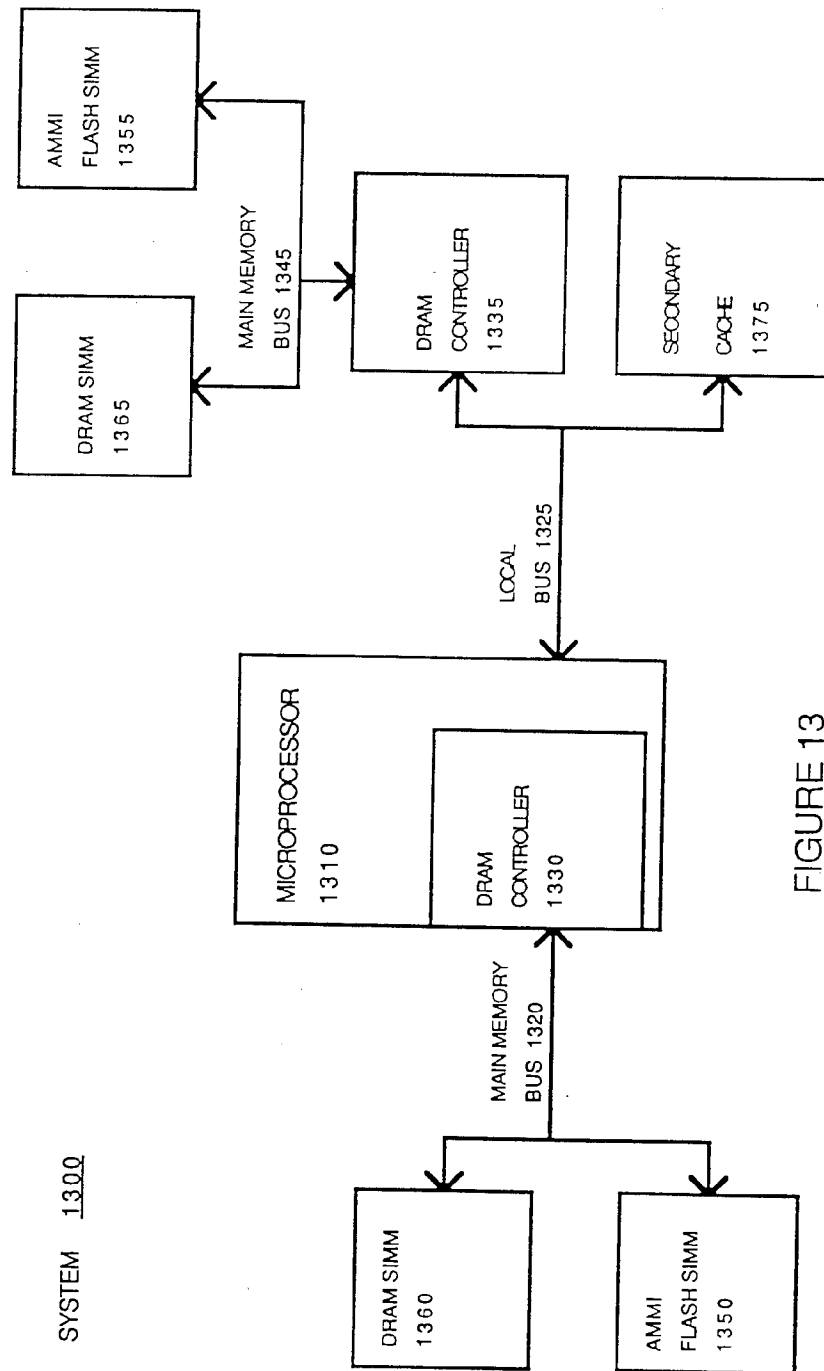
FIGURE 11



AMMI SINGLE IN-LINE
MEMORY MODULE (SIMM)
1200

FIGURE 12

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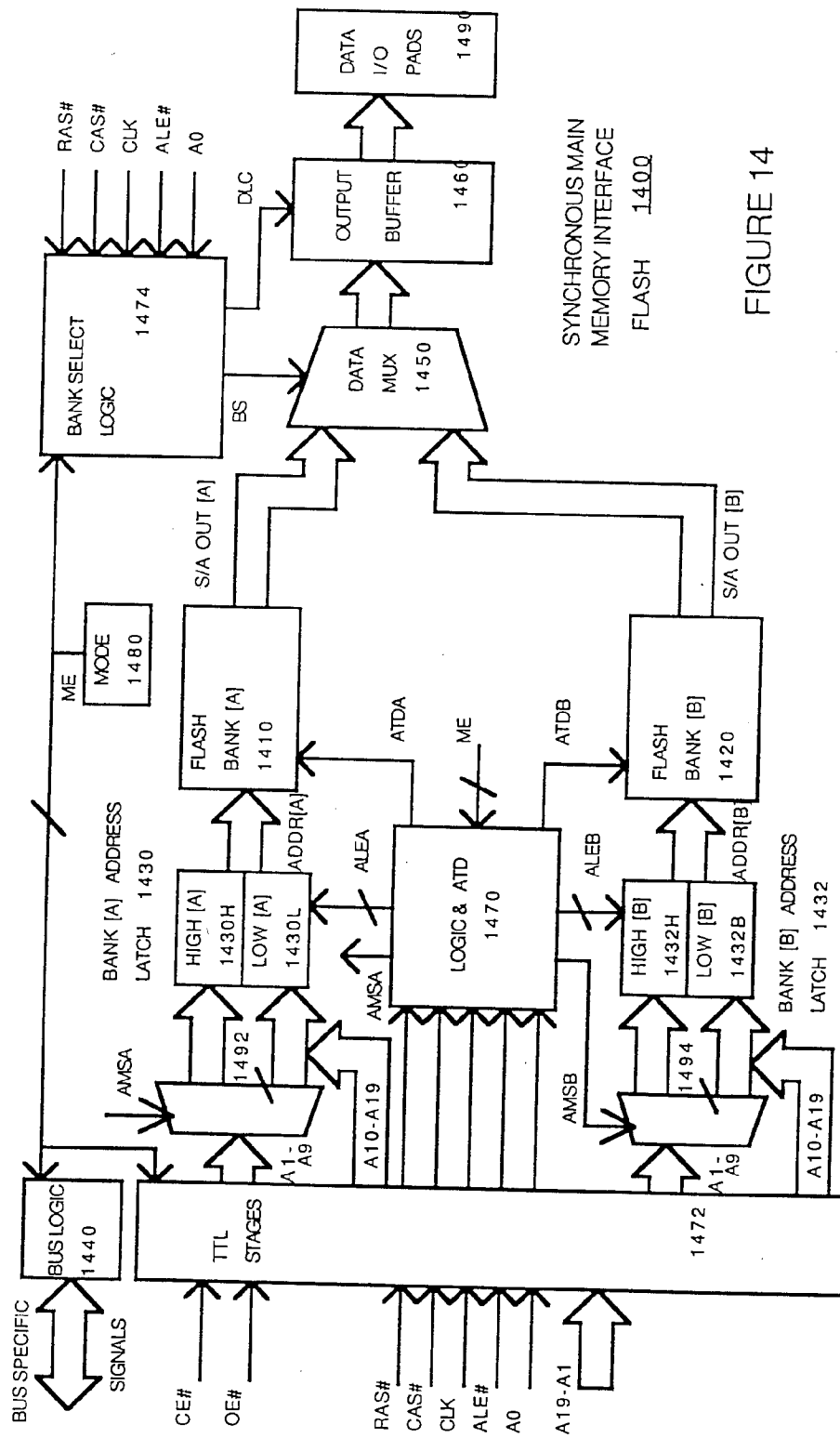


FIGURE 14

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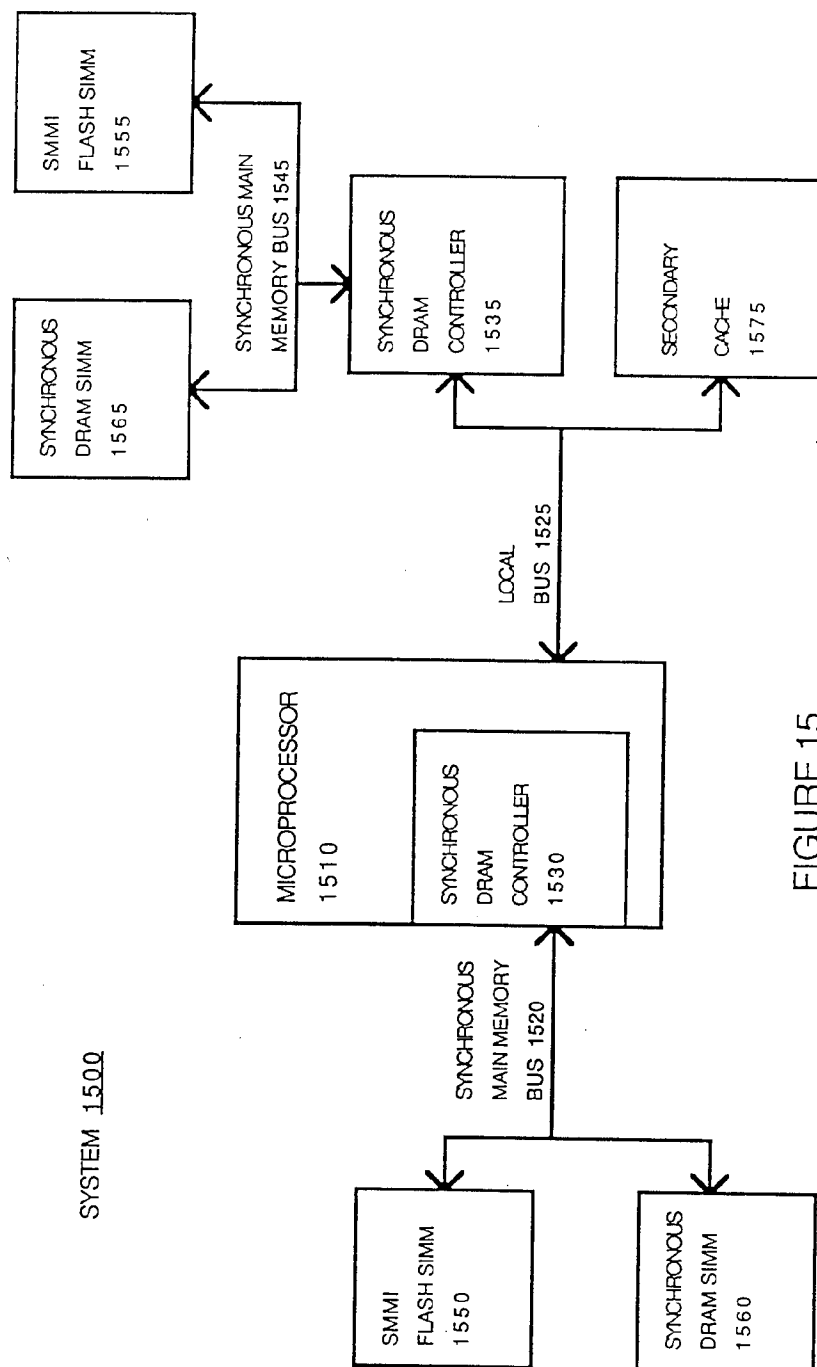


FIGURE 15

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添付公開書類

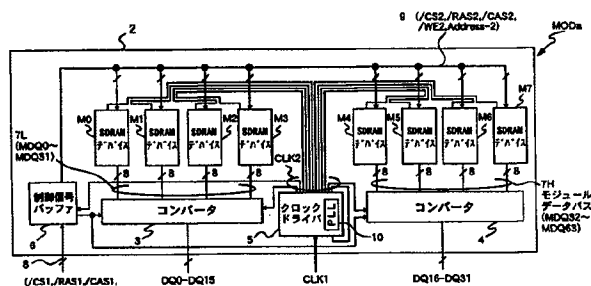
国際調査報告書

(54)Title: MEMORY MODULE AND DATA PROCESSING SYSTEM

(54)発明の名称 メモリモジュール及びデータ処理システム

(57) Abstract

A memory module enabling a high speed access. A memory module (MODa) has converters (3, 4) including register buffers (RBUFa0-RBUFa15) that are provided between data buses (7L, 7H) connected to a plurality of parallelly operated semiconductor memory devices (M0-M7) and external data input/output terminals (DQ0-DQ31). Because the bus connection between the inside and the outside of the memory module is buffered by the register buffers, the data buses on the mother board is separated from the data buses on the memory module. Thus, the data buses on the mother board no longer see the load of wiring after the buffer registers, so that the load that should be driven by the memory controller can be reduced, thus facilitating improvement in the data transfer frequency on the memory buses.



- M0 ... SDRAM device
- M1 ... SDRAM device
- M2 ... SDRAM device
- M3 ... SDRAM device
- M4 ... SDRAM device
- M5 ... SDRAM device
- M6 ... SDRAM device
- M7 ... SDRAM device
- 3 ... Converter
- 4 ... Converter
- 5 ... Clock driver
- 6 ... Control signal buffer
- 7H ... Module data bus (MDQ32-MDQ63)

(57)要約

本発明はアクセス動作の高速化を実現できるメモリモジュールを提供する。すなわち、並列動作される複数個の半導体記憶装置(M0～M7)が接続されたデータバス(7L, 7H)と外部データ入出力端子(DQ0～DQ31)との間にレジスタバッファ(RBUF a0～RBUF a15))を含むコンバータ(3、4)設けてメモリモジュール(M0Da)を構成する。メモリモジュール内部と外部とのバス接続は前記レジスタバッファでバッファリングされるから、マザーボード上のデータバスとメモリモジュール上のデータバスとは分離される。したがって、マザーボード上のデータバスからは前記バッファレジスタ以降の配線負荷が見えなくなり、メモリコントローラが駆動すべき負荷を低減でき、メモリバス上のデータ転送周波数を向上させることが容易になる。

PCTに基づいて公開される国際出願のパンフレット第一頁に掲載されたPCT加盟国を同定するために使用されるコード(参考情報)

AL	アルバニア	FI	フィンランド	LK	スリ・ランカ	SI	スロヴェニア
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BA	ボスニア・ヘルツェゴビナ	GE	グルジア	LV	ラトヴィア	TD	チャード
BB	バルバドス	GH	ガーナ	MC	モナコ	TG	トーゴ
BE	ベルギー	GM	ガンビア	MD	モルドヴァ	TJ	タジキスタン
BF	ブルキナ・ファソ	GN	ギニア	MG	マダガスカル	TM	トルクメニスタン
BG	ブルガリア	GW	ギニア・ビサウ	MK	マケドニア旧ユーゴスラヴィア共和国	TR	トルコ
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CG	コンゴ	IL	イスラエル	NE	ニジェール	VN	ヴェトナム
CH	スイス	IN	インド	NL	オランダ	YU	ユーゴスラビア
CI	コートジボワール	IS	アイスランド	NO	ノールウェー	ZW	ジンバブエ
CM	カメルーン	IT	イタリア	NZ	ニュー・ジーランド		
CN	中国	JP	日本	PL	ポーランド		
CU	キューバ	KE	ケニア	PT	ポルトガル		
CY	キプロス	KG	キルギスタン	RO	ルーマニア		
CZ	チェコ	KP	北朝鮮	RU	ロシア		
DE	ドイツ	KR	韓国	SD	スーダン		
DK	デンマーク	KZ	カザフスタン	SE	スウェーデン		
EE	エストニア	LC	セントルシア	SG	シンガポール		
ES	スペイン	LI	リヒテンシュタイン				

明 細 書

メモリモジュール及びデータ処理システム

5 技術分野

本発明は、メモリモジュール及びそれを用いたデータ処理システムに係り、殊に、メモリモジュールの高速インタフェース技術に関し、例えば S D R A M (Synchronous Dynamic Random Access Memory: シンクロナス・ダイナミック・ランダム・アクセス・メモリ) を用いた高速メモリモジュールに適用して有効な技術に関するものである。

背景技術

近年、マイクロプロセッサの高性能化にともなって、主記憶についても高速化の要求が高まってきている。特にマルチメディアパソコンなど大量の画像データの入出力を高速に行うシステムにおいて特に高速の主記憶に対する要求は高まる一方である。これに伴い、主記憶に用いられている D R A M (Dynamic Random Access Memory: ダイナミック・ランダム・アクセス・メモリ) に関しては、ファーストページモードから、より高速なデータ転送が可能な E D O (Extended Data-OUT) ページモード品への移行が進み、最近ではシステムクロックに同期して D R A M 内部をパイプライン動作させる事により 100MHz 程度のデータ転送率を実現した S D R A M も主記憶に使われ始めている。

一方、D R A M の応用分野は多岐にわたっており、ゲーム機などのように数個程度を使うものから、ワークステーションなどのように数 1 0 0 0 個使うシステムなど、様々なニーズに対応する必要が生じている。大容量を実装する場合にはメモリバスを複数設け、並列化する事により

システムとしてデータ転送速度を上げる事が可能である。しかし、その場合、メモリシステムを構成する増設単位は大きくなり、小容量のメモリしか必要としないシステムには適さないという問題があった。

5 小さな増設単位でも高いデータ転送速度を得るために、Rambus (ランバス) D R A M (以下 R D R A M と略す) と呼ばれる新しい高速 D R A M も提案され、一部小容量システムに使われている。これは、入出力回路とメモリバスを高速動作向けに新しくした D R A M である。すなわち、R D R A M は、コマンドによってアクセス制御されるプロトコル制御方式を採用するもので、予め決められた長さの packets に読み出しリクエストや書き込みリクエスト及びアドレス情報などを入れておき、それ
10 を R D R A M が解読してリード動作やライト動作を行う。R D R A M は、例えば第 20 図に示されるように、プロセッサボードなどのマザーボード上のメモリバス I O 0 ~ I O 1 5 に個々に搭載され、R D R A M 単位でその動作が選択される。前記 packets に入れられた制御情報とリード
15 又はライトのデータ情報は同一バス上に伝達される。このようなプロトコル制御方式による R D R A M は、500 M バイト / 秒のような高速でデータの入出力を行うことができる。

尚、R D R A M について記載された文献の例としては、500Mbyte/sec Data-Rate 512Kbits x9 DRAM Using a Novel I/O Interface (1992
20 Symposium on VLSI Circuits Digest of Technical Papers, pp.66-67)、日経エレクトロニクス (日経 B P 社 1992.3.30 発行 No.550) 第 197 ~ 209 頁がある。

このような背景のもと、種々のシステムによって D R A M への要求仕様は異なったものになりつつある。したがって、ある世代の D R A M に
25 対して、極めて多くの品種を開発する必要性が生じている。これによって、開発コストの上昇、品種構成に応じた製造面での対応、性能の異なる

種々の製品をテストイングするためのオーバヘッドなど、色々な問題が生じると予想される。

5 こうした状況下、パソコンやワークステーションなどDRAMが用いられるシステムの大半はメモリチップを複数個搭載したモジュール(メモリモジュール)の形で実装されており、メモリ単体の性能よりもメモリモジュールの性能が重要である。したがって、全てのメモリの中に高速動作のための複雑な入出力回路などを取り込まなくとも、メモリモジュールで必要な機能、性能が実現できれば大半のシステムの要求に応じることができる、ということが本発明者によって明らかにされた。本発明者は、上記の観点から、高性能化若しくは高速化を実現するメモリモジュールについて検討した。

10 前記SDRAMを用いて高速メモリモジュールを構成する場合には次の問題点のあることが本発明者によって明らかにされた。すなわち、現状のSDRAMを用いたジュールは、例えば第19図に示されるように、プロセッサボードのようなマザーボード上の64ビットのようなメモリバスIO0～IO63の所定の複数本の信号線毎に1個のSDRAMのデータ入出力端子を結合して64ビットのようなメモリバスとの間で並列的にデータの入出力が可能にされている。このように、複数のSDRAMを用いて所要バス幅のメモリバスとデータの入出力を行う構成では、SDRAM間におけるアクセスタイムなどの特性ばらつきを補償する回路手段がないと、高データ転送レートを実現することは困難である。また、データ入出力ビット数が×4構成のSDRAMを用いて64ビットバスに対応するには少なくとも16個のSDRAMが必要になり、メモリの増設単位が大きくなる。

25 一方、RDRAMを用いて高速メモリモジュールを構成する場合には次の問題のあることが本発明者によって見出された。第1に、RDR A

Mは、それ固有のプロトコル制御回路を個々のメモリチップが保有するから、全てのR D R A Mデバイスが一つのバスに共通接続される。このため、デバイスの入出力容量を相当小さくしないと、バスの負荷容量成分が大きくなり過ぎ、例えば3 2個のR D R A Mを実装するのも難しくなる。

第2に、R D R A Mでは各デバイス毎に高精度のクロック同期機構、例えば高精度のP L L (Phase Locked Loop) 又はD L L (Delayed Locked Loop) が必要であり、また、外部端子数の低減のために信号のビット幅を内部で6倍又は8倍にするように直並列変換を行う必要があり、内部バスのビット幅が大きく(例えば1 2 8ビット)、デバイスのチップサイズが大型化してします。例えば同じ記憶容量のS D R A Mに対して1 0 ~ 1 5 %程度チップが大型化する。

第3に、バス幅に対するデータが全て一つのR D R A Mデバイスから出力されるため、R D R A Mでは一つのデバイスに消費電力が集中し、デバイスの冷却に特別な考慮を払わなければならないと予想される。

第4には、R D R A Mではアクティブ電力を減らさないと周波数向上は難しい。システムに要求される性能向上に見合った低電力(低電圧化)を半導体集積回路の微細化技術でカバーするのは難しい。

第5には、高速になればなる程、実装系の電気特性の制約が大きく、実使用状態とテスト系の電気特性を整合させる工夫が必要になり、デバイステストのオーバーヘッドが大きくなると予想される。

本発明は、上記事情に鑑みてなされたものであり、その目的は、アクセス動作の高速化を実現できるメモリモジュールを提供することにある。

本発明の別の目的は実装するメモリに高速動作のための複雑な入出力回路や制御回路を取り込まなくても、高速メモリアクセスの為のプロ

トコルなど必要な機能や性能を容易に実現することができるメモリモジュールを提供することにある。

- 5 本発明の更に別の目的は、マイクロプロセッサの高速化に追従してデータ処理速度を容易に向上させることができるデータ処理装置を提供することにある。

本発明の上記並びにその他の目的と新規な特徴は本明細書の以下の記述と添付図面から明らかにされるであろう。

発明の開示

- 10 本願において開示される発明のうち代表的なものの概要を簡単に説明すれば下記の通りである。すなわち、並列動作される複数の半導体記憶装置が接続されたデータバスと外部データ入出力端子との間にレジスタバッファを設けてメモリモジュールを構成する。メモリモジュールはプロセッサボードなどのマザーボードに実装され、当該マザーボード上のメモリコントローラとメモリモジュール内の半導体記憶装置との間でのデータのやりとりはレジスタバッファを介して行われる。

- 15 これによれば、メモリモジュール内部と外部とのバス接続は前記レジスタバッファでバッファリングされるから、前記マザーボード上のデータバスとメモリモジュール上のデータバスとは分離される。したがって、マザーボード上のデータバスからは前記レジスタ以降の配線負荷が見えなくなり、メモリコントローラが駆動すべき負荷を低減できる。これにより、メモリバス上のデータ転送周波数を向上させることが容易になる。また、メモリモジュール上についても同様に信号線負荷が低減される。従来のメモリモジュールをマザーボードに実装すると、マザーボード上のメモリバスをメモリモジュール内への負荷の大きな分岐を途中に多数有することになる。上記によれば、メモリバス上におけるそのよ
- 20
- 25

うな負荷の分岐は実質的になくなる。したがって、メモリモジュール内のデータバス上では、分岐のない2地点間 (Point-to-Point) のデータ伝送が可能になるため、メモリモジュール内の動作周波数の向上も比較的容易である。

- 5 前記レジスタバッファを配線負荷の分離手段として位置付ける観点とは別の観点による本発明は、並列動作される複数個の半導体記憶装置が接続されたデータバスと外部端子との間に速度変換手段を設けてメモリモジュールを構成する。速度変換手段は、前記データバスから前記外部端子に与えるデータを所定ビット数単位で並列から直列に変換し、
- 10 前記外部端子から前記データバスに与えるデータを所定ビット数単位で直列から並列に変換する。前記マザーボード上のメモリコントローラと半導体記憶装置との間でのデータのやりとりは前記速度変換手段を介して行われる。この速度変換手段によれば、マザーボード上のメモリバスの信号線数は、メモリモジュール内のデータバスの信号線数よりも
- 15 少なくなる。したがって、マザーボードのメモリバス上の信号周波数よりも低い動作周波数の半導体記憶装置を用いることができる。また、例えばマザーボードのメモリバスをR D R A Mに対応した仕様とすると、一つのバスあたりに実装できる記憶容量を一桁以上向上させることができる。
- 20 また、メモリモジュールのインターフェースとは異なる制御手法でメモリモジュール内の半導体記憶装置を制御可能にする為の、プロトコル制御手段を追加してメモリモジュールを構成する。これにより、メモリモジュール内のバスのアクセス制御仕様 (メモリアクセスプロトコル) を、前記メモリバスのアクセス制御仕様と相異させることができる。し
- 25 たがって、速度変換手段とプロトコル制御手段の具体的な処理内容次第では、E D O D R A M、S D R A Mなど汎用の半導体記憶装置を用いな

がら、メモリモジュールではRDRAMやSyncLinkDRAMなどに適用されるプロトコルで動作させることができる。これにより、メモリモジュールに実装する半導体記憶装置毎に、プロトコルデコーダや内外の位相同期の為のPLL、DLLなどの複雑な制御手段を搭載する必要がなくなる。したがって、EDODRAM, SDRAM, RDRAM, 5 SyncLinkDRAMなど多種の半導体記憶装置を品種展開しなくとも、外部とのインタフェース仕様が異なる種々のメモリモジュールを容易に提供することができる。これは、各種仕様のメモリモジュールの為の半導体メモリを開発するコストの削減にもなる。

10 また、今後、データ転送の高速化のために位相変調、振幅変調、更にはQPSK (Quadrature Phase Shift Keying: 直交位相変調) に代表されるような多値変調などにより多重化された信号伝送が行なわれる場合を想定すると、メモリモジュールにそのためのエンコーダ/デコーダ手段(変復調手段)を搭載することができる。これにより、各半導体
15 記憶装置毎に、変復調の為の複雑な信号処理回路を設ける必要がなくなる。

更に、前記速度変換手段やプロトコル制御手段は、半導体記憶装置とは別の半導体集積回路で形成することができる。したがって、速度変換手段やプロトコル制御手段は、半導体記憶装置とは異なるプロセスで形成
20 できるため、より高速のCMOS論理プロセス、バイポーラプロセス、化合物半導体プロセスなど、マザーボードのメモリバス上でのデータ伝送に最適な半導体プロセスを使用することができる。

また、メモリコントローラとの間のデータ伝送に光伝送を用いれば、超高速のデータ伝送が可能である。これを考慮する時、外部からの光信号
25 を電気信号に変換し、メモリモジュール内部の電気信号を光信号に変換して外部に出力する光電変換手段をメモリモジュールに搭載するこ

とができる。メモリモジュール上にハーフミラーを設ければ光ファイバーによるバス結合が可能である。光の伝走路は大気中であってもよい。光の場合にはメモリモジュールの挿入による信号劣化が少ないため、一つの伝走路に接続できるメモリモジュールの数を飛躍的に増大させることが可能である。

上記で開示された発明のうち代表的なものによって得られる効果を整理すれば下記の通りである。

すなわち、マザーボード上のデータバスとメモリモジュール上のデータバスがレジスタバッファで分離されるため、マザーボード上のデータバスの負荷が最小になり、マザーボード上、メモリモジュール上ともに、より高いデータ転送速度を達成することが可能となる。

メモリコントローラとメモリのデータバス幅を一致させなくともデータのやりとりが可能になり、メモリコントローラのデータバス幅を小さくして高速で動作させ、一方メモリモジュール上のデータバス幅を大きくして低速で動作させることが可能となる。これにより、例えばメモリコントローラのピン数を削減することができる。

メモリチップの読み出し、書き込み、アドレス指定などの制御法をコントローラからの制御法と変えることができるため、見掛け上異なる仕様のメモリモジュールを構成することができる。これにより、各種メモリモジュール用の半導体記憶装置の品種の数を少なくすることができる。また、メモリチップ設計やテストに最適な仕様を選ぶことができるため、メモリモジュールのコストを低減することができる。

上記より、メモリモジュールに搭載すべき半導体記憶装置の入出力仕様に制約されずに、高速化に特化したインターフェース仕様でメモリコントローラあるいはマザーボードを設計することができ、これにより、設計の自由度が向上してデータ処理システムの性能向上を図り易くな

る。

図面の簡単な説明

第 1 図は本発明に係るメモリモジュールの一例を示すブロック図である。

第 2 図は第 1 図に示されるメモリモジュールのチップ配列と配線の概略を示す説明図である。

第 3 図は第 1 図に示されるメモリモジュールに含まれるコンバータの一例を示す論理回路図である。

第 4 図は第 1 図に示されるメモリモジュールを用いたデータ処理システムの部分的なシステム構成図である。

第 5 図は第 1 図に示されるメモリモジュールのアクセス動作タイミングの一例を示すタイミングチャートである。

第 6 図は第 1 図に示されるメモリモジュールに対して記憶容量を倍増したメモリモジュールのブロック図である。

第 7 図は R D R A M の代わりに S D R A M を用いて構成されたプロトコル制御方式のメモリモジュールのブロック図である。

第 8 図は第 7 図に示されるメモリモジュールのアクセス動作タイミングの一例を示すタイミングチャートである。

第 9 図は速度変換アダプタを採用したマザーボードの一例を示すブロック図である。

第 1 0 図は速度変換アダプタを採用した更に別のマザーボードの一例を示すブロック図である。

第 1 1 図は外部インタフェースを光で行うようにしたメモリモジュールの一例を示すブロック図である。

第 1 2 図は第 1 1 図に示されたメモリモジュールを用いたマザーボ

ードの一例を示すブロック図である。

第 1 3 図は外部インタフェースを多値変調信号で行うようにしたメモリモジュールの一例を示すブロック図である。

5 第 1 4 図は第 1 3 図のメモリモジュールを用いたマザーボードの一例を示すブロック図である。

第 1 5 図は冗長メモリデバイスを有するメモリモジュールの一例を示すブロック図である。

10 第 1 6 図は第 1 5 図のメモリモジュールの正規のメモリデバイスに欠陥が無い場合におけるメモリデバイスと外部データ入出力端子との接続態様を示すブロック図である。

第 1 7 図は第 1 5 図のメモリモジュールの正規のメモリデバイスに欠陥がある場合におけるメモリデバイスと外部データ入出力端子との接続態様の一例を示すブロック図である。

15 第 1 8 図は ECC 回路を有するメモリモジュールの一例を示すブロック図である。

第 1 9 図は S D R A M デバイスのデータ入出力端子を対応するメモリバスに直接接続する形式のメモリモジュールを概念的に示した説明図である。

20 第 2 0 図は R D R A M デバイスをメモリバスに接続した状態を示す説明図である。

発明を実施するための最良の形態

《レジスタバッファと速度変換》

25 先ず最初にレジスタバッファを用いて速度変換を行うようにしたメモリモジュールを説明する。

第 1 図には本発明に係るメモリモジュールの一例ブロック図が示さ

れる。メモリモジュールMOD aは、ガラスエポキシ樹脂基板の表面に
所要の配線パターンが形成されて成るような配線基板2に、例えば8個
のSDRAMデバイス(半導体記憶装置)M0~M7、コンバータ3,
4、クロックドライバ5、及び制御信号バッファ(タイミング制御回路)
5 6が実装されて、構成される。

前記SDRAMデバイスM0~M7は並列データ入出力ビット数が
夫々8ビット(×8)とされる。7Lで示されるものは下位32ビット
のモジュールデータバス、7Hで示されるものは上位32ビットのモジ
ュールデータバスである。4個のSDRAMデバイスM0~M3の各々
10 のデータ入出力端子はモジュールデータバス7Lの信号線に1対1対
応で結合され、4個のSDRAMデバイスM4~M7の各々のデータ入
出力端子は前記モジュールデータバス7Hの信号線に1対1対応で結
合される。第1図においてMDQ0~MDQ63は8個のSDRAMデ
バイスM0~M7全部のデータ入出力端子を意味する。

15 メモリモジュールMOD aのデータ入出力端子DQ0~DQ15,D
Q16~DQ31は合計32ビットとされる。前記コンバータ3は下位
側16ビットのデータ入出力端子DQ0~DQ15と前記下位32ビ
ットのモジュールデータバス7Lとの間でデータの直列・並列変換を行
う。同様に、前記コンバータ4は上位16ビットのデータ入出力端子D
20 Q16~DQ31と前記上位32ビットのモジュールデータバス7H
との間でデータの直列・並列変換を行う。第1図の例では、コンバータ
3,4は、モジュールデータバス7L,7H(MDQ0~MDQ63)
とデータ入出力端子DQ0~DQ31との間の速度変換手段を実現す
る。

25 前記制御信号バッファ6は、外部から供給されるアクセス制御情報8
を入力し、内部動作に合わせて内部アクセス制御情報9を各SDRAM

デバイスM0～M7に並列に供給する。外部アクセス制御情報8は、チップセレクト信号／CS1、ロウアドレスストローブ信号／RAS1、カラムアドレスストローブ信号／CAS1、及びライトイネーブル信号／WE1によって代表されるアクセス制御信号と、アドレス信号Address-1とを含む。内部アクセス制御情報9は、チップセレクト信号／CS2、ロウアドレスストローブ信号／RAS2、カラムアドレスストローブ信号／CAS2、及びライトイネーブル信号／WE2によって代表されるアクセス制御信号と、アドレス信号Address-2とを含む。

前記クロックドライバ5は外部から供給されるクロック信号CLK1を入力して内部クロック信号CLK2を生成する。クロックドライバ5から出力されるクロック信号CLK2の一つは帰還され、PLL（又はDLL）10を介して内部クロック信号CLK2を外部クロック信号CLK1に位相同期させる。内部クロック信号CLK2は外部クロック信号CLK1と実質的に同一のクロック信号と見なすことができる。前記内部クロック信号CLK2は、前記SDRAMデバイスM0～M7、コンバータ3、4及び制御信号バッファ6に供給され、それらの動作基準クロック信号とされる。

ここで前記SDRAMデバイスM0～M7それ自体の機能について簡単に説明する。前記アクセス制御信号／CS2、／RAS2、／CAS2、／WE2及びアドレス信号Address-2はクロック信号CLK2の立ち上がりエッジで参照される。信号／CS2はそのローレベルによってコマンド入力サイクルを開始する。コマンド入力サイクルによって始めてその他のアクセス制御信号及びアドレス信号が参照可能にされる。コマンドサイクルでは、信号／RAS2、／CAS2、／WE2及びアドレス信号Address-2の一部のレベルの組み合わせ

せが解読され、その解読結果に従ってSDRAMの動作内容が決定される。例えば、 $\text{RAS}2 = \text{"L"}$, $\text{CAS}2 = \text{"H"}$, $\text{WE}2 = \text{"H"}$ の組み合わせは、ロウアドレスストロブ・バンクアクティブコマンドとされ、ロウアドレス系に対する動作が行われる。 $\text{RAS}2 = \text{"H"}$,
5 $\text{CAS}2 = \text{"L"}$, $\text{WE}2 = \text{"H"}$ の組み合わせは、カラムアドレスリードコマンドとされ、ロウアドレス系動作の後にカラムアドレス系を介するリード動作を指示する。 $\text{RAS}2 = \text{"H"}$, $\text{CAS}2 = \text{"L"}$,
 $\text{WE}2 = \text{"L"}$ の組み合わせは、カラムアドレスライトコマンドとされ、ロウアドレス系動作の後にカラムアドレス系を介するライト動作を
10 指示する。

この例では、メモリモジュールMOD aに対する外部からのアクセス制御形態は、SDRAMデバイスに対するアクセス制御形態と基本的に同じとされる。但し、外部とメモリモジュール1との間のデータ転送速度を上げるため、メモリモジュール1は、クロック信号CLK1 (= CLK2)の立ち上がり立ち下りの両方のクロックエッジに同期して
15 データの入出力（書き込みと読み出し）を行うDDR（Double Data-Rate：ダブルデータレート）形式のSDRAMの制御仕様を有する。一方、SDRAMデバイスM0～M7は、前述の通り、クロック信号CLK2の立ち上がりエッジのみに同期して動作するSDR（Single
20 Data-Rate：シングルデータレート）形式のSDRAMの制御仕様で動作される。

前記メモリモジュールMOD aの外部データ入出力端子DQ0～DQ31は32ビット（メモリモジュール1が搭載されるマザーボード上のメモリバスのデータバス幅が32ビット）であるのに対し、メモリモ
25 ジュールMOD a上のモジュールデータバス7L, 7Hのバス幅は全体で64ビットである。マザーボードのメモリバス上でのデータ転送周波

数は、メモリモジュールMOD aのモジュールデータバス上でのデータ転送周波数の2倍とされる。これにより、データバス全体のデータ転送レート（データ転送周波数×データバスのビット幅）は、マザーボード上のメモリバスとメモリモジュールMOD a上のモジュールデータバスとの間で一致する。

第2図には前記メモリモジュールMOD aのチップ配列と配線の概略を示す。第2図に示される配線はその一部が省略されているが、実際には第1図と同じように形成されている。第2図の例では、SDRAMデバイスM0～M7、コンバータ3、4、クロックドライバ5、及び制御信号バッファ6は夫々別々に半導体集積回路化されている。第2図においてVDD及びVDDQは電源端子、VSS及びVSSQは接地端子である。特に電源端子VDDQ及び接地端子VSSQはデータ出力バッファへの給電に専用化された端子である。

第3図には前記コンバータ3の一例が示される。例えば1ビットの外部データ入出力端子DQ0はSDRAMデバイスM0の2ビットのデータ入出力端子MDQ0、MDQ1に対応され、それらの間にはレジスタバッファRBUF a0が配置されている。その他のデータ入出力端DQ1～DQ15についても同様のレジスタバッファRBUF a1～RBUF a15が配置されている。各レジスタバッファRBUF a0～RBUF a15の構成は相互に同一である。

前記レジスタバッファRBUF a0～RBUF a15は並直変換機能を有する。詳細な構成が例示されたレジスタバッファRBUF a0において、B1で示されるものは、データ入出力端子DQ0に結合された入力バッファである。前記入力バッファB1の出力には2個の入力レジスタREG1、REG3が結合され、入力レジスタREG1の出力は出力ゲートBC1を介してデータ入出力端子MDQ0に接続され、同様に、

入力レジスタ R E G 3 の出力は出力ゲート B C 3 を介してデータ入出力端子 M D Q 0 に接続される。

第 3 図において B C 5 で示されるものは前記データ入出力端子 D Q 0 に結合された出力バッファである。出力バッファ B C 5 とデータ入出力端子 M D Q 0 との間には、入力バッファ B 2、出力レジスタ R E G 2 及び出力ゲート B C 2 が直列配置され、出力バッファ B C 5 とデータ入出力端子 M D Q 1 との間には、入力バッファ B 3、出力レジスタ R E G 4 及び出力ゲート B C 4 が直列配置されている。前記レジスタ R E G 1 ~ R E G 4 及び出力バッファ B C 1 ~ B C 5 は、タイミングジェネレータ T G E N から出力される制御信号 $\phi 1 R$, $\phi 2 W$, $f a i 2 R$, $\phi 1 - 1 W$, $\phi 1 - 1 R$, $\phi 1 - 2 W$, $\phi 1 - 2 R$ によってラッチ動作と出力動作が制御される。前記タイミングジェネレータ T G E N は制御信号バッファ 6 から供給される制御信号とクロック信号 C L K 2 に基づいてそれら制御信号を生成する。タイミングジェネレータ T G E N による制御内容は後で詳述する。特に図示はしないが、コンバータ 4 も上記と同様に構成されている。

第 4 図には前記メモリモジュール M O D a を用いたプロセッサボード P C B の部分的なシステム構成図が示される。パーソナルコンピュータではプロセッサボードを P C ボードとも称する。プロセッサボードや P C ボードのような回路をマザーボードとも称する。

マザーボード P C B はマイクロプロセッサ M P U を中心に構成され、マイクロプロセッサ M P U が結合された C P U バス (システムバス) 1 1 には代表的に示されたメモリコントローラ B C O N T が結合されている。メモリコントローラ B C O N T は、メモリバス 1 2、周辺バス 1 3、グラフィックバス 1 4 などの動作速度の異なるバスを C P U バス 1 にインタフェースするためのバスアクセス制御を行うコントローラで

ある。前記メモリコントローラ B C O N T はバスコントローラ或いはインタフェースコントローラとも称される。グラフィックバス 1 4 にはグラフィックアクセラレータのようなグラフィックコントローラ G C O N T が結合されている。周辺バス 1 3 には図示を省略する I D E
5 (Integrated Device Electronics) コントローラなどが接続されている。メモリバス 1 2 には複数個の前記メモリモジュール M O D a が結合されている。マイクロプロセッサ M P U は前記複数個のメモリモジュール M O D a をメインメモリとしてワーク領域やデータの一時記憶領域に利用する。この例に従えば、マイクロプロセッサ M P U がメインメモ
10 りに割り当てられたアドレス空間をアクセスする時、メモリコントローラ B C O N T は、前記アクセス制御情報をメモリバス 1 2 に出力すると共に、書き込み動作では書き込みデータをメモリバス 1 2 に出力し、読出し動作ではメモリバス 1 2 に読出されたデータを取り込んでマイクロプロセッサ M P U に与える。複数個のメモリモジュール M O D a に対
15 するチップ選択信号 / C S 1 はメモリモジュール毎に固有の信号とされ、前記メモリコントローラ B C O N T から出力される。

第 5 図にはメモリモジュール M O D a のアクセス動作タイミングの一例が示される。第 5 図において、メモリコントローラ B C O N T は、最初、時刻 t 0 に同期させて、/ C S 1 = “L”、/ R A S 1 = “L”、
20 / C A S 1 = “H”、/ W E = “H” によってモジュールアクティブコマンドを発行すると共に、ロウアドレス信号 R A i を出力する。メモリモジュール M O D a の制御信号バッファ 6 は、時刻 t 0 において、前記 / C S 1 = “L”、/ R A S 1 = “L”、/ C A S 1 = “H”、/ W E = “H” を、クロック信号 C L K 2 (C L K 1) の立ち上がりエッジで
25 参照する。制御信号バッファ 6 はこれに応答して、/ C S 2 = “L”、/ R A S 2 = “L”、/ C A S 2 = “H”、/ W E = “H” によってメ

メモリチップアクティブコマンドとロウアドレス信号 RA_i をアクセス制御情報 9 として SDRAM デバイス $M_0 \sim M_7$ に並列的に出力する。SDRAM デバイス $M_0 \sim M_7$ は、時刻 t_1 に、メモリチップアクティブコマンドを認識して、ロウアドレス系の動作を開始する。

- 5 次にメモリコントローラ $BCONT$ は、時刻 t_2 に同期させて、 $/CS_1 = "L"$ 、 $/RAS_1 = "H"$ 、 $/CAS_1 = "L"$ 、 $/WE = "L"$ によりモジュールライトコマンドを発行すると共に、カラムアドレス信号 CA_i を出力する。更にメモリコントローラ $BCONT$ は、書き込みデータ $D(i)$ 、 $D(i+1)$ を出力する。このとき、メモリコントローラ $BCONT$ によるデータ $D(i)$ 、 $D(i+1)$ の転送は、前記ダブルデータレート
- 10 形式で、クロック信号 CLK_1 の立ち上がり（時刻 t_2 ）及び立ち下がり（時刻 t_3 ）に同期して行われる。メモリモジュール MOD_a の制御信号バッファ 6 は、時刻 t_4 において、前記メモリモジュールライトコマンドをクロック信号 CLK_2 (CLK_1) の立ち上がりエッジで参照し、
- 15 これに応答して、 $/CS_2 = "L"$ 、 $/RAS_2 = "H"$ 、 $/CAS_2 = "L"$ 、 $/WE = "L"$ によりメモリチップライトコマンドとカラムアドレス信号 CA_i をアクセス制御情報 9 として SDRAM デバイス $M_0 \sim M_7$ に並列的に出力する。また、制御信号バッファ 6 は、前記メモリチップライトコマンドに応答して、コンバータ 3、4 にライトデータ $D(i)$ 、 $D(i+1)$ の直列・並列変換動作を指示する。これにより、データ入出力端子 $DQ_0 \sim DQ_{31}$ に直列的に与えられたデータ $D(i)$ 、 $D(i+1)$ は、64 ビットの並列データとして時刻 t_4 にモジュールデータバス $7L$ 、 $7H$ に供給され、SDRAM デバイス $M_0 \sim M_7$ のデータ入出力端子 $MDQ_0 \sim MDQ_{63}$ を介して並列的に 8 個の SDRAM デバイス $M_0 \sim M_7$ に書き込まれる。
- 20
- 25

ここで前記直列・並列変換動作を更に説明する。例えば端子 DQ_0 に

最初に供給されるデータ $D(i)0$ はクロック信号 $CLK2$ ($CLK1$) の立ち上がりに同期してレジスタ $REG1$ に格納され、後続のデータ $D(i+1)0$ はクロック信号 $CLK2$ ($CLK1$) の立ち下がりに同期してレジスタ $REG3$ に取り込まれる。そしてクロック信号 $CLK2$ の次の立ち上がりに同期して双方のレジスタ $REG1$, $REG3$ の出力が開かれて、データ $D(i)0$, $D(i+1)0$ が端子 $MDQ0$, $MDQ1$ に向けて並列出力される。その他の端子 $DQ1 \sim DQ31$ に与えられるデータに関しても同じように直列・並列変換される。

書き込みの後、第5図の例では同一ロウアドレスに対するリード動作が行われる。即ち、メモリコントローラ $BCONT$ は、時刻 $t5$ に同期させて、 $/CS1 = "L"$ 、 $/RAS1 = "H"$ 、 $/CAS1 = "L"$ 、 $/WE = "H"$ によりモジュールリードコマンドを発行すると共に、カラムアドレス信号 CAj を出力する。メモリモジュール $MODa$ の制御信号バッファ6は、クロック信号 $CLK2$ ($CLK1$) の立ち上がりエッジに同期する時刻 $t5$ に、前記モジュールリードコマンドを参照する。制御信号バッファ6はこれに応答して、 $/CS2 = "L"$ 、 $/RAS2 = "H"$ 、 $/CAS2 = "L"$ 、 $/WE = "H"$ によってメモリチップリードコマンドとカラムアドレス信号 CAj をアクセス制御情報9としてSDRAMデバイス $M0 \sim M7$ に並列的に出力する。SDRAMデバイス $M0 \sim M7$ は、時刻 $t6$ に、メモリチップリードコマンドを認識して、カラムアドレス系の動作を開始する。カラムアドレス系の動作の結果、時刻 $t7$ にSDRAMデバイス $M0 \sim M7$ のデータ入出力端子 $MDQ0 \sim MDQ63$ からモジュールデータバス $7L$, $7H$ にリードデータ $D(j)$, $D(j+1)$ が並列的に出力される。この出力動作に同期して、制御信号バッファ6は、前記コンバータ3, 4にリードデータ $D(j)$, $D(j+1)$ の並列・直列変換動作を指示する。これにより、モジュールデー

タバス 7 L, 7 H 上の並列データ D (j), D (j+1) は 3 2 ビットずつデータ入出力端子 D Q 0 ~ D Q 3 1 から直列的に出力される。この出力動作は、前記ダブルデータレート形式により、クロック信号 C L K 1 の立ち上がりエッジ (時刻 t 8) 及び立ち下がりエッジ (時刻 t 9) の双方に同期して行われる。

ここで前記並列・直列変換動作を更に説明する。S D R A M デバイス M 0 ~ M 7 のデータ入出力端子 M D Q 0 ~ M D Q 6 3 から並列的に 6 4 ビットのデータがモジュールデータバス 7 L, 7 H に読出されたとき、例えば、データ入出力端子 M D Q 0, M D Q 1 から並列出力されたデータ D (j) 0, D (j+1) 0 は、クロック信号 C L K 2 (C L K 1) の立ち上がりに同期してレジスタ R E G 2, R E G 4 に並列的にラッチされる。そしてクロック信号 C L K 2 の次の立ち上がりに同期してレジスタ R E G 2 のデータ D (j) 0 がデータ入出力端子 D Q 0 から出力され、これに続くクロック信号 C L K 2 の立ち下がり同期してレジスタ R E G 4 のデータ D (j+1) 0 がデータ入出力端子 D Q 0 から出力される。その他の端子 M D Q 1 ~ M D Q 6 3 から供給されるデータに関しても同じように並列・直列変換される。

以上説明した図 1 のメモリモジュール M O D a によれば、コンバータ 3, 4 は外部からの書き込みデータに対しては直列・並列変換を行い、外部への読出しデータに対しては並列・直列変換を行い、内外のデータ転送レートに対する速度変換手段としての機能を有する。したがって、メモリモジュール M O D a の入出力は高速だがメモリバス 1 2 のバス幅は小さく、一方メモリモジュール M O D a のモジュールデータバス 7 L, 7 H に対してはそれよりも低速でバス幅が広い構成を実現することができる。したがって、マザーボード P C B のメモリバス上の信号周波数よりも低い動作周波数の S D R A M デバイス M 0 ~ M 7 を用いるこ

とができる。更に、メモリコントローラBCONTのピン数（パッケージの外部端子数）の削減にも役立つ。

前記マザーボードPCB上のメモリコントローラBCONTとメモリモジュールMODa内のSDRAMデバイスM0～M7との間でのデータのやりとりはレジスタバッファRBUF a 0～RBUF a 15を介して行われる。これによれば、メモリモジュールMODaの内部と外部とのバス接続は前記レジスタバッファRBUF a 0～RBUF a 15でバッファリングされるから、前記マザーボードPCB上のデータバスとメモリモジュールMODa上のモジュールデータバス7L, 7Hとは分離される。したがって、マザーボードPCB上のデータバスからは前記レジスタバッファRBUF a 0～RBUF a 15以降の配線負荷が見えなくなり、メモリコントローラBCONTが駆動すべき負荷を低減できる。これにより、メモリバス12上のデータ転送周波数を向上させることが容易になる。すなわち、メモリモジュールMODaの入出力容量成分を最小化することができる。

また、メモリモジュールMODaのモジュールデータバス7L, 7H上についても同様に信号線負荷が低減される。例えば第19図に示されるような従来のメモリモジュールをプロセッサボードに実装すると、プロセッサボード上のメモリバスにメモリモジュール内への負荷の大きな分岐を途中で多数有することになる。レジスタバッファRBUF a 0～RBUF a 15を有する上記メモリモジュールMODaの場合には、メモリバス12上におけるそのような負荷の分岐は実質的になくなる。したがって、メモリモジュールMODa内のモジュールデータバス7L, 7H上では、分岐のない2地点間のデータ伝送が可能になるため、メモリモジュールMODa内の動作周波数の向上も比較的容易になる。

第6図には第1図に対して記憶容量を倍増したメモリモジュールM

OD bの例が示される。このメモリモジュールMOD bは、SDRAM
デバイスM0 a～M7 aとM0 b～M7 bを有し、8個のSDRAMデ
バイスM0 a～M7 aはチップ選択信号／CS 2 aによってコマンド
サイクルが指示され、SDRAMデバイスM0 b～M7 bはチップ選択
5 信号／CS 2 bによってコマンドサイクルが指示される。チップ選択信
号／CS 2 a，／CS 2 bは、外部から供給されるチップ選択信号／C
S 1 a，／CS 1 bに対応され、チップ選択信号／CS 1 a，／CS 1
bは選択的に何れか一方がイネーブルにされる。これにより、第1図の
例ではメモリモジュール上に8個のSDRAMデバイスが搭載されて
10 いたのに対して、第6図のメモリモジュールは2倍の16個のSDRA
Mデバイスを搭載して利用することができる。尚、第6図では配線など
の図示を部分的に省略してあるが、上記相違点以外の構成は第1図と同
じである。

この構成においてモジュールデータバス7 L，7 Hの負荷成分（配線
15 抵抗及び寄生容量成分）は増えるが、外部データ入出力端子DQ 0～D
Q 3 1とモジュールデータバス7 L，7 Hとの間には前記コンバータ3，
4が配置されているので、コンバータ3，4に含まれる前記レジスタバ
ッファの作用により、SDRAMデバイスを2倍搭載した第6図の構成
でも、外部データ入出力端子DQ 0～DQ 3 1から見たときの電気特性
20 は第1図のメモリモジュールMOD aと実質的に同じである。この例で
はメモリモジュールMOD b上のモジュールデータバスに2組みのS
DRAMデバイスを接続しているが、これに限らず、さらに多数組のS
DRAMデバイスを接続しても構わない。

《レジスタバッファとプロトコル変換》

25 次に、レジスタバッファを用いた速度変換と共にプロトコル変換を行
うようにしたメモリモジュールを説明する。

第7図にはRDRAMの代わりにSDRAMを用いて Rambus (ランバス)などのプロトコル制御方式に準拠させたメモリモジュールの一例が示される。同図に示されるメモリモジュールMOD cは、ガラスエポキシ樹脂基板の表面に所要の配線パターンが形成されて成るような配線基板22に、例えば16個のSDRAMデバイス(半導体記憶装置)M0~M15及びプロトコルコンバータ23が実装されて成る。プロトコルコンバータ23は、特に制限されないが、1個の半導体集積回路として構成される。前記プロトコルコンバータ23は、クロックドライバ25、プロトコル制御回路26、並直変換機能を有するレジスタバッファRFUF0~RBUF15、入出力ゲートIOG0~IOG15及びタイミングジェネレータ28などを有する。

前記SDRAMデバイスM0~M15は並列データ入出力ビット数が夫々4ビット(×4)とされる。27で示されるものは64ビットのモジュールデータバスである。前記16個のSDRAMデバイスM0~M15のデータ入出力端子MDQ0~MDQ63はモジュールデータバス27の信号線に1対1対応で結合されている。

IOPO~IOP15はメモリモジュールMOD cの16個の入出力端子である。前記レジスタバッファRBUF c0及び入出力ゲートIOG0は、入出力端子IOP0と前記SDRAMデバイスM0の4ビットのデータ入出力端子MDQ0~MDQ3との間に配置されている。その他のレジスタバッファRBUF1~RBUF15と入出力ゲートIOG1~IOG15も同様に、入出力端子IOP1~IOP15と前記SDRAMデバイスM1~M15とに対応して配置されている。更にレジスタバッファRBUF0~RBUF15は64ビットのコマンドバス29を介してプロトコル制御回路26に接続されている。

第7図において代表的にその詳細が示された入出力ゲートIOG0

において、入出力端子 I O P 0 に直列的に与えられたデータは、入力バッファ B C 1 9 を介して、順次入力レジスタ R E G 1 1, R E G 1 3, R E G 1 5, R E G 1 7 にラッチされ、4 ビット単位で出力ゲート B C 1 1, B C 1 3, B C 1 5, B C 1 7 を介して出力される。一方、入出力ゲート I O G 0 から並列的に出力される 4 ビットのデータは出力レジスタ R E G 1 2, R E G 1 4, R E G 1 6, R E G 1 8 に並列的にラッチされ、ラッチされたデータは出力ゲート B C 1 2, B C 1 4, B C 1 6, B C 1 8 を介して順次直列的に出力バッファ B C 2 0 から入出力端子 I O P 0 に与えられる。その他のレジスタバッファ R B U F 1 ~ R B U F 1 5 も同様に構成される。このように構成されたレジスタバッファ R B U F 0 ~ R B U F 1 5 は、入出力端子 I O P 0 ~ I O P 1 5 から入力される 1 6 ビットのデータを 4 組の直列データ毎に 6 4 ビットの並列データに変換して、コマンドバス 2 9 や入出力ゲート I O G 0 ~ I O G 1 5 に与える。また、レジスタバッファ R B U F 0 ~ R B U F 1 5 は、入出力ゲート I O G 0 ~ I O G 1 5 から与えられる 6 4 ビットの並列データを 1 6 ビット単位の 4 組のデータに直列変換して、入出力端子 I O P 0 ~ I O P 1 5 から外部に出力させる。この例において、レジスタバッファ R B U F 0 ~ R B U F 1 5 は、モジュールデータバス 2 7 及びコマンドバス 2 9 と入出力端子 I O P 0 ~ I O P 1 5 との間の速度変換手段を実現する。

メモリモジュール M O D c に対する外部からのアクセス制御形態は、Rambus (ランバス) に代表されるようなプロトコル制御形態とされ、S D R A M デバイスに対するアクセス制御形態とは相違される。前記プロトコル制御回路 2 6 は入出力端子 I O P 0 ~ I O P 1 5 から供給されるコマンドパケットに含まれている読出しリクエストや書き込みリクエスト及びアドレス情報を解読して監視し、自らの動作が選択されたこ

とをその解読結果から認識すると、当該解読結果を用いて、前記 S D R A M デバイス M 0 ~ M 1 2 5 に対するアクセス制御情報 9 を出力する。このアクセス制御情報 9 は、第 1 図と同様の S D R A M デバイスにおけるシングルデータ転送レートに対応されるアクセス制御信号及びアドレス信号である。前記コマンドパッケージが書き込みリクエストの場合、当該コマンドパッケージの後に続く書き込みデータは入出力端子 I O P 0 ~ I O P 1 5 から入力される。

前記クロックドライバ 2 5 は外部から供給されるクロック信号 C L K 1 を入力して内部クロック信号 C L K 2 を生成する。この例では内部クロック信号 C L K 2 は外部クロック信号 C L K 1 に対して 2 分周されている。前記内部クロック信号 C L K 2 は、前記 S D R A M デバイス M 0 ~ M 1 5 及びプロトコル制御回路 2 6 などに供給され、それらの動作基準クロック信号とされる。前記 S D R A M デバイス M 0 ~ M 1 5 の機能は第 1 図で説明した内容と同じである。

前記タイミングジェネレータ 2 8 は、クロックドライバ 2 5 から前記クロック信号 C L K 1 , C L K 2 を入力すると共に、S D R A M デバイス M 0 ~ M 1 5 への外部データ入出力動作の指示をプロトコル制御回路 2 6 から入力して、前記レジスタバッファ R B U F c 0 ~ R B U F c 1 5 及び入出力ゲート I O G 0 ~ I O G 1 5 の制御信号を生成する。第 7 図において前記制御信号は、入力制御信号 $\phi 1 0 W$ 、ラッチ制御信号 $\phi 1 0 - 1 W$, $\phi 1 0 - 2 W$, $\phi 1 0 - 3 W$, $\phi 1 0 - 4 W$ 、ゲート出力信号 $\phi 1 1 W$ 、ゲート出力信号 $\phi 1 3 R$, ラッチ制御信号 $\phi 1 2 R$ 、ゲート出力信号 $\phi 1 2 - 1 R$, $\phi 1 2 - 2 R$, $\phi 1 2 - 3 R$, $\phi 1 2 - 4 R$ とされる。特に、プロトコル制御回路 2 6 は、入出力端子 I O P 0 ~ I O P 1 5 から供給されるコマンドパッケージを常に監視しなければならないため、スタンバイ状態でもレジスタバッファ R B U F c 0 ~ R

B U F c 1 5 は直列・並列変換動作を怠ることはない。

前記メモリモジュールMOD c も例えば第4図に示されるようなマザーボードP C Bに適用することができる。特にこの場合、前記メモリバス1 2はメモリモジュールMOD cの入出力端子I O P 0 ~ I O P 1 5をメモリコントローラB C O N Tにインタフェースされると共にクロック信号C L K 1をメモリモジュールMOD cに供給するための信号線を有すれば充分とされる。また、メモリコントローラB C O N Tは、Rambus (ランバス) に代表されるようなプロトコル制御方式でメモリモジュールMOD cをアクセス制御する機能を有する。換言すれば、メモリバス1 2にR D R A Mを実装したメモリモジュールを接続することも可能である。

第8図にはメモリモジュールMOD cのアクセス動作タイミングの一例が示される。第8図において、メモリコントローラB C O N Tは、時刻t 0 ~ t 1の期間にモジュールライトコマンドのようなコマンド packets をクロック信号C L K 1に同期してメモリバス1 2に出力する。その後、規定のタイミングに従って時刻t 2 ~ t 3の期間にクロック信号C L K 1に同期してモジュール書き込みデータをメモリバス1 2に出力する。

プロトコル制御方式によるアクセスをサポートするメモリモジュールMOD cは、メモリバス1 2上のコマンド packets を監視する。すなわち、入出力端子I O P 0 ~ I O P 1 5から供給される情報をレジスタバッファR B U F c 0 ~ R B U F c 1 5で並列信号に変換し、コマンドバス2 9を介してプロトコル制御回路2 6に供給している。プロトコル制御回路2 6は、それによって与えられる情報を解読し、自らのアクセスが指定されたかを判定する。コマンド packets には、メモリアccessモードを指定する情報と共にロウアドレス及びカラムアドレス情報な

どメモリサイクルに必要なアクセス制御情報が含まれている。

例えば、プロトコル制御回路 26 が、前記時刻 $t_0 \sim t_1$ の期間に供給されたコマンドパケットによりメモリモジュール MOD c の動作が指示されていることを認識すると、当該プロトコル制御回路 26 は、コマンドパケットの解読結果に従って、時刻 t_2 に同期させ、 $\neg CS_2 =$
5 “L”、 $\neg RAS_2 =$ “L”、 $\neg CAS_2 =$ “H”、 $\neg WE =$ “H” によってメモリチップアクティブコマンドとロウアドレス信号 RA_i をアクセス制御情報 9 として SDRAM デバイス $M_0 \sim M_7$ に並列的に出力する。SDRAM デバイス $M_0 \sim M_7$ は、クロック信号 CLK_2 の
10 立ち上がりエッジに同期して時刻 t_2 に、メモリチップアクティブコマンドを認識し、ロウアドレス系の動作を開始する。

その後、プロトコル制御回路 26 は、前記コマンドパケットの解読結果に従って、時刻 t_4 に同期させて、 $\neg CS_2 =$ “L”、 $\neg RAS_2 =$
“H”、 $\neg CAS_2 =$ “L”、 $\neg WE =$ “L” と、カラムアドレス信号
15 CA_i をアクセス制御情報 9 として SDRAM デバイス $M_0 \sim M_{15}$ に並列的に出力する。このとき、レジスタバッファ $RBUF_{c0} \sim RBUF_{c15}$ は、メモリコントローラ BCONT が時刻 $t_2 \sim t_3$ の期間に直列的に出力した書き込みデータ $D(i)$, $D(i+1)$, $D(i+2)$, $D(i+3)$ を 64 ビットの並列データに変換し、モジュールデータバス 27 に供給
20 している。これにより、SDRAM デバイス $M_0 \sim M_{15}$ には、データ入出力端子 $MDQ_0 \sim MDQ_{63}$ を介して 64 ビットのデータ $D(i)$, $D(i+1)$, $D(i+2)$, $D(i+3)$ が書き込まれる。

書き込みの後、第 8 図の例では同一ロウアドレスに対するリード動作が行われる。即ち、メモリコントローラ BCONT は、時刻 $t_5 \sim t_6$
25 に期間にクロック信号 CLK_1 に同期して供給されるモジュールリードコマンドとしてのパケットコマンドによってその動作が指示されて

- いることを認識すると、当該コマンドパケットの解読結果に従い、時刻 t_7 に同期させて、 $\text{CS}_2 = \text{"L"}$ 、 $\text{RAS}_1 = \text{"H"}$ 、 $\text{CAS}_1 = \text{"L"}$ 、 $\text{WE} = \text{"H"}$ によりメモリチップリードコマンドを出力すると共に、前記解読結果から得られたカラムアドレス信号 CA_j を SDRAM デバイス $\text{M}_0 \sim \text{M}_{15}$ に並列的に出力する。SDRAM デバイス $\text{M}_0 \sim \text{M}_{15}$ は、それに応答し、クロック信号 CLK_2 の立ち上がりエッジに同期する時刻 t_7 に、メモリチップリードコマンドを認識して、カラムアドレス系の動作を開始する。カラムアドレス系の動作の結果、時刻 t_8 に SDRAM デバイス $\text{M}_0 \sim \text{M}_{15}$ のデータ入出力端子 $\text{MDQ}_0 \sim \text{MDQ}_{63}$ からモジュールデータバス 27 にリードデータ $\text{D}(j)$ 、 $\text{D}(j+1)$ 、 $\text{D}(j+2)$ 、 $\text{D}(j+3)$ が並列的に出力される。この出力動作に同期して、プロトコル制御回路 26 は、前記レジスタバッファ $\text{RBUF}_c 0 \sim \text{RBUF}_c 15$ にリードデータ $\text{D}(j)$ 、 $\text{D}(j+1)$ 、 $\text{D}(j+2)$ 、 $\text{D}(j+3)$ の並列・直列変換動作を指示する。これにより、モジュールデータバス 27 上の並列データ $\text{D}(j)$ 、 $\text{D}(j+1)$ 、 $\text{D}(j+2)$ 、 $\text{D}(j+3)$ は 16 ビットづつ入出力端子 $\text{IOP}_0 \sim \text{IOP}_{15}$ から直列的に出力される。この出力動作は、前記ダブルデータレート形式と同じように、クロック信号 CLK_1 の立ち上がりエッジ及び立ち下がりエッジの双方に同期して行われる。
- 20 このように、上記メモリモジュール MOD_c は、データの直列・並列／並列・直列変換機構を構成するレジスタバッファ $\text{RBUF}_c 0 \sim \text{RBUF}_c 15$ とプロトコル制御回路 26 等を有するプロトコルコンバータ 23 を含むから、RDRAM の代わりに SDRAM デバイスを用いて Rambus (ランバス) に準拠するようなアクセスプロトコルをサポートすることができる。上記以外のメモリ仕様とモジュール仕様の組み合わせ
- 25 についても同様に実現することが可能である。したがって、メモリモジ

ユーラMODcに実装するSDRAMデバイスのような半導体記憶装置毎に、プロトコルデコーダや内外の位相同期の為のPLL、DLLなどの複雑な制御手段を搭載する必要はない。よって、EDODRAM, SDRAM, RDRAM, SycLinkDRAMなど多種の半導体記憶装置を品種展開しなくとも、外部とのインタフェース仕様の異なる種々のメモリモジュールを容易に提供することができる。これは、各種仕様のメモリモジュールの為の半導体メモリを開発するコストを削減可能にする。

また、データの直列・並列／並列・直列変換機構を構成するレジスタバッファRBUFc0～RBUFc15とプロトコル制御回路26等を有するプロトコルコンバータ23は、SDRAMデバイスM0～M15とは別の半導体集積回路で形成することができる。プロトコルコンバータ23は、SDRAMデバイスM0～M15とは異なるプロセスで形成できるため、より高速のCMOS論理プロセス、バイポーラプロセス、化合物半導体プロセスなど、マザーボードPCBのメモリバス上でのデータ伝送に最適な半導体プロセスを使用することができる。

《速度変換アダプタ》

前記速度変換技術はメモリバス上にも適用することができる。例えば、第9図に例示されるように、メモリコントローラBCONTとメモリバス12との接続を速度変換アダプタ30を介して行う。速度変換アダプタ30は、前記コンバータ3,4と同じようにレジスタバッファを用いた速度変換手段を備える。メモリコントローラBCONTと速度変換アダプタ30との間は16ビットのデータバス31に接続される。速度変換アダプタ30はデータバス31に直列的に伝達される16ビットのデータを32ビットのデータサイズに並列変換して32ビットのデータバス32に供給し、データバス32の32ビットデータを16ビット

に直列変換してデータバス 3 1 に直列的に伝達する。データバス 3 2 には例えば前記メモリモジュールMOD a が複数個結合されている。

メモリコントローラBCONTの動作周波数($4 \times f \text{ MHz}$)はメモリモジュールMOD a (1)~MOD a (n)の動作周波数($2 \times f \text{ MHz}$)の2倍とされる。前述の説明から明らかなように、メモリモジュールMOD a に搭載されているSDRAMデバイスM0~M7の2倍の周波数でメモリモジュールMOD a の入出力動作が行われる。従って、図9の例では、メモリコントローラBCONTの入出力の動作周波数はSDRAMデバイスM0~M7の動作周波数(f)の4倍になる。このようなシステム構成により、メモリコントローラBCONTは高速に動作するが、そのピン数を減らすことができる。

第10図の例はメモリバス12の間に速度変換アダプタ30を介在させた例であり、前段のメモリバス12Bは16ビットのデータバス31を備え、後段のメモリバス12Aは32ビットのデータバス32を備える。速度変換アダプタ30の機能は第9図と同じである。前段のメモリバス12Bには前記メモリモジュールMOD a が複数個結合されている。前記メモリバス12BにはメモリモジュールMOD a a とメモリデバイスMEMが結合されている。メモリモジュールMOD a a は前記メモリモジュールMOD a に対して入出力データが16ビットにされた点が相違される。メモリデバイスMEMはデータ入出力ビット数が16ビットのSDRAM等の半導体メモリである。

このシステム構成によれば、小容量だが高速に動作するメモリチップやメモリモジュールをメモリバス12Bに接続し、大容量だが速度の遅いメモリモジュールをメモリバス12Aに接続して、システムに最適な性能(アクセス速度、メモリ容量)を達成することができる。

前記速度変換アダプタ30を採用したシステムは、MOD b, MOD

cなどその他のメモリモジュールにも適用できることは言うまでもない。

《メモリモジュールの外部インタフェース方式》

第11図には外部インタフェースを光で行うようにしたメモリモジュールの一例が示される。第11図に示されるメモリモジュールMOD dは第1図のメモリモジュールMOD aに対して外部インタフェース信号を光とした点が相違される。すなわち、前記コンバータ3、4、クロックドライバ5及び制御信号バッファ6の外部インタフェース部分に光電変換回路3L、4L、5L、6Lが配置されている。前記光電変換回路3L、4L、5L、6Lは、ハーフミラー40を介して光バス41に接続される。8Lはアクセス制御情報8を伝達する光ファイバ、LDQ0～LDQ7はデータを伝達する光ファイバ、LCLKはクロック信号CLK1を伝達する光ファイバである。

光電変換回路3AL、4Lはピンフォトダイオード有する光レシーバと、フォトダイオードを有する光トランスミッタとを有する。光電変換回路5L、6Lはピンフォトダイオード有する光レシーバを備える。尚、第1図と同一回路ブロックには同一符号を付してその詳細な説明は省略する。

第12図にはメモリモジュールMOD dを用いたシステム構成例が示される。前記メモリコントローラBCONTと光バス41との接続は入出力アダプタ42を介して行われる。入出力アダプタ42は、前述と同様の速度変換機能を有するコンバータと光電変換回路の機能を有する。メモリモジュールMOD dへのデータ書き込みに際しては、入出力アダプタ42から送られる光信号の一部を所定のメモリモジュールMOD dに取り込む。メモリモジュールMOD dからのデータ読み出しに際しては、メモリモジュールMOD dからの光をハーフミラー40で反

射させて入出力アダプタ 4 2 の送信する。

5 ハーフミラー 4 0 は、この例のようにメモリモジュール M O D d 上に搭載しなくとも、マザーボード上に設置して、反射した光がメモリモジュール内の光電変換回路の受光／発光部分に照射されるように配置してもよい。また、プロセッサボード上の光の伝送媒体としては、光ファイバ 4 1 でも良いが、コンピュータの主記憶のように伝送距離が短く、光の強度の劣化が問題ない場合には、空中を伝送させてもよい。

10 このように、プロセッサボード上を光で伝送した場合、光は電気に比べて信号の減衰や劣化が少ないため従来の電気によるバス接続に比べて飛躍的に多数のメモリモジュール M O D d を実装することが可能になる。また、光同士は相互干渉がなく、また、他の電気信号との間の干渉もないため、メモリバスから他の信号にノイズが載ったり、あるいは逆に、他の信号からノイズを受けて誤動作するような危険性を少なくすることができる。また、従来のバスではマザーボード上の配線のインダクタンスがアンテナのように働き、電磁波を外部に放出する問題が起きやすかったが、光伝送とすることにより、そうした問題も回避することができる。

20 第 1 3 図には外部インタフェースを多値変調信号で行うようにしたメモリモジュールの一例が示される。第 1 3 図に示されるメモリモジュール M O D e は第 1 図のメモリモジュール M O D a に対して外部インタフェース信号を多値変調信号とした点が相違される。すなわち、前記コンバータ 3 , 4 の外部インタフェース部分に多値変復調回路 3 S , 4 S が配置されている。多値変復調方式には例えば Q P S K 等を採用できる。メモリバス 5 1 に含まれるデータバス S D Q 0 ~ S D Q 1 5 には多値変調信号が伝送される。これにより、各半導体記憶装置毎に、変復調の為の複雑な信号処理回路を設ける必要はない。尚、第 1 図と同一回路

ブロックには同一符号を付してその詳細な説明は省略する。

第14図にはメモリモジュールMOD eを用いたシステム構成例が示される。前記メモリコントローラBCONTとメモリバス51との接続は入出力アダプタ52を介して行われる。入出力アダプタ52は、前述と同様の速度変換機能を有するコンバータと多値変復調回路の機能を有する。メモリモジュールMOD eへのデータ書き込みに際しては、入出力アダプタ52から送られる変調信号は多値変復調回路3S, 4Sで復調されてコンバータに取り込まれる。メモリモジュールMOD eからのデータ読み出しに際しては、コンバータ3, 4の出力が多値変復調回路3S, 4Sで変調されてメモリバス51に供給される。

マザーボード上を多値変調信号でデータ伝送した場合、通常のデジタル信号に比べて伝送信号の周波数スペクトラムを狭めることが可能となる。例えば正弦波に近い伝送波形を用いることも可能になる。したがって、同じメモリバスを用いても、波形制御がやり易くなり高い周波数まで伝送することが可能となる。

《コンバータと冗長メモリデバイス》

第15図には冗長メモリデバイスを有するメモリモジュールMOD fの一例が示される。基本的な構成は第1図と同じであり、1個のSDRAMデバイスM8を余計に搭載している。そして、9個のSDRAMデバイスM0～M8の内のどの8個を前記モジュールデータバス7L, 7Hに接続するかを選択するセレクタ61が設けられ、前記セレクタ61による選択動作を決定する為の救済アドレス情報が設定される冗長プログラム回路60及び救済アドレス判定回路62が設けられている。冗長プログラム回路60は、救済すべきアドレス(不良ビットのアドレス)情報がプログラマブルに設定されると共に、その救済アドレスにおいて不良を有するSDRAMデバイスの番号情報(No.0～No.7)がプロ

グラマブルに設定される。一つの不良アドレスに対して不良 S D R A M デバイスの数は 1 個に制限される。冗長プログラム回路 6 0 はヒューズプログラム回路又は電氣的に書き込み可能な揮発性半導体メモリによって構成することができる。

- 5 前記救済アドレス判定回路 6 2 は、制御信号バッファ 6 に供給されるアドレス情報 A d d r e s s - 1 と前記冗長プログラム回路 6 0 に設定された救済すべきアドレスとを比較する。比較結果が一致の場合には、当該救済すべきアドレスと対の不良 S D R A M デバイスの番号に相当する S D R A M デバイスをセクタ 6 1 によってコンバータ 3 , 4 から
- 10 切り離し、これに代えて S D R A M デバイス M 8 がセクタ 6 1 によってコンバータ 3 , 4 に接続される。前記セクタ 6 1 、冗長プログラム回路 6 0 、救済アドレス判定回路 6 2 、コンバータ 3 , 4 、及び第 1 5 図には図示を省略したクロックドライバ 5 は、1 個の半導体集積回路化されている。その他の構成は第 1 図と同じであるからその詳細な説明は
- 15 省略する。

- 前記冗長プログラム回路 6 0 の初期状態においてセクタ 6 1 による選択状態は第 1 6 図のようにされる。S D R A M デバイス M 0 ~ M 7 の不良アドレスと不良デバイス番号に応じて冗長プログラム回路 6 0 がプログラムされる。例えば、第 1 7 図に例示されるように、あるアクセスアドレスに関する不良 S D R A M デバイスが M 4 であるとき、セクタ 6 1 は当該 S D R A M デバイス M 4 をモジュールデータバス 7 L から切り離し、これに代えて S D R A M デバイス M 8 をモジュールデータバス 7 L に接続する。
- 20

- これにより、不良のメモリチップを用いてメモリモジュールの良品を提供できる。これらはメモリチップの入出力がそのままモジュールの入出力端子に接続されている従来のメモリモジュールでは実現できない。
- 25

《コンバータとECC》

第18図にはECC回路を有するメモリモジュールの一例が示される。基本的な構成は第1図と同じであり、ECC回路70とSDRAMデバイスM8を余計に搭載している。ECC回路はコンバータ3, 4と
5 モジュールデータバスとの間に配置される。ECC回路は、コンバータ3, 4を通して並列化された64ビットのデータに対する誤り訂正コードを生成する。生成された誤り訂正コードはSDRAMデバイスM8に格納される。読出し動作時において、ECC回路70は、SDRAMデバイスM8から読出された前記誤り訂正コードを用いて、前記SDRAM
10 デバイスM0～M7から読出されたデータの誤り検出及び訂正を行い、その結果をコンバータ3, 4に与える。

これにより、信頼性の高いメモリモジュールを提供することができる。これらはメモリチップの入出力がそのままモジュールの入出力端子に接続されている従来のメモリモジュールでは実現できない。

15 以上本発明者によってなされた発明を実施例に基づいて具体的に説明したが、本発明はそれに限定されるものではなく、その要旨を逸脱しない範囲において種々変更可能であることは言うまでもない。例えばメモリモジュールに利用する半導体記憶装置それ自体の種類は上記の例に限定されず適宜変更可能である。メモリモジュールの入出力端子のビット数、メモリモジュールに実装する半導体記憶装置の数も適宜変更可能である。また、マザーボード上の入出力インターフェースはCMOS、
20 GTL (Gunning Transceiver Logic)、TTL等の何れの形式であっても本発明は適用可能である。SSTL (Stub Series Terminated Transceiver Logic) インターフェースの場合にはモジュールの入出力
25 端子とコンバータ3, 4との間にスタブ抵抗を配すればよい。

産業上の利用可能性

本発明は、パーソナルコンピュータやワークステーションのメモリモジュールなどに広く適用することができる。

請 求 の 範 囲

1. 配線基板と、前記配線基板に設けられた複数個の半導体記憶装置と、
並列動作される前記複数個の半導体記憶装置のデータ入出力端子が
5 個別に接続されるデータバスと、前記配線基板に設けられた複数個の
外部データ入出力端子と、前記データバスから前記データ入出力端子
に与えるデータを保持し、また、前記データ入出力端子から前記デー
タバスに与えるデータを保持するレジスタバッファと、を含んで成る
ものであることを特徴とするメモリモジュール。
- 10 2. 前記データバスのビット数 m は前記外部データ入出力端子のビット
数 n の整数倍であることを特徴とする請求の範囲第1項に記載のメ
モリモジュール。
3. 前記複数個の半導体記憶装置は外部から供給されるクロック信号に
同期動作され、前記外部データ入出力端子からのデータ入出力動作周
15 波数 f_1 は前記半導体集積回路のデータ入出力動作周波数 f_2 より
も高く、前記並列動作される複数個の半導体集積回路と前記データバ
スとの間での最高データ転送レート $n \times f_2$ と前記外部データ入出
力端子を介する入出力動作の最高データ転送レート $m \times f_1$ とが実
質的に等しいことを特徴とする請求の範囲第2項に記載のメモリモ
20 ジュール。
4. 前記データバスに接続されて並列動作される複数個の半導体記憶装
置を複数組有し、前記複数組の中の一組の半導体記憶装置が選択的に
動作可能にされて成るものであることを特徴とする請求の範囲第3
項に記載のメモリモジュール。
- 25 5. 予備の半導体記憶装置と、欠陥のある半導体記憶装置を前記予備の
半導体記憶装置に選択的に置き換える救済手段とを更に含んで成る

ものであることを特徴とする請求の範囲第4項に記載のメモリモジュール。

6. マイクロプロセッサと、マイクロプロセッサに結合されメモリバスを制御するメモリコントローラと、前記メモリバスに結合されたメモリモジュールと、を含んで成り、

上記メモリモジュールは、配線基板と、前記配線基板に設けられた複数個の半導体記憶装置と、並列動作される前記複数個の半導体記憶装置のデータ入出力端子が個別に接続されるデータバスと、前記配線基板に設けられた複数個の外部データ入出力端子と、前記データバスから前記データ入出力端子に与えるデータを一時的に保持し、また、前記データ入出力端子から前記データバスに与えるデータを一時的に保持するレジスタバッファと、を含んで成るものであることを特徴とするデータ処理システム。

7. 配線基板と、前記配線基板に設けられ外部からのクロック信号に同期動作される複数個の半導体記憶装置と、並列動作される前記複数個の半導体記憶装置のデータ入出力端子が個別に接続されるデータバスと、前記配線基板に設けられた複数個の外部端子と、前記データバスから前記外部端子に与えるデータを所定ビット数単位で並列から直列に変換し、前記外部端子から前記データバスに与えるデータを所定ビット数単位で直列から並列に変換する速度変換手段と、を含んで成るものであることを特徴とするメモリモジュール。

8. 前記速度変換手段は半導体集積回路化されて成るものであることを特徴とする請求の範囲第7項に記載のメモリモジュール。

9. メモリモジュールに対するアクセス制御情報がデータ入出力用とは異なる専用の外部端子を介して供給されるタイミング制御回路を有し、このタイミング制御回路は、前記アクセス制御情報に基づいて前

記複数個の半導体記憶装置に内部アドレス信号及び内部アクセス制御信号を並列的に供給するものであることを特徴とする請求の範囲第7項又は第8項に記載のメモリモジュール。

5 10. メモリモジュールに対するアクセス制御情報がデータ入出力用と兼用された外部端子から前記速度変換手段を介して供給されるプロトコル制御回路を有し、このプロトコル制御回路は、前記アクセス制御情報を解読して内部アドレス信号及び内部アクセス制御信号を生成し前記複数個の半導体記憶装置に並列的に供給するものであることを特徴とする請求の範囲第7項又は第8項に記載のメモリモジュール。

10 11. マイクロプロセッサと、前記マイクロプロセッサに結合されメモリバスを制御するメモリコントローラと、前記メモリバスに結合されたメモリモジュールとを含み、

15 前記メモリコントローラは、アクセス制御情報とアクセスデータとを前記メモリバスの異なる信号線に伝達して前記メモリモジュールをアクセス制御するものであり、

20 前記メモリモジュールは、配線基板と、前記配線基板に設けられ外部からのクロック信号に同期動作される複数個の半導体記憶装置と、並列動作される前記複数個の半導体記憶装置のデータ入出力端子が個別に接続されるデータバスと、前記配線基板に設けられた複数個の外部端子と、前記データバスから前記外部端子に与えるデータを所定ビット数単位で並列から直列に変換し、前記外部端子から前記データバスに与えるデータを所定ビット数単位で直列から並列に変換する速度変換手段と、を含んで成るものであることを特徴とするデータ処理システム。

25 12. マイクロプロセッサと、前記マイクロプロセッサに結合されメモ

リバスを制御するメモリコントローラと、前記メモリバスに結合されたメモリモジュールとを含み、

- 5 前記メモリモジュールは、配線基板と、前記配線基板に設けられ外部からのクロック信号に同期動作される複数個の半導体記憶装置と、並列動作される前記複数個の半導体記憶装置のデータ入出力端子が個別に接続されるデータバスと、前記配線基板に設けられた複数個の外部端子と、前記データバスから前記外部端子に与えるデータを所定ビット数単位で並列から直列に変換し、前記外部端子から前記データバスに与えるデータを所定ビット数単位で直列から並列に変換する速度変換手段と、該メモリモジュールに対するアクセス制御情報がデータ入出力用と兼用された外部端子から前記速度変換手段を介して供給されるプロトコル制御回路とを有し、
- 10

- 前記プロトコル制御回路は、前記アクセス制御情報を解読して内部アドレス信号及び内部アクセス制御信号を生成し前記複数個の半導体記憶装置に並列的に供給するものであり、
- 15

- 前記メモリコントローラは、アクセス制御情報とアクセスデータとを前記メモリバスの共通の信号線に伝達して前記メモリモジュールをアクセス制御するものであることを特徴とするデータ処理システム。

- 20 13. 前記メモリバスは、その途中に、前段から後段に与える情報を所定ビット数単位で直列から並列に変換し、後段から前段に与える情報を所定ビット数単位で並列から直列に変換する速度変換アダプタを有し、前記速度変換アダプタの後段のメモリバスに前記メモリモジュールが結合されて成るものであることを特徴とする請求の範囲第12項に記載のデータ処理システム。
- 25

14. 前記速度変換アダプタの前段のメモリバスにも前記メモリモジュールが結合されて成るものであることを特徴とする請求の範囲第13項に記載のデータ処理システム。

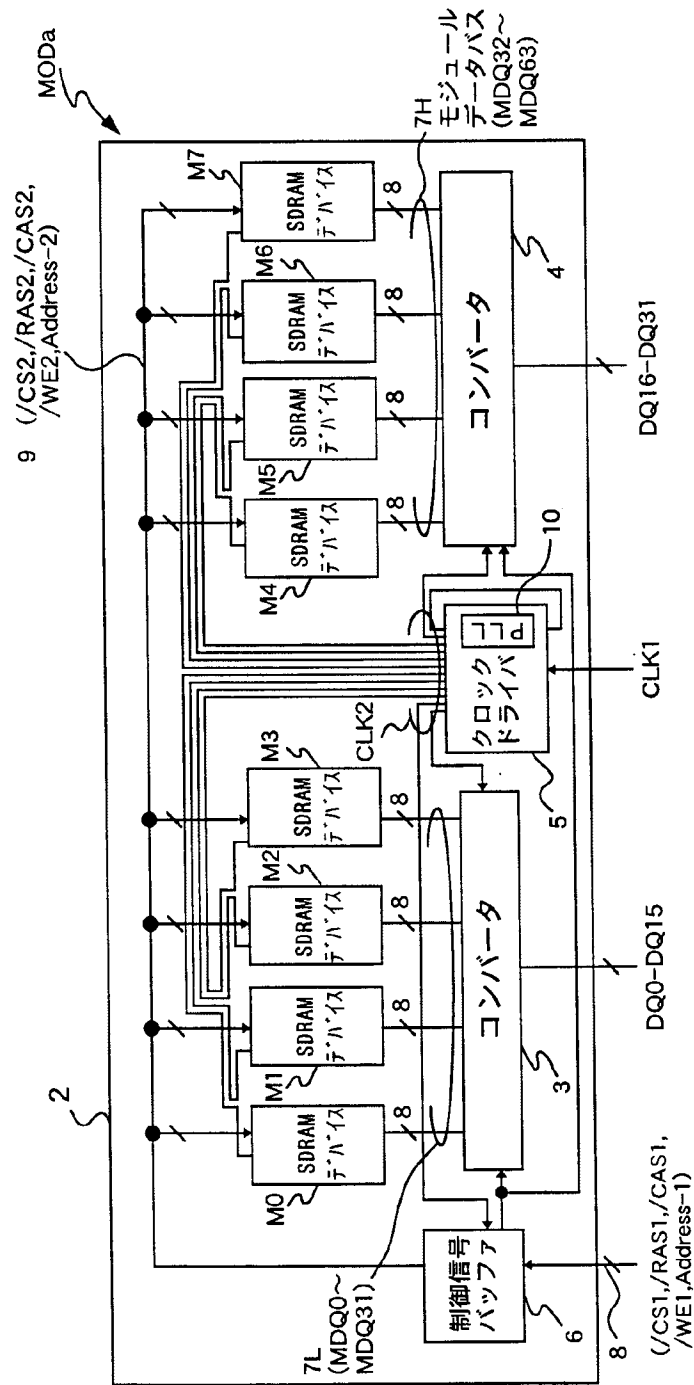
ジュールが結合されて成るものであることを特徴とする請求の範囲第 1 3 項に記載のデータ処理システム。

5 1 5 . 前記外部端子から供給される光信号を電気信号に変換し、前記外部端子から出力する信号を電気信号から光信号に変換する光電変換手段を更に備えて成るものであることを特徴とする請求の範囲第 7 項に記載のメモリモジュール。

10 1 6 . 前記外部端子の一部又は全部から供給される変調信号を復調し、前記一部又は全部の外部端子から出力する信号を変調する変復調手段を更に備えて成るものであることを特徴とする請求の範囲第 7 項に記載のメモリモジュール。

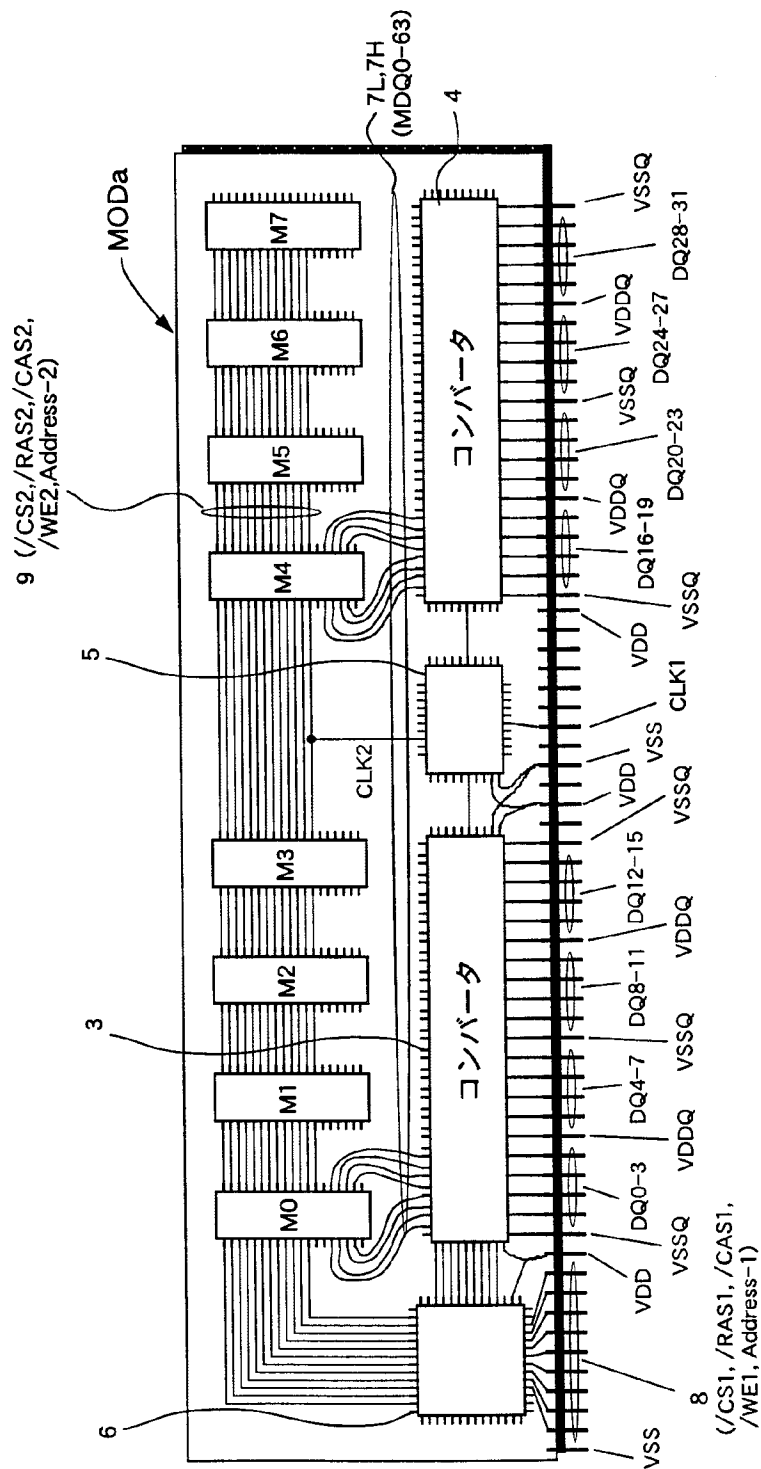
1 7 . 前記速度変換手段の前段又は後段に誤り訂正手段を設けると共に、前記誤り訂正手段で用いる誤り訂正コードを格納する半導体記憶装置を設けて成るものであることを特徴とする請求の範囲第 7 項に記載のメモリモジュール。

第1図



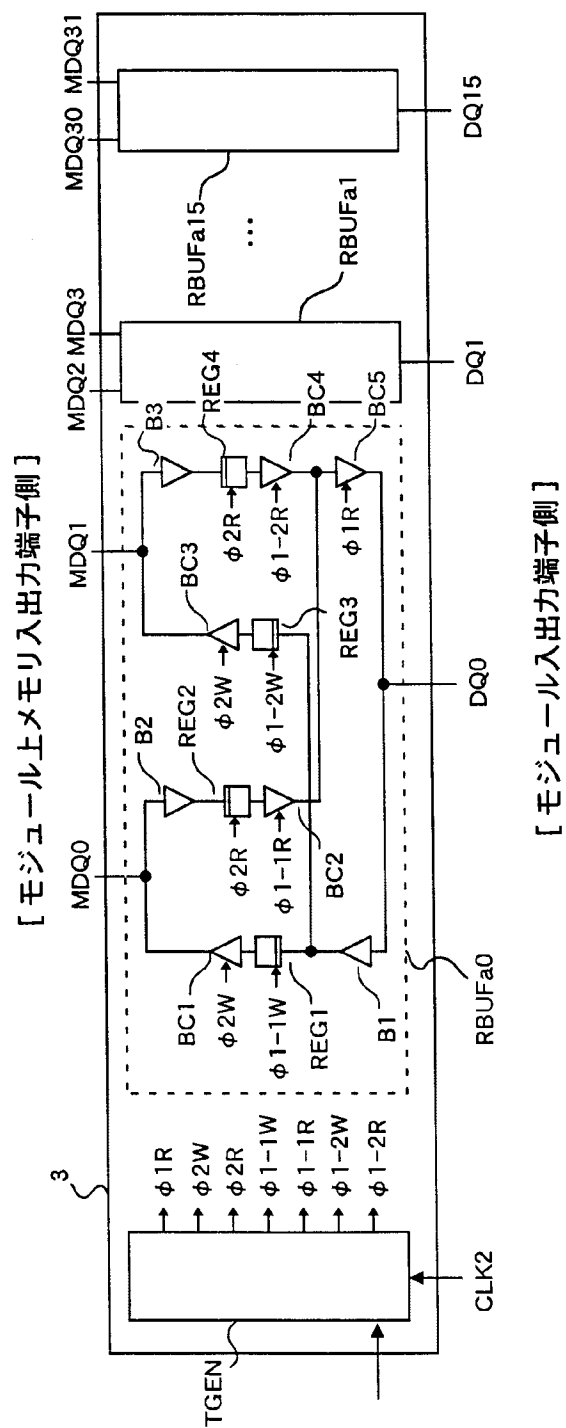
2 / 1 9

第 2 図



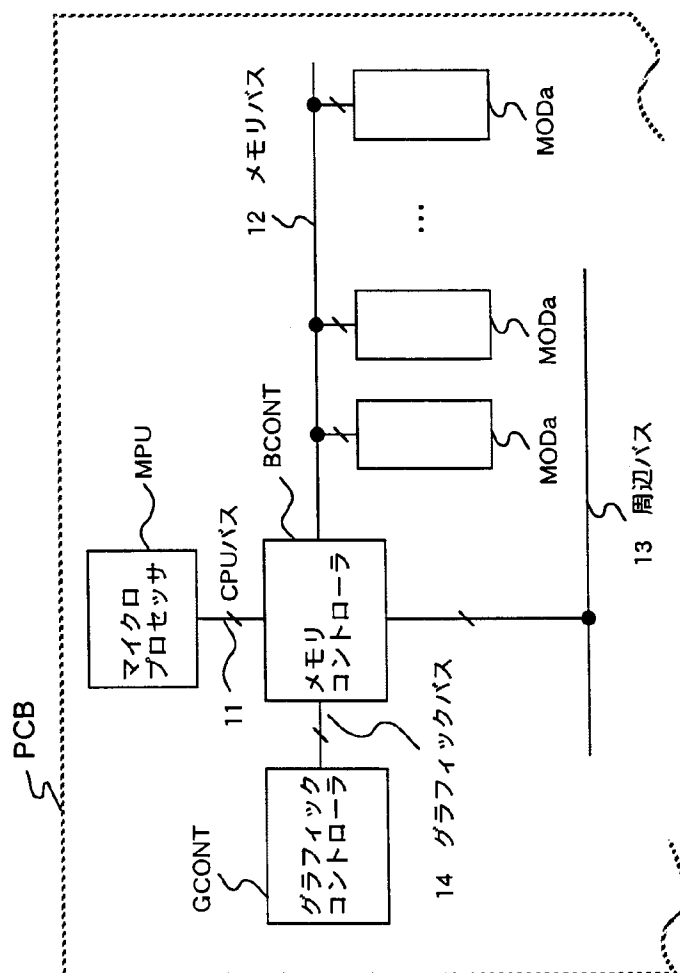
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第 3 図



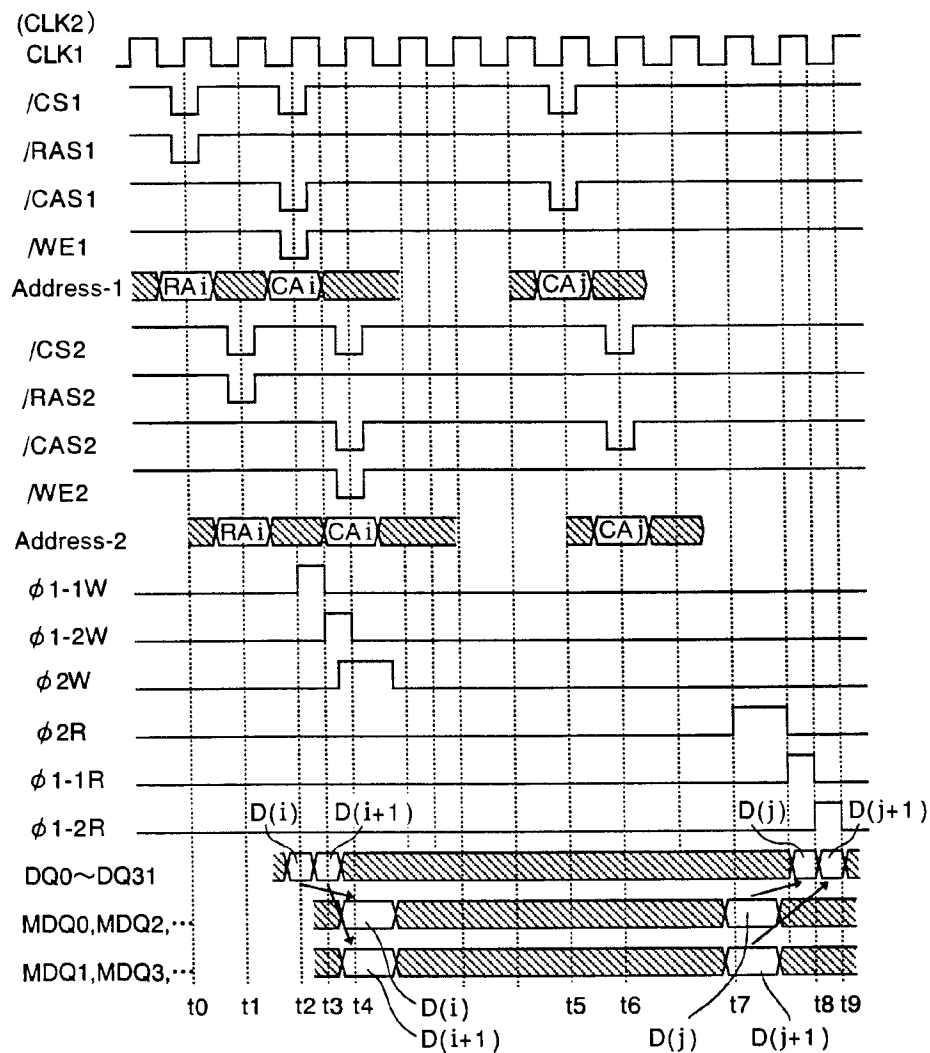
4/19

第4図



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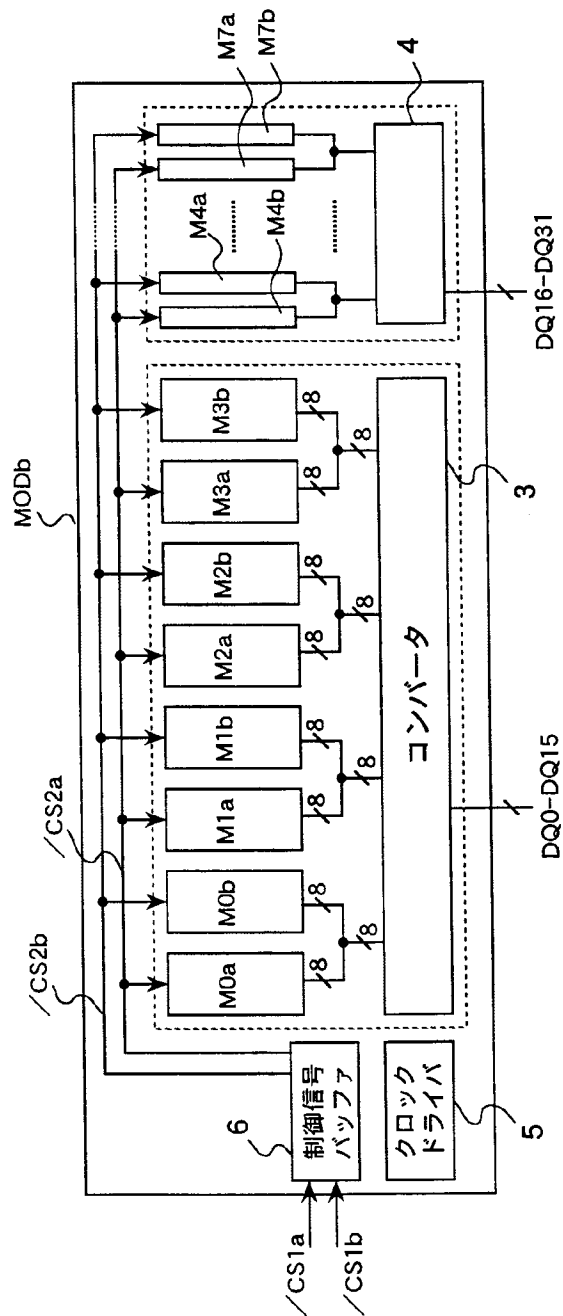
第5図



- t0: モジュールアクティブコマンド (from バスコントローラ)
 t1: メモリチップアクティブコマンド (from 制御信号バッファ)
 t2-t3: モジュールライトコマンド&ライトデータ (from バスコントローラ)
 t4: メモリチップライトコマンド&ライトデータ (from 制御信号バッファ)
 t5: モジュールリードコマンド (from バスコントローラ)
 t6: メモリチップリードコマンド (from 制御信号バッファ)
 t7: メモリチップリードデータ (from SDRAM)
 t8-t9: モジュールリードデータ (from メモリモジュール)

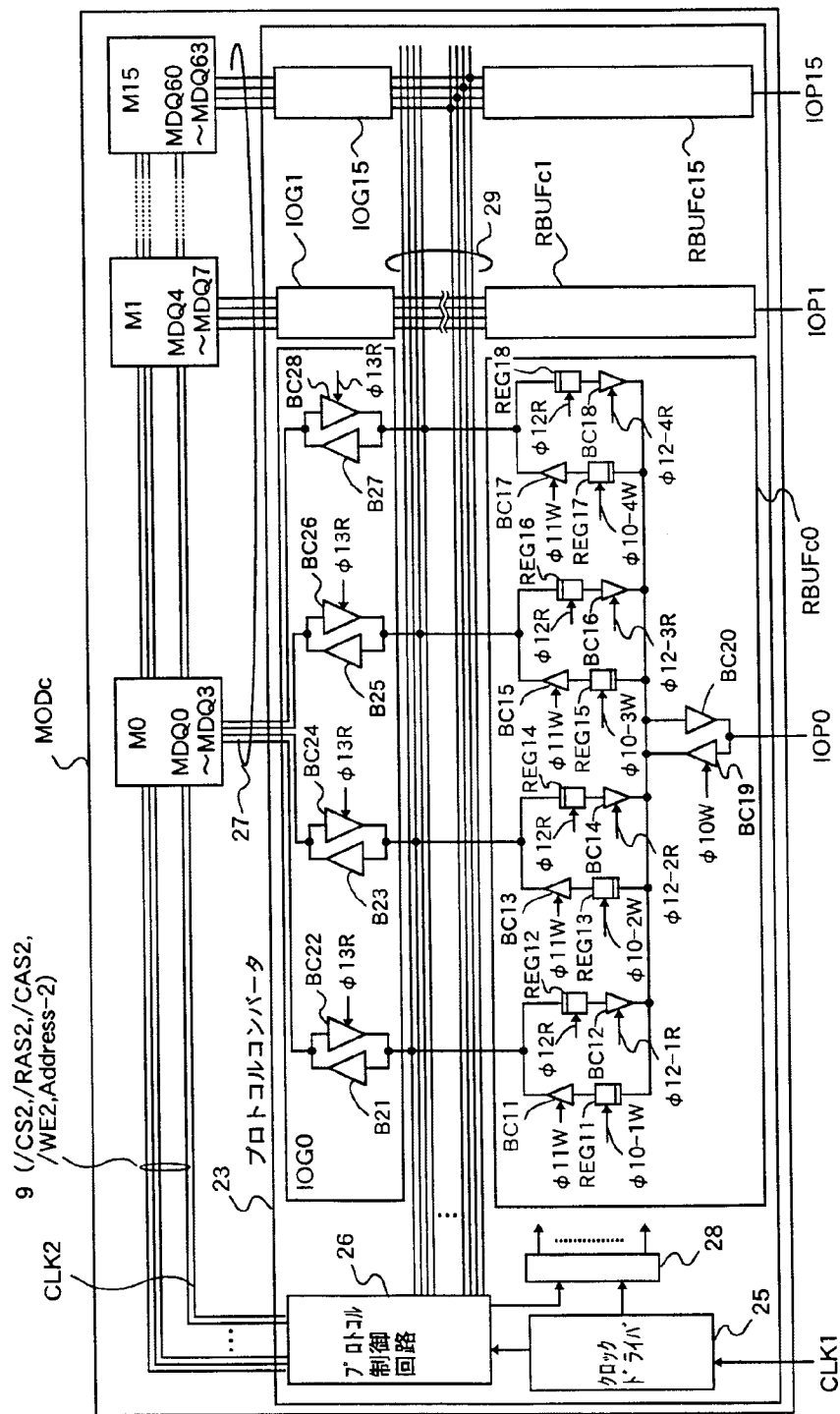
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第6図



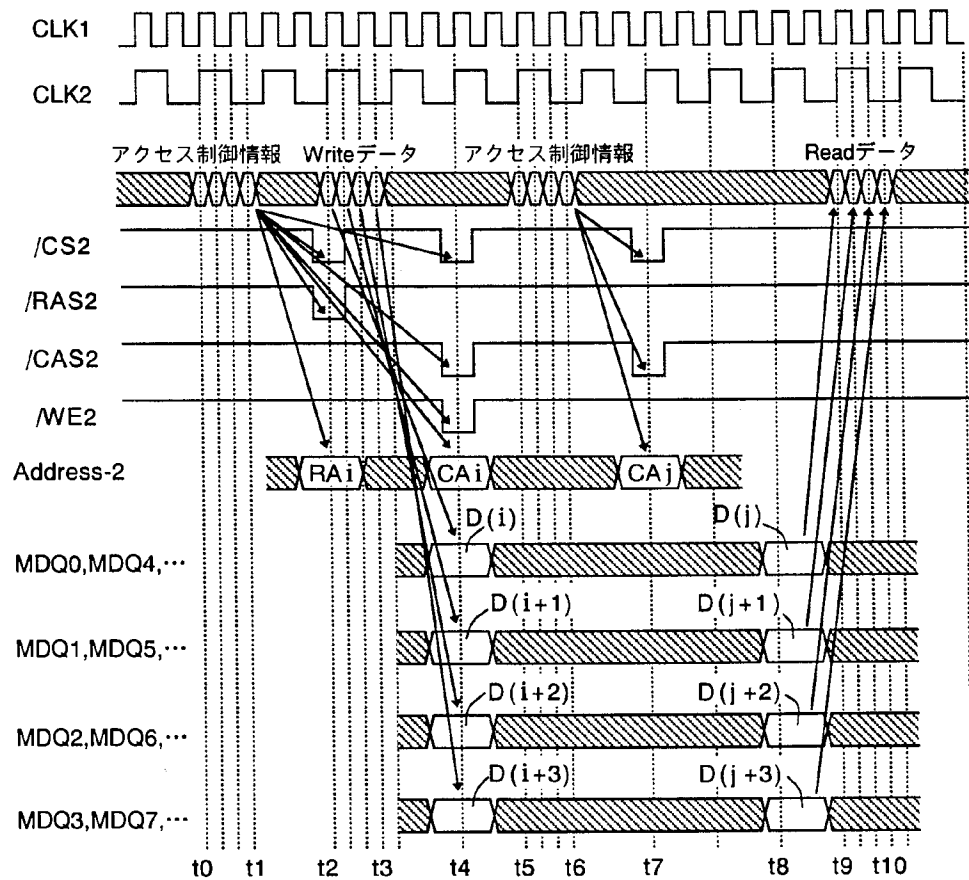
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第7図



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第8図



t0-t1: モジュールライトコマンド (from メモリコントローラ)

t2: メモリチップアクティブコマンド (from プロトコル制御回路)

t2-t3: モジュールライトデータ (from メモリコントローラ)

t4: メモリチップライトコマンド&ライトデータ (from プロトコル制御回路)

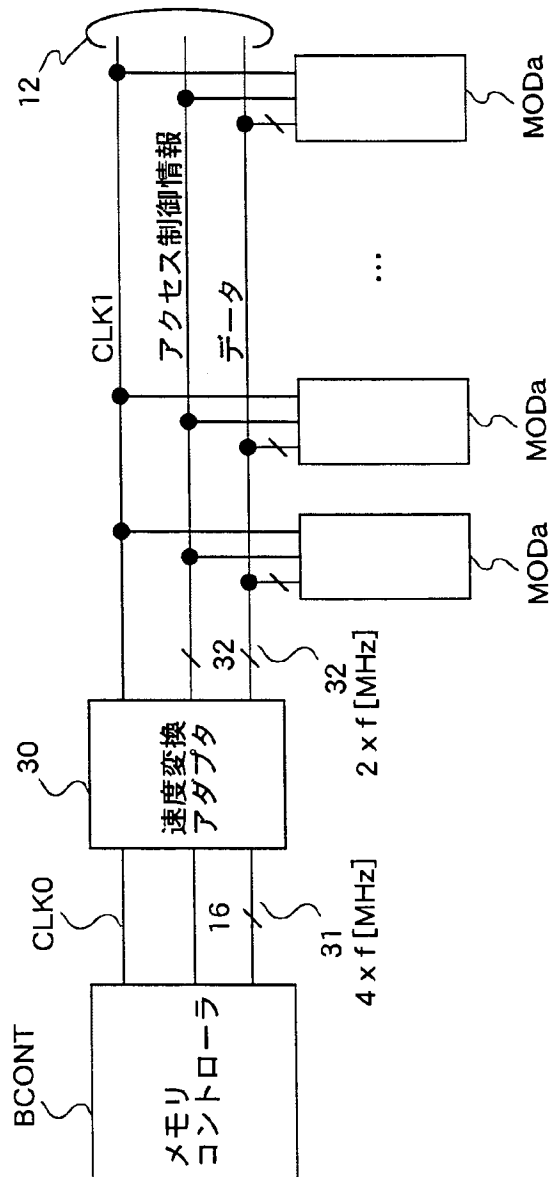
t5-t6: モジュールリードコマンド (from メモリコントローラ)

t7: メモリチップリードコマンド (from プロトコル制御回路)

t8: メモリチップリードデータ (from SDRAMデバイス)

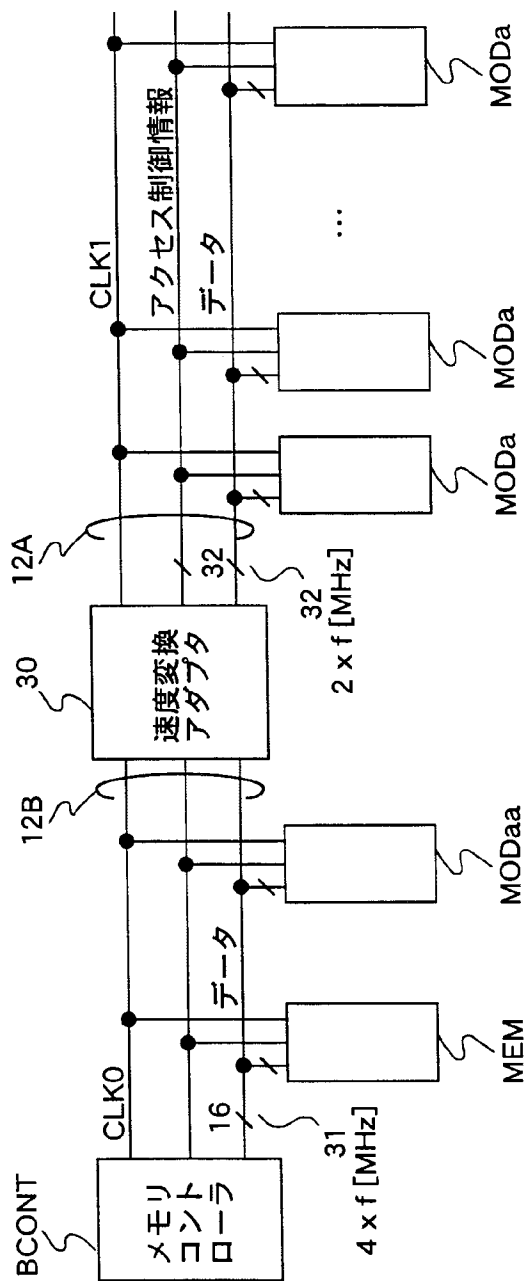
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第9図

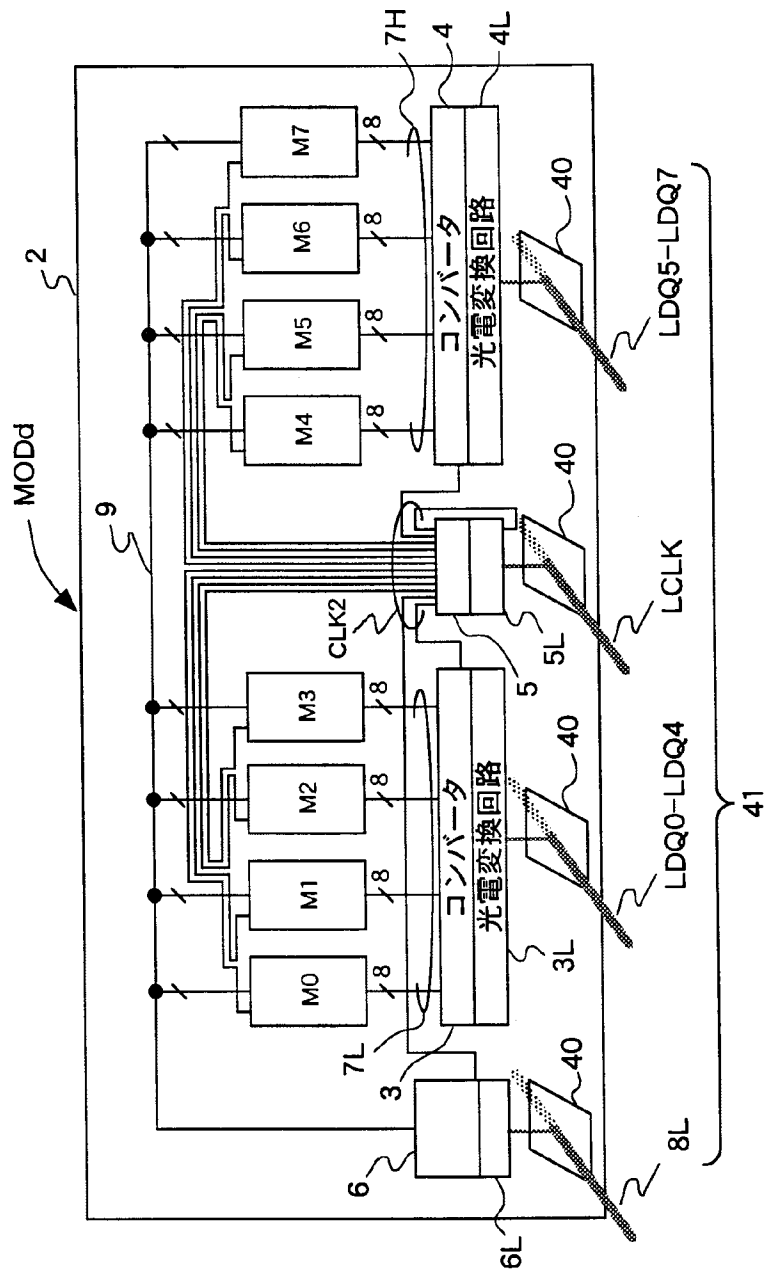


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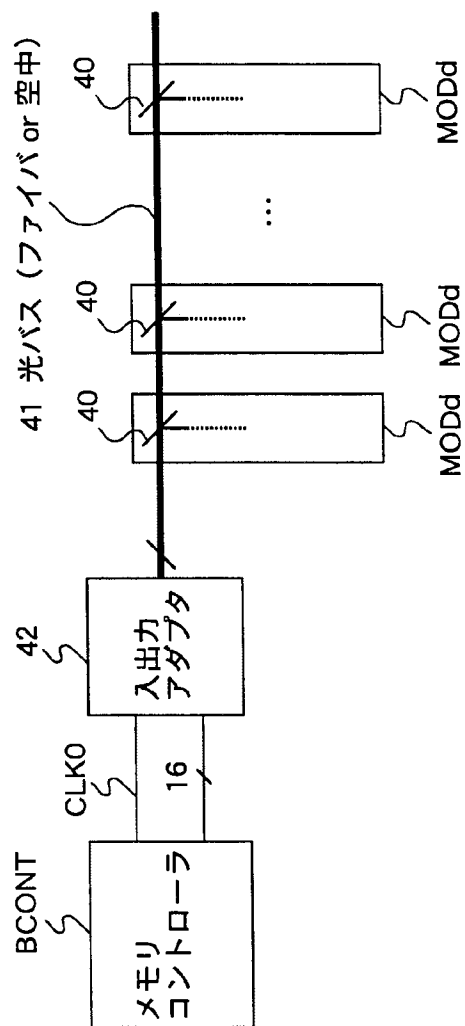


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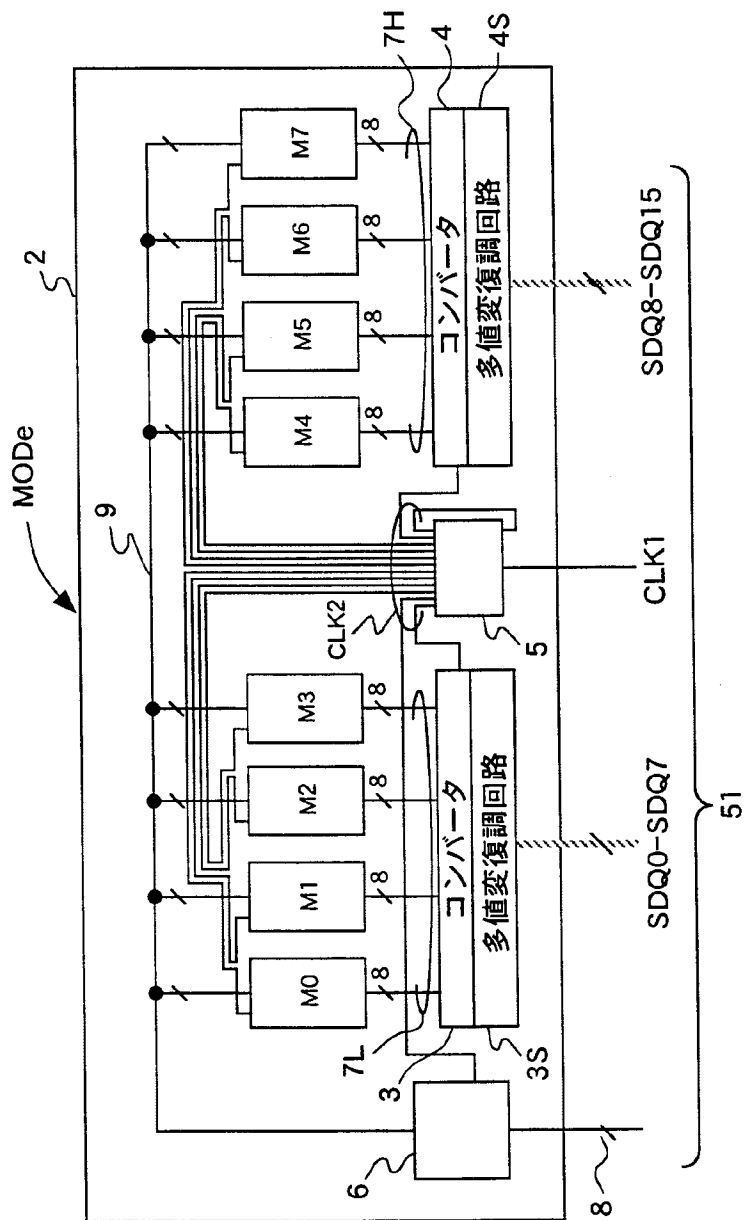


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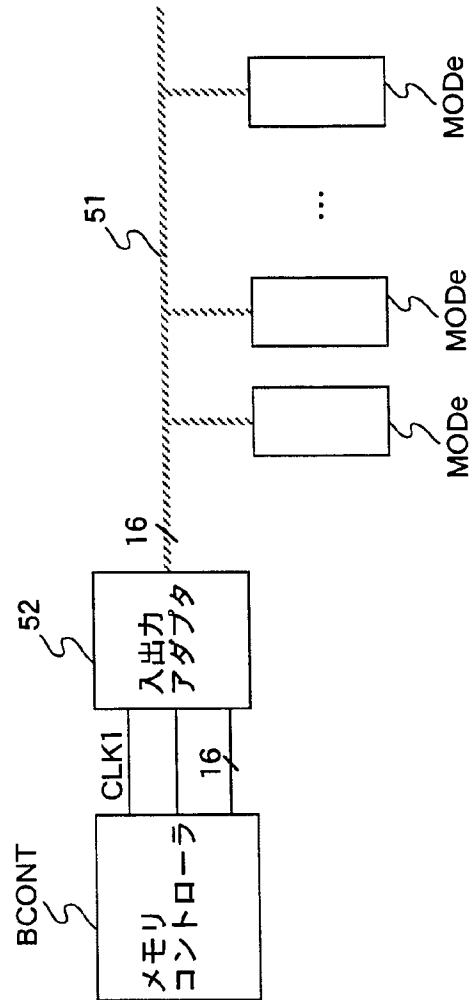
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第13図

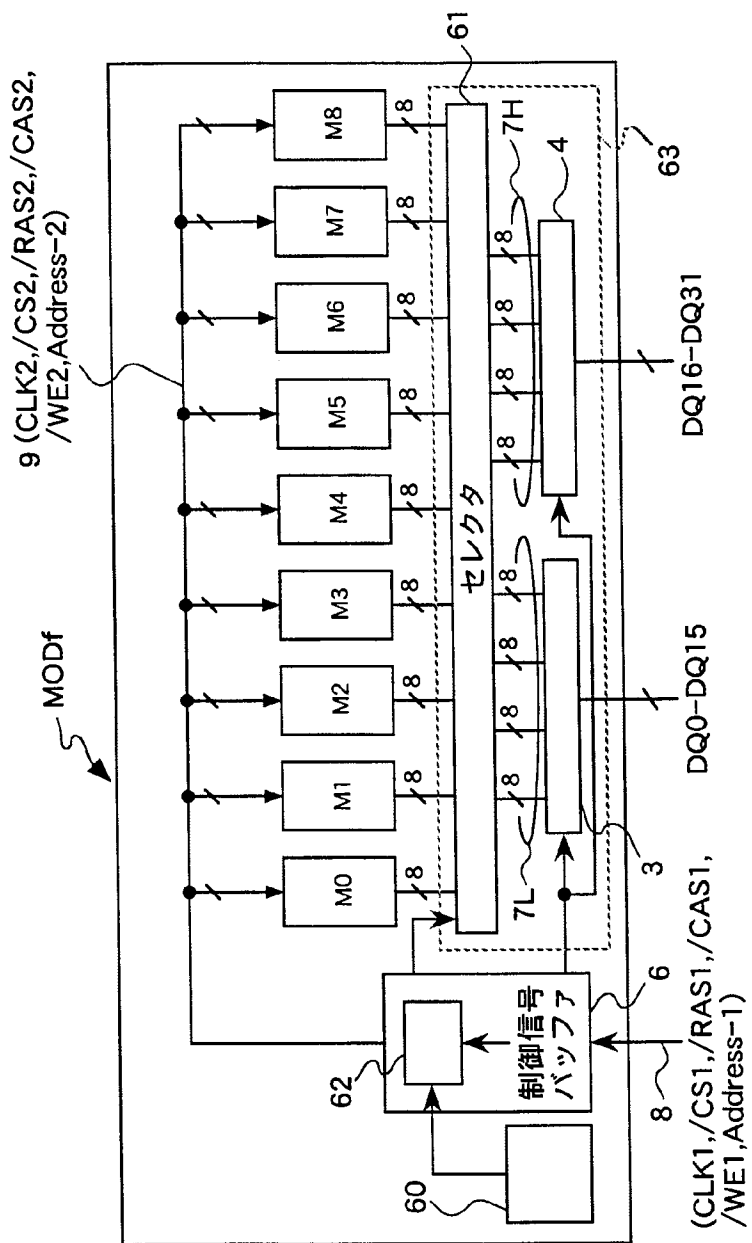


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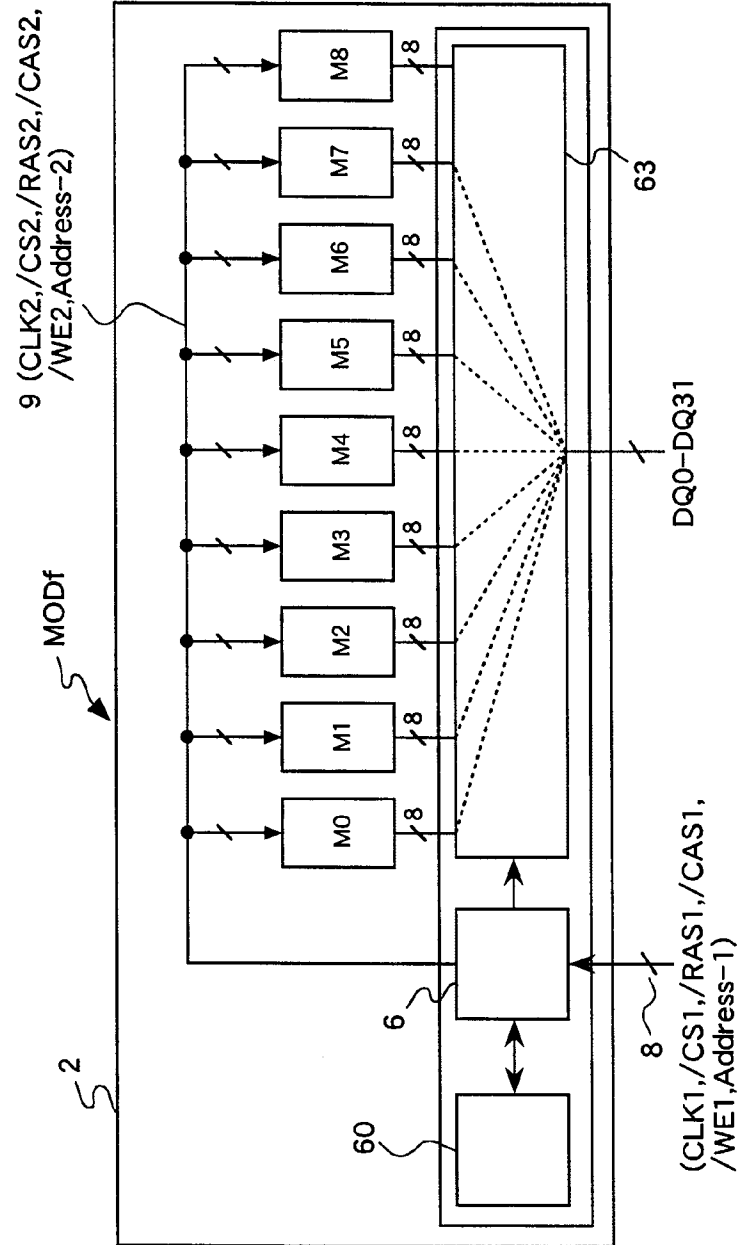
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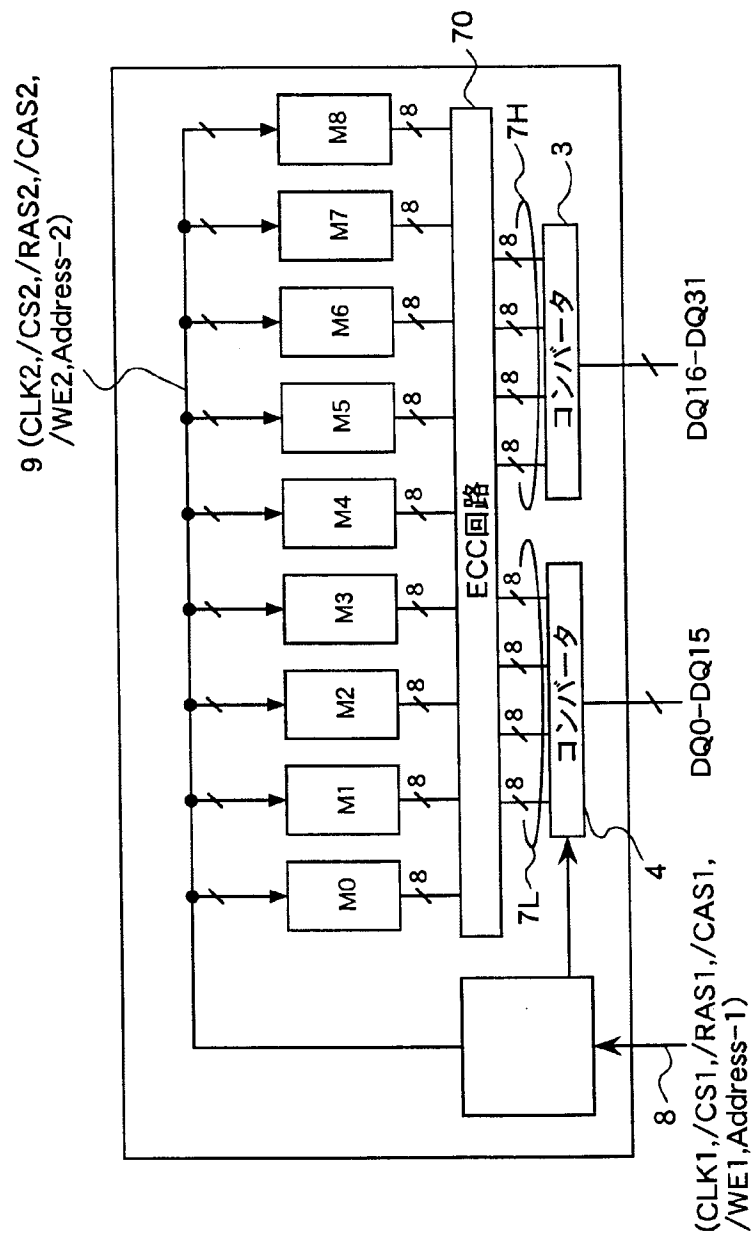
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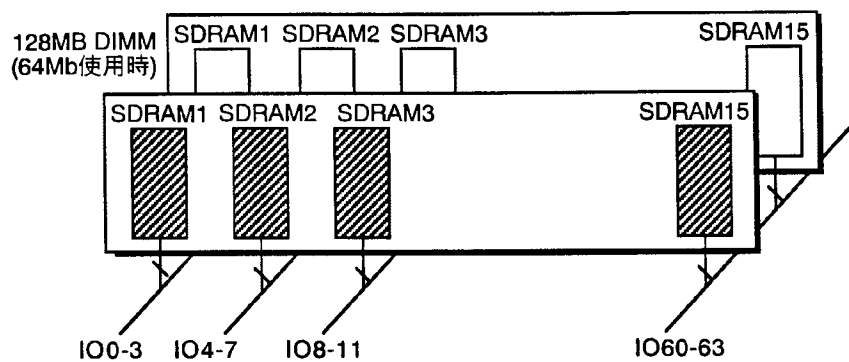
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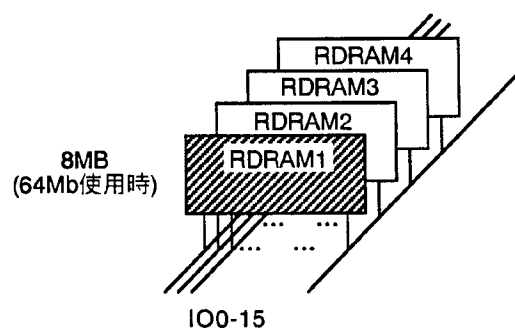


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第19図



第20図



INTERNATIONAL SEARCH REPORT

International application No.

PCT/JP98/02866

A. CLASSIFICATION OF SUBJECT MATTER
Int.Cl⁶ G06F12/06

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
Int.Cl⁶ G06F12/06Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched
Jitsuyo Shinan Koho 1922-1996 Toroku Jitsuyo Shinan Koho 1994-1998
Kokai Jitsuyo Shinan Koho 1971-1998 Jitsuyo Shinan Toroku Koho 1996-1998

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X Y A	JP, 62-121978, A (Hitachi ULSI Engineering Corp.), 3 June, 1987 (03. 06. 87) (Family: none)	1-2, 6 3-9, 11, 17 10, 12-16
Y A	JP, 8-77097, A (Ricoh Co., Ltd.), 22 March, 1996 (22. 03. 96) (Family: none)	3-5, 7-9, 17 10-12
Y	JP, 3-286234, A (Matsushita Electric Industrial Co., Ltd.), 17 December, 1991 (17. 12. 91) (Family: none)	4, 5
Y	JP, 1-171047, A (Fujitsu Ltd.), 6 July, 1989 (06. 07. 89) (Family: none)	5, 17
Y A	JP, 63-29357, B2 (Sony/Tektronix Corp.), 13 June, 1988 (13. 06. 88) (Family: none)	7-9 10-16

☐ Further documents are listed in the continuation of Box C.
 ☐ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier document but published on or after the international filing date

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"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

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"&" document member of the same patent family

Date of the actual completion of the international search
22 September, 1998 (22. 09. 98)Date of mailing of the international search report
6 October, 1998 (06. 10. 98)Name and mailing address of the ISA/
Japanese Patent Office

Authorized officer

Facsimile No.

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A. 発明の属する分野の分類 (国際特許分類 (IPC))
Int Cl⁸ G06F12/06

B. 調査を行った分野

調査を行った最小限資料 (国際特許分類 (IPC))
Int Cl⁸ G06F12/06

最小限資料以外の資料で調査を行った分野に含まれるもの

日本国実用新案公報 1922-1996年
日本国公開実用新案公報 1971-1998年
日本国登録実用新案公報 1994-1998年
日本国実用新案登録公報 1996-1998年

国際調査で使用した電子データベース (データベースの名称、調査に使用した用語)

C. 関連すると認められる文献

引用文献の カテゴリー*	引用文献名 及び一部の箇所が関連するときは、その関連する箇所の表示	関連する 請求の範囲の番号
X Y A	J P, 62-121978, A (日立超エル・エス・アイエンジニアリング) 3. 6月. 1987 (03. 06. 87) (ファミリーなし)	1-2, 6 3-9, 11, 17 10, 12-16
Y A	J P, 8-77097, A (株式会社リコー) 22. 3月. 1996 (22. 03. 96) (ファミリーなし)	3-5, 7-9, 17 10-12
Y	J P, 3-286234, A (松下電器産業株式会社) 17. 12月. 1991 (17. 12. 91) (ファミリーなし)	4, 5
Y	J P, 1-171047, A (富士通株式会社) 6. 7月. 1989 (06. 07. 89) (ファミリーなし) J	5, 17

☒ C欄の続きにも文献が列挙されている。

☐ パテントファミリーに関する別紙を参照。

* 引用文献のカテゴリー

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「E」 先行文献ではあるが、国際出願日以後に公表されたもの
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「P」 国際出願日前で、かつ優先権の主張の基礎となる出願

の日の後に公表された文献

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「&」 同一パテントファミリー文献

国際調査を完了した日

22. 09. 98

国際調査報告の発送日

06.10.98

国際調査機関の名称及びあて先

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特許庁審査官 (権限のある職員)

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(19) World Intellectual Property Organization
International Bureau



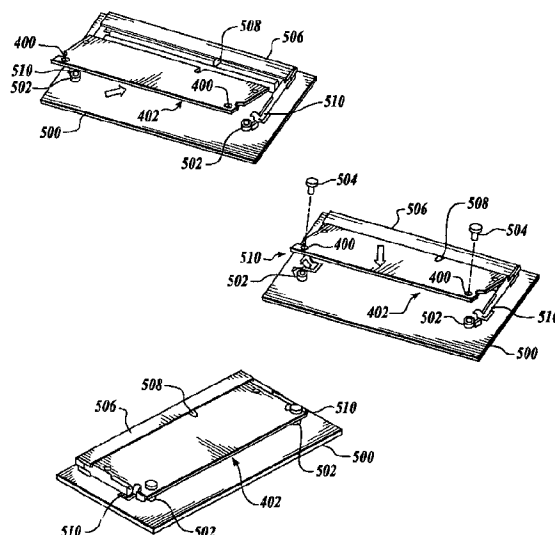
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- (21) International Application Number: PCT/US02/01575
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- (71) Applicant: HONEYWELL INTERNATIONAL INC. [US/US]; Law Department, 101 Columbia Road, P.O. Box 2245, Morristown, NJ 07962-2245 (US). Published: — without international search report and to be republished upon receipt of that report
- (72) Inventor: ANDERSON, David; 304 S. Market, Rock Port, MO 64482 (US). For two-letter codes and other abbreviations, refer to the "Guidance Notes on Codes and Abbreviations" appearing at the beginning of each regular issue of the PCT Gazette.

(54) Title: ENHANCED MEMORY MODULE ARCHITECTURE



(57) Abstract: A memory module architecture that supports Flash and static memory devices in addition to dynamic memory devices. The module architecture of the present invention preferably redefines standard application of chip select signals on existing module architectures to provide requisite signaling to support Flash and static RAM devices. Use of serial presence detect signaling features of standard memory modules is also modified to provide desired identity and parameters of such an enhanced module. A further aspect of the present invention provides for a support structure to immobilize the loose edge of the memory module opposite the electrical edge connector of the module to further enhance the module's resistance to vibration and mechanical shock by immobilizing the module with respect to rotation about the key in the socket connector.

WO 02/058069 A2

ENHANCED MEMORY MODULE ARCHITECTURE

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The invention relates to memory module architectures and in particular relates to an enhanced memory module configuration to provide additional flexibility in memory module applications in conjunction with PowerPC and other system applications.

2. Discussion of Related Art

[0002] Present-day computing systems often utilize standardized memory modules to add memory capacity to a system. A memory module is generally comprised of a number of memory chips mounted on a printed circuit card adapted for insertion into a standard carrier or socket connector. The module is easily added or removed from a system to permit flexible configuration of memory capacity. The module and socket adhere to a standardized pin assignment for exchange of signals between a system bus and the memory chips. Signals are routed from the carrier connector to the memory chips in accordance with the particular standard used for the module and the particular memory chips selected by the memory module manufacturer.

[0003] A number of standards are defined for such modules. Exemplary of one such standard is a so-called 144-pin Small Outline DIMM module (also referred to as "SO-DIMM") as defined by the JEDEC JESD21-C standard specification (available online at www.jedec.org). The 144-pin SO-DIMM provides a convenient, compact design for use in space-constrained applications such as portable computing devices or embedded systems. The module is a small printed circuit card with memory devices mounted thereon and one edge that provides electrical connectivity of up to 144 signals when the module is inserted into a mated connector on the main system board (i.e., the system motherboard). The printed circuit card signal traces connect standard signals assigned to the 144 pins of the module's edge connector to appropriate pins of the memory chips mounted thereon. A memory controller on the system board exchanges these standard memory control signals with the module through the mated connector.

[0004] In particular, the 144-pin standard module provides for memory chips to be mounted on one or both physical sides of the printed circuit board. A select signal is typically applied to the module by a CPU or memory controller to select the memory chips on one or the other side of the memory module for exchange of data and control signals. This select signal is often referred to as a "row select" signal in regards to the 144-pin standard configuration. When the "front" side is selected by the memory controller, only those chips respond to the exchange of signals with the system board. Similarly, when the "back" side is selected, only those memory chips respond to the memory controller signals.

[0005] Most memory modules and carriers define signal connections for standard DRAM memory devices (dynamic random access memory) mounted on the module. Such signal definitions include signals common to most DRAM devices including, in particular, SDRAMs (synchronous DRAM devices). Other forms of memory, in particular asynchronous memories, are not generally accommodated by the standard signal paths provided for on the standard DIMM memory module. For example, present standard DIMM modules such as the 144-pin SO-DIMM do not support so-called Flash memory devices. Neither are static RAMs (SRAM devices) generally supported by such standard DIMM modules.

[0006] This presents a particular problem to embedded applications where a wide variety of memory devices are often applied to resolve special problems in particular applications. For example, the non-volatile nature of Flash memory devices makes them particularly useful in many embedded applications for storage of computer program instructions while a standard SDRAM may be used for storage and retrieval of program data. Since no single standard memory module supports such a variety of memory devices, it is often necessary to design embedded applications for multiple memory module architectures and associated busses and/or to redesign the system as the memory requirements change or new memory components become available. For example, it is common in many such embedded applications to design a first memory bus for exchange of signals with synchronous memory components (i.e., SDRAMs) and a separate slower memory bus for asynchronous devices such as Flash or static RAMs. This added complexity adds cost and size to the design and complicates portability of a system design to multiple applications that may have different needs for memory configuration.

[0007] Furthermore, it is a problem for many such memory module designs to be used in embedded applications where there is significant vibration or mechanical shock. Standard DIMM modules insert into a mated socket at one edge. Some DIMM modules are designed to latch or lock on two other sides of the module. However, the edge opposing the socket is often unsupported and may lead to failures in applications of high vibration and mechanical shock. In particular it has been found that under certain common vibrational modes, the SO-DIMM module may twist about a pivot point at the key in the DIMM socket. Sufficient twisting due to such vibration can lead to unacceptable, premature DIMM module failure.

[0008] It is evident from the above discussion that a need exists for an improved memory module design that permits a wider variety of memory device types to be used in a standard DIMM socket on a common bus. It is further evident that a need exists for an improved mechanical design for a memory module that better withstands high vibration environments or environments that may entail significant mechanical shock.

SUMMARY OF THE INVENTION

[0009] The present invention solves the above and other problems, thereby advancing the state of the useful arts, by providing a modified memory module design that accommodates a wider variety of memory devices. In particular, the modified memory module architecture of the present invention redefines a number of standard signals to permit utilization of Flash and static memory components in a DIMM module while maintaining backward compatibility to standard DRAM and SDRAM DIMM modules. Still more specifically in one exemplary preferred embodiment, the present invention preferably utilizes row select signals ("CS(0:1)") of a standard 144-pin SO-DIMM module to provide requisite select and control signals for Flash and static memory devices.

[0010] Still more specifically, where a memory controller is capable of multiplexing both synchronous and asynchronous memory device accesses over the same signal paths, the present invention permits use of Flash and other special memory modules that are physically compatible with SO-DIMM configurations in conjunction with typical SDRAM (or other DRAM) SO-DIMM modules over the same multiplexed memory controller bus signals. For example, the Motorola family of embedded PowerPC microcomputers (MPC824x and others) includes such a memory controller -- specifically the MPC107 memory controller. The MPC106 or MPC107 memory controller device

provides such a memory controller feature and is integrated into the MPC824x chip die and available as a separate integrated circuit component. Systems using the MPC106 or MPC107 memory controller as a separate component may also benefit from the present invention. Further, other similarly featured memory controllers are commercially available and systems using such similarly featured memory controller may beneficially apply the features of the present invention.

[0011] In one particularly useful embodiment, the signals of a JEDEC JESD21-C 144-pin SO-DIMM standard memory module are modified to enable support of Flash and SRAM memory devices on the modified standard DIMM module. In particular, the chip select (row select) signals are modified to provide the requisite selection of Flash and SRAM memory components in addition to standard SDRAM (or DRAM) memory components. Setting of jumpers on the system board preferably indicate the presence of such additional memory device types in the SO-DIMM sockets of the system and adapt the signal exchange with a memory controller to conform to the signaling requirements of the special module. A single bus structure supports both synchronous and asynchronous memory device signals and couples the memory controller to multiple SO-DIMM sockets. Each DIMM socket may be populated with a Flash or SRAM special memory module or with a standard SDRAM or DRAM memory module. The design may then be easily ported to other applications requiring a different compliment of memory device types.

[0012] In this first exemplary preferred embodiment, a variety of memory configurations are achievable. In this first exemplary preferred embodiment, the system utilizes a DIMM module similar to that defines as the 144-pin SO-DIMM standard referenced above. In such memory modules, the module may be populated on one or both sides of the module printed circuit card with memory devices. All memory devices on one side typically respond to the same select signal. The select signal is usually referred to as a "row select" as noted above. As used herein, the terms "row" and "side" are used synonymously with respect to such SO-DIMM modules. Those skilled in the art will note however that it is not required that all memory devices on one side of a module respond to a single select signal. Rather, in accordance with the module standard, the physical place of the memory devices on the module is irrelevant with respect to the select signals. Use of the terms "row" and "side" as essentially synonymous is therefore not intended to limit the invention to a particular physical layout of memory modules. Rather, the terms

are both used to broadly refer to a first subset of memory devices on a module that respond to a first row select signal and a second subset of memory devices that respond to a second row select signal.

[0013] With two DIMM sockets in an exemplary system configuration, each socket may be populated with a standard SDRAM SO-DIMM where the DIMM is limited to single row configurations (i.e., SDRAM memory devices responding to only one row select of the module). Alternatively, only one DIMM socket may be populated with a double-row SO-DIMM module. The double-row DIMM may provide SDRAM devices on both rows, SDRAM on one row and Flash or other asynchronous devices on the other. In yet another alternative, a first single row DIMM may provide SDRAM memory devices and a second single row DIMM may provide Flash or other asynchronous memory devices. Jumper configurations on the system printed circuit board can then be used to route the row select signals as appropriate for the particular application.

[0014] In an alternative embodiment of the present invention, use of the serial presence detect ("SPD") signaling standard of DIMMs provides the required identity of the special modules supporting the extended range of memory device types. The SPD feature of memory modules provides for a simple serial interface (usually I²C protocol compliant) to return to the system certain identification and configuration information regarding the memory module. The memory controller of the system is then programmed to adapt its signaling as required for the sensed memory configuration. An application specific integrated circuit (ASIC) could be used in addition to a memory controller or in place of a memory controller to provide desired address processing. In general, a system CPU would preferably sense the configuration of memory modules in the sockets by reading and decoding the SPD information and program the memory controller and/or ASIC device to provide appropriate address decoding and timing according to the specific compliment of memory modules found in the system.

[0015] Another aspect of the present invention provides a support structure for the DIMM edge opposite the connector edge to enhance the module's resistance to vibration and mechanical shock. The present invention provides a support structure associated with the edge of a SIMM module opposite the connector edge of the DIMM. The support structure prevents a module inserted in the SO-DIMM socket from rotating or twisting within the socket connector. In particular, the present invention preferably prevents the module from twisting about an axis centered at the key of the module socket.

[0016] In one exemplary preferred embodiment representing the best presently known mode of practicing the invention, the module is adapted with holes at the corners, each overlaying a swaged extension nut or standoff on the motherboard or system board. The holes and associated nut or standoff are preferably positioned at the corners of the module opposite the connector edge of the module. The module is inserted in the mated connector and preferably secured with screws or other fasteners to the standoff or extension nut to immobilize the module and prevent undesired twisting of the module in the socket.

[0017] A first feature of the invention provides for a memory module comprising: a printed circuit assembly having connector pads at one edge of the assembly; and a plurality of memory devices mounted on the assembly and electrically coupled to the connector pads such that the printed circuit assembly is adapted to support both synchronous and asynchronous types of the memory devices.

[0018] Another aspect further provides that each of the plurality of memory devices is a synchronous dynamic random access memory device.

[0019] Another aspect further provides that each of the plurality of memory devices is a synchronous Flash memory device.

[0020] Another aspect further provides that each of the plurality of memory devices is an asynchronous Flash memory device.

[0021] Another aspect further provides that each of the plurality of memory devices is an asynchronous static random access memory device.

[0022] Another aspect further provides that each of the plurality of memory devices is an asynchronous fast static random access memory device.

[0023] Another aspect further provides that each of the plurality of memory devices is an asynchronous low power static random access memory device.

[0024] Another aspect further provides that the connector pads include: a first select signal connector pad that selects a first subset of memory devices mounted on the module when a signal is applied thereto such that the first subset of memory devices are synchronous memory devices; and a second select signal connector pad that selects a second subset of memory devices mounted on the module when a signal is applied thereto such that the second subset of memory devices are asynchronous memory devices.

[0025] Another aspect further provides that the connector pads include: a first select signal connector pad that selects a first subset of memory devices mounted on the

module when a signal is applied thereto such that the first subset of memory devices are synchronous dynamic random access memory devices; and a second select signal connector pad that selects a second subset of memory devices mounted on the module when a signal is applied thereto such that the second subset of memory devices are synchronous Flash memory devices.

[0026] A second feature of the invention provides for a system comprising: a system board; a memory bus adapted for exchanging signals between a memory controller and both synchronous and asynchronous memory devices; a memory controller on the system board coupled to the memory bus such that the memory controller is capable of generating signals for control of both synchronous and asynchronous memory devices and such that the memory controller is capable of multiplexing the signals on the memory bus; a first socket connector on the system board for receiving a first memory module such that the first socket connector is coupled to the memory controller through the memory bus; and a first memory module inserted in the first socket connector and electrically coupled to the memory controller such that the first memory module includes a plurality of synchronous or asynchronous memory devices.

[0027] Another aspect further provides for a jumper on the system board for configuring signals exchanged between the memory controller and the first memory module in accordance with the type of memory devices on the first memory module.

[0028] Another aspect further provides that the first memory module provides serial presence detect information used in conjunction with the memory controller to identify the type of memory devices included on the first memory module.

[0029] Another aspect provides for a second socket connector on the system board for receiving a second memory module such that the second socket connector is coupled to the memory controller through the memory bus; and a second memory module inserted in the second socket connector and electrically coupled to the memory controller such that the second memory module includes a plurality of synchronous or asynchronous memory devices.

[0030] Another aspect further provides for a jumper on the system board for configuring signals exchanged between the memory controller and the second memory module in accordance with the type of memory devices on the first memory module and on the second memory module.

[0031] Another aspect further provides that the first memory module provides serial presence detect information used in conjunction with the memory controller to identify the type of memory devices included on the first memory module, and that the second memory module provides serial presence detect information used in conjunction with the memory controller to identify the type of memory devices included on the second memory module.

[0032] A third feature of the invention provides for a system comprising: a system board; a socket connector on the system board for receiving a memory module such that the socket connector includes a key; a memory module having a connector edge inserted in the socket connector and having an opposing edge opposite the connector edge such that the memory module has a notch mated to the key when the memory module is inserted in the socket connector; and a memory module retainer adapted to substantially immobilize the opposing edge with respect to rotation about the key.

[0033] Another aspect further provides that the memory module retainer comprises: a nut affixed to the system board; a hole in the memory module along the opposing edge and aligned with the nut; and a screw inserted through the hole into the nut to substantially immobilize the opposing edge with respect to rotation about the key.

[0034] Another aspect provides that the nut is a swaged extension nut.

[0035] Another aspect provides that the memory module retainer comprises: a half card-cage affixed to the system board such that the half card-cage includes a channel for receiving the opposing edge of the memory module to substantially immobilize the opposing edge with respect to rotation about the key.

[0036] Another aspect provides that the memory module retainer comprises: a standoff pin affixed to the system board; a hole in the memory module along the opposing edge and aligned with the pin such that the standoff pin extends through the hole and locks when the memory module is completely inserted in the socket connector to substantially immobilize the opposing edge with respect to rotation about the key.

[0037] Another aspect provides that the memory module retainer comprises: a standoff pin affixed to the socket connector; a hole in the memory module aligned with the standoff pin such that the standoff pin extends through the hole and when the memory module is completely inserted in the socket connector to substantially immobilize the opposing edge with respect to rotation about the key.

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] Figure 1 is a block diagram of an exemplary preferred embodiment of a system using enhanced memory modules in accordance with the present invention.

[0039] Figure 2 is a schematic diagram showing relevant portions of an exemplary preferred embodiment of a DIMM socket connector with modified row select signals in accordance with the present invention.

[0040] Figure 3 is a block diagram depicting a typical system as presently practiced in the art with separate bus structures for support of different memory types.

[0041] Figure 4 is a dimensioned diagram of a memory module with holes for securing the module the system board in accordance with one exemplary preferred embodiment of he present invention.

[0042] Figure 5 is a sequence of diagrams depicting insertion of a memory module into a connector socket and securing the module to the system board in accordance with one exemplary preferred embodiment of he present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0043] While the invention is susceptible to various modifications and alternative forms, a specific embodiment thereof has been shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that it is not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

[0044] Figure 1 is a block diagram of a system 100 in which an embedded microcomputer interfaces with a variety of memory modules 116 and 118 via memory bus 150 and standard memory module socket connectors 114. Embedded microcomputer 102 preferably includes at least a processor 104 and a memory controller 106 to control and interact with memory modules 116 and 118. In one exemplary preferred embodiment of such a system, embedded microcomputer 102 may be a Motorola PowerPC MPC8245 or other similar members of that embedded microcomputer family. The MPC8245 includes a memory controller for interaction with, and control of, a variety of memory devices.

[0045] Figure 3 shows a system 300 using the same embedded microcomputer 102 adapted as known in the art to interact with standard SO-DIMM SDRAM (or DRAM) memory modules through a synchronous bus structure 350 while interacting with other memory devices (Flash 318 or SRAM 320) through a slower asynchronous memory

bus 352. As noted above, such a structure requires each system application to uniquely adapt to a particular memory configuration. The system cannot be easily re-used for different applications requiring different memory configurations.

[0046] Returning now to figure 1, by contrast and in accordance with the present invention, the memory control signals are all directed over the same multiplexed memory bus 150 to memory modules 116 and 118. The memory modules 116 and 118 are preferably adapted to SO-DIMM standard mechanical dimensions. As noted above, memory controller 106 is capable of multiplexing both synchronous memory control signals and asynchronous memory control signals over the same bus. Such a memory controller is embedded within an MPC8245 and other embedded microcomputers. Further, as noted above, those skilled in the art will readily recognize equivalent system architectures in which the memory controller element 106 is a separate and distinct integrated circuit component not embedded within the same chip die as the processor. For example, the MPC106 or MPC107 memory controller as well as other commercially available memory controller devices provide the desired multiplexing of synchronous and asynchronous memory signals over a common memory interface bus.

[0047] The memory bus signals are preferably directed through standard DIMM memory module connectors 114 to any of a variety of DIMM memory modules. For example, standard 144-pin SO-DIMM memory module 116 may provide standard DRAM or SDRAM memory devices. Flash/SRAM memory module 118 is adapted in accordance with the present invention to provide a 144-pin interface similar to that of the standard 144-pin DIMM 116. DIMM memory module 118 is adapted in accordance with the present invention to apply control signals from memory controller 106 to Flash or static memory devices (i.e., SRAM devices including low power SRAM, high-speed SRAM, etc.). Further, as noted above, a custom DIMM module may be utilized that provides SDRAM (or other DRAM) devices responding to a first row select signal and Flash or other asynchronous devices that respond to a second row select signal.

[0048] Other components may be embedded within microcomputer 102 including, for example, a DMA controller 108, integrated I/O peripheral device controllers 110, and external interface bus controllers 112 (i.e., PCI, PCMCIA, etc.). Those skilled in the art will recognize that the devices shown in figure 1 are intended as one exemplary embodiment of a system that may advantageously apply the enhanced memory architecture of the present invention.

[0049] As noted above, one standard DIMM configuration is defined by JEDEC specification JESD21-C as a 144-pin SO-DIMM memory module. This particular memory module is useful in a number of applications due to its small footprint. The small module is effective in portable systems including avionics systems and other embedded systems to conserve space required for memory expansion.

[0050] In one exemplary preferred embodiment of the invention, the signals of such a standard 144-pin DIMM are adapted to permit use of a wider variety of memory devices in the memory module configuration. Those skilled in the art will recognize that the features of the present invention may be easily adapted to other memory module configurations and standards. The exemplary preferred embodiment of the invention as a 144-pin SO-DIMM structure is not intended to limit the applicability of the invention to other memory modules. Rather, it is intended to express one particularly useful application for embedded and portable system applications.

[0051] The flexible memory architecture of the present invention allows a single system design to be applied to a number of different applications potentially requiring different memory configurations. By way of example, a first system may provide a large system such as an avionics Primary Flight Display. Such systems are often referred to as "Level A" or "mission critical" in the avionics and aerospace industries. Such a large system may use of a full 72-bit (64 + 8 check bits) ECC SDRAM module in a first DIMM module socket of the system to enhance system reliability through error correction. This memory module would preferably be a single row SO-DIMM module selected by a "front-side" select pin of a standard DIMM socket. As noted herein above, use of the term "side", "front-side" or "back-side" is to be understood as synonymous with "row" as defined for the standard 144-pin SO-DIMM module of an exemplary preferred embodiment. This SDRAM module would serve as storage for dynamic data and scratchpad memory for the system. Those skilled in the art will note that the number and size of memory devices presently commercially available and required to implement a 72-bit ECC module would consume both sides of the module. However, expected advancements and market demands will easily allow for such a memory module in a single row configuration. The application program instructions in such a system would preferably be stored in a Flash memory module inserted in the second module socket. The Flash module would also preferably be a single row module selected by a "back-side" select pin on the SO-DIMM standard socket. The width, depth, and timing characteristics

for Flash module memory accesses are preferably programmed into the memory controller by boot code operable at system startup and reset.

[0052] A second system application of the same system board could provide a very small system, such as a Frequency Control Head for a remote radio. The criticality of such a system application may be such that ECC reliability is not required on the data memory. Also the data memory and application size and performance requirements may be such that a 32-bit bus width is sufficient. In this case the data memory would preferably be 32-bit wide SDRAM totaling perhaps 4 megabytes in an SO-DIMM module populated on one side with SDRAM devices. The SDRAM devices are preferably selected by a "front-side" select pin of the DIMM socket. The application program instructions could be stored in 32-bits of SyncFlash™ (a trademarked technology of Micron Technology, Inc.) on the other side of the same module and responding to the "back-side" select signal. SyncFlash™ memory devices provide Flash memory features but present an addressing and control interface substantially similar to that of SDRAM. Both rows of this custom module may therefore be addressed as though containing standard SDRAM synchronous memory devices. Thus a single custom DIMM module may provide all the necessary memory for this small system application and still uses the same SO-DIMM standard socket.

[0053] A third system application of the same system board could provide the same Control Head using standard Flash and SRAM on different rows of a single custom SO-DIMM. Such a module might use 2 2Mx16-bit asynchronous Flash memory devices on one side for a total of 8MB selected by the second row select pin logically ANDed with the un-inverted highest used address bit. This ANDing can be accomplished by single-gate logic available from several logic vendors. The same custom SO-DIMM module might also use 4 1Mx8-bit asynchronous SRAM memory devices for a total of 4MB. This SRAM would preferably also be selected by the second row select signal ANDed with the inverted highest used address bit. In this preferred embodiment, both the synchronous Flash and the asynchronous SRAM would be selected by the second row ("back-side") select signal with the highest used address bit used to differentiate between the two types of memory.

[0054] The flexible memory architecture of the present invention allows all three systems to use the exact same system board design -- a desirable benefit to ease certification of systems in avionics and aerospace industries as well as other high-

reliability applications. All three systems can use the exact same software if the boot code configures the memory controller from SPD data retrieved from the module(s) (as discussed further herein below) -- another benefit to reliability certification requirements. Further, as memory device packaging evolves, a new SO-DIMM module can be easily adapted to the modified signaling of the present invention. The system board, CPU and memory controller need not be re-designed to utilize improved memory devices. Further, use of standard SO-DIMMs for all memory requirements of such systems helps minimize footprint requirements in space constrained applications.

[0055] The table below describes one exemplary preferred embodiment for the connection of standard memory controller signals generated by a typical memory controller to an enhanced DIMM in accordance with the present invention. The signals of the memory controller in the first three columns are expressed in the table as the signals generated by the memory controller element within a Motorola MPC8245 embedded microcomputer (i.e., the MPC107 memory controller). This memory controller is exemplary of a large number of memory controllers that may be useful in controlling memory modules of the present invention. Those skilled in the art will recognize a variety of other memory controller devices, both as separate devices and integrated within a microcomputer, that are capable of controlling a variety of synchronous and asynchronous memory devices over a common bus.

[0056] The first column indicates the signals used in connecting the memory controller to a standard SDRAM module. The second column indicates the signals used in connecting the memory controller to an asynchronous Flash or SRAM memory module. The third column indicates the signals used in connecting the memory controller to a burst/page Fast Flash or SRAM module. When a signal is not used for a particular module type, "NU" so designates an unused connection.

Synch. SDRAM Module	Asynch. FLASH/ SRAM Module	Burst/Page Fast Flash/ SRAM Module	Connector/Module Pin Numbers
D(0:31)	D(0:31)	D(0:31)	138,136,134,132,128,126,124,122,100, 98,96,94,90,88,86,84,54,52,50,48,44, 42,40,38,20,18,16,14,10,8,6,4
D(32:63)	D(32:63)	D(32:63)	137,135,133,131,127,125,123,121,99, 97,95,93,89,87,85,83,53,52,49,47,43,41,39,37,1 9,17,15,13,9,7,5,3
DQM_n(0:7)	NU	NU	23,25,115,117,24,26,116,118
MA(14)	XA(24)	XA(24)	Not currently supported, would probably be pin 73 on SDRAM module.
MA(13)	XA(23)	XA(23)	72*
MA(12)	XA(22)	XA(22)	70**
SDBA0	XA(11)	XA(11)	106
SDBA1	XA(20)	XA(20)	110
MA(0:11)	XA(0:10,21)	XA(0:10,21)	29,31,33,30,32,34,103,104,105,109,111,112)
CB(0:7)	XA(19:12)	XA(19:12)	57,59,77,79,58,60,78,80
RAS_n	NU	NU	65
CAS_n	NU	NU	66
WE_n	WE_n	WE_n	67
CKE(0:1)	NU	NU	62,68***
CLK(0:1)	NU	CLK	61,74****
SDA	SDA	SDA	141
SCL	SCL	SCL	142
DNU	OE_n	OE_n	73 (this is defined for EDO and FP DRAM modules)
S_n(0:1)	CE_n	CE_n and ADV_n	69,71***** (use pin 69 to select single-row SDRAM modules, pin 71 to select FLASH)
VDD	VDD	VDD	11,12,27,28,45,46,63,64,81,82,101,102, 113,114,129,130,143,144
VSS	VSS	VSS	1,2,21,22,35,36,55,56,75,76,91,92, 107,108,119,120,139,140
	RESET_n	RESET_n	72*

Notes regarding above table entries:

- * Seldom used pins, MA(13) not supported on 54-pin devices up to 256Mb. There is a reserved pin for higher density parts in this package.
- ** MA(12) is seldom used, supported ONLY for 256Mb parts.
- *** Sometimes used on single row modules (required by PC100/PC133). Loading on MPC824x not affected. Included here to allow possibility of use of COTS SDRAM modules.
- **** Second CLK is sometimes used in 1-Row modules to reduce loading and balance clock trees.
- ***** Sharing FLASH and SRAM select requires address decoding on the module.

[0057] Those skilled in the art will recognize other mappings of memory controller signals appropriate to control the above and other types of memory modules. The above mappings of signals are therefore intended as representative of one exemplary preferred embodiment of the invention to provide a flexible memory architecture for use of SO-DIMM modules. Further, as noted above, those skilled in the art will recognize pin assignments for signals that may be used for other memory module configurations and standards.

[0058] Figure 2 is a schematic drawing showing relevant portions of signals used in a first exemplary preferred embodiment of the present invention. Those skilled in the art will recognize from the tables above that numerous other signals required for accessing each type of memory device are removed from figure 2 to simplify the drawing.

[0059] Figure 2 shows a portion of a system having two sockets (connectors) for insertion of SO-DIMM memory modules. The first of the two sockets 202 may preferably be used for a standard SDRAM (or DRAM) SO-DIMM module to provide standard dynamic RAM features for program instruction and data storage and retrieval. The second socket 200 may preferably be reserved for a special module such as a Flash memory module or static RAM memory module adapted in accordance with the present invention for use in a modified 144-pin SO-DIMM socket.

[0060] In a first exemplary preferred embodiment as shown in figure 2, the row select signals (CS(0:1)) 204 are handled differently from the standards defined for the 144-pin SO-DIMM. In particular, the first signal of the row select signals (CS(0)) is preferably used in accordance with the standards defined for the module to select the front side (the first row) of the addressed module. CS(0) is therefore preferably connected to pin 69 of the standard SO-DIMM socket. In this exemplary preferred embodiment, standard SDRAM (or DRAM) SO-DIMMs are preferably limited to single-row configurations so that the second row select signal is not needed for SDRAM (or DRAM) addressing. Rather the second row select signal is provided by one of a plurality of configurable sources to select the Flash or asynchronous memory device in second socket 200 using pin 71 of the SO-DIMM standard socket. A first source of the second row select signal is the row select signal as normally generated by the memory controller (CS(1) to thereby permit use of any standard SO-DIMM module including double-row modules). Other sources for the second row select signal generation are programmable chip select signals generated by the memory controller of the MPC824x microcomputer

memory controller (or other similarly featured memory controller). Specifically, the F64CS control signal (generated by an MPC8240 with a separate address decoder ASIC) and the RCS(0:3) control signals (generated by an MPC8245) may be used as a second row select signal when an address is decoded as referencing the asynchronous memory device on a second module in the second SO-DIMM socket 200.

[0061] As noted above, in a first exemplary preferred embodiment, the various sources of signals used for the second row select signal are selected by configuring mutually exclusive jumpers (0 Ohm resistors) 273, 274 and 275. In particular, jumper 273 is inserted to enable selection of the second row by use of the CS(1) standard select signal generated by the memory controller. Jumper 274 enables use of the RCS(3) signal generated in response to properly decoded memory access addresses. Jumper 275 enables use of the F64CS signal generated in response to particular memory access addresses. Those skilled in the art will recognize a variety of other programmable chip select signals that may be used in a programmable memory controller to permit flexible generation of the second row select signal applied to the memory modules. The particular signals described in the tables above and shown in figure 2 for providing second row select signals to SO-DIMM modules are therefore merely exemplary of one preferred embodiment of the invention. Numerous other configurations will be apparent to those skilled in the art. The selected source for the second row select signal is then applied via path 210 to pin 71 of the SO-DIMM socket.

[0062] As noted above, several configurations of memory modules are permitted using the signaling features of the present invention as described above. In particular, the features of the present invention fully support use of standard SDRAM (or other DRAM) SO-DIMM modules. In other words, the features of the present invention retain "backward" compatibility to permit use of existing SDRAM (and other DRAM) SO-DIMM modules. In addition, the features of the present invention allow both SO-DIMM sockets of the system to be populated with single-row modules using memory devices other than DRAMs. For example, both modules may provide synchronous or asynchronous Flash or static RAM devices or combinations of the same. Further, special modules may be used providing SDRAM (or other DRAM) on the front side and Flash or SRAM on the other. Those skilled in the art will recognize a variety of memory configurations enabled by the signaling features of the present invention. Further, those skilled in the art will recognize that a similar jumper structure may be employed for the

first row select signal (CS(0)) preferably coupled to pin 69 of the SO-DIMM sockets 200 and 204. Such a structure would permit the added flexibility to allow both rows to utilize asynchronous devices if desired for a particular application.

[0063] As noted above, figure 2 suggests a first exemplary preferred embodiment wherein the particular configuration of memory devices used on the system is identified by jumpers. Those skilled in the art will recognize a jumper as a switch and therefore recognize a wide variety of equivalent switching means including for example DIP switches of various types and other mechanical and electrical switch devices and means.

[0064] In an alternate embodiment representing the best presently known mode of practicing the invention, an enhanced memory module in accordance with the present invention would identify its configuration in returned SPD data. Serial presence detect (SPD) information is usually stored in a read-only memory (i.e., PROM) on standard memory modules. The SPD information includes, among other things, timing, size and geometry parameters of the memory module. These parameters are used by a memory controller to adjust its memory signal timings and other parameters to match the requirements of the module. This SPD information is usually retrieved at boot time or system reset of the system board and is used by low level boot code to configure the memory controller.

[0065] In accordance with the best known mode of practicing the invention, an enhanced module of the present invention returns information sufficient to identify the type of memory elements on the module and other parameters related to control of memory access to the module by a memory controller. Those skilled in the art will recognize that official designations of SPD information bit and byte assignments and values are controlled by standards agencies (i.e., JEDEC in the exemplary preferred embodiment utilizing 144-pin SO-DIMM standards). In particular, JEDEC JESD21-C section 4.1.2 defines a standard content for the first 128 bytes of SPD information returned by a memory module. Byte 2 under this standard defines a memory type. At present 7 out of 255 possible memory type values are defined for existing memory modules. A new memory type would preferably be used in the exemplary preferred embodiment of the present invention to identify a memory module populated with Flash or SRAM memory devices (typically addressed as a flat memory space as distinct from a two dimensional array of bits organized according to column addresses and row addresses). With such a new memory type value defined, other standard fields of the SPD

information could be coded as appropriate for such "flat" memory devices to define any required memory addressing and timing parameters.

[0066] Another aspect of the present invention is the improved immobilization of the DIMM of the present invention to aid in resistance to vibration and shock related failures of the module. As noted above, testing results show that standard SO-DIMM sockets permit some rotation of the module about the mated keys used to help position the module within the socket. Over time, this rotation can wear the edge contacts (signal pads) to the point of premature failure. The invention provides for immobilizing the edge of the module opposite the connector edge that inserts into the socket. The increased immobilization of this edge helps prevent such wear to reduce the potential for premature failure of the module.

[0067] The invention broadly provides for any means for immobilizing the module to reduce the undesired rotation of the module within the socket. In particular, the present invention provides for securing the edge opposite the connector edge to the underlying system board on which the socket is also mounted. This serves to match the immobilization of the module with respect to the system board and that of the socket in which the module is inserted for use. The edge so immobilized is fastened to the system board by any of several fastening structures. Those skilled in the art will recognize a wide variety of fastening techniques and structures that may be equivalently employed to secure the edge of the memory module with respect to the socket and with respect to the system board.

[0068] Figures 4 and 5 depict one exemplary preferred embodiment of the invention to secure the memory module to the system board and thereby reduce movement of the module within the mated socket. Specifically, figure 4 is a dimensioned mechanical drawing showing mounting holes 400 positioned at the corners of a standard SO-DIMM module 402. The mounting holes are preferably positioned at the corners of the edge of the module opposite the connector edge. Figure 5 provides further detail of the mounting holes 400 in the DIMM module 402 mated with a connector 506 mounted on a system board 500. Specifically, figure 5 provides a sequence of three positions of the module 402 as it is mated with connector 506. Module 402 is first tilted at an angle and its edge connector is inserted into the mated slot of connector 506. A key notch in the module is positioned over the mated key 508 in the socket to ensure accurate positioning of the module within the socket. With the connector edge of the module so inserted in the

socket, the module is tilted downward deflecting by cam action retaining spring clips 510 of the socket on each side of the socket 506. When tilted down substantially parallel with the system board, spring clips 510 bias back to normal position to retain module 402 within socket 506. In accordance with the present invention, mounting screws 504 are inserted through mounting holes 400 of the module 402 and mated with standoff nuts 502 affixed to the system board.

[0069] In one exemplary preferred embodiment, the improvement of the present invention secures module 402 within socket 506 by inserting screws 504 through mounting holes 400 into nuts 502 on the system board. Those skilled in the art will recognize that the number and position of mounting holes 400 may be varied as a matter of design choice to achieve the desired effect of reducing the motion of the module within the mated socket.

[0070] Further, those skilled in the art will recognize that a screw and mated standoff nut type fastener structure is but one of a number of equivalent fastening techniques and structures useful to immobilize the module within the socket. Numerous equivalent examples are readily apparent to those skilled in the art as a matter of design choice. For example, standoff nuts 502 may be soldered components secured to the system board to simplify the manufacture of the system board. Further, other types of fastening systems may be employed. For example, a half card-cage with a channel for gripping the edge of the module opposite its connector edge may be affixed to the system board. The module is inserted in the socket and tilted down to "snap" into the side spring clips 510 as shown in figure 5. When in place, the module also snaps into the channel of the half card-cage structure on the system board. Another example uses an elastomer material between the module and the system board to reduce vibrations of the module within its connector socket. Another example provides for a mounting pin affixed to the system board such that the mounting pin enters through the mounting hole and snaps into a locked position to secure the module onto the mounting pin (and hence to the system board). Still another example provides for a pin associated with the socket connector such that the pin inserts through a hole in the module when the module is inserted in the socket. These and other alternative embodiments would be viewed as equivalent means for securing the module to the underlying system board to thereby immobilize the module with respect to the mated connector socket.

[0071] While the invention has been illustrated and described in the drawings and foregoing description, such illustration and description is to be considered as exemplary and not restrictive in character, it being understood that only the preferred embodiment and minor variants thereof have been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected.

CLAIMS

What is claimed is:

1. A memory module comprising:
a printed circuit assembly having connector pads at one edge of said assembly;
5 and
a plurality of memory devices mounted on said assembly and electrically coupled
to said connector pads wherein said printed circuit assembly is adapted to
support both synchronous and asynchronous types of said memory
devices.
- 10 2. The memory module of claim 1 wherein each of said plurality of memory
devices is a synchronous dynamic random access memory device.
3. The memory module of claim 1 wherein each of said plurality of memory
devices is a synchronous Flash memory device.
4. The memory module of claim 1 wherein each of said plurality of memory
15 devices is an asynchronous Flash memory device.
5. The memory module of claim 1 wherein each of said plurality of memory
devices is an asynchronous static random access memory device.
6. The memory module of claim 1 wherein each of said plurality of memory
devices is an asynchronous fast static random access memory device.
- 20 7. The memory module of claim 1 wherein each of said plurality of memory
devices is an asynchronous low power static random access memory device.
8. The memory module of claim 1 wherein said connector pads include:
a first select signal connector pad that selects a first subset of memory devices
mounted on said module when a signal is applied thereto wherein said first
25 subset of memory devices are synchronous memory devices; and
a second select signal connector pad that selects a second subset of memory
devices mounted on said module when a signal is applied thereto wherein
said second subset of memory devices are asynchronous memory devices.
9. The memory module of claim 1 wherein said connector pads include:
30 a first select signal connector pad that selects a first subset of memory devices
mounted on said module when a signal is applied thereto wherein said first
subset of memory devices are synchronous dynamic random access
memory devices; and

a second select signal connector pad that selects a second subset of memory devices mounted on said module when a signal is applied thereto wherein said second subset of memory devices are synchronous Flash memory devices.

- 5 10. A system comprising:
a system board;
a memory bus adapted for exchanging signals between a memory controller and both synchronous and asynchronous memory devices;
a memory controller on said system board coupled to said memory bus wherein
10 said memory controller is capable of generating signals for control of both synchronous and asynchronous memory devices and wherein said memory controller is capable of multiplexing said signals on said memory bus;
a first socket connector on said system board for receiving a first memory module wherein said first socket connector is coupled to said memory controller
15 through said memory bus; and
a first memory module inserted in said first socket connector and electrically coupled to said memory controller wherein said first memory module includes a plurality of synchronous or asynchronous memory devices.
- 20 11. The system of claim 10 further comprising:
a jumper on said system board for configuring signals exchanged between said memory controller and said first memory module in accordance with the type of memory devices on said first memory module.
- 25 12. The system of claim 10 wherein said first memory module provides serial presence detect information used in conjunction with said memory controller to identify the type of memory devices included on said first memory module.
13. The system of claim 10 wherein said first memory module includes a plurality of synchronous dynamic random access memory devices.
14. The system of claim 10 wherein said memory module includes a plurality of asynchronous Flash memory devices.
- 30 15. The system of claim 10 wherein said memory module includes both synchronous Flash memory devices and synchronous dynamic random access memory devices.

16. The system of claim 10 wherein said memory module includes both synchronous memory devices and asynchronous memory devices.

17. The system of claim 10 further comprising:

a second socket connector on said system board for receiving a second memory module wherein said second socket connector is coupled to said memory controller through said memory bus; and

a second memory module inserted in said second socket connector and electrically coupled to said memory controller wherein said second memory module includes a plurality of synchronous or asynchronous memory devices.

18. The system of claim 17 further comprising:

a jumper on said system board for configuring signals exchanged between said memory controller and said second memory module in accordance with the type of memory devices on said first memory module and on said second memory module

19. The system of claim 17

wherein said first memory module provides serial presence detect information used in conjunction with said memory controller to identify the type of memory devices included on said first memory module, and

wherein said second memory module provides serial presence detect information used in conjunction with said memory controller to identify the type of memory devices included on said second memory module.

20. The system of claim 17

wherein said first memory module includes a plurality of synchronous dynamic random access memory devices, and

wherein said second memory module includes a plurality of asynchronous Flash memory devices.

21. The system of claim 17

wherein said first memory module includes a plurality of synchronous dynamic random access memory devices, and

wherein said second memory module includes a plurality of asynchronous static random access memory devices.

22. The system of claim 17

wherein said first memory module includes a plurality of asynchronous Flash
memory devices, and

wherein said second memory module includes a plurality of asynchronous static
random access memory devices.

- 5 22. A system comprising:
a system board;
a socket connector on said system board for receiving a memory module wherein
said socket connector includes a key;
a memory module having a connector edge inserted in said socket connector and
10 having an opposing edge opposite said connector edge wherein said
memory module has a notch mated to said key when said memory module
is inserted in said socket connector; and
a memory module retainer adapted to substantially immobilize said opposing edge
with respect to rotation about said key.

- 15 23. The system of claim 22 wherein said memory module retainer comprises:
a nut affixed to said system board;
a hole in said memory module along said opposing edge and aligned with said nut;
and
a screw inserted through said hole into said nut to substantially immobilize said
20 opposing edge with respect to rotation about said key.

24. The system of claim 23 wherein said nut is a swaged extension nut.

25. The system of claim 22 wherein said memory module retainer comprises:
a half card-cage affixed to said system board wherein said half card-cage includes
a channel for receiving said opposing edge of said memory module to
25 substantially immobilize said opposing edge with respect to rotation about
said key.

26. The system of claim 22 wherein said memory module retainer comprises:
a standoff pin affixed to said system board;
a hole in said memory module along said opposing edge and aligned with said pin
30 wherein said standoff pin extends through said hole and locks when said
memory module is completely inserted in said socket connector to
substantially immobilize said opposing edge with respect to rotation about
said key.

27. The system of claim 22 wherein said memory module retainer comprises:
a standoff pin affixed to said socket connector;
a hole in said memory module aligned with said standoff pin wherein said
standoff pin extends through said hole and when said memory module is
completed inserted in said socket connector to substantially immobilize
said opposing edge with respect to rotation about said key.
28. A system comprising:
a system board;
a socket connector on said system board for receiving a memory module wherein
said socket connector includes a key;
a memory module having a connector edge inserted in said socket connector and
having an opposing edge opposite said connector edge wherein said
memory module has a notch mated to said key when said memory module
is inserted in said socket connector; and
memory module retainer means adapted to substantially immobilize said opposing
edge with respect to rotation about said key.
29. The system of claim 28 wherein said memory module retainer means
comprises:
nut means affixed to said system board;
a hole in said memory module along said opposing edge and aligned with said nut
means; and
screw means inserted through said hole into said nut means to substantially
immobilize said opposing edge with respect to rotation about said key.
30. The system of claim 29 wherein said nut means is a swaged extension nut.
31. The system of claim 28 wherein said memory module retainer means
comprises:
card-cage means affixed to said system board wherein said card-cage means
includes a channel for receiving said opposing edge of said memory
module to substantially immobilize said opposing edge with respect to
rotation about said key.
32. The system of claim 28 wherein said memory module retainer means
comprises:
pin means affixed to said system board;

a hole in said memory module along said opposing edge and aligned with said pin means wherein said pin means extends through said hole and locks when said memory module is completed inserted in said socket connector to substantially immobilize said opposing edge with respect to rotation about said key.

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33. The system of claim 28 wherein said memory module retainer means comprises:

pin means affixed to said socket connector;

a hole in said memory module aligned with said pin means wherein said pin means extends through said hole and when said memory module is completed inserted in said socket connector to substantially immobilize said opposing edge with respect to rotation about said key.

10

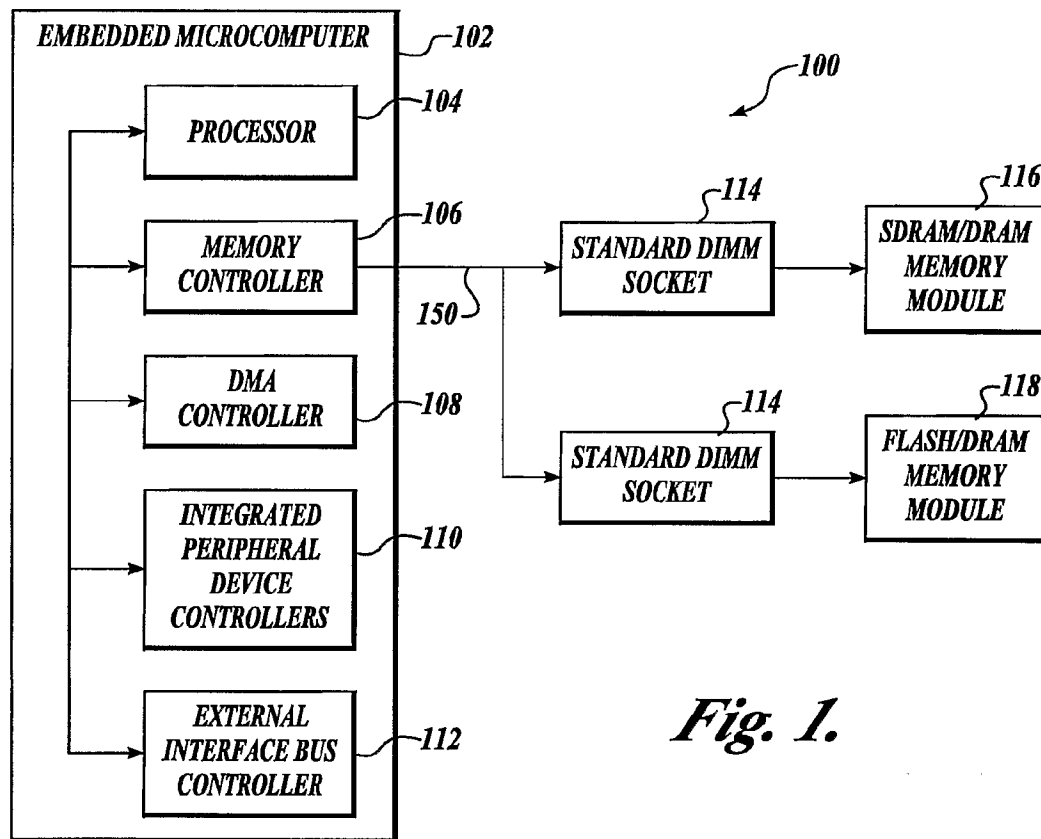
$\frac{1}{3}$ 

Fig. 1.

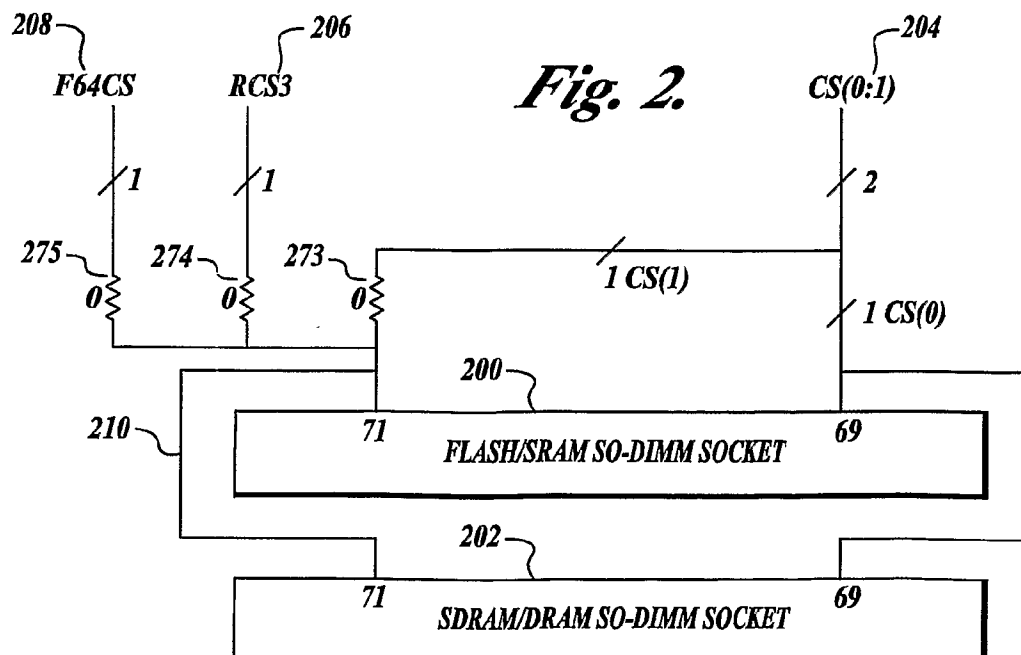
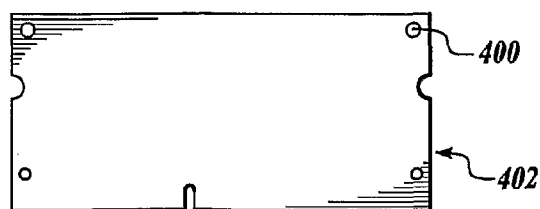
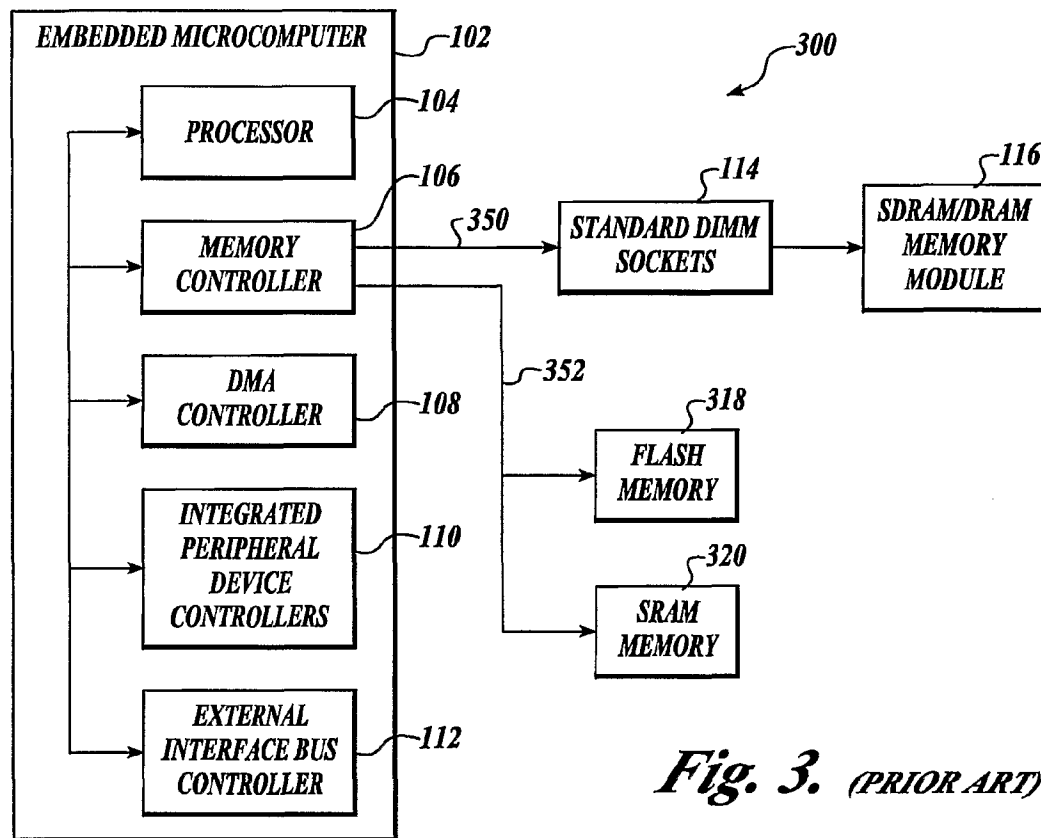
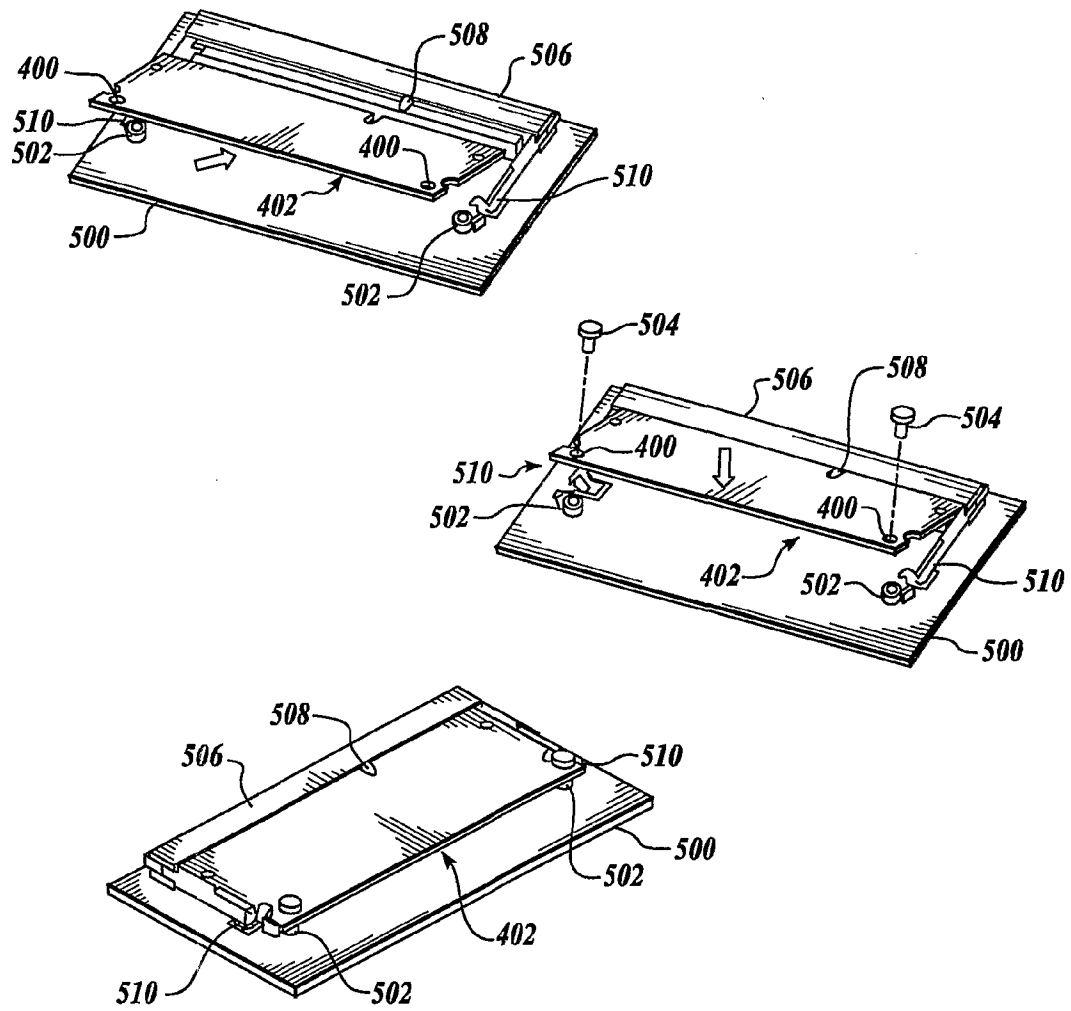


Fig. 2.

2/3

*Fig. 4.*

3/3

*Fig. 5.*

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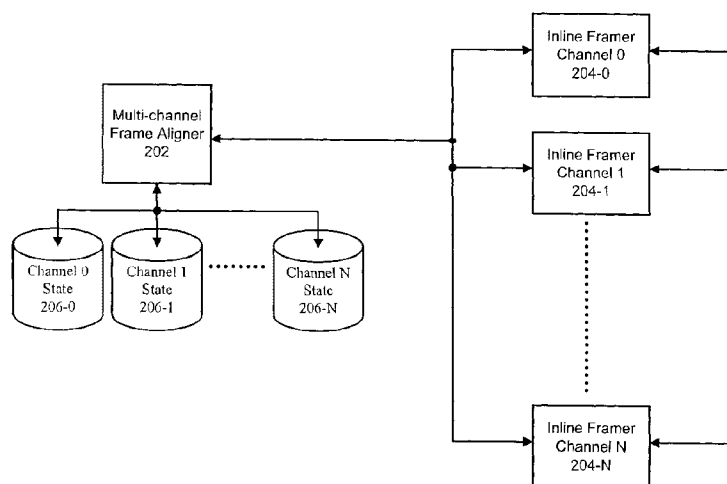
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(54) Title: SCALABLE MULTI-CHANNEL FRAME ALIGNER



(57) **Abstract:** An offline frame alignment circuit can simultaneously achieve frame alignment for a large number of TDM streams (i.e. channels) within a required amount of time (e.g. 15 ms for ESF). The Multi-channel Frame Aligner (MCFA) uses a high speed system clock independent of the individual line clocks to perform frame alignment for each of the channels. The MCFA includes a framer memory to store the alignment states of all possible framing bit candidates for all channels. The MCFA polls each channel to determine if frame alignment is requested, and if so, if data from the associated channel is available. A state machine in the MCFA compares the received data with the expected framing bits and adjusts the stored alignment states accordingly. All framing bit candidates are processed in parallel leading to fast alignment times. The MCFA architecture can be adapted for any of a plurality of channels and framing formats merely by adjusting the speed of the MCFA system clock and capacity and arrangement of data states in the framer memory.



For two-letter codes and other abbreviations, refer to the "Guidance Notes on Codes and Abbreviations" appearing at the beginning of each regular issue of the PCT Gazette.

SCALABLE MULTI-CHANNEL FRAME ALIGNER

INVENTORS

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FIELD OF THE INVENTION

This invention relates generally to frame alignment in a digital communications system, and more particularly, to a method and apparatus for performing frame alignment for a scalable plurality of communications channels.

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BACKGROUND OF THE INVENTION

The demand for telecommunications services continues to grow. With this growing demand for services comes the need for increased capacity, and telecommunications companies are scrambling to keep up. One way to handle increased capacity is to multiplex a plurality of communications channels onto higher-bandwidth channels. Thus, by adding a single high-bandwidth channel, many individual communications channels can be added and services can be increased dramatically.

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FIG. 1 is a block diagram illustrating a conventional communications system. As shown in FIG. 1, for transmission over a high-capacity transmission media 102 such as a fiber optic cable or trunk line (and generally over long distances), transmit side 104 includes components for multiplexing multiple DS0 channels (each capable of carrying a separate PSTN voice call, for example). As shown in this example, multiple DS0 channels are multiplexed onto a DS1 channel by channel banks 106, and multiple DS1 channels can be further multiplexed onto higher order channels (e.g. DS3 and higher) by multiplexers 108 for transmission over the media. Where necessary, further conversion and processing may be performed for different types of media (e.g. conversion to optical media and formats such as OC-3 to OC-192). Receive side 110 includes components 112, 114 for demultiplexing the received higher order channels onto DS1 channels and then further into DS0 channels, respectively.

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Generally, time division multiplexing (TDM) techniques are used to multiplex the multiple DS0 channels onto higher order channels such as DS1. In a time division multiplexed scheme, each DS1 bit stream is serially transmitted in frames, with each frame containing a plurality of bits for each DS0 channel. Frame synchronization or frame

alignment is the process that forces the transmitter and receiver to agree on the timing between the two systems, and thus allows the receiver to de-multiplex the individual DS0 channels from the received DS1 bit stream. Framing is performed on the transmit side by inserting one or more framing bits into the bit stream at predetermined intervals. A typical
5 example is found in the North American 24 channel DS1 extended superframe format (ESF). In DS1 ESF framing, one frame bit is inserted for every 192 data bits (i.e. 8 bits x 24 DS0 channels) at a frame rate of 8 kHz. A predetermined sequence of a frame bit pattern is transmitted at some or all of the Frame Bit positions. In the receiver, a frame alignment circuit detects this frame bit sequence and uses the frame alignment to recover the data bits
10 for the individual DS0 channels from the data stream.

According to ANSI T.107 (Digital Hierarchy- Formats and Specifications) Standards, an extended DS1 framing format (ESF) is defined in which the frame bits are used for cyclic redundancy check bits and a maintenance channel, as well as for frame synchronization. In the extended framing format, the framing bits containing the framing
15 pattern sequence (FPS) are sent only once in every fourth frame, every 772 bits. Six successive framing bits forming the FPS (e.g. the sequence 001011) are thus transmitted over 24 frames called superframes. The superframe interval repeats every 3 milliseconds.

The framing bits and their positions in the superframe are shown in Table 1.

Frame No.	Bit No.	FPS	CRC	FDL
1	0	-	-	M1
2	193	-	C1	-
3	386	-	-	M2
4	579	0	-	-
5	772	-	-	M3
6	965	-	C2	-
7	1158	-	-	M4
8	1351	0	-	-
9	1544	-	-	M5
10	1737	-	C3	-
11	1930	-	-	M6
12	2123	1	-	-
13	2316	-	-	M7
14	2509	-	C4	-
15	2702	-	-	M8
16	2895	0	-	-
17	3088	-	-	M9
18	3281	-	C5	-
19	3474	-	-	M10
20	3667	1	-	-
21	3860	-	-	M11
22	4053	-	C6	-
23	4246	-	-	M12
24	4439	1	-	-

Table 1. DS1 Extended Superframe Format.

FPS - Frame Pattern Sequence

CRC - Cyclic Redundancy Check

FDL - Facility Data Link

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Thus, frame alignment requires detection of the frame pattern sequence (FPS) in each of the above bit positions spaced apart by 772 bits in a data stream. Since the data stream is subjected to bit errors during transmission, however, reliable frame detection may require the detection of framing bits over several frames or superframes. A further requirement is that the frame aligner should reliably align in the presence of data errors that may also affect the frame bit pattern. A common requirement for ESF is that frame alignment should be achieved within 15 ms. Since there is always a finite probability of random data mimicking the frame alignment pattern, the requirement is usually stated in terms of finding a frame alignment within a certain amount of time and probability of

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correctness, for example, within 15 ms and a 99% probability that no data errors are present. A confirmation process often follows the frame alignment process, where the frame alignment is confirmed by additional checks before being accepted as the correct alignment. A common confirmation technique used with the ESF format is to calculate the CRC with
5 the provisional alignment and compare the calculated CRC with the actual CRC bits C1-C6 transmitted over the DS1 frame bits.

Once frame alignment is achieved, the framer continues to check the frame alignment pattern for correctness. If errors exceed a certain threshold, an Out of Frame (OOF) signal is declared and a search for a fresh alignment is started. It is often observed
10 that OOF conditions are declared due to burst errors subjected on the data stream and not due to a genuine change in the frame alignment. To avoid frequent frame alignments that can cause a disruption in the data processing, an offline frame aligner is often used. The inline framer, on detecting an OOF condition, requests the offline framer for a fresh search for frame alignment. The inline framer continues with the old alignment. If the offline
15 frame aligner detects the same alignment as earlier, the inline framer will not be disturbed. Thus only genuine changes or loss in alignment will affect the inline framer.

Conventional circuits used to implement inline and offline framers suffer from many problems. For example, to detect the frame pattern sequence (FPS) spread over 4632 bits, the data is clocked one bit at a time through a shift register, and a check is made at each bit
20 position for the frame pattern sequence in bit positions spaced at 772 bits. A large shift register capable of storing several frames would be required to reliably and quickly detect the frame pattern sequence. However, implementing large shift registers on integrated circuits requires substantial chip area and leads to other difficulties.

The above problems are exacerbated when multiple channels are required. For
25 example, handling multiple channels using conventional approaches requires completely duplicating the amount of circuitry required for one channel for each of the multiple channels. A scalable approach would be preferred.

SUMMARY OF THE INVENTION

30 The present invention relates to an architecture and methodology for performing frame alignment that is scalable for any of a plurality of multiplexed channels, and is adaptable for any of a variety of framing formats.

In accordance with an aspect of the invention, an offline frame alignment circuit can simultaneously achieve frame alignment for a large number of TDM streams (i.e. channels) within a required amount of time (e.g. 15 ms for ESF). The Multi-channel Frame Aligner (MCFA) uses a high speed system clock independent of the individual line clocks to
5 perform frame alignment for each of the channels. The MCFA includes a framer memory to store the alignment states of all possible framing bit candidates for all channels. The MCFA polls each channel to determine if frame alignment is requested, and if so, if data from the associated channel is available. A state machine in the MCFA compares the received data with the expected framing bits and adjusts the stored alignment states
10 accordingly. All framing bit candidates are processed in parallel leading to fast alignment times. The MCFA architecture can be adapted for any of a plurality of channels and framing formats merely by adjusting the speed of the MCFA system clock and capacity and arrangement of data states in the framer memory.

15 **BRIEF DESCRIPTION OF THE DRAWINGS**

These and other aspects and features of the present invention will become apparent to those ordinarily skilled in the art upon review of the following description of specific embodiments of the invention in conjunction with the accompanying figures, wherein:

FIG. 1 is a block diagram illustrating a multiplexing scheme in a conventional
20 communications system;

FIG. 2 is a block diagram of an example frame alignment architecture in accordance with an embodiment of the present invention;

FIG. 3 illustrates a framing scheme in a data stream that can be detected by the frame alignment architecture of the present invention;

25 FIG. 4 is a functional block diagram illustrating an example frame alignment method that can be implemented by the frame alignment architecture of the present invention;

FIG. 5 is a structural block diagram illustrating an example implementation of the frame alignment architecture of the present invention in accordance with an ESF framing scheme;

30 FIG. 6 is a state transition diagram illustrating an example of a state transition scheme that can be implemented by the frame aligner of the present invention;

FIG. 7 illustrates an example of data structures that can be used to implement storage of channel and data state information for a plurality of channels for which frame alignment can be performed by the frame aligner of the present invention;

FIG. 8 is a block diagram further illustrating an example of a frame alignment engine that can be included in the frame alignment architecture of the present invention
5 illustrated in FIG. 5;

FIG. 9 is a state diagram illustrating an example of the processing flow in the frame alignment engine illustrated in FIG. 8 in accordance with an embodiment of the present invention; and

FIG. 10 is a block diagram further illustrating an example of a RAM interface block that can be included in the frame alignment engine illustrated in FIG. 8 in accordance with an embodiment of the present invention.
10

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention will now be described in detail with reference to the drawings, which are provided as illustrative examples of the invention so as to enable those skilled in the art to practice the invention. Notably, the implementation of certain elements of the present invention may be accomplished using software, hardware or any combination thereof, as would be apparent to those of ordinary skill in the art, and the figures and
20 examples below are not meant to limit the scope of the present invention. Moreover, where certain elements of the present invention can be partially or fully implemented using known components, only those portions of such known components that are necessary for an understanding of the present invention will be described, and detailed descriptions of other portions of such known components will be omitted so as not to obscure the invention.
25 Further, the present invention encompasses present and future known equivalents to the known components referred to herein by way of illustration.

FIG. 2 is a block diagram illustrating an example architecture including a Multi-channel Frame Aligner (MCFA) in accordance with an embodiment of the present invention.

As shown in FIG. 2, MCFA 202 communicates with inline framers 204 for each channel 1 to N. MCFA 202 further maintains a state table 206 for each channel 1 to N. In operation, inline framers 204 continuously monitor the data stream for a corresponding channel for an out of frame (OOF) condition. When such a condition occurs, framers 204
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request MCFA 202 to detect the presence of the framing pattern sequence (FPS) in the corresponding channel 1 to N. After receiving a request from a framer 204, MCFA 202 begins to collect data from the requesting channel. MCFA 202 then compares the collected data with the FPS to update a state maintained for that channel, and when frame alignment is detected, it is reported back to the requesting framer 204.

An advantage of the architecture of FIG. 2 is that the single MCFA 202 can simultaneously perform frame alignment for a plurality of channels 1 to N. In other words, even though channels 1 to N and framers 204 are operating independently of each other, the MCFA 202, by maintaining state tables for all of them, can identify and maintain a frame alignment state for each channel simultaneously and independently. A further advantage is that the number of channels that can be processed by the single MCFA 202 can be easily extended by adding additional state tables, and also perhaps by increasing a processor clock speed, as will be described in more detail below. Further, the present architecture can be used to perform frame alignment for a variety of framing formats, such as DS1 SF, DS1 DDS, DS1 SLC-96, ITU-T G.704 based E1, which formats can be either pre-programmed or adjusted dynamically.

FIG. 3 further illustrates the data stream for a given channel. As shown in FIG. 3, within the data the entire FPS (comprising FPS bits FPS_1 to FPS_n) will be repeated once per FPS interval. Generally, the FPS interval will coincide with an extended frame for the given frame format (e.g. six frames = an extended frame in the ESF format). FIG. 3 shows m iterations (FPS Iteration 1 to FPS Iteration m) of the FPS in a data stream for a given channel. For an unaligned data stream, the location of the FPS in the data stream is unknown. Accordingly, each bit of the FPS occurs at an arbitrary FPS bit offset within the data stream. However, it is known that a FPS bit will occur repeatedly at that same FPS bit offset after a known FPS bit interval in the data stream. For example, in an ESF framing scheme, a FPS bit will occur once every 772 bits in a data stream. Accordingly, for a given number of bits in an FPS bit interval, an FPS bit will occur at a constant FPS bit offset in each consecutive FPS bit interval of the data stream. It should be further apparent that an entire FPS will be received after n FPS bit intervals of the data stream have been received, although perhaps not beginning with the first FPS bit in the sequence.

As should be thus apparent from the above, one way to perform frame alignment is to determine the FPS bit offset, given an arbitrary starting point of collecting data from a data stream for a given channel. It should be further apparent that it may not be sufficient to

simply collect data corresponding to one iteration of the FPS, because the starting bit in the FPS will be an arbitrary one of the FPS bits FPS_1 to FPS_n . Accordingly, MCFA 202 collects data for m iterations of an FPS sequence, where m is selected to yield a frame alignment determination within a given probability of correctness (e.g. 99%).

5 It should be noted that the FPS need not be comprised of single bits at each bit offset, but that the FPS may include multiple bits at each bit offset.

A general operation of MCFA 202 for detecting the FPS bit offset in a data stream for a given channel will now be described with reference to FIG. 4. Although the following describes a frame alignment operation for only one channel for ease of illustration, it is an aspect of the invention that MCFA 202 includes the ability to simultaneously perform frame alignment for up to all channels associated with the MCFA 202 (e.g. by maintaining state tables for all channels). Such aspects of the invention will be described in more detail in following sections of the present specification.

When MCFA 202 receives a request for frame alignment from an inline framer 204, it will set up a state table for the corresponding channel. As shown in FIG. 4, the table 402 includes p entries, one for each bit in a frame bit interval (e.g. 772 for an ESF framing scheme). Each of the entries is initialized to 0, as is the bit offset counter.

Data received from the corresponding channel is examined one bit at a time (in an example where the FPS is comprised of individual bits at each frame bit interval). The comparator 404 compares the received data bit to the FPS bit indicated by the FPS bit interval counter 406. The FPS bit interval counter 406 contains the value of the entry in table 402 pointed to by the bit offset counter 408 plus one (i.e. the next expected bit in the FPS sequence 410). For example, when the first bit is received at each bit offset, the value in the entry corresponding to each bit will be zero, so the received bit is compared to the first FPS bit. Whenever the received data bit matches the expected FPS bit, the entry in state table 402 corresponding to the value of the bit offset counter is incremented. Otherwise, the entry is reset to 0. For example, if the first bit received (i.e. Bit 0) matches the first FPS bit (i.e. 0 plus one), the Bit 0 entry in the table 402 is incremented to 1.

After each received data bit is processed, the bit offset counter 408 is incremented, and after p data bits are received and processed, the bit offset counter is reset. Thus, when the p -th data bit (i.e. Bit $p-1$) is received, the value of the bit offset counter 408 will be $p-1$. Accordingly, if the p -th received data bit matches the first FPS bit, the Bit $p-1$ entry in the table 402 is incremented to 1. After the p -th data bit is processed, the bit offset counter 408

resets to 0. The next received data bit will thus be compared to the next FPS bit in the sequence after the value in the table entry for Bit 0, and if it matches, the Bit 0 entry in table 402 will be incremented by 1. Otherwise, the Bit 0 entry in table 402 will be cleared (i.e. reset to 0).

5 MCFA 202 thus, in this example of the invention, maintains a running total of the number of sequential matches with the FPS at each bit (or collection of bits) in an FPS bit interval. When the value of any entry in table 402 reaches n, this indicates that one entire FPS was present at the FPS bit offset corresponding to that entry, and the bit offset is reported to the inline framer.

10 It should be noted, however, that it may take data corresponding to several FPS intervals to be received and processed (i.e. several multiples of, or FPS bit matches greater than, n) for only a single entry to exhibit a frame alignment match due to bit errors and mimics, for example. It should be further noted that the state maintained for a channel can be different than a running total, and may implement other state transition schemes, as will
15 be described in more detail below. It should be even further noted that each channel may simultaneously use different framing schemes, or that all channels may use the same framing scheme, and the MCFA 202 can include the ability to simultaneously detect the same or different framing sequences in a number of channels using the same or different framing schemes.

20 The present invention will now be described in more detail in connection with an example implementation illustrated in FIG. 5. In this example of the invention, ESF is assumed as the framing scheme for all channels. However, those skilled in the art will understand how to extend the principles of the invention to other and/or simultaneously different framing schemes after being taught by the foregoing example, and so the example
25 should not be construed as limiting.

 The frame alignment architecture illustrated in FIG. 5 can be implemented, for example, in a portion of an integrated circuit (i.e. channel bank), perhaps along with, or in addition to, other channel and frame processing circuitry, which integrated circuit can be included in a physical layer device coupled to T1 lines, for example. As should be apparent,
30 such a frame alignment architecture and physical layer device can find uses in a variety of applications in association with various points of aggregation including a central office, a digital loop carrier, channel bank, DSU/CSU, Digital Access Cross Connects(DACS), PBX, multiplexors etc.

As shown in FIG. 5, in this example of the invention, multi-channel frame alignment engine (MFAE) 502 is adapted to perform off-line frame alignment for 28 inline framers 504-1 to 504-28. As further shown in FIG. 5, inline framer 504 includes an OOF Detector 506, frame registers 508, serial-in-parallel-out (SIPO) block 510 and line counter 512.

5 Each inline framer 504 receives a receive line data signal, *rx_line_data* containing a DS1 time division multiplexed signal. Associated with each line data signal is an *rx_line_clk* signal that is recovered from the corresponding line data signal using a clock recovery circuit (not shown). At each active edge of *rx_line_clk*, the *rx_line_data* signal is serially clocked into serial-in-parallel-out (SIPO) register 510. Each SIPO 510 can be
 10 comprised, for example, of two registers of 16 bits each so that while one register is being written, the other is available for reading. SIPO 510 generates a *sipo_full* signal once all 16 bits are written into the register.

Line counter 512 is comprised of a simple counter that increments from 0 to 771 on each *rx_line_clk* pulse. It resets to zero after 772 pulses have been received, or when a
 15 *frame_req* signal has been generated by OOF detector 506.

OOF Detector 506, when in frame alignment, monitors the framing bits for bit errors. If the frame bit errors exceed the OOF criteria, a *frame_req* signal is generated, requesting a fresh frame realignment. For example, given a current frame alignment (i.e. a detected FPS at a given bit offset), OOF detector 506 monitors the continued presence of
 20 the FPS at the expected bit offsets. If a certain number of bits in the data stream do not match the FPS at the expected bit offsets over a certain number of repetitions of the FPS, an out of frame condition is assumed. The *frame_req* signal also resets the line counter 512 to 0.

Frame registers 508 store the frame location (i.e. the bit offset for the FPS) for the
 25 associated channel. Frame registers 508 also latch the *frame_req* signal generated by the OOF detector 506 so that it can be detected by MFAE 502.

Multi-channel Frame Aligner Engine (MFAE) 502 continuously polls all channels sequentially for *frame_req* and *sipo_full* conditions. This is achieved through the FAE control bus 520 using the *fae_addr[4:0]*, *fae_rd* and *fae_wr* signals. These signals
 30 generated by MFAE 502 and, generally, all signals on buses 520, 522 and 524, are synchronous to the high speed system clock *fae_clk*. In order to read or write into an inline framer 504, MFAE 502 uses the word *fae_addr[4:0]* to output the address of the channel along with a *fae_rd* or *fae_wr* signal for one clock cycle on bus 520. Data is read out from

an addressed inline framer 504 on read bus 522 via the word *fae_data_in*[15:0]. Data can be written to an inline framer 504 on write bus 524 via the word *fae_data_out*[9:0].

An active *frame_req* signal generated by an inline framer 504 causes MFAE 502 to initiate a search for frame alignment for that channel. The signal is also used to initialize a frame bit counter associated with each channel in the MFAE 502 and maintained in framer RAM. This counter is used to keep track of all 772 frame bit candidates. Because the *frame_req* signal also resets the line counter 512, the frame bit counter in the MFAE 502 is thus synchronized to the line counter 512 in the corresponding inline framer 504. Once the request for frame alignment is registered for a channel, polling for the *sipo_full* for that channel is started. An active *sipo_full* condition indicates that 16 bits of data are available for processing. The data bits are read out from the requesting inline framer 504 by addressing the channel through *fae_addr* and simultaneously asserting *fae_rd* on control bus 520. The addresses on the *fae_addr* lines are compared to a hard-wired *fae_id* signal in each of the inline framers 504. The *fae_id* signal in each framer 504 has a unique value ranging from binary 0 to binary 27. If the address on *fae_addr* matches that of *fae_id* for a channel, the data bits stored in SIPO 510 for the channel are driven on read bus 522 via the *fae_data_in* word.

The *sipo_full* signal is asserted by each inline framer 504 once every 16 cycles of the associated *rx_line_clk*. Since the nominal *rx_line_clock* rate is 1.544 MHz for DS1, the *sipo_full* signal will be independently asserted by each inline framer 504 every 10.36 microseconds. To accommodate jitter and frequency variations, which is estimated to be a maximum of 2 UI in 16 clocks, the MFAE 502 should be able to read out data from the SIPO 510 at least once every 9.06 microseconds to avoid SIPO overflow and data loss. In one example implementation of the invention that will be described in more detail below, processing of 16 bits of data by MFAE 502 takes 17 *fae_clk* cycles. In the worst case, all 28 channels may request for frame alignment, and so 28 x 17 *fae_clk* cycles may need to elapse before the next *sipo_full* indication from the same inline framer 504 can be serviced. Thus, the maximum *fae_clk* period can be estimated as $9.06 / (28 \times 17) = 19.03$ nanoseconds. In other words, the *fae_clk* must have a frequency greater than 52.5 MHz in order to process data for all channels in such a worst case scenario without losing data. If the number of channels to be handled is 84 rather than 27 (as in a DS3 framing scheme, for example), then the *fae_clk* frequency will have to be greater than 158 MHz.

Referring back to FIG. 5, MFAE 502 further communicates with framer RAM 514 and state machines 516-1 and 516-2. Two state machines operate in parallel in this example of the invention for reasons that will be explained in more detail below. However, the invention is not limited to this example, and one or more than two parallel operating state machines may be implemented. The following will describe the operation of one state machine 516, which description will also apply to the other state machine. Further, those skilled in the art will be able to implement the logical structure of state machine 516 after being taught by the following descriptions and the attached drawings.

State machine 516 receives a framing bit candidate's previous state from framer RAM 514 and updates the previous state based on a comparison between the current received data bit and the FPS bit expected to be received (i.e. the next bit in the FPS sequence corresponding to the stored state). For example, if the current stored state indicates that three bits in the FPS sequence have been correctly and sequentially received at that bit position (i.e. a sequence of 001 has already been received in the last three FPS bit intervals at that bit position), then the next expected bit should have a value of 0. Thus, state machine 516 will receive the current stored state and the current received data bit. If the current received data bit has a value of 0, the state machine will output an updated state indicating that four bits in the FPS sequence have been correctly and sequentially received at that bit position. Otherwise, it will output an updated state indicating that the comparison failed.

An example of a state transition scheme that can be implemented by state machines 516 is shown in FIG. 6. The state transition diagram is shown for a single bit position in a single channel. As shown in FIG. 6, in this example of the invention, the state for a given bit offset can transition from states S0 to S23. The states S18 to S23 correspond to frame aligned states.

Initially on receipt of a *frame_req* signal, the default state for each bit position of that channel is S0. The state machine 516 advances the stored state by one state each time a bit received at that bit position corresponds to the expected bit pattern in the FPS (e.g. the ESF Frame pattern sequence of 001011). For example, immediately after starting to collect data for a channel, MFAE 502 causes state machine 516 to start looking for the first bit in the ESF sequence, which is 0. Accordingly, when the first bit for each of the 772 possible bit positions is received, if the value of the received bit is 1 rather than 0, the state for that bit position stays at S0; otherwise, the state for that bit position advances to S1. In the next

FPS bit interval of the received data for that channel (i.e. after all other 772 bits have been received and processed for that channel), the next expected bit in the FPS is 0, regardless of whether the state was advanced to S1 or remained at S0. When each subsequent bit is received, state machine 516 also receives the stored state for the corresponding bit offset
 5 (either S0 or S1). As shown in FIG. 6, a received data bit of 0 will cause the state machine to advance the corresponding bit position to the next state from both states S0 and S1. Accordingly, if the next received bit for the bit position has a value of 0, the state for that bit position transitions to state S1 or S2 (depending on whether the last received bit had a value of 0 also); otherwise, the state returns to S0 (from either S0 or S1).

10 After all 6 bits in the FPS have been received correctly in sequence at a given bit position, the stored state for that bit position will have been advanced by state machine 516 to S6. However, in one example of the invention shown in FIG. 6, a further 12 bits having the sequence of 001011001011 must be received correctly to declare a frame alignment at that bit position. Thus, a total of 18 bits must be received correctly and the state for that bit
 15 position must have advanced to S18 in order to declare a frame alignment (indicated by the signal SF from state S18). This is because the number of frames required for a 99 % correct probability of alignment is 17 for a frame word length $n = 1$ and frame content $\alpha = 0.129$ %. (Frame content α is defined as the fraction of frame alignment word to the total frame capacity. Since there are 6 FPS bits over a frame of 4632 bits in the ESF mode,
 20 $\alpha = 6 \times 100 / 4632 = 0.129$ %).

Since each ESF frame bit is separated by 772 bits, the frame length refers to the number of 772 bit frames as sub frames (i.e. the FPS bit interval). Each one of the 772 bits can be a framing bit candidate. Hence state machine 516 compares each received bit with the next FPS bit corresponding to the state stored for the previous 772 frame bit candidate
 25 (e.g. if the stored state is S0, S6, S12 or S18, the first FPS bit is compared with the received bit). This is done in parallel by state machines 516-1 and 516-2 for all 772 bit candidates by maintaining 772 states in framer RAM 514. Assuming the worst case condition where the *frame_req* signal is generated at a bit position one clock after the frame bit position, and that the frame sequence starts at 010110 instead of 001011, a maximum of 24 sub-frames may
 30 be required to be received and processed for declaring a frame alignment with the necessary probability of correctness. The frame bit interval corresponds to 772 line clock periods, which is equal to 0.5 ms. This leads to a time (referred to as MART - maximum average reframe time) of 12 ms for frame alignment, which is well within the desired time of 15 ms.

FIG. 6 further illustrates how the state machines 516 can implement a state transition scheme that leads to more rapid frame alignment. In principle, the state transition scheme illustrated in FIG. 6 does always not require a return to state S0 when an unexpected bit is encountered at every other state. Rather, the state returns to a previous state in accordance with a strength of the match with the FPS at that state. For example, as shown in FIG. 6, when the state for a bit position has advanced to S6 (i.e. an entire sequence of the FPS has been received at that bit position), a subsequent non-matching bit causes the state to be returned to S1. However, when the state for a bit position has advanced to S18 (i.e. three entire sequences of the FPS have been consecutively received at that bit position), a subsequent non-matching bit causes the state to only be returned to S7. Thus, only two additional sequences of the FPS need be consecutively received at that bit position for a frame alignment to be declared.

It should be apparent that the state transition scheme illustrated in FIG. 6 is only an example, and that other schemes may be implemented. For example, a strict state transition scheme where all non-matching conditions require a return to state S0 may be implemented. Alternatively, more forgiving state transition schemes may be implemented wherein only a return to a state corresponding to a previous FPS interval is required upon a non-matching condition, for example. Moreover, it should be further apparent that the number of states and the number of iterations of the FPS that must be received for frame alignment can be changed depending on the framing scheme used.

Framer RAM 514 is, for example, a single port burst mode RAM that is organized as 5432 words, each word having 32 bits. This desired capacity is determined from the number of channels that can be aligned in this example implementation (i.e. 28), as well as the number of possible data states that are stored for each framing bit candidate (i.e. 24), as will become more apparent from the descriptions below. It should be noted, however, that the desired capacity of the RAM can be easily designed for different state transition schemes, numbers of channels, and framing schemes, for example, after those skilled in the art are taught by the following descriptions for the present example implementation. It should be further noted that, although a single RAM is used to store the alignment states for all channels in this example implementation, that this is not necessary for the invention.

An example of a data structure 700 that can be stored in RAM 514 for maintaining channel frame alignment state information is shown in FIGs. 7A to 7C. As shown in FIG. 7A, block 702 comprises the first twenty-eight words in RAM 514 which are used to store

each channel's state variables. An example of the structure of a data word 706 that can be used to store one channel's state variables is shown in FIG. 7B. As shown in FIG. 7B, the state variables maintained for each channel include *frm_align_on*, *frm_status*[1:0], *frm_ctr*[3:0], *frm_bit_ctr*[8:0], *init_ram* and *ram_addr_ctr*[12:0]. The state variable

5 *frm_align_on* is used to store the current frame alignment status for that channel (e.g. a value of 1 indicates that the channel is in frame alignment). The state variable *frm_status*[1:0] is used to store the number of FPS patterns detected in a data stream (i.e. the number of entries having a state of 24; this can be more than one due to the possibility of mimics in the data stream). The state variable *frm_ctr*[3:0] is used to store the number of

10 superframes that have been processed from a data stream. The state variable *frm_bit_ctr*[8:0] is used to store the number of bits received for a channel, modulo 772. The state variable *init_ram* is used to store a value of 1 for the first 772 bits received for a channel after a *frame_req* signal has been received. This is used to cause the state machines to use a hard-wired initial state rather than the stored state so that the RAM need not be re-

15 written with zeroes. The state variable *ram_addr_ctr*[12:0] is used to store the current RAM index into the data states for the associated data channel from the beginning offset for that channel.

As further shown in FIG. 7A, the remainder of the RAM is divided into 28 blocks 704-0 to 704-27, each block 704 having 193 words for storing the data states for every bit in

20 a respective one of the 28 channels. An example of a data state word structure 708 is shown in FIG. 7C. As shown in FIG. 7C, four data states each having 5 bits are stored in one word. Thus 772 data states (193 x 4) are stored in memory for each channel. Each data state represents the current ESF frame alignment state of that candidate bit. Since 5 bits are stored for each data bit, up to 32 state conditions can be stored. However, in an example of

25 the invention where the state machines 516 described above are implemented, only 24 state conditions (i.e. S0 to S23) are stored, with a stored value of 0 corresponding to the state S0 up to a stored value of 23 corresponding to state S23.

Although FIGs. 7A through 7C illustrate an example of data structures that can be used to implement storage of alignment states for a predetermined number of channels (e.g.

30 28) and a predetermined framing format (e.g. ESF), those skilled in the art will be able to understand how to implement storage of alignment states for variable numbers of channels and/or framing formats after being taught by the present example, and so the invention is not limited to these illustrative descriptions.

FIG. 8 is a block diagram illustrating an example of MFAE 502 in more detail. As shown in FIG. 8, MFAE 502 includes an engine controller 802, a data buffer 804, a multiplexer 806, a RAM interface 808, a frame detection block 810 and a channel counter 812.

5 Generally, engine controller 802 receives and operates at the *fae_clk* to coordinate the task of performing frame alignment for each of inline framers 504 that has requested alignment. Engine controller 802 cycles through each of the 28 channels in accordance with channel counter 812 (which cycles from 1 to 28). For each channel, engine controller 802 determines whether frame alignment has been requested and if data is available (e.g. by
10 checking the *frame_req* and *sipo_full* signals for that channel). If so, sixteen bits of data are read from the inline framer on *fae_data_in[15:0]* from read bus 522 and latched into buffer 804 (e.g. a 16 bit shift register). The received data bits are then serially shifted out two bits at a time. Even bits are diverted to state machine 516-1 and odd bits to state machine 516-2 using the multiplexer 806. Thus for every clock cycle *fae_clk*, engine controller 802 causes
15 the two state machines 516-1 and 516-2 to receive the previous state in *state_in[4:0]* from the RAM interface 808 and the data candidate bit from data buffer 804.

Frame detection block 810 monitors the state machines for an indication that a frame alignment has been detected, and generates a *frame_sync* signal to the associated channel in that case. A *frame_bit_ctr* maintained in the framer RAM 514 for each channel increments
20 from 0 to 771, which tracks which candidates bit or bits frame detection block 810 provides along with *frame_sync* signal.

Engine controller 802 can be implemented substantially by a state machine. An example of the operation of engine controller 802 is further illustrated in FIG. 9. Those skilled in the art will be able to implement engine controller 802 based on the following and
25 foregoing operational descriptions.

As shown in FIG. 9, in sequence, engine controller 802 may cycle through up to seven MFAE states for each of the 28 channels in 17 cycles of the *fae_clk* per channel. In the FAE_RD state, the engine controller 802 reads the SIPO status and data from the channel indicated by channel counter 812. Only if a valid *frame_req* and *sipo_full* is
30 received as determined in the FAE_CH state, will the engine controller cycle through the remaining states for that channel. Otherwise, the cycle is aborted, the channel counter is incremented (e.g. from 0 to 27, modulo 28) and the FAE_RD state commences for the next channel.

If valid *frame_req* and *sipo_full* signals are received, engine controller 802 causes RAM interface 808 to fetch the channel's previous state variables from the RAM 514's state variable area 702 during the ST_RD state. In the ST_RAM_RD state, engine controller 802 causes the data states for the current 16 bits to be fetched from the data state locations 704
 5 for the associated channel in RAM 514. In the ST_PROCESS state, engine controller 802 causes the ESF state processing to be performed by state machines 516-1 and 516-2 two bits at a time, so as to reduce the number of clock cycles needed to process all 16 bits. Hence two instances of ESF state machines are used in this example of the invention. The updated states from the ESF state machine are stored back into the RAM in the ST_RAM_WR state.
 10 Finally, in the ST_WR state, the channel state variables are stored back into the channel state location in the RAM 514. If frame alignment is achieved, then frame detection block 810 writes the status and frame bit location back into the inline framer 504 via the write bus 524 and word *frame_loc*. As shown in FIG. 8, engine controller 802 can allow for some overlap of functions between states, thus allowing all 16 bits of received data to be processed in 17
 15 *fae_clks*.

An example of RAM Interface 808 is further shown in FIG. 10. As shown in FIG. 10, it includes address multiplexer 1002, data state address generator 1004, RAM address lookup table 1006, input buffer 1008, output buffer 1010, RAM pointer 1012, buffer counter 1014, input multiplexers 1016, 1018 and output multiplexers 1020, 1022. Generally, RAM
 20 interface 808 receives the MFAE states generated by engine controller 802, along with the current channel being processed as indicated by channel counter 812, and controls the writing and reading of channel and data states between framer RAM 514 and state machines 516-1 and 516-2 based on those received inputs. It should be noted that RAM interface 808 preferably also includes circuitry for reading, updating and writing channel state variables
 25 from and to the channel state area 702 of RAM 514. However, such circuitry is not shown here for ease of illustration of the present invention.

Referring to FIG. 10, it can be seen that the 13 bit wide *ram_addr* signals can address either the channel state area or the data state area in the framer RAM 514. More particularly, address multiplexer 1002 selects the channel address generated by channel
 30 counter 812 during the ST_RD and SR_WR states (i.e. for addressing the appropriate channel in channel state area 702) and the data state address generator 1004 in other states (i.e. for addressing the appropriate set of data states in data state areas 704). The channel counter increments from 0 to 27 and indicates the current channel or framer being

processed. The data state address generator 1004 generates the address for the individual data states located in the data state area of the RAM. The generated address will be a combination of the index into the appropriate data state area obtained from the *ram_addr_ctr* of the channel state word stored in the RAM as shown in FIG. 7B and the starting address for the channel, *ram_start_addr*, obtained from RAM address look up table 1006. The data state address generator 1004 will increment the address by one word for every four candidate data bits that are processed. Once 193 words have been read out for a particular channel, the value of *ram_addr_ctr* will be reset to zero and the value of *ram_start_addr* from the look up table will be used as the starting address.

10 In this example implementation of the invention, 16 data bits for each channel are processed at a time. Accordingly, 4 words from the framer RAM 514 are read out in one burst during the ST_RAM_RD cycle. The data state address generation is repeated two times -- once in the ST_RAM_RD and once in the ST_RAM_WR cycles. In the ST_RAM_WR cycle the current data states for the 16 bits are written back into the data state area of the memory from the state machines 516-1 and 516-2. In the last MFAE state, ST_WR, the current *ram_addr_ctr* is stored back into the channel state word as shown in FIG. 7B. Thus for the next cycle of the channel, the data state address generator 1004 will continue to generate addresses from the last accessed location.

RAM interface 808 also generates signals for controlling the reading and writing of data states to RAM 514 depending upon the current MFAE state as provided by the engine controller 802. The RAM output enable signal, *ram_oe*, is kept permanently enabled. The RAM memory enable signal, *ram_me*, is driven during ST_RD, ST_WR, ST_RAM_RD, and ST_RAM_WR states. The RAM write enable signal, *ram_we*, is kept at logic 1 during ST_RAM_WR and ST_WR states when data is being written into the RAM.

25 The four data state words read out from the RAM beginning at the address generated by the data state address generator 1004 are stored into a 16 location deep and 5 bit wide register array, input buffer 1008, during the ST_RAM_RD state. Four individual 5 bit data states are arranged in each 32 bit data word as shown in FIG. 7C. As shown, the lower 5 bits of each byte in each 32 bit data word contain the data state for a respective candidate bit (e.g. bit 0 to 771 for each channel). The data states read from the RAM during the ST_RAM_RD state are written into a location in input buffer 1008 in accordance with the value of the RAM pointer 1012. The pointer is incremented by 4 every clock during the ST_RAM_RD state. Thus, the first four data states read from the RAM (i.e. from the first

data state word addressed by the data state address generator 1014) are written into a first column of the input buffer 1008 on the first clock, the next four data states read from the RAM (i.e. from the second data state word addressed by the data state address generator 1014) are written into a second column of the input buffer 1008 on the second clock, and so on.

Data is read out from input buffer 1008 in a first-in-first-out fashion two data states at a time. Input multiplexers 1016, 1018 redirect five bits of data from an appropriate location in input buffer 1008 into either state machine 516-1 or 516-2. All even data bit states (i.e. states corresponding to Bit 0, Bit 2, Bit 4, etc.) are sent to state machine 516-1 through the *state_in_0[4:0]* signals, while all odd data bit states (i.e. states corresponding to Bit 1, Bit 3, Bit 5, etc.) are sent to state machine 516-2 through the *state_in_1[4:0]* signals. The reading of the data states out to the state machines is controlled by *st_buffer_ctr* 1014 which increments from 0 to 7 during the ST_RAM_RD and ST_RAM_WR states. For example, when *st_buffer_ctr* is 0, input multiplexer 1016 selects a first entry in a first column (i.e. entry 0) to be provided from input buffer 1008 on the *state_in_0[4:0]* signals, while input multiplexer 1018 selects a second entry in the first column (i.e. entry 1) to be provided from input buffer 1008 on the *state_in_1[4:0]* signals. Next, when *st_buffer_ctr* is 1, input multiplexer 1016 selects a second entry in a first column (i.e. entry 3) to be provided from input buffer 1008 on the *state_in_0[4:0]* signals, while input multiplexer 1018 selects a fourth entry in the first column (i.e. entry 4) to be provided from input buffer 1008 on the *state_in_1[4:0]* signals. Then, when *st_buffer_ctr* increments to 2, input multiplexer 1016 selects a first entry in a second column (i.e. entry 4) to be provided from input buffer 1008 on the *state_in_0[4:0]* signals, while input multiplexer 1018 selects a second entry in the second column (i.e. entry 5) to be provided from input buffer 1008 on the *state_in_1[4:0]* signals. It should be apparent that this scheme can allow for data to be written into certain locations of input buffer 1008 while data is read out from certain other locations of input buffer 1008.

The processed data states from the two state machines 516-1 and 516-2 are received and written back into RAM 514 by RAM interface 808 in substantially the reverse manner as described above. The processed data states are supplied by state machines 516-1 and 516-2 on the *state_out0* and *state_out1* signals, respectively. These are stored in output buffer 1010 via output multiplexers 1020, 1022. Output multiplexers 1020, 1022 cause processed data states received via the *state_out0* signals to be written into the even locations

in the columns of output buffer 1010 while data states received on the *state_out1* signals are written into odd locations in the columns of output buffer 1010. During the ST_RAM_WR state, four five-bit data states at a time are written back into RAM 514. The *ram_ptr* points to the next location to be read from output buffer 1010 and increments by 4 every clock
 5 cycle. The four data states are formatted into one 32 bit word as shown in FIG. 7C and written back into the data state area of the RAM 514.

It should be noted that the above descriptions of the timings of operations of various components of RAM interface 808 with respect to the MFAE processing states received from engine controller 802 are given for ease of illustration. However, those skilled in the
 10 art will understand that more complex timing schemes may need to be implemented for proper sequence and/or pipelining of operations.

Frame detection block 810 monitors the *frame_sync* signals from both state machines 516-1 and 516-2 and latches the *frame_bit_ctr* value into a *frame_location[9:0]* register if a valid *frame_sync* signal is received. To avoid framing to a pattern that is
 15 random data mimicking the frame alignment pattern, frame detection block 810 is inactive until a minimum number of frames (e.g. 24) have been processed for the channel (as indicated by the *frm_ctr* word in the channel state variable, for example).

After the requisite number of frames have been processed for a given channel, frame detection block 810 will begin operation after each set of 16 data bits are received from the
 20 associated channel. At that time, three scenarios are possible:

1. No *frame_sync* signal has been generated. The search for a frame alignment is thus continued in the subsequent frames until the first frame alignment is found. This state is saved by setting *frm_status[1:0]* equal to a value of 0.

2. Only one *frame_sync* was detected in the entire range of 772 candidate bits. In
 25 this case, frame detection block 810 will set *frm_status* to a value of 1 and assert the *frm_align* signal. *Frame_location[9:0]* will indicate the position of the frame bit with respect to the *frame_req* signal.

3. More than one *frame_sync* signals is detected within a frame period of 772 bits. This indicates the presence of one or more mimic pattern and is indicated by *frm_status*
 30 having a value of 2. In this case the frame search is continued in subsequent frames until the mimic pattern dies out and only one frame alignment is obtained.

During each cycle in the ST_WR state, a check is made on the *frm_status* signals. If it has a value of 1, indicating a successful alignment, the *frame_location* value is written

through the common bus *fae_data_out[9:0]* using the *fae_addr[4:0]* and *fae_wr* signals to the corresponding inline framer 504. The inline framer can then use this value to decode all other relevant data bits in the frame.

5 Although the present invention has been particularly described with reference to the preferred embodiments thereof, it should be readily apparent to those of ordinary skill in the art that changes and modifications in the form and details may be made without departing from the spirit and scope of the invention. It is intended that the appended claims include such changes and modifications.

What is claimed is:

1. An apparatus for simultaneously determining a respective frame alignment in each of a plurality of channels, the apparatus comprising:

- 5 means for simultaneously storing a frame alignment state for a plurality of frame bit candidates for each of the plurality of channels;
 means for receiving data from the plurality of channels;
 means for updating the stored frame alignment state for each of the plurality of frame bit candidates in accordance with the received data; and
10 means for declaring the respective frame alignment in accordance with the updated frame alignment state for a certain one of the plurality of frame bit candidates for each of the plurality of channels.

2. An apparatus according to claim 1, wherein the updating means includes:

- 15 means for retrieving the frame alignment state for certain of the frame bit candidates associated with the received data;
 means for comparing the received data with expected framing data based on the retrieved frame alignment states;
 means for advancing the retrieved frame alignment states upon a successful
20 comparison of the received data with the expected framing data;
 means for withholding advancing of the retrieved frame alignment states upon a failed comparison of the received data with the expected framing data; and
 means for replacing the retrieved frame alignment states with the updated frame alignment states in the storing means.

25

3. An apparatus according to claim 2, wherein the withholding means is operative to reset the retrieved frame alignment states to less advanced states in accordance with current frame alignment strengths indicated by the retrieved frame alignment states.

30

4. An apparatus according to claim 1, further comprising :

- means for polling the plurality of channels for frame alignment requests; and

means for controlling the operation of the receiving means, the updating means and the declaring means for certain of the plurality of channels based on the polled frame alignment requests.

5 5. An apparatus according to claim 4, wherein each of the plurality of channels includes an inline framer, the inline framer comprising:

 means for generating the frame alignment request; and

 means for processing channel data using an old frame alignment simultaneously while the respective frame alignment is being determined.

10

 6. An apparatus according to claim 1, wherein the respective frame alignment for a first one of the plurality of channels is in accordance with a first framing scheme, and wherein the respective frame alignment for a second one of the plurality of channels is in accordance with a second framing scheme, the first and second framing schemes both being
15 one of DS1 SF, DS1 DDS, DS1 SLC-96, and ITU-T G.704 based E1.

 7. An apparatus according to claim 1, wherein the respective frame alignment for a first one of the plurality of channels is in accordance with a first framing scheme, and wherein the respective frame alignment for a second one of the plurality of channels is in
20 accordance with a second framing scheme, the first and second framing schemes each being a different one of DS1 SF, DS1 DDS, DS1 SLC-96, and ITU-T G.704 based E1.

 8. A method for simultaneously determining a respective frame alignment in each of a plurality of channels, the apparatus comprising:

25 simultaneously storing a frame alignment state for a plurality of frame bit candidates for each of the plurality of channels;

 receiving data from the plurality of channels;

 updating the stored frame alignment state for each of the plurality of frame bit candidates in accordance with the received data; and

30 declaring the respective frame alignment in accordance with the updated frame alignment state for a certain one of the plurality of frame bit candidates for each of the plurality of channels.

9. A method according to claim 8, wherein the updating step includes:
retrieving the frame alignment state for certain of the frame bit candidates
associated with the received data;
comparing the received data with expected framing data based on the
retrieved frame alignment states;
5 advancing the retrieved frame alignment states upon a successful comparison
of the received data with the expected framing data;
withholding advancing of the retrieved frame alignment states upon a failed
comparison of the received data with the expected framing data; and
10 replacing the retrieved frame alignment states with the updated frame
alignment states.

10. A method according to claim 9, wherein the withholding step includes resetting
the retrieved frame alignment states to less advanced states in accordance with current
15 frame alignment strengths indicated by the retrieved frame alignment states.

11. A method according to claim 8, further comprising :
polling the plurality of channels for frame alignment requests; and
20 controlling performance of the receiving step, the updating step and the
declaring step for certain of the plurality of channels based on the polled frame alignment
requests.

12. A method according to claim 11, wherein each of the plurality of channels
25 includes an inline framer, the method further comprising:
processing channel data at the inline framer using an old frame alignment
simultaneously while the respective frame alignment is being determined.

13. A method according to claim 8, wherein the respective frame alignment for a
30 first one of the plurality of channels is in accordance with a first framing scheme, and
wherein the respective frame alignment for a second one of the plurality of channels is in
accordance with a second framing scheme, the first and second framing schemes both being
one of DS1 SF, DS1 DDS, DS1 SLC-96, and ITU-T G.704 based E1.

14. A method according to claim 8, wherein the respective frame alignment for a first one of the plurality of channels is in accordance with a first framing scheme, and wherein the respective frame alignment for a second one of the plurality of channels is in accordance with a second framing scheme, the first and second framing schemes each being
5 a different one of DS1 SF, DS1 DDS, DS1 SLC-96, and ITU-T G.704 based E1.

15. An apparatus, comprising:
a frame alignment engine coupled to a bus for commonly receiving data from
10 a plurality of channels, the received data from each of the plurality of channels including a framing pattern embedded at a frame bit offset with respect to an arbitrary starting bit of a stream of data respectively associated with each of the plurality of channels;
a memory coupled to the frame alignment engine, the memory having a plurality of channel state entries respectively corresponding to the plurality of channels,
15 each of the channel state entries including a counter referenced to the arbitrary starting bit, the memory further having a plurality of data state entries for each of the plurality of channels, the plurality of data state entries respectively corresponding to a plurality of frame bit candidates, the frame alignment engine being responsive to data received from a requesting one of the channels so as to retrieve the counter for the requesting channel, and
20 to retrieve certain of the data state entries corresponding to the received data from the requesting channel based on the counter; and
a state machine coupled to the frame alignment engine, the state machine including a comparator that is adapted to compare the data received from the requesting channel with the framing pattern and to update the certain data state entries based on the
25 comparison.

16. An apparatus according to claim 15, wherein each of the plurality of channels includes an inline framer, the inline framer comprising:
a frame register for storing an old frame bit offset;
30 an out of frame detector coupled to the frame register for processing the respective stream of data using the old frame bit offset simultaneously while the respective frame bit offset is being determined.

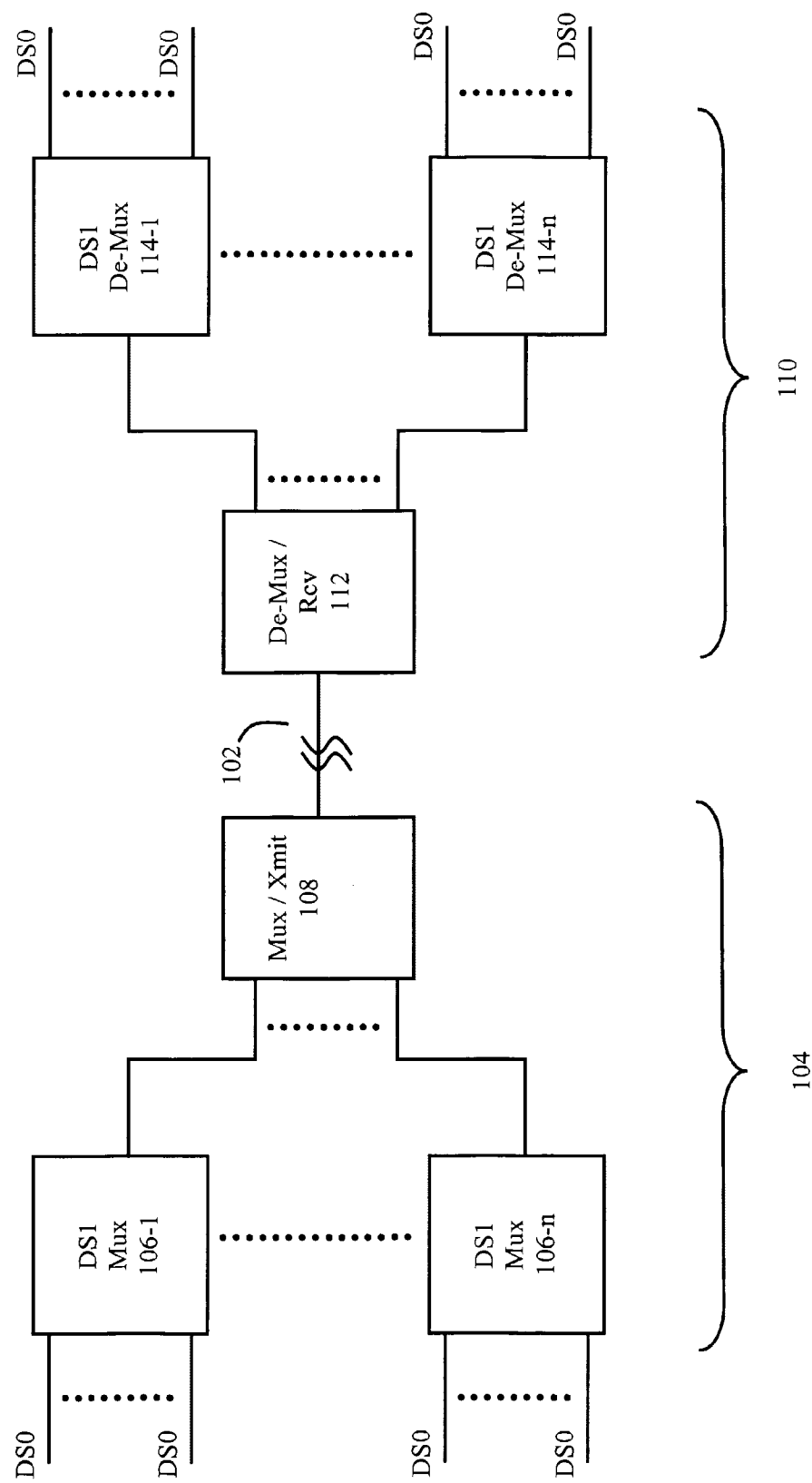


FIG. 1
(PRIOR ART)

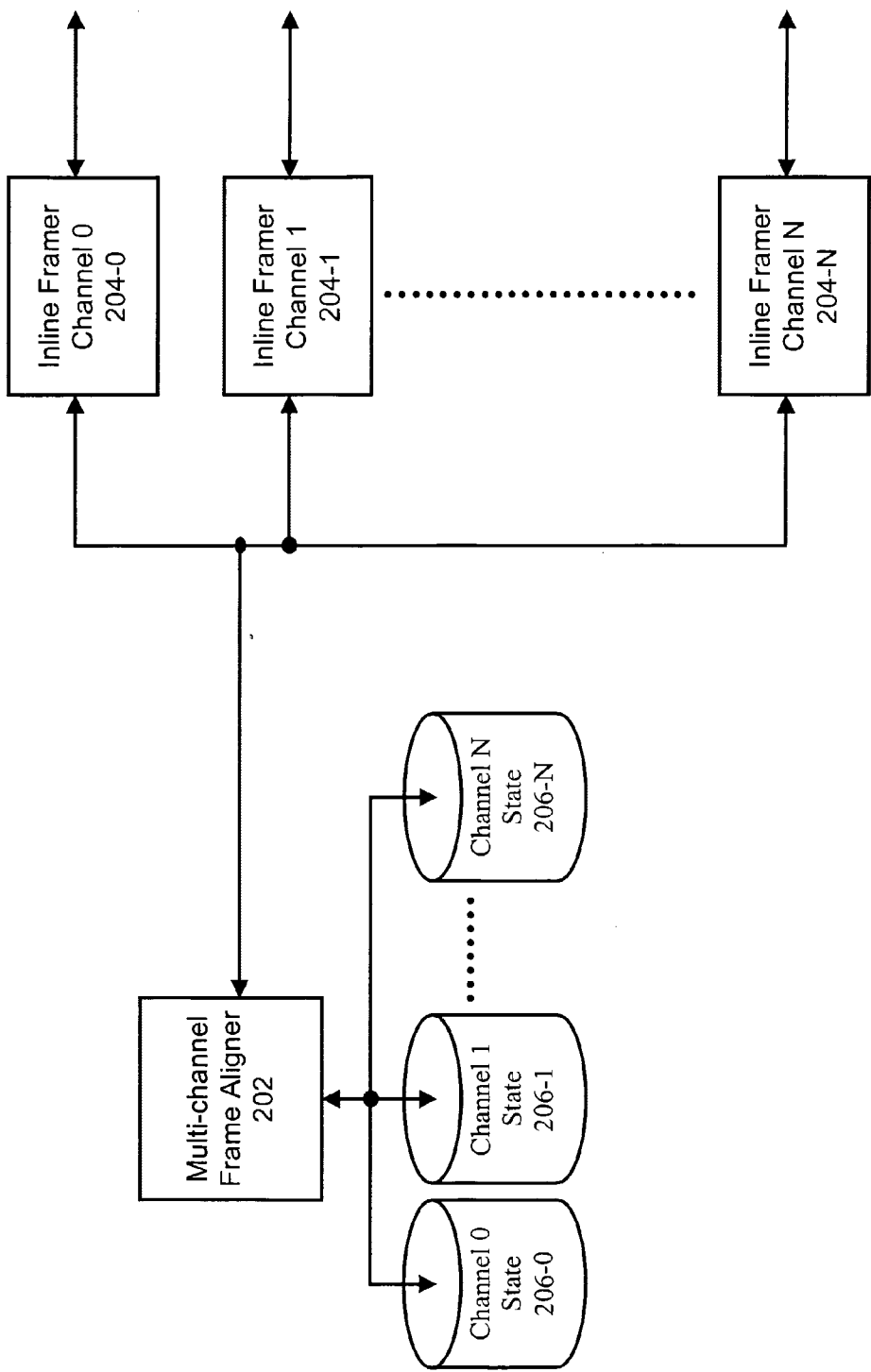


FIG. 2

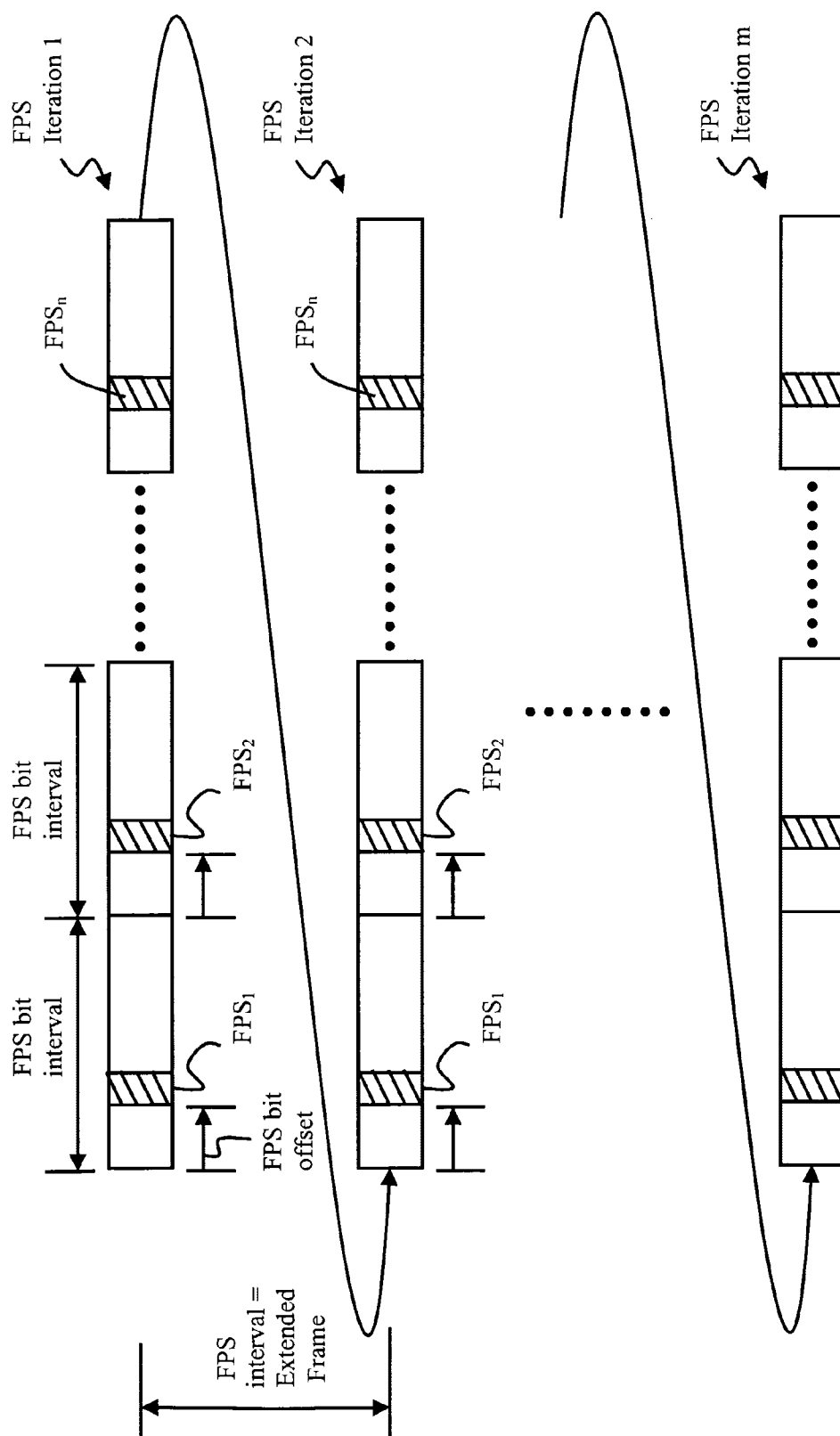


FIG. 3

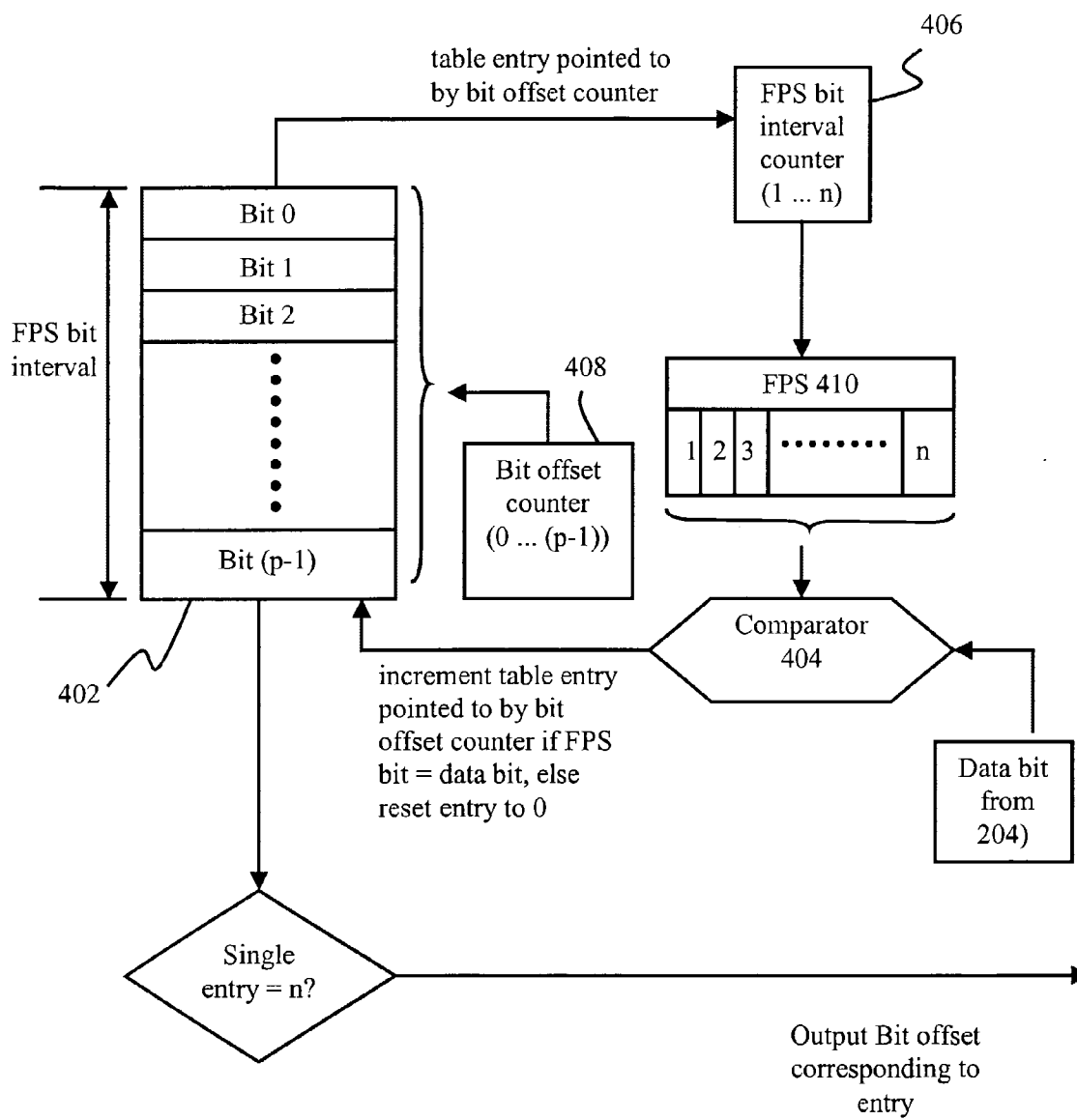


FIG. 4

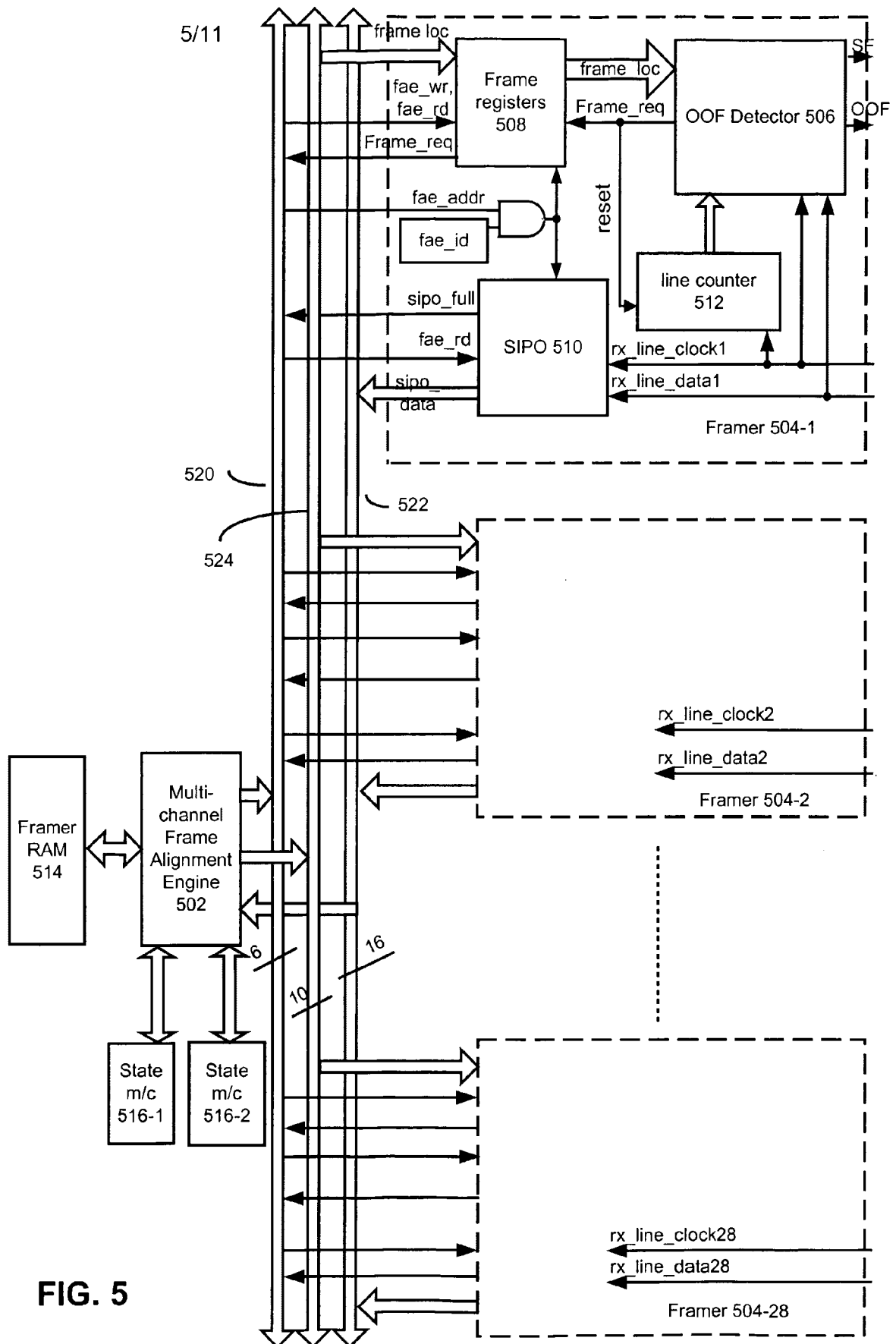


FIG. 5

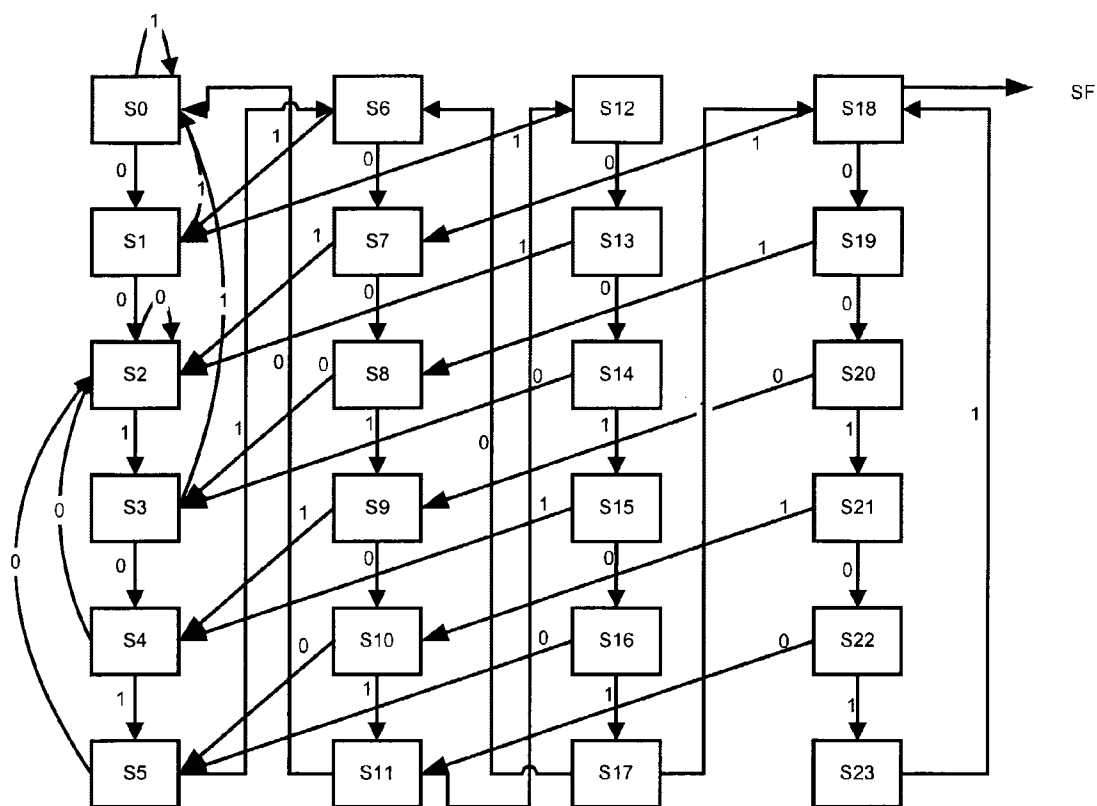
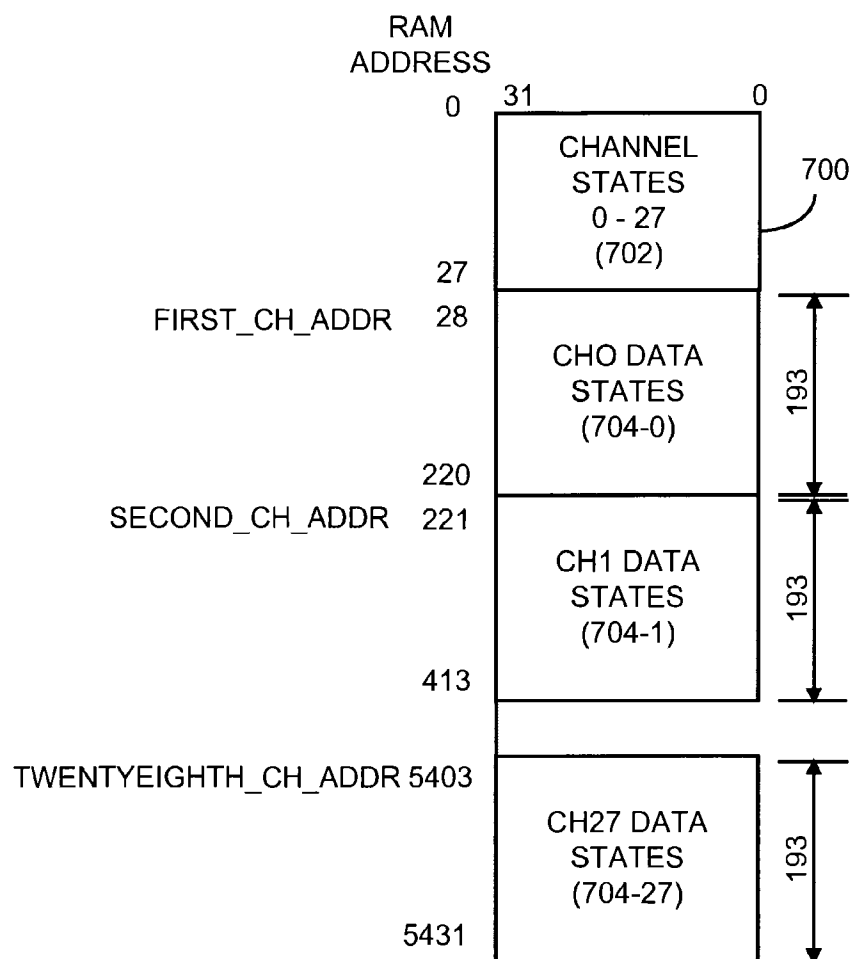


FIG. 6

7/11

**FIG. 7A**

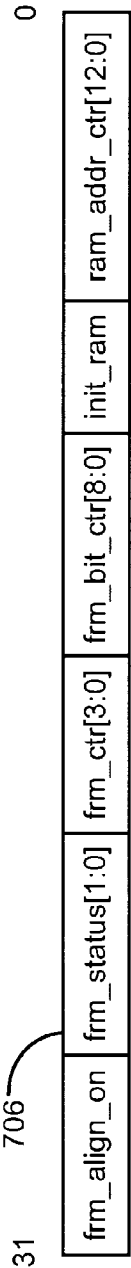


FIG. 7B

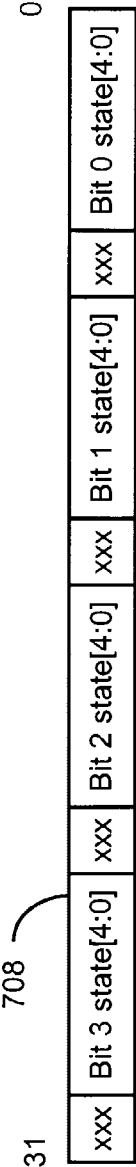


FIG. 7C

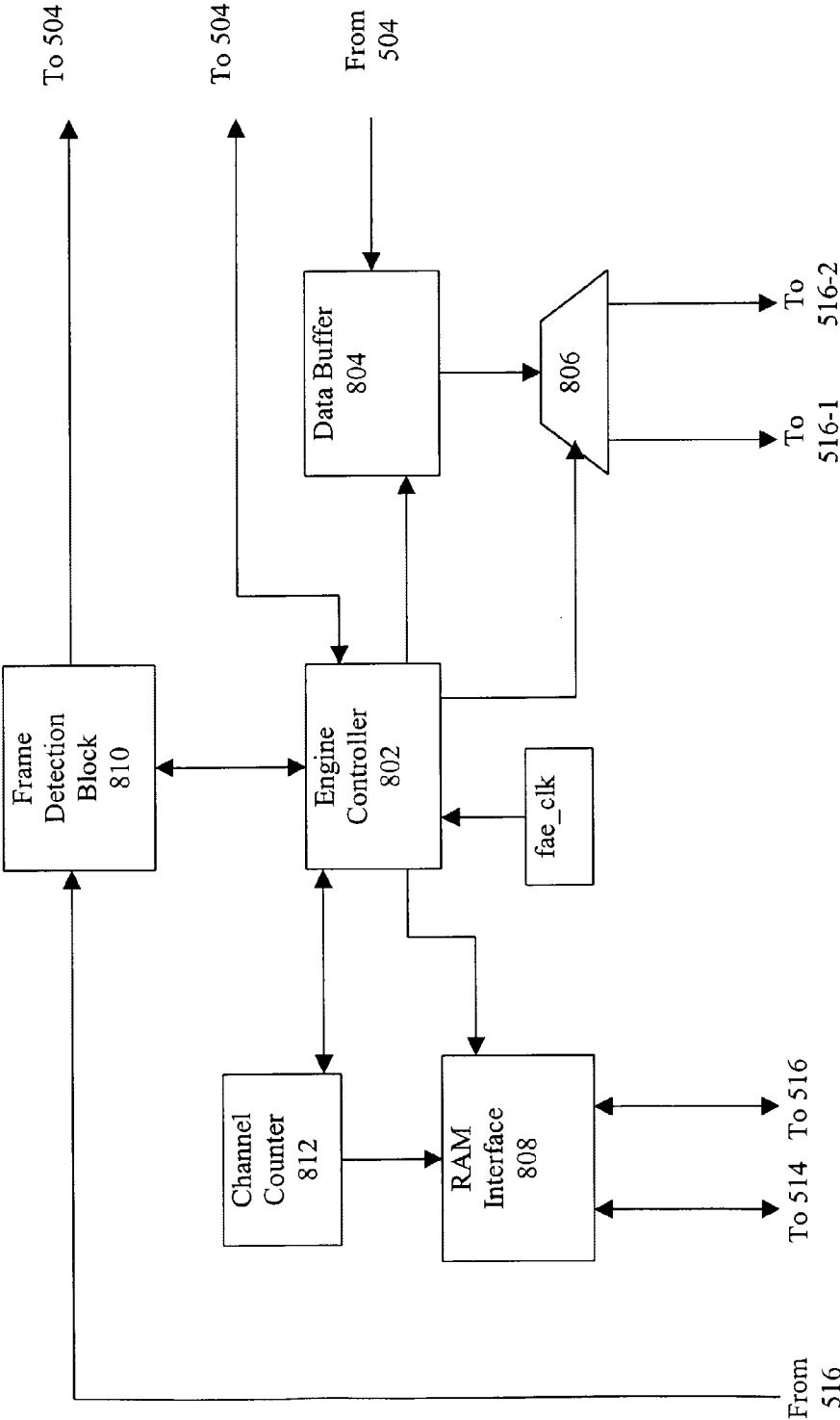


FIG. 8

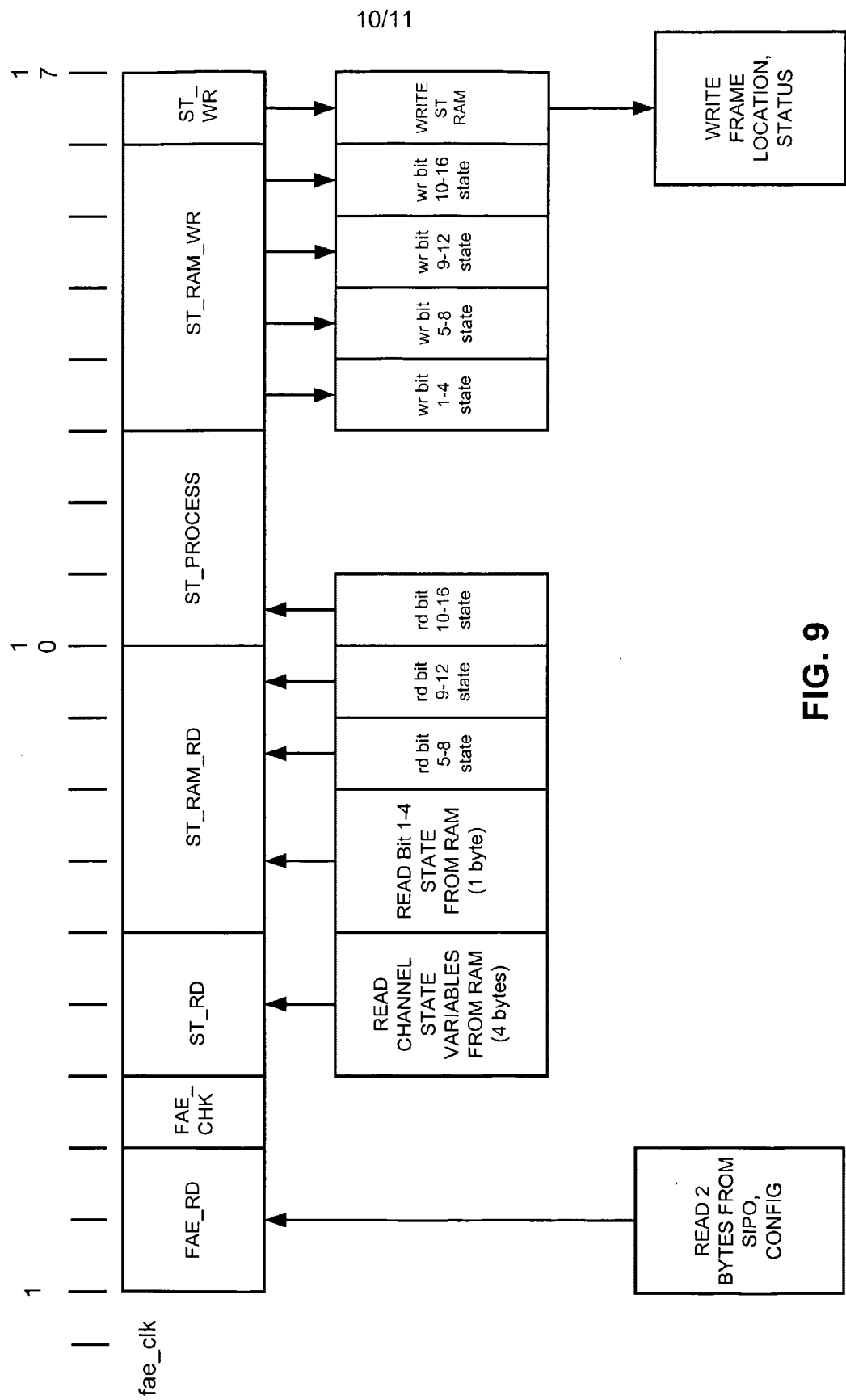
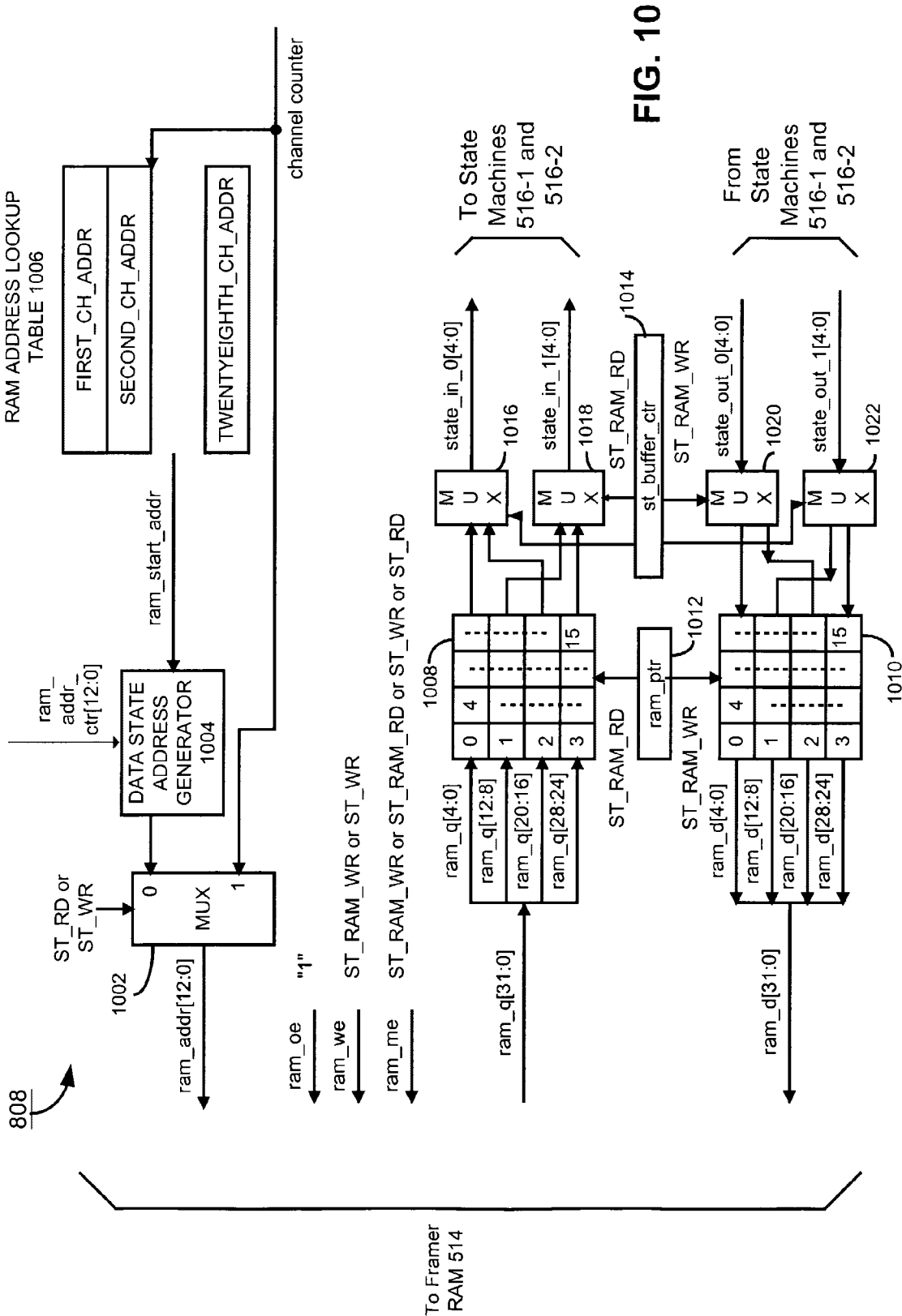


FIG. 9



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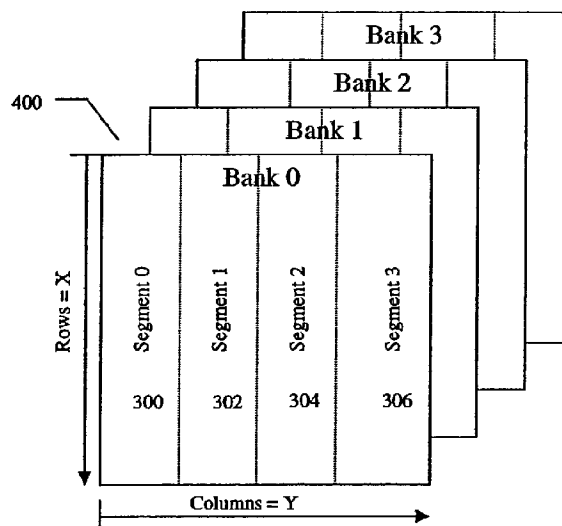
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(54) Title: **SYNCHRONOUS FLASH MEMORY WITH VIRTUAL SEGMENT ARCHITECTURE**



(57) Abstract: An improved memory device and architecture has been detailed that enhances a Flash memory device that has an SDRAM compatible interface. The memory device employs a virtual paging scheme that allows the memory to have an efficient Flash internal structure, while logically re-mapping this architecture externally to a compatible virtual SDRAM architecture. This allows for access and operation of the improved memory device with a compatible SDRAM controller device, while Flash specific functions can be performed with an SDRAM command sequence. Internal to the memory, memory array banks are divided into four equal segments by row range and logically re-mapped by placing the segments virtually beside each other. This forms a virtual memory bank structure of equivalent rows and columns as a comparable SDRAM device. Additionally, the improved memory device may also have an extended interface that allows for direct access to the internal Flash memory architecture without logical abstraction.



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SYNCHRONOUS FLASH MEMORY WITH VIRTUAL SEGMENT ARCHITECTURE

Technical Field of the Invention

5 The present invention relates generally to non-volatile memory devices and in particular the present invention relates to synchronous non-volatile flash memory with virtual segment architecture.

Background of the Invention

Memory devices are typically provided as internal storage areas in the computer.
10 The term memory identifies data storage that comes in the form of integrated circuit chips. There are several different types of memory used in modern electronics, one common type is RAM (random-access memory). RAM is characteristically found in use as main memory in a computer environment. RAM refers to read and write memory; that is, you can both write data into RAM and read data from RAM. This is in contrast to ROM, which permits you
15 only to read data. Most RAM is volatile, which means that it requires a steady flow of electricity to maintain its contents. As soon as the power is turned off, whatever data was in RAM is lost.

Computers almost always contain a small amount of read-only memory (ROM) that holds instructions for starting up the computer. Unlike RAM, ROM cannot be written to.
20 An EEPROM (electrically erasable programmable read-only memory) is a special type non-volatile ROM that can be erased by exposing it to an electrical charge. Like other types of ROM, EEPROM is traditionally not as fast as RAM. EEPROM comprise a large number of memory cells having electrically isolated gates (floating gates). Data is stored in the memory cells in the form of charge on the floating gates. Charge is transported to or removed from
25 the floating gates by programming and erase operations, respectively.

Yet another type of non-volatile memory is a Flash memory. A Flash memory is a type of EEPROM that can be erased and reprogrammed in blocks instead of one byte at a time. Many modern PCs have their BIOS stored on a flash memory chip so that it can easily be updated if necessary. Such a BIOS is sometimes called a flash BIOS. Flash memory is
30 also popular in modems because it enables the modem manufacturer to support new protocols as they become standardized.

A typical Flash memory comprises a memory array, which includes a large number of memory cells. Each of the memory cells includes a floating gate field-effect transistor capable of holding a charge. The cells are usually grouped into blocks. Each of the cells within a block can be electrically programmed in a random basis by charging the floating gate. The charge can be removed from the floating gate by a block erase operation. The data in a cell is determined by the presence or absence of the charge in the floating gate.

Both RAM and ROM random access memory devices have memory cells that are typically arranged in an array of rows and columns. During operation, a row (page) is accessed and then memory cells can be randomly accessed on the page by providing column addresses. This access mode is referred to as page mode access. To read or write to multiple column locations on a page requires the external application of multiple column addresses. To increase access time, a burst mode access has been implemented. The burst mode uses an internal column address counter circuit to generate additional column addresses. The address counter begins at an externally provided address and advances in response to an external clock signal or a column address strobe signal.

A synchronous DRAM (SDRAM) is a type of DRAM that can run at much higher clock speeds than conventional DRAM memory. SDRAM synchronizes itself with a CPU's bus and is capable of running at 100 MHZ or 133 MHZ, about three times faster than conventional FPM (Fast Page Mode) RAM, and about twice as fast EDO (Extended Data Output) DRAM and BEDO (Burst Extended Data Output) DRAM. An extended form of SDRAM that can transfer a data value on the rising and falling edge of the clock signal is called double data rate SDRAM (DDR SDRAM, or simply, DDR). SDRAM's can be accessed quickly, but are volatile. Many computer systems are designed to operate using SDRAM, but would benefit from non-volatile memory. A synchronous Flash memory has been designed that allows for a non-volatile memory device with an SDRAM interface. Although knowledge of the function and internal structure of a synchronous Flash memory is not essential to understanding the present invention, a detailed discussion is included in United States Patent Application Serial No. 09/627,682 filed July 28, 2000 and titled, "Synchronous Flash Memory," which is commonly assigned and incorporated by reference.

In general, the goal of synchronous Flash is to mimic the architecture of SDRAM. It has an SDRAM interface which is compatible to SDRAM for read operation to the synchronous Flash memory. Programming, erasing, block protection and other flash specific

function differ from SDRAM and are performed with a three cycle SDRAM command sequence. Unfortunately, the general internal architecture of conventional SDRAM memory is not the most efficient architecture for non-volatile memories and for Flash memory specifically.

For the reasons stated above, and for other reasons stated below which will become apparent to those skilled in the art upon reading and understanding the present specification, there is a need in the art for a non-volatile memory device that can operate in a manner similar to SDRAM operation and maintain an internal architecture that is more applicable to non-volatile memory device access and power efficiencies.

10 Summary of the Invention

The above-mentioned problems with memory devices and other problems are addressed by the present invention and will be understood by reading and studying the following specification.

In one embodiment, a synchronous flash memory device comprises a first memory array with a first memory array dimensionality, a control circuit, and a synchronous memory interface, wherein the control circuit logically adapts the first memory array dimensionality to a second memory array dimensionality.

In another embodiment, a synchronous non-volatile memory device comprises a memory array with a plurality of array banks, each array bank having a first number of rows and a first number of columns, a control circuit, and a synchronous memory interface, wherein, for each array bank of the plurality of array banks, the control circuit logically adapts the first number of rows and the first number of columns to a second number of rows and a second number of columns.

25 In a further embodiment, a synchronous flash memory device comprises a first memory array with a first memory array dimensionality, a control circuit, wherein the control circuit logically adapts the first memory array dimensionality to a second memory array dimensionality, and a synchronous memory interface. Wherein the synchronous memory interface comprises, an address interface, a data interface, and a control interface.

30 In yet another embodiment, a synchronous flash memory device comprises a first memory array with a plurality of array banks, each having a first row dimension and a first column dimension, a control circuit, wherein the control circuit logically adapts the first row

dimension and the first column dimension of each of the plurality of banks to a second row dimension and a second column dimension, and a synchronous memory interface. Wherein the synchronous memory interface comprises, an address interface, an extended address interface, a data interface, and a control interface.

5 A method of operating a synchronous flash memory device comprises dividing a memory array with a first memory array dimensionality into a plurality of sections, and logically adapting the plurality of sections of the memory array to form an emulated virtual second memory array dimensionality.

 Another method of operating a synchronous flash memory device comprises
10 dividing a memory array with a first memory array dimensionality into a plurality of array banks, dividing each array bank of the plurality of array banks into a plurality of segments, logically adapting the plurality of array banks and plurality of segments of the first memory array dimensionality of the memory array to form an emulated virtual second memory array dimensionality, wherein accessing the emulated virtual second memory array dimensionality
15 occurs through a synchronous memory interface, and allowing access to the first memory array dimensionality of the memory array through the synchronous memory interface and an extended interface.

 In yet another method of logically mapping a synchronous non-volatile memory device comprises forming a first memory array with a first plurality of banks, dividing each
20 bank of the first plurality of banks into a plurality of segments, and logically mapping the plurality of segments to appear as a second memory array with a second plurality of segments.

 In a further embodiment, a synchronous flash memory device comprises a memory array with a first set of four array banks, such that each array bank has a first number
25 of rows and a first number of columns and each array bank is divided into four segments by row range, a latch circuit, a control circuit, and an SDRAM compatible synchronous memory interface. Wherein the control circuit logically maps the latch circuit and the four segments in each of the four array banks to appear as an emulated virtual SDRAM memory device of an equal memory size with a second virtual set of four array banks, such that the first number
30 of rows and the first number of columns of each array bank of the memory array map to a second number of rows, that is one fourth the first number of rows, and a second number of

columns, that is four times the first number of columns, of the second virtual set of four array banks of the emulated virtual SDRAM memory device.

In another method of making a synchronous flash memory device comprises forming a non-volatile memory array with a plurality of array banks, each array bank having a first number of rows and a first number of columns, forming a control circuit, and forming a synchronous interface, wherein the control circuit logically adapts the first number of rows and the first number of columns of each array bank of the plurality of array banks to a second number of rows and a second number of columns.

In another embodiment, a system comprises a synchronous memory controller, and a synchronous flash memory device. Wherein the synchronous flash memory device comprises, a memory array with a first memory array dimensionality, a control circuit, and a synchronous memory interface, wherein the control circuit logically adapts the first memory array dimensionality to a second memory array dimensionality.

Brief Description of the Drawings

Figure 1 illustrates a prior art SDRAM memory array with four banks of memory within the array.

Figure 2 illustrates a virtual synchronous flash memory array of the present invention with four banks of memory within the array.

Figures 3 and 4 illustrate a virtual synchronous Flash memory array of the present invention subdivided into segments and the segments being mapped into a virtual SDRAM memory array.

Figure 5 illustrates a block diagram example of a 4Mx32 virtual synchronous Flash memory, showing address, control and data interfaces.

Figure 6 is a diagram of a memory address for a virtual synchronous Flash, showing bank address, additional virtual synchronous Flash addresses, row address, and column address.

Figure 7 illustrates a bank of virtual synchronous Flash memory array of the present invention showing mapping of a segment to other segment positions within the bank.

Figure 8 is a block diagram of a virtual synchronous Flash memory device coupled to an external processor.

Detailed Description of the Invention

In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific preferred embodiments in which the inventions may be practiced. These
5 embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, and it is to be understood that other embodiments may be utilized and that logical, mechanical and electrical changes may be made without departing from the spirit and scope of the present invention. The following detailed description is, therefore, not to be taken in a
10 limiting sense, and the scope of the present invention is defined only by the claims.

A conventional synchronous Flash memory matches the internal architecture and simulates the operation of an SDRAM memory quite closely. In contrast, a virtual synchronous flash memory of the present invention changes the internal architecture, while maintaining a compatible conventional SDRAM interface. This allows it to better match the
15 efficiencies of the underlying memory technology. The resulting memory structure of a virtual synchronous Flash internal memory array is then logically reassembled to simulate an SDRAM configuration for the purposes of interfacing to a standard SDRAM compatible memory controller, processor, or other compatible external device. The virtual synchronous Flash memory of the present invention allows for additional memory address pins to be added
20 to the external address interface so as to enable external devices with the appropriate capability to access the internal architecture of the virtual synchronous Flash memory directly.

Figure 1 is a simplified diagram of a prior art SDRAM memory array architecture. SDRAM memory arrays generally consist of a sequence of memory array banks that have
25 rows and columns. The SDRAM memory arrays, upon being accessed with a bank, row, and column address, load an internal data latch circuit with a defined number of bits. The SDRAM memory array 100 of Figure 1 has four array banks 102, 104, 106, and 108. Each of the four banks 102, 104, 106, and 108 in turn has X rows 110 and Y columns 112. When a bank and row are selected by an input row address, the contents of a row of memory cells are
30 placed on column bit lines coupled to sense amplifiers (not shown) and the bit values of the memory cells are detected. The selected memory array row, and therefore the contents of the

sense amplifiers, is generally called the active row page. The values detected by the sense amplifiers are latched in latch circuit 114. An input column address to the SDRAM, which defines the column page, selects from the latches of the latch circuit 114, and thus from the columns of the active row page, the Z number of bit values to be latched by an internal data buffer (not shown). The Z data bits are held in the internal data buffer for transfer from the SDRAM memory device through the data interface (not shown). An example of a common SDRAM memory is a 4Mx32 SDRAM, which has 4 banks, 4,096 rows, 256 columns, and 32 bits. However, SDRAM memories come in multiple densities and bus widths, for example 8Mx8, 4Mx16, 4Mx32, 2Mx32, and 8Mx16 to list a few.

An SDRAM memory read operation generally starts with an "ACTIVE" command that sets the bank and row address, by reading in an address placed on the address interface to the SDRAM memory device. The "ACTIVE" command thus activates the active row page and places it on the internal sense amplifiers to be read and latched by internal latch circuits. The column address is read in from the address interface by a following "READ" command that sets the column page by selecting columns from the latch circuits to place in an internal data buffer in the SDRAM memory device. The SDRAM command sequence is then finished on the next clock cycle with a data word transfer from the internal data buffer.

In an SDRAM compatible command sequence, before any READ or WRITE commands can be issued to a bank within the synchronous flash memory, a row in that bank must be "opened." This is accomplished via an ACTIVE command, which is used to open (or activate) a row in a particular array bank for a subsequent access. The value on the BA0, BA1 inputs selects the bank, and the address provided on inputs A0-A11 selects the row. This row remains active for accesses until the next ACTIVE command, power-down or RESET.

A subsequent ACTIVE command to a different row in the same bank can be issued without having to close a previous active row, provided the minimum time interval between successive ACTIVE commands to the same bank is met. A subsequent ACTIVE command to another bank can also be issued while the first bank is being accessed, which results in a reduction of total row access overhead.

After an ACTIVE command to activate a row in the memory, the READ command is used to initiate a read access to an active row. The value on the BA0, BA1 inputs selects the bank, and the address provided on inputs A0-A7 selects the starting column

location. Read data appears on data outputs, the DQs (not shown) of the data interface (not shown) subject to the logic level on the data mask (DQM) input that is sometimes used to mask data DQ outputs and values not needed for the current memory access operation.

During READ bursts, the valid data-out element from the starting column address
5 is available following a specified latency after the READ command. In SDRAM compatible memory, each subsequent data-out element is valid by the next positive clock edge. Upon completion of a burst, assuming no other commands have been initiated, all data outputs go as to a high impedance state. A full page burst continues until terminated. (At the end of the page, it wraps to column 0 and continue.) Data from any READ burst may be truncated with
10 a subsequent READ command, and data from a fixed-length READ burst may be immediately followed by data from a subsequent READ command. In either case, a continuous flow of data can be maintained. The first data element from the new burst follows either the last element of a completed burst, or the last desired data element of a longer burst that is being truncated. A READ command can be initiated on any clock cycle following a
15 previous READ command. Full-speed, random read accesses within a page can be performed, or each subsequent READ may be performed to a different bank.

A fixed-length or full-page READ burst can be truncated with either ACTIVE
TERMINATE (may or may not be bank specific) or BURST TERMINATE (not bank
specific) commands. Data from any READ burst may also be truncated with a subsequent
20 WRITE command (WRITE commands must be preceded by WRITE SETUP), and data from a fixed-length READ burst may be immediately followed by data from a subsequent WRITE command (subject to bus turnaround limitations). The WRITE may be initiated on the clock edge immediately following the last (or last desired) data element from the READ burst, provided that I/O contention can be avoided.

25 As stated above, a virtual synchronous flash memory of the present invention changes the internal architecture, while maintaining a compatible conventional SDRAM interface. This allows it to better match the efficiencies of the underlying memory technology. Figure 2 details a simplified diagram of a memory array of a virtual synchronous Flash of the present invention. The resulting memory structure of a virtual synchronous
30 Flash internal memory array is then logically reassembled to simulate an SDRAM configuration for the purposes of interfacing to a standard SDRAM compatible memory controller, processor, or other compatible external device. The virtual synchronous Flash

memory of the present invention allows for additional memory address pins to be added to the external address interface so as to enable external devices with the appropriate capability to access the internal architecture of the virtual synchronous Flash memory directly.

To accomplish this goal, the virtual synchronous Flash memory of the present invention internally reduces the effective column length of a memory array bank to one-fourth the column length of a corresponding SDRAM memory device. In addition, the number of rows of a memory array bank is increased to four times the number of rows of a corresponding SDRAM memory device, maintaining the same overall memory density. It is noted that other internal memory array configurations are possible and would be apparent to those skilled in the art.

The virtual synchronous Flash memory array 200 of Figure 2 is shown with four array banks 202, 204, 206, and 208. Each of the four banks 202, 204, 206, and 208 in the memory array in turn has $4(X)$ rows 210 and $Y/4$ columns 212. When a bank and row are selected by an input row address, the contents of a row of memory cells is placed on column bit lines coupled to sense amplifiers (not shown) and the bit values of the memory cells are detected. The selected memory array row, and contents of the sense amplifiers, is called the active row page. The values detected by the sense amplifiers are latched in by latch circuit 214. An input column address to the virtual synchronous Flash, which defines the column page, selects from the latches of the latch circuit 214, and thus from the columns of the active row page, the Z number of bit values to be latched by an internal data buffer (not shown). The Z data bits are held in the internal data buffer for transfer from the virtual synchronous Flash memory device through the data interface (not shown). From the above example, a $4M \times 32$ SDRAM, which has 4 banks, 4,096 rows, 256 columns, and 32 bits, would correspond to a $4M \times 32$ virtual synchronous Flash that internally has 4 banks, 16,384 rows, 64 columns, and 32 bits.

A process for logically reassembling the internal rows and columns of the virtual synchronous Flash of Figure 2 to map to a comparable conventional SDRAM configuration is shown in Figures 3 and 4. In Figure 3, each of the four banks 308, 310, 312, and 314 of the internal memory array 316 of the virtual synchronous Flash of Figure 2, which has $4(X)$ rows and $Y/4$ columns, is shown in Figure 3. As detailed in Figure 2, each of the virtual synchronous Flash internal memory array banks is divided by row range into four segments 300, 302, 304, and 306. The number of rows in each separate segment of Figure 3 is equal to

the overall X number of rows in the SDRAM that the virtual synchronous Flash is compatible with. The segments 300, 302, 304, and 306 of each separate bank of the virtual synchronous Flash are then logically mapped by arranging them in a virtual manner side by side into a corresponding virtual memory bank 400 that is X rows long and Y columns wide, as shown
5 in Figure 4, matching a corresponding compatible SDRAM device. It is noted that other methods of logically mapping a virtual synchronous Flash memory to match an SDRAM are possible and would be apparent to those skilled in the art with the benefit of the present disclosure.

In the example previously given of a 4Mx32 memory device, a 4Mx32 virtual
10 synchronous Flash using the technique of Figures 3 and 4 internally has 4 banks, 16,384 rows, 64 columns, and 32 bits. Each bank is logically divided into 4 segments of 4,096 rows and 64 columns each. The segments of each bank are logically mapped to form a virtual bank of 4,096 rows and 256 columns. This procedure can be used on each bank to allow the 4Mx32 virtual synchronous Flash to simulate a 4Mx32 synchronous SDRAM compatible
15 memory device, which has 4 banks, 4,096 rows, 256 columns, and 32 bits.

Figure 5 is a simplified external diagram of a 4Mx32 virtual synchronous Flash memory device 500 of an embodiment of the present invention, showing external connections, major interfaces, and extended virtual synchronous Flash address lines. The virtual synchronous Flash memory device 500 of Figure 5 includes bank address lines BA0-
20 BA1 502, address lines A0-A15 504, data interface 506, and control line interface 508. The bank address lines BA0-BA1 502 select the desired memory array bank for a subsequent access. The data interface 506 transfers data to and from the virtual synchronous Flash memory device 500. The control line interface 508 contains the signal lines necessary to control the virtual synchronous Flash memory device 500 in its operation, such as, but not
25 limited to, CS, DQS, clock, CAS, RAS, and WE. Depending on the SDRAM command being issued the values placed on the address lines A0-A15 504 select the row or column of the memory array for an access. The address lines A0-A15 are additionally split into two separate subsections, A0-A11 510 and A12-A15 512. The addresses in subsection A0-A11 510 allow standard SDRAM compatible row and column memory address accesses to the
30 virtual synchronous Flash memory device 500. The additional address lines in subsection A12-A15 512 are not necessary for SDRAM compatible memory access, but if used by an extended capability SDRAM control device, address lines A12-A15 512 allow access to

extended virtual synchronous Flash capabilities and internal architecture. The virtual synchronous Flash uses address lines A12-A13 for access to the extended internal Flash memory array row addresses. Address lines A14-A15 are used to set the extended internal Flash memory array segment address.

5 A virtual synchronous Flash of the present invention is compatible with conventional SDRAM interfaces and commands, such as detailed above. In addition, virtual synchronous Flash supports extended operations to allow Flash specific operations and to allow direct access to the extended rows and columns of the internal memory array architecture. Direct access to the extended rows and columns without logical re-mapping of
10 the rows and columns to a compatible SDRAM configuration is enabled by the additional address interface lines optionally fitted to the virtual synchronous Flash memory device, such as A12-A15 512 of Figure 5.

 An example of a virtual synchronous Flash address space 600 is shown in Figure 6. The virtual synchronous Flash address space of Figure 6 matches the address space of the
15 SDRAM it is compatible with and contains a row address 606, a column address 608, a word address 610, and bank address 612. In addition, the virtual synchronous Flash address space 600 contains extended addresses for segment addressing 602 and row addressing 604 that allow for the direct access of the virtual synchronous Flash without logical mapping by compatible host controllers.

20 In the example 4Mx32 virtual synchronous Flash of Figure 5, address lines A12-A13 are used for extended row accesses into four times as many physical rows (16,384 rows) that a virtual synchronous Flash has as compared to a compatible conventional 4Mx32 SDRAM (4,096 rows). Address lines A14-A15 are used for selecting which segment section of four (i.e., 300, 302, 304, and 306 of Figure 3) of the virtual synchronous Flash's currently
25 active memory array bank (i.e., 308 of Figure 3) are accessed in the operation.

 A virtual synchronous Flash memory device's capability to set its segment address allows for full segment association within the bank as shown in Figure 7. In Figure 7 is shown a bank of virtual synchronous Flash memory 700 with four segments 702, 704, 706, and 708 that has been logically re-mapped to an SDRAM configuration. A row of the
30 segment zero 710 is shown being logically repositioned by an access that includes an extended virtual synchronous Flash segment address to appear as if it was placed in one of the other three segments of the current memory bank (segment one 712, segment two 714, or

segment three 716). The ability of full segment association allows any row selected in a bank to be addressed to any of the four segments in that bank. This has the effect of logically shifting the address space by that number of segments for a subsequent memory access.

In operation, a virtual synchronous Flash memory reduces the active page of the internal memory array bank by one-fourth. This reduces the number of sense amplifiers activating by one-fourth, saving power and complexity. In the example 4Mx32 virtual synchronous Flash, the memory device only activates 2K sense amplifiers versus 8K sense amplifiers for an SDRAM or synchronous Flash memory device. In addition, because sixteen segments are available for the virtual synchronous Flash memory device versus four segments for an SDRAM, when each of the four banks has their respective four segments activated, there is a higher hit rate for READs without activating new rows. The increased granularity of a virtual synchronous Flash memory device decreases power usage, and in addition, decreases memory access latency. Therefore effectively increasing bandwidth to the virtual synchronous Flash over that of a conventional synchronous Flash component.

Figure 8 shows an illustration of a virtual synchronous Flash memory system, wherein a 4Mx32 virtual synchronous Flash memory device 800 of an embodiment of the present invention is coupled to an external processor 802. It is noted that the virtual synchronous Flash memory system of Figure 8 is only shown as an example, and other systems and embodiments of the present invention can include multiple types of other integrated circuits (i.e., a memory controller, a field programmable gate array (FPGA), a volatile memory device, an application specific integrated circuit (ASIC), etc.). Systems containing memory devices are well known in the prior art and the following description is intended only to be an overview of their operation and provide an example of their operation with an embodiment of the present invention.

In the system of Figure 8, address values for the virtual synchronous Flash memory 800 are received from the processor 802 on the external address bus connections 804 and bank address lines 806. The received address values are stored internal to the memory device and utilized to select the memory cells in the internal memory array (not shown). In a conventional SDRAM compatible operation, the processor 802, with an ACTIVE command, inputs a row and bank address on address lines A0-A11 808 and bank address lines BA0 and BA1 806 activating an internal virtual row page (not shown) in the virtual synchronous Flash memory device 800. The row and bank address is followed by a READ command with a

column address placed on address lines A0-A8 (a subset of address lines A0-A11 808) by the processor 802. Internal to the virtual synchronous Flash memory device 800, data values from the bank segments (not shown) are captured in latch circuits, logically re-mapped to form an SDRAM compatible data format, and readied for transfer from the virtual
5 synchronous Flash memory device 800. Data transfer from the virtual synchronous Flash memory device 800 begins on the following clock cycle received and transmitted on the bi-directional data word interface 810 to the processor 802. Control of the virtual synchronous Flash memory device 800 for operations is actuated by the internal control circuitry (not shown). The control circuitry operates in response external control signals received from the
10 processor 802 on control signal external interface connections 820 and to internal events.

Extended access to the virtual synchronous Flash memory device 800 is accomplished utilizing the extended virtual synchronous Flash address lines, A12-A15 814. The processor 802, with an ACTIVE command, inputs a row on address lines A0-A11 808, extended row and segment address lines A12-A15 814, and bank address lines BA0 and BA1
15 806. The ACTIVE command directly activates a bank and internal row page and/or segment of the internal memory array (not shown) in the virtual synchronous Flash memory device 800 for further operations. The row, extended row, and bank address is followed by a READ command with a column address placed on address lines A0-A8 (a subset of address lines A0-A15 804) by the processor 802. Internal to the virtual synchronous Flash memory device
20 800, data values from the bank segments (not shown) are readied for transfer from the virtual synchronous Flash memory device 800 by being copied into internal latch circuits (not shown). Data transfer from or to the virtual synchronous Flash memory device 800 begins on the following clock cycle received and transmitted on the bi-directional data word interface 810 to the processor 802. Control of the virtual synchronous Flash memory device 800 for
25 operations is actuated by the internal control circuitry (not shown). The control circuitry operates in response external control signals received from the processor 802 on control signal external interface connections 820 and to internal events.

Conclusion

30 An improved memory device and architecture has been detailed that enhances a Flash memory device that has an SDRAM compatible READ interface. The memory device employs a virtual paging scheme that allows for the architecture of the memory to implement

an efficient Flash memory structure internally. Externally, the memory logically maps the internal Flash architecture to an SDRAM compatible interface and virtual architecture, allowing for memory access and operation with a compatible SDRAM controller device. Programming, erasing, block protection, and other Flash specific functions differ from SDRAM and are performed with an SDRAM command sequence.

In one embodiment, a memory device has four times as many rows in a memory array bank as a comparable SDRAM device, but only one fourth as many columns. This reduces the number of sense amplifiers activating, therefore saving power and complexity in the memory device. Internal to the memory, memory array banks are divided into four equal segments by row range and logically re-mapped by placing the segments virtually beside each other. This logically re-mapped memory bank forming a virtual memory bank structure of equivalent rows and columns as the comparable SDRAM device.

In another embodiment, a memory device with an extended memory interface with additional address lines was detailed, allowing for direct access to the internal Flash memory architecture without the abstraction of logical re-mapping the internal memory array. Additionally, the extended interface allows for full bank association and mapping of a segment row of an activated virtual row page to any segment of the bank for a subsequent memory access.

More segments are available for the memory device versus a comparable SDRAM. Therefore, a memory has a higher hit rate for READs without necessitating activation of further new rows. This increases the memory array granularity and has the effect of decreasing power and latency. The improved latency effectively increases the available bandwidth of the memory device.

Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that any arrangement, which is calculated to achieve the same purpose, may be substituted for the specific embodiment shown. This application is intended to cover any adaptations or variations of the present invention. Therefore, it is manifestly intended that this invention be limited only by the claims and the equivalents thereof.

What is claimed is:

1. A synchronous flash memory device comprising:
a first memory array with a first memory array dimensionality;
a control circuit; and
a synchronous memory interface, wherein the control circuit logically adapts the first memory array dimensionality to a second memory array dimensionality.
2. The synchronous flash memory device of claim 1, wherein the memory array further comprises a plurality of array banks.
3. The synchronous flash memory device of claim 2, wherein the plurality of array banks each have a first row dimension and a first column dimension.
4. The synchronous flash memory device of claim 3, wherein the control circuit logically adapts the first row dimension and the first column dimension of each array bank to a second row dimension and a second column dimension.
5. The synchronous flash memory device of claim 4, wherein the second row dimension and the second column dimension are the same as a SDRAM memory device or a DDR memory device.
6. The synchronous flash memory device of claim 2, wherein each array bank of the plurality of array banks further comprise a plurality of segments.
7. The synchronous flash memory device of claim 6, wherein each array bank of the plurality of array banks comprise four segments.
8. The synchronous flash memory device of claim 1, wherein the synchronous memory interface further comprises one of a SDRAM or a DDR compatible interface.

9. A synchronous non-volatile memory device comprising:
 - a memory array with a plurality of array banks, each array bank having a first number of rows and a first number of columns;
 - a control circuit; and
 - a synchronous memory interface, wherein, for each array bank of the plurality of array banks, the control circuit logically adapts the first number of rows and the first number of columns to a second number of rows and a second number of columns.
10. The synchronous flash memory device of claim 9, wherein the plurality of array banks equal four array banks.
11. The synchronous flash memory device of claim 9, wherein each array bank of the plurality of array banks further comprise a plurality of segments.
12. The synchronous flash memory device of claim 11, wherein the plurality of segments are divided by a row range.
13. The synchronous flash memory device of claim 11, wherein each array bank of the plurality of array banks comprises four segments.
14. The synchronous flash memory device of claim 9, wherein the second number of rows and the second number of columns matches that of a SDRAM or a DDR memory.
15. A synchronous flash memory device comprising:
 - a first memory array with a first memory array dimensionality;
 - a control circuit, wherein the control circuit logically adapts the first memory array dimensionality to a second memory array dimensionality; and
 - a synchronous memory interface, wherein the synchronous memory interface comprises,
 - an address interface,
 - a data interface, and
 - a control interface.

16. The synchronous flash memory device of claim 15, wherein the address interface further comprises an extended address interface, wherein the extended address interface allows extended access to the synchronous flash memory device.
17. The synchronous flash memory device of claim 16, wherein the extended address interface further allows access to the first memory dimensionality.
18. A synchronous flash memory device comprising:
 - a first memory array with a plurality of array banks, each having a first row dimension and a first column dimension;
 - a control circuit, wherein the control circuit logically adapts the first row dimension and the first column dimension of each of the plurality of banks to a second row dimension and a second column dimension; and
 - a synchronous memory interface, wherein the synchronous memory interface comprises,
 - an address interface,
 - an extended address interface,
 - a data interface, and
 - a control interface.
19. The synchronous flash memory device of claim 18, wherein the extended address interface further allows access to the first row dimension.
20. The synchronous flash memory device of claim 18, wherein the extended address interface further allows access to the first column dimension.
21. The synchronous flash memory device of claim 18 wherein the extended address interface further comprises an additional address interface to the first row dimension of the synchronous flash memory device.

22. The synchronous flash memory device of claim 21, wherein the additional address interface has two address lines for access into the first row dimension of the synchronous flash memory device.
23. The synchronous flash memory device of claim 18, wherein the plurality of array banks further comprise a plurality of segments, and wherein the extended address interface further comprises an additional address interface to select a segment of an active bank of the synchronous flash memory device.
24. The synchronous flash memory device of claim 23, wherein the additional address interface has two address lines for selection of the segment of the active bank of the synchronous flash memory device.
25. The synchronous flash memory device of claim 23, wherein the additional address interface allows associating an active row of a selected segment with any other segment in the active bank of the synchronous flash memory device.
26. A method of operating a synchronous flash memory device comprising:
dividing a memory array with a first memory array dimensionality into a plurality of sections; and
logically adapting the plurality of sections of the memory array to form an emulated virtual second memory array dimensionality.
27. The method of claim 26, wherein dividing the memory array into a plurality of sections further comprises dividing the memory array into a plurality of array banks.
28. The method of claim 27, wherein dividing the memory array into the plurality of array banks further comprises dividing the memory array into a plurality of array banks such that the array banks have a first row dimension and a first column dimension.
29. The method of claim 28, wherein logically adapting the plurality of sections of the

memory array to form the emulated virtual second memory array dimensionality further comprise logically adapting the first row dimension and the first column dimension of each array bank to a second row dimension and a second column dimension.

30. The method of claim 29, wherein the second row dimension and the second column dimension of each array bank of the plurality of array banks are the same as a SDRAM memory device or a DDR memory device.
31. The method of claim 27, wherein dividing the memory array into the plurality of array banks further comprises dividing each array bank into a plurality of segments.
32. The method of claim 26, further comprising coupling to one of a SDRAM or a DDR compatible interface.
33. A method of operating a synchronous flash memory device comprising:
dividing a memory array with a first memory array dimensionality into a plurality of array banks;
dividing each array bank of the plurality of array banks into a plurality of segments;
logically adapting the plurality of array banks and plurality of segments of the first memory array dimensionality of the memory array to form an emulated virtual second memory array dimensionality, wherein accessing the emulated virtual second memory array dimensionality occurs through a synchronous memory interface; and
allowing access to the first memory array dimensionality of the memory array through the synchronous memory interface and an extended interface.
34. The method of claim 33, wherein the extended interface is an extended address interface.
35. The method of claim 33, wherein the extended interface operates in conjunction with the synchronous memory interface.

36. The method of claim 33, wherein the extended interface allows access to the first row dimension of each array bank.
37. The method of claim 36, wherein the extended interface allows extended access to the first column dimension of each array bank of the plurality of array banks.
38. The method of claim 34, wherein the extended address interface further comprises two address lines to allow access into the first row dimension of the synchronous flash memory device.
39. The method of claim 36, wherein the extended interface further comprises an additional address interface to allow selection of a segment of an active bank of the synchronous flash memory device.
40. The method of claim 39, wherein forming the additional address interface further comprises forming two address lines for selection of the segment of the active bank of the synchronous flash memory device.
41. The method of claim 39, wherein the additional address interface allows associating an active row of a selected segment with any other segment in the active bank of the synchronous flash memory device.
42. A method of logically mapping a synchronous non-volatile memory device comprising:
forming a first memory array with a first plurality of banks;
dividing each bank of the first plurality of banks into a plurality of segments; and
logically mapping the plurality of segments to appear as a second memory array with a second plurality of segments.
43. A synchronous flash memory device comprising:
a memory array with a first set of four array banks, such that each array bank has a

first number of rows and a first number of columns and each array bank is divided into four segments by row range;

a latch circuit;

a control circuit; and

an SDRAM compatible synchronous memory interface, wherein the control circuit logically maps the latch circuit and the four segments in each of the four array banks to appear as an emulated virtual SDRAM memory device of an equal memory size with a second virtual set of four array banks, such that the first number of rows and the first number of columns of each array bank of the memory array map to a second number of rows, that is one fourth the first number of rows, and a second number of columns, that is four times the first number of columns, of the second virtual set of four array banks of the emulated virtual SDRAM memory device.

44. A method of making a synchronous flash memory device comprising:
forming a non-volatile memory array with a plurality of array banks, each array bank having a first number of rows and a first number of columns;
forming a control circuit; and
forming a synchronous interface, wherein the control circuit logically adapts the first number of rows and the first number of columns of each array bank of the plurality of array banks to a second number of rows and a second number of columns.
45. A system comprising:
a synchronous memory controller; and
a synchronous flash memory device, wherein the synchronous flash memory device comprises,
a memory array with a first memory array dimensionality,
a control circuit, and
a synchronous memory interface, wherein the control circuit logically adapts the first memory array dimensionality to a second memory array dimensionality.
46. The system of claim 45, wherein the synchronous memory controller further

comprises one of a processor, an ASIC, or an FPGA.

47. The system of claim 45, wherein the synchronous memory controller further comprises a SDRAM or a DDR compatible controller.
48. The system of claim 45, wherein the synchronous memory controller further comprises an extended virtual synchronous Flash compatible controller.
49. The system of claim 45, wherein the synchronous memory controller further comprises an extended synchronous Flash compatible controller.

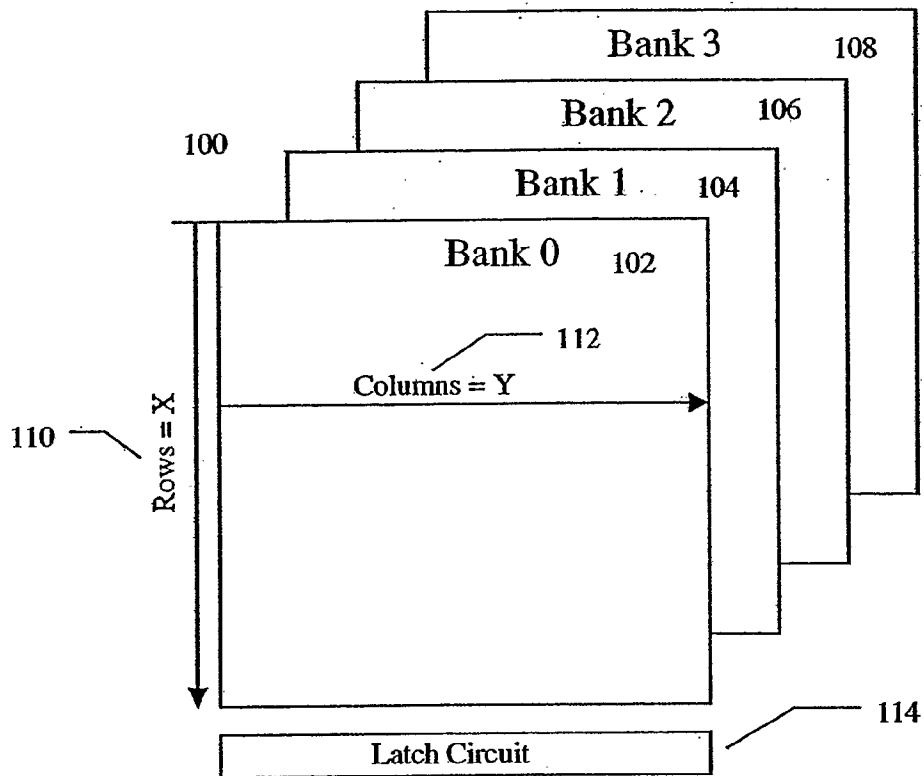


FIG. 1
Prior Art

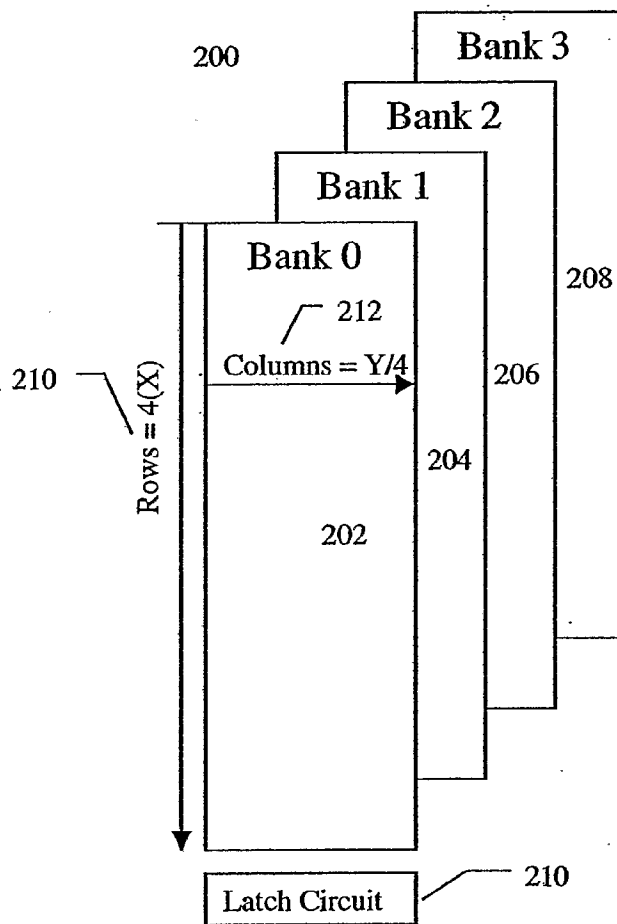


FIG. 2

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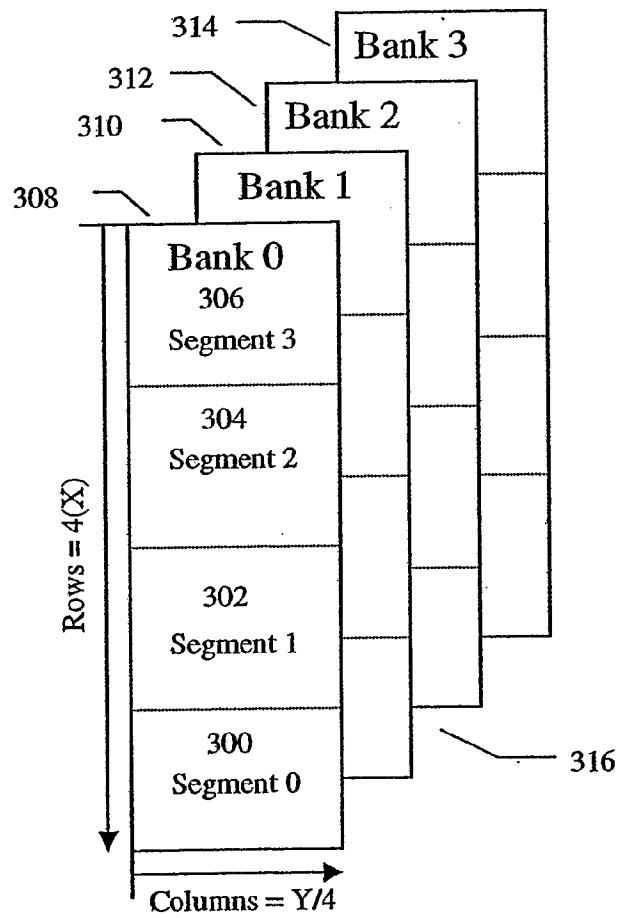


FIG. 3

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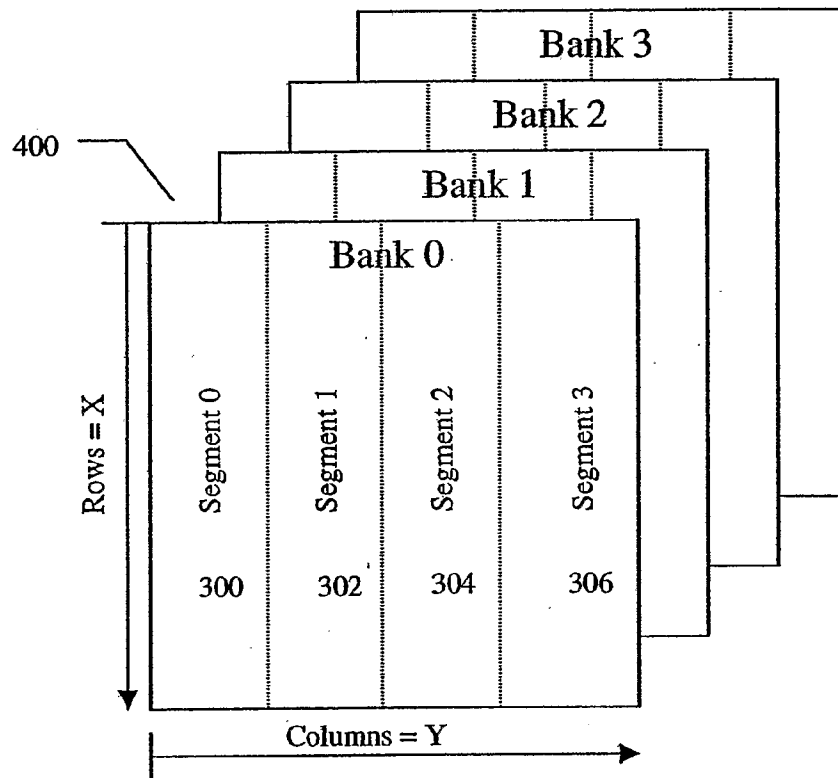


FIG. 4

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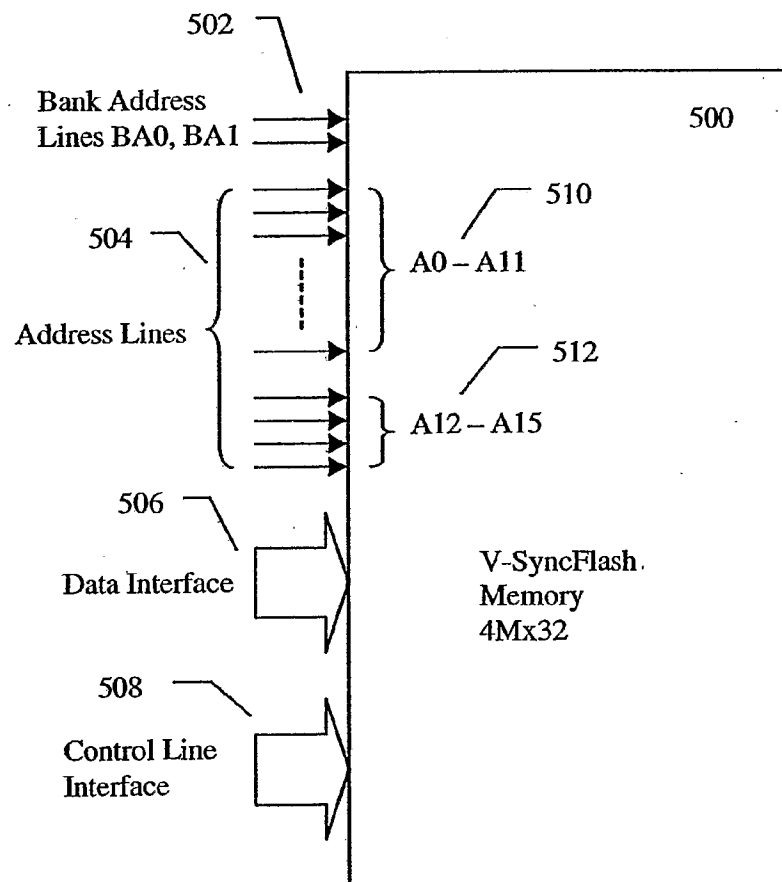
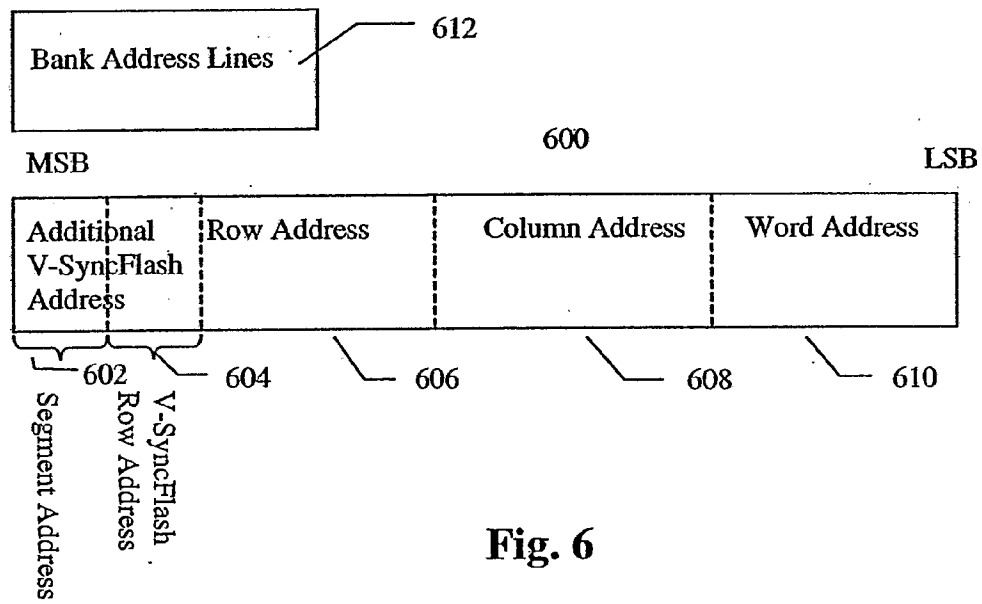


FIG. 5

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**Fig. 6**

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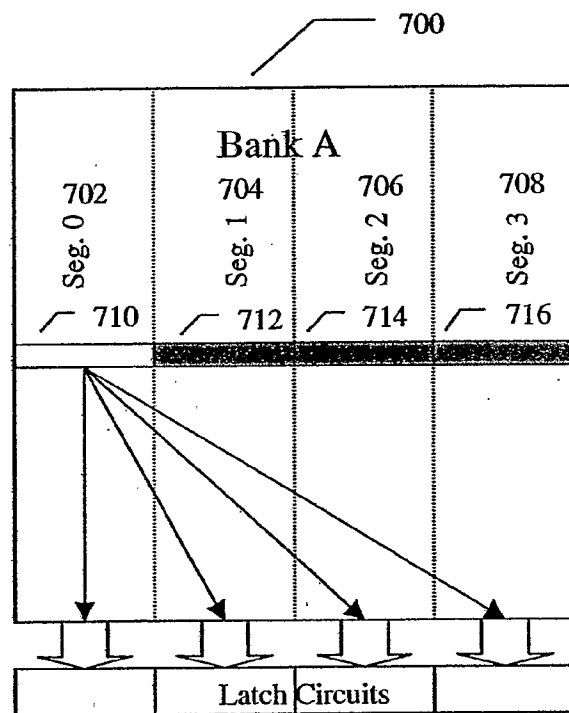


FIG. 7

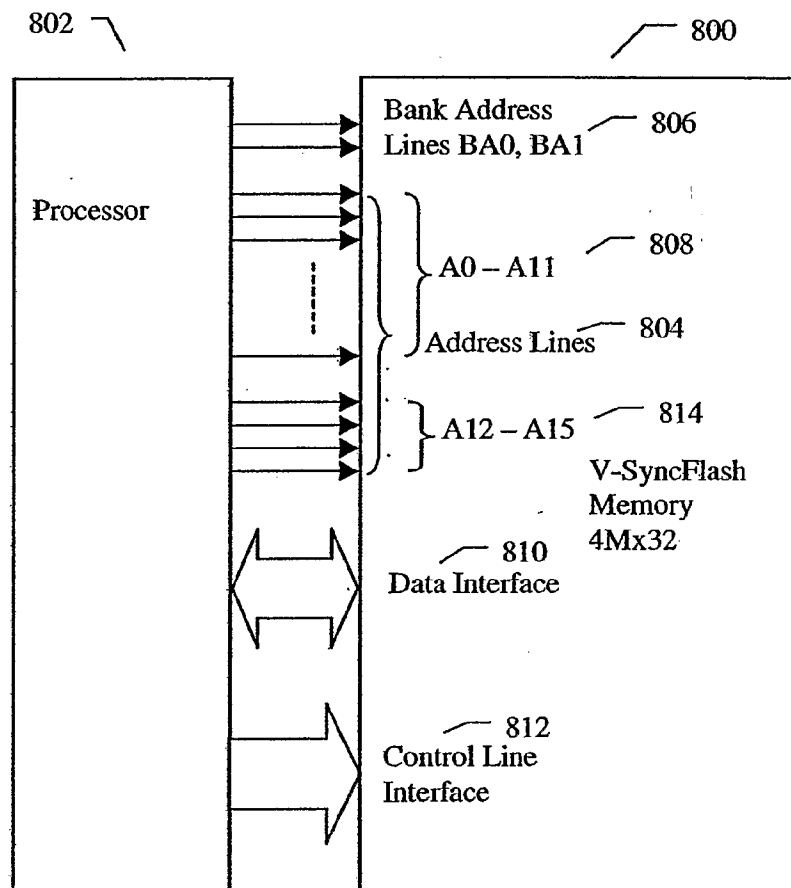


FIG. 8

INTERNATIONAL SEARCH REPORT

International Application No

PCT/US 02/25583

A. CLASSIFICATION OF SUBJECT MATTER
IPC 7 G11C16/08 G11C8/06

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC 7 G11C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal, WPI Data, PAJ

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category *	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 6 246 626 B1 (ROOHPARVAR FRANKIE F) 12 June 2001 (2001-06-12) column 3, line 53 -column 9, line 25; figures 1A,19	1-49
X	EP 1 063 652 A (SILICON STORAGE TECH INC) 27 December 2000 (2000-12-27) abstract paragraph '0021! - paragraph '0022! figures 2-4	1-49
X	US 6 026 465 A (PASHLEY RICHARD D ET AL) 15 February 2000 (2000-02-15) abstract column 20, line 60 -column 29, line 17; figures 1,3,8,12,14,15 -/--	1-49

☒ Further documents are listed in the continuation of box C.

☒ Patent family members are listed in annex.

* Special categories of cited documents:

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'X' document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

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Date of the actual completion of the international search

2 December 2002

Date of mailing of the international search report

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INTERNATIONAL SEARCH REPORT

International Application No

PCT/US 02/25583

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Information on patent family members

International Application No

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METHOD AND APPARATUS FOR SUPPLEMENTARY COMMAND BUS**Field of Invention**

The invention relates to integrated circuits and semiconductor systems, and more
5 particularly, to accessing data.

Background of the Invention

Many electronic systems use memory to store information. Memory devices include
an interface to communicate with other elements of the system. An ordinary interface
10 provides several inputs and outputs, including various bus connections to accommodate
multi-bit transfers in parallel. In particular, memory devices typically include a command
bus and an address bus that transmit command and address information from a controller to
the memory device.

For example, many existing DRAM devices include a command/address bus having
15 three command signals, ordinarily row address strobe (RAS), column address strobe (CAS),
and write enable (WE), and several address signals, including row/column address (A[X:0]),
bank address (BA[1:0]), and chip select (CS). The command signals decode yielding several
commands, such as ACTIVATE ROW, COLUMN SELECT with WRITE/READ,
PRECHARGE ROW, PRECHARGE ALL, AUTO REFRESH, SELF REFRESH, WAKE,
20 LOAD MODE, and LOAD EXTENDED MODE.

Certain access processes saturate the command/address bus, limiting the
throughput/bandwidth of the memory device. For example, referring to Figure 9 illustrating
a simplified memory access process, if the addressing patterns dictate all page misses, each
burst transfer operation 900 of data to/from the memory devices requires one ACTIVATE
25 command 910 for the row address 911 and one READ or WRITE command 912
corresponding to the column address data 913. Bank selection data is transferred via a bank
selection bus 924. A PRECHARGE command 914 for the selected bank 915 is then asserted
to close the selected bank. If the burst size does not allow time for three commands,
however, the data transfer rate is throttled. A common example is a burst size of four data
30 bus transfers 916A-D with the data bus 920 running at double the rate of the
command/address bus 922. This is a simplified, illustrative example, for memory accesses
typically utilize interleaved commands to the various banks to optimize the operation of the
memory system, but it serves to illustrate that each data bus burst transfer operation 900

provides two command/address slots 910, 912. Because three commands are required to effect the transfer, however, the data bus efficiency may be reduced to two-thirds of its capacity.

5 **Summary of the Invention**

An electronic system according to various aspects of the present invention includes a memory having a location-specific command interface and a general command interface. The memory communicates with other components in the system via a main command bus configured to transfer address-specific commands and a supplementary command bus
10 configured to transfer general commands. Commands may be received by the memory simultaneously at the respective interfaces. For example, a PRECHARGE command may be received on the general command interface while a memory access is received on the location-specific interface.

15 **Brief Description of the Drawings**

A more complete understanding of the present invention may be derived by referring to the detailed description and claims when considered in connection with the following illustrative Figures, which may not be to scale. In the following Figures, like reference numbers refer to similar elements throughout the Figures.

20 Figure 1 is a block diagram of an electronic system according to various aspects of the present invention;

Figure 2 is a block diagram of a memory interface and a bus system;

Figure 3 is a block diagram of a main control bus and a supplementary control bus;

Figure 4 is a timing diagram for consecutive memory accesses;

25 Figure 5 is a block diagram of an alternative embodiment of a main control bus and a supplementary control bus;

Figure 6 is another alternative embodiment of a main control bus and a supplementary control bus;

Figure 7 is a timing diagram for providing a secondary command indicator;

30 Figure 8 is a timing diagram for providing a tertiary command indicator; and

Figure 9 is a timing diagram for a simplified, illustrative prior art memory access.

Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the

figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

Detailed Description

5 The present invention may be described in terms of functional block components and various processing steps. Such functional blocks may be realized by any number of hardware and software components configured to perform specified functions. For example, the present invention may employ various components, e.g., memory elements, interface elements, logic elements, buses, package connections, and the like, which may carry out a
10 variety of functions under the control of one or more processors or other control devices. In addition, the present invention may be practiced in conjunction with any number of storage systems and data transmission media and protocols and the systems described are merely exemplary applications of the invention. Further, the present invention may employ any number of conventional techniques for data transmission, signaling, data processing, bus
15 control, and the like.

 Referring to Figure 1, an electronic system 100 according to various aspects of the present invention may include a processor 102 and a memory system 104. The present electronic system 100 comprises a system using a memory, such as a personal computer system. The electronic system 100 may comprise, however, any suitable electronic system,
20 such as a communication system, computing system, entertainment system, control system, portable electronic device, audio component, appliance, or factory control system, and the various components may differ according to the particular system and environment. The processor 102 generally controls operation of the electronic system 100, and may comprise any appropriate processor or controller, such as an Intel, Motorola, or Advanced Micro
25 Devices microprocessor. In some applications, the processor 102 may be replaced by other devices, such as logic circuits or ASICs, or omitted altogether.

 The memory system 104 comprises a storage system for storing data. The memory system 104 may comprise any appropriate memory system for storing data and transferring data between the memory system 104 and the processor 102 or another component. In the
30 present embodiment, the memory system 104 includes one or more memory modules 210A, B, a memory controller 212, and a bus system 106. The memory controller 212 controls access to, including data transfers to and from, the memory modules 210, and may perform further functions and operations as well. The memory controller 212 may comprise any

suitable memory controller for controlling access to the memory modules 210. In some embodiments, the memory controller 212 may be omitted and/or have its functions performed by other system components.

5 The memory modules 210 may comprise any system for storing data, such as a ROM, SRAM, DRAM, SDRAM, or any other suitable storage system. In the present embodiment, the memory modules 210 comprise double data rate synchronous dynamic random access memory (DDR SDRAM) modules from Micron. For example, each memory module 210 suitably comprises a high-speed CMOS SDRAM internally configured as four banks of DRAMs. The memory modules 210, however, may comprise any appropriate
10 memory or configuration, such as an individual memory chip, a multi-component device, or another type of storage device altogether. The memory modules 210 interface with the rest of the electronic system 100 through any suitable interface, such as a plurality of connections comprising pins, solder, conductive connections, optical couplings, or any other suitable coupling.

15 The bus system 106 connects the components of the memory system 104. The memory system 104, including the bus system 106, may utilize any appropriate signals, including optical or electric signals. The bus system 106 may comprise any suitable medium for transferring information, such as a plurality of wires, optical fibers, or other communication medium. In the present embodiment, the medium comprises an electronic
20 bus, such as a serial bus or multi-bit bus, to which the memory system 104 and other components may be attached.

Communications with the memory system 104 may be controlled in any suitable manner and by any suitable component, such as using the processor 102, the memory controller 212, or other circuit or system to control communications. In the present
25 embodiment, communications with the memory module 210 are suitably controlled by the memory controller 212.

The bus 106 and the connected components may communicate using any appropriate communication techniques and/or protocols. For example, the components may communicate via the bus 106 using a selected conventional protocol, such as those described
30 in specifications for Micron SDRAMs. The bus 106 is also suitably configured to communicate with the memory modules 210. For example, referring to Figure 2, each memory module 210 includes an interface 250 having a data bus connection 320, address bus connection 324, main command bus connection 326, and supplementary command bus

connection 328. The bus 106 may be configured accordingly to facilitate communication, such as including a data bus 220, an address bus 224, a main control bus 226, and a supplementary control bus 228.

5 The memory interface 250 and the bus 106 provide one or media for transferring information to and from the memory module 210. Each bus component 220, 224, 226, 228 ordinarily transfers a selected type of information, i.e., address selection information on the address bus 224, data on the data bus 220, and command signals on the main control bus 226 and the supplementary control bus 228. Further, each bus may be suitably configured to transfer the relevant information. For example, each bus may be configured to handle a
10 selected number of bits.

More particularly, the data bus interface 320 of the present embodiment comprises a multibit, such as 4-, 8-, or 16-bit, connection for transferring information to and from the data bus 220. The address bus interface 324 of the present embodiment comprises a 13-bit connection for, among other things, receiving row and column address information to select
15 one location out of the memory array 210 in a selected bank. The data bus 220 and address bus 224 are suitably configured to interface with the memory module 210 via the corresponding interfaces 320, 324.

Signals on the main control bus 226 and the supplementary control bus 228 define commands, and may further identify the chips and/or banks to which the commands apply.
20 The commands may be any set of commands for operation of the memory module 210. In the present embodiment, the supplementary control bus 228 is used to provide a first selected subset of commands and the main control bus 226 to provide a second selected subset or, alternatively, the entire set of commands. For example, the supplementary control bus 228 may be used for general commands, i.e., commands that relate to large blocks of memory
25 locations (like the entire module or an entire bank). The main control bus 226 handles location-specific commands, i.e., that relate to specific individual locations or groups of locations in the memory module 210, as well as other commands in various embodiments. The memory location relating to the location-specific command may be indicated in any suitable manner, such as by signals on the address bus 224. Thus, the main control bus 226
30 is used for commands using row and/or column information, typically provided via the address bus 224, and the supplementary control bus 228 is used for commands that are not associated with row or column information, and thus do not utilize the address bus 224.

Command signals may be divided between the main control bus 226 and the supplementary control bus 228 in any suitable manner; similarly, the main control bus 226 and the supplementary control bus 228 may be configured in any suitable manner to accommodate the command signals. For example, in one embodiment illustrated in Figure 3, the main control bus 226 comprises a main command bus 410 having three main command bits (M-CMD), a main chip select bus bit 412 having one bit (M-CS), and a main bank selection bus 414 having two bits (M-BA). The three main command bits are suitably designated as a row address strobe (RAS), a column address strobe (CAS), and a write enable (WE). In the present embodiment, the main bank selection bus 222 comprises a two-bit bus which defines the bank to which a command is to be applied, and the one-bit main chip select bus 412 identifies the relevant chip for the corresponding operation.

Similarly, the supplementary control bus 228 may be configured in any suitable manner to accommodate the relevant command set. For example, the supplementary control bus 228 suitably comprises a supplementary command bus 416 (S-CMD), a supplementary chip select bus 418 (S-CS), and a supplementary bank selection bus 420 (S-BA). In one embodiment, the supplementary control bus 228 solely facilitates PRECHARGE operations. PRECHARGE operations may comprise operations to deactivate open rows in a particular bank or all banks of the memory, though the PRECHARGE operation may comprise any appropriate process for preparing a memory or portion of memory for access or restoring a memory to a selected state following an access.

In the present dedicated PRECHARGE embodiment, the supplementary command bus 416 is a one-bit signal such that assertion of the supplementary command bit signals the memory module 210 to execute a PRECHARGE command on the bank 310A indicated by the supplementary chip select bus 418 and supplementary bank selection bus 420. Alternatively, because PRECHARGE is the only command controlled by the supplementary control bus 228, the supplementary command bus 416 may be omitted altogether, so that the supplementary chip select bus operates as the PRECHARGE signal. Activation of the supplementary chip select causes a PRECHARGE operation on the indicated chip in the bank designated by the supplementary bank selection bus 420.

Thus, the ACTIVE and subsequent READ or WRITE commands may be asserted using the main control bus 226 in conjunction with the address bus 224 to retrieve or store data. The next ACTIVE command or other command may then be asserted via the main control bus 226 on another bank 310B concurrently with assertion of a PRECHARGE

command on the original bank 310A via the supplementary control bus 228. Accordingly, the PRECHARGE commands may be asserted without using the main control bus 226, thus relieving congestion on the main control bus 226 and tending to improve the performance of the memory module 210.

5 More particularly, an electronic system 100 and method according to various aspects of the present invention may use the main control bus 226 to activate rows and select columns for memory access operations and the supplementary control bus 228 to close banks of memory. For example, referring to Figure 4, an initial memory access may be performed by identifying the relevant bank on the main bank selection bus 414, the relevant chip on the
10 main chip select bus 412, and the relevant memory row on the address bus 224, and providing the ACTIVE signal on the main command bus 410 at a first time T1. At substantially the same time, the supplementary control bus 228 receives a NOP command to prevent any unwanted command from being registered on the supplementary control bus 228.

15 Following the initial memory access, a second memory access operation may be initiated at a second memory location on a different bank. Accordingly, the second memory location may be identified on the main chip select bus 412, main bank select bus 414, and address bus 224, and the ACTIVE and READ or WRITE commands may be successively provided via the main command bus 410 at times T3 and T4. Meanwhile, the supplementary
20 control bus 228 may provide a PRECHARGE command to a previously accessed bank to close the still-active row from the preceding memory access operation. For example, the previously accessed bank and chip are identified on the supplementary bank selection bus 420 and supplementary chip select bus 418, respectively, at time T3. The PRECHARGE command is provided on the supplementary command bus 416 to initiate the precharge of
25 the previously accessed bank while the second bank is being accessed via the main control bus 226.

 The command set may be apportioned and/or shared by the main control bus 226 and the supplementary control bus 228 in any suitable manner. For example, the commands may be shared such that all of the commands requiring the address bus 224 are asserted via the
30 main control bus 226, and all of the commands that do not require the address bus 224 may be asserted via either the supplementary control bus 228 or the main control bus 226. Thus, the device providing the commands, such as the memory controller 212, may select which control bus 226, 228 on which to provide a command according to any selected criteria.

Alternatively, the commands may be restricted to assertion on either the main control bus 226 or the supplementary control bus 228, or the commands may be apportioned such that one set of commands is asserted only on the main control bus 226, a second set is asserted only on the supplementary control bus 228, while a third set may be asserted on either bus 226, 228.

Similarly, memory module interface 250, the main control bus 226, and the supplementary control bus 228 may be configured in any suitable manner to facilitate assertion of the commands as apportioned or shared. In the embodiment in which all commands not requiring the address bus 224 may be asserted on either the main control bus 226 or the supplementary control bus 228, the respective control buses 226, 228 are suitably configured with sufficient bits to accommodate the relevant command set. For example, referring to Figure 5, the memory module 210 may accommodate nine commands such that three commands require the address bus 224 (such as ACTIVE, READ, and WRITE) and six commands do not (such as DESELECT, NO OPERATION, BURST TERMINATE, PRECHARGE, AUTO REFRESH, and LOAD MODE REGISTER). To accommodate all nine commands, the main control bus 226 may be configured with four command bits. Similarly, the supplementary control bus 228 is suitably configured with three command bits to accommodate the six commands that do not use the address bus 224.

In alternative embodiments, the number of pins may be reduced by reconfiguring the command set and the main and supplementary command buses 226, 228 and the interface 250. For example, the commands may be apportioned such that commands not requiring the address bus 224 are asserted only on the supplementary control bus 228. Thus, in the present embodiment, the main control bus 226 may be configured to handle only three commands (ACTIVE, READ, and WRITE), requiring only two command bits, as illustrated in Figure 6.

In another embodiment, the number of bits required by the supplementary control bus 228 may be reduced by packetizing various commands. Packetizing comprises sending consecutive signals for a single command, such as for lower priority commands. For example, referring to Figure 7, the supplementary control bus 228 may be configured with two supplementary command bits 416 as well as the supplementary chip select 418 and supplementary bank select bits 420. The supplementary command bits 416 may receive one of four main commands. Three commands are suitably executable commands, such as PRECHARGE, AUTO REFRESH, and NOP. The fourth command may be a secondary

command indicator 620, referred to in this embodiment as EXTEND1. EXTEND1 designates a secondary command 622 following, for example, in the next cycle. Thus, if EXTEND1 is transmitted, the next cycle contains the secondary command 622. The secondary commands may comprise any appropriate commands, such as SELF REFRESH, WAKE, or LOAD MODE. In the present embodiment, the secondary commands comprise commands that are lower priority than the main commands. The main and secondary command sets may be selected, however, according to any desired criteria.

Further, referring to Figure 8, the secondary command 622 may indicate another command following, such as in the next cycle, by asserting a tertiary command indicator 724 (EXTEND2). The tertiary command 726 may then be asserted in the following cycle. Additional command capability may be provided using additional sub-command indicators if desired. Thus, many low priority commands may be added without adding additional command bits.

It should be appreciated that the particular implementations shown and described herein are illustrative of the invention and its best mode and are not intended to otherwise limit the scope of the present invention in any way. Indeed, for the sake of brevity, conventional signal processing, data transmission, and other functional aspects of the systems (and components of the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical communication system.

The present invention has been described above with reference to a preferred embodiment. However, those skilled in the art having read this disclosure will recognize that changes and modifications may be made to the preferred embodiment without departing from the scope of the present invention. These and other changes or modifications are intended to be included within the scope of the present invention, as expressed in the following claims.

Claims

1. An electronic system, comprising:
- 5 a processor;
- a memory having more than one memory location; and
- a bus connecting the processor to the memory, wherein the bus comprises:
- 10 a data bus for transmitting data;
- an address bus for identifying a first memory location;
- a main command bus for transferring a first command that relates to the first memory location; and
- a supplementary command bus for transferring a second command, wherein the second command does not relate to the first memory location.
- 15
2. An electronic system according to claim 1, wherein the second command is a PRECHARGE command, and wherein the supplementary command bus comprises a one-bit bus.
- 20
3. An electronic system according to claim 1, wherein:
- the main command bus transfers location-specific commands; and
- the supplementary command bus transfers only general commands.
4. An electronic system according to claim 1, wherein:
- 25 the second command is a secondary command indicator; and
- the supplementary command bus transfers a third command after the second command.
5. An electronic system, comprising:
- 30 a processor;
- a memory having more than one memory location; and
- a bus connecting the processor to the memory, wherein the bus comprises:

a main command bus configured to transfer an address-specific command; and
a supplementary command bus configured to transfer a general command.

5

6. An electronic system according to claim 5, wherein the general command is a PRECHARGE command.

7. An electronic system according to claim 5, wherein:
the general command is a secondary command indicator; and
the supplementary command bus is further configured to transfer a third command after the second command.

10

8. A memory having an interface, wherein the interface comprises:
a location-specific command interface configured to receive location-specific commands; and
a general command interface configured to receive only one or more general commands.

15

9. A memory according to claim 8, wherein the general command interface is configured to receive a PRECHARGE command.

20

10. A memory according to claim 8, wherein the location-specific command interface is configured to receive location-specific commands and general commands.

25

11. A memory having an interface, wherein the interface comprises a general command interface configured to receive a general command.

12. A memory according to claim 11, wherein the general command is a PRECHARGE command.

30

13. A memory according to claim 11, wherein the memory includes more than one bank, and the interface further comprises a general bank select interface configured to receive bank address information for the general command.

5 14. A memory including a one-bit PRECHARGE input dedicated to receiving a PRECHARGE command.

15. A memory according to claim 14, wherein the memory includes more than one bank, further including a general bank select interface configured to receive bank
10 address information for the PRECHARGE command.

16. A memory system, comprising:
a memory controller; and
a memory in communication with the memory controller, wherein the
15 memory includes an interface, comprising:
a main control interface for receiving a location-specific command; and
a supplementary control interface for receiving a
general command.

20 17. A memory system according to claim 16, wherein the general command is a PRECHARGE command.

18. A memory system according to claim 16, wherein:
25 the general command is a secondary command indicator; and
the supplementary control interface is configured to transfer a secondary command after the secondary command indicator.

19. An electronic system, comprising:
30 a processor; and
a memory system connected to the processor, comprising:
a memory controller connected to the processor; and

a memory connected to the memory controller and having an interface, comprising:

an address interface for receiving an address signal;

a main control interface, comprising:

5 a main command bus for receiving a first command signal relating to a memory location specified by the address signal; and

10 a main bank select bus for receiving a first bank select signal specifying a first bank corresponding to the first command signal; and

a supplementary control interface, comprising:

15 a supplementary command bus for receiving a second command signal, wherein the second command signal corresponds to a general command; and

a supplementary bank select bus for receiving a second bank signal specifying a second bank corresponding to the second command signal.

20

20. A method of accessing a memory, comprising:

in a first time slot, requesting activation of a first row;

in a second time slot, requesting an access of a memory location in the first row;

25 in a third time slot, requesting activation of a second row and requesting closure the first row.

21. A method according to claim 20, further comprising:

30 providing a main command bus, wherein the requesting activation of the second row occurs on the main command bus; and

providing a supplementary command bus, wherein the requesting closure of the first row occurs on the supplementary command bus.

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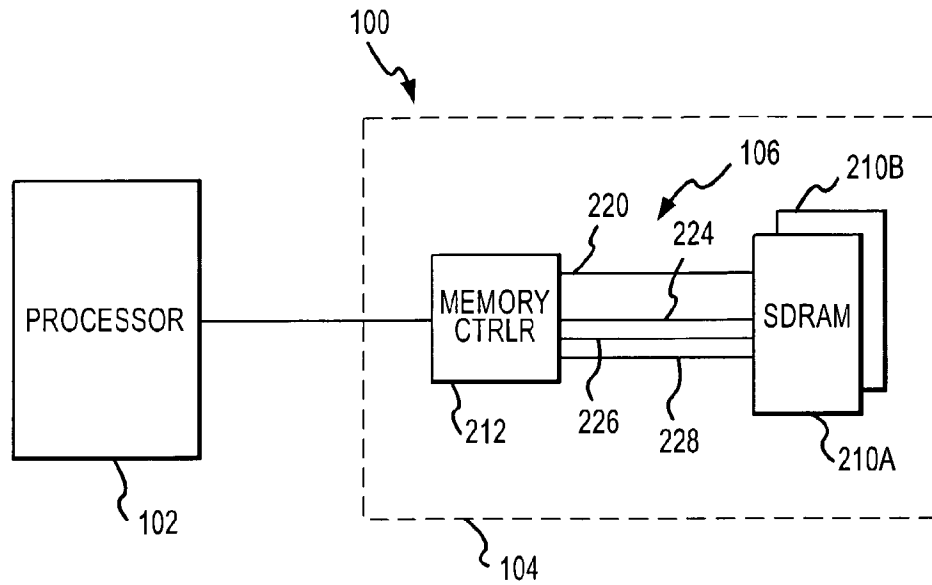


FIG. 1

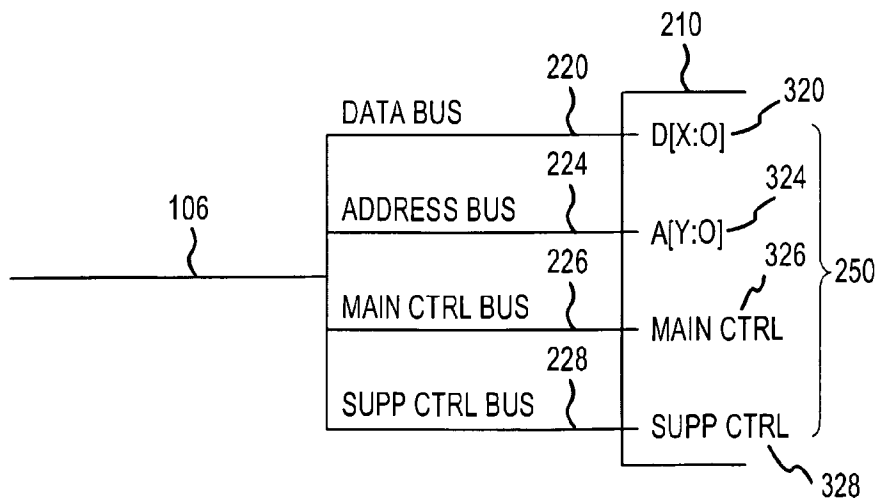


FIG. 2

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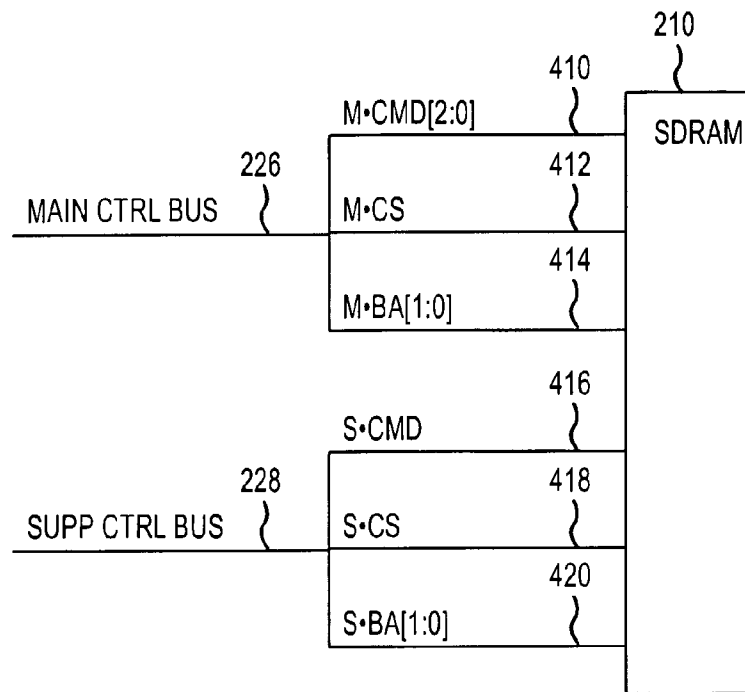


FIG.3

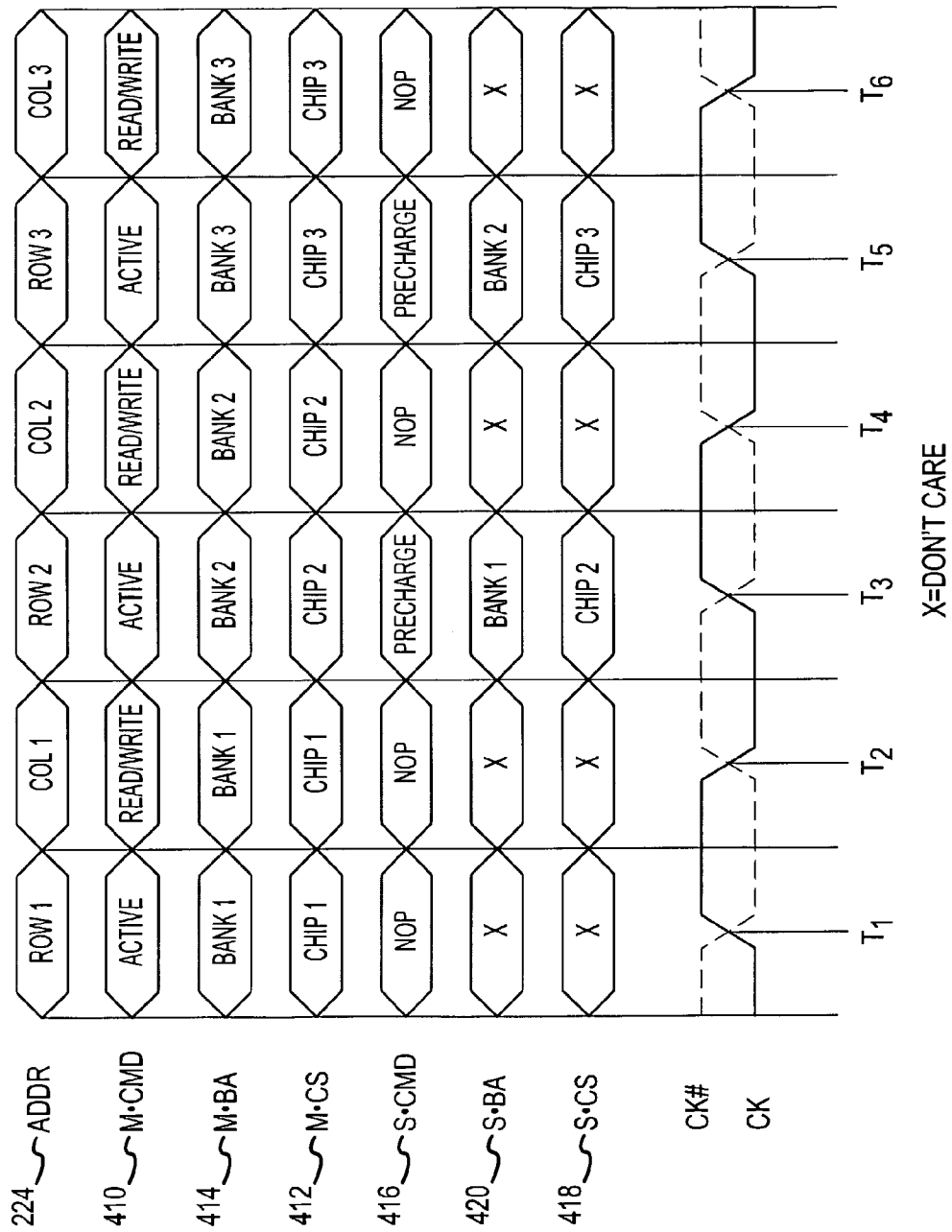


FIG. 4

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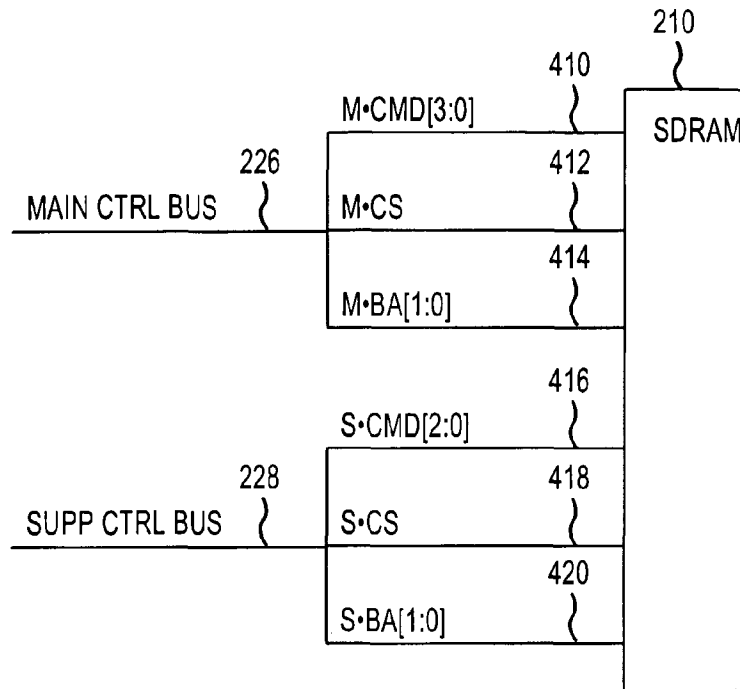


FIG. 5

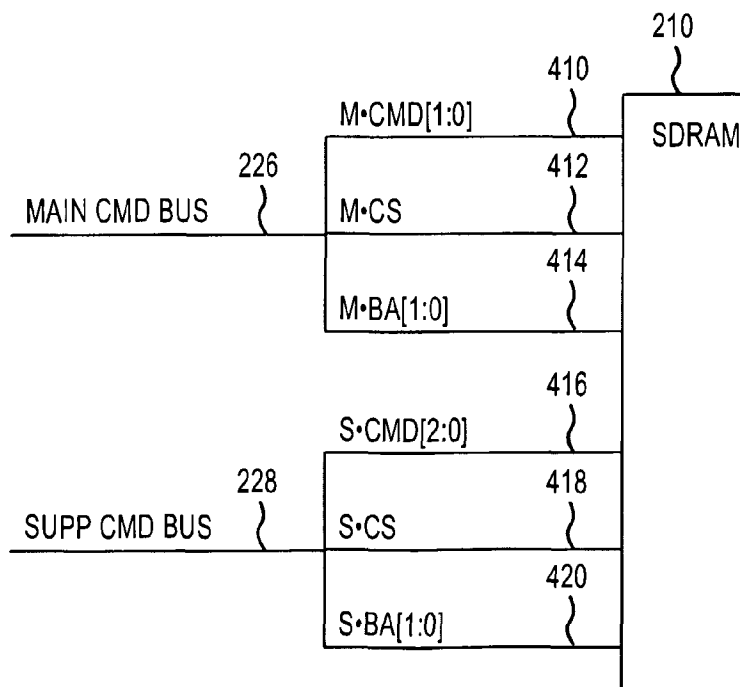


FIG. 6

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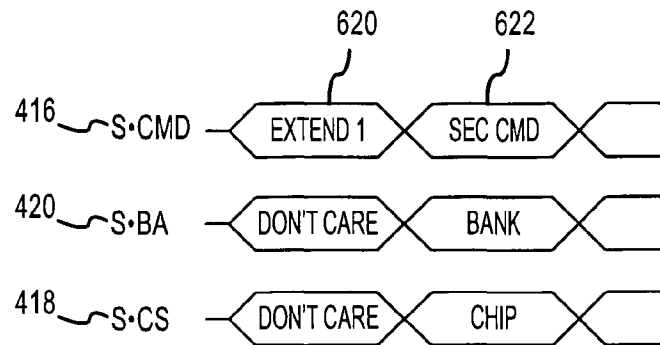


FIG. 7

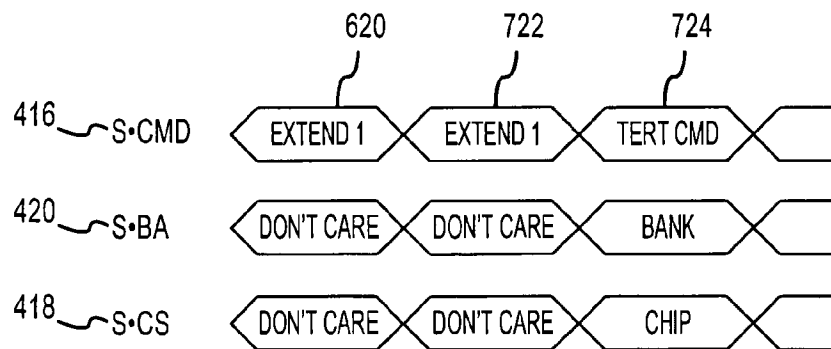
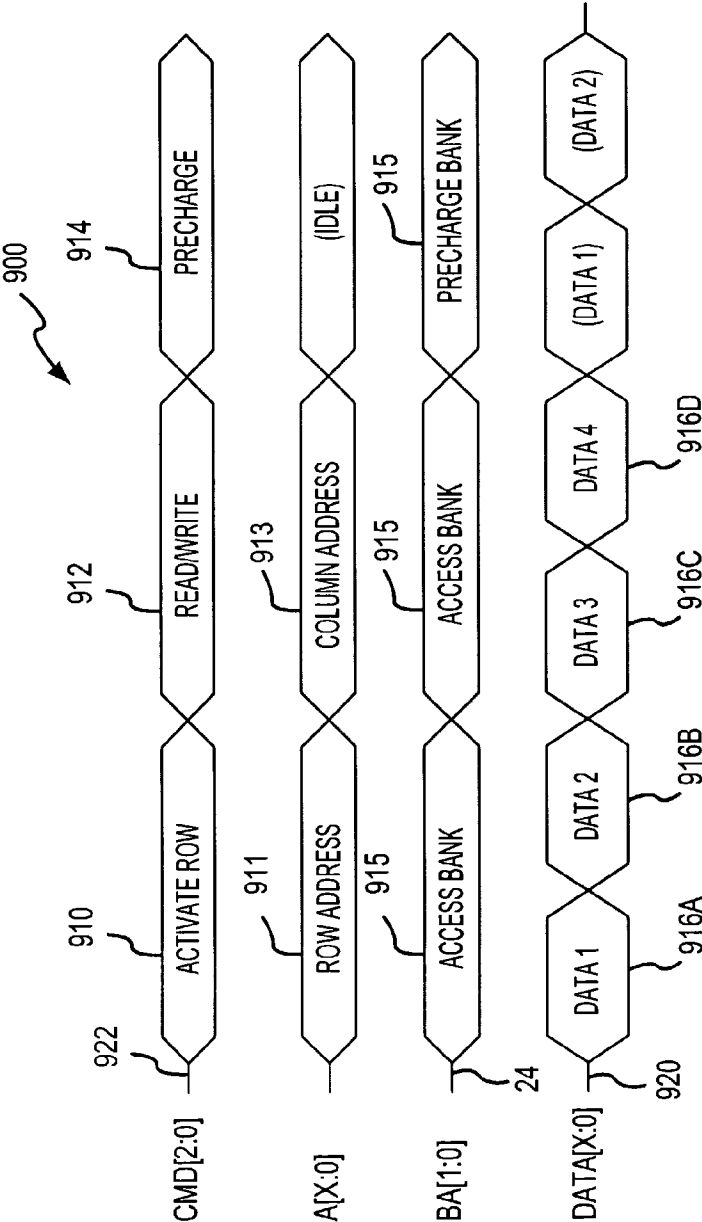


FIG. 8



(PRIOR ART)
FIG.9

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(54) Title: COMMAND CONTROLLING DIFFERENT OPERATIONS IN DIFFERENT CHIPS

(57) Abstract: Abstract of the Disclosure In some embodiments, the invention includes a chip having a register to include an operation type signal. The chip also includes control circuitry to receive a first command and in response to the first command to cause the chip to perform a first operation if the operation type signal has a first value and to cause the chip to perform a second operation if the operation type signal has a second value. The chip may be a memory chip in a memory system. Other embodiments are described and claimed.



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COMMAND CONTROLLING DIFFERENT OPERATIONS IN DIFFERENT CHIPS

BACKGROUND

TECHNICAL FIELD

The present invention relates to semiconductor chips and, more particularly, to
5 chips in which a command causes different operations in different chips.

BACKGROUND ART:

Conductors are typically terminated through resistive structures, such as resistive
materials and transistors. The terminations have been positioned in various places.
10 Examples of the location of terminations include on motherboards or other circuit boards
or cards. Other examples of locations of terminations include on a chip that transmits a
signal and/or on a chip that receives a signal. Terminations on the chip are commonly
referred to as on die terminations (ODTs).

Some Dynamic Random Access Memories (DRAM) chips have registers referred
15 to as a mode register set (MRS) which includes bits to control functions of the DRAM
chips. One or more bits in the MRS may control particular functions of the DRAM chips.
Extended MRSs (such as EMRS0, EMRS1 etc.) may provide additional bits for use.

DRAM chips are typically included in memory modules, some of which are dual
in-line memory modules (DIMMS). The term rank refers to a group of memory chips that
20 are selected together, for example, through a shared chip select signal or through some
other way. Some modules include only one rank per module and some include more than
one rank per module. Some ranks include chips on more than one module.

In some two rank memory systems, for best electrical performance, the ODT is
enabled on the DRAM chips that are not receiving the data for write operations.

25 The JEDEC Solid State Technology Association has provided standards for DDR2
(dual data rate 2 or double data rate 2) DRAM chips. In DDR2 DRAM chips, the ODTs
are controlled by using discrete signals per rank. In some two rank-1 DIMM systems, two
ODT pins are used on the controller and the DIMM. Two pins may also used per DRAM
chip to support stacking solutions. As used in this disclosure, the word "pin" means
30 traditional pins or other connections to die pads.

In GDDR3 (graphics DDR3) and also DDR3/DDR2 proposals, the command bus may be shared between two ranks. The DRAM can monitor or snoop the command bus for a write transaction. If it sees a write cycle and no CS# asserted then, it can turn its ODT on. The write command may target the other rank. (CS# are rank or device specific).

5 The term "2N timing" refers to a situation in which it takes an extra clock cycle for sampling as compared to a 1N timing. The sampling may be of a signal, such as an address signal. Snooping may work well for 1N timings, but not for 2N timings (with an additional clock cycle before sampling) because snooping may cause additional clock cycles in turnaround cycles and in leadoff write cycles.

10

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will be understood more fully from the detailed description given below and from the accompanying drawings of embodiments of the invention which, however, should not be taken to limit the invention to the specific embodiments described,

15 but are for explanation and understanding only.

FIG. 1 is a block diagram representation of a system including a memory controller and a memory module including memory chips according to some embodiments of the invention.

FIG. 2 is a block diagram representation of further details of the memory controller and memory chips of FIG. 1 according to some embodiments of the invention.

20

FIG. 3 is a representation of a register according to some embodiments of the invention.

FIG. 4 is a block diagram representation of further details of a memory chip of FIGS. 1 and 2 according to some embodiments of the invention.

FIG. 5 is a block diagram representation of a system including a memory controller and a memory module according to some embodiments of the invention.

25

FIG. 6 is a block diagram representation of a system including a memory controller and memory modules according to some embodiments of the invention.

FIG. 7 is a timing diagram illustrating operation of the memory controller and memory chips of FIG. 5 according to some embodiments of the invention.

30

FIG. 8 is a block diagram representation of the memory controller of FIGS. 1, 2, 5, and 6 being included in the same die as a processor.

FIG. 9 is a block diagram representation of the memory controller of FIGS. 1, 2, 5, and 6 being included in the same die as a memory controller hub of a chipset.

FIG. 10 is a system showing a program to provide values to the registers of FIG. 2.

DETAILED DESCRIPTION

Referring to FIG. 1, a memory controller 12 provides various signals including command signals (hereinafter called commands) to a memory chip MC0 and to a memory chip MC1 on a support 16. A memory module 18 includes chips MC0 and MC1 and a support 16 and may include additional chips not illustrated in FIG. 1. Various signals, including commands, pass through interconnects 14 between memory controller 12 and MC0 and MC1. Data, addresses, commands, and chip select signals may be on different conductors or may be shared on at least some of the same conductors. Signals of different types may be carried on different interconnects or the signals may be fully or partially packetized or time multiplexed over the same conductors in interconnects 14.

Chip MC0 responds to a particular command by performing one operation and chip MC1 responds to the same command by performing another operation. For example, chip MC0 may respond to a command, for example, called command A, by performing a write operation while chip MC1 responds to command A by turning on an ODT. Stated another way, chip MC0 may interpret command A as being a write command while chip MC1 interprets command A as being a turn on ODT command. In some embodiments, there is a second command, for example, called command B, for which chip MC0 responds by turning on an ODT while MC1 responds by performing a write operation. As another example, chip MC0 may respond to a command, for example, called command C, by performing a read operation while chip MC1 responds to command C by performing a refresh operation. Likewise, there may be a second command, for example, called command D, for which chip MC0 responds by performing a refresh operation while MC1 responds by performing a read operation. Still other different operations may be performed in response to a command. Note that the labels A, B, C, and D are arbitrary.

There are many ways in which chips MC0 and MC1 may be implemented. FIG. 2 illustrates some example implementations (embodiments), but the invention is not limited to the details of FIG. 2 and the following description of it. The invention may be implemented in various embodiments that are different from FIG. 2 and the description of

it. Further, the embodiments of FIG. 2 may include additional components not illustrated in FIG. 2.

Referring to FIG. 2, memory controller 12 is coupled to chips MC0 and MC1 through a control bus 20, a data bus 22, and an address bus 24. Chip select (CS) signals are provided to chips MC0 and MC1 through buses 26 and 28. Chip select signals are sometimes active low and designated CS#. In the case of FIG. 2, the chip select signals are CS0# for chip MC0 and CS1# for chip MC1. Chip select signals are sometimes referred to as device select signals.

The commands referred to in this disclosure may be provided from memory controller 12 to chips MC0 and MC1 through various ways. In some embodiments, the commands described herein are provided only through control bus 20. In other embodiments, the commands are provided through a combination of control bus 20 and address bus 24. In still other embodiments, the data bus is also used to communicate commands. Control circuitry 32 in chip MC0 and control circuitry 42 receive the commands from controller 12, although the signal including the commands may be modified or otherwise changed between the controller 12 and controller circuitry 32 and 42.

Chip MC0 includes a register 30 and chip MC1 includes a register 40. Registers 30 and 40 include operation type signals to control how chips MC0 and MC1 interpret at least some commands. The operation type signals may be in one or more than one bits. As an example, registers 30 and 40 each may be included in a MRS or EMRS using previously unused bits or new bits. However, it is not required that registers 30 and 40 be in an MRS or EMRS.

As an example (called example 1), an operation type signal in register 30 has a first value because, for example, a particular bit in register 30 has a first voltage state. The same operation type signal in register 40 has a second value because the corresponding bit in register 40 has a second voltage state. In example 1, there are commands called commands A and B. Chip MC0 responds to command A as a write command, meaning that control circuitry 32 causes certain data bits on data bus 22 to be stored in core 36 of chip MC0. Chip MC1 responds to command A by turning on ODT 48. More particularly control circuitry 42 turns on ODT 48. Conversely, chip MC0 responds to command B by turning on ODT 38 and chip MC1 responds to command B as a write command. More

particularly, in response to command B, control circuitry 32 turns on ODT 38 and control circuitry 42 causes data on bus 22 to be stored in core 46. In some embodiments, if the values of the operation type signals in registers 30 and 40 were reversed, then chip MC0 would respond to command A by turning on ODT 38 and to command B by performing a write operation; and chip MC1 would respond to command A by performing a write operation and to command B by performing an ODT operation.

As another example (called example 2), there are commands called commands C and D. An operation type signal in registers 30 and 40 could be set with particular values such that chip MC0 responds to command C as a read command and to a command D as a refresh command, whilst chip MC1 responds to command C as a refresh command and to command D as a read command. The values of the operation type signal can be set by the states (voltages) of one or more bits in registers 30 and 40. There can be only one or more than one operation type signal in a register. In some embodiments, chips MC0 and MC1 have the features of both examples 1 and 2, whilst in other embodiments, chips MC0 and MC1 have the features of example 1, but not example 2, and in still other embodiments, chips MC0 and MC1 have the features of example 2, but not example 1.

FIG. 3 illustrates a register with one or more bits for “write or ODT on” and one or more bits for “read or refresh.” The register of FIG. 3 may represent register 30 or 40, although registers 30 and 40 are not required to include bits for both “write or ODT on” and “read or refresh.” In FIG. 3, different bits are shown to control “write or ODT on” and “read or refresh.” Alternatively, the same bit(s) can control both. For example, if the operation type signal has a first value, then the chip may respond to command A as a write request and command C as a read request, and if the operation type signal as a second value, the chip responds to command B as an ODT on command and command D as a refresh request. Even other information can be encoded in the operation type signal.

In examples 1 and 2, the first state could be a logical high voltage and the second state a logical low voltage, but that is not required. For example, the first state could be a logical low voltage and the second state could be a logical high voltage. In other embodiments, more than two states (voltages) can be included in a register bit(s).

There are various ways in which an ODT may be constructed and this invention is not limited to any particular way. FIG. 4 illustrates an implementation of ODT 38 in chip MC0, but the invention is not limited to these details. Indeed, various other types of ODT

may be used. Further, some embodiments do not include any ODT. Referring to FIG. 4, resistive structures R1-D ... RN-D are between a data node 50 and switches Sw1-D ... SwN-D, respectively, which in turn are coupled to a supply voltage node VDDQ.

Resistive structures R1-S ... RM-S are between data node 50 and switches Sw1-S ...

5 SwM-S, respectively, which in turn are coupled to a ground voltage node VSSQ. The numbers "N" and "M" may be the same or different and may have different values in different implementations. Control circuitry 32 controls the switches and hence whether ODT 38 is "on." The number of switches closed may determine the amount of resistance in the termination. For example, a command A1 may cause one level of resistance in the
10 ODT and a command A2 may cause another level of resistance. In FIG. 4, data node 50 is between a pin (pad) and an input buffer 52. Some chips do not include traditional pins, but may include pads.

In current typical implementations, more than one memory chip is included in a module. The chips are typically grouped in ranks. For example, FIG. 5 illustrates
15 memory controller 12 being coupled to memory chips in a memory module 64, which has two ranks of chips supported by support 62. A first rank (rank 0) includes memory chips MC0-0 MC0-N (where N is not necessarily the same as in FIG. 4). A second rank (rank 1) includes memory chips MC1-0 MC1-N. The two ranks may have the same number of chips or a different number. FIG. 5 illustrates each chip in a rank being
20 selected by the same chip select signal (CS0 for the chips of rank 0 and CS1 for the chips of rank 1). The chip select signal could be provided on its own multi-drop bus as illustrated in FIG. 5 or through other ways such as a multiplexed or packetized signal.

In FIG. 6, memory controller 12 is coupled to a first memory module 68 with a first rank of chips (MC0-0 ... MC0-N) and a second rank of chips (MC1-0 ... MC1-N)
25 supported by a support 66 and to a second memory module 74 with a third rank of chips (MC2-0 ... MC2-N) and a fourth rank of chips (MC3-0 ... MC3-N) supported by a support 72. Chip select signals CS0, CS1, CS2, and CS3 are used to select the chips in the corresponding ranks. Accordingly, chip select signals CS0 is used for all the chips of the first rank (rank 0), CS1 is used for all the chips of the second rank (rank 1) and so forth.

30 The chip select signals in FIG. 6 could include a "#" symbol.

Various alternatives may be used. For example, in some embodiments, each module has only one rank. In this example, FIG. 6 could be modified so that all the chips

in module 68 were in one rank and all the chips in module 74 were in another rank. As another alternative, each module could have two ranks, but the system would include only two ranks. In this example, FIG. 6 could be modified so that chips MC0-0 ... MC0-N and MC2-0 ... MC2-N would be in one rank and chips MC1-0 ... MC1-N and MC3-0 ... MC3-N would be in the other rank. There could be more than two ranks per module and more than four ranks per system.

As illustrated in FIGS. 5 and 6, each of the memory chips are on the same side of supports 62, 66, and 72. In practice, the chips of one rank may be on one side of the support and the chips of another rank may be on the other side of support, or some other arrangement may be used. In some embodiments, chips from the same rank are on different sides of the support.

FIG. 7 provides an illustrative timing diagram for operation of the system of FIG. 5, although the invention is not limited to these details.

During clock cycle 0, command A is issued with CS0# and CS1# asserted.

During clock cycle 3, the ODT in each of chips MC1-0 ... MC1-N (rank 1) is turned on after a delay of tAOND.

During clock cycles 4-7, data is written to chips MC0-0 ... MC0-N (rank 0) on the bus after a delay of tWL. As an example, a DDR3 memory chip may be used with a burst length (BL) of 8 and data double pumped (also called dual data rate or double data rate), meaning data is transmitted or received once each half clock cycle. Note that there is some time between the time the ODT is turned on in the chips of rank 1 and the time data is on the bus for the chips of rank 0.

During clock cycle 5, command B is issued with CS0# and CS1# asserted.

During clock cycle 8, the ODT in chips MC1-0 ... MC1-N (rank 1) are turned off automatically (without a separate command) after a fixed delay based on a burst length of 8. (In some other embodiments, the ODTs are not turned off automatically.) During clock cycle 8, the ODT in chips MC0-0 ... MC0-N (rank 0) is turned on after a delay tAOND in response to command B.

During clock cycles 9-12, data is written to chips MC1-0 ... MC1-N (rank 1) on the bus after a delay of tWL.

During clock cycle 13, the ODT in chips MC0-0 ... MC0-N (rank 0) are turned off automatically after a fixed delay based on a burst length of 8.

In some embodiments, if only one CS# is asserted then the second rank may ignore the command.

Although memory controller 12 is illustrated in different figures, it is not necessary that a memory controller according to the different embodiments of the invention have the ability to operate in each of the different systems.

FIG. 8 illustrates memory controller 12 as being in a processor 84, such as a microprocessor. FIG. 9 illustrates memory controller 12 as being in a memory controller hub 86 in a chipset.

Although FIGS. 2, 5 and 6 are illustrated with multi-drop bus configurations, point to point interconnects may be used. For example, interconnect 14 in FIG. 1 could be multi-drop, point-to-point or a combination of them. A command that is responded to differently by different chips can be received simultaneously by the chips or it can be received a different times by the chips.

Although the memory chips MC0 and MC1 are illustrated, the invention may be used with chips whose primary purpose is not memory storage.

The command may include multiple bits in parallel, in series or in a combination of parallel and series. The command may be packetized with other signals.

The states of the bits in registers 30 and 40 may be set at boot up (turning on the system that includes chips MC0 and MC1) or after boot up. For example, the state might be changed depending on various things that could happen after boot up. In some embodiments (different from those of the example 1), the state of the bits in registers 30 and 40 are changed so that the chips MC0 and MC1 respond to the same command as being sometimes a write command and other times a turn on ODT command. In this case, simply using command A would be sufficient and command B would not be used as it is in example 1.

FIG. 10 is the same as FIG. 2, except that system circuitry 90 interfaces between a program 92 in memory 94. In some embodiments, program 92 may be BIOS, an operating system or other software. Among other things, program 92 may control at least part of the contents of registers 30 and 40 including how chips MC0 and MC1 respond to commands, such as in examples 1 and 2. In other embodiments, program 92 does not control the contents of those parts of registers 30 and 40 that control how chips MC0 and MC1 respond to the commands, such as in examples 1 and 2. For example, memory controller

12 may decide what is to be those contents of registers 30 and 40 based on internal logic of controller 12 independent of control of program 92.

By using a command to control the ODTs, it is not necessary to have a pin (pad) to control the ODTs, although there may be a pin to control the ODTs in some embodiments. In some embodiments, the ODTs are controlled only by commands, not pins. This results in reduction of pins compared to a system with pins used to control the ODTs. In other embodiments, the ODTs are controlled only by pins. In still other embodiments, the ODTs are controlled by commands sometimes and by pins at other times.

By using commands to control ODTs, it does not matter whether a 1N or 2N timing is used. As noted, a 2N timing involves one extra clock cycle to sample a signal.

The term “control” means to at least partially control. The term “cause” means to at least partially cause.

In the disclosure and claims, the use of the word “first” is just for convenience in distinguishing. It does not imply a temporal, spatial, or importance order. For example, reference to a first operational type signal does not mean the first operational type signal is first in a temporal, spatial, or importance order with respect to a second operational type signal. Likewise, reference to a first command does not mean that the first command is first in a temporal, spatial, or importance order with respect to other commands. Similarly, the words “second,” “third,” and “fourth” are merely labels without implying a temporal, spatial, or importance order.

An embodiment is an implementation or example of the invention. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” or “other embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances of “an embodiment,” “one embodiment,” or “some embodiments” are not necessarily all referring to the same embodiments.

If the specification states a component, feature, structure, or characteristic “may,” “might,” or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or

claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

The invention are not restricted to the particular details described herein. Indeed, many other variations of the foregoing description and drawings may be made within the scope of the present invention. Accordingly, it is the following claims including any
5 amendments thereto that define the scope of the invention.

CLAIMS

What is claimed is:

1. A chip comprising:
a register to include an operation type signal;
5 control circuitry to receive a first command and in response to the first command to cause the chip to perform a first operation if the operation type signal has a first value and to cause the chip to perform a second operation if the operation type signal has a second value.
2. The chip of claim 1, further comprising:
10 a memory core; and
an on die termination, and wherein the first operation is to turn on the on die termination and the second operation is to cause data to be written into the memory core.
3. The chip of claim 1, wherein the control circuitry is to receive a second
15 command and to cause the chip to perform the second operation if the operation type signal has the first value and to cause the chip to perform the first operation if the operation type signal has the second value.
4. The chip of claim 1, wherein the operation type signal is a first operation
20 type signal and the register also is to include a second operation type signal and the control circuitry is to receive a third command and to cause the chip to perform a third operation if the second operation type signal has a first value and to cause the chip to perform a fourth operation if the second operation type signal has a second value.
5. A system comprising:
a first chip to provide a first command; and
a second chip and a third chip to receive the first command, wherein the first chip
25 responds to the first command by performing a first operation and wherein the second chip responds to the first command by performing a second operation.
6. The system of claim 5, wherein the first chip is a memory controller, the
second chip is in a first rank of memory chips and the second chip is in a second rank of
memory chips.

30

7. The system of claim 5, wherein the first and second chips further comprising:

a memory core; and

an on die termination, and wherein the first operation is to turn on the on die
5 termination and the second operation is to cause data to be written into the memory core.

8. A system comprising:

a first chip to provide a first command; and

a second chip including a register to include an operation type signal having a first
value; and

10 a third chip including a register to include an operation type signal having a second
value, and wherein in response to the first command and the operation type signals, the
second chip performs a first operation and the third chip performs a second operation.

9. The system of claim 8, wherein the first chip is a memory controller, the
second chip is in a first rank of memory chips and the second chip is in a second rank of
15 memory chips.

10. The system of claim 8, wherein in response to receiving a second
command, the second chip performs the second operation and the third chip performs the
first operation.

11. The system of claim 8, wherein in response to receiving a third command,
20 the second chip performs a third operation and the third chip performs a fourth operation.

12. The system of claim 8, wherein the command is simultaneously received by
the second and third chips.

13. The system of claim 8, wherein the command is received at different times
by the second and third chips.

25 14. A system comprising:

a memory controller to provide a first command;

a first group of memory chips each including a register to include an operation type
signal having a first value; and

a second group of memory chips each including a register to include an operation
30 type signal having a second value, and wherein in response to the first command and the
values of the operation type signals, the chips of the first group each perform a first
operation and the chips of the second group each perform a second operation.

15. The system of claim 14, wherein each chip in the first group responds to a second command by performing the second operation and wherein each of the chips in the second group responds to the second command by performing the first operation.

16. The system of claim 14, wherein the operation type signals are a first
5 operation type signal and the registers are also to include a second operation type signal and wherein in response to the a third command and the values of the second operation type signals, the chips of the first group of each perform a third operation and the chips of the second group each perform a fourth operation.

17. The system of claim 14, wherein the first operation is a write operation and
10 the second operation is to turn on an on die termination in the chip performing the operation.

18. The system of claim 14, wherein the first operation is to turn on an on die termination in the chip performing the first operation and the second operation is a write operation.

19. The system of claim 14, wherein the first operation is a read operation and
15 the second operation is a refresh operation.

20. The system of claim 14, wherein the first operation is a refresh operation and the second operation is a read operation.

21. The system of claim 14, wherein the first group of memory chips is in a
20 first rank and the second group of memory chips is in a second rank.

22. The system of claim 14, further including system circuitry to interface between a program and the memory controller to control the state of the at least one bit in the chips.

23. The system of claim 14, wherein the program includes BIOS.

24. The system of claim 14, wherein the program includes an operating system.
25

25. The system of claim 14, wherein the memory controller provides signals to the memory chips to set the operation type signals in the registers.

26. The system of claim 25, wherein a value of the operation type signal is set when the chip becomes operational.

27. The system of claim 25, wherein a value of the operation type signal may
30 change after the chip becomes operational.

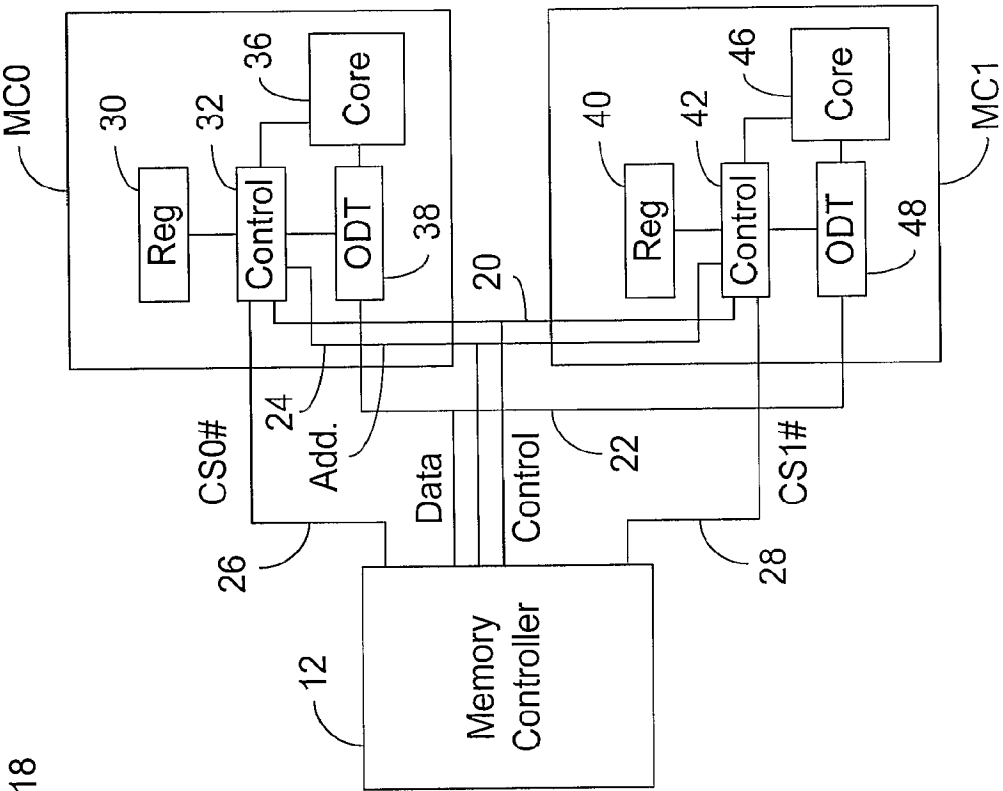


FIG. 2

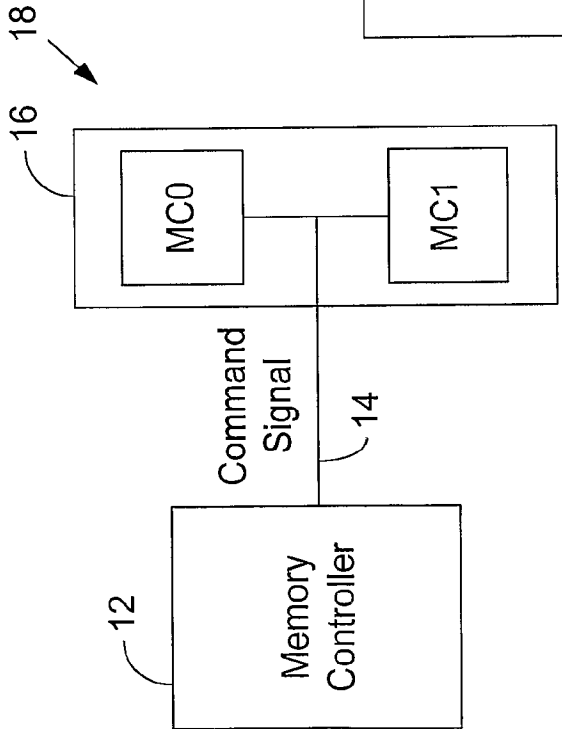


FIG. 1

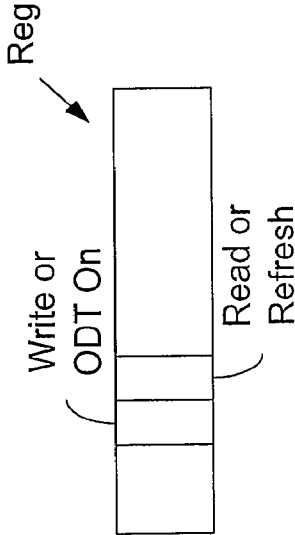


FIG. 3

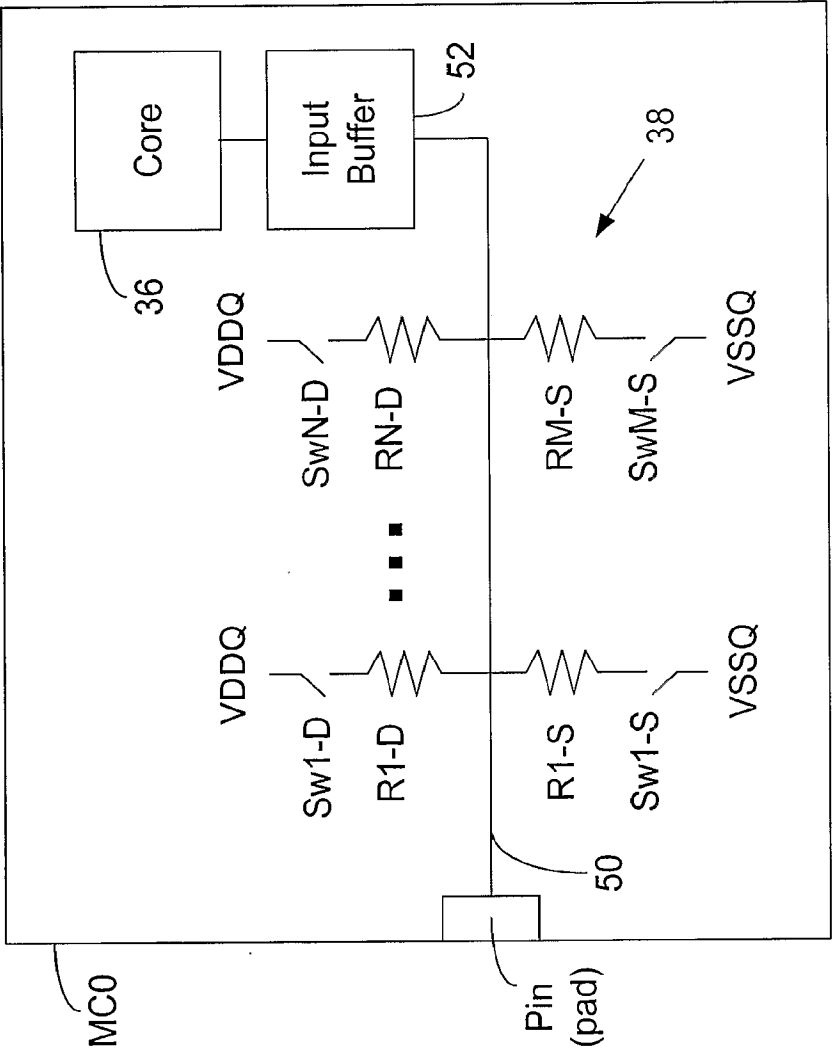


FIG. 4

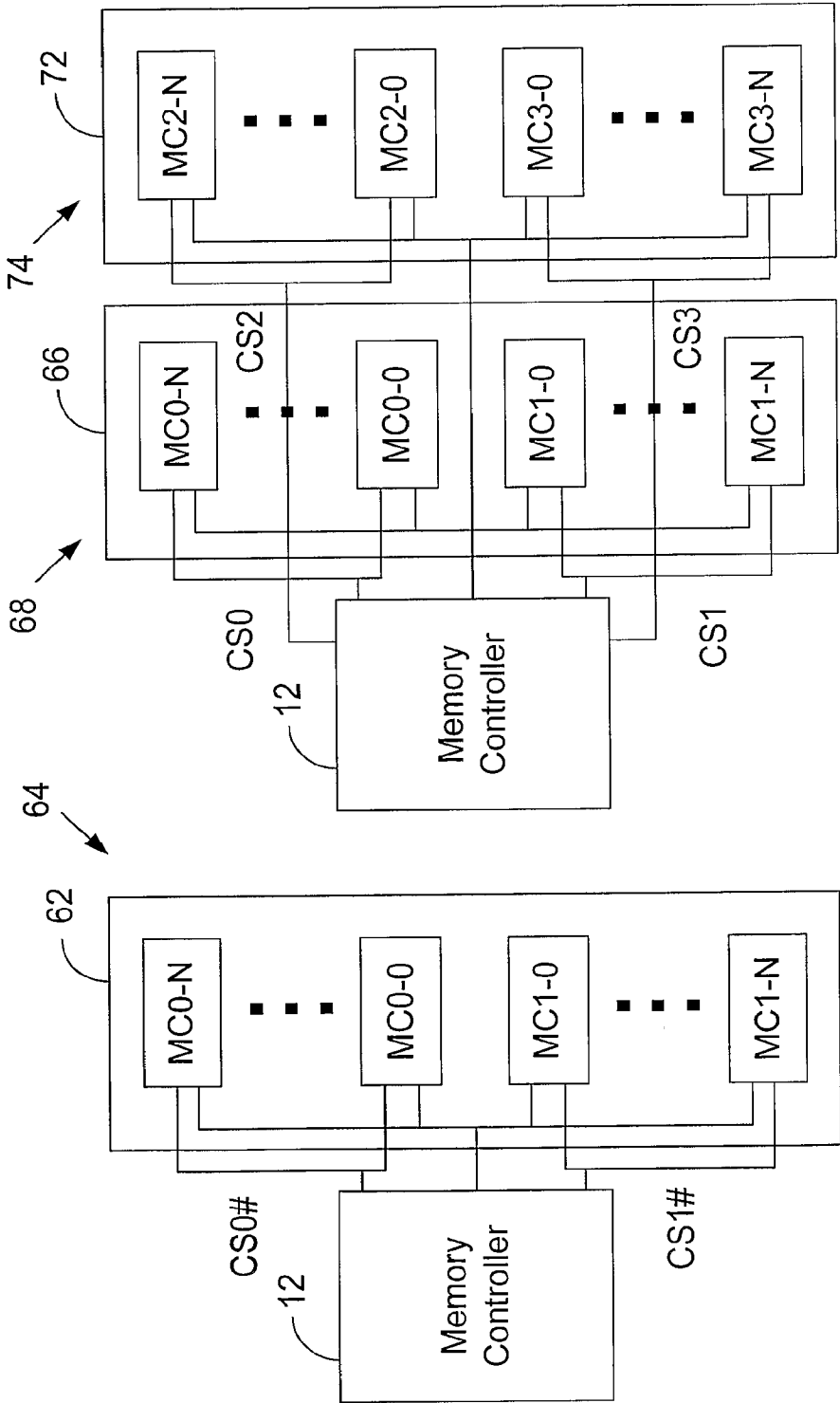


FIG. 5

FIG. 6

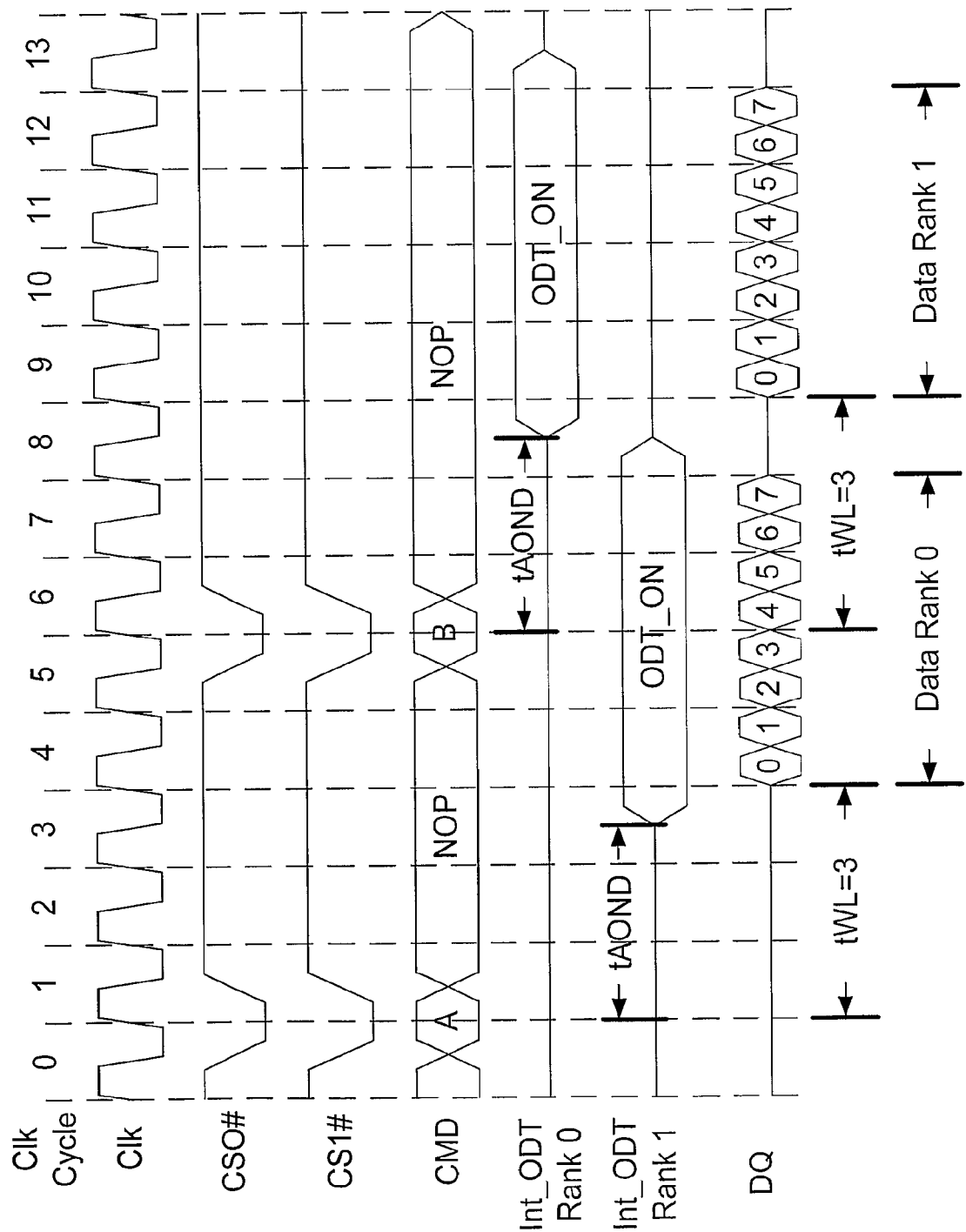


FIG. 7

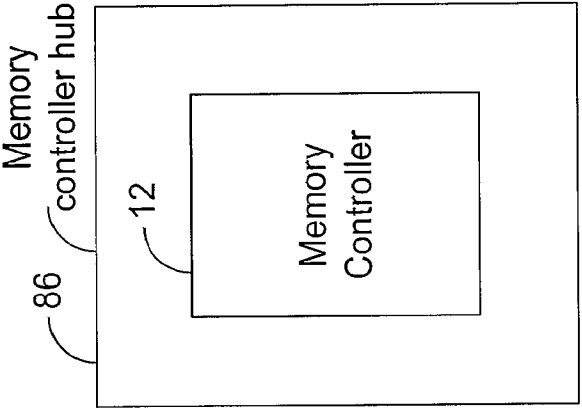


FIG. 9

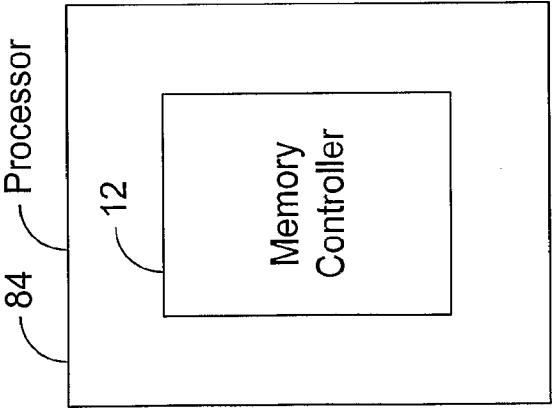


FIG. 8

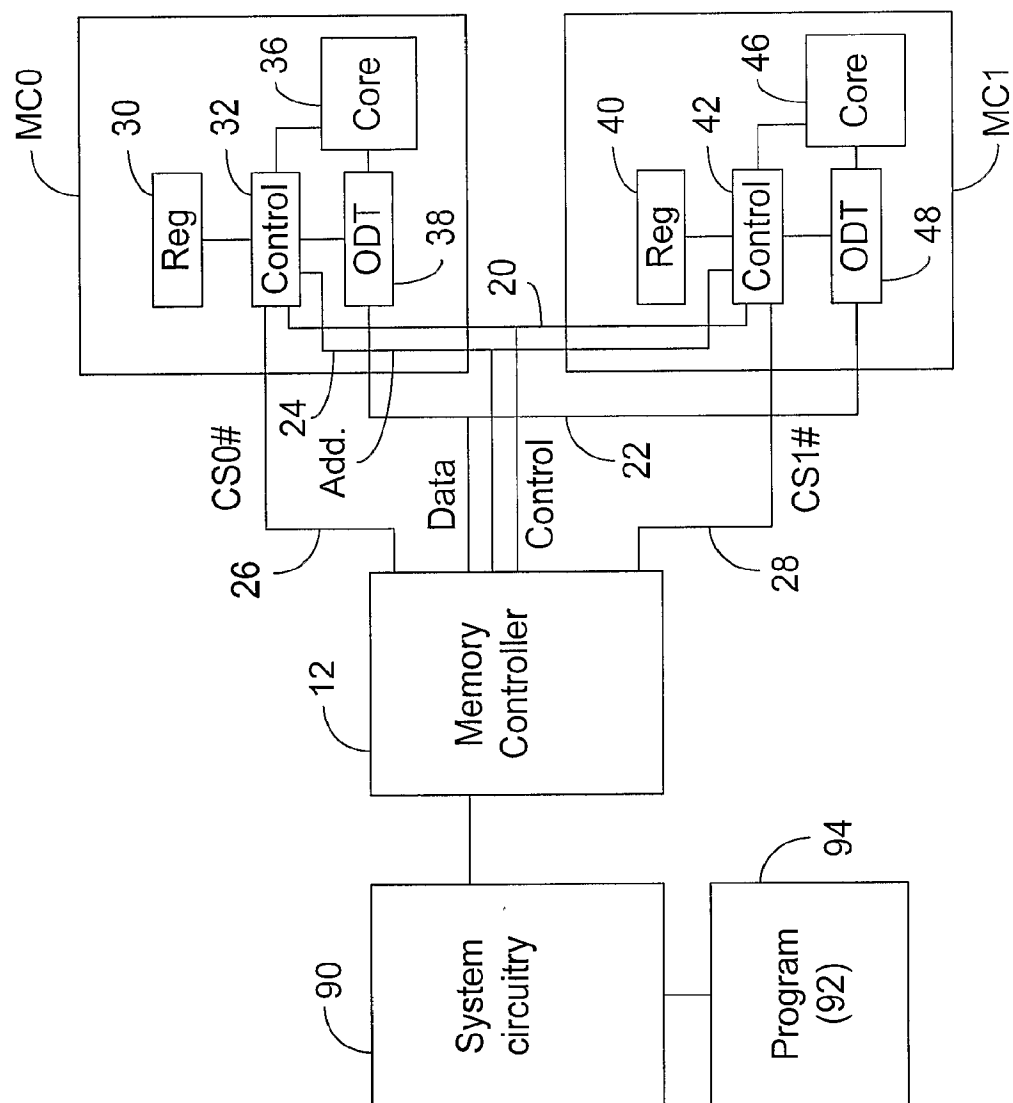


FIG. 10

PATENT COOPERATION TREATY

From the INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

Delaney, Karoline A.
KNOBBE, MARTENS, OLSON & BEAR, LLP
2040 Main Street, 14th Floor
Irvine, CA 92614
ETATS-UNIS D'AMERIQUE

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

Date of mailing (day/month/year)		27 September 2010 (27-09-2010)
Applicant's or agent's file reference NETL048VPC	FOR FURTHER ACTION See paragraphs 1 and 4 below	
International application No. PCT/US2010/040826	International filing date (day/month/year) 1 July 2010 (01-07-2010)	
Applicant NETLIST, INC.		

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

Filing of amendments and statement under Article 19:

The applicant is entitled, if he so wishes, to amend the claims of the International Application (see Rule 46):

When? The time limit for filing such amendments is normally two months from the date of transmittal of the International Search Report.

Where? Directly to the International Bureau of WIPO, 34 chemin des Colombettes
1211 Geneva 20, Switzerland, Facsimile No.: (41-22) 338.82.70

For more detailed instructions, see the notes on the accompanying sheet.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.

3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:

- ☐ the protest together with the decision thereon has been transmitted to the International Bureau together with the applicant's request to forward the texts of both the protest and the decision thereon to the designated Offices.
☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.

4. Reminders

Shortly after the expiration of **18 months** from the priority date, the international application will be published by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau as provided in Rules 90bis.1 and 90bis.3, respectively, before the completion of the technical preparations for international publication.

The applicant may submit comments on an informal basis on the written opinion of the International Searching Authority to the International Bureau. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established. These comments would also be made available to the public but not before the expiration of 30 months from the priority date.

Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for entry into the national phase before those designated Offices.

In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months.

See the Annex to Form PCT/IB/301 and, for details about the applicable time limits, Office by Office, see the *POT Applicant's Guide*, National Chapters.

Name and mailing address of the International Searching Authority



European Patent Office, P.B. 5818 Patentlaan 2
NL-2280 HV Rijswijk
Tel. (+31-70) 340-2040
Fax: (+31-70) 340-3016

Authorized officer

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Tel: +49 (0)89 2399-8003

10-10-1964
 10-10-1964

SEP 29 2010

INFORMANT

NOTES TO FORM PCT/ISA/220

These Notes are intended to give the basic instructions concerning the filing of amendments under article 19. The Notes are based on the requirements of the Patent Cooperation Treaty, the Regulations and the Administrative Instructions under that Treaty. In case of discrepancy between these Notes and those requirements, the latter are applicable. For more detailed information, see also the *PCT Applicant's Guide*.

In these Notes, "Article", "Rule", and "Section" refer to the provisions of the PCT, the PCT Regulations and the PCT Administrative Instructions, respectively.

INSTRUCTIONS CONCERNING AMENDMENTS UNDER ARTICLE 19

The applicant has, after having received the international search report and the written opinion of the International Searching Authority, one opportunity to amend the claims of the international application. It should however be emphasized that, since all parts of the international application (claims, description and drawings) may be amended during the international preliminary examination procedure, there is usually no need to file amendments of the claims under Article 19 except where, e.g. the applicant wants the latter to be published for the purposes of provisional protection or has another reason for amending the claims before international publication. Furthermore, it should be emphasized that provisional protection is available in some States only (see *PCT Applicant's Guide*, Annex B).

The attention of the applicant is drawn to the fact that amendments to the claims under Article 19 are not allowed where the International Searching Authority has declared, under Article 17(2), that no international search report would be established (see *PCT Applicant's Guide*, International Phase, paragraph 296).

What parts of the international application may be amended?

Under Article 19, only the claims may be amended.

During the international phase, the claims may also be amended (or further amended) under Article 34 before the International Preliminary Examining Authority. The description and drawings may only be amended under Article 34 before the International Examining Authority.

Upon entry into the national phase, all parts of the international application may be amended under Article 28 or, where applicable, Article 41.

When?

Within 2 months from the date of transmittal of the international search report or 16 months from the priority date, whichever time limit expires later. It should be noted, however, that the amendments will be considered as having been received on time if they are received by the International Bureau after the expiration of the applicable time limit but before the completion of the technical preparations for international publication (Rule 46.1).

Where not to file the amendments?

The amendments may only be filed with the International Bureau and not with the receiving Office or the International Searching Authority (Rule 46.2).

Where a demand for international preliminary examination has been/is filed, see below.

How?

Either by cancelling one or more entire claims, by adding one or more new claims or by amending the text of one or more of the claims as filed.

A replacement sheet or sheets containing a complete set of claims in replacement of all the claims previously filed must be submitted.

Where a claim is cancelled, no renumbering of the other claims is required. In all cases where claims are renumbered, they must be renumbered consecutively in Arabic numerals (Section 205(a)).

The amendments must be made in the language in which the international application is to be published.

What documents must/may accompany the amendments?

Letter (Section 205(b)):

The amendments must be submitted with a letter.

The letter will not be published with the international application and the amended claims. It should not be confused with the "Statement under Article 19(1)" (see below, under "Statement under Article 19(1)").

The letter must be in English or French, at the choice of the applicant. However, if the language of the international application is English, the letter must be in English; if the language of the international application is French, the letter must be in French.

NOTES TO FORM PCT/ISA/220 (continued)

The letter must indicate the differences between the claims as filed and the claims as amended. It must, in particular, indicate, in connection with each claim appearing in the international application (it being understood that identical indications concerning several claims may be grouped), whether

- (i) the claim is unchanged;
- (ii) the claim is cancelled;
- (iii) the claim is new;
- (iv) the claim replaces one or more claims as filed;
- (v) the claim is the result of the division of a claim as filed.

The following examples illustrate the manner in which amendments must be explained in the accompanying letter:

1. [Where originally there were 48 claims and after amendment of some claims there are 51]:
"Claims 1 to 29, 31, 32, 34, 35, 37 to 48 replaced by amended claims bearing the same numbers; claims 30, 33 and 36 unchanged; new claims 49 to 51 added."
2. [Where originally there were 15 claims and after amendment of all claims there are 11]:
"Claims 1 to 15 replaced by amended claims 1 to 11."
3. [Where originally there were 14 claims and the amendments consist in cancelling some claims and in adding new claims]:
"Claims 1 to 6 and 14 unchanged; claims 7 to 13 cancelled; new claims 15, 16 and 17 added." or
"Claims 7 to 13 cancelled; new claims 15, 16 and 17 added; all other claims unchanged."
4. [Where various kinds of amendments are made]:
"Claims 1-10 unchanged; claims 11 to 13, 18 and 19 cancelled; claims 14, 15 and 16 replaced by amended claim 14; claim 17 subdivided into amended claims 15, 16 and 17; new claims 20 and 21 added."

"Statement under article 19(1)" (Rule 46.4)

The amendments may be accompanied by a statement explaining the amendments and indicating any impact that such amendments might have on the description and the drawings (which cannot be amended under Article 19(1)).

The statement will be published with the international application and the amended claims.

It must be in the language in which the international application is to be published.

It must be brief, not exceeding 500 words if in English or if translated into English.

It should not be confused with and does not replace the letter indicating the differences between the claims as filed and as amended. It must be filed on a separate sheet and must be identified as such by a heading, preferably by using the words "Statement under Article 19(1)."

It may not contain any disparaging comments on the international search report or the relevance of citations contained in that report. Reference to citations, relevant to a given claim, contained in the international search report may be made only in connection with an amendment of that claim.

Consequence if a demand for international preliminary examination has already been filed

If, at the time of filing any amendments and any accompanying statement, under Article 19, a demand for international preliminary examination has already been submitted, the applicant must preferably, at the time of filing the amendments (and any statement) with the International Bureau, also file with the International Preliminary Examining Authority a copy of such amendments (and of any statement) and, where required, a translation of such amendments for the procedure before that Authority (see Rules 55.3(a) and 62.2, first sentence). For further information, see the Notes to the demand form (PCT/IPEA/401).

If a demand for international preliminary examination is made, the written opinion of the International Searching Authority will, except in certain cases where the International Preliminary Examining Authority did not act as International Searching Authority and where it has notified the International Bureau under Rule 66.1bis(b), be considered to be a written opinion of the International Preliminary Examining Authority. If a demand is made, the applicant may submit to the International Preliminary Examining Authority a reply to the written opinion together, where appropriate, with amendments before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later (Rule 43bis.1(c)).

Consequence with regard to translation of the international application for entry into the national phase

The applicant's attention is drawn to the fact that, upon entry into the national phase, a translation of the claims as amended under Article 19 may have to be furnished to the designated/elected Offices, instead of, or in addition to, the translation of the claims as filed.

For further details on the requirements of each designated/elected Office, see the *PCT Applicant's Guide*, National Chapters.

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference NETL048VPC	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US2010/040826	International filing date (day/month/year) 01/07/2010	(Earliest) Priority Date (day/month/year) 16/07/2009
Applicant NETLIST, INC.		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 4 sheets.



It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:



the international application in the language in which it was filed



a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

b. ☐

This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐

With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐

Certain claims were found unsearchable (See Box No. II)

3. ☐

Unity of invention is lacking (see Box No III)

4. With regard to the **title**,



the text is approved as submitted by the applicant



the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,



the text is approved as submitted by the applicant



the text has been established, according to Rule 38.2(b), by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority

6. With regard to the **drawings**,

a. the figure of the **drawings** to be published with the abstract is Figure No. 3a



as suggested by the applicant



as selected by this Authority, because the applicant failed to suggest a figure



as selected by this Authority, because this figure better characterizes the invention

b. ☐

none of the figures is to be published with the abstract

INTERNATIONAL SEARCH REPORT

International application No.

PCT/US2010/040826

Box No. IV Text of the abstract (Continuation of item 5 of the first sheet)

A memory system and method utilizing one or more memory modules is provided. The memory module (400) includes a plurality of memory devices (412) and a controller (430) configured to receive control information (440) from a system memory controller (420) and to produce module control signals (442). The memory module further includes a plurality of circuits, for example byte-wise buffers (416), which are configured to selectively isolate the plurality of memory devices from the system memory controller. The circuits are operable, in response to the module control signals, to drive write data from the system memory controller to the plurality of memory devices and to merge read data from the plurality of memory devices to the system memory controller. The circuits are distributed at corresponding positions separate from one another.

INTERNATIONAL SEARCH REPORT

International application No

PCT/US2010/040826

A. CLASSIFICATION OF SUBJECT MATTER
 INV. G11C5/02 G11C5/06

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
 G11C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X A	US 2006/262586 A1 (SOLOMON JEFFREY C [US] ET AL) 23 November 2006 (2006-11-23) the whole document	1-10, 18-44 11-17
X A	US 2008/162790 A1 (IM JEON-TAEK [KR]) 3 July 2008 (2008-07-03) paragraphs [0069] - [0088]; figures 11,12	1-5, 10-21, 23-31, 34-44 6-9, 22, 32, 33
X	US 2009/103387 A1 (SHAU JENG-JYE [US]) 23 April 2009 (2009-04-23) paragraphs [0026] - [0032]	1, 18, 28, 39

☐ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier document but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art.

"&" document member of the same patent family

Date of the actual completion of the international search

31 August 2010

Date of mailing of the international search report

27/09/2010

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
 NL - 2280 HV Rijswijk
 Tel. (+31-70) 340-2040,
 Fax: (+31-70) 340-3016

Authorized officer

Trifonov, Antoniy

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2010/040826

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2006262586 A1	23-11-2006	US 2009201711 A1 US 2010128507 A1	13-08-2009 27-05-2010
US 2008162790 A1	03-07-2008	CN 101221809 A KR 20080062174 A	16-07-2008 03-07-2008
US 2009103387 A1	23-04-2009	NONE	

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

see form PCT/ISA/220

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY (PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/US2010/040826

International filing date (day/month/year)
01.07.2010

Priority date (day/month/year)
16.07.2009

International Patent Classification (IPC) or both national classification and IPC
INV. G11C5/02 G11C5/06

Applicant
NETLIST, INC.

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☒ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. **FURTHER ACTION**

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

3. For further details, see notes to Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
D-80298 Munich
Tel. +49 89 2399 - 0
Fax: +49 89 2399 - 4465

Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Trifonov, Antony

Telephone No. +49 89 2399-7168



**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2010/040826

Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))
3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:
 - a. (means)
 - ☐ on paper
 - ☐ in electronic form
 - b. (time)
 - ☐ in the international application as filed
 - ☐ together with the international application in electronic form
 - ☐ subsequently to this Authority for the purposes of search
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims	
	No: Claims	<u>1-44</u>
Inventive step (IS)	Yes: Claims	
	No: Claims	<u>1-44</u>
Industrial applicability (IA)	Yes: Claims	<u>1-44</u>
	No: Claims	

2. Citations and explanations

see separate sheet

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2010/040826

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item V.

- 1 Reference is made to the following documents:
 - D1 US 2006/262586 A1 (SOLOMON JEFFREY C [US] ET AL) 23 November 2006 (2006-11-23)
 - D2 US 2008/162790 A1 (IM JEON-TAEK [KR]) 3 July 2008 (2008-07-03)
 - D3 US 2009/103387 A1 (SHAU JENG-JYE [US]) 23 April 2009 (2009-04-23)
- 2 A plurality of claims in the present application attempt to define the subject-matter by merging a plurality of data signals from the memory devices and driving the merged data signal to the computer system memory controller. However, the term merging is not correctly used since an impression could be conveyed of some kind of "twisting" of the data connections or even driving/multiplexing signals from different ranks into one single data line, e.g. as shown on Fig. 4B lines 452 (0:3&4-7) being driven onto the lines 450 (0:3). For the purpose of further examination therefore the merging will be considered as nothing but a "logical" combination of two independent transmission data circuits, each driving its own connection lines (in the case of Fig. 4B, the left half of 416 drives all and only lines 0-3 and the right half drives only lines 4-7).
- 3 Independent claim 39 specifies a "load reducing switching circuit". Since no technical features for supporting said "load reducing" are included in the claim, this term will not be considered for further examination.
- 4 The present application does not meet the criteria of Article 33(1) PCT, for the following reasons:
 - 4.1 Document D1 discloses a memory module comprising:
 - at least one printed circuit board (10);
 - a plurality of memory devices mechanically coupled to the at least one printed circuit board (32);

- a control circuit (a part of 40, e.g. 122 generating control signals for switching circuits 120) mechanically coupled to the at least one printed circuit board, the control circuit configurable to receive control signals from a system memory controller and to transmit module control signals to the plurality of memory devices (see Fig. 9A); and

- a plurality of data transmission circuits (120, 120a, 120b) mechanically coupled to the at least one printed circuit board and distributed at corresponding positions relative to the at least one printed circuit board, the plurality of data transmission circuits configurable to be operatively coupled to the system memory controller and configurable to receive module control signals from the control circuit,

wherein at least one first data transmission circuit (see 120a on Fig. 4A and par. [0052], wherein the line 102a represents a set of data lines coupled to the plurality of memory devices 30) of the plurality of data transmission circuits is operatively coupled to at least two memory devices of the plurality of memory devices (30), at least one second data transmission circuit (120b) of the plurality of data transmission circuits is operatively coupled to at least two memory devices of the plurality of memory devices (30),

wherein the at least one first data transmission circuit is configurable to respond to the module control signals by selectively allowing or inhibiting data transmission between the system memory controller and at least one selected memory device of the at least two memory devices operatively coupled to the at least one first data transmission circuit, and the at least one second data transmission circuit is configurable to respond to the module control signals by selectively allowing or inhibiting data transmission between the system memory controller and at least one selected memory device of the at least two memory devices operatively coupled to the at least one second data transmission circuit (the circuits 120a and 120b isolate the data lines 102a and 102b of respectively ranks 32a and 32b from the common data line(s) 112, see also par. [0054]).

D2 discloses also (see Fig. 12) a memory module comprising a printed circuit board (510), a plurality of memory devices (511), a control circuit (513, 514, 516) and a plurality of data transmission circuits (515) connected as required in claim 1 in the present application.

Claim 1 is therefore not new in the sense of Article 33(2) PCT.

- 4.2 Since the data transmission circuits (or equivalent "switching circuits") of D1 and D2 are used for:
- isolating one or more of the loads of the memory devices from the computer system (see D1: pars.[0040, 41] and D2: par. [0072]),
 - for bidirectional data transfer between a memory controller and memory devices (i.e. for read and write operations), and for
 - "merging" read data (see Figs. 1 and 4A in D1, wherein the circuit 40 comprises a plurality of sub-circuits for multiplexing/de-multiplexing of lines (102) onto common lines (112). If the memory devices' output bus is 4 bit wide and the module output bus 64 bit, the data from the memory devices can be considered as "merged" on the memory module data bus),
- the subject-matter of claims 18, 28 and 39 is also not new compared to the prior art of D1 and D2.
- 4.3 The memory devices in D1 can have different configurations (par. [0108]), e.g. 128Mx8bit or 128Mx4bit and the memory module can operate as a data path rank buffer (par. [0055]). Accordingly, the data transmission circuit of D1 comprises a byte-wise buffer as required in claim 25 in the present application. This and the remaining technical features of claim 25 could therefore be found in D1.
- 5 The remaining dependent claims 2-17, 19-24, 26, 27, 29-38, 40-44 do not contain any features which, in combination with the features of any claim to which they refer, meet the requirements of the PCT in respect of novelty and/or inventive step (Article 33(2) and (3) PCT).

Re Item VII.

- 6 The independent claims are not in the two-part form in accordance with Rule 6.3(b) PCT.
- 7 Reference signs are not inserted in parentheses in the claims in accordance with Rule 6.2(b) PCT.

- 8 Contrary to the requirements of Rule 5.1(a)(ii) PCT, the relevant background art disclosed in documents D1 - D3 is not mentioned in the description, nor are these documents identified therein.

Re Item VIII.

- 9 The various definitions of the invention given in the independent claims are such that the claims as a whole are not concise so that the requirements of Article 6 PCT are not met.

Possible steps after receipt of the international search report (ISR) and written opinion of the International Searching Authority (WO-ISA)

General information	For all international applications filed on or after 01/01/2004 the competent ISA will establish an ISR. It is accompanied by the WO-ISA. Unlike the former written opinion of the IPEA (Rule 66.2 PCT), the WO-ISA is not meant to be responded to, but to be taken into consideration for further procedural steps. This document explains about the possibilities.
Amending claims under Art. 19 PCT	Within 2 months after the date of mailing of the ISR and the WO-ISA the applicant may file amended claims under Art. 19 PCT directly with the International Bureau of WIPO. The PCT reform of 2004 did not change this procedure. For further information please see Rule 46 PCT as well as form PCT/ISA/220 and the corresponding Notes to form PCT/ISA/220.
Filing a demand for international preliminary examination	In principle, the WO-ISA will be considered as the written opinion of the IPEA. This should, in many cases, make it unnecessary to file a demand for international preliminary examination. If the applicant nevertheless wishes to file a demand this must be done before expiry of 3 months after the date of mailing of the ISR/ WO-ISA or 22 months after priority date, whichever expires later (Rule 54bis PCT). Amendments under Art. 34 PCT can be filed with the IPEA as before, normally at the same time as filing the demand (Rule 66.1 (b) PCT). If a demand for international preliminary examination is filed and no comments/amendments have been received the WO-ISA will be transformed by the IPEA into an IPRP (International Preliminary Report on Patentability) which would merely reflect the content of the WO-ISA. The demand can still be withdrawn (Art. 37 PCT).
Filing informal comments	After receipt of the ISR/WO-ISA the applicant may file informal comments on the WO-ISA directly with the International Bureau of WIPO. These will be communicated to the designated Offices together with the IPRP (International Preliminary Report on Patentability) at 30 months from the priority date. Please also refer to the next box.
End of the international phase	At the end of the international phase the International Bureau of WIPO will transform the WO-ISA or, if a demand was filed, the written opinion of the IPEA into the IPRP, which will then be transmitted together with possible informal comments to the designated Offices. The IPRP replaces the former IPER (international preliminary examination report).
Relevant PCT Rules and more information	Rule 43 PCT, Rule 43bis PCT, Rule 44 PCT, Rule 44bis PCT, PCT Newsletter 12/2003, OJ 11/2003, OJ 12/2003



US 20060262586A1

(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2006/0262586 A1****Solomon et al.**(43) **Pub. Date: Nov. 23, 2006**(54) **MEMORY MODULE WITH A CIRCUIT PROVIDING LOAD ISOLATION AND MEMORY DOMAIN TRANSLATION**(76) Inventors: **Jeffrey C. Solomon**, Irvine, CA (US);
Jayesh R. Bhakta, Cerritos, CA (US)

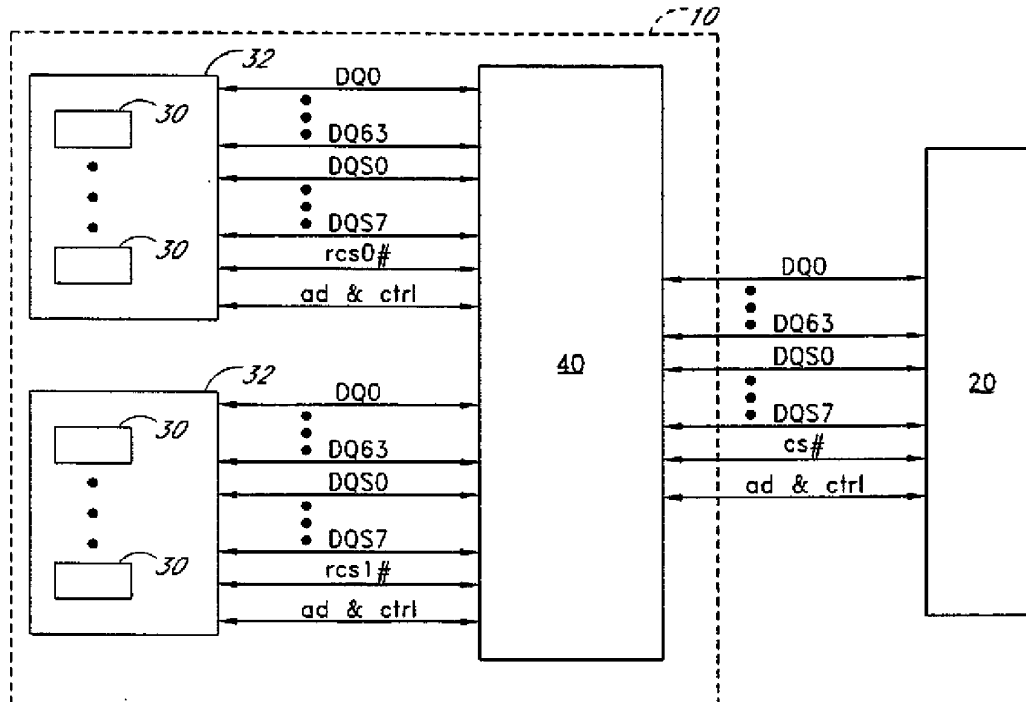
Correspondence Address:

KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614 (US)**Publication Classification**(51) **Int. Cl.**
G11C 5/06 (2006.01)
(52) **U.S. Cl.** **365/63**(21) Appl. No.: **11/335,875**(22) Filed: **Jan. 19, 2006****Related U.S. Application Data**

(63) Continuation-in-part of application No. 11/173,175, filed on Jul. 1, 2005, and which is a continuation-in-part of application No. 11/075,395, filed on Mar. 7, 2005.

(57) **ABSTRACT**

A memory module includes a plurality of memory devices and a circuit. Each memory device has a corresponding load. The circuit is electrically coupled to the plurality of memory devices and is configured to be electrically coupled to a memory controller of a computer system. The circuit selectively isolates one or more of the loads of the memory devices from the computer system. The circuit comprises logic which translates between a system memory domain of the computer system and a physical memory domain of the memory module.



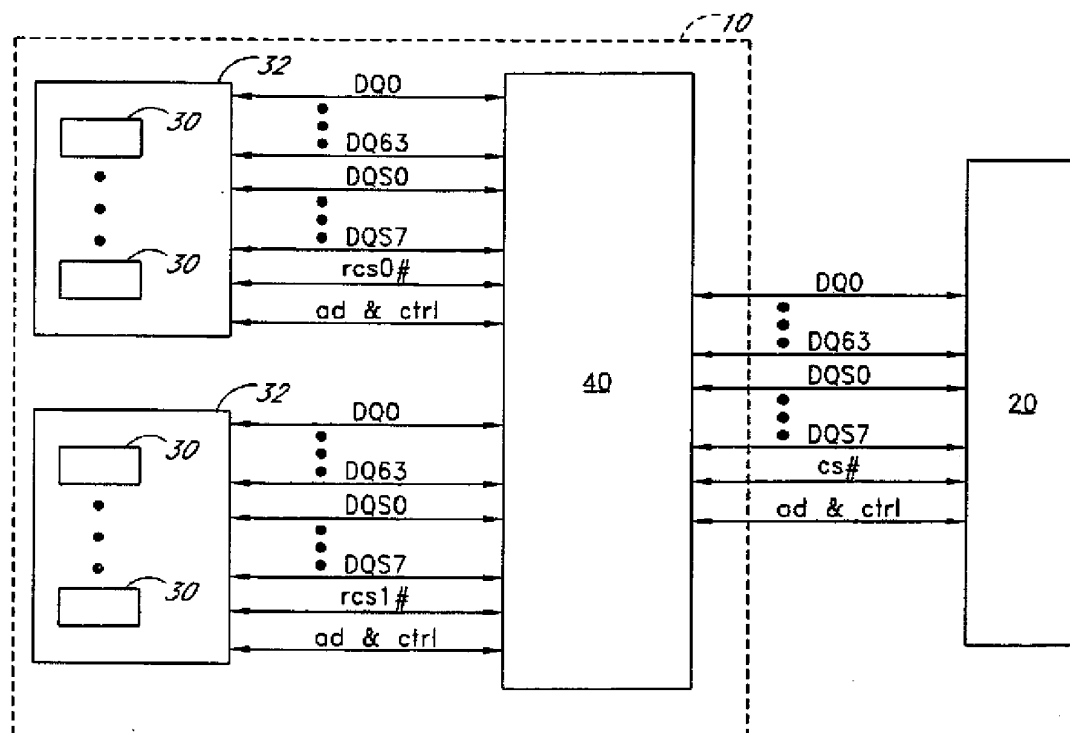
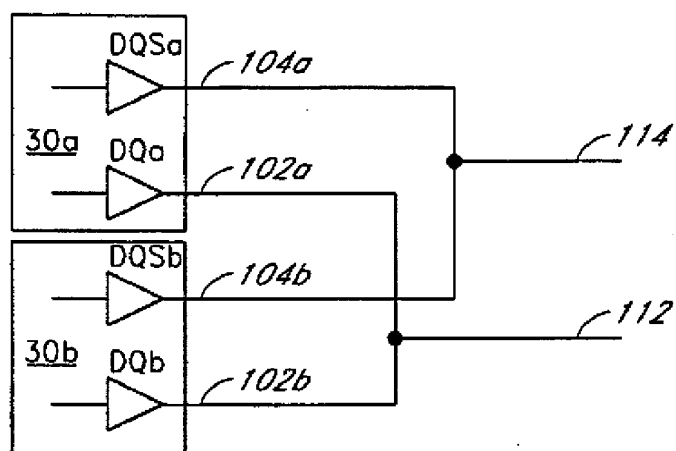
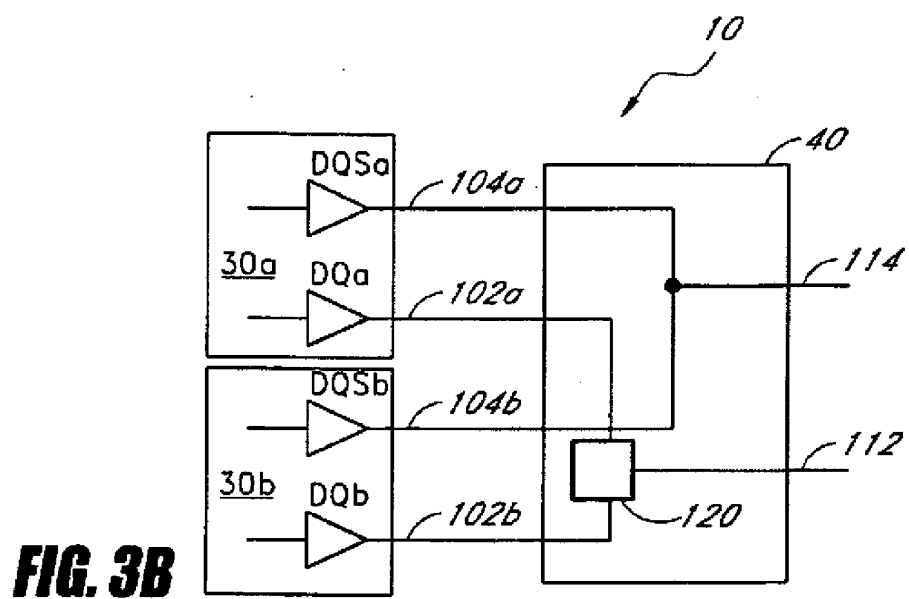
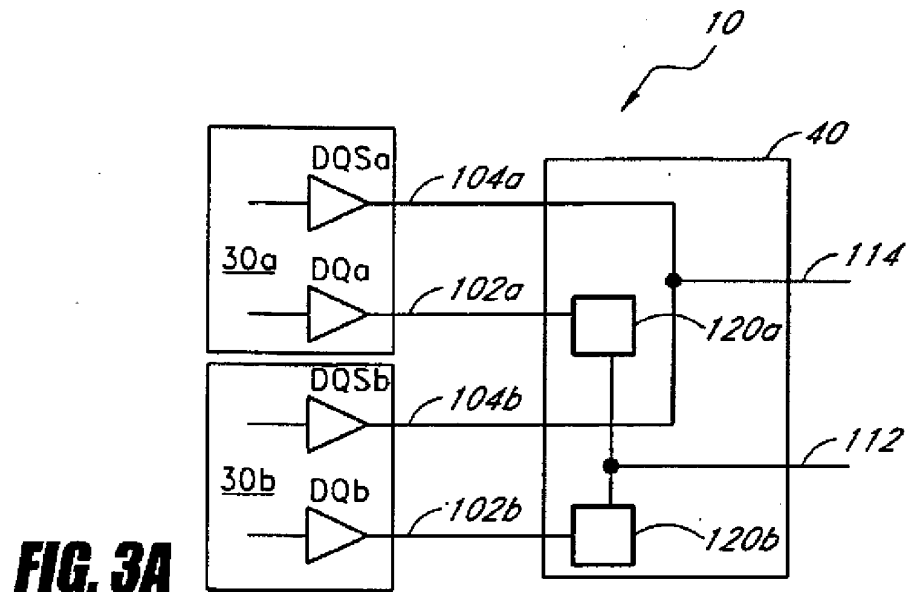
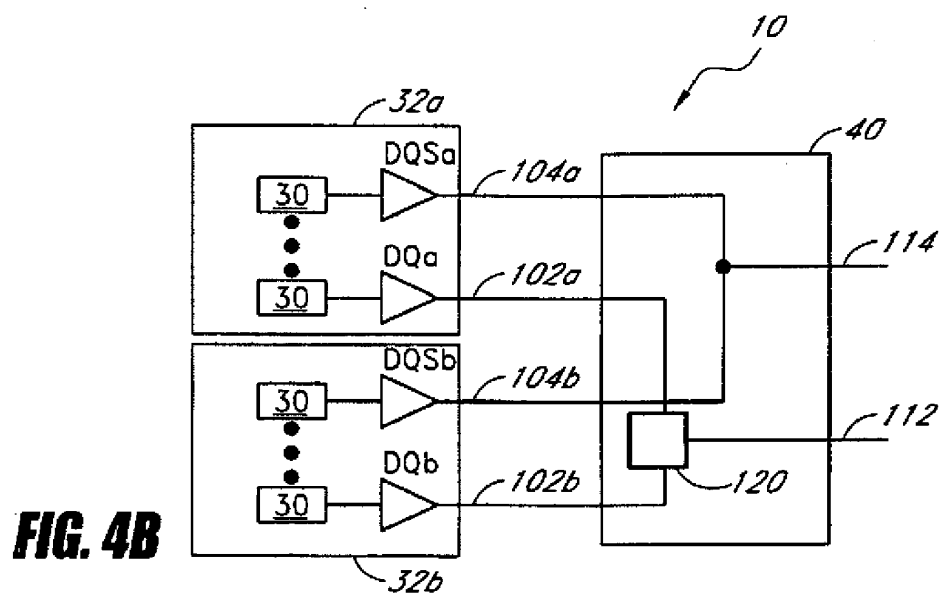
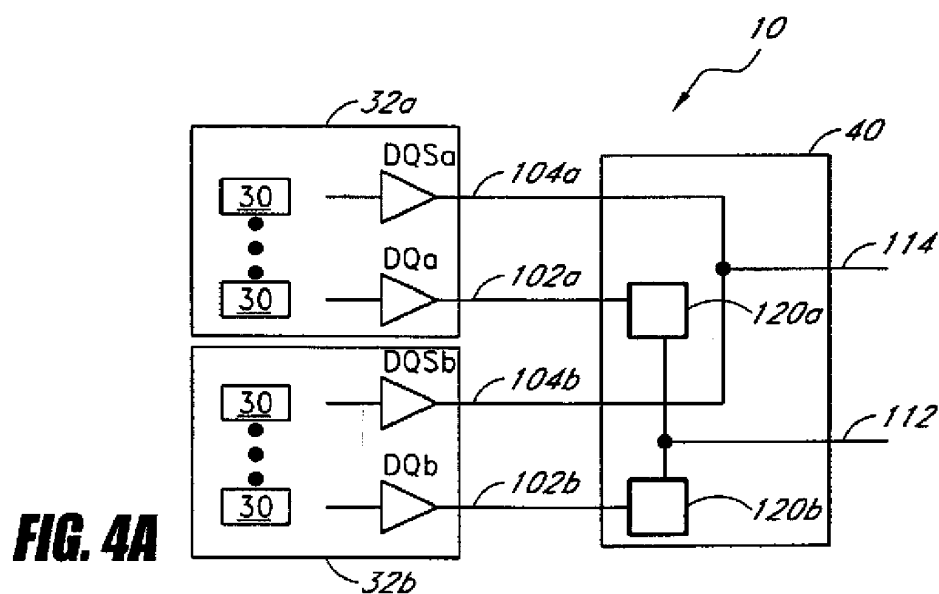


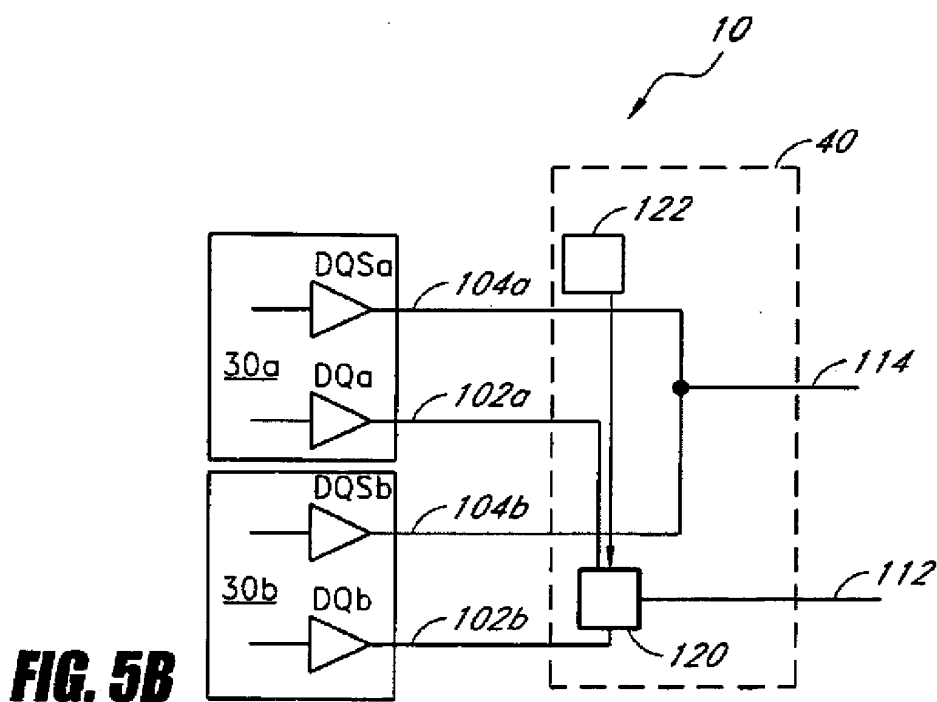
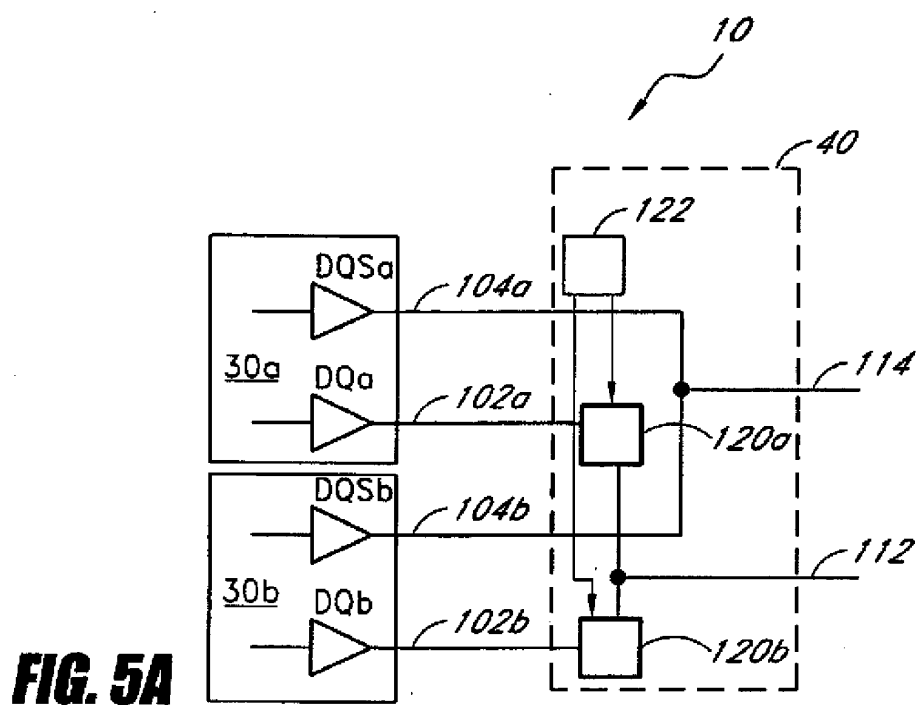
FIG. 1

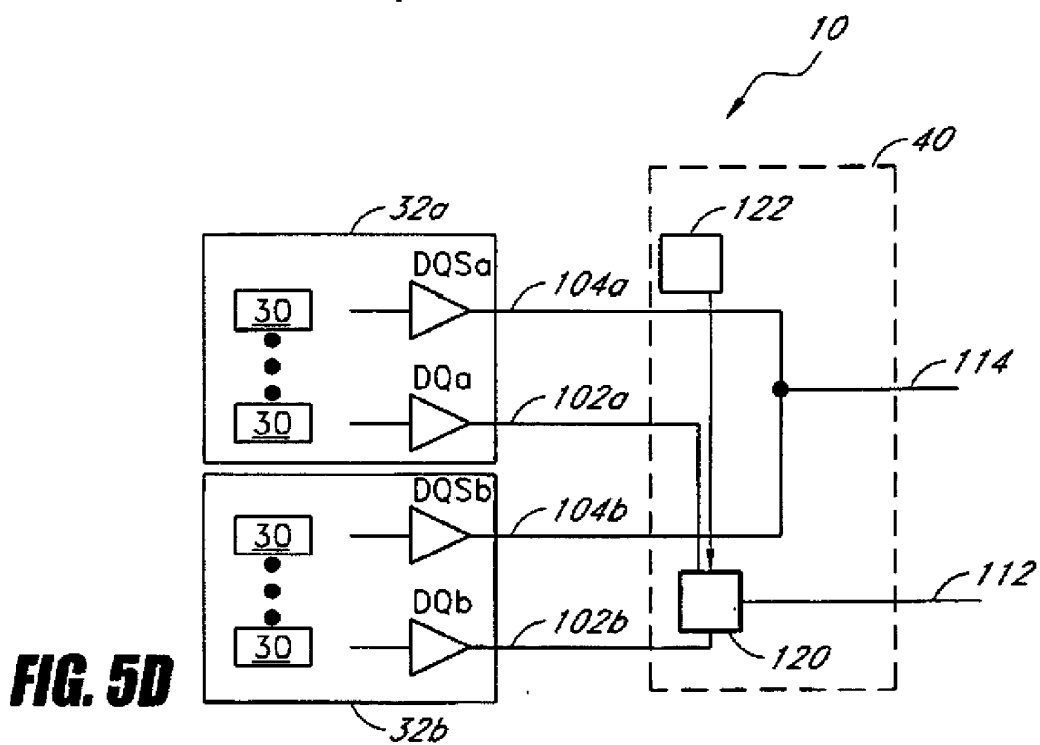
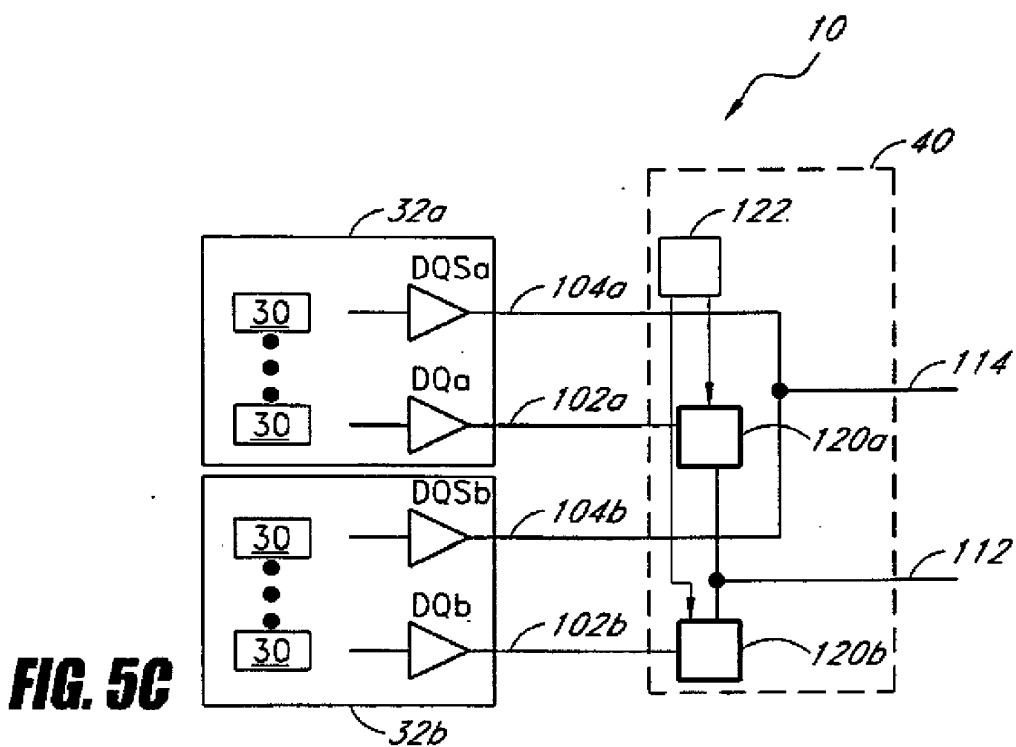
FIG. 2











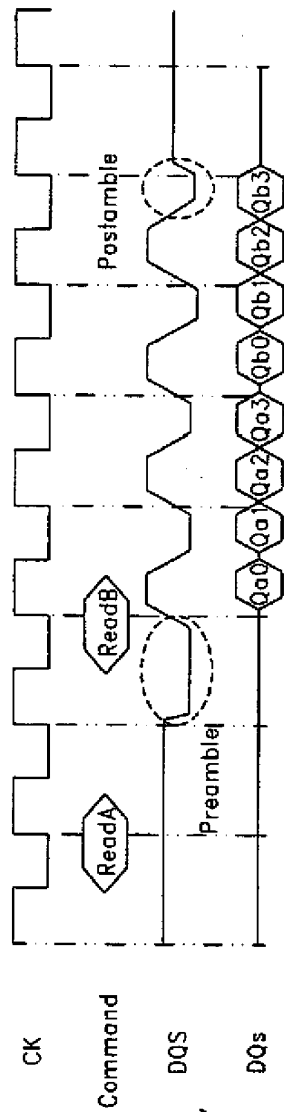


FIG. 6A

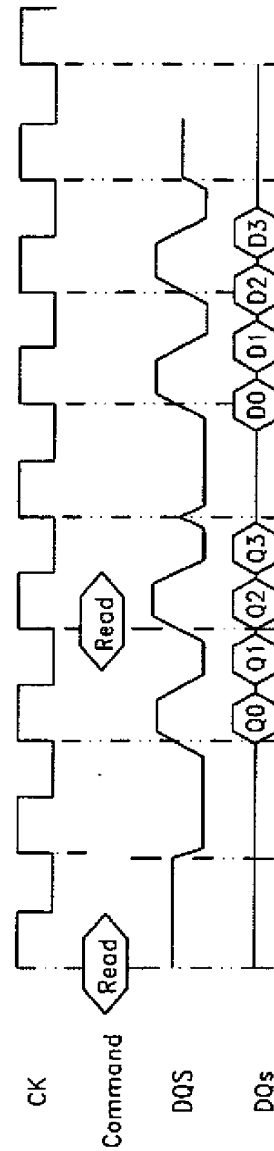


FIG. 6B

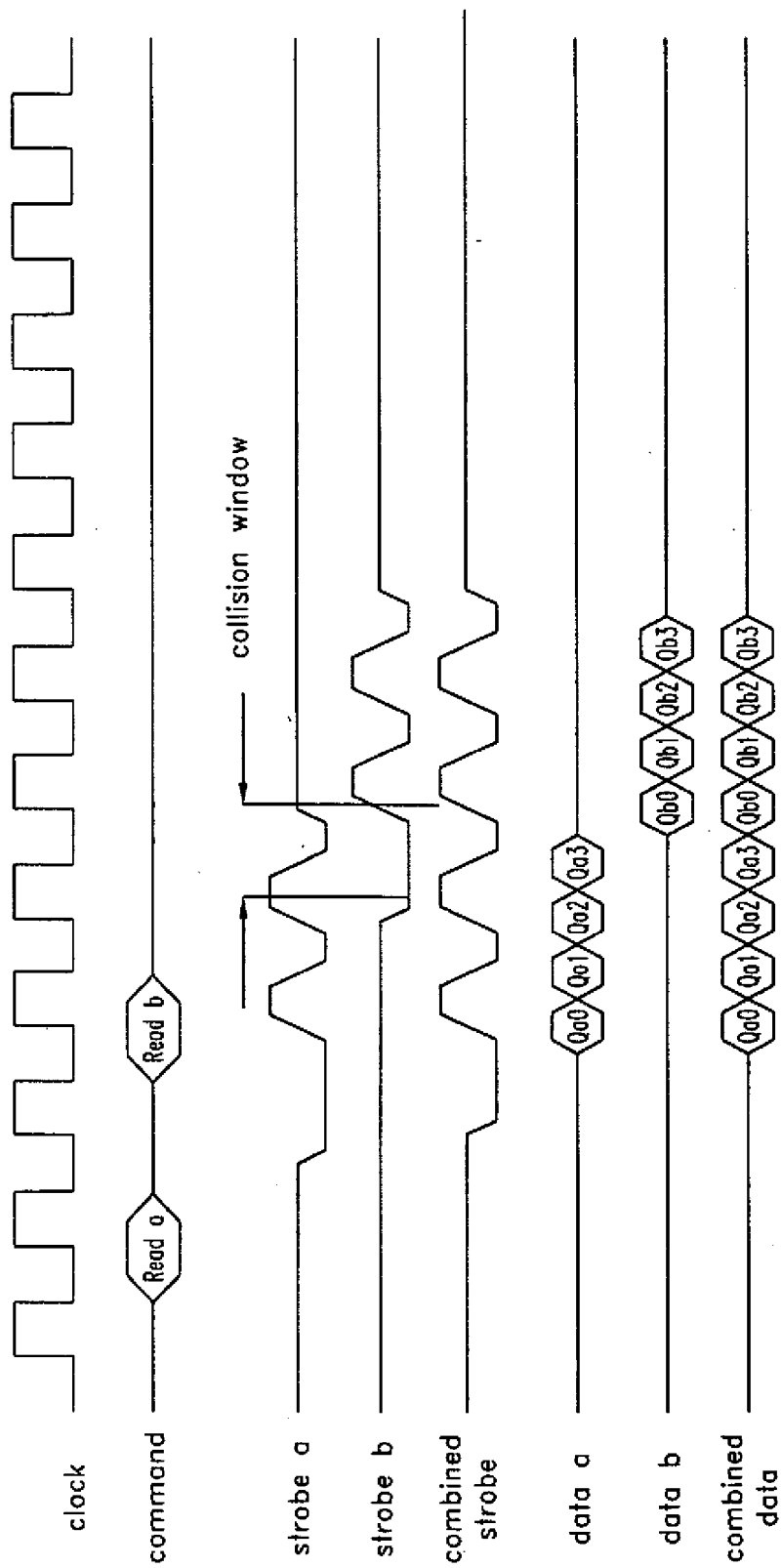
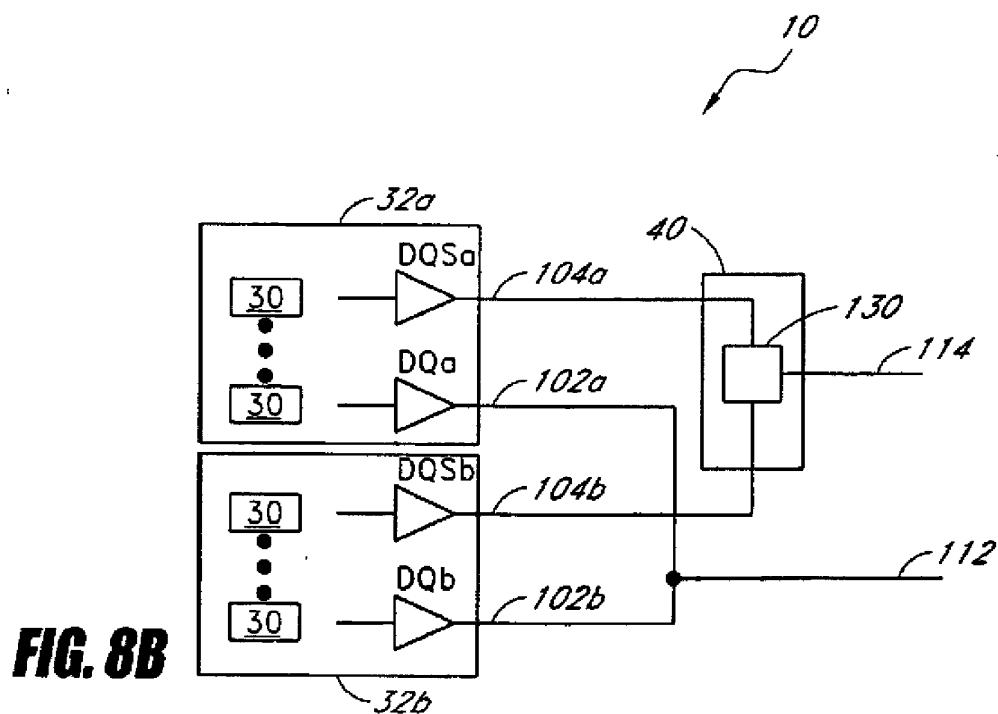
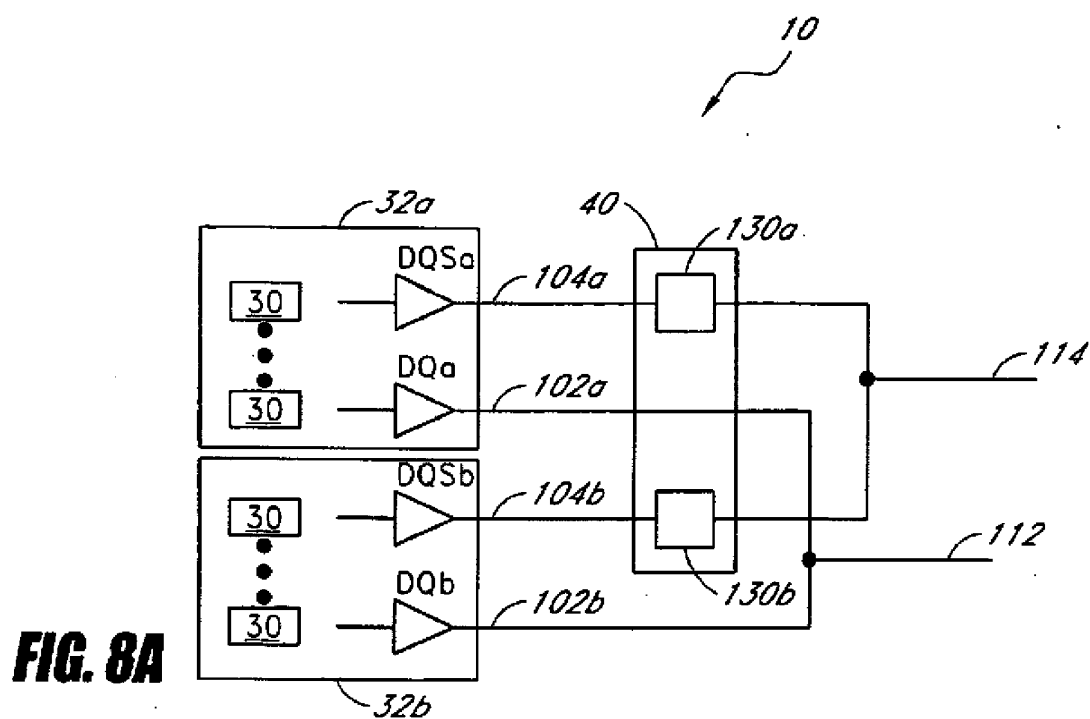
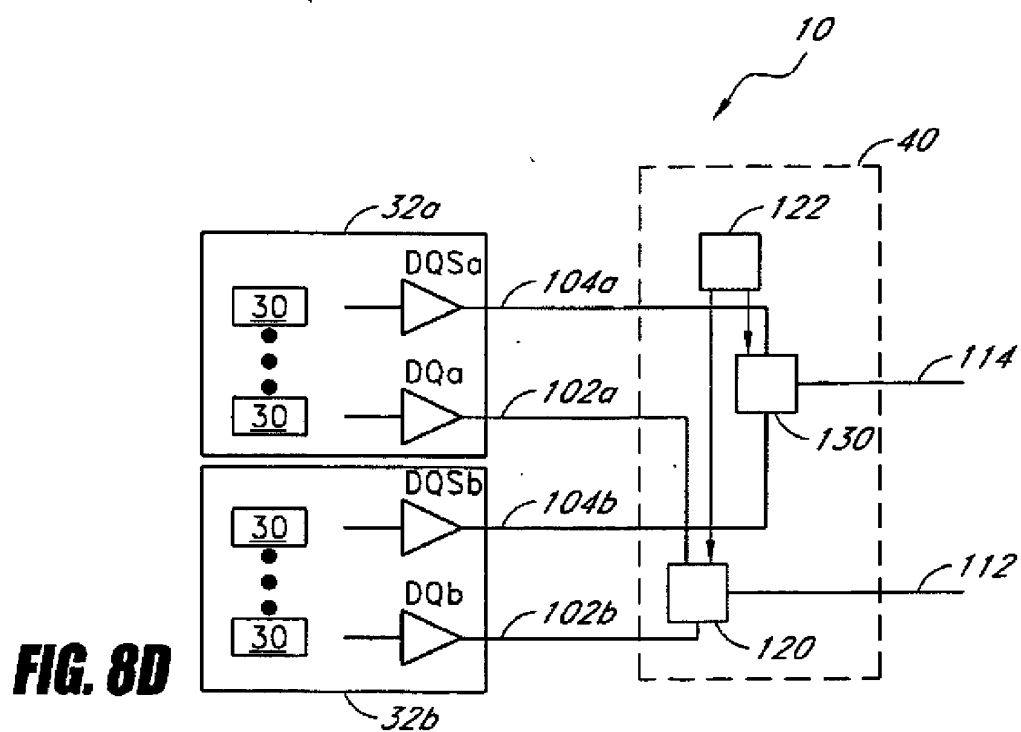
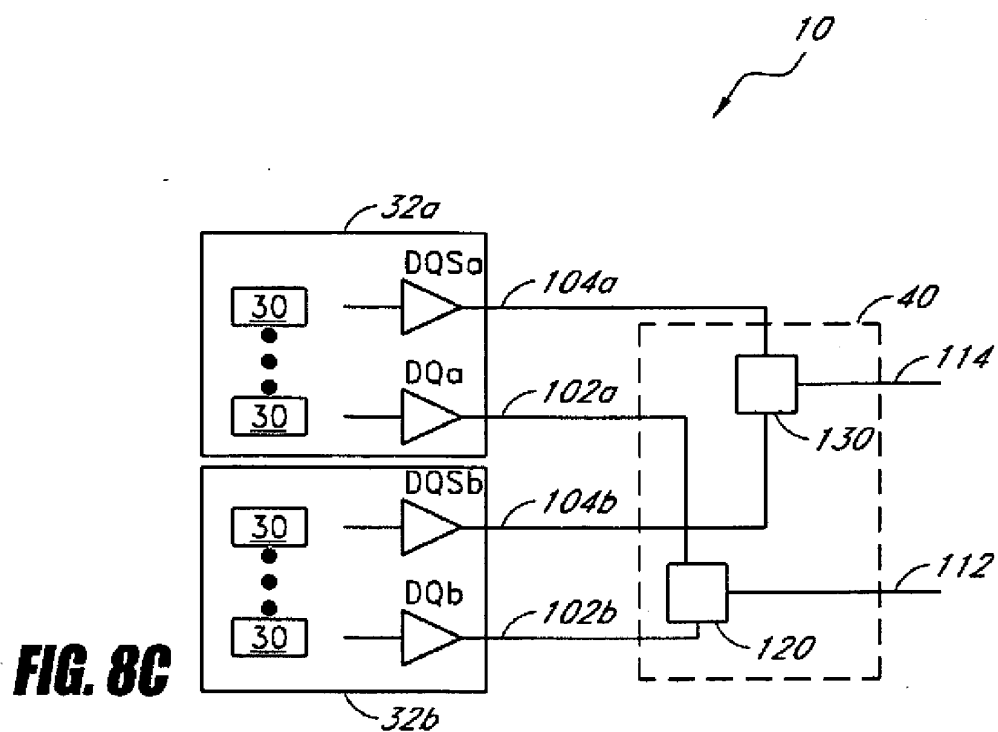


FIG. 7





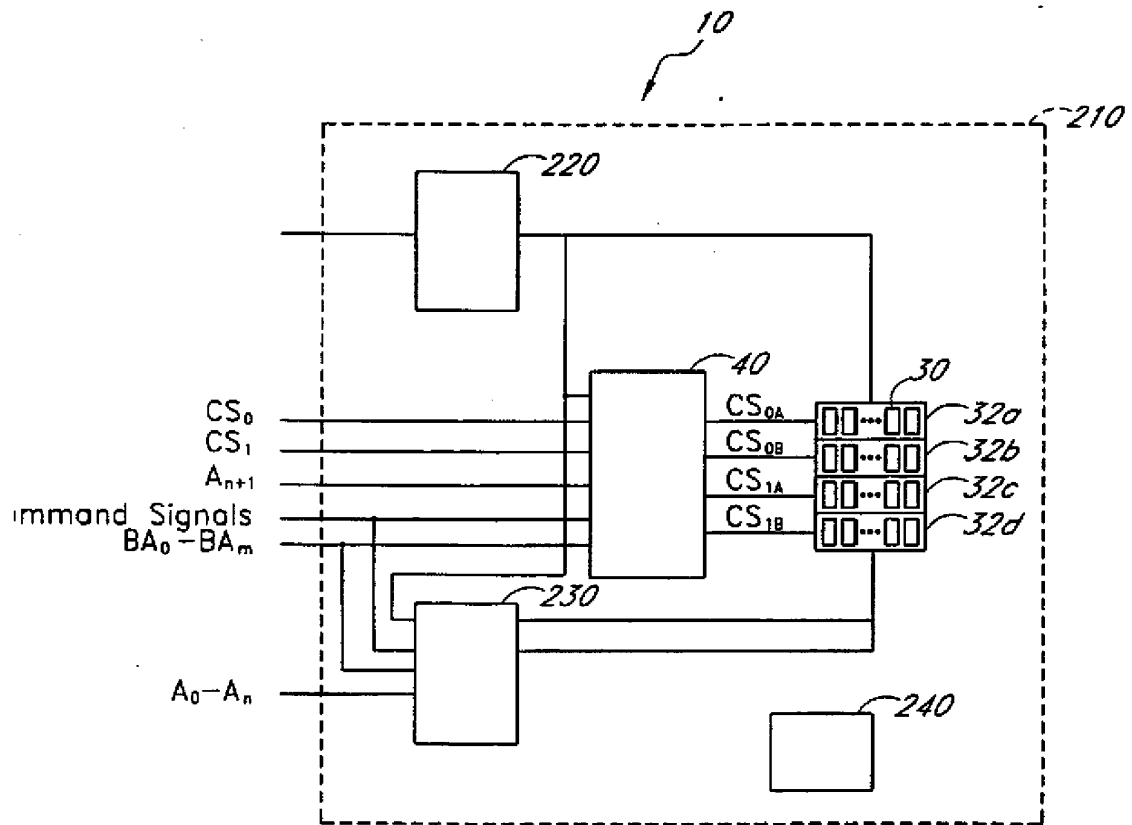


FIG. 9A

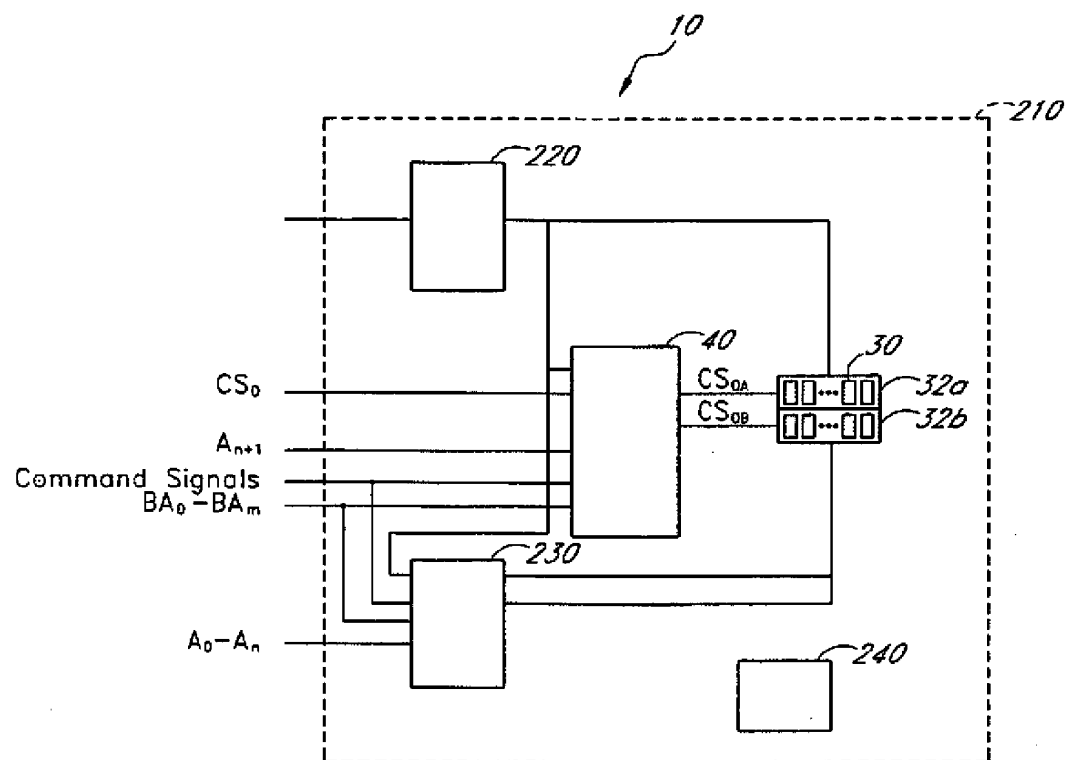


FIG. 9B

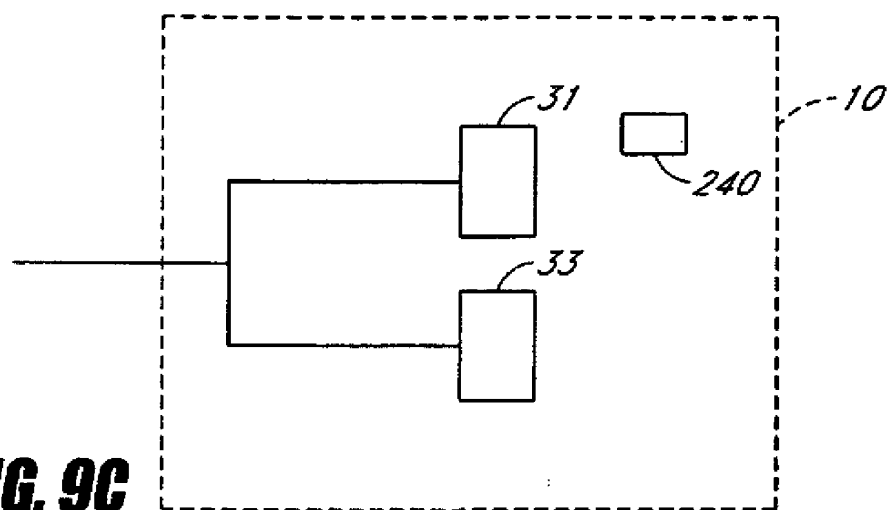
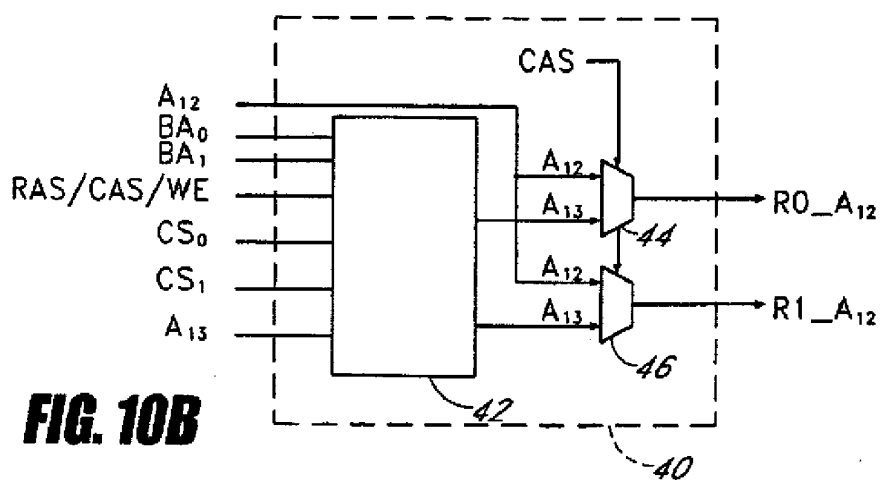
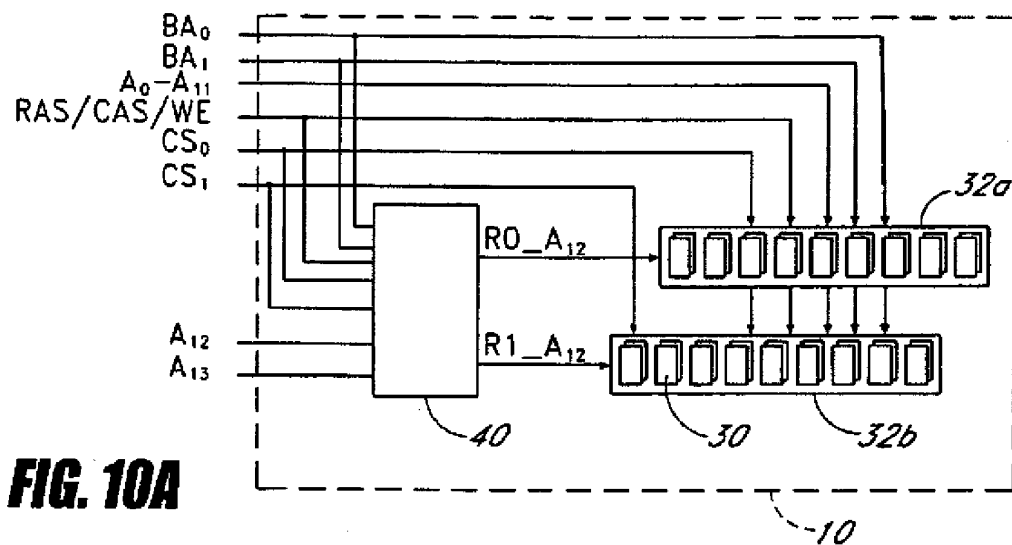
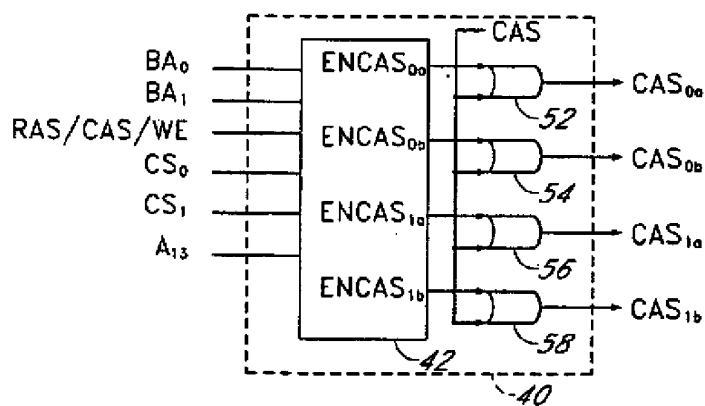
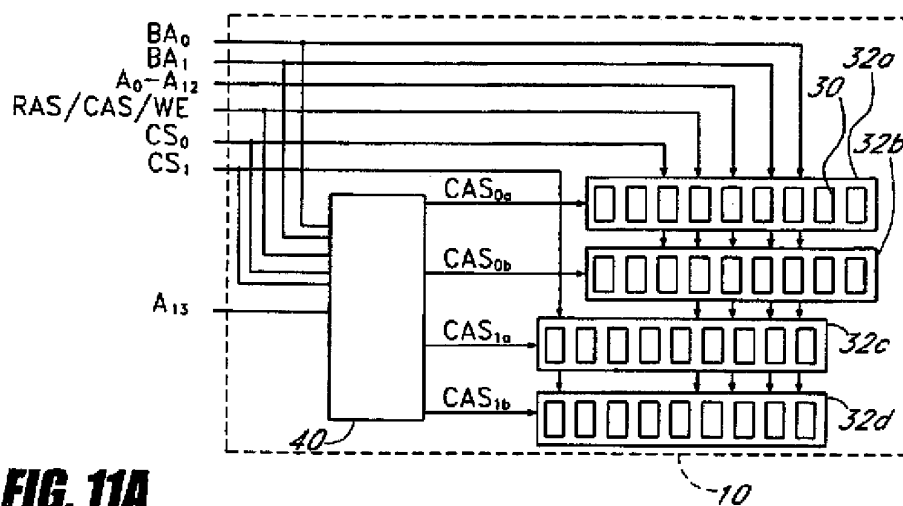


FIG. 9C





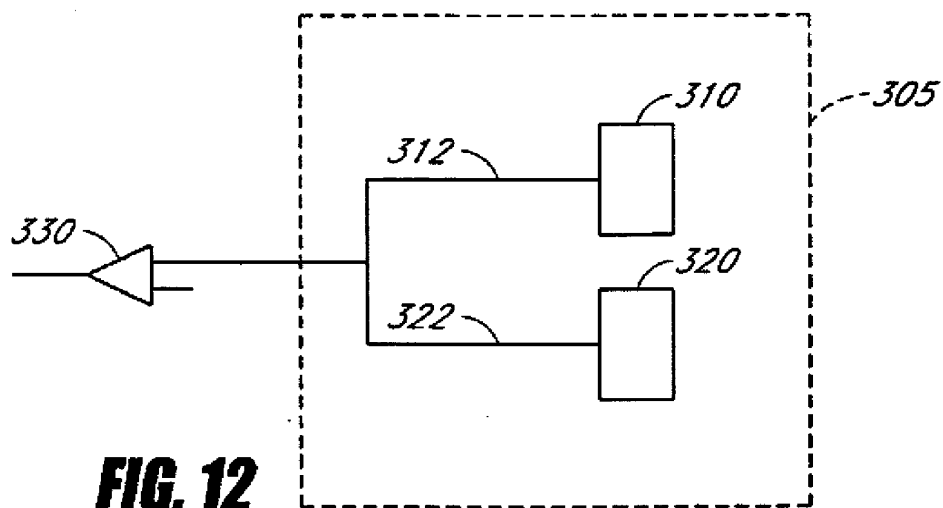


FIG. 12

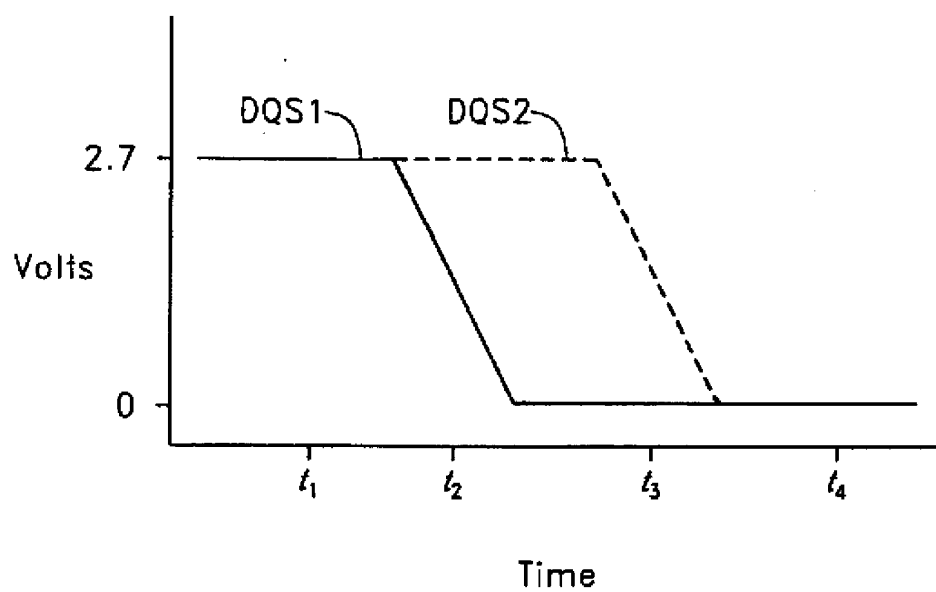


FIG. 13

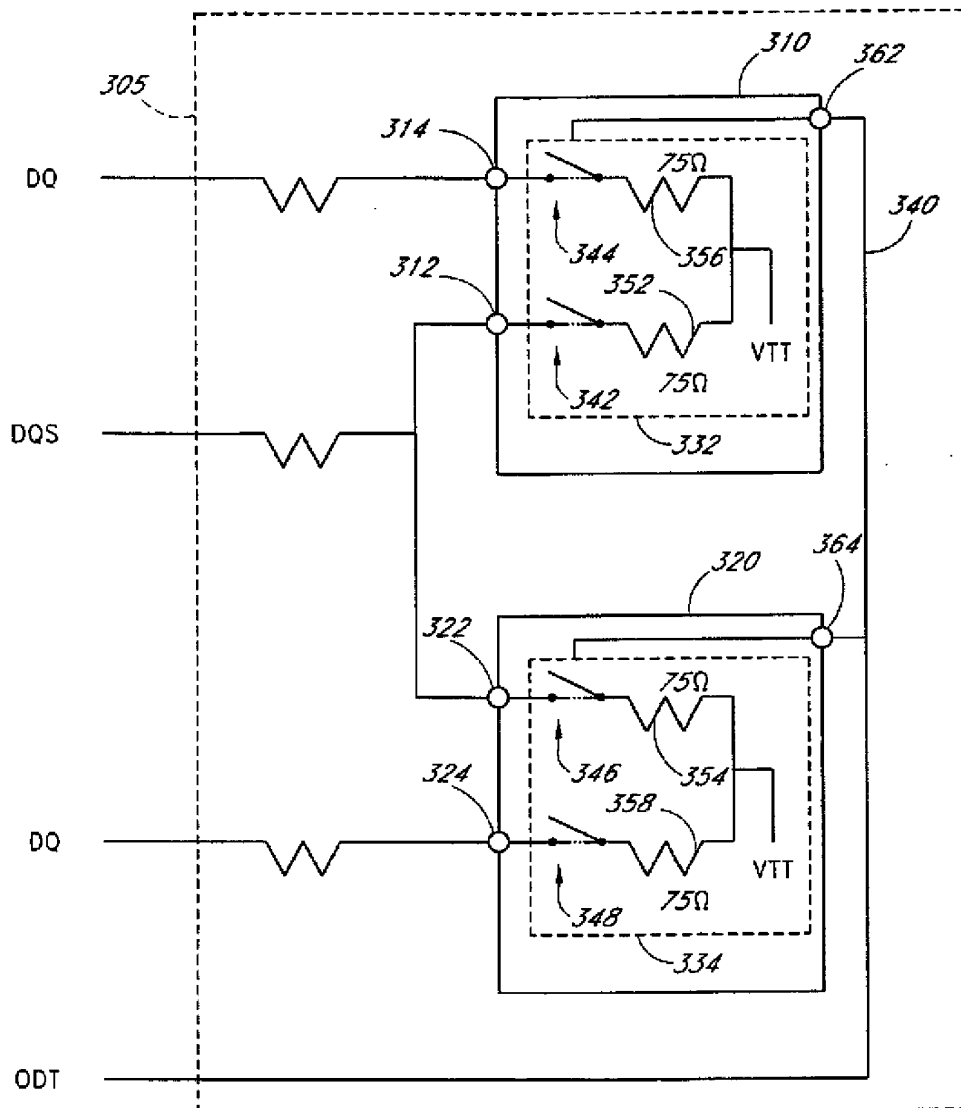


FIG. 14

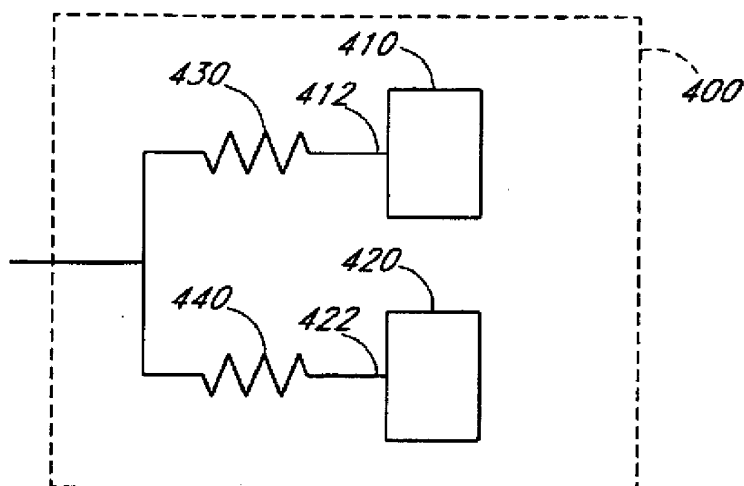


FIG. 15A

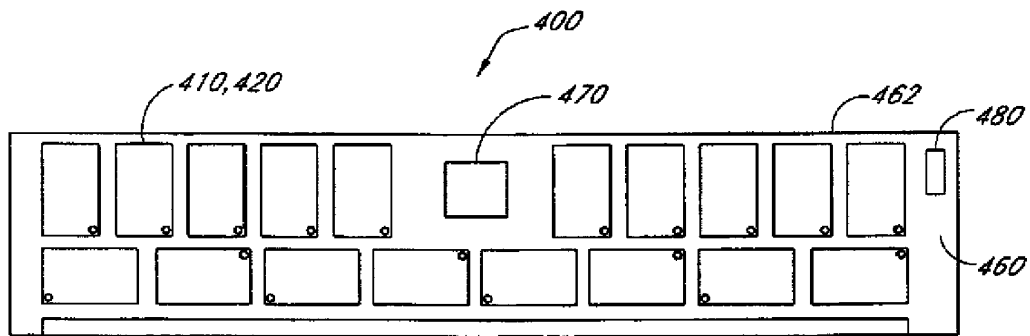


FIG. 16A

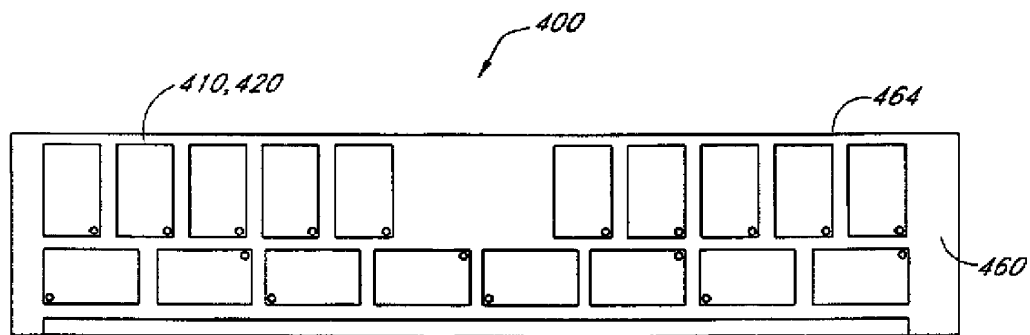


FIG. 16B

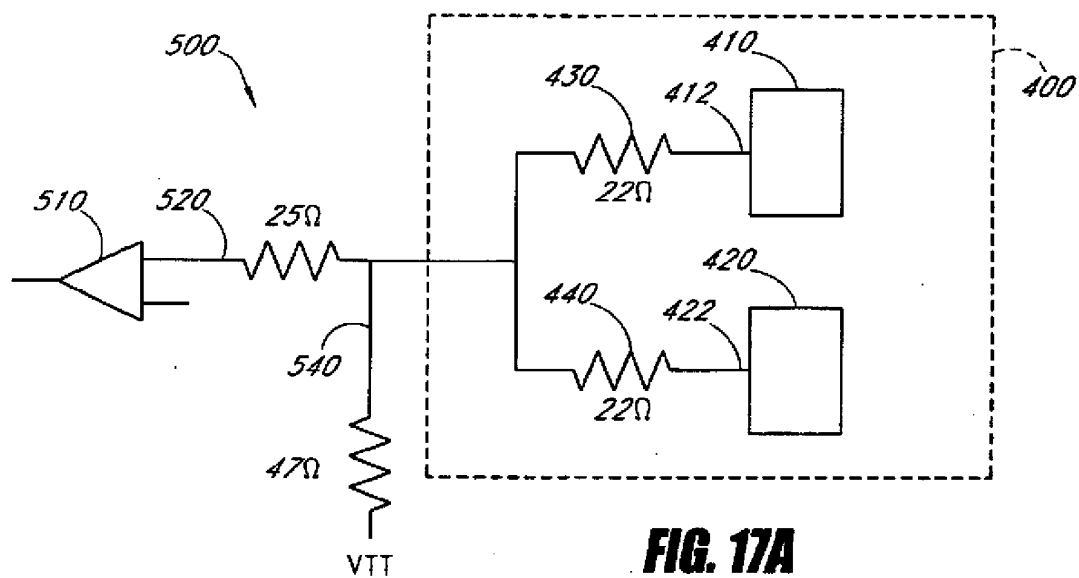


FIG. 17A

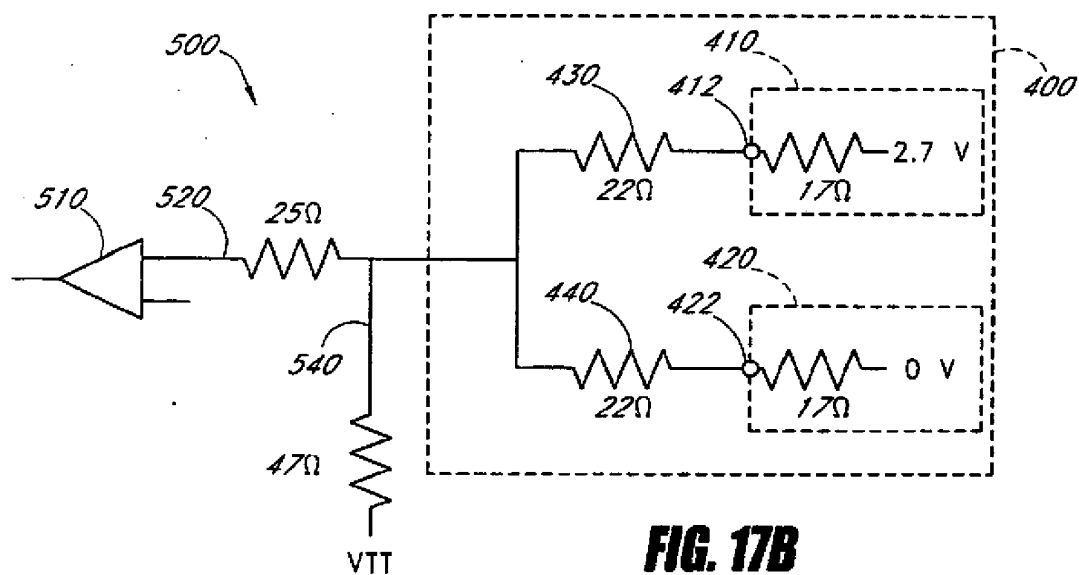


FIG. 17B

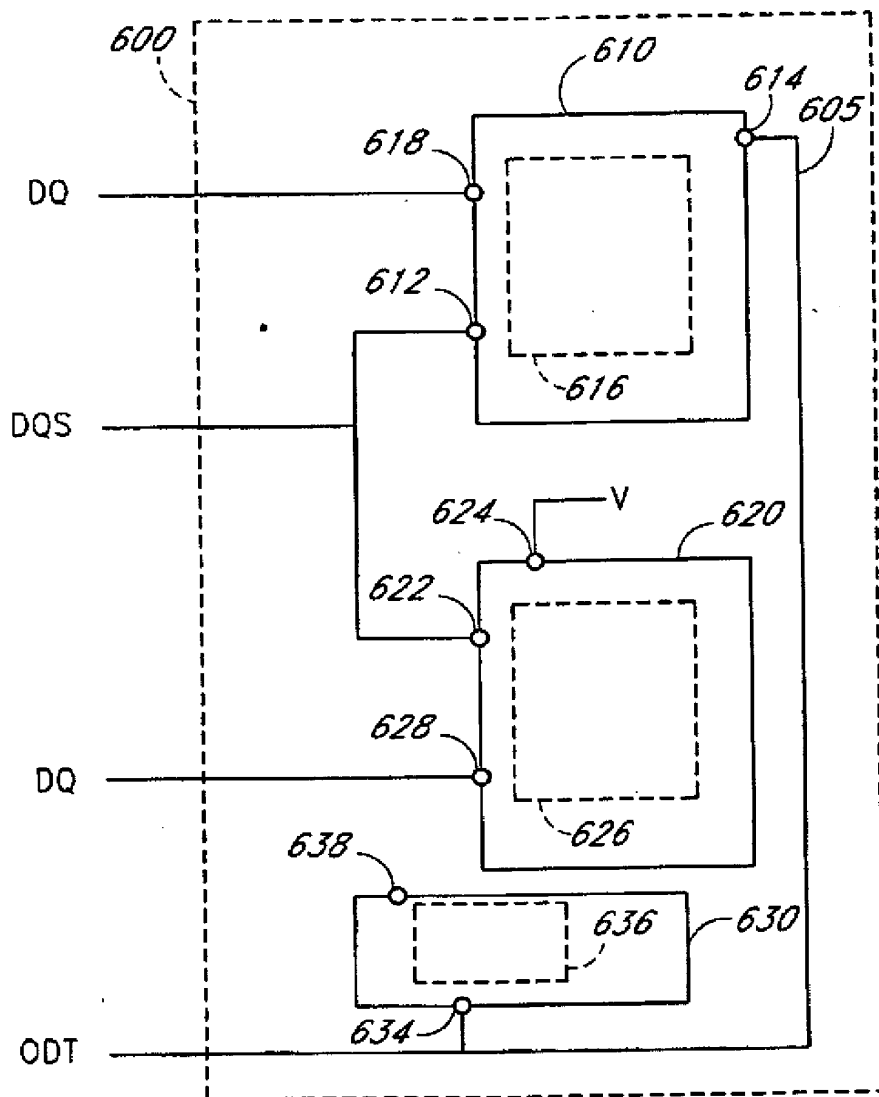


FIG. 18

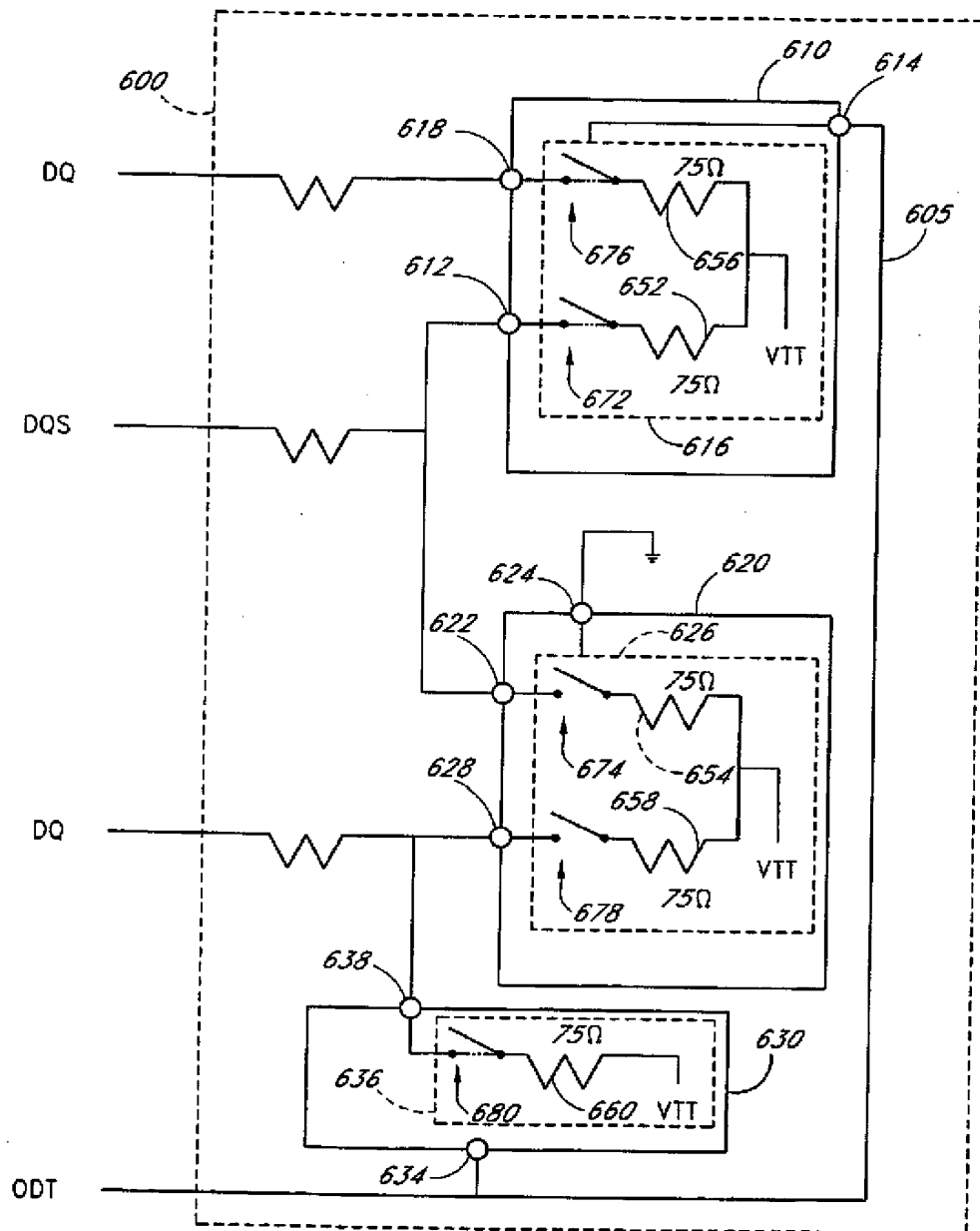


FIG. 19

MEMORY MODULE WITH A CIRCUIT PROVIDING LOAD ISOLATION AND MEMORY DOMAIN TRANSLATION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation-in-part of U.S. patent application Ser. No. 11/173,175, filed Jul. 1, 2005, which claims the benefit of U.S. Provisional Application No. 60/588,244, filed Jul. 15, 2004 and which is a continuation-in-part of U.S. patent application Ser. No. 11/075,395, filed Mar. 7, 2005, which claims the benefit of U.S. Provisional Application No. 60/550,668, filed Mar. 5, 2004, U.S. Provisional Application No. 60/575,595, filed May 28, 2004, and U.S. Provisional Application No. 60/590,038, filed Jul. 21, 2004. U.S. patent application Ser. Nos. 11/173,175 and 11/075,395, and U.S. Provisional Application Nos. 60/588,244, 60/550,668, 60/575,595, and 60/590,038 are each incorporated in its entirety by reference herein. The present application also claims the benefit of U.S. Provisional Application No. 60/645,087, filed Jan. 19, 2005, which is incorporated in its entirety by reference herein.

BACKGROUND OF THE INVENTION

[0002] 1. Field of the Invention

[0003] The present invention relates generally to memory modules of a computer system, and more specifically to devices and methods for improving the performance, the memory capacity, or both, of memory modules.

[0004] 2. Description of the Related Art

[0005] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the processor of the computer system. Memory modules typically have a memory configuration with a unique combination of rows, columns, and banks which result in a total memory capacity for the memory module.

[0006] For example, a 512-Megabyte memory module (termed a "512-MB" memory module, which actually has 2^{29} or 536,870,912 bytes of capacity) will typically utilize eight 512-Megabit DRAM devices (each identified as a "512-Mb" DRAM device, each actually having 2^{29} or 536,870,912 bits of capacity). The memory cells (or memory locations) of each 512-Mb DRAM device can be arranged in four banks, with each bank having an array of 2^{24} (or 16,777,216) memory locations arranged as 2^{13} rows and 2^{11} columns, and with each memory location having a width of 8 bits. Such DRAM devices with 64M 8-bit-wide memory locations (actually with four banks of 2^{27} or 134,217,728 one-bit memory cells arranged to provide a total of 2^{26} or 67,108,864 memory locations with 8 bits each) are identified as having a "64 Mb×8" or "64M×8-bit" configuration, or as having a depth of 64M and a bit width of 8. Furthermore, certain commercially-available 512-MB memory modules are termed to have a "64M×8-byte" configuration or a "64M×64-bit" configuration with a depth of 64M and a width of 8 bytes or 64 bits.

[0007] Similarly, a 1-Gigabyte memory module (termed a "1-GB" memory module, which actually has 2^{30} or 1,073,

741,824 bytes of capacity) can utilize eight 1-Gigabit DRAM devices (each identified as a "1-Gb" DRAM device, each actually having 2^{30} or 1,073,741,824 bits of capacity). The memory locations of each 1-Gb DRAM device can be arranged in four banks, with each bank having an array of memory locations with 2^{14} rows and 2^{11} columns, and with each memory location having a width of 8 bits. Such DRAM devices with 128M 8-bit-wide memory locations (actually with a total of 2^{27} or 134,217,728 memory locations with 8 bits each) are identified as having a "128 Mb×8" or "128M×8-bit" configuration, or as having a depth of 128M and a bit width of 8. Furthermore, certain commercially-available 1-GB memory modules are identified as having a "128M×8-byte" configuration or a "128M×64-bit" configuration with a depth of 128M and a width of 8 bytes or 64 bits.

[0008] The commercially-available 512-MB (64M×8-byte) memory modules and the 1-GB (128M×8-byte) memory modules described above are typically used in computer systems (e.g., personal computers) which perform graphics applications since such "×8" configurations are compatible with data mask capabilities often used in such graphics applications. Conversely, memory modules with "×4" configurations are typically used in computer systems such as servers which are not as graphics-intensive. Examples of such commercially available "×4" memory modules include, but are not limited to, 512-MB (128M×4-byte) memory modules comprising eight 512-Mb (128 Mb×4) memory devices.

[0009] The DRAM devices of a memory module are generally arranged as ranks or rows of memory, each rank of memory generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "×64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "×72" organization.

[0010] The memory capacity of a memory module increases with the number of memory devices. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks. For example, a memory module with four ranks has double the memory capacity of a memory module with two ranks and four times the memory capacity of a memory module with one rank. Rather than referring to the memory capacity of the memory module, in certain circumstances, the memory density of the memory module is referred to instead.

[0011] During operation, the ranks of a memory module are selected or activated by address and command signals that are received from the processor. Examples of such address and command signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support one-rank and two-rank memory modules. By only supporting one-rank and two-rank memory modules, the memory density that can be incorporated in each memory slot is limited.

[0012] Various aspects of the design of a memory module impose limitations on the size of the memory arrays of the memory module. Certain such aspects are particularly important for memory modules designed to operate at higher frequencies. Examples of such aspects include, but are not limited to, memory device (e.g., chip) densities, load fan-out, signal integrity, available rank selects, power dissipation, and thermal profiles.

SUMMARY OF THE INVENTION

[0013] In certain embodiments, a memory module comprises a plurality of memory devices and a circuit. Each memory device has a corresponding load. The circuit is electrically coupled to the plurality of memory devices and is configured to be electrically coupled to a memory controller of a computer system. The circuit selectively isolates one or more of the loads of the memory devices from the computer system. The circuit comprises logic which translates between a system memory domain of the computer system and a physical memory domain of the memory module.

[0014] In certain embodiments, a method of using a memory module with a computer system is provided. The method comprises providing a memory module having a plurality of memory devices and a circuit. The circuit is electrically coupled to the plurality of memory devices. The circuit is configured to be electrically coupled to a computer system. Each memory device has a corresponding load. The method further comprises electrically coupling the memory module to the computer system. The method further comprises activating the circuit to selectively isolate at least one of the loads of the memory devices from the computer system and to translate between a system memory domain of the computer system and a physical memory domain of the memory module.

[0015] In certain embodiments, a memory module is connectable to a computer system. The memory module comprises a first memory device having a first data signal line and a first data strobe signal line. The memory module further comprises a second memory device having a second data signal line and a second data strobe signal line. The memory module further comprises a common data signal line connectable to the computer system. The memory module further comprises a device electrically coupled to the first data signal line, to the second data signal line, and to the common data signal line. The device selectively electrically couples the first data signal line to the common data signal line and selectively electrically couples the second data signal line to the common data signal line. The device comprises logic which translates between a system memory domain of the computer system and a physical memory domain of the memory module.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 schematically illustrates an example memory module in accordance with certain embodiments described herein.

[0017] FIG. 2 schematically illustrates a circuit diagram of two memory devices of a conventional memory module.

[0018] FIGS. 3A and 3B schematically illustrate example memory modules having a circuit which selectively isolates one or both of the DQ data signal lines of the two memory devices from the computer system in accordance with certain embodiments described herein.

[0019] FIGS. 4A and 4B schematically illustrate example memory modules having a circuit which selectively isolates one or both of the DQ data signal lines of the two ranks of memory devices from the computer system in accordance with certain embodiments described herein.

[0020] FIGS. 5A-5D schematically illustrate example memory modules having a circuit comprising a logic element and one or more switches operatively coupled to the logic element in accordance with certain embodiments described herein.

[0021] FIG. 6A shows an exemplary timing diagram of a gapless read burst for a back-to-back adjacent read condition from one memory device.

[0022] FIG. 6B shows an exemplary timing diagram with an extra clock cycle between successive read commands issued to different memory devices for successive read accesses from different memory devices.

[0023] FIG. 7 shows an exemplary timing diagram in which the last data strobe of memory device "a" collides with the pre-amble time interval of the data strobe of memory device "b."

[0024] FIGS. 8A-8D schematically illustrate circuit diagrams of example memory modules comprising a circuit which multiplexes the DQS data strobe signal lines from one another in accordance with certain embodiments described herein.

[0025] FIG. 9A schematically illustrates an example memory module with four ranks of memory devices compatible with certain embodiments described herein.

[0026] FIG. 9B schematically illustrates an example memory module with two ranks of memory devices compatible with certain embodiments described herein.

[0027] FIG. 9C schematically illustrates another example memory module in accordance with certain embodiments described herein.

[0028] FIG. 10A schematically illustrates an exemplary memory module which doubles the rank density in accordance with certain embodiments described herein.

[0029] FIG. 10B schematically illustrates an exemplary circuit compatible with embodiments described herein.

[0030] FIG. 11A schematically illustrates an exemplary memory module which doubles number of ranks in accordance with certain embodiments described herein.

[0031] FIG. 11B schematically illustrates an exemplary circuit compatible with embodiments described herein.

[0032] FIG. 12 schematically illustrates an exemplary memory module in which a data strobe (DQS) pin of a first memory device is electrically connected to a DQS pin of a second memory device while both DQS pins are active.

[0033] FIG. 13 is an exemplary timing diagram of the voltages applied to the two DQS pins due to non-simultaneous switching.

[0034] FIG. 14 schematically illustrates another exemplary memory module in which a DQS pin of a first memory device is connected to a DQS pin of a second memory device.

[0035] FIG. 15 schematically illustrates an exemplary memory module in accordance with certain embodiments described herein.

[0036] FIGS. 16A and 16B schematically illustrate a first side and a second side, respectively, of a memory module

with eighteen 64M×4 bit, DDR-1 SDRAM FBGA memory devices on each side of a 184-pin glass-epoxy printed circuit board.

[0037] FIGS. 17A and 17B schematically illustrate an exemplary embodiment of a memory module in which a first resistor and a second resistor are used to reduce the current flow between the first DQS pin and the second DQS pin.

[0038] FIG. 18 schematically illustrates another exemplary memory module compatible with certain embodiments described herein.

[0039] FIG. 19 schematically illustrates a particular embodiment of the memory module schematically illustrated by FIG. 18.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

Load Isolation

[0040] FIG. 1 schematically illustrates an example memory module 10 compatible with certain embodiments described herein. The memory module 10 is connectable to a memory controller 20 of a computer system (not shown). The memory module 10 comprises a plurality of memory devices 30, each memory device 30 having a corresponding load. The memory module 10 further comprises a circuit 40 electrically coupled to the plurality of memory devices 30 and configured to be electrically coupled to the memory controller 20 of the computer system. The circuit 40 selectively isolates one or more of the loads of the memory devices from the computer system. The circuit 40 comprises logic which translates between a system memory domain of the computer system and a physical memory domain of the memory module 10.

[0041] As used herein, the term “load” is a broad term which includes, without limitation, electrical load, such as capacitive load, inductive load, or impedance load. As used herein, the term “isolation” is a broad term which includes, without limitation, electrical separation of one or more components from another component or from one another. As used herein, the term “circuit” is a broad term which includes, without limitation, an electrical component or device, or a configuration of electrical components or devices which are electrically or electromagnetically coupled together (e.g., integrated circuits), to perform specific functions.

[0042] Various types of memory modules 10 are compatible with embodiments described herein. For example, memory modules 10 having memory capacities of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, as well as other capacities, are compatible with embodiments described herein. Certain embodiments described herein are applicable to various frequencies including, but not limited to 100 MHz, 200 MHz, 400 MHz, 800 MHz, and above. In addition, memory modules 10 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the memory module 10 comprises a printed circuit board on which the memory devices 30 are mounted, a plurality of edge connectors configured to be electrically coupled to a corresponding plurality of contacts of a module slot of the computer system, and a plurality of electrical conduits

which electrically couple the memory devices 30 to the circuit 40 and which electrically couple the circuit 40 to the edge connectors. Furthermore, memory modules 10 compatible with embodiments described herein include, but are not limited to, single in-line memory modules (SIMMs), dual in-line memory modules (DIMMs), small-outline DIMMs (SO-DIMMs), unbuffered DIMMs (UDIMMs), registered DIMMs (RDIMMs), fully-buffered DIMM (FBDIMM), rank-buffered DIMMs (RBDIMMs), mini-DIMMs, and micro-DIMMs.

[0043] Memory devices 30 compatible with embodiments described herein include, but are not limited to, random-access memory (RAM), dynamic random-access memory (DRAM), synchronous DRAM (SDRAM), and double-data-rate DRAM (e.g., SDR, DDR-1, DDR-2, DDR-3). In addition, memory devices 30 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with embodiments described herein. Memory devices 30 compatible with embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μBGA), mini-BGA (mBGA), and chip-scale packaging (CSP). Memory devices 30 compatible with embodiments described herein are available from a number of sources, including but not limited to, Samsung Semiconductor, Inc. of San Jose, Calif., Infineon Technologies AG of San Jose, Calif., and Micron Technology, Inc. of Boise, Id. Persons skilled in the art can select appropriate memory devices 30 in accordance with certain embodiments described herein.

[0044] In certain embodiments, the plurality of memory devices 30 comprises a first number of memory devices 30. In certain such embodiments, the circuit 40 selectively isolates a second number of the memory devices 30 from the computer system, with the second number less than the first number.

[0045] In certain embodiments, the plurality of memory devices 30 are arranged in a first number of ranks. For example, in certain embodiments, the memory devices 30 are arranged in two ranks, as schematically illustrated by FIG. 1. In other embodiments, the memory devices 30 are arranged in four ranks. Other numbers of ranks of the memory devices 30 are also compatible with embodiments described herein.

[0046] In certain embodiments, the circuit comprises a logic element selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD). In certain embodiments, the logic element of the circuit 40 is a custom device. Sources of logic elements compatible with embodiments described herein include, but are not limited to, Lattice Semiconductor Corporation of Hillsboro, Oreg., Altera Corporation of San Jose, Calif., and Xilinx Incorporated of San Jose, Calif. In certain embodiments, the logic element comprises various discrete electrical elements, while in certain other embodiments, the logic element comprises one or more integrated circuits.

[0047] In certain embodiments, the circuit 40 further comprises one or more switches which are operatively coupled to the logic element to receive control signals from the logic

element. Examples of switches compatible with certain embodiments described herein include, but are not limited to, field-effect transistor (FET) switches, such as the SN74AUC1G66 single bilateral analog switch available from Texas Instruments, Inc. of Dallas, Tex.

[0048] FIG. 2 schematically illustrates a circuit diagram of two memory devices 30a, 30b of a conventional memory module showing the interconnections between the DQ data signal lines 102a, 102b of the memory devices 30a, 30b and the DQS data strobe signal lines 104a, 104b of the memory devices 30a, 30b. Each of the memory devices 30a, 30b has a plurality of DQ data signal lines and a plurality of DQS data strobe signal lines, however, for simplicity, FIG. 2 only illustrates a single DQ data signal line and a single DQS data strobe signal line for each memory device 30a, 30b. The DQ data signal lines 102a, 102b and the DQS data strobe signal lines 104a, 104b are typically conductive traces etched on the printed circuit board of the memory module. As shown in FIG. 2, each of the memory devices 30a, 30b has their DQ data signal lines 102a, 102b electrically coupled to a common DQ line 112 and their DQS data strobe signal lines 104a, 104b electrically coupled to a common DQS line 114. The common DQ line 112 and the common DQS line 114 are electrically coupled to the memory controller 20 of the computer system. Thus, the computer system is exposed to the loads of both memory devices 30a, 30b concurrently.

[0049] In certain embodiments, the circuit 40 selectively isolates the loads of at least some of the memory devices 30 from the computer system. The circuit 40 of certain embodiments is configured to present a significantly reduced load to the computer system. In certain embodiments in which the memory devices 30 are arranged in a plurality of ranks, the circuit 40 selectively isolates the loads of some (e.g., one or more) of the ranks of the memory module 10 from the computer system. In certain other embodiments, the circuit 40 selectively isolates the loads of all of the ranks of the memory module 10 from the computer system. For example, when a memory module 10 is not being accessed by the computer system, the capacitive load on the memory controller 20 of the computer system by the memory module 10 can be substantially reduced to the capacitive load of the circuit 40 of the memory module 10.

[0050] As schematically illustrated by FIGS. 3A and 3B, an example memory module 10 compatible with certain embodiments described herein comprises a circuit 40 which selectively isolates one or both of the DQ data signal lines 102a, 102b of the two memory devices 30a, 30b from the common DQ data signal line 112 coupled to the computer system. Thus, the circuit 40 selectively allows a DQ data signal to be transmitted from the memory controller 20 of the computer system to one or both of the DQ data signal lines 102a, 102b. In addition, the circuit 40 selectively allows one of a first DQ data signal from the DQ data signal line 102a of the first memory device 30a or a second DQ data signal from the DQ data signal line 102b of the second memory device 30b to be transmitted to the memory controller 20 via the common DQ data signal line 112 (see, e.g., triangles on the DQ and DQS lines of FIGS. 3A and 3B which point towards the memory controller). While various figures of the present application denote read operations by use of DQ and DQS lines which have triangles pointing towards the memory controller, certain embodiments described herein are also compatible with write operations

(e.g., as would be denoted by triangles on the DQ or DQS lines pointing away from the memory controller).

[0051] For example, in certain embodiments, the circuit 40 comprises a pair of switches 120a, 120b on the DQ data signal lines 102a, 102b as schematically illustrated by FIG. 3A. Each switch 120a, 120b is selectively actuated to selectively electrically couple the DQ data signal line 102a to the common DQ signal line 112, the DQ data signal line 102b to the common DQ signal line 112, or both DQ data signal lines 102a, 102b to the common DQ signal line 112. In certain other embodiments, the circuit 40 comprises a switch 120 electrically coupled to both of the DQ data signal lines 102a, 102b, as schematically illustrated by FIG. 3B. The switch 120 is selectively actuated to selectively electrically couple the DQ data signal line 10a to the common DQ signal line 112, the DQ data signal line 102b to the common DQ signal line 112, or both DQ signal lines 102a, 102b to the common DQ signal line 112. Circuits 40 having other configurations of switches are also compatible with embodiments described herein. While each of the memory devices 30a, 30b has a plurality of DQ data signal lines and a plurality of DQS data strobe signal lines, FIGS. 3A and 3B only illustrate a single DQ data signal line and a single DQS data strobe signal line for each memory device 30a, 30b for simplicity. The configurations schematically illustrated by FIGS. 3A and 3B can be applied to all of the DQ data signal lines and DQS data strobe signal lines of the memory module 10.

[0052] In certain embodiments, the circuit 40 selectively isolates the loads of ranks of memory devices 30 from the computer system. As schematically illustrated in FIGS. 4A and 4B, example memory modules 10 compatible with certain embodiments described herein comprise a first number of memory devices 30 arranged in a first number of ranks 32. The memory modules 10 of FIGS. 4A and 4B comprises two ranks 32a, 32b, with each rank 32a, 32b having a corresponding set of DQ data signal lines and a corresponding set of DQS data strobe lines. Other numbers of ranks (e.g., four ranks) of memory devices 30 of the memory module 10 are also compatible with certain embodiments described herein. For simplicity, FIGS. 4A and 4B illustrate only a single DQ data signal line and a single DQS data strobe signal line from each rank 32.

[0053] The circuit 40 of FIG. 4A selectively isolates one or more of the DQ data signal lines 102a, 102b of the two ranks 32a, 32b from the computer system. Thus, the circuit 40 selectively allows a DQ data signal to be transmitted from the memory controller 20 of the computer system to the memory devices 30 of one or both of the ranks 32a, 32b via the DQ data signal lines 102a, 102b. In addition, the circuit 40 selectively allows one of a first DQ data signal from the DQ data signal line 102a of the first rank 32a and a second DQ data signal from the DQ data signal line 102b of the second rank 32b to be transmitted to the memory controller 20 via the common DQ data signal line 112. For example, in certain embodiments, the circuit 40 comprises a pair of switches 120a, 120b on the DQ data signal lines 102a, 102b as schematically illustrated by FIG. 4A. Each switch 120a, 120b is selectively actuated to selectively electrically couple the DQ data signal line 102a to the common DQ data signal line 112, the DQ data signal line 102b to the common DQ data signal line 112, or both DQ data signal lines 102a, 102b to the common DQ data signal line 112. In certain other

embodiments, the circuit 40 comprises a switch 120 electrically coupled to both of the DQ data signal lines 102a, 102b, as schematically illustrated by FIG. 4B. The switch 120 is selectively actuated to selectively electrically couple the DQ data signal line 102a to the common DQ data signal line 112, the DQ data signal line 102b to the common DQ data signal line 112, or both DQ data signal lines 102a, 102b to the common DQ data signal line 112. Circuits 40 having other configurations of switches are also compatible with embodiments described herein.

[0054] In the example embodiments schematically illustrated by FIGS. 3A, 3B, 4A, and 4B, the circuit 40 comprises a logic element which is integral with and comprises the switches 120 which are coupled to the DQ data signal lines and the DQS data strobe signal lines. In certain such embodiments, each switch 120 comprises a data path multiplexer/demultiplexer. In certain other embodiments, the circuit 40 comprises a logic element 122 which is a separate component operatively coupled to the switches 120, as schematically illustrated by FIGS. 5A-5D. The one or more switches 120 are operatively coupled to the logic element 122 to receive control signals from the logic element 122 and to selectively electrically couple one or more data signal lines to a common data signal line. Example switches compatible with embodiments described herein include, but are not limited to field-effect transistor (FET) switches, such as the SN74AUC1G66 single bilateral analog switch available from Texas Instruments, Inc. of Dallas, Tex. Example logic elements 122 compatible with certain embodiments described herein include, but are not limited to, programmable-logic devices (PLD), application-specific integrated circuits (ASIC), field-programmable gate arrays (FPGA), custom-designed semiconductor devices, and complex programmable-logic devices (CPLD). Example logic elements 122 are available from Lattice Semiconductor Corporation of Hillsboro, Oreg., Altera Corporation of San Jose, Calif., and Xilinx Incorporated of San Jose, Calif.

[0055] In certain embodiments, the load isolation provided by the circuit 40 advantageously allows the memory module 10 to present a reduced load (e.g., electrical load, such as capacitive load, inductive load, or impedance load) to the computer system by selectively switching between the two ranks of memory devices 30 to which it is coupled. This feature is used in certain embodiments in which the load of the memory module 10 may otherwise limit the number of ranks or the number of memory devices per memory module. In certain embodiments, the memory module 10 operates as having a data path rank buffer which advantageously

isolates the ranks of memory devices 30 of the memory module 10 from one another, from the ranks on other memory modules, and from the computer system. This data path rank buffer of certain embodiments advantageously provides DQ-DQS paths for each rank or sets of ranks of memory devices which are separate from one another, or which are separate from the memory controller of the computer system. In certain embodiments, the load isolation advantageously diminishes the effects of capacitive loading, jitter and other sources of noise. In certain embodiments, the load isolation advantageously simplifies various other aspects of operation of the memory module 10, including but not limited to, setup-and-hold time, clock skew, package skew, and process, temperature, voltage, and transmission line variations.

[0056] For certain memory module applications that utilize multiple ranks of memory, increased load on the memory bus can degrade speed performance. In certain embodiments described herein, selectively isolating the loads of the ranks of memory devices 30 advantageously decreases the load on the computer system, thereby allowing the computer system (e.g., server) to run faster with improved signal integrity. In certain embodiments, load isolation advantageously provides system memory with reduced electrical loading, thereby improving the electrical topology to the memory controller 20. In certain such embodiments, the speed and the memory density of the computer system are advantageously increased without sacrificing one for the other.

[0057] In certain embodiments, load isolation advantageously increases the size of the memory array supported by the memory controller 20 of the computer system. The larger memory array has an increased number of memory devices 30 and ranks of memory devices 30 of the memory module 10, with a corresponding increased number of chip selects. Certain embodiments described herein advantageously provide more system memory using fewer chip selects, thereby avoiding the chip select limitation of the memory controller.

[0058] An exemplary section of Verilog code corresponding to logic compatible with a circuit 40 which provides load isolation is listed below in Example 1. The exemplary code of Example 1 corresponds to a circuit 40 comprising six FET switches for providing load isolation to DQ and DQS lines.

EXAMPLE 1

[0059]

```
//===== declarations
reg      rasN_R, casN_R, weN_R;
wire     actv_cmd_R, pch_cmd_R, pch_all_cmd_R, ap_xfr_cmd_R_R;
wire     xfr_cmd_R, mrs_cmd_R, rd_cmd_R;
//----- DDR 2 FET
reg      brs0N_R; // registered chip sel
reg      brs1N_R; // registered chip sel
reg      brs2N_R; // registered chip sel
reg      brs3N_R; // registered chip sel
wire     sel;
wire     sel_01;
wire     sel_23;
wire     rd_R1;
wire     wr_cmd_R, wr_R1;
```

-continued

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reg          rd_R2,rd_R3,rd_R4,rd_R5;
reg          wr_R2,wr_R3,wr_R4,wr_R5;
reg          enfet1,enfet2,enfet3,enfet4,enfet5,enfet6;
wire         wr_01_R1,wr_23_R1;
reg          wr_01_R2,wr_01_R3,wr_01_R4;
reg          wr_23_R2,wr_23_R3,wr_23_R4;
wire         rodt0_a,rodt0_b;

//----- logic
always @(posedge clk_in)
begin
    brs0N_R <= brs0_in_N; // cs0
    brs1N_R <= brs1_in_N; // cs1
    brs2N_R <= brs2_in_N; // cs2
    brs3N_R <= brs3_in_N; // cs3
    rasN_R <= brras_in_N;
    casN_R <= brcas_in_N;
    weN_R <= bwe_in_N;
end

assign sel = ~brs0N_R | ~brs1N_R | ~brs2N_R | ~brs3N_R;
assign sel_01 = ~brs0N_R | ~brs1N_R;
assign sel_23 = ~brs2N_R | ~brs3N_R;
assign actv_cmd_R = !rasN_R & !casN_R & !weN_R; // activate cmd
assign pch_cmd_R = !rasN_R & !casN_R & !weN_R; // pchg cmd
assign xfr_cmd_R = rasN_R & !casN_R; // xfr cmd
assign mrs_cmd_R = !rasN_R & !casN_R & !weN_R; // mdr reg set cmd
assign rd_cmd_R = rasN_R & !casN_R & !weN_R; // read cmd
assign wr_cmd_R = rasN_R & !casN_R & !weN_R; // write cmd

//-----
assign rd_R1 = sel & rd_cmd_R; // rd cmd cyc 1
assign wr_R1 = sel & wr_cmd_R; // wr cmd cyc 1

//-----
always @(posedge clk_in)
begin
    rd_R2 <= rd_R1;
    rd_R3 <= rd_R2;
    rd_R4 <= rd_R3;
    rd_R5 <= rd_R4;
    rd0_o_R6 <= rd0_o_R5;
    wr_R2 <= wr_R1;
    wr_R3 <= wr_R2;
    wr_R4 <= wr_R3;
    wr_R5 <= wr_R4;
end

//-----
assign wr_01_R1 = sel_01 & wr_cmd_R; // wr cmd cyc 1 for cs 2 & cs3
assign wr_23_R1 = sel_23 & wr_cmd_R; // wr cmd cyc 1 for cs 2 & cs3
always @(posedge clk_in)
begin
    wr_01_R2 <= wr_01_R1;
    wr_01_R3 <= wr_01_R2;
    wr_01_R4 <= wr_01_R3;
    wr_23_R2 <= wr_23_R1;
    wr_23_R3 <= wr_23_R2;
    wr_23_R4 <= wr_23_R3;
end

assign rodt0_ab = (rodt0) // odt cmd from sys
| (wr_23_R1) // wr 1st cyc to other mks (assume single dimm per channel)
| (wr_23_R2) // wr 2nd cyc to other mks (assume single dimm per channel)
| (wr_23_R3) // wr 3rd cyc to other mks (assume single dimm per channel)
;

assign rodt1_ab = (rodt1) // odt cmd from sys
| (wr_01_R1) // wr 1st cyc to other mks (assume single dimm per channel)
| (wr_01_R2) // wr 2nd cyc to other mks (assume single dimm per channel)
| (wr_01_R3) // wr 3rd cyc to other mks (assume single dimm per channel)
;

//-----
always @(posedge clk_in)
begin
    if (
        | (rd_R2) // pre-am rd
        | (rd_R3) // 1st cyc of rd brst (cl3)
        | (rd_R4) // 2nd cyc of rd brst (cl3)
        | (wr_R1) // pre-am wr
        | (wr_R2) // wr brst 1st cyc
        | (wr_R3) // wr brst 2nd cyc
    ) begin

```

-continued

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        enfet1 <= 'b1;           // enable fet
        enfet2 <= 'b1;           // enable fet
        enfet3 <= 'b1;           // enable fet
        enfet4 <= 'b1;           // enable fet
        enfet5 <= 'b1;           // enable fet
        enfet6 <= 'b1;           // enable fet
    end
else
    begin
        enfet1 <= 'b0;           // disable fet
        enfet2 <= 'b0;           // disable fet
        enfet3 <= 'b0;           // disable fet
        enfet4 <= 'b0;           // disable fet
        enfet5 <= 'b0;           // disable fet
        enfet6 <= 'b0;           // disable fet
    end
end
end

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Back-to-Back Adjacent Read Commands

[0060] Due to their source synchronous nature, DDR SDRAM (e.g., DDR1, DDR2, DDR3) memory devices operate with a data transfer protocol which surrounds each burst of data strobes with a pre-ambble time interval and a post-ambble time interval. The pre-ambble time interval provides a timing window for the receiving memory device to enable its data capture circuitry when a known valid level is present on the strobe signal to avoid false triggers of the memory device's capture circuit. The post-ambble time interval provides extra time after the last strobe for this data capture to facilitate good signal integrity. In certain embodiments, when the computer system accesses two consecutive bursts of data from the same memory device, termed herein as a "back-to-back adjacent read," the post-ambble time interval of the first read command and the pre-ambble time interval of the second read command are skipped by design protocol to increase read efficiency. FIG. 6A shows an exemplary timing diagram of this "gapless" read burst for a back-to-back adjacent read condition from one memory device.

[0061] In certain embodiments, when the second read command accesses data from a different memory device than does the first read command, there is at least one time interval (e.g., clock cycle) inserted between the data strobes of the two memory devices. This inserted time interval allows both read data bursts to occur without the post-ambble time interval of the first read data burst colliding or otherwise interfering with the pre-ambble time interval of the second read data burst. In certain embodiments, the memory controller of the computer system inserts an extra clock cycle between successive read commands issued to different memory devices, as shown in the exemplary timing diagram of FIG. 6B for successive read accesses from different memory devices.

[0062] In typical computer systems, the memory controller is informed of the memory boundaries between the ranks of memory of the memory module prior to issuing read commands to the memory module. Such memory controllers can insert wait time intervals or clock cycles to avoid collisions or interference between back-to-back adjacent read commands which cross memory device boundaries, which are referred to herein as "BBARX."

[0063] In certain embodiments described herein in which the number of ranks 32 of the memory module 10 is doubled or quadrupled, the circuit 40 generates a set of output address and command signals so that the selection decoding is transparent to the computer system. However, in certain such embodiments, there are memory device boundaries of which the computer system is unaware, so there are occasions in which BBARX occurs without the cognizance of the memory controller 20 of the computer system. As shown in FIG. 7, the last data strobe of memory device "a" collides with the pre-ambble time interval of the data strobe of memory device "b," resulting in a "collision window."

[0064] FIGS. 8A-8D schematically illustrate circuit diagrams of example memory modules 10 comprising a circuit 40 which multiplexes the DQS data strobe signal lines 104a, 104b of two ranks 32a, 32b from one another in accordance with certain embodiments described herein. While the DQS data strobe signal lines 104a, 104b of FIGS. 8A-8D correspond to two ranks 32a, 32b of memory devices 30, in certain other embodiments, the circuit 40 multiplexes the DQS data strobe signal lines 104a, 104b corresponding to two individual memory devices 30a, 30b.

[0065] FIG. 8A schematically illustrates a circuit diagram of an exemplary memory module 10 comprising a circuit 40 in accordance with certain embodiments described herein. In certain embodiments, BBARX collisions are avoided by a mechanism which electrically isolates the DQS data strobe signal lines 104a, 104b from one another during the transition from the first read data burst of one rank 32a of memory devices 30 to the second read data burst of another rank 32b of memory devices 30.

[0066] In certain embodiments, as schematically illustrated by FIG. 8A, the circuit 40 comprises a first switch 130a electrically coupled to a first DQS data strobe signal line 104a of a first rank 32a of memory devices 30 and a second switch 130b electrically coupled to a second DQS data strobe signal line 104b of a second rank 32b of memory devices 30. In certain embodiments, the time for switching the first switch 130a and the second switch 130b is between the two read data bursts (e.g., after the last DQS data strobe of the read data burst of the first rank 32a and before the first DQS data strobe of the read data burst of the second rank 32b). During the read data burst for the first rank 32a, the first switch 130a is enabled. After the last DQS data strobe

of the first rank 32a and before the first DQS data strobe of the second rank 32b, the first switch 130a is disabled and the second switch 130b is enabled.

[0067] As shown in FIG. 8A, each of the ranks 32a, 32b otherwise involved in a BBARX collision have their DQS data strobe signal lines 104a, 104b selectively electrically coupled to the common DQS line 114 through the circuit 40. The circuit 40 of certain embodiments multiplexes the DQS data strobe signal lines 104a, 104b of the two ranks 32a, 32b of memory devices 30 from one another to avoid a BBARX collision.

[0068] In certain embodiments, as schematically illustrated by FIG. 8B, the circuit 40 comprises a switch 130 which multiplexes the DQS data strobe signal lines 104a, 104b from one another. For example, the circuit 40 receives a DQS data strobe signal from the common DQS data strobe signal line 114 and selectively transmits the DQS data strobe signal to the first DQS data strobe signal line 104a, to the second DQS data strobe signal line 104b, or to both DQS data strobe signal lines 104a, 104b. As another example, the circuit 40 receives a first DQS data strobe signal from the first rank 32a of memory devices 30 and a second DQS data strobe signal from a second rank 32b of memory devices 30 and selectively switches one of the first and second DQS data strobe signals to the common DQS data strobe signal line 114.

[0069] In certain embodiments, the circuit 40 also provides the load isolation described above in reference to FIGS. 1-5. For example, as schematically illustrated by FIG. 8C, the circuit 40 comprises both the switch 120 for the DQ data signal lines 102a, 102b and the switch 130 for the DQS data strobe signal lines 104a, 104b. While in certain embodiments, the switches 130 are integral with a logic element of the circuit 40, in certain other embodiments, the switches 130 are separate components which are operatively coupled to a logic element 122 of the circuit 40, as schematically illustrated by FIG. 8D. In certain such embodiments, the control and timing of the switch 130 is performed by the circuit 40 which is resident on the memory module 10. Example switches 130 compatible with embodiments described herein include, but are not limited to field-effect transistor (FET) switches, such as the SN74AUC1G66 single bilateral analog switch available from Texas Instruments, Inc. of Dallas, Tex., and multiplexers, such as the SN74AUC2G53 2:1 analog multiplexer/demultiplexer available from Texas Instruments, Inc. of Dallas, Tex.

[0070] The circuit 40 of certain embodiments controls the isolation of the DQS data strobe signal lines 104a, 104b by monitoring commands received by the memory module 10 from the computer system and producing "windows" of operation whereby the appropriate switches 130 are activated or deactivated to enable and disable the DQS data strobe signal lines 104a, 104b to mitigate BBARX collisions. In certain other embodiments, the circuit 40 monitors the commands received by the memory module 10 from the computer system and selectively activates or deactivates the switches 120 to enable and disable the DQ data signal lines 102a, 102b to reduce the load of the memory module 10 on the computer system. In still other embodiments, the circuit 40 performs both of these functions together.

Command Signal Translation

[0071] Most high-density memory modules are currently built with 512-Megabit ("512-Mb") memory devices wherein each memory device has a 64Mx8-bit configuration. For example, a 1-Gigabyte ("1-GB") memory module with error checking capabilities can be fabricated using eighteen such 512-Mb memory devices. Alternatively, it can be economically advantageous to fabricate a 1-GB memory module using lower-density memory devices and doubling the number of memory devices used to produce the desired word width. For example, by fabricating a 1-GB memory module using thirty-six 256-Mb memory devices with 64Mx4-bit configuration, the cost of the resulting 1-GB memory module can be reduced since the unit cost of each 256-Mb memory device is typically lower than one-half the unit cost of each 512-Mb memory device. The cost savings can be significant, even though twice as many 256-Mb memory devices are used in place of the 512-Mb memory devices. For example, by using pairs of 512-Mb memory devices rather than single 1-Gb memory devices, certain embodiments described herein reduce the cost of the memory module by a factor of up to approximately five.

[0072] Market pricing factors for DRAM devices are such that higher-density DRAM devices (e.g., 1-Gb DRAM devices) are much more than twice the price of lower-density DRAM devices (e.g., 512-Mb DRAM devices). In other words, the price per bit ratio of the higher-density DRAM devices is greater than that of the lower-density DRAM devices. This pricing difference often lasts for months or even years after the introduction of the higher-density DRAM devices, until volume production factors reduce the costs of the newer higher-density DRAM devices. Thus, when the cost of a higher-density DRAM device is more than the cost of two lower-density DRAM devices, there is an economic incentive for utilizing pairs of the lower-density DRAM devices to replace individual higher-density DRAM devices.

[0073] FIG. 9A schematically illustrates an exemplary memory module 10 compatible with certain embodiments described herein. The memory module 10 is connectable to a memory controller 20 of a computer system (not shown). The memory module 10 comprises a printed circuit board 210 and a plurality of memory devices 30 coupled to the printed circuit board 210. The plurality of memory devices 30 has a first number of memory devices 30. The memory module 10 further comprises a circuit 40 coupled to the printed circuit board 210. The circuit 40 receives a set of input address and command signals from the computer system. The set of input address and command signals correspond to a second number of memory devices 30 smaller than the first number of memory devices 30. The circuit 40 generates a set of output address and command signals in response to the set of input address and command signals. The set of output address and command signals corresponds to the first number of memory devices 30.

[0074] In certain embodiments, as schematically illustrated in FIG. 9A, the memory module 10 further comprises a phase-lock loop device 220 coupled to the printed circuit board 210 and a register 230 coupled to the printed circuit board 210. In certain embodiments, the phase-lock loop device 220 and the register 230 are each mounted on the printed circuit board 210. In response to signals received from the computer system, the phase-lock loop device 220 transmits clock signals to the plurality of memory devices 30, the circuit 40, and the register 230. The register 230

receives and buffers a plurality of command signals and address signals (e.g., bank address signals, row address signals, column address signals, gated column address signals, chip-select signals), and transmits corresponding signals to the appropriate memory devices 30. In certain embodiments, the register 230 comprises a plurality of register devices. While the phase-lock loop device 220, the register 230, and the circuit 40 are described herein in certain embodiments as being separate components, in certain other embodiments, two or more of the phase-lock loop device 220, the register 230, and the circuit 40 are portions of a single component. Persons skilled in the art are able to select a phase-lock loop device 220 and a register 230 compatible with embodiments described herein.

[0075] In certain embodiments, the memory module 10 further comprises electrical components which are electrically coupled to one another and are surface-mounted or embedded on the printed circuit board 210. These electrical components can include, but are not limited to, electrical conduits, resistors, capacitors, inductors, and transistors. In certain embodiments, at least some of these electrical components are discrete, while in other certain embodiments, at least some of these electrical components are constituents of one or more integrated circuits.

[0076] In certain embodiments, the printed circuit board 210 is mountable in a module slot of the computer system. The printed circuit board 210 of certain such embodiments has a plurality of edge connections electrically coupled to corresponding contacts of the module slot and to the various components of the memory module 10, thereby providing electrical connections between the computer system and the components of the memory module 10.

[0077] In certain embodiments, the plurality of memory devices 30 are arranged in a first number of ranks 32. For example, in certain embodiments, the memory devices 30 are arranged in four ranks 32a, 32b, 32c, 32d, as schematically illustrated by FIG. 9A. In certain other embodiments, the memory devices 30 are arranged in two ranks 32a, 32b, as schematically illustrated by FIG. 9B. Other numbers of ranks 32 of the memory devices 30 are also compatible with embodiments described herein.

[0078] As schematically illustrated by FIGS. 9A and 9B, in certain embodiments, the circuit 40 receives a set of input command signals (e.g., refresh, precharge) and address signals (e.g., bank address signals, row address signals, column address signals, gated column address signals, chip-select signals) from the memory controller 20 of the computer system. In response to the set of input address and command signals, the circuit 40 generates a set of output address and command signals.

[0079] In certain embodiments, the set of output address and command signals corresponds to a first number of ranks in which the plurality of memory devices 30 of the memory module 10 are arranged, and the set of input address and command signals corresponds to a second number of ranks per memory module for which the computer system is configured. The second number of ranks in certain embodiments is smaller than the first number of ranks. For example, in the exemplary embodiment as schematically illustrated by FIG. 9A, the first number of ranks is four while the second number of ranks is two. In the exemplary embodiment of FIG. 9B, the first number of ranks is two while the second number of ranks is one. Thus, in certain embodiments, even though the memory module 10 actually has the first number of ranks of memory devices 30, the memory module 10

simulates a virtual memory module by operating as having the second number of ranks of memory devices 30. In certain embodiments, the memory module 10 simulates a virtual memory module when the number of memory devices 30 of the memory module 10 is larger than the number of memory devices 30 per memory module for which the computer system is configured to utilize. In certain embodiments, the circuit 40 comprises logic (e.g., address decoding logic, command decoding logic) which translates between a system memory domain of the computer system and a physical memory domain of the memory module 10.

[0080] In certain embodiments, the computer system is configured for a number of ranks per memory module which is smaller than the number of ranks in which the memory devices 30 of the memory module 10 are arranged. In certain such embodiments, the computer system is configured for two ranks of memory per memory module (providing two chip-select signals CS_0 , CS_1) and the plurality of memory modules 30 of the memory module 10 are arranged in four ranks, as schematically illustrated by FIG. 9A. In certain other such embodiments, the computer system is configured for one rank of memory per memory module (providing one chip-select signal CS_0) and the plurality of memory modules 30 of the memory module 10 are arranged in two ranks, as schematically illustrated by FIG. 9B.

[0081] In the exemplary embodiment schematically illustrated by FIG. 9A, the memory module 10 has four ranks of memory devices 30 and the computer system is configured for two ranks of memory devices per memory module. The memory module 10 receives row/column address signals or signal bits (A_0 - A_{n+1}), bank address signals (BA_0 - BA_m), chip-select signals (CS_0 and CS_1), and command signals (e.g., refresh, precharge, etc.) from the computer system. The A_0 - A_n row/column address signals are received by the register 230, which buffers these address signals and sends these address signals to the appropriate ranks of memory devices 30. The circuit 40 receives the two chip-select signals (CS_0 - CS_1) and one row/column address signal (A_{n+1}) from the computer system. Both the circuit 40 and the register 230 receive the bank address signals (BA_0 - BA_m) and at least one command signal (e.g., refresh, precharge, etc.) from the computer system.

Logic Tables

[0082] Table 1 provides a logic table compatible with certain embodiments described herein for the selection among ranks of memory devices 30 using chip-select signals.

TABLE 1

State	CS_0	CS_1	A_{n+1}	Command	CS_{0A}	CS_{0B}	CS_{1A}	CS_{1B}
1	0	1	0	Active	0	1	1	1
2	0	1	1	Active	1	0	1	1
3	0	1	x	Active	0	0	1	1
4	1	0	0	Active	1	1	0	1
5	1	0	1	Active	1	1	1	0
6	1	0	x	Active	1	1	0	0
7	1	1	x	x	1	1	1	1

Note:

1. CS_0 , CS_1 , CS_{0A} , CS_{0B} , CS_{1A} , and CS_{1B} are active low signals.
2. A_{n+1} is an active high signal.
3. 'x' is a Don't Care condition.
4. Command involves a number of command signals that define operations such as refresh, precharge, and other operations.

[0083] In Logic State 1: CS₀ is active low, A_{n+1} is non-active, and Command is active. CS_{0A} is pulled low, thereby selecting Rank 0.

[0084] In Logic State 2: CS₀ is active low, A_{n+1} is active, and Command is active. CS_{0B} is pulled low, thereby selecting Rank 1.

[0085] In Logic State 3: CS₀ is active low, A_{n+1} is Don't Care, and Command is active high. CS_{0A} and CS_{0B} are pulled low, thereby selecting Ranks 0 and 1.

[0086] In Logic State 4: CS₁ is active low, A_{n+1} is non-active, and Command is active. CS_{1A} is pulled low, thereby selecting Rank 2.

[0087] In Logic State 5: CS₁ is active low, A_{n+1} is active, and Command is active. CS_{1B} is pulled low, thereby selecting Rank 3.

[0088] In Logic State 6: CS₁ is active low, A_{n+1} is Don't Care, and Command is active. CS_{1A} and CS_{1B} are pulled low, thereby selecting Ranks 2 and 3.

[0089] In Logic State 7: CS₀ and CS₁ are pulled non-active high, which deselects all ranks, i.e., CS_{0A}, CS_{0B}, CS_{1A}, and CS_{1B} are pulled high.

[0090] The "Command" column of Table 1 represents the various commands that a memory device (e.g., a DRAM device) can execute, examples of which include, but are not limited to, activation, read, write, precharge, and refresh. In certain embodiments, the command signal is passed through to the selected rank only (e.g., state 4 of Table 1). In such embodiments, the command signal (e.g., read) is sent to only one memory device or the other memory device so that data is supplied from one memory device at a time. In other embodiments, the command signal is passed through to both associated ranks (e.g., state 6 of Table 1). In such embodiments, the command signal (e.g., refresh) is sent to both memory devices to ensure that the memory content of the memory devices remains valid over time. Certain embodiments utilize a logic table such as that of Table 1 to simulate a single memory device from two memory devices by selecting two ranks concurrently.

[0091] Table 2 provides a logic table compatible with certain embodiments described herein for the selection among ranks of memory devices 30 using gated CAS signals.

[0092] In certain embodiments in which the density bit is a row address bit, for read/write commands, the density bit is the value latched during the activate command for the selected bank.

Serial-Presence-Detect Device

[0093] Memory modules typically include a serial-presence detect (SPD) device 240 (e.g., an electrically-erasable-programmable read-only memory or EEPROM device) comprising data which characterize various attributes of the memory module, including but not limited to, the number of row addresses the number of column addresses, the data width of the memory devices, the number of ranks, the memory density per rank, the number of memory devices, and the memory density per memory device. The SPD device 240 communicates this data to the basic input/output system (BIOS) of the computer system so that the computer system is informed of the memory capacity and the memory configuration available for use and can configure the memory controller properly for maximum reliability and performance.

[0094] For example, for a commercially-available 512-MB (64M×8-byte) memory module utilizing eight 512-Mb memory devices each with a 64M×8-bit configuration, the SPD device contains the following SPD data (in appropriate bit fields of these bytes):

[0095] Byte 3: Defines the number of row address bits in the DRAM device in the memory module [13 for the 512-Mb memory device].

[0096] Byte 4: Defines the number of column address bits in the DRAM device in the memory module [11 for the 512-Mb memory device].

[0097] Byte 13: Defines the bit width of the primary DRAM device used in the memory module [8 bits for the 512-Mb (64M×8-bit) memory device].

[0098] Byte 14: Defines the bit width of the error checking DRAM device used in the memory module [8 bits for the 512-Mb (64M×8-bit) memory device].

[0099] Byte 17: Defines the number of banks internal to the DRAM device used in the memory module [4 for the 512-Mb memory device].

TABLE 2

CS*	RAS*	CAS*	WE*	Density Bit	A ₁₀	Command	CAS0*	CAS1*
1	x	x	x	x	x	NOP	x	x
0	1	1	1	x	x	NOP	1	1
0	0	1	1	0	x	ACTIVATE	1	1
0	0	1	1	1	x	ACTIVATE	1	1
0	1	0	1	0	x	READ	0	1
0	1	0	1	1	x	READ	1	0
0	1	0	0	0	x	WRITE	0	1
0	1	0	0	1	x	WRITE	1	0
0	0	1	0	0	0	PRECHARGE	1	1
0	0	1	0	1	0	PRECHARGE	1	1
0	0	1	0	x	1	PRECHARGE	1	1
0	0	0	0	x	x	MODE REG SFT	0	0
0	0	0	1	x	x	REFRESH	0	0

[0100] In a further example, for a commercially-available 1-GB (128M×8-byte) memory module utilizing eight 1-Gb memory devices each with a 128M×8-bit configuration, as described above, the SPD device contains the following SPD data (in appropriate bit fields of these bytes):

[0101] Byte 3: Defines the number of row address bits in the DRAM device in the memory module [14 for the 1-Gb memory device].

[0102] Byte 4: Defines the number of column address bits in the DRAM device in the memory module [11 for the 1-Gb memory device].

[0103] Byte 13: Defines the bit width of the primary DRAM device used in the memory module [8 bits for the 1-Gb (128M×8-bit) memory device].

[0104] Byte 14: Defines the bit width of the error checking DRAM device used in the memory module [8 bits for the 1-Gb (128M×8-bit) memory device].

[0105] Byte 17: Defines the number of banks internal to the DRAM device used in the memory module [4 for the 1-Gb memory device].

[0106] In certain embodiments, the SPD device 240 comprises data which characterize the memory module 10 as having fewer ranks of memory devices than the memory module 10 actually has, with each of these ranks having more memory density. For example, for a memory module 10 compatible with certain embodiments described herein having two ranks of memory devices 30, the SPD device 240 comprises data which characterizes the memory module 10 as having one rank of memory devices with twice the memory density per rank. Similarly, for a memory module 10 compatible with certain embodiments described herein having four ranks of memory devices 30, the SPD device 240 comprises data which characterizes the memory module 10 as having two ranks of memory devices with twice the memory density per rank. In addition, in certain embodiments, the SPD device 240 comprises data which characterize the memory module 10 as having fewer memory devices than the memory module 10 actually has, with each of these memory devices having more memory density per memory device. For example, for a memory module 10 compatible with certain embodiments described herein, the SPD device 240 comprises data which characterizes the memory module 10 as having one-half the number of memory devices that the memory module 10 actually has, with each of these memory devices having twice the memory density per memory device. Thus, in certain embodiments, the SPD device 240 informs the computer system of the larger memory array by reporting a memory device density that is a multiple of the memory devices 30 resident on the memory module 10. Certain embodiments described herein advantageously do not require system level changes to hardware (e.g., the motherboard of the computer system) or to software (e.g., the BIOS of the computer system).

[0107] FIG. 9C schematically illustrates an exemplary memory module 10 in accordance with certain embodiments described herein. The memory module 10 comprises a pair of substantially identical memory devices 31, 33. Each memory device 31, 33 has a first bit width, a first number of banks of memory locations, a first number of rows of memory locations, and a first number of columns of

memory-locations. The memory module 10 further comprises an SPD device 240 comprising data that characterizes the pair of memory devices 31, 33. The data characterize the pair of memory devices 31, 33 as a virtual memory device having a second bit width equal to twice the first bit width, a second number of banks of memory locations equal to the first number of banks, a second number of rows of memory locations equal to the first number of rows, and a second number of columns of memory locations equal to the first number of columns.

[0108] In certain such embodiments, the SPD device 240 of the memory module 10 is programmed to describe the combined pair of lower-density memory devices 31, 33 as one virtual or pseudo-higher-density memory device. In an exemplary embodiment, two 512-Mb memory devices, each with a 128M×4-bit configuration, are used to simulate one 1-Gb memory device having a 128M×8-bit configuration. The SPD device 240 of the memory module 10 is programmed to describe the pair of 512-Mb memory devices as one virtual or pseudo-1-Gb memory device.

[0109] For example, to fabricate a 1-GB (128M×8-byte) memory module, sixteen 512-Mb (128M×4-bit) memory devices can be used. The sixteen 512-Mb (128M×4-bit) memory devices are combined in eight pairs, with each pair serving as a virtual or pseudo-1-Gb (128M×8-bit) memory device. In certain such embodiments, the SPD device 240 contains the following SPD data (in appropriate bit fields of these bytes):

[0110] Byte 3: 13 row address bits.

[0111] Byte 4: 12 column address bits.

[0112] Byte 13: 8 bits wide for the primary virtual 1-Gb (128M×8-bit) memory device.

[0113] Byte 14: 8 bits wide for the error checking virtual 1-Gb (128M×8-bit) memory device.

[0114] Byte 17: 4 banks.

[0115] In this exemplary embodiment, bytes 3, 4, and 17 are programmed to have the same values as they would have for a 512-MB (128M×4-byte) memory module utilizing 512-Mb (128M×4-bit) memory devices. However, bytes 13 and 14 of the SPD data are programmed to be equal to 8, corresponding to the bit width of the virtual or pseudo-higher-density 1-Gb (128M×8-bit) memory device, for a total capacity of 1-GB. Thus, the SPD data does not describe the actual-lower-density memory devices, but instead describes the virtual or pseudo-higher-density memory devices. The BIOS accesses the SPD data and recognizes the memory module as having 4 banks of memory locations arranged in 2^{13} rows and 2^{12} columns, with each memory location having a width of 8 bits rather than 4 bits.

[0116] In certain embodiments, when such a memory module 10 is inserted in a computer system, the computer system's memory controller then provides to the memory module 10 a set of input address and command signals which correspond to the number of ranks or the number of memory devices reported by the SPD device 240. For example, placing a two-rank memory module 10 compatible with certain embodiments described herein in a computer system compatible with one-rank memory modules, the SPD device 240 reports to the computer system that the memory module 10 only has one rank. The circuit 40 then receives a

set of input address and command signals corresponding to a single rank from the computer system's memory controller, and generates and transmits a set of output address and command signals corresponding to two ranks to the appropriate memory devices 30 of the memory module 10.

[0117] Similarly, when a two-rank memory module 10 compatible with certain embodiments described herein is placed in a computer system compatible with either one- or two-rank memory modules, the SPD device 240 reports to the computer system that the memory module 10 only has one rank. The circuit 40 then receives a set of input address and command signals corresponding to a single rank from the computer system's memory controller, and generates and transmits a set of output address and command signals corresponding to two ranks to the appropriate memory devices 30 of the memory module 10.

[0118] Furthermore, a four-rank memory module 10 compatible with certain embodiments described herein simulates a two-rank memory module whether the memory module 10 is inserted in a computer system compatible with two-rank memory modules or with two- or four-rank memory modules. Thus, by placing a four-rank memory module 10 compatible with certain embodiments described herein in a module slot that is four-rank-ready, the computer system provides four chip-select signals, but the memory module 10 only uses two of the chip-select signals.

[0119] In certain embodiments, the circuit 40 comprises the SPD device 240 which reports the CAS latency (CL) to the memory controller of the computer system. The SPD device 240 of certain embodiments reports a CL which has one more cycle than does the actual operational CL of the memory array. In certain embodiments, data transfers between the memory controller and the memory module are registered for one additional clock cycle by the circuit 40. The additional clock cycle of certain embodiments is added to the transfer time budget with an incremental overall CAS latency. This extra cycle of time in certain embodiments advantageously provides sufficient time budget to add a buffer which electrically isolates the ranks of memory devices 30 from the memory controller 20. The buffer of certain embodiments comprises combinatorial logic, registers, and logic pipelines. In certain embodiments, the buffer adds a one-clock cycle time delay, which is equivalent to a registered DIMM, to accomplish the address decoding. The one-cycle time delay of certain such embodiments provides sufficient time for read and write data transfers to provide the functions of the data path multiplexer/demultiplexer. Thus, for example, a DDR2 400-MHz memory system in accordance with embodiments described herein has an overall CAS latency of four, and uses memory devices with a CAS latency of three. In still other embodiments, the SPD device 240 does not utilize this extra cycle of time.

Memory Density Multiplication

[0120] In certain embodiments, two memory devices having a memory density are used to simulate a single memory device having twice the memory density, and an additional address signal bit is used to access the additional memory. Similarly, in certain embodiments, two ranks of memory devices having a memory density are used to simulate a single rank of memory devices having twice the memory density, and an additional address signal bit is used to access the additional memory. As used herein, such simulations of

memory devices or ranks of memory devices are termed as "memory density multiplication," and the term "density transition bit" is used to refer to the additional address signal bit which is used to access the additional memory by selecting which rank of memory devices is enabled for a read or write transfer operation.

[0121] For example, for computer systems which are normally limited to using memory modules which have a single rank of 128Mx4-bit memory devices, certain embodiments described herein enable the computer system to utilize memory modules which have double the memory (e.g., two ranks of 128Mx4-bit memory devices). The circuit 40 of certain such embodiments provides the logic (e.g., command and address decoding logic) to double the number of chip selects, and the SPD device 240 reports a memory device density of 256Mx4-bit to the computer system.

[0122] In certain embodiments utilizing memory density multiplication embodiments, the memory module 10 can have various types of memory devices 30 (e.g., DDR1, DDR2, DDR3, and beyond). The circuit 40 of certain such embodiments utilizes implied translation logic equations having variations depending on whether the density transition bit is a row, column, or internal bank address bit. In addition, the translation logic equations of certain embodiments vary depending on the type of memory module 10 (e.g., UDIMM, RDIMM, FBDIMM, etc.). Furthermore, in certain embodiments, the translation logic equations vary depending on whether the implementation multiplies memory devices per rank or multiplies the number of ranks per memory module.

[0123] Table 3 A provides the numbers of rows and columns for DDR-1 memory devices, as specified by JEDEC standard JESD79D, "Double Data Rate (DDR) SDRAM Specification," published February 2004, and incorporated in its entirety by reference herein.

TABLE 3A

	128-Mb	256-Mb	512-Mb	1-Gb
Number of banks	4	4	4	4
Number of row address bits	12	13	13	14
Number of column address bits for "x4" configuration	11	11	12	12
Number of column address bits for "x8" configuration	10	10	11	11
Number of column address bits for "x16" configuration	9	9	10	10

[0124] As described by Table 3A, 512-Mb (128Mx4-bit) DRAM devices have 2^{13} rows and 2^{12} columns of memory locations, while 1-Gb (128Mx8-bit) DRAM devices have 2^{14} rows and 2^{11} columns of memory locations. Because of the differences in the number of rows and the number of columns for the two types of memory devices, complex address translation procedures and structures would typically be needed to fabricate a 1-GB (128Mx8-byte) memory module using sixteen 512-Mb (128Mx4-bit) DRAM devices.

[0125] Table 3B shows the device configurations as a function of memory density for DDR2 memory devices.

TABLE 3B

	Number of Rows	Number of Columns	Number of Internal Banks	Page Size (x4s or x8s)
256 Mb	13	11	4	1 KB
512 Mb	14	11	4	1 KB
1 Gb	14	11	8	1 KB
2 Gb	15	11	8	1 KB
4 Gb	16	11	8	1 KB

[0126] Table 4 lists the corresponding density transition bit for the density transitions between the DDR2 memory densities of Table 3B.

TABLE 4

Density Transition	Density Transition Bit
256 Mb to 512 Mb	A ₁₃
512 Mb to 1 Gb	BA ₂
1 Gb to 2 Gb	A ₁₄
2 Gb to 4 Gb	A ₁₅

Other certain embodiments described herein utilize a transition bit to provide a transition from pairs of physical 4-Gb memory devices to simulated 8-Gb memory devices.

[0127] In an example embodiment, the memory module comprises one or more pairs of 256-Mb memory devices, with each pair simulating a single 512-Mb memory device. The simulated 512-Mb memory device has four internal banks while each of the two 256-Mb memory devices has four internal banks, for a total of eight internal banks for the pair of 256-Mb memory devices. In certain embodiments, the additional row address bit is translated by the circuit 40 to the rank selection between each of the two 256-Mb memory devices of the pair. Although there are eight total internal banks in the rank-converted memory array, the computer system is only aware of four internal banks. When the memory controller activates a row for a selected bank, the circuit 40 activates the same row for the same bank, but it does so for the selected rank according to the logic state of the additional row address bit A₁₃.

[0128] In another example embodiment, the memory module comprises one or more pairs of 512-Mb memory devices, with each pair simulating a single 1-Gb memory device. The simulated 1-Gb memory device has eight internal banks while each of the two 512-Mb memory devices has four internal banks, for a total of eight internal banks for the pair of 512-Mb memory devices. In certain embodiments, the mapped BA₂ (bank 2) bit is used to select between the two

banks of 512-Mb memory devices to preserve the internal bank geometry expected by the memory controller of the computer system. The state of the BA₂ bit selects the upper or lower set of four banks, as well as the upper and lower 512-Mb rank.

[0129] In another example embodiment, the memory module comprises one or more pairs of 1-Gb memory devices, with each pair simulating a single 2-Gb memory device. Each of the two 1-Gb memory devices has eight internal banks for a total of sixteen internal banks, while the simulated 2-Gb memory device has eight internal banks. In certain embodiments, the additional row address bit translates to the rank selection between the two 1-Gb memory devices. Although there are sixteen total internal banks per pair of 1-Gb memory devices in the rank-converted memory array, the memory controller of the computer system is only aware of eight internal banks. When the memory controller activates a row of a selected bank, the circuit 40 activates the same row for the same bank, but it does so for the selected rank according to the logic state of the additional row address bit A₁₄.

[0130] The circuit 40 of certain embodiments provides substantially all of the translation logic used for the decoding (e.g., command and address decoding). In certain such embodiments, there is a fully transparent operational conversion from the "system memory" density domain of the computer system to the "physical memory" density domain of the memory module 10. In certain embodiments, the logic translation equations are programmed in the circuit 40 by hardware, while in certain other embodiments, the logic translation equations are programmed in the circuit 40 by software. Examples 1 and 2 provide exemplary sections of Verilog code compatible with certain embodiments described herein. As described more fully below, the code of Examples 1 and 2 includes logic to reduce potential problems due to "back-to-back adjacent read commands which cross memory device boundaries or "BBARX." Persons skilled in the art are able to provide additional logic translation equations compatible with embodiments described herein.

[0131] An exemplary section of Verilog code compatible with memory density multiplication from 512 Mb to 1 Gb using DDR2 memory devices with the BA₂ density transition bit is listed below in Example 2. The exemplary code of Example 2 corresponds to a circuit 40 which receives one chip-select signal from the computer system and which generates two chip-select signals.

EXAMPLE 2

[0132]

```

always @(posedge clk_in)
begin
    rs0N_R <= rs0_in_N; // cs0
    rasN_R <= ras_in_N;
    casN_R <= cas_in_N;
    weN_R <= we_in_N;
end
// Gated Chip Selects
assign pos0a_1 = (~rs0_in_N & ~ras_in_N & ~cas_in_N) // ref.md reg set
               | (~rs0_in_N & ras_in_N & cas_in_N) // ref exit, pwr dn

```

-continued

```

| (~rst0_in_N & ~ras_in_N & cas_in_N & ~we_in_N & a10_in) // pchg all
| (~rst0_in_N & ~ras_in_N & cas_in_N & ~we_in_N & ~a10_in & ~ba2_in) // pchg single bnk
| (~rst0_in_N & ~ras_in_N & cas_in_N & we_in_N & ~ba2_in) // activate
| (~rst0_in_N & ras_in_N & ~cas_in_N & ~ba2_in) // xfr
;
assign pcs0b_1 = (~rst0_in_N & ~ras_in_N & ~cas_in_N) // ref.md reg set
| (~rst0_in_N & ras_in_N & cas_in_N) // ref exit, pwr dn
| (~rst0_in_N & ~ras_in_N & cas_in_N & ~we_in_N & a10_in) // pchg all
| (~rst0_in_N & ~ras_in_N & cas_in_N & ~we_in_N & ~a10_in & ba2_in) // pchg single bnk
| (~rst0_in_N & ~ras_in_N & cas_in_N & we_in_N & ba2_in) // activate
| (~rst0_in_N & ras_in_N & ~cas_in_N & ba2_in) // xfr
;
//-----
always @(posedge clk_in)
begin
    a4_r <= a4_in ;
    a5_r <= a5_in ;
    a6_r <= a6_in ;
    a10_r <= a10_in ;
    ba0_r <= ba0_in ;
    ba1_r <= ba1_in ;
    ba2_r <= ba2_in ;
    q_mrs_cmd_cyc1 <= q_mrs_cmd ;

end
//=====
// determine the cas latency
//=====
assign q_mrs_cmd_r = (!rasN_R & !casN_R & !weN_R)
& (!ba0_r & !ba1_r)
; // md reg set cmd

always @(posedge clk_in)
if (~reset_N) // lmr
cl3 <= 1'b1 ;
else if (q_mrs_cmd_cyc1) // load mode reg cmd
begin
cl3 <= (~a6_r & a5_r & a4_r) ;
end
always @(posedge clk_in)
if (~reset_N) // reset
cl2 <= 1'b0 ;
else if (q_mrs_cmd_cyc1) // load mode reg cmd
begin
cl2 <= (~a6_r & a5_r & ~a4_r) ;
end
always @(posedge clk_in)
if (~reset_N) // reset
cl4 <= 1'b0 ;
else if (q_mrs_cmd_cyc1) // load mode reg cmd
begin
cl4 <= (a6_r & ~a5_r & ~a4_r) ;
end
always @(posedge clk_in)
if (~reset_N) cl5 <= 1'b0 ;
else if (q_mrs_cmd_cyc1) // load mode reg cmd
begin
cl5 <= (a6_r & ~a5_r & a4_r) ;
end
end
assign pre_cyc2_enfet = (wr_cmd_cyc1 & acs_cyc1 & cl3) // wr brst cl3 preamble
;
assign pre_cyc3_enfet = (rd_cmd_cyc2 & cl3) // rd brst cl3 preamble
| (wr_cmd_cyc2 & cl3) // wr brst cl3 1st pair
| (wr_cmd_cyc2 & cl4) // wr brst cl4 preamble
;
assign pre_cyc4_enfet = (wr_cmd_cyc3 & cl3) // wr brst cl3 2nd pair
| (wr_cmd_cyc3 & cl4) // wr brst cl4 1st pair
| (rd_cmd_cyc3 & cl3) // rd brst cl3 1st pair
| (rd_cmd_cyc3 & cl4) // rd brst cl4 preamble
;
assign pre_cyc5_enfet = (rd_cmd_cyc4 & cl3) // rd brst cl3 2nd pair
| (wr_cmd_cyc4 & cl4) // wr brst cl4 2nd pair
| (rd_cmd_cyc4 & cl4) // rd brst cl4 1st pair
;
// dq
assign pre_dq_cyc = pre_cyc2_enfet
| pre_cyc3_enfet

```

-continued

```

        | pre_cyc4_enfet
        | pre_cyc5_enfet
        ;
assign    pre_dq_nycyc = enfet_cyc2
        | enfet_cyc3
        | enfet_cyc4
        | enfet_cyc5
        ;

// dqs
assign    pre_dqsa_cyc = (pre_cyc2_enfet & ~ba2_r)
        | (pre_cyc3_enfet & ~ba2_cyc2)
        | (pre_cyc4_enfet & ~ba2_cyc3)
        | (pre_cyc5_enfet & ~ba2_cyc4)
        ;
assign    pre_dqsb_cyc = (pre_cyc2_enfet & ba2_r)
        | (pre_cyc3_enfet & ba2_cyc2)
        | (pre_cyc4_enfet & ba2_cyc3)
        | (pre_cyc5_enfet & ba2_cyc4)
        ;
assign    pre_dqsa_nycyc = (enfet_cyc2 & ~ba2_cyc2)
        | (enfet_cyc3 & ~ba2_cyc3)
        | (enfet_cyc4 & ~ba2_cyc4)
        | (enfet_cyc5 & ~ba2_cyc5)
        ;
assign    pre_dqsb_nycyc = (enfet_cyc2 & ba2_cyc2)
        | (enfet_cyc3 & ba2_cyc3)
        | (enfet_cyc4 & ba2_cyc4)
        | (enfet_cyc5 & ba2_cyc5)
        ;
always @(posedge clk_in)
begin
    acs_cyc2 <= acs_cyc1 ;    // cs active
    ba2_cyc2 <= ba2_r ;
    ba2_cyc3 <= ba2_cyc2 ;
    ba2_cyc4 <= ba2_cyc3 ;
    ba2_cyc5 <= ba2_cyc4 ;
    rd_cmd_cyc2 <= rd_cmd_cyc1 & acs_cyc1;
    rd_cmd_cyc3 <= rd_cmd_cyc2 ;
    rd_cmd_cyc4 <= rd_cmd_cyc3 ;
    rd_cmd_cyc5 <= rd_cmd_cyc4 ;
    rd_cmd_cyc6 <= rd_cmd_cyc5 ;
    rd_cmd_cyc7 <= rd_cmd_cyc6 ;
    wr_cmd_cyc2 <= wr_cmd_cyc1 & acs_cyc1;
    wr_cmd_cyc3 <= wr_cmd_cyc2 ;
    wr_cmd_cyc4 <= wr_cmd_cyc3 ;
    wr_cmd_cyc5 <= wr_cmd_cyc4 ;

end
always @(negedge clk_in)
begin
    dq_nycyc <= dq_cyc;
    dqs_nycyc_a <= dqs_cyc_a;
    dqs_nycyc_b <= dqs_cyc_b;

end
// DQ FET enables
assign    enq_fet1 = dq_cyc | dq_nycyc ;
assign    enq_fet2 = dq_cyc | dq_nycyc ;
assign    enq_fet3 = dq_cyc | dq_nycyc ;
assign    enq_fet4 = dq_cyc | dq_nycyc ;
assign    enq_fet5 = dq_cyc | dq_nycyc ;
// DQS FET enables
assign    ens_fet1a = dqs_cyc_a | dqs_nycyc_a ;
assign    ens_fet2a = dqs_cyc_a | dqs_nycyc_a ;
assign    ens_fet3a = dqs_cyc_a | dqs_nycyc_a ;
assign    ens_fet1b = dqs_cyc_b | dqs_nycyc_b ;
assign    ens_fet2b = dqs_cyc_b | dqs_nycyc_b ;
assign    ens_fet3b = dqs_cyc_b | dqs_nycyc_b ;

```

[0133] Another exemplary section of Verilog code compatible with memory density multiplication from 256 Mb to 512 Mb using DDR2 memory devices and gated CAS signals with the row A₁₃ density transition bit is listed below in Example 3. The exemplary code of Example 3 corresponds to a circuit 40 which receives one gated CAS signal

from the computer system and which generates two gated CAS signals.

EXAMPLE 3

[0134]

```

// latched a13 flags cas0, banks 0-3
always @(posedge clk_in)
    if (actv_cmd_R & ~rstN_R & ~bnk1_R & ~bnk0_R) // activate
        begin
            l_a13_00 <= a13_r;
        end
    if (actv_cmd_R & ~rstN_R & ~bnk1_R & bnk0_R) // activate
        begin
            l_a13_01 <= a13_r;
        end
    if (actv_cmd_R & ~rstN_R & bnk1_R & ~bnk0_R) // activate
        begin
            l_a13_10 <= a13_r;
        end
    if (actv_cmd_R & ~rstN_R & bnk1_R & bnk0_R) // activate
        begin
            l_a13_11 <= a13_r;
        end
    end

// gated cas
assign cas_i = ~(casN_R);
assign cas0_o = (~rasN_R & cas_i)
    | (rasN_R & ~l_a13_00 & ~bnk1_R & ~bnk0_R & cas_i)
    | (rasN_R & ~l_a13_01 & ~bnk1_R & bnk0_R & cas_i)
    | (rasN_R & ~l_a13_10 & bnk1_R & ~bnk0_R & cas_i)
    | (rasN_R & ~l_a13_11 & bnk1_R & bnk0_R & cas_i);
assign cas1_o = (~rasN_R & cas_i)
    | (rasN_R & l_a13_00 & ~bnk1_R & ~bnk0_R & cas_i)
    | (rasN_R & l_a13_01 & ~bnk1_R & bnk0_R & cas_i)
    | (rasN_R & l_a13_10 & bnk1_R & ~bnk0_R & cas_i)
    | (rasN_R & l_a13_11 & bnk1_R & bnk0_R & cas_i);

assign pcas0_N = ~cas0_o;
assign pcas1_N = ~cas1_o;
assign rd0_o_R1 = rasN_R & cas0_o & weN_R & ~rstN_R; // mk0 rd cmd cyc
assign rd1_o_R1 = rasN_R & cas1_o & weN_R & ~rstN_R; // mk1 rd cmd cyc
assign wr0_o_R1 = rasN_R & cas0_o & ~weN_R & ~rstN_R; // mk0 wr cmd cyc
assign wr1_o_R1 = rasN_R & cas1_o & ~weN_R & ~rstN_R; // mk1 wr cmd cyc

always @(posedge clk_in)
    begin
        rd0_o_R2 <= rd0_o_R1;
        rd0_o_R3 <= rd0_o_R2;
        rd0_o_R4 <= rd0_o_R3;
        rd0_o_R5 <= rd0_o_R4;
        rd1_o_R2 <= rd1_o_R1;
        rd1_o_R3 <= rd1_o_R2;
        rd1_o_R4 <= rd1_o_R3;
        rd1_o_R5 <= rd1_o_R4;
        wr0_o_R2 <= wr0_o_R1;
        wr0_o_R3 <= wr0_o_R2;
        wr0_o_R4 <= wr0_o_R3;
        wr0_o_R5 <= wr0_o_R4;
        wr1_o_R2 <= wr1_o_R1;
        wr1_o_R3 <= wr1_o_R2;
        wr1_o_R4 <= wr1_o_R3;
        wr1_o_R5 <= wr1_o_R4;
    end

always @(posedge clk_in)
    begin
        if (
            (rd0_o_R2 & ~rd1_o_R4) // pre-am rd if no ped on mk 1
            | rd0_o_R3 // 1st cyc of rd brst
            | rd0_o_R4 // 2nd cyc of rd brst
            | (rd0_o_R5 & ~rd1_o_R2 & ~rd1_o_R3) // post-rd cyc if no ped on mk 1
        )
    end

```

-continued

```

        | (wr0_o_R1)
        | wr0_o_R2 | wr0_o_R3
        | (wr0_o_R4)
        | wr1_o_R1 | wr1_o_R2 | wr1_o_R3 | wr1_o_R4
    |
    |
        en_fet_a <= 'b1;
    else
        en_fet_a <= 'b0;
    end
    always @(posedge clk_in)
    begin
        if (
            (rd1_o_R2 & ~rd0_o_R4)
            | rd1_o_R3
            | rd1_o_R4
            | (rd1_o_R5 & ~rd0_o_R2 & ~rd0_o_R3)
            | (wr1_o_R1)
            | wr1_o_R2 | wr1_o_R3
        | (wr1_o_R4)
        | wr0_o_R1 | wr0_o_R2 | wr0_o_R3 | wr0_o_R4
        )
            en_fet_b <= 'b1;
        else
            en_fet_b <= 'b0;
        end
    end

```

// pre-am wr
 // wr hst 1st & 2nd cyc
 // post-wr cyc (chgef9)
 // rank 1 (chgef9)
 // enable fet
 // disable fet
 // (chgef8)
 // post-wr cyc (chgef9)
 // rank 0 (chgef9)
 //

[0135] In certain embodiments, the chipset memory controller of the computer system uses the inherent behavioral characteristics of the memory devices (e.g., DDR2 memory devices) to optimize throughput of the memory system. For example, for each internal bank in the memory array, a row (e.g., 1 KB page) is advantageously held activated for an extended period of time. The memory controller, by anticipating a high number of memory accesses or hits to a particular region of memory, can exercise this feature to advantageously eliminate time-consuming pre-charge cycles. In certain such embodiments in which two half-density memory devices are transparently substituted for a single full-density memory device (as reported by the SPD device 240 to the memory controller), the memory devices advantageously support the “open row” feature.

[0136] FIG. 10A schematically illustrates an exemplary memory module 10 which doubles the rank density in accordance with certain embodiments described herein. The memory module 10 has a first memory capacity. The memory module 10 comprises a plurality of substantially identical memory devices 30 configured as a first rank 32a and a second rank 32b. In certain embodiments, the memory devices 30 of the first rank 32a are configured in pairs, and the memory devices 30 of the second rank 32b are also configured in pairs. In certain embodiments, the memory devices 30 of the first rank 32a are configured with their respective DQS pins tied together and the memory devices 30 of the second rank 32b are configured with their respective DQS pins tied together, as described more fully below. The memory module 10 further comprises a circuit 40 which receives a first set of address and command signals from a memory controller (not shown) of the computer system. The first set of address and command signals is compatible with a second memory capacity substantially equal to one-half of the first memory capacity. The circuit 40 translates the first set of address and command signals into a second set of address and command signals which is compatible with the first memory capacity of the memory module 10 and which is transmitted to the first rank 32a and the second rank 32b.

[0137] The first rank 32a of FIG. 10A has 18 memory devices 30 and the second rank 32b of FIG. 10A has 18 memory devices 30. Other numbers of memory devices 30 in each of the ranks 32a, 32b are also compatible with embodiments described herein.

[0138] In the embodiment schematically illustrated by FIG. 10A, the memory module 10 has a width of 8 bytes (or 64 bits) and each of the memory devices 30 of FIG. 10A has a bit width of 4 bits. The 4-bit-wide (“×4”) memory devices 30 of FIG. 10A have one-half the width, but twice the depth of 8-bit-wide (“×8”) memory devices. Thus, each pair of “×4” memory devices 30 has the same density as a single “×8” memory device, and pairs of “×4” memory devices 30 can be used instead of individual “×8” memory devices to provide the memory density of the memory module 10. For example, a pair of 512-Mb 128M×4-bit memory devices has the same memory density as a 1-Gb 128M×8-bit memory device.

[0139] For two “×4” memory devices 30 to work in tandem to mimic a “×8” memory device, the relative DQS pins of the two memory devices 30 in certain embodiments are advantageously tied together, as described more fully below. In addition, to access the memory density of a high-density memory module 10 comprising pairs of “×4” memory devices 30, an additional address line is used. While a high-density memory module comprising individual “×8” memory devices with the next-higher density would also utilize an additional address line, the additional address lines are different in the two memory module configurations.

[0140] For example, a 1-Gb 128M×8-bit DDR-1 DRAM memory device uses row addresses A₁₃-A₀ and column addresses A₁₁ and A₉-A₀. A pair of 512-Mb 128M×4-bit DDR-1 DRAM memory devices uses row addresses A₁₂-A₀ and column addresses A₁₂, A₁₁, and A₉-A₀. In certain embodiments, a memory controller of a computer system utilizing a 1-GB 128M×8 memory module 10 comprising pairs of the 512-Mb 128M×4 memory devices 30 supplies the address and command signals including the extra row

address (A_{13}) to the memory module 10. The circuit 40 receives the address and command signals from the memory controller and converts the extra row address (A_{13}) into an extra column address (A_{12}).

[0141] FIG. 10B schematically illustrates an exemplary circuit 40 compatible with embodiments described herein. The circuit 40 is used for a memory module 10 comprising pairs of "x4" memory devices 30 which mimic individual "x8" memory devices. In certain embodiments, each pair has the respective DQS pins of the memory devices 30 tied together. In certain embodiments, as schematically illustrated by FIG. 10B, the circuit 40 comprises a programmable-logic device (PLD) 42, a first multiplexer 44 electrically coupled to the first rank 32a of memory devices 30, and a second multiplexer 46 electrically coupled to the second rank 32b of memory devices 30. In certain embodiments, the PLD 42 and the first and second multiplexers 44, 46 are discrete elements, while in other certain embodiments, they are integrated within a single integrated circuit. Persons skilled in the art can select an appropriate PLD 42, first multiplexer 44, and second multiplexer 46 in accordance with embodiments described herein.

[0142] In the exemplary circuit 40 of FIG. 10B, during a row access procedure (CAS is high), the first multiplexer 44 passes the A_{12} address through to the first rank 32, the second multiplexer 46 passes the A_{12} address through to the second rank 34, and the PLD 42 saves or latches the A_{13} address from the memory controller. In certain embodiments, a copy of the A_{13} address is saved by the PLD 42 for each of the internal banks (e.g., 4 internal banks) per memory device 30. During a subsequent column access procedure (CAS is low), the first multiplexer 44 passes the previously-saved A_{13} address through to the first rank 32a as the A_{12} address and the second multiplexer 46 passes the previously-saved A_{13} address through to the second rank 32b as the A_{12} address. The first rank 32a and the second rank 32b thus interpret the previously-saved A_{13} row address as the current A_{12} column address. In this way, in certain embodiments, the circuit 40 translates the extra row address into an extra column address in accordance with certain embodiments described herein.

[0143] Thus, by allowing two lower-density memory devices to be used rather than one higher-density memory device, certain embodiments described herein provide the advantage of using lower-cost, lower-density memory devices to build "next-generation" higher-density memory modules. Certain embodiments advantageously allow the use of lower-cost readily-available 512-Mb DDR-2 SDRAM devices to replace more expensive 1-Gb DDR-2 SDRAM devices. Certain embodiments advantageously reduce the total cost of the resultant memory module.

[0144] FIG. 11A schematically illustrates an exemplary memory module 10 which doubles number of ranks in accordance with certain embodiments described herein. The memory module 10 has a first plurality of memory locations with a first memory density. The memory module 10 comprises a plurality of substantially identical memory devices 30 configured as a first rank 32a, a second rank 32b, a third rank 32c, and a fourth rank 32d. The memory module 10 further comprises a circuit 40 which receives a first set of address and command signals from a memory controller (not shown). The first set of address and command signals is

compatible with a second plurality of memory locations having a second memory density. The second memory density is substantially equal to one-half of the first memory density. The circuit 40 translates the first set of address and command signals into a second set of address and command signals which is compatible with the first plurality of memory locations of the memory module 10 and which is transmitted to the first rank 32a, the second rank 32b, the third rank 32c, and the fourth rank 32d.

[0145] Each rank 32a, 32b, 32c, 32d of FIG. 11A has 9 memory devices 30. Other numbers of memory devices 30 in each of the ranks 32a, 32b, 32c, 32d are also compatible with embodiments described herein.

[0146] In the embodiment schematically illustrated by FIG. 11A, the memory module 10 has a width of 8 bytes (or 64 bits) and each of the memory devices 30 of FIG. 11A has a bit width of 8 bits. Because the memory module 10 has twice the number of 8-bit-wide ("x8") memory devices 30 as does a standard 8-byte-wide memory module, the memory module 10 has twice the density as does a standard 8-byte-wide memory module. For example, a 1-GB 128Mx8-byte memory module with 36 512-Mb 128Mx8-bit memory devices (arranged in four ranks) has twice the memory density as a 512-Mb 128Mx8-byte memory module with 18 512-Mb 128Mx8-bit-memory devices (arranged in two ranks).

[0147] To access the additional memory density of the high-density memory module 10, the two chip-select signals (CS_0 , CS_1) are used with other address and command signals to gate a set of four gated CAS signals. For example, to access the additional ranks of four-rank 1-GB 128Mx8-byte DDR-1 DRAM memory module, the CS_0 and CS_1 signals along with the other address and command signals are used to gate the CAS signal appropriately, as schematically illustrated by FIG. 11A. FIG. 11B schematically illustrates an exemplary circuit 40 compatible with embodiments described herein. In certain embodiments, the circuit 40 comprises a programmable-logic device (PLD) 42 and four "OR" logic elements 52, 54, 56, 58 electrically coupled to corresponding ranks 32a, 32b, 32c, 32d of memory devices 30.

[0148] In certain embodiments, the PLD 42 comprises an ASIC, an FPGA, a custom-designed semiconductor device, or a CPLD. In certain embodiments, the PLD 42 and the four "OR" logic elements 52, 54, 56, 58 are discrete elements, while in other certain embodiments, they are integrated within a single integrated circuit. Persons skilled in the art can select an appropriate PLD 42 and appropriate "OR" logic elements 52, 54, 56, 58 in accordance with embodiments described herein.

[0149] In the embodiment schematically illustrated by FIG. 11B, the PLD 42 transmits each of the four "enabled CAS" ($ENCAS_{0a}$, $ENCAS_{0b}$, $ENCAS_{1a}$, $ENCAS_{1b}$) signals to a corresponding one of the "OR" logic elements 52, 54, 56, 58. The CAS signal is also transmitted to each of the four "OR" logic elements 52, 54, 56, 58. The CAS signal and the "enabled CAS" signals are "low" true signals. By selectively activating each of the four "enabled CAS" signals which are inputted into the four "OR" logic elements 52, 54, 56, 58, the PLD 42 is able to select which of the four ranks 32a, 32b, 32c, 32d is active.

[0150] In certain embodiments, the PLD 42 uses sequential and combinatorial logic procedures to produce the gated

CAS signals which are each transmitted to a corresponding one of the four ranks 32a, 32b, 32c, 32d. In certain other embodiments, the PLD 42 instead uses sequential and combinatorial logic procedures to produce four gated chip-select signals (e.g., CS_{0a}, CS_{0b}, CS_{1a}, and CS_{1b}) which are each transmitted to a corresponding one of the four ranks 32a, 32b, 32c, 32d.

Tied Data Strobe Signal Pins

[0151] For proper operation, the computer system advantageously recognizes a 1-GB memory module comprising 256-Mb memory devices with 64Mx4-bit configuration as a 1-GB memory module having 512-Mb memory devices with 64Mx8-bit configuration (e.g., as a 1-GB memory module with 128Mx8-byte configuration). This advantageous result is desirably achieved in certain embodiments by electrically connecting together two output signal pins (e.g., DQS or data strobe pins) of the two 256-Mb memory devices such that both output signal pins are concurrently active when the two memory devices are concurrently enabled. The DQS or data strobe is a bi-directional signal that is used during both read cycles and write cycles to validate or latch data. As used herein, the terms "tying together" or "tied together" refer to a configuration in which corresponding pins (e.g., DQS pins) of two memory devices are electrically connected together and are concurrently active when the two memory devices are concurrently enabled (e.g., by a common chip-select or CS signal). Such a configuration is different from standard memory module configurations in which the output signal pins (e.g., DQS pins) of two memory devices are electrically coupled to the same source, but these pins are not concurrently active since the memory devices are not concurrently enabled. However, a general guideline of memory module design warns against tying together two output signal pins in this way.

[0152] FIGS. 12 and 13 schematically illustrate a problem which may arise from tying together two output signal pins. FIG. 12 schematically illustrates an exemplary memory module 305 in which a first DQS pin 312 of a first memory device 310 is electrically connected to a second DQS pin 322 of a second memory device 320. The two DQS pins 312, 322 are both electrically connected to a memory controller 330.

[0153] FIG. 13 is an exemplary timing diagram of the voltages applied to the two DQS pins 312, 322 due to non-simultaneous switching. As illustrated by FIG. 13, at time t₁, both the first DQS pin 312 and the second DQS pin 322 are high, so no current flows between them. Similarly, at time t₄, both the first DQS pin 312 and the second DQS pin 322 are low, so no current flows between them. However, for times between approximately t₂ and approximately t₃, the first DQS pin 312 is low while the second DQS pin 322 is high. Under such conditions, a current will flow between the two DQS pins 312, 322. This condition in which one DQS pin is low while the other DQS pin is high can occur for fractions of a second (e.g., 0.8 nanoseconds) during the dynamic random-access memory (DRAM) read cycle. During such conditions, the current flowing between the two DQS pins 312, 322 can be substantial, resulting in heating of the memory devices 310, 320, and contributing to the degradation of reliability and eventual failure of these memory devices.

[0154] A second problem may also arise from tying together two output signal pins. FIG. 14 schematically

illustrates another exemplary memory module 305 in which a first DQS pin 312 of a first memory device 310 is electrically connected to a second DQS pin 322 of a second memory device 320. The two DQS pins 312, 322 of FIG. 14 are both electrically connected to a memory controller (not shown). The DQ (data input/output) pin 314 of the first memory device 310 and the corresponding DQ pin 324 of the second memory device 320 are each electrically connected to the memory controller by the DQ bus (not shown). Typically, each memory device 310, 320 will have a plurality of DQ pins (e.g., eight DQ pins per memory device), but for simplicity, FIG. 14 only shows one DQ pin for each memory device 310, 320.

[0155] Each of the memory devices 310, 320 of FIG. 14 utilizes a respective on-die termination or "ODT" circuit 332, 334 which has termination resistors (e.g., 75 ohms) internal to the memory devices 310, 320 to provide signal termination. Each memory device 310, 320 has a corresponding ODT signal pin 362, 364 which is electrically connected to the memory controller via an ODT bus 340. The ODT signal pin 362 of the first memory device 310 receives a signal from the ODT bus 340 and provides the signal to the ODT circuit 332 of the first memory device 310. The ODT circuit 332 responds to the signal by selectively enabling or disabling the internal termination resistors 352, 356 of the first memory device 310. This behavior is shown schematically in FIG. 14 by the switches 342, 344 which are either closed (dash-dot line) or opened (solid line). The ODT signal pin 364 of the second memory device 320 receives a signal from the ODT bus 340 and provides the signal to the ODT circuit 334 of the second memory device 320. The ODT circuit 334 responds to the signal by selectively enabling or disabling the internal termination resistors 354, 358 of the second memory device 320. This behavior is shown schematically in FIG. 14 by the switches 346, 348 which are either closed (dash-dot line) or opened (solid line). The switches 342, 344, 346, 348 of FIG. 14 are schematic representations of the operation of the ODT circuits 332, 334, and do not signify that the ODT circuits 332, 334 necessarily include mechanical switches.

[0156] Examples of memory devices 310, 320 which include such ODT circuits 332, 334 include, but are not limited to, DDR2 memory devices. Such memory devices are configured to selectively enable or disable the termination of the memory device in this way in response to signals applied to the ODT signal pin of the memory device. For example, when the ODT signal pin 362 of the first memory device 310 is pulled high, the termination resistors 352, 356 of the first memory device 310 are enabled. When the ODT signal pin 362 of the first memory device 310 is pulled low (e.g., grounded), the termination resistors 352, 356 of the first memory device 310 are disabled. By selectively disabling the termination resistors of an active memory device, while leaving the termination resistors of inactive memory devices enabled, such configurations advantageously preserve signal strength on the active memory device while continuing to eliminate signal reflections at the bus-die interface of the inactive memory devices.

[0157] In certain configurations, as schematically illustrated by FIG. 14, the DQS pins 312, 322 of each memory device 310, 320 are selectively connected to a voltage VTT through a corresponding termination resistor 352, 354 internal to the corresponding memory device 310, 320. Similarly,

in certain configurations, as schematically illustrated by FIG. 14, the DQ pins 314, 324 are selectively connected to a voltage VTT through a corresponding termination resistor 356, 358 internal to the corresponding memory device 310, 320. In certain configurations, rather than being connected to a voltage VTT, the DQ pins 314, 324 and/or the DQS pins 312, 322 are selectively connected to ground through the corresponding termination resistors 352, 354, 356, 358. The resistances of the internal termination resistors 352, 354, 356, 358 are selected to clamp the voltages so as to reduce the signal reflections from the corresponding pins. In the configuration schematically illustrated by FIG. 14, each internal termination resistor 352, 354, 356, 358 has a resistance of approximately 75 ohms.

[0158] When connecting the first memory device 310 and the second memory device 320 together to form a double word width, both the first memory device 310 and the second memory device 320 are enabled at the same time (e.g., by a common CS signal). Connecting the first memory device 310 and the second memory device 320 by tying the DQS pins 312, 322 together, as shown in FIG. 14, results in a reduced effective termination resistance for the DQS pins 312, 322. For example, for the exemplary configuration of FIG. 14, the effective termination resistance for the DQS pins 312, 322 is approximately 37.5 ohms, which is one-half the desired ODT resistance (for 75-ohm internal termination resistors) to reduce signal reflections since the internal termination resistors 352, 354 of the two memory devices 310, 320 are connected in parallel. This reduction in the termination resistance can result in signal reflections causing the memory device to malfunction.

[0159] FIG. 15 schematically illustrates an exemplary memory module 400 in accordance with certain embodiments described herein. The memory module 400 comprises a first memory device 410 having a first data strobe (DQS) pin 412 and a second memory device 420 having a second data strobe (DQS) pin 422. The memory module 400 further comprises a first resistor 430 electrically coupled to the first DQS pin 412. The memory module 400 further comprises a second resistor 440 electrically coupled to the second DQS pin 422 and to the first resistor 430. The first DQS pin 412 is electrically coupled to the second DQS pin 422 through the first resistor 430 and through the second resistor 440.

[0160] In certain embodiments, the memory module 400 is a 1-GB unbuffered Double Data Rate (DDR) Synchronous Dynamic RAM (SDRAM) high-density dual in-line memory module (DIMM). FIGS. 16A and 16B schematically illustrate a first side 462 and a second side 464, respectively, of such a memory module 400 with eighteen 64Mx4-bit, DDR-1 SDRAM FBGA memory devices on each side of a 184-pin glass-epoxy printed circuit board (PCB) 460. In certain embodiments, the memory module 400 further comprises a phase-lock-loop (PLL) clock driver 470, an EEPROM for serial-presence detect (SPD) data 480, and decoupling capacitors (not shown) mounted on the PCB in parallel to suppress switching noise on VDD and VDDQ power supply for DDR-1 SDRAM. By using synchronous design, such memory modules 400 allow precise control of data transfer between the memory module 400 and the system controller. Data transfer can take place on both edges of the DQS signal at various operating frequencies and

programming latencies. Therefore, certain such memory modules 400 are suitable for a variety of high-performance system applications.

[0161] In certain embodiments, the memory module 400 comprises a plurality of memory devices configured in pairs, each pair having a first memory device 410 and a second memory device 420. For example, in certain embodiments, a 128Mx72-bit DDR SDRAM high-density memory module 400 comprises thirty-six 64Mx4-bit DDR-1 SDRAM integrated circuits in FBGA packages configured in eighteen pairs. The first memory device 410 of each pair has the first DQS pin 412 electrically coupled to the second DQS pin 422 of the second memory device 420 of the pair. In addition, the first DQS pin 412 and the second DQS pin 422 are concurrently active when the first memory device 410 and the second memory device 420 are concurrently enabled.

[0162] In certain embodiments, the first resistor 430 and the second resistor 440 each has a resistance advantageously selected to reduce the current flow between the first DQS pin 412 and the second DQS pin 422 while allowing signals to propagate between the memory controller and the DQS pins 412, 422. In certain embodiments, each of the first resistor 430 and the second resistor 440 has a resistance in a range between approximately 5 ohms and approximately 50 ohms. For example, in certain embodiments, each of the first resistor 430 and the second resistor 440 has a resistance of approximately 22 ohms. Other resistance values for the first resistor 430 and the second resistor 440 are also compatible with embodiments described herein. In certain embodiments, the first resistor 430 comprises a single resistor, while in other embodiments, the first resistor 430 comprises a plurality of resistors electrically coupled together in series and/or in parallel. Similarly, in certain embodiments, the second resistor 440 comprises a single resistor, while in other embodiments, the second resistor 440 comprises a plurality of resistors electrically coupled together in series and/or in parallel.

[0163] FIGS. 17A and 17B schematically illustrate an exemplary embodiment of a memory module 400 in which the first resistor 430 and the second resistor 440 are used to reduce the current flow between the first DQS pin 412 and the second DQS pin 422. As schematically illustrated by FIG. 17A, the memory module 400 is part of a computer system 500 having a memory controller 510. The first resistor 430 has a resistance of approximately 22 ohms and the second resistor 440 has a resistance of approximately 22 ohms. The first resistor 430 and the second resistor 440 are electrically coupled in parallel to the memory controller 510 through a signal line 520 having a resistance of approximately 25 ohms. The first resistor 430 and the second resistor 440 are also electrically coupled in parallel to a source of a fixed termination voltage (identified by VTT in FIGS. 17A and 17B) by a signal line 540 having a resistance of approximately 47 ohms. Such an embodiment can advantageously be used to allow two memory devices having lower bit widths (e.g., 4-bit) to behave as a single virtual memory device having a higher bit width (e.g., 8-bit).

[0164] FIG. 17B schematically illustrates exemplary current-limiting resistors 430, 440 in conjunction with the impedances of the memory devices 410, 420. During an exemplary portion of a data read operation, the memory controller 510 is in a high-impedance condition, the first

memory device 410 drives the first DQS pin 412 high (e.g., 2.7 volts), and the second memory device 420 drives the second DQS pin 422 low (e.g., 0 volts). The amount of time for which this condition occurs is approximated by the time between t_2 and t_3 of FIG. 13, which in certain embodiments is approximately twice the tDQSQ (data strobe edge to output data edge skew time, e.g., approximately 0.8 nanoseconds). At least a portion of this time in certain embodiments is caused by simultaneous switching output (SSO) effects.

[0165] In certain embodiments, as schematically illustrated by FIG. 17B, the DQS driver of the first memory device 410 has a driver impedance R_1 of approximately 17 ohms, and the DQS driver of the second memory device 420 has a driver impedance R_2 of approximately 17 ohms. Because the upper network of the first memory device 410 and the first resistor 430 (with a resistance R_3 of approximately 22 ohms) is approximately equal to the lower network of the second memory device 420 and the second resistor 440 (with a resistance R_4 of approximately 22 ohms), the voltage at the midpoint is approximately $0.5 \times (2.7 - 0) = 1.35$ volts, which equals VTT, such that the current flow across the 47-ohm resistor of FIG. 17B is approximately zero.

[0166] The voltage at the second DQS pin 422 in FIG. 17B is given by $V_{DQS2} = 2.7 \times R_4 / (R_1 + R_2 + R_3 + R_4) = 0.59$ volts and the current flowing through the second DQS pin 422 is given by $I_{DQS2} = 0.59 / R_4 = 34$ milliamps. The power dissipation in the DQS driver of the second memory device 420 is thus $P_{DQS2} = 34 \text{ mA} \times 0.59 \text{ V} = 20$ milliwatts. In contrast, without the first resistor 430 and the second resistor 440, only the 17-ohm impedances of the two memory devices 410, 420 would limit the current flow between the two DQS pins 412, 422, and the power dissipation in the DQS driver of the second memory device 420 would be approximately 107 milliwatts. Therefore, the first resistor 430 and the second resistor 440 of FIGS. 17A and 17B advantageously limit the current flowing between the two memory devices during the time that the DQS pin of one memory device is driven high and the DQS pin of the other memory device is driven low.

[0167] In certain embodiments in which there is overshoot or undershoot of the voltages, the amount of current flow can be higher than those expected for nominal voltage values. Therefore, in certain embodiments, the resistances of the first resistor 430 and the second resistor 440 are advantageously selected to account for such overshoot/undershoot of voltages.

[0168] For certain such embodiments in which the voltage at the second DQS pin 422 is $V_{DQS2} = 0.59$ volts and the duration of the overdrive condition is approximately 0.8 nanoseconds at maximum, the total surge is approximately $0.59 \text{ V} \times 1.2 \text{ ns} = 0.3 \text{ V}\cdot\text{ns}$. For comparison, the JEDEC standard for overshoot/undershoot is 2.4 V·ns, so certain embodiments described herein advantageously keep the total surge within predetermined standards (e.g., JEDEC standards).

[0169] FIG. 18 schematically illustrates another exemplary memory module 600 compatible with certain embodiments described herein. The memory module 600 comprises a termination bus 605. The memory module 600 further comprises a first memory device 610 having a first data strobe pin 612, a first termination signal pin 614 electrically

coupled to the termination bus 605, a first termination circuit 616, and at least one data pin 618. The first termination circuit 616 selectively electrically terminating the first data strobe pin 612 and the first data pin 618 in response to a first signal received by the first termination signal pin 614 from the termination bus 605. The memory module 600 further comprises a second memory device 620 having a second data strobe pin 622 electrically coupled to the first data strobe pin 612, a second termination signal pin 624, a second termination circuit 626, and at least one data pin 628. The second termination signal pin 624 is electrically coupled to a voltage, wherein the second termination circuit 626 is responsive to the voltage by not terminating the second data strobe pin 622 or the second data pin 628. The memory module 600 further comprises at least one termination assembly 630 having a third termination signal pin 634, a third termination circuit 636, and at least one termination pin 638 electrically coupled to the data pin 628 of the second memory device 620. The third termination signal pin 634 is electrically coupled to the termination bus 605. The third termination circuit 636 selectively electrically terminates the data pin 628 of the second memory device 620 through the termination pin 638 in response to a second signal received by the third termination signal pin 634 from the termination bus 605.

[0170] FIG. 19 schematically illustrates a particular embodiment of the memory module 600 schematically illustrated by FIG. 18. The memory module 600 comprises an on-die termination (ODT) bus 605. The memory module 600 comprises a first memory device 610 having a first data strobe (DQS) pin 612, a first ODT signal pin 614 electrically coupled to the ODT bus 605, a first ODT circuit 616, and at least one data (DQ) pin 618. The first ODT circuit 616 selectively electrically terminates the first DQS pin 612 and the DQ pin 618 of the first memory device 610 in response to an ODT signal received by the first ODT signal pin 614 from the ODT bus 605. This behavior of the first ODT circuit 616 is schematically illustrated in FIG. 14 by the switches 672, 676 which are selectively closed (dash-dot line) or opened (solid line).

[0171] The memory module 600 further comprises a second memory device 620 having a second DQS pin 622 electrically coupled to the first DQS pin 612, a second ODT signal pin 624, a second ODT circuit 626, and at least one DQ pin 628. The first DQS pin 612 and the second DQS pin 622 are concurrently active when the first memory device 610 and the second memory device 620 are concurrently enabled. The second ODT signal pin 624 is electrically coupled to a voltage (e.g., ground), wherein the second ODT circuit 626 is responsive to the voltage by not terminating the second DQS pin 622 or the second DQ pin 624. This behavior of the second ODT circuit 626 is schematically illustrated in FIG. 14 by the switches 674, 678 which are opened.

[0172] The memory module 600 further comprises at least one termination assembly 630 having a third ODT signal pin 634 electrically coupled to the ODT bus 605, a third ODT circuit 636, and at least one termination pin 638 electrically coupled to the DQ pin 628 of the second memory device 620. The third ODT circuit 636 selectively electrically terminates the DQ pin 628 of the second memory device 620 through the termination pin 638 in response to an ODT signal received by the third ODT signal pin 634 from the

ODT bus 605. This behavior of the third ODT circuit 636 is schematically illustrated in FIG. 19 by the switch 680 which is either closed (dash-dot line) or opened (solid line).

[0173] In certain embodiments, the termination assembly 630 comprises discrete electrical components which are surface-mounted or embedded on the printed-circuit board of the memory module 600. In certain other embodiments, the termination assembly 630 comprises an integrated circuit mounted on the printed-circuit board of the memory module 600. Persons skilled in the art can provide a termination assembly 630 in accordance with embodiments described herein.

[0174] Certain embodiments of the memory module 600 schematically illustrated by FIG. 19 advantageously avoid the problem schematically illustrated by FIG. 12 of electrically connecting the internal termination resistances of the DQS pins of the two memory devices in parallel. As described above in relation to FIG. 14, FIGS. 18 and 19 only show one DQ pin for each memory device for simplicity. Other embodiments have a plurality of DQ pins for each memory device. In certain embodiments, each of the first ODT circuit 616, the second ODT circuit 626, and the third ODT circuit 636 are responsive to a high voltage or signal level by enabling the corresponding termination resistors and are responsive to a low voltage or signal level (e.g., ground) by disabling the corresponding termination resistors. In other embodiments, each of the first ODT circuit 616, the second ODT circuit 626, and the third ODT circuit 636 are responsive to a high voltage or signal level by disabling the corresponding termination resistors and are responsive to a low voltage or signal level (e.g., ground) by enabling the corresponding termination resistors. Furthermore, the switches 672, 674, 676, 678, 680 of FIG. 18 are schematic representations of the enabling and disabling operation of the ODT circuits 616, 626, 636 and do not signify that the ODT circuits 616, 626, 636 necessarily include mechanical switches.

[0175] The first ODT signal pin 614 of the first memory device 610 receives an ODT signal from the ODT bus 605. In response to this ODT signal, the first ODT circuit 616 selectively enables or disables the termination resistance for both the first DQS pin 612 and the DQ pin 618 of the first memory device 610. The second ODT signal pin 624 of the second memory device 620 is tied (e.g., directly hard-wired) to the voltage (e.g., ground), thereby disabling the internal termination resistors 654, 658 on the second DQS pin 622 and the second DQ pin 628, respectively, of the second memory device 620 (schematically shown by open switches 674, 678 in FIG. 19). The second DQS pin 622 is electrically coupled to the first DQS pin 612, so the termination resistance for both the first DQS pin 612 and the second DQS pin 622 is provided by the termination resistor 652 internal to the first memory device 610.

[0176] The termination resistor 656 of the DQ pin 618 of the first memory device 610 is enabled or disabled by the ODT signal received by the first ODT signal pin 614 of the first memory device 610 from the ODT bus 605. The termination resistance of the DQ pin 628 of the second memory device 620 is enabled or disabled by the ODT signal received by the third ODT signal pin 634 of the termination assembly 630 which is external to the second memory device 620. Thus, in certain embodiments, the first

ODT signal pin 614 and the third ODT signal pin 634 receive the same ODT signal from the ODT bus 605, and the termination resistances for both the first memory device 610 and the second memory device 620 are selectively enabled or disabled in response thereto when these memory devices are concurrently enabled. In this way, certain embodiments of the memory module 600 schematically illustrated by FIG. 19 provides external or off-chip termination of the second memory device 620.

[0177] Certain embodiments of the memory module 600 schematically illustrated by FIG. 19 advantageously allow the use of two lower-cost readily-available 512-Mb DDR-2 SDRAM devices to provide the capabilities of a more expensive 1-GB DDR-2 SDRAM device. Certain such embodiments advantageously reduce the total cost of the resultant memory module 600.

[0178] Certain embodiments described herein advantageously increase the memory capacity or memory density per memory slot or socket on the system board of the computer system. Certain embodiments advantageously allow for higher memory capacity in systems with limited memory slots. Certain embodiments advantageously allow for flexibility in system board design by allowing the memory module 10 to be used with computer systems designed for different numbers of ranks (e.g., either with computer systems designed for two-rank memory modules or with computer systems designed for four-rank memory modules). Certain embodiments advantageously provide lower costs of board designs.

[0179] In certain embodiments, the memory density of a memory module is advantageously doubled by providing twice as many memory devices as would otherwise be provided. For example, pairs of lower-density memory devices can be substituted for individual higher-density memory devices to reduce costs or to increase performance. As another example, twice the number of memory devices can be used to produce a higher-density memory configuration of the memory module. Each of these examples can be limited by the number of chip select signals which are available from the memory controller or by the size of the memory devices. Certain embodiments described herein advantageously provide a logic mechanism to overcome such limitations.

[0180] Various embodiments of the present invention have been described above. Although this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting. Various modifications and applications may occur to those skilled in the art without departing from the true spirit and scope of the invention.

What is claimed is:

1. A memory module comprising:

- a plurality of memory devices, each memory device having a corresponding load; and
- a circuit electrically coupled to the plurality of memory devices and configured to be electrically coupled to a memory controller of a computer system, the circuit selectively isolating one or more of the loads of the memory devices from the computer system, the circuit comprising logic which translates between a system

memory domain of the computer system and a physical memory domain of the memory module.

2. The memory module of claim 1, wherein the circuit comprises a logic element selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

3. The memory module of claim 2, wherein the circuit further comprises one or more switches operatively coupled to the logic element to receive control signals from the logic element.

4. The memory module of claim 1, wherein the plurality of memory devices has a first number of memory devices, and the circuit selectively isolates the loads of a second number of the memory devices from the computer system, the second number less than the first number.

5. The memory module of claim 4, wherein the first number of memory devices are configured in a plurality of ranks, and wherein the second number of memory devices are configured as one rank of the plurality of ranks.

6. The memory module of claim 1, wherein the logic translates address signals in the system memory domain to address signals in the physical memory domain.

7. The memory module of claim 1, wherein the system memory domain is compatible with a first number of chip selects, and the physical memory domain is compatible with a second number of chip selects equal to twice the first number of chip selects.

8. The memory module of claim 1, wherein the system memory domain is compatible to a first memory density, and the physical memory domain is compatible with a second memory density equal to twice the first memory density.

9. The memory module of claim 1, wherein the plurality of memory devices are arranged in two or more ranks of memory devices.

10. The memory module of claim 9, wherein the circuit comprises a data path multiplexer/demultiplexer which isolates the ranks of memory devices from one another.

11. The memory module of claim 1, wherein the plurality of memory devices comprises a pair of memory devices, each memory device of the pair having a first memory capacity, the pair of memory devices configured to simulate a memory device having a second memory capacity equal to twice the first memory capacity.

12. The memory module of claim 1, wherein the memory module comprises a dual inline memory module.

13. The memory module of claim 12, wherein the memory module comprises a rank-buffered dual inline memory module.

14. The memory module of claim 1, wherein the memory module comprises a DDR 2 SDRAM memory array.

15. The memory module of claim 1, wherein the circuit selectively isolates the loads of the memory devices from the computer system in response to a command or address signal received from the computer system.

16. A method of using a memory module with a computer system, the method comprising:

providing a memory module having a plurality of memory devices and a circuit electrically coupled to the plurality of memory devices, the circuit configured to be electrically coupled to a computer system, each memory device having a corresponding load;

electrically coupling the memory module to the computer system; and

activating the circuit to selectively isolate at least one of the loads of the memory devices from the computer system and to translate between a system memory domain of the computer system and a physical memory domain of the memory module.

17. The method of claim 16, wherein the plurality of memory devices comprises a first number of memory devices and activating the circuit selectively isolates the loads of a second number of memory devices from the computer system, the second number less than the first number.

18. The method of claim 17, wherein the first number of memory devices are arranged in two or more ranks of memory devices, and wherein the second number of memory devices are arranged as one rank of the two or more ranks.

19. The method of claim 18, wherein the circuit comprises a data path multiplexer/demultiplexer which isolates the ranks of memory devices from one another.

20. The method of claim 16, wherein the plurality of memory devices comprises a pair of memory devices, each memory device of the pair having a first memory capacity, the pair of memory devices configured to simulate a memory device having a second memory capacity equal to twice the first memory capacity.

21. The method of claim 16, wherein the logic translates address signals in the system memory domain to address signals in the physical memory domain.

22. The method of claim 16, wherein the system memory domain is compatible with a first number of chip selects, and the physical memory domain is compatible with a second number of chip selects equal to twice the first number of chip selects.

23. The method of claim 16, wherein the system memory domain is compatible to a first memory density, and the physical memory domain is compatible with a second memory density equal to twice the first memory density.

24. The method of claim 16, wherein the memory module comprises a dual inline memory module.

25. A memory module connectable to a computer system, the memory module comprising:

a first memory device having a first data signal line and a first data strobe signal line;

a second memory device having a second data signal line and a second data strobe signal line;

a common data signal line connectable to the computer system; and

a device electrically coupled to the first data signal line, to the second data signal line, and to the common data signal line, the device selectively electrically coupling the first data signal line to the common data signal line and selectively electrically coupling the second data signal line to the common data signal line, the device comprising logic which translates between a system memory domain of the computer system and a physical memory domain of the memory module.

26. The memory module of claim 25, wherein the device comprises a logic element.

27. The memory module of claim 26, wherein the device further comprises one or more switches selectively electri-

cally coupling the first data signal line to the common data signal line and selectively electrically coupling the second data signal line to the common data signal line, the one or more switches operatively coupled to the logic element to receive control signals from the logic element.

28. The memory module of claim 26, wherein the logic element is selected from the group consisting of: an application-specific integrated circuit, a custom-programmable logic device, and a field-programmable gate array.

29. The memory module of claim 25, wherein the memory module further comprises a common data strobe signal line connectable to the computer system, the device electrically coupled to the first data strobe signal line, to the second data strobe signal line, and to the common data strobe signal line,

the device selectively electrically coupling the first data strobe signal line to the common data strobe signal line and selectively electrically coupling the second data strobe signal line to the common data strobe signal line.

30. The memory module of claim 29, wherein the device comprises a logic element and one or more switches selectively electrically coupling the first data strobe signal line to the common data strobe signal line and selectively electrically coupling the second data strobe signal line to the common data strobe signal line, the one or more switches operatively coupled to the logic element to receive control signals from the logic element.

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(19) **United States**(12) **Patent Application Publication**
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G06F 13/28 (2006.01)(52) **U.S. Cl.** 711/103(57) **ABSTRACT**

A NAND flash memory in which a command/address pin is separated from a data input/output pin. The NAND flash memory includes a memory cell array used for storing data, a command/address pin through which a command and an address are received for transmitting data in the memory cell array, and a data input/output pin through which data are transmitted in the memory cell array. The command/address pin is separated from the data input/output pin in the NAND flash memory. Data input/output speed is increased. Furthermore, the NAND flash memory can perform a bank interleaving operation with a minimal delay time.

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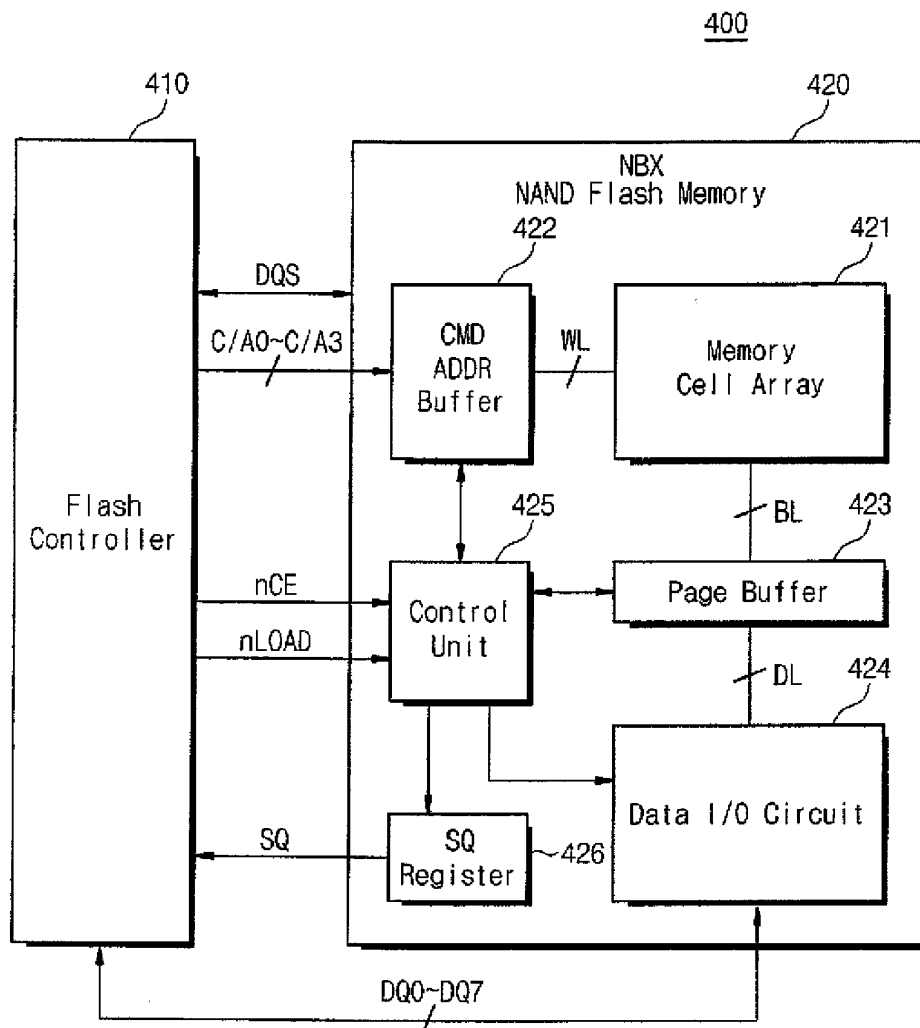


Fig. 1

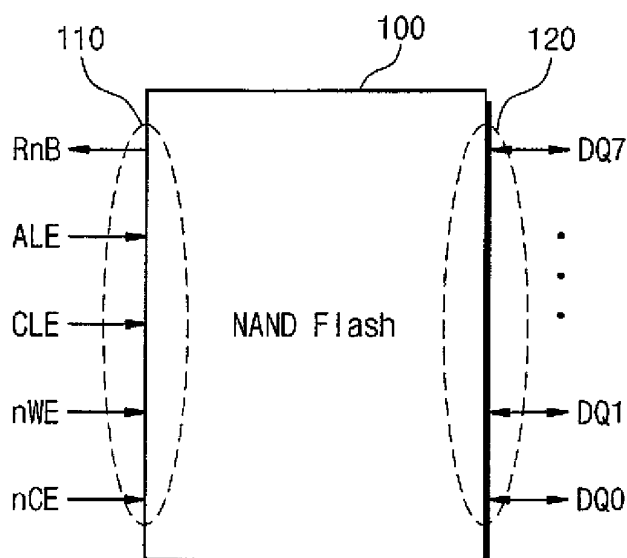


Fig. 2

Pin Name	Description
RnB	Ready, Busy Status Out
nRE	Read Enable
nCE	Chip Enable
Vcc	Power Supply
Vss	Ground
CLE	Command Latch Enable
ALE	Address Latch Enable
nWE	Write Enable
nWP	Write Protection
DQ0~DQ7	Data Input & Output

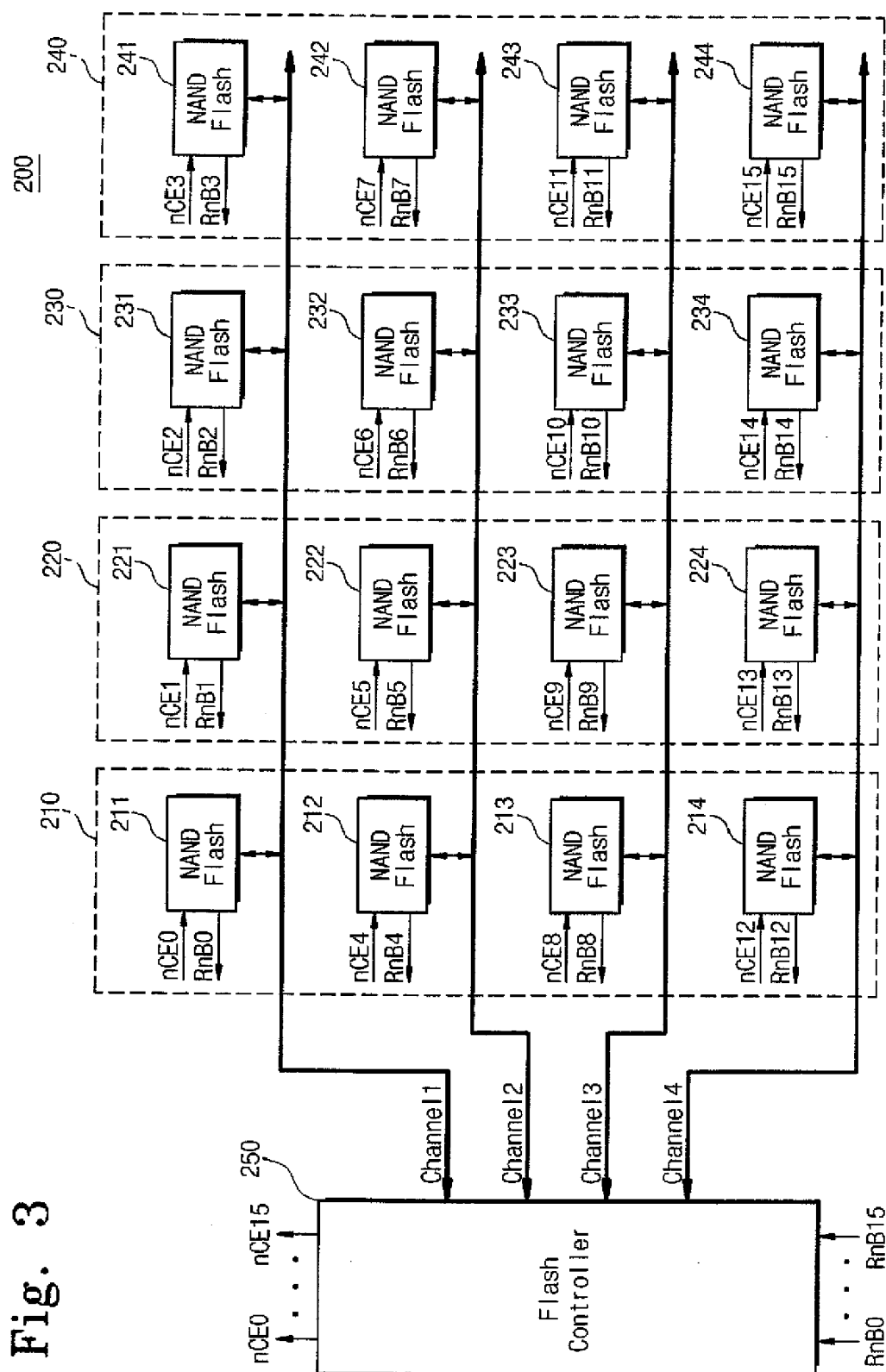


Fig. 4

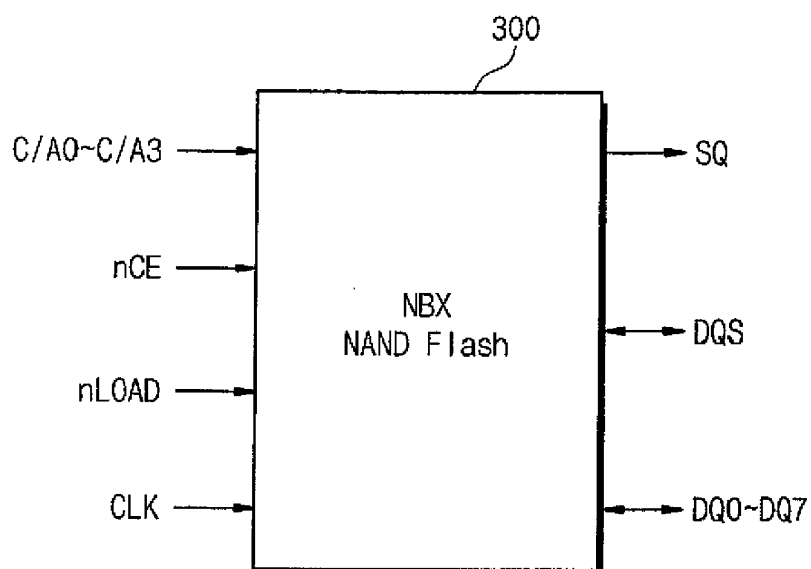


Fig. 5

Pin Name	Description
C/A0~C/A3	Command/Address Mux
nLOAD	Load Command or Address
nCE	Chip Enable
CLK	Clock
Vcc	Power Supply
Vss	Ground
SQ	Status Out
DQS	Strobe for Data I/O
DQ0~DQ7	Data Input & OutPut

Fig. 6

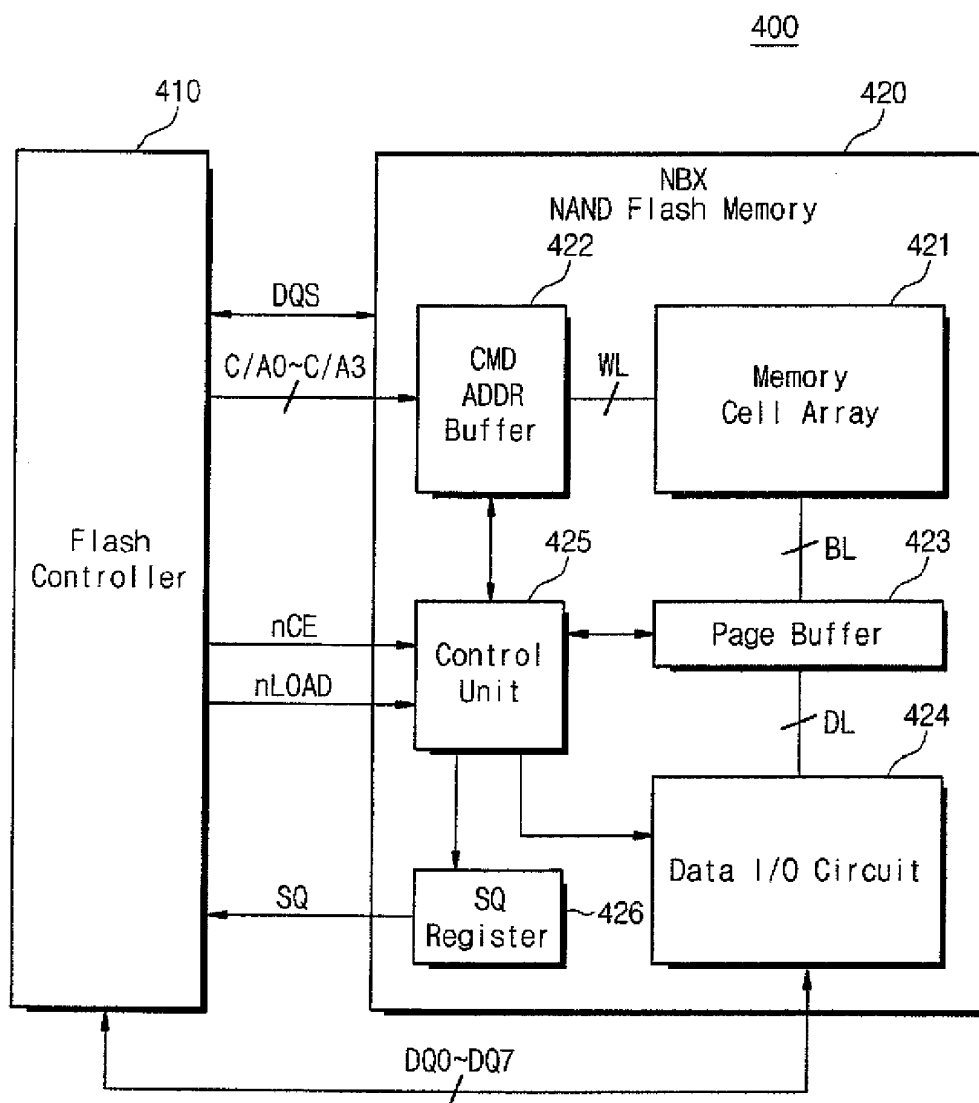


Fig. 7

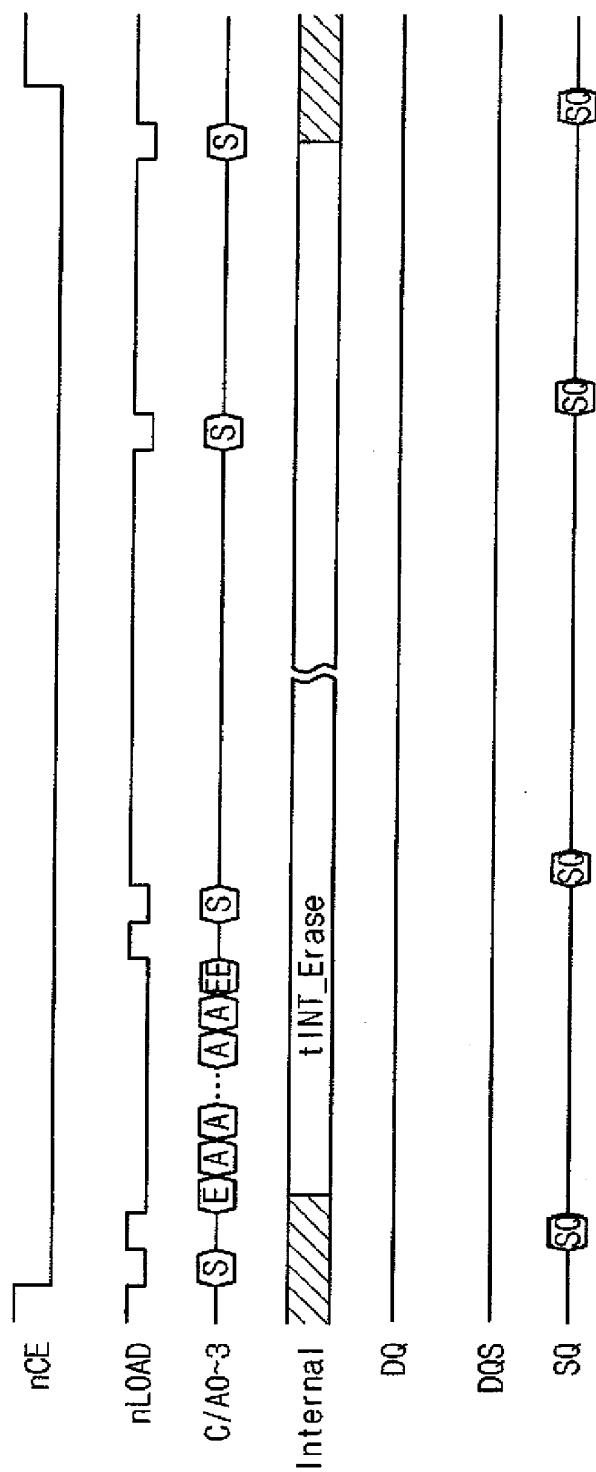


Fig. 8

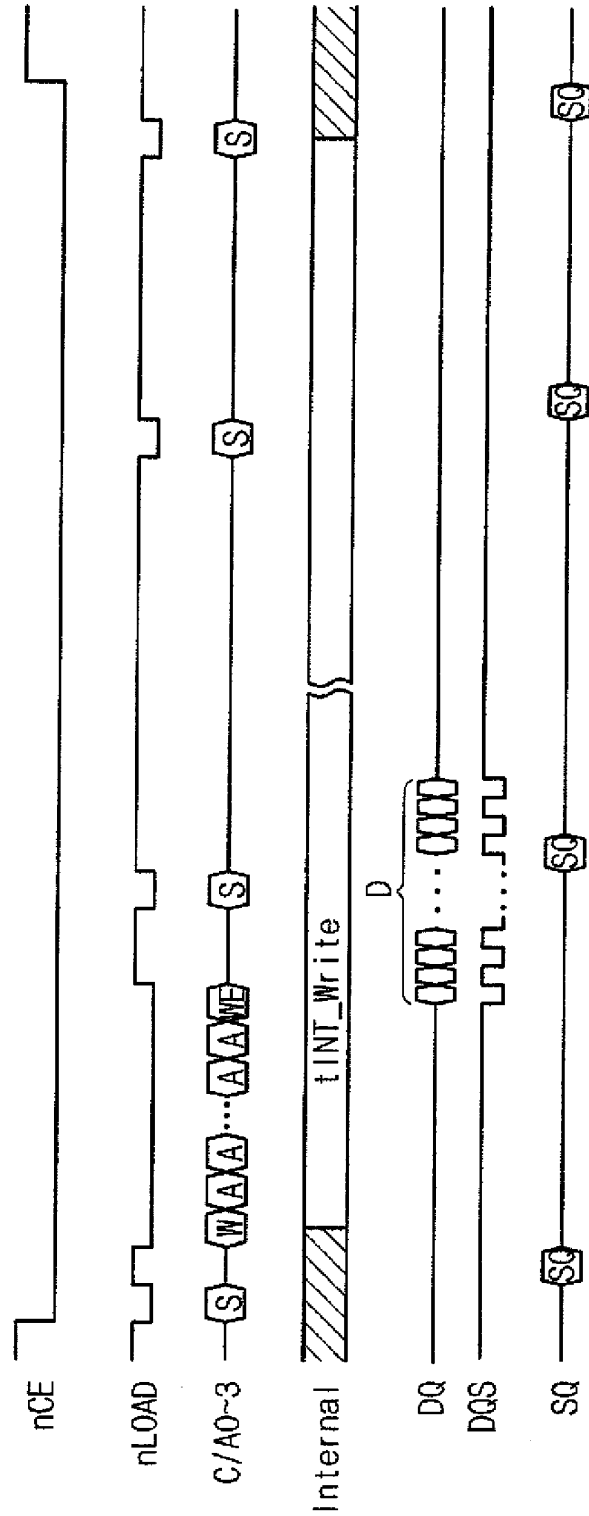


Fig. 9

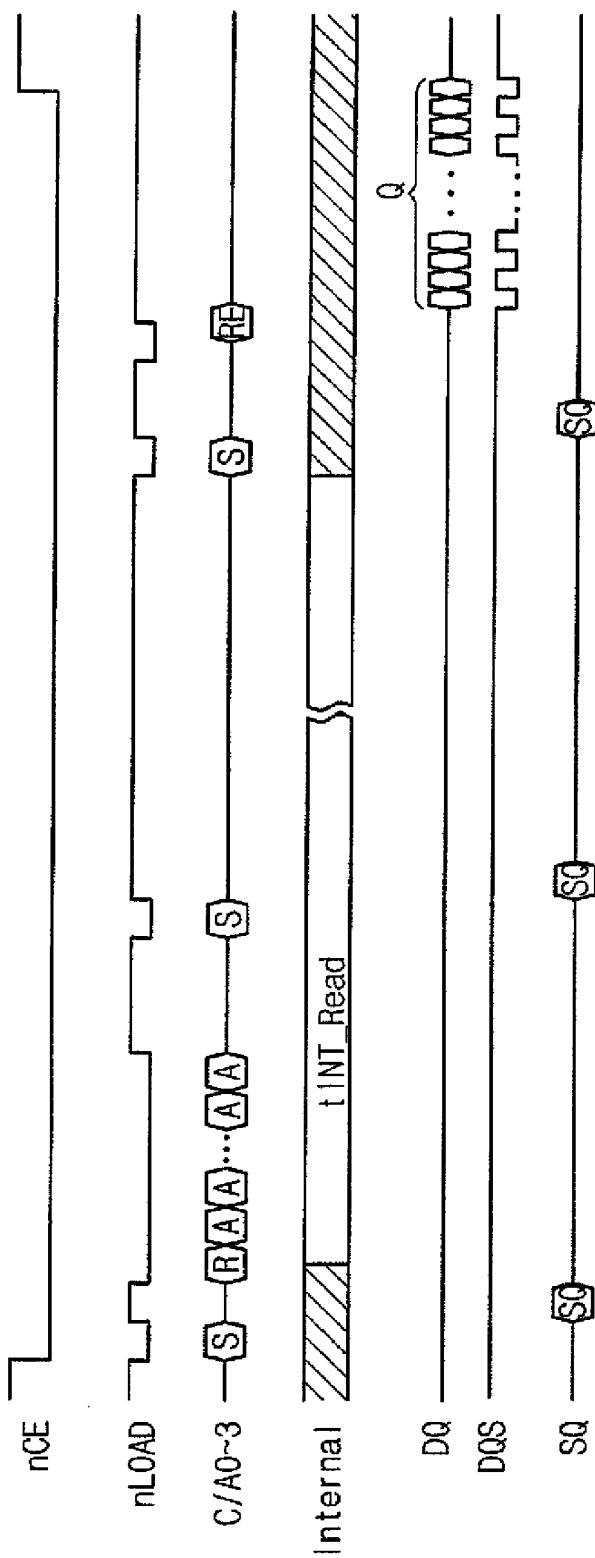


Fig. 10

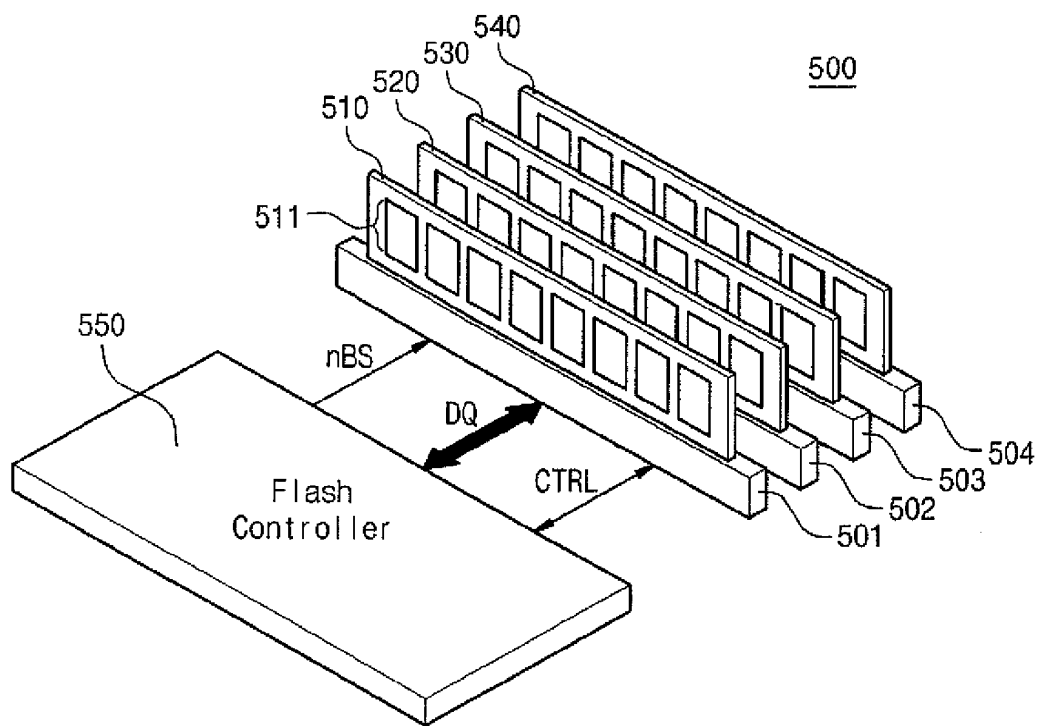


Fig. 11

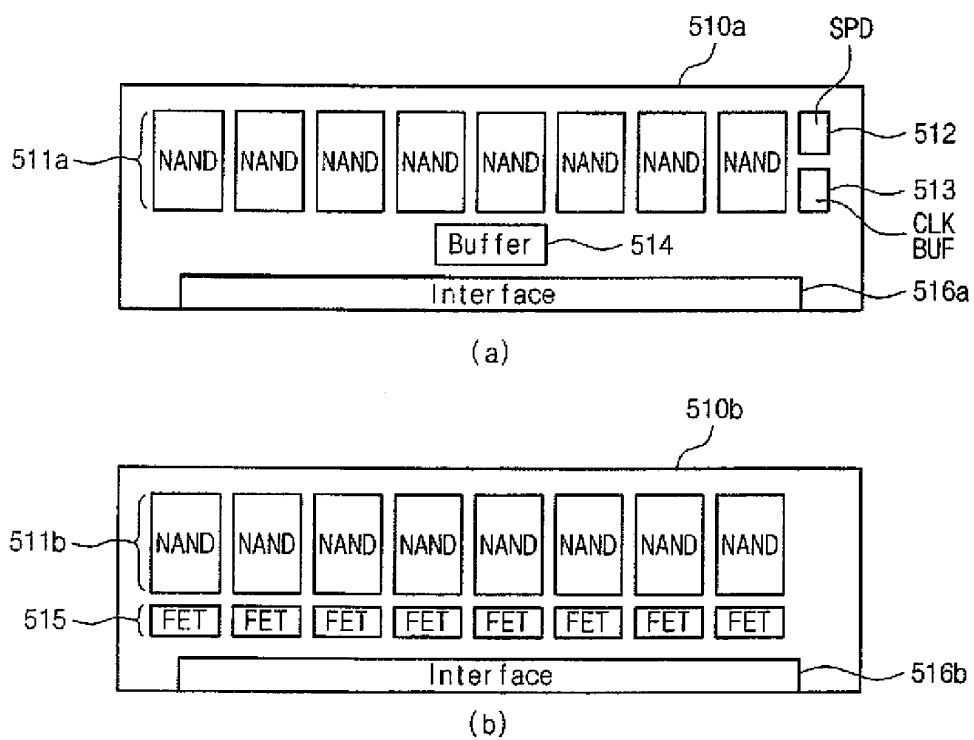


Fig. 12

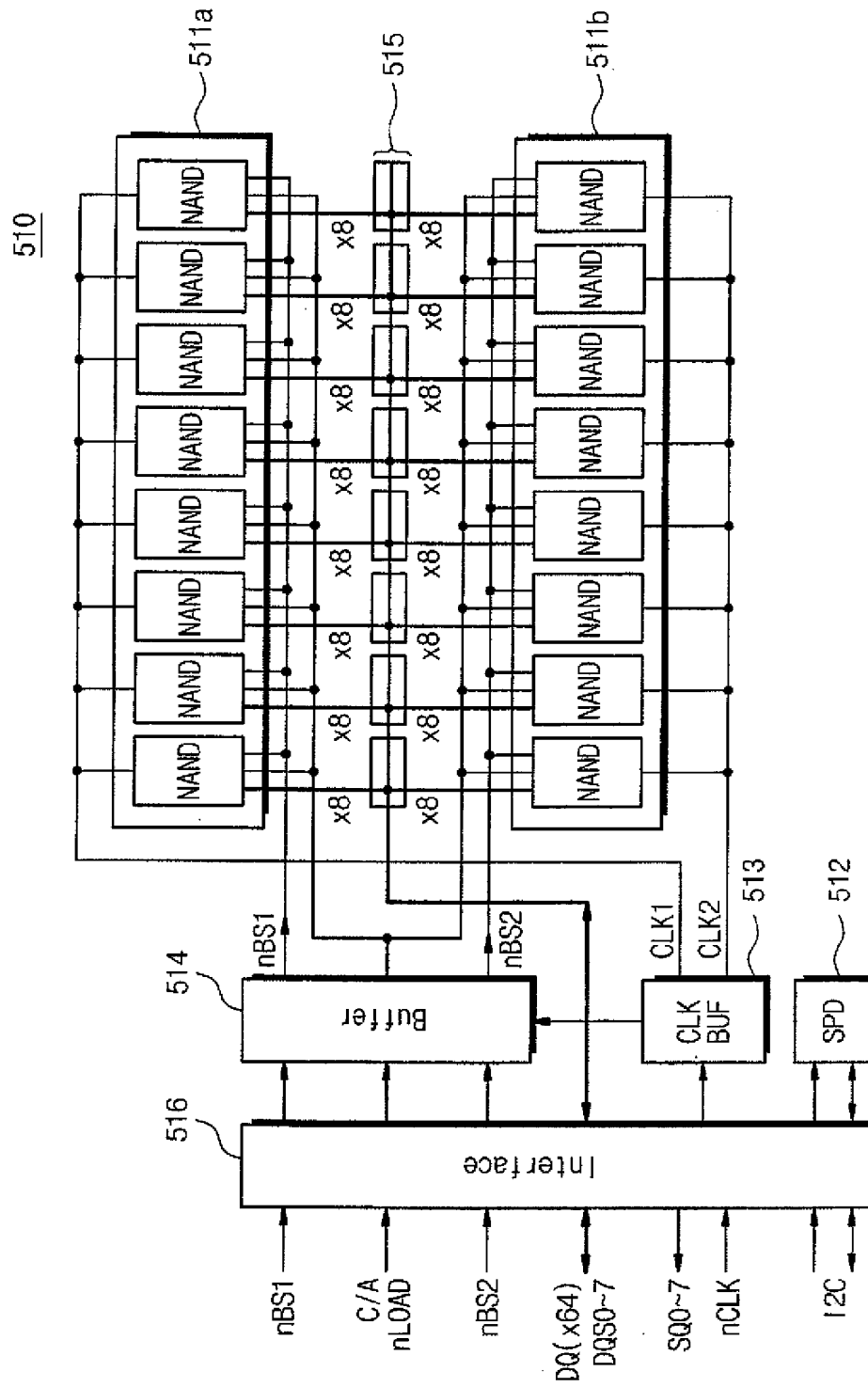


Fig. 13

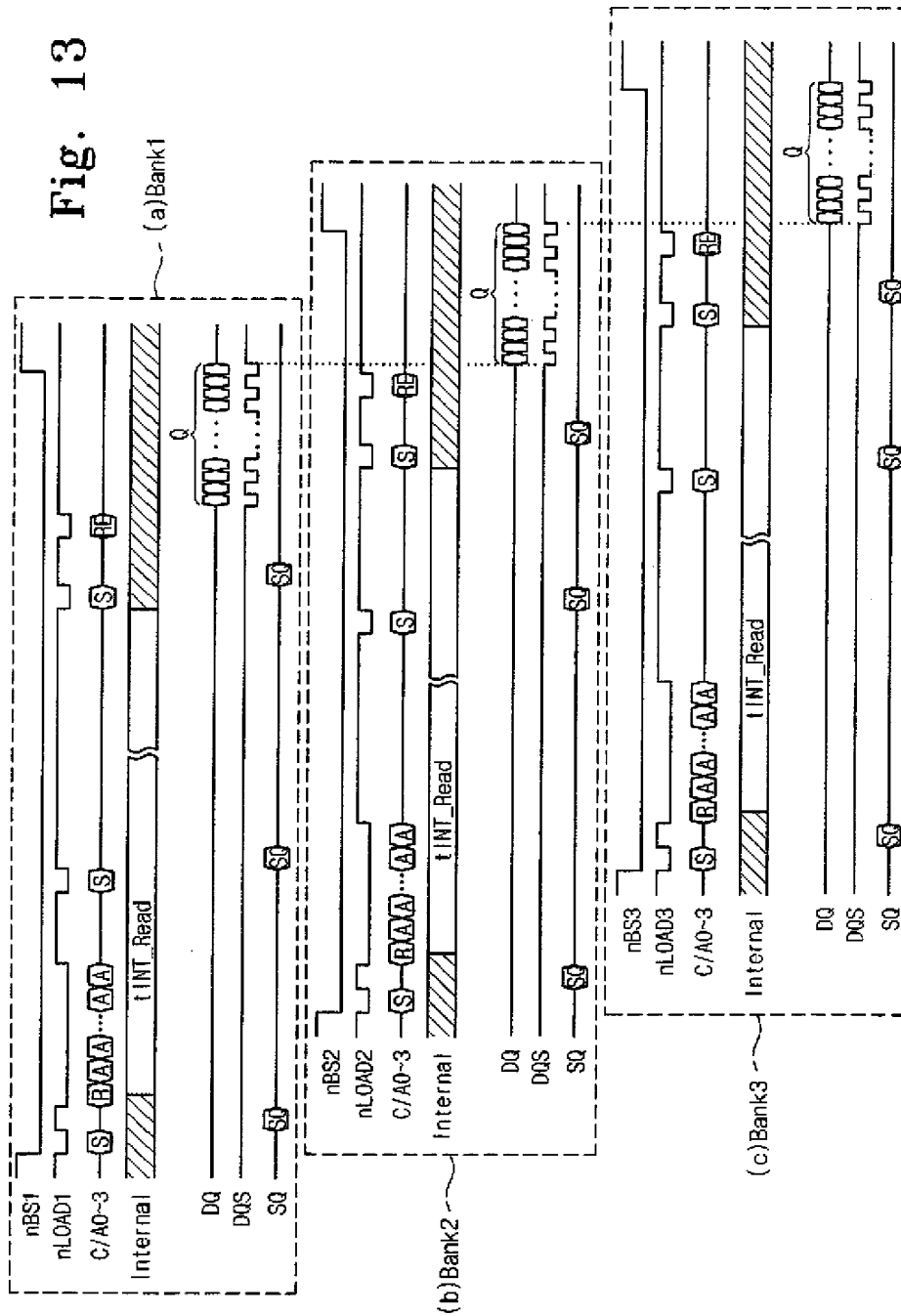


Fig. 14

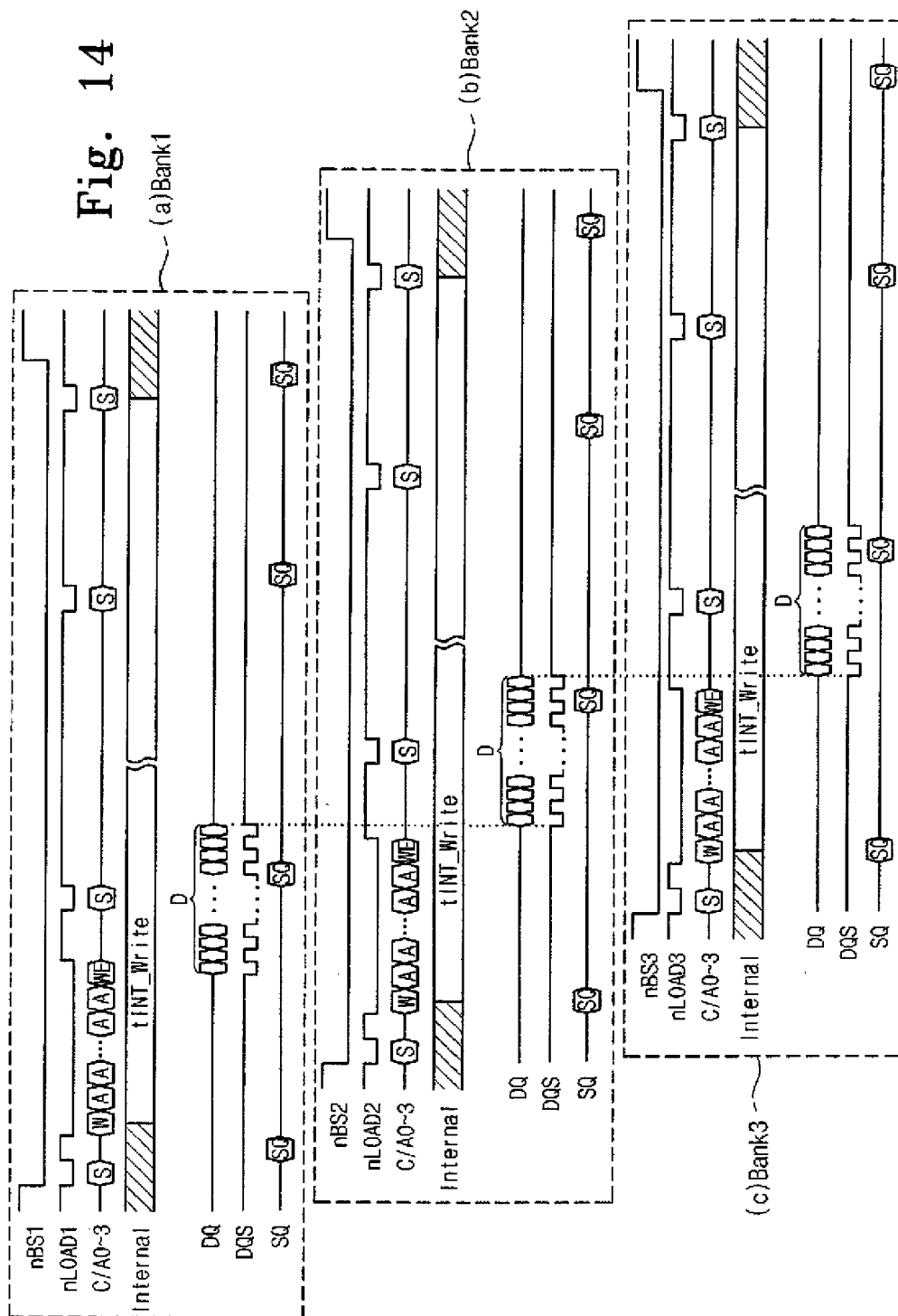


Fig. 15

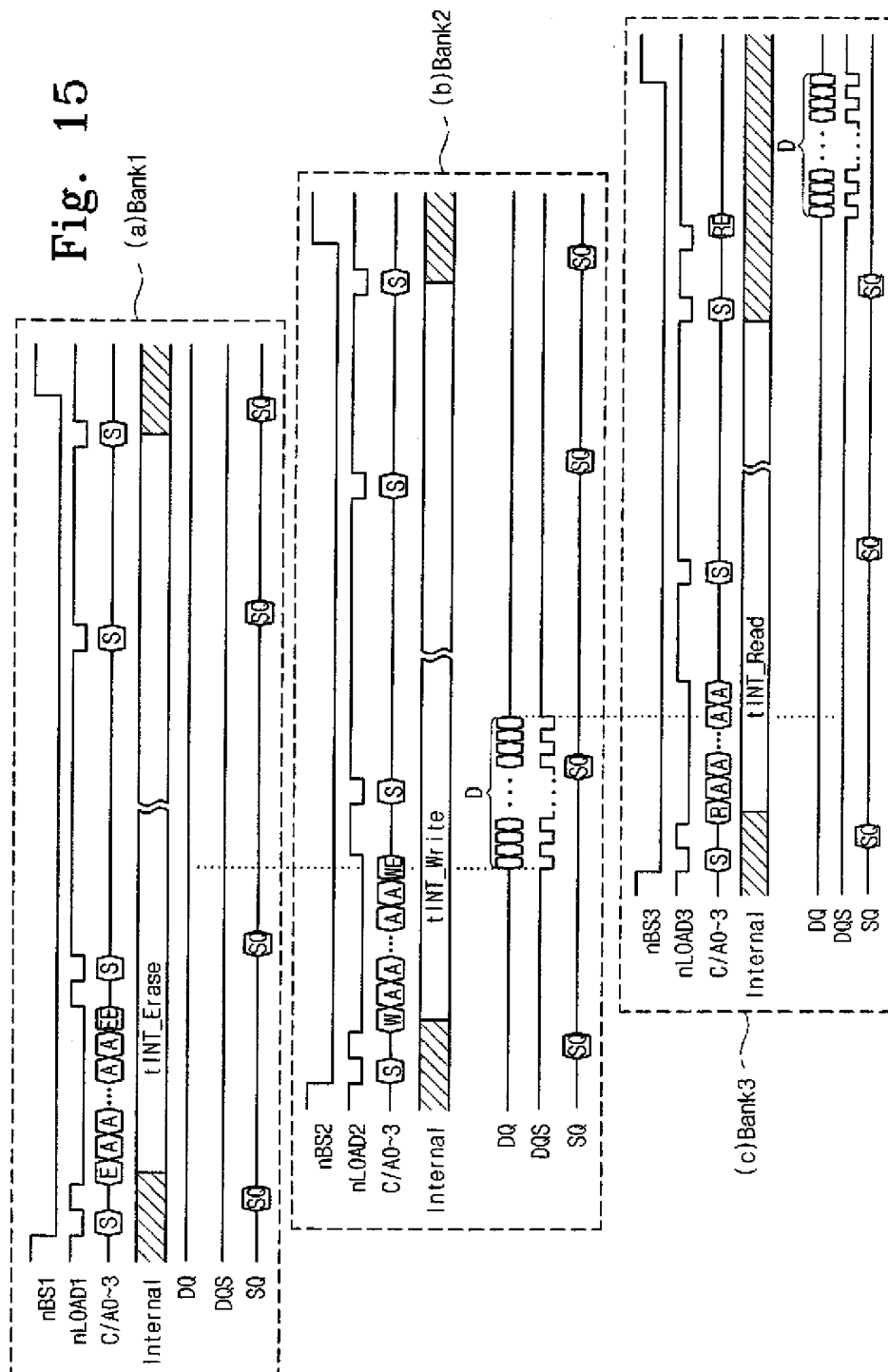


Fig. 16

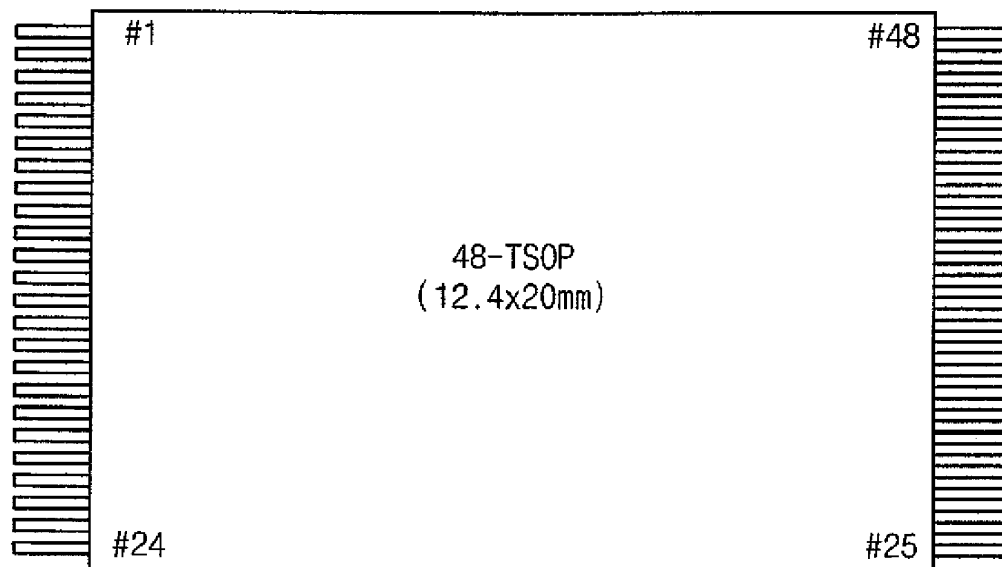


Fig. 17

Pin	Name	Description
#1	C/A3	Command/Address Mux
#2	C/A2	Command/Address Mux
#3	C/A1	Command/Address Mux
#4	C/A0	Command/Address Mux
#5	nLOAD	Load Command or Address
#6	nCE	Chip Enable
#7	CK	Clock
#8	VSS	Ground for Core
#9	NC	No Connection
#10	DQ7	Data Input & OutPut
#11	DQ6	Data Input & OutPut
#12	DQ5	Data Input & OutPut
#13	DQ4	Data Input & OutPut
#14	VDDQ	Power Supply for DQ
#15	DQS	Strobe for Data Input & Output
#16	VSSQ	Ground for DQ
#17	DQ3	Data Input & OutPut
#18	DQ2	Data Input & OutPut
#19	DQ1	Data Input & OutPut
#20	DQ0	Data Input & OutPut
#21	VDD	Power Supply for DQ
#22	SQ	Status Out(Not DDR, Clock to Data Out)
#23	nCKE	Clock Enable(Clock Stop When High)
#24	WP	Write Protection

NAND FLASH MEMORY HAVING C/A PIN AND FLASH MEMORY SYSTEM INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 U.S.C. §119 to Korean Patent Application No. 10-2006-0137629, filed on Dec. 29, 2006, the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] 1. Technical Field

[0003] The present disclosure relates to a memory device, and more particularly, to a NAND flash memory having a command/address (C/A) pin and a flash memory system including the same.

[0004] 2. Discussion of the Related Art

[0005] Semiconductor memory devices are storage devices for storing data. Semiconductor memory devices can be classified as a random access memory (RAM) and a read only memory (ROM). A RAM is a volatile memory device that requires power to maintain stored data. A ROM is a nonvolatile memory device that can maintain stored data even when not powered.

[0006] Examples of RAMs include a dynamic RAM (DRAM) and a static RAM (SRAM). Examples of ROMs include a programmable ROM (PROM), an erasable programmable ROM (EPROM), an electrically erasable programmable ROM (EEPROM), and a flash memory. Examples of flash memories include a NOR flash memory and a NAND flash memory. NAND flash memories are widely used for mobile communication terminals, portable media players, digital cameras, and mobile storage media.

[0007] FIG. 1 illustrates a conventional NAND flash memory 100, and FIG. 2 is a table providing descriptions of pins of the NAND flash memory 100. Referring to FIG. 1, the NAND flash memory 100 includes control pins such as RnB, ALE, CLE, nWE, and nCE pins that are formed on a first surface 110 of the NAND flash memory 100. The NAND flash memory 100 further includes data pins such as DQ0 through DQ7 pins formed on a second surface 120.

[0008] The pin structure of the NAND flash memory 100 shown in FIG. 1 is a structure for a thin small outline package (TSOP). However, since the control pins are formed on one surface of the NAND flash memory 100, board structure for the NAND flash memory 100 is complicated. Furthermore, when a memory module is formed using a plurality of NAND flash memories having the pin structure illustrated in FIG. 1, the structure of a printed circuit board (PCB) for the memory module is complicated.

[0009] FIG. 3 is a block diagram illustrating a flash memory system 200 having a multi-bank architecture using the NAND flash memory 100 of FIG. 1. Referring to FIG. 3, the flash memory system 200 may include a flash controller 250, first bank 210, second bank 220, third bank 230, and fourth bank 240.

[0010] Each of the banks 210, 220, 230, and 240 includes four NAND flash memories. For example, the first bank 210 includes four NAND flash memories 211, 212, 213, and 214. The second bank 220 includes four NAND flash memories 221, 222, 223, and 224. The third bank 230 includes four NAND flash memories 231, 232, 233, and 234. The fourth

bank 240 includes four NAND flash memories 241, 242, 243, and 244. The flash controller 250 is connected to the banks 210, 220, 230, and 240 through four channels 1 through 4. Here, each of the channels 1 through 4 connects corresponding NAND flash memories of the banks 210, 220, 230, and 240. For example, channel 1 connects NAND flash memories 211, 221, 231, and 241 of the banks 210, 220, 230, and 240, respectively. Similarly, channel 2 connects NAND flash memories 212, 222, 232, and 242. Channel 3 connects NAND flash memories 213, 223, 233, and 243. Channel 4 connects NAND flash memories 214, 224, 234, and 244.

[0011] The controller 250 performs a bank interleaving operation using chip enable signals nCE0-nCEX (where X is a positive integer). In so doing, the controller 250 receives as many enable signals and read and busy signals RnB0-RnBX as there are NAND flash memories. As used herein, bank interleaving is a data reading or writing operation performed between banks of a memory system in which two or more banks share a common channel. For example, in a bank interleaving operation, the flash controller 250 reads data from and/or writes data to the NAND flash memories 211, 221, 231, and 241 connected to channel 1 while moving between the NAND flash memories 211, 221, 231, and 241.

[0012] As described above, as many chip enable signals nCE0 to nCEX and ready and busy signals RnB0 to RnBX are used as the number of flash memory chips for a bank interleaving operation. Therefore, when the flash memory system 200 uses all the four channels 1 to 4, sixteen chip enable signals nCE0 to nCE15 and sixteen read and busy signals RnB0 to RnB15 are used for a bank interleaving operation. Accordingly, the structure of the flash memory system 200 becomes more complicated as the numbers of banks and flash memory chips increase.

[0013] A conventional NAND flash memory receives address and command signals through a data input/output (DQ) pin. Therefore, when address and command signals are input to the NAND flash memory, data cannot be input to or output from the NAND flash memory. This results in a data delay time. Data input/output is especially delayed when a bank interleaving operation is performed.

[0014] Furthermore, when data is written to or read from a cell array of a conventional NAND flash memory, RnB signals are generated. In this case, a flash controller cannot perform any operation until the writing/reading operation is completed. This reduces the performance of a flash memory system.

SUMMARY OF THE INVENTION

[0015] Exemplary embodiments of the present invention provide a NAND flash memory having fast data input/output.

[0016] Exemplary embodiments of the present invention provide a flash memory system that performs a bank interleaving operation while minimizing delay time.

[0017] Exemplary embodiments of the present invention provide NAND flash memories including a memory cell array used for storing data, a command/address pin through which a command and an address are received for inputting data to or outputting data from the memory cell array, and a data input/output pin through which data are input to or output from the memory cell array.

[0018] In some exemplary embodiments, the NAND flash memory further includes a status register receiving a status read command through the command/address pin and providing an operational status of the NAND flash memory to a flash

controller. The flash controller sends the status read command to the NAND flash memory when the NAND flash memory operates, before the NAND flash memory operates, or after the NAND flash memory operates. The status register sends a status signal SQ to the flash controller to inform the flash controller whether the NAND flash memory is internally operational. The flash controller controls the internal operation of the NAND flash memory in response to the status signal SQ.

[0019] In some exemplary embodiments, the data is input/output through the data input/output pin depending upon the toggling of a data strobe signal DOS. The data is input/output through the data input/output pin by a DDR (double data rate) transmission method. The NAND flash memory further includes a command/address buffer receiving the command and address transmitted through command/address pin. The NAND flash memory further includes a control unit controlling the reception of the command and address. The control unit receives a chip enable signal nCE and a load signal nLOAD from a flash controller and controls the reception of the command and address.

[0020] In some exemplary embodiments of the present invention, there are provided flash memory systems including a flash controller and a flash memory module formed of a plurality of NAND flash memories. Each of the NAND flash memories includes a memory cell array used for storing data, a command/address pin through which a command and an address are received from the flash controller for inputting/outputting data to/from the memory cell array and a data input/output pin through which data are input to and output from the memory cell array.

[0021] In some exemplary embodiments, each of the NAND flash memories further includes a status register receiving a status read command through the command/address pin and providing an operational status of the NAND flash memory to the flash controller. The flash controller sends the status read command to the NAND flash memory when the NAND flash memory operates, before the NAND flash memory operates or after the NAND flash memory operates. The status register sends a status signal SQ to the flash controller to inform the flash controller whether the NAND flash memory is internally operational. The flash controller controls the internal operation of the NAND flash memory in response to the status signal SQ. The data input/output through the data input/output pin is performed according to a toggling of a data strobe signal DQS. The data input/output through the data input/output pin is performed by a DDR transmission method. Each of the NAND flash memories further includes a command/address buffer receiving the command and address transmitted through command/address pin. Each of the NAND flash memories further includes a control unit controlling the reception of the command and address. The control unit receives a chip enable signal nCE and a load signal nLOAD from the flash controller and controls the reception of the command and address.

BRIEF DESCRIPTION OF THE FIGURES

[0022] The above and other features of the present disclosure will become more apparent by describing in detail exemplary embodiments of the present invention with reference to the attached drawings in which:

[0023] FIG. 1 illustrates a conventional NAND flash memory;

[0024] FIG. 2 is a table illustrating pins of the NAND flash memory of FIG. 1;

[0025] FIG. 3 is a block diagram illustrating a flash memory system with a multi-bank architecture using the NAND flash memory of FIG. 1;

[0026] FIG. 4 illustrates an NBX NAND flash memory according to an exemplary embodiment of the present invention;

[0027] FIG. 5 is a table providing descriptions of pins of the NBX NAND flash memory of FIG. 4, according to an exemplary embodiment of the present invention;

[0028] FIG. 6 is a block diagram of an NBX NAND flash memory system illustrating the NBX NAND flash memory of FIG. 4, according to an exemplary embodiment of the present invention;

[0029] FIG. 7 is a timing diagram showing how data are erased from the NBX NAND flash memory of FIG. 6, according to an exemplary embodiment of the present invention;

[0030] FIG. 8 is a timing diagram for explaining how data are written into the NBX NAND flash memory of FIG. 6, according to an exemplary embodiment of the present invention;

[0031] FIG. 9 is a timing diagram for explaining how data are read from the NBX NAND flash memory of FIG. 6, according to an exemplary embodiment of the present invention;

[0032] FIG. 10 is a diagram illustrating an NBX NAND flash memory system according to an exemplary embodiment of the present invention;

[0033] FIG. 11 is a block diagram illustrating an NBX flash module of the NBX NAND flash memory system of FIG. 10, according to an exemplary embodiment of the present invention;

[0034] FIG. 12 is a block diagram illustrating an internal structure of the NBX flash module of FIG. 11, according to an exemplary embodiment of the present invention;

[0035] FIGS. 13 through 15 are timing diagrams illustrating a bank interleaving operation of the NBX flash memory system of FIG. 10, according to an exemplary embodiment of the present invention;

[0036] FIG. 16 illustrates a thin small outline package (TSOP) of an NBX NAND flash memory according to an exemplary embodiment of the present invention; and

[0037] FIG. 17 is a table illustrating pins of the TSOP of FIG. 16, according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0038] Exemplary embodiments of the present invention will be described below in more detail with reference to the accompanying drawings. The present invention may, however, be embodied in different forms and should not be construed as limited to the exemplary embodiments set forth herein.

[0039] FIG. 4 illustrates a NAND flash memory 300 according to an exemplary embodiment of the present invention, and FIG. 5 is a table illustrating pins of the NBX NAND flash memory 300 according to an exemplary embodiment of the present invention. The NAND flash memory 300 includes additional command/address pins C/A0 to C/A3. The NAND flash memory 300 further includes a data strobe pin DQS in addition to data input/output pins DQ0 to DQ7 for input/output data synchronization. Input signals of the NAND flash

memory 300 are synchronized with a clock (CLK) signal. The NAND flash memory 300 of FIG. 4 will now be referred to as a NAND bus scalable (NBX) NAND flash memory.

[0040] Referring to FIG. 5, the command/address pins C/A0 to C/A3 are used for inputting command and address signals. Although four command/address pins C/A0 to C/A3 are shown in FIG. 5, the number of command/address pins C/A0 to C/A3 can be increased. The nLOAD pin is used for loading a command or address signal to the NAND flash memory 300. The nCE pin is a chip enable pin. The CLK pin is a clock pin. The Vcc pin is a power voltage pin, and the Vss pin is a ground voltage pin.

[0041] The SQ pin indicates which operation is being performed or is completed in the NBX NAND flash memory 300. For example, a high SQ signal indicates that an operation is completed in the NBX NAND flash memory 300 and the NBX NAND flash memory 300 is ready for the next operation. A low SQ signal indicates that an operation is being performed in the NBX NAND flash memory 300. The SQ signal is output in synchronization with a CLK signal. As described above, the DQS pin is a data strobe pin used for data input/output, and the DQ0 to DQ7 pins are data input/output pins.

[0042] FIG. 6 is a block diagram of an example of the NBX NAND flash memory of FIG. 4, according to an exemplary embodiment of the present invention. The NBX NAND flash memory 420 is electrically connected to a flash controller

410. The NBX NAND flash memory 420 receives address and command signals from the flash controller 410 through a C/A pin and receives data from and sends data to the flash controller 410 through a data input/output pin.

[0043] Referring to FIG. 6, the NBX NAND flash memory 420 includes a memory cell array 421, a command address buffer 422, a page buffer 423, a data input/output circuit 424, a control unit 425, and an SQ register 426.

[0044] The memory cell array 421 includes a plurality of memory blocks (not shown). Each of the memory blocks includes a plurality of pages (not shown). Each of the pages includes a plurality of memory cells that share a word line. The size of each page may be 512 bytes, 2 KB, or 4 KB. The size of page varies according to the type of the flash memory. In the NBX NAND flash memory 420, data are erased in units of a block, and data are read and written in units of a page.

[0045] The command address buffer 422 is connected to the memory cell array 421 through a word line WL. The command address buffer 422 receives a command or address signal from the flash controller 410. The control unit 425 receives control signals nCE and nLOAD and controls the command address buffer 422.

[0046] The command signal is input to the NBX NAND flash memory 420 when the control signals nCE and nLOAD are low. Table 1 shows examples of command and control signals used in the NBX NAND flash memory system 400. In Table 1, nCKE denotes a clock enable signal.

TABLE 1

Command (4bits)	Code	nCKE	nCE	nLOAD	DQS	DQ	Clock Cycle
Chip Deselect	—	L	H	X	High-Z	High-Z	L → H
Clock Stop	—	H	X	X	High-Z	X	X(after n cycles)
Power On	—	H	H	X	High-Z	High-Z	Stead State After n cycles
Read	0h	L	L	L	High-Z	High-Z	L → H, 1 st cycle: Read in (after that address in)
Read Enable	1h	L	L	L	Toggle	Data out	L → H, 1 st cycle: Read enable (after that read data out)
Program (Write)	2h	L	L	L	High-Z	High-Z	L → H, 1 cycle: Program Command (after that address in)
Program Enable	3h	L	L	L	Toggle	Data in	L → H, 1 st cycle: command in (after that program the page)
Page Read Stop	4h	L	L		High-Z	High-Z	L → H, 1 cycle
Page Write Stop	5h	L	L		High-Z	High-Z	L → H, 1 cycle
ID Read	6h	L	L		Toggle	Data Out	L → H, 1 st cycle: cmd in
Register Configuration	7h	L	L		Toggle	Data In	L → H, 1 st cycle: cmd in (after that configuration for n cycles)
Status Read	8h	—	—		High-Z	High-Z	L → H, 1 st cycle: cmd in, 2 nd cycle: status data out
Reserved	9h-Eh						
Reset	Fh	L	L		High-Z	High-Z	After n cycles

[0047] Referring to Table 1, Read is a command for starting a read operation. Read Enable is a command for reading data. Write is a command for starting a write operation. Write Enable or Program Enable is a command for writing data into the memory cell array 421.

[0048] Status Read is a command for checking the operation status of the NBX NAND flash memory 420. Reset is a command for initializing the NBX NAND flash memory 420.

[0049] For example, a read command is input to the NBX NAND flash memory 420 in one clock cycle when nCE and nLOAD signals are low. Next, an address is input to the NBX NAND flash memory 420. Other commands are input to the NBX NAND flash memory 420 in the same manner.

[0050] Referring again to FIG. 6, the NBX NAND flash memory 420 includes the SQ register 426. The SQ register 426 stores the status of the NBX NAND flash memory 420. When a status read command is input, the SQ register 426 sends a status signal SQ to the flash controller 410 under the control of the control unit 425.

[0051] The status signal SQ is sent to the flash controller 410 to inform the flash controller 410 of the status of the NBX NAND flash memory 420. For example, when the NBX NAND flash memory 420 does not perform any operation, the status signal SQ is high, and when the NBX NAND flash memory 420 performs an operation, the status signal SQ is low. The status signal is generated in synchronization with a CLK signal. Referring to Table 1, after a status read command is input and one cycle passes, a status signal SQ is output. The flash controller 410 can provide the status signal SQ to the flash memory at anytime.

[0052] An address is input from the first clock cycle after a command is input. The bit size of the address is determined by an nLOAD signal. Erase, write, and read operations of the NBX NAND flash memory 420 of FIG. 6 will now be described with reference to FIGS. 7 through 9.

[0053] FIG. 7 is a timing diagram illustrating how data are erased from the NBX NAND flash memory 420 of FIG. 6, according to an exemplary embodiment of the present invention. Referring to FIG. 7, when an nCE signal and an nLOAD signal are low, a command or an address is input through a C/A pin.

[0054] Referring to FIG. 7, S denotes a status read command, E denotes an erase command, A denotes an address, and EE denotes an erase enable command. The status read command S can be input regardless of whether an erase operation is performed. When the status read command S is input, the NBX NAND flash memory 420 generates a status signal SQ after one clock cycle.

[0055] The NBX NAND flash memory 420 starts a read operation in response to an erase command. When an erase command is input, an address A is input through the C/A pin. After the address is completely input, an erase enable command EE is input. The NBX NAND flash memory 420 erases data from a memory block corresponding to the address A in response to the erase enable command EE.

[0056] During the erase operation, the NBX NAND flash memory 420 generates a status signal SQ in response to a status read command S to indicate its status (erase status). After the erase operation, the NBX NAND flash memory 420 generates a status signal SQ in response to a status read command S to indicate its status (end of erase operation).

[0057] FIG. 8 is a timing diagram for explaining how data are written into the NBX NAND flash memory 420 of FIG. 6, according to an exemplary embodiment of the present inven-

tion. Referring to FIG. 8, when an nCE signal and an nLOAD signal are low, a command or an address is input through a C/A pin.

[0058] In FIG. 8, W denotes a write command, A denotes an address, and WE denotes a write enable command. Here, a status read command S can be input regardless of whether a write operation is performed. When the status read command S is input, the NBX NAND flash memory 420 generates a status signal SQ after one clock cycle.

[0059] The NBX NAND flash memory 420 starts a write operation in response to a write command W. After a write command W and an address A are input, a write enable command WE is input. After that, the NBX NAND flash memory 420 receives data through a data input/output (DQ) pin according to the toggling of a data strobe (DQS) signal. Data are input to or output from the NBX NAND flash memory 420 in synchronization with the DQS signal. Referring to FIG. 8, data are transferred twice per one period of the DQS signal by a double data rate (DDR) transmission method.

[0060] After the data are received, the NBX NAND flash memory 420 performs programming on a page corresponding to the address A using the received data. During the write operation, the NBX NAND flash memory 420 generates a status signal SQ in response to a status read command S to indicate its status (write status). After the write operation, the NBX NAND flash memory 420 generates a status signal SQ in response to a status read command S to indicate its status (end of write operation).

[0061] FIG. 9 is a timing diagram for explaining how data are read from the NBX NAND flash memory 420 of FIG. 6, according to an exemplary embodiment of the present invention. In FIG. 9, R denotes a read command, A denotes an address, and RE denotes a read enable command. A status read command S can be input regardless of whether a read operation is performed. When the status read command S is input, the NBX NAND flash memory 420 generates a status signal SQ after one clock cycle.

[0062] The NBX NAND flash memory 420 starts a read operation in response to a read command R. After a read command R and an address A are input, the NBX NAND flash memory 420 reads data from a page corresponding to the address A and stores the data in the page buffer 423 (refer to FIG. 6). The NBX NAND flash memory 420 outputs a status signal SQ in response to a status read command S indicating that the NBX NAND flash memory 420 is ready for data transmission.

[0063] The NBX NAND flash memory 420 outputs the data stored in the page buffer 423 in response to a read enable command RE. The NBX NAND flash memory 420 outputs the data through a DQ pin according to the toggling of a DQS signal. Data are output from the NBX NAND flash memory 420 in synchronous with the DQS signal. Referring to FIG. 9, data are read from the NBX NAND flash memory 420 at double data rate (DDR).

[0064] As explained above, the NBX NAND flash memory of the present invention includes separate C/A and DQ pins. Furthermore, the NBX NAND flash memory includes the SQ register to generate a status signal SQ whenever the flash controller sends a status read command S. Since the CA pin and the DQ pin are separate, read and write speeds can be increased. Furthermore, command and address buses can have a width different from that of a data bus.

[0065] FIG. 10 is a diagram illustrating an NBX NAND flash memory system 500 according to an exemplary embodi-

ment of the present invention. Referring to FIG. 10, the NBX NAND flash memory system 500 includes a flash controller 550 and a plurality of NBX flash modules 510, 520, 530, and 540. In the exemplary embodiment of FIG. 10, four NBX flash modules are illustrated.

[0066] The NBX flash modules 510, 520, 530, and 540 are connected to flash sockets 501, 502, 503, and 504, respectively, and the flash sockets 501, 502, 503, and 504 are connected to the flash controller 550 through a data bus (DQ bus). The NBX flash modules 510, 520, 530, and 540 receive a bank selection (nBS) signal from the flash controller 550. Furthermore, data (DQ) and control (CTRL) signals are transmitted between the flash controller 550 and the NBX flash modules 510, 520, 530, and 540. Each of the NBX flash modules 510, 520, 530, and 540 includes a plurality of NBX NAND flash memories. In the exemplary embodiment shown in FIG. 10, each of the NBX flash modules 510, 520, 530, and 540 includes sixteen NBX NAND flash memories 511.

[0067] The flash controller 550 controls write, read, and erase operations of the NBX NAND flash memories of the NBX flash modules 510, 520, 530, and 540. Furthermore, the flash controller 550 performs bank interleaving between banks using the nBS signal.

[0068] When it is necessary to update SQ registers of the NBX NAND flash memories of the NBX flash modules 510, 520, 530, and 540, the flash controller 550 can transmit a register value that can be commonly used for all the NBX flash modules 510, 520, 530, and 540. This function of the flash controller 550 is referred to as a broadcasting function. After simultaneously enabling nBS signals for all banks, the flash controller 550 provides a register configuration command (refer to Table 1). The register configuration command is simultaneously provided to each of the NBX NAND flash memories through a C/A pin. Then, the flash controller 550 writes a register value to an SQ register using the C/A pin.

[0069] FIG. 11 is a block diagram illustrating an NBX flash module of the NBX NAND flash memory system 500 of FIG. 10, according to an exemplary embodiment of the present invention. In FIG. 11, reference numeral 510a denotes a front surface of the NBX flash module 510 of the NBX NAND flash memory system 500, and reference numeral 510b denotes a back surface of the NBX flash module 510.

[0070] Referring to FIGS. 11(a) and (b), eight NBX NAND flash memories 511a are formed in the front surface 510a of the NBX flash module 510, and eight NBX NAND flash memories 511b are formed on the back surface 510b of the NBX flash module 510. The eight NBX NAND flash memories 511a form one bank, and the eight NBX NAND flash memories 511b form another bank. For example, one NBX flash module includes two banks. When each of the NBX NAND flash memories 511a and 511b has an 8-bit bus width, one bank has a 64-bit bus width.

[0071] An electrically erasable programmable read only memory (EEPROM) 512 for status presence detection (SPD), a clock buffer 513, a buffer 514 for C/A and control signals, and an interface 516a are formed on the front surface 510a of the NBX flash module 510. The EEPROM 512 is used to store data (SPD data) necessary for the flash controller 550 (refer to FIG. 10) to access the NBX flash module 510. The clock buffer 513 is used for distributing a clock signal to the NBX NAND flash memories 511a and 511b. The buffer 514 is used for distributing command, address, and control signals to the NBX NAND flash memories 511a and 511b.

[0072] Fast enable transfer (PET) switches 515 and an interface 516b are formed on the back surface 510b of the NBX flash module 510. The FET switches 515 are used to reduce loads on data buses, thereby enabling high-speed data transmission. Since a DDR transmission method is used for the NBX NAND flash memories 511a and 511b, the FET switches 515 are used to facilitate the use of the DDR transmission method. Each of the FET switches 515 can latch 10 bits: 8 bits for data (DQ), 1 bit for a DQS signal, and 1 bit for a SQ signal. When assembled, the interfaces 516a and 516b of the NBX flash module 510 are connected to the flash socket 510 (refer to FIG. 10).

[0073] FIG. 12 is a block diagram illustrating an internal structure of the NBX flash module 510 of FIG. 11, according to an exemplary embodiment of the present invention. As illustrated in FIG. 11, the NBX flash module 510 includes the NBX NAND flash memories 511a and 511b (first and second banks), the EEPROM 512 for SPD, the clock buffer 513, the buffer 514 for C/A and control signals, the FET switches 515, and the interface 516.

[0074] The interface 516 receives a command, an address, data, first and second bank selection signals nBS1 and nBS2, and an nLOAD signal from the flash controller 550 (refer to FIG. 10). Here, the nBS1 signal is used for selecting the first bank 511a, and the nBS2 signal is used for selecting the second bank 511b.

[0075] The nBS1 signal is provided as a chip enable signal nCE for the NBX NAND flash memories of the first bank 511a. The nBS2 signal is provided as a chip nCE signal for the NBX NAND flash memories of the second bank 511b. The buffer 514 for C/A and control signals is used to temporarily store a command, an address, and an nLOAD signal and sends them to a selected bank.

[0076] The clock buffer 513 receives a clock signal nCLK and generates first and second clock signals CLK1 and CLK2. The first clock signal CLK1 is provided to the first bank 511a, and the second clock signal CLK2 is provided to the second bank 511b. Furthermore, the clock buffer 513 provides a clock signal for the buffer 514 for C/A and control signals. The buffer 514 is operated in synchronization with a clock signal.

[0077] The EEPROM 512 for SPD stores SPD information necessary for the flash controller to access the NBX flash module 510. For example, the EEPROM 512 for SPD stores information about internal delay time necessary for writing data to or reading data from the NBX NAND flash memories of the first and second banks 511a and 511b. In read or write operation, the flash controller 550 reads the internal delay time information from the EEPROM 512 and outputs signals and data according to the read information for read or write operation.

[0078] NBX NAND flash memories included in a given bank have the same page size. Information about this page size is stored in the EEPROM 512. The page size of a NBX NAND flash memory may vary from one bank to another bank and/or from one NBX flash module to another NBX flash module. The EEPROM 512 for SPD stores this page information. When accessing a NBX flash module, the flash controller 550 reads the page information from the EEPROM 512 and operates according to the read page information.

[0079] Banks included in one NBX flash module can have different storage capacities. In addition, NBX NAND flash memories included in one bank can have different storage capacities. The EEPROM 512 for SPD stores information

about the bank capacities of banks. When accessing a NBX flash module, the flash controller 550 reads the bank capacity information from the EEPROM 512 and operates according to the read bank capacity information. That is, the flash controller 550 reads SPD information from the EEPROM 512 and sets itself using the SPD information.

[0080] FIGS. 13 through 15 are timing diagrams for explaining a bank interleaving operation of the NBX NAND flash memory system 500 of FIG. 10, according to an exemplary embodiment of the present invention. Bank interleaving is a read, write, or erase operation that is performed alternately on banks. FIG. 13 illustrates an interleaving operation between banks for reading data from the banks. FIG. 14 illustrates an interleaving operation between banks for writing data to the banks. FIG. 15 illustrates an interleaving operation between banks for erasing, writing, and reading.

[0081] Referring to FIG. 13, a first bank is enabled by a first bank selection signal nBS1. For example, when the first bank selection signal nBS1 is at a low level, data are read from the first bank. Reading data from the first bank starts in response to a read command R. After the read command R and an address A is input, data are read from a selected page and transmitted to a page buffer. When data corresponding to one page is stored in the page buffer, an NBX NAND flash memory outputs a status signal SQ in response to a status read command S to indicate that it is ready to output data.

[0082] The NBX NAND flash memory outputs the data stored in the page buffer in response to a read enable command RE. Here, the NBX NAND flash memory outputs the data through a data input/output pin DQ according to the toggling of a data strobe signal DQS.

[0083] When a second bank selection signal nBS2 is changed to a low level during the reading operation for the first bank, data are read from a second bank. Similarly, when a third bank selection signal nBS3 is changed to a low level during the reading operations for the first and second banks, data are read from a third bank. The reading operations for the second and third banks are performed in the same manner as that for the first bank.

[0084] After data are output from the first bank, data are output from the second bank without a delay time. Similarly, after data are output from the second bank, data are output from the first bank.

[0085] Referring to FIG. 14, when a first bank selection signal nBS1 is at a low level, data are written into a first bank. Writing data into the first bank starts in response to a write command W. After the write command W and an address A are input, a write enable command WE is input. An NBX NAND flash memory receives data through a data input/output pin DQ according to the toggling of a data strobe signal DQS. The received data are stored in a page buffer.

[0086] After data are stored in the page buffer, the NBX NAND flash memory performs internal programming on a page of the address A using the stored data. The NBX NAND flash memory outputs a status signal SQ in response to a status read command S to indicate the end of the write operation.

[0087] When second and third bank selection signals are changed to a low level during the write operation for the first bank, data are written into second and third banks. The write operations for the second and third banks are performed in the same manner as that for the first bank.

[0088] Referring to FIG. 14, after data are completely loaded from a flash controller to a page buffer of the first bank, data are loaded from the flash controller to a page buffer of the

second bank without a delay time. Similarly, after the data are loaded to the page buffer of the second bank, data are loaded from the flash controller to a page buffer of the third bank.

[0089] Referring to FIG. 15, an erase operation is performed on a first bank, a write operation is performed on a second bank, and a read operation is performed on a third bank. During the erase operation for the first bank, the write operation for the second bank is performed. In addition, during the erase and write operations for the first and second banks, the read operation for the third bank is performed.

[0090] Since a conventional NAND flash memory uses a data input/output pin DQ for receiving command and address signals as well as data, a data stream in a data bus or channel is often interrupted. Furthermore, a conventional NAND flash memory module has a long data delay time during a bank interleaving operation. However, as illustrated in FIGS. 13 through 15, the NBX flash module of exemplary embodiments of the present invention can perform an interleaving operation for data input/output without a data delay time, so that data can be read from and written to the NBX flash module at a high speed.

[0091] FIG. 16 illustrates a thin small outline package (TSOP) of an NBX NAND flash memory according to an exemplary embodiment of the present invention, and FIG. 17 is a table providing descriptions of pins of the TSOP of FIG. 16, according to an exemplary embodiment of the present invention. Referring to FIG. 16, the NBX NAND flash memory has 48 pins. The 1st to 24th pins are formed on the left side, and the 25th to 48th pins are formed on the right side. Referring to FIG. 17, all control functions are assigned to the left-side pins. According to exemplary embodiments of the present invention, the number of pins can be decreased, and the NBX NAND flash memory can be designed to have all controls pins on one side. Therefore, the NBX NAND flash memory can have a simple structure.

[0092] As described above, in the NAND flash memory of exemplary embodiments of the present invention, the C/A pin is separated from the data input/output pin. Therefore, data input/output speed can be improved. Furthermore, the NAND flash memory can perform a bank interleaving operation with less delay time.

[0093] The above-disclosed subject matter is to be considered illustrative, and not restrictive. It will be understood by those of ordinary skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention.

What is claimed is:

1. A NAND flash memory comprising:
 - a memory cell array storing data;
 - a command/address pin through which a command and an address are received for transmitting data; and
 - a data input/output pin through which data are transmitted in the memory cell array.
2. The NAND flash memory of claim 1, further comprising a status register receiving a status read command through the command/address pin and providing an operational status of the NAND flash memory to a flash controller.
3. The NAND flash memory of claim 2, wherein the flash controller sends the status read command to the NAND flash memory.
4. The NAND flash memory of claim 2, wherein the status register sends a status signal SQ to the flash controller to inform the flash controller whether the NAND flash memory is internally operational.

5. The NAND flash memory of claim 4, wherein the flash controller controls the internal operation of the NAND flash memory in response to the status signal SQ.

6. The NAND flash memory of claim 1, wherein the data transmitted through the data input/output pin is transmitted according to a toggling of a data strobe signal DQS.

7. The NAND flash memory of claim 6, wherein the data transmitted through the data input/output pin is transmitted by a DDR (double data rate) transmission method.

8. The NAND flash memory of claim 1, further comprising a command/address buffer receiving the command and address received through command/address pin.

9. The NAND flash memory of claim 8, further comprising a control unit controlling the reception of the command and address.

10. The NAND flash memory of claim 9, wherein the control unit receives a chip enable signal nCE and a load signal nLOAD from a flash controller and controls the reception of the command and address.

11. A flash memory system comprising:

a flash controller; and

a flash memory module comprising a plurality of NAND flash memories.

wherein each of the NAND flash memories comprises:

a memory cell array storing data;

a command/address pin through which a command and an address are received from the flash controller for transmitting data in the memory cell array; and

a data input/output pin through which data is transmitted in the memory cell array.

12. The flash memory system of claim 11, wherein each of the NAND flash memories further comprises a status register

receiving a status read command through the command/address pin and providing an operational status of the NAND flash memory to the flash controller.

13. The flash memory system of claim 12, wherein the flash controller sends the status read command to the NAND flash memory.

14. The flash memory system of claim 12, wherein the status register sends a status signal SQ to the flash controller to inform the flash controller whether the NAND flash memory is internally operational.

15. The flash memory system of claim 14, wherein the flash controller controls the internal operation of the NAND flash memory in response to the status signal SQ.

16. The flash memory system of claim 11, wherein the data transmitted through the data input/output pin is transmitted according to a toggling of a data strobe signal DQS.

17. The flash memory system of claim 16, wherein the data transmitted through the data input/output pin is transmitted by a DDR (double data rate) transmission method.

18. The flash memory system of claim 1, wherein each of the NAND flash memories further comprises a command/address buffer receiving the command and address received through command/address pin.

19. The flash memory system of claim 18, wherein each of the NAND flash memories further comprises a control unit controlling the reception of the command and address.

20. The flash memory system of claim 19, wherein the control unit receives a chip enable signal nCE and a load signal nLOAD from the flash controller and controls the reception of the command and address.

* * * * *



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MEMORY SYSTEMS****Publication Classification**(75) **Inventor: Jeng-Jye Shau, Palo Alto, CA (US)**

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(57)

ABSTRACT

The present invention provides memory system architectures developed to increase the capacity of memory systems. Typically applications including the main memory of computers. Memory systems of the present invention can achieve capacities larger than prior art systems by one or two orders of magnitudes without significant degradation in performance while using system interfaces that are compatible with existing memory systems with no or minimal modifications.

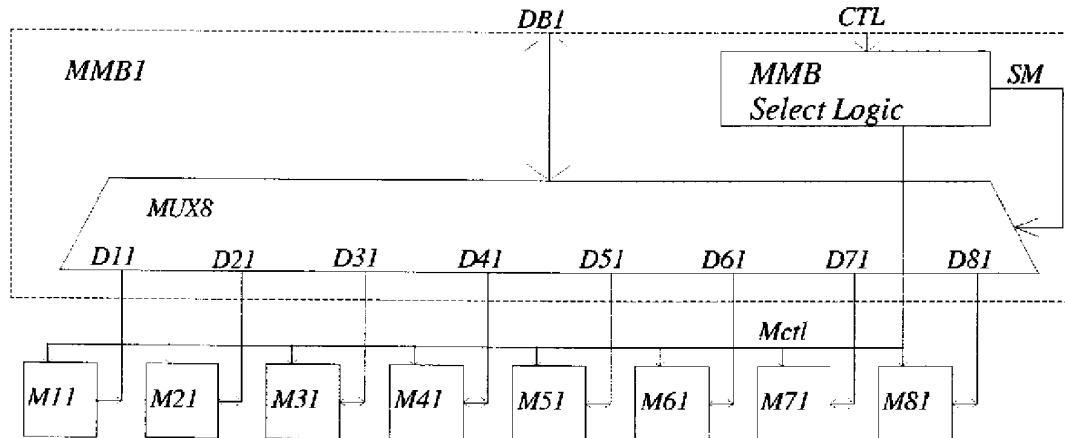


FIG. 1(a) Prior art

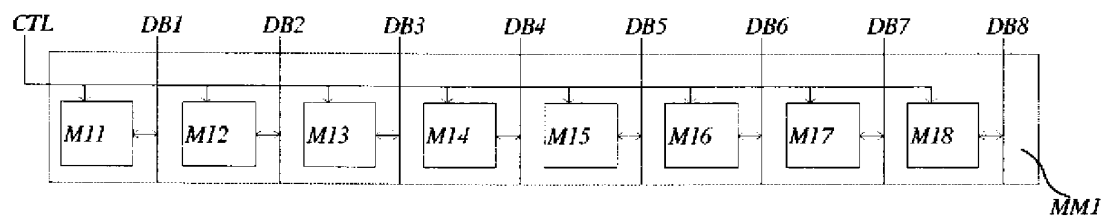


FIG. 1(b) Prior art

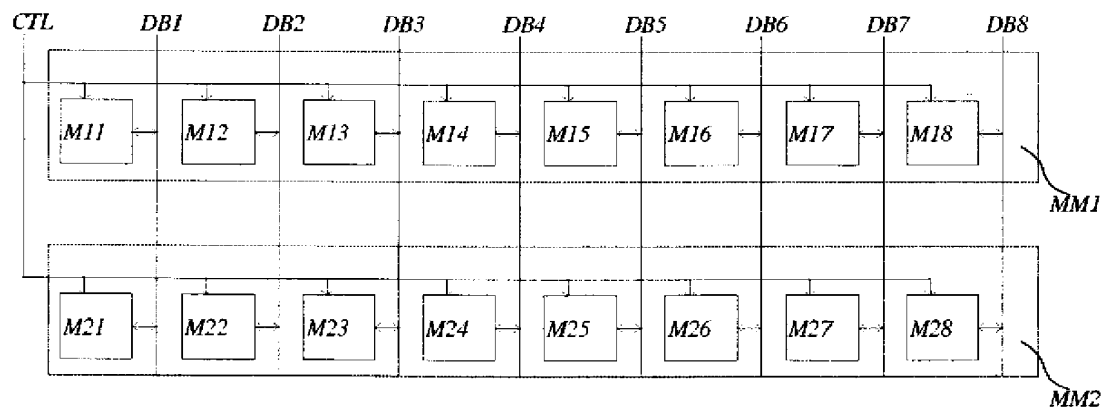


FIG. 1(c) Prior art

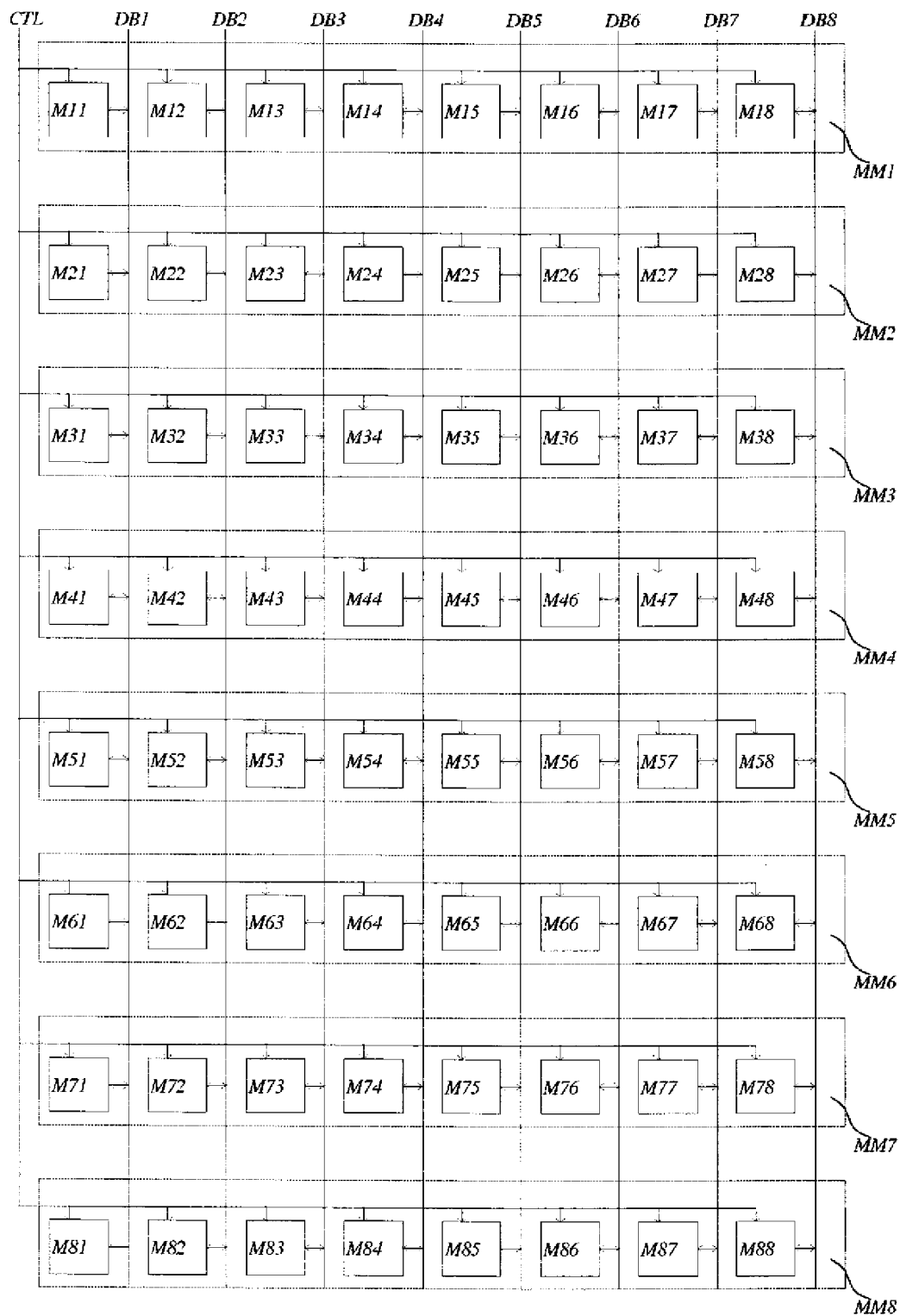


FIG. 2(a) Prior art

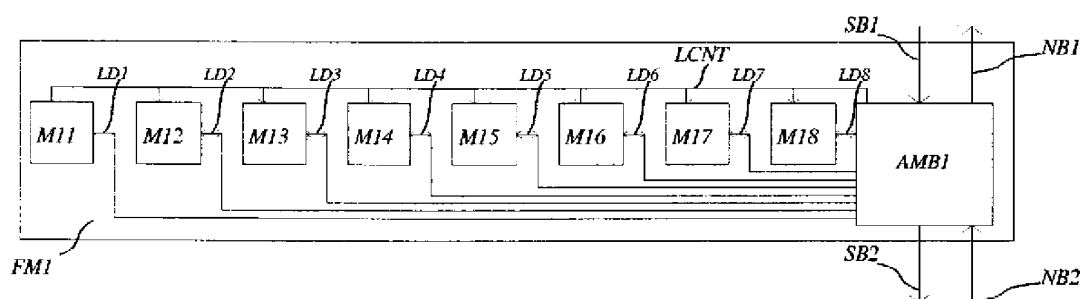


FIG. 2(b) Prior art

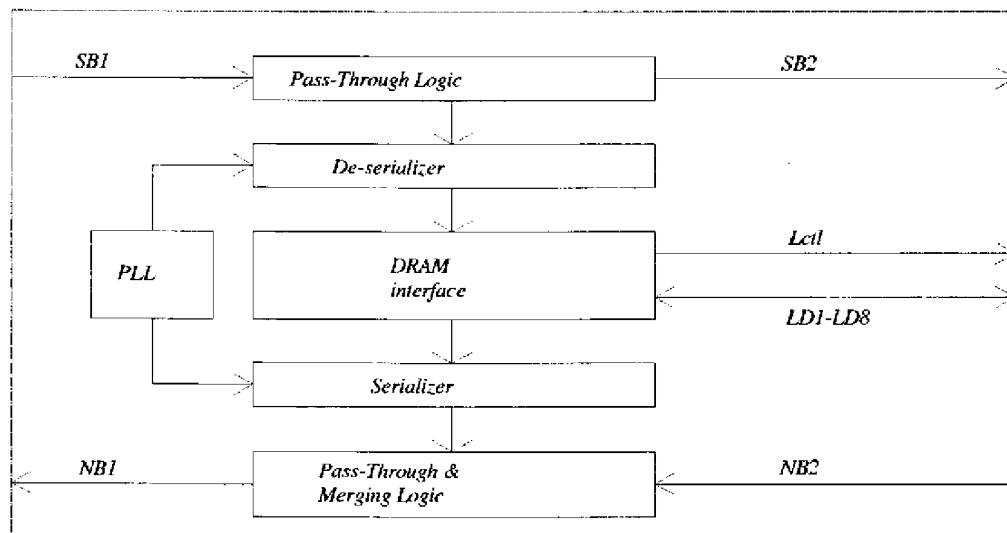


FIG. 2(c) Prior art

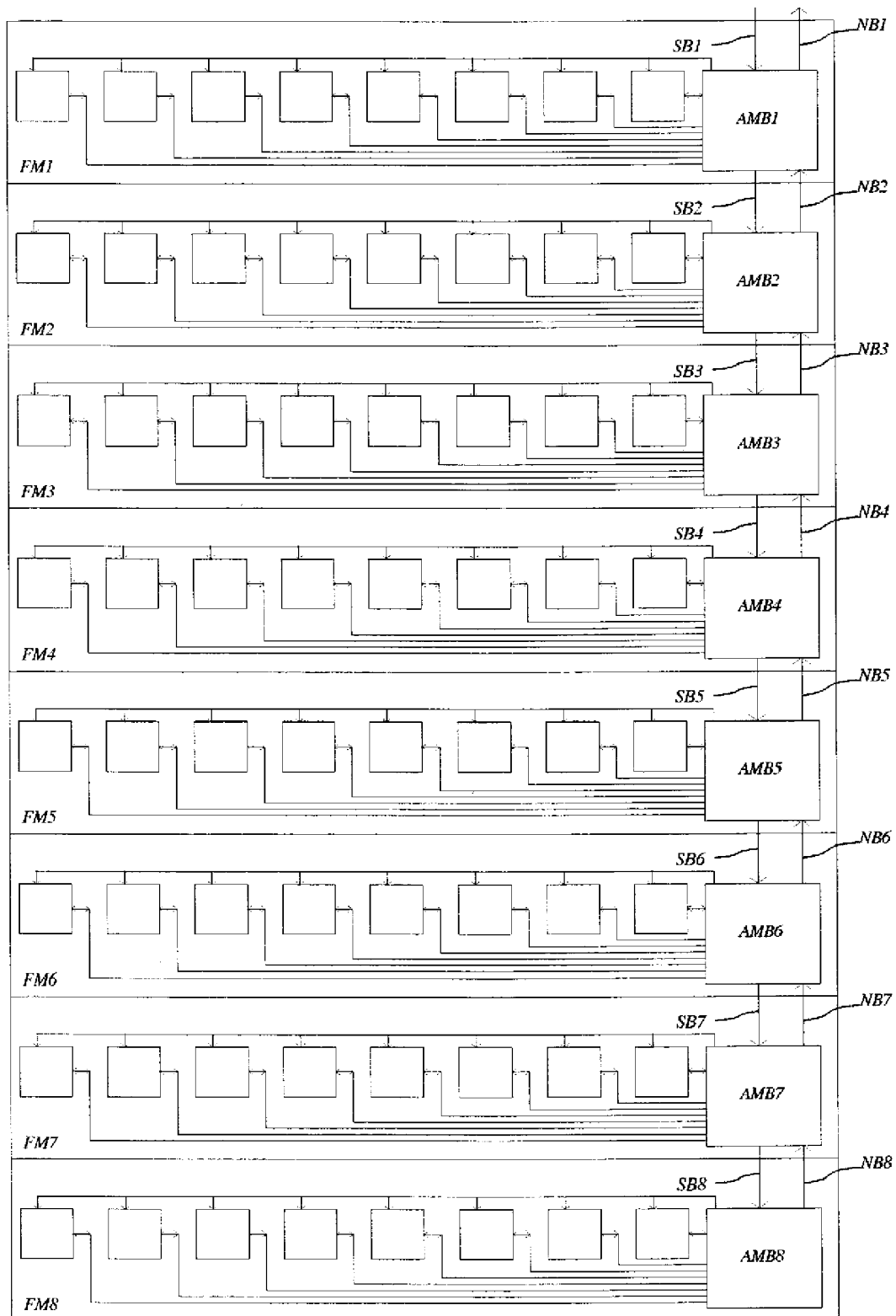


FIG. 3(a)

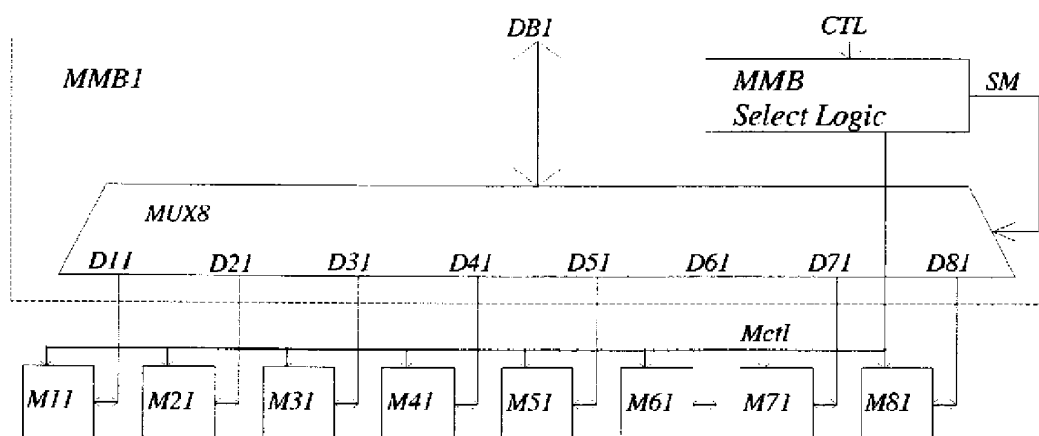


FIG. 3(b)

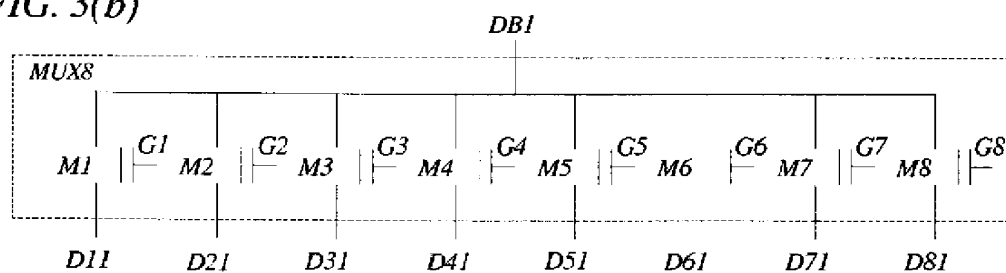


FIG. 3(c)

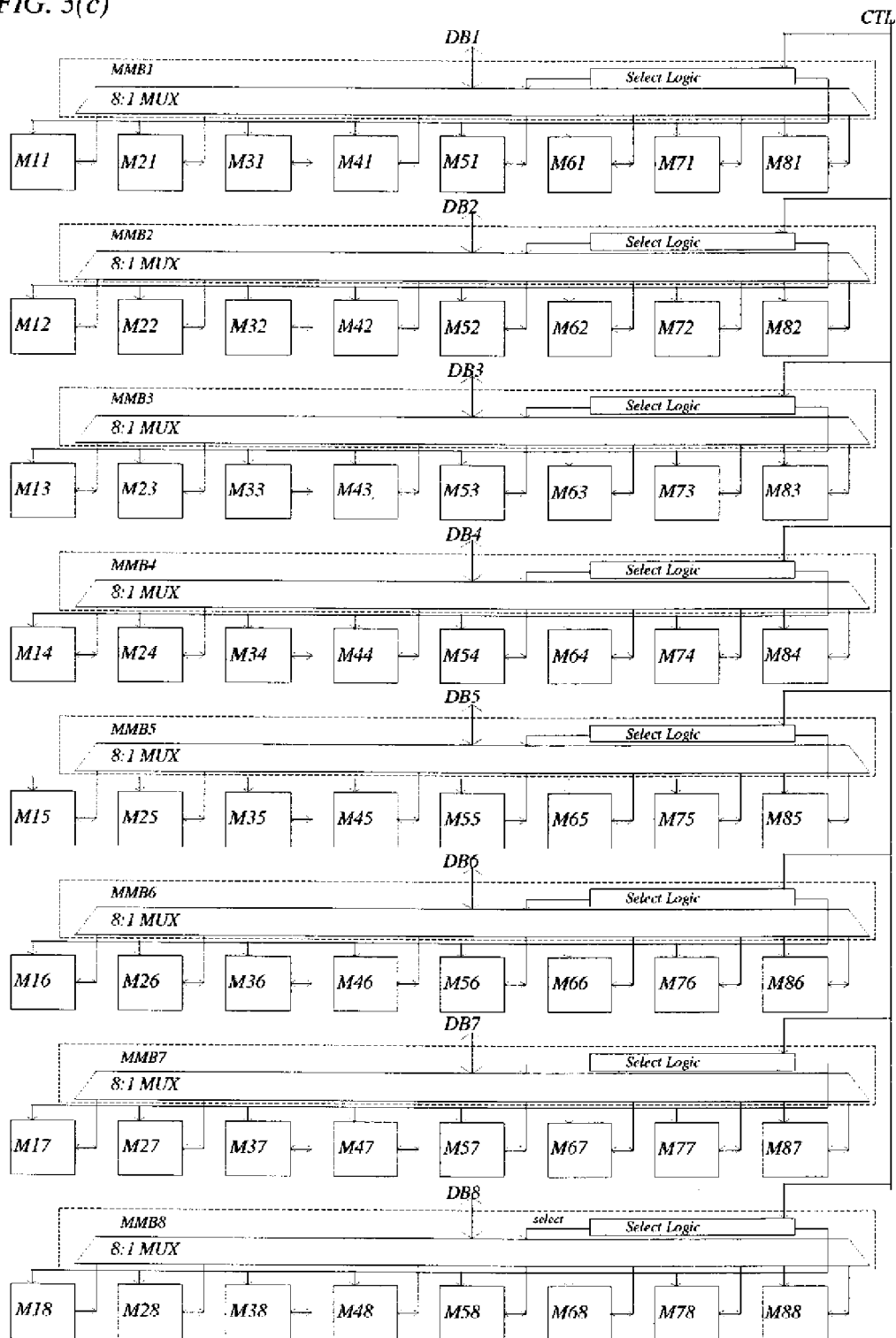


FIG. 4(a)

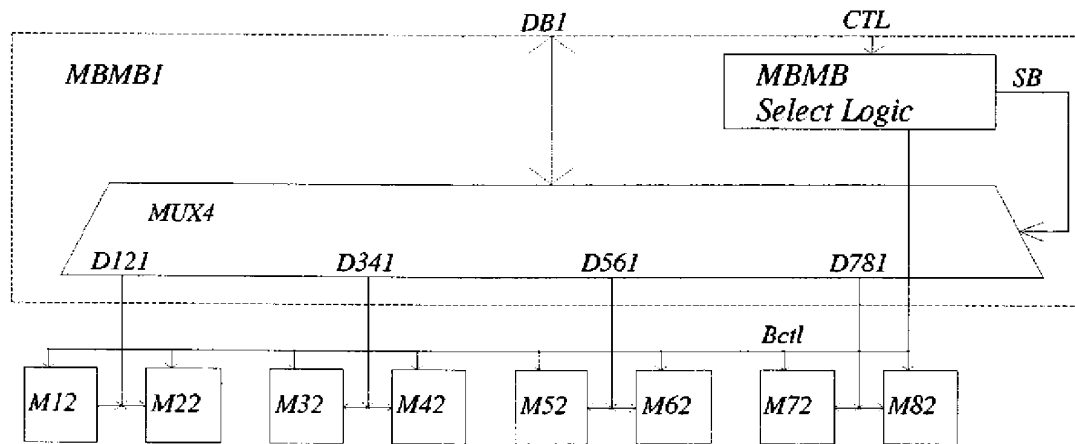


FIG. 4(b)

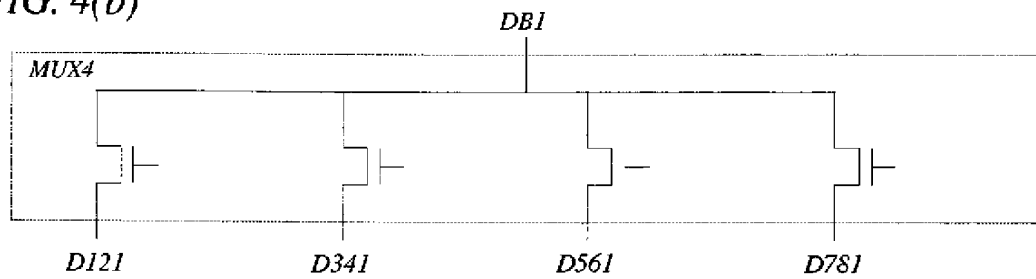
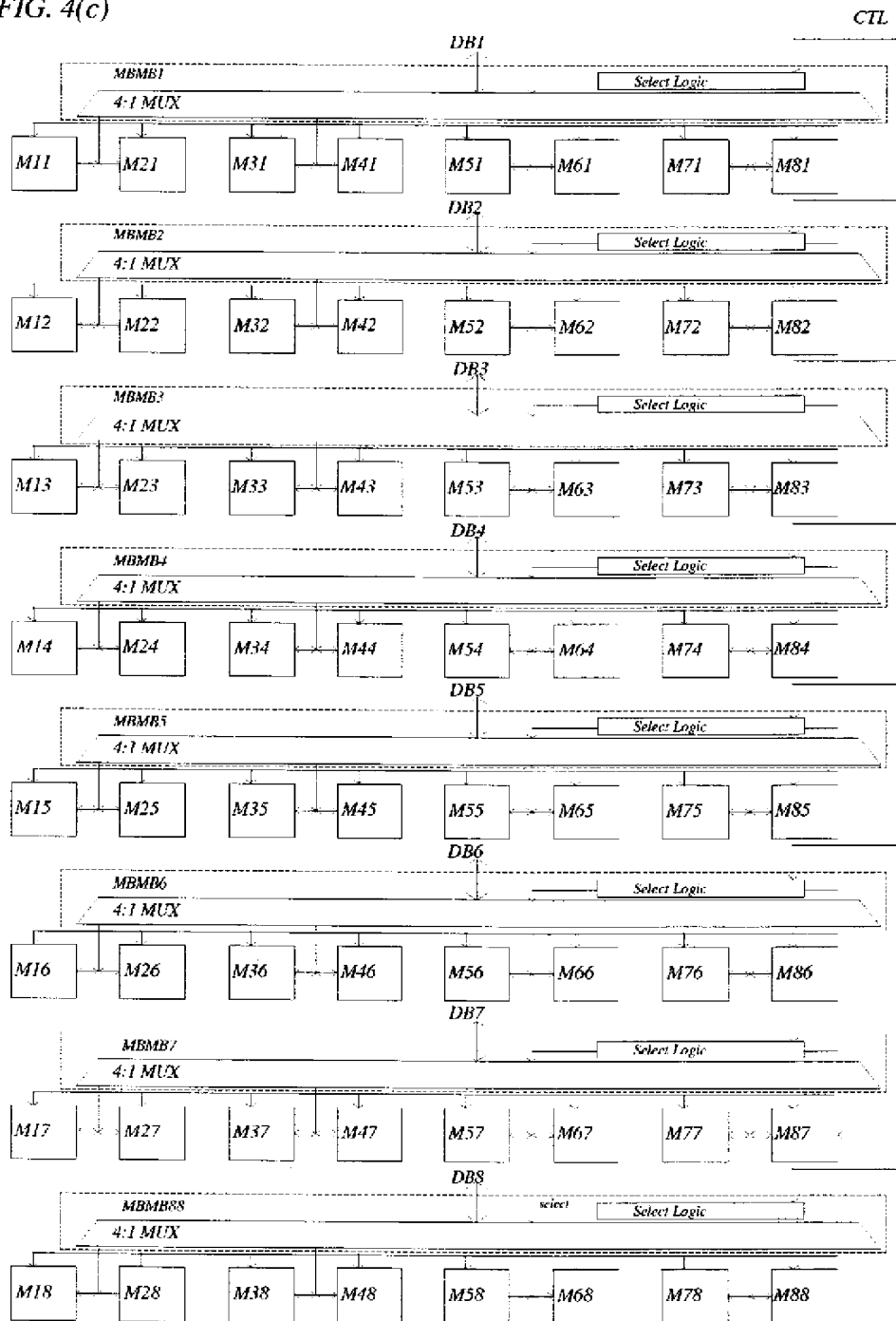


FIG. 4(c)



HIGH PERFORMANCE HIGH CAPACITY MEMORY SYSTEMS

BACKGROUND OF THE INVENTION

[0001] The present invention relates to structures and methods designed to increase the capacity of high performance memory systems.

[0002] The present invention is applicable to most types of memories such as dynamic random access memory (DRAM), static random access memory (SRAM), nonvolatile memories, etc. Among the wide varieties of possible applications, the most well known applications are the main memory in computers. We will focus on computer main memory using double data rate version 2 (DDR2) dynamic random access memory (DRAM) as examples to demonstrate the basic principles of the present invention. The scope of the present invention is certainly not limited to particular types of memory or particular types of applications used in our examples.

[0003] A "memory system" defined in this patent application is board level circuits supporting memory operation of memory chips. A "memory module" is defined as a sub-circuit of a memory system. A "system level signal" is defined as an electrical signal used to communicate with circuits external to a memory system. A "chip level signal" is defined as an electrical signal used to communicate with memory chips.

[0004] It is well known that the performance of a computer is strongly dependent on both the performance as well as the capacity of its main memory. Ideally, a computer wants to have high performance system memory at as large capacity as possible. In reality, high performance and high capacity have conflicting requirements that can become limiting factors. We will discuss key factors on those limitations using typical personal computer memory systems as examples.

[0005] The most common memory chip used for computer system memory is DRAM. Table 1 lists typical chip level

interface signals for a current art 1 G (2³⁰) bit DDR2 synchronized DRAM integrated circuit chip.

TABLE 1

Standard 1G-bit DDR2 DRAM Interface signals		
Name	Type	Descriptions
DQ0-DQ7	In/out	8-bit data Bidirectional bus
DQS, DQS#	In/out	Bidirectional data strobe
DM	input	Input data mask
A0-A12	input	Addresses
BA0-BA2	input	Bank addresses
CK, CK#	input	Differential clocks
CKE	input	Clock enable
CS#	input	Chip select
RAS#, CAS#, WE#	input	Command inputs; alone with CS# define commands
ODT	input	On-die termination
Vref	input	Reference voltage
VDD, VDDQ, VDDL, VSS, VSSQ, VSSL	power	Power and ground lines for core, I/O, and DLL

[0006] DRAM chips are typically mounted on small printed circuit board (PCB) called Single-In-line Memory Module (SIMM) or Dual-In-line Memory Module (DIMM); a DIMM is equivalent to two SIMM modules placed into one PCB utilizing both sides of the circuit board. The SIMM or DIMM memory modules provide the flexibility to expand the capacity of computer main memory. The memory controller in chipset typically has the flexibility to support 8 SIMM or 4 DIMM modules. A personal computer typically starts with one installed DIMM module while providing additional empty sockets. A user who wants to improve the performance of computer can insert additional modules into the expandable sockets. To support such expandable memory systems, personal computers typically support a system level memory interface with signals listed in Table 2.

TABLE 2

Standard personal computer system memory interface signals		
Name	Type	Descriptions
DQ0-DQ63	In/out	64-bit data Bidirectional bus, supported by eight 8-bit data bus. 8 more data (DQ64-DQ71) can be added for parity or error correction code (ECC).
DQS0-DQS7, DQS0#-DQS7#	In/out	Bidirectional data strobe, one pair for each 8-bit data bus. One more pair (DQS8, DQS8#) can be added for parity or ECC.
DM0-DM7	input	Input data mask. One for each 8-bit data bus. One more (DM8) can be added for parity or ECC.
A0-A13	input	Addresses, may have more or less address bits.
BA0-BA2	input	Bank addresses, may have only two bank address bits.
CK, CK#	input	Differential clocks, may have separated clocks for different modules
CKE0-CKE7	input	Clock enable, one fore each memory module
CS#0-CS#7	input	Chip select signals, one for each memory module.
RAS#, CAS#, WE#	input	Command inputs.
ODT0-ODT7	input	On-die termination, one for each memory module
RESET#	input	Reset
PAR_IN	input	Parity bit for address and control
PAR_ERR	output	Parity error found in address and control
SCL, SA0-SA2	input	EEPROM clock and addresses
SDA	In/out	EEPROM data
Vref	input	Reference voltage

TABLE 2-continued

Standard personal computer system memory interface signals		
Name	Type	Descriptions
VDD, VDDQ, VDDL, VDE, VSS, VSSQ, VSSL	power	Power and ground lines for core, I/O, and DLL

[0007] If we draw all these signals in our figures, the resulting figures will be very busy, making it less clear in demonstrating the key points of the present invention. Therefore, in our figures the interface signals are simplified into two groups, namely data signals and control signals. Data signals (DB) are signals directly related to data transfers while following the same signal transfer protocols, including the data bus (DQ), data strobe (DQS and #DQS), and input data mask (DM) signals. Control signals (CTL) are signals used to determine operation states of the memory chips, including the addresses, bank addresses, clocks signals (CK, CK#, CKE), chip select signal (CS#), and command inputs (RAS#, CAS#, WE#). We will not show DC or slow signals such as power lines, reference voltage signals, EEPROM signals, and on-die-termination signals because those connections are not related to the key factors of the present invention. To facilitate clear understanding of the present invention, there is no need to show those details that are well known to people skilled in the art; we will focus on the key elements related to the present invention—the data and control signals of memory chips. For simplicity, the optional parity/ECC data signals are also not included in our discussion because a person with ordinary skill in the art would understand how to apply the present invention on the parity/ECC signals upon disclosure of our examples. The simplified representations of memory interface signals used in our discussions are listed in Table 3.

TABLE 3

Simplified representation of memory interface signals		
meaning	representation	Corresponding signals in Table 2
Data signal bus 1	DB1	DQ0-DQ7, DQS0, DQS#0, DM0
Data signal bus 2	DB2	DQ8-DQ15, DQS1, DQS#1, DM1
Data signal bus 3	DB3	DQ16-DQ23, DQS2, DQS#2, DM2
Data signal bus 4	DB4	DQ24-DQ31, DQS3, DQS#3, DM3
Data signal bus 5	DB5	DQ32-DQ39, DQS4, DQS#4, DM4
Data signal bus 6	DB6	DQ40-DQ47, DQS5, DQS#5, DM5
Data signal bus 7	DB7	DQ48-DQ55, DQS6, DQS#6, DM6
Data signal bus 8	DB8	DQ56-DQ63, DQS7, DQS#7, DM7
Control signals	CTL	A0-A13, BAO-BA2, CK, CK#, CS#0-CS#7, CKE0-CKE7, RAS#, CAS#, WE#
Not shown		DQ64-DQ71, DQS8, DQS#8, DM8, ODT0-ODT8, RESET#, PAR_IN, PAR_ERR, SCL, SA0-SA2, Vref, VDD, VDDQ, VDDL, VDE, VSS, VSSQ, VSSL

[0008] Using the simplified representations in Table 3, the architectures of typical prior art memory systems can be illustrated by FIGS. 1(a-c). FIGS. 1(a) is the simplified schematic block diagrams for a typical prior art memory module (MM1). This memory module comprises a plurality of memory chips (M11-M18) that shares the same control signals (CTL). The data signals of memory chips are connected

in parallel; the first memory chip (M11) supports data signal bus 1 (DB1); the second memory chip (M12) supports data signal bus 2 (DB2); the third memory chip (M13) supports data signal bus 3 (DB3); the fourth memory chip (M14) supports data signal bus 4 (DB4); the fifth memory chip (M15) supports data signal bus 5 (DB5); the sixth memory chip (M16) supports data signal bus 6 (DB6); the seventh memory chip (M17) supports data signal bus 7 (DB7); the eighth memory chip (M18) supports data signal bus 8 (DB8). The width of module level data bus is therefore the combined width of all memory chips (M11-M18) on the same module (MM1). We will call such connection as “parallel data connection” in the following discussions.

[0009] A common prior art method to increase the capacity of a memory system is to use DIMM modules instead of SIMM modules. FIG. 1(b) shows the simplified schematic block diagram for a DIMM module. A DIMM module comprises one additional memory module (MM2) that is typically placed on the other side of the same print circuit board used to place the first memory module (MM1). The memory chips (M21-M28) of the second memory module (MM2) are connected in the same way as that of the first memory module (MM1). Since both memory modules (MM1, MM2) share the same data signals (DB1-DB8) in a shared bus structure, each memory module must use different chip select signals (part of CTL but not shown separately in figures for simplicity) to avoid driver conflicts; typically, different modules are also connected to different clock enable signals (not shown). Other than chip enable and clock enable signals, typically all other control signals are the same for all memory modules. The two memory modules (MM1, MM2) on the same DIMM module often can share most of signal lines so that the increase in loading is typically less than twice of a single module. Using DIMM module is therefore an efficient prior art method to increase the capacity of memory systems.

[0010] If we want to have larger capacity than a DIMM module, we need to add more memory modules to the system. FIG. 1(c) shows the simplified schematic block diagram for a memory system that has 6 additional memory modules. The memory chips (M31-M38) of the third memory module (MM3) are connected in the same way as that of the first memory module (MM1). The memory chips (M41-M48) of the fourth memory module (MM4) are connected in the same way as that of the first SIMM module (MM1). The memory chips (M51-M58) of the fifth memory module (MM5) are connected in the same way as that of the first memory module (MM1). The memory chips (M61-M68) of the sixth memory module (MM6) are connected in the same way as that of the first SIMM module (MM1). The memory chips (M71-M78) of the seventh memory module (MM7) are connected in the same way as that of the first memory module (MM1). The memory chips (M81-M88) of the eighth memory module (MM8) are connected in the same way as that of the first

SIMM module (MM1). All the memory modules in the same system share the same data signals (DB1-DB8) in a shared bus structure. Therefore, each memory module must use different chip select signals (part of CTL but not shown separately in figures for simplicity) to avoid driver conflicts; typically, different modules are also connected to different clock enable signals (not shown). Other than chip enable and clock enable signals, typically all other control signals are the same for all memory modules.

[0011] The capacity of the memory system in FIG. 1(c) is four times the capacity of the memory system in FIG. 1(b). However, when the number of memory modules is increased, the loading on the shared data signals (DB1-DB8) and control signals (CTL) also increases. The "Loading" on a signal is the non-ideal factors that can slow down signals performances such as leakage currents, parasitic capacitances, inductances, or resistances. The loadings for the system in FIG. 1(c) are about four times that of the system in FIG. 1(b). Increase in loading typically means degradation in performance and/or stability. This problem is especially significant for prior art DDR2 synchronized DRAM with data rate higher than 600 millions of bits per second (MPS) per pin. DDR2 DRAM uses Stub Series Terminated Logic (SSTL) buses with on-chip terminal resistors so that each memory chip (even when it is not active) is sinking currents through terminal resistors, making it impractical to connect large number of prior art memory modules. It is well known that using multiple DDR2 DIMM modules would degrade performance significantly, especially at data rate higher than 600 millions of bits per second (MPS) per pin. Increasing capacity by adding more and more prior art memory modules is therefore not practical. It is therefore strongly desirable to provide methods that can increase the capacity of a memory system without increasing the loading of data and control signals.

[0012] One prior art solution to solve the loading problem is to use phase locked loop (PLL) to generate local clock signals, and use buffers to generate local control signals. Such methods reduce the loading on control signals, but the loading problems in data signals are not solved.

[0013] Another prior art solution for the loading problem is the JEDEC standard "Fully Buffered DIMM" (FBDIMM) approach. An FBDIMM uses an integrated circuit (IC) chip called "Advanced Memory Buffer (AMB)" to control all the interface signals to all memory chips on the module. The loadings on memory chip data and control signals are therefore completely isolated from other memory modules. FIG. 2(a) is a simplified schematic block diagrams for an FBDIMM (FM1). The memory chips (M11-M18) on the FBDIMM (FM1) are arranged in parallel data connection while the data signals (LD1-LD8) and control signals (LCTL) of the memory chips are internal signals controlled by an advanced memory buffer (AMB1). FIG. 2(b) is a simplified schematic block diagram for prior art AMB. The inputs of an AMB come from south bound signal transfer lanes (SB1) that typically comprise 10 pairs of high speed differential signal transfer lines. Currently, each pair of the differential signal transfer lines is capable of transferring signals at 4.8 billion bits per second (GPS). The input signals on SB1 are latched and analyzed by pass-through logic circuits. If the inputs request operations to other FBDIMM, the input signals are passed to the next FBDIMM through another south bound signal transfer lanes (SB2). If the inputs request operations on the same FBDIMM, the input signals are sent to a de-serializer, then to a DRAM interface logic circuitry that translates

the input signals into control signals (LCTL) to memory chips. The data (LD1-LD8) signals returned from memory chips on the same module received by the DRAM interface are sent to a serializer. The serializer converts the data into proper format and sends the output data to pass-through and merging (P&M) circuits. The P&M logic circuits transfer outputs through north bound signal transfer lanes (NB1) that typically comprise 14 pairs of high speed differential signal transfer lines. Output signals from other FBDIMM modules from another north bound signal transfer lanes (NB2) are also latched and processed by the P&M circuits before sending to NB1. Those high speed signal transfer lanes (SB1, SB2, NB1, NB2) are synchronized by phase-locked loop (PLL) circuits. FIG. 2(b) is a simplified block diagram emphasizing features related to key points of the present invention. Please refer to the data sheets of existing AMB products such as Intel 6400 or NEC P720901 for further details. Those existing AMB products are typically complex high cost integrated circuits (IC) comprise more than 600 interface signals.

[0014] To increase the capacity of an FBDIMM system, multiple FBDIMM modules (FM1-FM8) are connected in daisy-chained bus architecture as illustrated in FIG. 2(c). The system input (SB1) is connected to the south bound signal transfer lanes (SB1) of the first module (FM1). The system output is connected to the north bound signal transfer lanes (NB1) of the first module (FM1). The inputs to the second module (FM2) are supported by south bound signal transfer lanes (SB2) that are provided by AMB1 in FM1. The outputs from the module (FM2) are supported by north bound signal transfer lanes (NB2) to AMB1 in FM1. The inputs to the third module (FM3) are supported by south bound signal transfer lanes (SB3) that are provided by AMB2 in FM2. The outputs from the module (FM3) are supported by north bound signal transfer lanes (NB3) to AMB2 in FM2. The inputs to the forth module (FM4) are supported by south bound signal transfer lanes (SB4) that are provided by AMB3 in FM3. The outputs from the module (FM4) are supported by north bound signal transfer lanes (NB4) to AMB3 in FM3. The inputs to the fifth module (FM5) are supported by south bound signal transfer lanes (SB5) that are provided by AMB4 in FM4. The outputs from the module (FM5) are supported by north bound signal transfer lanes (NB5) to AMB4 in FM4. The inputs to the sixth module (FM6) are supported by south bound signal transfer lanes (SB6) that are provided by AMB5 in FM5. The outputs from the module (FM6) are supported by north bound signal transfer lanes (NB6) to AMB5 in FM5. The inputs to the seventh module (FM7) are supported by south bound signal transfer lanes (SB7) that are provided by AMB6 in FM6. The outputs from the module (FM7) are supported by north bound signal transfer lanes (NB7) to AMB6 in FM6. The inputs to the eighth module (FM8) are supported by south bound signal transfer lanes (SB8) that are provided by AMB7 in FM7. The outputs from the module (FM8) are supported by north bound signal transfer lanes (NB8) to AMB7 in FM7. The capacity of the memory system in FIG. 2(c) is the same as that of the memory system in FIG. 1(c) while the loadings on all data and controls signals are about the same of a single module in FIG. 1(a). In addition, the loading on all signals lines remain the same no matter how many FBDIMM modules are connected in the memory system, effectively solving the loading problems. However, the memory access latency is increase by the need to transfer signals serially through the AMBs connected in daisy chain architecture. For example, if we want to access the memory chips in the seventh module (FM7), we need to

add 7 south bound signal transfer cycles, 7 north bound signal transfer cycles, plus delays caused by AMB logic processing as the overhead in timing. The worst delay time increases linearly with the number of FBDIMM modules linked in the daisy chain, limiting the capability to increase capacity. In addition, the FBDIMM modules are by far more expensive than conventional memory modules, and they are not compatible with conventional memory interfaces, limiting their application on high cost server or work stations. FBDIMM saves power by isolating memory chips in different modules, but the power consumed by overhead in AMB is significant.

[0015] It is therefore highly desirable to provide other solutions that can increase total capacity of memory systems without the drawbacks of existing solutions such as FBDIMM approaches.

SUMMARY OF THE INVENTION

[0016] The primary objective of this invention is, therefore, to provide high capacity memory systems without increasing the loading of data signals. The other primary objective of this invention is to achieve the above objective with minimum overhead in performance and in cost. Another objective is to achieve the above objectives while using interfaces that are compatible with conventional memory systems. These and other objectives are achieved by using multiplexing to isolate loadings on data signals. The resulting memory systems are capable of achieving high capacity with basically the same performance and power of a single conventional memory. The interface signals also can be compatible with conventional memory systems.

[0017] While the novel features of the invention are set forth with particularity in the appended claims, the invention, both as to organization and content, will be better understood and appreciated, along with other objects and features thereof, from the following detailed description taken in conjunction with the drawing.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIGS. 1(a-c) are simplified schematic block diagrams for prior art conventional memory systems;

[0019] FIGS. 2(a-c) are simplified schematic block diagrams for prior art FBDIMM systems;

[0020] FIG. 3(a) is a simplified schematic block diagram for one example of the Multiplexed Memory Buffer (MMB) module of the present invention;

[0021] FIG. 3(b) is a simplified symbolic diagram for the bidirectional multiplexer in FIG. 3(a);

[0022] FIG. 3(c) is a simplified schematic block diagram for one example of the MMB memory system of the present invention;

[0023] FIG. 4(a) is a simplified schematic block diagram for one example of the Multiplexed Bus Memory Buffer (MBMB) module of the present invention;

[0024] FIG. 4(b) is a simplified symbolic diagram for the bidirectional multiplexer in FIG. 4(a); and

[0025] FIG. 4(c) is a simplified schematic block diagram for MBMB one example of the memory system of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0026] FIG. 3(a) is a simplified schematic block diagram for one example of the Multiplexed Memory Buffer (MMB)

module of the present invention. In this example, the MMB memory module (MMB1) comprises 8 memory chips (M11, M21, M31, M41, M51, M61, M71, M81). Comparing to the prior art memory module in FIG. 1(a), the key difference is that the memory chips (M11-M18) in the prior art memory module is arranged in parallel data connection to support a complete set of system data signals (DB1-DB8). In contrast, the memory chips (M11, M21, M31, M41, M51, M61, M71, M81) in memory modules of the present invention is arranged to support a sub set (DB1) of the system data signals, while the first memory chip (M11) supports DB1, the second memory chip (M21) supports DB1, . . . , and the eighth memory chip (M81) also supports DB1. In other words, all those memory chips (M11, M21, M31, M41, M51, M61, M71, M81) are arranged to support the same data signals (DB1). The functions of those memory chips are equivalent to the functions of the memory chips in one vertical column of the prior art memory system in FIG. 1(c). Therefore, we call such architecture as "vertical data connection". We will call the memory chips (M11, M21, M31, M41, M51, M61, M71, M81) in a MMB module as an "MMB group". Under vertical data connection, at any given time no more than one of the memory chips in the MMB group is allowed to access the system data signal (DB1) under normal operation conditions, making it possible to isolate the loadings of different chips by multiplexing. As shown in FIG. 3(a), the chip level data signals (D11, D21, D31, D41, D51, D61, D71, D81) are connected to the branch entries of bidirectional multiplexers (MUX8), while the system level data signals (DB1) are connected to the root entries of the bidirectional multiplexers (MUX8). FIG. 3(a) uses the symbolic view of a multiplexer to represent a plurality of bi-directional multiplexers because we need one bi-directional multiplexer for each bit of system level data signal (DB1). An MMB select logic circuitry analyzes the system control signal (CTL) and calculates the select signals (SM) for the bidirectional multiplexers (MUX8). This MMB select logic circuitry also serves as buffers to provide chip level control signals (Mctl) to memory chips.

[0027] Since data signals of memory chips are typically bi-direction signals (with possible exceptions such as input data masks), the multiplexers (MUX8) in MMB modules actually need to have both multiplexing and de-multiplexing functions. We will call such circuitry as "bidirectional multiplexer" in our discussions. A person with ordinary skill in circuit design would be able to design bidirectional multiplexers in wide varieties of configurations. FIG. 3(b) shows one of the simplest implementations of bidirectional multiplexers useful for applications of the present invention. For this example, the chip level data signals (D11, D21, D31, D41, D51, D61, D71, D81) are connected to the sources of MOS transistors (M1-M8), while the drains of those transistors are all connected to the same system level data signal (DB1). By controlling the gate signals (G1-G8) we can select chip level signals that are allowed to communicate with the system level signal, and isolate the loadings on unselected signals. There are many other ways to implement bidirectional multiplexers. A typical example is to use a pair of p-channel and n-channel pass gate transistors to control one entry. Combinational logic gates also can form equivalent circuitry. The scope of the present invention is not limited by particular implementations of the detailed circuit designs. A "bidirectional multiplexer" defined in the present invention is a circuitry that provides multiplexing as well as de-multiplex-

ing functions for bidirectional signal communication: A "bidirectional multiplexer" has one "root entry" and a plurality of "branch entries". Using FIG. 3(b) as an example, the transistor sources connected to signals D11, D21, D31, D41, D51, D61, D71, D81 are "branch entries" while the transistor drains connected to signal DB1 is the "root entry" defined in this patent application. In our definition, bidirectional multiplexers used in the present invention must be able to isolate loadings on unselected data signals. "Isolate loadings from a signal" means significantly reduce the effective loading caused by the signal. During normal operation conditions, one or no branch entry of a bidirectional multiplexer is selected to communicate with the "root entry" while the loadings of unselected branch entries are isolated from the root entry. However "bidirectional multiplexer" used for the present invention allows exceptions. For example, we may want to simultaneously select multiple entries in special modes. For another example, during the time to switch from one entry to another entry, we may have both entries turned on for a short period of time. We also want to have the capability to turn off all branch entries. Therefore, unlike the strictly defined logic function of multiplexers, the bidirectional multiplexers used by the present invention does not always guaranteed to have only one selected entry at all time.

[0028] FIG. 3(c) is the simplified schematic block diagram for an MMB memory system that has the same capacity as the prior art memory system in FIG. 1(c). In this example, the memory system comprises 8 MMB modules (MMB1-MMB8). Each MMB module comprises 8 memory chips. Each MMB module is equipped with eight-entry bidirectional multiplexers. Each MMB module support one set of the system level data signals: MMB1 supports DB1, MMB2 supports DB2, MMB3 supports DB3, MMB4 supports DB4, MMB5 supports DB5, MMB6 supports DB6, MMB7 supports DB7, and MMB8 supports DB8. This MMB memory system has the same interface signals, the same capacity, and the same functions as the prior art system in FIG. 1(c); while the loading is equivalent to the loading of one prior art module in FIG. 1(a). Using such architecture is therefore able to support roughly 8 times more capacity than the architecture in FIG. 1(c).

[0029] It is well known that a properly controlled bidirectional multiplexer is able to isolate the loadings on unselected branches. The bidirectional multiplexer itself introduces additional loading, but such loading can be designed to be insignificant relative to overall loading. The bidirectional multiplexer also introduced additional delay, but such additional delay can be designed to be insignificant relative to overall delay. The selection logic signal (SM) of the bidirectional multiplexer (MUX8) is determined from system level control signals (CTL) by the MMB Select logic circuitry. The MMB Select logic circuitry can isolate the loading seen by the system level control signals (CTL), but it also introduces additional delays. However, the buffer delay can be designed to be insignificant. In many cases, we may not need to buffer the control signals. The logic function of the MMB Select logic circuitry is similar to DRAM data bus control logic circuits that are well known to the industry. An MMB is certainly by far less complex than a prior art AMB. Upon disclosure of the present invention, a person with ordinary skill in the art will certainly be able to design the MMB in wide varieties of ways so that there is no need to discuss in further details.

[0030] The MMB memory systems have many advantages comparing to prior art systems. It has identical functions and identical interface signals (DB1-DB8, CTL) as the prior art system in FIG. 1(c). MMB systems can be fully compatible with existing systems with no or minimal modifications. While the loadings on the data and control signals are equivalent to the loadings of a single module in FIG. 1(a) plus small overhead added by the MMB circuits, the MMB overhead typically can be designed to be insignificant relative to the system loading. Using MMB architectures, it is very common to be able to increase system capacity by 4 to 16 times or more. The timing overhead is typically much less than that of FBDIMM systems. The MMB systems are by far more cost efficient than prior art AMB systems. The power consumed by MMB systems is by far less than prior art systems with equivalent capacities.

[0031] While specific embodiments of the invention have been illustrated and described herein, it is realized that other modifications and changes will occur to those skilled in the art. Upon disclosure of the present invention, those skilled in the art will be able to develop wide varieties of circuits to implement the elements of the present invention. For example, there are many ways in designing the bidirectional multiplexer and supporting selection logic circuits. For another example, the chip select signals connected to memory chips in the same MMB group can be defined in many different ways. If each memory chip in the same MMB group has separated chip select signal, then the function of an MMB system is equivalent to the function of many conventional modules. If all the memory chips in the same MMB group are connected to the same chip select signal, then the function of a MMB group is equivalent to a memory chip of the combined capacity of all memory chips in the group. We certainly can use combinations of the above two chip selection methods. For another example, we can modify the data signal connection methods to define a variation of the MMB architecture called "Multiplexed Bus Memory Buffer" (MBMB) architecture as illustrated by FIGS. 4(a-c).

[0032] For the MMB example in FIG. 3(a), each entry of a bidirectional multiplexer is connected to a single memory chip. For MBMB modules, each entry of a bidirectional multiplexer can be shared by multiple memory chips. The MBMB example in FIG. 4(a) illustrates the option when each entry of a multiplexer is shared by two memory chips. Memory chips M11 and M21 are sharing the same data signals (D121) in a bus structure, memory chips M31 and M41 are sharing another set of data signals (D341) in a bus structure, Memory chips M51 and M61 are sharing the same data signals (D561) in a bus structure, while memory chips M71 and M81 are sharing another set of data signals (D781) in a bus structure. Using such configuration, we only need 4-entry bidirectional multiplexers (MUX4) instead of 8-entry bidirectional multiplexers. FIG. 4(b) shows one of the simplest implementation of bidirectional multiplexer useful for applications of the present invention. For this example, the shared data entries (D121, D341, D561, D781) are connected to the sources of MOS transistors (M12, M34, M56, M78), while the drains of those transistors are all connected to the same system level data signal (DB1). By controlling the gate signals (G12, G34, G56, G78) we can select chip level signals that are allowed to communicate with the system level signal, and isolate the loadings on unselected signals.

[0033] FIG. 4(c) is the simplified schematic block diagram for an MBMB memory system that has the same capacity as

the prior art memory system in FIG. 1(c). In this example, the memory system comprises 8 MBMB modules (MBMB1-MBMB8). Each MBMB module comprises 8 memory chips. Each MBMB is equipped with four-entry bidirectional multiplexers to select one set of data signals from one of the eight memory chips in the same MBMB module (with the helps of chip select signals that are not shown separately), while every pair of memory chips share one entry of the MBMB bidirectional multiplexer. The MBMB system in FIG. 4(c) can serve the same function as the prior art system in FIG. 1(c) as well as the MMB system in FIG. 3(c). The signal loadings of the MBMB system are equivalent to that of two memory modules in FIG. 1(b), which is higher than the loading of the MMB system in FIG. 3(a). In the mean time, MBMB modules are more cost efficient than MMB modules due to less entries in bidirectional multiplexers and lower pin counts in MMB chips. The optimum selection is determined by system requirements.

[0034] While specific embodiments of the invention have been illustrated and described herein, it is realized that other modifications and changes will occur to those skilled in the art. For example, each entry of MBMB multiplexer certainly can support more than 2 memory chips by trading higher loading to achieve lower costs. Different number of memory chips can be connected to different entries of multiplexers. The number of branch entries of each bidirectional multiplexer can be any number larger or equal to 2, not limited to 4 or 8 entries. We certainly can connect more modules to the MMB or MBMB systems. It is also possible to link MMB or MBMB modules with FBDIMM architectures to achieve very large capacity.

[0035] The present invention is a board level architecture developed to increase the total capacity of memory systems while isolating the loading of data signals by multiplexing. Comparing to prior art memory modules, the loadings of an MMB system of the present invention are equivalent to a prior art SIMM module. The variation of MMB system called MBMB system allows multiple memory chips to share the same entry of a bidirectional multiplexer in a bused connection. When each entry of a bidirectional multiplexer is shared by two memory chips, the equivalent loadings are about the same as a prior art DIMM module. Using MMB or MBMB architectures, we can achieve memory capacity much higher than prior art memory systems without significant degradation in system performance. The memory systems of the present invention can be fully compatible with prior art memory systems. The costs of MMB or MBMB systems are by far lower than the cost of prior art FBDIMM systems.

[0036] Prior art memory systems typically fit one memory module into one printed circuit board. That is not necessary the case for memory modules of the present invention. We often fit multiple modules into a single printed circuit board. It is even possible to fit the whole memory system into a single printed circuit board. The memory systems of the present invention can have identical system level interface as prior art systems. It is therefore possible to design printed circuit boards of the present invention that can use existing DIMM sockets with no or minimal modifications. The printed circuit boards of the present invention sometimes do not use all the interface signals on a conventional DIMM socket, and sometimes we may need more signals such as chip select signals and clock enable signals in other sockets. We may need to use additional board level connectors or small modifications in

board interface to design circuit boards of the present invention that fit into prior art DIMM sockets.

[0037] A "memory system" is defined as board level circuits supporting memory operations. A "memory module" is defined as separable sub circuits of a memory system. A "system level signal" is defined as an electrical signal used to communicate with circuits external to a memory system. A "chip level signal" is defined as an electrical signal used to communicate with memory chips. The "Loading" on a signal is the non-ideal factors that can slow down performances such as leakage currents, parasitic capacitances, inductances, or resistances. A "bidirectional multiplexer" defined in the present invention is a circuitry that provides multiplexing as well as de-multiplexing functions for bidirectional signal communication; A "bidirectional multiplexer" has one "root entry" and a plurality of "branch entries"; During normal operation conditions, one or no branch entry of a bidirectional multiplexer is selected to communicate with the "root entry" while the loadings of unselected branch entries are isolated from the root entry; However "bidirectional multiplexer" allows exceptions, such as transitional operations or special mode operations, to have conditions when multiple branch entries are selected simultaneously. "Isolate loadings from a signal" means significantly reduce the effective loading caused by the signal. An "IC chip" is defined as packed integrated circuit or integrated circuit bare die that is ready to be placed on printed circuit board. A "memory chip" is defined as packaged IC memories or bare die memory integrated circuit that is ready to be placed on printed circuit board.

[0038] While specific embodiments of the invention have been illustrated and described herein, it is realized that other modifications and changes will occur to those skilled in the art. It is therefore to be understood that the appended claims are intended to cover all modifications and changes as fall within the true spirit and scope of the invention.

What is claimed is:

1. A memory system or a memory module comprising:

A plurality of integrated circuit memory chips placed on printed circuit boards;

System level data signals for data communication to circuits external to said memory system or memory module;

Chip level data signals for data communication to said memory chips;

Integrated circuit chip(s) comprising a plurality of bidirectional multiplexers;

Wherein a plurality of system level data signals are connected to the root entries of said bidirectional multiplexers, while the chip level data signals supporting said system level data signals are connected to the branch entries of said bidirectional multiplexers for selective isolation of loadings.

2. The memory chips in claim 1 are dynamic random access memory chips.

3. The dynamic random access memory chips in claim 2 are synchronized dynamic random access memory integrated circuit with data transfer rate higher than 600 million bits per second per signal.

4. The dynamic random access memory chips in claim 2 supports double data rate operations.

5. The memory system in claim 1 is compatible with JEDEC standard DIMM interface with no or minimal modifications.

6. The memory system in claim 1 is compatible with JEDEC standard SIMM interface with no or minimal modifications.

7. One branch entry of the bidirectional multiplexers in claim 1 is connected to one data signal of one memory chip.

8. One branch entry of the bidirectional multiplexers in claim 1 is shared by data signals from multiple memory chips.

9. A method to manufacture a memory system or a memory module comprising the steps of:

Placing a plurality of integrated circuit memory chips on printed circuit board(s);

Providing system level data signals for data communication to circuits external to said memory system or memory module;

Providing chip level data signals for data communication to said memory chips;

Providing integrated circuit chip(s) comprising a plurality of bidirectional multiplexers;

Wherein a plurality of system level data signals are connected to the root entries of said bidirectional multiplexers, while the chip level data signals supporting said system level data signals are connected to the branch entries of said bidirectional multiplexers for selective isolation of loadings.

10. The method in claim 9 comprising the step of placing a plurality of memory chips on printed circuit board(s) using dynamic random access memory chips.

11. The method in claim 10 comprising the step of placing a plurality of dynamic random access memory chips on printed circuit board(s) using synchronized dynamic random access memory with data transfer rate higher than 600 million bits per second per signal.

12. The method in claim 9 comprising the step of placing a plurality of dynamic random access memory chips on printed circuit board(s) using dynamic random access memory chips that supports double data rate operations.

13. The method in claim 9 provides a memory system that is compatible with JEDEC standard DIMM interface with no or minimal modifications.

14. The method in claim 9 provides a memory system that is compatible with JEDEC standard SIMM interface with no or minimal modifications.

15. The method in claim 9 comprises the step of connecting one data signal of one memory chip to one branch entry of the bidirectional multiplexers in claim 9.

16. The method in claim 9 comprises the step of connecting data signals from multiple memory chips to share one branch entry of the bidirectional multiplexers in claim 9.

* * * * *

PATENT COOPERATION TREATY

From the
INTERNATIONAL PRELIMINARY EXAMINING AUTHORITY

To:

KNOBBE, MARTENS, OLSON & BEAR, LLP
2040 Main Street, 14th Floor
Irvine, CA 92614
ETATS-UNIS D'AMERIQUE

PCT

WRITTEN OPINION OF THE INTERNATIONAL PRELIMINARY EXAMINING AUTHORITY (PCT Rule 66)

Date of mailing (day/month/year)	24.10.2011
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Applicant's or agent's file reference NETL.048VPC	REPLY DUE within <u>1</u> month(s) from the above date of mailing
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International application No. PCTUS2010/040826	International filing date (day/month/year) 01.07.2010	Priority date (day/month/year) 16.07.2009
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International Patent Classification (IPC) or both national classification and IPC INV. G11C5/02
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Applicant Netlist, Inc.

1. ☒ The written opinion established by the International Searching Authority:
 ☒ is ☐ is not
 considered to be a written opinion of the International Preliminary Examining Authority.
2. This second opinion contains indications relating to the following items:
 - ☒ Box No. I Basis of the opinion
 - ☐ Box No. II Priority
 - ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
 - ☐ Box No. IV Lack of unity of invention
 - ☒ Box No. V Reasoned statement under Rule 66.2(a)(ii) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement
 - ☐ Box No. VI Certain documents cited
 - ☒ Box No. VII Certain defects in the international application
 - ☒ Box No. VIII Certain observations on the international application
3. The applicant is hereby **invited to reply** to this opinion.

 When? See the time limit indicated above. The applicant may, before the expiration of that time limit, request this Authority to grant an extension, see Rule 66.2(e).
 How? By submitting a written reply, accompanied, where appropriate, by amendments, according to Rule 66.3. For the form and the language of the amendments, see Rules 55.3 and 66.8.
 Also: For the examiner's obligation to consider amendments and/or arguments, see Rule 66.4bis. For an informal communication with the examiner, see Rule 66.6. For an additional opportunity to submit amendments, see Rule 66.4.
 If no reply is filed, the international preliminary examination report will be established on the basis of this opinion.
4. The final date by which the international preliminary report on patentability (Chapter II of the PCT) must be established according to Rule 69.2 is: 16.11.2011

Name and mailing address of the international preliminary examining authority:	Authorized Officer
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**KNOBBE, MARTENS,
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**WRITTEN OPINION OF THE INTERNATIONAL
PRELIMINARY EXAMINING AUTHORITY**

International application No.
PCT/US2010/040826

Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
- ☒ the international application in the language in which it was filed
 - ☐ a translation of the international application into ,
which is the language of a translation furnished for the purposes of:
 - ☐ international search (under Rules 12.3(a) and 23.1(b))
 - ☐ publication of the international application (under Rule 12.4(a))
 - ☐ international preliminary examination (under Rules 55.2(a) and/or 55.3(a) and (b))
2. With regard to the elements of the international application, this opinion has been established on the basis of
(replacement sheets which have been furnished to the receiving Office in response to an invitation under Article 14 are referred to in this opinion as "originally filed"):

Description, Pages

1-26 as originally filed

Claims, Numbers

1-55 filed with telefax on 14-05-2011

Drawings, Sheets

1-13 filed with the letter of

19-08-2010

- ☐ a sequence listing - see Supplemental Box Relating to Sequence Listing.
3. ☐ The amendments have resulted in the cancellation of:
- ☐ the description, pages
 - ☐ the claims, Nos.
 - ☐ the drawings, sheets/figs
 - ☐ the sequence listing (*specify*):
4. ☐ This opinion has been established as if (some of) the amendments listed below had not been made, since either they are considered to go beyond the disclosure as filed, or they were not accompanied by a letter indicating the basis for the amendments in the application as filed, as indicated in the Supplemental Box (Rules 70.2(c) and (c-bis)):
- ☐ the description, pages
 - ☐ the claims, Nos.
 - ☐ the drawings, sheets/figs
 - ☐ the sequence listing (*specify*):
5. ☐ This opinion has been established:
- ☐ taking into account the rectification of an obvious mistake authorized by or notified to this Authority under Rule 91 (Rule 66.1(d-bis)).
 - ☐ without taking into account the rectification of an obvious mistake authorized by or notified to this Authority under Rule 91 (Rule 66.4bis).

**WRITTEN OPINION OF THE INTERNATIONAL
PRELIMINARY EXAMINING AUTHORITY**

International application No.
PCT/US2010/040826

6. ☐ Supplementary international search report(s) from Authority(ies) have been received and taken into account in drawing up this opinion (Rule 45bis.8(b) and (c)).

Box No. V Reasoned statement under Rule 66.2(a)(II) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Claims	<u>1-5, 10-17, 19-26, 29-33, 35-38, 43, 46-51, 55</u>
Inventive step (IS)	Claims	<u>1-5, 10-17, 19-27, 29-33, 35-38, 43, 45-51, 55</u>
Industrial applicability (IA)	Claims	

2. Citations and explanations:

see separate sheet

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item V.

1 Reference is made to the following documents:

- D1 US 2006/262586 A1 (SOLOMON JEFFREY C [US] ET AL) 23 November 2006 (2006-11-23)
- D2 US 2008/162790 A1 (IM JEON-TAEK [KR]) 3 July 2008 (2008-07-03)
- D3 US 2009/103387 A1 (SHAU JENG-JYE [US]) 23 April 2009 (2009-04-23)

2 The term "merge" or "merging", as used in a plurality of claims in the present application, is not clear (Article 6 PCT).

3 The present application does not meet the criteria of Article 33(1) PCT, for the following reasons:

3.1 Document D2 discloses (see Figs. 11, 12) a memory module comprising:

- at least one printed circuit board (510, 520, 530, 540);
- a plurality of memory devices (511a) mechanically coupled to the at least one printed circuit board;
- a control circuit (516a, 516b) mechanically coupled to the at least one printed circuit board, the control circuit configurable to receive control signals from a system memory controller (550) and to transmit module control signals to the plurality of memory devices (see Fig. 12); and
- a plurality of data transmission circuits (fast enable transfer switches 515) mechanically coupled to the at least one printed circuit board and distributed at corresponding positions relative to the at least one printed circuit board, the plurality of data transmission circuits configurable to be operatively coupled to the system memory controller (through the interface 516) and configurable to receive module control signals from the control circuit (DQS and SQ signals, see par. [0072]),

wherein at least one first data transmission circuit of the plurality of data transmission circuits is operatively coupled to at least two memory devices of the plurality of memory devices (see Fig. 12, wherein one first circuit 515 is coupled to at least two memory devices, 511a and 511b), at least one second data transmission circuit (one second circuit 515) of the plurality of data transmission circuits is operatively coupled to at least two memory devices of the plurality of memory devices.

The plurality of data transmission circuits are obviously configured to response to the module control signals, since they have multiplexing and latching functionalities, e.g. the fast enable transfer switches (FET switches) are controlled by the control circuit to store/latch and multiplex data from/to at least two memory devices. The FETs are therefore responsible for inhibition/driving of data to/from the memory devices (i.e. the FETs are coupled to the read/write paths of the memory devices).

The independent claims 1, 21, 37 are therefore not new in the sense of Article 33(2) PCT.

- 3.2 Independent method claims 31 and 48 contain each, respectively defined in method terms, the technical features of at least one of the independent apparatus claims 1, 21, 37 and additionally the following definitions:
- the data transmission circuits comprising a byte-wise data buffer; and
 - the data transmission circuits being load reducing switching circuits.
- Since the additional definitions are directly disclosed in D2 (see par. [0072]), the subject-matter of claims 31 and 48 is not new compared to the prior art of D2.
- 4 The technical features of dependent claims 2-5, 10 can be directly found in D2 (see par. [0072 and Figs. 11,12]).
- 5 The alignment of the data transmission circuits, the connector and the memory devices in D2 is shown on Figs. 10 and 11 and is done in accordance with the requirements of claims 11-17 in the present application.
- 6 The technical features of claims 19, 20, 22-26, 29, 30, 35, 36, 38, 43, 46, 47, 49-51, 55 can be directly found in D2, see par. [0072].
- 7 Since the FET switches in D2 switch data to/from at least two memory devices, these are considered as driven by the controller circuit in accordance with the currently running operation, as required by dependent claims 32 and 33.
- 8 The data transmission circuits and the memory devices in D2 are not specified as having different bit widths as required by claims 6-9 in the present application.
- 9 A tristate buffer, as defined in claims 18, 28, 34, 54, is not disclosed in the prior art of D2.

- 10 The memory module in D2 does not comprise DRAM memory devices, as required in claims 27, 45 in the present application.
- 11 The technical features of dependent claims 39-42, 44, 52, 53 are not disclosed in D2.

Re Item VII.

- 12 The independent claims are not in the two-part form in accordance with Rule 6.3(b) PCT.
- 13 Reference signs are not inserted in parentheses in the claims in accordance with Rule 6.2(b) PCT.
- 14 Contrary to the requirements of Rule 5.1(a)(ii) PCT, the relevant background art disclosed in documents D1 - D2 is not mentioned in the description, nor are these documents identified therein.

Re Item VIII.

- 15 The various definitions of the invention given in the independent claims are such that the claims as a whole are not concise so that the requirements of Article 6 PCT are not met.

PATENT COOPERATION TREATY

From the
INTERNATIONAL PRELIMINARY EXAMINING AUTHORITY

To:

KNOBBE, MARTENS, OLSON & BEAR, LLP
2040 Main Street, 14th Floor
Irvine, CA 92614
ETATS-UNIS D'AMERIQUE

PCT

NOTIFICATION OF TRANSMITTAL OF THE INTERNATIONAL PRELIMINARY REPORT ON PATENTABILITY

(PCT Rule 71.1)

Date of mailing
(day/month/year)

28.11.2011

Applicant's or agent's file reference
NETL.048VPC

IMPORTANT NOTIFICATION

International application No.
PCT/US2010/040826

International filing date (day/month/year)
01.07.2010

Priority date (day/month/year)
16.07.2009

Applicant
Netlist, Inc.

1. The applicant is hereby notified that this International Preliminary Examining Authority transmits herewith the international preliminary report on patentability and its annexes, if any, established on the international application.
2. A copy of the report and its annexes, if any, is being transmitted to the International Bureau for communication to all the elected Offices.
3. Where required by any of the elected Offices, the International Bureau will prepare an English translation of the report (but not of any annexes) and will transmit such translation to those Offices.
4. **REMINDER**

The applicant must enter the national phase before each elected Office by performing certain acts (filing translations and paying national fees) within 30 months from the priority date (or later in some Offices) (Article 39(1)) (see also the reminder sent by the International Bureau with Form PCT/IB/301).

Where a translation of the international application must be furnished to an elected Office, that translation must contain a translation of any annexes to the international preliminary report on patentability. It is the applicant's responsibility to prepare and furnish such translation directly to each elected Office concerned.

For further details on the applicable time limits and requirements of the elected Offices, see Volume II of the PCT Applicant's Guide.

The applicant's attention is drawn to Article 33(5), which provides that the criteria of novelty, inventive step and industrial applicability described in Article 33(2) to (4) merely serve the purposes of international preliminary examination and that "any Contracting State may apply additional or different criteria for the purposes of deciding whether, in that State, the claimed inventions is patentable or not" (see also Article 27(5)). Such additional criteria may relate, for example, to exemptions from patentability, requirements for enabling disclosure, clarity and support for the claims.

Name and mailing address of the international
preliminary examining authority:



European Patent Office
D-80298 Munich
Tel. +49 89 2399 - 0
Fax: +49 89 2399 - 4465

Authorized Officer

Slater, Sharon

Tel. +49 89 2399-2565



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DEC 05 2011

KNOBBE, MARTENS,
OLSON & BEAR, LLP

PATENT COOPERATION TREATY

PCT

INTERNATIONAL PRELIMINARY REPORT ON PATENTABILITY

(Chapter II of the Patent Cooperation Treaty)

(PCT Article 36 and Rule 70)

Applicant's or agent's file reference NETL048VPC	FOR FURTHER ACTION		See Form PCT/PEA/416
International application No. PCT/US2010/040826	International filing date (day/month/year) 01.07.2010	Priority date (day/month/year) 16.07.2009	
International Patent Classification (IPC) or national classification and IPC INV. G11C5/02			
Applicant Netlist, Inc.			
1. This report is the international preliminary examination report, established by this International Preliminary Examining Authority under Article 35 and transmitted to the applicant according to Article 36. 2. This REPORT consists of a total of <u>6</u> sheets, including this cover sheet. 3. This report is also accompanied by ANNEXES, comprising: a. ☒ (sent to the applicant and to the International Bureau) a total of <u>26</u> sheets, as follows: ☒ sheets of the description, claims and/or drawings which have been amended and/or sheets containing rectifications authorized by this Authority, unless those sheets were superseded or cancelled, and any accompanying letters (see Rules 46.5, 66.8, 70.16, 91.2, and Section 607 of the Administrative Instructions). ☐ sheets containing rectifications, where the decision was made by this Authority not to take them into account because they were not authorized by or notified to this Authority at the time when this Authority began to draw up this report, and any accompanying letters (Rules 66.4bis, 70.2(e), 70.16 and 91.2). ☐ superseded sheets and any accompanying letters, where this Authority either considers that the superseding sheets contain an amendment that goes beyond the disclosure in the international application as filed, or the superseding sheets were not accompanied by a letter indicating the basis for the amendments in the application as filed, as indicated in item 4 of Box No. I and the Supplemental Box (see Rule 70.16(b)). b. ☐ (sent to the International Bureau only) a total of (indicate type and number of electronic carrier(s)) , containing a sequence listing, in electronic form only, as indicated in the Supplemental Box Relating to Sequence Listing (see paragraph 3bis of Annex C of the Administrative Instructions).			
4. This report contains indications relating to the following items: ☒ Box No. I Basis of the report ☐ Box No. II Priority ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability ☐ Box No. IV Lack of unity of invention ☒ Box No. V Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement ☐ Box No. VI Certain documents cited ☒ Box No. VII Certain defects in the international application ☒ Box No. VIII Certain observations on the international application			
Date of submission of the demand		Date of completion of this report	
14.05.2011		28.11.2011	
Name and mailing address of the international preliminary examining authority:  European Patent Office D-80298 Munich Tel. +49 89 2399 - 0 Fax: +49 89 2399 - 4465		Authorized officer Trifonov, Antoniy Telephone No. +49 89 2399-7168	



**INTERNATIONAL PRELIMINARY REPORT
ON PATENTABILITY**

International application No.
PCT/US2010/040826

Box No. I Basis of the report

1. With regard to the **language**, this report is based on
- ☒ the international application in the language in which it was filed
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of:
 - ☐ international search (under Rules 12.3(a) and 23.1(b))
 - ☐ publication of the international application (under Rule 12.4(a))
 - ☐ international preliminary examination (under Rules 55.2(a) and/or 55.3(a) and (b))
2. With regard to the **elements*** of the international application, this report is based on *(replacement sheets which have been furnished to the receiving Office in response to an invitation under Article 14 are referred to in this report as "originally filed" and are not annexed to this report):*

Description, Pages

1-26 as originally filed

Claims, Numbers

1-57 filed with telefax on 15-11-2011

Drawings, Sheets

1-13 filed with the letter of 19-08-2010

- ☐ a sequence listing - see Supplemental Box Relating to Sequence Listing.
3. ☐ The amendments have resulted in the cancellation of:
- ☐ the description, pages
 - ☐ the claims, Nos.
 - ☐ the drawings, sheets/figs
 - ☐ the sequence listing (*specify*):
 - ☐ any table(s) related to sequence listing (*specify*):
4. ☐ This report has been established as if (some of) the amendments annexed to this report and listed below had not been made, since either they are considered to go beyond the disclosure as filed, or they were not accompanied by a letter indicating the basis for the amendments in the application as filed, as indicated in the Supplemental Box (Rules 70.2(c) and (c-bis)):
- ☐ the description, pages
 - ☐ the claims, Nos.
 - ☐ the drawings, sheets/figs
 - ☐ the sequence listing (*specify*):
5. ☐ This report has been established:
- ☐ taking into account the rectification of an obvious mistake authorized by or notified to this Authority under Rule 91 (Rules 66.1(d-bis) and 70.2(e)).
 - ☐ without taking into account the rectification of an obvious mistake authorized by or notified to this Authority under Rule 91(Rules 66.4bis and 70.2(e)).

**INTERNATIONAL PRELIMINARY REPORT
ON PATENTABILITY**

International application No.
PCT/US2010/040826

6. ☐ Supplementary international search report(s) from Authority(ies) has/have been received and taken into account in establishing this report (Rule 45bis.8(b) and (c)).

Box No. V Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims	<u>6-12, 18, 21-36, 41, 42, 45, 54, 56, 57</u>
	No: Claims	<u>1-5, 13-17, 19, 20, 37-40, 43, 44, 46-53, 55</u>
Inventive step (IS)	Yes: Claims	<u>6-12, 18, 21-36, 41, 42, 45, 54, 56, 57</u>
	No: Claims	<u>1-5, 13-17, 19, 20, 37-40, 43, 44, 46-53, 55</u>
Industrial applicability (IA)	Yes: Claims	<u>1-57</u>
	No: Claims	

2. Citations and explanations (Rule 70.7):

see separate sheet

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item V.

1 Reference is made to the following documents:

D1 US 2006/262586 A1 (SOLOMON JEFFREY C [US] ET AL) 23 November 2006 (2006-11-23)

D2 US 2008/162790 A1 (IM JEON-TAEK [KR]) 3 July 2008 (2008-07-03)

D3 US 2009/103387 A1 (SHAU JENG-JYE [US]) 23 April 2009 (2009-04-23)

2 The present application does not meet the criteria of Article 33(1) PCT, for the following reasons:

2.1 Document D2 discloses (see Figs. 6, 11, 12) a memory module comprising:

- at least one printed circuit board (510, 520, 530, 540);
- a plurality of memory devices (511a on Fig. 11, 420 on Fig. 6) mechanically coupled to the at least one printed circuit board;
- a control circuit (425, 516a, 516b) mechanically coupled to the at least one printed circuit board, the control circuit configurable to receive control signals from a system memory controller (410, 550) and to transmit module control signals to the plurality of memory devices (see Figs. 6, 12); and
- a plurality of data transmission circuits. The data transmission circuits in D2 are shown on Figs. 6 (page buffer 423) and 12 (fast enable transfer switches 515) and are mechanically coupled to the at least one printed circuit board and distributed at corresponding positions relative to the at least one printed circuit board. The plurality of data transmission circuits are further operatively coupled to the system memory controller (410) through the interface 516, receive module control signals from the control circuit (see the control unit 425 on Fig. 6), and are also used to latch (see pars. [0063, 0072]) the data written into or read out of the memory devices (e.g. the data transferred through the interface circuit 424). For the reason that they are used to latch both the input/output data and are obviously time controlled by the control circuit 425, the data transmission circuits are considered to include read/write buffer- and drive/inhibit data- transmission functionality.

Further, at least one first data transmission circuit of the plurality of data transmission circuits is operatively coupled to at least two memory devices of the plurality of memory devices (see Fig. 12, wherein one first circuit 515 is

coupled to at least two memory devices, 511a and 511b), at least one second data transmission circuit (one second circuit 515) of the plurality of data transmission circuits is operatively coupled to at least two memory devices of the plurality of memory devices.

Independent claim 1 is therefore not new in the sense of Article 33(2) PCT.

- 2.2 Independent claim 21 paraphrases and generalises the subject-matter of claim 1 by renaming the "data transmission circuits" in "circuits" and by removing the previously defined first and second data transmission circuits. On the other side claim 21 specifies that each of the plurality of circuits and the respectively coupled to each of them memory devices have different bit widths (understand that the respective read/write I/O data paths connected to the data transmission circuits have "bit widths") and thereby provides a feature, which is not disclosed in D2.
The subject-matter of independent claim 21 is therefore new over the prior art of D2.

- 2.3 Independent claim 31 corresponds to claim 21 in method terms and is also considered new over the prior art of D2.

- 2.4 Independent claims 37 and 48 respectively correspond to independent claims 21 and 31. The difference between the subject-matter of these claims is in that the plurality of data transmission circuits (in claim 21) or simply circuits (in claim 31) are replaced by a switching circuit (in claim 37) respectively a data load switching circuit (in claim 48). For a single switching circuit, which is coupled to a plurality of memory devices is, however, obvious to have a greater bit width than a single memory device (see in D2, wherein all the switches 515 build a switching circuit, which is coupled to all memory devices having each lower bit width).

Accordingly, the technical feature in claims 21 and 31, which makes a difference over the closest prior art of D2, is not considered to provide the same or even any difference between claims 37, 48 and said prior art.

The subject-matter of independent claims 37 and 48 is therefore not new over the prior art of D2.

- 3 The data transmission circuits and the memory devices in D2 are not specified as having different bit widths as required by claims 6-9, 21-36 in the present application.
- 4 A tristate buffer, as defined in claims 18, 41, 42, 54, is not disclosed in the prior art of D2.
- 5 The memory module in D2 does not comprise SDRAM memory devices, as required by claims 10-12, 45, 56, 57 in the present application.
- 6 The features of the remaining dependent claims can either directly be found in the prior art or concern an obvious modification of the prior art and add nothing new or inventive to the independent claims.

Re Item VII.

- 7 The independent claims are not in the two-part form in accordance with Rule 6.3(b) PCT.
- 8 Reference signs are not inserted in parentheses in the claims in accordance with Rule 6.2(b) PCT.
- 9 Contrary to the requirements of Rule 5.1(a)(ii) PCT, the relevant background art disclosed in documents D1 - D2 is not mentioned in the description, nor are these documents identified therein.

Re Item VIII.

- 10 The various definitions of the invention given in the independent claims are such that the claims as a whole are not concise so that the requirements of Article 6 PCT are not met.

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KNOB NPB.

002/025

Knobbe Martens Olson & Bear LLP*Intellectual Property Law*

2040 Main Street
Fourteenth Floor
Irvine, CA 92614
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November 14, 2011

VIA FACSIMILE (49.89.2933.4465)

International Preliminary Examining Authority
Attn: Authorized Officer Antony Trifonov
THE EUROPEAN PATENT OFFICE
D-80298 Munich
GERMANY

Re: **WRITTEN REPLY, UNDER RULE 66.3, TO WRITTEN OPINION OF
THE INTERNATIONAL PRELIMINARY EXAMINING AUTHORITY**
Title: **SYSTEM AND METHOD UTILIZING DISTRIBUTED BYTE-WISE
BUFFERS ON A MEMORY MODULE**
Application No.: **PCT/US2010/040826**
Applicant: **NETLIST, Inc.**
Our Reference No.: **NETL.048VPC**

Dear Examiner Trifonov:

This letter is in response to the "Written Opinion of the International Preliminary Examining Authority" mailed October 24, 2011. With this response, the Applicant submits the following amendments and remarks.

The Applicant provides herewith an amended claim set on Replacement Pages 27-35 including Replacement Claims 1-57 which replace Claims 1-55 as previously filed on May 13, 2011. Additionally, for the Examiner's convenience, the Applicant includes a marked-up copy of the amended claim set showing the changes as compared to the previously-filed claims (Appendix A). The Applicant has amended Replacement Claims 10, 11, 15, 21, 25, 31, 33, 35-37, 40, 48, 50, 51, and 55, and has added new Claims 56 and 57.

The amendment to Claim 10 and the new Claims 56 and 57 recite that "the plurality of memory devices comprise one or more synchronous dynamic random access memory devices" and are supported by the application as originally filed, including but not limited to paragraph [0046].

The amendments to Claim 11 clarify the claim by using proper antecedent basis and do not add new matter to Claim 11. The amendments to Claim 50 clarify the claim by including the word "the" and do not add new matter to Claim 50.

The amendment to Claim 15 corrects a typographical error in its dependency and does not add new matter to Claim 15.

At least some of the amendments to Claims 21, 31, 37, and 48 recite the relationship between the bit width of the circuits and the bit widths of the memory devices and are supported by the application as originally filed, including but not limited to paragraph [0056].

San Diego
858-707-4000

San Francisco
415-954-4114

Los Angeles
310-551-3450

Riverside
951-781-9231

Seattle
206-405-2000

Washington, DC
202-640-6400

Silicon Valley
650-752-1100

ation: 15.11.2011 01:33:10 - 15.11.2011 01:38:04. This page 2 of 25 was completed at 15.11.2011 01:33:31
Received at the EPO on Nov 15, 2011 01:38:04. Page 2 of 25

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003/025

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Attn: Authorized Officer Antony Trifonov

November 14, 2011

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At least some of the amendments to Claims 21, 31, 37, and 48 and the amendments to Claims 25, 33, 35, 36, 40, 51, and 55 remove the terms "merge," "merges," and "merged" and instead refer to data being received. These amendments are supported by the application as originally filed, including but not limited to paragraphs [0055], [0056], and [0064].

The Applicant acknowledges that the Written Opinion has been established on the basis of previously-filed Claims 1-55. The Written Opinion indicates that each of these claims has industrial applicability. The Written Opinion also appears to indicate that (i) Claims 6-9, 18, 27, 28, 34, 39-42, 44, 45, and 52-54 also satisfy the criteria of novelty and inventive step, and that (ii) Claims 1-5, 10-17, 19-27, 29-33, 35-38, 43, 45-51, and 55 do not. The Applicant respectfully traverses each of the pending rejections. However, as discussed, the Applicant has amended certain Replacement Claims in order to advance prosecution.

Claim 1

Claim 1 was rejected as not being new in the sense of Article 33(2) PCT in view of U.S. Patent Application Publication No. 2008/0162790 ("D2"). The Applicant respectfully submits, however, that D2 fails to identically disclose every feature of Claim 1, so Claim 1 satisfies the criteria of novelty and inventive step over D2.

For example, D2 fails to disclose or render obvious, among other things:

a plurality of data transmission circuits ... configurable to receive module control signals from the control circuit, ...

wherein the at least one first data transmission circuit is configurable to respond to the module control signals by selectively driving or inhibiting data transmission between the system memory controller and at least one selected memory device..., and wherein the at least one second data transmission circuit is configurable to respond to the module control signals by selectively driving or inhibiting data transmission between the system memory controller and at least one selected memory device

(emphasis added).

In section 3.1, the Written Opinion identifies the "interfaces 516a and 516b" of D2 as the control circuit, the "fast enable transfer switches 515" of D2 as the data transmission circuits, and the DQS and SQ signals of D2 as the "module control signals." However, nowhere does D2 disclose or suggest that the "fast enable transfer switches 515" are configurable to respond to the DQS and SQ signals "by selectively driving or inhibiting data transmission between the system memory controller and at least one selected memory device," as is recited by Claim 1.

The only discussion of the operation of the "fast enable transfer switches 515" is found in paragraph [0072] of D2, which merely discloses that the "fast enable transfer switches 515" are "used to reduce loads on data buses, thereby enabling high-speed data transmission" and "are used to facilitate the use of the DDR transmission method" (D2 at ¶[0072]). Instead of responding to the DQS or SQ signals, the "fast enable transfer switches 515" of D2 appear to only pass the received DQS signals onto the "NBX NAND flash memories 511a and 511b" to which the switches 515 are connected (see, e.g., Fig. 12 of D2). Therefore, the "fast enable transfer switches 515" cannot be interpreted to be the data

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Knobbe Martens Olson & Bear LLP

Attn: Authorized Officer Antony Trifonov
November 14, 2011
Page -3-

transmission circuits recited by Claim 1, and the DQS signals cannot be interpreted to be the "module control signals" recited by Claim 1.

Furthermore, the SQ signals of D2 also cannot be interpreted to be the "module control signals" of Claim 1. The SQ signals are produced by the memory devices and are not received by the switches 515 from the interface 516a, 516b (see, e.g., the single-headed arrow for SQ0~7 pointing away from the interface 516 in Fig. 12 of D2). Therefore, D2 does not disclose or suggest all the features of Claim 1.

In addition, the Written Opinion fails to identify any disclosure of D2 to correspond to "the plurality of data transmission circuits each comprising at least one write buffer configured to drive data onto at least two data paths to at least two corresponding memory devices of the plurality of memory devices and at least one read buffer configured to drive data from the corresponding at least two memory devices onto a data line to the system memory controller," as recited by Claim 1 (emphasis added).

While D2 describes at paragraph [0072] that "each of the FET switches 515 can latch 10 bits: 8 bits for data (DQ), 1 bit for a DQS signal, and 1 bit for a SQ signal," this passage does not disclose or suggest that the FET switches 515 comprise "at least one write buffer...and at least one read buffer," as recited by Claim 1. Instead, this passage is merely referring to the well-known technique of latching data lines by pulling them to their last valid logic level to keep the data bus stable (also known as "bus holding" or "bus keeping"). This interpretation is bolstered by Figs. 8 and 13 of D2, which show timing diagrams for the operation of the module of D2. These timing diagrams show DQ and DQS being toggled substantially immediately after the write enable (WE) command (see, Fig. 8 of D2) and substantially immediately after the read enable (RE) command (see, Fig. 13 of D2). None of these timing diagrams show the values of DQ or DQS being latched and then outputted some clock cycles later. Thus, the disclosure of D2 is not consistent with the interpretation being relied upon by the Written Opinion in rejecting Claim 1 that "the FETs are therefore responsible for inhibition/driving of data to/from the memory devices."

At least for the above reasons, the Applicant respectfully submits that D2 does not identically disclose or render obvious every feature of Claim 1, so Claim 1 is patentably distinguished from D2.

Independent Claims 21, 31, 37, and 48

Claims 21, 31, 37, and 48 were also rejected for not being new in view of D2. The Applicant respectfully submits, however, that these claims are patentably distinguished from D2, at least for reasons similar to those set forth with respect to Claim 1.

Claim 21 has been amended to recite "a plurality of circuits each having a first bit width and operatively coupled to at least two corresponding memory devices of the plurality of memory devices, the at least two corresponding memory devices each having a second bit width smaller than the first bit width." The Applicant submits that this feature is not disclosed or suggested by D2, so at least for this additional reason, Claim 21 is patentably distinguished over D2.

Claim 31 has been amended to recite that "the received data signals each having a bit width less than a bit width of the data transmission circuit." The Applicant submits that this

11/14/2011 17:41 FAX

KNOB NPB.

003/025

Knecht Martin Olson & Bear LLP

Attn: Authorized Officer Antony Trifonov
November 14, 2011
Page -4-

feature is not disclosed or suggested by D2, so at least for this additional reason, Claim 31 is patentably distinguished over D2.

Claim 37 has been amended to recite "a switching circuit configured to transmit data to and from the plurality of memory devices, the switching circuit having a first bit width and ... a memory device of the plurality of memory devices has a second bit width less than the first bit width." The Applicant submits that this feature is not disclosed or suggested by D2, so at least for this additional reason, Claim 37 is patentably distinguished over D2.

Claim 48 has been amended to recite "the load-reducing switching circuit having a first bit width ..., the memory devices each having a second bit width less than the first bit width." The Applicant submits that this feature is not disclosed or suggested by D2, so at least for this additional reason, Claim 48 is patentably distinguished over D2.

Dependent Claims 2-20, 22-30, 32-36, 38-47, and 49-55

In the Written Opinion, dependent Claims 6-9, 18, 27, 28, 34, 39-42, 44, 45, and 52-54 were found to satisfy the novelty and inventive step criteria, and dependent Claims 2-5, 10-17, 19, 20, 22-26, 29, 30, 32, 33, 35, 36, 38, 43, 46, 47, 49-51, and 55 were rejected in view of D2. Each of Claims 2-5, 10-17, 19, 20, 22-26, 29, 30, 32, 33, 35, 36, 38, 43, 46, 47, 49-51, and 55 is dependent upon one of Claims 1, 21, 31, 37, and 48. Therefore, each of Claims 2-5, 10-17, 19, 20, 22-26, 29, 30, 32, 33, 35, 36, 38, 43, 46, 47, 49-51, and 55 is patentably distinguished from the references for at least the reasons set forth with respect to its corresponding independent claim, based on its dependency and for its additional features.

New Dependent Claims 56 and 57

New dependent Claim 56 is dependent from Claims 31 and 48 and new dependent Claim 57 is dependent upon Claim 37. Therefore, each of Claims 56 and 57 is patentably distinguished from the references for at least the reasons set forth with respect to its corresponding independent claim, based on its dependency and for its additional features.

CONCLUSION

The Applicant submits that the present claims meet the requirements of PCT Articles 33(2) and 33(3). Nevertheless, if any undeveloped issues remain or if any issues require clarification, the Examiner is invited to contact the undersigned.

Sincerely,



Daniel E. Altman

Enclosures

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WHAT IS CLAIMED IS:

1. A memory module comprising:
 - at least one printed circuit board;
 - a plurality of memory devices mechanically coupled to the at least one printed circuit board;
 - a control circuit mechanically coupled to the at least one printed circuit board, the control circuit configurable to receive control signals from a system memory controller and to transmit module control signals to the plurality of memory devices; and
 - a plurality of data transmission circuits mechanically coupled to the at least one printed circuit board and distributed at corresponding positions relative to the at least one printed circuit board, the plurality of data transmission circuits configurable to be operatively coupled to the system memory controller and configurable to receive module control signals from the control circuit, the plurality of data transmission circuits each comprising at least one write buffer configured to drive data onto at least two data paths to at least two corresponding memory devices of the plurality of memory devices and at least one read buffer configured to drive data from the corresponding at least two memory devices onto a data line to the system memory controller,
- wherein at least one first data transmission circuit of the plurality of data transmission circuits is operatively coupled to a first at least two memory devices of the plurality of memory devices, at least one second data transmission circuit of the plurality of data transmission circuits is operatively coupled to a second at least two memory devices of the plurality of memory devices,
- wherein the at least one first data transmission circuit is configurable to respond to the module control signals by selectively driving or inhibiting data transmission between the system memory controller and at least one selected memory device of the first at least two memory devices operatively coupled to the at least one first data transmission circuit, and wherein the at least one second data transmission circuit is configurable to respond to the module control signals by selectively driving or inhibiting data transmission between the system memory controller and at least one

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selected memory device of the second at least two memory devices operatively coupled to the at least one second data transmission circuit.

2. The memory module of Claim 1, wherein the at least one first data transmission circuit comprises a byte-wise buffer.

3. The memory module of Claim 2, wherein the at least one second data transmission circuit comprises a byte-wise buffer.

4. The memory module of Claim 1, wherein the at least one first data transmission circuit has the same bit width as does each of the first at least two memory devices to which the at least one first data transmission circuit is operatively coupled.

5. The memory module of Claim 4, wherein the at least one second data transmission circuit has the same bit width as does each of the second at least two memory devices to which the at least one second data transmission circuit is operatively coupled.

6. The memory module of Claim 1, wherein the at least one first data transmission circuit has a first bit width and each of the first at least two memory devices to which the at least one first data transmission circuit is operatively coupled has a second bit width which is less than the first bit width.

7. The memory module of Claim 6, wherein the at least one first data transmission circuit is operatively coupled to two memory devices per rank, and the first bit width is twice the second bit width.

8. The memory module of Claim 6, wherein the at least one second data transmission circuit has a third bit width and each of the second at least two memory devices to which the at least one second data transmission circuit is operatively coupled has a fourth bit width which is less than the third bit width.

9. The memory module of Claim 8, wherein the at least one second data transmission circuit is operatively coupled to two memory devices per rank, and the third bit width is twice the fourth bit width.

10. The memory module of Claim 1, wherein the plurality of memory devices comprise one or more synchronous dynamic random access memory devices.

11. The memory module of Claim 10, wherein the corresponding positions of the at least one first data transmission circuit and the at least one second data transmission circuit are along an edge of the at least one printed circuit board such that the at least one first data transmission circuit is located substantially between the edge and at least some of the first at

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least two memory devices operatively coupled to the at least one first data transmission circuit and the at least one second data transmission circuit is located substantially between the edge and at least some of the second at least two memory devices operatively coupled to the at least one second data transmission circuit.

12. The memory module of Claim 11, wherein the at least one printed circuit board comprises a connector comprising a plurality of electrical contacts which are positioned on the edge of the at least one printed circuit board and are configured to be releasably coupled to corresponding contacts of a computer system socket.

13. The memory module of Claim 1, wherein a package containing the at least one first data transmission circuit is at a location spaced from a location of a package containing the at least one second data transmission circuit.

14. The memory module of Claim 13, wherein a position of the at least one first data transmission circuit is generally aligned with one or more of the first at least two memory devices operatively coupled to the at least one first data transmission circuit.

15. The memory module of Claim 14, wherein the at least one first data transmission circuit and the first at least two memory devices operatively coupled to the at least one first data transmission circuit are positioned generally along a line that is substantially perpendicular to an edge of the at least one printed circuit board.

16. The memory module of Claim 13, wherein a position of the at least one first data transmission circuit is generally offset from a line defined by the positions of the first at least two memory devices operatively coupled to the at least one first data transmission circuit.

17. The memory module of Claim 16, wherein the line is substantially perpendicular to an edge of the at least one printed circuit board and the at least one first data transmission circuit is offset from the line in a direction generally along the edge.

18. The memory module of Claim 1, wherein the at least one write buffer comprises at least one tristate buffer and the at least one read buffer comprises at least one tristate buffer.

19. The memory module of Claim 1, wherein the at least one write buffer regenerates data signals from the system memory controller and wherein the at least one read buffer regenerates data signals from the memory devices.

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20. The memory module of Claim 1, wherein read data that is selectively driven by the first data transmission circuit is transmitted through the at least one read buffer of the first data transmission circuit, and wherein write data that is driven by the first data transmission circuit is transmitted through the at least one write buffer of the first data transmission circuit, and wherein read data that is selectively driven by the second data transmission circuit is transmitted through the at least one read buffer of the second data transmission circuit, and wherein write data that is selectively driven by the second data transmission circuit is transmitted through the at least one write buffer of the second data transmission circuit.

21. A memory module comprising:

a plurality of memory devices;

a controller configured to receive control information from a system memory controller and to produce module control signals; and

a plurality of circuits each having a first bit width and operatively coupled to at least two corresponding memory devices of the plurality of memory devices, the at least two corresponding memory devices each having a second bit width smaller than the first bit width and comprising at least one write buffer and at least one read buffer, the plurality of circuits configured to selectively isolate the plurality of memory devices from the system memory controller, wherein the circuits are operable, in response to the module control signals, to actively drive write data from the system memory controller to the plurality of memory devices through the at least one write buffer and to receive and drive read data from the plurality of memory devices to the system memory controller through the at least one read buffer, wherein the circuits are distributed at corresponding positions separate from one another.

22. The memory module of Claim 21, wherein the plurality of circuits is contained in a plurality of packages at locations spaced from one another.

23. The memory module of Claim 22, wherein the plurality of memory devices, the controller, and the plurality of circuits are mechanically coupled to a printed circuit board having an edge, wherein the packages are positioned along the edge and between the edge and the plurality of memory devices.

24. The memory module of Claim 21, wherein the circuits comprise byte-wise buffers.

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25. The memory module of Claim 21, wherein the each circuit of the plurality of circuits comprises:

a data terminal configurable to be operatively coupled to the system memory controller;

a first memory terminal configurable to be operatively coupled to a first group of the plurality of memory devices; and

a second memory terminal configurable to be operatively coupled to a second group of the plurality of memory devices;

wherein, when the circuit actively drives write data, the data terminal is operatively coupled to one of the first memory terminal and the second memory terminal; and

wherein, when the circuit receives and drives read data, one of the first memory terminal and the second memory terminal is operatively coupled to the data terminal.

26. The memory module of Claim 21, wherein the memory module is a dual in-line memory module.

27. The memory module of Claim 21, wherein the plurality of memory devices comprise one or more synchronous dynamic random access memory devices.

28. The memory module of Claim 21, wherein the at least one write buffer of each of the plurality of circuits comprises at least one tristate buffer and the at least one read buffer of the plurality of circuits comprises at least one tristate buffer.

29. The memory module of Claim 21, wherein the at least one write buffer of each of the plurality of circuits regenerates data signals from the system memory controller.

30. The memory module of Claim 21, wherein the at least one read buffer of each of the plurality of circuits regenerates data signals from the memory devices.

31. A method of operating a memory module comprising a plurality of memory devices, the method comprising:

providing a data transmission circuit on a data line between a computer system memory controller and the plurality of memory devices of the memory module, the data transmission circuit comprising a byte-wise data buffer comprising at least one write buffer and at least one read buffer;

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during a write operation, enabling the data transmission circuit to drive a data signal from the computer system memory controller on one of a plurality of paths to the memory devices of the memory module through the at least one write buffer; and

during a read operation, enabling the data transmission circuit to receive a plurality of data signals from the memory devices of the memory module and to drive the received data signals to the computer system memory controller through the at least one read buffer, the received data signals each having a bit width less than a bit width of the data transmission circuit.

32. The method of Claim 31, further comprising providing a controller configured to receive control information from the computer system memory controller and to produce at least one module control signal in response to the control information, wherein enabling the data transmission circuit to drive a data signal comprises transmitting the at least one module control signal from the controller to the data transmission circuit.

33. The method of Claim 31, further comprising providing a controller configured to receive control information from the computer system memory controller and to produce at least one module control signal in response to the control information, wherein enabling the data transmission circuit to receive the plurality of data signals comprises transmitting the at least one module control signal from the controller to the data transmission circuit.

34. The method of Claim 31, wherein the at least one write buffer comprises at least one tristate buffer and the at least one read buffer comprises at least one tristate buffer.

35. The method of Claim 31, wherein, during a write operation, the byte-wise data buffer regenerates the data signal from the computer system memory controller and drives the regenerated data signal to the memory devices of the memory module, and wherein, during a read operation, the byte-wise data buffer regenerates the received data signals and drives the regenerated received data signals to the computer system memory controller.

36. The method of Claim 31, wherein during a write operation, the data signal is regenerated and driven by the at least one write buffer to the memory devices of the memory module, and during a read operation, the received data signals are regenerated and driven by the at least one read buffer to the computer system memory controller.

37. A memory module comprising:
a plurality of memory devices;

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a controller configured to receive control information from a system memory controller and produce module control signals; and

a switching circuit configured to transmit data to and from the plurality of memory devices, the switching circuit having a first bit width and comprising at least one write buffer and at least one read buffer, the switching circuit configured to isolate the plurality of memory devices from the system memory controller, wherein the switching circuit is operable, in response to the module control signals, to drive write data from the system memory controller to the plurality of memory devices through the at least one write buffer and to receive read data from the plurality of memory devices, wherein a memory device of the plurality of memory devices has a second bit width less than the first bit width, the received read data transmitted to the system memory controller through the at least one read buffer.

38. The memory module of claim 37, wherein the plurality of memory devices comprises a first group of the plurality of memory devices in a first rank and a second group of the plurality of memory devices in a second rank.

39. The memory module of claim 38, wherein the plurality of memory devices further comprises a third group of the plurality of memory devices in a third rank and a fourth group of the plurality of memory devices in a fourth rank, and where data lines of the first rank are connected to data lines of the first rank and to the switching circuit, and data lines of the second rank are connected to data lines of the fourth rank and to the switching circuit.

40. The memory module of claim 37, wherein the switching circuit comprises:

a data terminal for coupling to the system memory controller;

a first memory terminal coupled to the first group of the plurality of memory devices; and

a second memory terminal coupled to the second group of the plurality of memory devices;

wherein, when the switching circuit drives write data, the data terminal is coupled to one of the first memory terminal and the second memory terminal through the at least one write buffer; and

wherein, when the switching circuit receives read data from the plurality of memory devices, one of the first memory terminal and the second memory terminal is coupled to the data terminal through the at least one read buffer.

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41. The memory module of claim 40, wherein the at least one write buffer further comprises:

a first tristate buffer configured to conditionally drive the first memory terminal; and

a second tristate buffer configured to conditionally drive the second memory terminal.

42. The memory module of claim 41, wherein the at least one read buffer further comprises a multiplexer configured to receive data signals from the first memory terminal and data signals from the second memory terminal and supply data signals selected from the data signals received from the first memory terminal and the data signals received from the second memory terminal to the at least one read buffer.

43. The memory module of claim 37, wherein the memory module is a dual in-line memory module.

44. The memory module of claim 38, wherein the memory module is configured to combine the first rank and the second rank into one logical memory rank.

45. The memory module of claim 37, wherein the memory devices comprise synchronous dynamic random access memories.

46. The memory module of claim 37, wherein the switching circuit is configured to reshape a signal waveform.

47. The memory module of claim 37, wherein the controller comprises a register for latching address and control signals from the memory controller.

48. A method of operating a memory module comprising a plurality of memory devices, the method comprising:

providing a load-reducing switching circuit on a data line between a computer system memory controller and the plurality of memory devices of the memory module, the load-reducing switching circuit having a first bit width;

during a write operation, enabling the load-reducing switching circuit to drive a data signal from the computer system memory controller on one of a plurality of paths through at least one write buffer of the load-reducing switching circuit to the memory devices of the memory module; and

during a read operation, enabling the load-reducing switching circuit to receive a plurality of data signals from the memory devices of the memory module,

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the memory devices each having a second bit width less than the first bit width, and driving the received data signals through at least one read buffer of the load-reducing switching circuit to the computer system memory controller.

49. The method of claim 48, wherein the enabling of the load-reducing switching circuit comprises extracting control information from the computer system memory controller to provide an enable control signal to the load-reducing switching circuit.

50. The method of claim 48, wherein during a write operation, the enabling of the load-reducing switching circuit to drive a data signal from the computer system memory controller on one of a plurality of paths to the memory devices of the memory module comprises regenerating the data signal using the at least one write buffer.

51. The method of claim 48, wherein during a read operation, the enabling of the load-reducing switching circuit to receive the plurality of data signals from the memory devices of the memory module and driving the received data signals to the computer system memory controller comprises performing a multiplex function on data signals from the memory devices of the memory module.

52. The method of claim 48, further comprising combining two or more physical memory ranks into one logical memory rank.

53. The method of claim 52, wherein the two or more physical memory ranks are accessible with a single chip select signal from the computer system memory controller.

54. The method of claim 48, wherein the at least one write buffer comprises at least one tristate buffer and the at least one read buffer comprises at least one tristate buffer.

55. The method of claim 48, wherein during a read operation, the enabling of the load-reducing switching circuit to receive the plurality of data signals from the memory devices of the memory module and driving the received data signals to the computer system memory controller comprises regenerating the received data signals using the at least one read buffer.

56. The method of Claim 31 or Claim 48, wherein the plurality of memory devices comprises one or more synchronous dynamic random access memory devices.

57. The memory module of Claim 37, wherein the plurality of memory devices comprises one or more synchronous dynamic random access memory devices.

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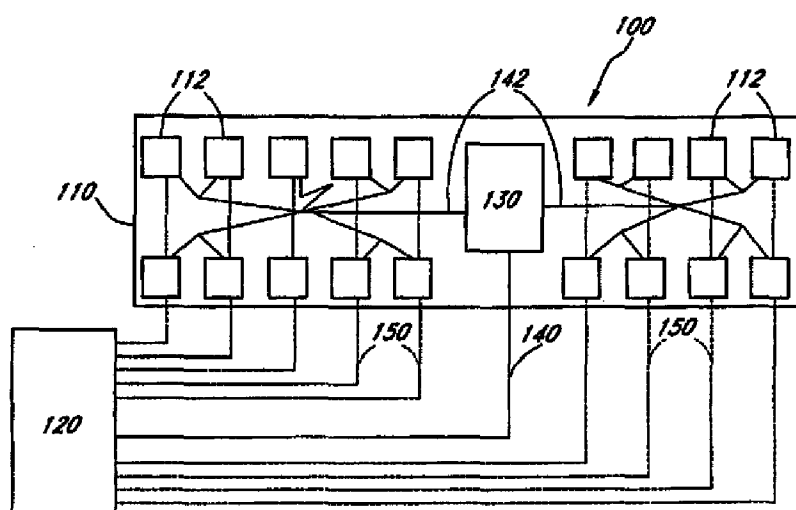
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12. 08. 2010

FIG. 1A
(Prior Art)

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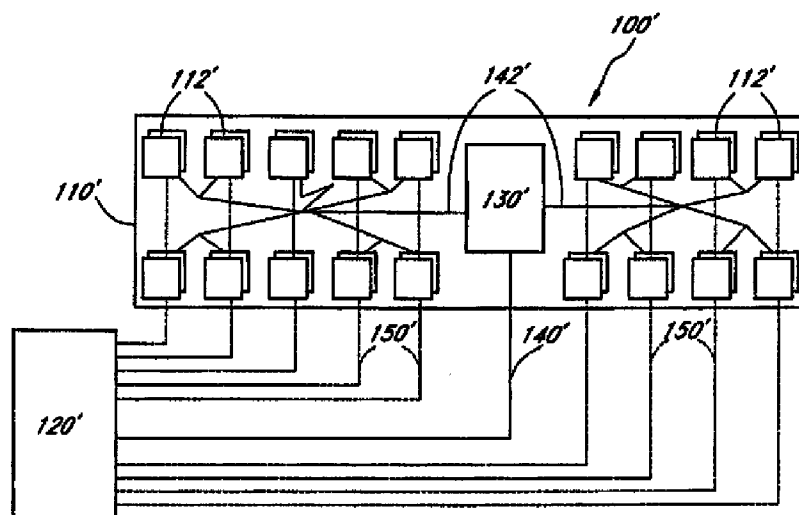


FIG. 1B
(Prior Art)

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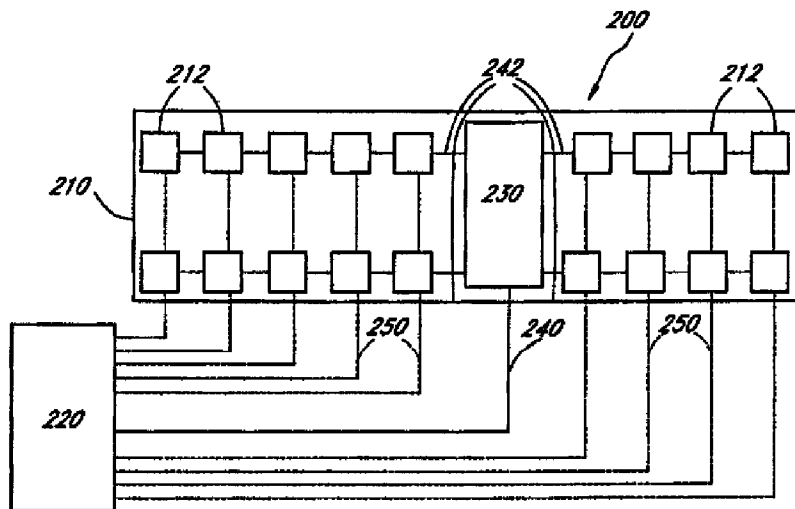


FIG. 2A
(Prior Art)

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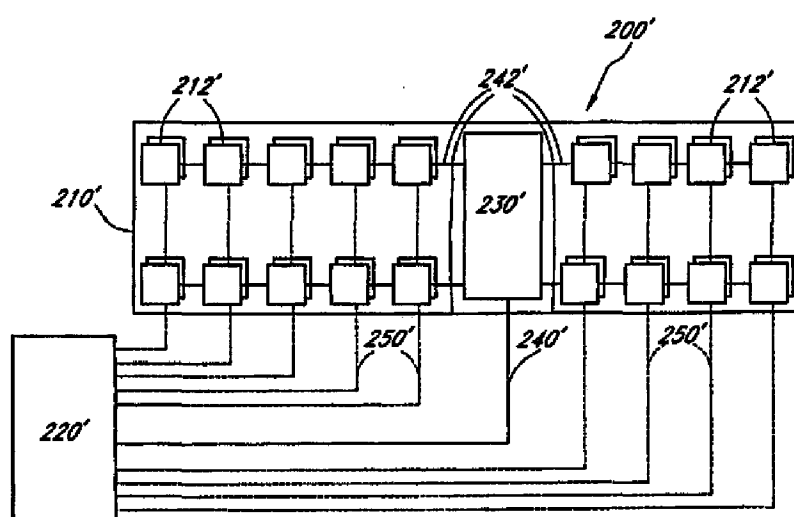


FIG. 2B
(Prior Art)

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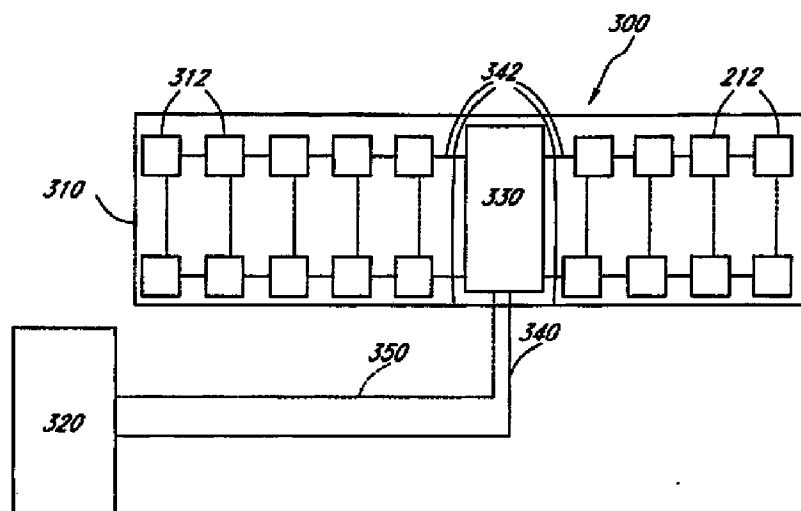


FIG. 2C
(Prior Art)

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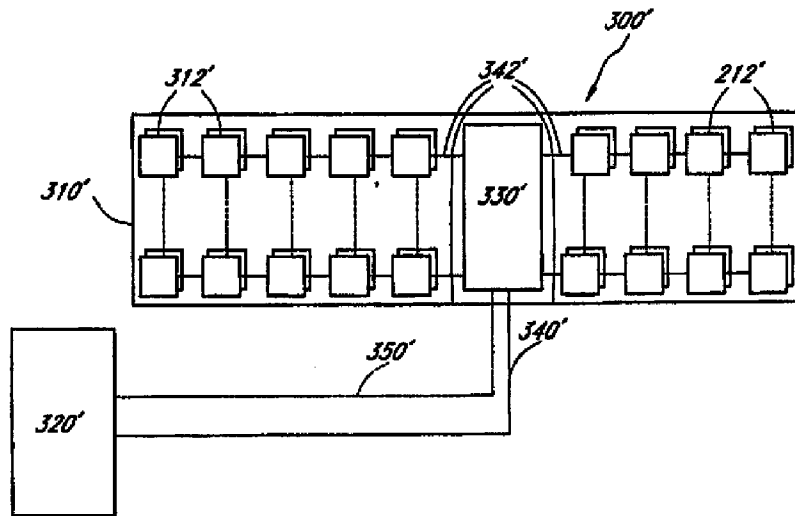
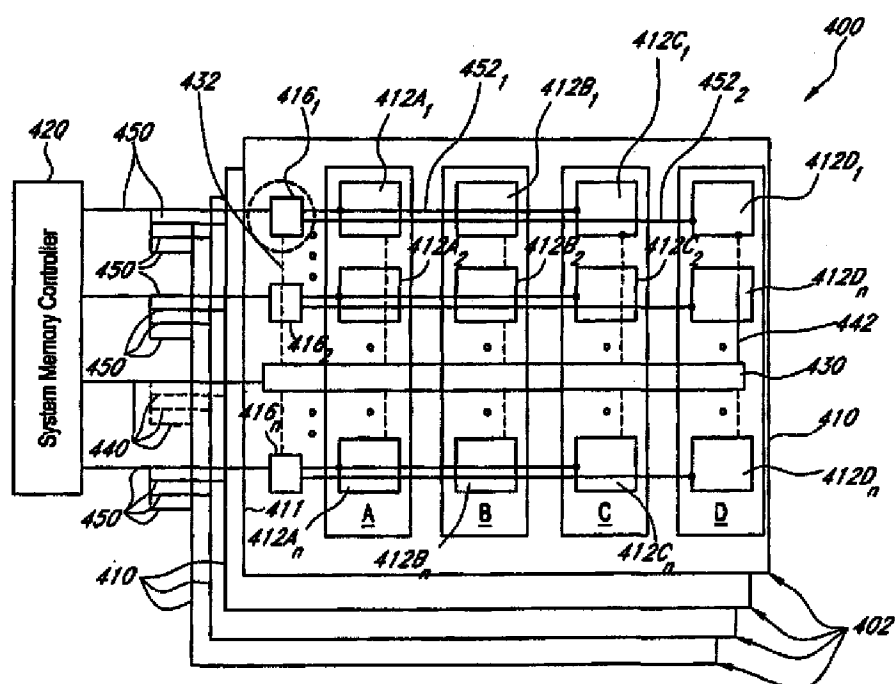


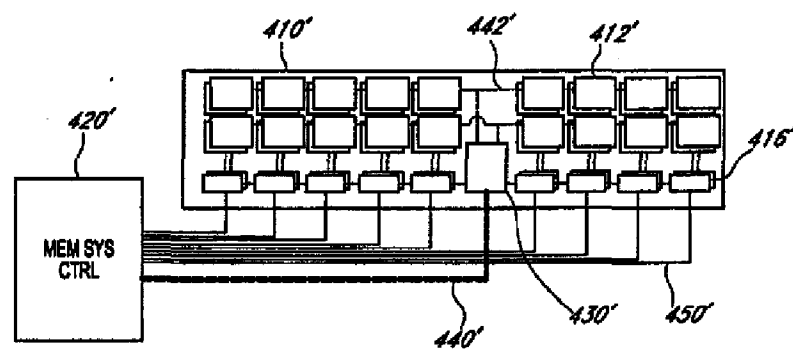
FIG. 2B
(Prior Art)

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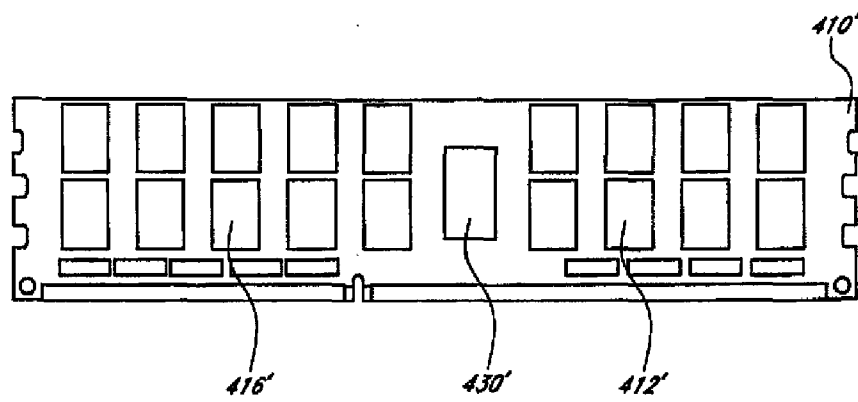
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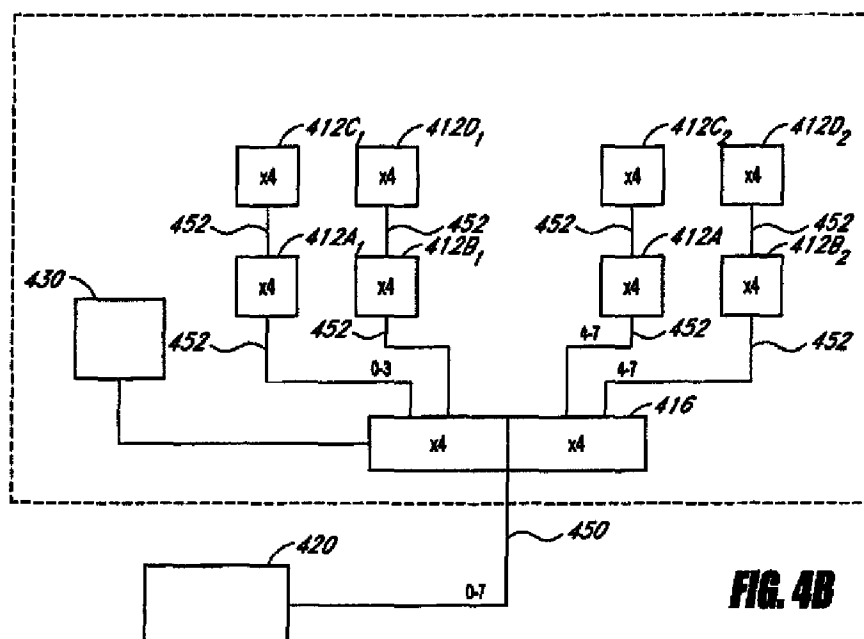
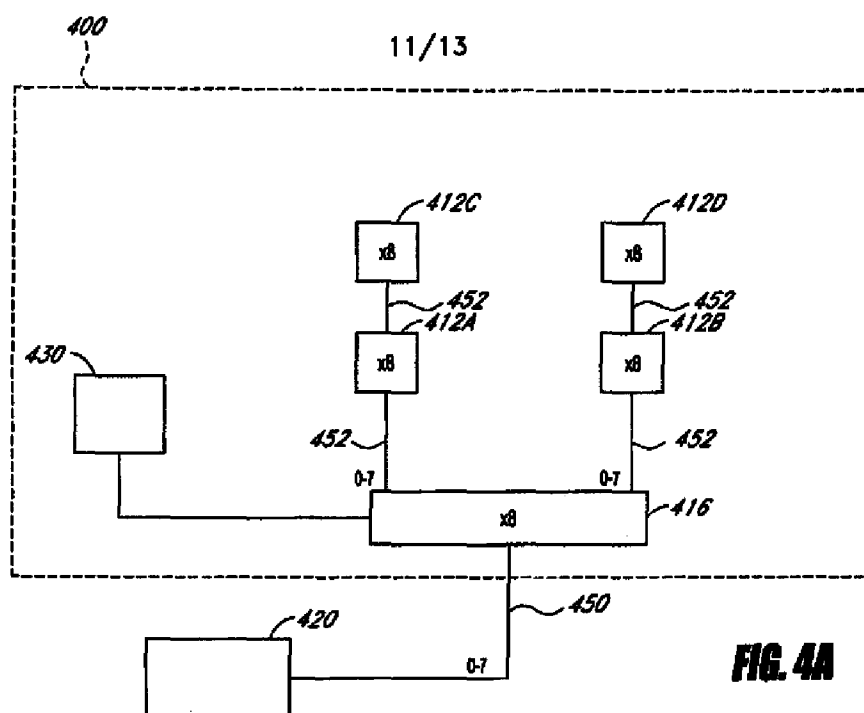
**FIG. 3A**

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**FIG. 3C**

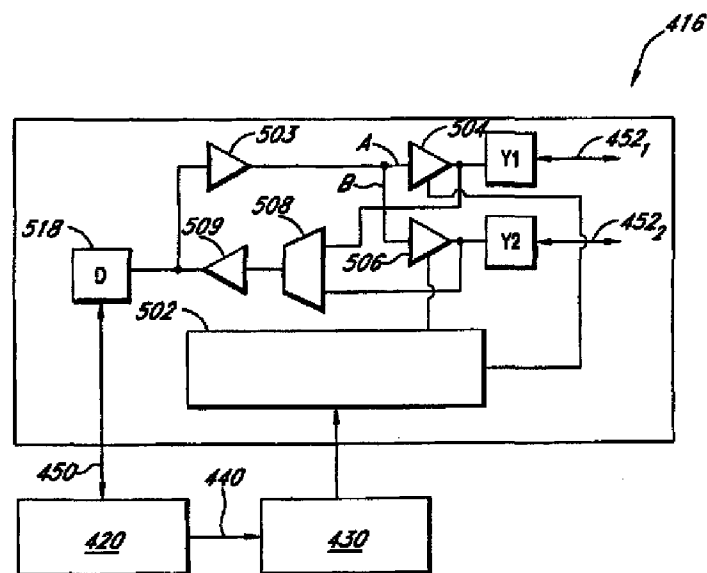
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**FIG. 3D**

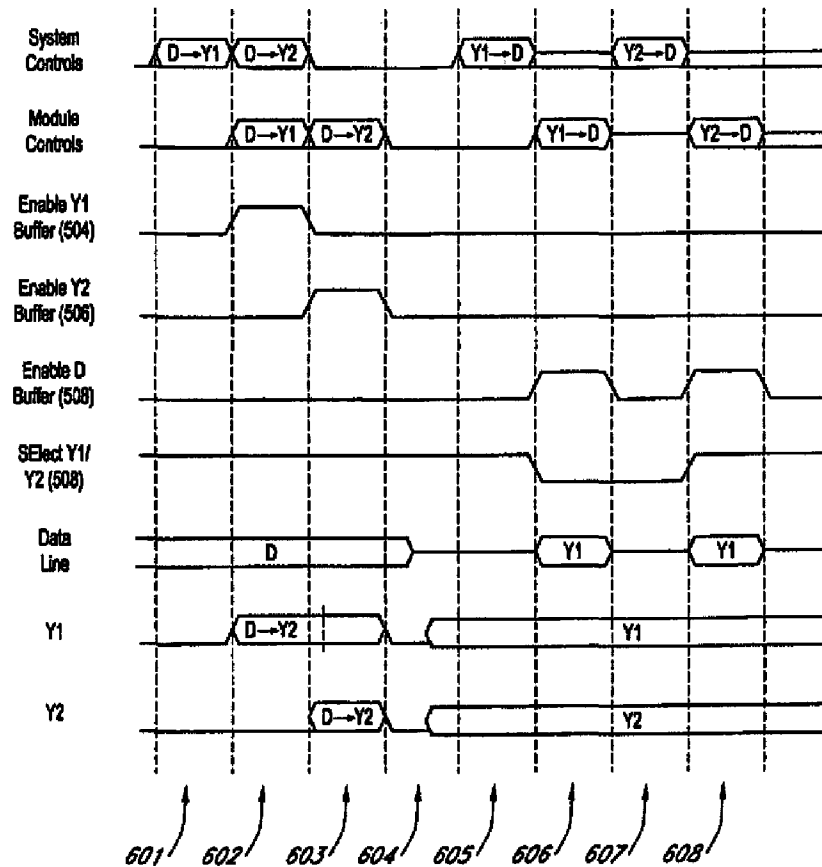


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**FIG. 5**

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**FIG. 6**

PATENT COOPERATION TREATY

From the INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

Christensen, Michael R.
KNOBBE, MARTENS, OLSON & BEAR LLP
2040 Main Street, 14th Floor
Irvine, CA 92614
ETATS-UNIS D'AMERIQUE

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

Date of mailing (day/month/year)		30 January 2013 (30-01-2013)
Applicant's or agent's file reference NETL071WO	FOR FURTHER ACTION See paragraphs 1 and 4 below	
International application No. PCT/US2011/059209	International filing date (day/month/year) 3 November 2011 (03-11-2011)	
Applicant NETLIST, INC.		

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

Filing of amendments and statement under Article 19:

The applicant is entitled, if he so wishes, to amend the claims of the International Application (see Rule 46):

When? The time limit for filing such amendments is normally two months from the date of transmittal of the International Search Report.

Where? Directly to the International Bureau of WIPO, 34 chemin des Colombettes
1211 Geneva 20, Switzerland, Facsimile No.: (41-22) 338.82.70

For more detailed instructions, see PCT Applicant's Guide, International Phase, paragraphs 9.004 - 9.011.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.

3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:

☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.

☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.

4. Reminders

The applicant may submit comments on an informal basis on the written opinion of the International Searching Authority to the International Bureau. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established. Following the expiration of 30 months from the priority date, these comments will also be made available to the public.

Shortly after the expiration of **18 months** from the priority date, the international application will be published by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before completion of the technical preparations for international publication (Rules 90*bis*.1 and 90*bis*.3).

Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for entry into the national phase before those designated Offices.

In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months.

For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide*, National Chapters.

Name and mailing address of the International Searching Authority



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Received Orange County Docketing

FEB 05 2013

Knobbe, Martens, Olson & Bear LLP

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference NETL071WO	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US2011/059209	International filing date (day/month/year) 03/11/2011	(Earliest) Priority Date (day/month/year) 03/11/2010
Applicant NETLIST, INC.		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 6 sheets.

☒ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed
☐ a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (See Box No. II)

3. ☒ **Unity of invention is lacking** (see Box No. III)

4. With regard to the **title**,

- ☒ the text is approved as submitted by the applicant
☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

- ☒ the text is approved as submitted by the applicant
☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority

6. With regard to the **drawings**,

- a. the figure of the **drawings** to be published with the abstract is Figure No. 2
☒ as suggested by the applicant
☐ as selected by this Authority, because the applicant failed to suggest a figure
☐ as selected by this Authority, because this figure better characterizes the invention
 b. ☐ none of the figures is to be published with the abstract

INTERNATIONAL SEARCH REPORT

International application No.
PCT/US2011/059209

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☒ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

- ☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.
- ☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.
- ☒ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT

International application No

PCT/US2011/059209

A. CLASSIFICATION OF SUBJECT MATTER

INV. G11C5/06 G11C7/10 G11C7/18
 ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
 G11C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, IBM-TDB, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2008/025137 A1 (RAJAN SURESH NATARAJAN [US] ET AL) 31 January 2008 (2008-01-31) paragraph [0051]; figures 2b,2c -----	1-28
A	US 2006/233012 A1 (SEKIGUCHI TOMONORI [JP] ET AL) 19 October 2006 (2006-10-19) paragraph [0138]; figure 16 -----	1-28
X	EP 1 816 570 A2 (RAMBUS INC [US]) 8 August 2007 (2007-08-08) paragraph [0019] - paragraph [0022]; figures 2,3 paragraph [0109] - paragraph [0112]; figure 15 -----	29-34



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

- "A" document defining the general state of the art which is not considered to be of particular relevance
 "E" earlier application or patent but published on or after the international filing date
 "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
 "O" document referring to an oral disclosure, use, exhibition or other means
 "P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

17 January 2013

Date of mailing of the international search report

30/01/2013

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
 NL - 2280 HV Rijswijk
 Tel. (+31-70) 340-2040,
 Fax: (+31-70) 340-3016

Authorized officer

Lindquist, Jim

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2011/059209

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2008025137 A1	31-01-2008	US 2008025137 A1	31-01-2008
		US 2009285031 A1	19-11-2009
US 2006233012 A1	19-10-2006	CN 1841551 A	04-10-2006
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		US 6034918 A	07-03-2000
		US 6035365 A	07-03-2000
		US 6038195 A	14-03-2000

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2011/059209

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
		US 6044426 A	28-03-2000
		US 6049846 A	11-04-2000
		US 6067592 A	23-05-2000
		US 6070222 A	30-05-2000
		US 6085284 A	04-07-2000
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		WO 9116680 A1	31-10-1991

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-28

To arrange die die interconnects (e.g. data paths) to interconnect the control die to a plurality of array dies, to minimize loads.

2. claims: 29-34

Modes to operate the data path control circuit, to either transport data bits or command/address information, thus maximizing the use of the data path.

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

see form PCT/ISA/220

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY (PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/US2011/059209

International filing date (day/month/year)
03.11.2011

Priority date (day/month/year)
03.11.2010

International Patent Classification (IPC) or both national classification and IPC
INV. G11C5/06 G11C7/10 G11C7/18

Applicant
NETLIST, INC.

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☒ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☒ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
D-80298 Munich
Tel. +49 89 2399 - 0
Fax: +49 89 2399 - 4465

Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Lindquist, Jim

Telephone No. +49 89 2399-2727



**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2011/059209

Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))
3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:
 - a. (means)
 - ☐ on paper
 - ☐ in electronic form
 - b. (time)
 - ☐ in the international application as filed
 - ☐ together with the international application in electronic form
 - ☐ subsequently to this Authority for the purposes of search
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2011/059209

Box No. IV Lack of unity of invention

1. ☒ In response to the invitation (Form PCT/ISA/206) to pay additional fees, the applicant has, within the applicable time limit:
- ☒ paid additional fees
 - ☐ paid additional fees under protest and, where applicable, the protest fee
 - ☐ paid additional fees under protest but the applicable protest fee was not paid
 - ☐ not paid additional fees
2. ☐ This Authority found that the requirement of unity of invention is not complied with and chose not to invite the applicant to pay additional fees.
3. This Authority considers that the requirement of unity of invention in accordance with Rule 13.1, 13.2 and 13.3 is
- ☐ complied with
 - ☒ not complied with for the following reasons:
see separate sheet
4. Consequently, this report has been established in respect of the following parts of the international application:
- ☐ all parts.
 - ☒ the parts relating to claims Nos. 1-34

Box No. V Reasoned statement under Rule 43*bis*.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims	
	No: Claims	<u>1-34</u>
Inventive step (IS)	Yes: Claims	
	No: Claims	<u>1-34</u>
Industrial applicability (IA)	Yes: Claims	<u>1-34</u>
	No: Claims	

2. Citations and explanations

see separate sheet

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2011/059209

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

V. Citations and Explanations

1 The following documents are cited in this Written Opinion:

D1 US 2008/025137 A1 (RAJAN SURESH NATARAJAN [US] ET AL) 31
January 2008 (2008-01-31)

D2 EP 1 816 570 A2 (RAMBUS INC [US]) 8 August 2007 (2007-08-08)

2 The subject matter of claim 1 lacks novelty so that the criteria of Articles 33(1) and (2) PCT are not met.

In comparison to claim 1, D1 (fig.2B, para.0051) discloses a memory package, comprising:

a plurality of array dies (206A-D) having data ports;

at least a first die interconnect (Data) and a second die interconnect (Data), the first die interconnect (Data) in electrical communication with at least one data port of a first array die (206A) of the plurality of array dies and at least one data port of a second array die (206C) of the plurality of array dies and not in electrical communication with the data ports of at least a third array die (206B,206D) of the plurality of array dies, the second die interconnect (Data) in electrical communication with at least one data port of the third array die (206B) and not in electrical communication with the data ports of the first array die (206A) and the data ports of the second array die (206C); and

a control die (202) comprising at least a first data conduit (right Data connection to 202) configured to transmit a data signal to the first die interconnect (right Data) and to not transmit the data signal to the second die interconnect (left Data), and at least a second data conduit (left Data connection to 202) configured to transmit the data signal to the second die interconnect (left Data) and to not transmit the data signal to the first die interconnect (right Data).

- 3 Independent claims 11 and 20 contains substantially the same features as claim 1 only differently worded. The subject-matter of these claims therefore lacks novelty for the same reasons as those set out here above.
- 4 The subject-matter of claim 29 lacks novelty so that Articles 33(1) and (2) PCT are not met.

In comparison to claim 29, D2 (para.0109-0112, fig.2,3,15) discloses a memory module comprising:

a register device (19) configured to receive command/address signals (20) from a memory control hub (19) and to generate data path control signals (18); and

a plurality of DRAM packages (15,16,17), each DRAM package comprising:

a control die (para.0111, in part. p.16,l.53) comprising a plurality of command/address buffers and a data path control circuit configured to control command/address time slots and data bus time slots (para.0020), the control die configured to receive data signals from the register device, and command/address signals from the register device (para.0021); and

a plurality of DDR DRAM dies (150-157, para.0109) operatively coupled to the control die to receive the data signals from the control die,

wherein the memory module is selectively configurable into at least two operational modes comprising:

a first operational mode (para.0111) in which the register device generates the data path control signals, and the control die uses the data path control signals to operate the data path control circuit (para.0111); and

a second operational mode (0020) in which the control die operate the data path control circuit to provide the command/address signals to the plurality of DDR DRAM dies without decoding the command/address signals (para.0111).

- 5 The features of the dependent claims are either known from D1 or D2, or derivable therefrom in an obvious manner.

VII. Certain Defects

- 5.1 The independent claims are not drafted in two-part form delimited against D1, contrary to Rule 6.3.(b) PCT.
- 5.2 Reference signs are not inserted in the claims contrary to Rule 6.2.(b) PCT.
- 5.3 Contrary to the requirements of Rule 5.1(a)(ii) PCT, documents D1 and D2 are not identified in the description and the relevant background art disclosed therein is not discussed.
- 5.4 With regard to the phrase "hereby incorporated by reference" throughout the description, according to Rule 4.14 PCT, if there is a wish to treat the international application as a continuation-in-part of another application, this application must be indicated in the PCT request of the international application.
- 5.5 The description also contains references to documents not retrievable by the public, e.g. application numbers. These references must either be replaced by retrievable references or they must be removed.
- 5.6 And finally, the reference to the spirit of the present invention in page 37 is not allowed, since it wrongly makes the impression that the extent of protection is expanded beyond the contents of the claims (see Chapter III, 4.3a of the PCT Preliminary Examination Guidelines).

VIII. Certain Observations

- 6 The set of claims contains too many independent claims. The various definitions of the invention given in the independent claims 1, 11, 20 and 29 are such that the claims as a whole are not clear and concise, contrary to Article 6 PCT, since it is not clear for which matter protection is sought. To meet the criteria of this article the claims may only include one independent claim in each category, with dependent claims as appropriate (Rule 6.4(a)-(c) PCT).
- 7 Dependent claims 30-32 are not clear, because the meaning of the term "rank multiplication" is not clear. This term must be clarified to make any positive contribution to the patentability of the previous subject-matter.

Possible steps after receipt of the international search report (ISR) and written opinion of the International Searching Authority (WO-ISA)

General information	For all international applications filed on or after 01/01/2004 the competent ISA will establish an ISR. It is accompanied by the WO-ISA. Unlike the former written opinion of the IPEA (Rule 66.2 PCT), the WO-ISA is not meant to be responded to, but to be taken into consideration for further procedural steps. This document explains about the possibilities.
Amending claims under Art. 19 PCT	Within 2 months after the date of mailing of the ISR and the WO-ISA the applicant may file amended claims under Art. 19 PCT directly with the International Bureau of WIPO. The PCT reform of 2004 did not change this procedure. For further information please see Rule 46 PCT as well as form PCT/ISA/220 and the corresponding Notes to form PCT/ISA/220.
Filing a demand for international preliminary examination	In principle, the WO-ISA will be considered as the written opinion of the IPEA. This should, in many cases, make it unnecessary to file a demand for international preliminary examination. If the applicant nevertheless wishes to file a demand this must be done before expiry of 3 months after the date of mailing of the ISR/ WO-ISA or 22 months after priority date, whichever expires later (Rule 54bis PCT). Amendments under Art. 34 PCT can be filed with the IPEA as before, normally at the same time as filing the demand (Rule 66.1 (b) PCT). If a demand for international preliminary examination is filed and no comments/amendments have been received the WO-ISA will be transformed by the IPEA into an IPRP (International Preliminary Report on Patentability) which would merely reflect the content of the WO-ISA. The demand can still be withdrawn (Art. 37 PCT).
Filing informal comments	After receipt of the ISR/WO-ISA the applicant may file informal comments on the WO-ISA directly with the International Bureau of WIPO. These will be communicated to the designated Offices together with the IPRP (International Preliminary Report on Patentability) at 30 months from the priority date. Please also refer to the next box.
End of the international phase	At the end of the international phase the International Bureau of WIPO will transform the WO-ISA or, if a demand was filed, the written opinion of the IPEA into the IPRP, which will then be transmitted together with possible informal comments to the designated Offices. The IPRP replaces the former IPER (international preliminary examination report).
Relevant PCT Rules and more information	Rule 43 PCT, Rule 43bis PCT, Rule 44 PCT, Rule 44bis PCT, PCT Newsletter 12/2003, OJ 11/2003, OJ 12/2003

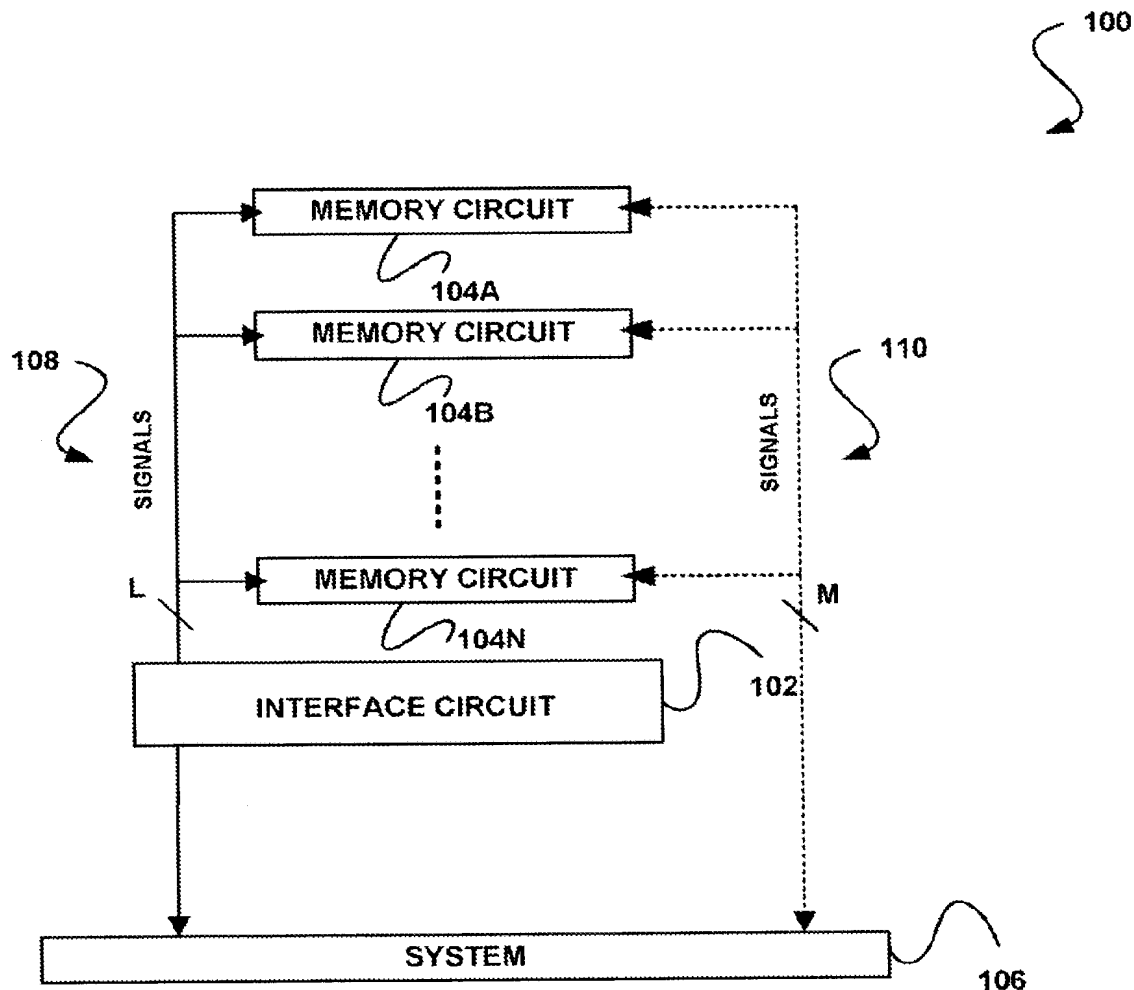


US 20080025137A1

(19) **United States**(12) **Patent Application Publication****Rajan et al.**(10) **Pub. No.: US 2008/0025137 A1**(43) **Pub. Date: Jan. 31, 2008**(54) **SYSTEM AND METHOD FOR SIMULATING
AN ASPECT OF A MEMORY CIRCUIT**(22) Filed: **Jul. 31, 2006****Publication Classification**(75) Inventors: **Suresh Natarajan Rajan**, San Jose, CA (US); **Keith R. Schakel**, San Jose, CA (US); **Michael John Sebastian Smith**, Palo Alto, CA (US); **David T. Wang**, San Jose, CA (US); **Frederick Daniel Weber**, San Jose, CA (US)(51) **Int. Cl.**
G11C 8/00 (2006.01)(52) **U.S. CL.** **365/239**(57) **ABSTRACT**

A system and method are provided for simulating an aspect of a memory circuit. Included is an interface circuit that is in communication with a plurality of memory circuits and a system. Such interface circuit is operable to interface the memory circuits and the system for simulating at least one memory circuit with at least one aspect that is different from at least one aspect of at least one of the plurality of memory circuits. In accordance with various embodiments, such aspect may include a signal, a capacity, a timing, and/or a logical interface.

Correspondence Address:

ZILKA-KOTAB, PC**P.O. BOX 721120****SAN JOSE, CA 95172-1120**(73) Assignee: **METARAM, INC.**(21) Appl. No.: **11/461,427**

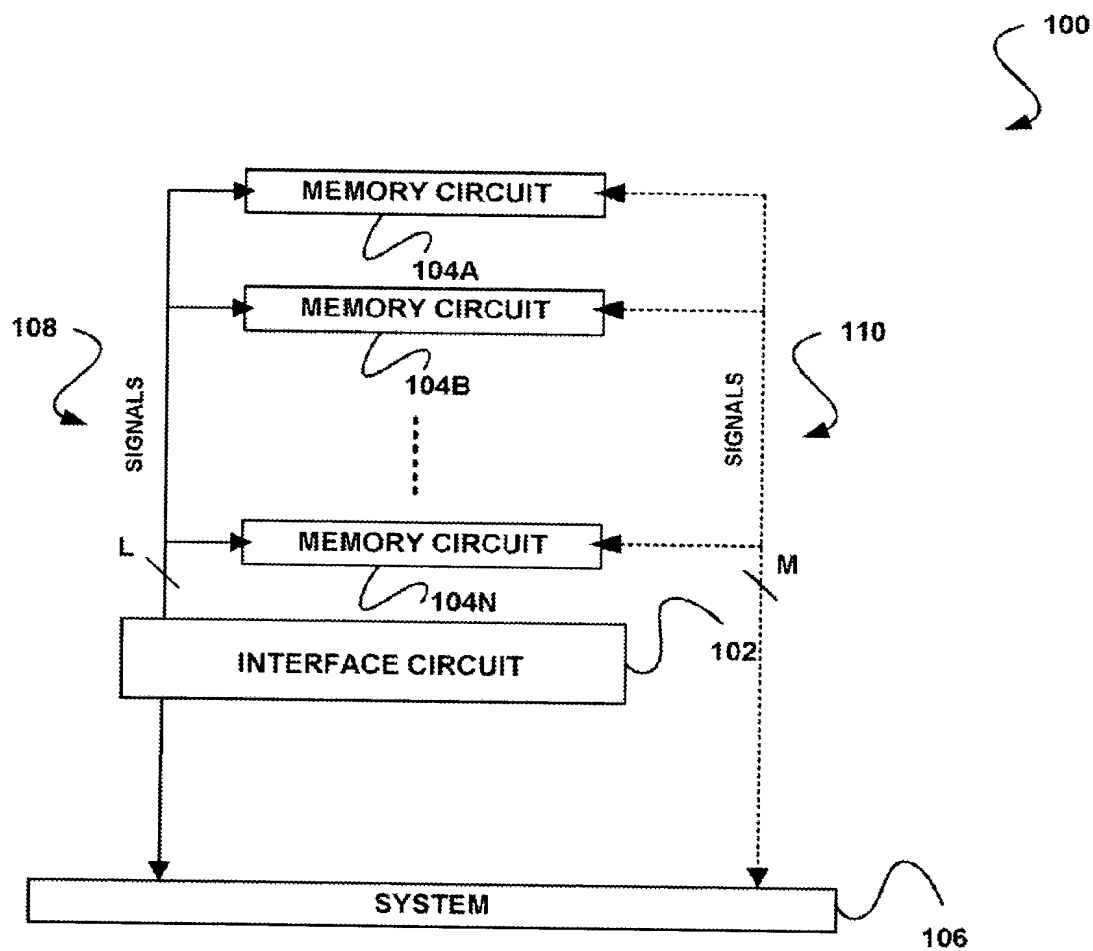


FIGURE 1

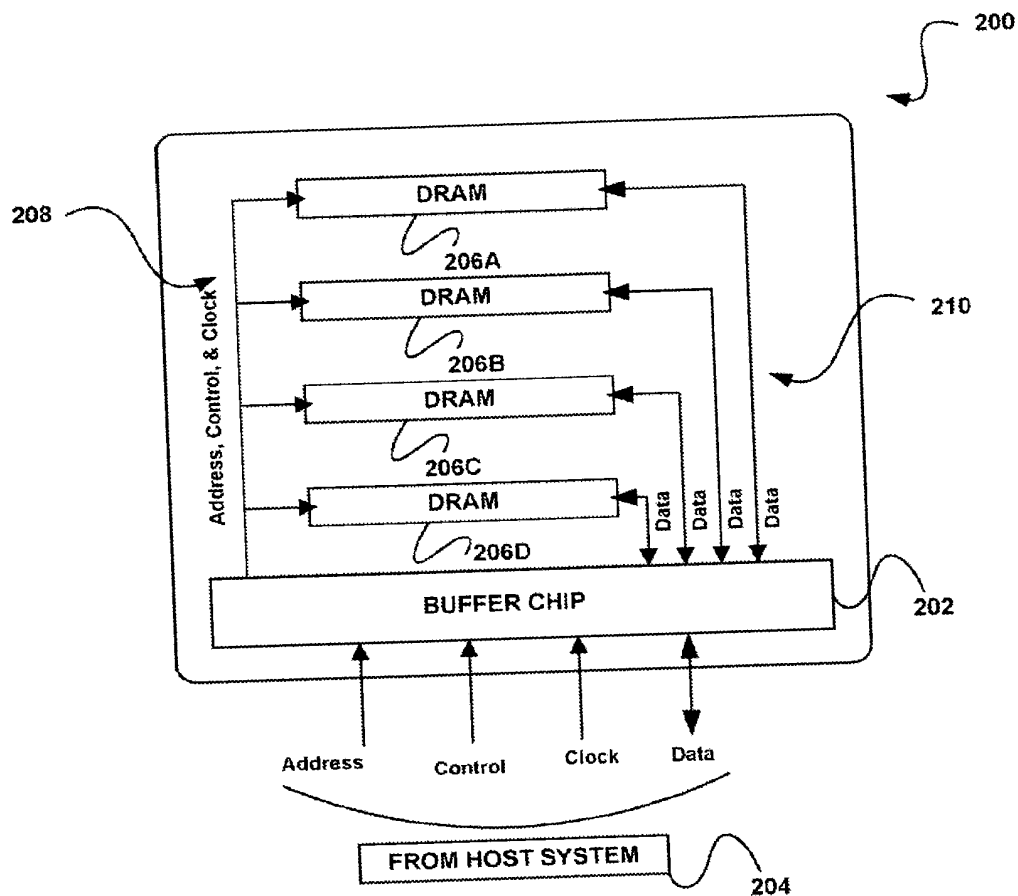


FIGURE 2A

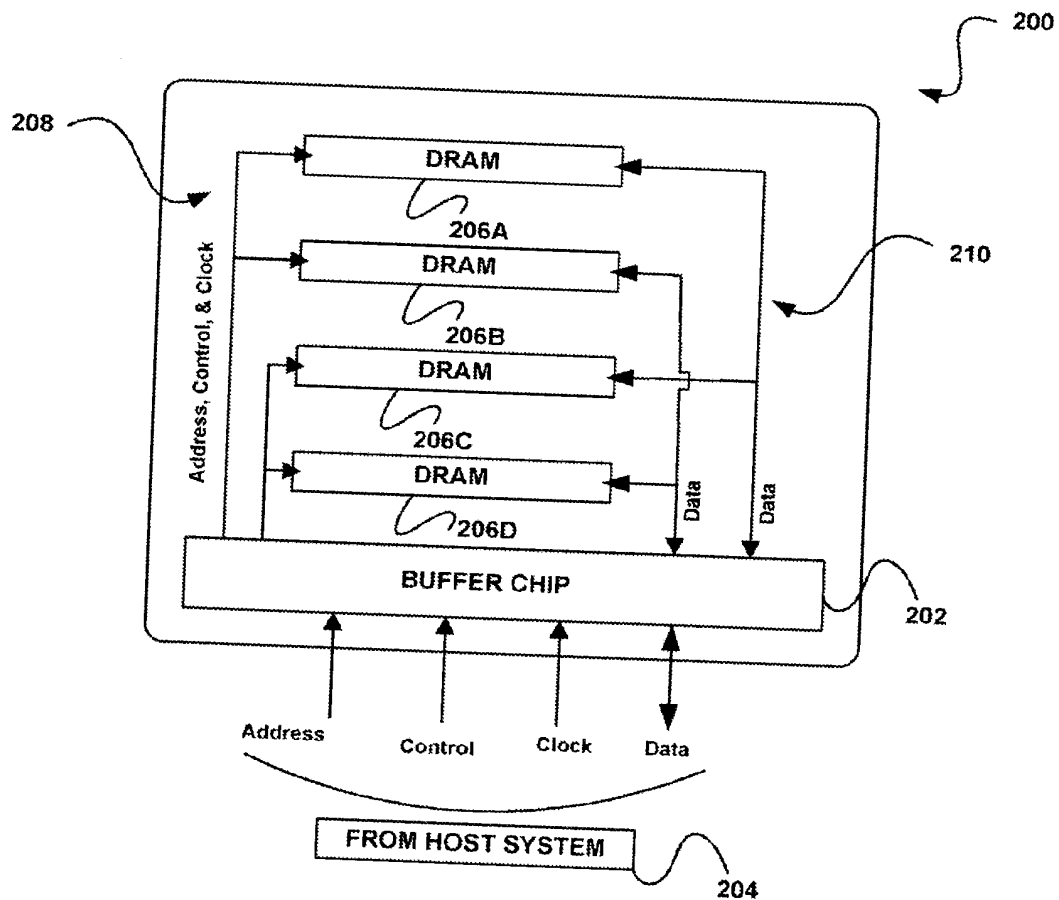


FIGURE 2B

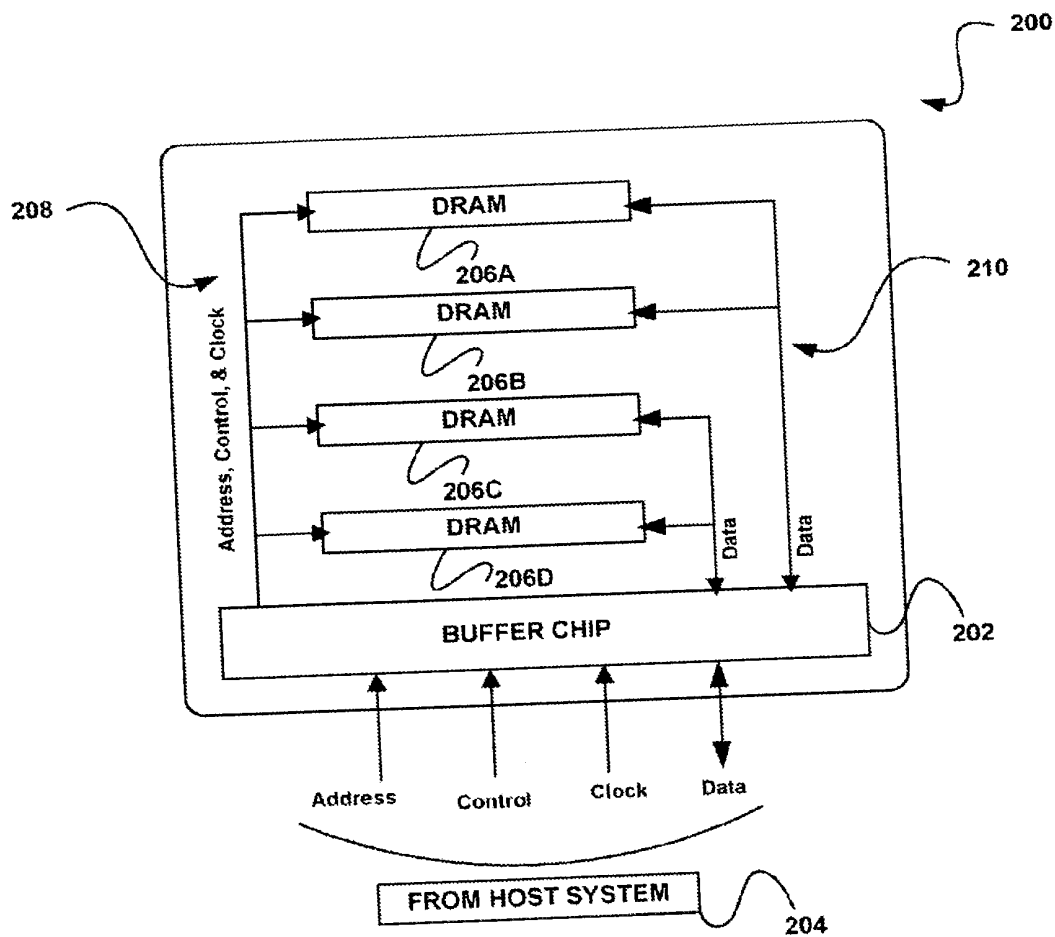


FIGURE 2C

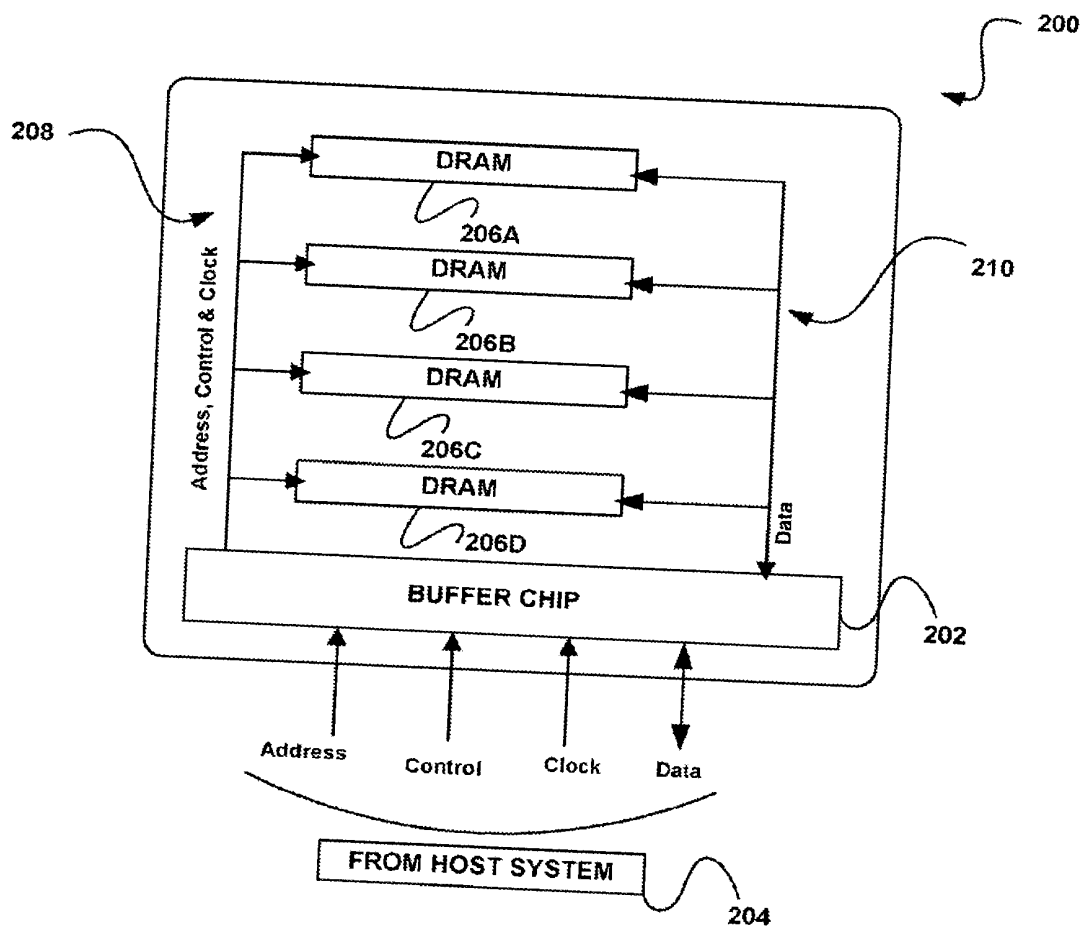


FIGURE 2D

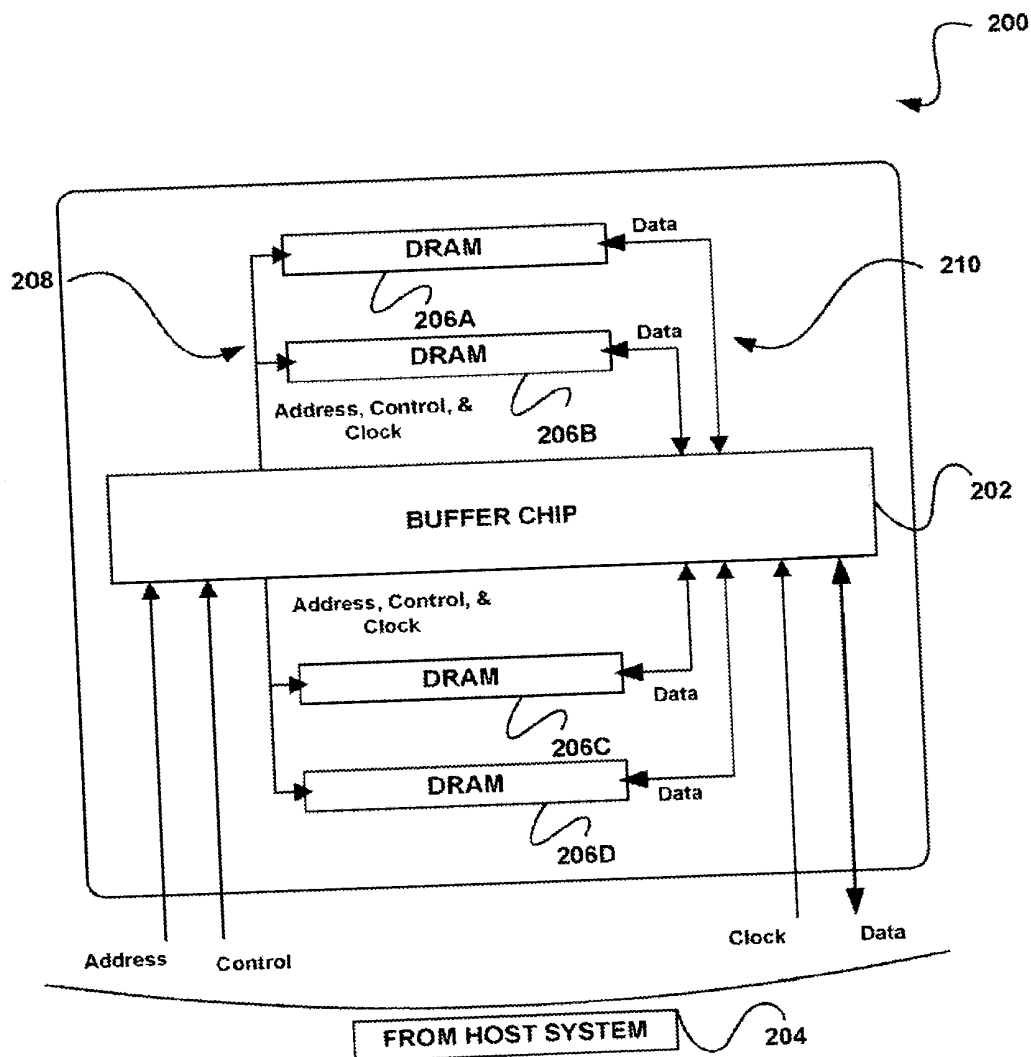


FIGURE 2E

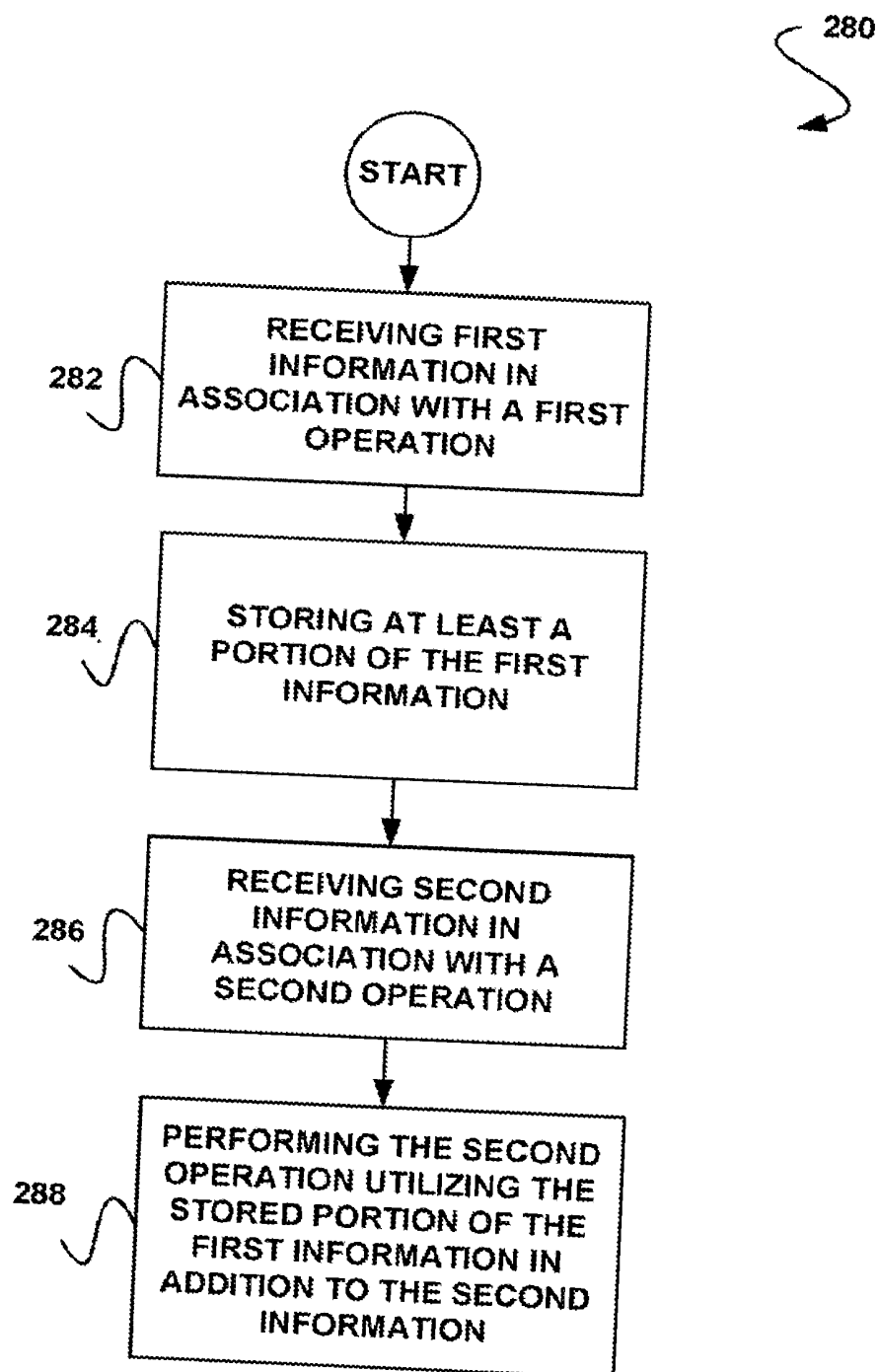


FIGURE 2F

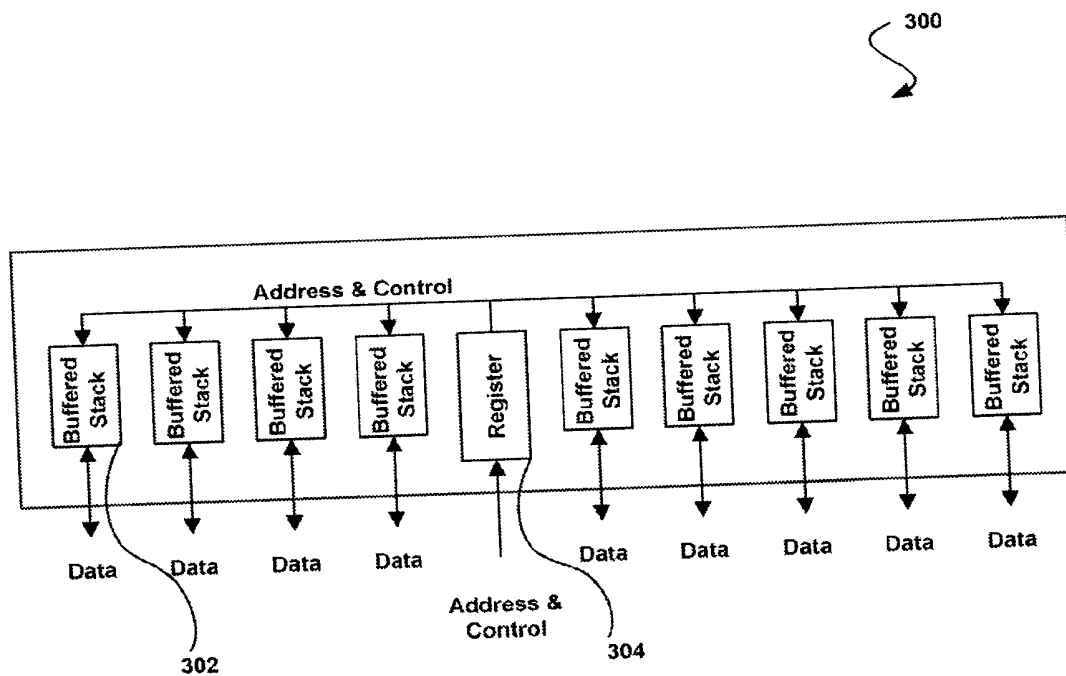


FIGURE 3

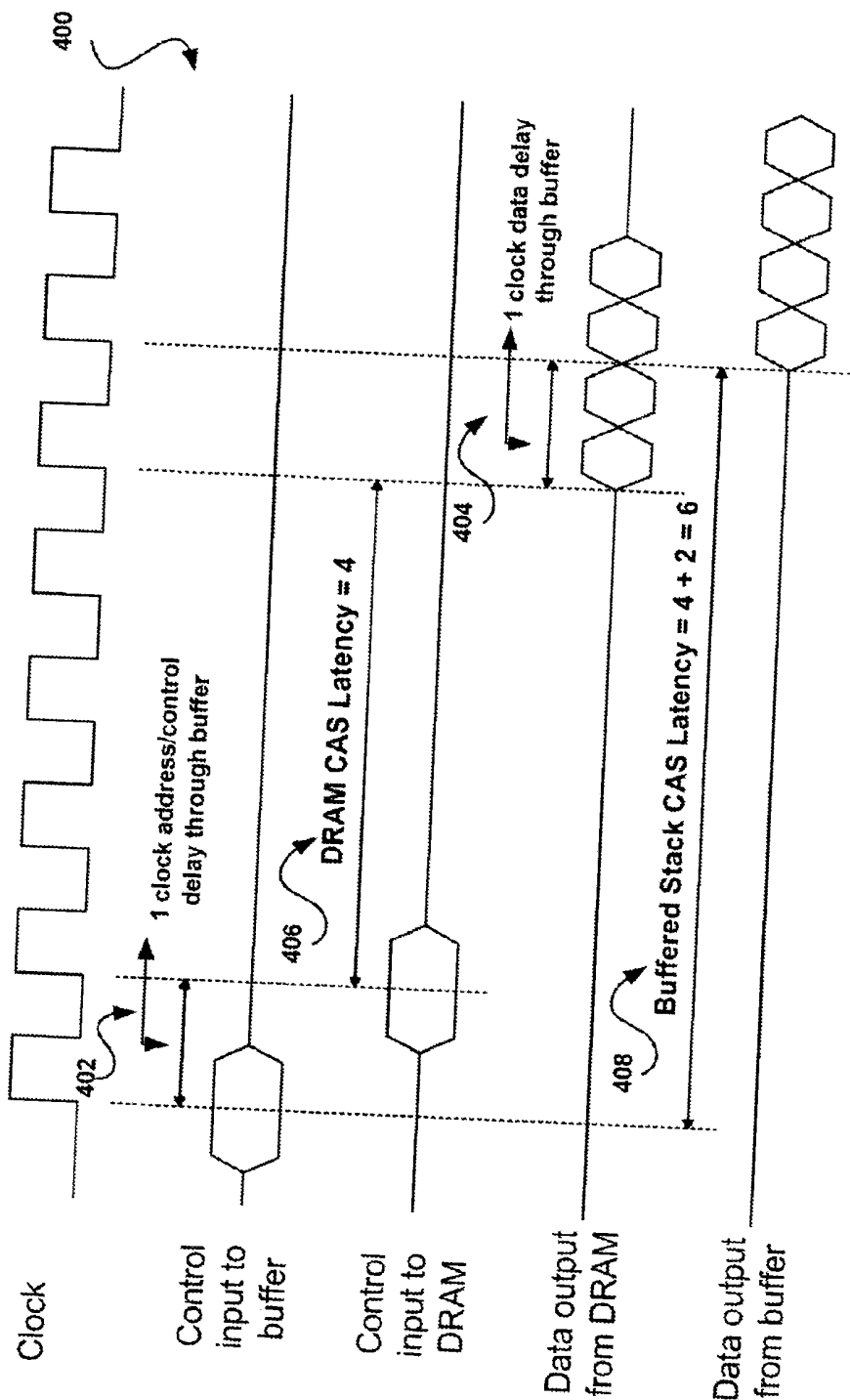


FIGURE 4

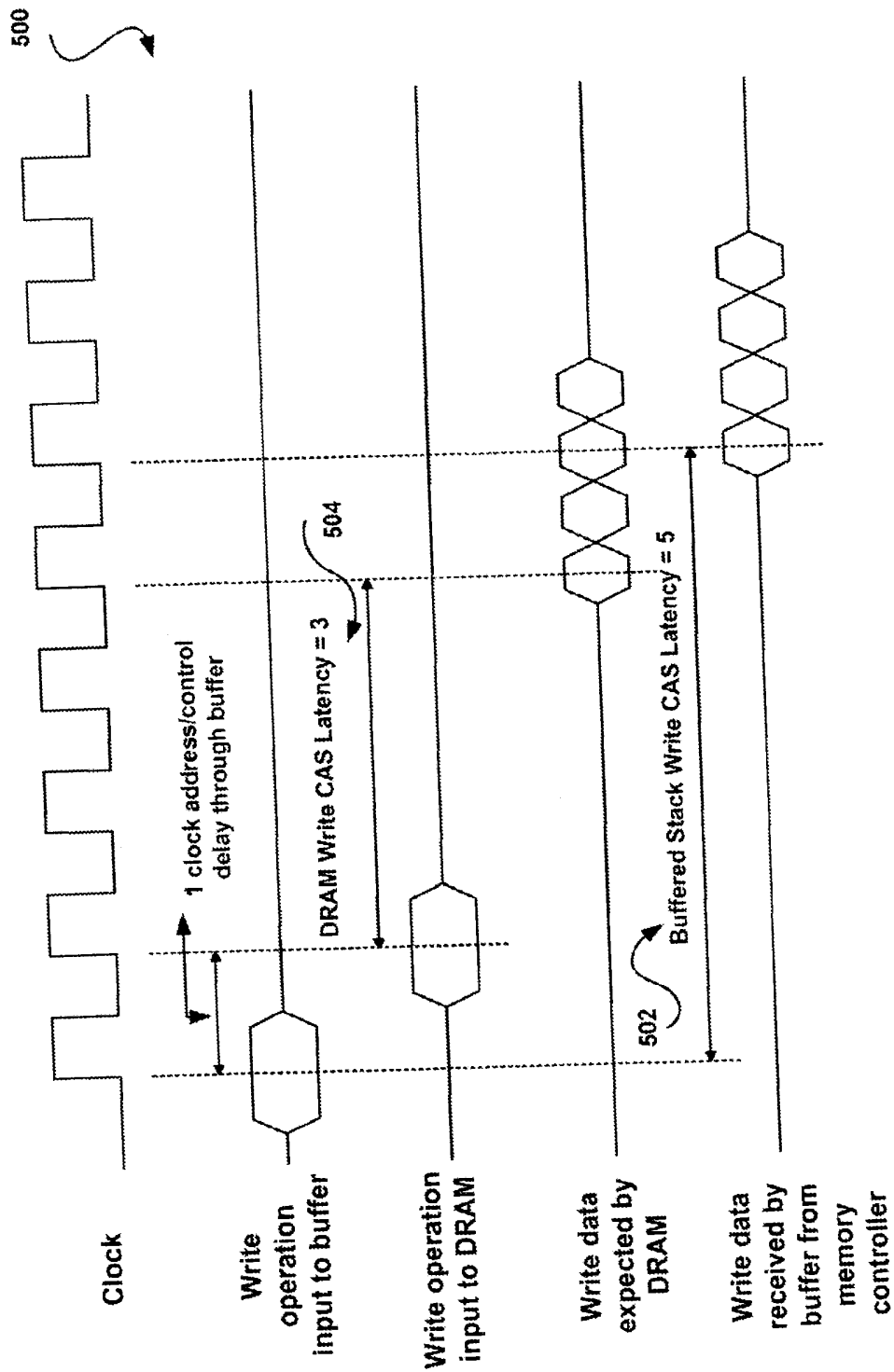


FIGURE 5

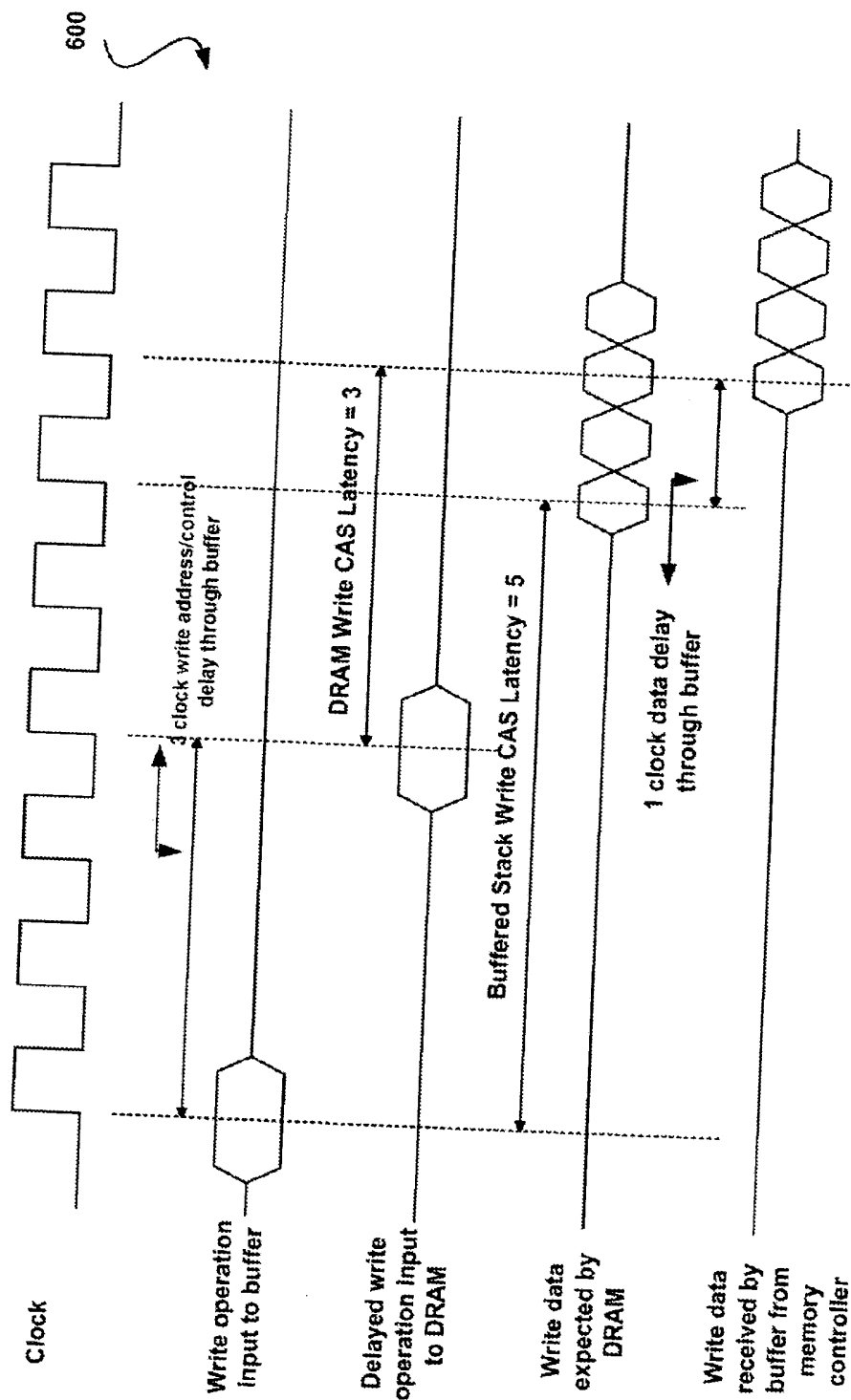


FIGURE 6

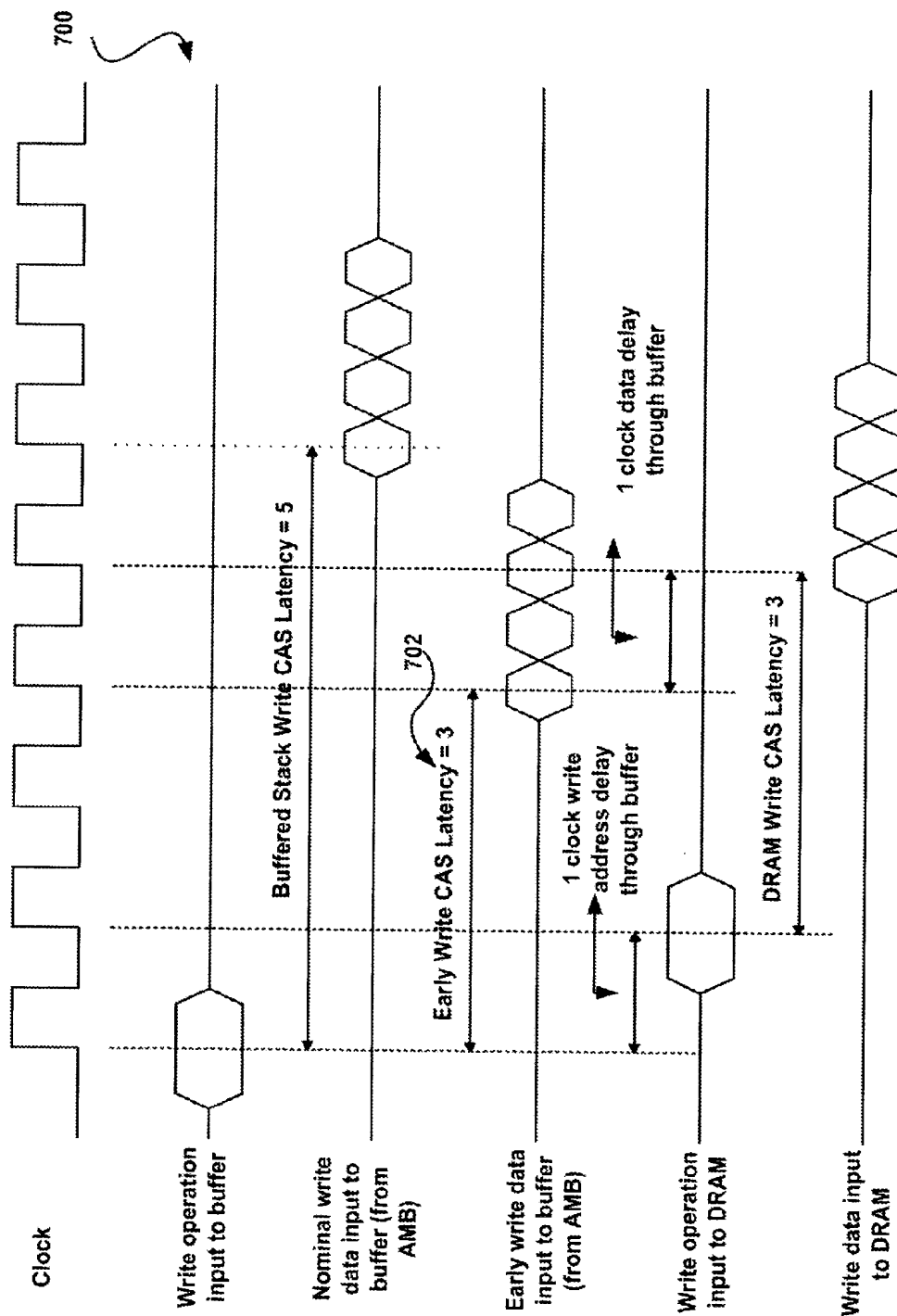


FIGURE 7

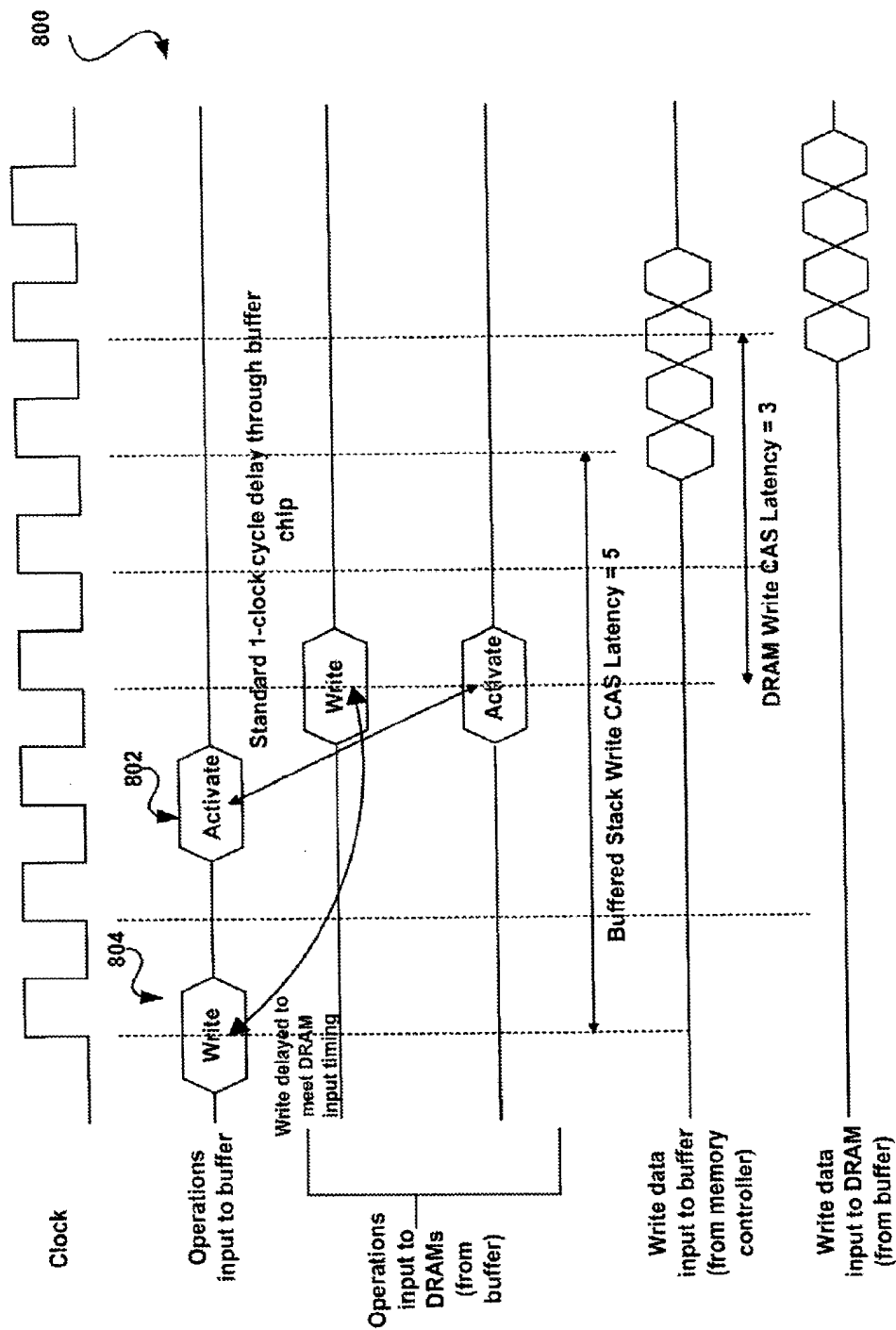


FIGURE 8

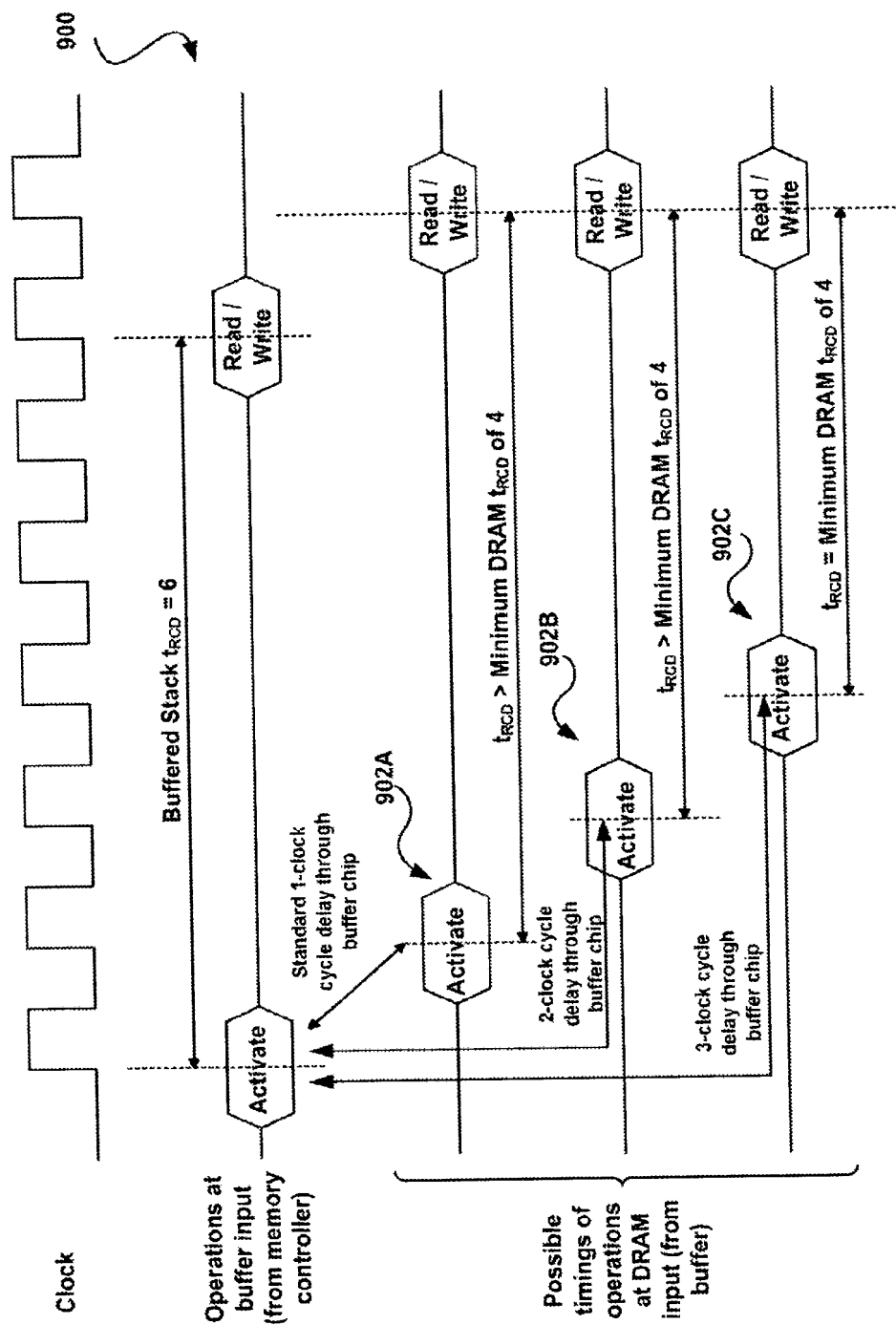


FIGURE 9A

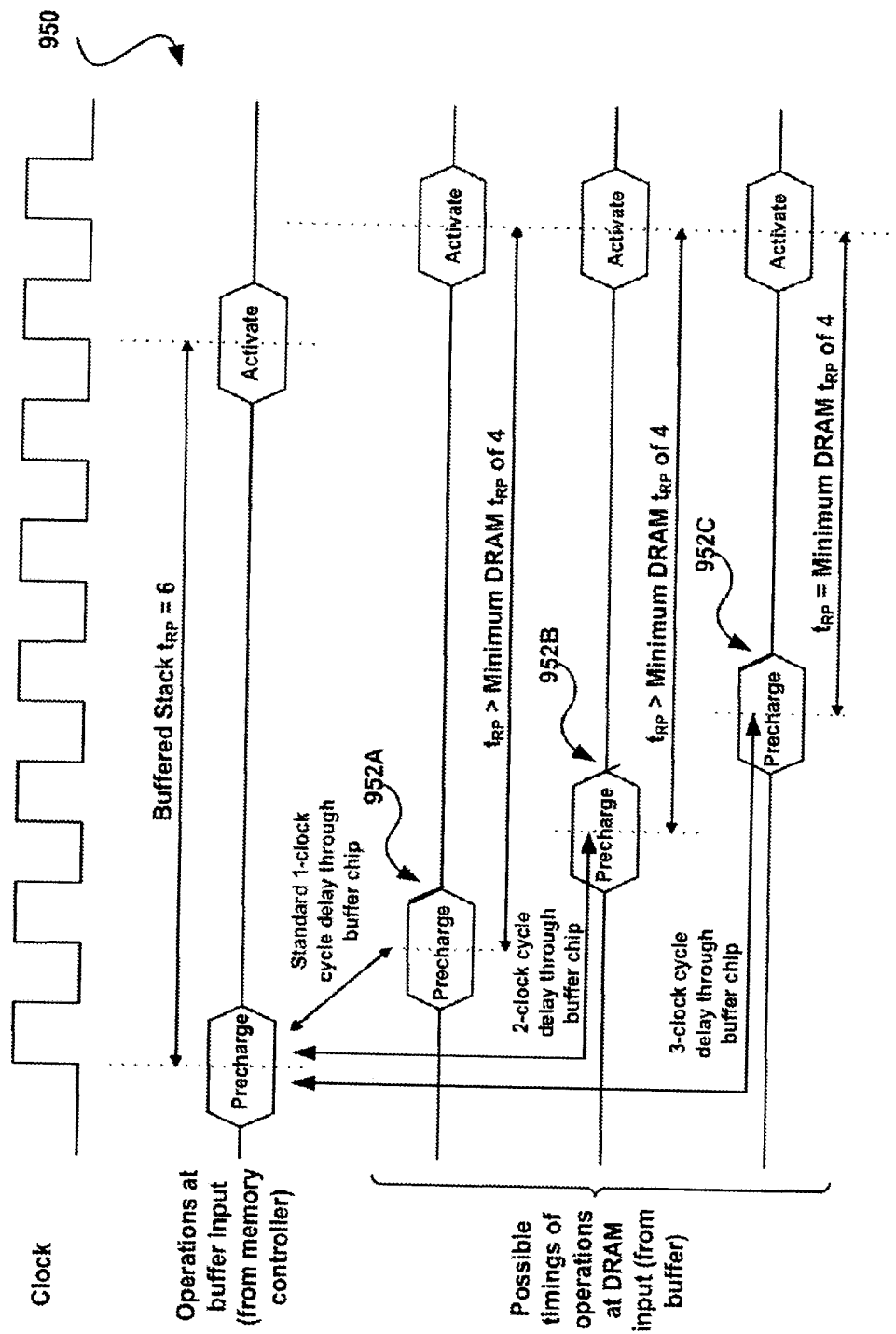


FIGURE 9B

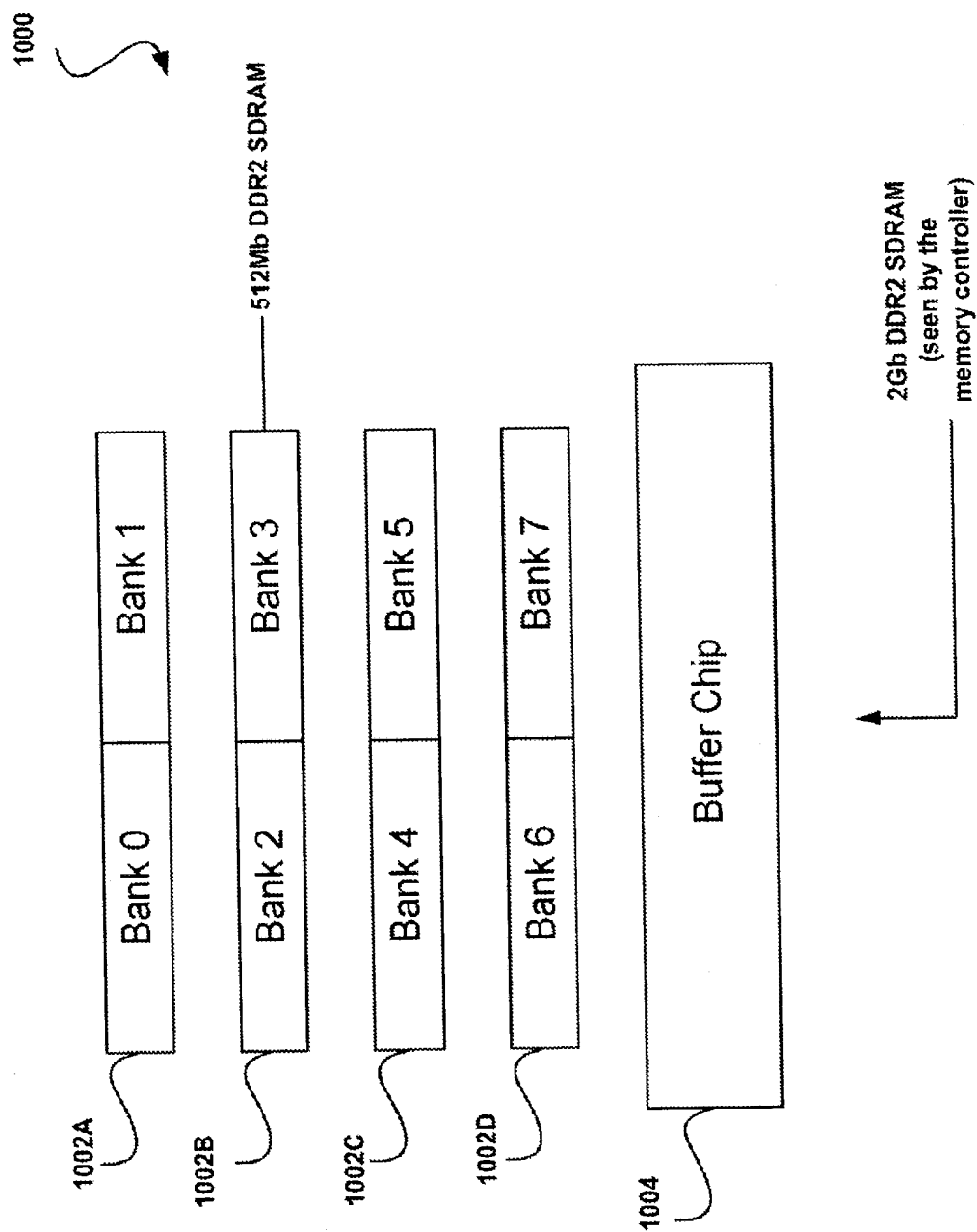


FIGURE 10

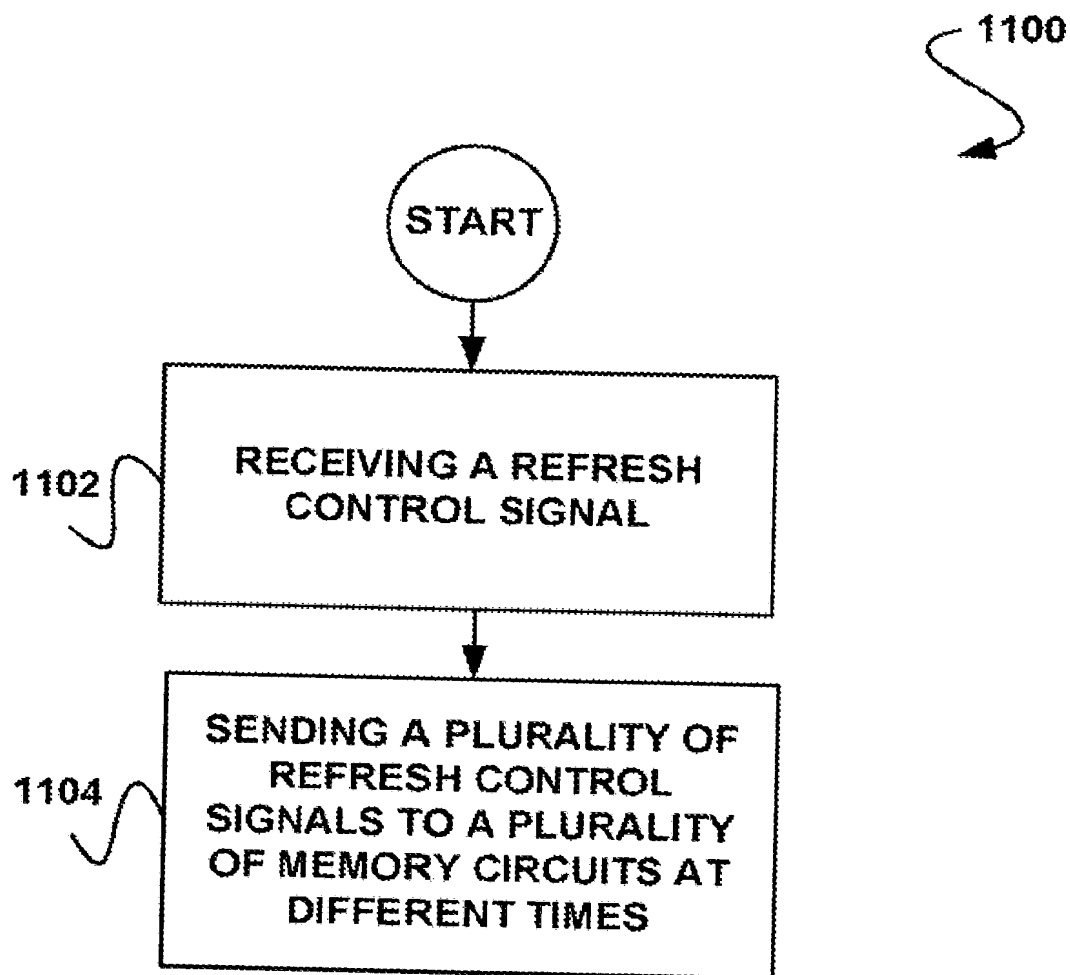


FIGURE 11

SYSTEM AND METHOD FOR SIMULATING AN ASPECT OF A MEMORY CIRCUIT

FIELD OF THE INVENTION

[0001] The present invention relates to memory, and more particularly to multiple-memory circuit systems.

BACKGROUND

[0002] The memory capacity requirements of computers in general, and servers in particular, are increasing rapidly due to various trends such as 64-bit processors and operating systems, multi-core processors, virtualization, etc. However, other industry trends such as higher memory bus speeds and small form factor machines, etc. are reducing the number of memory module slots in such systems. Thus, a need exists in the industry for large capacity memory circuits to be used in such systems.

[0003] However, there is also an exponential relationship between a capacity of monolithic memory circuits and a price associated therewith. As a result, large capacity memory modules may be cost prohibitive. To this end, the use of multiple smaller capacity memory circuits is a cost-effective approach to increasing such memory capacity.

SUMMARY

[0004] A system and method are provided for simulating an aspect of a memory circuit. Included is an interface circuit that is in communication with a plurality of memory circuits and a system. Such interface circuit is operable to interface the memory circuits and the system for simulating at least one memory circuit with at least one aspect that is different from at least one aspect of at least one of the plurality of memory circuits. In accordance with various embodiments, such aspect may include a signal, a capacity, a timing, and/or a logical interface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 illustrates a multiple memory circuit framework, in accordance with one embodiment.

[0006] FIGS. 2A-2E show various configurations of a buffered stack of dynamic random access memory (DRAM) circuits with a buffer chip, in accordance with various embodiments.

[0007] FIG. 2F illustrates a method for storing at least a portion of information received in association with a first operation for use in performing a second operation, in accordance with still another embodiment.

[0008] FIG. 3 shows a high capacity dual in-line memory module (DIMM) using buffered stacks, in accordance with still yet another embodiment.

[0009] FIG. 4 shows a timing design of a buffer chip that makes a buffered stack of DRAM circuits mimic longer column address strobe (CAS) latency DRAM to a memory controller, in accordance with another embodiment.

[0010] FIG. 5 shows the write data timing expected by DRAM in a buffered stack, in accordance with yet another embodiment.

[0011] FIG. 6 shows write control signals delayed by a buffer chip, in accordance with still yet another embodiment.

[0012] FIG. 7 shows early write data from an advanced memory buffer (AMB), in accordance with another embodiment.

[0013] FIG. 8 shows address bus conflicts caused by delayed write operations, in accordance with yet another embodiment.

[0014] FIGS. 9A-B show variable delays of operations through a buffer chip, in accordance with another embodiment.

[0015] FIG. 10 shows a buffered stack of four 512 Mb DRAM circuits mapped to a single 2 Gb DRAM circuit, in accordance with yet another embodiment.

[0016] FIG. 11 illustrates a method for refreshing a plurality of memory circuits, in accordance with still yet another embodiment.

DETAILED DESCRIPTION

[0017] FIG. 1 illustrates a multiple memory circuit framework 100, in accordance with one embodiment. As shown, included are an interface circuit 102, a plurality of memory circuits 104A, 104B, 104N, and a system 106. In the context of the present description, such memory circuits 104A, 104B, 104N may include any circuit capable of serving as memory.

[0018] For example, in various embodiments, one or more of the memory circuits 104A, 104B, 104N may include a monolithic memory circuit. For instance, such monolithic memory circuit may take the form of dynamic random access memory (DRAM). Such DRAM may take any form including, but not limited to synchronous (SDRAM), double data rate synchronous (DDR DRAM, DDR2 DRAM, DDR3 DRAM, etc.), quad data rate (QDR DRAM), direct RAM-BUS (DRDRAM), fast page mode (FPM DRAM), video (VDRAM), extended data out (EDO DRAM), burst EDO (BEDO DRAM), multibank (MDRAM), synchronous graphics (SGRAM), and/or any other type of DRAM. Of course, one or more of the memory circuits 104A, 104B, 104N may include other types of memory such as magnetic random access memory (MRAM), intelligent random access memory (IRAM), distributed network architecture (DNA) memory, window random access memory (WRAM), flash memory (e.g. NAND, NOR, or others, etc.), pseudostatic random access memory (PSRAM), wetware memory, and/or any other type of memory circuit that meets the above definition.

[0019] In additional embodiments, the memory circuits 104A, 104B, 104N may be symmetrical or asymmetrical. For example, in one embodiment, the memory circuits 104A, 104B, 104N may be of the same type, brand, and/or size, etc. Of course, in other embodiments, one or more of the memory circuits 104A, 104B, 104N may be of a first type, brand, and/or size; while one or more other memory circuits 104A, 104B, 104N may be of a second type, brand, and/or size, etc. Just by way of example, one or more memory circuits 104A, 104B, 104N may be of a DRAM type, while one or more other memory circuits 104A, 104B, 104N may be of a flash type. While three or more memory circuits 104A, 104B, 104N are shown in FIG. 1 in accordance with one embodiment, it should be noted that any plurality of memory circuits 104A, 104B, 104N may be employed.

[0020] Strictly as an option, the memory circuits 104A, 104B, 104N may or may not be positioned on at least one dual in-line memory module (DIMM) (not shown). In various embodiments, the DIMM may include a registered DIMM (R-DIMM), a small outline-DIMM (SO-DIMM), a fully buffered-DIMM (FB-DIMM), an un-buffered DIMM,

etc. Of course, in other embodiments, the memory circuits 104A, 104B, 104N may or may not be positioned on any desired entity for packaging purposes.

[0021] Further in the context of the present description, the system 106 may include any system capable of requesting and/or initiating a process that results in an access of the memory circuits 104A, 104B, 104N. As an option, the system 106 may accomplish this utilizing a memory controller (not shown), or any other desired mechanism. In one embodiment, such system 106 may include a host system in the form of a desktop computer, lap-top computer, server, workstation, a personal digital assistant (PDA) device, a mobile phone device, a television, a peripheral device (e.g. printer, etc.). Of course, such examples are set forth for illustrative purposes only, as any system meeting the above definition may be employed in the context of the present framework 100.

[0022] Turning now to the interface circuit 102, such interface circuit 102 may include any circuit capable of indirectly or directly communicating with the memory circuits 104A, 104B, 104N and the system 106. In various optional embodiments, the interface circuit 102 may include one or more interface circuits, a buffer chip, etc. Embodiments involving such a buffer chip will be set forth hereinafter during reference to subsequent figures. In still other embodiments, the interface circuit 102 may or may not be manufactured in monolithic form.

[0023] While the memory circuits 104A, 104B, 104N, interface circuit 102, and system 106 are shown to be separate parts, it is contemplated that any of such parts (or portions thereof) may or may not be integrated in any desired manner. In various embodiments, such optional integration may involve simply packaging such parts together (e.g. stacking the parts, etc.) and/or integrating them monolithically. Just by way of example, in various optional embodiments, one or more portions (or all, for that matter) of the interface circuit 102 may or may not be packaged with one or more of the memory circuits 104A, 104B, 104N (or all, for that matter). Different optional embodiments which may be implemented in accordance with the present multiple memory circuit framework 100 will be set forth hereinafter during reference to FIGS. 2A-2E, and 3 et al.

[0024] In use, the interface circuit 102 may be capable of various functionality, in the context of different embodiments. More illustrative information will now be set forth regarding such optional functionality which may or may not be implemented in the context of such interface circuit 102, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. For example, any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0025] For instance, in one optional embodiment, the interface circuit 102 interfaces a plurality of signals 108 that are communicated between the memory circuits 104A, 104B, 104N and the system 106. As shown, such signals may, for example, include address/control/clock signals, etc. In one aspect of the present embodiment, the interfaced signals 108 may represent all of the signals that are communicated between the memory circuits 104A, 104B, 104N and the system 106. In other aspects, at least a portion of signals 110 may travel directly between the memory circuits

104A, 104B, 104N and the system 106 or component thereof [e.g. register, advanced memory buffer (AMB), memory controller, or any other component thereof, where the term component is defined hereinbelow]. In various embodiments, the number of the signals 108 (vs. a number of the signals 110, etc.) may vary such that the signals 108 are a majority or more ($L > M$), etc.

[0026] In yet another embodiment, the interface circuit 102 may be operable to interface a first number of memory circuits 104A, 104B, 104N and the system 106 for simulating at least one memory circuit of a second number. In the context of the present description, the simulation may refer to any simulating, emulating, disguising, transforming, converting, and/or the like that results in at least one aspect (e.g. a number in this embodiment, etc.) of the memory circuits 104A, 104B, 104N appearing different to the system 106. In different embodiments, the simulation may be electrical in nature, logical in nature, protocol in nature, and/or performed in any other desired manner. For instance, in the context of electrical simulation, a number of pins, wires, signals, etc. may be simulated, while, in the context of logical simulation, a particular function may be simulated. In the context of protocol, a particular protocol (e.g. DDR3, etc.) may be simulated.

[0027] In still additional aspects of the present embodiment, the second number may be more or less than the first number. Still yet, in the latter case, the second number may be one, such that a single memory circuit is simulated. Different optional embodiments which may employ various aspects of the present embodiment will be set forth hereinafter during reference to FIGS. 2A-2E, and 3 et al.

[0028] In still yet another embodiment, the interface circuit 102 may be operable to interface the memory circuits 104A, 104B, 104N and the system 106 for simulating at least one memory circuit with at least one aspect that is different from at least one aspect of at least one of the plurality of the memory circuits 104A, 104B, 104N. In accordance with various aspects of such embodiment, such aspect may include a signal, a capacity, a timing, a logical interface, etc. Of course, such examples of aspects are set forth for illustrative purposes only and thus should not be construed as limiting, since any aspect associated with one or more of the memory circuits 104A, 104B, 104N may be simulated differently in the foregoing manner.

[0029] In the case of the signal, such signal may refer to a control signal (e.g. an address signal; a signal associated with an activate operation, precharge operation, write operation, read operation, a mode register write operation, a mode register read operation, a refresh operation; etc.), a data signal, a logical or physical signal, or any other signal for that matter. For instance, a number of the aforementioned signals may be simulated to appear as fewer or more signals, or even simulated to correspond to a different type. In still other embodiments, multiple signals may be combined to simulated another signal. Even still, a length of time in which a signal is asserted may be simulated to be different.

[0030] In the case of protocol, such may, in one exemplary embodiment, refer to a particular standard protocol. For example, a number of memory circuits 104A, 104B, 104N that obey a standard protocol (e.g. DDR2, etc.) may be used to simulate one or more memory circuits that obey a different protocol (e.g. DDR3, etc.). Also, a number of memory circuits 104A, 104B, 104N that obey a version of protocol (e.g. DDR2 with 3-3-3 latency timing, etc.) may be

used to simulate one or more memory circuits that obey a different version of the same protocol (e.g. DDR2 with 5-5-5 latency timing, etc.).

[0031] In the case of capacity, such may refer to a memory capacity (which may or may not be a function of a number of the memory circuits **104A**, **104B**, **104N**; see previous embodiment). For example, the interface circuit **102** may be operable for simulating at least one memory circuit with a first memory capacity that is greater than (or less than) a second memory capacity of at least one of the memory circuits **104A**, **104B**, **104N**.

[0032] In the case where the aspect is timing-related, the timing may possibly relate to a latency (e.g. time delay, etc.). In one aspect of the present embodiment, such latency may include a column address strobe (CAS) latency, which refers to a latency associated with accessing a column of data. Still yet, the latency may include a row address to column address latency (tRCD), which refers to a latency required between the row address strobe (RAS) and CAS. Even still, the latency may include a row precharge latency (tRP), which refers a latency required to terminate access to an open row, and open access to a next row. Further, the latency may include an activate to precharge latency (tRAS), which refers to a latency required to access a certain row of data between an activate operation and a precharge operation. In any case, the interface circuit **102** may be operable for simulating at least one memory circuit with a first latency that is longer (or shorter) than a second latency of at least one of the memory circuits **104A**, **104B**, **104N**. Different optional embodiments which employ various features of the present embodiment will be set forth hereinafter during reference to FIGS. **2A-2E**, and **3** et al.

[0033] In still another embodiment, a component may be operable to receive a signal from the system **106** and communicate the signal to at least one of the memory circuits **104A**, **104B**, **104N** after a delay. Again, the signal may refer to a control signal (e.g. an address signal; a signal associated with an activate operation, precharge operation, write operation, read operation; etc.), a data signal, a logical or physical signal, or any other signal for that matter. In various embodiments, such delay may be fixed or variable (e.g. a function of the current signal, the previous signal, etc.). In still other embodiments, the component may be operable to receive a signal from at least one of the memory circuits **104A**, **104B**, **104N** and communicate the signal to the system **106** after a delay.

[0034] As an option, the delay may include a cumulative delay associated with any one or more of the aforementioned signals. Even still, the delay may result in a time shift of the signal forward and/or back in time (with respect to other signals). Of course, such forward and backward time shift may or may not be equal in magnitude. In one embodiment, this time shifting may be accomplished by utilizing a plurality of delay functions which each apply a different delay to a different signal. In still additional embodiments, the aforementioned shifting may be coordinated among multiple signals such that different signals are subject to shifts with different relative directions/magnitudes, in an organized fashion.

[0035] Further, it should be noted that the aforementioned component may, but need not necessarily take the form of the interface circuit **102** of FIG. **1**. For example, the component may include a register, an AMB, a component positioned on at least one DIMM, a memory controller, etc.

Such register may, in various embodiments, include a Joint Electron Device Engineering Council (JEDEC) register, a JEDEC register including one or more functions set forth herein, a register with forwarding, storing, and/or buffering capabilities, etc. Different optional embodiments which employ various features of the present embodiment will be set forth hereinafter during reference to FIGS. **4-7**, and **9A-B** et al.

[0036] In a power-saving embodiment, at least one of a plurality of memory circuits **104A**, **104B**, **104N** may be identified that is not currently being accessed by the system **106**. In one embodiment, such identification may involve determining whether a page [i.e. any portion of any memory (s), etc.] is being accessed in at least one of the plurality of memory circuits **104A**, **104B**, **104N**. Of course, any other technique may be used that results in the identification of at least one of the memory circuits **104A**, **104B**, **104N** that is not being accessed.

[0037] In response to the identification of the at least one memory circuit **104A**, **104B**, **104N**, a power saving operation is initiated in association with the at least one memory circuit **104A**, **104B**, **104N**. In one optional embodiment, such power saving operation may involve a power down operation and, in particular, a precharge power down operation. Of course, however, it should be noted that any operation that results in at least some power savings may be employed in the context of the present embodiment.

[0038] Similar to one or more of the previous embodiments, the present functionality or a portion thereof may be carried out utilizing any desired component. For example, such component may, but need not necessarily take the form of the interface circuit **102** of FIG. **1**. In other embodiments, the component may include a register, an AMB, a component positioned on at least one DIMM, a memory controller, etc. One optional embodiment which employs various features of the present embodiment will be set forth hereinafter during reference to FIG. **10**.

[0039] In still yet another embodiment, a plurality of the aforementioned components may serve, in combination, to interface the memory circuits **104A**, **104B**, **104N** and the system **106**. In various embodiments, two, three, four, or more components may accomplish this. Also, the different components may be relatively configured in any desired manner. For example, the components may be configured in parallel, serially, or a combination thereof. In addition, any number of the components may be allocated to any number of the memory circuits **104A**, **104B**, **104N**.

[0040] Further, in the present embodiment, each of the plurality of components may be the same or different. Still yet, the components may share the same or similar interface tasks and/or perform different interface tasks. Such interface tasks may include, but are not limited to simulating one or more aspects of a memory circuit, performing a power savings/refresh operation, carrying out any one or more of the various functionalities set forth herein, and/or any other task relevant to the aforementioned interfacing. One optional embodiment which employs various features of the present embodiment will be set forth hereinafter during reference to FIG. **3**.

[0041] Additional illustrative information will now be set forth regarding various optional embodiments in which the foregoing techniques may or may not be implemented, per the desires of the user. For example, an embodiment is set forth for storing at least a portion of information received in

association with a first operation for use in performing a second operation. See FIG. 2F. Further, a technique is provided for refreshing a plurality of memory circuits, in accordance with still yet another embodiment. See FIG. 11.

[0042] It should again be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0043] FIGS. 2A-2E show various configurations of a buffered stack of DRAM circuits 206A-D with a buffer chip 202, in accordance with various embodiments. As an option, the various configurations to be described in the following embodiments may be implemented in the context of the architecture and/or environment of FIG. 1. Of course, however, they may also be carried out in any other desired environment (e.g. using other memory types, etc.). It should also be noted that the aforementioned definitions may apply during the present description.

[0044] As shown in each of such figures, the buffer chip 202 is placed electrically between an electronic host system 204 and a stack of DRAM circuits 206A-D. In the context of the present description, a stack may refer to any collection of memory circuits. Further, the buffer chip 202 may include any device capable of buffering a stack of circuits (e.g. DRAM circuits 206A-D, etc.). Specifically, the buffer chip 202 may be capable of buffering the stack of DRAM circuits 206A-D to electrically and/or logically resemble at least one larger capacity DRAM circuit to the host system 204. In this way, the stack of DRAM circuits 206A-D may appear as a smaller quantity of larger capacity DRAM circuits to the host system 204.

[0045] For example, the stack of DRAM circuits 206A-D may include eight 512 Mb DRAM circuits. Thus, the buffer chip 202 may buffer the stack of eight 512 Mb DRAM circuits to resemble a single 4 Gb DRAM circuit to a memory controller (not shown) of the associated host system 204. In another example, the buffer chip 202 may buffer the stack of eight 512 Mb DRAM circuits to resemble two 2 Gb DRAM circuits to a memory controller of an associated host system 204.

[0046] Further, the stack of DRAM circuits 206A-D may include any number of DRAM circuits. Just by way of example, a buffer chip 202 may be connected to 2, 4, 8 or more DRAM circuits 206A-D. Also, the DRAM circuits 206A-D may be arranged in a single stack, as shown in FIGS. 2A-2D.

[0047] The DRAM circuits 206A-D may be arranged on a single side of the buffer chip 202, as shown in FIGS. 2A-2D. Of course, however, the DRAM circuits 206A-D may be located on both sides of the buffer chip 202 shown in FIG. 2E. Thus, for example, a buffer chip 202 may be connected to 16 DRAM circuits with 8 DRAM circuits on either side of the buffer chip 202, where the 8 DRAM circuits on each side of the buffer chip 202 are arranged in two stacks of four DRAM circuits.

[0048] The buffer chip 202 may optionally be a part of the stack of DRAM circuits 206A-D. Of course, however, the buffer chip 202 may also be separate from the stack of DRAM circuits 206A-D. In addition, the buffer chip 202 may be physically located anywhere in the stack of DRAM circuits 206A-D, where such buffer chip 202 electrically sits between the electronic host system 204 and the stack of DRAM circuits 206A-D.

[0049] In one embodiment, a memory bus (not shown) may connect to the buffer chip 202, and the buffer chip 202 may connect to each of the DRAM circuits 206A-D in the stack. As shown in FIGS. 2A-D, the buffer chip 202 may be located at the bottom of the stack of DRAM circuits 206A-D (e.g. the bottom-most device in the stack). As another option, and as shown in FIG. 2E, the buffer chip 202 may be located in the middle of the stack of DRAM circuits 206A-D. As still yet another option, the buffer chip 202 may be located at the top of the stack of DRAM circuits 206A-D (e.g. the top-most device in the stack). Of course, however, the buffer chip 202 may be located anywhere between the two extremities of the stack of DRAM circuits 206A-D.

[0050] The electrical connections between the buffer chip 202 and the stack of DRAM circuits 206A-D may be configured in any desired manner. In one optional embodiment: address, control (e.g. command, etc.), and clock signals may be common to all DRAM circuits 206A-D in the stack (e.g. using one common bus). As another option, there may be multiple address, control and clock busses. As yet another option, there may be individual address, control and clock busses to each DRAM circuit 206A-D. Similarly, data signals may be wired as one common bus, several busses or as an individual bus to each DRAM circuit 206A-D. Of course, it should be noted that any combinations of such configurations may also be utilized.

[0051] For example, as shown in FIG. 2A, the stack of DRAM circuits 206A-D may have one common address, control and clock bus 208 with individual data busses 210. In another example, as shown in FIG. 2B, the stack of DRAM circuits 206A-D may have two address, control and clock busses 208 along with two data busses 210. In still yet another example, as shown in FIG. 2C, the stack of DRAM circuits 206A-D may have one address, control and clock bus 208 together with two data busses 210. In addition, as shown in FIG. 2D, the stack of DRAM circuits 206A-D may have one common address, control and clock bus 208 and one common data bus 210. It should be noted that any other permutations and combinations of such address, control, clock and data busses may be utilized.

[0052] These configurations may therefore allow for the host system 204 to only be in contact with a load of the buffer chip 202 on the memory bus. In this way, any electrical loading problems (e.g. bad signal integrity, improper signal timing, etc.) associated with the stacked DRAM circuits 206A-D may (but not necessarily) be prevented, in the context of various optional embodiments.

[0053] FIG. 2F illustrates a method 280 for storing at least a portion of information received in association with a first operation for use in performing a second operation, in accordance with still yet another embodiment. As an option, the method 280 may be implemented in the context of the architecture and/or environment of any one or more of FIGS. 1-2E. For example, the method 280 may be carried out by the interface circuit 102 of FIG. 1. Of course, however, the method 280 may be carried out in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0054] In operation 282, first information is received in association with a first operation to be performed on at least one of a plurality of memory circuits (e.g. see the memory circuits 104A, 104B, 104N of FIG. 1, etc.). In various embodiments, such first information may or may not be received coincidentally with the first operation, as long as it is

associated in some capacity. Further, the first operation may, in one embodiment, include a row operation. In such embodiment, the first information may include address information (e.g. a set of address bits, etc.).

[0055] For reasons that will soon become apparent, at least a portion of the first information is stored. Note operation 284. Still yet, in operation 286, second information is received in association with a second operation. Similar to the first information, the second information may or may not be received coincidentally with the second operation, and may include address information. Such second operation, however, may, in one embodiment, include a column operation.

[0056] To this end, the second operation may be performed utilizing the stored portion of the first information in addition to the second information. See operation 288. More illustrative information will now be set forth regarding various optional features with which the foregoing method 280 may or may not be implemented, per the desires of the user. Specifically, an example will be set forth illustrating the manner in which the method 280 may be employed for accommodating a buffer chip that is simulating at least one aspect of a plurality of memory circuits.

[0057] In particular, the present example of the method 280 of FIG. 2F will be set forth in the context of the various components (e.g. buffer chip 202, etc.) shown in the embodiments of FIGS. 2A-2E. It should be noted that, since the buffered stack of DRAM circuits 206A-D may appear to the memory controller of the host system 204 as one or more larger capacity DRAM circuits, the buffer chip 202 may receive more address bits from the memory controller than are required by the DRAM circuits 206A-D in the stack. These extra address bits may be decoded by the buffer chip 202 to individually select the DRAM circuits 206A-D in the stack, utilizing separate chip select signals to each of the DRAM circuits 206A-D in the stack.

[0058] For example, a stack of four $\times 4$ 1 Gb DRAM circuits 206A-D behind a buffer chip 202 may appear as a single $\times 4$ 4 Gb DRAM circuit to the memory controller. Thus, the memory controller may provide sixteen row address bits and three bank address bits during a row (e.g. activate) operation, and provide eleven column address bits and three bank address bits during a column (e.g. read or write) operation. However, the individual DRAM circuits 206A-D in the stack may require only fourteen row address bits and three bank address bits for a row operation, and eleven column address bits and three bank address bits during a column operation.

[0059] As a result, during a row operation in the above example, the buffer chip 202 may receive two address bits more than are needed by each DRAM circuit 206A-D in the stack. The buffer chip 202 may therefore use the two extra address bits from the memory controller to select one of the four DRAM circuits 206A-D in the stack. In addition, the buffer chip 202 may receive the same number of address bits from the memory controller during a column operation as are needed by each DRAM circuit 206A-D in the stack.

[0060] Thus, in order to select the correct DRAM circuit 206A-D in the stack during a column operation, the buffer chip 202 may be designed to store the two extra address bits provided during a row operation and use the two stored address bits to select the correct DRAM circuit 206A-D during the column operation. The mapping between a system address (e.g. address from the memory controller, including the chip select signal(s)) and a device address (e.g.

the address, including the chip select signals, presented to the DRAM circuits 206A-D in the stack) may be performed by the buffer chip 202 in various manners.

[0061] In one embodiment, a lower order system row address and bank address bits may be mapped directly to the device row address and bank address inputs. In addition, the most significant row address bit(s) and, optionally, the most significant bank address bit(s), may be decoded to generate the chip select signals for the DRAM circuits 206A-D in the stack during a row operation. The address bits used to generate the chip select signals during the row operation may also be stored in an internal lookup table by the buffer chip 22 for one or more clock cycles. During a column operation, the system column address and bank address bits may be mapped directly to the device column address and bank address inputs, while the stored address bits may be decoded to generate the chip select signals.

[0062] For example, addresses may be mapped between four 512 Mb DRAM circuits 206A-D that simulate a single 2 Gb DRAM circuits utilizing the buffer chip 202. There may be 15 row address bits from the system 204, such that row address bits 0 through 13 are mapped directly to the DRAM circuits 206A-D. There may also be 3 bank address bits from the system 204, such that bank address bits 0 through 1 are mapped directly to the DRAM circuits 206A-D.

[0063] During a row operation, the bank address bit 2 and the row address bit 14 may be decoded to generate the 4 chip select signals for each of the four DRAM circuits 206A-D. Row address bit 14 may be stored during the row operation using the bank address as the index. In addition, during the column operation, the stored row address bit 14 may again be used with bank address bit 2 to form the four DRAM chip select signals.

[0064] As another example, addresses may be mapped between four 1 Gb DRAM circuits 206A-D that simulate a single 4 Gb DRAM circuits utilizing the buffer chip 202. There may be 16 row address bits from the system 204, such that row address bits 0 through 14 are mapped directly to the DRAM circuits 206A-D. There may also be 3 bank address bits from the system 204, such that bank address bits 0 through 3 are mapped directly to the DRAM circuits 206A-D.

[0065] During a row operation, row address bits 14 and 15 may be decoded to generate the 4 chip select signals for each of the four DRAM circuits 206A-D. Row address bits 14 and 15 may also be stored during the row operation using the bank address as the index. During the column operation, the stored row address bits 14 and 15 may again be used to form the four DRAM chip select signals.

[0066] In various embodiments, this mapping technique may optionally be used to ensure that there are no unnecessary combinational logic circuits in the critical timing path between the address input pins and address output pins of the buffer chip 202. Such combinational logic circuits may instead be used to generate the individual chip select signals. This may therefore allow the capacitive loading on the address outputs of the buffer chip 202 to be much higher than the loading on the individual chip select signal outputs of the buffer chip 202.

[0067] In another embodiment, the address mapping may be performed by the buffer chip 202 using some of the bank address signals from the memory controller to generate the individual chip select signals. The buffer chip 202 may store

the higher order row address bits during a row operation using the bank address as the index, and then may use the stored address bits as part of the DRAM circuit bank address during a column operation. This address mapping technique may require an optional lookup table to be positioned in the critical timing path between the address inputs from the memory controller and the address outputs, to the DRAM circuits 206A-D in the stack.

[0068] For example, addresses may be mapped between four 512 Mb DRAM circuits 206A-D that simulate a single 2 Gb DRAM utilizing the buffer chip 202. There may be 15 row address bits from the system 204, where row address bits 0 through 13 are mapped directly to the DRAM circuits 206A-D. There may also be 3 bank address bits from the system 204, such that bank address bit 0 is used as a DRAM circuit bank address bit for the DRAM circuits 206A-D.

[0069] In addition, row address bit 14 may be used as an additional DRAM circuit bank address bit. During a row operation, the bank address bits 1 and 2 from the system may be decoded to generate the 4 chip select signals for each of the four DRAM circuits 206A-D. Further, row address bit 14 may be stored during the row operation. During the column operation, the stored row address bit 14 may again be used along with the bank address bit 0 from the system to form the DRAM circuit bank address.

[0070] In both of the above described address mapping techniques, the column address from the memory controller may be mapped directly as the column address to the DRAM circuits 206A-D in the stack. Specifically, this direct mapping may be performed since each of the DRAM circuits 206A-D in the stack, even if of the same width but different capacities (e.g. from 512 Mb to 4 Gb), may have the same page sizes. In an optional embodiment, address A[10] may be used by the memory controller to enable or disable auto-precharge during a column operation. Therefore, the buffer chip 202 may forward A[10] from the memory controller to the DRAM circuits 206A-D in the stack without any modifications during a column operation.

[0071] In various embodiments, it may be desirable to determine whether the simulated DRAM circuit behaves according to a desired DRAM standard or other design specification. A behavior of many DRAM circuits is specified by the JEDEC standards and it may be desirable, in some embodiments, to exactly simulate a particular JEDEC standard DRAM. The JEDEC standard defines control signals that a DRAM circuit must accept and the behavior of the DRAM circuit as a result of such control signals. For example, the JEDEC specification for a DDR2 DRAM is known as JESD79-2B.

[0072] If it is desired, for example, to determine whether a JEDEC standard is met, the following algorithm may be used. Such algorithm checks, using a set of software verification tools for formal verification of logic, that protocol behavior of the simulated DRAM circuit is the same as a desired standard or other design specification. This formal verification is quite feasible because the DRAM protocol described in a DRAM standard is typically limited to a few control signals (e.g. approximately 15 control signals in the case of the JEDEC DDR2 specification, for example).

[0073] Examples of the aforementioned software verification tools include MAGELLAN supplied by SYNOPSYS, or other software verification tools, such as INCISIVE supplied by CADENCE, verification tools supplied by JASPER, VERIX supplied by REAL INTENT, 0-IN supplied by

MENTOR CORPORATION, and others. These software verification tools use written assertions that correspond to the rules established by the DRAM protocol and specification. These written assertions are further included in the code that forms the logic description for the buffer chip. By writing assertions that correspond to the desired behavior of the simulated DRAM circuit, a proof may be constructed that determines whether the desired design requirements are met. In this way, one may test various embodiments for compliance with a standard, multiple standards, or other design specification.

[0074] For instance, an assertion may be written that no two DRAM control signals are allowed to be issued to an address, control and clock bus at the same time. Although one may know which of the various buffer chip/DRAM stack configurations and address mappings that have been described herein are suitable, the aforementioned algorithm may allow a designer to prove that the simulated DRAM circuit exactly meets the required standard or other design specification. If, for example, an address mapping that uses a common bus for data and a common bus for address results in a control and clock bus that does not meet a required specification, alternative designs for buffer chips with other bus arrangements or alternative designs for the interconnect between the buffer chips may be used and tested for compliance with the desired standard or other design specification.

[0075] FIG. 3 shows a high capacity DIMM 300 using buffered stacks of DRAM circuits 302, in accordance with still yet another embodiment. As an option, the high capacity DIMM 300 may be implemented in the context of the architecture and environment of FIGS. 1 and/or 2A-F. Of course, however, the high capacity DIMM 300 may be used in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0076] As shown, a high capacity DIMM 300 may be created utilizing buffered stacks of DRAM circuits 302. Thus, a DIMM 300 may utilize a plurality of buffered stacks of DRAM circuits 302 instead of individual DRAM circuits, thus increasing the capacity of the DIMM. In addition, the DIMM 300 may include a register 304 for address and operation control of each of the buffered stacks of DRAM circuits 302. It should be noted that any desired number of buffered stacks of DRAM circuits 302 may be utilized in conjunction with the DIMM 300. Therefore, the configuration of the DIMM 300, as shown, should not be construed as limiting in any way.

[0077] In an additional unillustrated embodiment, the register 304 may be substituted with an AMB (not shown), in the context of an FB-DIMM.

[0078] FIG. 4 shows a timing design 400 of a buffer chip that makes a buffered stack of DRAM circuits mimic longer CAS latency DRAM to a memory controller, in accordance with another embodiment. As an option, the design of the buffer chip may be implemented in the context of the architecture and environment of FIGS. 1-3. Of course, however, the design of the buffer chip may be used in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0079] In use, any delay through a buffer chip (e.g. see the buffer chip 202 of FIGS. 2A-E, etc.) may be made transparent to a memory controller of a host system (e.g. see the

host system 204 of FIGS. 2A-E, etc.) utilizing the buffer chip. In particular, the buffer chip may buffer a stack of DRAM circuits such that the buffered stack of DRAM circuits appears as at least one larger capacity DRAM circuit with higher CAS latency.

[0080] Such delay may be a result of the buffer chip being located electrically between the memory bus of the host system and the stacked DRAM circuits, since most or all of the signals that connect the memory bus to the DRAM circuits pass through the buffer chip. A finite amount of time may therefore be needed for these signals to traverse through the buffer chip. With the exception of register chips and advanced memory buffers (AMB), industry standard protocols for memory [e.g. (DDR SDRAM), DDR2 SDRAM, etc.] may not comprehend the buffer chip that sits between the memory bus and the DRAM. Industry standard protocols for memory [e.g. (DDR SDRAM), DDR2 SDRAM, etc.] narrowly define the properties of chips that sit between host and memory circuits. Such industry standard protocols define the properties of a register chip and AMB but not the properties of the buffer chip 202, etc. Thus, the signal delay through the buffer chip may violate the specifications of industry standard protocols.

[0081] In one embodiment, the buffer chip may provide a one-half clock cycle delay between the buffer chip receiving address and control signals from the memory controller (or optionally from a register chip, an AMB, etc.) and the address and control signals being valid at the inputs of the stacked DRAM circuits. Similarly, the data signals may also have a one-half clock cycle delay in traversing the buffer chip, either from the memory controller to the DRAM circuits or from the DRAM circuits to the memory controller. Of course, the one-half clock cycle delay set forth above is set forth for illustrative purposes only and thus should not be construed as limiting in any manner whatsoever. For example, other embodiments are contemplated where a one clock cycle delay, a multiple clock cycle delay (or fraction thereof), and/or any other delay amount is incorporated, for that matter. As mentioned earlier, in other embodiments, the aforementioned delay may be coordinated among multiple signals such that different signals are subject to time-shifting with different relative directions/magnitudes, in an organized fashion.

[0082] As shown in FIG. 4, the cumulative delay through the buffer chip (e.g. the sum of a first delay 402 of the address and control signals through the buffer chip and a second delay 404 of the data signals through the buffer chip) is j clock cycles. Thus, the buffer chip may make the buffered stack appear to the memory controller as one or more larger DRAM circuits with a CAS latency 408 of $i+j$ clocks, where i is the native CAS latency of the DRAM circuits.

[0083] In one example, if the DRAM circuits in the stack have a native CAS latency of 4 and the address and control signals along with the data signals experience a one-half clock cycle delay through the buffer chip, then the buffer chip may make the buffered stack appear to the memory controller as one or more larger DRAM circuits with a CAS latency of 5 (i.e. $4+1$). In another example, if the address and control signals along with the data signals experience a 1 clock cycle delay through the buffer chip, then the buffer chip may make the buffered stack appear as one or more larger DRAM circuits with a CAS latency of 6 (i.e. $4+2$).

[0084] FIG. 5 shows the write data timing 500 expected by a DRAM circuit in a buffered stack, in accordance with yet another embodiment. As an option, the write data timing 500 may be implemented in the context of the architecture and environment of FIGS. 1-4. Of course, however, the write data timing 500 may be carried out in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0085] Designing a buffer chip (e.g. see the buffer chip 202 of FIGS. 2A-E, etc.) so that a buffered stack appears as at least one larger capacity DRAM circuit with higher CAS latency may, in some embodiments, create a problem with the timing of write operations. For example, with respect to a buffered stack of DDR2 SDRAM circuits with a CAS latency of 4 that appear as a single larger DDR2 SDRAM with a CAS latency of 6 to the memory controller, the DDR2 SDRAM protocol may specify that the write CAS latency is one less than the read CAS latency. Therefore, since the buffered stack appears as a DDR2 SDRAM with a read CAS latency of 6, the memory controller may use a write CAS latency of 5 (see 502) when scheduling a write operation to the buffered stack.

[0086] However, since the native read CAS latency of the DRAM circuits is 4, the DRAM circuits may require a write CAS latency of 3 (see 504). As a result, the write data from the memory controller may arrive at the buffer chip later than when the DRAM circuits require the data. Thus, the buffer chip may delay such write operations to alleviate any of such timing problems. Such delay in write operations will be described in more detail with respect to FIG. 6 below.

[0087] FIG. 6 shows write operations 600 delayed by a buffer chip, in accordance with still yet another embodiment. As an option, the write operations 600 may be implemented in the context of the architecture and environment of FIGS. 1-5. Of course, however, the write operations 600 may be used in any desired environment. Again, it should also be noted that the aforementioned definitions may apply during the present description.

[0088] In order to be compliant with the protocol utilized by the DRAM circuits in the stack, a buffer chip (e.g. see the buffer chip 202 of FIGS. 2A-E, etc.) may provide an additional delay, over and beyond the delay of the address and control signals through the buffer chip, between receiving the write operation and address from the memory controller (and/or optionally from a register and/or AMB, etc.), and sending it to the DRAM circuits in the stack. The additional delay may be equal to j clocks, where j is the cumulative delay of the address and control signals through the buffer chip and the delay of the data signals through the buffer chip. As another option, the write address and operation may be delayed by a register chip on a DIMM, by a AMB, or by the memory controller.

[0089] FIG. 7 shows early write data 700 from an AMB, in accordance with another embodiment. As an option, the early write data 700 may be implemented in the context of the architecture and environment of FIGS. 1-5. Of course, however, the early write data 700 may be used in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0090] As shown, an AMB on an FB-DIMM may be designed to send write data earlier to buffered stacks instead of delaying the write address and operation, as described in reference to FIG. 6. Specifically, an early write latency 702 may be utilized to send the write data to the buffered stack.